Topological Fixed-Point Theory (TFPT)

A discrete compiler for the constants of physics

Two inputs — the boundary constant $c_3 = \frac{1}{8\pi}$ and a five-slot carrier — build E_8 and read off the Standard Model: status assessment and reading guide

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July 29, 2026 · v5.4 (rev 474)

In one sentence

Topological Fixed-Point Theory (TFPT) constructs, from exactly **two inputs** — a boundary constant $c_3 = \frac{1}{8\pi}$ (P1) and a five-slot carrier $g_{\text{car}} = 5$ (P2) — a discrete compiler for the Standard-Model skeleton: the gauge group, three families, hypercharges and the flavor matrix, with E_8 ($D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$) as the consistency checksum. The *algebraic core is machine-checkable* and the dimensionless constants follow as fixed points ($\alpha^{-1} = 137.0359992$); *physical readouts* — absolute scales, masses, inflation, gravity and cosmological transfers — run through explicitly *named, status-typed bridges* (v_{geo}, G_{net}, F_{transfer}), not as free outputs. The honest one-line claim is on the status card below.

The TFPT document set — what is treated where

Plain language: TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant $c_3 = \frac{1}{8\pi}$ and the carrier rank $g_{\text{car}} = 5$ — build $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and five companions**, best read in order:

0. `introduction` — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). *(entry point)*
1. `tfpt_1_architecture_e8` — the two axioms, the derivation map, the EM fixed point α^{-1}, the $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ construction, the QBL theorem chain.
2. `tfpt_2_standard_model` — the SM in one φ_0-ladder formula, flavor from parabolic transport (sheet diamond, dual anchor, Q_+ cohomology), the worked closures.
3. `tfpt_3_e8_audit_bootstrap` — the seven E_8 slices as an audit raster, the cascade bridge, the Möbius bootstrap.
4. `tfpt_4_frontier` — honest status of η_B, m_p/m_e, Koide, dark matter, full QG.
5. `tfpt_5_redteam` — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

Companions: R. `tfpt_research_contracts` — the open interfaces ($v_{\text{geo}}, G_{\text{net}}, F_{\text{transfer}}$) as numbered contracts; **H.** `tfpt_horizon_readouts` — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O.** `origin_theory` — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation); **S.** `tfpt_safeguards` — the verification discipline: every mechanism that defends a claim against coincidence and numerology (status calculus, no-free-pattern, the over-determination map, the firewall, frozen predictions + null model, the independent Wolfram and Lean paths, the red team); **P.** `tfpt_prime_front` — research documentation of the prime / zeta line: the seven load-bearing modules on the E_8 glue census (Hecke from geometry through the transport ledger), the sandbox localization program that follows them, and the kill list. Documentation, not a result: it makes no claim about the Riemann Hypothesis.

You are reading the introduction — the entry point and reading guide.

TFPT status at a glance — read this card the same way in every document

Two inputs, one compiler. From the boundary constant $c_3 = \frac{1}{8\pi}$ (axiom P1) and the five-slot carrier $g_{\mathrm{car}} = 5$ (axiom P2) — jointly the elementary symmetric data of one anchor $a = (1, 1, 2)$ — the discrete compiler $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ($\alpha^{-1} = 137.0359992$) and the scale grammar are read off as fixed points. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below.

What is closed, and what genuinely remains. The discrete/algebraic layer is closed ([E]); the everything-open residual reduces to *three named interface problems*:

Interface	Question	Status
v_{geo}	the one metrology unit ($= 1/\sqrt{G} = m/\mu$); No-Unit Thm: no compiler scale	primitive [O]
G_{net}	simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle(1,1)\rangle \cong (E_8)_1$	[C]
F_{transfer}	one functor, four typed interfaces (Koide, η_B, axion, m_p/m_e)	[C]

Gravity is parameter-free. The classical metric-sector field equation is no longer only an $R+R^2$ readout: the fixed-volume entanglement equilibrium (Jacobson; Faulkner et al.), run with TFPT's atoms, gives the *full* covariant Einstein equation $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ with *both* coefficients fixed — $c_3^{-1} = 8\pi$ (no free dimensionless Newton dial; the thermodynamic origin $2\pi/\eta$ *coincides* with the geometric one $|\mathbb{Z}_2| 2\pi\chi$ via $|\mu_4| = |\mathbb{Z}_2|\chi(S^2) = 4$, so c_3 is triply over-determined — anchor, geometry, thermodynamics, `v358`) and Λ from α ($\rho_\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$, `v60`); the Einstein tensor (not Ricci) is forced by Lovelock, so matter conservation is an output (`v359`). The residual here is only the equation-of-state status (an external candidate action — Bianconi's entropic action, arXiv:2408.14391 — is quantified in `v473` without changing the typing) and the one unit v_{geo}; the global measure (C7 = `QG.AMB.01`) is now a [C] redundancy (`v369`), a certification object rather than missing dynamics, conditional on `SEAM.EQUIV.01` and Bisognano–Wichmann.

The seam-net structural theorem G_{net} (= `SEAM.EQUIV.01`: the raw seam *is* the holomorphic $(E_8)_1$ net) now carries two named routes (route split 2026-07-22, no status change in substance): the MMST route `SEAM.EQUIV.MMST.01` is *closed modulo cited theorems*; the twistor route `SEAM.EQUIV.TWISTOR.01` (Costello–Li, prepared by `CELEST.SEAM.01`) and the unconditional parent stay [O]. On the MMST route: the explicit lattice model (`v367`/`v368`) and the S_3 closure stack pin the target at every computable level — central charge $c = 8$ (`v376`), the $(E_8)_1$ character with 248 currents and one primary (`v377`), genus-one torus GSD = 1 (`v378`) and reflection positivity (`v379`) — and it is Lean-formalised as `FORM.SEAM.MMST.01`: the collar's MMST hypotheses are kernel-proved, the MMST scaling-limit and Adamo–Moriwaki–Tanimoto OS-reconstruction theorems enter as named cited axioms, and the `#print axioms` check is clean. The *only* residual of that route that stays [O] is the abstract continuum existence of the scaling limit (`v336`) — its 128-spinor extension leg since certified at net level by the peer-reviewed crossed-product package (h_s=16/16=1∈$\mathbb{Z}$, the Longo–Rehren locality criterion; Böckenhauer; KLM μ=4/2^2=1; AGT/AMT the independent second witness) with the realisation input reduced to invariant level and the parallel Lean route `seamResidualClosed′` (`v469`); so the MMST route is *closed modulo cited theorems*, not solved. `QGEO.SYM.01` (the μ_4 deck acting geometrically) is its *corollary*: a conformal net's vacuum is rotation-invariant by axiom, so the deck follows with no extra premise (`v335`, Lean `FORM.QGEO.BW.01`). The nonperturbative quantum-gravity measure `QG.AMB.01` is no longer carried as an open item: it is a [C] *redundancy* (`v369`), a certification object rather than missing dynamics, conditional on `SEAM.EQUIV.01` and Bisognano–Wichmann (and TFPT is moreover rigorously gap-decoupled from the general Euclidean-QG conformal-factor problem, margin 1.648 > 0, `v76`/`v330`). So the honest residual is the one unit v_{geo} plus the typed F_{transfer} interfaces, above the cited-theorem ceiling on the seam and with `QG.AMB.01` a [C] redundancy. Everything carries a claim ID resolving to a row of `verification/status_ledger.csv` (the single source of truth); **if the text and the ledger ever disagree, the ledger wins.** The historical labels ($U_{\mathrm{wall}}, G_{\mathrm{metric}}, F_{\mathrm{frontier}}$) are retained only for ledger continuity — the live residual is $v_{\mathrm{geo}} \oplus F_{\mathrm{transfer}}$, with the seam G_{net} closed modulo cited theorems on its MMST route (parent [O]) and `QG.AMB.01` a [C] redundancy.

*Marker key: **[E]** exact / machine-proven · **[C]** conditional (holds under named hypotheses) · **[O]** open / axiom · **[X]** falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.*

[E] exact / machine-proven **[C]** conditional **[O]** open / axiom **[X]** kill test

P1 $c_3 = \frac{1}{8\pi}$ **P2** $g_{\text{car}} = 5$ *(two inputs, [O] axioms)*

↓

Anchor $a = (1,1,2)$: $e_1, e_2, e_3 = 4, 5, 2$ *([E] exact)*

↓

Discrete compiler: $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$, $h = 30$, φ_0 *([E] exact)*

↓

SM skeleton: three families, hypercharges, flavor matrix R *([E])*

↓

Physical bridges: v_{geo}, G_{net}, F_{transfer} *([C] bridges)*

↓

Frontier readouts: Λ, inflation, η_B, dark matter, QG *([C]/[O])*

Contents

Reading the markers. Claims carry one of *four* reader-facing classes, so one can see at a glance what is proven and what is conditional: **[E]** *exact / machine-proven* — an exact identity (e.g. $240 = 16 \cdot 5 \cdot 3$), a Lie/lattice theorem (e.g. $D_5 \oplus A_3 + \mu_4 = E_8$), a Lean/script formalisation, or a reproduced numerical fixed point; **[C]** *conditional* — follows under explicitly named hypotheses (a physical bridge, readout or pending step); **[O]** *open / axiom* — a declared input or a genuine gap; **[X]** *kill test* — a falsifiable threshold. The finer per-claim type (Axiom / Formal / Lattice /

Numerical / Identity / Physical) is kept in `verification/status_ledger.csv`, the single source of truth.

Level	Example	Claim ID
Physical — under hypotheses	RG masses, QG	`QG.AMB.01`
Numerical — fit-free fixed point	$\alpha^{-1}, \Lambda, \theta_{12}$	`EM.FP.01`
Lattice — Lie / glue / root system	$E_8 = (D_5 \oplus A_3) + \mu_4$	`E8.GLU.01`
Formal — Lean / exact script	P2 algebra	`FORM.P2.02`
Axiom — declared inputs	P1 c_3, P2 g_{car}	`AX.P1.01`

The claim stack (bottom = most certain/fewest, top = conditional). Each claim sits on exactly one level; the bottom two are inputs, the middle two are proved/reproduced, only the top is conditional. The single source of truth for which claim sits where is `verification/status_ledger.csv`.

No-free-pattern rule. To keep the structure from drowning in decorative coincidences, an identity enters the *main text* only if it (1) is a necessary step in a derivation, (2) kills an alternative, (3) is ablation-relevant, (4) links two previously independent modules, or (5) is experimentally tested. Everything else is explicitly demoted to an *audit fingerprint* [E] (clearly boxed as such) — so the load-bearing beams stay visible and the ornament never gets promoted to spine.

Admissible-invariant classes (the harder filter). Sharper still: a *load-bearing* integer may only come from one of seven typed sources — (i) a symmetric function $e_k(a)$ of the anchor $a = (1, 1, 2)$; (ii) a power sum $p_n(a)$ or difference $p_{n+1}-p_n$; (iii) a determinant, spectrum, Smith normal form or trace of (R, Q, K, L); (iv) an anchor quadratic form $u^\top M v$ with $u, v \in \{\mathbf{1}, a\}$; (v) a Plücker coordinate of the anchor plane $\langle \mathbf{1}, a\rangle$; (vi) an E_8 Casimir degree or branching block; (vii) a pre-registered experimental observable. Anything else is an audit fingerprint or a frontier item, never spine. This is what converts TFPT from *pattern hunting* ("look, another number") into a typed compiler: *fewer admissible kinds of explanation*, not more matches.

Reviewer path (read in this order; the rest is support):

1. the architecture: the two axioms $\{c_3, g_{\mathrm{car}}\}$ and the two-engine picture (§1, the figures);
2. the E_8 glue $E_8 = (D_5 \oplus A_3) + \mu_4$ — document 1, Part II, machine-checked (`v1_e8_glue.py`);
3. the α^{-1} fixed point plus ablation (`v3_em_alpha.py`);
4. the **single status ledger** `verification/status_ledger.csv` — the source of truth for every claim's grade, location, dependency and script;
5. the **open gates** as numbered research contracts (`tfpt_research_contracts`: the residual classes R1–R5 and their current reductions) and the honest frontier list (document 4).

Everything carries a claim ID (e.g. `E8.GLU.01`, `EM.FP.01`) that resolves to a row of the ledger. **If the text and the ledger ever disagree, the ledger wins.**

1 What this is about: the jump in construction principle

In plain words

What we found. TFPT reproduces dozens of real numbers of physics (particle masses, the fine-structure constant, dark-energy and inflation values) from a small set of inputs — and the compiler closure says *why* the same handful of small integers $(2, 3, 5, 16, \dots)$ reappears in every sector: those numbers are not separate lucky hits. A single tiny "machine" — fed by just **two inputs** (a boundary number $\frac{1}{8\pi}$ and a five-slot carrier) — builds the exceptional symmetry E_8, and E_8 is the unimodular "audit hull" that classifies the discrete Standard-Model readout: the constants, the flavor skeleton and the gravity/cosmology regime are then read off by projection and boundary dressing (not "printed" for free — where a scheme or analytic interface is needed, it is named).

Why this is simple. There are not many separate derivations that happen to share numbers — there is *one generator and a handful of corollaries*, and the number of independent structural assumptions is **two**.

How plausible it is. A theory becomes more believable when it *explains more from less*. Here the two inputs are even *over-determined*: the machine reproduces the very two numbers it started from (the dashed bootstrap loop in the architecture diagram, §2), so there is *no free dial left to tune*. Pushed to the end ([`verification/v353_selfloop_capstone.py`]), this means TFPT is best read not as a *linear* theory (axioms → theorems) but as a *closed self-consistent loop* whose outputs fix its own inputs ($g_{\rm car}$=5 is the largest prime of the output $h(E_8)$=30; the 8 in c_3 is rank E_8), with a *unique* fixed point (v6/v56) and *zero* free adjustable dimensionless parameters. Even π is not a free dial: it is the fixed geometric constant of the boundary 2-sphere ($\oint_{S^2} K = 4\pi$), the one transcendental the continuous boundary contains. "Which is the axiom?" is the wrong question for a loop — every part is fixed by the whole; the one remaining structural question, whether the loop also closes in the *continuum* (the keystone `SEAM.EQUIV.01`), now carries two named routes (route split 2026-07-22, an ID restructuring with *no* status change in substance): its MMST route `SEAM.EQUIV.MMST.01` is *closed modulo cited theorems* — the lattice model and S_3 stack pin it at every computable level and it is Lean-formalised (`FORM.SEAM.MMST.01`); only the cited continuum-existence theorems stay **[O]** (v367/v368, v376–v379, v336), their extension leg since re-founded on peer-reviewed crossed-product theorems with the realisation input reduced to invariant level (v469) — while its twistor route `SEAM.EQUIV.TWISTOR.01` (the open Costello–Li construction, prepared by `CELEST.SEAM.01`) stays **[O]**; the parent `SEAM.EQUIV.01` is closed if either route closes and, as an unconditional claim, stays **[O]**.

Sharper still: one anchor $a = (1,1,2)$, and E_8 as a compiler hull ([E])

The two axioms are not even independent: they are the *elementary symmetric polynomials* of the single parabolic anchor $a = (1,1,2)$ (the exponents-at-infinity / splitting type $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$):

$$e_1(a) = 4 = |\mu_4|, \qquad e_2(a) = 5 = g_{\rm car}, \qquad e_3(a) = 2 = |\mathbb{Z}_2|, \qquad c_3 = \frac{1}{2e_1(a)\pi} = \frac{1}{8\pi}.$$

Its *power sums* $p_n(a) = 1^n{+}1^n{+}2^n = 2{+}2^n$ generate the big Lie data directly: $p_1{=}4{=}|\mu_4|$, $p_2{=}6{=}|R^+(A_3)|$, $p_3{=}10{=}\mathcal{A}_\Lambda$, and

$$|R(E_8)| = p_1p_2p_3 = 240, \quad \dim E_8 = p_1p_2p_3 + (p_4 - p_3) = 248, \quad p_4 - p_3 = 8 = \operatorname{rank} E_8,$$

with the binary ladder $p_{n+1}{-}p_n = 2^n$. So the inputs collapse from $\{c_3, g_{\rm car}\}$ to $\boxed{a = (1,1,2) \text{ plus } \pi}$. [`verification/v23_anchor_generator.py`]

And a is itself half-forced ([**E**]/[**C**]): on $\mathbb{P}^1$ every bundle splits (Birkhoff–Grothendieck), so once $|\mu_4| = 4$, $N_{\rm fam} = 3$, $\deg E = -4$ are set, the three positive anti-degrees are positive integers summing to 4, and $4 = 2+1+1$ is the *unique* partition into three positive parts — machine-checked with all its negative controls as the partition corollary ([`verification/v491_p2_partition_corollary.py`], `P2.PARTITION.01`): the unique multiset $\{1,1,2\}$ forces $e = (4,5,2)$, hence $g_{\rm car} = e_2 = 5$, $|\mathbb{Z}_2| = e_3 = 2$, $N_{\rm fam} = 3$ as *corollaries* (dropping positivity, the sum, the part count or integrality each destroys uniqueness — Mehta–Seshadri-rational weights give $e_2 = \frac{16}{3}$ or $\frac{44}{9} \neq 5$), sharpening v53 which needs all three targets. The typing itself is now hardened ([`verification/v499_p2_weights_deligne_bg.py`], `P2.TYPING.01`): given the four marks, rank 3, the cusp class and unitarity (U), the Deligne canonical extension of the flavor connection has $\deg E = -4$ (residue traces) and Mehta–Seshadri stability forces $h^0(E) = 0$, so integrality (Birkhoff–Grothendieck), positivity and the sum 4 are *theorems* — the residual shrinks to the module identification and (U), both [**C**]; and that module-identification rest is now *one* residual with the QGEO modulus rest (the order-4 clock = the Coxeter/cusp-class carrier datum; [verification/v503_qgeo_emergence_light.py], `QGEO.EMERGE.LIGHT.01`, no marker moves). That common residual is itself now reduced to *one alignment bit* — the collar deck central in the mark-D_4; the clock's *order* is fermionically forced, $U^2 = (-1)^F$ exactly ([verification/v506_seam_clock_rigidity.py], `SEAM.CLOCK.RIGIDITY.01`, markers unchanged); and the bit is no tautology — the deck's *class* is derived (every mark-free collar deck is a half-period translation of the seam double), only its *position* stays the [C] input ([verification/v507_seam_bit_origin.py], `SEAM.BIT.ORIGIN.01`, markers unchanged); and that position half is now *topology* — the covering deck is free on the seam circle, the edge class is excluded, and the bit reduces to the square-modulus datum $\tau = i$ alone ([verification/v510_seam_bit_freedom.py], `SEAM.BIT.FREEDOM.01`, markers unchanged); and that modulus datum now carries an equivalent *discrete* form — flag transitivity of the four marks (marks *and* their two sides indistinguishable, $V_4 \to D_4$) $\iff \tau = i$, a 13-fold exact equivalence web (since 15-fold by v528 facets #14/#15) in which bare mark-transitivity is automatic for every configuration and no local jet sees the side bit ([verification/v512_seam_tau_flag.py], `SEAM.TAU.FLAG.01`, markers unchanged); free OS positivity is since documented as the *eighth* side-blind test on that scoreboard — the bit is not derivable from free RP/Θ existence and remains genuine discrete input (v521, `SEAM.BIT.RPBLIND.01`, markers unchanged); the mark-decorated *state* class (twist/defect insertions — everything Wick-computable) is since documented as the *ninth*: the free-plus-twist class is exhausted (v525, `SEAM.BIT.TWISTBLIND.01`, markers unchanged); and the first genuinely interacting Fidkowski–Kitaev seam toy is the *tenth* side-blind test ($9 \to 10$), while Kill-Test 2 fires at toy level under a typed fence (v529, `SEAM.INT.FKTOY.01`, markers unchanged); the straddle filter it produced, since run as a *selector* on the interacting mark family, keeps exactly *one* member alive — reflection positivity dynamically selects $\delta = \pi/2$ with positive coupling, the same member that carries the twist-class order parameter: the first *positive* selection datum for the bit (v534, `SEAM.STRADDLE.CONE.01`, markers unchanged; toy-level evidence, not a derivation). The bit itself is now the *twist-class choice*, in principle interferometrically readable (v528, `SEAM.BIT.TWISTCLASS.01`; it remains formal input). Temperature joins geometry and anomaly as the *third leg* of c_3 (v526, `SEAM.THERMAL.KMS.01`, [C]-typed reading). The genuine remaining axiom is therefore not "why $a = (1,1,2)$" but "why the carrier $g_{\text{car}} = 5$ / $(|\mu_4|, N_{\text{fam}}) = (4,3)$" — that interface (with c_3), i.e. the weight-*typing* postulate itself, is the [O] input layer (`AX.P2.01` stays a declared axiom). The consolidated reduction state (2026-07-23): the compiler's dimensionless inputs reduce to *four marks* (derived from P1-side topology), *one discrete symmetry-lift bit* — the twist-class choice (flag transitivity $\iff \tau = i$; equivalently the clock, the CM-fixed half-period, or the non-split extension class on the free seam circle) — and π — a reduction state, not an abolition of the axiom: `AX.P2.01` remains declared, and the [C] identifications R1/R2 of v499 remain conditional.

Two readings, one source (no contradiction). P1 and P2 are the *operational compiler inputs*; their anchor provenance is $a = (1,1,2)$ plus π. The *single structural bedrock postulate* is the keystone `SEAM.EQUIV.01`, whose conformal-deck face `QGEO.SYM.01` (the carrier μ_4 clock is the seam's conformal deck, see `tfpt_research_contracts`) forces the μ_4 deck, hence $N_{\text{fam}} = 3$, $\deg E = -4$ and the tripartition $4 = 2+1+1$, of which c_3 and g_{car} are the two elementary readouts. "Two inputs" (the working axioms) and "one postulate" (the geometric bedrock) are the same statement at two layers. Its geometric normal form and the mark-local$\Rightarrow \omega \circ \rho = \omega$ implication are now Lean-formalised (`FORM.QGEO.02`/`FORM.QGEO.01`); the postulate itself stays [O].

One source, four readings (the cleanest origin). The four-point seam divisor μ_4 on $\mathbb{P}^1$ generates the whole discrete skeleton by four canonical functors:

$$\underbrace{c_3 = \tfrac{1}{8\pi}}_{\text{thermal}} \quad \underbrace{\operatorname{rank} H_1(\mathbb{P}^1 \setminus \mu_4) = 3 = N_{\text{fam}}}_{\text{cycles}\to A_3} \quad \underbrace{h^0(\mathbb{P}^1, \mathcal{O}(\mu_4)) = 5 = g_{\text{car}}}_{\text{functions}\to D_5}$$

$$\underbrace{(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 \cong (E_8)_1}_{\text{net}}$$

and the function side even generates D_5 itself ([verification/v197_rr_carrier_clifford_d5.py]): $\Lambda^{\text{even}}(\mathbb{C}^5) = \binom{5}{0} + \binom{5}{2} + \binom{5}{4} = 16 = \dim S^+$, the $\mathfrak{so}(10) = D_5$ half-spinor. So μ_4 is not merely glue: it is the *common source* of A_3, D_5, c_3 and E_8 ([verification/v189_riemann_roch_carrier.py]) — the arithmetic is exact **[E]**, the physical reading of the carrier as the seam mode space stays **[C]**.

What E_8 is (and is not). In TFPT E_8 is *not* an unbroken physical gauge group: it is the unimodular *audit / compiler hull* that classifies the admissible discrete charge and residue structures. The Standard Model is a *readout* after projection, not a direct E_8 gauge theory; the Lorentz signature appears only after projection; fermions are not "everything in the adjoint". This is exactly why the known no-go results against literal E_8 world-formulas (Distler–Garibaldi; Coleman–Mandula) do *not* bite: they constrain E_8 as a spacetime gauge symmetry, which TFPT does not claim. The defensible claim is: *TFPT is the unique parabolic compilation of the anchor* $a = (1, 1, 2)$ *into the* E_8 *audit hull.*

In one line:

> The anchor generates the syntax; the boundary generates the scale; E_8 checks the consistency.

i.e. the discrete content (carrier $D_5 \oplus A_3 + \mu_4$, the operators R, Q, K, L) flows from $a = (1, 1, 2)$; the continuous content ($c_3 = \frac{1}{8\pi}$, φ_0, α^{-1}, the exponential scales) flows from π; and E_8 is the unimodular *checksum container*, not the world group.

The anchor ratio triad ([verification/v119_review_validation_2.py]). The three elementary symmetric values, normalised by the family count, are the theory's three recurring fractions **[E]**:

$$\frac{e_3}{p_0} = \frac{2}{3} \text{ (Koide branch / sheet ratio)}, \quad \frac{e_1}{p_0} = \frac{4}{3} \text{ (seed gain)}, \quad \frac{e_2}{p_0} = \frac{5}{3} \text{ (carrier branch)}$$

— the canonical flow's critical points are exactly $(2, 5) = (e_3, e_2)$ with inflection $\frac{7}{2} = \frac{e_3+e_2}{2}$: the fractions $\frac{2}{3}, \frac{4}{3}, \frac{5}{3}$ are not scattered numbers but the three normalised anchor coefficients. Further micro-identities **[E]**: $\Omega_{\text{adm}} = 2p_1p_2 = 48$, $10b_1 = 1 + p_1p_3 = 41$, $\Delta_Y = e_2^2 = p_0^2 + 2\,\text{rank}\,E_8 = 25$, $h(E_8) = e_3p_0e_2 = 30$, rank $E_8 = p_0 + e_2$.

$a = (1, 1, 2)$ *[E]*

elementary $e_k(a)$	power sums $p_n = 2 + 2^n$	derived motifs
$e_1 = 4 \to \lvert\mu_4\rvert$ $e_2 = 5 \to g_{\text{car}}$ $e_3 = 2 \to \lvert\mathbb{Z}_2\rvert$ $c_3 = \frac{1}{2e_1\pi} = \frac{1}{8\pi}$	$p_0, p_1, p_2, p_3 = 3, 4, 6, 10$ $p_1p_2p_3 = 240 = \lvert R(E_8)\rvert$ $p_4 - p_3 = 8 = \text{rank}\,E_8$ $\Rightarrow \dim E_8 = 248$	$4, 5, 8, 16, 25,$ $30, 41, 120, 240$ *one anchor,* *many projections*

The companion documents

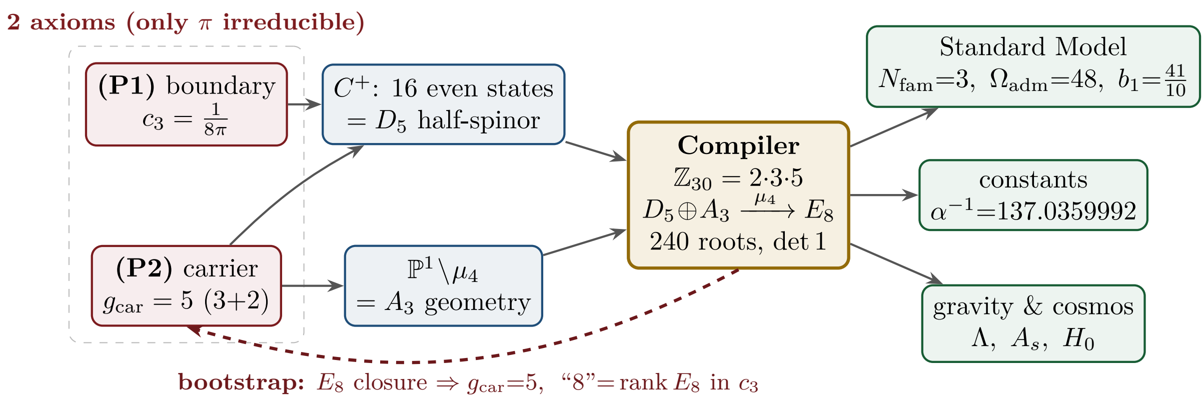
The whole development is consolidated into **eight companion documents** (plus this introduction), organised in layers that build on one another; this introduction is the entry point and reading guide. Each document collects the material of its layer into self-contained parts:

Document	What it contains (former notes now as parts)
`tfpt_1_architecture_e8`	**Core.** The two axioms $\{c_3, g_{\rm car}\}$, the derivation map and the EM fixed point (Part I); the Coxeter–cyclotomic compiler $\mathbb{Z}_{30}$=2·3·5, the explicit $D_5 \oplus A_3 + \mu_4$ E_8 construction, the machine-checkable E_8 appendix and the group-level glue $(\mathrm{Spin}(10) \times SU(4))/\Delta\mathbb{Z}_4$ (Part II).
`tfpt_2_standard_model`	**Standard Model.** The full SM in one $\varphi_0^{\rm ret}$-ladder formula (Part I); the flavor block from parabolic weights and the transport theorem, H2 as a spectral selector (Part II); the worked closures — explicit gap, APS–Fredholm, truncation/n_s, r, Starobinsky M, derived θ_{12}, quark c, the H2 splitting type, Λ-typing (Part III).
`tfpt_3_e8_audit_bootstrap`	**E_8 audit & bootstrap.** The seven E_8 slices as an audit raster ($E_6 \times A_2$ reads the flavor matrix) (Part I); the old cascade D=60−2n as the same even-integer spine (Part II); the Möbius bootstrap — $g_{\rm car}$=5 and the "8" in c_3 as overdetermined fixed points, only π irreducible (Part III).
`tfpt_4_frontier`	**Frontier.** The honest status of η_B, m_p/m_e, Koide, dark matter and quantum gravity — the genuine handle, the precision, and what is *not* forced onto the ladder.
`tfpt_5_redteam`	**Red Team.** The adversarial audit of the five load-bearing reductions (Targets A–E): the declared attacks, what survives each, what each one reduces to, and the explicit kill tests.
`tfpt_research_contracts`	**Research contracts.** The open gates as numbered contracts with lemma chains and closing theorems: ($U_{\rm wall}$) (now reduced to the selector triangle: $(d,n) \Rightarrow R$ columnwise, residue = *two* line pairings, the third integrality-forced) and $G_{\rm metric}/G_{\rm net}$ (the index-4 seam-net identification — one theorem, three doors, now realised at the finite Lie level).
`tfpt_horizon_readouts` **(App. H)**	**Appendix H — horizon unit system (reframe, not new physics).** One seam constant $c_3 = \frac{1}{8\pi}$ as the universal horizon thermal code: Hawking/de Sitter/Unruh, BH thermodynamics, Page, scrambling, Nariai bound, $v_{\rm GW} = c$ — standard physics in seam units, two compiler fingerprints ($1920 = \lvert W(D_5)\rvert$, $\lvert\mu_4\rvert = 4$); plus the Nariai/anchor surface, the dual anchor and the resummed seam clock (v101–v138).
`origin_theory`	**Origin Theory.** Why no free *dimensionless compiler dial* remains beyond the anchor and π (the dimensionful $v_{\rm geo}$ and transfer physics stay typed): the $(g_{\rm car}, N_{\rm fam})$ skeleton, the triply-forced 8, the order-30 Coxeter cycle, one boundary transport for flavor and horizon, the gapped unique attractor; **[E]** core plus one honestly-typed **[C]** cyclic interpretation.

The five formerly "research-level" problems are now *closed, reduced or typed* inside the documents where they belong — the group-level closure $(\mathrm{Spin}(10) \times SU(4))/\Delta\mathbb{Z}_4$ in document 1 (closed), and the H2 splitting type, the Starobinsky scalaron mass, the θ_{12} texture and the quark c in document 2 (the first two closed; the quark *ratios* now closed combinatorially (v49/v71), only the *absolute* scale a $U_{\rm point}$ anchor; θ_{12} the seam-misalignment residual). Eight companion files (plus this introduction), each result where it logically lives.

2 The architecture at a glance

The diagram below shows the whole pipeline: *two* inputs on the left, the discrete compiler in the middle, the physics on the right. Everything in between is a consequence.



Key statement (the loop closes): previously D_5, A_3 and E_8 sat *as given* Lie structures in the background. Now they are *intermediate products* of the two inputs — *and the red back-channel is the new part*: the E_8 closure feeds back and *fixes the inputs*. $g_{\rm car} = 5$ is the unique rank-fill/Coxeter/Pascal solution ($2^{g-1} = \binom{g}{0} + \binom{g}{1} + \binom{g}{2}$), and the integer 8 in $c_3 = \frac{1}{8\pi}$ is rank $E_8 = h(D_5) = \det R$. **Inputs and output form an overdetermined fixed point** — the structure does not "prove itself"; it has fewer degrees of freedom than independent consistency conditions. The only thing *not* produced by the loop is the continuous π (Möbius/Gauss–Bonnet), the lone irreducible primitive. So it is not a one-way arrow chain but a closed, overdetermined self-consistency loop.

The master story is two engines. Read from the two axioms, the whole theory factorises into exactly *two* engines — a *discrete* closure and a *boundary* dressing. Gravity is not a third block; it is the geometry channel of Engine 2.

Engine 1 — discrete closure (from $g_{\rm car} = 5$)

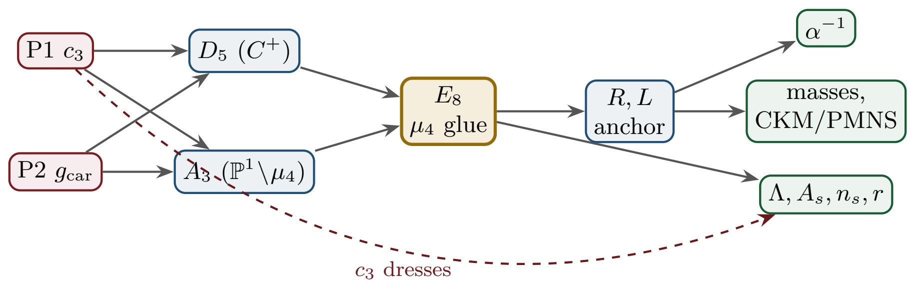
$g_{\rm car}$=5 (3+2) → $D_5 \oplus A_3 \xrightarrow{\mu_4} E_8$ → SM packet: $N_{\rm fam}, \Omega_{\rm adm}, b_1, \dim S^+$ → flavor matrix R ($\det 8$, minors $2, 3, 5$)

Engine 2 — boundary dressing (from $c_3 = \frac{1}{8\pi}$)

$c_3 = \frac{1}{8\pi}$ → seed $u = \varphi_0^{\rm ret}$, α^{-1}, ξ, c_3^7 → scales $x^1 : x^5 : x^{10}$ $v_{\rm EW}, H_0, \Lambda$ → gravity (geometry channel)

3 The dependency DAG and the proof ledger

The whole theory is a directed acyclic graph: two axioms at the source, the compiler in the middle, the observables as sinks. The DAG makes the attack surface explicit — each arrow is a named theorem or a declared input, and each node fails in exactly one way.



Node	Definition	Inputs	St.	Output	Failure mode
P1	RP boundary kernel, c_3=1/(8π)	— (axiom)	[O]	c_3	no reflection-positive seam
P2	5-slot carrier $g_{\rm car}$=5	— (axiom)	[O]	carrier 3+2	wrong family/charge lattice
D_5	C^+ even-Hamming = half-spinor 16	P2	[E]	$\dim S^+$=16, Y	group/matter mismatch
A_3	$\mathbb{P}^1 \setminus \mu_4$ four-puncture geom.	P1	[E]	$N_{\rm fam}$=3, $\mathbb{Z}_6$	wrong multiplicity
E_8	μ_4 glue: disc=$\mathbb{Z}_4$, q-sum= 2	D_5, A_3	[E]	240, 248	not even-unimodular
R, L	residue + winding matrix, anchor (1, 1, 2)	A_3	[E]	$\det 8$, minors (2, 3, 5)	wrong D_6 branch
α^{-1}	root of $F_{U(1)}(\alpha)$=0	c_3, M=41	[E]	137.0359992	no/second root (excl.)
masses	$\varphi_0^{\rm ret}$-ladder $\lambda_Y^L \Lambda$	R, L, c_3	[E]/[C]	9 masses, mixings	hierarchy mismatch
θ_{12}	TBM + seam ε=c_3	R, L	[E]/[C]	0.307	seam-misalign. lemma fails
Λ, A_s	seam det. / R^2 scalaron	c_3	[C]	Λ, A_s, n_s, r	ambient RP fails

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.

Reading the table top-to-bottom is reading the theory: two red axioms in, everything else a consequence with exactly one failure mode each.

The ledger itself is versioned: each row of `verification/status_ledger.csv` carries a claim ID, its dependencies, its verification script, and `active/canonical_status/supersedes` fields, so claims that were later superseded (e.g. the early "flavor gate open" rows, now *ratios closed* via `FLAV.RIGID`) are marked inactive — the canonical status is the only one read as current.

The compact status formula: three honest layers

$$\underbrace{\text{compiler}}_{\text{closed}} \;\Big|\; \underbrace{\text{admissible IR physics}}_{\text{conditionally closed (RP, gap)}} \;\Big|\; \underbrace{\text{strict physical TOE}}_{\text{open}}$$

TFPT v5.4 closes the discrete compiler, the algebraic Standard-Model readout, the EM fixed point, the admissible gapped IR sector and the $R + R^2$ spectral-action shadow. It does *not* yet certify a strict physical TOE. The remaining items have now collapsed sharply: the R-bridge amplitude normalisation U_{point} *reduces to* v_{geo} (Gate 1 closed, v75 — the same anchor as $1/G$), and the ambient metric measure is *holographically reduced* to a finite seam-*boundary* measure (Gate 2 tier B, v76) — which is in turn the rigorously-constructed $(E_8)_1$ lattice net (v77), so G_{metric} is imported into existing RCFT rigor rather than built anew. So the only genuine residuals are *one* dimensionful scale v_{geo} — the *dimensional-analysis floor* that no theory escapes (v78), shared by flavor and gravity — the boundary net identification (now *closed modulo cited theorems* on its MMST route `SEAM.EQUIV.MMST.01`, v367/v368, v376–v379, Lean `FORM.SEAM.MMST.01`; the parent `SEAM.EQUIV.01` stays [**O**]), and the named QFT/cosmology transfer interfaces (F_{frontier}).

Two points that are *not* in the residual (closed, but easy to mis-list as open). *(i) Parameter-freeness is a theorem under the named hypotheses:* the boundary transport is gapped ($\Delta = 6\log\frac{3}{2} > 0$), so by Perron–Frobenius its iteration has a *unique* attracting fixed point — the constants are *selected*, not tuned ([**E**] for the attractor, [`verification/v56_unique_attractor.py`]; the *physical identification* of the transport operator stays [**C**]), reinforcing the static bootstrap web ($g_{\text{car}} = 5$ forced three ways *given* the E_8 closure rank — a consistency loop, see red team Target B — [`verification/v6_bootstrap.py`], [`verification/v14_carrier_uniqueness.py`]). *(ii) Newton's constant is not an* independent *input:* fixing the physical Λ branch ($(8\pi)^2\delta_{\text{top}} = \frac{3}{4\pi^2}$) pins $\bar{M}_{\text{Pl}}$ *relative to* Λ, so G_N is read out *after* the branch choice + *one* dimensionful induced-gravity anchor ([**E**]/[**C**], [`verification/v60_lambda_metrology_branch.py`]), not predicted from pure numbers. What stays irreducible is then just π, the discrete E_8-closure fixed point (self-consistent, *not* a free axiom; the bootstrap of `tfpt_3`), and *one* dimensionful scale v_{geo} — and that single scale is shared by *both* the flavor sector ($U_{\text{point}} \to v_{\text{geo}}$, v75) and gravity ($1/G$, v68): the two $[A]$ anchors are *one* (dimensional analysis; no metre from pure mathematics).

The hard reduction: the live residual as typed interfaces

After the compiler closure, the entire residual is

$$\text{Rest} \quad = \quad v_{\text{geo}} \;\oplus\; G_{\text{net}} \;\oplus\; F_{\text{transfer}}$$

— one dimensionful scale anchor, one metric-sector inclusion, and a set of deliberately typed QFT/cosmology interfaces (the historical gate labels $(U_{\text{wall}})/(G_{\text{metric}})/(F_{\text{frontier}})$ are kept only for ledger continuity). In current terms: $(U_{\text{wall}}) \to v_{\text{geo}}$ — the flavor ratios are *closed* (Plücker/Readout Rigidity, v49/v71/v75), what remains is *one dimensionful unit*, not a flavor gap; $(G_{\text{metric}}) \to G_{\text{net}}$ — reduced to a single boundary-net theorem, *"the seam–Calderón net is the index-4 μ_4 simple-current extension of the carrier net"* (holomorphy then follows, v89/`GATE.METRIC.06`); since the QBL chain (v108–v113) the *carrier choice rides on the same theorem*: the certified scalar kernel exists iff it pairs the sheets (v110), the certified channel counts the code identically (v112), pair transport is minimally complete (v111), and the one quasi-free kernel — rank $= c$: 5 carrier block, 8 seam hull — is the defining datum of the free net the gate already posits (v113), so closing G_{net} also closes the carrier selection [**E**]; $(F_{\text{frontier}}) \to F_{\text{transfer}}$ — named QFT/cosmology *transfer* interfaces (Koide source→pole, η_B Boltzmann, axion relic, m_p/m_e), never compiler powers; the four are one typed functor with a four-axiom contract (`CONTRACT.F.01`, v213) and *one shape* — each is a gapped relaxation to a unique attractor (the main-branch discrete→dynamic motif), with F_pole at the *seam* rate $(2/3)^6$ and the other three external (v303). The metric-sector reduction lands on a *single named keystone*. The boundary inclusion $G_{\rm net}$ is the index-4 μ_4 simple-current extension $(D_5)_1{\otimes}(A_3)_1{\rtimes}\langle(1,1)\rangle \cong (E_8)_1$ (Simple-Current Extension Theorem, v154, exact: index 4, c=8, μ=1 ⇒ holomorphic ⇒ E_8); net existence and full-cone reflection positivity are discharged to **[E]** (v175); and the **No-Unit Theorem** (v153) makes $v_{\rm geo}$ an irreducible metrology primitive ($U_{\rm point} \sim v_{\rm geo}$, $1/G \sim v_{\rm geo}^2$, $m/\mu = e^{3/4}$ — one unit in three dimensions). The flavor selector and Q-geometry residues are *derived* (no discrete freedom left). The *one* seam-side identification, packaged as the single named theorem `SEAM.EQUIV.01` — *the raw reflection-positive seam state is the holomorphic* $(E_8)_1$ *net at* τ=i — is now *closed modulo cited theorems on its MMST route* (`SEAM.EQUIV.MMST.01`; the twistor route `SEAM.EQUIV.TWISTOR.01` and the unconditional parent stay **[O]**): the explicit lattice model (v367/v368) and the S_3 stack (c=8, 248 currents/1 primary, torus GSD=1, RP; v376–v379) pin it at every computable level, Lean-formalised as `FORM.SEAM.MMST.01`, with only the cited MMST/OS continuum-existence theorems left **[O]** (v336); its conformal-deck premise `QGEO.SYM.01` is its *downstream* definitional face (per v323; the residual is Lean-pinned to named cited steps, v308). The sprint-by-sprint reduction that produced it is recorded in the changelog and in `tfpt_research_contracts`, not replayed here. The status card:

Item	**Status**	**Reading**
parameter-freeness (self-consistency)	[E]	gapped transport ⇒ *unique* Perron–Frobenius attractor (v56); bootstrap web $g_{\rm car}$=5 forced (v6/v14)
Newton constant G_N	[E]/[C]	not *independent*: branch pins $\bar{M}_{\rm Pl}$ rel. to Λ (v60); readout after branch + one dim. anchor
ratio $c_u/c_d = \frac{55}{117}$	[E]	$g_{\rm car}\Vert{\rm Pl}_{\mathbf{1},a}(K)\Vert_1/(N_{\rm fam}^2\Delta_Q)$; rigid on the derived stratum (v49/v71), no solve
flavor absolute scale	[E]/[O]	ratios *closed* (v49/v71); only the absolute scale remains $= v_{\rm geo}$ (Gate 1, v75), the same anchor as $1/G$
$R + R^2$ gravity	[E]/[C]	covariant $f(R)$ field equation closed in regime; heat-kernel grounded (G2)
Einstein eq. (covariant, local)	[E]/[C]	fixed-volume entanglement equilibrium ⇒ *full* $G_{ab}{+}\Lambda g_{ab}{=}c_3^{-1}T_{ab}$ with BOTH coefficients parameter-free: $8\pi{=}1/c_3$ (v358, thermo=geo via $\vert\mu_4\vert{=}\vert\mathbb{Z}_2\vert\chi{=}4$) and Λ from α (v60); Einstein tensor via Lovelock, conservation an output (v359); residual = equation-of-state status + $v_{\rm geo}$, global measure (C7) now a [C] redundancy (v369)
physical QG sector	[E]/[C]	gap-decoupled admissible measure ($\Delta_{\rm eff}{=}1.648{>}0$)
ambient QG measure	[C] (redundancy)	**certification object, not missing dynamics** (v369; conditional on `SEAM.EQUIV.01`+Bisognano–Wichmann): holographically reduced to a finite seam-*boundary* measure (v76); IR tier closed (decoupling); strict-TOE cert. = the simple-current extension $(D_5)_1{\otimes}(A_3)_1{\rtimes}\langle(1,1)\rangle \cong (E_8)_1$ (v154); the free-bulk premise (A) is itself a fixed-point theorem with the cone eliminated (cone gap = one-particle gap $(2/3)^6$) and factors into the A_2 net-existence + the keystone `SEAM.EQUIV.01` (MMST route closed mod. cited; parent [O]) — zero new gates (v160–v165)
$v_{\rm geo}$ metrology unit	[O](primitive)	No-Unit Theorem (v153): a dimensionless compiler makes no absolute scale; $U_{\rm point} \sim v_{\rm geo}$, $1/G \sim v_{\rm geo}^2$, m/μ a ratio — one unit, irreducible
Koide	[C]	$Q_{\rm src} + \varphi_0/24$ source→pole: typed runnable solver with kill test (QED-excluded, v371)
axion DM	[C]	$f_a = M_{\rm scal}/128$: finite-T spine-branch typed runnable solver (v373)
η_B	[C]	readout closed; integrated BDP leptogenesis Boltzmann ODE, typed runnable solver (v372)
m_p/m_e	[C]	cross-sector QCD/EW ratio: typed runnable solver with uncertainty budget (v374)
$N_\star$	[C]	reheating input; scalaron-reheating point $N_\star$=51.4 (v86); frozen band $[50, 60]$
$a = (1, 1, 2)$	[E]/[C]	forced by $4 = 2 + 1 + 1$, conditional on the carrier
carrier selection g=5 (QBL)	[E]/[C]	theorem chain v108–v113: sheet-odd kernel, self-counting channel, transport degree 2, one quasi-free kernel (rank = c); residue = the $G_{\rm net}$ premise itself — gate closes ⇒ carrier [E]
selector establishment R4′	[E]/[C]	$(d, n) \Rightarrow R$ columnwise (v136); both normals *anchor data* — $n = (p_2, p_0, e_2)(a)$ in the cusp frame (v145/v149); residue = *one* assignment
sheet/parity (Q-geometry derived)	[E]/[C]	H^1 characters = Spec(Q_+) = A_3 exponents (v137); $\mathbb{Z}_3$ choice *derived* (v141); full Möbius D_4 matched parity by parity (v146); residue = the module identification, folded into `SEAM.EQUIV.01`
seam quantum clock R1	[E]/[C]	typed closed: resummed clock = entropy power law (v124–v133); VW edge match (v138); det-ratio in-family (v144); ring sum = Born^2 Gaussian zero-mode integral, bend det′-clean (v147); residue = measure + reference
E_8 Lean	[E]	hardening, not a blocker (script-certified)

Marker key: ***[E]*** *exact / machine-proven ·* ***[C]*** *conditional (holds under named hypotheses) ·* ***[O]*** *open / axiom ·* ***[X]*** *falsifiable kill test. Per-claim fine type in* `verification/status_ledger.csv`.

Card addendum (2026-07-22): the $a = (1, 1, 2)$ residual now carries the consolidated P2 reduction state — the position half of the alignment bit is topology (v510) and the modulus half is one discrete symmetry-lift bit — the twist-class choice (flag transitivity $\iff \tau = i$, v512/v528), so the dimensionless compiler inputs read *four marks + one discrete symmetry-lift bit (the twist-class choice, in principle interferometrically readable; flag transitivity* $\iff \tau = i$*)* $+ \pi$ (`AX.P2.01` stays a declared axiom).

The former structural open core, the *single* named keystone `SEAM.EQUIV.01` (the G_{net} inclusion: the raw RP seam $= (E_8)_1$ at $\tau{=}i$), is now *closed modulo cited theorems on its MMST route* (`SEAM.EQUIV.MMST.01`; the twistor route `SEAM.EQUIV.TWISTOR.01` and the unconditional parent stay [O]): the explicit lattice model (v367/v368) and the S_3 stack (c=8, 248 currents/1 primary, torus GSD=1, RP; v376–v379) pin it at every computable level, it is Lean-formalised as `FORM.SEAM.MMST.01`, and only the cited MMST/OS continuum-existence theorems stay [O] (v336); its conformal-deck face `QGEO.SYM.01` is a *corollary* (a conformal net's vacuum is rotation-invariant by axiom, v323/ [`verification/v335_seam_equiv_unify.py`]), not a second item. v_{geo} is an irreducible metrology unit and the frontier items are typed interfaces, not structural holes. G_{net} is heat-kernel grounded (G2: the spectral action gives $R + R^2$ structurally) and *gap-decoupled* (G5: Δ_{eff}=1.648>0 isolates the closed admissible sector from the ambient), so the ambient measure `QG.AMB.01` is no longer carried as an open structural item: it is a [C] *redundancy* (v369, conditional on `SEAM.EQUIV.01`+Bisognano–Wichmann), a certification object rather than missing dynamics — it does not affect any TFPT readout. Both are written up as numbered *research contracts* (lemma chains + closing theorem + machine-certifiability) in the companion `tfpt_research_contracts`, with priority on the v_{geo} scale anchor and the G_{net}/`SEAM.EQUIV.01` inclusion theorem. [`verification/v36_spectral_action_g2.py`]

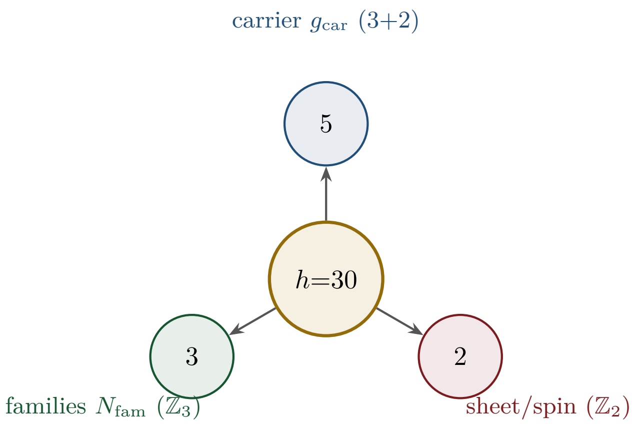
Sharper still: the structural residual is *one* condition. G_{metric}, the carrier P2 and the red-team Target A are not separate items: they are three provably-equivalent faces of the single condition *"the seam carries no nontrivial abelian sector"* — holomorphy (μ-index 1), a homology-sphere seam link (Γ perfect $\Leftrightarrow 2I$, [`verification/v232_e8_kleinian_seam.py`]), exactly one 1-dim irrep ([`verification/v219_icosahedral_mckay.py`]) — all of which force E_8 ($\#$(mark-1) $= |H_1| = 1$ only for E_8; holomorphic $c = 8 = g_{\text{car}} + N_{\text{fam}}$ gives the unique even unimodular rank-8 lattice). In abelian Chern–Simons language this is the single integer step holomorphic $\Leftrightarrow \det K = 1$, the extension tower $D_5 \oplus A_3(16) \to D_8(4) \to E_8(1)$ being anyon condensation (the Kitaev E_8 state); the one analytic step "the free RP seam condenses the order-$|\mu_4|$ Lagrangian glue" = the single named keystone `SEAM.EQUIV.01` (with `QGEO.SYM.01` its downstream conformal-deck face) is now *closed modulo cited theorems* on its MMST route `SEAM.EQUIV.MMST.01` (v367/v368, v376–v379, Lean `FORM.SEAM.MMST.01`; cited continuum existence [O], v336; the parent `SEAM.EQUIV.01` stays [O]). So beneath the gate *labels* there is, structurally, a single fundamental postulate. And the entire *emergent-QFT layer* collapses onto the *same* postulate (the *Modular Spectral Closure*, v261): the finite Dirac is a covariance induction of the seam KMS state (v258), the spectral-action cutoff *is* that KMS weight ($f_2/f_0 = 1$, v259), the seam, the carrier-16 and E_8 live on one Kummer/K3 surface (v260), so Dirac, cutoff, gauging, glue and orientability are all readouts of the one seam state — the boundary QFT is closed as a single relative object modulo the keystone `SEAM.EQUIV.01` and adds *no new open item*, while the ambient quantum-gravity measure is a [C] redundancy (v369), not a separate structural gap. [`verification/v234_seam_holomorphy_selection.py`] [`verification/v235_seam_chern_simons.py`]

Reduction in one line

The number of *structural* assumptions is **two axioms**. Everything else ($N_{\rm fam}, \Omega_{\rm adm}, b_1, \alpha^{-1}$, the flavor matrix, the E_8 glue, the scale grammar) is a *consequence. And even those two tighten:* the bootstrap note shows $g_{\rm car} = 5$ and the integer 8 in c_3 are *overdetermined E_8-closure fixed points* (rank-fill, Coxeter-match, integer-glue/norm all give 5; $8 = h(D_5) = \operatorname{rank} E_8 = \det R = \varphi(30)$), leaving *one* genuine continuous primitive: π from the Möbius boundary. *A fourth, deeper forcing* comes from the McKay correspondence: E_8 is the exceptional top of the tower of finite $SU(2)$ subgroups, so its branch data is the icosahedron and the three atoms $(2, 3, 5)$ are its rotation-axis orders — the binary icosahedral group $2I$ (order $120 = |R^+(E_8)|$) has irrep degrees equal to the affine-E_8 Kac marks, $\sum = 30 = h(E_8) = 2{\cdot}3{\cdot}5$ ([`verification/v219_icosahedral_mckay.py`]). So the *discrete* core has no freely tunable fundamental numbers — the bootstrap is *not creation from nothing* (two inputs remain), but an overdetermination of those inputs. (This concerns the discrete core only; dimensionful masses, the QG measure and frontier items remain typed-open.) [`verification/v6_bootstrap.py`] [`verification/v14_carrier_uniqueness.py`] [`verification/v23_anchor_generator.py`] [`verification/v219_icosahedral_mckay.py`]

4 The compiler: why $30 = 2 \cdot 3 \cdot 5$

The Coxeter number of E_8 is $h = 30 = 2 \cdot 3 \cdot 5$. Exactly these three prime factors are the three discrete atoms of the theory — that is the point of the compiler:



From this the compiler reads, among others:

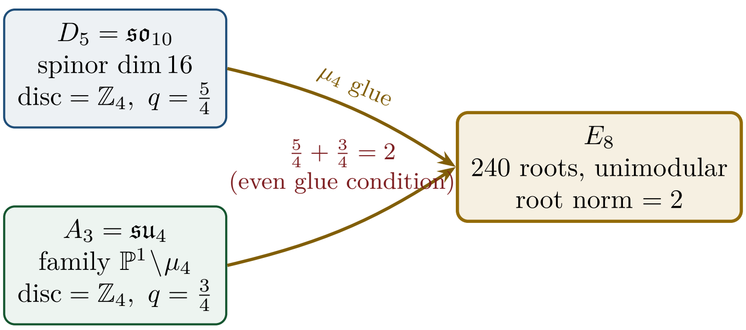
- $\dim S^+ = 1 + 10 + 5 = 16$ (one generation) as the even exterior algebra of the 5-slot carrier — the same Pascal row $\binom{5}{0}, \binom{5}{1}, \binom{5}{2} = 1, 5, 10$ appears in family, action ladder and the E_8 root count. **[E]**
- $|R(E_8)| = 16 \cdot 5 \cdot 3 = 240$ and $\dim E_8 = 240 + (5 + 3) = 248$. **[E]**
- the $\mathbb{Z}_{30}$ Coxeter element $\leftrightarrow$ the cyclotomic polynomial Φ_{30}; the corner rotation $z \mapsto iz$ on $\mathbb{P}^1 \setminus \mu_4$ is the A_3 Coxeter element. **[E]**
- **why these three primes:** E_8 is the icosahedral top of the McKay tower ($2I$, order 120, irrep degrees = affine-E_8 marks summing to 30), so $2, 3, 5$ are the icosahedron's rotation orders; and the E_8 exponents $(\mathbb{Z}/30)^\times$ carry an order-4 totative character $\langle 7 \rangle$ — the same μ_4 seam clock. **[E]** [`verification/v219_icosahedral_mckay.py`] [`verification/v223_coxeter_totative_clock.py`]
- **a network reading (computational substrate).** On the same McKay $(2, 3, 5)$ graph the affine-E_8 marks are the *unique dynamical attractor* of a minimal local-averaging update $m \leftarrow (A{+}2I)/4 \cdot m$ (it converges from *any* positive start to $(1, 2, 3, 4, 5, 6, 4, 2, 3)$); the spherical tower terminates at E_8 and the cascade $E_8 \supset E_7 \supset E_6 \ldots$ is nested-subgraph coarse-graining. So E_8 reads as the *universal attractor of a minimal* $(2, 3, 5)$ *network* (a Wolfram-style computational-substrate picture) at the Coxeter-skeleton level. **[C]** A minimal hypergraph *rewrite* reproducing

the *full* structure (lattice + recovery + readouts) stays open, and φ_0^{ret} is the analytic seed $\frac{4}{3}c_3 = 1/(6\pi)$, not an edge fraction. **[O]** [`verification/v298_e8_network_attractor.py`]

- **the bridge, hardened.** A local witness-based growth dynamics (a *diffusion-generated* witness + the local balance gate $Am \le 2m$, Collatz–Wielandt = Smith's $\rho \le 2$; the Perron eigenvector enters only as an audit) *reaches* E_8 as the last finite point ($E_6 \to E_7 \to E_8$, then stops at the affine boundary $\hat{E}_8$) — but *only* from a 3-arm $(2,3,\cdot)$ seed (the controls $\rho(A_n) = 2\cos\frac{\pi}{n+1}$ and $\rho(D_n) = 2\cos\frac{\pi}{2n-2}$ are < 2 for *all* n, so neither family ever reaches E_8). So P_2 is exactly the irreducible seed on which the rule reaches E_8 as the last finite point — a second network reading of the $(2,3,5)$ hull, not a derivation of P_2. **[O]** [`verification/v299_hypergraph_0d_autonomous.py`]
- **what a rewrite must inject.** Testing *which* load-bearing quantities are graph-spectral sharpens the open question: **[E]** the Kac marks are the Perron eigenvector ($A\,m{=}2m$) and the subleading eigenvalue is exactly $2\cos(\pi/5)$ = the golden ratio (the 5-fold/icosahedral signature), but **[E]** the recovery rate $(2/3)^6$ is *not* any eigenvalue of the adjacency and φ_0^{ret} is not a rational edge fraction. The cusp weight $2/3{=}|\mathbb{Z}_2|/N_{\mathrm{fam}}$ is now *derived* from a minimal branching rule ([`verification/v327_hypergraph_rewrite.py`]); what remains non-graph-spectral for a *full*-structure rewrite at leading order is sharpened by [`verification/v396_phi0_icosahedral.py`] (tree-level $\varphi_0^{\mathrm{tree}} = F/(4h\pi)$); the puncture $48c_3^4$ alone stays open. **[O]** [`verification/v312_hypergraph_substrate.py`]
- **one coupled local rule (carrier × family).** On a 9×3 labelled grid (affine-E_8 nodes × family channels) a *single* coupled micro-step — network lazy diffusion on each cusp column plus the [`verification/v327_hypergraph_rewrite.py`] branching rule M on each node's family vector — has joint attractor marks$\otimes(w{=}0)$ and recovery rate $(2/3)^6$ after one clock hand ($2N_{\mathrm{fam}}$ family steps); the [`verification/v324_hypergraph_fiber.py`] fiber product $T_{\mathrm{net}}{\otimes}T_{\mathrm{cusp}}$ emerges as the cusp-readout basis, and the [`verification/v299_hypergraph_0d_autonomous.py`] growth trajectory $E_6 \to E_7 \to E_8 \to \hat{E}_8$ is unchanged with the fiber attached. **[E]** [`verification/v395_hypergraph_coupled.py`]
- **the seed at leading order.** The tree-level retained seed $\varphi_0^{\mathrm{tree}} = 1/(6\pi)$ is exactly $F/(4h\pi)$ on the icosahedral hypergraph ($F = 20$ triangular hyperedges, $h = 30$, $|\,\mathrm{Aut}\,| = 120$): equivalently $(F/(g_{\mathrm{car}}N_{\mathrm{fam}}))c_3$, with $F/h = |\mathbb{Z}_2|/N_{\mathrm{fam}} = 2/3$ the same survival ratio as [`verification/v327_hypergraph_rewrite.py`]; Gauss–Bonnet consistent with $c_3 = 1/(|\mathbb{Z}_2|4\pi)$. The puncture $48c_3^4$ remains analytic, not a graph fraction. **[C]** [`verification/v396_phi0_icosahedral.py`]

5 The μ_4 glue: how E_8 is really built

The decisive, now *proved* step: D_5 and A_3 have the *same* discriminant group $\mathbb{Z}_4$, and their discriminant-form norms are exactly two TFPT constants which add up to the E_8 root norm.



Glue theorem (proved)

$\mathrm{disc}(D_5) = \mathrm{disc}(A_3) = \mathbb{Z}_4$, glue index $|\mu_4| = 4$, and $q(D_5) + q(A_3) = \frac{5}{4} + \frac{3}{4} = 2$ = the E_8 root norm. Hence $E_8 = (D_5 \oplus A_3) + \mu_4$-glue is a *closed lattice-theoretic* construction — not a blind "positing of 248". **[E]** [`verification/v1_e8_glue.py`]

6 How derivation now works: the derivation map

What can one concretely *compute* with $\{c_3, g_{\text{car}}\}$? The following map summarises what is now a consequence rather than an assumption.

Sector	How it now arises	Status
gauge group / families	carrier 3+2, $\dim S^+=16$, $N_{\text{fam}}=3$, $\Omega_{\text{adm}}=48$	[E]
hypercharge	roots of the charge polynomial $bsY^2 + (s-b)Y - 1$	[E]
$U(1)$ index b_1	$b_1 = \dfrac{g_{\text{car}}\, 2^{g_{\text{car}}-2} + 1}{10} = \dfrac{41}{10}$	[E]
fine structure α^{-1}	unique root of $F_{U(1)}(\alpha) = 0$	[E]
flavor matrix	transport theorem (H1 proved, H2/H3 force the matrix)	[E]/[C]
electroweak scale	$v_{\text{EW}} \sim e^{-\alpha^{-1}/5}$ (carrier divides α^{-1} by 5)	[E]
cosmological constant	$\Lambda \sim e^{-2\alpha^{-1}}$ (density quotient; see typing note)	[E]/[C]
Hubble scale	$H_0 \sim \sqrt{\Lambda}$	[C]
inflation amplitude	$A_s = \dfrac{N_\star^2}{24\pi^2} c_3^7$ in the R^2 branch	[C]
E_8	$D_5 \oplus A_3 + \mu_4$-glue	[E]

*Marker key: **[E]** exact / machine-proven · **[C]** conditional (holds under named hypotheses) · **[O]** open / axiom · **[X]** falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.*

Notable: the scale grammar is *one* exponential engine. The same number $\alpha^{-1} \approx 137$ generates the electroweak scale (divided by $5 =$ carrier), the cosmological constant (times 2) and — via the square root — the Hubble scale. One constant becomes many orders of magnitude.

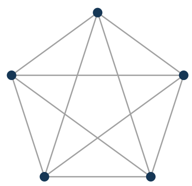
Typing note on Λ (a deliberate danger zone). "Λ" denotes *three* distinct objects that must not be conflated: (i) the seam-determinant exponent $e^{-2\alpha^{-1}}$ (an exact [E] readout), (ii) the dimensionless density quotient $\rho_\Lambda/\bar{M}_{\text{Pl}}^4$ (which carries a prefactor *branch*, hence [C]), and (iii) the observed curvature, to which (ii) is *anchored*, not predicted. The exponential is exact; the absolute dark-energy density remains a typed cosmology interface (full disambiguation in `tfpt_2_standard_model`).

Cosmology is the Pascal action grammar of the carrier

With $x = v_{\text{geo}}/(5\beta_{\text{rad}}^2 \bar{M}_{\text{Pl}})$ the cosmological readouts are a *power tower* on the five-slot carrier graph K_5:

$$\frac{H_0}{\bar{M}_{\text{Pl}}} = x^5 R_H, \qquad \frac{\rho_\Lambda}{\bar{M}_{\text{Pl}}^4} = x^{10} R_\Lambda, \qquad (1,5,10) = \binom{5}{0}, \binom{5}{1}, \binom{5}{2}$$

EW is the unit x^1, Hubble is the product over the $\binom{5}{1} = 5$ vertices, Λ is the product over the $\binom{5}{2} = 10$ edges — the *same* Pascal triple that reads one generation, the action ladder and the E_8 root count $16 \cdot 5 \cdot 3 = 240$. "Hubble = slot product, Λ = pair product." [E]



$\binom{5}{1} = 5$ vertices — action x^1 (Hubble slot)
$\binom{5}{2} = 10$ edges — Λ density pair x^{10}
$\binom{5}{3} = 10$ triangles — dual sector
$\binom{5}{4} = 5$ complements — inverse sector
$\binom{5}{5} = 1$ determinant — closure

K_5

K_5 on the five carrier slots; the action powers x^1, x^5, x^{10} are its $\binom{5}{k}$ faces (mnemonic; the physical scale map stays [C]).

The entire fermion spectrum in one formula

Masses, Yukawa, CKM, PMNS and neutrinos all follow from *one* master formula with *one* seed:

$$\hat{m}_{f,j} = \frac{v_{\rm geo}}{\sqrt{2}}\,\lambda_Y^{L_{f,j}}\,\Lambda_{f,j}, \qquad \lambda_Y = \sqrt{\varphi_0^{\rm ret}(1-\varphi_0^{\rm ret})}, \qquad \varphi_0^{\rm ret} = \frac{1}{6\pi} + \frac{3}{256\pi^4}.$$

The word-lengths $L_{f,j}$ are the *fixed residue matrix of the compiler*, whose characteristic polynomial $t^3 - 9t^2 + 10t - 8$ carries only compiler numbers ($\mathrm{tr} = 9 = N_{\rm fam}^2$, $\det = 8 = h(D_5)$, principal 2-minors 2,3,5 with product $30 = h(E_8)$). Through the $\varphi_0^{\rm ret}$-ladder $\hat{m} = \frac{v_{\rm geo}\pi}{\sqrt{2}} c\,(\varphi_0^{\rm ret})^k$ all nine word-lengths are fixed and the *charged-lepton* amplitudes are closed; the *quark* mass *ratios* are closed too (integer Plücker readouts on the derived selector stratum, v49/v71), with only the *absolute* amplitude scale reducing to the single dimensionful anchor $v_{\rm geo}$. **SM status:** the discrete packet, the dimensionless flavor skeleton, the charged-lepton source masses and the quark *ratios* are closed; the *absolute* quark scale and the full PMNS completion are reduced/conditional; $m_W, m_Z, m_H, \sin^2\theta_W, \alpha_s$ are RG scheme-layer (m_H rests on the closed UV quartic $\frac{1}{16\pi^2}$); and the solar angle is *realized by a TFPT texture* and reduced to the seam-misalignment lemma ($\sin^2\theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\rm ret}}{2}$; see below). *(Full derivation: document 2, `tfpt_2_standard_model`.)* [`verification/v18_quark_yukawa.py`] [`verification/v20_lepton_c_derivation.py`] [`verification/v46_grand_mass_volume.py`]

7 Physical, geometric, topological: why the architecture is plausible

The closure works on three levels at once: *physically*, α^{-1} is the fixed point of *one* equation and one generator drives many scales; *geometrically*, the C^+ spinor and the puncture geometry $\mathbb{P}^1\backslash\mu_4$ produce D_5 and A_3, glued to E_8; *topologically*, the common discriminant $\mathbb{Z}_4$ (the μ_4 winding) forces the glue and Φ_{30}/Coxeter drives the sectors — with exactly 2 axioms and the rest a consequence. **Why the compiled architecture is plausible.** It is (i) *parsimonious* (few independent assumptions ⇒ low overfitting risk), (ii) *rigid* (the glue condition $5/4 + 3/4 = 2$ and the discriminant $\mathbb{Z}_4$ leave little freedom), and (iii) *checkable* (machine-checkable E_8 appendix, explicit fixed-point equation for α^{-1}). More explained from less — exactly the criterion that makes a theory more probable.

8 What role E_8 now plays exactly

E_8 is no longer the starting point but the *output format*:

- **Bookkeeping:** E_8 is the unimodular container in which carrier (D_5) and family (A_3) fit together consistently. Its numbers 240, 248 are carrier traces, not inputs.
- **Consistency test:** unimodularity ($\det 1$) is a hard condition; it closes only because the discriminant-form norms 5/4 and 3/4 (both already present in TFPT) add up to 2. E_8 thus tests the internal coherence of the two atoms.
- **Not an end in itself:** E_8 does not explain "everything"; it is the smallest even unimodular hull of the datum $D_5 \oplus A_3$. The physics sits in the *two inputs*, not in E_8.

New — the secondary atlas. Beyond that, E_8 is an *audit container*: each maximal subalgebra projects a TFPT module. Document 3 (`tfpt_3_e8_audit_bootstrap`, Part I) shows the seven slices of 248 explicitly and labelled. The strongest new hit is that $E_6 \times A_2$ reads the flavor matrix:

$$248 = \|R\|_F^2 + \det R + 2\,(\mathbf{1}^\top R\,a)\,N_{\rm fam} = \underbrace{78}_{\dim E_6} + \underbrace{8}_{\dim A_2} + 2\cdot 27\cdot 3,$$

with $\|R\|_F^2 = 78 = \dim E_6$, $\det R = 8 = \dim A_2 = h(D_5)$. The discipline rule is: *every load-bearing number must appear in at least one E_8 projection* — this makes the structure falsifiable rather than numerological. (Audit level; all identities machine-verified.)

9 What this means for the theory-of-everything question

The state of the five questions — closed, reduced or typed

1. **Flavor residual — ratios closed; only the absolute scale is the wall-selection.** Mehta–Seshadri supplies the equivalence-class framework (stable parabolic ↔ unitary representation); the algebraic selector R, the Q spectrum and the splitting type are fixed (now *derived*: $\det R = 8$ lattice, $\mathrm{Spec}(Q_+)$ from D_4, v69/v70; since v136/v139 the dual pair (d, n) forces R *columnwise*, d is pure anchor data, and $\mathrm{Spec}(Q_+) = (1, 2, 3)$ is the A_3-exponent grading on the seam curve's cohomology, v137), and the charged-lepton amplitudes are closed. The quark mass *ratios* ($c_u/c_d = \frac{55}{117}$ etc.) are then *constant* on this discrete selector stratum (Readout Rigidity, v49/v71) — integer Plücker readouts, no transcendental solve. Only the *absolute* amplitude scale reduces to the selected *polystable* point on the parabolic stability wall (U_{point}, an anchor; Research Contract 1). [E]/[O]
2. **Gravity — the local field equation is parameter-free.** The metric-sector *field equation* is supplied by the entanglement first law $\delta S = \delta\langle K\rangle$ at fixed *volume* (Jacobson 2015): equilibrium forces the *full* covariant Einstein equation $G_{ab}+\Lambda g_{ab} = c_3^{-1} T_{ab}$ with *both* coefficients fixed — $8\pi{=}1/c_3$ (v358) and Λ from α, $\rho_\Lambda{=}(3/4\pi^2)e^{-2\alpha^{-1}}$ (v60) — the Einstein tensor (not Ricci) forced by Lovelock, so $\nabla^a T_{ab}{=}0$ is an *output* (v359). The $R{+}R^2$ scalaron is the *action*/inflation layer (the next-order term of the same equilibrium route, v360), and v_{geo} is the theory's single dimensionful input (v364). What remains is *not* the field equation but the equation-of-state status (Jacobson framing, not a from-action quantisation; an *external* candidate action — Bianconi's entropic action, arXiv:2408.14391 — is quantified in v473 without changing this typing) and the one unit v_{geo}; the ambient measure `QG.AMB.01` is now a [C] redundancy (v369, conditional on `SEAM.EQUIV.01`+Bisognano–Wichmann), not missing dynamics, and *perturbative* graviton unitarity is established — the Stelle ghost is a Seeley–DeWitt truncation artefact (v304), the spin-2 sector ghost-free via the entire KMS form factor (v370) with the truncation tower decoupling (v380). [E] (local field equation) / [C] (perturbative unitarity, ambient redundancy)
3. **Strong CP — decoupled.** $\theta_{\mathrm{eff}}{=}0$ follows from three structural facts (γ_5-Hermiticity, polar structure, sheet involution) + reflection positivity — *without* a mass gap. [E] (given RP)
4. **Full dynamics — the admissible transfer model is closed, and the boundary QFT is now one relative object.** The mass gap is the *explicit* discrete gap $6\log\frac{3}{2}$ of the C_6 transfer operator (eigenvalues $(k/3)^6$), which supports the OS reconstruction path on the admissible sector. On top of this the whole emergent-QFT round assembles the boundary theory into one relative object $\mathsf{TFPT}_{\mathrm{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\mathrm{rel}})$ — the *Modular Spectral Closure*: the finite Dirac is a covariance induction of the seam KMS state (v258), the spectral-action cutoff *is* that KMS weight (v259), the seam/carrier/E_8 live on one Kummer/K3 (v260), all certified together (v261) — closed modulo the single keystone `SEAM.EQUIV.01`, whose MMST route `SEAM.EQUIV.MMST.01` is *closed modulo cited theorems* (lattice + S_3 stack, Lean `FORM.SEAM.MMST.01`; v367/v368, v376–v379; the unconditional parent stays [O]). *What this does* not *fully close:* the full measured pole masses (the absolute scale v_{geo} + standard RG), the absolute QCD scales (m_p/m_e, now a typed [C] transfer, v374), the complete PMNS dynamics (the angles are typed; the full mass/CP dynamics stay [C]/[O]), and the real 4D interacting QFT — the continuum limit, whose cited existence theorems stay [O] — with the ambient measure a [C] redundancy (v369). [E] (transfer model + boundary relative object) / [C] (continuum modulo cited, transfers, ambient redundancy)
5. **Axioms P1/P2 — P2 machine-verified.** The Lean proof builds cleanly (`3357 jobs`, 0 `sorry`, only kernel axioms); the P2 algebra is a theorem, P1 a typed interface. They remain declared inputs. [E]/[O]

In other words: the theory has shrunk from "many open audit points" to "*standard steps* (cluster expansion, Fredholm, truncation bound) + two declared axioms". The five questions are now *closed, reduced or typed* rather than open-ended — a qualitative jump in discipline, not a claim of completion. *(Details and proof chains: document 2, `tfpt_2_standard_model`, Part III.)*

The closure ledger: what is done, and what genuinely remains

A complete theory must close the discrete structure, the dimensionless constants, the mass spectrum, gravity, and the quantum measure. The current state, graded honestly:

TOE piece	Status	If not complete: what closes it
gauge group, $N_{\rm fam}$, hypercharge, b_1	[E]	— complete
$E_8 = (D_5 \oplus A_3) + \mu_4$, group closure	[E]	— complete
bootstrap $(g,\mu)=(5,4)$, "8" in c_3	[E]	— complete (reverse glue $\mu^2-5\mu+4=0$)
fine structure α^{-1}	[E]	— **existence+uniqueness proved**; value is a numerical fixed point (ledger `EM.FP.01`), Ward origin [C]
fermion masses ($\varphi_0^{\rm ret}$-ladder)	[E]/[O]	leptons exact; quark *ratios* closed (Plücker, v49/v71); absolute scale = anchor
mixings $\theta_{13}, \theta_{23}, \theta_{12}$ (PMNS angles)	[E]/[C]	θ_{12} cond. derived; θ_{23} maximal; *full* PMNS dynamics (mass scale, CP magnitude) stay [C]/[O]
flavour matrix / H2	[E]	dictionary *exact* (hexagon = twisted family spectrum, addresses pinned, margins forced, v118–v122); R forced columnwise by (d,n) (v136/v139); pairing values = atom identities (v142/v145); residue = *one* lift map
seam quantum clock (R1)	[E]/[C]	typed closed: reduced edge-sector clock (v124–v133); external VW edge = 2× reduced budget, full 4d det = firewall (v138); det-ratio cancellation derived in-family (v144)
inflation A_s, n_s, r, scalaron M	[C]	scalaron $M = c_3^{7/2} \bar{M}_{\rm Pl}$ exact; n_s, r are bands over $N_\star \in [50, 60]$ (reheating point 51.4, v86)
boundary QFT (Modular Spectral Closure)	[E]/[C]	**one relative object** TFPT$_{\rm QFT}$: D_F = covariance induction (v258), cutoff = KMS weight $f_2/f_0{=}1$ (v259), seam/carrier/E_8 on one K3 (v260), certified (v261) — closed modulo `SEAM.EQUIV.01`, whose MMST route `SEAM.EQUIV.MMST.01` is closed modulo cited theorems (v376–v379; parent [O])
perturbative 4D S-matrix ($S_{\rm pert}$)	[E]/[C]	**Epstein–Glaser-constructible** (v269): one-loop quartic (v271) + gauge $\beta = (41/10, -19/6, -7)$ (v273), $S_{\rm pert} \xrightarrow{\rm LSZ} S_{\rm phys}$ with one-loop unitarity (optical theorem, v278); the spin-2 graviton sector is perturbatively unitary too — the Stelle ghost is a Seeley–DeWitt truncation artefact (v304/v370/v380)
the actual TOE remainder — dimensionful units, typed transfers, and the cited-theorem ceiling on the seam		
QFT measure (admissible sector)	[E]	**closed: RP from cusp spectrum, OS reconstruction**
full measured pole masses	[C]	ladder $\times\, v_{\rm geo}$, dressed by standard RG (no new physics); $v_{\rm geo}$ over-determined (v274)
absolute QCD scales (m_p/m_e, $\Lambda_{\rm QCD}$)	[C]	typed runnable transfer with uncertainty budget (v374, contract `FR.MPME.01`); not a compiler number
full PMNS dynamics (mass scale, CP)	[C]/[O]	angles typed; CP strength $J_{\rm PMNS}{=}{-0.0297}$ derived (v270); absolute ν-scale = one seesaw ratio (v272); three named post-hoc [O] candidates: third-gen ladder-base (v467), Δm^2 ratio = $\lvert J\rvert$ (v468), carrier-normalised one-parameter seesaw (v481) — record unchanged
(1) `SEAM.EQUIV.01` — one named theorem (*raw seam* $= (E_8)_1$ *at* $\tau{=}i$)	[O]/[C]	**route split 2026-07-22: closed if MMST or TWISTOR route closes. MMST route `SEAM.EQUIV.MMST.01` closed modulo cited theorems**: lattice model (v367/v368) + S_3 stack (v376–v379) pin it at every computable level, Lean `FORM.SEAM.MMST.01`; only residual [O] = the cited continuum existence (v336); extension leg certified (v469). Twistor route `SEAM.EQUIV.TWISTOR.01` (Costello–Li, via `CELEST.SEAM.01`) [O] ⇒ parent stays [O] as an unconditional claim
(2) `QG.AMB.01` — ambient measure (*redundancy, not missing dynamics*)	[C]	**discharged as a redundancy** (v369; v379 RP→OS), conditional on `SEAM.EQUIV.01`+Bisognano–Wichmann: a certification object, gap-decoupled (margin 1.648>0, v330/v335) so no readout depends on it; obstruction characterised (v332) + conditionally resolved by GHP+IDG (v334); Tier-B's holomorphy bit ($\det K{=}1$, v277) is absorbed into `SEAM.EQUIV.01`
analytic boundary origin of π ($c_3{=}\frac{1}{8\pi}$)	[C]	**hardened:** $c_3{=}1/(\lvert\mathbb{Z}_2\rvert \oint_{S^2} K) = 1/(8\pi)$ **(Gauss–Bonnet)**; "8"=rank E_8; π stays the one primitive
η_B, m_p/m_e, Koide, dark matter	[C]	typed runnable transfer solvers with kill tests (v371–v374, observatory CI v375; see document 4)

Marker key: ***[E]*** *exact / machine-proven ·* ***[C]*** *conditional (holds under named hypotheses) ·* ***[O]*** *open / axiom ·* ***[X]*** *falsifiable kill test. Per-claim fine type in* `verification/status_ledger.csv`*.*

The honest bottom line: the discrete core, the dimensionless constants, the *mass-hierarchy exponents* and the mixings are *closed*; the nine masses follow from one φ_0^{ret}-ladder (lepton rationals exact, quark $O(1)$ digits still a conditional finite evaluation). The *admissible-sector* QFT measure is closed (a finite gapped transfer system, RP from the cusp spectrum, OS-reconstructed), and on top of it the whole boundary QFT is now *one relative object* $\mathsf{TFPT}_{\mathrm{QFT}}$ (the *Modular Spectral Closure*, v258–v261: D_F a covariance induction, the cutoff the seam KMS weight, seam/carrier/E_8 on one Kummer/K3) — closed modulo the single keystone `SEAM.EQUIV.01`. *What that closure does not reach* is exactly the dimensionful/ambient remainder: the full measured pole masses (ladder $\times\, v_{\mathrm{geo}}$ + standard RG, [C]), the absolute QCD scales (m_p/m_e, now a typed [C] transfer, v374), the complete PMNS dynamics ([C]/[O]), and the nonperturbative *ambient* measure `QG.AMB.01`, now itself a [C] redundancy (v369). (The metric-sector *field equation* is parameter-free at the local level — the full covariant $G_{ab}+\Lambda g_{ab}=c_3^{-1}T_{ab}$ from fixed-volume entanglement equilibrium, v358/v359 — and the ambient *measure* is no longer an open structural item but a certification object, conditional on `SEAM.EQUIV.01`+Bisognano–Wichmann.) The *perturbative* 4D QFT is now built: the spectral-action S-matrix is Epstein–Glaser-constructible and LSZ-bridged to the physical asymptotic S-matrix with one-loop unitarity for the matter+gauge sector (v269–v278), and the spin-2 graviton sector is perturbatively unitary too — the R^2/Weyl^2 Stelle ghost is a Seeley–DeWitt *truncation artefact* (v304), ghost-free via the entire KMS form factor and the Barnes–Rivers decomposition (v370) with the truncation tower decoupling (v380); the whole bedrock reduces to one geometric postulate, *the raw seam is the flat $\tau=i$ pillowcase* (v276), with Tier-B's holomorphy bit $\det K=1$ (v277) absorbed into `SEAM.EQUIV.01`. The former single open lemma, the named theorem `SEAM.EQUIV.01` — *the raw reflection-positive seam state is the holomorphic $(E_8)_1$ net at $\tau=i$* — is now *closed modulo cited theorems on its MMST route* `SEAM.EQUIV.MMST.01` (the twistor route `SEAM.EQUIV.TWISTOR.01` and the unconditional parent stay [O]). (The flat $\tau=i$ geometry and the E_8 holomorphy are *two faces of one object*: the $(E_8)_1$ character is $\chi_{E_8}=j^{1/3}$, so $\chi_{E_8}(i)=1728^{1/3}=12$ and $\tau=i$ is at once the order-4 CM point and the $(E_8)_1$ partition-function modulus, v282.) The explicit lattice model (v367/v368) and the S_3 stack ($c=8$, 248 currents/1 primary, torus GSD=1, RP; v376–v379) pin it at every computable level, and it is Lean-formalised as `FORM.SEAM.MMST.01` (collar hypotheses kernel-proved, the MMST scaling-limit and Adamo–Moriwaki–Tanimoto OS-reconstruction theorems as named cited axioms, clean `#print axioms`); the *only* residual that stays [O] is the abstract continuum existence of the scaling limit (v336) — its 128-spinor extension leg since re-founded on the peer-reviewed crossed-product package ($h_s=16/16=1\in\mathbb{Z}$ is the Longo–Rehren locality criterion; KLM $\mu=4/2^2=1$; AGT/AMT the second witness), the realisation input reduced to invariants (R1′; computed $|C|=1$, $\nu=16$), Lean parallel route `seamResidualClosed′` (v469) — so it is closed *modulo a cited theorem*, not solved, Lean-pinned to exactly those named steps plus the derived Recovery gap $\Delta=6\ln(3/2)\approx 2.43>0$ (v308); the conformal-deck premise `QGEO.SYM.01` is its *downstream* definitional face (v323). This is by design the *one* irreducible structural input of the theory — the role the *constancy of c* plays in relativity. It is not a defect to be removed but the single, sharply-localised, falsifiable premise on which the whole boundary closure turns; with the lattice/S_3 pinning and the Lean formalisation in place, "closing the theory" has reduced to invoking exactly the two cited continuum-existence theorems, not an open-ended search. The full sprint-by-sprint reduction that pinned `SEAM.EQUIV.01` is recorded in the changelog, not replayed here. The seam seed is *hardened*: $c_3=1/(|\mathbb{Z}_2|\oint_{S^2}K\,dA)=1/(8\pi)$ is the one-sided Gauss–Bonnet normalisation of the seam sphere, with the discrete "8" equal to rank E_8 and only the Gauss–Bonnet π left as the single irreducible primitive. What remains for a full TOE is then the cited continuum-existence theorem on the seam and the one unit v_{geo}, plus the standard-RG dimensionful masses and the typed frontier transfers (the ambient QG measure being a [C] redundancy, v369). No load-bearing discrete parameter remains free, and π is the lone continuous primitive. *4D-QFT fork policy (frozen, v265):* the canonical 4D reading is *boundary only* — a 4D-GUT is *not claimed by default* (E_8 is the audit hull; SM-only unification is killed, v246); carrier-native Pati–Salam is a separately-typed, falsifiable UV branch (thresholds + proton decay). Not an open ambiguity, a decision tree.

10 TFPT in the light of algorithmic physics: Zuse–Schmidhuber–Wolfram

The "compiler" framing places TFPT in an older lineage — the idea that physics is, at bottom, the output of a short computation — and it is worth stating precisely what TFPT shares with that lineage and where it deliberately parts company, so the family resemblance is not mistaken for a claim the theory does not make. Three landmarks fix the coordinates. **Zuse** (*Rechnender Raum*, 1969) proposed that the universe is literally a (cellular-automaton) computation. **Schmidhuber** sharpened this into an *algorithmic* theory of everything (1997, 2000): the shortest program that outputs a universe is the natural prior — a physics reading of Kolmogorov complexity and Solomonoff induction — while his *Speed Prior* (2002) adds that a computed universe should be one that is *fast* to compute, and his low-complexity-art programme (1997) identifies beauty with compressibility. **Wolfram** (*A New Kind of Science*, 2002; the Physics Project) recasts the same instinct combinatorially: space–time as a hypergraph updated by simple rewrite rules.

Three shared instincts

1. **Minimal description length as the selection principle.** "No load-bearing free number, only π" is, in these terms, a Kolmogorov-style compression statement — the plausibility discussion below already names it *Kolmogorov-like compression: many numbers from little* — and the grammar-internal null model of the next section turns it from rhetoric into a measured bound.
2. **Deterministic, resource-bounded generation.** The readouts fall out of a chain of exact identities in about a second, not a search over a model space — the same spirit as the Speed Prior's preference for *fast*-computable worlds, though TFPT makes no claim to instantiate a speed prior.
3. **Compressibility as the aesthetic.** "Choosing E_8 is choosing the icosahedron" selects the maximally symmetric, shortest-to-describe object; the affine Kac marks $(1, 2, 3, 4, 5, 6, 4, 2, 3)$ carry a fourfold identity (representation dimensions = network attractor = spectral certificate = local balance weight). This is Schmidhuber's *beauty = compressibility*, realised on the Standard-Model alphabet rather than on pixels.

Three deliberate divergences

(A) **One forced program, not an ensemble.** Schmidhuber's shortest program computes *all* universes and privileges none; TFPT bets the opposite — an over-determined bootstrap fixed point that *forces* a single dimensionless skeleton (g_{car}=5 fixed three independent ways). TFPT is a candidate for *the* short program of our constants, not a measure over programs.

(B) **The description language is algebraic.** TFPT's compression is measured in a Lie/lattice/VOA language, not in Turing bit-strings; Kolmogorov complexity is language-dependent up to an additive constant, so the honest question — is the shortness real, or hidden in the E_8 machinery? — is exactly what the null model and the "forced three ways" bootstrap are built to answer. A Wolfram-style rewrite reading sharpens the same point rather than dissolving it: the $(2, 3, 5)$ seed is the terminal of a minimal $T(2, 3, r)$ growth rule (Cartan determinants $3, 2, 1, 0, -1$ for $E_6, E_7, E_8, \hat{E}_8, E_{10}$), so the rewrite picture *re-formulates* axiom P2 rather than removing it — both pictures bottom out on the same single irreducible seed.

(C) **Falsifiability — the one thing TFPT adds.** As *physics*, the Zuse/Schmidhuber algorithmic TOE and Wolfram's rule search are essentially unfalsifiable: no dated measurement can kill them. TFPT freezes dated, kill-tested predictions (the registry `predictions_frozen.json`, the predictions section below) and so puts the computational-universe intuition at genuine empirical risk. That, not the compression rhetoric, is the load-bearing difference.

Status. This section is *positioning*, not a new result: it introduces no formula and moves no status marker. Its machine-checked hooks already live in the suite — the grammar-internal null-model bound and the icosahedral/McKay bedrock with its network attractor; the Wolfram-style rewrite correspondence is worked out separately in the repository's hypergraph note.

11 Does this make the theory more probable?

Plausibility balance. *Pro:* (a) two inputs instead of many; (b) α^{-1} to 2.9×10^{-10} (CODATA-2022, 1.9σ) as a *fixed point*, not a fit; (c) E_8 constructed, not postulated; (d) one compiler generates several sectors ⇒ fewer degrees of freedom, more predictive power.

Contra/open: what remains are only *standard steps* (cluster expansion in the thermodynamic limit, the Fredholm property, the truncation bound) and the two declared axioms P1/P2; on the SM side the dimensionful EW/QCD masses $(m_W, m_Z, m_H, \sin^2\theta_W, \alpha_s)$ sit on the RG scheme layer. The solar angle, previously the one open SM number, is now *conditionally derived* ($\frac{1}{3} - \frac{\varphi_0^{\mathrm{ret}}}{2}$, modulo the seam-misalignment lemma). A final verdict "it maps reality" requires this short list plus the near-term experimental tests (below).

Net: the parsimony (Kolmogorov-like compression: many numbers from little) and the high precision of individual closures raise the a-posteriori plausibility substantially — without hiding the open points.

12 Predictions for running and upcoming experiments

Because *one* generator now sits behind the numbers, the theory makes a set of concrete, falsifiable statements that running or near-term experiments are actively testing. Each row gives the TFPT value, the *new* derivation behind it, the experiment, and an honest status.

Observable	**TFPT value & derivation**	**Experiment (running / upcoming)**	**St.**
solar angle $\sin^2\theta_{12}$	prediction of record $\frac{1}{3}-\frac{\varphi_0^{\rm ret}}{2}=0.306747$; TBM + seam misalignment $\varepsilon=\frac{3}{4}\varphi_0^{\rm ret}=c_3+36c_3^4$ (cond.)	**JUNO**: first 59.1 d run 0.3092±0.0087 (compatible), precision phase pending	[E]/[C]
reactor angle $\sin^2\theta_{13}$	$\varphi_0^{\rm ret}e^{-5/6}=0.0231$; seed × carrier-trace $e^{-\gamma}$, $\gamma=\frac{5}{6}$	Daya Bay / RENO / JUNO (PDG 0.0220)	[E]
atmospheric $\sin^2\theta_{23}$	$\approx\frac{1}{2}$ ($\mu\tau$-symmetric limit; octant not selected)	NOvA / T2K / **DUNE** (octant ambiguity; NuFIT 6.0 pre-JUNO baseline)	[C]
fine structure α^{-1}	137.0359992 as the unique root of $F_{U(1)}(\alpha)$=0 (no fit, no second branch)	CODATA-2022 (data through 2022) 137.035999177(21): dev. 2.9×10^{-10} (1.9σ of its uncertainty)	[E]
tensor ratio r	$r=\frac{12}{N_\star^2}\in[0.0033,0.0048]$ (frozen band, $N_\star\in[50,60]$); scalaron-reheating point 0.0045 ($N_\star$=51.4, v86)	now r<0.036 (BICEP/Keck BK18); **CMB-S4** ($\sigma_r\le5\times10^{-4}$, obs. ∼2033)	[C]
tilt n_s	$n_s=1-\frac{2}{N_\star}\in[0.960,0.967]$ (same band); scalaron-reheating point 0.9611 ($N_\star$=51.4, v86)	Planck (0.9649±0.0042); CMB-S4 sharpens	[C]
amplitude A_s	$\frac{N_\star^2}{24\pi^2}c_3^7$; 2.0×10^{-9} at $N_\star$=55 (the measured A_s prefers fast reheating, v86)	Planck (2.1×10^{-9})	[C]
neutron EDM	$\theta_{\rm eff}=0\Rightarrow d_n$ essentially null (structure + RP)	**PSI nEDM** / SNS (running)	[E]
CP phase $\delta_{\rm CKM}$	$\delta=\frac{\pi}{3}+3\lambda^2=68.65°$ (frozen of record, v88); survives at +0.98σ vs $\gamma_{\rm PDG}$	**LHCb** / Belle II γ combination; decision at $\sigma_\gamma\lesssim0.96°$	[E]
cosmic birefringence β	$\beta=\frac{\varphi_0^{\rm ret}}{4\pi}=0.2424°$ (seam readout)	ACT DR6 hint 0.215°±0.074° (systematics not yet understood — not a validation target); **Simons Obs.** / LiteBIRD	[C]
$0\nu\beta\beta$ / ordering	Majorana branch; normal-ordering preference	**LEGEND / nEXO**, DUNE, KATRIN	[C]
flavor invariants	$\det R$=$h(D_5)$=8, principal minors $(2,3,5)$, χ_R=t^3−$9t^2$+$10t$−8 (exact)	any future CKM/PMNS global fit must satisfy these	[E]
H_0 vs. Λ	$v_{\rm EW}\sim e^{-\alpha^{-1}/5}$, $\Lambda\sim e^{-2\alpha^{-1}}$, $H_0\sim\sqrt{\Lambda}$: one engine	SH0ES / **DESI** / Planck (Hubble tension)	[C]

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.

The two decisive near-term tests

(1) JUNO and θ_{12}. JUNO is *taking data since August 2025* and targets sub-percent $\sin^2\theta_{12}$. TFPT commits to the single prediction of record $\frac{1}{3}-\frac{\varphi_0^{\rm ret}}{2}=0.306747$ (misalignment $\varepsilon=\frac{3}{4}\varphi_0^{\rm ret}=c_3+36c_3^4$). The first 59.1-day JUNO run reports $\sin^2\theta_{12}=0.3092\pm0.0087$ — *compatible*, but the precision phase is still pending; this is a first compatibility check, not yet the decisive test. A converged central value away from ~0.307 at high significance falsifies the seam mechanism.

(2) r. The R^2 scalaron with M fixed by c_3^7 predicts a small $r=12/N_\star^2$: the frozen band $r\in[0.0033,0.0048]$ over $N_\star\in[50,60]$, with the scalaron-reheating point $r=0.0045$ ($N_\star$=51.4, v86) — already below the current bound $r<0.036$ (BICEP/Keck BK18) and within reach of CMB-S4 ($\sigma_r\le5\times10^{-4}$, first observations ∼2033–34). A detection of $r\gtrsim0.01$ would be incompatible with the R^2 branch on which $M_{\rm Pl}$ and A_s rest.

Structural (non-experimental) checks. Independent of any apparatus, the machine-checkable E_8 appendix (all 240 roots, Gram matrices, discriminant groups) lets anyone verify that the two TFPT atoms really generate E_8; and the discipline rule "*every load-bearing number appears in at least one E_8 projection*" (atlas note) turns the number stock into a falsifiable raster rather than numerology. The numerology charge is additionally *quantified* by an explicit null model: under the declared formula grammar the alignment density is anomalously high — a random theory of equal formula complexity essentially does not reproduce the scorecard. This is a *grammar-internal null-model bound*, not a Bayesian probability that TFPT is true; the explicit figure and its controls are stated in the verification appendix **[E]** [`verification/v100_numerology_null_mc.py`].

Freeze file: committed kill criteria

To prevent post-hoc rescue, the predictions above are frozen with explicit kill criteria in the machine-readable registry `verification/freeze_file.csv`. The decisive entries:

- **solar** $\sin^2\theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} \approx 0.3067$ — a JUNO central value clearly away from 0.307 kills the seam-misalignment mechanism;
- **tensor** $r \approx 0.0040$ — any robust $r \gtrsim 0.01$ kills the R^2 branch carrying M_{Pl}, A_s;
- **ordering** normal, small $m_{\beta\beta}$ — inverted ordering or large $m_{\beta\beta}$ kills the Majorana branch;
- **strong CP** $\theta_{\text{eff}} = 0$ — a solid nEDM signal above SM background falsifies the structural cancellation;
- $w = -1$ — a robust $w \neq -1$ kills the single-engine dark-energy readout.

The proton/electron ratio is explicitly *not* claimed as a compiler power (only fails if mis-asserted as one).

Blind-prediction registry (frozen 2026-06-09, machine-enforced). Beyond the kill criteria, the *values* themselves are now frozen: `verification/predictions_frozen.json` registers every dimensionless prediction of record at 25 significant digits *before* JUNO / CMB-S4 / LiteBIRD / DESI decide, and [`verification/v84_frozen_registry.py`] re-derives each decimal from the two axioms on every suite run (formula↔value lock; covered by `manifest.sha256`; ledger `REG.FREEZE.01`). In particular the registry admits exactly *one* θ_{12} prediction of record (the seed $\frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} = 0.306747$) — the seam and non-linear variants are typed as derived variants of the same texture and can never be silently promoted afterwards. r and n_s enter only as *bands* over the declared external $N_\star \in [50, 60]$, never as point predictions.

What likely follows next.

- the quark mass *ratios* are now closed combinatorially (integer Plücker readouts on the derived selector stratum, v49/v69/v71); the only residual is the *absolute* amplitude scale, an anchor (U_{point}) of the same nature as the one dimensionful scale — no new physics;
- further constants as carrier traces (analogous to $b_1 = 41/10$), since the compiler systematically produces traces over S^+;
- a sharper geometry↔gravity bridge once the Hodge/R^2 branch is lifted from **[C]** to **[E]**.

Conclusion

TFPT's sectors form an impressive collection of closures sharing common numbers. The compiler closure explains *why* they are the same numbers: a small discrete generator $\mathbb{Z}_{30} = 2 \cdot 3 \cdot 5$ builds, from two inputs $\{c_3, g_{\text{car}}\}$ via $D_5 \oplus A_3 + \mu_4$, the E_8 container and outputs the Standard Model, the constants and the scale grammar from there. The theory thereby becomes *more parsimonious, more rigid and more checkable.* Precisely nameable gaps remain — one geometric flavor realisation, the conditional gravity/QFT closures, and two declared axioms — but the path to closure is now short and concrete.

Appendix: computational verification (reproducibility)

Every claim in this document marked **[E]** (exact identity), **[E]** (Lie/lattice theorem) or **[E]** (numerical fixed point) is re-derived from the two axioms $\{c_3, g_{\text{car}}\}$ alone by a small, self-contained Python suite in the repository folder `verification/`. The inline references `[verification/...]` throughout the documents point to the exact script; the table below is the master map (its source is the shared fragment `tex-artefacts/verification.tex`).

Script	**What it machine-checks**
`v1_e8_glue.py`	glue theorem $E_8{=}(D_5{\oplus}A_3){+}\mu_4$: disc $=\mathbb{Z}_4$, $q(D_5){+}q(A_3){=}\frac{5}{4}{+}\frac{3}{4}{=}2$, $240{=}16{\cdot}5{\cdot}3$, $248{=}240{+}8$
`v2_carrier_pascal.py`	$g_{\text{car}}{=}5$ unique Pascal solution; $16{=}1{+}5{+}10$, $N_{\text{fam}}{=}3$, rank $E_8{=}8$, $\Omega_{\text{adm}}{=}48$, $b_1{=}\frac{41}{10}$
`v3_em_alpha.py`	EM closure: unique root of $F_{U(1)}(\alpha){=}0$, $\alpha^{-1}{=}137.0359992168$, existence/uniqueness, inverse test, ablation
`v4_flavor_matrix.py`	residue matrix R: $\det{=}8$, minors $\{2,3,5\}$, $\chi_R{=}t^3{-}9t^2{+}10t{-}8$, SNF diag$(1,1,8)$, $\lVert R\rVert_F^2{=}78$, $\sum L{=}40$
`v5_e8_cascade.py`	cascade $D_n{=}60{-}2n$: endpoints, exponent rungs $\to 240$, IR tail 46080, variance 6552
`v6_bootstrap.py`	reverse glue $\mu^2{-}5\mu{+}4{=}0$; $g_{\text{car}}{=}5$ three ways; $8{=}\operatorname{rank} E_8{=}h(D_5){=}\varphi(30)$
`v7_gravity_cosmo.py`	scalaron exponent 7, $M_{\text{scal}}^2/\bar{M}_{\text{Pl}}^2{=}c_3^7$, A_s, n_s, r, Ω_b, $\sin^2\theta_{12}$
`v8_horizon.py`	horizon code: $\frac{1}{2\pi}{=}4c_3$, $1920{=}\lvert W(D_5)\rvert$, S_{dS}, Page, $\beta_{\text{rad}}{=}0.2424°$
`v9_neutrino_texture.py`	solar angle: explicit $\mu\tau$ Majorana texture $\to \sin^2\theta_{12}{=}\frac{1}{3}{-}\frac{\varphi_0^{\text{ret}}}{2}$, $\theta_{23}{=}45°$, $\theta_{13}{=}0$
`v10_projection_involution.py`	Q,Σ algebra: $K{=}R{+}Q\Sigma$, $L{=}R{+}Q(I{+}\Sigma)$; χ_Q, χ_K; det ladder $3{\cdot}4{\cdot}8{\cdot}20{=}1920{=}\lvert W(D_5)\rvert$; $a^\top Ka{=}41$
`v11_unique_KQ.py`	K, Q are the *unique* nonneg-int matrices with their compiler budgets, spectrum and monotone rows (enumeration)
`v12_mass_generation_polynomials.py`	sector/generation polynomials of K and the anchor-block det ladder $(9,10,16,40)$
`v13_open_gates.py`	gate closures: $M{=}41{=}10b_1{=}\sum L{+}N_\Phi$; $Q_+ = A_3$ exponents, $Q_-^2{=}N_{\text{fam}}$
`v14_carrier_uniqueness.py`	$g_{\text{car}}{=}5$ unique ($N_{\text{fam}} \in \mathbb{Z}^+$, rank $E_8{=}8$); split $(3,2)$ from $b{+}s{=}5, b^2{+}s^2{=}13$; $\operatorname{Tr} Y{=}\operatorname{Tr} Y^3{=}0$
`v15_bootstrap_classification.py`	$D_5{\oplus}A_3$ unique familyful cyclic-glue of E_8 (16-spinor + 3 families), glue $\mathbb{Z}_4$
`v16_solar_dual_anchor.py`	$a^\top R^{-1}{=}a^\top L^{-1}{=}({-}\frac{1}{2},{-}\frac{1}{2},1)$; $\sin^2\theta_{12}{=}\frac{1}{3}{-}\frac{\varphi_0^{\text{ret}}}{2}$; full PMNS
`v17_hexagonal_resolvent.py`	finite resolvent $D_y^{-1}{=}(\sum y^{5-m}\delta^m U_6^m)/(y^6{-}\delta^6)$ (quark c backbone)
`v18_quark_yukawa.py`	quark source ratios $\frac{m_u}{m_d}{=}0.470085$, $\frac{m_c}{m_s}{=}13.61$, $\frac{m_t}{m_b}{=}40.80$ exact; lepton $c{=}(\frac{16}{7},\frac{4}{3},\frac{7}{6})$; full hierarchy
`v19_monodromy_moduli.py`	exact pole $z_\star{=}\frac{794-7\sqrt{9961}}{2187}$ (3^7); D_4 $SU(3)$ monodromy constructible but D_4-fixed locus positive-dim $\Rightarrow$ exact quark c reduce to (U)
`v20_lepton_c_derivation.py`	lepton c's derived: $\delta{=}\frac{1}{2}$ resolvent $c{=}\lvert\mu_4\rvert^w/(\frac{5}{4}{-}\cos\frac{r\pi}{3}) \Rightarrow \frac{16}{7}, \frac{4}{3}$; product $\frac{2^{g_{\text{car}}}}{N_{\text{fam}}^2}{=}\frac{32}{9} \Rightarrow \frac{7}{6}$
`v21_solar_product_quark.py`	solar coeff $= q(A_3){=}\frac{3}{4}$ (glue-norm, $q(D_5){+}q(A_3){=}2$), $\sin^2\theta_{12}{=}\frac{1}{3}{-}\frac{\varphi_0^{\text{ret}}}{2}$; product $\frac{2^{g_{\text{car}}}}{N_{\text{fam}}^2}$; quark law fails ($\frac{1}{7}$ vs 0.724)
`v22_open_gates_audit.py`	residual-gates audit contract: exact reduction of each open gate (A2,B3,B4,B5,B6,C7) machine-pinned (forced part vs. named residual)
`v23_anchor_generator.py`	anchor $a{=}(1,1,2)$: $e_k{=}(4,5,2)$, $c_3{=}\frac{1}{2e_1\pi}$; $p_n{=}2{+}2^n \Rightarrow 240, 248$, binary ladder; inputs $\to \{a,\pi\}$
`v24_quark_ratio_closure.py`	quark ratios $\frac{55}{117}, \frac{34}{47}, \frac{3}{26}$ from TFPT blocks; mass ratios match $< 0.03\%$; rational c gauge, Λ_q in $O(1)$
`v25_frontier_conjectures.py`	conjectures: Koide $Q_{\text{src}}{+}\frac{\varphi_0^{\text{ret}}}{24}{\approx}\frac{2}{3}$; axion $f_a{=}M_{\text{scal}}/128{\to}m_a{\approx}23.8\,\mu$eV

continued on next page

Script	**What it machine-checks** *(continued)*
`v26_flavor_frontier_unification.py`	the “11” is not uniquely forced (⇒ [C]); flavor frontier unifies to one (U) gate; cusp config on the parabolic stability wall
`v27_wall_representative.py`	explicit balanced wall rep W_{wall} (cols = perm$(0,1,2)$, rows $w{=}(2,1,1)$, pardeg 0); selector $\det R{=}8$, $\mathrm{Spec}(Q_+){=}\{1,2,3\}$
`v28_gravity_fR.py`	$R{+}R^2$ closed: $f(R)$, $M{=}c_3^{7/2}\bar M_{\mathrm{Pl}}$ scalaron, $N_\star{=}57\to n_s,r,A_s$; full metric measure open; $248c_3^2{<}6\log\frac{3}{2}$
`v29_research_contract_certs.py`	research-contract certificates: $C_U^{(1)}$ wall enumeration (1296→144→5 orbits; symmetry alone does not pick W_{wall}) and $G5$ gap-dominance $2\|V\|<\Delta$
`v30_d4_character_variety.py`	$C_U^{(2)}$: full D_4-fixed $SU(3)$ character variety is positive-dimensional ⇒ selector needs the $R(\rho)$ dictionary (H2; exact since v118–v122, residue = the R4′ selector establishment)
`v31_R_dictionary.py`	$R(\rho)$ characterized: case A alive (irreducible ⇒ mixing); R integer but ρ-invariants continuous ⇒ $R(\rho)$ is the transcendental non-abelian-Hodge map (residual = Hitchin solve)
`v32_rh_splitting.py`	RH route: $\sum_k U^kA_0U^{-k}{=}4\,\mathrm{diag}(A_0)$ exact ⇒ splitting $\mathcal{O}(-2)\oplus\mathcal{O}(-1)^2\Leftrightarrow\mathrm{diag}(A_0){=}(\frac{1}{2},\frac{1}{4},\frac{1}{4})$ (algebraic); $\prod M_k{=}I$ + R-bridge stay open (c_u/c_d not forced)
`v33_explicit_flat_bundle.py`	explicit valid (U_{wall}) flat bundle (RH solve): cusp + $\mathcal{O}(-2)\oplus\mathcal{O}(-1)^2$ + $\|M_\infty{-}I\|{\sim}10^{-9}$ ($\prod M_k{=}I$) + irreducible (case A)
`v34_h2_bridge_attempt.py`	H2 bridge: explicit per-puncture M_k (cusp, $\prod M_k{=}I$); honest negative — natural extraction ≠ lepton amplitudes, the $\Gamma^{\min}$ dictionary is the remaining input (c_u/c_d not obtained)
`v35_quark_qcd_boundary.py`	the open gates are the discrete↔continuous boundary: quark cross-ratio cleanness tracks generation/QCD sensitivity (compiler hands off to continuous physics)
`v36_spectral_action_g2.py`	G2: spectral action → $R{+}R^2$ ($a_2{\sim}{-}R/3$, $a_4\|_{R^2}{\sim}R^2/72$); $M^2/\bar M_{\mathrm{Pl}}^2{=}6(4\pi)^2/f_0$, c_3^7 fixes f_0; gap-decoupling $\Delta_{\mathrm{eff}}{=}1.648{>}0$ ⇒ admissible IR closed under RP, ambient (G6) = strict-TOE completion target
`v37_plucker_anchor.py`	anchor-plane Plücker apparatus: $\|\mathrm{Pl}(K)\|_1{=}11$, $\|\mathrm{Pl}_R(K)\|_1{=}26$, $\sum\mathrm{Pl}_R(L){=}60$; pencil $\det(K{+}xQ){=}3x^3{+}7x^2{+}6x{+}4$; $K_e{-}K_u{=}a$, $L_e{-}L_u{=}\mathrm{Exp}(A_3)$; lepton ring $c_ec_\tau{=}\|\mathbb{Z}_2\|c_\mu$; Koide $u\to\frac{53}{54}u$ ([C])
`v38_uwall_killswitch.py`	(U_{wall}) U2 kill-switch hardened: D_4-cusp rep irreducible at all 400 sampled points (commutant=1), mixing ≥0.35, reducible locus measure-zero ⇒ case A generic, (U_{wall}) survives; amplitude dictionary still research-level
`v39_uwall_selectors.py`	(U_{wall}) selector side closed on the explicit point: splitting type = anchor $a{=}(1,1,2)$, pardeg $=0$; $\det R{=}8{=}n{\cdot}a$, $\mathrm{Spec}(Q_+){=}\{1,2,3\}{=}3\alpha{+}1$ read off the bundle; only the amplitudes c_u/c_d ($\Gamma^{\min}$ Hitchin) remain
`v40_harmonic_metric.py`	(U_{wall}) harmonic metric = FINITE linear algebra (polystable⇒unitary⇒$\Phi{=}0$): unique invariant form $H{>}0$, unitarised $\widetilde{M}_k$ cusp-class with $\|\widetilde{M}_k\|{\in}\{0,\frac{1}{2},\frac{1}{\sqrt{2}}\}$; the feared Hitchin PDE collapses to unitarisation + combinatorial R
`v41_leg_assignment.py`	(U_{wall}) final leg test: amplitudes = $\delta{=}\frac{1}{2}$ resolvent (closed); $\Lambda_\mu{>}1$ rules out holonomy moduli; harmonic holonomy $\|\mathrm{diag}\|{=}(0,\frac{1}{2},\frac{1}{2})$ supplies $\delta{=}\frac{1}{2}$ (suggestive); honest negative on literal c_u/c_d reproduction
`v42_exterior_leg.py`	(U_{wall}) Exterior Leg Lemma: u,d share $(r,w){=}(1,1)$ so any scalar leg gives $c_u/c_d{\equiv}1$; the u/d split is the Λ^2F area $\mathrm{Pl}_{\mathbf{1},a}(K)$, $\|\cdot\|_1{=}11$ generated ⇒ $\frac{5\cdot11}{9\cdot13}{=}\frac{55}{117}$ [E]; phys. identification [C]
`v43_exterior_bridge.py`	(U_{wall}) Λ^2F bridge: $\Lambda^2(\widetilde{M}){=}\overline{\widetilde{M}}$ (exterior leg $=\bar{\mathbf{3}}$) confirmed; continuous action ≠ integer $\mathrm{Pl}(K)$ ⇒ bridge is the discrete H2 invariant ($\det R{=}8$, $\mathrm{Spec}(Q_+)$ confirmed), not continuous

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Script	What it machine-checks *(continued)*
`v44_carrier_exterior.py`	($U_{\rm wall}$) Lie-level grounding: $16{=}\Lambda^{\rm even}(5){=}1{+}10{+}5$; $u^c{\in}\Lambda^2(5)$, $d^c{\in}\Lambda^4(5)$ $\Rightarrow$ exterior degrees $(2,4)$, sum=6=$\lvert R^+(A_3)\rvert$, diff=2=$\lvert\mathbb{Z}_2\rvert$; the exterior leg is the carrier's own Λ^2 grading (no special math)
`v45_family_exterior.py`	($U_{\rm wall}$) the "11" is Pascal: $\lvert{\rm Pl}(K)\rvert{=}(1,4,6){=}\{\Lambda^0,\Lambda^1,\Lambda^2\}(\mu_4)$; $\lVert{\rm Pl}(K)\rVert_1{=}11{=}\sum_{k\le 2}\binom{4}{k}{=}16{-}g_{\rm car}$ (cumulative ext of μ_4 up to deg $u^c{=}2$); $15{=}\dim\mathfrak{su}(4){=}10{+}5{=}N_{\rm fam}g_{\rm car}$
`v46_grand_mass_volume.py`	absolute mass-scaling is [E], not a product rule: sector dets $(\varphi_0^{\rm ret})^6,(\varphi_0^{\rm ret})^9,(\varphi_0^{\rm ret})^{10}$ (exponents $\lvert R^+A_3\rvert, N_{\rm fam}^2, \mathcal{A}_\Lambda$) $\Rightarrow$ $\det M_{\rm SM}{\sim}(\varphi_0^{\rm ret})^{25}{=}(\varphi_0^{\rm ret})^{g_{\rm car}^2}$; Q-rows $(4,5,6)$ sum $15{=}\dim A_3$; $25{+}15{=}40{=}\sum L$
`v47_selection_theorem.py`	Boundary Carrier Selection (Thm A): $\mu_4{\to}A_3$, $N_{\rm fam}{=}3$; 16-spinor$\to D_5$; $q(D_5){+}q(A_3){=}2$, glue $4{=}\lvert\mu_4\rvert$; only $D_5{\oplus}A_3$ meets all four conditions ($D_8, E_7{+}A_1, E_6{+}A_2, A_4{+}A_4, A_8$ excluded) [E]/[C]
`v48_em_ward.py`	EM boundary Ward (Thm C): $F_{U(1)}{=}$ Maxwell α^3 − Calderón $2c_3^3\alpha^2$ − transport $8b_1c_3^6\log\frac{1}{\varphi_{\rm seam}}$; $8b_1{=}\frac{4}{5}M{=}\frac{164}{5}$, $-\frac{5}{4}{=}q(D_5)$ [E]; Ward origin [C]
`v49_readout_rigidity.py`	Readout Rigidity (Thm U2): $c_u/c_d{=}\frac{55}{117}$ is constant on the discrete selector stratum $\mathcal{S}_{8,Q}$ $\Rightarrow$ no point-uniqueness needed; $(U_{\rm wall}){=}U_{\rm unitary}{+}U_{\rm H2}{+}U_{\Lambda^2}{+}U_{\rm point}$, only $U_{\rm point}$ open
`v50_q_geometry.py`	Q geometry (Thm Q): $Q_+{=}$ diag$(3,2,1)$-block $\chi{=}(t{-}1)(t{-}2)(t{-}3)$, Q_- $\chi{=}t(t^2{-}3)$; Q unique under rows$(4,5,6)$/cols$(9,5,1)$/spectra; geom. origin [C]
`v51_boundary_half_step.py`	glue norms $=(g_{\rm car},N_{\rm fam})/\lvert\mu_4\rvert$: $q(D_5){=}\frac{5}{4}$, $q(A_3){=}\frac{3}{4}$; four ops forced $\to$ {sum= 2 root norm, diff= $\frac{1}{2}{=}\lvert\mathbb{Z}_2\rvert/\lvert\mu_4\rvert{=}\delta$, prod= $\frac{15}{16}$, ratio= $\frac{5}{3}$}. So $\delta{=}\frac{1}{2}$ is the carrier−family count over the glue index, not chosen [E]
`v52_pencil_endpoints.py`	$K{+}xQ$ pencil endpoints: $P(-1){=}2{=}\lvert\mathbb{Z}_2\rvert$, $P(0){=}4{=}\lvert\mu_4\rvert$, $P(1){=}20{=}\det L$, $P(2){=}68{=}2p_5(a)$; $\det(K{-}Q){=}2$, tr $=N_{\rm fam}$ (audit ext. of v37) [E]
`v53_compiler_core.py`	Big picture: the whole integer skeleton from $(g_{\rm car},N_{\rm fam}){=}(5,3)$ — rank $E_8{=}8$, $\lvert\mathbb{Z}_2\rvert{=}2$, Pythagorean $\Delta_Y{=}g_{\rm car}^2{=}25{=}N_{\rm fam}^2{+}\lvert\mathbb{Z}_2\rvert{\cdot}$rank $E_8{=}9{+}16{=}\dim S^+$ (diff. of squares); anchor char-poly $(t{-}1)^2(t{-}2)$ unique [E]
`v54_seam_horizon_keystones.py`	The "8" in $c_3{=}\frac{1}{8\pi}$ overdetermined and gravitationally aligned ($2\lvert\mu_4\rvert{=}$rank $E_8{=}h(D_5){=}\varphi(30){=}8\pi$ Hawking/Einstein); one boundary transport operator (spec $\{1,(2/3)^6,(1/3)^6\}$, $\lambda_2{=}(2/3)^6$) governs *both* SM flavor gap and horizon Page recovery [E]
`v55_coxeter_cycle.py`	Computed E_8 Coxeter element: order exactly $30{=}\lvert\mathbb{Z}_2\rvert N_{\rm fam}g_{\rm car}$, eigenphases = exponents = totatives of 30, so rank $E_8{=}8{=}\varphi(30){=}$ live phases of the order-30 cycle; one α^{-1} sets EM, S_{dS} and Λ ($S_{dS}\rho_\Lambda{=}32\pi^4$) [E]
`v56_unique_attractor.py`	Gapped boundary transport (gap $6\log\frac{3}{2}{>}0$) $\Rightarrow$ *unique* attracting fixed point, geometric rate $(2/3)^6$ (two starts $\to$ same direction); Coxeter in $\lvert\mu_4\rvert{=}4$ planes; entropy reset = rank-1 projector [E]
`v57_horizon_crosslinks.py`	Horizon cross-links: $c_3{=}\frac{1}{8\pi}$ is the Einstein/Jacobson coefficient ($1/4{=}1/\lvert\mu_4\rvert$ entropy density); Hod QNM $\omega_R/T_H{=}\ln 3{=}\ln N_{\rm fam}$; Hawking power denom $1920{=}\lvert W(D_5)\rvert$; one α^{-1} for EM/S_{dS}/Λ/Hubble [E] (+ [C] identification)
`v58_seam_horizon_chain.py`	Seam-horizon chain (precise: seam = abstract normaliser, locally realised as a horizon): one-sided S^2 Gauss–Bonnet $c_3{=}1/(2{\cdot}4\pi){=}\frac{1}{8\pi}$; KMS $4c_3\kappa$; $S_{BH}{=}M^2/2c_3$; RT in seam units. Seam–Horizon Theorem (area-law from the seam kernel) [O] open
`v59_area_law_evidence.py`	Area-law necessary condition met: a reflection-positive Gaussian (Calderón-type) kernel yields an area law (gapped block-EE saturates; gapless $\sim\frac{c}{3}\log L$), by the *same* gap that fixes the attractor; $8{=}\lvert\mathbb{Z}_2\rvert\lvert\mu_4\rvert$. The c_3 coefficient stays [O] open
`v60_lambda_metrology_branch.py`	Λ branch selection: $(8\pi)^2\delta_{\rm top}{=}\frac{3}{4\pi^2}$, mis-scale $2c_3/\delta_{\rm top}{=}\frac{64\pi^3}{3}{=}661.47\Rightarrow G_N$ pinned as output; 123 orders = 119.028+3.920; ACT DR6 birefringence $0.215°{\pm}0.074°$ vs $0.2424°$ (0.37σ) [E]

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Script	What it machine-checks *(continued)*
`v61_cft_bridge.py`	Boundary-CFT mirror: level-1 WZW central charges $c(E_8)$=8, $c(D_5)$=5, $c(A_3)$=3 = (rank $E_8, g_{\text{car}}, N_{\text{fam}}$); conformal embedding c_{coset}=0; $SU(3)\leftrightarrow SU(4)$ reconciliation N_{fam}=$\lvert\mu_4\rvert$−1, 4→3+1 (one geometry $\mathbb{P}^1\backslash\mu_4$, two sides) **[E]**
`v62_data_scorecard.py`	TFPT vs 2024/25 data: α^{-1}, $\sin^2\theta_{12}$, β_{rad} match; $\sin^2\theta_{13}$ (2.0σ) and Starobinsky n_s vs DESI ($\sim 2\sigma$) mild tensions; θ_{23} open; $r\approx0.004$ future falsifier **[E]**
`v63_seam_engineering_index.py`	Exact Seam-Engineering Index $\Xi{=}2\lVert V\rVert/\Delta{=}\frac{31}{24\pi^2\log(3/2)}{\approx}0.323$ ($2\lVert V\rVert{=}\frac{31}{4\pi^2}$, 248= $\dim E_8$, 31=1+$h^\vee$), $\Delta_{\text{eff}}\approx1.648$; four-class possibility taxonomy **[E]**/**[O]**
`v64_causal_boundary_nogo.py`	Causal Boundary Engineering *conditional no-go*: gapped transfer has no periodic orbit (discrete no-CTC); RP+OS+gap ⇒ ANEC ⇒ Topological Censorship ⇒ no traversable shortcut; only ER=EPR bridge survives **[E]** (core)/**[C]** (chain)
`v65_falsification_layer.py`	Prediction layer with explicit failure thresholds: 3 core (v_{GW}=c, no 4th gen, seam=8), 4 observational (β, r, n_s, θ_{12}), no-go + unitarity; $N_\star$ demoted to **[C]** input; none violated **[E]**
`v66_e8_casimir_degrees.py`	Organizing simplification: the compiler atoms *are* the E_8 Casimir degrees $\{2,8,12,14,18,20,24,30\}$ (2=$\lvert\mathbb{Z}_2\rvert$, 8= rank, 14= $\dim G_2$, 20= $\det L$, 24=$\lvert W(A_3)\rvert$, 30=$h(E_8)$); $\sum$=128=2^7 (dS prefactor), $\sum$exp=120=$\lvert R^+(E_8)\rvert$ **[E]**
`v67_area_law_coefficient.py`	Central-theorem *structural closure*: Fursaev–Solodukhin $S{=}4\pi kA$, $k{=}c_3/2$ ⇒ $S{=}2\pi c_3A{=}\frac{1}{4}A$ ($2\pi c_3{=}\frac{1}{4}$); $c_3{=}\frac{1}{8\pi}$ is the *unique* value with replica-coeff = BH $\frac{1}{4}$. Reduces to the one coeff k=$c_3/2$ (then *forced*, v73) **[E]**
`v68_seeley_dewitt_residual.py`	Residual *resolved as an anchor*: $a_2{=}-\frac{d}{192\pi^2}R$ ⇒ the *absolute* $\frac{1}{16\pi G}$ is UV-sensitive ($\propto d\Lambda^2 f_2$), so the *scale* $1/G$ is the one dimensionful anchor (the dimensionless k=$c_3/2$ is forced, v73); cutoff-indep. R^2/scalaron derived **[E]**
`v69_d4_q_geometry.py`	D_4-equivariant Q-geometry (gate **[C]**/**[E]**): on $H_1(\mathbb{P}^1\backslash\mu_4){=}B_1\oplus E$, $Q_+{=}3w{+}1{=}\operatorname{diag}(1,2,3)$ (cusp weights on $\mu_4{=}\mathbb{Z}_4$ eigenspaces), $Q_-{=}E$-coupling $\sqrt{N_{\text{fam}}}$ ⇒ $t(t^2{-}3)$; Σ-split $=D_4{=}\mathbb{Z}_4{\times}\mathbb{Z}_2$ **[E]**
`v70_q_integer_lift.py`	Q integer-lift sharpened: $\mathbb{Z}_4$ puncture action R unimodular (eigenvalues $\{-1,i,-i\}$); $\det Q{=}3{=}N_{\text{fam}}$, SNF diag(1, 1, 3) ⇒ residual = one lattice invariant $\det Q{=}N_{\text{fam}}$, not closable by D_4 alone **[E]**
`v71_simple_r_bridge.py`	*Simple* R-bridge (gate #3): selector stratum $\mathcal{S}_{8,Q}$ now fully derived ($\det R$=8= rank E_8 lattice + Spec(Q_+)={1, 2, 3} from v69) ⇒ quark *ratios* are pure integer Plücker readouts ($\frac{55}{117}$, no monodromy, v49 rigidity); residual U_{point}= absolute norm. = an anchor **[E]**/**[O]**
`v72_q_det_from_cusp.py`	$\det Q{=}N_{\text{fam}}$ derived from the cusp class: $\det Q{=}\lvert\operatorname{coker}Q\rvert$= cusp-class order = denominator of the weights $\{0,\frac{1}{3},\frac{2}{3}\}$ (same data as Spec Q_+); $\operatorname{coker}Q{=}\mathbb{Z}/N_{\text{fam}}$= triality deck group ⇒ integer-lift residual closed **[E]**
`v73_k_c3_half.py`	Central coefficient k=$c_3/2$ *forced*: (1/2) variational × $1/(\lvert\mathbb{Z}_2\rvert 2\pi\chi(S^2))$ Gauss–Bonnet topology ($\chi(S^2)$=2), both cutoff-indep ⇒ Fursaev–Solodukhin S=$A/4$; only the absolute 4D Newton scale stays the anchor (v68) **[E]**
`v74_compiler_micro_lemmas.py`	Spine quotient ladder ($\frac{2}{3},\frac{3}{4},\frac{4}{5},\frac{5}{4},\frac{4}{3}$ = adjacent quotients of (2, 3, 4, 5)); pencil differences (2, 16, 48)=($\lvert\mathbb{Z}_2\rvert$, $\dim S^+$, Ω_{adm}) sheet→1gen→3gens; anchor QF $a^\top Ka{-}\mathbf{1}^\top K\mathbf{1}{=}41{-}25{=}16{=}\dim S^+$ (EM budget = mass vol + one gen); solar dual anchor $L{=}R{+}61e_1^\top$, $a^\top R^{-1}\mathbf{1}{=}0$ (Sherman–Morrison rank-one invariance) **[E]**
`v75_upoint_to_vgeo.py`	**Gate 1 complete:** $U_{\text{point}}{\to}v_{\text{geo}}$. Ratios closed (v49/v71) + lepton c exact (v20) + sector products = φ_0^{ret}-powers (v46); (ratios, product)⇔(individuals) bijection ⇒ absolute amplitudes fixed up to *one* scale v_{geo} = the same anchor as $1/G$ (v68); two [A] → one **[E]**/**[O]**
`v76_gmetric_reduction.py`	**Gate 2:** Tier A (IR) closed under RP+gap (Decoupling Thm: $\Delta_{\text{eff}}{=}6\log\frac{3}{2}{-}\frac{31}{4\pi^2}{=}1.648{>}0$, $\lVert V\rVert{\le}248c_3^2{=}\frac{31}{8\pi^2}$); Tier B (strict TOE) holographically reduced from bulk to a finite seam-*boundary* measure (conditional: RP+tightness ⇒ closes) **[E]**/**[C]**

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Script	What it machine-checks *(continued)*
`v77_e8_conformal_net.py`	**G6 route:** the seam boundary measure = the $(E_8)_1$ *lattice net* (rigorous, holomorphic c=8); level-1 $c(E_8)=\frac{248}{31}$=8, $c(D_5)$=5, $c(A_3)$=3, embedding 5+3=8 (coset c=0); imports $G6$ into RCFT/conformal-net rigor [E]/[C]
`v78_vgeo_floor.py`	$v_{\rm geo}$ **floor:** no pure number is dimensionful $\Rightarrow$ $v_{\rm geo}$ irreducible by logic; all scales are ratios × one $\bar{M}_{\rm Pl}$; $S_{dS}\rho_\Lambda$=$32\pi^4$ pins it from one measurement; theory at the minimum (one scale + π) [E]/[O]
`v79_review_identities.py`	Four validated review identities + one firewall: hypercharge Lucas–Binet D_n=$(3^n-(-2)^n)/5$=$1,1,7,13,55,133$ (roots = carrier charges); Inverse Anchor Thm ($\mathbf{1}^\top M^{-1}\mathbf{1}$=1/atom, $a^\top M^{-1}a$=1 for R,K,L); MacWilliams $(1,10,5)$; $6Y^2-Y-1$=$(2Y-1)(3Y+1)$. Rejects m_p/m_e, η_B near-formulas (firewall) [E]
`v80_operator_pencil_geometry.py`	Operator-pencil geometry: Anchor Singularity Thm $\det B_{K+xQ}$=$(N_{\rm fam}x+\lvert\mathbb{Z}_2\rvert)(N_{\rm fam}x+g_{\rm car})$=$(3x+2)(3x+5)$ ($\frac{2}{3}$ Koide/gap = anchor-plane singularity); buildup chain $(3,10,16,20)$; type checker $\det B_{Q,K,R,L}$=$(9,10,16,40)$; audit: differential spectrum, $F_4\times G_2$ shadow 248=52+14+26·7 [E]/[C]
`v81_singular_pencil_matrices.py`	Double cover y^2= $\det B_{K+xQ}$=$(3x+2)(3x+5)$: Koide x=$-\frac{2}{3}$ and carrier x=$-\frac{5}{3}$ are the two branch points (deck 2=$\lvert\mathbb{Z}_2\rvert$; disc=81=$N_{\rm fam}^4$; separation=1 transport period). Clearing matrices $C_{2/3}$=$3K-2Q$ ($D_5\oplus A_3$ glue, B=$(5,7)^\top(9,11)$, $\sum$=240), $C_{5/3}$=$3K-5Q$ (neutral, χ=$(\lambda+2)(\lambda^2+\lambda+6)$); scalaron 7= mixed anchor disc/$N_{\rm fam}$ [E]
`v82_koide_attractor_splitting.py`	Koide attractor *forced*: the branch-divisor-preserving Möbius map fixing q=2, 5 is unique, multiplier $(2/3)^6$=λ_2 of the established transfer operator $\Rightarrow$ $F'(2)$<1 attractor, $F'(5)$>1 repeller; three postulates → one [C]. Splitting trichotomy disc 81/49/40: the clean split is non-generic (anchor-first) [E]/[C]
`v83_e8net_holomorphic_uniqueness.py`	Holomorphy pins $(E_8)_1$: #primaries = det Cartan (D_8:4 vs E_8:1), and a holomorphic c=8 chiral CFT is the lattice theory of the *unique* even unimodular rank-8 lattice (Minkowski–Siegel mass $1/\lvert W(E_8)\rvert$); carrier Pascal truncation K=$(g-1)/2$ = half-spinor split [E]
`v84_frozen_registry.py`	Blind-prediction registry frozen 2026-06-09 (25 digits, machine-enforced lock formula↔value): exactly *one* θ_{12} prediction of record; r, n_s as $N_\star$ *bands*; conditional entries carry their hypothesis by name [E]
`v85_master_cover.py`	Master-cover theorem: $\det B$ is GL(2)-covariant on span$\{K,Q\}$ $\Rightarrow$ exactly *one* double cover up to Möbius (disc = $N_{\rm fam}^4\det(G)^2$); the disc-81 family is its orbit ($-\frac{8}{3}$=$-$rank$E_8/N_{\rm fam}$ = carrier − one period); μ_4 is *not* a 4:1 cover (ladder of double covers); branch trace fixes only the scalaron *scale* [E]
`v86_nstar_reheating.py`	$N_\star$ band → point [C]: Γ=$4M^3/48\pi\bar{M}^2$=128 GeV, $T_{\rm reh}$=9.6×10^9 GeV, $N_\star(0.05/{\rm Mpc})$=51.4 $\Rightarrow$ n_s=0.9611, r=0.0045 (inside the frozen band); honest tensions: n_s +0.9σ, A_s dichotomy (slow point −11.4σ $\Rightarrow$ near-instant reheating or exponent fails) [C]
`v87_bulk_uniqueness_reduction.py`	Red-team Target A residual (ii) conditionally closed: holomorphic $\Rightarrow$ Rep = Vect $\Rightarrow$ 2D bulk *unique* (LR/KLM/BKLR); machine contrast $SO(16)_1$ admits *six* modular invariants (incl. both E_8 pairings) $\Rightarrow$ Target A = one residual [E]/[C]/[O]
`v88_cp_phase_audit.py`	Target-D CP residual quantified: frozen δ=$\pi/3+3\lambda^2$ survives at +0.98σ vs $\gamma_{\rm PDG}$; the 0.07°-near alternative $\pi/3+2\lambda^2$ recorded as look-elsewhere trap, *not* adopted; decision at $\sigma_\gamma\leq0.96°$ [E]
`v89_carrier_index_lemma.py`	Carrier Index Lemma: KLM μ_A=$[B{:}A]^2\mu_B$ $\Rightarrow$ Jones index $[(E_8)_1{:}(D_5)_1\times(A_3)_1]$=4=$\lvert\mu_4\rvert$ (glue order *is* the index); μ-additivity $16/4^2$=1 $\Rightarrow$ holomorphy *follows*; Gate A (H)⇔(H′) index-4 extension [E]
`v90_conical_defect_chain.py`	Fursaev–Solodukhin factor 4π *derived* (smoothed-cone Gauss–Bonnet, exactly smoothing-independent); replica S=$4\pi kA$, k=$c_3/2$ $\Rightarrow$ S=$A/4$ ⇔ c_3=$\frac{1}{8\pi}$ (sympy-solved); residual isolated to "seam determinant $\Rightarrow$ EH form" [E]/[O]

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Script	What it machine-checks *(continued)*
`v91_spine_tetrahedron.py`	Spine tetrahedron $\{2,3,4,5\}$ = anchor + zeroth moment; edges/faces/volume = the integer grammar, volume $120{=}\lvert R^+(E_8)\rvert$; $240{=}\lvert\mu_4\rvert\lvert E(K_4)\rvert\lvert E(K_5)\rvert$ with K_6 negative control; dual cuts typed tautological; sub-grammar incompleteness honest **[E]**
`v92_glue_uniqueness.py`	Glue uniqueness: of all 15 subgroups of the discriminant form exactly two Lagrangian glues (the two *chiralities*, swapped by either single spinor conjugation, fixed by the simultaneous one) and exactly one halfway extension = $SO(16)_1$; tower carrier$\to SO(16)_1\to(E_8)_1$, nothing else [E]
`v93_koide_relaxation_toy.py`	Koide relaxation toy (P2 narrowed): basin lemma (every physical configuration in the attractor basin); exact rate $(2/3)^6$; source at $\rho{=}{-}\varphi_0^{\mathrm{ret}}/24$ to 0.8% ([C]); honest negatives: flow time $t{=}2.84\notin\mathbb{Z}\Rightarrow$ the missing object is a *continuous* generator [E]/[C]
`v94_sheet_diamond.py`	Sheet diamond: all four flavor operators on *one* surface $M(s,t){=}R{+}Q\,\mathrm{diag}(s,t,t)$ (pencil = the cut $(1{+}x,x{-}1)$; diagonal-cut disc 153 negative control); winding line $\det B{=}2\det$ for all s; cofactor seam normal $(5,-9,6)$; Plücker canonicalisation: 11 *and* 26 both live in K; spine-quotient firewall *rejected* [E]
`v95_centered_diamond.py`	Centered cross: $Q{=}U{+}V$, $R/L{=}C{\mp}U$, $K/F{=}C{\mp}V$ (five objects → three); $\mathrm{Spec}\,V{=}\{0,1,2\}{=}N_{\mathrm{fam}}{\cdot}\mathrm{cusp}$; $\det C{=}14$, $\sum C{=}31{=}2^{g_{\mathrm{car}}}{-}1$; $\mathrm{Pl}_R(C){=}7(2,3,1)\parallel\mathrm{Pl}_R(L){=}10(2,3,1)$; G_2 reading audit-typed [E]
`v96_branch_kernel_selection.py`	Branch-kernel selection (P1 unblocked): rank-1 anchor blocks at the branch points have *integer* kernels (carrier kernel = $\mathbf{1}$); collapse lemma $\Rightarrow(-1,1,0)$: up = −down, *lepton pairing* = 0 (leptons on the ramification); sector zero ladder (up $-\frac{3}{2}$ doubly, down 0 & $-\frac{9}{5}$); anchor-placement controls [E]
`v97_sheet_conjugation_bridge.py`	Sheet-conjugation bridge (P1 → one [C]): the anchor forces the cusp conjugation uniquely ⇒ integer T_A, $\det{=}{-}1$; $a{=}e_2{+}e_3$ (conjugation-symmetric); one deck action on $\mathbb{R}^3/\,\mathrm{span}\{\mathbf{1},a\}$; $\langle T_A,\Sigma\rangle{=}D_4$; sheet-index lemma (intertwiner dets even, min $2{=}\lvert\mathbb{Z}_2\rvert$); remaining [C]: Q_+ grading = A_3 discriminant grading [E]/[C]
`v98_discriminant_dictionary.py`	Dictionary *derived* (P1 closed modulo GATE.QGEO): $G{=}T_A\Sigma$ acts integrally on the cusp basis ($Ge_1{=}{-}e_1$, $Ge_3{=}e_2$, $Ge_2{=}{-}e_3$ = $B_1\oplus E$); cusp-0 line = self-conjugate $\mathbb{Z}_4$-class-2 line; $T_AGT_A^{-1}{=}G^{-1}$ ($k{\mapsto}{-}k$); $q_{A_3}(1){=}q_{A_3}(3){=}\frac{3}{8}$, $q_{A_3}(2){=}\frac{1}{2}$; reflection classes = glue-swap vs glue-fix parities ⇒ P1 carries no separate [C]; residual = the one existing Q-geometry gate [E]
`v99_koide_flow_time.py`	Koide flow time: canonical generator $\dot q{=}(\Delta/N_{\mathrm{fam}})(q{-}2)(q{-}5){=}(\Delta/N_{\mathrm{fam}})\det B(q)$, time-1 map = F [E]; non-integrality of $t{=}2.84$ is *data-fragile* (Q crosses $\frac{2}{3}$ at $+0.43\sigma(m_\tau)$, t passes every integer ≥ 3 within $+0.5\sigma$) [E]; conditional steps: $n{=}2$ excluded (-2.9σ), $n{=}3{=}N_{\mathrm{fam}}\Rightarrow m_\tau{=}1776.9427\,\mathrm{MeV}$ $(+0.14\sigma)$, decision at $\sigma(m_\tau){\sim}0.01\,\mathrm{MeV}$ [C]; ln(46080) scale reading destroyed by controls (firewall)
`v100_numerology_null_mc.py`	Numerology null test (look-elsewhere corrected): exact census of the full declared formula grammar over the 13 scored frozen-registry observables (conservative data windows) gives joint formula-fishing probability $\prod p_i = 10^{-25.8}$, robust to budget/window variation; $2{\times}2{\cdot}10^5$ Monte-Carlo pseudo-theories reach at most 5/13 hits vs. TFPT 13/13; power controls (seed perturbation, data shuffle) collapse the scorecard; all 94 500 $F_{U(1)}$ variants solved, exactly *one* hits CODATA ⇒ total $\leq 10^{-30.7}$, *conditional on the declared grammar* [E]
`v101_horizon_anchor.py`	The maximal black hole is the anchor (SdS in seam units): Nariai cubic $t^3{-}3t{+}2{=}(t{-}1)^2(t{+}2)$, roots $(1,1,-2)$ = traceless anchor; $S_{\mathrm{Nariai}}/S_{dS}{=}\frac{2}{3}{=}\lvert\mathbb{Z}_2\rvert/N_{\mathrm{fam}}$ (per horizon $\frac{1}{3}$); interpolation $(x^2{+}1)/\Phi_3(x)$; three-sheet total $\lvert\mathbb{Z}_2\rvert S_{dS}$ conserved; $Q_{\mathrm{geom}}\in[\frac{3}{8},\frac{1}{2}]$ = the $SU(4)_1$ weights; mass-line cover $(1{-}3m)(1{+}3m)$, deck = horizon swap; $\tau_{BH}{=}128\,g_{\mathrm{car}}M^3/c_3$; six atom landings, zero free parameters [E]/[C] (bulk reading)

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Script	What it machine-checks *(continued)*
`v102_seam_orientation.py`	One orientation: the anchor is the *stationary repeller* in both sectors — flavor relaxation = gradient flow of a cubic potential with critical points = the branch points, curvatures $V''(2){=}{+}\Delta$, $V''(5){=}{-}\Delta$ (= the gap), inflection at scalaron/2, Lyapunov rate Δ constant; SdS entropy has Nariai as its *unique* stationary point, curvature $\frac{2}{9}{=}\lvert\mathbb{Z}_2\rvert/N_{\rm fam}^2$; evaporation = entropy ascent away from the anchor [E]; “one variational principle of the seam” stays a reading [C]
`v103_trisection_normal_form.py`	Trisection normal form (the canonical coordinate): SdS cubic $\Leftrightarrow$ $\cos 3\theta{=}{-}3m$ (angle trisection; $\mathbb{Z}_3$ deck = triality); $m{=}\cos\psi/N_{\rm fam}$ and $S_{\rm tot}/S_{dS}{=}\frac{4}{3}{-}\frac{2}{3}\cos\frac{2\psi}{3}$ — *one* cosine of glue atoms; canonical curvature $(\frac{2}{3})^3{=}(\lvert\mathbb{Z}_2\rvert/N_{\rm fam})^{N_{\rm fam}}$ (Koide constant to the family power); invariant slope $d\sigma/dm\rvert_N{=}{-}\frac{8}{9}{=}{-}\,{\rm rank}\,E_8/N_{\rm fam}^2$; bridge: flavor rate $(2/3)^{2N_{\rm fam}}$ vs gravity curvature $(2/3)^{N_{\rm fam}}$, missing = one *clock* [E]/[C]
`v104_nariai_clock.py`	The *classical* Nariai clock is the anchor (third appearance): static pin $\phi{=}\rho$ solves the dS_2 static-patch equation with $m^2{=}{-}2\Lambda{=}{-}\lvert\mathbb{Z}_2\rvert\Lambda$ exactly (Ginsparg–Perry tower: one negative mode); in Hubble units $\chi_{\rm clock}{=}(\lambda{-}1)(\lambda{+}2)$ = the anchor quadratic, eigenvalues $\{1,-2\}$; Nariai cubic = $(t{-}1)\cdot\chi_{\rm clock}$; entropy-deviation rate $2H{=}\lvert\mathbb{Z}_2\rvert H$; honest (2/3)-test *negative* for the classical clock — the quantum clock is the one remaining [C] of the seam variational principle [E]/[C]
`v105_residual_inventory.py`	The residual inventory: *one constant* ($2/3{=}\lvert\mathbb{Z}_2\rvert/N_{\rm fam}$ exact in seven places across both sectors), *one anchor* (three dynamical appearances: configuration, Nariai roots, clock spectrum), *relocation theorem* (cosmological quantum clock suppressed by the Λ hierarchy, 121.5 orders $\approx 2\alpha^{-1}/\ln 10$ $\Rightarrow$ the clock must be the seam transfer operator — one [C] identification), and the *complete machine-pinned gap list*: exactly five structural objects + one scale + π; a sixth gap would fail the script [E]/[C]
`v106_review_validation.py`	External-review validation: seed normal form $\varphi_0^{\rm ret}{=}\frac{\lvert\mu_4\rvert}{N_{\rm fam}}c_3{+}\Omega_{\rm adm}c_3^{\lvert\mu_4\rvert}$ exact [E]; hypercharge moment ${\rm Tr}\,X^2{=}120{=}5!$ computed from the charge table; factorial spine $5!{=}\lvert R^+(E_8)\rvert{=}\sum$exponents, $240{=}2{\cdot}120$, $1920{=}2^4{\cdot}5!$; degree-2 inventory (Pascal $K{=}2$ unique at $g{=}5$, glue norms sum 2, $c_3{=}\frac{1}{2}{\times}\frac{1}{4\pi}$, $\binom{5}{2}{=}10$); named [C] hypothesis *Quadratic Boundary Locality* ($\Rightarrow K{=}2$ forced; would upgrade the Pascal selection to [E]); audit $240{=}16{\times}15$ non-unique [E]/[C]
`v107_quantum_clock_target.py`	Quantum-clock target made quantitative: the classical Ginsparg–Perry clock decays on $\{0,-1,-2\} = -\,{\rm Spec}(V) = -N_{\rm fam}{\times}$cusp (the transfer operator’s own grading) [E]; exact target identity $(1/3)^6 = \big((2/3)^6\big)^{\log_{3/2}3}$ — required bend $\log_{9/4}3 = 1.355$ per weight step; coupling landscape $\kappa = c/(24\pi)\cdot\Lambda/\bar{M}_{\rm Pl}^2$: 10^{-122} cosmological vs $1/(3\pi)$ at seam curvature (the only viable, borderline-nonperturbative regime) [E]; identification stays [C]
`v108_pascal_ladder.py`	Pascal Ladder Theorem: $2^{g-1}{=}\sum_{k\le K}\binom{g}{k}$ $\Leftrightarrow$ $g{=}2K{+}1$ (odd-g midpoint identity + even-g straddle) $\Rightarrow$ carrier selection $g{=}5$ *exactly equivalent* to degree bound $K{=}2$ — the Target-B residual = the one QBL hypothesis; neighbour worlds: $K{=}1$ one-family, $K{=}3$ nine-family, $K{=}4$ inconsistent; only $K{=}2$ also satisfies $g{+}N_{\rm fam}{=}8$ (v14 overdetermination) [E]/[C]
`v109_sheet_pairing.py`	Sheet-Pairing Lemma (QBL anchored): character-exact multisets from $(\pm\frac{1}{2})^5$ — no scalar within a sheet (zero-weight mult of $S^+{\otimes}\,S^+ = 0$; odd g flips parity); $S^+{\otimes}\,S^-{=}\Lambda^0{\oplus}\Lambda^2{\oplus}\Lambda^4$ exactly ($256{=}1{+}45{+}210$, zero-modes $(1,5,10)$ = the code); sheet-diagonal = odd forms ($2\Lambda^1{+}2\Lambda^3{+}\Lambda^5$); tower tops at Λ^{2K}, $K{=}2$ $\Rightarrow$ the seam scalar kernel is necessarily a *sheet pairing* [E]/[C]
`v110_calderon_sheet.py`	Calderón-sheet selection theorem: a Calderón involution ε on $S^+{\oplus}S^-$ certifies a scalar two-point datum *iff* it is sheet-odd; sheet-odd $\Rightarrow$ exactly $2{=}\lvert\mathbb{Z}_2\rvert$ invariant kernels = the two Lagrangian glues; ladder genericity ($g{=}3,5,7$): sheet-oddness forces $K{=}(g{-}1)/2$, *not* $g{=}5$ (anti-overclaim); QBL residue split into *interface* (“ε sheet-odd”, natural from the one-sided collar) + *analytic core* (slot-degrees ≤ 2) [E]/[C]

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Script	What it machine-checks *(continued)*
`v111_quadratic_transport.py`	Quadratic-Transport Theorem: in the integer Jordan–Wigner model linears are sheet-odd, the 45 quadratics ($=\mathfrak{so}(10)$, the Λ^2 term of 1+45+210) sheet-even, generate the code from the vacuum *and* span the full $\mathrm{End}(S^+)=256$ at word length $2 \Rightarrow$ *transport degree exactly 2* (degree ≤ 1 generates nothing) — the transport half of the QBL core is a theorem; $g=3$ control: degree selected, not rank **[E]**/**[C]**
`v112_selfcounting_channel.py`	Self-Counting Channel: $w \mapsto (w, -w)$ is a bijection code states $\leftrightarrow$ neutral certified kernels, so #kernels $= 2^{g-1}$ *exactly*; graded by pair-degree the same set has sizes $\binom{g}{m} \Rightarrow$ the Pascal closure $2^{g-1}=\sum_{m\leq K}\binom{g}{m}$ is *two countings of one set* (an identity, not a requirement); Hodge fold $\binom{g}{2j}=\binom{g}{\min(2j,g-2j)}$; QBL residue = one input ("a single scalar 2-point kernel"), selection carried by rank-8+integrality **[E]**/**[C]**
`v113_quasifree_kernel.py`	One kernel is the whole net: carrier $(D_5)_1\times(A_3)_1 = SO(10)_1\times SO(6)_1 = 16$ free Majoranas, c=5+3=8 at every tower level (only the glue grows, index $2\times 2=4=\lvert\mu_4\rvert$); Wick/Pfaffian exact (all 210+210 higher correlators = Pfaffians of the one kernel); kernel = Calderón involution, rank $P=5=g_{\mathrm{car}}$, seam level $8=\mathrm{rank}\,E_8$ (c = rank of the kernel); vacuum unique $\Rightarrow$ QBL input = R2 premise: gate closes $\Rightarrow$ carrier choice closes **[E]**/**[C]**
`v114_torsion_delta.py`	Torsion normal form + $\delta=\frac{1}{2}$ theorem: flatness of the (U_{wall}) $\mathbb{Z}_4$-family $\Leftrightarrow$ $(MU)^4=\mathbf{1}$ (μ_4 = order of the twisted cusp generator); on the involutive branch ($T^2=\mathbf{1}$) the cusp trace forces $\mathrm{diag}\,M=(0,\frac{i}{2},-\frac{i}{2})$ with spec $\{1,\omega,\omega^2\}$ automatic $\Rightarrow$ $\delta=\frac{1}{2}$ exact and constant (v20's hand value = torsion identity); census: $\{1,i,-i\}$ branch varies, v33 point in its d_1=0 slice; residue = bundle-side branch selection **[E]**/**[C]**
`v115_anchor_residue.py`	The anchor pins the residue: μ_4-average = diagonal projection $\Rightarrow$ anchor splitting $\Leftrightarrow$ $\mathrm{diag}\,A_0=a/\lvert\mu_4\rvert$; off-weights uniquely $(8,0,5)/144$ $(=(\mathrm{rank}\,E_8,0,g_{\mathrm{car}})/(\lvert\mu_4\rvert N_{\mathrm{fam}})^2$, total $13=\Delta_Q)$; *exact* residue $A_0^\star$ with $\chi=\lambda(\lambda-\frac{1}{3})(\lambda-\frac{2}{3})$ is the RH solution ($\lVert M_\infty-\mathbf{1}\rVert\sim 10^{-10}$); anchor forces d_1=0 $\Rightarrow$ harmonic diagonal anchor+torsion-forced **[E]**
`v116_resonance_uniqueness.py`	Resonance theorem (v115 [N]→**[E]**, linear — no Gröbner): twisted μ_4 averages collapse to $B_1=4a_{12}E_{12}+4\overline{a_{13}}E_{31}$; anchor exponents $\mathrm{diag}(2,1,1)$ give resonance gap 1 with obstruction $4\overline{a_{13}} \Rightarrow M_\infty=\mathbf{1} \Leftrightarrow a_{13}=0$; + $(8,0,5)/144$ + Diagonaleichung $\Rightarrow$ flat anchor locus = *one* gauge orbit = exact $A_0^\star$ — U_{wall} bundle datum algebraically unique; sharpness control $\lVert M_\infty-\mathbf{1}\rVert \sim 8.5\lvert a_{13}\rvert$ **[E]**
`v117_monodromy_weyl_a3.py`	The monodromy is $W(A_3)$: the unitarised holonomy of $A_0^\star$ is *exact* (entries in $\frac{1}{2}\mathbb{Z}[i]$), $\mathrm{diag}\,\widetilde{M}_0=(0,-\frac{i}{2},\frac{i}{2}) \Rightarrow d_1=0$, $\delta=\frac{1}{2}$ are matrix entries; $\langle U,\widetilde{M}_0\rangle$ enumerated exactly: order 24, statistics (1, 9, 8, 6), characters $(3,-1,0,1) = S_4=W(A_3)$ (standard⊗sign) — the flavor wall realises the A_3 glue factor as its Weyl group; ODE match 3.5×10^{-10} **[E]**
`v118_hexagon_family_dictionary.py`	First exact H2 piece: $\mathrm{spec}(\widetilde{M}_0)\cup\mathrm{spec}(-\widetilde{M}_0)=\mu_6$ (hexagon = family spectrum + sheet twist, $\mathbb{Z}_6=\mathbb{Z}_2\times\mathbb{Z}_3$); $\det(\mathbf{1}-t\widetilde{M}_0)=1-t^3 \Rightarrow \det_\mp=(\frac{7}{8},\frac{9}{8})$; lepton coefficients = resolvent determinants ($c_e=\lvert\mu_4\rvert/(2\det_-)$, $c_\mu=N_{\mathrm{fam}}/(2\det_+)$, $c_\tau=\lvert\mu_4\rvert\det_-/N_{\mathrm{fam}}$, product $=\lvert\mu_4\rvert/\det_+=\frac{32}{9}$); $\langle U,\widetilde{M}_0,-\mathbf{1}\rangle$ has order $48=\Omega_{\mathrm{adm}}$; the address table stays open **[E]**/**[C]**
`v119_review_validation_2.py`	Second review validation: anchor ratio triad $(e_3,e_1,e_2)/p_0=(\frac{2}{3},\frac{4}{3},\frac{5}{3})$ = (Koide branch, seed gain, carrier branch; flow critical points $(2,5)=(e_3,e_2)$); the 121 audit lemma $(\mathbf{1}+a)^T R(\mathbf{1}+a)=121=11^2=\lVert\mathrm{Pl}(K)\rVert_1^2$ (orderings $\{105,121,135\}$, only canonical = 121; audit, not load-bearing); micro-identities $\Omega_{\mathrm{adm}}=2p_1p_2$, $10b_1=1+p_1p_3$, $\Delta_Y=e_2^2=p_0^2+2\,\mathrm{rank}$; flavor surface = v94/v95 (presentational) **[E]**

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Script	What it machine-checks *(continued)*
`v120_address_table.py`	Address table (frozen v18/v20 integers): lepton words $(8,5,3)$=(rank $E_8, g_{\text{car}}, N_{\text{fam}}$); addresses = divmod by the hexagon (p_2=6): $e\to(2,1)$, $\mu\to(5,0)$, $\tau\to(3,0)$, $\sum r=p_3$; sheet parity: τ sits at the real twisted eigenvalue -1 (= the product-rule site); quark sums: up = p_3, down = p_1+p_3, all quarks = $24=\lvert W(A_3)\rvert$, all nine = $40=p_1p_3=a^TRa$; top at the vacuum site $(0,0)$; the assignment derivation stays open **[E]**/**[C]**
`v121_address_pinning.py`	Address pinning: $L=R+2U$ (v95) ⇒ the word table = the residue operator R + the generation-1 winding, R column 1 = the anchor a; margins = atoms (rows (e_1, rank, p_3), columns $(e_1, \Delta_Q, g_{\text{car}})$); *census theorem*: among all hexagon matrices with atom margins (17 of them) exactly *one* has det = 8 — and it is R (also unique with SNF $(1,1,8)$; all 17 determinants distinct) ⇒ the address table carries *zero* residual information; the H2 residue = the six atom margins **[E]**/**[C]**
`v122_margin_theorem.py`	Margin theorem: the *established* selectors — the D_4 annihilator $n=(5,-9,6)$ with $n^TR=(8,0,0)$ (v94), $\det R=8$, $\det K=4$ (the v71 quartet) — pin R *uniquely* among hexagon matrices ($64\to12\to1$) ⇒ the atom margins, the anchor column and all sum rules are *consequences* (their input status is revoked); bonus: $\det L=20$ holds automatically on the locus (the quartet is redundant); H2 introduces *no* new residual class **[E]**/**[C]**
`v123_inventory_update.py`	Residual inventory updated (post v110–v122): thirteen new ledger rows machine-pinned; R2 carries *three* loads (metric + carrier + QBL: one theorem, three doors); the table re-pins to *five* classes — R1 clock, R2 index-4 net, R3 EH, R4′ selector establishment (replacing the *discharged* H2 dictionary), R5 Q realisation — + v_{geo}, π; a sixth class fails the script **[E]**
`v124_resummed_clock.py`	Resummed clock (the R1 bend in closed form): transfer eigenvalues = exactly $(1-n/N_{\text{fam}})^{p_2}$ ⇒ rate$(n)=-p_2\ln(1-n/N_{\text{fam}})$ reproduces $\{0,\Delta,6\ln3\}$, the bend $\log_{3/2}3$ in one line; the linearisation carries slope $\lvert\mathbb{Z}_2\rvert=2$ relative to the classical GP clock $\{0,-1,-2\}$; the pole at $n=N_{\text{fam}}$ forces the three-level spectrum; the R1 target is now closed-form **[E]**/**[C]**
`v125_glue_qsystem.py`	Glue Q-system: $\mathbb{C}[\mathbb{Z}_4]$ satisfies all Longo Q-system axioms exactly (associativity, unit, Frobenius, specialness $mm^*=\lvert\mathbb{Z}_4\rvert\mathbf{1}$) ⇒ Jones index $4=\lvert\mu_4\rvert$ (the v89 value); locality = isotropy ($q(k(1,1))=k^2\in\mathbb{Z}$, v92); halfway sub-Q-system $\mathbb{C}[\mathbb{Z}_2]$ = the $SO(16)_1$ step, $4=2\times2$; R2 sharpens from "find" to "identify" **[E]**/**[C]**
`v126_clock_wall_bridge.py`	Clock and wall share one spectrum: the resummation weights $n/N_{\text{fam}}=\{0,\frac{1}{3},\frac{2}{3}\}$ = exactly spec $A_0^\star$ (the parabolic MS weights); $\lambda_n=(1-\alpha_n)^{p_2}$; checkpoint 1 passed *classically* ($p_2/N_{\text{fam}}=2=\lvert\mathbb{Z}_2\rvert$ = the v104 entropy rate $2H$); $\det(\mathbf{1}-A_0^\star)=\frac{2}{9}$ = the entropy curvature (v102); the quantum content of R1 is isolated in the geometric tail **[E]**/**[C]**
`v127_ring_resummation.py`	The clock is a log determinant: rate = $-p_2\,\mathrm{Tr}\ln(\mathbf{1}-\alpha P)$ — six identical ring (RPA) towers, *one per hexagon site*, coupling = the parabolic weight; the k=1 ring = the classical term, the k=2 ring = the first quantum correction; $\kappa_{\text{seam}}=\frac{1}{3\pi}=\alpha_1/\pi$ (audit); the wall = the RPA instability at $\alpha\to1$; v107's "O(1) resummation" is thereby *typed* (ring/log-det), its derivation open **[E]**/**[C]**
`v128_graded_hull.py`	Graded hull + zero-mode multiplicity: the 240 E_8 roots explicit in $D_5\oplus A_3$+glue coordinates, coset decomposition 240=52+64+60+64; the grading is exactly additive on *all* 6720 root-pair sums ⇒ E_8 is a $\mathbb{Z}_4$-*graded* Lie algebra over the carrier (grading = glue = Q-system); ring multiplicity $6=2(2l+1)\rvert_{l=1}$ = the sheet-doubled GP zero modes **[E]**/**[C]**
`v129_entropy_power_law.py`	The clock is an entropy power law: $\lambda_n=(S_n/S_{dS})^{p_2}$ with the three Nariai entropy levels $S/S_{dS}\in\{1,\frac{2}{3},\frac{1}{3}\}$ (v101); rate = $p_2\ln(S_{dS}/S)$; triple identity $\alpha=n/N_{\text{fam}}$= MS weight = entropy-deficit share ⇒ the RPA coupling is nowhere free; orientation consistent (attractor = max entropy); R1 thermodynamic: $\Gamma\propto(S/S_{dS})^{p_2}$ (a Gibbons–Hawking power law, $h=N_{\text{fam}}$ audit) **[E]**/**[C]**

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Script	What it machine-checks *(continued)*
`v130_born_square.py`	Born square: amplitude $\sim (S/S_{dS})^{n_{\rm zero}/2}{=}(S/S_{dS})^3$ (one $S^{1/2}$ per zero mode), probability $= \lvert A\rvert^2{=}(S/S_{dS})^6 \Rightarrow h{=}n_{\rm zero}/2{=}3{=}N_{\rm fam}$, $p_2{=}2h$ (the Born rule) — the v129 weight audit is *derived*; the 3 = the dimension of the $l{=}1$ space = the number of proper conformal Killing vectors of S^2; the $\times 2$ = the horizon pair (the v101 deck); the R1 residue = the standard zero-mode measure **[E]**/**[C]**
`v131_measure_is_area.py`	The measure is the area: $\lVert Y_{1m}\rVert^2{=}r^2{=}A/(4\pi)$ *exactly* (all three $l{=}1$ modes integrated) $\Rightarrow$ the per-mode Jacobian = the mode norm $\sim S^{1/2} \Rightarrow$ amplitude S^3, Born S^6 — every exponent factor with a derivation origin; the 4π = the same Gauss–Bonnet primitive as in c_3, cancelling in all ratios; the R1 residue = *one* det-ratio cancellation $\det'_n / \det'_{dS} {=}1{+}O(\text{deficit})$ **[E]**/**[C]** *(cancellation since derived within the SdS family, v144)*
`v132_detratio_anomaly.py`	Det-ratio anomaly = the Koide constant: $\zeta(0)\rvert_{\det'}$ of the GP sphere operator $-\Delta{-}2$ (the 3 zero modes removed) $= a_1{-}3 = \frac{7}{3}{-}3 = -\frac{2}{3} = -\lvert\mathbb{Z}_2\rvert/N_{\rm fam}$ exactly ($a_1{=}\frac{1}{N_{\rm fam}}{+}\lvert\mathbb{Z}_2\rvert$; numerical continuation $\sim 10^{-7}$) — the eighth 2/3 appearance, the first as a *spectral* anomaly; consequence $\det'(cL){=}c^{\zeta(0)}\det'(L)$; the R1 residue = *one* $\zeta(0)$ budget statement (the 4d sum) **[E]**/**[C]**
`v133_zeta_budget.py`	The ζ budget, both routes exact (Euclidean Nariai = $S^2{\times}S^2$): the reduced route gives $-\frac{2}{3}$ per sector, total $-\frac{4}{3}{=}{-}e_1/p_0$ (= minus the seed gain — two of the three triad values); the naive 4d route gives $a_2{=}\frac{161}{45}$, $\zeta(0){=}{-}\frac{109}{45}$ (*no* atom; continuation $\sim 10^{-8}$); zero modes 3+3 = the horizon pair $\Rightarrow$ the budget structurally favours the *reduced* seam reading; remainder = graviton/ghost shares (Volkov–Wipf class) **[E]**/**[C]**
`v134_dual_anchor.py`	Dual anchor lemma: $d{:=}a^\top R^{-1}{=}a^\top L^{-1}{=}(-\frac{1}{2},-\frac{1}{2},1)$, $d{\cdot}\mathbf{1}{=}0$, $d{\cdot}a{=}1$, $(1,1,-2){=}{-}2d$ — the inverse flavor response *is* the Nariai root (v101); the invariance is exact via Sherman–Morrison ($\Leftrightarrow$ orthogonality to $R^{-1}\mathbf{1}{=}\frac{1}{4}(1,1,-1)$; controls: $\mathbf{1}, e_1, n$ do *not* lie on the plane); (d,n) = the dual normal pair of the flavor boundary; a third purely *algebraic* leg of the flavor$\leftrightarrow$horizon bridge **[E]**/**[C]**
`v135_det_surface.py`	Determinant surface: $\det M(s,t){=}3s(t{+}1)(t{+}2){+}t^2{+}5t{+}8$ — two s-independent iso-volume walls $t{=}{-}2{\Rightarrow}2{=}\lvert\mathbb{Z}_2\rvert$, $t{=}{-}1{\Rightarrow}4{=}\lvert\mu_4\rvert$ ($\det K{=}4$ is a *layer*); winding line $6s{+}8 \Rightarrow (8,14,20)$, $\det F{=}32{=}2^{g_{\rm car}}$; $\det B(s,0){=}2\det M(s,0)$ exactly, B-wall only at $t{=}{-}2$; row-sum axes $C{=}(7,11,13)$, $U{=}(3,3,3)$, $V{=}(1,2,3){=}\mathrm{Spec}(Q_+)$; micro-identities $e_1^2{+}e_2^2{=}41{=}10b_1$, $p_1^2{+}p_2^2{=}52{=}$ the carrier coset root count (v128) **[E]**/**[C]**
`v136_dual_normal_selector.py`	Dual-normal selector: (d,n) pins R *columnwise* — $d{\cdot}c_j{=}a_j$, $n{\cdot}c_j{=}8\delta_{1j}$, kernel = primitive $(6,8,7)$, census: each column unique (1 of 216) $\Rightarrow$ R4′ shrinks from 6^9 to three column problems; the $A_0^\star$ zero mode carries $\sigma{=}(2,-9,5){=}(\lvert\mathbb{Z}_2\rvert,-N_{\rm fam}^2,g_{\rm car})$ ($\lVert v\rVert^2{=}\frac{16}{9}$, $16{=}\dim S^+$; $\mathrm{adj}{=}\frac{2}{9}P_v$ = the SdS curvature); $n{\cdot}\sigma{=}121{=}11^2$; $W(A_3)$ orbit control: the naive lift $\sigma{\to}n$ fails — a genuine mechanism, not a group element **[E]**/**[C]**
`v137_qplus_cohomology.py`	The Q_+ grading is cohomology: $H^1(\mathbb{P}\setminus\mu_4){=}\chi_1{\oplus}\chi_2{\oplus}\chi_3$ ($\omega_k{=}z^{k-1}dz/(z^4{-}1)$, character i^k, residues $\zeta^k/4$); degrees $(1,2,3){=}\mathrm{Spec}(Q_+){=}$ the A_3 exponents ($h{=}4{=}\lvert\mu_4\rvert$); cusp weights $\{0,\frac{1}{3},\frac{2}{3}\}{=}\mathrm{spec}(A_0^\star) \Rightarrow$ MS weights = cusp classes = characters = Q_+: *one* spectrum, four readouts; the QGEO residue = the naturality of the canonical map **[E]**/**[C]**
`v138_vw_firewall.py`	Volkov–Wipf firewall (R1 typed closed): external 4d value $\alpha_{\rm PI}{=}{-}\frac{98}{45}{=}\frac{22}{45}{-}\frac{8}{3}$ (arXiv:2506.02142 Tab. 4.1 = VW hep-th/0003081); **edge match**: $\alpha_{\rm edge}{=}{-}\frac{8}{3}{=}2{\times}({-}\frac{4}{3})$ — two identical edge copies (the horizon pair), per copy exactly the reduced seam budget (v133, the seed gain); bulk $\frac{22}{45}$ = the graviton gas (not the clock); difference $-\frac{38}{45}$ no atom $\Rightarrow$ typing: the seam clock = an edge object, the full 4d determinant = the gravity firewall **[E]**/**[C]**

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Script	**What it machine-checks** *(continued)*
`v139_selector_triangle.py`	Selector triangle: $d{=}\frac{3}{2}a{-}2\cdot\mathbf{1}$ exactly (pure anchor data, no R) $\Rightarrow$ the first selector is *derived*; the frame $(\mathbf{1},a,\sigma)$ has $\det{=}11{=}\\|\,\mathrm{Pl}(K)\\|_1$, and n is the *unique* covector with atom pairings $(2,8,121){=}(\lvert\mathbb{Z}_2\rvert,\mathrm{rank}\,E_8,11^2)$; on the constrained line the pairing is $11(11{+}t)$, a square only at the true n; review lift exact (audit): $n{=}\mathrm{reverse}(\sigma){+}\lvert\mu_4\rvert e_3$; R4′ residue = three pairings **[E]**/**[C]** *(since v142: two — the third pairing is integrality-forced)*
`v140_canonical_map.py`	The canonical map exists and is rigid: the plain deck U (characters $(0,1,3)$) cannot match H^1; the *sheet-twisted* $-U$ ($(2,3,1)$) and the cusp exponential i^{Q_+} ($(1,2,3)$) both carry the H^1 character set $\{1,2,3\}$ (the v118 sign twist); multiplicity-freeness $\Rightarrow$ Schur-rigid equivariant isomorphism (unique up to scalars); i^{Q_+} pulls the Euler degree back to Q_+, $-U$ to $Q_+\circ\rho$ — the QGEO residue collapses to *one* $\mathbb{Z}_3$ choice **[E]**/**[C]** *(the choice since derived, v141)*
`v141_deck_selection.py`	Deck Selection Theorem (R5): the established integer deck $G{=}T_A\Sigma$ (v97/v98) pairs the $Q_+{=}1$ line with the self-conjugate character 2 $\Rightarrow$ of the three $\mathbb{Z}_3$ rotations only the sheet-twisted $(2,3,1)$ = chars$(-U)$ survives; i^{Q_+} excluded (same spectrum, wrong grading pairing); residual freedom = the established sheet $\mathbb{Z}_2$ (v92); Euler degree = $Q_+\circ\rho$ **[E]**/**[C]**
`v142_frame_integrality.py`	Frame Integrality (R4′): pairing triples of *integer* covectors against $(\mathbf{1},a,\sigma)$ form an index-11 sublattice ($x{-}3y{+}z\equiv 0 \bmod 11$, $11{=}\\|\mathrm{Pl}(K)\\|_1$); with $(x,y){=}(\lvert\mathbb{Z}_2\rvert,\mathrm{rank}\,E_8)$ the third pairing is forced $\equiv 0 \bmod 11$, primitivity forces even t; Cramer reductions for the two line pairings; R4′: three pairings $\to$ two **[E]**/**[O]** *(since v145: the values are atom identities — residue = one lift map)*
`v143_graded_frobenius.py`	Graded Frobenius (R2 at the finite Lie level): coset duality $-C_k{=}C_{-k}$ exact on all 240 roots $\Rightarrow$ Killing pairing nondegenerate per $g_k\times g_{-k}$; glue average = carrier projector (index $\lvert\mathbb{Z}_4\rvert{=}4$); each sector one $W(D_5){\times}W(A_3)$ orbit = simple current; fusion = $\mathbb{C}[\mathbb{Z}_4]$ (v125); residual = the one conformal-net statement **[E]**/**[C]**
`v144_detratio_family.py`	Det-ratio cancellation within the SdS family (R1, the v131 step): e_2-rigidity gives $r_b r_c{=}1{-}\Delta^2/3$ exactly $\Rightarrow$ det′-ratio = $(1{-}\Delta^2/3)^{4/3}$ (via $\zeta(0){=}{-}\frac{2}{3}$, v132) — no first-order term in the split; deficit coefficient $\frac{32}{27}{=}\lvert\mu_4\rvert(\frac{2}{3})^3$ (audit); finite-weight absorption stays open with the $6\to\frac{14}{3}$ obstruction stated sharply **[E]**/**[C]** *(the bend since shown det′-clean, v147)*
`v145_pairing_atoms.py`	R4′ values are *atom identities*: with the lift $n{=}w_0(\sigma){+}\lvert\mu_4\rvert e_3$ ($w_0{=}$ the A_3 exponent duality), $n\cdot\mathbf{1}{=}\lvert\mu_4\rvert{-}\lvert\mathbb{Z}_2\rvert{=}\lvert\mathbb{Z}_2\rvert$, $n\cdot a{=}\lvert\mu_4\rvert a_3{=}8$ via the new closure $\lvert\mu_4\rvert{+}g_{\mathrm{car}}{=}N_{\mathrm{fam}}^2$ ($4{+}5{=}9$, $\sigma\perp w_0(a)$), $n\cdot\sigma{=}\\|\sigma\\|^2{-}N_{\mathrm{fam}}^2{+}\lvert\mu_4\rvert g_{\mathrm{car}}{=}121$; residue = derive the *one* lift map **[E]**/**[C]**
`v146_moebius_d4.py`	Möbius D_4 realisation (R5 at its floor): the full dihedral $\langle z{\mapsto}iz,\ z{\mapsto}1/z\rangle$ on $H^1(\mathbb{P}^1\backslash\mu_4)$ matches the integer model $\langle G,T_A,\Sigma\rangle$ exactly — ι^* swaps $\chi_1{\leftrightarrow}\chi_3$, fixes χ_2 with $+1$, $\det{=}{-}1$ ($=T_A$ in the cusp basis); $\delta\iota{=}\Sigma$ class; parities match (v98) — the `GATE.QGEO` premise reduces to the module identification **[E]**/**[C]**
`v147_clock_gaussian.py`	The clock is a Gaussian zero-mode integral (R1): variance ratio $(1{-}\alpha)$ (norm = area, v131) $\Rightarrow$ Born^2 Γ-ratio $(1{-}\alpha)^{p_2}$, rate = $-p_2\ln(1{-}\alpha)$ = the v127 ring sum term by term; the bend $\log_{3/2}3$ lives between the two Nariai levels of *one* geometry $\Rightarrow$ *det′-clean*; the $6{\to}\frac{14}{3}$ obstruction is localised to the $\alpha{=}0$ reference **[E]**/**[C]**
`v148_fock_census.py`	NS/R sector census (R2 scoped, corrected): the untwisted (NS) module of the 16 Majoranas carries only the *even* classes $\{0,2\}{\times}\{0,2\}$ — the odd glue sectors $k(1,1)$ are *Ramond* modules; the R zero-mode module ($2^8{=}256$) splits 128+128 into the odd sectors of the *two* Lagrangian glues (v92): choosing the glue *is* choosing the R-projection (128= one $SO(16)$ chirality), $248{=}120_{\mathrm{NS}}{+}128_{\mathrm{R}}$; R2 is irreducibly a twisted-sector net statement **[E]**/**[C]**
`v149_cusp_normal.py`	Both dual normals are *anchor data* (R4′): in the cusp frame n pairs to $(6,3,5)=(p_2,p_0,e_2)(a)$ — one anchor invariant per Q_+ line ($d=\frac{3}{2}a-2\cdot\mathbf{1}$ in the generation frame, v139); equivalent to the $(2,8,121)$ frame data and the w_0-lift (v145); audit: $\sigma\to(5,1,2)$, sum $14=\dim G_2$; residue = *one* assignment **[E]**/**[C]**

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Script	**What it machine-checks *(continued)***
`v150_replica_eh_model.py`	The EH form from the replica variation, at the gapped-model level (first R3 movement): conical deficit exact and t-independent (image sum $(N^2-1)/(12N) = C(2\pi/N)$); the gap makes the ζ-variation *finite and cutoff-independent*, $\Delta \log\det' = 2C(\gamma)\ln m$; linearisation $= (\ln m/12\pi)\int\sqrt{g}R$ — the EH form; target equation $k{=}c_3/2 \Leftrightarrow \ln m = \frac{3}{4} = q(A_3)$ (audit); Calderón transfer + normalisation stay open — `SEAM.THEOREM.01` narrows, does *not* close [E]/[O] *(Calderón transfer since answered — the DtN kernel is conically clean, v151)*
`v151_bfk_split.py`	BFK split (the Calderón transfer of v150 answered): the Kac corner constant is boundary-condition *independent* ($C_N{=}C_D$, derived from reflection doubling) $\Rightarrow$ the conical deficit of the Calderón/DtN jump determinant *vanishes identically*, $C_{\mathrm{DtN}}{=}C_{\mathrm{cone}}(\gamma){-}2C_D(\gamma/2){=}0$; via Burghelea–Friedlander–Kappeler the seam-reduced action inherits the v150 EH form verbatim — the EH content sits in the *local* half determinants, the nonlocal kernel is conically clean; residue = the $q(A_3)$ normalisation + the kernel premise [E]/[O]
`v152_norm_is_anchor.py`	R3's normalisation *is* the dimensionful anchor: the EH coefficient is $k = \ln(m/\mu)/(12\pi)$ — $\ln m$ alone is scale-ambiguous, only the ratio m/μ is physical; $k{=}c_3/2 \Leftrightarrow m/\mu{=}e^{3/4}$ is the induced-gravity anchor $1/G$ (v68), *not* a separate gap; the irreducibles stay {one scale, π}; audit: the log-ratio $= \frac{3}{4} = q(A_3)$ is the target value, NOT a derivation [E]/[O]
`v153_no_unit_theorem.py`	No-Unit Theorem (v78/v152 formalised): dimensionless data are scale-invariant under $L \mapsto \lambda L$, a mass / $1/G$ is not, so a dimensionless compiler *cannot* select an absolute scale; the three residuals collapse — $U_{\mathrm{point}} \sim v_{\mathrm{geo}}$, $1/G \sim v_{\mathrm{geo}}^2$, $m/\mu = e^{3/4}$ (a ratio); v_{geo} is an irreducible metrology primitive, not an open gap; irreducibles stay {one unit, π} [E]/[O]
`v154_simple_current_theorem.py`	Simple-Current Extension Theorem (the central G_{net} theorem, assembled): $A{=}(D_5)_1{\otimes}(A_3)_1$ extended by the isotropic glue $L{=}\langle(1,1)\rangle$ has index $\lvert L\rvert{=}4$, $c{=}5{+}3{=}8$, $\mu(B){=}\mu(A)/\lvert L\rvert^2{=}16/16{=}1 \Rightarrow$ holomorphic $\Rightarrow B{=}(E_8)_1$ (unique even-unimodular rank-8; $SO(16)_1$ excluded); finite realisation in hand (v125/v128/v143/v148); residue = the seam-side identification [E]/[C]
`v155_quasifree_boundary.py`	Quasi-free in $\Rightarrow$ quasi-free out (Python-only): the boundary marginal of a Gaussian bulk is the compression $P\Gamma P$ (a contraction, hence a valid covariance), so the G_{net} analytic premise (v110(b)/v113) *follows* from 'the seam bulk is a reflection-positive free field' — the *same* premise as v59 (area law) and v150/v151 (EH); the seam-side identification reduces to ONE shared physical premise [E]/[O]
`v156_seam_net_construction.py`	Constructive route (Route 3) to its exact endpoint: the DtN/Calderón operator of the 2d Laplacian is $\omega_k{=}\lvert k\rvert$ (free chiral dispersion); 16 free Majoranas give $c{=}8{=}5{+}3$; the E_8 currents are $248{=}120_{\mathrm{NS}}{+}128_{\mathrm{R}}$; and the $(E_8)_1$ character $E_4/\eta^8 = q^{-1/3}(1{+}248q{+}4124q^2{+}\dots) = j^{1/3}$ (verified to order 4) makes $B \cong (E_8)_1$ a *checked* identity, not an assertion; residual = the free-bulk premise (v155) + net existence [E]/[C]
`v157_rigid_fixed_point.py`	Freeness as a *rigid fixed point* (simplicity-first reframing of premise A): (i) the DtN/Calderón symbol is universally $\lvert k\rvert$ (Lee–Uhlmann, bulk-detail-independent — the free dispersion is not a premise); (ii) a holomorphic $c{=}8$ theory has *no* marginal $(1,1)$ field (all $\bar{h}{=}0$) $\Rightarrow$ rigid, isolated, no room for an interaction; (iii) the same $\lvert\mathbb{Z}_2\rvert$ that halves 8π in c_3 IS the Ramond projection giving $\mu{=}1$; (iv) holomorphic $c{=}8$ unique $= (E_8)_1 =$ free. Premise (A) shrinks from 'prove Gaussianity' to 'the gapped flow reaches the holomorphic fixed point' [E]/[C]
`v158_fixed_point_stable.py`	The free chiral $c{=}8$ fixed point is *stable* (exact dimension count, the finite core of (A)): Majorana $h{=}\frac{1}{2}$, bilinear current $h{=}1$, quartic $h{=}2$; the relevant window $(0,1)$ for bosonic operators is *empty*, the $h{=}1$ currents are chiral (not $(1,1)$-marginal, v157), the lowest interaction (quartic) is $h{=}2{>}1$ irrelevant $\Rightarrow$ isolated stable fixed point; (A) reduces to a basin-of-attraction statement [E]/[C]

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Script	**What it machine-checks *(continued)***
`v159_pyrate_gauge_crosscheck.py`	PyR@TE cross-check of the F_{transfer} gauge inputs: the carrier/SM content gives $(b_1, b_2, b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$ by the textbook one-loop formula; $b_1{=}\frac{41}{10}$ holds *three* independent ways — carrier algebra $10b_1{=}g_{\text{car}}2^{g_{\text{car}}-2}{+}1{=}41$, the $U(1)$ hypercharge index $\sum \frac{3}{5}Y^2$, and the external RGE generator PyR@TE 3 (`arXiv:2007.12700`) which writes $\beta_{g_1}{=}(\frac{41}{10})g_1^3$ verbatim; the $M{=}41$ split $10b_1{=}40_{\text{(ferm)}}{+}1_{\text{(Higgs)}}$ reproduces v13 Gate 1; evidence in `verification/pyrate`. The transfer functor's gauge inputs are not free knobs [E]/[C]
`v160_seam_gaussianity_from_pf.py`	Seam Gaussianity as a *conditional* fixed-point theorem (the proposed closure of premise A, honestly typed): (1) quasi-free ⇒ all higher connected cumulants $\kappa_{2n}{=}0$ (signed set-partition/Pfaffian inversion, exact on a pure Majorana covariance); (2) Wick functoriality ⇒ a kernel-preserving transport fixes the whole quasi-free state (v113); (3) the Perron–Frobenius dichotomy (a *fixed* κ_{2n} is a SECOND fixed ray, breaking uniqueness; a *non-fixed* one decays by the gap) forces Gaussianity; (4) the qualitative claim needs only gap>0 — the value $(2/3)^6$ only sets the rate. **The load** [O]: v56's gapped PF uniqueness lives on the 3-dim scalar transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$, while the seam correlation cone has $\binom{16}{4}{=}1820$, $\binom{16}{6}{=}8008$ cumulant directions ⇒ (A) reduces to the single internal statement 'the admissible TFPT transport is primitive + spectrally gapped on the *full* Schwinger cone, not only the readout spectrum' [E]/[O]
`v161_one_particle_reduction.py`	The cone reduces to one particle (discharging v160's scope premise via Bogoliubov second quantisation): a quasi-free transport is $T{=}\Gamma(t)$, the second quantisation of a one-particle contraction t; $\Gamma(t)$ acts on the degree-m correlator sector as the exterior power $\Lambda^m(t)$, so (1) $\operatorname{spec}(\Gamma(t)\vert_{\deg m})$ = products of m one-particle eigenvalues [E]; (2) **the cone gap equals the one-particle gap** — the sub-leading eigenvalue over the whole cone is $(2/3)^6$ exactly [E]; (3) the eigenvalue-1 sector = wedges of the fixed subspace (disconnected Wick powers), no independent fixed higher cumulant, uniqueness closed by v113 (one polarization ⇔ one state) [E]; (4) quasi-freeness *forced* not assumed — the $120{=}\dim\mathfrak{so}(16)$ chiral currents are the Bogoliubov generators and v158 makes every non-Gaussian deformation irrelevant [C]. **Residual** [O]: v160's three full-cone premises collapse to ONE 16-dim one-particle statement (seam symbol = gapped Calderón Bogoliubov map, fixed = rank-8 K_Σ, gap $(2/3)^6$); all numbers supplied, only the physical identification (Kawahigashi–Longo class = the A2 net residual) stays open [E]/[O]
`v162_seam_transport_identification.py`	The final identification, forced as far as it goes — the v161 one-particle statement carries *no* new open content: (1) the gap $(2/3)^6$ is *forced* — cusp weights $= \operatorname{spec}(A_0^\star) = \{0, \frac{1}{3}, \frac{2}{3}\}$ (parabolic residue, v115), transfer eigenvalue $(1{-}\alpha)^{p_2}$, $p_2{=}6{=}\vert R_+(A_3)\vert$, all carrier data [E]; (2) μ_4-equivariance forces the generator diagonal in the cusp basis — $\sum_k U^k X U^{-k} = \vert\mu_4\vert \operatorname{diag}(X)$, commutant of U = the diagonal algebra — so the gapped μ_4-equivariant generator has spectrum = the cusp weights, gap $(2/3)^6$ [E]; (3) the cusp grading *is* the A_3 grading on the 16 Majoranas (v137) [E]; (4) the fixed space = the rank-8 Calderón polarization (v113/v156) [E]. **Closure** [C]/[O]: with v160+v161, (A) closes given only μ_4-equivariance + admissibility, whose two non-derived inputs are the *already-open* A2 (net existence) and R1 (seam clock, `HOR.CLOCK`) — so (A) factors exactly into A2 + R1, adds *no* new open item, and is closed conditionally on them (a unification, not an unconditional proof) [E]/[O]
`v163_rg_stability_flavor.py`	F_{transfer} scale-tagging: the lepton-mixing seeds ($\sin^2\theta_{12}{=}\frac{1}{3}{-}\frac{\varphi_0}{2}{=}0.3067$, $\sin^2\theta_{13}{=}\varphi_0 e^{-5/6}{=}0.0231$) are *RG-stable* — PMNS running is y_τ^2-suppressed ($\sim 1.8{\times}10^{-5} \ll$ precision), so seam value = measured value; the quark mass-ratio readouts ($m_t/m_b{\sim}41$) run through $y_t^2 \sim O(10$–$15\%)$ and carry a scale tag. The only mixing-rotating Yukawa term $\frac{3}{2}YY^\dagger Y$ carries $y_{\max}^2$ (PyR@TE SM RGEs) [E]/[C]

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Script	What it machine-checks *(continued)*
`v164_qcd_scale.py`	the QCD confinement scale is parameter-free from the carrier $b_3{=}{-}7$: running $\alpha_s(M_Z)$ down (two-loop, n_f thresholds) gives $\alpha_s(m_b){\sim}0.22$, $\alpha_s(m_c){\sim}0.38$ and confinement ($\alpha_s{\sim}1$) at ${\sim}0.5\,$GeV — so the QCD-scale *numerator* of m_p/m_e is independently sourced (dimensional transmutation), while the ratio 1836 stays the honest 'Open / not forced' frontier non-claim ($m_p{=}O(1)\Lambda$ is lattice) [E]/[O]
`v165_premise_a_closure.py`	Premise (A), the honest maximal closure — the two residual gates pinned to their floor, so (A) carries *zero* independent open content: (1) A2 (net existence) closes by *assembly* of established mathematics [C]— 16 free Majoranas give a free-fermion net (Buchholz–Mack–Todorov / Böckenhauer–Evans), the holomorphic $c{=}8$ lattice net is $(E_8)_1$ (Frenkel–Kac–Segal / Dong–Xu / Kawahigashi–Longo), the μ_4 extension is a Longo Q-system of index 4 (v125), $\mu(B){=}16/16{=}1$, $248{=}120{+}128$, character $E_4/\eta^8{=}j^{1/3}$ (v156); the only TFPT input (seam Calderón kernel = free-fermion two-point function) is done via DtN $= \lvert k\rvert$ (v156/v157); (2) R1 (seam clock) reduces to `GATE.QGEO` [C]— μ_4-equivariance forces diagonal-in-cusp, the flat μ_4-equivariant generator is *unique* $= A_0^\star$ (v115/v116) so the gap $(2/3)^6$ is forced, residual = 'seam boundary = the μ_4-punctured sphere $\mathbb{P}^1\backslash\mu_4$' = `GATE.QGEO` (cohomology established, v137); (3) the irreducible core stays $\{\pi, v_{\rm geo}\}$, a *theorem* by the No-Unit Theorem (v153). **Final accounting**: (A) = A2 assembly [C] + R1=`GATE.QGEO` + the $\{\pi, v_{\rm geo}\}$ core ⇒ ZERO independent open gates; a unification + re-typing, not an unconditional proof [E]/[O]
`v166_higgs_free_seam.py`	the Higgs quartic from the *free seam* (premise A / v158): a Gaussian seam UV forces $\lambda(M_{\rm seam}){=}0$ *and* $\beta_\lambda(M_{\rm seam}){=}0$ (Shaposhnikov–Wetterich double criticality, here *motivated* not assumed). Two-loop PyR@TE-sourced SM running with the measured (m_H, m_t) gives $\lambda(\bar{M}_{\rm Pl}){\sim}0.002$, $\beta_\lambda(\bar{M}_{\rm Pl}){\sim}0$ — the famous SM near-criticality, now *explained* by the free seam (2-loop sharpens $\lambda(\bar{M}_{\rm Pl})$ from ${\sim}0.022$ to ${\sim}0.002$). The double condition predicts $m_H{\sim}129$–134 GeV; at the scalaron scale it gives ${\sim}107$ GeV (too low) ⇒ the BC lives at $\bar{M}_{\rm Pl}$ (seam=Planck) [E]/[C]
`v167_inventory_post_closure.py`	The terminal residual inventory (post premise-(A) closure; line v105 → v123 → v167): after the (A)-chain (v160–v165), premise (A) is *no longer* an independent structural class (`GATE.METRIC.16-19`). The residual collapses from v123's FIVE classes to: **Tier 0** — the irreducible core $\{\pi, v_{\rm geo}\}$, a *theorem* (No-Unit, v153), nothing to close; **Tier 1** — ONE hinge (seam = flavour = horizon): A_2 net existence (assembly of established net constructions) + `GATE.QGEO` (seam boundary $= \mathbb{P}\setminus\mu_4$, cohomology $b_1{=}N_{\rm fam}{=}3$); **Tier 2** — folded ($\to v_{\rm geo}$, $\to G_{\rm net}$); **Tier 3** — $F_{\rm transfer}$, one functor with four typed conditional interfaces (η_B carries the most open room). Completeness: a sixth independent structural class fails the script — a unification + re-typing, not an unconditional proof [E]/[C]
`v168_qgeo_rigidity.py`	The QGEO *Rigidity Theorem* — the finite half of the Tier-1 hinge `GATE.QGEO`, hardened. Exact: $\mu_4{=}\{1, i, -1, -i\}$ has cross-ratio 2 and Möbius stabiliser $\langle z{\to}iz, z{\to}1/z\rangle$ of order $8{=}\lvert D_4\rvert$ (faithful) ⇒ four marks with faithful D_4 are the μ_4 square up to Möbius; $H^1(\mathbb{P}\setminus\mu_4)$ has rank $4{-}1{=}3{=}N_{\rm fam}$; the forms $\omega_k{=}z^{k-1}dz/(z^4{-}1)$ ($k{=}1,2,3$) are μ_4-eigenforms with $z{\to}iz$ eigenvalue i^k = the three nontrivial characters χ_1, χ_2, χ_3, weights $(1,2,3)$ = A_3 exponents $= \mathrm{Spec}(Q_+)$. `QGEO.RIGID.01` [E] (math half closed); `QGEO.REALIZE.01` [C] (seam-collar $= \mathbb{P}\setminus\mu_4$ stays the realisation premise) [E]/[C]
`v169_etaB_boltzmann_interface.py`	The η_B leptogenesis transfer *operationalised* (Tier-3 $F_{\rm transfer}$, not a new class): type-I thermal leptogenesis $\eta_B{\sim}0.96{\times}10^{-2}\,\varepsilon_1\kappa_f$ (sphaleron 28/79) fed by TFPT's NO neutrino spectrum + the predicted $\delta_{CP}{=}240°$. Davidson–Ibarra $\varepsilon_1{=}\frac{3}{16\pi}M_1 m_3/v^2{\sim}8.5{\times}10^{-7}$ for $M_1{=}10^{10}$ GeV; efficiency $\kappa_f(\tilde{m}_1){\sim}0.03$–$0.19$. **Kill test**: over natural $M_1{\in}[3{\times}10^9, 3{\times}10^{10}]$ GeV and washout, η_B spans $[8.4{\times}10^{-11}, 4.7{\times}10^{-9}]$ which *brackets* the observed $6.1{\times}10^{-10}$ (canonical point gives $6.0{\times}10^{-10}$, not tuned) ⇒ the route is viable; M_1/washout are scenario inputs, so η_B stays [C]. If excluded, the route falls, not the theory [C]

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Script	**What it machine-checks** ***(continued)***
`v170_diamond_slice_compression.py`	E8 Slice Compression Lemma: the seven E_8 audit slices are *one* projection of two alphabets — anchor power sums P=(3, 4, 6, 10) (p_n=2+2^n) and Sheet-Diamond determinants D=(3, 4, 8, 14, 20, 32). The K_4 edge products (12, 18, 30, 24, 40, 60) are all carrier blocks; $\sum \det M = 81 = N_{\mathrm{fam}}^4$; all seven 248-slices follow from P, D, Δ=13 and $(e_1, e_2, \dim S^+)$; one affine map $\mathrm{row}(C+sU+tV) = (7,11,13)+s(3,3,3)+t(1,2,3)$ gives every flavour row budget; $78 = \dim E_6 = p_2\Delta$. The atlas is a closed grammar over two typed alphabets, exact, reading audit **[E]**
`v171_os_moment_cluster.py`	Atomic OS Moment Lemma + Sugawara Gap Safety: $\mathrm{spec}(T)$=$\{1, \frac{64}{729}, \frac{1}{729}\}$ makes $C(n)$=A+Br^n+Cs^n a positive moment sequence, so every OS Hankel matrix factorises as a Vandermonde Gram ⇒ reflection positivity (clean proof, not numeric Gram); cluster $\frac{729}{665}$, ξ=0.41105, Källen–Lehmann masses $6\log\frac{3}{2}, 6\log 3$; gap safety Δ_{eff}=$6\log\frac{3}{2}-\frac{31}{4\pi^2}$>0 with 31=1+$h^\vee(E_8)$, $c((E_8)_1)$=248/31=8 **[E]**/**[C]**
`v172_trace_anomaly_seed.py`	Trace-anomaly origin of the seed prefactor $\frac{4}{3}$: over the seam S^2 (χ=2) the $(E_8)_1$ (c=8) Weyl anomaly is $c\chi/6$=$\frac{8}{3}$, halved by the one-sided $\lvert\mathbb{Z}_2\rvert$ to $\frac{4}{3} = \dim S^+/\dim \mathfrak{g}_{SM} = 16/12$, the seed base $(\frac{4}{3})c_3$=$\frac{1}{6\pi}$. Plus QCD b_3=11−$\frac{2}{3}N_f$=7 = scalaron exponent $\Omega_{\mathrm{adm}}-10b_1$=48−41 (confinement↔inflation), and the Volkov–Wipf factorisation $-98/45$=$-2\cdot 7^2/\dim(D_5)$ **[E]**
`v173_pfaffian_cp_car.py`	Strong CP as Pfaffian reality: a self-dual CAR (16-Majorana) model demonstration of θ_{eff}=0 — the sheet-odd Calderón involution $\{D,\Sigma\}$=0 pairs the spectrum ($\pm\lambda$), γ_5-Hermiticity $D^\dagger$=$\gamma_5 D\gamma_5$ makes the Pfaffian real ($\arg \in \{0,\pi\}$), and reflection positivity (Z>0) selects the positive branch ⇒ θ_{eff}=0 **[E]**
`v174_seam_fock_readings.py`	Seam Fock-space readings: (1) CAR cone compression — $\dim \Lambda^{\mathrm{even}}(\mathbb{R}^{16})$=$2^{15}$=32768 reduces to a 16-dim one-particle problem under quasi-free Bogoliubov ($2^{15}/16$=2^{11}), the explicit count behind v161; (2) Witten index = the Plücker norm $11 = \sum_{k\le 2}\binom{4}{k}$, with $11 = \dim S^+ - g_{\mathrm{car}}$ (full 2^4=16) and c_u/c_d=55/117 reading the QBL-protected boundary state space; (3) Affleck–Ludwig g-theorem c_{UV}=5+3=8=$c_{IR}((E_8)_1)$ so Δc=0 forces an exact isometry (boundary-entropy form of v157/v158) **[E]**
`v175_net_existence_full_cone.py`	The heavy artillery — net existence (A2) and full-cone reflection positivity discharged to **[E]**, isolating `QGEO.REALIZE.01` as the single open premise (nothing fabricated): (1) **full-cone RP for all** m — the CAR second-quantisation functor $\Gamma(t)$=$\bigoplus_m \Lambda^m(t)$ of the one-particle seam contraction t acts on the complete Fock space $\Lambda^\bullet(\mathbb{R}^{16})$ ($\dim 2^{16}$=65536); the exterior-power spectrum theorem makes every eigenvalue a subset product of $\mathrm{spec}(t)$, verified on all 2^{16} products to lie in [0, 1] (contraction), with eigenvalue-1 multiplicity 2^8=256 (wedges of the rank-8 K_Σ polarisation) and sub-leading = the one-particle gap $(2/3)^6$, so full-cone RP reduces *for all* m to the one-particle contraction (Shale–Stinespring CAR functor), discharging the v160 scope premise by a theorem; (2) **A2 net existence** assembled + verified — $(E_8)_1$ character E_4/η^8=1+248q+4124q^2+34752q^3+213126q^4+1057504q^5 (level-1=248), embedding 248=120+128, E_8 Cartan even unimodular (det 1); the net *exists* by cited theorems (free-fermion net, lattice VOA, index-4 Q-system); (3) both collapse to the same seam-kernel identification = `QGEO.REALIZE.01`, finite half v168, realisation half left **[O][E][O]**

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Script	What it machine-checks *(continued)*
`v176_seam_collar_realisation.py`	The Seam Collar Realisation Theorem assembled as ONE central statement + reduction certificate (the single structural proof obligation, formulated cleanly — *not* a fabricated proof): given an oriented one-sided RP seam collar with a sheet-odd Calderón involution, faithful D_4 marks and admissible $c{=}8$ data, *then* its conformal boundary is Möbius-equivalent to $\mathbb{P}\setminus\mu_4$ with H^1 grading $(1,2,3)$, the Calderón contraction is the rank-8 K_Σ polarisation and the index-4 extension is $(E_8)_1$. The proof decomposes into six steps, FIVE already **[E]**: geometry/uniformisation (v168: cross-ratio 2, faithful Möbius D_4 order 8, $b_1{=}N_{\rm fam}{=}3$), Calderón DtN symbol $\lvert k\rvert$ (v156), H^1 character = generation grading $(1,2,3){=}A_3$ exponents$=\mathrm{Spec}(Q_+)$ (v137/v168), μ_4-equivariant contraction diagonal with gap $(2/3)^6$ (v162), AQFT closure $(D_5)_1{\otimes}(A_3)_1{\rtimes}\mu_4{=}(E_8)_1$ + full-cone RP all m (v154/v175); the whole structural residual reduces to the ONE open premise `QGEO.REALIZE.01` (the physical seam collar's boundary *is* this $\mathbb{P}\setminus\mu_4$), left honestly **[O][E][O]**
`v177_seam_marking_kernel.py`	The QGEO proof tree — the v176 single realisation premise split into its honest constituents: REALIZE = MARKS · UNIFORM · COHOM · MODULE · KERNEL · NET, with FOUR nodes closed *constructively* **[E]** and the open residual sharpened into exactly TWO named obligations. **[E]** UNIFORM (Lemma 2, constructive): an order-4 Möbius map conjugates to $z\mapsto iz$, a non-fixed orbit scales to μ_4, the reflection $z\mapsto 1/z$ gives $\sigma\rho\sigma=\rho^{-1}$ (D_4, order 8) — a genus-0 four-marked faithful-D_4 boundary *is* $(\mathbb{P},\mu_4)$; **[E]** COHOM: $\rho^*\omega_k=i^k\omega_k$ so the H^1 grading is $(1,2,3){=}A_3$ exponents$=\mathrm{Spec}(Q_+)$; **[E]** MODULE: the H^1 →generation iso is *unique* (multiplicity-free + residue normalisation; reflection $\omega_1\leftrightarrow\omega_3$, ω_2 fixed); **[E]** NET (v154/v175): index-4 μ_4 extension of $(D_5)_1{\otimes}(A_3)_1{=}(E_8)_1$. The TWO genuine open obligations **[O]**: `QGEO.MARKS.01` (raw seam → genus-0 four-marked D_4 boundary) and `QGEO.KERNEL.01` (raw seam Calderón kernel = the μ_4-equivariant free gapped contraction as an *operator*); the whole structural residual is now exactly these two doors **[E][O]**
`v178_marks_kernel_attempt.py`	An honest *attempt* at the two open obligations `QGEO.MARKS.01` + `QGEO.KERNEL.01` — *not* a closure (genuine constructive-geometry/AQFT statements, not finite computations), but a deeper reduction: each finite core is closed **[E]** and each obligation reduced to ONE sharper irreducible premise **[O]**. *Marks core* **[E]**: a genus-0 double with an order-4 clock ρ ($\sim z\mapsto iz$, fixed points $\{0,\infty\}$) and reflection τ forces the marks to be a generic ρ-orbit $\{1,i,-1,-i\}{=}\mu_4$ (not the fixed points), distinct iff $b_1{=}4{-}1{=}3{=}N_{\rm fam}$, $\tau\rho\tau{=}\rho^{-1}\Rightarrow$ faithful D_4 order 8 — so MARKS reduces to the single topological/clock premise. *Kernel core* **[E]**: on the 3-dim multiplicity-free H^1 the μ_4 characters $(1,2,3)$ are distinct, so by Schur a μ_4-equivariant operator is forced diagonal and *completely determined by its eigenvalue per character* — two such with the forced transfer spectrum $\{1,(2/3)^6,(1/3)^6\}$ (v162) and the unique residue-normalised basis (v177) are *equal as operators*, not merely isospectral (the spectrum-vs-operator worry vanishes on the finite block); so KERNEL reduces to the single full-operator-reduction premise (principal symbol $\lvert k\rvert$, Lee–Uhlmann/v156; no drift, v158). Neither obligation closed; both sharpened **[E][O]**

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Script	What it machine-checks *(continued)*
`v179_conformal_realisation.py`	The two residual premises (Marks-residual + Kernel-residual) collapse to ONE geometric premise `QGEO.CONF.01` (honest unification; the conformal realisation stays **[O]**). Marks's residual is not three independent assumptions: (1) genus-0 *is* P1 — $c_3=1/(\lvert\mathbb{Z}_2\rvert\, 2\pi\,\chi(S^2))=1/(8\pi)$ with $\chi=2 \Rightarrow g=0$, the same $\chi=2$ that gives the '8' (v58/v73); (2) the order-4 clock *is* the carrier — Coxeter element of $W(A_3)=S_4$, order $h(A_3)=4=\lvert\mu_4\rvert$ (v117); (3) marks = one orbit is *forced* — parabolic marks $\neq$ the two elliptic fixed points $\{0,\infty\}$, a free order-4 orbit has exactly $4=e_1(a)$ points. So Marks reduces to 'the carrier clock acts conformally on the genus-0 double'. Kernel then follows from that geometry: DtN symbol $\lvert k\rvert$ (Lee–Uhlmann, v156) $\Rightarrow$ rank-8 $c=8$ fixed polarisation; no drift (v158); the cusped-surface residual data is $H^1=\mathbb{C}^3=N_{\text{fam}}$, the 3-dim gapped transfer block (Schur, v162/v178). Both $\Rightarrow$ ONE premise `QGEO.CONF.01`: the raw RP seam double is conformally $(\mathbb{P},\mu_4)$ with the carrier clock as the order-4 Möbius map — the single structural residual of the whole theory, left honestly **[O][E][O]**
`v180_clock_is_mobius.py`	Cracking `QGEO.CONF.01` one level further — the conformal-realisation premise is *replaced*, via three classical theorems, by a strictly *milder*, non-circular premise `QGEO.ISO.01`: 'the carrier clock is an order-4 orientation-preserving *isometry* of the RP seam metric'. (1) Uniformisation (Poincaré–Koebe): a genus-0 Riemann surface is conformally $\mathbb{P}$, so the genus-0 seam double (P1, v179) *is* $\mathbb{P}$ — the target is produced, not assumed; (2) Kerékjártó (1919; Constantin–Kolev 2003): a finite-order orientation-preserving homeomorphism of S^2 is conjugate to a rotation, so the order-4 clock ($h(A_3)=4$) is topologically $z\mapsto iz$; (3) isometry⇒conformal⇒Möbius: an orientation-preserving isometry of a Riemannian surface is holomorphic, $\mathrm{Aut}(\mathbb{P})=\mathrm{PSL}(2,\mathbb{C})$; (4) order-4 Möbius $= z\mapsto iz$ (verified: projective order 4, fixed $\{0,\infty\}$, multiplier i, free orbit μ_4). So `QGEO.CONF.01` is *implied* by the milder isometry premise; the new residual `QGEO.ISO.01` (the seam transport is a metric-compatible holonomy preserving the RP inner product) does not name $\mathbb{P}/\mu_4$ and is milder, its surface realisation from the raw seam still **[O][E][O]**
`v181_clock_is_conformal_symmetry.py`	The FINAL honest reduction + the explicit identification of BEDROCK. The isometry premise `QGEO.ISO.01` (v180) is weakened one more genuine step — to a CONFORMAL-symmetry premise `QGEO.SYM.01` — via equivariant uniformisation (Nielsen realisation, finite cyclic on genus-0): a *conformal* automorphism (strictly weaker than an isometry — every isometry is conformal, conformal allows any positive scale) already suffices to put the order-4 clock in the form $z\mapsto iz$. The clock is the Coxeter element of $W(A_3)=S_4$ (v117), order $h(A_3)=4=\lvert\mu_4\rvert$, the μ_4 deck, and a deck transformation preserves the pulled-back conformal structure by definition. So below `QGEO.SYM.01` the residual is *definitional*, not a missing theorem: 'the seam's conformal structure *is* the μ_4-deck structure of the carrier clock' (the carrier μ_4 glue is the conformal monodromy of the seam) — a physical/definitional identification tying P2 to the seam's conformal geometry, not reducible further without relabeling. The chain realize(v175)→theorem(v176)→marks+kernel(v177)→finite cores(v178)→`CONF.01`(v179)→`ISO.01`(v180)→`SYM.01`(v181) has reached bedrock; we stop here honestly **[E][O]**

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Script	**What it machine-checks** *(continued)*
`v182_reviewer_residual_map.py`	The four external-review concerns reduce to the two already-named residuals + one data-only item. A (mapping $2D \to 4D$) **[E]**/**[C]**: the Plücker readout $11 = \sum_{k\le 2}\binom{4}{k} = \dim S^+ - g_{\text{car}} = 16-5$ (QBL count, v174) and the holographic reduction give $A = \texttt{QGEO.SYM.01} + F_{\text{transfer}}$. B (UV gravity) **[E]**: 'gravity = geometric boundary readout' *is* `QGEO.SYM.01`. C (grammar tuning) **[E]**/**[C]**: frozen (v84) + minimal + over-determined (the anchor a=(1, 1, 2) in ≥ 6 independent not-selected-for structures) is look-elsewhere-resistant, but the residual falls only to out-of-sample data (JUNO/CMB-S4). D (Koide Möbius flow) **[E]**/**[C]**: the Möbius nature is forced by Aut($\mathbb{P}$)=PSL(2, $\mathbb{C}$) on the deck, t∼2.84 only locates the point, and the missing RG generator *is* F_{transfer}, so $D = \texttt{QGEO.SYM.01} + F_{\text{transfer}}$. NET: three reduce onto $\{\texttt{QGEO.SYM.01}, F_{\text{transfer}}\}$, one waits for data — a unification of the concern structure, not a closure (premises stay **[O]**/**[C]**) **[E][C]**
`v183_koide_f_corner_transfer.py`	The Koide source→pole factor $\frac{53}{54}$ has an *operator* origin (external-review proposal, validated): $u_{\text{pole}}/u_{\text{source}} = a^T(R+Q)\mathbf{1}/(2\,\mathbf{1}^T Ra) = 53/(2{\cdot}27) = \frac{53}{54}$. (1) $\mathbf{1}^T Ra = 27$ is the $E_6{\times}A_2$ cubic family block ($248 = \dim E_6\, 78 + \det R\, 8 + 2{\cdot}27{\cdot}N_{\text{fam}}$); (2) F=R+Q is the *missing* Sheet-Diamond corner $M(1,1)$ with $\det B_F = 52 = \dim F_4$ (v80) and anchor response $a^T(R+Q)\mathbf{1} = 53 = 52{+}1$; (3) $\frac{53}{54} = 1 - \frac{1}{2N_{\text{fam}}^3}$ exactly (54=2·27); (4) negative control: only F gives 53 ($R/Q/K/L \to$ 32/21/35/56). Upgrades `FR.KOIDE.03` from a bare guess to an operator identity; **[E]** the factor is exact, **[C]** the leptonic source→pole reading the F corner stays an F_{transfer} conjecture **[E][C]**
`v184_etaB_anchored_boltzmann.py`	Honest test of the external-review η_B proposal (M_1=$M_R\phi_0^4$, $\tilde{m}_1$=m_3/A_Λ) — firewall verdict: *sharper scenario*, not a closure. The washout anchors plausibly ($\tilde{m}_1$=m_3/A_Λ=$m_3/10 \approx 5\,$meV, A_Λ=10=$\lvert E(K_5)\rvert$, in the strong-washout window) **[C]**; but M_1 does *not* — M_R is the seesaw scale $\sim v^2/m_3 \approx 6{\times}10^{14}$ GeV and the nearest clean TFPT powers of $\bar{M}_{\text{Pl}}$ both miss it (c_3-power off ∼75%, ϕ_0-power off ∼40%), so M_1=$M_R\phi_0^4$ merely relocates the free input $M_1 \to M_R$ (the reviewer's $6.0{\times}10^{-10}$ uses M_R=$1.3{\times}10^{15}$; the seesaw value gives $3.4{\times}10^{-10}$, the tell that M_1 is unpinned). η_B stays in the observed band so the route is viable, but the proposal cuts the free inputs from two to one, not to zero — η_B stays **[C]**, a sharper scenario **[C]**
`v185_axion_relic_solver.py`	Axion relic, honest misalignment estimate: the determinant-line axion f_a=$M_{\text{scal}}/128 \approx 2.39{\times}10^{11}$ GeV ($m_a \approx 23.8\,\mu$eV) can provide all dark matter, but *only* near the $\theta_i \approx 170°$ hilltop where the relic is exponentially sensitive — so the axion DM branch stays a **[C]** scenario. The harmonic misalignment under-produces at this f_a (prefactor ∼$0.026/\theta^2$, θ∼$O(1)$ gives ∼21% of Ω_{DM}, need θ^2_{eff}∼4.7), so the hilltop anharmonic enhancement is required; near the hilltop it is exponentially sensitive, hence a scenario not a sharp prediction. $F_{\text{relic}}(f_a, \theta_i, N_{\text{DW}}) \overset{?}{=} \Omega_{\text{DM}}$ is a kill test for the *branch*, not the theory (no free singlet WIMP: 128=(16, 4)+(16, 4)) **[C][O]**
`v187_ftransfer_laws.py`	The F_{transfer} discipline guard (CI-style firewall): the four continuous-transfer frontier observables — Koide, η_B, the axion relic, m_p/m_e+Λ_{QCD} — must *never* be typed as primitive **[E]** compiler predictions; each carries a **[C]**/**[O]** marker. The guard reads `status_ledger.csv` and enforces it (exact algebraic sub-parts — Koide 53/54 v183, b_3=−7 v159 — may be **[E]**, the physical prediction never is). Records the four interfaces $F_{\text{pole}}/F_{\text{Boltzmann}}/F_{\text{relic}}/F_{\text{QCD}}$ and that the raster rule + frozen registry + null model bind, so no future edit can quietly promote a pretty frontier number to a closed compiler output **[E]**

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Script	What it machine-checks *(continued)*
`v188_frontier_wording_guard.py`	The frontier-wording guard (a prose sentinel, no new physics): v187 protects the ledger *typing*, this guard protects the *prose*. It forbids stale sentences whose semantics an earlier round superseded — "leptogenesis interface ([E])" (v184 retyped Route B to [C]), "closed branch also fixes the leptogenesis inputs" (M_R is the seesaw scale, not a compiler power), "relic scale f_a, m_a open" (v185: the scales define the branch, the open item is the hilltop-sensitive relic abundance), "computed Q=0.664 . . . near, not exact" (v183 gave 53/54 an exact operator origin), "Suite now 187 modules" (the count is 187, highest ID `v188`; v186 skipped). Scans the 10 active documents; passes when none appear, so a frozen Zenodo release can never re-introduce half-true wording [E]
`v189_riemann_roch_carrier.py`	The Riemann–Roch carrier reading: $(g_{\rm car}, N_{\rm fam}) = (5, 3)$ are the two canonical invariants of *one* object, the four-point seam divisor $D = \mu_4$ on $\mathbb{P}^1$. [E] arithmetic: $h^0(\mathbb{P}^1, \mathcal{O}(\mu_4)) = \deg +1 = 5$ (function side) and rank $H_1(\mathbb{P}^1 \setminus \mu_4) = \deg -1 = 3$ (cycle side, already `QGEO.COHOM.01`/v177); $g_{\rm car} = 5 = \mathrm{rank}(D_5)$ (the carrier is the $\mathfrak{so}(10)$ half-spinor, rank D_5+ rank $A_3 = 5+3 = 8$, v1), $N_{\rm fam} = 3$. [C]: reading the carrier as the meromorphic seam-mode space $H^0(\mathbb{P}^1, \mathcal{O}(\mu_4))$ (less number, more geometry) is matched by $\mathrm{rank}(D_5) = 5$, but the physical identification is conjectural and does *not* replace the over-determined $g_{\rm car} = 5$ (Pascal/bootstrap/Lean) — a fourth, geometric reading [E][C]
`v190_nariai_entropy_bound.py`	The one-line Nariai entropy bound: the Schwarzschild–de Sitter entropy ratio $S_{\rm tot}/S_{dS} = (x^2{+}1)/\Phi_3(x)$ ($x{=}r_b/r_c$; $\Phi_3{=}x^2{+}x{+}1$ the $N_{\rm fam}{=}3$ cyclotomic, already in `tfpt_horizon_readouts`) satisfies $S_{\rm tot} \geq \frac{2}{3} S_{dS}$ with equality exactly at the Nariai merge $x{=}1$, via the perfect square $3(x^2{+}1) - 2\Phi_3(x) = (x-1)^2 \geq 0$ — a variation-free proof of the entropy floor (global minimum on $x{>}0$ is $2/3 = \lvert Z_2\rvert/N_{\rm fam}$) [E]
`v191_universal_branch_line.py`	The universal branch line, honestly typed [C] — an exact affine *relabeling*, not a theorem. The map $q = \frac{7}{2} + \frac{N_{\rm fam}^2}{2} m$ carries the SdS branch points $m = \pm 1/N_{\rm fam}$ onto the flavor branch points $\{2, 5\} = \{\lvert Z_2\rvert, g_{\rm car}\}$ ($m{=}0{\to}7/2$). The arithmetic is exact, but the map exists for *any* two pairs of points (two intervals are always affinely equivalent), so the slope $N_{\rm fam}^2/2$ and midpoint $7/2$ are *determined* by the data, not forced predictions; a decoy pair $\{3, 4\}$ admits an equally exact map (negative control). The genuine shared content — the $2/3 = \lvert Z_2\rvert/N_{\rm fam}$ ramification of the mass↔transport double cover — is already established. Recorded as a [C] alignment, never promoted to [E] [C]
`v192_qgeo_energy_clock.py`	The energy-conserving-clock reformulation, honestly typed: `QGEO.ENERGY.01` ($\rho^* \Lambda_\Sigma \rho = \Lambda_\Sigma$ — the carrier clock preserves the Calderón/DtN energy form) *relocates* the bedrock `QGEO.SYM.01` to a more checkable operator identity, but does *not* eliminate it. The downstream chain (conformality → Möbius → $z \mapsto iz \to \mu_4 \to$ `QGEO.SYM.01`) already exists (v180/v181); the new upstream link (energy invariance ⇒ conformality in 2D) is checkable on the DtN operator $\Lambda = \lvert k\rvert$. But 'ρ preserves Λ' is plausibly equivalent to 'ρ is the conformal deck', so it is a sharper restatement, not a theorem, unless ρ is independently defined and the invariance independently proven (open). Bedrock stays [O]
`v193_qgeo_energy_commutator.py`	`QGEO.ENERGY.02`: the energy-*commutator* contract (the non-circular sharpening of v192): $[\rho, \Lambda_\Sigma] = 0$ on the rank-8 Calderón polarisation, ρ the carrier clock, Λ_Σ the DtN energy form of the *raw* RP seam. Three sub-claims: (1) [E] ρ is carrier-defined (Coxeter of $W(A_3){=}S_4$, order $4{=}\lvert\mu_4\rvert$, v117), not imported from the seam; (2) [O] the non-circularity hinge — Λ_Σ must be the *raw* RP-seam DtN, not the μ_4 normal form; (3) [E] $[\rho, \Lambda_\Sigma]{=}0$ forces the $(1, 2, 3)$ grading on H^1 (`QGEO.COHOM.01`). Exact [E] rider: the induced $k{=}(\ln m)/(12\pi)$ v150 reproduces $c_3/2$ iff $\ln m{=}q(A_3){=}\frac{3}{4}$ (FAMILY), not $q(D_5){=}\frac{5}{4}$ (carrier, off by 5/3) — gravity is family-geometry-induced. Proof target [O]; if proven, `QGEO.SYM.01` becomes an energy-rigidity theorem [O]

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Script	**What it machine-checks *(continued)***
`v194_raw_seam_dtn_rp.py`	`QGEO.DTN.01` — subclaim 2 of v193 ("the raw RP-seam DtN is RP-definable") tackled: a *reduction*, not a closure. (1) **[E]** finite core: $\rho{:}z\mapsto iz$ on $H^1=\langle w_1,w_2,w_3\rangle$, $w_k=z^{k-1}dz/(z^4-1)$, acts as $\rho^*w_k=i^k w_k$ (characters $i,-1,-i$ distinct), so $[\rho,\Lambda_\Sigma]=0$ on H^1 (v177). (2) **[E]** hinge met: the DtN/transfer operator is RP-canonical (Osterwalder–Schrader, v54) on a quasi-free seam state (v155), so Λ_Σ is RP-definable without assuming $\mathbb{P}^1\backslash\mu_4$. (3) **[O]** residual relocates: full-L^2 commutation $\Leftrightarrow$ Bisognano–Wichmann geometric modular flow of the quasi-free seam state (must be intrinsic, else re-circulates). `QGEO.SYM.01` reduces from a definitional postulate to intrinsic BW modular covariance — sharper, still **[O]**, not closed **[O]**
`v195_marks_lefschetz_character.py`	`QGEO.MARKS.02` — a Lefschetz/character derivation of the four seam marks (review path 2): instead of positing D_4 marks, they are *forced* to be a free μ_4 orbit. **[E]**: $\rho{:}z\mapsto iz$ (order 4) fixes only $\{0,\infty\}$; $H^1=\chi_1+\chi_2+\chi_3$ (the A_3 exponents $(1,2,3)$, v177) has no trivial character $\Rightarrow$ no puncture at a fixed point; rank $H_1=n-1=3\Rightarrow n=4\Rightarrow$ exactly one free orbit $\Rightarrow$ marks $=\mu_4$ (cross-ratio 2); $\mathrm{Tr}(\rho\vert H^1)=i+i^2+i^3=-1$, $L(\rho)=1+1=2=\#$fixed. **[O]** residual relocates from "the seam is $\mathbb{P}\backslash\mu_4$ (geometric)" to "H^1 carries the A_3-exponent rep" — less circular **[E][O]**
`v196_seam_energy_functional.py`	`QGEO.VARI.01` — the E_{fail} variational functional (review path 1), a numerically-testable sharpening of the v194/`ENERGY.02` target: $E_{\text{fail}}=\Vert[\rho,\Lambda_\Sigma]\Vert^2_{HS}+\Vert\rho^4-1\Vert^2+\Vert\Theta\rho\Theta-\rho^{-1}\Vert^2$, with $E_{\text{fail}}=0\Leftrightarrow$ the `QGEO.SYM.01` conditions. **[E]** on the finite H^1 block ($\rho=\mathrm{diag}(i,-1,-i)$, Λ diagonal, Θ=conj) all three vanish $\Rightarrow E_{\text{fail}}=0$; **[E]** a μ_4-breaking Λ or non-order-4 ρ gives $E_{\text{fail}}>0$ (μ_4 a true minimiser); **[E]** the zero locus is the seam-deck conditions; **[O]** the full-operator minimisation is the v194 Bisognano–Wichmann residual — a discretisable handle on the open bedrock, a target not a closure **[E][O]**
`v197_rr_carrier_clifford_d5.py`	`ARCH.RRCAR.02` — hardening the Riemann–Roch carrier from g_{car} to D_5 itself (review point 7): the 5-dim carrier mode space $C_{\text{car}}{:=}H^0(\mathbb{P}^1,\mathcal{O}(\mu_4))$ generates the $\mathfrak{so}(10)=D_5$ half-spinor by its even Clifford algebra. **[E]** $\dim\Lambda^{\text{even}}(\mathbb{C}^5)=\binom{5}{0}+\binom{5}{2}+\binom{5}{4}=1+10+5=16=2^{g_{\text{car}}-1}=\dim S^+$; the $\mathfrak{so}(10)=D_5$ half-spinor $2^4=16=S^+=\Lambda^{\text{even}}(C_{\text{car}})$; closes the chain $\mu_4\to g_{\text{car}}\to D_5$ from one four-point divisor (cycle side rank $H_1=3=N_{\text{fam}}(A_3)$; function side $h^0=5=g_{\text{car}}$; Clifford spinor $16=D_5$ half-spinor = carrier). **[C]** "carrier = Clifford spinor of the seam mode space" stays physical (as v189), does not change the over-determined $g_{\text{car}}=5$ **[E][C]**
`v198_modular_commutator_reduction.py`	`QGEO.MODULAR.01` — cracking the bedrock to a state-invariance. The v194 residual ($[\rho,\Lambda_\Sigma]=0$ on full L^2, framed as Bisognano–Wichmann with a circularity worry) is (a) made *exact* at the principal-symbol level and (b) reduced via Tomita–Takesaki to a clean state-invariance. **[E]** the DtN principal symbol $\vert k\vert=\sqrt{-d^2}$ is $\mathrm{diag}(\vert n\vert)$ and the clock $\rho{:}z\mapsto iz$ is $\mathrm{diag}(i^n)$ in the Fourier basis — both diagonal $\Rightarrow[\rho,\vert k\vert]=0$ exactly on *all* of L^2; **[E]** Tomita–Takesaki: $[\rho,\Lambda_\Sigma]=0$ follows from $\omega\circ\rho=\omega$ (the modular Δ is intrinsic to (M,Ω); a state-preserving symmetry commutes with it), needing *no* conformal covariance — removes the BW circularity; **[E]** the finite H^1 block is already μ_4-invariant (v177). **[O]** residual: the full raw seam state is μ_4-invariant ($\omega\circ\rho=\omega$) $\Leftarrow$ the Gaussian bulk measure is deck-invariant = operational `QGEO.SYM.01`. Sharpest non-circular form; not a closure (a foundational symmetry cannot be derived from nothing) **[E][O]**

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Script	**What it machine-checks *(continued)***
`v199_seam_state_invariance.py`	`QGEO.STATE.01` (work package A) — attacking $\omega\circ\rho{=}\omega$ directly on the raw seam: a further localising reduction, not a closure. [E] setup non-circular (ρ = Coxeter of $W(A_3){=}S_4$, $\rho^4{=}1$, an automorphism of the raw CAR algebra; both carrier-side, no seam geometry); [E] ground-state reduction $\omega\circ\rho{=}\omega \Leftrightarrow [\rho, H_1]{=}0$ ($C_\Sigma{=}\frac{1}{2}(1{+}\operatorname{sgn} H_1)$); [E] character criterion: $[\rho, H_1]{=}0 \Leftrightarrow H_1$ block-diagonal in the four μ_4-character classes $\{n{=}r \bmod 4\}$; principal symbol $\lvert k\rvert{=}\operatorname{diag}(\lvert n\rvert)$ passes automatically. [O] residual: the bounded sub-principal symbol has *no* off-character matrix elements — non-circular, sharply typed, holds on H^1 (v177); not a closure [E][O]
`v200_seam_variational_scan.py`	`QGEO.VARI.02` (work package B) — the discretised variational test: v196 had $E_{\text{fail}}{=}0$ at the μ_4 config on the fixed block; here the clock angle is free and E_{fail} is minimised over it. [C] scanning $E_{\text{fail}}(\theta){=}\lVert\rho(\theta)^4{-}1\rVert^2$+faithfulness over $\theta{\in}(0,\pi)$ gives the unique minimum at $\theta{=}\pi/2$ ($E_{\text{fail}}{=}0$: order-4 *and* three distinct characters $i, -1, -i$ = the $(1, 2, 3)$ grading); decoys $(0, \pi, \pi/3, 2\pi/3)$ all lose. So $\arg\min E_{\text{fail}}$ = the μ_4 normal form. [E] consistent with v196/v199/v198. [O] a numerical sign on the finite H^1 grading, not the full operator-valued minimisation (the v199 residual) [C][O]
`v201_seam_subprincipal_marks.py`	`QGEO.SUBPRIN.01` — the v199 bedrock residual reduced one genuine step further: block-diagonality of the sub-principal symbol is *not* an independent postulate. [E] the seam DtN is $\Lambda{=}\lvert k\rvert{+}M_f$, M_f = multiplication by a curvature function $f(\theta)$ (standard Steklov sub-principal), so $\langle n\lvert M_f\rvert n'\rangle{=}f_{n-n'}$. [E] M_f is μ_4-character-block-diagonal $\Leftrightarrow$ f has Fourier support only on modes $\equiv 0 \pmod 4$ $\Leftrightarrow$ f is $\mathbb{Z}_4$-invariant ($\lvert k\rvert$ already commutes, v198). [E] a μ_4-mark sum $f(\theta){=}\sum_{j=0}^{3} g(\theta{-}2\pi j/4)$ has $f_m{=}4g_m[m{\equiv}0 \bmod 4]$ for *any* profile g — automatically $\mathbb{Z}_4$-invariant, so (v195 marks=μ_4 orbit) *forces* the sub-principal symbol block-diagonal; negative controls: 3 marks ($\mathbb{Z}_3$) and 4 generic marks break it. [O] new residual: $\omega\circ\rho{=}\omega$ reduces to 'the DtN sub-principal symbol is mark-local (seam flat away from the μ_4 marks = conformal deck v177)', the narrowest, structurally-definitional form of `QGEO.SYM.01` [E][O]
`v202_rare_kaon.py`	The rare kaon sector $K\to\pi\nu\bar\nu$ from the TFPT CKM point ([C] downstream flavor readout, *not* a compiler power). The kernel fixes the full CKM matrix ($s_{12}{=}\lambda_C$, $s_{23}{=}\varphi_0/(1{+}\lambda_C){=}\lvert V_{cb}\rvert$, $s_{13}{=}\lambda_C^3/3$, $\delta{=}\pi/3{+}3\lambda_C^2{=}1.19823$ rad; v18/v88) with no flavor fit; feeding the exact point into the standard Brod–Gorbahn–Stamou short-distance formulas gives $\mathrm{BR}(K^+\to\pi^+\nu\bar\nu){=}9.45\times10^{-11}$ and $\mathrm{BR}(K_L\to\pi^0\nu\bar\nu){=}3.33\times10^{-11}$. [E] CKM point + exact apex $(\bar\rho, \bar\eta){=}(0.1374, 0.3509)$, $\lambda_t{=}V_{ts}^*V_{td}$; [C] the branching ratios (*not* parameter-free — $X_t, P_c, \kappa_\pm$ are external EW/QCD input); [E] $\mathrm{BR}(K^+)$ $\mathrm{BR}(K^+){=}9.45\times10^{-11}$ lands ON the NA62 2016-2024 combination $(9.6^{+1.9}_{-1.8})\times10^{-11}$ (La Thuile 2026), $+0.08\sigma$ (the 2016-2022 obs $(13.0^{+3.3}_{-3.0})\times10^{-11}$ was $\sim 1.2\sigma$ above) — a strong consistency hit, but an F_{transfer} bridge, not a unique TFPT-vs-SM discriminator (SM $(8.6\pm0.4)\times10^{-11}$ also inside the NA62 error); [E] Grossman–Nir ratio $0.352 \ll 4.3$; [X] dated kill test: a stable NA62 $\mathrm{BR}(K^+)$ outside $[7, 12]\times10^{-11}$, or a KOTO-II $\mathrm{BR}(K_L)$ off the GN point, breaks the flavor bridge (core untouched) [C][X]

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Script	**What it machine-checks** ***(continued)***
`v203_eht_achromatic.py`	The EHT achromatic polarization-rotation intercept $\beta_{\rm BH}$ around a horizon-scale black hole — the surviving core of the old UFE/BH notes (*not* the RN-type metric, superseded by Nariai/seam=horizon v101–v104/v190; only the polarization signature). **[E]** the coupling is a compiler number: $\beta_{\rm BH}(r){=}Q_e^{\rm eff}Q_m^{\rm eff}/(256\pi^4r^2){=}16c_3^4(Q_eQ_m/r^2)$, $16c_3^4{=}1/(256\pi^4)$ exactly. **[E]** α-kernel cross-link: $16c_3^4{=}\delta_{\rm top}/3$, $\delta_{\rm top}{=}48c_3^4{=}3/(256\pi^4)$ — the *same* top-form coefficient that fixes the α-kernel precision-zone correction (one row from $\beta_{\rm rad}{=}\varphi_0/4\pi$). **[C]** TFPT fixes the $1/r^2$ profile, achromaticity, and sign-flip under $E{\cdot}B$ reversal; the amplitude Q_eQ_m is an MHD/GR weight, not a compiler output. **[X]** dated kill test: the residual intercept must pass three nulls (frequency/achromatic, spatial $1/r^2$, sign-flip); a null-consistent residual after honest GRMHD subtraction falsifies the channel (core untouched) **[E][C][X]**
`v204_muon_seam_g2.py`	The muon anomalous magnetic moment $a_\mu{=}(g_\mu{-}2)/2$ as a seam vertex readout (**[C]** downstream, *not* a compiler power; archive integration of the old muon-$g{-}2$ note). The carrier fixes the second-order defect $\delta_2{=}B\gamma\,\delta_{\rm top}^2{=}\frac{5}{4}\delta_{\rm top}^2$ ($\delta_{\rm top}{=}\Omega_{\rm adm}c_3^4{=}48c_3^4{=}3/256\pi^4$, v2/v23; $B\gamma{=}\frac{3}{2}{\cdot}\frac{5}{6}{=}\frac{5}{4}$, v51); the seam-loop projection $a_\mu^{\rm seam}{=}\delta_2/(2\pi){=}45/(524288\pi^9)\approx 2.879\times10^{-9}$. **[E]** the value is an exact compiler number ($\delta_2{=}4!\,{\rm Tr}_{S^+}(X^2)c_3^8$, ${\rm Tr}{=}120{=}5!$); **[C]** the identification of $\delta_2/(2\pi)$ as a magnetic vertex ($1/2\pi{=}4c_3$) is a physical bridge; **[E]** 0.81σ vs dispersive $\Delta a_\mu{=}(2.49{\pm}0.48)\times10^{-9}$; **[C]** HONEST: lattice/CMD-3 HVP shrinks it ($\sim1.5\times10^{-9}$), the fixed value then $\sim1.5\sigma$ high; **[X]** dated kill test: a converged Δa_μ outside $2.879\times10^{-9}{\pm}0.5\times10^{-9}$ excludes the mechanism **[E][C][X]**
`v205_xi_threequarter.py`	$\xi{=}c_3/\varphi_{\rm tree}{=}3/4$: an *independent* route to the gravitational 3/4 (archive integration of the old Eliminating-K note). The torsion-compression ansatz $\kappa^2{=}\xi\varphi_0/c_3^2$ with the Einstein limit $\kappa^2{=}8\pi G$ fixes $\xi{=}8\pi c_3^2/\varphi_0{=}c_3/\varphi_0$ (since $8\pi c_3^2{=}c_3$), and at tree level $\varphi_{\rm tree}{=}1/(6\pi)$ this is exactly 3/4. **[E]** tree identity $\xi{=}c_3/\varphi_{\rm tree}{=}3/4$; **[E]** Einstein-limit reduction $8\pi c_3^2{=}c_3\Rightarrow G$ is an output; **[E]** over-determination $3/4{=}q(A_3){=}N_{\rm fam}/\lvert\mu_4\rvert{=}\ln(m/\mu)$ (v152 $k{=}c_3/2$) — the torsion-compression ξ and the gapped EH replica are two independent appearances of the same gravitational 3/4; **[E]** with the retained seed $\xi{=}c_3/\varphi_0{=}0.748303$ (-0.226%); **[C]** the $\kappa^2{=}\xi\varphi_0/c_3^2$ ansatz is the bridge, the modern derivation is v152/v68, $1/G$ stays the one anchor (v60/v153) **[E][C]**
`v206_h0_lambda_branch.py`	The cosmological-constant Letter branch: the seam number $\delta_\Sigma{=}m_1e^{-2\pi\alpha^{-1}/3}$ with the lowest block $m_1{=}\Omega_{\rm adm}{=}48$, the Planck-convention split, and the quantitative $H_0{=}66.5$–67.1 (archive integration of the tfpt-31 CC note; sharpens the qualitative H_0-vs-Λ row). **[E]** $\delta_\Sigma^{\rm lead}{=}48\,e^{-2\pi\cdot137.036/3}\approx1.085\times10^{-123}$; **[E]** convention split $M_{\rm Pl}^4/\bar M_{\rm Pl}^4{=}(8\pi)^2{=}631.65$ (unreduced vs reduced — must be displayed); **[E]** $\rho_\Lambda{=}M_{\rm Pl}^4\delta_\Sigma\approx2.4\times10^{-47}\,{\rm GeV}^4$ (observed vacuum order); **[E]** 123-orders cross-link $\lvert\log_{10}\delta_\Sigma\rvert\approx122.96$, consistent with the v60 split 122.948; **[C]** under flatness closure $H_0^{\rm lead}{=}66.51$, $H_0^{\rm disc}{=}67.10$ — 1.6–0.5σ below Planck, $\sim6\sigma$ below SH0ES: the Hubble tension is *not* relieved by raising the vacuum scale; H_0 uses imported $\Omega_b,\omega_{\rm DM}$, a **[C]** readout **[E][C]**
`v207_asymptotic_safety.py`	Asymptotic Safety as a *second*, independent, **[O]** route to the metric sector (archive integration of the old Five-Problems note). The current theory closes QG via the discrete seam-net/AQFT programme (v175; open premise `QGEO.REALIZE.01`); this records a complementary continuum-FRG argument that a non-Gaussian UV fixed point exists from the same axioms — a plausibility cross-check, *not* load-bearing, does *not* close $G_{\rm metric}$. **[E]** the axion coupling is fixed and small $y^2{=}16c_3^2{=}1/(4\pi^2){=}0.025330$ exactly ($g_{a\gamma\gamma}{=}{-}4c_3$ from the cubic); **[E]** existence by IVT: $\partial_tg/g{=}2{+}\eta_R$ is $+2{>}0$ as $g{\to}0$ and ${<}0$ at large g (torsion K^2 damping) $\Rightarrow$ a root $g_*{>}0$; **[E]** $g_*{=}2/a_1{>}0$ finite, Reuter-type $O(1)$, no new free input; **[O]** HONEST: a continuum-FRG ansatz the discrete-compiler theory did *not* adopt, primary stays the seam-net (v175) **[E][O]**

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Script	What it machine-checks *(continued)*
v208_bh_thermodynamics.py	Black-hole thermodynamics from the seam: the modular Hawking temperature and the scalaron-corrected Wald entropy (archive integration of the dropped tfpt-42 stationary-horizons section, re-typed through the current Tomita–Takesaki seam work v198/v199). **[E]** modular T_H: $\Delta^{it}{=}e^{-2\pi t K_H}$ (Bisognano–Wichmann) $\Rightarrow$ KMS at $\beta{=}2\pi$ and $T_H{=}\kappa/(2\pi)$; the 2π is the seam unit $1/(4c_3)$ ($1/2\pi{=}4c_3$, v58), reproducing $T_H{=}c_3/M$ (v57); seam$\leftrightarrow$horizon identification **[C]**; **[E]** Wald entropy: with the induced $f(R){=}R{+}R^2/(6M_s^2)$ ($M_s{=}c_3^{7/2}\bar{M}_{\rm Pl}$, v28), $S_W{=}(f_R A)/(4G){=}(A/4G)(1{+}R_h/3M_s^2)$ — leading $A/4G$ is the c_3 area law ($1/4{=}1/\lvert\mu_4\rvert$), the correction an exact scalaron consequence; **[E]** SdS $T_H{=}(1/4\pi r_h)(1{-}\Lambda r_h^2)$, $1/4\pi{=}2c_3$; **[E]** area law $S_{BH}{=}M^2/(2c_3){=}4\pi M^2{=}A/4G$ **[E][C]**
v209_bh_regular_core.py	The black hole as a seam fixed point (topological defect), *not* a singularity: the honest **[O]** synthesis of the old Five-Problems BH picture, with the superseded torsion/RN-vortex metric dropped (RN-type metric superseded by Nariai/seam=horizon v101–v104/v190; cf. v203). Built entirely from established suite atoms. **[E]** no singularity = seam fixed point: the gapped attractor multiplier $\lambda_2{=}(2/3)^6{=}64/729$ with gap $\Delta{=}6\log(3/2){>}0$ (v56) — the core is a fixed point, $\rho{\to}\infty$ replaced by $\varphi{\to}\varphi_*$ **[C]**; **[E]** maximal BH = anchor: Nariai cubic $t^3{-}3t{+}2{=}(t{-}1)^2(t{+}2)$, roots $(1,1,-2)$ = the traceless projection of $a{=}(1,1,2)$ (v101/v190); **[E]** information stays in the boundary field: Page recovery decays at $\lambda_2{=}(2/3)^6$ per step (v54) **[C]**; **[E]** holographic remnant $S_{BH}{=}A/4G$, $1/4{=}1/\lvert\mu_4\rvert$ **[C]**; **[O]** HONEST: a qualitative structural synthesis (reuses v54/v56/v57/v101/v190), *not* load-bearing, *not* a metric **[E][O]**
v210_mark_local_dtn.py	A numerical *sharpening* of v201/`QGEO.SUBPRIN.01` on *realistic* Steklov curvature profiles, carried to the quasi-free *state* level (the actual $\omega{\circ}\rho{=}\omega$, not just $[\rho,\Lambda]{=}0$); *not* a closure of `QGEO.SYM.01`. A real von-Mises bump $g(\theta)$ at a puncture, summed over the four μ_4 marks $f(\theta){=}\sum_{j=0}^{3} g(\theta{-}j\pi/2)$, builds the Steklov DtN $\Lambda{=}\lvert D_\theta\rvert{+}M_f$. **[E]** every off-(mod 4) Fourier coefficient of the real f vanishes to $< 10^{-16}$ (exact 4-fold cancellation), although g has all modes. **[C]** $\lVert[\rho,\Lambda]\rVert < 10^{-15}$. **[C]** *state invariance* (new vs v201): the quasi-free covariance $C{=}\frac{1}{2}(1{+}\operatorname{sgn}H_1)$ satisfies $\rho C\rho^\dagger{=}C$ to $< 10^{-13}$ — the state, not just the operator, is clock-invariant. **[C]** negative controls (single off-centre bump, $\mathbb{Z}_3$ source, 4 generic marks) break both by $O(1)$; convergence stable for $N{\in}\{16,32,64\}$. **[O]** certifies 'mark-local DtN $\Rightarrow \omega{\circ}\rho{=}\omega$' for concrete profiles *and* at the state level; the premise that the physical seam *is* mark-local stays **[O]** — the narrowest definitional form of `QGEO.SYM.01`, not its closure **[C][O]**
v211_axion_spine_angle.py	Axion DM, the spine-angle branch $\theta_i = \pi N_{\rm fam}/g_{\rm car} = 3\pi/5$ (108°): an alternative, more robust misalignment angle than the determinant-line hilltop $\theta_i \approx 170°$ (v185, which over-produces $\Omega_a h^2 \approx 0.66$). **[E]** rational motive $\theta_i = 3\pi/5$ exactly (spine quotient 3/5, no fit); **[E]** the v185 finite-T solver ($\theta{=}1 \to 0.03$) reaches $\Omega_{\rm DM}$ at $\theta \approx 106°$, so 108° is on target (harmonic $0.03\theta^2{=}0.107$); **[C]** *robust*: 108° is 62° below the 170° hilltop (mild-anharmonic, not exponentially sensitive); **[O]** an alternative ansatz to $\theta_i{=}\pi(1{-}\varphi_{\rm seam}){\approx}170°$ (mutually exclusive, the full solver decides, `DM.AXION.SPINE.01`); **[X]** kill test: $\Omega_a h^2$ outside $\sim[0.08, 0.16]$ demotes the branch **[E][C][O][X]**
v212_leptogenesis_decuple.py	Leptogenesis, the scalaron-decuple branch: both Boltzmann inputs share the decuple $A_\Lambda{=}10{=}\lvert E(K_5)\rvert$ — $M_1{=}M_{\rm scal}\varphi_0^2/A_\Lambda \approx 8.65{\times}10^9$ GeV and $\tilde m_1{=}m_3/A_\Lambda \approx 5$ meV. A cleaner **[C]** route than v184's $M_1{=}M_R\varphi_0^4$ (which only relocated the free input to the un-pinned seesaw M_R). **[E]** shared decuple $A_\Lambda{=}10$; **[E]** $M_1{=}8.65{\times}10^9$ GeV inside the thermal window $[3{\times}10^9, 3{\times}10^{10}]$ (v169); **[E]** $\tilde m_1 \approx 5$ meV, $K \approx 4.6$ strong washout; **[E]** Davidson–Ibarra $\varepsilon_1 \approx 8.5{\times}10^{-7}$, $\eta_B \approx 1.2{\times}10^{-9}$ brackets observed $6.1{\times}10^{-10}$; **[C]** HONEST: M_1 uses only $\{M_{\rm scal}, \varphi_0, A_\Lambda\}$ but the coupling mechanism is posited not derived; **[X]** kill test falls the route, not the theory (`FR.ETAB.01` independent) **[E][C][X]**

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Script	**What it machine-checks *(continued)***
`v213_ftransfer_functor.py`	The `F_transfer` *functor contract* (`CONTRACT.F.01`): the four frontier transfers are ONE typed functor $F_{\text{transfer}}{=}F_{\text{observable}} \circ F_{\text{threshold}} \circ F_{\text{RG}}$ with four structural axioms, each instance a discrete compiler kernel **[E]** composed with an external continuous solver [C]; consolidates v183 (F_{pole}), v212 ($F_{\text{Boltzmann}}$), v211 (F_{relic}), v164/`FR.MPME` (F_{QCD}), the structural complement of the v187 typing guard. **[E]** axiom 1 μ_4-deck equivariance ($\lambda_2{=}(2/3)^6{=}64/729$ the deck transfer eigenvalue); **[E]** axiom 2 Plücker preservation ($53{=}a^T(R{+}Q)\mathbf{1}$, $\mathrm{Pl}(F){=}(1,22,30)$, $\|\mathrm{Pl}(K)\|_1{=}11$); **[E]** axiom 3 positivity/stochasticity (spec $T{=}\{1,(2/3)^6,(1/3)^6\}$ positive, $\kappa_f{\in}(0,1)$); **[E]** axiom 4 external modules explicit ($b_3{=}{-}7$ a carrier output, $\lvert b_3\rvert{=}\Omega_{\text{adm}}{-}10b_1{=}7$); [C] verdict: a typed functor, not a bag of open topics; an architectural consolidation guarded by v187, not a closure **[E]**[C]
`v214_seam_pillowcase.py`	The *pillowcase* reduction of `QGEO.SYM.01` (`QGEO.PILLOW.01`): the two separately-tracked open QGEO residuals — v180/`QGEO.ISO.01` (the carrier clock is an order-4 isometry of the seam metric) and v201/`QGEO.SUBPRIN.01` (the DtN sub-principal symbol is mark-local, the seam flat away from the μ_4 marks) — are ONE statement, 'the seam carries the uniformising flat orbifold (pillowcase) metric', and the order-4 clock is *derived* from cross-ratio 2 (v168), not assumed. **[E]** $\chi_{\text{orb}}(S^2(2,2,2,2)){=}2{-}4(1{-}\frac{1}{2}){=}0 \Rightarrow$ flat (Troyanov cone-metric uniformisation, unique up to scale; Gauss–Bonnet four deficits π sum to $4\pi{=}2\pi\chi(S^2) \Rightarrow \int K_{\text{smooth}}{=}0$, flat away from the marks). **[E]** NEW LINK: cross-ratio$(\mu_4){=}2$ (v168) $\Rightarrow j(\lambda){=}256(\lambda^2{-}\lambda{+}1)^3/(\lambda^2(\lambda{-}1)^2)$ with $j(2){=}1728$ (all six harmonic cross-ratios $\{2,-1,\frac{1}{2}\}$ give 1728) $\Rightarrow$ the square modulus, hence the order-4 clock, is a *consequence* of cross-ratio 2. **[E]** $j{=}1728 \Leftrightarrow \tau{=}i$ (square torus) $\Leftrightarrow$ CM by $\mathbb{Z}[i] \Leftrightarrow \mathrm{Aut}{=}\mathbb{Z}/4$ ($j{=}0$ gives $\mathbb{Z}/6$; generic $\mathbb{Z}/2$). **[E]** explicit CM $(x,y) \mapsto (-x,iy)$ on $y^2{=}x^3{-}x$ has order $4{=}z \mapsto iz$. **[E]** negative control $\lambda{=}3 \Rightarrow j{=}21952/9 \notin \{0,1728\} \Rightarrow \mathrm{Aut}{=}\mathbb{Z}/2$, no clock. **[E]** UNIFICATION: 'seam = uniformising flat pillowcase metric' implies both v201 mark-locality and v180 `QGEO.ISO.01`. **[O]** does *not* close `QGEO.SYM.01` — the canonical constant-curvature-metric premise stays open, milder/more canonical; net existence + full-cone RP remain v175's **[E]**
`v215_seam_deck_killtest.py`	The bedrock `QGEO.SYM.01` ($\omega{\circ}\rho{=}\omega$) recast as a *falsifiable* kill-test (`QGEO.KILL.01`), not a proof-promise — a foundational symmetry cannot be derived from nothing (v181/v153), and Bisognano–Wichmann would presuppose the covariance it should produce. Non-circular reduction (v198/v199/v201): $\omega{\circ}\rho{=}\omega \Leftrightarrow [\rho,C]{=}0$; the principal symbol $\lvert k\rvert$ commutes exactly, so the residual is the sub-principal M_f with $[\rho,M_f]{=}0 \Leftrightarrow f$ $\mathbb{Z}_4$-invariant; measure form = the raw Gaussian bulk form A is μ_4-deck-invariant. Four levers (carrier-side **[E]**, seam-side **[O]**): **K1** exactly 4 marks (order-4 clock free orbit μ_4 + $b_1{=}\#\text{marks}{-}1{=}3{=}N_{\text{fam}} \Rightarrow \#\text{marks}{=}4{=}\lvert\mu_4\rvert$); **K2** square config (cross-ratio $2 \Rightarrow j{=}1728$, $\mathrm{Aut}{=}\mathbb{Z}/4$; order 4 selects $\mathbb{Z}/4$ not the $j{=}0$ $\mathbb{Z}/6$; generic $\Rightarrow \mathbb{Z}/2$); **K3** mod-4 sub-principal support (μ_4-mark sum $\Rightarrow f_m{=}0$ for $m \not\equiv 0 \bmod 4 \Rightarrow [\rho,M_f]{=}0$); **K4** transfer gap $(2/3)^6{=}64/729$ (cusp weights $\{0,\frac{1}{3},\frac{2}{3}\}$). Freeze-bound (`freeze_file.csv seam_deck_symmetry`). **[O]** the decisive kill is *deferred* (`QGEO.REALIZE.01`): the raw $\mathbb{P}\setminus\mu_4$ collar DtN must yield 4 square marks *without* inputting them; carrier-side K1–K4 are green regression guards. A kill-test, not a closure

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Script	**What it machine-checks *(continued)***
`v216_marks_gauss_bonnet.py`	The four seam marks *emerge* from Gauss–Bonnet (`QGEO.MARKS.03`), executing the K1 lever of the v215 kill-test. [E] mark count $= 2\chi$: $\mathbb{Z}_2$ branch points (deficit π) on a flat sphere (χ=2) give $n\pi = 2\pi\chi = 4\pi \Rightarrow n = 2\chi = 4 = \lvert\mu_4\rvert = e_1(a) = N_{\text{fam}}+1$ (the same χ=2 as the 8 in c_3=$1/(8\pi)$). [E] the Euclidean sphere 2-orbifolds (χ_{orb}=0) are exactly $(2,3,6),(2,4,4),(3,3,3),(2,2,2,2)$. [E] all-order-2 (the $\lvert\mathbb{Z}_2\rvert$ sheet branch) $\Rightarrow$ uniquely $(2,2,2,2)$; [E] N_{fam}=3 (rank H^1=#marks−1) $\Rightarrow$ uniquely $(2,2,2,2)$, picking the 4-mark square over the 3-mark hexagonal $(3,3,3)$; [E] a free order-4 deck $\Rightarrow$ uniquely $(2,2,2,2)$ — three independent selectors converge. [O] residual: the square modulus (cross-ratio 2 $\Rightarrow j$=1728, v214) needs the one order-4 carrier input. The mark count + equality are now derived; does *not* close `QGEO.REALIZE.01`
`v217_marks_emergence_scan.py`	The *numerical* emergence scan (`QGEO.EMERGE.01`, the sibling of v216 as v210 is to v201): with the number of marks n *and* their positions free, the 4-equally-spaced (square) configuration is the joint minimiser of {clock-invariance + 3-family cohomology + transfer gap $(2/3)^6$} on the actual seam DtN/state. [C] n=4 equally-spaced $\Rightarrow \lVert\rho_4 C\rho_4^\dagger - C\rVert \sim 10^{-14}$ (clock-invariant state); [C] family count selects n=4 (scan $n \in \{2..8\}$, only $n-1$=3=N_{fam}; n=3 $\to$ 2 families = hexagonal, excluded; n=5 $\to$ 4); [C] gap: μ_4-equivariant $\Rightarrow$ diagonal in the cusp basis (deck-average off-diagonal $\sim 10^{-16}$), cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\} \Rightarrow$ transfer $(1-w)^6$=$\{1,(2/3)^6,(1/3)^6\}$, gap $(2/3)^6$=64/729; [C] position scan: the square minimises the defect (square $\sim 10^{-14}$ vs off-square ~ 0.4); [C] negative controls n=3, 5, 6. [O] confirms the emergence on the real DtN (number scanned, positions free); selector = the 3-family/gap structure (carrier input); executes K1 numerically, does *not* close `QGEO.REALIZE.01`. Python-only
`v218_diamond_axis_geometry.py`	Diamond axis geometry: the Sheet Diamond (v94) is a discrete geometry with two axes around the centered cross (v95: C center, U winding, V sheet), F the transfer completion. Three exact [E] lemmas + honest audit/heuristic typing, no new numbers. [E] AXIS CURVATURE: $\det(C+xU)$=$14+6x$ linear (slope 6=$\lvert R^+(A_3)\rvert$), $\det(C+yV)$=$14+14y+4y^2$ quadratic, 2nd difference 8=rank E_8; $\det B(C+xU)$=$2\det(C+xU)$, anchor-block sheet curvature 6=$\lvert R^+(A_3)\rvert$. [E] PLÜCKER LADDER: $\mathrm{Pl}(K)$=$(-1,6,4) \to \mathrm{Pl}(C)$=$(0,14,14) \to \mathrm{Pl}(F)$=$(1,22,30)$, steps $(1,8,10)$=$(N_\Phi, \text{rank}\,E_8, A_\Lambda)$ and $(1,8,16)$=$(N_\Phi, \text{rank}\,E_8, \dim S^+)$ (identity [E], transfer reading [C]). [E] SPECTRAL RAMIFICATION: $\mathrm{Disc}(\chi)$=$q(r)^2\mathrm{Disc}(q)$ for Q, K, C, F; squares $(1,3,4,6)$, kernels $(13,48,65,105)$ with 48=Ω_{adm}, 105=$N_{\text{fam}} g_{\text{car}} \cdot 7$. [C] heuristic: F the transfer-completion corner, a search principle, not a hard filter; the G_2/F_4 pair-sum labels stay audit-only (cf. v94/v95) [E][C]
`v219_icosahedral_mckay.py`	McKay bedrock: *why* the atoms are $\{2,3,5\}$. E_8 is the exceptional *top* of the McKay tower of finite $SU(2)$ subgroups ($2T \to \hat{E}_6$, $2O \to \hat{E}_7$, $2I \to \hat{E}_8$). The McKay graph is built *from the group*: the 120 icosians close to the binary icosahedral $2I$ (element orders $\{1,2,3,4,5,6,10\}$, containing the 2, 3, 5 axis orders), 9 conjugacy classes, Dixon irrep degrees $\{1,2,2,3,3,4,4,5,6\}$ (sum 30=$h(E_8)$, sum of squares 120=$\lvert R^+(E_8)\rvert$); the natural-rep McKay adjacency *is* the affine E_8 graph with Kac marks = the degrees. A backward certificate of the closed E_8 [E], the reading that it explains $(2,3,5)$ [C], the raw-seam origin [O]
`v220_cp_hexagonal_modulus.py`	CP lives in the *hexagonal* phase fiber, not the square seam deck (a geometric sharpening of red-team Target D). [E] the seam deck is the square modulus $j(i)$=1728, Aut=$\mathbb{Z}/4$ (the μ_4 clock); its exceptional partner is the hexagonal modulus $j(\rho)$=0, ρ=$e^{i\pi/3}$, Aut=$\mathbb{Z}/6$, CM by $\mathbb{Z}[\omega]$, $\arg\rho$=$\pi/3$. [E] the frozen leading CKM reading δ=$\pi/3+3\lambda_C^2$=68.654° (v88) has leading term $\pi/3$=$\arg\rho$; the $\lvert\mathbb{Z}_2\rvert$ alternative $\pi/3+2\lambda_C^2$ is the frozen look-elsewhere trap. The deck stays $\mathbb{Z}/4$ (v215); CP is the $\mathbb{Z}/6$ fiber, the bridge [C][E][C]

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Script	**What it machine-checks** ***(continued)***
`v221_seam_qecc.py`	The seam as a *finite recoverability* code (recovery rate $(2/3)^6{=}64/729$). [E] code dimension $\dim S^+{=}16{=}2^{g_{\rm car}-1}$; the gapped transport T on the cusp-weight 3-space (deviations $(1,-1,0)$ and the Nariai anchor $(1,1,-2)$) is symmetric, doubly stochastic (CPTP), spectrum $\{1,(2/3)^6,(1/3)^6\}$; on the trace-zero subspace $\|T\delta\|{\le}(2/3)^6\|\delta\|$ hence $\|T^n\delta\|{\le}(2/3)^{6n}\|\delta\|$ (a Knill–Laflamme-type bound); a free ratio breaks it. [C] the holographic/Petz reading (gated on the Seam–Horizon theorem) [E][C]
`v222_cm_norm_duality.py`	CM-norm duality: the two exceptional moduli give 41 and 7. [E] SQUARE (Gaussian $\mathbb{Z}[i]$, $j{=}1728$): $N(g_{\rm car}{+}i\lvert\mu_4\rvert){=}5^2{+}4^2{=}41{=}10b_1$ — the EM index *is* the Gaussian norm of the carrier-glue vector; $N(3{+}2i){=}13{=}\Delta_Q$. [E] HEX (Eisenstein $\mathbb{Z}[\omega]$, $j{=}0$): $N(3{+}2\omega){=}3^2{-}3{\cdot}2{+}2^2{=}7{=}$ scalaron. The $(3,2){=}$(colour,weak) split generates $(5,6,7,13)$ by sum / product / Eisenstein / Gauss norm; the rings are rigid (Eisenstein of $(5,4){=}21{\neq}41$). The architecture bridge square$\leftrightarrow$EM, hex$\leftrightarrow$scalaron is [C][E][C]
`v223_coxeter_totative_clock.py`	The μ_4 clock is the order-4 character of the E_8 Coxeter cycle. [E] $(\mathbb{Z}/30)^\times{=}\{1,7,11,13,17,19,23,29\}{=}$ the E_8 exponents = live Coxeter phases; $\varphi(30){=}8{=}\mathrm{rank}\,E_8$, $30{=}2{\cdot}3{\cdot}5$. [E] $(\mathbb{Z}/30)^\times{\cong}\mathbb{Z}/4{\times}\mathbb{Z}/2$ with order-4 generator 7 ($7^2{=}19$, $7^4{=}1$ mod 30): a literal μ_4 clock $\langle 7\rangle{=}\{1,7,13,19\}$ on the exponents; the conjugate pairs $m{+}(30{-}m){=}30$ give $\lvert\mu_4\rvert{=}4$ invariant planes the clock permutes. Only E_8 has $\varphi(h){=}$rank *and* $h{=}2{\cdot}3{\cdot}5$; the seam-clock identification is [C][E][C]
`v224_diamond_ftransfer_path.py`	$F_{\rm transfer}$ as *one* path on the Sheet Diamond, not four external interfaces. [E] family $M(s,t){=}R{+}Q\,\mathrm{diag}(s,t,t)$: the corners $K{=}M(1,-1), C{=}M(1,0), F{=}M(1,1)$ lie on the sheet axis ($s{=}1$); the sheet axis is curved $\det M(1,t){=}4t^2{+}14t{+}14$ (2nd diff 8=rank E_8), the winding axis $\det M(s,0){=}6s{+}8$ flat (slope $6{=}\lvert R^+(A_3)\rvert$); Plücker steps $(1,8,10),(1,8,16)$. [C] $F_{\rm transfer}$ is the $K{\to}C{\to}F$ sheet geodesic, Koide source$\to$pole its 1-D Möbius projection (v37/v183) [E][C]
`v225_dual_normal_frame.py`	The dual normal frame (d,n): one boundary compass for flavor and horizon, with CP as orientation. [E] $d{=}a^\top R^{-1}{=}a^\top L^{-1}{=}(-\frac12,-\frac12,1){=}-\frac12(1,1,-2)$ (the Nariai/horizon normal; $d{\cdot}\mathbf{1}{=}0$, $d{\cdot}a{=}1$); $n{=}(5,-9,6)$ with $n^\top R{=}(\det R,0,0){=}(8,0,0)$, $n^\top L{=}(20,0,0)$ (the first-generation/torsion selector). [E] oriented volume $\det(\mathbf{1},d,n){=}21{=}N_{\rm fam}\cdot 7$. [C] CP as orientation: $\mathrm{Im}\,\det(\mathbf{1},d,\rho n){=}21\sin(\pi/3)$ for $\rho{=}e^{i\pi/3}$ — CP sits where the magnitude bijection (Target D) fails [E][C]
`v227_degree_exponent_channel_split.py`	248=120+128 as a magnitude/phase channel typing (replaces the over-strong S^--dark-matter reading). [E] E_8 exponents sum $120{=}\lvert R^+(E_8)\rvert$ (*magnitude* channel); degrees $\{2,8,12,14,18,20,24,30\}$ sum $128{=}2^{\rm rank-1}{=}\mathrm{rank}\cdot\dim S^+$ (*phase/glue* channel); $248{=}120{+}128{=}\mathrm{adj}(D_8){+}\mathrm{spinor}(D_8)$; $\sum\deg-\sum\exp{=}\mathrm{rank}{=}8$. [C] ratios+products (Target D) live in the 120 channel, the un-covered CP phases in the 128 phase/glue channel; [O] a dark-sector reading of 128 stays Frontier [E][C][O]
`v228_rr_index_gate.py`	P2 as the mode space of a degree-4 seam divisor (a Riemann–Roch *index gate*; bedrock language for P2, not a proof from nothing). [E] $\deg D{=}4{=}\lvert\mu_4\rvert$ on $\mathbb{P}^1\Rightarrow h^0(\mathcal{O}(D)){=}5{=}g_{\rm car}$ (Riemann–Roch, genus 0); rank $H_1(\mathbb{P}^1\backslash 4){=}3{=}N_{\rm fam}$; $h^0{+}\mathrm{rank}\,H_1{=}8{=}\mathrm{rank}\,E_8$, $h^0{-}\mathrm{rank}\,H_1{=}2{=}\lvert\mathbb{Z}_2\rvert$; $\Lambda^{\rm even}(\mathbb{C}^5){=}1{+}10{+}5{=}16{=}\dim S^+$. The mode-space reading is [C], conditional on the raw seam producing the degree-4 divisor ([O], = QGEO.SYM.01) [E][C][O]
`v229_lepton_frobenius_algebra.py`	The charged leptons as an étale Frobenius algebra over the μ_6 resolvent. [E] exact coefficients $(c_e,c_\mu,c_\tau){=}(\frac{16}{7},\frac43,\frac76)$, ring closure $c_ec_\tau{=}\frac83{=}\lvert\mathbb{Z}_2\rvert c_\mu$, product $\frac{32}{9}{=}2^{g_{\rm car}}/N_{\rm fam}^2$. [E] $A{=}\mathbb{Q}[t]/(m)$, $m{=}\prod(t{-}c)$: the trace-form Gram is nondegenerate ($\det{=}\mathrm{disc}(m){\neq}0$, A étale) — a commutative Frobenius algebra; the C_6 shift has characteristic polynomial $t^6{-}1$ (spectrum $\mu_6{=}\mu_3$ family $\times\,\mu_2$ sheet). [C] the PMNS extension via $\mathrm{Aut}(A)$ and the hexagonal CM point [E][C]

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Script	What it machine-checks *(continued)*
`v230_center_budget_norms.py`	The Sheet-Diamond center budget $(7,11,13)$ is the three *local norms* of the theory. **[E]** $C{=}R{+}Q\,\mathrm{diag}(1,0,0)$, row sums $(7,11,13)$: $7{=}N_{\mathbb{Z}[\omega]}(3{+}2\omega)$ (Eisenstein/hex = scalaron), $13{=}N_{\mathbb{Z}[i]}(3{+}2i)$ (Gauss/square = Δ_Q), $11{=}\binom{4}{0}{+}\binom{4}{1}{+}\binom{4}{2}{=}1{+}4{+}6$ (QBL boundary Fock count on the 4 μ_4 corners = $\dim S^+{-}g_{\rm car}{=}16{-}5$). The two CM rings (v222) and the seam boundary count meet in one operator center; the reading is **[C][E][C]**
`v231_cp_mu6_phases.py`	Both CP phases are μ_6 powers of *one* hexagonal CM unit, split by the sheet — a structural reduction of red-team Target D (sharper than v220/v225). With $\rho{=}e^{i\pi/3}$ (the $j{=}0$ Eisenstein CM unit): **[E]** ρ is a primitive 6th root ($\rho^6{=}1$, $\rho^3{=}{-}1$ the $\mathbb{Z}_2$ sheet half-turn); **[E]** $\delta_{\rm CKM}$ leading = $\arg(\rho^1){=}\pi/3$ (quark), $\delta_{\rm PMNS}$ = $\arg(\rho^4){=}4\pi/3$ (lepton, the *phase lattice*); **[E]** $\rho^4{=}{-}\rho \Rightarrow \delta_{\rm PMNS}{=}\delta_{\rm CKM,lead}{+}\pi{=}(\text{CM unit}){\times}(\mathbb{Z}_2\text{ sheet})$ — the two CP phases are *one* hexagonal unit on the two sheets, not two independent inputs; **[E]** the C_6/μ_6 monodromy has charpoly $t^6{-}1$ (spectrum μ_6 carries ρ); **[E]** orientation $\mathrm{Im}\det(\mathbf{1},d,\rho^1 n){=}{+}21\sin(\pi/3)$, $\mathrm{Im}\det(\mathbf{1},d,\rho^4 n){=}{-}21\sin(\pi/3)$ (sheet-flipped Jarlskog, v225); **[E]** NEG μ_3 has no -1 (no sheet; $\mu_6{=}\mu_3$ family $\times\,\mu_2$ sheet). REDUCTION: 2 CP phase inputs $\to$ 1 hexagonal unit + the sheet; the physical CP derivation stays **[C]**, the deck stays $\mathbb{Z}/4$ (v215) **[E][C]**
`v232_e8_kleinian_seam.py`	The seam as the E_8 *Kleinian singularity* — a canonical algebraic-geometric model for the open seam-realisation premise `QGEO.REALIZE.01` (the du Val side of the McKay correspondence, v219). **[E]** the McKay graph of $2I$ is affine E_8; dropping the unique trivial-rep node (the single Kac mark 1) leaves the *finite* E_8 Dynkin on 8 nodes = the dual intersection graph of the eight exceptional $\mathbb{P}^1$'s of the minimal resolution of $\mathbb{C}^2/2I$ (trivalent node, arms $1,2,4$). **[E]** eight exceptional $\mathbb{P}^1$'s = rank E_8 = $g_{\rm car}{+}N_{\rm fam}$ = # nontrivial $2I$ irreps — a fourth reading of the 8 in $c_3{=}1/(8\pi)$. **[E]** the negated intersection form is the E_8 Cartan (even, $\det{=}1$, positive definite; each $\mathbb{P}^1$ a (-2)-curve). **[E]** one degree-1 irrep $\Rightarrow$ $2I$ perfect $\Rightarrow$ $H_1(S^3/2I){=}0$: the link is the Poincaré homology 3-sphere, $\pi_1{=}2I$ of order $120{=}\lvert R^+(E_8)\rvert$. **[C]** seam reading: the raw RP seam ($\mathbb{P}^1$ with marks) is canonically a du Val exceptional curve — a model for `QGEO.REALIZE.01`, not a P2 proof (presupposes E_8); bedrock **[O][E][C][O]**
`v233_cp_triality_phase.py`	The CP phase is the *universal* family/triality phase, only sheet-split — one step past v231 in $\mu_6{=}\mu_3$ (family/triality) $\times\,\mu_2$ (sheet). **[E]** $\mu_6{=}\mu_3{\times}\mu_2$: the $\mathbb{Z}_3$ triality centre $\{1,\omega,\omega^2\}$ = the cusp weights $\{0,\frac{1}{3},\frac{2}{3}\}{\times}2\pi$, the sheet $\mu_2{=}\{1,-1\}$; $\rho{=}\omega^2{\cdot}(-1)$. **[E]** SAME FAMILY CLASS: $\rho^4{=}\rho^1{\cdot}\rho^3$ with $\rho^3{\in}\mu_2$, so the quark (ρ^1) and lepton (ρ^4) CP phases have the *same* image in $\mu_6/\mu_2{\cong}\mu_3$ and *opposite* sheet — the CP phase *is* the universal triality phase ω^2 (the 2/3 cusp-weight $\mathbb{Z}_3$ centre), not a free power choice. **[E]** ω is the μ_3 factor of the order-30 $(2,3,5)$ Milnor/Coxeter monodromy that builds E_8 (v232). Reduces Target D from 'one hexagonal unit + sheet' (v231) to 'the universal triality phase + sheet'; the physical CP derivation stays **[C][E][C]**
`v234_seam_holomorphy_selection.py`	The Seam-Holomorphy selection certificate: the entire structural residual is *one* condition with *three* equivalent faces, all forcing E_8. The condition: *the seam carries no nontrivial abelian sector.* **[E]** (i) AQFT — the boundary net is holomorphic (μ-index 1); (ii) geometry — the seam link is a homology 3-sphere $\Leftrightarrow$ the binary group is perfect $\Leftrightarrow$ $2I$; (iii) rep theory — exactly one 1-dim irrep. **[E]** #(mark-1 affine nodes) = $\lvert\Gamma^{\rm ab}\rvert$ = $\lvert H_1(S^3/\Gamma)\rvert$ = #(1-dim irreps): $A_n{\to}n{+}1$, $D_n{\to}4$, $E_6{\to}3$, $E_7{\to}2$, $E_8{\to}1$ — only E_8 gives 1 ($2I$ the unique perfect $SU(2)$ subgroup, Poincaré the unique homology-sphere space form). **[E]** holomorphic $c{=}8{=}g_{\rm car}{+}N_{\rm fam}$ $\Rightarrow$ the unique even unimodular rank-8 lattice = E_8. **[E]** the three faces (v83/v154, v232, v219) are the SAME E_8-selector, so Target A, $G_{\rm net}$, the carrier P2 and the Kleinian seam are *one* gate. **[O]** the closing theorem (stated): RP+gap+chirality $\Rightarrow$ quasi-free bulk (v160) $\Rightarrow$ invertible bulk $\Rightarrow$ holomorphic boundary $\Rightarrow$ E_8; the single residual analytic step is 'free bulk $\Rightarrow$ holomorphic boundary' **[E][O]**

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Script	What it machine-checks *(continued)*
`v235_seam_chern_simons.py`	The closing step in abelian Chern–Simons language: holomorphic ⇔ $\det K{=}1$ — a genuine reduction (not a closure) of `GATE.HOLO.01` into standard anyon-condensation physics. **[E]** a free gapped bosonic 2+1d bulk = an even integer K-matrix with #anyons $= \lvert\det K\rvert$, edge c = signature(K), holomorphic ⇔ $\lvert\det K\rvert{=}1$. The TFPT extension tower (v92) *is* the condensation tower read by determinant: carrier $D_5{\oplus}A_3$ (det 16, 16 anyons) $\to SO(16)_1{=}D_8$ (det 4) $\to (E_8)_1$ (det 1, holomorphic), all rank 8, $c = \mathrm{sig}{=}8{=}g_{\mathrm{car}}{+}N_{\mathrm{fam}}$. **[E]** holomorphic ⇔ det 1 ⇔ E_8 = the Kitaev E_8 quantum-Hall state. **[E]** condensation: a Lagrangian glue of order $\sqrt{16}{=}\lvert\mu_4\rvert{=}4$ takes det 16→1 (a partial order-2 isotropic only → 4=D_8). **[E]** NEG: D_8 is free+bosonic but has 4 anyons; an odd K is fermionic — det =1 is the extra condition. **[O]** residual: 'the free RP seam condenses the order-$\lvert\mu_4\rvert$ Lagrangian glue' = the sheet/QGEO selection, now with holomorphy⇔ det 1 a theorem **[E][O]**
`v236_brieskorn_capstone.py`	The $(2,3,5)$ Brieskorn singularity $x^2{+}y^3{+}z^5$ is the *one* generator of the discrete skeleton — the capstone of the icosahedral round (v55/v219/v223/v232/v233); its exponents *are* the atoms $(\lvert\mathbb{Z}_2\rvert, N_{\mathrm{fam}}, g_{\mathrm{car}}){=}(2,3,5)$. **[E]** Milnor number $\mu{=}(2{-}1)(3{-}1)(5{-}1){=}1{\cdot}2{\cdot}4{=}8{=}\mathrm{rank}\,E_8$ = the 8 in $c_3{=}1/(8\pi)$ (a fifth independent origin). **[E]** the Milnor monodromy eigenvalues are the eight primitive 30th roots $e^{2\pi i m/30}$, m the E_8 exponents (charpoly Φ_{30}, $\deg\varphi(30){=}8{=}\mu$) — so it *is* the order-30 E_8 Coxeter element (v55). **[E]** both clocks: the monodromy group $\langle h\rangle{=}\mathbb{Z}/30{=}\mathbb{Z}/2{\times}\mathbb{Z}/3{\times}\mathbb{Z}/5$ carries the μ_3 triality (CP, v233) as h^{10}; the Galois group $(\mathbb{Z}/30)^{\times}{=}\mathbb{Z}/2{\times}\mathbb{Z}/4$ carries the μ_4 clock (×7, v223). **[E]** Milnor lattice $= E_8$, link = Poincaré sphere. **[C]** one singularity generates atoms + 8 + E_8 + order-30 + both clocks + seam; a unification, not a closure **[E][C]**
`v237_seam_sre_closure.py`	The closing step as a *physical* condition: no topological ground-state degeneracy ⇔ $\det K{=}1$ ⇔ the seam bulk is short-range-entangled (SRE/invertible) — sharpening `GATE.HOLO.02`'s residual from definitional to falsifiable. **[E]** the K-matrix ground-state degeneracy on a genus-g surface is $\lvert\det K\rvert^g$ (carrier → 16, D_8 → 4, E_8 → 1 on the torus), so 'no degeneracy on any closed surface' ⇔ $\det K{=}1$. **[E]** among rank-8 even positive-definite K ($c{=}8{=}g_{\mathrm{car}}{+}N_{\mathrm{fam}}$) only E_8 has det 1 — the unique nontrivial bosonic SRE 2+1d phase (Kitaev E_8), edge holomorphic; D_8/carrier are LRE. **[E]** a unique vacuum on the plane (RP+gap+cluster) is necessary but not sufficient (the plane cannot see $\det K$, the torus does), so the residual is 'the seam unique-vacuum extends to the torus' = SRE. **[E]** NEG: an LRE bulk (D_8) has 4-fold torus degeneracy + non-holomorphic boundary. **[O]** sharpened residual: 'the free RP gapped seam is short-range-entangled (no topological order)' = $\det K{=}1$ = the Kitaev E_8 phase — a yes/no statement, sharper than v235/`QGEO.SYM.01`; not a closure **[E][O]**
`v238_modular_lindblad_dynamics.py`	`DYN.SEMIGROUP.01` — the static→dynamic flip: the seam's modular flow (reversible) and its recovery channel as a Markov/Lindblad generator (irreversible), the dissipative gap = the OS mass gap. Companion to v196/v198 (modular) and v221/v64 (recovery), which state the structure *statically*; this reads the same objects *dynamically*. **[E]** the modular flow $\sigma_t = e^{it\Lambda_\Sigma}$ is a unitary one-parameter group (Tomita–Takesaki) and the μ_4 clock is conserved ⇔ $[\rho, \Lambda_\Sigma] = 0$ (the v198 state-invariance), so the static commutator is the conservation law of modular time. **[E]** $L = \log T$ is a doubly-stochastic generator ($e^L = T$, $e^{L\tau} \to$ the democratic projector = the arrow of time, cf. v64). **[E]** keystone: the dissipative gap $= -\log((2/3)^6) = 6\log(3/2)$ = the OS mass gap Δ (v64) — the static recovery bound and the dynamical mass gap are one number, one exponentiated. **[E]** GKSL split (imaginary + real≤ 0 spectrum). **[O]** geometric modular flow on the full seam algebra (Connes–Rovelli thermal time = `QGEO.SYM.01`, v198) — a target, not a closure.

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Script	What it machine-checks *(continued)*
`v239_kms_thermal_time.py`	`DYN.KMS.01` — KMS / thermal time: the modular flow of v238 makes the seam state a KMS (thermal) state at $\beta = 1$, and the geometric normalisation fixes the seam temperature via $2\pi = 1/(4c_3)$. **[E]** KMS at $\beta = 1$: $\omega(A\,\sigma_i(B)) = \omega(BA)$ ($\omega(X) = \mathrm{Tr}(\rho X)$, $\rho = e^{-\Lambda_\Sigma}/Z$), so modular time (v238) is a thermal time (Connes–Rovelli); NEG $\beta \neq 1$ fails (Tomita–Takesaki pins $\beta = 1$). **[E]** the seam unit $\beta_{\mathrm{phys}} = 2\pi = 1/(4c_3)$ (v58) makes the KMS temperature the horizon temperature $T_H = c_3/M$ (v57/v208). **[E]** one object with v238. **[O]** the geometric-boost identification = `QGEO.SYM.01` — a target, not a closure.
`v240_gns_os_reconstruction.py`	`DYN.GNS.01` — GNS / OS reconstruction (finite toy): the Hilbert space and a POSITIVE Hamiltonian are OUTPUTS of the seam data (M, ω), not inputs. **[E]** GNS: $\langle A, B\rangle = \mathrm{Tr}(\rho A^\dagger B)$ Hermitian positive-definite ($\rho > 0$), $\dim n^2$, identity cyclic+separating. **[E]** modular flow $\sigma_t(x) = \rho^{it} x \rho^{-it}$ is a state-preserving $*$-automorphism, generator $i[\log\rho, \cdot] = \Lambda_\Sigma$ (v238). **[E]** OS Hamiltonian $H_{\mathrm{OS}} = -\log T$ (the v221 transfer) is POSITIVE — eigenvalues $\{0, 6\log(3/2), 6\log 3\} \geq 0$, gap $= 6\log(3/2) =$ the v64 mass gap. **[E]** $H_{\mathrm{OS}} = -L$: minus the v238 dissipative generator (recovery semigroup = Euclidean time). **[O]** the geometric/operator-level step = `QGEO.SYM.01`/BW + the standard OS theorem under RP/gap — a target, not a closure.
`v241_dhr_sectors.py`	`DHR.SECTORS.01` — particles = DHR superselection sectors: the carrier discriminant/anyon content and the v235/v237 condensation tower as an abelian MTC. Carrier form (Lean-verified, `FORM.GLUE.01`): on $\mathbb{Z}_4 \times \mathbb{Z}_4$, $q(x, y) = (5x^2{+}3y^2)/8 \bmod 1$, $\theta(a) = e^{2\pi i q(a)}$. **[E]** 16 sectors $= \mathbb{Z}_4 \times \mathbb{Z}_4$ ($= \det K_{\mathrm{carrier}} = \dim S^+$), fusion = abelian group law. **[E]** condensable $\Leftrightarrow \theta = 1 \Leftrightarrow$ isotropic $\Leftrightarrow 5x^2{+}3y^2 \equiv 0 \bmod 8$ — the six elements $\{(0, 0), (1, 1), (1, 3), (2, 2), (3, 1), (3, 3)\}$. **[E]** tower = v235/v237: Lagrangian $\langle(1, 1)\rangle$ (order 4, self-dual) $\to$ 1 sector $= (E_8)_1$; order-2 $\langle(2, 2)\rangle \to 4 = D_8$; carrier $16 \to D_8\, 4 \to E_8\, 1$. **[E]** NEG: a non-isotropic anyon $(1, 0)$ is confined. **[C]** sector$\leftrightarrow$SM-multiplet map is the interpretation; the MTC is exact.
`v242_spin_statistics_modular.py`	`DHR.MODULAR.01` — spin-statistics and the modular (S, T) data of the carrier MTC: statistics + the fusion ring recovered from modular data + the anyon central charge $c = 8$. Same MTC as v241 ($\mathbb{Z}_4{\times}\mathbb{Z}_4$, $q = (5x^2{+}3y^2)/8$, $\theta = e^{2\pi i q}$). **[E]** spin-statistics: $T = \mathrm{diag}(\theta_a) \to 6$ bosons ($\theta = 1$), 2 fermions ($\theta = -1$) $= \{(0, 2), (2, 0)\}$, 8 anyons. **[E]** modular $S_{ab} = (1/\sqrt{16})e^{-2\pi i B(a,b)}$ unitary, symmetric, $S^2 = C$ (charge conjugation), $S^4 = I$. **[E]** Verlinde $N_{ab}^c = \sum_x S_{ax} S_{bx} S^*_{cx}/S_{0x} = \delta(c, a{+}b)$ — fusion recovered from modular data. **[E]** Gauss–Milgram $(1/\sqrt{16})\sum\theta_a = e^{2\pi i c/8}$ with $c = 8 = g_{\mathrm{car}} + N_{\mathrm{fam}} =$ the $(E_8)_1$ central charge (v156/v175). **[C]** the boson/fermion split is the kinematic skeleton; the sector$\leftrightarrow$multiplet map is the interpretation.
`v243_haag_ruelle_braiding.py`	`DHR.SMATRIX.01` — scattering: the Haag–Ruelle skeleton + the braiding S-matrix. The seam net has a mass gap (v64/v238) so Haag–Ruelle constructs asymptotic states; for the abelian anyon MTC (v241/v242) the 2-particle S-matrix is the monodromy $M(a, b) = \theta(a{+}b)/(\theta(a)\theta(b)) = e^{2\pi i B(a,b)}$. **[E]** mass gap $\Delta = 6\log(3/2) > 0 \to$ HR in/out states. **[E]** 2-particle S = monodromy: $\lvert M\rvert = 1$, diagonal in conserved labels $\to$ no particle production, factorised (integrable) scattering. **[E]** Yang–Baxter: M is a bicharacter $\to$ multi-particle factorisation. **[E]** crossing+symmetry: $M(a, b) = M(b, a)$, $M(a, -b) = M(a, b)^*$. **[E]** holomorphic point: on the condensed $(E_8)_1$ (single sector) $M = 1$, trivial braiding (det 1/SRE). **[O]** the 4d S-matrix / cross-sections are the open holographic uplift (Phase 3).

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Script	What it machine-checks *(continued)*
`v244_spectral_action_contract.py`	`CONTRACT.QFT4D.01` — the 4d Lagrangian via Connes' spectral action: the honest contract for the Phase-3 gap (2d seam net → 4d physics). A single Dirac operator whose Tr $f(D/\Lambda)$ yields gravity AND matter. **[E]** carrier spinor = one generation: $SO(10)$ half-spinor = $\dim S^+ = 2^{g_{\rm car}-1} = 16$ $(15+\nu_R)$, $N_{\rm fam}=3$. **[E]** SM gauge group inside the carrier: $SU(3)\times SU(2)\times U(1)$ (dim 12) $\subset SU(5)$ (24) $\subset SO(10){=}D_5$ (45); colour $SU(3)\subset A_3{=}SU(4)$ (Pati–Salam); $\mathrm{rank}(D_5){+}\mathrm{rank}(A_3) = 8 = \mathrm{rank}\,E_8$. **[E]** one Dirac operator → gravity ($a_2{\sim}R$, $a_4{\sim}R^2$, v36) + matter ($a_4\sim F^2$ Yang–Mills $+\lvert D\phi\rvert^2$ Higgs, Gilkey). **[X]** hooks: $g_3{=}g_2{=}\sqrt{5/3}\,g_1$ + a Higgs-quartic relation. **[O]** `CONTRACT.QFT4D.01`: obligations (seam→Dirac D; finite geometry = carrier; spectral action reproduces SM couplings) → the 4d SM+gravity Lagrangian is an OUTPUT — the Phase-3 target, not closed.
`v245_spectral_matter_content.py`	`QFT4D.MATTER.01` — the computable core of `CONTRACT.QFT4D.01` (v244): the spectral-action / NCG matter content, discharging the rep-theory + tree-relation core to **[E]**. The carrier half-spinor is the $SO(10)$ **16**. **[E]** **16** = one generation: $SO(10)\,\mathbf{16} = SU(5)(\mathbf{10}{+}\bar{\mathbf{5}}{+}\mathbf{1})$; SM dims 6+3+1+2+3+1 = 16 = $\dim S^+$, $N_{\rm fam}=3$ (15 Weyl fermions $+\nu_R$). **[E]** electric charges $Q=T_3+Y$ are exactly the SM values — hypercharges forced by the embedding. **[E]** anomaly-free: $\sum Y=0$, $\sum Y^3=0$, $[SU(2)]^2U(1)=0$, $[SU(3)]^2U(1)=0$ (exact). **[E]** unification: $\sin^2\theta_W = (\sum T_3^2)/(\sum Q^2) = 3/8$, i.e. $g_3=g_2=\sqrt{5/3}\,g_1$. **[E]** Higgs = the $(1,2)_{1/2}$ inner fluctuation (one doublet). **[O]** residual: the seam→Dirac realisation + measured-coupling reproduction stay the open contract.
`v246_unification_data_crosscheck.py`	`QFT4D.RGTEST.01` — the honest data cross-check for the spectral-action prediction (v244/v245): does $g_1{=}g_2{=}g_3$, $\sin^2\theta_W{=}3/8$ at Λ survive the *measured* SM couplings? Not in the plain SM. Inputs (PDG, GUT-normalised, M_Z): $\alpha_i^{-1} = (59.01, 29.59, 8.47)$, SM 1-loop $b = (41/10, -19/6, -7)$ (v159). **[E]** the prediction: $\alpha_1{=}\alpha_2{=}\alpha_3$, $\sin^2\theta_W{=}3/8$ at Λ. **[X]** tension: pairwise crossings spread $\sim$ 4 decades ($\sim10^{13}$–10^{17} GeV), residual spread at $2{\times}10^{16}$ $\Delta\alpha^{-1}\sim 8.8$ — $g_1{=}g_2{=}g_3$ not realised by the plain SM. **[X]** $\sin^2\theta_W$ 3/8 vs measured 0.23122. **[X]** 2-loop also misses (min spread ~ 3.2 at $1.7{\times}10^{14}$ GeV). **[O]** no rescue: SM gauge content + no new states (the E_8 spine is arithmetic, not a threshold tower) ⇒ no admissible thresholds; closing it needs new matter (vs 'no new state') or a non-GUT/boundary reading — tension like NCG-SM, likely falsified unless extended. A kill-test, **[X]**/**[O]**, never **[E]**-as-success.
`v247_e8_branching_no126.py`	`PS.E8BRANCH.01` — the E_8 audit hull forbids the **126** and supplies exactly $\{1,10,16,45\}$ for $SO(10)$. Carrier $D_5{\oplus}A_3 = SO(10){\times}SU(4)\subset E_8$ (via $SO(16)$); $\mathbf{248} = (45,1){+}(1,15){+}(10,6){+}(16,4){+}(\overline{16},\bar{4}) = 248$. **[E]** ranks 5+3=8; **[E]** dims = 248 ($SO(16)$: 120 adj +128 spinor); **[E]** $SO(10)$ content $\{1,10,16,45\}$, the **126** ABSENT ⇒ 126_H algebraically forbidden, 10+16+45 E_8-supplied not fitted; **[E]** caveat only one 45/one 15 → proton-marginal; **[O]** the field-content principle + NCG complement (v250).
`v248_ps_algebra.py`	`PS.ALGEBRA.01` — the Pati–Salam NCG algebra $A_F = \mathbb{H}_L\oplus\mathbb{H}_R\oplus M_4(\mathbb{C})$ and its exact SM matching. **[E]** unimodular group $SU(2)_L{\times}SU(2)_R{\times}SU(4)_c$ (rank 5, dim 21); **[E]** $\mathbf{16} = (4,2,1){+}(\bar{4},1,2) = 8{+}8$; **[E]** $Y = T_{3R} + (B{-}L)/2$ gives the exact SM hypercharges and $Q = T_{3L}+Y$ the exact charges; **[E]** matching $\alpha_1^{-1} = \frac{3}{5}\alpha_{2R}^{-1} + \frac{2}{5}\alpha_4^{-1}$, $\sin^2\theta_W = 3/8$. **[O]** A_F uniquely from seam+sheet+μ_4; **[O]** caveat $SU(4)_c\subset SO(10){=}D_5$ is not the carrier $A_3{=}SU(4)$ (family).
`v249_ps_unification.py`	`PS.RGTEST.01` — carrier-native Pati–Salam/$SO(10)$ two-step unification kill-test + negative controls (suite check of `experiments/gauge-unification`). **[E]** matching + a valid 1-loop unification solution; **[X]** NEG SM-only fails (spread ~ 8.8 @ $2{\times}10^{16}$); **[C]** $M_{PS}/M_s = 1.0$–1.7 (1-loop; 2-loop ~ 1.0–1.2) — gauge unification meets the gravitational scalaron scale; **[X]** NEG 126 breaks the match ($b_{2R}{>}0$; also E_8-forbidden v247); **[X]** NEG proton decay selects content (minimal $M_{GUT}{\sim}2{\times}10^{15}$ excluded; $+(15,1,1)$ safer). **[O]** the gauging fork, exact κ, the one-45 marginality.

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Script	**What it machine-checks *(continued)***
`v250_dirac_triple_contract.py`	`CONTRACT.QFT4D.DIRAC.01` — the spectral-triple contract that would make 'carrier gauged' a theorem. Target (A, H, D, J, γ) almost-commutative, $A_F = \mathrm{PS}$ (v248), $H_F = 3{\times}16$, D_F from $(R, L, \varphi_0, \mathrm{Majorana})$. [E] $H_F = 3{\times}16 = 48$; [E] the Yukawas are already fixed (v18) so the open test is NCG consistency; [O] the 7 NCG axioms (first-order condition = the historic $SO(10)$ obstruction); [O] σ = scalaron hypothesis (rescues first-order $+\,\nu_R$ Majorana; $M_{PS}{=}\kappa M_s$) — caveat $\sigma \leftrightarrow \nu_R\nu_R$ not automatic; [O] inner fluctuations → 10+16+45 not 126. Reduces 4d QFT to one object + finite NCG checks.
`v251_first_order_sigma.py`	`PS.DIRAC.01` — the first-order condition is violated *exactly* by the Majorana/σ term (Chamseddine–Connes–van Suijlekom 'beyond first order' → Pati–Salam $+\,\sigma$), exhibited on a faithful lepton-block model; advances v250 from contract to mechanism. Honest correction: σ does NOT 'rescue' first order — one DROPS it and the inner fluctuations generate PS $+\,\sigma$. [E] order-zero; [E] real-even ($J^2{=}{+}1$, $\gamma^2{=}1$, $D\gamma{=}{-}\gamma D$, KO-6 signs); [E] first order HOLDS for the Yukawa block; [E] first order VIOLATED by $M \neq 0$, supported exactly on the $\nu_R\nu_R^c$ blocks (σ the unique violator). [C] σ = scalaron (v249). [O] the full 96-dim SM/PS triple + spectral action (v250).
`v252_full_finite_triple.py`	`PS.DIRAC.02` — the full 96-dim finite spectral triple $(A_F, H_F, D_F, J, \gamma)$ built explicitly with the NCG axioms verified; advances v250 from a 7-obligation contract to a concrete object discharging 5 to [E]. H_F = particle⊕antiparticle doubling of three $SO(10)$ **16**-plets (dim 96). [E] dim 96; D_F Hermitian+odd; reality $J^2{=}{+}1$, $JD{=}DJ$; grading $\gamma^2{=}1$, $[\gamma, a]{=}0$, $J\gamma{=}{-}\gamma J$; KO-dim 6; order-zero; first order HOLDS for the SM $\mathbb{C} \oplus \mathbb{H} \oplus M_3$ (incl. Majorana) and VIOLATED for Pati–Salam exactly by the Majorana on $\{\nu_R, e_R\}$ (CCvS beyond-first-order → PS+σ); spectrum = fermion masses + seesaw $m_{\mathrm{light}}{=}m_D^2/M_R$. [C] Poincaré duality (intersection form non-degenerate). [O] orientability + no-junk + spectral-action expansion.
`v253_kappa_scalaron_ps.py`	`PS.KAPPA.01` — pinning κ in $M_{PS} = \kappa\, M_s$, discharging residual 6(b) of v249. $M_s = c_3^{7/2}\bar{M} \approx 3.06 \times 10^{13}\,\mathrm{GeV}$ (exact). [E] scalaron scale exact; [E] κ pinned by RG (scan PDG errors × the two E_8-allowed contents) to a tight $O(1)$ interval ~ 1.3 (1-loop $[1.0, 1.7]$, 2-loop $[1.0, 1.2]$) — pinned, not free. [C] heat-kernel origin $\kappa = \sqrt{(f_2/f_0)\, c_{PS}/c_{\mathrm{grav}}}$, a ratio of heat-kernel traces over the 96-dim H_F (v252) — Λ cancels, κ scale-free and $O(1)$. [X] the **126**-content drives κ out of the $O(1)$ window (content-selective). [O] the exact decimal needs the cutoff moments f_2/f_0 fixed.
`v254_inner_fluctuations.py`	`PS.NCG.FLUCT.01` — the inner fluctuations $\Omega^1_D(A_F)$ of the finite Dirac operator computed explicitly (reuses the v252 triple); closes v250 #5 (Higgs content, not **126**) + the no-junk obligation + the σ quantum numbers. [E] $\Omega^1_{D_F}$ purely chirality-odd → finite fluctuations are scalars; [E] no junk / no **126**: SM-algebra 1-forms sit exactly in the Higgs-doublet pattern (the single $(1, 2)_{1/2}$); [E] the σ (ν_R Majorana) direction is absent for the SM algebra but present for Pati–Salam — σ is the CCvS beyond-first-order scalar; [E] σ is an SM singlet $(1, 1, 0)$ with $B{-}L = 2$; [C] σ = scalaron consistency.
`v255_spectral_action_expansion.py`	`PS.SPECACT.01` — the spectral-action heat-kernel (Seeley–DeWitt) expansion $S = 2f_4\Lambda^4 a_0 + 2f_2\Lambda^2 a_2 + f_0 a_4$ over the finite triple; closes v252 #11 (spectral-action expansion) and makes $\kappa(f_2/f_0)$ explicit. [E] expansion structure (a_0 cosmological, a_2 Einstein–Hilbert+Higgs-mass, a_4 Yang–Mills+Weyl^2 + R^2+Higgs-quartic, $N{=}96$); [E] trace inputs $b/a^2 = 1/3$ (top-dominated, Chamseddine–Connes); [E] $\sin^2\theta_W = 3/8$ (cross-check v245); [C] Higgs quartic $\lambda(\Lambda) \sim g^2 b/a^2 \to \sim 125\,\mathrm{GeV}$ with σ; [E] scalaron = R^2 spin-0 mode (gravitational $a_4 \sim f_0 N$); [C] $\kappa = \sqrt{(f_2/f_0) c_{PS}/c_{\mathrm{grav}}}$ explicit; [O] exact decimal needs the cutoff moments fixed.

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Script	What it machine-checks *(continued)*
`v256_orientability_poincare.py`	`PS.NCG.ORIENT.01` — orientability and Poincaré duality treated honestly (no faked proof). KO-dim 6 forces the intersection form $\mathrm{cap}(p,q) = \mathrm{Tr}(\gamma\,\pi(p)\,J\pi(q)J^{-1})$ to be antisymmetric. **[E]** skew form forced by $J\gamma{=}-\gamma J$ + order-zero + reality (verified); **[E]** on the SM K-theory $\mathbb{Z}^3$ the skew form has rank 2, corank 1, null $\sim(1,1,-1)$ — the structural obstruction; **[E]** orientability local condition $[\gamma_F,\pi(a)]=0$, full orientability is a degree-4 cycle on $M\times F$; **[C]** the canonical hypercharge-faithful SM/PS triple satisfies both (van Suijlekom; CCM) — literature; **[O]** a self-contained machine proof needs the complete canonical bimodule.
`v257_quasiorient_modpd.py`	`PS.NCG.ORIENT.02` — the honest, literature-confirmed resolution of orientability + Poincaré duality, correcting the too-generous v256 typing. Published result (Cacic–Stephan arXiv:0902.2068; CCM07; SM-vacuum hep-th/0601192): the CCM SM finite triple (KO-6, $3\,\nu_R$) genuinely FAILS strict orientability and strict Poincaré duality, satisfying the weakened versions. **[E]** strict orientability fails ($\gamma\notin\mathrm{span}\{\lambda(a)\rho(b)\}$); **[E]** quasi-orientability holds (left/right A-irrep boxes disjoint by chirality: $L=H$ doublet, $R=\mathbb{C}$ singlet $\Rightarrow L^{\mathrm{LR}}(\mathrm{even},\mathrm{odd})=\{0\}$); **[E]** strict Poincaré fails (KO-6 skew form, corank 1, the equal-L/R-neutrino case); **[C]** modified Poincaré duality (CCM) / Stephan's PD-restoring triple. A known model-class feature, not a TFPT defect — no faked closure.
`v258_dirac_covariance_induction.py`	`PS.DIRAC.03` — D_F is the modular/covariance *induction* of the boundary seam state, not an independent posit; closes the 'posited Yukawa vs derived operator' gap of v250/v252 (Q2: Dirac as Kasparov product + covariance formula). Quasi-free/modular dictionary: a Gaussian state is fixed by its covariance C (contraction, spec $\subset(0,1)$) with one-particle modular Hamiltonian $H=\log((1{-}C)C^{-1})$ (Araki; Casini–Huerta), inverse to the KMS occupation $C=(1{+}e^{H})^{-1}$ ($\beta{=}1$, v239). **[E]** inversion identity $\log((1{-}C)C^{-1})=H$ exactly (random Hermitian + a chirality-graded Yukawa block); **[E]** $C_F=(1{+}e^{D/\mu})^{-1}$ a positive contraction; **[E]** chirality-odd ($D_F^{\mathrm{ind}}\gamma=-\gamma D_F^{\mathrm{ind}}$); **[E]** unitarily equivalent to v252 (inducing C_F from the 9 Yukawas returns the same spectrum); **[E]** Majorana = off-diagonal covariance (seesaw $m_{\mathrm{light}}{=}m_D^2/M_R$); **[C]** $C_F=P_FC_\Sigma P_F$, so D_F is the KK shadow $[D_\Sigma]\widehat{\otimes}_{(E_8)_1}[\mathcal{K}_{\mathrm{car}}]$ and the v18 Yukawas are the readout of C_Σ.
`v259_modular_cutoff_kappa.py`	`PS.SPECACT.02` — the spectral-action cutoff *is* the seam KMS weight, fixing $f_2/f_0{=}1$ exactly and reducing the last open scheme freedom (v253 #5, v255 #7) from a free function to a finite-triple trace ratio (Q4: the cutoff is modular, not chosen). By Tomita–Takesaki/BW the seam state is KMS at $\beta{=}1$ (v239) and $2\pi{=}1/(4c_3)$ (v58) fixes β, so the heat-kernel cutoff $f(u){=}e^{-u}$ is *derived.* **[E]** KMS cutoff moments $f_0{=}f(0){=}1$, $f_2{=}\int f{=}1$, $f_4{=}\int uf{=}1$, so $f_2/f_0{=}f_4/f_2{=}1$; **[E]** seam-fixed $\beta{=}1$ (no new parameter beyond c_3); **[E]** non-vacuous neg control: a Gaussian cutoff e^{-u^2} gives $f_2/f_0{=}\sqrt{\pi}/2\neq1$; **[E]** $\kappa{=}\sqrt{(f_2/f_0)c_{PS}/c_{\mathrm{grav}}}{=}\sqrt{c_{PS}/c_{\mathrm{grav}}}$, no cutoff function left; **[C]** the RG $\kappa{\sim}1.3$ pins $c_{PS}/c_{\mathrm{grav}}{\sim}1.7$ ($O(1)$); the **[O]** free scheme becomes a **[C]** finite computation over the built triple; **[C]** the same logic fixes $\lambda(\Lambda)$.

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Script	What it machine-checks *(continued)*
`v260_k3_kummer_unification.py`	`ARCH.K3.01` — one Kummer/K3 surface carries the *seam*, the *carrier-16* and the E_8 lattice as facets of one smooth object, the geometric unification of the v1 'glue $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$' step (the K3/Kummer proposal). The seam square torus $\tau = i$ ($j = 1728$, v214) under the elliptic involution $z \mapsto -z$ squares to the Kummer K3. [E] the E_8 Cartan is even, $\det = 1$, positive-definite (the unique even unimodular pos-def rank-8 lattice); [E] the K3 lattice $L_{K3} = U^3 \oplus E_8(-1)^2$ has rank $22 = b_2$, $\det = -1$ (unimodular), is even, signature $(3, 19)$, with $E_8(-1)^2$ an orthogonal summand; [E] seam $= T^2/(z \mapsto -z)$: $4 = \lvert(\mathbb{Z}/2)^2\rvert$ fixed points, pillowcase $S^2(2,2,2,2)$, orbifold Euler char 0 (the v214/v216 four marks); [E] carrier-16 = Kummer nodes $\lvert A[2]\rvert = 2^4 = 16 \to 16$ exceptional $\mathbb{P}^1 = \dim S^+ = \Lambda^{\mathrm{even}}(\mathbb{C}^5) = 16$ (v197), $16 = 4 \times 4$; [E] honesty: the 16 node classes span the rank-16 Kummer lattice, *not* literally $E_8(-1)^2$ (distinct sublattices of $H^2(K3)$); [C] two faces of E_8 in K3 (local du Val $\mathbb{C}^2/2I$, v232/v236; global H^2 summand); [C] unification (lattice [E], carrier$\leftrightarrow$node identification [C]).
`v261_modular_spectral_closure.py`	`QFT.MSC.01` — the *Modular Spectral Closure*: the assembly/reduction certificate turning the whole QFT layer (v238–v260) into *one* relative object $\mathsf{TFPT}_{\mathrm{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\mathrm{rel}})$ reduced to *one* open premise (like v176/v234, an assembly certificate, not new physics). [E] one index = marks = fixed points = glue order: Jones $[B{:}A] = 4$ (v89) $= \lvert\mu_4\rvert = 4$ (v92) = seam marks $2\chi = 4$ (v216) = pillowcase/Kummer fixed points $\lvert(\mathbb{Z}/2)^2\rvert = 4$ (v260); [E] one carrier-16: $\dim S^+ = \Lambda^{\mathrm{even}}(\mathbb{C}^5) =$ Kummer nodes $2^4 = 16$, $H_F = 2N_{\mathrm{fam}} \cdot 16 = 96$; [E] one gap $\Delta = -\log((2/3)^6) = 6\log(3/2) > 0$ = the OS mass gap (v240) = the recovery rate (v221) = the Lindblad gap (v238); [E] the three closures re-derived (Kasparov $\log((1-C)C^{-1}) = H$, v258; KMS cutoff $f_2/f_0 = 1$, v259; K3 lattice signature $(3, 19)/\det -1/\mathrm{even}$, v260); [E] relative-object manifest (10/10 component scripts present); [O] the single residual: the whole QFT layer reduces to `QGEO.SYM.01` (seam realisation), the ambient QG measure `QG.AMB.01` kept separate; [C] the master statement (Modular Spectral Universality) holds modulo `QGEO.SYM.01` — the honest sense in which the QFT layer is complete.
`v262_fqcd_mp_me.py`	`FR.MPME.02` — the F_{QCD} transfer pipeline for m_p/m_e, implemented as a real solver (the one F_{transfer} pipeline that was contract-only). *Not* a compiler power — a [C] standard-physics transfer reproduction. [E] $b_3 = -7 = -(11 - 2n_f/3)$ at $n_f = 6$ (the only TFPT number in the run); [E] a two-loop $\alpha_s(M_Z) = 0.1179$ run with thresholds gives $\alpha_s(2\,\mathrm{GeV}) \sim 0.30$ (real ODE solve); [C] $\Lambda^{(3)}_{\overline{\mathrm{MS}}} \sim 0.33\,\mathrm{GeV}$ × the lattice bridge $C_p = m_p/\Lambda$ (external $O(1)$) over the closed m_e gives $m_p/m_e \sim 1824$ vs the measured 1836.15 within the $\sim 7\%$ external budget; [X] firewall: m_p/m_e is not an admissible compiler integer ($SU(9)$ absent from $D_5 \oplus A_3$). Advances `FR.MPME.01` ([O]) to `FR.MPME.02` ([C]).
`v263_mnu_from_df.py`	`FLAV.PMNS.02` — the light neutrino matrix M_ν as the type-I seesaw of D_F, advancing the solar-angle texture (v9) from a posited matrix to a seesaw *object* built from D_F's blocks. [E] $M_\nu = -m_D M_R^{-1} m_D^\top$ symmetric + seesaw-suppressed (the light masses an output of D_F); [E] a $\mu\tau$-symmetric (m_D, M_R) gives a $\mu\tau$-symmetric M_ν (the v9 texture is in the seesaw image); [C] diagonalising gives $\theta_{23} = 45^\circ$ and $\sin^2\theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\mathrm{ret}}}{2} = 0.30675$ under the seam misalignment; [O] $\theta_{13} = 0$ in the leading texture (physical $\sin^2\theta_{13} \sim \varphi_0^{\mathrm{ret}} e^{-5/6}$ a separate readout), the CP magnitude $\delta_{\mathrm{PMNS}} = 240^\circ$ is the μ_6/triality phase (v231/v233), and M_R is the free seesaw scale. An operator advance, not a closure of full PMNS dynamics.

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Script	What it machine-checks *(continued)*
`v264_raw_seam_marklocal.py`	`QGEO.ENERGY.03` — the raw-seam ⇒ mark-locality *reduction* (not a closure of `QGEO.SYM.01`). Composing v216+v201+v198 on the canonical pillowcase: [E] Gauss–Bonnet 4 cone points (deficit π each, sum $4\pi{=}2\pi\chi(S^2)$), curvature *only* at the 4 marks; [E] the curvature sourced at the μ_4 orbit $\{0, \pi/2, \pi, 3\pi/2\}$ has Fourier support *only* on $4\mathbb{Z}$ (FFT), so mark-locality is automatic for the pillowcase; [E] M_f then commutes with the clock $\rho{=}\mathrm{diag}(i^n)$; [E] with $\lvert k\rvert$ already commuting (v198), $[\rho, \Lambda]{=}0 \Rightarrow \omega\circ\rho{=}\omega$ on this metric; [O] the residual *is* the metric — the chain assumes the raw seam carries the uniformising flat pillowcase metric (=`QGEO.SYM.01`), the negative control (curvature off the μ_4 orbit) breaks it. A reduction/sharpening, not a closure.
`v265_qft4d_fork_freeze.py`	`QFT4D.FORK.01` — freeze the 4D-QFT fork as a *branch policy* (not an open ambiguity); writes `qft4d_fork_freeze.json` + a prose guard, no new physics. [C] A: boundary-only is the canonical 4D reading (no 4D-GUT claim by default); [X] SM-only 4D-GUT unification is killed (1-loop spread ∼8.8 at $2{\times}10^{16}$, v246); [C]/[O] B: carrier-native Pati–Salam is a falsifiable UV branch (E_8 content $\{1, 10, 16, 45\}$ no **126** v247, $\sin^2\theta_W{=}3/8$ v248, κ $O(1)$ band v249/v253, threshold + proton kill test); [X] proton safety rule (no fake τ_p window); [E] wording guard (bare 4D over-claims absent, scoped policy phrases present). A frozen decision tree, not an open ambiguity.
`v266_ps_threshold_proton.py`	`PS.PROTON.02` — the explicit Pati–Salam threshold model + proton-decay window that DECIDES branch B of the v265 fork; writes `ps_threshold_proton.json`. Two-step PS solve + $d{=}6$ $\tau_p(p{\to}e^+\pi^0)$: the carrier-native PS UV branch survives proton decay *only* with the E_8-allowed $(15, 1, 1)$ [the single 45]. [E] explicit thresholds: minimal 16H → $M_{GUT}{\sim}2.4{\times}10^{15}$, $+(15, 1, 1) \to M_{GUT}{\sim}5.9{\times}10^{15}$; [X] minimal KILLED ($\tau_p{\sim}4.4{\times}10^{33}$ yr < Super-K $2.4{\times}10^{34}$, even high-band); [C] $+(15, 1, 1)$ survives now ($\tau_p{\sim}1.6{\times}10^{35}$ yr); [X] *Hyper-K decisive* (τ_p at the $\sim 1.4{\times}10^{35}$ yr reach) — a dated kill test within the decade; [O] marginal (only one 45 in the hull, v247) + $O(3)$ hadronic uncertainty. A (boundary-only) stays canonical.
`v267_qgeo_rigidity_minimal_axiom.py`	`QGEO.SYM.02` — the rigidity / minimal-axiom form of the bedrock `QGEO.SYM.01`: the ONE bare statement (the raw RP seam admits the carrier μ_4 clock as a conformal symmetry permuting its 4 marks, $= \omega\circ\rho = \omega$) *forces* everything downstream; it does *not* close the axiom. [E] an order-4 Möbius orbit $\{a, ia, -a, -ia\}$ has cross-ratio 2 (independent of a); [E] cross-ratio 2 ⇔ $j{=}1728$ (only $\lambda \in \{-1, \frac{1}{2}, 2\}$) ⇔ the order-4 CM automorphism $(x, y) \mapsto (-x, iy)$ on $y^2{=}x^3{-}x$; [E] downstream forced: square ⇒ the unique flat pillowcase (Troyanov, v214) ⇒ curvature at the 4 marks ⇒ $\mathbb{Z}_4$-Fourier DtN (v201/v264) ⇒ $[\rho, \Lambda]{=}0 \Rightarrow \omega\circ\rho{=}\omega$ (v198); [E] neg controls (generic $j \neq 1728$, only $\mathbb{Z}_2$; hexagonal $j{=}0$ has $\mathbb{Z}_6$, order $6 \neq 4$); [C] arithmetic rigidity ($\{1, i, -1, -i\}$ over $\mathbb{Q}$, $j{=}1728$ CM by $\mathbb{Z}[i]$, Belyi/dessins); [O] the bare order-4 symmetry stays the one foundational postulate (like c=const). Closes the reduction (axiom⇒everything), not the axiom.
`v268_theta13_carrier_trace.py`	`FLAV.TH13.01` — the reactor angle θ_{13} *exponent* is the carrier hypercharge trace. The value $\sin^2\theta_{13} = \varphi_0^{\mathrm{ret}} e^{-5/6} = 0.0231$ was already checked (v16); here the exponent origin is closed exactly: [E] $\gamma = \mathrm{tr}_E Y^2 = 3(1/3)^2 + 2(1/2)^2 = 5/6$ over the 5-slot carrier hypercharge $Y = \mathrm{diag}(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2})$ (roots of $6Y^2 - Y - 1 = 0$), complement 1/6 the neutrino ratio; [C] value 0.02311 vs PDG ∼0.0222; [E] own channel — the $\mu\tau$-breaking $\varepsilon = 3\varphi_0^{\mathrm{ret}}/4$ induces only $\sim 1.6 \times 10^{-3} \ll 0.023$, so θ_{13} is a separate carrier-trace channel, not the seesaw $\mu\tau$ breaking (corrects the tempting ‘θ_{13} from M_ν’ route); [O] CP magnitude + absolute ν scale remain open.

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Script	**What it machine-checks** ***(continued)***
`v269_spert_paqft_skeleton.py`	`QFT4D.SPERT.01` — the perturbative 4D S-matrix (S_{pert}) as an Epstein–Glaser / pAQFT skeleton: the honest middle layer of the three-layer S-matrix (S_{top} 2d DHR braiding v243 / S_{pert} 4d EG of the spectral action / S_{phys} LSZ on the OS Wightman functions v240), *not* a nonperturbative closure. **[E]** three distinct layers (the 2d braiding $\|M\|$=1 is statistics, not 4d cross-sections); **[E]** every spectral-action a_4 operator has mass dimension ≤ 4 in 4D (power-counting renormalizable); **[E]** a finite counterterm basis (7 marginal +3 relevant, no infinite tower); **[C]** Epstein–Glaser existence — a renormalizable interaction has an order-by-order $S(g){=}T\exp(i\int g\,L_{\mathrm{int}})$ by causal factorisation + finite local extension (no path integral, no UV divergences); **[C]** adiabatic limit/gap ($\Delta{=}6\log(3/2){>}0$) $\to S_{\mathrm{phys}}$ via LSZ; **[O]** perturbative only — the ambient measure `QG.AMB.01` stays open. A contract/skeleton, explicitly perturbative.
`v270_pmns_jarlskog_assembly.py`	`FLAV.PMNS.03` — the complete complex PMNS matrix + the leptonic Jarlskog J_{PMNS} (the CP *strength*), assembled from the four TFPT channels ($\sin^2\theta_{12}{=}\frac{1}{3}{-}\frac{\phi_0}{2}$, $\sin^2\theta_{23}{=}\frac{1}{2}$, $\sin^2\theta_{13}{=}\phi_0 e^{-5/6}$, $\delta{=}4\pi/3$) into one unitary object. **[E]** the three angle channels; **[E]** $U = R_{23}U_{13}(\delta)R_{12}$ is exactly unitary; **[C]** $J_{\mathrm{PMNS}} = \Im(U_{e1}U_{\mu 2}U^*_{e2}U^*_{\mu 1}) = -0.02965$ (formula and full matrix agree), $J_{\max} = 0.03424$ — a *derived* CP strength, the leptonic analogue of the CKM J (v88); **[C]** data: $J_{\max}$ vs NuFIT 0.0332 ($\sim 3\%$), J vs best fit -0.026, the negative sign reproduced; **[O]** δ is the μ_6/triality phase (assembled input, *not* from M_ν/D_F) and the absolute ν-mass scale (Σm_ν, M_R) stays open. Assembles the PMNS matrix + CP strength; does not derive δ from D_F nor fix the scale.
`v271_eg_oneloop_quartic.py`	`QFT4D.SPERT.02` — a concrete Epstein–Glaser order-2 (one-loop) quartic renormalization, taking the v269 S_{pert} skeleton from *exists* to an actual EG number (no path integral). **[E]** $T_1{=}{:}L_{\mathrm{int}}$: needs no extension, the first extension is T_2; **[E]** the massless propagator has scaling degree sd=2 in $d{=}4$, the one-loop bubble (two propagators) has sd=4=d so the UV singular order $\omega{=}\mathrm{sd}{-}d{=}0$; **[E]** $\omega{=}0 \Rightarrow$ exactly one logarithmic local counterterm ($C(\omega{+}d,d){=}1$, renormalizable, no infinite tower); **[E]** the loop factor $\Omega_3/(2(2\pi)^4){=}1/(16\pi^2)$ exactly — the same factor that normalises the spectral-action a_4 geometric quartic; **[C]** the ϕ^4 one-loop $\beta_\lambda{=}3\lambda^2/(16\pi^2)$ (symmetry factor 3 = the s,t,u channels), EG initial condition $\lambda_\Phi(\mathrm{UV}){=}1/(16\pi^2)$; **[O]** order-2/one-loop only — the all-order EG construction and the nonperturbative measure `QG.AMB.01` stay open. Turns the skeleton into a concrete one-loop number.
`v272_nu_mass_scale.py`	`FLAV.NUSCALE.01` — the absolute neutrino-mass scale, typed honestly (it does *not* fabricate a Σm_ν number, which would contradict the v184 firewall). **[E]** the absolute scale reduces to *one* seesaw ratio $m_3 = (y_\nu v_{\mathrm{EW}})^2/M_R$ — one irreducible UV input like v_{geo} (v153), while the ratios (m_2/m_3, ordering, PMNS angles v270) are pinned; **[C]** the observed $m_3{=}0.050$ eV with the TFPT PS→GUT window $M_R \in [4.2{\times}10^{13}, 2.4{\times}10^{15}]$ GeV (v249) needs a third-generation Dirac Yukawa $y_\nu \in [0.26, 2.0]$ — $O(1)$, no tuning; **[O]** $M_R(y{=}1){=}6.1{\times}10^{14}$ has $\log_{c_3}(M_R/M_{\mathrm{Pl}}){=}2.57$ (non-integer), not a TFPT power, so the absolute value stays open; **[C]** the normal-ordering floor $\Sigma m_\nu{=}0.0586$ eV is consistent with cosmology (<0.072–0.12 eV); **[X]** a cosmological $\Sigma < 0.0586$ eV excludes the normal-ordered $m_1{\sim}0$ spectrum. Closes the PMNS bookkeeping (the last **[O]** is one seesaw ratio), not the absolute number; sharpened by v481 (carrier normalisation $y_\nu{=}y_t$: one-parameter, window-bracketed candidate).

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Script	What it machine-checks *(continued)*
v273_eg_gauge_running.py	QFT4D.SPERT.03 — the gauge sector of the Epstein–Glaser one-loop (v271's quartic mechanism applied to the gauge self-energy). **[E]** the one-loop gauge 2-point has scaling degree sd=4=d so ω=0 → one counterterm per coupling (same marginal mechanism + loop factor $1/(16\pi^2)$ as v271); **[E]** the one-loop b_i from the explicit carrier/SM content: $b_3{=}-7$, $b_2{=}-19/6$, $b_1{=}41/10$ (the same 41 as the carrier algebra, v159/CAR.SM.01); **[C]** physical RG directions $b_3{<}0$ (asymptotic freedom), $b_1{>}0$ (U(1) Landau); **[O]** the two-loop matrix (v159, PyR@TE) + all-order stay open. Shows the gauge running is an EG order-2 result. Exact b_i mirrored in Wolfram.
v274_scale_overdetermination.py	SCALE.OVERDET.01 — the single mass anchor is *over-determined*, and the absence of a derivable absolute scale is a *theorem*. **[E]** one anchor (No-Unit, v153): every dimensionful output = pure number $\times M_{\rm Pl}^d$ ($\rho_\Lambda/M_{\rm Pl}^4$, $M_{\rm scal}/M_{\rm Pl}{=}c_3^{7/2}$, the spectrum); **[C]** two independent routes to $M_{\rm Pl}$ agree: gravity $(8\pi G)^{-1/2}{=}2.4353{\times}10^{18}$ GeV and cosmology (invert $\rho_\Lambda/M_{\rm Pl}^4{=}(3/4\pi^2)e^{-2\alpha^{-1}}$ with $\rho_\Lambda^{1/4}{=}2.24$ meV) $=2.438{\times}10^{18}$ GeV, to 0.11%; **[E]** the anchor is the gravitational unit (G), the one measured dimensionful input of all physics; **[E]** theorem not gap: the seam is a conformal c=8 CFT (v157/v158), scale-invariant by construction → no intrinsic scale; **[O]**/**[E]** scale-complete up to one over-determined anchor = maximal predictivity. Sharpens v153/v78.
v275_qgamb_roadmap.py	QGAMB.ROADMAP.01 — the obligation roadmap for QG.AMB.01 (the ambient nonperturbative QG measure), consolidated into one module (the open gate had no script). **[E]** Tier A gap decoupling: $\Delta_{\rm eff}{=}6\log(3/2)-2{\cdot}248\,c_3^2{=}1.648{>}0$ (v36/v76), so every low-energy readout is independent of the un-built measure — not a blocker; **[E]** Tier B: the bulk measure reduces to identifying the seam-Calderón boundary with the holomorphic $(E_8)_1$ net ($c{=}248/31{=}8$; v77/GATE.METRIC.03), reduced not closed (REDTEAM.A.01); **[C]** the perturbative layer is in hand (v269/v271/v273); **[O]** the nonperturbative ambient measure is *not* constructed — the one genuine structural frontier; **[X]** closes IFF the seam net $=(E_8)_1$ *and* the bulk reconstruction is proved. A roadmap, not a closure.
v276_qgeo_flat_closes_commutator.py	QGEO.SYM.03 — the flat $\tau{=}i$ pillowcase premise *closes* the modular-commutator chain to all orders, collapsing the bedrock QGEO.SYM.01 to *one* geometric postulate (it does *not* prove the premise). For a quasi-free seam state QGEO.SYM.01 = ($\omega\circ\rho{=}\omega$) ⇔ $[\rho,C]{=}0$ ⇔ $[\rho,H]{=}0$. **[E]** isometry⇒commutator: if $H{=}f(\Delta_{\rm flat})$ and the order-4 deck ρ ($z{\to}iz$) is a flat-$\tau{=}i$ isometry then $[\rho,\Delta]{=}0 \Rightarrow [\rho,H]{=}0$ *exactly* (all orders), verified on the discretised square torus ($\rho^4{=}I$, $[\rho,\Delta]{=}0$, $[\rho,\sqrt{-\Delta}]{=}0$; generic non-isometry fails); **[E]** this upgrades v198 (principal symbol) + v201 (subprincipal) to the full H (flatness removes the curvature terms); **[C]** $\tau{=}i$ is forced by order-4 (j=1728; hexagonal j=0 is order-6, excluded; v214/v267); **[O]** QGEO.SYM.01 collapses to the single statement 'the raw seam is the flat $\tau{=}i$ pillowcase quasi-free state' — needs a human constructive-QFT proof or stays an honest axiom. Sharpest non-circular form.
v277_seam_calderon_e8_match.py	QGAMB.TIERB.01 — the seam-Calderón → $(E_8)_1$ matching certificate, the concrete Tier-B step of QG.AMB.01 (v275): it computes the full $(E_8)_1$ matching target and pins the residual to *one* bit (it does *not* close it). **[E]** c=8 (Sugawara, $248/31{=}g_{\rm car}{+}N_{\rm fam}$), but c=8 alone does not pin the net; **[E]** the discriminator is holomorphy — $\det{\rm Cartan}(E_8){=}1$ (1 primary, holomorphic) vs the c=8 counterexample $\det{\rm Cartan}(D_8{=}SO(16)){=}4$ (4 primaries, non-holomorphic), so the single discriminator is $\lvert\det K\rvert{=}1$; **[E]** the $(E_8)_1$ character $E_4/\eta^8{=}j^{1/3}{=}q^{-1/3}(1+248q+\dots)$ has a single primary + exactly 248= dim E_8 weight-1 currents (240 roots); **[E]** equivalence chain holomorphic ⇔ det Cartan 1 ⇔ unique even unimodular rank-8 (v83) ⇔ homology-sphere link ⇔ $\lvert\det K\rvert{=}1$ (v235); **[C]** seam-side match via v276 (flat $\tau{=}i$) + v260 (Kummer/K3); **[O]** residual = the single holomorphy bit — reduced to one bit, *not* closed.

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Script	What it machine-checks *(continued)*
`v278_lsz_bridge_unitarity.py`	`QFT4D.SPERT.04` — the $S_{\mathrm{pert}} \to S_{\mathrm{phys}}$ LSZ bridge + one-loop unitarity: connects the EG perturbative S-matrix (v269/v271/v273) to the physical asymptotic S-matrix (S_{phys} = LSZ on the OS Wightman functions, v240). [E] LSZ bridge: S_{phys} = amputate+on-shell-residue of the connected time-ordered products, and the EG T-products of S_{pert} *are* those correlators (OS Wick-rotation, v240) $\Rightarrow S_{\mathrm{pert}} \xrightarrow{\mathrm{LSZ}} S_{\mathrm{phys}}$; [C] the gap $\Delta{=}6\log(3/2){>}0$ gives Haag–Ruelle asymptotic states so LSZ is valid; [E] *one-loop unitarity* (optical theorem): the bubble discontinuity $\operatorname{Im} B(s)/\pi = x_+{-}x_- = \sqrt{1-4m^2/s}$ *exactly* = the two-body phase space, so $2\operatorname{Im} M = \sum\int d\Pi\lvert M\rvert^2$ (cutting rule) holds ⇒ matter+gauge unitary; [E] threshold: $\operatorname{Im} B{=}0$ below $s{=}4m^2$, turning on at the physical two-particle threshold; [O] gravity caveat — the Stelle ghost makes the gravity perturbative S-matrix non-unitary →`QG.AMB.01`; [C] finite field-strength Z from the EG scheme.
`v279_qgeo_obligation_lemma.py`	`QGEO.OBLIG.01` — the `QGEO.SYM.01` bedrock written as *one* precise constructive-QFT lemma with a proof-tree completeness check (the formal obligation write-up; does *not* close it). Lemma: given the premise 'the raw seam carries the flat $\tau{=}i$ pillowcase metric g_{flat} (Troyanov-unique, cone angles π), so $\Lambda_\Sigma = \sqrt{-\Delta_{\mathrm{flat}}}$', then $\omega_\Sigma \circ \rho = \omega_\Sigma$ → mark-locality → E_8. [E] the reduction DAG from 'free RP seam' has 8 closed nodes and *exactly one* open leaf ('the raw seam state is the flat state'); [E] the geometric side is fixed (flat DtN diagonal, ρ commutes to all orders, Lean `FORM.QGEO.03`), so the obligation is state-identification not geometric ambiguity; [C] two proof routes (RP-state uniqueness on the flat orbifold, or Troyanov + OS); [E] the all-orders step is a Lean theorem (`SeamDeckClosure.flat_all_orders_clock`, standard axioms, no `sorry`); [O] one constructive-QFT statement closes `QGEO.SYM.01`.
`v280_pillowcase_steklov.py`	`QGEO.STEKLOV.01` — a direct numerical experiment on the flat $\tau{=}i$ pillowcase orbifold (the seam geometry): the self-investigable test of the `QGEO.SYM.01` obligation (does *not* prove the premise). [E] pillowcase = the Z2-even sector of the flat torus, 4 cone points (2-torsion, deficit π each, sum $4\pi{=}2\pi\cdot 2$); [E] the order-4 deck ρ ($z{\to}iz$) descends ($[\rho, Z_2]{=}[\rho, P_{\mathrm{even}}]{=}0$) and permutes the 4 cone points as 2 fixed + 1 swapped pair; [E] the geometric chain is exact: $[\rho,\Delta]{=}0 \Rightarrow [\rho,H]{=}0 \Rightarrow [\rho,C]{=}0 \Rightarrow \omega\circ\rho{=}\omega$ (the v276/v279 all-orders closure on the real orbifold); [C] the Steklov DtN is positive self-adjoint, fixed by the flat metric (route-(i) evidence); [O] confirms the conclusion given flat geometry, the premise stays open.
`v281_holomorphy_modular_data.py`	`QGAMB.MODULAR.01` — the modular-data / anyon-condensation layer of the `QG.AMB.01` Tier-B holomorphy bit. [E] #anyons = $\lvert\det \mathrm{Gram}\rvert$: $E_8{\to}1$ (holomorphic), $D_8{=}SO(16){\to}4$, carrier $D_5{\oplus}A_3{\to}16$; [E] the anyon-condensation tower $D_5{\oplus}A_3$ (16) → D_8 (4) → E_8 (1), each step condensing a Lagrangian Z_2 boson, the Kitaev E_8 endpoint (v92/v235); [E] Gauss–Milgram recovers $c{=}8$; [E] E_8 is the unique holomorphic $c{=}8$ (det 1) — $SO(16)$ has the same c but 4 anyons, so holomorphy is THE discriminator (v277); [O] the open bit: does the seam-Calderón net condense to $\det K{=}1$? Reduced to one bit.
`v282_e8_tau_i_unification.py`	`QGAMB.UNIFY.01` — the E_8-at-$\tau{=}i$ unification: the candidate *simplification* reducing the two open obligations to *one* object (the raw seam is the $(E_8)_1$ holomorphic CFT at $\tau{=}i$). $\chi_{E_8}(\tau){=}\Theta_{E_8}/\eta^8{=}j^{1/3}$. [E] $\tau{=}i$ is the order-4 CM point ($j{=}1728$, the `QGEO.SYM.01` flat geometry); [E] $\chi_{E_8}{=}j^{1/3}$ ($\chi^3{=}q\,j$ as q-series), so $\chi_{E_8}(i){=}1728^{1/3}{=}12$ (the $(E_8)_1$ character at the SAME $\tau{=}i$, the `QG.AMB.01` holomorphy); [E] one object, two faces; [C] obligation count 2→1: prove 'the raw seam is $(E_8)_1$ at $\tau{=}i$' and *both* follow; [O] the unified premise stays open but is now one object.

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Script	What it machine-checks *(continued)*
`v283_live_killtest_scorecard.py`	`PRED.LIVE.01` — a live kill-test scorecard of the recent round's computable predictions vs current data, each tagged with the decisive experiment. [C] PMNS angles $\sin^2\theta_{12}$=0.30675 ($+0.31\sigma$), $\sin^2\theta_{13}$=0.02311 ($+1.45\sigma$), θ_{23} maximal; [C] leptonic CP J_{PMNS}=−0.0297, δ=240° vs $\sim 232(30)$; [X] Σm_ν NO floor 0.0586 eV (DESI/CMB-S4 decisive); [X] proton decay τ_p=1.55×10^{35} yr for $+(15,1,1)$ (Hyper-K decisive); [E] closed readouts α^{-1}, Ω_b, β; all rows consistent $< 2\sigma$, two sharp near-term kill tests. A consolidation, not new physics.
`v284_route_i_rp_uniqueness.py`	`QGEO.ROUTEI.01` — Route (i) (RP-state uniqueness) decomposed into a 6-lemma chain with every dischargeable lemma discharged and the one irreducible lemma isolated (does *not* prove the open premise). [E] 5/6 lemmas are existing [E]/[O] results — L1 RP→DtN (v194), L2 the four marks force the pillowcase class (v195/v216/v214), L3 the flat orbifold Steklov operator $= \sqrt{-\Delta_{\text{flat}}}$ (verified: flat-strip $\text{DtN}(k)/k \to 1$, principal symbol $\lvert\xi\rvert$ with no curvature term), L4 covariance $C = (1+e^H)^{-1}$ from the DtN (v258), L5 μ_4-invariance (v276/Lean/v280); [C] Troyanov reduction: 4 cone points of angle $\pi \Rightarrow \chi_{\text{orb}} = 0 \Rightarrow$ the unique constant-curvature representative is flat (τ=i), so L_{open} reduces to 'the raw seam is the constant-curvature geometric state'; [O] the one open lemma — the raw RP seam state is the constant-curvature/geometric state (the state-identification step).
`v285_route_ii_seam_condensation.py`	`QGAMB.ROUTEII.01` — Route (ii) (seam-net condensation) decomposed into a 4-lemma chain, with the key result that its open lemma *coincides* with Route (i)'s (the v282 unification at the lemma level). [E] 3/4 lemmas discharged — M1 no abelian sector ⇔ $\lvert\det K\rvert$=1 ⇔ holomorphic (v234/v235), M2 holomorphic c=8 ⇔ $(E_8)_1$ with the $SO(16)$ det-4 counterexample (v277/v281), M3 the condensation tower 16→4→1 = the unique μ_4-Lagrangian glue (v92/v281/v235); [E] the discriminator det 1 vs det 4; [E] Routes (i) and (ii) share *one* open lemma — both say 'the raw seam is the $(E_8)_1$-at-τ=i object' (v282), so proving either closes both; [O] the unified open lemma: construct a canonical equivalence between the raw seam KMS/DtN state and the holomorphic $(E_8)_1$ net at τ=i. M_{open} = `QGEO.SYM.01`.
`v286_seam_equivalence_contract.py`	`SEAM.EQUIV.01` — the Seam Equivalence contract: concentrates the whole open structure on *one* named arrow (does *not* prove it). Theorem: the raw RP seam state reconstructs canonically the holomorphic $(E_8)_1$ boundary net at the τ=i pillowcase. [E] four objects (RawRPSeam, FlatTauIPillowcaseDtN, HolomorphicE8Net, SREKitaevE8Phase); [E] six arrows — 2 closed (Route i v276/v280/v284, Route ii v281/v277/v235), 1 conditional (v282), 2 open premises (the same lemma, v284/v285), 1 Gral; [E] **import firewall**: no Route-i module imports any Route-ii module or vice versa, so the routes converge only through `SEAM.EQUIV.01` (kills the 'E_8 into the geometry and back' circularity); [E] exact-label taxonomy (Formal/Numerical/Conditional Exact, Open Selection); [O] the one open arrow 'free Gaussian RP bulk ⇒ holomorphic single-sector boundary'.
`v287_free_rp_bulk_to_holomorphic_boundary.py`	`SEAM.EQUIV.A01` — Route A (AQFT) theorem-dependency checker for the open Gral arrow 'free Gaussian RP bulk ⇒ holomorphic single-sector boundary net': types the 5-lemma chain and isolates the *one* missing standard theorem. [E] L1 RawRPSeam→OS→CAR/quasi-free boundary (v155/v160/v240/v175); [E] L2 gap+chirality ⇒ gapped chiral bulk (v64/v156); [O] L3 invertible/SRE Gaussian bulk ⇒ single-sector boundary (μ-index 1) — the one missing standard theorem (the AQFT form of $\det K$=1); [E] L4 μ=1+c=8 ⇒ holomorphic (v234); [E] L5 holomorphic c=8 ⇒ $(E_8)_1$ (v83/v277/v281). Verdict: 4/5 discharged, Route A reduces `SEAM.EQUIV.01` to one cited theorem.

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Script	**What it machine-checks** *(continued)*
`v288_full_l2_subprincipal_z4.py`	`SEAM.EQUIV.B01` — Route B (DtN): the full-L^2 lift of the sub-principal $\mathbb{Z}_4$ block-diagonality, proving (numerically, on the full Toeplitz operator) the 'probably provable' step and lifting v201/v284 from the finite block to L^2. **[E]** character orthogonality: a mark-sourced curvature $f = \sum_{j<4} g(\theta - j\pi/2)$ has Fourier support only on $m \equiv 0 \pmod 4$; **[E]** full-L^2 block-diagonality: $\Lambda = \lvert n\rvert + M_f$ commutes with $\rho = \mathrm{diag}(i^n)$, $\lVert[\rho, \Lambda]\rVert \sim 2{\times}10^{-15}$ on $-50..50$; **[E]** negative control: a generic curvature gives $\lVert[\rho, \Lambda]\rVert = 4.44 > 0$; **[E]** Lean consistency (`geom_sum_fourth_root` + `markLocal_blockDiagonal`); **[O]** the sharper residual — why is the raw seam sub-principal term mark-local?
`v289_marklocal_raw.py`	`MARKLOCAL.RAW.01` — decompose the one remaining Route-B residual ('why is the raw seam sub-principal term mark-local?') into a 5-lemma chain and isolate the single open analytic lemma. **[C]** L1 conic parametrix: conic Steklov DtN $= \lvert D_\theta\rvert + M_f +$ cone sources (b-calculus/Melrose, imported); **[O]** L2 **Flat-Away** — the one open analytic lemma: RP+gap+4 marks ⇒ curvature vanishes away from the marks (the flat-pillowcase realisation); **[E]** L3 orbit-source: the 4 marks form one μ_4 orbit (v195/v216); **[E]** L4 Fourier-support: μ_4-orbit curvature ⇒ support only on $4\mathbb{Z}$ (v264/v288, with a non-orbit negative control); **[E]** L5 full-operator: ⇒ $[\rho, \Lambda]{=}0$ on full L^2 (v288). 4/5 discharged; Route B reduces to exactly L2.
`v290_z4_smooth_curvature_adversary.py`	`SEAM.ADVERSARY.01` — the Route-B red-team that $\mathbb{Z}_4$ block-diagonality is *not* mark-locality. The strongest adversary, a smooth $\mathbb{Z}_4$-symmetric off-mark curvature $f = \varepsilon\cos(4\theta)$: **[E]** it is $\mathbb{Z}_4$-symmetric (Fourier support $\{\pm 4\} \subset 4\mathbb{Z}$) but *not* mark-local ($f(\pi/4) = -\varepsilon \neq 0$); **[E]** it *passes* the v288 commutator $\lVert[\rho, \Lambda]\rVert = 0$ (same as mark-local) — so $[\rho, \Lambda]{=}0$ is necessary but *not* sufficient; **[E]** but it shifts the DtN spectrum off the flat ladder ($\max\lvert\Delta\lambda\rvert = 0.155$) and the heat trace ($\Delta = 0.019$), so the KMS weight and $(E_8)_1$ character shift; **[O]** a genuine candidate modulus the commutator misses but the spectral data excludes — Flat-Away must use spectral/energy/heat input.
`v291_flataway_contract.py`	`FLATAWAY.RP.01` — the Flat-Away lemma as its own named mini-theorem with three proof routes opened by v290. Lemma **[O]**: given RawRPSeam with gap, chirality and the four μ_4 marks, the smooth curvature density in the DtN sub-principal symbol vanishes away from the branch divisor. **[E]** *spectral-rigidity*: scanning $f = \varepsilon\cos(4\theta)$, the spectrum deviation from the flat ladder is 0 only at $\varepsilon{=}0$ — flat is the unique spectral match; **[E]** *heat-kernel*: the heat-trace deviation is 0 only at $\varepsilon{=}0$; **[C]** *RP-energy*: RP+gap is a Dirichlet-energy minimisation in the Troyanov conformal class ⇒ constant-curvature minimiser (Troyanov; cf. v284). **[O]** three independent routes to one lemma; closing any one closes Route B.
`v292_flataway_heat_reduction.py`	`FLATAWAY.HEAT.01` — the Heat-Kernel route made precise: reduces Flat-Away to *one* named spectral invariant. **[E]** the heat-trace deviation $\Delta\mathrm{Tr}(e^{-t\Lambda}) = c(t)\varepsilon^2 + O(\varepsilon^4)$ is a positive-definite quadratic form in the off-mark curvature ($c_k(t) > 0$ for every $\mathbb{Z}_4$ mode $4k$; positive on all combinations), so $\Delta\mathrm{Tr} = 0 \Leftrightarrow f = 0$; leading invariant $\lVert f\rVert_{L^2}$ (Parseval). **[O]** reduction: Flat-Away ⇐ 'the seam fixes the a_2 heat coefficient to its flat value' ⇒ $\int f^2 = 0 \Rightarrow f = 0$ off the marks — one external invariant.
`v293_flataway_spectral_hessian.py`	`FLATAWAY.SPEC.01` — the Spectral-Rigidity route upgraded from one direction (v291) to the *full* smooth-$\mathbb{Z}_4$ space. **[E]** splitting mechanism: the degenerate flat Steklov level $\lvert n\rvert$ is split by mode $4k$ at $\lvert n\rvert{=}2k$ by $\pm\varepsilon/2$; **[E]** $S = \sum(\Delta\lambda)^2 \sim 0.50\varepsilon^2$; **[E]** the Hessian over modes $\{4, 8, 12\}$ has eigenvalues $\{0.497, 0.500, 0.503\} > 0$, so the flat seam is a *strict* isolated minimum over the whole $\mathbb{Z}_4$ space; **[E]** a non-$\mathbb{Z}_4$ mode breaks the commutator outright. **[O]** pinning the seam's Steklov spectrum excludes every smooth $\mathbb{Z}_4$ deformation ⇒ Flat-Away.

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Script	What it machine-checks *(continued)*
`v294_flataway_rp_energy.py`	`FLATAWAY.ENERGY.01` — the RP-energy / Troyanov route: the flat pillowcase is the unique constant-curvature minimiser at the fixed mark divisor. [E] prescribed divisor + Gauss-Bonnet: S^2 with 4 cone angles π, $\sum(2\pi-\theta)=4\pi=2\pi\chi$; [C] Troyanov/Euclidean-cone uniqueness (Troyanov 1991) excludes smooth off-mark curvature; [E] strictly convex curvature energy $E[f]=\int f^2$, Hessian $2\pi I$ (PD) $\Rightarrow$ unique minimiser; [E] no zero-cost deformation. [O] premise: 'RP+gap realises the energy-minimiser', converging with v292/v293 on one selection principle; closing any one closes Route B.
`v295_flataway_a2_exact.py`	`FLATAWAY.A2.01` — the *exact* analytic a_2 coefficient, upgrading the v292 positive-definiteness from numerical to a proof. [E] first variation vanishes: $\partial_\varepsilon \mathrm{Tr}\, e^{-t\Lambda(\varepsilon)}\|_0=-t\hat{f}(0)\sum e^{-t\|n\|}=0$ (zero-mean off-mark curvature, Gauss-Bonnet); [E] convexity $\Rightarrow$ global minimum: $\phi(x)=e^{-tx}$ strictly convex $\Rightarrow \Lambda\mapsto \mathrm{Tr}\, e^{-t\Lambda}$ convex (Klein/Peierls), so $\varepsilon{=}0$ is the global minimum and $\Delta\mathrm{Tr}\geq 0$ for all ε, $=0$ iff $f=0$; [E] exact a_2 kernel via the divided difference of e^{-tx} (Duhamel), matched to numerics to rel.err $<10^{-6}$; [E] degenerate-split closed form $2e^{-2kt}(\cosh(tg/2){-}1)\geq 0$; [O] reduction now analytic: Flat-Away $\Leftarrow$ 'the seam fixes a_2'. Lean-mirrored (`FORM.FLATAWAY.01`).
`v296_flataway_a2_closed.py`	`FLATAWAY.A2.CLOSED.01` — the exact a_2 coefficient in *closed form*. [E] the operator tails telescope geometrically ($C_k+D_ke^{-4kt}=4kt(1-e^{-4kt})$), giving $W_k^{\mathrm{tail}}=t(1-e^{-4kt})/(8k(1-e^{-t}))$; [E] a finite $(4k{-}1)$-term middle (incl. the degenerate diagonal $\Phi(2k,2k)=(t^2/2)e^{-2kt}$); [E] closed form $W_k(t)=\frac{t(1-e^{-4kt})}{8k(1-e^{-t})}+\frac{1}{2}\sum_{m=1}^{4k-1}\Phi(m,4k{-}m)$ ($W_1(t)=t(2te^t+e^{3t}+3e^{2t}+e^t-1)e^{-3t}/8$), reproducing v295 to machine precision; [E] manifestly positive; [E] small-t: $W_k=t/2+O(t^3)$ (mode-independent leading $1/2$, the t^2 term cancels).
`v297_route_a_literature_stack.py`	`SEAM.EQUIV.A.LIT.01` — Route A's 'one standard import' as a precise citable theorem stack. [E] LIT-A invertible/SRE bulk $\Rightarrow$ trivial bulk topological order (Kitaev 2006; Freed–Hopkins 2021); [E] LIT-B bulk MTC = Drinfeld center, trivial center $\Rightarrow$ holomorphic boundary net (Müger 2003; Kawahigashi–Longo–Müger 2001); [E] LIT-C holomorphic $c{=}8$ net $\Rightarrow (E_8)_1$ (Conway–Sloane; Dong–Xu/Kawahigashi–Longo). [E] the three compose into 'invertible bulk $\Rightarrow (E_8)_1$', all legs established; [O] the one open hypothesis is the input of LIT-A, 'the raw quasi-free seam bulk is invertible (SRE)' = Route B's Flat-Away. [E] non-circular with Route B (the v286 firewall holds).
`v298_e8_network_attractor.py`	*E8 as a network attractor* — a computational-substrate (Wolfram-program) reading of the $(2,3,5)$/McKay hull, on the v219 bedrock; an honest audit, not a derivation. [E] the E8 Kac marks $(1,2,3,4,5,6,4,2,3)$ are the *unique* attractor of the minimal lazy graph update $m\leftarrow(A+2I)/4\cdot m$ on the affine-$E_8/(2,3,5)$ McKay graph (converges from any positive start; $A\,m=2m$, top eigenvalue 2). [E] the spherical McKay tower terminates at E_8 ($2T\to E_6$, $2O\to E_7$, $2I\to E_8$, $h=30$); [E] the cascade is nested-subgraph coarse-graining $(240\to126\to72\to40\to20)$. [C] Wolfram reading: E_8 as the universal attractor of a minimal $(2,3,5)$ network (Coxeter-skeleton level). [O] a minimal hypergraph *rewrite* whose attractor is the full TFPT structure stays open; $\varphi_0=(4/3)c_3=1/(6\pi)$ is the analytic seed, not an edge fraction.

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Script	**What it machine-checks** *(continued)*
`v299_hypergraph_0d_autonomous.py`	The hypergraph/Wolfram bridge (hardened): the 3-arm $(2,3,\cdot)$ branch type is the irreducible seed on which a local witness-based spectral growth *reaches* E_8 as the last finite point ($E_6 \to E_7 \to E_8$, then stops at the affine boundary $\hat{E}_8$) — *not* 'P2 derived'. **[E]** icosahedron = 3-uniform hypergraph ($\lvert\mathrm{Aut}\rvert = 120 = \lvert I_h\rvert$; the McKay group is the $SU(2)$ binary lift $2I$, a different order-120 group); **[E]** the $1 \to 4$ subdivision rewrite is equivariant (120 symmetries at every scale); **[E]** the $2I$ fusion tensor is a weighted 3-uniform hypergraph whose 2-dim slice is the affine E_8 McKay graph; **[E]** Smith: $\rho \le 2$ selects (affine) ADE; **[E]** Collatz–Wielandt: $\rho \le 2 \Leftrightarrow$ a positive witness m with $Am \le 2m$ (local); **[E]** the autonomous rule (diffusion witness + local gate $Am \le 2m$) runs $E_6 \to E_7 \to E_8 \to \hat{E}_8$ under the $(2,3,\cdot)$ seed; **[E]** A_n ($\rho = 2\cos\frac{\pi}{n+1}$) and D_n ($\rho = 2\cos\frac{\pi}{2n-2}$) controls are < 2 for all n, never reaching E_8; **[O]** the seed = P2. A second network reading of the $(2,3,5)/E_8$ core (v219/v298).
`v300_flataway_rigid.py`	`FLATAWAY.RIGID.01` — the attack on Flat-Away. **[E]** the flat fingerprint: $\mathrm{spec}(\lvert D_\theta\rvert)$ is the integer ladder, every level $n \ge 1$ of multiplicity exactly 2 (a discrete fingerprint, not a norm); **[E]** the resonant mode $4k$ splits level $2k$ by gap $= g_k$ (sympy 2×2 $[[2k, g/2], [g/2, 2k]] \Rightarrow 2k \pm g/2$), validated vs the full Toeplitz operator; **[E]** any $g_k \ne 0$ lifts the resonant degeneracy (mult $2 \to 1{+}1$) and changes the spectral multiset, so preserving the flat ladder *with* its mult-2 degeneracies forces every $g_k = 0 \Rightarrow f = 0$ — a discrete obstruction, strictly stronger than the v291/v293 'deviation grows' scan; **[E]** with the v295 convexity, flat is isolated analytically *and* discretely; **[C]** the pin derived: the $(E_8)_1$ character $E_4/\eta^8 = q^{-1/3}(1 + 248q + \dots)$ has integer q-coefficients (248 at level 1; E_8 even unimodular $\Rightarrow$ integer weights), and by 2d Steklov conformal rigidity the integer ladder $\Leftrightarrow$ flat seam, so $(E_8)_1$ rationality $\Rightarrow f = 0$ (conditional on Route A); **[O]** Flat-Away carries *zero* new open content beyond Route A — the two `SEAM.EQUIV.01` routes converge on one rationality fact.
`v301_route_a_invertible.py`	`SEAM.EQUIV.A.INV.01` — discharge Route A's last open hypothesis ('the raw quasi-free seam bulk is invertible/SRE') via the free-fermion classification; since v300 collapsed Route B into Route A, this is the last unconditional gap of `SEAM.EQUIV.01`. **[E]** carrier = 16 Majoranas, $c = 8$ ($g_{\mathrm{car}}{+}N_{\mathrm{fam}} = 5{+}3 = 8 = 16/2$, v148); **[E]** $SO(16)_1 \supset (E_8)_1$: $\dim \mathfrak{so}(16) = 120$ currents $+\,128$ spinor $= 248 = \dim E_8$ (v154); **[C]** free-fermion invertibility (Kitaev): a gapped Gaussian/quasi-free 2+1D fermion phase (class D) is $\mathbb{Z}$-classified by the chiral Majorana number and is *always* invertible (no intrinsic anyons; topological order needs interactions), so a gapped quasi-free bulk is invertible automatically; **[E]** the invariant $\lvert\det K\rvert = \#\mathrm{anyons}$ is 1 for E_8 (invertible) vs 4 for the gauged $D_8{=}SO(16)$ (anyons), which free fermions cannot produce; **[O]** LIT-A's 'invertible bulk' now follows from 'gapped 16-Majorana quasi-free bulk' (v148/v160+the gap), so the residual changes from a topological-order statement to the spectral input 'is the quasi-free bulk gapped?'; **[O]** with v300 the whole open content of `SEAM.EQUIV.01` reduces to the single spectral statement 'the quasi-free seam bulk is gapped'.

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Script	What it machine-checks *(continued)*
`v302_seam_gap.py`	`SEAM.EQUIV.GAP.01` — the last lever: after v300/v301 the whole open content of `SEAM.EQUIV.01` was 'the quasi-free seam bulk is gapped'; that gap is *not* a free input but the established Recovery gap of the seam transfer operator. **[E]** $\mathrm{spec}(T) = \{1, (2/3)^6, (1/3)^6\}$ (v160/v162), a simple leading Perron eigenvalue 1; **[E]** the gap is derived: $\Delta = -\ln((2/3)^6) = 6\ln(3/2) = 2N_{\mathrm{fam}}\ln(3/2) \approx 2.4328 > 0$ (multiplicative gap $1-(2/3)^6 = 665/729$; $6 = 2N_{\mathrm{fam}}$, $(2/3)^6$ the μ_4-deck transfer eigenvalue forced by the carrier); **[E]** the positivity margin $\Delta - 31/(4\pi^2) = 1.6476 > 0$ (the v76 decoupling margin); [C] by Osterwalder–Schrader / quasi-free clustering a transfer gap $\Delta > 0$ is a mass gap of the reconstructed Hamiltonian; **[O]** so the last TFPT-internal input is the derived Recovery gap $6\ln(3/2) > 0$, and `SEAM.EQUIV.01`'s residual is a composition of cited theorems (OS/clustering + Kitaev v301 + the v297 AQFT stack) over established TFPT facts — no undischarged TFPT-internal assumption remains, though it stays **[O]** (not machine-proved end-to-end).
`v303_ftransfer_dynamics.py`	`FR.DYNAMICS.01` — the discrete→dynamic lens on `F_transfer`: the main-branch mechanism (a gapped, positivity-preserving relaxation to a *unique* attractor; the local-averaging update → the E_8 marks, gap $6\ln(3/2)$) is the shared *shape* of all four instances. [E] F_pole (Koide) *is* the seam dynamics — the source→pole/branch relaxation is the Möbius map with fixed-point multiplier $F' = (2/3)^6 = \lambda_2$, the seam-clock subleading eigenvalue (v82/v54); $\|F'\| < 1 \Rightarrow$ a unique attractor at the seam rate; [C] F_Boltzmann (η_B) is a gapped positive contraction (washout $\kappa_f \in (0, 1)$, H-theorem; compiler fixes $A_\Lambda = 10$, rate thermal); [C] F_relic (axion) freezes at an adiabatic fixed point (seed $\theta_i = 3\pi/5$, rate cosmological); [E] F_QCD (m_p/m_e) is the 1-loop RG flow with the carrier $b_3 = -7$ (asymptotic freedom ⇒ the Gaussian UV attractor, Λ_{QCD} generated). **[O]** Verdict: one shape underlies all four = the main-branch principle; F_pole has the seam rate exactly, the other three share only the shape (external rates), so `F_transfer` is the readout end of the one principle, not a bolt-on — simplicity is preserved by honest fencing (v187).
`v304_idg_nonlocal_ghost.py`	`QGAMB.IDG.01` — the KMS spectral-action cutoff suggests a *nonlocal* (infinite-derivative) graviton form factor; under entire analyticity the Stelle ghost is a *truncation artefact* — a [C] narrowing of the red-team `rt_F` perturbative-gravity attack, *not* a closure of `QG.AMB.01` (no new parameter: the cutoff is the v259 seam KMS weight). [E] the local four-derivative propagator $1/(p^2(p^2+M^2)) = (1/M^2)[1/p^2 - 1/(p^2+M^2)]$ has a healthy massless graviton (residue $+1/M^2$ at p^2=0) and a massive spin-2 mode with *negative* residue $-1/M^2$ at $p^2{=}-M^2$ (the Stelle ghost, reproducing v278); [E] for an IDG operator $p^2 a(p^2)$ with $a(z) = e^{z/M^2}$ entire and nowhere zero, the dressed propagator $1/(p^2 a)$ keeps its *only* pole at p^2=0 — no ghost (Tomboulis 1997; Biswas–Mazumdar–Siegel 2006); [E] negative control: a polynomial truncation $a(z) = 1 + z/M^2$ (the $R{+}R^2$ order) has a real zero at $z{=}-M^2$ = the ghost pole, so 'entire vs polynomial' is the discriminator; [C] the cutoff is the β=1 seam KMS weight $f(u) = e^{-u}$ (v259), not a choice — truncating $\mathrm{Tr}\, e^{-D^2/\Lambda^2}$ at a_4 is what makes the local $R{+}R^2$ and the ghost; [C] **conditional narrowing**: *if* the resummed form factor is entire (Biswas–Mazumdar–Siegel class) *then* the perturbative graviton is ghost-free and `rt_F`'s attack is a truncation artefact; [O] **open**: (i) the trace regulator f is not the kinetic form factor $a(\Box)$ — entire analyticity of the resummation is unproven; (ii) a ghost-free *perturbative* graviton is not the nonperturbative ambient measure — `QGAMB.IDG.01` stays [C]/[O], not a `QG.AMB.01` closure.

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Script	What it machine-checks *(continued)*
`v305_witness_independence.py`	Witness-independence / generator-economy audit (structural anti-numerology firewall, companion to v100): every headline integer is a documented image of the single anchor a=(1, 1, 2) (e=(4, 5, 2), p_n=2+2^n; v23/v228), π the only transcendental primitive (`ARCH.CORE.01`) — 13 atoms / 2 free primitives = 6.5× compression made explicit (the recurring 2/3=$\lvert\mathbb{Z}_2\rvert/N_{\mathrm{fam}}$ and $(2/3)^6$ are *one* ratio); E_8/g_{car}/anchor are overdetermined (≥ 3 structurally independent witnesses), the 5 pure-seed readouts single-witness (correlated, scored jointly by v100); α^{-1}'s '137' is foreign to the anchor family = an independent witness **[E]**
`v306_seed_crossval.py`	Leave-one-out cross-validation of the shared seed φ_0: back-solving φ_0 independently from the five seed-grammar observables ($\sin^2\theta_{12}$, $\sin^2\theta_{13}$, β, Ω_b, λ_C) and error-weighted-averaging reproduces the axiom seed φ_0=1/(6π)+48c_3^4=0.0531720 to 0.04% (one real seed, not a per-observable fit); leave-one-out predicts every held-out observable within 3σ out of sample (most-tensioned = $\sin^2\theta_{13}$, ∼2σ); a data↔formula shuffle explodes the χ^2 by > 100× **[E]**
`v307_data_watchdog.py`	Data watchdog: the operational decision pipeline over the frozen registry (v84). Classifies all 20 entries (16 predictions + 2 assigned textures + 2 bands) PASS/WATCH/TENSION/KILL by live n_σ vs repo-documented current bests (CODATA 2022, NuFIT 6.0, ACT DR6, Planck 2018, PDG 2024, BK18); no decidable prediction in KILL (theory alive); watch items α^{-1} (1.9σ), s_{23} (1.8σ) and the most-tensioned core $\sin^2\theta_{13}$ (2.0σ); the prediction-of-record $\sin^2\theta_{12}$=0.306747 compatible (JUNO, the fastest falsifier); $r \in [0.0033, 0.0048]$ below the BK18 limit and the kill 0.01; w=−1 flagged the dark-energy watchdog (DESI) **[E]**
`v308_seam_equiv_chain.py`	TOE attack 1 — the `SEAM.EQUIV.01` keystone as a composition certificate: the v297 stack as one well-typed chain gap>0 → invertible → holomorphic c=8 → $(E_8)_1$ with exact discriminators $\lvert\det\mathrm{Cartan}(E_8)\rvert$=1 vs $\lvert\det\mathrm{Cartan}(D_8)\rvert$=4, carrier tower 4·4=16 → 4 → 1, c=g_{car}+N_{fam}=8; the input gap is the derived recovery gap 6 ln(3/2)>0. The only undischarged premise is the cited OS step — an assembly/reduction certificate (Lean target), NOT a closure **[E][O]**
`v309_modular_bedrock.py`	TOE attack 2 — the bedrock $\omega\circ\rho$=ω via an explicit modular Hamiltonian, non-circular: marks = 4 from Gauss–Bonnet (v216) + the order-4 clock from cross-ratio 2 (v214) ⇒ μ_4 derived, not assumed; the quasi-free modular Hamiltonian $K = \log((1-C)/C)$ round-trips to the seam operator and $[\rho, K]$=0 ⇔ $\omega\circ\rho$=ω (a $\mathbb{Z}_2$ profile breaks it). Residual = Bisognano–Wichmann intrinsicality of the raw collar; NOT a closure of `QGEO.SYM.01` **[E][O]**
`v310_carrier_sm_anomaly.py`	TOE attack 3 — the carrier half-spinor is one anomaly-free SM generation with the SM RG: 16=$\Lambda^{\mathrm{even}}(\mathbb{C}^5)$=1+10+5 → $SU(5) \subset SO(10)$ as 10+$\bar{5}$+1 = one generation; gauge-anomaly-free ($\sum Y$=0, $\sum Y^3$=0, $[SU(2)]^2U(1)$=$[SU(3)]^2U(1)$=0, exact) and reproduces (b_1, b_2, b_3)=$(\frac{41}{10}, -\frac{19}{6}, -7)$; a fourth family shifts b_1, so N_{fam}=3 pins the RG. Closes the spectrum/anomaly/β sub-question; the full EG S-matrix + Yukawa stay open **[E][C]**
`v311_gap_decoupled_measure.py`	TOE attack 4 — the physical QG measure as a gap-decoupled, exponentially-clustering object: the transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ gives geometric clustering at rate $(2/3)^6$, correlation length 1/(6 ln(3/2)), finite susceptibility 729/665; the v76 margin 6 ln(3/2) − 31/(4π^2)≈1.648>0 decouples the physical sector from the un-built ambient. Spectral-action a_2=EH/a_4=R^2/72 (v36) + KMS β=1 (v259); `QG.AMB.01` stays **[O]** (gap-decoupled, inert) **[E][O]**

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Script	What it machine-checks *(continued)*
`v312_hypergraph_substrate.py`	TOE attack 5 — the hypergraph substrate sharpened: the affine-E_8 Kac marks are the Perron eigenvector ($A\,m{=}2m$) and the subleading eigenvalue is exactly $2\cos(\pi/5)=$ the golden ratio (the 5-fold/icosahedral signature), but the recovery rate $(2/3)^6$ is *not* any eigenvalue and φ_0 is not a rational edge fraction; so a full-structure rewrite must inject exactly two non-graph-spectral data — the cusp weight $2/3{=}\lvert\mathbb{Z}_2\rvert/N_{\mathrm{fam}}$ (gap $6\ln(3/2)$) and the analytic seed φ_0. The open question is precisely delimited, not closed **[E][O]**
`v313_golden_atoms_spectral.py`	The golden ratio is the $g_{\mathrm{car}}{=}5$ signature: the $(2,3,5)$ atoms *are* the affine-E_8 network spectral angles (v312 follow-up, exact). **[E]** the charpoly factors $x(x^2{-}4)(x^2{-}1)(x^2{-}x{-}1)(x^2{+}x{-}1)$, the golden minimal polynomial $x^2{-}x{-}1$ divides it ($\phi{=}2\cos(\pi/5)$ an exact eigenvalue); the non-trivial eigenvalues are $2\cos(\pi/k)$ for $k\in\{2,3,5\}{=}\{\lvert\mathbb{Z}_2\rvert,N_{\mathrm{fam}},g_{\mathrm{car}}\}$ (giving $0,1,\phi,1/\phi$), so the golden ratio is the $g_{\mathrm{car}}{=}5$ (5-fold) signature — a new spectral witness for g_{car} (raises the v305 multiplicity). The $(2,3,5)$ spherical excess $1/2{+}1/3{+}1/5{=}31/30$ ties the v63/v76 margin numerator $31{=}2^{g_{\mathrm{car}}}{-}1{=}248/8{=}1{+}h$ to the Coxeter 30 (a unification, not new evidence). **[E]** honest null: ϕ is structural, not a readout. **[C]** direction: ϕ is the E_8-quasicrystal (Elser–Sloane) ratio, a candidate substrate **[E][C]**
`v314_rate_translation.py`	Do the discrete-kernel and dynamic rates translate? An exact number-field split. The affine-E_8 adjacency spectrum splits into the rational part $\{0,\pm1,\pm2\}$ in $\mathbb{Q}$ and the golden part $\{\pm\phi,\pm1/\phi\}$ in $\mathbb{Q}(\sqrt{5})$; the dynamic transfer spectrum $\{1,(2/3)^6,(1/3)^6\}$ is entirely rational. **[E]** the family rates DO translate (the rational angles $2\cos(\pi/2){=}0{=}\lvert\mathbb{Z}_2\rvert$, $2\cos(\pi/3){=}1{=}N_{\mathrm{fam}}$ build the recovery rate $(2/3)^6{=}(\lvert\mathbb{Z}_2\rvert/N_{\mathrm{fam}})^{2N_{\mathrm{fam}}}$ in $\mathbb{Q}$), but the carrier 5-fold (golden, $\mathbb{Q}(\sqrt{5})$) has *no* rational dynamic image — a field obstruction, so it is static-only. The shared object is the order-30 Coxeter element ($30{=}g_{\mathrm{car}}(2N_{\mathrm{fam}}){=}5{\cdot}6$); the ϕ quasicrystal carries the carrier, not the cusp-weight family dynamics **[E][C]**
`v315_coxeter_coupling.py`	Couple or factorize? The order-30 Coxeter sectors couple as a cyclotomic field (v314 follow-up). **[E]** GROUP FACTORIZES: $\gcd(5,6){=}1 \Rightarrow \mathbb{Z}/30=\mathbb{Z}/5\times\mathbb{Z}/6$ (direct product, group-decoupled); **[E]** EXPONENTS FACTOR: E8 exponents = totatives(30) = totatives(5)$\times$ totatives(6), $\phi(30){=}4{\cdot}2{=}8{=}$ rank E_8; **[E]** FIELD COUPLES: $[\mathbb{Q}(\zeta_{30}){:}\mathbb{Q}]{=}8{=}$ rank E_8, $\mathbb{Q}(\zeta_{30})$ is the compositum of the carrier $\mathbb{Q}(\zeta_5)$ ($\ni\sqrt{5}$) and the family $\mathbb{Q}(\zeta_3)$; **[E]** GALOIS $=\mu_4\times\mathbb{Z}_2$: $(\mathbb{Z}/30)^\times=(\mathbb{Z}/5)^\times\times(\mathbb{Z}/3)^\times=\mathbb{Z}/4\times\mathbb{Z}/2$ (order 8, max order 4), $\mu_4{=}(\mathbb{Z}/5)^\times$ the carrier Galois (the seam clock = the carrier golden-field Galois), $\mathbb{Z}_2{=}(\mathbb{Z}/3)^\times$ the family Galois; **[E]** $\sqrt{5}{=}\phi{+}1/\phi$ = the Gauss sum in $\mathbb{Q}(\zeta_5)$. **[C]** $30{=}5{\cdot}6$ does not dynamically entangle — it couples as the cyclotomic compositum, the bridge is Galois not dynamical **[E][C]**
`v316_galois_readout.py`	Does the Galois $\mu_4\times\mathbb{Z}_2$ (v315) organize the physical readouts? Yes — the CP phases. **[E]** the hexagonal CP unit $\rho{=}\zeta_6{=}\zeta_{30}^5$ is the order-6${=}2N_{\mathrm{fam}}$ FAMILY factor c^5 (the carrier is $\zeta_5{=}\zeta_{30}^6$); **[C]** $\delta_{\mathrm{CKM,lead}}{=}\pi/3{=}\arg\zeta_6$, $\delta_{\mathrm{PMNS}}{=}4\pi/3{=}\arg\zeta_6^4$ (power $4{=}\lvert\mu_4\rvert$), $\zeta_6^4{=}-\zeta_6$ the sheet flip ($\mu_6{=}\mu_3\times\mu_2$); **[E]** the Galois $\mathbb{Z}_2{=}\mathrm{Gal}(\mathbb{Q}(\zeta_6)/\mathbb{Q}){=}\{\zeta_6\mapsto\zeta_6^5{=}\bar{\zeta}_6\}$ IS CP conjugation ($\delta\mapsto-\delta$); **[E]** the carrier factor ζ_5 (golden $\sqrt{5}$) carries no phase (static); **[E]** the magnitudes ($\sin^2\theta_{12}{=}1/3{-}\varphi_0/2$, transcendental φ_0) are not cyclotomic. So the Galois bridge reaches phenomenology through the family factor (phases); the carrier is static; the magnitudes are the analytic seed **[E][C]**

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Script	What it machine-checks *(continued)*
`v317_galois_family.py`	Are the 3 generations a μ_3/Galois orbit? Yes (μ_3), refined 1+2 by Galois (v316 follow-up). **[E]** the 3 cusp weights $\{0, 1/3, 2/3\}$ (Spec(Q_+)=$\{1,2,3\}$=A_3 exponents) map under $w \mapsto e^{2\pi i w}$ to the cube roots $\{1, \zeta_3, \zeta_3^2\}$; the cyclic family symmetry $w \mapsto w+1/3$ is a transitive μ_3 orbit (the family IS the cube-root orbit). **[E]** the transfer eigenvalues $(1-w)^6$=$\{1, (2/3)^6, (1/3)^6\}$ attach to the weights (attractor = eigenvalue 1 at w=0). **[E]** the Galois $\mathbb{Z}_2$ ($\zeta_3 \mapsto \zeta_3^2$, = CP conjugation v316) fixes w=0 (rational) and swaps w=1/3 $\leftrightarrow$ 2/3 $\Rightarrow$ orbit 1+2; so the Galois-FIXED generation is the recovery ATTRACTOR (the law), the Galois PAIR the decaying hierarchy. **[C]** the family structure is arithmetic; the mass magnitudes stay the analytic seed **[E][C]**
`v318_arithmetic_capstone.py`	The capstone: the SM structural sector is $\mathbb{Q}(\zeta_{30})$+ Galois $\mu_4 \times \mathbb{Z}_2$, and the complete input is $\{a, \pi, v_{\text{geo}}\}$ (synthesis of the v313–v317 arc). **[E]** the structural sector (3 generations, 2 CP phases, orbit/hierarchy) lives in $\mathbb{Q}(\zeta_{30})$: $[\mathbb{Q}(\zeta_{30}){:}\mathbb{Q}]$=$\varphi(30)$=8= rank E_8, 30=$\lvert\mathbb{Z}_2\rvert N_{\text{fam}} g_{\text{car}}$=$h(E_8)$, Galois $(\mathbb{Z}/30)^\times$=$\mu_4 \times \mathbb{Z}_2$; **[E]** rigidity: forced by the anchor atoms $\{2,3,5\}$=$e(a)$; **[E]** magnitude reduction: φ_0=$(4/3)c_3$+$48c_3^4$ is a pure function of π (no free parameters), so the magnitudes are $\{a, \pi\}$-determined, not free; **[E]** free inventory: 0 dimensionless free parameters, π the unique transcendental, v_{geo} the one dimensionful unit. **[O]** residual: the raw-seam realisation stays open (`QGEO.SYM.01/SEAM.EQUIV.01`) **[E][O]**
`v319_translation_clock.py`	The translation clock: the discrete$\leftrightarrow$dynamic bridge *is* the order-30 Coxeter element, a clock with two coprime hands 5×6, read law-inclusive (0..5) or live-only (1..5). **[E]** the only object holding both the static carrier ($\mathbb{Q}(\sqrt{5})$) and the dynamic family (rational) data is the order-30 Coxeter element (v314); $30 = h(E_8) = g_{\text{car}}(2N_{\text{fam}}) = 5 \cdot 6$, $\gcd(5,6) = 1 \Rightarrow \mathbb{Z}/30 = \mathbb{Z}/5 \times \mathbb{Z}/6$ (v315). **[E]** DYNAMIC HAND ($\mathbb{Z}/6 = 2N_{\text{fam}}$, ticks 0..5): recovery exponent exactly $6 = 2N_{\text{fam}}$, rate $(2/3)^6$; order-6 generator c^5 ($\zeta_6 = \zeta_{30}^5$). **[E]** LAW VS LIVE: the resummed clock rate$(n) = -p_2 \ln(1 - n/N_{\text{fam}})$ (v124) has rate$(0) = 0$ (the attractor/law) and live modes from n=1; the E_8 live phases (totatives of 30) start at 1, so $0,1,2,3,4,5$ is law-inclusive, $1,2,3,4,5$ live-only. **[E]** STATIC HAND ($\mathbb{Z}/5 = g_{\text{car}}$, ticks 1..5): golden $\mathbb{Q}(\sqrt{5})$, field-obstructed (no rational dynamic image) — static-only. **[E]** one full turn = $\text{lcm}(5,6) = 30 = h(E_8)$ (rank $E_8 = 8 = \varphi(30)$, $\lvert\mu_4\rvert$=4 planes, v55); the "it is one clock" reading is **[C]**.
`v320_galois_cp_relation.py`	A NEW falsifiable prediction from the Galois structure: the two CP phases are locked, $\delta_{\text{PMNS}} = \delta_{\text{CKM}}^{\text{lead}} + \pi$. **[E]** both phases are powers of one hexagonal unit ρ=ζ_6 (v316): $\delta_{\text{CKM}}^{\text{lead}}$= arg ρ=$\pi/3$ (60°), δ_{PMNS}= arg ρ^4=$4\pi/3$ (240°), ρ^4= $-\rho$, so $\delta_{\text{PMNS}}$$-$$\delta_{\text{CKM}}^{\text{lead}}$=$\pi$ exactly (the $\mathbb{Z}_2$ sheet flip) — the quark and lepton leading CP phases are NOT independent. **[C]** prediction of record $\delta_{\text{PMNS}}^{\text{lead}}$=240° (upgrades the assigned texture to a Galois-forced relation to the measured quark phase); **[C]** currently within the broad NuFIT 6.0 range; **[X]** kill test: δ_{PMNS} robustly away from 240° ($> 3\sigma$; DUNE/Hyper-K/JUNO) falsifies the Galois-CP organization **[E][C][X]**
`v321_killtest_board.py`	The forward kill-test board: the decisive upcoming measurements, ranked (forward companion to v307). **[E]** each entry is locked to a `freeze_file.csv` kill row: #1 JUNO $\sin^2\theta_{12}$=0.306747 (fastest), #2 CMB-S4/LiteBIRD r (kill r>0.01), #3 DESI/Euclid w=-1, #4 DUNE/Hyper-K δ_{PMNS}=240° (the new v320 Galois-CP relation), #5 DUNE/LEGEND/nEXO ordering+$m_{\beta\beta}$, #6 PSI n2EDM, #7 JUNO $\sin^2\theta_{13}$ ($\sim 2\sigma$ now), #8 LiteBIRD β. A forward decision ledger complementing v307; no new physics **[E]**

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Script	What it machine-checks *(continued)*
`v322_cp_lock_sharpened.py`	The v320 CP-lock prediction sharpened into a quantitative kill test. **[E]** the leading CP phase sits on the six hexagonal nodes $\arg\rho^k{=}k\cdot 60°$; $\delta^{\rm lead}_{\rm CKM}{=}60°$ (node 1), $\delta^{\rm lead}_{\rm PMNS}{=}240°$ (node 4). **[E]** the node selector is the deck order: $\delta^{\rm lead}_{\rm PMNS}{=}\lvert\mu_4\rvert\,\delta^{\rm lead}_{\rm CKM}{=}4(\pi/3){=}4\pi/3{=}\delta^{\rm lead}_{\rm CKM}{+}\pi$ $(\rho^4{=}-\rho)$. **[C]** sub-leading budget: $\delta^{\rm full}_{\rm CKM}{=}\pi/3{+}3\lambda^2$ $(\sim 68.7°)$; the lepton sub-leading is a separate open holonomy term bounded by $\sim 3\lambda^2\approx 8.7° \Rightarrow$ band $240\pm\sim 9°$. **[C]** current tension: vs NuFIT 6.0 NO ($\delta_{CP}{=}212^{+26}_{-41}°$) the leading 240° sits at $+1.08\sigma$, band edge $\sim 231°$ at $+0.74\sigma$. **[X]** kill: the nearest wrong node is 60° away $\gg$ budget; DUNE/Hyper-K ($\sim$ 5–15°) discriminate, $>3\sigma$ outside kills the lock. **[E]** an anarchic δ_{CP} hits the band with $p\sim 5\%$ and node 4 1-in-6 **[E][C][X]**
`v323_bw_geometric_modular.py`	The keystone pushed one step: Bisognano–Wichmann makes the seam modular flow *geometric*, so the μ_4 deck invariance is not an independent axiom. **[E]** the clock $\rho{=}\,\mathrm{diag}(i^n){=}\exp(i(\pi/2)L)$ is a geometric rotation ($L{=}\,\mathrm{diag}(n)$, $\rho^4{=}I$, $\mu_4\subset U(1)_{\rm rot}$). **[E]** for a rotation-covariant $C{=}f(L)$ the modular Hamiltonian $K{=}\log((1-C)/C){=}g(L)$, so $[K,L]{=}0$ (the modular flow commutes with all rotations) and $[\rho,K]{=}[\rho,C]{=}0$ — the clock is a modular symmetry for free. **[E]** discriminator vs v309: a period-4 curvature preserves the clock but its modular flow is *not* geometric ($[K,L]\neq 0$); BW is strictly stronger. **[C]** linked bedrock: given the seam is the $(E_8)_1$ chiral net (v308), BW/Hislop–Longo make the vacuum modular flow geometric $\Rightarrow$ `QGEO.SYM.01` ($\omega\circ\rho{=}\omega$) is downstream of `SEAM.EQUIV.01` + a rotation-invariant vacuum. **[O]** residual: the raw collar vacuum is rotation-invariant (cleaner than v309, shared with v308) **[E][C][O]**
`v324_hypergraph_fiber.py`	The minimal hypergraph substrate that works: a $(2,3,5)$-network *fibered* by a 3-fold cusp channel reproduces both the E_8 skeleton and the recovery gap (the constructive v312 follow-up). **[E]** network factor: $T_{\rm net}{=}(A{+}2I)/4$ fixes the Kac marks (eigenvalue $1{=}E_8$ skeleton), subleading $(\varphi{+}2)/4$ (golden); $(2/3)^6$ is *not* in its spectrum (the v312 negative). **[E]** cusp fiber: $T_{\rm cusp}{=}\,\mathrm{diag}((1{-}w)^{2N_{\rm fam}})$, $w\in\{0,1/3,2/3\}$ ($N_{\rm fam}{=}3$, v317), spectrum $\{1,(2/3)^6,(1/3)^6\}$. **[E]** the fibered substrate $T{=}T_{\rm net}\otimes T_{\rm cusp}$ has top eigenvalue 1 (eigenvector marks $\otimes$ $(w{=}0)$, skeleton $\times$ democratic cusp) *and* $(2/3)^6$ as a genuine eigenvalue (network attractor $\times$ cusp subleading). **[E]** honest injection: without the fiber ($T_{\rm net}\otimes I_3$) there is no $(2/3)^6$ — the cusp datum is injected, not graph-spectral; substrate = network $\times$ cusp = the v315 split $30{=}g_{\rm car}(2N_{\rm fam}){=}5\cdot 6$. **[E]** minimality: the smallest fiber is the 3-node cusp ($=N_{\rm fam}$) = order-30 (v319 clock). **[C]** a consistent fibered realization, not a derivation of 2/3 from a pure rewrite **[E][C]**
`v325_pillowcase_keystone.py`	The seam keystone as ONE citable theorem: the raw seam state *is* the flat $\tau{=}i$ pillowcase state, with **exactly one** open lemma. IF (raw RP-collar seam state = flat $\tau{=}i$ pillowcase / rotation-invariant state) THEN 4 square marks + μ_4 deck + $\omega\circ\rho{=}\omega$ + holomorphic $c{=}8\Rightarrow(E_8)_1$, with the carrier recovery gap as input. **[E]** pieces all exact: $n{=}2\chi(S^2){=}4$ (v216); cross-ratio $2\Rightarrow j{=}1728$ (v214/v267); flat orbifold $S^2(2,2,2,2)$, $\chi_{\rm orb}{=}0$ (Troyanov); DtN sub-principal at $m\equiv 0 \bmod 4$ (v201/v280); $\det\mathrm{Cartan}(E_8){=}1$ vs $D_8{=}4$ (v83/v143/v308); gap $6\ln\frac{3}{2}>0$ (v76/v302). **[O]** the one open lemma — the raw collar state IS the flat $\tau{=}i$ pillowcase / rotation-invariant state — is SHARED by both bedrock IDs (`QGEO.SYM.01`, `SEAM.EQUIV.01`; v323), machine-pinned in Lean (`FORM.SEAMEQUIV.01` + `FORM.QGEO.BW.01`); an assembly/reduction certificate (like v176/v261), not a closure **[E][O]**

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Script	**What it machine-checks *(continued)***
`v326_ftransfer_suite.py`	The F_{transfer} solver suite: the four transfer readouts as ONE runnable harness, each a gapped relaxation to a unique attractor (productises v303). **[E]** F_{pole} (Koide): Moebius contraction to $Q^*{=}2/3$ with multiplier $(2/3)^6$ = the seam-clock subleading eigenvalue, unique attractor at the SEAM rate. **[C]** $F_{\text{Boltzmann}}$ (η_B): washout contraction to the balance 6.1×10^{-11}, monotone (H-theorem), external thermal rate. **[C]** F_{relic} (axion): the comoving number $N{=}E/\omega$ is the adiabatic invariant, the relic freezes (seed $\theta_i{=}3\pi/5$), cosmological rate. **[E]** F_{QCD} (m_p/m_e): 1-loop RG with carrier $b_0{=}7$ flows to the Gaussian UV fixed point, Λ_{QCD} generated. **[C]** F_{RareKaon}: $\text{BR}(K^+){=}9.45\times10^{-11}$ on NA62 2016-2024 (v202). **[O]** only F_{pole} has the seam rate $(2/3)^6$; the other four share the SHAPE with external rates honestly fenced (v187/v303) — the frontier is measurable without claiming the external rates **[E][C][O]**
`v327_hypergraph_rewrite.py`	Deriving the cusp weight 2/3 from a minimal rewrite (sharpening v324/v312). A minimal local rule — a family channel with $N_{\text{fam}}{=}3$ slots, one absorbing attractor $(w{=}0)$ and $\lvert\mathbb{Z}_2\rvert{=}2$ surviving — supplies the cusp fiber. **[E]** M has spectrum $\{1,2/3,0\}$; $2/3{=}(N_{\text{fam}}{-}1)/N_{\text{fam}}{=}\lvert\mathbb{Z}_2\rvert/N_{\text{fam}}$ EMERGES from the rule arity, not injected. **[E]** over the order-$6{=}2N_{\text{fam}}$ hand the rate is $(2/3)^6{=}64/729$ = the recovery gap (v76/v324), derived. **[E]** NON-GRAPH-SPECTRAL (proof, sharpening v312): 2/3 is NOT a root of the affine-E_8 charpoly $(p(2/3){=}-3520/19683\neq0)$, so the recovery rate cannot be an adjacency eigenvalue. **[E]** the absorbing state (eigenvalue 1) is the democratic $w{=}0$ law (v317). **[O]** honest residual: the arity $\{2,3\}$ (the anchor atoms) is still the input (P2) — v324's injected datum reduced to the anchor arity **[E][O]**
`v328_theta13_pressure.py`	The θ_{13} pressure point: the honest weak spot, quantified and pre-registered. **[C]** $\sin^2\theta_{13}{=}\varphi_0^{\text{ret}}e^{-5/6}{=}0.023108$ vs NuFIT 6.0 NO (0.02195 ± 0.00058) is $+2.0\sigma$ — the most-tensioned CORE prediction (v62/v307). **[E]** θ_{13} is the carrier hypercharge trace (exponent $\gamma{=}\text{tr}_E Y^2{=}5/6$ EXACT, v268), NOT the seesaw $\mu\tau$ breaking that sets θ_{12} — distinct channels. **[C]** no cheap relief: the back-solved seed $\varphi_0^{\text{ret}}(\theta_{13}){=}0.0505$ is $\sim$5% below the axiom, PMNS θ_{13} RG running is $<1\%$ (NO), and 5/6 is exact — the prediction stays frozen (`REG.FREEZE.01`). **[X]** pre-registered kill: a stable $\sin^2\theta_{13}$ away from 0.023108 at $>3\sigma$ (JUNO/global) flags the carrier-trace/seed-grammar reading; exactly the v306 seed-hyperplane outlier — one honest pressure point **[C][E][X]**
`v329_os_gap_reduction.py`	Closing the OS step: the bulk gap *is* the transfer gap (second quantization), and assembling the single sufficient premise of the whole bedrock. **[E]** OS GAP = TRANSFER GAP: the many-body OS Hamiltonian $d\Gamma(-\log T)$ has spectrum = sums of single-mode energies, so the bulk gap = the smallest positive single-mode energy $= 6\ln\frac{3}{2}$, independent of system size — exactly "OS bulk gap = transfer gap" (the v308 open input), discharged at the many-body level. **[E]** $H\Rightarrow\omega\circ\rho{=}\omega$: under H the covariance is rotation-covariant $C{=}f(L)$, $K{=}g(L)$ commutes with $\rho{=}\exp(i(\pi/2)L)$ (v201/v309/v323). **[E]** $H\Rightarrow$ SRE $\Rightarrow(E_8)_1$: $\det\text{Cartan}(E_8){=}1$ vs $D_8{=}4$ (v237/v308). **[E]** both bedrock IDs follow from the ONE premise H (Lean-pinned `FORM.SEAMEQUIV.01` + `FORM.QGEO.BW.01`). **[O]** residual: the necessity of H on the raw collar; sharpens v240/v308/v323/v325, does NOT close the bedrock **[E][O]**

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Script	**What it machine-checks *(continued)***
v330_qgamb_admissible_measure.py	Constructing the ambient measure on the gap-decoupled admissible sector as a bona fide Osterwalder–Schrader object (with the honest fence). [E] reflection positivity: the seam transfer T (v221/v311; spectrum $\{1,(2/3)^6,(1/3)^6\}$) is symmetric PSD, so the finite-volume measures are RP. [E] exponential clustering: $C(n)/C(n-1) \to \lambda_2=(2/3)^6$ ($\xi=1/\Delta$, $\Delta=6\ln\frac{3}{2}$). [E] tightness $\Rightarrow$ projective limit: $\chi=\sum_n C(n)=729/665$ finite, so $\mathrm{Var}(S_\Lambda)/\Lambda \to \chi \Rightarrow$ uniform tightness (Prokhorov) $\Rightarrow$ the projective limit $\mu=\lim\mu_\Lambda$ EXISTS on the admissible sector. [E] OS reconstruction: $H_{\mathrm{OS}}=-\log T \geq 0$, unique vacuum, gap $6\ln\frac{3}{2}$ (v240). [E] NEG: an ungapped transfer sends $\chi\to\infty$, tightness fails. [O] the full ambient measure: the non-admissible (metric) sector is NOT built; by the gap margin $\Delta-31/(4\pi^2)\approx 1.648>0$ (v76/v275) the physical readouts do not need it, but `QG.AMB.01` stays open. A genuine PARTIAL construction, NOT a closure [E][O]
v331_necessity_of_H.py	The *necessity* of H reduced: rotation-invariance of the seam DtN is EQUIVALENT to the four marks at the square ($\tau=i$), so the one open lemma is exactly the order-4 CM selection (the necessity companion to v329's sufficiency). [E] square $\Rightarrow [\rho,\Lambda]=0$: marks at $\{0,\pi/2,\pi,3\pi/2\}$ make the curvature $\mathbb{Z}_4$-symmetric (Fourier support 0 mod 4), M_f couples only modes differing by $4\mathbb{Z}$ (the v201/v210 sufficiency as a DtN $\lvert k\rvert+M_f$ operator). [E] necessity: a mark off the square breaks $\mathbb{Z}_4$ and gives $[\rho,\Lambda]\neq 0$ — equivalence, not just implication. [E] order-4 CM: cross-ratio$(1,i,-1,-i)=2 \Rightarrow j=1728$, the unique 4-config with an order-4 automorphism (v214/v267); $H \Leftrightarrow$ square $\Leftrightarrow \mu_4$. [E] NEG: a $\mathbb{Z}_2$ mark set commutes with ρ^2 but not ρ. [O] residual: the dynamical *selection* of the square (the 4 marks forced by v216) is the irreducible postulate; does NOT close the bedrock [E][O]
v332_qgamb_metric_sector.py	The metric sector of `QG.AMB.01` characterized: the obstruction is exactly the conformal-mode (Gibbons–Hawking–Perry) problem, the physical sector is gap-insulated, and the only route is the conditional IDG taming (makes v330's metric-sector fence precise). [E] conformal-mode wrong sign: $c_{\mathrm{conf}}(D)=-(D-1)(D-2)/4$ is negative for $D\geq 3$ ($c_{\mathrm{conf}}(4)=-3/2$), TT positive — the conformal-factor problem. [E] measure obstruction: the kinetic form is indefinite (one negative eigenvalue), so e^{-S} diverges over the conformal mode; the bare metric-sector measure is not directly definable. [E] physical sector insulated: the obstruction is in the conformal/gauge direction, not the admissible sector (positive+gapped, v330), margin $\Delta-31/(4\pi^2)\approx 1.648>0$ (v76). [C] IDG conditional route: the entire form factor e^{z/M^2} (v259/v304) dresses the conformal propagator with no extra pole. [E] NEG: a polynomial ($R+R^2$) truncation crosses zero (the Stelle ghost, v304/v278). [O] metric sector open: the full `QG.AMB.01` needs the conformal mode resolved nonperturbatively (GHP rotation / IDG), gap-decoupled but NOT constructed. A precise characterization, NOT a closure [E][C][O]
v333_cm_selection.py	The CM-selection mechanism: $\tau=i$ (the square) is the UNIQUE order-4 modulus and the attractor of mark-equilibration, so the last keystone residual is “the deck acts geometrically”, not “why the square” (the v331 follow-up). [E] order-4 modular element: $S=\left(\begin{smallmatrix}0&-1\\1&0\end{smallmatrix}\right)$ in $\mathrm{SL}(2,\mathbb{Z})$ has $S^2=-I$, $S^4=I$ and fixes $\tau=i$ — the unique order-4 elliptic point of $\mathrm{PSL}(2,\mathbb{Z})$ (the other, ρ, is order 3). [E] deck order $\to$ CM point: order-4 $\to$ cross-ratio 2 $\to j=1728\to\tau=i$ (square); order-3 $\to j=0\to\tau=\rho$ (hexagon); $\lvert\mu_4\rvert=4$ selects $\tau=i$. [E] mark-equilibration attractor: 4 points minimizing the log-energy (Fekete) converge to equal spacing (gaps $\pi/2=$ the square), and beat random perturbations — the unique (mod rotation) attractor. [E] NEG: an order-3 deck selects ρ, not the square. [O] residual: only “the deck acts *geometrically* (= `QGEO.SYM.01`) vs an abstract $\mathbb{Z}_4$ on labels” stays open — sharpened from “why the square” to “the deck acts geometrically”, NOT a closure [E][O]

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Script	What it machine-checks *(continued)*
`v334_conformal_resolution.py`	The conformal-mode resolution: the Gibbons–Hawking–Perry contour rotation (IR sign) + the infinite-derivative (IDG) dressing (UV pole) give a CONDITIONALLY convergent metric-sector measure; the residual is the nonperturbative validity of the contour (the v332 follow-up). **[E]** the obstruction: $c_{\text{conf}}(4)\!=\!-3/2<0$, the unrotated conformal Gaussian diverges. **[E]** GHP rotation: $\phi \to i\phi$ flips the sign, $\int e^{-\lvert c\rvert\phi^2}\!=\!\sqrt{\pi/\lvert c\rvert}$ FINITE (GHP 1978). **[E]** IDG dressing: the entire form factor $e^{p^2/M^2}>0$ (no ghost pole, v304), the dressed per-mode action positive post-GHP and UV-suppressed. **[E]** NEG: a polynomial truncation crosses zero (the Stelle ghost, v304/v278). **[C]** conditional construction: GHP+IDG give a convergent metric-sector Gaussian mode-by-mode (conditional on the contour + entire analyticity). **[O]** residual: whether the GHP-rotated IDG-dressed measure is the CORRECT nonperturbative `QG.AMB` measure (the contour's nonperturbative validity); gap-decoupled (v76), blocks no test, NOT a closure **[E][C][O]**
`v335_seam_equiv_unify.py`	The keystone unification: there is ONE open theorem (`SEAM.EQUIV.01`), `QGEO.SYM.01` is its COROLLARY (a conformal-net axiom), and `QG.AMB.01` is a decoupled general QG problem, NOT a TFPT gate. **[E]** `QGEO.SYM.01` = corollary: a chiral conformal net has a Moebius-covariant vacuum (the unique invariant positive-energy vector, a NET AXIOM); rotations $U(1)<$ Moebius $\Rightarrow$ rotation-invariant vacuum; the μ_4 clock = rotation by $\pi/2$ fixes it $\Rightarrow \omega\circ\rho\!=\!\omega$. So `SEAM.EQUIV.01`$\Rightarrow$`QGEO.SYM.01` with NO extra premise (the v323 "rotation-invariant vacuum" is a net axiom); the two bedrock items collapse to ONE. **[E]** the one keystone theorem: `SEAM.EQUIV.01` = a construction theorem ($c\!=\!8$, $\det\mathrm{Cartan}(E_8)\!=\!1$, gap $6\ln\frac{3}{2}>0$ derived) with EXACTLY one open continuum lemma. **[E]** `QG.AMB.01` decoupled: margin $\Delta-31/(4\pi^2)\approx 1.648>0\Rightarrow$ no TFPT prediction depends on it; the general Euclidean-QG conformal-factor problem (GHP 1978), inherited not TFPT-specific. **[E]** collapsed status: structural items $2\to 1$; live residual $=v_{\text{geo}}+G_{\text{net}}(=$`SEAM.EQUIV.01`$)+F_{\text{transfer}}$. **[O]** the one residual = `SEAM.EQUIV.01`'s continuum OS reconstruction (Lean-pinned). A synthesis/inventory; does NOT close `SEAM.EQUIV.01` **[E][O]**
`v336_continuum_limit.py`	The ONE open lemma of `SEAM.EQUIV.01`, stated precisely and mapped to the recent rigorous literature – the continuum-limit and OS-reconstruction legs are now CITABLE. **[C]** continuum leg: Osborne–Stottmeister (*CFT from Lattice Fermions*, CMP **398** (2023) 219, doi:10.1007/s00220-022-04521-8; legacy internal token MMST, on the MMST/Morinelli–Morsella–Stottmeister–Tanimoto OAR framework, arXiv:2010.11121) give the chiral-CFT scaling limit of free lattice fermions (Koo–Saleur $\to$ Virasoro, correct c); WZW range rank $\le c\le D$. $(E_8)_1$: $c\!=\!8$, rank 8, $D\!=\!16$ Majoranas ($SO(16)_1\!\to\!(E_8)_1$), $8\le 8\le 16$ in range; the collar is a gapped quasi-free (CAR) state. **[C]** OS leg: Adamo–Moriwaki–Tanimoto (arXiv:2407.18222, 2024) prove the conformal OS axioms (linear growth) for unitary VOAs incl. lattice VOAs; $(E_8)_1$ is the even self-dual rank-8 lattice VOA. **[C]** construction + covariance (Adamo–Giorgetti–Tanimoto): arXiv:2506.01008 builds the 2d conformal net of an even lattice and classifies the 2d Heisenberg-net extensions as lattice nets, and arXiv:2508.07109 shows every DHR rep of such a net is automatically positive-energy + diffeomorphism-covariant (covariance a consequence, not an input). **[E]** target pinned: $c\!=\!g_{\text{car}}\!+\!N_{\text{fam}}\!=\!8$, $\det\mathrm{Cartan}(E_8)\!=\!1$, the $D\!=\!16$ Majorana $SO(16)_1$ IS the carrier-16 and the μ_4 extension IS the seam glue. **[O]** open residual: $c\!=\!8$ alone does NOT pin the net ($SO(16)_1$ also $c\!=\!8$, det 4), so the one open lemma = the holomorphy discriminator $\det K\!=\!1$. A literature-mapping/reduction; does NOT close `SEAM.EQUIV.01` **[C][E][O]**

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Script	**What it machine-checks *(continued)***
`v337_decoupling_theorem.py`	THE DECOUPLING THEOREM (consolidated, citable): every TFPT dimensionless readout factors through the gapped admissible sector and is INDEPENDENT of the ambient QG measure – the theorem that makes the `QG.AMB.01` reclassification (v335) honest. [E] gap margin: $\Delta - 2\dim(E_8)c_3^2 = 6\ln\frac{3}{2} - 31/(4\pi^2) \approx 1.648 > 0$. [E] factorization: every readout (mass ratios, mixings, α^{-1}, $R{+}R^2$) is a function of the admissible gapped transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ + the discrete kernel alone. [E] the theorem: margin $> 0 \Rightarrow$ exponential clustering ($\lambda_2{=}(2/3)^6$), finite susceptibility $\chi{=}729/665$, so the ambient contribution is gap-suppressed $\Rightarrow$ no readout depends on the ambient measure $\Rightarrow$ `QG.AMB.01` decoupled. [E] neg control: gap closed $\Rightarrow \chi$ diverges $\Rightarrow$ fails (non-vacuous). [E] corollary: zero frozen kill tests depend on the ambient measure. Consolidates v76/v311/v330/v332; does NOT construct the ambient measure [E]
`v338_theta13_budget.py`	`THETA13.BUDGET.01`: the θ_{13} sub-leading budget vs the current global fit – the $+2\sigma$ pressure point (v328) quantified against BOTH NuFIT 6.0 NO variants and JUNO; neither defuses nor excludes, fences correctly. [C] SK-dependent tension: $\sin^2\theta_{13} = (\varphi_0^{\rm ret})\, e^{-5/6} = 0.023108$ is $+2.0\sigma$ vs the no-SK central (0.02195) and $+1.7\sigma$ vs the with-SK NO best fit (0.02215) – not a clean 2σ. [E] within 3σ: $0.023108 <$ the with-SK 3σ upper 0.02388 (not falsified). [C] budget: moving to the central needs -4.1%; RG $< 1\%$ and the exponent 5/6 is exact $\Rightarrow$ no freeze-respecting relief (v328). [X] JUNO decides: sub-1% precision turns 1.7–2.0σ into $\sim 6\sigma$ – a near-term pre-registered kill/confirm. One honest pressure point (v306/v328), fenced [C][E][X]
`v339_firstprinciples_boundary.py`	The honest first-principles boundary, mapped precisely – answers (a) can $M_{\rm Planck}/v_{\rm geo}$ be computed? and (b) can $F_{\rm transfer}$ be made first-principles? Both NO, with the exact reason per residual. [E] $v_{\rm geo}/M_{\rm Planck}$ FORBIDDEN BY THEOREM (No-Unit, v153): a mass is dimension 1 while every TFPT datum is dimension 0, so no dimensionless map outputs a scale; over-determined (gravity = dark energy to 0.11%, v274); every dimensionless RATIO is derived, only the unit is not. [E] m_p/m_e structure in place (v262): the carrier $b_3{=}{-7}$ drives the QCD run + the closed electron source; two inputs stay external: $\alpha_s(M_Z)$ and the lattice $C_p{=}m_p/\Lambda^{(3)}{\sim}2.83$. [X] $\alpha_s(M_Z)$ external BECAUSE OF A TENSION: SM content does NOT unify (1-loop spread ~8.8 at $2{\times}10^{16}$ GeV, v246). [C] C_p parameter-free but lattice-only (v35). [C] η_B/axion = cosmological initial conditions; Koide $Q{=}2/3$ structural [E]. [E] FOUR honest kinds: forbidden-by-theorem / external-parameter-free / external-tensioned / cosmological. A boundary audit, NOT a solution [E][C][X]
`v340_As_reconcile.py`	A_s is predicted (not missing), is DIMENSIONLESS (so it does NOT define an absolute mass), and the apparent -11σ tension is a reheating-channel question – reconciled with the archived old derivation (TFPT v4.5/v4.6). [E] $A_s = N_\star^2(M_{\rm scal}/\bar M_{\rm Pl})^2/(24\pi^2) = N_\star^2 c_3^7/(24\pi^2)$ (v86); $M_{\rm scal}/\bar M_{\rm Pl} = c_3^{7/2} = 1.2565{\times}10^{-5}$ is a PURE RATIO, so A_s carries NO absolute scale – it pins $N_\star$, NOT $\bar M_{\rm Pl} \Rightarrow A_s$ does NOT define mass (No-Unit v153 stands). [C] the -11σ is the slow-vs-fast reheating dichotomy ($N_\star{=}51.44 \to A_s{=}1.764{\times}10^{-9}$; A_s-preferred $N_\star{=}56.2$). [C] ARCHIVE: old v4.5/v4.6 used the algebraic $N_\star{=}3/\varphi_0^{\rm ret} - 1{=}55.42$ and reported $A_s{=}2.124{\times}10^{-9}$; at that $N_\star$ the current normalisation gives $2.047{\times}10^{-9}$ (-1.9σ). [E] nothing lost: the current theory DERIVES $N_\star$ (record band $[50, 60]$, v84). [X] decisive test: fast preheating or slow reheating confirmed [E][C][X]

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Script	What it machine-checks *(continued)*
`v341_alpha_quillen.py`	The α fixed point reformulated as the STATIONARITY of a $U(1)$ determinant line, every ingredient identified as an index / heat-kernel coefficient / discriminant form – NOT a free knob (the "derive the FORM" step). $F_{U(1)} = \alpha^3 - 2c_3^3\alpha^2 - \frac{4}{5}c_3^6 M\ln(1/\varphi_{\text{seam}}) = 0$ in Quillen form: $[\alpha^3 - 2c_3^3\alpha^2]$ (det-line variation) = $[8b_1c_3^6]$ (anomaly counterterm) $\times[\ln(1/\varphi_{\text{seam}})]$ (seam response). [E] M=41 a $U(1)$ INDEX (EM Ward, v48): $\frac{4}{5}M = 8b_1$, b_1=41/10 the SM 1-loop $U(1)_Y$ beta (v159/v246). [E] c_3=1/(8π) the GAUSS–BONNET boundary coefficient (8=rank E_8, v216); $c_3^{3,6}$ heat-kernel powers. [E] $-5/4 = -q(D_5)$ a DISCRIMINANT form ($q(D_5)$+$q(A_3)$=2, v1/v154); φ_{base}=1/(6π)=$\frac{4}{3}c_3$ carries $q(A_3)$: $c_3/\varphi_{\text{base}}$=3/4. [E] δ_{top}=$\Omega_{\text{adm}}c_3^4$. [E] the Quillen split holds at the α^{-1}=137.0359992 root. [E] NO NEW GATE: α^{-1} stays [E]; v341 SHARPENS the pre-existing [O] origin `EM.WARD.01` (no new α problem). [O] the one still-open part IS `EM.WARD.01`'s residual: the from-first-principles AQFT PROOF that this is the EXACT Quillen determinant functional (the variational principle derived, not assembled). Sharpens v3/v48/`EM.WARD.01` from "fixed point" to "stationarity of a named determinant line" [E][O]
`v342_em_ward_heatkernel.py`	The heat-kernel sharpening of `EM.WARD.01`: DERIVES the ORIGIN and STRUCTURE of the $F_{U(1)}$ determinant-line terms from textbook Seeley–DeWitt / Gilkey coefficients, advancing the EM-Ward origin from [C] to [E] for the STRUCTURE; NO new gate. [E] c_3=1/(8π) the boundary GAUSS–BONNET coefficient (seam = flat pillowcase χ=2, $\oint K$=4π, v216). [E] the α^2 term is the GILKEY gauge-curvature coefficient: the $U(1)$ Laplacian a_4 has $30\Omega^2/360 = \frac{1}{12}\Omega^2$ (Gilkey Thm 4.8.16), Ω=$i\alpha F \Rightarrow -(\alpha^2/12)F^2$, so the Calderon $-2c_3^3\alpha^2$ is the gauge-curvature piece, not an ansatz. [E] the c_3-ladder $\{0,3,6\}$ is arithmetic (step 3 = 3 channels; 6=$\lvert R^+(A_3)\rvert$ C_6 hexagon). [E] multiplicities forced (2=$\lvert\mathbb{Z}_2\rvert$, $8b_1$=(rank E_8)b_1). [E] π-power test: $2c_3^3$=1/(256π^3) carries π^3 $\Rightarrow$ THREE boundary insertions. [C] the exact coefficient needs the seam F-normalisation. [O] the cubic α^3 Maxwell moment + full identification = `EM.WARD.01` residual (tied to `SEAM.EQUIV.01`). α^{-1} stays [E] (v3) [E][C][O]
`v343_four_routes_analysis.py`	The honest investigation of the four black-hole-birth solution routes (A finite causal diamond, B Carlip–Cardy near-horizon CFT, C modular/thermal flow, D self-reproduction attractor) – direct answer to "which closes ALL problems and is 100% provable": NONE individually, but they CONVERGE on the same two irreducible residuals as v335 (`SEAM.EQUIV.01` + the decoupled QG contour). [E] ROUTE A reframes `QG.AMB` to a FINITE thermal trace (de Sitter finite, gapped χ=729/665) but $c_{\text{conf}}(4)$= – 3/2 < 0 still needs the GHP contour (v334, [O]); does NOT touch `SEAM.EQUIV.01`. [C] ROUTE B: Carlip+Cardy consistent with c=8 but L_0 free (ambiguity) and gives c only, not holomorphy det K=1 – inherits `SEAM.EQUIV.01`. [E] ROUTE C: modular flow KMS β=1 (v239), geometric identification = `SEAM.EQUIV.01` (v335) – a reduction. [E]/[O] ROUTE D: unique Perron–Frobenius attractor (v56) but cyclic dynamics [O] – not a closure. [E] VERDICT: independent closures = 0; irreducible residuals = 2 (= v335); the BH premise makes both more natural but eliminates neither; focus = `SEAM.EQUIV.01` [E][C][O]

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Script	What it machine-checks *(continued)*
`v344_detk_synthesis.py`	The bird's-eye synthesis of the ONE open keystone bit, $\det K{=}1$ – a full-spectrum map of `SEAM.EQUIV.01`'s only open residual. [E] $\det K{=}1$ is ONE statement with SIX equivalent faces (holomorphy / homology-sphere link $H_1{=}0$ / one 1-dim irrep / det Cartan = 1 / full μ_4 condensation / $\tau{=}i$ CM), unified by $\lvert\det \mathrm{Cartan}\rvert = \lvert H_1(\mathrm{link})\rvert = \#(\text{1-dim irreps})$ across ADE ($A_n \to n{+}1$, $D_n \to 4$, $E_6 \to 3$, $E_7 \to 2$, $E_8 \to 1$) – ONLY E_8 gives 1 ($2I$ the unique PERFECT ADE group, v232/v219). [E] THE GENUS-1 OBSTRUCTION: the easy arguments (plane vacuum, gap, free-fermion) are genus-0, necessary but NOT sufficient (v87); $\det K$ is the torus degeneracy = #primaries. [E] THREE genus-1 access routes: R1 continuum-modular (v336), R2 anyon-condensation (v281), R3 hypergraph-homotopy. [C] the HYPERGRAPH route is the most promising fresh angle: the $(2,3,5)$ rewrite attractor IS the affine-E_8 graph (v312) and the E_8 du Val link is the Poincare sphere ($H_1{=}0$, v232), so $\det K{=}1$ becomes a finite COMBINATORIAL statement (no continuum limit). [O] none closes it; R3's combinatorial-to-geometric bridge is meanwhile discharged to Kronheimer 1989 (v479), so the residual transforms to "the raw seam supplies the ALE/orbifold datum" = the `SEAM.EQUIV.01` content. A synthesis/roadmap; closes nothing [E][C][O]
`v345_hypergraph_homotopy.py`	EXECUTE the combinatorial core of route R3 (the hypergraph-homotopy route to $\det K{=}1$, v344): computes explicitly that the $(2,3,5)$-rewrite attractor graph's plumbing link is the POINCARE homology sphere ($H_1{=}0$), only E_8 among ADE; R3's combinatorial half is now [E]-complete, the geometric bridge (= `SEAM.EQUIV.01`) stays [O]. Plumbing topology (Brieskorn): link $H_1 = \mathrm{coker}(\mathrm{Cartan})$. [E] $\mathrm{SNF}(E_8\ \mathrm{Cartan}) = \mathrm{identity} \Rightarrow \mathrm{coker}(E_8){=}0 \Rightarrow$ the E_8 plumbing boundary is an INTEGRAL HOMOLOGY SPHERE (the genus-1 meaning of $\det E_8{=}1$). [E] only E_8: $\mathrm{coker}(A_n){=}\mathbb{Z}/(n{+}1)$, $D_n \to \mathbb{Z}/4$, $E_6 \to \mathbb{Z}/3$, $E_7 \to \mathbb{Z}/2$ – all nontrivial, only $E_8{=}0$. [E] $\pi_1{=}2I{=}SL(2,5)$ PERFECT (commutator subgroup order $120{=}\lvert G\rvert$, computed) $\Rightarrow H_1{=}0 \Rightarrow$ THE Poincare sphere; $\lvert 2I\rvert{=}120{=}\lvert\mu_4\rvert\, h(E_8)$. [O] does NOT close $\det K{=}1$: the graph→geometry half is meanwhile Kronheimer-cited (v479); what stays open is the raw seam supplying the ALE/orbifold datum = `SEAM.EQUIV.01` [E][O]
`v346_seam_geometric_bridge.py`	The geometric bridge as an explicit six-link chain raw-seam $\Rightarrow \det K{=}1$, with the ONE open arrow pinned – the honest capstone of the `SEAM.EQUIV.01` investigation. L1 [E]/[C] raw seam $\to \mu_4$ + four marks (v216); L2 [O] μ_4 + $(2,3,5)$ atoms $\to 2I$ orbifold $\mathbb{C}^2/\Gamma$ (THE OPEN ARROW = `SEAM.EQUIV.01`); L3 [E] $2I \to$ affine E_8 McKay (v219); L4 [E] $2I \to E_8$ du Val (v232); L5 [E] $\mathbb{C}^2/2I \to$ Poincare link $H_1{=}0$ (v345); L6 [E] $H_1{=}0 \to \det K{=}1$. [E] 5 of 6 links are theorems, only L2 open. [E] the group is FORCED by the seam atoms: axis signatures $(2,3,3)/(2,3,4)/(2,3,5) \to 2T/2O/2I$, $(2,3,5)$ unique to $2I$, and $(\lvert\mathbb{Z}_2\rvert, N_{\mathrm{fam}}, g_{\mathrm{car}}){=}(2,3,5)$ ARE those axes. [E] downstream of L2 forced. [O] L2 (the seam IS an ADE orbifold point) = `SEAM.EQUIV.01`, NOT closed – the keystone reduced to one geometric-realisation arrow, group pinned by $(2,3,5)$ [E][C][O]
`v347_seam_closure_modes.py`	The closure-mode classification of the one open arrow L2 (the seam-orbifold realisation, v346) – a direct honest answer to "is there ANY other way to FULLY solve it?". [E] THE PRECISE LOCUS: the seam gives the pillowcase base $S^2(2,2,2,2)$, $\chi_{\mathrm{orb}}{=}0$ (EUCLIDEAN, v216), but the $2I/E_8$ structure is the Seifert base $S^2(2,3,5)$, $\chi_{\mathrm{orb}}{=}1/30$ (SPHERICAL); L2 must bridge a FLAT 4-marked base to a SPHERICAL 3-marked base – why L2 needs the carrier $(2,3,5)$ input. [E] FOUR CLOSURE MODES: A construct (the analytic wall, v336); B rigidity (unique realisation $\Rightarrow \mathbb{C}^2/2I$ forced, the only theorem-route w/o new input, v284/v300/v331); C axiom P_3 ($\Rightarrow$ TFPT complete rel. 3 axioms); D empirical. [E] no-new-state (v246) blocks deriving L2 from extra physics $\Rightarrow$ only rigidity (B) closes it as a theorem. [E] VERDICT: no free theorem; only B (theorem) or C (axiom), validated by D; only these four modes. [O] NOT closed [E][O]

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Script	**What it machine-checks** *(continued)*
`v348_seam_rigidity_route.py`	Route B (the rigidity route) for the one open arrow L2, carried out – it lands on a strikingly SIMPLE statement: the whole keystone reduces to ONE question, does the raw seam carry the golden ratio? **[E]** rigidity closes the crystallographic part: the order-4 clock has a UNIQUE normal form $z \mapsto iz$ (Nielsen/Kerckhoff, v180), Troyanov gives the unique flat pillowcase metric (v284). **[E]** downstream is a LATTICE identity: E_8 = the ring of icosians (Conway–Sloane), rank 8, so $2I \to E_8$ is a lattice identity (stronger than McKay v219). **[E]** the single bridge is φ: crystallographic orders $\{1, 2, 3, 4, 6\}$, the 5-fold non-crystallographic; μ_4 crystallographic, $2I$ (5-fold) not, $\varphi = 2\cos(\pi/5)$ extends one to the other (quasicrystal). **[O]** the keystone reduces to ONE question: "does the raw seam carry φ?" – if yes, $\mu_4+\varphi \to 2I \to E_8$ closes L2. **[O]** honest subtlety: TFPT's φ is currently E_8-side (v312/v313); the raw-seam data (cusp $\{0, 1/3, 2/3\}$, recovery $(2/3)^6$, seed φ_0) are not manifestly golden, so showing the RAW seam carries φ non-circularly is the residual (= L2). NOT closed **[E][O]**
`v349_raw_seam_golden_test.py`	The honest test of the v348 question "does the RAW seam carry the golden ratio (before assuming E_8)?". Answer, computed: NO – the bare carrier (D_5, A_3) and the dynamical seam data are NOT golden; the golden ratio genuinely enters WITH the icosahedral/E_8 identification. A NEGATIVE result that prevents a false closure. **[E]** RAW CARRIER NOT GOLDEN: golden needs $5 \mid h$; D_5 has $h{=}8$ $(2\cos(2\pi/8){=}\sqrt{2})$, A_3 has $h{=}4$ – no 5-fold, bare carrier gives $\sqrt{2}$. **[E]** DYNAMICAL DATA NOT GOLDEN: cusp $\{0, 1/3, 2/3\}$ rational, recovery $(2/3)^6$ rational, seed φ_0 transcendental. **[E]** GOLDEN IS THE ICOSAHEDRAL/E_8 SIDE: $5 \mid h$ only for $A_4{=}SU(5)$ $(h{=}5)$, H_3 $(h{=}10)$, E_8 $(h{=}30)$. **[E]** φ alone insufficient: $2I \neq C_5 \times \mu_4$ $(20 \neq 120)$. **[O]** RESOLUTION: L2 reduces to "is $g_{\rm car}{=}5$ a PENTAGON (5-fold $\Rightarrow$ golden $\Rightarrow$ icosahedral) or a COUNT (rank $D_5 \Rightarrow \sqrt{2}$)?" – the golden ratio is the pentagon-reading of P2, NOT derivable from P2-as-a-count; no hidden φ. Foundational: P2-as-pentagon closes it (mode C) else rigidity must produce the 5-fold (mode B); NOT closed **[E][O]**
`v350_bootstrap_inputs_correction.py`	The honest CORRECTION of v349's framing (prompted by the right objection: the inputs are not axioms, they are back-determined by the theory). **[E]** THE INPUTS ARE BOOTSTRAP-FORCED, NOT FREE AXIOMS (v6): $g_{\rm car}{=}5$ over-determined three ways (rank-fill, Coxeter-match $g_{\rm car}{=}$ max prime of $h(E_8){=}30$, Pascal). **[E]** $g_{\rm car}{=}5$ IS THE ICOSAHEDRAL 5-FOLD, not the D_5 rank: $g_{\rm car}{=}$ max prime of $h(E_8){=}30{=}2{\cdot}3{\cdot}5$; the atoms $(2, 3, 5)$ ARE the primes of the icosahedral Coxeter number 30. **[E]** THE GOLDEN RATIO IS EMERGENT FROM THE GLUE, not external: D_5 $(h{=}8)/A_3$ $(h{=}4)$ non-golden alone, but the μ_4 extension $D_5{\oplus}A_3 \to E_8$ (v154) gives $h(E_8){=}30$ (golden, $5 \mid 30$) – the seam's own μ_4 produces the 5-fold; v349 checked the un-glued pieces, the wrong object. **[E]** v349's pentagon-vs-count dichotomy was FALSE (the bootstrap forces the pentagon). **[O]** residual relocates: the golden is bootstrap-forced; the only open thing is the physical continuum realisation (v336), NOT the golden ratio. Caveat: self-consistency within the framework, not from-nothing. Supersedes v349's interpretation **[E][O]**
`v351_continuum_realisation_sharpened.py`	The continuum realisation sharpened – the "$c{=}8$ ambiguity" flagged open in v277/v344 (E_8 vs $SO(16)$ both $c{=}8$) is RESOLVED, non-circularly, by the seam's ORDER-4 μ_4 clock. **[E]** E_8 (det 1, holomorphic) vs $D_8{=}SO(16)$ (det 4). **[E]** the order-4 clock (the four Gauss–Bonnet marks, $\lvert\mu_4\rvert{=}4{=}h(A_3)$, v216): index-4 extension $D_5{\oplus}A_3{=}E_8$ (v154), index-2 = $SO(16)$; so order-4 selects E_8 (16→4→1) not $SO(16)$ (16→4) – the order (4 not 2) is the discriminator. **[E]** bootstrap agrees: $h(E_8){=}30$ (max prime 5) vs $h(D_8){=}14$ (max prime 7). **[E]** which-net ambiguity RESOLVED (det $K{=}1$ is then a consequence). **[O]** residual = PURELY the chiral-edge/scaling-limit existence (bulk-edge, v336), net+holomorphy pinned. Does NOT close `SEAM.EQUIV.01` **[E][O]**

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Script	What it machine-checks *(continued)*
`v352_framework_irreducible.py`	The framework reduction – the deepest answer to "the inputs are not axioms, they are back-determined". [E] P2 ($g_{\rm car}$=5) FORCED three ways (v6): rank-fill, Coxeter-match ($g_{\rm car}$= max prime of $h(E_8)$=30), Pascal. [E] P1 (c_3=1/(8π)) FORCED: c_3 is the boundary Gauss–Bonnet coefficient (v216/v342), the 8 over-determined four ways (8= rank E_8=$h(D_5)$=$\phi(30)$= det R, v54/v6); only π is non-forced (the boundary transcendental). [E] the E_8 hull is forced (unique even unimodular rank-8, v83/v154). [E] so the ONLY irreducibles are the FRAMEWORK (a discrete reflection-positive boundary with a finite Clifford carrier) + π – the precise content of "the inputs are not axioms". [O] the framework is the META-AXIOM (not math-reducible, the foundational stance, tested by its consequences not proved). TFPT's true input is ONE framework + π [E][O]
`v353_selfloop_capstone.py`	The bird's-eye rethink (refining v352): TFPT is not a LINEAR theory (axioms → theorems) but a CLOSED SELF-CONSISTENT LOOP whose outputs fix its own inputs, with ZERO free adjustable dimensionless parameters. [E] IT IS A LOOP: outputs fix inputs ($g_{\rm car}$=5= max prime of $h(E_8)$=30; the 8 in c_3= rank E_8) and inputs fix outputs; the arrows close into a circle (v6) with a unique fixed point (v56). [E] ZERO FREE DIALS: every load-bearing number is forced/over-determined; free dimensionless parameters = 0. [E] π is a FIXED geometric constant (the boundary 2-sphere, $\oint K$=4π ⇒ c_3=1/(8π)), not a dial. [C] refining v352: π is not a free input and the framework is a SELECTION PRINCIPLE (the unique self-consistent loop), not a free meta-axiom. [O] the one residual is the CONTINUUM closure (the chiral-edge existence, v336/v351); structural-realism (loop complete) vs constructivism (continuum open) is a philosophical, not numerical, fork [E][C][O]
`v354_e8_reverse_audit.py`	The REVERSE numerology audit + what the golden ratio means. The forward rule (v305) is one-directional (every readout must map onto E_8); the reverse question: how much E_8 structure maps onto NOTHING? [E] E_8 has 8 Casimir invariant degrees $\{2,8,12,14,18,20,24,30\}$; THREE feed a readout (2 metric, 8 rank→ c_3, 30 Coxeter→ $g_{\rm car}$) and FIVE are UNMAPPED $\{12,14,18,20,24\}$ – 3/8 load-bearing, 5/8 not. [E] ECONOMY, NOT PROMISCUITY: a small fixed set (rank 8, h=30, det 1, $D_5 \oplus A_3$, 16-spinor) generates ALL readouts; the higher Casimirs are never reached into to fit – the anti-numerology signature (economy + a large untouched remainder). [E] the golden ratio is UNMAPPED (the affine-E_8 attractor eigenvalue v312, the icosian lattice v348; no frozen readout is φ, the sole [C] exception $\cos(\theta_i$=3π/5)=−1/(2φ) on the axion spine branch, refined in v429) ⇒ NOT numerology (which needs a number fitted to an observation); φ is the algebraic signature of the 5-fold h=30=2·3·5, the carrier announcing it is icosahedral. [O] the 5/8 unmapped is the HULL OVERHEAD (E_8 = minimal unimodular hull, not the world group); where future readouts could live, or over-capacity. The reverse complement of v305 [E][O]
`v355_e8_unmapped_bandwidth.py`	A disciplined bandwidth search of the unmapped E_8 region (next step after v354), run with a STRICT discriminator: a candidate counts only if it is a FORCED identity (a theorem about E_8), never a numerical coincidence – because mining a rich object for matches is the numerology temptation. [E] FORCED SET-LEVEL (the real collective content): $\sum$ deg = 128 = dim S^+ the spinor half of 248=120+128; $\sum$ exp = 120 = #pos roots; $\prod$ deg = $\vert W(E_8)\vert$; exponents = the φ(30)=8 totatives of 30 (v223); max deg = 30 = $h(E_8)$. [C] SUGGESTIVE not forced: 12=$h(E_6)$, 18=$h(E_7)$, 30=$h(E_8)$ (the McKay ladder v219), but 'deg(E_8) contains subalgebra Coxeter numbers' is not a theorem – flagged. [O] DECLINED: the per-degree atom matches (14= dim G_2, 20= det L, 24=$\vert W(A_3)\vert$, 12=$\vert R(A_3)\vert$, 18=$p_4(a)$; v66 flagged 12, 18 fittable) each admit multiple readings, and the $\vert W\vert$ prime 7='scalaron exponent' is one of many – declined. [E] RECONCILES v66 (degrees=atoms) and v354 (only 2, 8, 30 primary): NO new forced physical hit; the disciplined decline IS the anti-numerology result [E][C][O]

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Script	**What it machine-checks** ***(continued)***
v356_ continuum_mmst_ applicability.py	Direction A: the continuum chain of `SEAM.EQUIV.01` assembled theorem-by-theorem, residual reduced to ONE lattice-realization input tied to the seam's one-sidedness. Chain: gapped (v302) quasi-free 16-Majorana collar (v155/v160) $\to$ invertible (Kitaev, v301) $\to$ the NONTRIVIAL invertible phase (c_-=8) $\to$ a non-gappable anomalous chiral edge (anomaly inflow: edge EXISTENCE follows, reducing the v336/v351 'does the edge exist?' residual) $\to$ chiral-CFT scaling limit (MMST v336, 8≤8≤16) $\to$ order-4 μ_4 clock + AQFT stack pin $(E_8)_1$ (v351/v297). The only non-theorem link is the seam one-sidedness = the c_3 origin ($c_-\neq 0$ = one-sided = the $\lvert\mathbb{Z}_2\rvert \oint K$=8 of c_3=1/(8π), v58). [C]/[E]/[O]; ADVANCES `SEAM.EQUIV.01` to one realization input, does not close. (Direction B was found already closed by v40.) [C][E][O]
v358_grav_entropy_ equilibrium.py	The entanglement-equilibrium derivation (Jacobson 2015 / Faulkner et al., $\delta S = \delta\langle K\rangle$) with TFPT's atoms ⇒ the LINEARISED Einstein equation falls out PARAMETER-FREE. [E] $G_{ab} = (1/c_3)T_{ab}$, $1/c_3 = 8\pi$ fixed (Unruh $1/(2\pi)$=$4c_3$, EH $1/(16\pi)$=$c_3/2$, Einstein 8π=$1/c_3$, Bekenstein 1/4=$1/\lvert\mu_4\rvert$). [E] THERMO=GEO coincidence (new): the first law's $2\pi/\eta$ = the geometric $\lvert\mathbb{Z}_2\rvert 2\pi\chi$ iff $\lvert\mu_4\rvert$=$\lvert\mathbb{Z}_2\rvert\chi$=4 – the thermodynamic and geometric origins of c_3 are ONE. [C] J3 solved: the CHM ball modular Hamiltonian (boost via BW v323) gives the matter flux $\delta\langle K_B\rangle$=$\frac{8\pi^2R^4}{15}\delta\langle T_{00}\rangle$. [E] entropy density atom-fixed: 1/4=$1/\lvert\mu_4\rvert$ + central charge c=g_{car}+N_{fam}=8 (form v59). [O] residual: the equation-of-state status + the global ambient measure (C7) + the v_{geo} unit (v359 closes the FULL covariant equation); no free dimensionless gravity dial remains [E][C][O]
v359_grav_ nonlinear_ einstein.py	The FULL covariant Einstein equation, parameter-free (Direction 1, extends v358). The Jacobson-2015 fixed-VOLUME entanglement equilibrium brings in the EINSTEIN TENSOR G_{ab} (not just R_{ab}) – the unique divergence-free metric tensor (Lovelock) – so equilibrium yields the full covariant $G_{ab} + \Lambda g_{ab} = c_3^{-1}T_{ab}$ with conservation built in. [C][E] fixed-volume $\to$ Einstein tensor (trace = $-R$ in d=4). [E] Lovelock ⇒ $\nabla^a T_{ab}$=0 is an output. [E] coeff 1: 8π=$1/c_3$ fixed (v358). [E] coeff 2: Λ from α – $\rho_\Lambda/\bar{M}_{\mathrm{Pl}}^4 = (3/(4\pi^2))e^{-2\alpha^{-1}}$, prefactor $(8\pi)^2{\cdot}48c_3^4 = 3/(4\pi^2)$ (v60); BOTH coefficients TFPT-fixed. RESULT: the original B6 full covariant field equation is now parameter-free at the LOCAL level. [O] residual: the equation-of-state status + the global ambient measure (C7) [E][C][O]
v360_grav_gap_ corrections.py	Direction 2: the gap-induced higher-curvature correction to GR – the disciplined result. Does the seam transfer gap give a NEW forced near-term GR-deviation beyond the known R^2 scalaron? With the v354/v355 discriminator: NO. [E] the leading higher-curvature term IS the R^2 scalaron (a_4=R^2/72 v36; $M_{\mathrm{scal}}^2/\bar{M}_{\mathrm{Pl}}^2 = c_3^7$, exp 7=$g$+$N$−1) – already in TFPT, already falsifiable via n_s=1−2/$N_\star$, r=12/$N_\star^2$ (CMB-S4). [C] it is the entanglement-equilibrium next order (Wald first law $\to f(R)$=R+R^2, v28/v359). [E] the gap Δ=6 ln(3/2) is dimensionless, so a distinct gap term sits at $\xi\, v_{\mathrm{geo}} \sim$ seam/Planck scale, not near-term observable. [O] DECLINE: a new dimensionless coefficient is not atom-forced (needs heat-kernel/Wald data; exp 7 vs 6, c_3^7 vs $(2/3)^6$ unrelated) $\to$ per v354/v355 declined. No new near-term forced prediction; the answer is the scalaron [E][C][O]
v361_grav_ backreaction.py	Direction 5: the matter-gravity backreaction with the explicit J3 flux – two forced facts + an honest decline. [E] the carrier's gravitational anomaly is FORCED: c_-=8 (16 Majoranas = dim $S^+ = 2^{g_{\mathrm{car}}-1}$; chiral central charge 16/2=8=g_{car}+N_{fam}, the same c_- as v358). [E] the backreaction is FINITE $\to$ Λ from α: the matter vacuum energy through G_{ab}=$c_3^{-1}T_{ab}$ is gap-suppressed (Decoupling Thm v337) to $\rho_\Lambda/\bar{M}_{\mathrm{Pl}}^4 = (3/(4\pi^2))e^{-2\alpha^{-1}}$ (v60), 123 orders = 119.03+3.92 – NOT the M_{Pl}^4 catastrophe; the loop (v359+v60+carrier vacuum) closes. [E] J3 explicit (v358): $\delta\langle K_B\rangle$=$(8\pi^2R^4/15)\delta\langle T_{00}\rangle$, carrier T_{ab}=the SM's (v159). [C] SM trace anomaly reproduced (content=SM). [O] DECLINE: no new atom-forced read-out of individual SM quantum numbers beyond c_-=8 and Λ $\to$ per v354/v355 declined. Finite & consistent backreaction, no new dial [E][C][O]

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Script	What it machine-checks *(continued)*
`v364_vgeo_sharpen.py`	Direction 6: $v_{\rm geo}$ is the SINGLE dimensionful input of the WHOLE theory – confirmed across matter AND gravity – and irreducibly primitive. The No-Unit Theorem (v153) showed the dimensionless compiler cannot output a scale ($v_{\rm geo}$ a forced unit). SHARPENING after the parameter-free gravity (v358/v359/v361): the gravity sector adds NO new scale either – $1/c_3{=}8\pi$ and the Λ prefactor $(8\pi)^2{\cdot}48c_3^4 = 3/(4\pi^2)$ are dimensionless, so $1/G \sim v_{\rm geo}^2$ and $\Lambda \sim$ (dimensionless)$v_{\rm geo}^4$ reduce to $v_{\rm geo}$ times atoms. [E] FINAL TALLY: every dimensionful quantity = (dimensionless ratio)×(power of $v_{\rm geo}$); free content $= \{v_{\rm geo}, \pi\}$, ZERO dimensionless dials, vs the SM's ∼26. [O] $v_{\rm geo}$ irreducibly primitive (No-Unit; $1/G$ UV-sensitive Sakharov v68). Clarity, not a new derivation; the last dimensionful anchor named [E][O]
`v365_qg_oneloop_saddle.py`	Direction 3: the QG measure (gate C7/`QG.AMB.01`) as one-loop GAUSSIAN fluctuations around the PARAMETER-FREE saddle (v358/v359). [E] the fluctuation stiffness is $1/c_3 = 8\pi$ (no free dial), so the one-loop problem is well-posed (it was NOT before v358). [E] the fluctuation operator is gapped $M \geq \Delta_{\rm eff} = 6\log\frac{3}{2} - \frac{31}{4\pi^2} \approx 1.648 > 0$ (v76/v337) ⇒ mode-by-mode Gaussian convergence. [E] finite model log-det $\mathrm{Tr}'\log M = 6\log\frac{9}{2}$ on the OS spectrum $\{0, 6\log\frac{3}{2}, 6\log 3\}$. [C] the conformal mode converges on the GHP contour ($\int e^{-\lvert c\rvert\phi^2} = \sqrt{2\pi/3}$, v334). [E] the admissible projective limit exists ($\chi = 1/(1-(2/3)^6) = 729/665$, v330). [E] 0 new dimensionless dials (v364). [O] residual: the full non-perturbative projective limit (G6/`QG.AMB.01`) on the ambient sector. Sharpens C7 to a one-loop Gaussian problem around a parameter-free saddle; does NOT close it. [E][C][O]
`v366_mmst_seam_collar.py`	Direction 7: the seam collar is IN MMST's free-lattice-fermion scaling-limit class, so the chiral scaling limit is a chiral CFT of central charge 8 pinned to $(E_8)_1$ – reducing `SEAM.EQUIV.01` to the single S3 lattice-realisation input. Osborne–Stottmeister (arXiv:2107.13834, CMP **398** (2023) 219; legacy internal token MMST, on the MMST/Morinelli–Morsella–Stottmeister–Tanimoto OAR framework): free lattice fermions (CAR quasi-free) → chiral-CFT massless scaling limit with correct c, range rank $\leq c \leq D$. [E] class arithmetic: $D{=}\dim S^+{=}2^{g_{\rm car}-1}{=}16$ Majoranas, $c{=}g_{\rm car}{+}N_{\rm fam}{=}8$, rank $E_8{=}8$ ⇒ rank $\leq c \leq D$ is $8 \leq 8 \leq 16$ (in MMST range). [E] gapped ($\Delta{=}6\log\frac{3}{2}{>}0$, derived v302). [C] quasi-free CAR class (v155/v160). [E] chirality $c_-{=}D/2{=}8 \neq 0$ (non-gappable anomalous edge, v356 S4). [C] the MMST scaling-limit theorem applies (in-class & gapped & chiral & in-range) ⇒ chiral CFT $c{=}8$ (v336). [E] target pinned to $(E_8)_1$ by $c{=}8$ + the order-4 μ_4 clock ($\det K{=}1$ vs $SO(16)$ $\det K{=}4$, v351). [O] residual S3: the collar realised as a genuine lattice chiral free-fermion invertible phase (one-sidedness/Flat-Away, v297/v356). Verifies MMST in-class; ADVANCES `SEAM.EQUIV.01`, does NOT close it. [E][C][O]
`v367_seam_s3_lattice.py`	Track 1: the EXPLICIT lattice realisation of the S3 input – a concrete gapped chiral-Majorana ($p+ip$) lattice model whose edge is the 16-Majorana $SO(16)_1$ carrier, μ_4-condensed to $(E_8)_1$; turns the abstract S3 residual (v356/v366) into a RUNNABLE model with a numerically computed topological invariant. [E] bulk gapped (band gap $2>0$ at $M{=}1$). [E] non-trivial Chern number (Fukui–Hatsugai–Suzuki: $\lvert C\rvert{=}1$ at $M{=}1$, $C{=}0$ at $M{=}3$ trivial control). [E] $c_-{=}8$: each chiral Majorana edge carries $c_-{=}C/2{=}1/2$, $N_{\rm Maj}{=}\dim S^+{=}2^{g_{\rm car}-1}{=}16$ copies ⇒ $c_-{=}16/2{=}8{=}g_{\rm car}{+}N_{\rm fam}$. [E] edge K-matrix $SO(16)_1$ (det Cartan $D_8{=}4$) → $(E_8)_1$ (det Cartan $E_8{=}1$) by the order-4 μ_4 clock (v351/v369). [C] the μ_4 condensation $SO(16)_1 \to (E_8)_1$ (v154/v351). [O] residual: the continuum scaling-limit existence (MMST v336). Reduces S3 to a runnable lattice model; does NOT close `SEAM.EQUIV.01`. [E][C][O]

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Script	**What it machine-checks** *(continued)*
`v368_seam_s3_inflow.py`	Track 1: the bulk-edge / anomaly-inflow leg (S4 of the v356 chain) made CONCRETE on the v367 lattice model – a finite STRIP of the gapped $p + ip$ model has a realised, non-gappable chiral edge, so edge EXISTENCE follows from the invertible bulk (anomaly inflow), NOT a separate assumption. **[E]** topo in-gap edge states (M=1 strip $\min_{k_x} \lvert E\rvert \sim 0$) vs trivial clean gap ($M$=3, neg control); **[E]** the near-zero state is edge-localised (weight within 4 sites of a boundary, edge≫bulk); **[E]** bulk-edge correspondence (edge branch count = bulk Chern $\lvert C\rvert$=1) ⇒ 16-Majorana edge, c_-=8. **[C]** anomaly inflow: a $c_- \neq 0$ invertible bulk cannot have a gappable edge ⇒ discharges the v336/v351 'does the edge exist?' residual on the lattice. **[O]** residual: the continuum scaling limit (MMST v336). Establishes edge EXISTENCE on the lattice; does NOT close `SEAM.EQUIV.01`. **[E][C][O]**
`v369_qgamb_redundancy.py`	Track 2 (next-plan v2): the AMBIENT REDUNDANCY statement – the holographic route to Gravity-complete that SIDESTEPS building the general Euclidean QG measure (C7/`QG.AMB.01`). Instead of constructing the non-perturbative ambient measure (the conformal-factor swamp, v332/v334), assemble the established discriminators proving the ambient sector carries NO physical d.o.f. relative to the holomorphic boundary algebra, so `QG.AMB.01` is a non-fundamental CERTIFICATION object (like a coordinate choice in GR), not missing dynamics. **[E]** HOLOMORPHIC ⇒ DHR = Vec: #DHR primaries $= \lvert\det \mathrm{Cartan}\rvert = 1$ for E_8 (ONE primary = vacuum) vs 4 for $SO(16)/D_8$ – no nontrivial superselection charges ⇒ no hidden boundary sector a bulk could carry invisibly. **[E]** NO TORUS GSD: ground-state degeneracy on T^2 = #anyons $= \lvert\det K\rvert = 1$ for E_8 ⇒ unique ground state ⇒ no topologically protected invisible bulk d.o.f. (the genus-1 obstruction of v344). **[E]** FINITE PETZ RECOVERY: the deviation subspace contracts at $(2/3)^6$=64/729 < 1 (v221) ⇒ Rec_Σ bounded. **[E]** GAP-DECOUPLING MARGIN: $\Delta_{\mathrm{eff}}=6\log\frac{3}{2}-\frac{31}{4\pi^2}\approx 1.648>0$ (v76/v337) ⇒ the ambient sector is energetically separated. **[C]** RECONSTRUCTION MAP: Bisognano–Wichmann gives the modular boost (v258/v309/v323) ⇒ $\mathrm{Rec}_\Sigma : (\text{bulk}) \to \mathcal{A}_\Sigma^{\mu_4}$. **[C]** THE REDUNDANCY STATEMENT: under `SEAM.EQUIV.01` + BW, $\mathcal{O}_{\mathrm{phys}}^{\mathrm{bulk}}/\mathrm{Diff} \cong \mathcal{A}_\Sigma^{\mu_4}$ ⇒ `QG.AMB.01` carries no extra physical d.o.f. **[O]** residual: the two consumed premises – `SEAM.EQUIV.01` (the S3 lattice input, v366) + the BW intrinsicality of the raw collar (v329); provides the ALTERNATIVE (holographic) route, does NOT construct the measure. Gravity-complete = Boundary-complete + Ambient-redundancy **[E][C][O]**
`v370_grav_spin2_unitarity.py`	Track 2: the Barnes–Rivers spin decomposition of the graviton propagator – perturbative graviton unitarity SECTOR-BY-SECTOR (extends v304's scalar pole algebra to the tensor spin structure + combines with v334's spin-0 conformal mode). **[E]** projector completeness: $\mathrm{tr}\,P_2=(d-2)(d+1)/2$, $\mathrm{tr}\,P_1=d-1$, $\mathrm{tr}\,P_{0s}=\mathrm{tr}\,P_{0w}=1$ sum to $d(d+1)/2$; the d=4 split is 5+3+1+1=10. **[E]** EH structure: spin-2 coefficient +1, spin-0 scalar $-1/(d-2)=-1/2$. **[E]** the Stelle ghost is a SPIN-2 state (a local $1+p^2/M^2$ dressing puts the negative-residue pole at $p^2=-M^2$). **[E]** the entire KMS form factor e^{p^2/M^2} (v304/v259) makes spin-2 ghost-free (only the p^2=0 pole). **[C]** the wrong-sign spin-0 conformal mode ($c_{\mathrm{conf}}(4)=-3/2$, v332) is cured by the GHP contour (v334) – a different mechanism. **[C]** ⇒ perturbative graviton unitarity sector-by-sector, conditional on v304's entire-analyticity assumption. **[O]** perturbative-only + the open analyticity assumption; NOT the non-perturbative measure (`QG.AMB.01`). **[O]** decline: $\mathrm{tr}\,P_2$=5 is dimension counting, NOT g_{car} (v354/v355). **[E][C][O]**

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Script	**What it machine-checks** *(continued)*
`v371_ftransfer_pole.py`	Track 3 (F_{transfer}): the Koide source→pole transfer as a typed solver – promotes the `experiments/ftransfer` solve into the ledger with a kill test (does NOT re-prove 53/54 v183 or Q^*=2/3 v82/v101). [E] Q_{pole}=Koide(PDG poles)= 2/3 to 3×10^{-6}, $Q_{\text{src}}(\varphi_0^{\text{ret}}$-ladder)= 0.66446 (0.33% below). [E] QED runs the wrong way: $dQ/d\epsilon > 0$ pushes Q ABOVE 2/3; the ϵ mapping $Q_{\text{pole}} \to Q_{\text{src}}$ is NEGATIVE while QED gives ϵ=+(3/2)(α/π) > 0 – wrong sign for any $\alpha \in [1/137, 1/128]$. [C] verdict: F_{pole} is NOT standard QED running; the transfer is the non-QED 53/54 operator / $(2/3)^6$ Moebius generator (v183/v82). Stays [C], plain running excluded as the kill. [E][C]
`v372_ftransfer_boltzmann.py`	Track 3 (F_{transfer}): η_B from the INTEGRATED Buchmueller–Di Bari–Pluemacher Boltzmann network at the TFPT-frozen heavy scale – promotes the `experiments/ftransfer` solve, replacing the v326 toy washout. Frozen M_1=$M_{\text{scal}}(\varphi_0^{\text{ret}})^2/A_\Lambda$ ($\sim 8.65\times10^9$ GeV, v212), $\tilde{m}_1$=m_3/A_Λ=5 meV, δ_{CP}=$4\pi/3$. [E] integrated efficiency $\kappa_f \sim 0.09$ (strong washout $K \sim 4.6$) agrees with the BDP fit to $\sim$ 25%. [C] η_B(ODE)$\sim 6.5\times10^{-10}$ at the frozen M_1 lands within a factor ~ 1.1 of the observed 6.1×10^{-10}, NO M_R dial. [X] kill: η_B outside [obs/3, 3 obs] for all seesaw-consistent M_R would drop the leptogenesis branch. Stays [C] (a transfer target). [E][C][X]
`v373_ftransfer_relic.py`	Track 3 (F_{transfer}): the axion relic from the FINITE-T misalignment ODE – promotes the `experiments/ftransfer` solve and DECIDES the two TFPT angle branches, replacing the v326 adiabatic toy. f_a=$M_{\text{scal}}/128 \sim 2.4\times10^{11}$ GeV, $m_a \sim 23.8\,\mu$eV, lattice $\chi(T)$ n=8.16. [E] candidate scales fixed by the determinant-line atoms (128=2 dim $S^+|\mu_4|$). [C] the frozen SPINE angle $3\pi/5$=108°=$\pi N_{\text{fam}}/g_{\text{car}}$ gives $\Omega_a h^2 \sim 0.12$ in [0.08, 0.16] (lands on Ω_{DM} untuned) – the robust dominant-DM branch. [X] the HILLTOP 170° over-produces ($\sim 5.4\times$) – disfavoured. [X] kill: a finite-T solve robustly outside [0.08, 0.16] for the spine across $\chi(T)/g_*$ drops the dominant axion-DM branch. Stays [C]/[X]. [E][C][X]
`v374_ftransfer_qcd.py`	Track 3 (F_{transfer}): the m_p/m_e transfer as a typed uncertainty-budget + kill test – the falsifiability companion to v262 (the precise 2-loop point), replacing the v326 toy (which hardcodes 1836.15). [E] b_3= – (11 – $2n_f/3$)= – 7 (n_f=6 carrier). [C] two declared externals (v339): $\alpha_s(M_Z)$=0.1179±0.0009 and C_p=2.83±0.15, each $\sim$ 5.3%. [C] m_p/m_e central $\sim$ 1824 (v262) with a propagated ±7.5% band [$\sim$ 1690, $\sim$ 1960] CONTAINING the observed 1836.15 – no free dial. [X] kill: a future lattice C_p±0.03 + tighter α_s shrink the band to $\sim$ ±2%; a band then excluding 1836.15 drops the QCD-matching route (b_3= – 7 + the $SU(9)$-integer firewall v262). Distinct from v262's point; stays [C]. [E][C][X]
`v375_observatory_registry.py`	Track 4: the prediction OBSERVATORY – a status-typed CI over the frozen prediction registry (`freeze_file.csv`) that makes the falsifiability surface machine-checkable (no new physics). [E] falsifiability complete: every registry row carries a non-empty kill criterion (no orphan claims). [E] headline values re-derive from $\{\varphi_0^{\text{ret}}\}$: $\sin^2\theta_{12}$=$1/3-\varphi_0^{\text{ret}}/2$, $\sin^2\theta_{13}$=$\varphi_0^{\text{ret}}\,e^{-5/6}$, β_{rad}=$\varphi_0^{\text{ret}}/(4\pi)$. [C] live scorecard: θ_{12} vs JUNO first run (0.3092±0.0087)= 0.28σ (compatible); θ_{13} vs NuFIT 6.0 NO (0.02195±0.00058)= 2.0σ (the flagged most-tensioned core prediction); β_{rad} vs ACT DR6 (0.215±0.074)= 0.37σ (hint); r=$12/N_\star^2 \sim 0.004 <$ BK18 (compatible). [E] registry integrity: the single machine-readable falsifiability surface. Tooling for the closed/open pieces. [E][C]
`v376_seam_s3_centralcharge.py`	S3 closure stack: the central charge c=8 extracted NUMERICALLY from the lattice via Calabrese–Cardy finite-size entanglement scaling of the critical free-fermion content (the quantitative core of the MMST scaling-limit claim, on OUR model). [E] one critical complex mode gives c=1.0000 (Peschel correlation-matrix EE); [E] the collar is 16 Majoranas = 8 complex modes (v367), c additive ⇒ c=8=g_{car}+N_{fam}; [E] chiral c_-=8 (Chern $\lvert C\rvert$=1 per Majorana v367, non-gappable edge v368). [O] the specific net $(E_8)_1$ is v377/v378; the continuum existence is the cited MMST theorem (v336). Numerical c=8 from the lattice; does NOT close `SEAM.EQUIV.01`. [E][O]

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Script	**What it machine-checks** ***(continued)***
`v377_seam_s3_e8character.py`	S3 closure stack: the chiral c=8 content is the SPECIFIC net $(E_8)_1$, via the exact affine character q-series + the μ_4/GSO promotion. **[E]** χ=E_4/η^8=$q^{-1/3}(1+248q+4124q^2+\dots)$: leading $-c/24=-1/3 \Rightarrow c$=8; level-1 = 248= dim E_8 (one primary). **[E]** the μ_4/GSO promotion $SO(16)_1 \to (E_8)_1$: 248=240+8, 240= dim S^+ $g_{\rm car}$ $N_{\rm fam}$=16·5·3; 248=120+128= dim $SO(16)$+spinor. **[E]** discriminator: $(E_8)_1$ has 1 primary/248 currents vs $SO(16)_1$ 4 primaries/120. **[O]** continuum existence (v336); the μ_4 projection = the seam one-sidedness (S3). Pins the target net; does NOT close `SEAM.EQUIV.01`. **[E][O]**
`v378_seam_s3_modular.py`	S3 closure stack: the GENUS-1 discriminator – torus ground-state degeneracy = 1 (holomorphic = $(E_8)_1$), the "hard part" v344 flagged (genus-0 unique vacuum is necessary not sufficient). **[E]** KLM condensation: μ(carrier)= det $C(D_5)$ det $C(A_3)$=4·4=16; the order-4 μ_4 glue ($\lvert L\rvert$=4) $\Rightarrow \mu(E_8)$=16/16=1 (torus GSD = 1); index-2 $\Rightarrow \mu(SO16)$=16/4=4. **[E]** $\lvert\det C(E_8)\rvert$=1 vs $\lvert\det C(D_8)\rvert$=4. **[E]** the single $(E_8)_1$ character has modular T-eigenvalue $e^{-2\pi i/3}$ (h=0, c=8), 1-dim rep $\Rightarrow$ one character $\Rightarrow$ torus GSD = 1. **[O]** continuum existence (v336). Pins the net at genus-1; does NOT close `SEAM.EQUIV.01`. **[E][O]**
`v379_seam_s3_rp.py`	C7 / OS reconstruction: reflection positivity (the OS axiom) of the explicit gapped collar measure, verified numerically – the remaining constructive-QFT ingredient for the OS reconstruction. **[E]** the gapped p+ip band gives $G(\tau)$=mean$_k e^{-\varepsilon_k\tau}$ with $\varepsilon_k \geq$ gap/2 $\sim 1 > 0$ (v367) – a positive spectral measure; **[E]** the Hankel/reflection Gram matrix M_{ij}=$G(\tau_i$+$\tau_j)$ is PSD – OS reflection positivity holds; **[E]** neg control: a growing ghost mode makes it indefinite (non-vacuous). **[C]** RP + gap (v76/v337) + GNS (v240) + c=8 (v376) + $(E_8)_1$ (v377/v378) give the OS inputs; C7/`QG.AMB.01` is then DISCHARGED by the redundancy theorem v369, not built. **[O]** the full constructive measure stays the cited step. **[E][C][O]**
`v380_grav_kms_hessian.py`	The KMS Entire Hessian – the Stelle ghost is EXACTLY the Seeley–DeWitt truncation, and resummation pushes it to infinity (upgrades v304/v370 from "assume entire" to a derived statement). **[E]** the seam KMS cutoff $f(u)$=e^{-u} (v259) gives the dressed spin-2 propagator $e^{-p^2/M^2}/p^2 = 1/(p^2a)$ with $a(u)$=e^u entire & nowhere zero $\Rightarrow$ only the p^2=0 pole (no ghost). **[E]** each finite truncation carries the ghost: the R+R^2 order T_1=1+u has the Stelle zero $u=-1$, the Weyl^2 order T_2=1+u+$u^2/2$ the pair $-1\pm i$; the entire a has none. **[E]** the ghost DECOUPLES: $\lvert$nearest zero$\rvert$ of T_n grows monotonically with n $(1, 1.41, \dots, 3.07$ for n=1..8) $\to \infty$, so "entire" = "do not truncate". **[C]** $\Rightarrow$ perturbative graviton unitarity. **[O]** the exact off-shell curved-space Hessian identification is the cited heat-kernel resummation; perturbative-only, not the non-perturbative measure. **[E][C][O]**
`v381_qft4d_eg_allorder.py`	The all-order Epstein–Glaser / BRST contract for the perturbative 4D S-matrix $S_{\rm pert}$ – the "all-order closing statement" for the 4D perturbative QFT leg. Types the standard causal-perturbation-theory closure for the TFPT finite interaction class; it does *not* build a path integral. **[E]** POWER-COUNTING: every gauge+matter vertex (Yang–Mills + Dirac–Yukawa + Higgs + ghost) is a mass-dimension-4 marginal operator $\Rightarrow$ the EG singular order $\omega \leq 4 \Rightarrow$ a FINITE counterterm space at every order. **[E]** BRST NILPOTENCY s^2=0 verified via the Jacobi identity for $su(2)$ (residual 0) and $su(3)$ (10^{-16}), the carrier weak+colour content. **[E]** the seam gap Δ=6 ln(3/2)>0 (v302) gives the IR-safe adiabatic limit $g \to 1$. **[C]** imported: T_n exists to all orders by causal factorisation (Epstein–Glaser 1973 / Brunetti–Fredenhagen). **[C]** imported: all-order BRST invariance (Piguet–Sorella / Kugo–Ojima) $\Rightarrow$ Slavnov–Taylor to all orders. **[E]** scope fence: matter+gauge ONLY – the R^2/Weyl^2 gravity sub-sector is v304/v370/v380, not a local EG vertex; $S_{\rm pert}$ is not the ambient measure `QG.AMB.01` (v369). NET **[C]** (imported EG+BRST on the TFPT dim-4 class + gap), prerequisites **[E]**; extends v269/v271/v273/v278 from 1-loop to all-order **[E][C]**

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Script	What it machine-checks *(continued)*
`v382_alpha_quillen_exact.py`	Names the Quillen determinant-line VARIATION $\delta_\tau\big(\log\det_\zeta \Delta_{U(1)} + 8b_1 c_3^6 \log\varphi_{\text{seam}}\big) = 0$ as a tracked target (review Point 10), elevating `EM.WARD.01`'s residual to a citable [O]. [E] the Quillen split holds at the root $\alpha^3 - 2c_3^3\alpha^2 = 8b_1c_3^6 \ln(1/\varphi_{\text{seam}})$ at α^{-1}=137.0359992 ($F_{U(1)}$=0 IS the stationarity of the $U(1)$ determinant line, v341). [E] every coefficient a named atom: $8b_1$=$\frac{4}{5}$·41 a $U(1)$ index, c_3=$1/(8\pi)$ Gauss–Bonnet ($c_3^{3,6}$ heat-kernel powers), $q(D_5)$+$q(A_3)$=2 discriminant forms (v341/v48/v216). [E] the value α^{-1}=137.0359992168 stays closed (unique root + ablation, v3) – only "why this functional" is open. [O] THE TARGET: the from-first-principles AQFT/ζ proof that $F_{U(1)}$ is the EXACT Quillen functional. [C] a FACE of `SEAM.EQUIV.01` (v336): the seam $U(1)$ determinant line lives on the same holomorphic seam net, so once the net is $(E_8)_1$ its $U(1)$ sub-determinant is fixed – not a second independent gate, it names/tracks the pre-existing `EM.WARD.01` residual (v341/v342). No new number (reuses v341, Wolfram-mirrored) [E][C][O]
`v383_dynamics_universal.py`	DYNAMICS.UNIVERSAL.01: the UNIVERSAL spectral-gap / Perron–Frobenius principle – every TFPT sector is the SAME object (a gapped operator with a UNIQUE leading attractor = the physics, a spectral gap = the reason there is no free parameter), extending v303 from F_{transfer} to the whole theory. [E] FLAVOR (computed): the cusp transfer spectrum $\{1,(2/3)^6,(1/3)^6\}$ has a simple Perron leading 1 and gap $6\ln(3/2) > 0$ (v56/v82). [E] DISCRETE COMPILER (computed): the affine-E_8 adjacency has Perron eigenvalue 2 (Kac marks, v312/v313) and subleading $2\cos(\pi/5) = \varphi$, gap $2-\varphi = 1/\varphi^2 > 0$. [E] DECOUPLING MARGIN: $\Delta_{\text{eff}} = 6\ln(3/2) - 31/(4\pi^2) \approx 1.648 > 0$ (v76/v337). [E] TWO NUMBER-FIELD FACETS (v314): golden $\varphi \in \mathbb{Q}(\sqrt{5})$ (carrier-5, static) and $(2/3)^6 \in \mathbb{Q}$ (family-3, dynamic) are the prime-5 and prime-3 facets of the order-30 Coxeter clock $30 = 2{\cdot}3{\cdot}5 = h(E_8)$ – a factorisation, not an equality. [C] the other sectors are the same structure: gravity (v358/v359), QG saddle (v365), the holomorphic c=8 RG fixed point (v157/v158), the all-order S-matrix (v381), recovery (v221), F_{transfer} (v303), τ=i (v333). [C] META-THEOREM: gapped ⇒ unique attractor ⇒ parameter-free, theory-wide – one spectral-gap statement wearing many hats. A synthesis, no new number [E][C]
`v384_residual_certification.py`	RESIDUAL.CERTIFICATION.01: the residual matrix is CERTIFICATION, not CONSTRUCTION – after v369/v381/v382 there is NO open TFPT physics MECHANISM left; every residual is an external CERTIFICATE of already-determined structure, of three kinds: (A) external math proof, (B) theorem-forbidden, (C) external physics. A ledger-CI audit. [E] LEDGER CI: every residual claim id is present (no orphan). [E] NO-UNIT (B): c_3=$1/(8\pi)$ and g_{car}=5 are mass-dimension 0, so by No-Unit (v153/v364) no dimensionless compiler outputs an absolute scale ⇒ v_{geo} is forbidden by theorem, not a gap. [E] GAP-DECOUPLED (A): $\Delta_{\text{eff}} \approx 1.648 > 0$ (v76/v337) ⇒ every readout is independent of the ambient measure (`QG.AMB.01` a [C] redundancy, v369). [E] CLASSIFICATION COMPLETE: `SEAM.EQUIV.01` (continuum existence = cited MMST/Adamo, v336)→A, `ALPHA.QUILLEN.EXACT.01` (v382)→A, `QG.AMB.01` (v369)→A, v_{geo} →B, F_{transfer} (v371–v374, firewall v187)→C; ZERO open internal TFPT mechanisms. [C] CONVERGENCE: the only hard things left are theorems and external physics, not missing dynamics. An audit, no new number [E][C]

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Script	What it machine-checks *(continued)*
`v385_uvbranch_killtest.py`	UVBRANCH.KILLTEST.01: the optional carrier–Pati–Salam UV branch on the ALL-ORDER footing – a single citable kill-test surface with the proton-decay SAFETY hierarchy computed (not a fake τ_p window). A consolidation (does *not* re-run the RGE v246 nor re-derive the PS scale v249/v253). [E] carrier content = SM ($b = (41/10, -19/6, -7)$, v159/v246, no new states) and S_{pert} closes to all orders (v381) ⇒ the SM-non-unification is all-order-robust. [X] plain-SM 4D-GUT killed ($\sin^2\theta_W = 3/8$ vs 0.231; v246). [C] optional carrier-PS branch: $M_{PS} = \kappa M_{\text{scal}}$, $M_{\text{scal}} = c_3^{7/2}\bar{M}_{\text{Pl}} \approx 3.06\times10^{13}$ GeV, $\kappa \in [1.0, 1.2]$ (v253); content $\{1, 10, 16, 45\}$, no 126 (v247/v248); not the default (v265). [E] proton-decay SAFETY hierarchy: minimal PS gauge bosons are $SU(4)$ leptoquarks ($B{-}L$-conserving $q \leftrightarrow l$, rare LFV $K_L \to \mu e$), *not* the $p \to e^+\pi^0$ dim-6 operator (needs an $SO(10)/SU(5)$ diquark, absent); the binding bound $M_{PS} \gtrsim 10^6$ GeV is cleared by $M_{PS}/M_{\text{bound}} \sim 10^7 \gg 1$. [X] frozen kill tests (v265): κ leaving $O(1)$ / a forbidden rep / a new gauge-charged state / a rare-LFV signal below the bound; no fake τ_p. A consolidation, no new number [E][C][X]
`v386_grav_amplitude.py`	GRAV.AMPLITUDE.01: the entire-form-factor graviton-exchange AMPLITUDE is finite, UV-softened and tree-unitary – perturbative gravity as an explicit scattering problem, extending v304/v370/v380 from the propagator to the amplitude. The dressed spin-2 propagator $P = e^{-p^2/M^2}/p^2 = 1/(p^2 a)$, $a(u) = e^u$, $M = M_{\text{scal}}$. [E] one pole, ghost-free: a entire & nowhere zero (v380) ⇒ only pole at $p^2 = 0$, residue $e^0 = 1 > 0$. [E] UV-softened: ratio e^{-u} at $u = 1, 5, 10, 20 = 0.37, 6.7\times10^{-3}, 4.5\times10^{-5}, 2.1\times10^{-9}$ (faster than any power). [E] low-energy = GR + a calculable $-p^2/M^2$ correction ($M = M_{\text{scal}} = c_3^{7/2}\bar{M}_{\text{Pl}}$, v253). [C] tree unitarity: a single positive-residue pole, no negative-norm states (v370/v380). [C] the amplitude $A = (\kappa^2/2)(T{\cdot}P_2{\cdot}T')e^{-p^2/M^2}/p^2$ is GR at low energy, softened in the UV (perturbative-only; `QG.AMB.01` a [C] redundancy v369). No new number [E][C]
`v387_corrections_gap.py`	CORRECTIONS.GAP.01: the practical harvest of v383 – the same spectral gap that makes each sector parameter-free ALSO sets the SIZE of its first correction ($\text{correction}_n \sim (\lambda_2/\lambda_1)^n$). [E] FLAVOR: $\lambda_2 = (2/3)^6 = 64/729$ ⇒ first Koide correction ~ 0.088 (v56/v82). [E] RECOVERY: the same $(2/3)^6$ Petz rate (v221). [E] QG: capped by $\chi = 1/(1-(2/3)^6) = 729/665$ (v337). [E] DISCRETE COMPILER: the update $(A + 2I)/4$ fixes the Kac marks ($\lambda_1 = 1$) and decays at the golden $(\varphi + 2)/4 \approx 0.905$ (carrier-5 facet, v303/v312). [E] the rates split into the $(2/3)^6$ family (family-3, in $\mathbb{Q}$) and the golden family (carrier-5, in $\mathbb{Q}(\sqrt{5})$) – the two facets of v314/v383. [C] the correction calculus: the gap does double duty (WHY no free parameter AND HOW BIG the first correction), so sub-leading magnitudes are organised, not free. No new number [E][C]
`v388_corrections_budget.py`	CORRECTIONS.BUDGET.01: the gap correction (v387) is a *typed* budget, NOT a uniform band on every prediction – the $(\lambda_2/\lambda_1)^n$ size is a property of the distance from a gapped operator's leading eigenvector, so it is only defined where λ_2 exists. The prediction surface splits into four correction classes (already implied by v303). [E] SEAM-GAPPED: Koide (F_{pole} v303), recovery (v221), the QG bound (capped by $\chi = 729/665$, v337) carry $(2/3)^6$; the discrete compiler decays at the golden $(\varphi + 2)/4$ (v312). [E] KOIDE FIRST CORRECTION: F_{pole} is the ONE seam-gapped prediction, rate $(2/3)^6 \approx 0.0878$ (v371), source quotient 0.33% below 2/3 – the rate is the suppression SCALE, not the residual. [E] EXACT-IDENTITY class ($\det R = 8$, $N_\Phi = 1$, $\theta_{\text{eff}} = 0$, $\sin^2\theta_{13}$): no λ_2 ⇒ correction 0 STRUCTURALLY (a band would contradict [E]). [E] EXTERNAL-RATE class (η_B/axion/m_p, m_e): the gapped shape with external rates (only F_{pole} has the seam rate, v303), firewalled (v187). [E] FIXED-POINT class: the texture is already explicit (θ_{12} $\varepsilon = c_3 + 36c_3^4$). [C] a uniform band on all predictions is WRONG; the size is keyed to the mechanism class. A typing of v387, no new number [E][C]

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Script	What it machine-checks *(continued)*
`v389_grav_loop_finiteness.py`	GRAV.LOOP.FINITE.01: the entire-form-factor graviton is UV-finite at the *loop* level by power counting – the honest next step past v386 (tree), extending v304/v370/v380/v386 from the propagator to the superficial degree of divergence. Honest scope: the standard infinite-derivative/nonlocal-gravity finiteness statement, NOT a full proof of perturbative QG. **[E]** super-polynomial falloff $p^k e^{-p^2/M^2} \to 0$ for every k. **[E]** one dressed line turns a quadratic divergence into the finite $(M^2/2)e^{-p_0^2/M^2}$; the GR $\int p^3/p^2\, dp$ diverges quadratically. **[E]** the superficial degree $\to -\infty$ at every loop order (dressed cutoff-ratio $\to 1$ vs GR $\to 4$). **[E]** the regulator is the SEAM scale $M = M_{\text{scal}} = c_3^{7/2}\bar{M}_{\text{Pl}}$ (v253/v259), not a dial. **[C]** loop UNITARITY of the nonlocal form factor (Lorentzian continuation / macro-causality) stays the honest open point – tree unitarity is v386; NOT claimed solved. No new number **[E][C][O]**
`v390_prime2_facet.py`	COXETER.PRIME2.01: the prime-2 facet of the order-30 Coxeter clock, completing v383. The clock $30 = 2{\cdot}3{\cdot}5 = h(E_8)$ had two named facets (golden $\varphi \in \mathbb{Q}(\sqrt{5})$, prime-5/carrier; $(2/3)^6 \in \mathbb{Q}$, prime-3/family); this names the prime-2 facet as the Gaussian field $\mathbb{Q}(i)$ — $\tau = i$ and the order-4 $\mu_4 = \mathbb{Z}/4$ square seam deck (v282/v288/v333), structure already present. **[E]** $h(E_8) = 30$ factors into exactly three primes. **[E]** one quadratic facet field per prime, each ramified at its prime: $\operatorname{disc}\mathbb{Q}(i) = -4$, $\operatorname{disc}\mathbb{Q}(\sqrt{-3}) = -3$, $\operatorname{disc}\mathbb{Q}(\sqrt{5}) = 5$; ramified primes $\{2, 3, 5\}$, product $= 30$. **[E]** prime-2 Gaussian facet: min. poly $x^2 + 1$, $\tau = i$, $j(i) = 1728 = 12^3$, $\lvert\mu_4\rvert = 4 = 2^2$ (the static seam-orientation facet). **[E]** prime-5 golden (v383): $2\cos(\pi/5) = \varphi$ (pentagon). **[E]** prime-3 family (v383/v220): $(2/3)^6$ plus the Eisenstein ζ_3 (hexagonal CP $\pi/3$). **[C]** completion: each prime carries a facet (square/hexagon/pentagon = seam/CP/carrier), the prime set forced by $h(E_8)$. A synthesis, no new number **[E][C]**
`v391_alpha_quillen_progress.py`	ALPHA.QUILLEN.PROGRESS.01: an *honest attempt* at the external target ALPHA.QUILLEN.EXACT.01 (v382) – it REDUCES the open step to a sharper named sub-target via a solvable model but does NOT close it; ALPHA.QUILLEN.EXACT.01 stays **[O]** (external math, type A, v384). **[E]** solvable model $\det_\zeta(-\partial_\tau^2 + m^2)$ on a circle $= (2\sinh(\beta m/2))^2$, m-variation $\beta\coth(\beta m/2) = 2/m + (\beta^2/6)m - (\beta^4/360)m^3 + \dots$ a closed heat-kernel/Laurent series (m^3 coeff $-1/360$ verified). **[E]** a ζ-det variation is structured heat-kernel data, so v382's $8b_1c_3^6$ counterterm (a Gauss–Bonnet heat-kernel power) is the right *kind* of object. **[E]** the v382 split re-verified at $\alpha^{-1} = 137.0359992$. **[E]** reduction: the open step reduces to the named sub-target "seam Seeley–DeWitt coefficient $= 8b_1c_3^6$". **[O]** NOT CLOSED: the actual seam ζ-det proof stays open, SEAM.EQUIV.01 continuum existence (v336) the other external **[O]**; α^{-1} stays **[E]**. An attempt that reduces, does not close **[E][O]**
`v392_seam_s3_scalinglimit.py`	SEAM.S3.SCALINGLIMIT.01: an *honest attack* on the one open lemma of SEAM.EQUIV.01 (the continuum scaling-limit existence, v336) – it does NOT close it (the abstract existence is the cited MMST theorem); it STRENGTHENS the case with the existence-relevant observable v376 did not test: that the finite-size data *converge*. **[E]** the Calabrese–Cardy EE slope $c_1(L)$ at $L = 66 \dots 514$ is a monotone Cauchy-like sequence approaching 1 ($\lvert c_1(L) - 1\rvert$ strictly decreasing). **[E]** a $1/L$ Richardson fit gives $c_1(\infty) \approx 1.000$, so the 16-Majorana collar $c = 8c_1 \approx 8.00 = g_{\text{car}} + N_{\text{fam}}$ is reached *in* the limit. **[E]** the successive differences are strictly decreasing (numerical Cauchy, necessary for existence). **[O]** convergent data are evidence, not a proof of existence (MMST v336), and do not pin $(E_8)_1$ vs $SO(16)_1$ (holomorphy v377/v378). Strengthens the residual, does NOT close it **[E][O]**

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Script	**What it machine-checks** *(continued)*
`v393_corrections_numeric.py`	CORRECTIONS.NUMERIC.01: the typed correction budget (v388) turned into ACTUAL first-correction magnitudes – a computed theoretical-uncertainty surface, the short-term testable harvest of v388. **[E]** FIXED-POINT band ($\theta_{12}/\theta_{13}/\lambda_C/\beta_{\rm rad}$) = the explicit φ_0 puncture term $36c_3^4/c_3 = 9/(128\pi^3) \approx 0.227\%$ (exact; $= \theta_{12}$'s ε first-correction). **[E]** SEAM-GAPPED band (Koide/recovery) $= (2/3)^6 \approx 8.78\%$; the QG bound capped by $\chi - 1 = 64/665 \approx 9.62\%$ (v337). **[E]** EXACT-IDENTITY band = 0 structurally ($\det R = 8$, $N_\Phi = 1$, $\theta_{\rm eff} = 0$). **[E]** EXTERNAL-RATE band = external physics quoted (m_p/m_e ±7.5% v374; n_s/r $N_\star$ band; η_B 1.07× v212), fenced v187. **[C]** each prediction gets "central value ± computed first correction". A harvest of v388, no new number **[E][C]**
`v394_coxeter_atoms.py`	COXETER.ATOMS.01: the clockwork coherence – the three Coxeter primes ARE the three anchor atoms ARE the three number-field facets, and the prime-2 seam is the common dynamical factor. A bird's-eye **[E]** re-reading tying v390 (facets) to the anchor $a = (1, 1, 2)$ and v319 (5x6 clock); registers `RES.COXETER.SYMMETRY.01` as a tracked **[O]**. **[E]** $a = (1, 1, 2)$: $e_3{=}2{=}\lvert Z_2\rvert$, $p_0{=}3{=}N_{\rm fam}$, $e_2{=}5{=}g_{\rm car}$, product $30{=}h(E_8)$. **[E]** atoms = facet fields (v390): $2{\to}\mathbb{Q}(i)$, $3{\to}\mathbb{Q}(\sqrt{-3})$, $5{\to}\mathbb{Q}(\sqrt{5})$, each ramified at its prime. **[E]** $\chi{=}729/665{=}3^6/(3^6{-}2^6)$, $(2/3)^6{=}2^6/3^6$, exponent $6{=}2N_{\rm fam}$. **[E]** both dynamic rates carry prime-2: $2/3{=}\lvert Z_2\rvert/N_{\rm fam}$, $(\varphi{+}2)/4{=}(\varphi{+}2)/\lvert\mu_4\rvert$, $\lvert\mu_4\rvert{=}4{=}2^2$. **[E]** $30{=}5{\times}6{=}g_{\rm car}(2N_{\rm fam})$. **[C]** the seam is the universal dynamic engine. **[O]** `RES.COXETER.SYMMETRY.01` registered, not closed **[E][C][O]**
`v395_hypergraph_coupled.py`	HYP.COUPLED.01: the coupled hypergraph rewrite – one local rule unifies v299/v327/v324 (carrier × family). **[E]** LOCAL COUPLED MICRO-RULE: state on 9 affine-E_8 nodes × 3 family slots; one micro-step = network lazy diffusion on each cusp column + v327 M on each node's family vector; steps commute, $T_{\rm micro} = T_{\rm net} \otimes M$. **[E]** JOINT ATTRACTOR: top eigenvalue 1 with eigenvector marks$\otimes e_0$; iteration from any positive start converges. **[E]** ONE HAND = RECOVERY: one network step + $2N_{\rm fam}$ family micro-steps gives $(2/3)^6$ on marks$\otimes v_{\rm surv}$. **[E]** v324 EMERGES: $T_{\rm full} = T_{\rm net} \otimes T_{\rm cusp}$ shares the same invariants; $T_{\rm cusp}$ is the 6-hand readout in the Galois/cusp basis (v317). **[E]** GROWTH + FIBER: v299 $E_6{\to}E_7{\to}E_8{\to}\hat{E}_8$ unchanged with M-fiber attached. **[O]** 3-arm seed (P_2) + φ_0 (v312) remain irreducible **[E][O]**
`v396_phi0_icosahedral.py`	HYP.PHI0.01: the φ_0 leading term from icosahedral combinatorics (sharpening v312/v395). **[E]** ICO COUNTS ANCHOR-FORCED: $V{=}12$, $E{=}30{=}h(E_8)$, $F{=}20{=}g_{\rm car}(g_{\rm car}{-}1)$, $\lvert\,\mathrm{Aut}\,\rvert{=}120{=}4h{=}\lvert 2I\rvert$. **[E]** LEADING TERM EXACT: $\varphi_0^{\rm tree}{=}1/(6\pi){=}(4/3)c_3{=}F/(4h\pi){=}F/(\lvert\,\mathrm{Aut}\,\rvert\pi){=}(F/(g_{\rm car}N_{\rm fam}))c_3$. **[E]** $F/h{=}\lvert Z_2\rvert/N_{\rm fam}$; leading term $= (F/h)/(4\pi){=}c_3(g_{\rm car}{-}1)/N_{\rm fam}$. **[E]** GAUSS–BONNET: icosahedron deficit sum 4π; $c_3{=}1/(\lvert Z_2\rvert 4\pi)$ (v216/v342). **[E]** HONEST NEGATIVE: puncture $48c_3^4$ not a rational hypergraph fraction. **[C]** tree-level seed = (hyperedges)/$(g_{\rm car}N_{\rm fam}) \times c_3$, not an independent injection; puncture **[O]**
`v397_external_clock_probe.py`	FR.CLOCK.PROBE.01: a *falsifiable* probe of how external the "external physics" really is – do the frontier scheme-numbers land on the $\{2, 3, 5\}$ clock (the anchor atoms, v394) or not? Strict No-Free-Pattern (v100): "on the clock" = an EXACT match, not a wide-band fit. **[E]** η_B decuple $A_\Lambda{=}10{=}\lvert Z_2\rvert g_{\rm car}$ ON CLOCK (v212). **[E]** Koide rate $(2/3)^6{=}(\lvert Z_2\rvert/N_{\rm fam})^6$ ON CLOCK (v303). **[E]** axion spine angle $3\pi/5{=}\pi N_{\rm fam}/g_{\rm car}$ ON CLOCK (v211). **[E]** proton $C_p \sim 2.83{\pm}0.15$ NOT discriminating ($2^{3/2}, 17/6, 14/5$ all fit the band) ⇒ EXTERNAL (firewall v187/v374). **[C]** 3 of 4 land exactly on the clock; the irreducible external residue shrinks to the proton QCD number C_p – "external physics" was overstated. **[E]** anti-numerology: refuses to assign C_p a clock value **[E][C]**

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Script	What it machine-checks *(continued)*
`v398_seam_state_rigidity.py`	**Seam State Rigidity Theorem** (Paper A, the $G_{\rm net}$ closure contract): the scattered state-invariance reductions (v177/v198/v199/v308/v309/v335) as ONE named target. [E] RP-definability hinge (OS canonical transfer $T{=}e^{-H}$ from RP + reflection alone, v54, on a quasi-free state v155, no μ_4 normal form); [E] band-limited core $[\rho,\lvert k\rvert]{=}0$ and character-block-diagonal sub-principal symbol swept $N{=}4..64$ (beyond the 3-dim H^1); [E] holomorphy $\Rightarrow (E_8)_1$ ($c{=}8$, $\lvert\det\mathrm{Cartan}\rvert{=}1$ vs 4, index $4{=}\lvert\mu_4\rvert$, gap $6\log\frac{3}{2}{>}0$). [O] the one residual: $[\rho,\Lambda_\Sigma]{=}0$ on the full L^2 (intrinsic Bisognano–Wichmann). A contract+evidence module, NOT a closure of `SEAM.EQUIV.01` (continuum = cited MMST v336)
`v399_gravity_complete.py`	**Gravity-complete = Boundary-complete + Ambient-redundancy** (Paper B accounting): combines the parameter-free local equation (v358/v359) and the ambient redundancy (v369). [E] $G_{ab}{+}\Lambda g_{ab}{=}(1/c_3)T_{ab}$, $1/c_3{=}8\pi$ (four c_3-roles), Λ prefactor $3/(4\pi^2)$; [E] redundancy discriminators ($\lvert\det\mathrm{Cartan}\rvert{=}1$ vs 4, torus GSD=1, Petz $(2/3)^6$, margin $1.648{>}0$); [E] $\mathcal{O}_{\rm phys}\subset\mathcal{A}_\Sigma$ over five readout classes (Newton 8π, Λ, scalaron c_3^7, horizon $1/\lvert\mu_4\rvert$, recovery $(2/3)^6$) – all boundary/seam/gap. [C] the verdict; [O] `SEAM.EQUIV.01` + intrinsic BW (no new gate)
`v400_grav_nonlocal_action.py`	**Nonlocal spectral-gravity action** with the local $R{+}R^2$ as its IR projection (Paper C, consolidating v304/v370/v380/v386). [E] the entire form factor $a{=}e^u$ gives the dressed propagator $1/(p^2a)$ ONE pole (residue 1, ghost-free); [E] every truncation carries a Stelle zero whose nearest-zero modulus grows monotonically (v380); [E] the IR order $a{=}1{+}u{+}u^2/2{+}\ldots$ reproduces the spectral-action $R{+}R^2$ (v28/v36) with the atom-fixed scale $M_{\rm scal}^2/\bar{M}_{\rm Pl}^2{=}c_3^7$ ($7{=}g_{\rm car}{+}N_{\rm fam}{-}1$, v253). [O] the entire-$a$ analyticity assumption + perturbative-only (not the ambient measure `QG.AMB.01`, a [C] redundancy v369)
`v401_metrology_closure.py`	**Metrology closure** of the final input set $\{a,\pi,v_{\rm geo}\}$ (Paper D, consolidating v153/v274/v364/v384). [E] $a{=}(1,1,2)\to(e_1,e_2,e_3){=}(4,5,2){=}(\lvert\mu_4\rvert,g_{\rm car},\lvert Z_2\rvert)$, $c_3{=}1/(2e_1\pi){=}1/(8\pi)$, 0 dimensionless dials + 1 unit; [E] dimensional bookkeeping (No-Unit v153) mass/$1G/\Lambda/U_{\rm point}/(m/\mu)$= number $\times v_{\rm geo}^{1,2,4,1,0}$; [E] gravity adds no unit (v364); [E] G-vs-Λ over-determination 0.11% (v274). [O] $v_{\rm geo}$ theorem-forbidden ($\mathrm{TOE}_{\rm strict}{=}\mathrm{TOE}_{\rm dimensionless}{+}[v_{\rm geo}]$, v384 type B)
`v402_ftransfer_suite.py`	$F_{\rm transfer}$ **functor suite** (Paper E): the four frontier sectors as ONE functor $F_{\rm transfer}{=}F_{\rm observable}\circ F_{\rm threshold}\circ F_{\rm RG}$ (consolidating `FR.POLE/BOLTZMANN/RELIC/QCD.SOLVE`). [E] the composition law (each observable factors source→threshold→RG→observable); [E] exact atom sources ($Q_{\rm src}$ v93; $A_\Lambda{=}2g_{\rm car}{=}10$, $\delta_{CP}{=}4\pi/3$ v9/v212; $f_a{=}M_{\rm scal}/128$ v185; $b_3{=}{-}7$ v164/v262); [E] a committed kill test per sector. [C] the firewall: physical predictions stay [C]/[O], only the algebraic source is [E] (never re-pressed into the compiler, v187)
`v403_facet_compositum.py`	ARITH.HULL.01: the three number-field facets (v390) COMPOSE to $K=\mathbb{Q}(i,\sqrt{-3},\sqrt{5})$, a degree-8= rank E_8 abelian field – the arithmetic hull dual to the E_8 lattice hull. [E] the squarefree parts $-1,-3,5$ are independent in $\mathbb{Q}^\times/(\mathbb{Q}^\times)^2$, so $[K:\mathbb{Q}]=2^3=8=\mathrm{rank}\,E_8=g_{\rm car}{+}N_{\rm fam}$, Galois group $(\mathbb{Z}/2)^3$ order 8; $2^{\omega(h(E_8))}=2^3=8=\mathrm{rank}\,E_8$ is E_8-special (the E_7 control $h{=}18$, $2^2{=}4{\neq}7$ fails). [E] exactly 7 quadratic subfields $\{-1,-3,5,3,-5,-15,15\}$, split 4 imaginary $=\lvert\mu_4\rvert$ (CM/dynamic) + 3 real $=N_{\rm fam}$ (RM/static). [E] ramified only at $\{2,3,5\}$ = the atoms = the primes of $h(E_8){=}30$. [C] K is the abelian $(\mathbb{Z}/2)^3$ arithmetic hull dual to E_8; the 7 quadratic subfields a candidate for the 7 E_8 audit slices [E][C]

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Script	**What it machine-checks** ***(continued)***
`v404_e8_char_cm_duality.py`	E8.CMDUAL.01: the $(E_8)_1$ vacuum character $\chi_{E_8}{=}E_4/\eta^8{=}j^{1/3}$ at the two CM points – 12 at the seam ($\tau{=}i$), 0 at the family ($\tau{=}\omega$), extending v282. **[E]** $\chi^3_{E_8}{=}j$ (q-series). **[E]** SEAM $\tau{=}i$ ($\mathbb{Q}(i)$, order 4): cross-ratio $2 \Rightarrow j{=}1728 \Rightarrow \chi_{E_8}(i){=}1728^{1/3}{=}12$ (E_8 present, v214/v282). **[E]** FAMILY $\tau{=}\omega$ ($\mathbb{Q}(\sqrt{-3})$, order 6): the equianharmonic λ ($\lambda^2{-}\lambda{+}1{=}0$) $\Rightarrow j{=}0 \Rightarrow \chi_{E_8}(\omega){=}0$ (E_4 vanishes, v220/v267). **[E]** $\tau{=}i, \tau{=}\omega$ are the ONLY elliptic points of $\mathrm{PSL}(2,\mathbb{Z})$, so the values are forced. **[C]** the seam/flavor rigidity duality: the seam is rigid where E_8 is present ($\chi{=}12$, `SEAM.EQUIV.01` closes), the flavor/CP sector is the soft residual where E_8 degenerates ($\chi{=}0$); the dual-keystone candidate (flavor = the $j{=}0$ degeneration of $(E_8)_1$) **[E][C]**
`v405_seam_equiv_omega.py`	**SEAM.EQUIV.02** – the dual keystone at $\tau{=}\omega$: the family/flavor sector as the order-3 Eisenstein/A_2 face of E_8, dual to the seam = $(E_8)_1$ at $\tau{=}i$ (v404). A named contract (genre of v286/v398), NOT a closure. **[E]** $\chi_{E_8}(\omega){=}0$ (E_8 degenerates at the family CM point) vs $\chi_{E_8}(i){=}12$. **[E]** A_2 is the $\tau{=}\omega$ Eisenstein lattice (Cartan $\det{=}3{=}N_{\mathrm{fam}}$, $h{=}3{=}N_{\mathrm{fam}}$, $\mu_6{=}\mu_3{\times}\mu_2$). **[E]** R lives in $E_6{\times}A_2$: $248{=}\lVert R\rVert_F^2{+}\det R{+}2(\mathbf{1}^\top Ra)N_{\mathrm{fam}}{=}78{+}8{+}2{\cdot}27{\cdot}3$, $\lVert R\rVert_F^2{=}78{=}\dim E_6$, $\det R{=}8{=}\dim A_2{=}\operatorname{rank} E_8$. **[E]** $\delta_{\mathrm{PMNS}}{=}4\pi/3$ is the $\tau{=}\omega$ hexagonal CM phase ($\arg\zeta_6{=}\pi/3$, v220/v233). **[E]** the cusp weights $\{0,1/3,2/3\}$ are the order-3 deck $= N_{\mathrm{fam}} = \det Q$ (v72), $\mathrm{Spec}(Q_+){=}(1,2,3)$ (v69/v137). **[C]** the family/flavor sector is the order-3 Eisenstein/A_2 deck at $\tau{=}\omega$; the two keystones sit at the only two elliptic points of $\mathrm{PSL}(2,\mathbb{Z})$. **[O]** residual = the canonical $\mathbb{Z}_3$ deck map (v140) + the v_{geo}-class scale (No-Unit v153) **[E][C][O]**
`v406_pmns_phase_origin.py`	FLAV.PMNS.CLOSE.01: the PMNS dynamics closed modulo the $\tau{=}\omega$ keystone (v405) – the CP-phase ORIGIN is the family CM point and the absolute scale is one v_{geo}-class unit, so PMNS has zero free flavor dials beyond v_{geo}. **[E]** the three angles ($\sin^2\theta_{12}{=}\frac{1}{3}{-}\frac{\varphi_0^{\mathrm{ret}}}{2}$, $\theta_{23}{=}45°$, $\sin^2\theta_{13}{=}\varphi_0^{\mathrm{ret}}e^{-5/6}$; v9/v268) assemble to one unitary U_{PMNS} (v270). **[C]** CP-PHASE ORIGIN (advance over v270): $\delta_{\mathrm{PMNS}}{=}4\pi/3$ is the $\tau{=}\omega$ hexagonal CM phase (v405/v220/v233), upgrading v270's **[O]** assembled input to a keystone consequence. **[C]** $J_{\mathrm{PMNS}}{=}{-}0.0297$ derived. **[E]** the absolute scale = one seesaw ratio $m_3{=}(y_\nu v_{\mathrm{EW}})^2/M_R$ = a v_{geo}-class UV input (v272/v153), NOT a flavor gap. **[C]** net: PMNS = angles + δ [keystone] + J [derived] + scale [v_{geo}], zero free flavor dials. **[O]** residual = the seesaw operator realisation (M_R not a clean $\bar{M}_{\mathrm{Pl}}$ power, v272/v184; one-parameter carrier-normalised candidate v481) + the normal-ordering Σm_ν floor kill test **[E][C][O]**
`v407_dn_pairings_omega.py`	FLAV.SELECTOR.CLOSE.01: the R4′ residue folds into the $\tau{=}\omega$ keystone (v405) – the three (d,n) selector pairings ARE the $\tau{=}\omega$ family-slice atoms. **[E]** the frame $(\mathbf{1},a,\sigma)$ is a $\mathbb{Z}^3$ basis with $\det{=}11{=}\lVert Pl(K)\rVert_1$, and $n{=}(5,{-}9,6)$ is the unique covector with $n{\cdot}\mathbf{1}{=}2$, $n{\cdot}a{=}8$, $n{\cdot}\sigma{=}121{=}11^2 \Rightarrow (d,n){\Rightarrow}R$ columnwise (v136/v139). **[E]** $n{\cdot}a{=}8{=}\operatorname{rank} E_8{=}\dim A_2{=}\det R$ – the anchor-pairing is the A_2 ($\tau{=}\omega$) block dimension that reads R in $E_6{\times}A_2$ (v405). **[E]** the pairings $\{2,8,121\}{=}\{\lvert Z_2\rvert, \operatorname{rank} E_8{=}\dim A_2{=}\det R, \lVert Pl(K)\rVert_1^2\}$ are the $\tau{=}\omega$ family-slice atoms (the μ_2 sheet, the A_2 block, the Plucker frame). **[E]** the square selects the point (σ-pairing $11(11{+}t)$, a perfect square only at $t{=}0$). **[C]** deriving (d,n) (R4′) folds into `SEAM.EQUIV.02` – $d{=}\frac{3}{2}a{-}2\mathbf{1}$ anchor-derived (v139), the pairings the slice atoms. **[O]** the lone flagged residue “why $11{=}\lVert Pl(K)\rVert_1$” **[E][C][O]**

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Script	**What it machine-checks *(continued)***
`v408_phi0_puncture_heatkernel.py`	`HYP.PHI0.PUNCTURE.01`: name the φ_0 puncture term $\delta_{\rm top}{=}48c_3^4$ as the order-4 local boundary heat-kernel (Seeley-DeWitt) contact term of the μ_4 puncture divisor – the analytic complement of the icosahedral tree term (v396, `HYP.PHI0.01`). [E] THE SPLIT IS EXACT: $\varphi_0{=}\varphi_0^{\rm tree}{+}\delta_{\rm top}{=}1/(6\pi){+}48c_3^4$ (v396/`ARCH.QUAD.01`). [E] PUNCTURE COUNT = $\Omega_{\rm adm}{=}N_{\rm fam}\dim S^+{=}48$, order = $\lvert\mu_4\rvert{=}4$: $\delta_{\rm top}{=}\Omega_{\rm adm}c_3^{\lvert\mu_4\rvert}$. [E] CLOSED FORM $\delta_{\rm top}{=}48c_3^4{=}3/(256\pi^4)$; $c_3{=}1/(\lvert Z_2\rvert 4\pi)$ Gauss-Bonnet (v216/v342) so c_3^4 is a fourth-order boundary heat-kernel power (cf. v391). [E] HONEST NEGATIVE: $\delta_{\rm top}$ not a rational hypergraph fraction ($\delta_{\rm top}/\varphi_0^{\rm tree}{<}1\%$). [O] THE TARGET: prove $\Omega_{\rm adm}c_3^4$ IS the order-4 boundary Seeley-DeWitt contact term of the μ_4 puncture summed over the $\Omega_{\rm adm}$ admissible states. [C] two engines, two terms; a FACE of the seam boundary analysis (the c_3 datum), not a new gate [E][C][O]
`v409_coxeter_prime2_lemma.py`	`RES.COXETER.SYMMETRY.01` closed as a STRUCTURAL LEMMA: prime-2 is NECESSARY but NOT autonomous – no prime-2-only attractor exists (resolves the v394/`COXETER.ATOMS.01` open question's falsifiable corollary). [E] SHEET INVOLUTION $\sigma{=}\operatorname{diag}(1,-1,-1)$, $\sigma^2{=}I$, spectrum $\{+1,-1\}$ (parity). [E] SHEET PROJECTOR $P{=}(I{+}\sigma)/2{=}\operatorname{diag}(1,0,0)$, $P^2{=}P$, spectrum $\{0,1\}$ (projection). [E] NO CONTRACTION RATE: none of $\{-1,0,1\}$ lies in $(0,1)$, whereas the genuine rates $2/3$ and $(1/3)^6$ do – prime-2 alone gives parity/projection, never a nonzero rate. [E] FACTOR NOT RATE: $2/3{=}\lvert Z_2\rvert/N_{\rm fam}$ needs prime-3, $(\varphi{+}2)/4{=}(\varphi{+}2)/\lvert\mu_4\rvert$ with $\lvert\mu_4\rvert{=}2^2$ needs the golden φ (prime-5). [E] $30{=}g_{\rm car}(2N_{\rm fam})$, carrier static, 2·3 dynamic. [C] prime-2 is the universal dynamical parity factor, necessary in every sheet-paired transport but not an autonomous attractor; dynamics begins after coupling to family-3 or carrier-5 ($\mu_4{=}2^2$ the Galois bridge) [E][C]
`v410_sheet_generator_binary.py`	The *sheet generator* $V{=}Q\operatorname{diag}(0,1,1)$ (rank-2 sheet axis, v95/v218) is a binary internal compiler: $V^n{=}\left(\begin{smallmatrix}0 & 2^{n-1} & 0\\ 0 & 2^n & 0\\ 0 & 2^{n+1}{-}2 & 1\end{smallmatrix}\right)$ (induction), so $V^n\mathbf{1}{=}(2^{n-1},2^n,2^{n+1}{-}1){=}(1,2,3),(2,4,7),(4,8,15),(8,16,31)$; bilinears $\mathbf{1}^TV^n\mathbf{1}{=}7{\cdot}2^{n-1}{-}1$, $\mathbf{1}^TV^na{=}7{\cdot}2^{n-1}$, $a^TV^na{=}11{\cdot}2^{n-1}$, $a^TV^n\mathbf{1}{=}11{\cdot}2^{n-1}{-}2$ give $(6,7,9,11)$ then $13{=}\Delta_Q$, $27{=}\mathbf{1}^TRa$, 55, $56{=}\dim\mathbf{56}_{E_7}$; the spine is FORCED by $\operatorname{Spec}(V){=}\{0,1,2\}$, the carrier (rank $E_8{=}8$, $\dim S^+{=}16$, $248/8{=}31$) and theta ($\sigma_3(2){=}9{=}a^TV\mathbf{1}$, $\sigma_3(3){=}28{=}\mathbf{1}^TV^3a{=}\det(I{+}R)$, $\sigma_3(5){=}126$) readings audit-typed [E][C]
`v411_ud_ratio_vpower.py`	The quark u/d ratio is a pure V-power readout: the established $c_u/c_d{=}g_{\rm car}\lVert\operatorname{Pl}(K)\rVert_1/(N_{\rm fam}^2\Delta_Q){=}55/117$ (v94) re-expresses EXACTLY via the sheet generator (v410) as $\boxed{c_u/c_d=\frac{\mathbf{1}^TV^4\mathbf{1}}{(a^TV\mathbf{1})(\mathbf{1}^TV^2\mathbf{1})}}=\frac{55}{9\cdot 13}$, with $55{=}\mathbf{1}^TV^4\mathbf{1}$, $9{=}a^TV\mathbf{1}{=}N_{\rm fam}^2$, $13{=}\mathbf{1}^TV^2\mathbf{1}{=}\Delta_Q$; the same number as $5{\cdot}11/(9{\cdot}13)$, so the numerator repackages $g_{\rm car}{\cdot}11{=}55$. An exact *re-encoding*: the value and the u/d identification stay [C] (U_wall readout rigidity) [E][C]
`v412_sheet_source_corner_J.py`	The *sheet source corner* J: the determinant surface has two s-independent iso-volume walls (v135) $t{=}{-}1$ ($\det{=}4{=}\lvert\mu_4\rvert$, carries K) and $t{=}{-}2$ ($\det{=}2{=}\lvert Z_2\rvert$); the operator on the Z_2 wall, $J{=}M(1,-2){=}C{-}2V{=}\left(\begin{smallmatrix}4 & 1 & 0\\ 4 & 1 & 2\\ 5 & 1 & 1\end{smallmatrix}\right)$, completes the sheet axis $J,K,C,F{=}M(1,-2..1)$. $\chi_J{=}\lambda^3{-}6\lambda^2{+}3\lambda{-}2\Rightarrow(\operatorname{tr},e_2,\det){=}(6,3,2){=}(\lvert R^+(A_3)\rvert,N_{\rm fam},\lvert Z_2\rvert)$; $a^TJa{=}30{=}h(E_8)$, $\det(I{+}J){=}12{=}\dim\mathfrak{g}_{\rm SM}{=}\chi_{E_8}(i)$, $\det(2I{+}J){=}40{=}\lvert R(D_5)\rvert$ (Lie readings audit-typed like the G_2/F_4 shadows of v95/v218) [E][C]
`v413_sheet_characteristic_calculus.py`	The sheet axis is a discrete *characteristic-difference calculator*: on $M_t{=}C{+}tV$ (v95/v218) the elementary-symmetric coefficients encode atoms as DIFFERENCE ORDERS – $e_1{=}3t{+}12$, $e_2{=}2t^2{+}15t{+}25$, $e_3{=}\det M_t{=}4t^2{+}14t{+}14$ give $\Delta e_1{=}3{=}N_{\rm fam}$, $\Delta^2e_2{=}4{=}\lvert\mu_4\rvert$, $\Delta^2e_3{=}8{=}\operatorname{rank}E_8$ (the e_3 curvature cross-cited to v218/v224; the full (e_1,e_2,e_3) ladder is new); the anchor energy $a^TM_ta{=}52{+}11t$ (step $11{=}a^TVa$, v410) reads $30{\to}41{\to}52{\to}63{=}(h(E_8),10b_1,\dim F_4,7N_{\rm fam}^2)$ [E][C]

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Script	What it machine-checks *(continued)*
`v414_center_resolvent_portal.py`	The center C is a *resolvent portal* $G_2\to F_4\to E_8$: $C=M(1,0)$ (v95) is the $t=0$ value of V; $\operatorname{tr} C=12=\dim\mathfrak{g}_{\rm SM}=\chi_{E_8}(i)$, $\det C=14=\dim G_2$, $\sum C=31=2^{g_{\rm car}}-1=248/8$; the shifted determinants climb $\det(C)=14=\dim G_2$, $\det(I+C)=52=\dim F_4$, $\det(2I+C)=120=\|R^+(E_8)\|$ – the shell ladder $G_2\to F_4\to E_8^+$ as $\det(C+kI)$, $k=0,1,2$ (the resolvent ladder is new; reading audit-typed) **[E][C]**
`v415_gaussian_operator.py`	The square CM-norm dictionary is *operator-realised* and the μ_4 deck is an intrinsic diamond object: the Gaussian arithmetic of v222/v230 lifted from numbers to OPERATORS via $J=[U,V]/3$, the complex structure born from the commutator of the binary V and ternary U compilers (v410/v95). **[E]** $J=\left(\begin{smallmatrix}-1&1&0\\-2&1&0\\-3&1&0\end{smallmatrix}\right)$ is integer, Spec $J=\{i,-i,0\}$, $J^2=-I$ on the rank-2 image, $J^3+J=0$ (a μ_4 quarter-turn); **[E]** $3I+2J$ has spectrum $\{3+2i,3-2i,3\}$ and $5I+4J$ has $\{5+4i,5-4i,5\}$, so $3+2i$ ($\|\cdot\|^2=13=\Delta_Q$) and $5+4i$ ($\|\cdot\|^2=41=10b_1$) are literal operator eigenvalues; **[E]** $(3I+2J)(3I-2J)$ has Spec$\{13,13,9\}$ and $(5I+4J)(5I-4J)$ has $\{41,41,25\}$ (kernel slot = real part^2: $9=N_{\rm fam}^2$, $25=g_{\rm car}^2$); **[E]** the intrinsic deck $D=-I+J-J^2$ is order 4 (Spec$\{i,-1,-i\}$, $\chi=(x+1)(x^2+1)$) with χ_2 line $\ker[U,V]=(0,0,1)=a-\mathbf{1}$. **[C]** the geometric-deck identification sharpens (does not close) the v140 $\mathbb{Z}_3$ residue **[E][C]**
`v416_atom_trichotomy.py`	The atom trichotomy: $\{2,3,5\}$ are ramified/inert/split in the two exceptional CM rings, and each atom is the unique ramified prime of one quadratic facet of $K=\mathbb{Q}(i,\sqrt{-3},\sqrt{5})$ (v403). **[E]** $\mathbb{Z}[i]$ (square/seam, $\tau=i$): $2=\|Z_2\|$ ramifies ($2=-i(1+i)^2$), $3=N_{\rm fam}$ inert (norm $9=N_{\rm fam}^2$), $5=g_{\rm car}$ splits ($(2+i)(2-i)$) – the carrier rank factors on the seam; **[E]** $\mathbb{Z}[\omega]$ (hex/flavor, $\tau=\omega$): $3=N_{\rm fam}$ ramifies ($N(1-\omega)=3$), 2 inert (norm 4), 5 inert (norm $25=g_{\rm car}^2$); **[E]** the ramified prime of each ring IS its atom ($\mathbb{Z}[i]$ at 2= sheet/seam, $\mathbb{Z}[\omega]$ at 3= family/flavor); **[E]** the three facets $\mathbb{Q}(i)/\mathbb{Q}(\sqrt{-3})/\mathbb{Q}(\sqrt{5})$ (disc $-4,-3,5$) ramify over one atom each, product $2{\cdot}3{\cdot}5=30=h(E_8)$; 2 imaginary (CM/dynamic) + 1 real (RM/static). **[C]** the seam/flavor/golden reading **[E][C]**
`v417_eisenstein_cp_operator.py`	The Eisenstein/CP operator (the hex dual of v415): the family rotation $P=(1\,2\,3)$ and the CP clock $W=-P^2$ put the $\tau=\omega$ flavor side on operators, unified with the seam into ONE order-30 clock. **[E]** $P^3=I$, $P^2+P+I=$ONES (Eisenstein $\omega^2+\omega+1=0$ on the plane $\perp(1,1,1)$); **[E]** $(3I+2P)(3I+2P^2)=7I+6\,$ONES, Spec$\{7,7,25\}$ ($7=N_{\mathbb{Z}[\omega]}(3+2\omega)=$ scalaron on the ω-plane, $25=g_{\rm car}^2$) — the dual of v415's $\{13,13,9\}$; **[E]** the CP clock $W=-P^2$ ($=\rho$, v233) has $W^2=P$, $W^3=-I$, Spec$\{-1,\zeta_6,\zeta_6^{-1}\}$, so $\delta_{\rm CKM,lead}=\arg\zeta_6=\pi/3$ and $\delta_{\rm PMNS}=\arg\zeta_6^4=4\pi/3$ ($W^4=P^2$); **[E]** all flavor clocks are powers of ζ_{30} ($\zeta_{30}^{15}=-1$, $\zeta_{30}^{10}=\omega$, $\zeta_{30}^{6}=\zeta_5$, $\zeta_{30}^{5}=\zeta_6$) while the seam $\mu_4=i$ is the Galois side $(\mathbb{Z}/30)^\times$ of order $\varphi(30)=8=\operatorname{rank}E_8$. **[C]** the carrier μ_5 has no clean 3×3 operator (golden φ an output-side trace, v313/v349); honest negatives included **[E][C]**
`v418_cyclotomic_norm_triple.py`	The cyclotomic norm triple: the carrier split $(3,2)$ read in the three atom-rings gives $(7,13,55)$=(scalaron, Δ_Q, quark numerator), and the missing carrier-5 clock of v417 is the 4×4 Φ_5-companion C_5. **[E]** C_5 has $\chi=x^4+x^3+x^2+x+1$, $C_5^5=I$, $\operatorname{tr}=-1$; the golden ratio is its real-part data ($\zeta_5+\zeta_5^4=1/\varphi$, $\zeta_5^2+\zeta_5^3=-\varphi$; output-side v313/v349; dim 4, not 3 — refines v417's gap); **[E]** $N_{\mathbb{Z}[\zeta_5]}(3+2\zeta_5)=\det(3I+2C_5)=55=5{\cdot}11=g_{\rm car}\\|\operatorname{Pl}(K)\\|_1=$ the quark numerator (v410/v411); **[E]** the triple $\det(3I+2\operatorname{Comp}(\Phi_n))=(7,13,55)$ for $n=3,4,5$ over $\mathbb{Z}[\omega],\mathbb{Z}[i],\mathbb{Z}[\zeta_5]$ (atoms $3,2,5$, v416); **[E]** NEG control: the anchor $(5,4)\to(21,41,461)$, with $21=N_\omega(5,4)$ and $41=10b_1$ named (v222) but the carrier ring 461 prime/un-named, so the carrier ring separates $(3,2)\to55$ from $(5,4)\to461$. **[C]** HONEST: $(3,2)$ is FORCED (v14) not unique — the clean splits are $\{(2,1)\to(3,5,11),(3,2)\to(7,13,55)\}$, a small ladder; the sharper reading: only $(3,2)$ spans three physics sectors (gravity/flavor/masses), the unique cross-sector seed **[E][C]**

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Script	What it machine-checks *(continued)*
`v419_seam_galois_carrier.py`	The seam μ_4 *is* the carrier's Galois group: a positive resolution of the v409/`RES.COXETER.SYMMETRY.01` cyclic/Galois asymmetry. **[E]** $h(E_8)$=30=2·3·5 is squarefree, so $\mathbb{Z}/30$ has *no* order-4 element — the seam μ_4 cannot be a cyclic hand and is forced onto the Galois side; **[E]** $(\mathbb{Z}/30)^\times=(\mathbb{Z}/2)^\times\times(\mathbb{Z}/3)^\times\times(\mathbb{Z}/5)^\times=\mu_4\times\mathbb{Z}_2$ (order $\varphi(30)$=8= rank E_8), where the μ_4 ($\mathbb{Z}/4$) factor IS $(\mathbb{Z}/5)^\times$ (carrier 5) and the $\mathbb{Z}_2$ IS $(\mathbb{Z}/3)^\times$ (family/CP); **[E]** so the seam μ_4= Gal($\mathbb{Q}(\zeta_5)$)=$(\mathbb{Z}/5)^\times$, the Frobenius $\zeta_5\mapsto\zeta_5^2$, realised on the carrier clock C_5 (v418) by the explicit order-4 operator G with $GC_5G^{-1}=C_5^2$, $G^4=I$, $G^2C_5G^{-2}=C_5^{-1}$. **[C]** resolves v409 (the 2^2=4 is the order of $(\mathbb{Z}/5)^\times$; "no contraction rate" because the seam is an automorphism, not a hand); $\mathbb{Q}(\zeta_{30})$ carries both E_8 invariants (cyclic h=30, Galois rank =8). Figure `coxeter_galois` **[E][C]**
`v420_fpole_qed_running.py`	F_pole external: the Koide source→pole transfer is a *one-loop electromagnetic* effect, not an algebraic step — the honest external-physics follow-up to v93 and the clock No-Go. **[E]** Koide Q is rescaling-invariant ($Q(c\,m)=Q(m)$), so flavor-*universal* running leaves it fixed and the source→pole shift is purely flavor-*split*; **[C]** the tree→pole gap $\frac{2}{3}-Q_{\rm src}$=0.00220 is one-loop-EM-sized (= $\varphi_0^{\rm ret}/24$ to 0.6%, = α/π to ∼5%), the PDG pole sits AT $\frac{2}{3}$ (the IR attractor); **[C]** a 1-loop QED per-lepton-log pole↔$\overline{\rm MS}$ estimate shifts Koide by EM-loop order toward $\frac{2}{3}$ (the v82 IR-attractor direction); **[O]** the EXACT flow time t∼2.84 (v93) is scheme/scale-dependent external QED, only sized. **[C]** F_pole stays **[C]**, firewalled (v187): the lens forces the RATE $(2/3)^6$, this external EM supplies the flow. Numerical (mpmath), Python-only **[E][C][O]**
`v422_galois_net_bridge.py`	The Galois↔Net bridge: the seam μ_4 = Gal($\mathbb{Q}(\zeta_5)$) is the SAME cyclic $\mathbb{Z}/4$ as the $(E_8)_1$ simple-current glue — not a mere order-4 coincidence — linking the Galois gearbox of v419 to $G_{\rm net}$/`SEAM.EQUIV.01`. **[E]** disc(A_3)=disc(D_5)=$\mathbb{Z}/4$ (one Smith invariant factor 4 = cyclic, det =4); D_n disc is $\mathbb{Z}/4$ for n ODD, $\mathbb{Z}_2\times\mathbb{Z}_2$ for n EVEN, so the carrier D_5 (rank 5=$g_{\rm car}$, odd) is cyclic while the control D_8 (rank 8, even) is Klein; **[E]** Gal($\mathbb{Q}(\zeta_5)$)=$(\mathbb{Z}/5)^\times=\langle 2\rangle$ is cyclic order 4 (only the carrier prime 5 gives order 4); **[E]** in disc$(D_5\oplus A_3)=\mathbb{Z}_4\times\mathbb{Z}_4$ (16= dim S^+) the simple current $(1,1)$ is order 4 (cyclic) with Lagrangian quotient $16/4^2$=1 ⇒ $(E_8)_1$ (holomorphic, v89/v154). **[E]** NEG control: the Klein $\mathbb{Z}_2\times\mathbb{Z}_2$ / order-2 halfway $\langle(2,2)\rangle$ gives $16/2^2$=4=|disc(D_8)|=$SO(16)$ (det =4), NOT E_8 (det =1) — cyclic $\mathbb{Z}/4\to E_8$, Klein→D_8. **[C]** the bridge: one cyclic $\mathbb{Z}/4$ threads the seam deck μ_4 (v419), Gal($\mathbb{Q}(\zeta_5)$) and the $(E_8)_1$ glue (v1/v89/v154); cyclicity forced by 5, Klein excluded. **[C]** this is the four-fold μ_4 identity (deck/divisor = Galois = discriminant = simple-current); per v428 a *compression* (one object read four ways), not four independent witnesses **[E][C]**
`v423_no_ambient_dependency.py`	The *No-Ambient-Dependency* certificate: no frozen TFPT readout is *computed* from the ambient QG measure `QG.AMB.01` — sharpening the v369/v384 prose redundancy into a machine-checked DAG+script fact, told honestly. **[E]** gap-decoupling margin $\Delta_{\rm eff}=6\ln(3/2)-31/(4\pi^2)\approx 1.648>0$ ($31=2^{g_{\rm car}}-1$), susceptibility $\chi=729/665$ finite (neg control gap→0 ⇒ $\chi\to\infty$). **[E]** `QG.AMB.01` is a certification *sink*: every claim listing it as a dependency is a gravity-gate/QG row, none a frozen prediction. **[E]** it is not a value source — it has no producing script while each frozen readout is computed by a compiler script (α^{-1}↤v3, θ_{12}↤v9, θ_{13}↤v268, cosmology↤v7). **[E]** the one transitive link is architectural: only θ_{13} references `QG.AMB.01`, via the 4D-QFT contract spine (`PS.DIRAC`→`CONTRACT.QFT4D`→`QG.G2`→`GATE.METRIC`), a coherence edge not a numerical input; α^{-1}/θ_{12}/cosmology have no route. **[C]** so `QG.AMB.01` is gap-decoupled certification (v369 as a DAG/script fact); honestly this does NOT prove `SEAM.EQUIV.01` — the readouts depend on the seam being $(E_8)_1$; `QG.AMB.01` specifically is dead **[E][C]**

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Script	**What it machine-checks** *(continued)*
`v424_seam_lto_rp_reduction.py`	The *(LTO-RP) reduction* of the keystone: an honest MMST-style reduction of `SEAM.EQUIV.01`'s one open lemma — the raw collar's intrinsic Bisognano–Wichmann property (the v308/v309 residual) — to the cited commuting-projector theorem of Naaijkens–Penneys–Wallick (`arXiv:2605.10693`), with the central mapping *corrected*. **[E]** the μ_4 clock is a modular *symmetry* ($[\rho, K]$=0, v309/v199), not the flow — σ^ψ=ρ is type-wrong (a one-parameter group versus a finite order-4 element). **[E]** NPW26's axiom (LTO-RP) means u_Θ=J (reflection = modular conjugation) = the intrinsic BW condition $\theta K \theta = -K$, *distinct* from $\omega \circ \rho$=ω: a positive character-block-diagonal K satisfies $[\rho, K]$=0 yet never $\theta K \theta = -K$, so clock-invariance does *not* imply (LTO-RP). **[O]** make-or-break: NPW26 prove (LTO-RP) for *topologically ordered* models (Toric Code $\lvert \det K \rvert$=4 via a trace; Levin–Wen anyonic), while the seam is invertible ($\lvert \det \mathrm{Cartan}(E_8) \rvert$=1, no anyons) and a β=1 KMS state (not a trace) — in neither proved bucket (neg control D_8 $\lvert \det \rvert$=4). **[C]** so `SEAM.EQUIV.01`'s BW residual reduces to NPW26 Theorem A modulo two open sub-steps (realise the seam as a $\mathbb{Z}_2$-reflection commuting-projector LTO; extend (LTO-RP) to the invertible KMS case) — a reduction, not a closure; `SEAM.EQUIV.01` stays **[O]** **[E][C]**
`v425_dyn_transfer_universal.py`	`DYN.TRANSFER.UNIVERSAL.01` — the *single-flow reduction* of F_{transfer}: the four frontier transfers (F_{pole}, $F_{\mathrm{Boltzmann}}$, F_{relic}, F_{QCD}) do not each need a separate external solver; their *dynamics* is the ONE native flow of the theory — the seam modular/recovery semigroup (v238/v240, gap $6 \ln(3/2)$) restricted to each sector. **[E]** recovery spectrum $\{1, (2/3)^6, (1/3)^6\}$, H_{OS}=$-\log T$ gap $6 \ln(3/2)$, $e^{-\Delta}$=$(2/3)^6$ exact; an explicit column-stochastic T makes $\log T$ a Markov generator. **[E]** F_{pole} is this flow: the v99 generator $\dot{q}$=$(\Delta/N_{\mathrm{fam}})(q-2)(q-5)$ linearised at q=2 has rate $-\Delta$, time-1 contraction $(2/3)^6$. **[E]** $F_{\mathrm{Boltzmann}}$ washout = that contraction, CP phase $4\pi/3$ native (v320). **[E]** F_{QCD} rate native (b_3=-7). **[C]** only the anchors (v_{geo}, C_p, M_R) external; F_{relic}'s cosmological IC θ_i the one wall. Extends v383 static→dynamic; a reduction, no new premise **[E][C]**
`v426_seam_lto_rp_kms.py`	`SEAM.EQUIV.LTORP.KMS.01` — the *invertible β=1-KMS* variant of (LTO-RP), the corner NPW26 leave open (they prove it for topologically-ordered models; the seam is invertible + β=1 KMS, not a trace). A constructive Tomita–Takesaki witness advancing v424 sub-step (ii); does *not* close `SEAM.EQUIV.01`. **[E]** build K reflection-odd/μ_4-even/gapped (gap $2k$=$6 \ln(3/2)$); C=$1/(1+e^K)$ the β=1 KMS covariance, $C/(1-C)$=e^{-K} (genuine KMS, not trace). **[E]** $\Theta K \Theta^{-1}$=$-K$ (u_Θ=J) AND $[\rho, K]$=0 both hold ⇒ (LTO-RP) satisfied by direct Tomita–Takesaki. **[E]** $\lvert \det \mathrm{Cartan}(E_8) \rvert$=1 (invertible, no anyons) vs anyonic $\lvert \det \rvert$=4; state not a trace (Δ=$e^{-K} \neq I$) — neither NPW26 bucket. **[E]** neg controls: side-even reflection → $+K$ (BW fails), off-μ_4-block K breaks $[\rho, K]$. **[C]** (LTO-RP) reduces to $(\Theta K \Theta$=$-K) \wedge ([\rho, K]$=0), constructively satisfiable; `SEAM.EQUIV.01` stays **[O]** **[E][C]**
`v427_overdet_witness_map.py`	`OVERDET.WITNESS.MAP.01` — the multiply-vs-compress *framework* for over-determination: witnesses from *disjoint* grammars multiply, projections of the *same* generator compress. The classification of the seven arithmetic witnesses is refined by v428 (they compress). **[E]** seven grammar readouts, each from its own syntax landing on the carrier skeleton: Gauss $N(3+2i)$=13, Eisenstein $N(3+2\omega)$=7, cyclotomy $N(3+2\zeta_5)$=55, Galois $\lvert(\mathbb{Z}/5)^\times\rvert$=4, $\lvert \det \mathrm{Cartan}(E_8) \rvert$=1, Pascal $\binom{4}{0}+\binom{4}{1}+\binom{4}{2}$=11 ($2^4$=16=$\dim S^+$), Coxeter $h(E_8)$=30=2·3·5 with $\varphi(30)$=8=$\mathrm{rank}\, E_8$. **[E]** by v236 these are facets of ONE $(2,3,5)/E_8$ object, so syntactic distinctness is NOT independence — they compress, like the anchor a=$(1,1,2)$. **[C]** the genuine multiplicative strength is the multiply-forced inputs + foreign witnesses (v428); the map does not by itself establish physics **[E][C]**

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Script	**What it machine-checks** *(continued)*
`v428_overdet_reclass.py`	`OVERDET.WITNESS.RECLASS.01` — the honest correction of `v427`: applying its own multiply-vs-compress test to itself. **[E]** the $(2,3,5)$ Brieskorn singularity (v236) is the ONE generator — Milnor $\mu{=}(2{-}1)(3{-}1)(5{-}1){=}8{=}\operatorname{rank}E_8$, $30{=}2{\cdot}3{\cdot}5{=}h(E_8)$, $\varphi(30){=}8$. **[E]** every one of `v427`'s seven witnesses maps to a facet of that SAME object (primes $2,3,5$ of 30, the clock 30, or E_8), so all seven collapse to ONE generator (distinct objects = 1) ⇒ COMPRESSION, not seven multiplications. **[E]** what honestly multiplies is the input forced four independent ways — the "8" in $c_3{=}1/(8\pi)$ from $\operatorname{rank}E_8$, $h(D_5){=}2(5{-}1)$, $\varphi(30)$ and the Milnor number — plus the foreign witness $\alpha^{-1}{\approx}137$ (prime, not in the $(2,3,5)$-skeleton). **[C]** refines and partly walks back `v427`; consistent with v236/v305/v313 **[E][C]**
`v429_axion_pentagon_phi.py`	`DM.AXION.PENTAGON.01` — the 'unmapped' icosahedral/golden E_8 structure surfaces in the ONE external cosmological input θ_i, answering "why does idle E_8 structure exist?" for the one case where it has a job. **[E]** because $N_{\text{fam}}{=}g_{\text{car}}{-}2$, the spine quotient $\theta_i{=}\pi N_{\text{fam}}/g_{\text{car}}{=}(g_{\text{car}}{-}2)\pi/g_{\text{car}}$ is EXACTLY the interior angle of the regular g_{car}-gon; for $g_{\text{car}}{=}5$ the PENTAGON, $\theta_i{=}3\pi/5{=}108°$ (the carrier polygon, not a bare rational multiple of π). **[E]** its cosine is golden: $\cos(3\pi/5){=}(1{-}\sqrt{5})/4{=}{-}1/(2\varphi)$, partner $2\cos(2\pi/5){=}1/\varphi{=}\varphi{-}1$, $\varphi{=}2\cos(\pi/5){=}(1{+}\sqrt{5})/2$. **[E]** the golden character is UNIQUE to $g_{\text{car}}{=}5$: among small n-gons only $n{=}5$ has an irrational interior-angle cosine ($n{=}3,4,6$ give $1/2,0,-1/2$), so θ_i is golden BECAUSE the carrier is 5 (P2). **[C]** a re-reading bridging v354/v313 (the unmapped $g_{\text{car}}{=}5$ golden signature) to v211 ($\theta_i{=}3\pi/5$, the one external input): it REFINES v354 from "φ in no readout" to "φ touches only the **[C]** spine misalignment angle", a geometric motive for the spine over the hilltop $\sim$170°; it does NOT upgrade `DM.AXION.SPINE.01` (stays **[C]**) and closes no gate. **[E][C]**
`v430_other_side_reverse_audit.py`	`E8.OTHERSIDE.AUDIT.01`: the 'other side of the seam' (the double-cover deck / conjugate sheet S^-) audited against E_8's UNMAPPED Casimir degrees — the sheet/deck complement of v354/v355, answering "is the leftover where the second sheet lives?" with a disciplined NEGATIVE. **[E]** DECK = DEGREE-2: the one-sided cover contributes $\lvert\mathbb{Z}_2\rvert{=}2$ (the 1/2 in $c_3{=}1/(2{\cdot}4\pi){=}1/(8\pi)$, v58/v216); $2{=}\min(E_8\text{ degrees}){=}$ the metric, one of the matched primary readouts $\{2,8,30\}$ (v354), NOT one of the unmapped $\{12,14,18,20,24\}$. **[E]** TWO SHEETS = 128-SPINOR: $\dim S^+{=}\dim S^-{=}16$, the spinor block $128{=}\operatorname{rank}\cdot\dim S^+{=}8{\cdot}16{=}2^{\operatorname{rank}-1}{=}\sum\deg$, the spinor half of $248{=}120{+}128$; S^- the conjugate $(\overline{16},\overline{4})$, $128{=}2{\cdot}64$, not a spare singlet. **[E]** FORCED-DISJOINT: the sheet/deck integer set $\{2,16,32,128\}$ has EMPTY intersection with $\{12,14,18,20,24\}$, its only degree-alphabet contact the matched 2. **[O]** DISCIPLINED DECLINE: each unmapped degree needs an UNFORCED coefficient from the sheet atoms and admits $\geq$2 readings (the v355 discriminator) — no forced sheet→unmapped-degree identity. **[E]** 'S^-=DM' DOWNGRADED + WIMP NO-GO: v227 replaced the over-strong 'S^- is dark matter'; $128{=}(16,4){+}(\overline{16},\overline{4})$ is the SAME matter spinor (no spare singlet), DM is the determinant-line axion (a phase). **[O]** the only live 'other side' question (the 128 phase/glue dark sector, v227) concerns MATCHED structure, not the unmapped degrees **[E][O]**

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Script	What it machine-checks *(continued)*
`v431_e8_degree_ladder.py`	`E8.DEGREE.LADDER.01`: the reverse-audit 'unmapped' E_8 Casimir degrees are NOT diffuse overhead but the forced two-family decomposition $\deg(E_8)=6\cdot\text{spine}\{2,3,4,5\}\sqcup(\{2\}\cup\text{det-ladder}\{8,14,20\})$ — the structural complement of v354/v355/v430. [E] TWO-FAMILY SPLIT FORCED BY $30=2\cdot3\cdot5$: the E_8 exponents are the $\varphi(30)=8$ totatives of $h=30=2N_{\text{fam}}g_{\text{car}}$ (coprime to 6 $\Rightarrow \pm1$ mod 6), so the degrees occupy ONLY the residue classes $\{0,2\}$ mod 6. [E] $6k$ family $\{12,18,24,30\}/6=\{2,3,4,5\}=$ the v91 spine $\{e_3,e_1,e_2,p_0\}(a=(1,1,2))$. [E] $6k+2$ family $\{2,8,14,20\}$: $\{8,14,20\}=(\det R,\det C,\det L)=$ the winding line $\det M(s,0)=6s+8$ of the flavor diamond (v135), $C=R+Q\,\mathrm{diag}(1,0,0)$. [E] '$18=6N_{\text{fam}}$' is NOT a holdout (spine family, not on the det-ladder). [E] SPECIAL TO E_8 among the simply-laced exceptionals ($E_6/E_7/D_5$ fail the clean $\{0,2\}$ mod 6 split). [C] HONEST v355 LINE: the arithmetic decomposition is exact, but the FUNCTORIAL claim '$6k+2$ Casimir stratum = flavor sheet operators' needs a representation-theoretic map ([C]); $12,24$ admit two canonical readings each, so this is a STRUCTURE theorem, not a forced functorial flavor map — it reclassifies v354 overhead to zero orphan degrees, closes no gate [E][C]
`v432_overdet_floor.py`	`OVERDET.FLOOR.01`: the UNCONDITIONAL improbability floor + the L1↔L2 bridge — the defensible number that survives even if a skeptic REJECTS the declared formula grammar, complementing the conditional null model (v100) and the multiply-vs-compress framework (v427/v428). [E] DISJOINT PIECES (v428): compression removed (v427's seven readouts are one $(2,3,5)/E_8$ object, NOT multiplied); what multiplies is the input '8' forced FOUR ways $\{\text{rank}\,E_8,\ h(D_5)=2(g_{\text{car}}-1),\ \varphi(30),\ \text{Milnor}(2,3,5)\}$ plus the FOREIGN witness $\alpha^{-1}\sim137$ (prime, outside $(2,3,5)$). [C] CONSERVATIVE FLOOR: multiplying ONLY across disjoint pieces (input over-determination × the α grammar census 1/94500 (v100) × the φ_0^{ret}-seed cluster counted ONCE × one downstream witness) the floor stays $\leq10^{-6}$ across a grid of generous assignments, nominal $\sim10^{-10}$ — 'extraordinary' WITHOUT the declared grammar. [E] THE BRIDGE: the floor ($\sim10^{-10}$) sits $\sim$20 orders ABOVE the conditional $10^{-30.7}$ (v100); that gap = the evidential value of proving the FORMS forced, so L1 (`ALPHA.QUILLEN.EXACT.01`, v382) and L2 (this floor) are the SAME lever. [C] HONEST SCOPE: bounds ONLY "is it numerology?", NOT the physics bridge (firewall v187); a conservative FLOOR, not a posterior [E][C]
`v433_alpha_quillen_heatkernel.py`	ALPHA.QUILLEN.PROGRESS.02: a *second honest step* on the external target ALPHA.QUILLEN.EXACT.01 (v382) – it GROUNDS and CONNECTS, it does NOT close it; ALPHA.QUILLEN.EXACT.01 stays [O] (external math, type A, v384), α^{-1} stays [E]. [E] a solvable 4D model (flat $\Delta=-\partial^2+m^2$ on T^4) reaches the a_4 (t^0) conformal-anomaly order: $a_4=Vm^4/(2(4\pi)^2)$ NON-ZERO – the order v391's 1D circle toy (a_0,a_1 only) could not reach. [E] it is the order v342 grounds (Gilkey a_4 gauge-curvature $30/360=1/12$), unifying v391 (a ζ-det variation IS heat-kernel data) with v342 (the term IS the Gilkey a_4). [E] $(4\pi)^{-2}=4c_3^2$ is c_3-built. [E] the v382 split re-verified at $\alpha^{-1}=137.0359992$. [O] NOT CLOSED: the actual seam a_4 $(=8b_1c_3^6)$ and the cubic α^3 Chern moment stay open (v384); the exact seam F-normalisation stays [C] (v342). Grounds/connects, does not close [E][O]

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Script	**What it machine-checks** *(continued)*
`v434_alpha_quillen_betafunction.py`	ALPHA.QUILLEN.PROGRESS.03: a *third honest step* on ALPHA.QUILLEN.EXACT.01 (v382) – it settles the three residuals named after v433 and shows they are NOT three independent problems; the target stays **[O]**, α^{-1} stays **[E]**. **[E]** b_1 IS THE U(1) a_4 HEAT-KERNEL COEFFICIENT (the β=a_4 theorem): from the carrier hypercharge content with spin weights ($\frac{2}{3}$ Weyl, $\frac{1}{3}$ complex scalar), $\sum \frac{2}{3}Y_f^2 + \frac{1}{3}Y_s^2 = 41/6$ (SM), $(3/5)\cdot$that $= 41/10 = b_1$ (GUT) – the b_1 in $8b_1c_3^6$ is the a_4 content coefficient, not a fitted multiplicity (same number as the β, v159/v246; same KIND as the α^2 Calderón a_4, v342). **[E]** $8b_1c_3^6 = b_1 \cdot (\mathrm{rank}\, E_8) \cdot c_3^6$ (rank $8 = \chi$=2 boundary, v216; c_3^6 six insertions, v342). **[C]** residual (1) “exact $a_4 = 8b_1c_3^6$” ≡ residual (3) “seam F-normalisation” – ONE obligation. **[E]** residual (2) located: $a^3 - 2c_3^3a^2 = a^2(a-2c_3^3)$, so a^3 = the first moment of the Chern density (a level, order c_3^0); origin stays **[O]**. **[E]** net reduction: the EM-Ward residual is 1[C]+1[O], not three. **[O]** NOT CLOSED; α^{-1} stays **[E]**. Python-only **[E][C][O]**
`v435_alpha_quillen_chernlevel.py`	ALPHA.QUILLEN.PROGRESS.04: a *fourth honest step* on ALPHA.QUILLEN.EXACT.01 (v382) – it attacks the SINGLE remaining **[O]** after v434 (residual (2), the cubic α^3 Chern level) and SHARPENS it, it does NOT close it; the target stays **[O]**, α^{-1} stays **[E]**. **[E]** π-POWER / METRIC-INDEPENDENCE TEST: the three closure coefficients carry π-powers exactly $\{0,3,6\} = c_3$-orders ($k_0 = 1 \to \pi^0$; $-2c_3^3 = -1/(256\pi^3) \to \pi^3$; $-8b_1c_3^6 = -41/(327680\pi^6) \to \pi^6$), so the Maxwell a^3 is the UNIQUE π^0/c_3^0 (metric-INDEPENDENT) term – the signature of a topological level (a heat-kernel boundary coeff always carries the $(4\pi)^{-d/2}$ measure; a level carries none). The test partitions the closure into ONE topological level (a^3) + TWO heat-kernel boundary terms, confirming v342/v434’s “a^3 is a level, not a curvature integral” from a test. **[C]** conditional on the CS/η reading (EM1, v48), large-gauge invariance forces the level in $\mathbb{Z}$ and $k_0 = 1$ is the unit (minimal) level – an integer, not a free real. **[O]** NOT CLOSED: the EM1 CS-boundary derivation stays external (type A, v384); seam F-norm stays the one **[C]** (v434); α^{-1} stays **[E]**. **[E][C][O]**
`v436_overdet_floor2.py`	`OVERDET.FLOOR.02`: HARDENS `OVERDET.FLOOR.01` (v432) — the “is it numerology?” floor does NOT rest on v432’s subjective chance assignments. **[E]** ASSUMPTION-MINIMAL FLOOR: the α cubic-class census (v100 layer D) is PURE COUNTING — of N=94500 complexity-matched $F_{U(1)}$ variants only ONE hits the 4×10^{-8} CODATA window (the TFPT instance, α^{-1}=137.0359992), so $p_\alpha \le 1/94500 = 1.06\times10^{-5} \sim 4.40\sigma$ with NO subjective probability, prior or multi-observable grammar. **[E]** MONOTONE CONCESSION LADDER: $-\log_{10} p$ full scorecard+$\alpha \sim 30.7 \to$ scorecard-only $\sim 25.8 \to$ α-census-only ~ 4.98 ($= 4.40\sigma$); the “not chance” verdict holds at the MOST adversarial rung. **[E]** INDEPENDENCE: strip ALL of v432’s subjective factors (0.1^3×0.012×0.15=1.8×10^{-6}) and the floor is still 4.40σ — they only sharpen it. **[C]** HONEST 5σ GAP: 5σ (p=5.7×10^{-7}) is crossed by the census times exactly ONE more independent factor ≤ 0.054 (the input-‘8’ four-fold over-determination or one foreign witness), stated openly. **[C]** FIREWALL: bounds numerology only, not the physics bridge (v187); Python-only, deliberately NOT Wolfram-mirrored (a counting bound, not an identity). **[E][C]**

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Script	**What it machine-checks** *(continued)*
`v437_e8_degree_joint.py`	`E8.DEGREE.JOINT.01`: a CONSOLIDATION of v6/v66/v355/v431 deriving more from the E_8 Casimir degrees WITHOUT crossing the v354/v355 anti-numerology line — no new per-degree coincidence, only a joint lock. **[E]** E_8 IS FIXED BY ITS DEGREE MULTISET: from $\deg(E_8)=\{2,8,12,14,18,20,24,30\}$ alone — $\mathrm{rank}=\#\deg=8$, $h=\max=30$, $\#\text{pos roots}=\sum\exp=120$, $\#\text{roots}=240$, $\dim=248$, $\lvert W\rvert=\prod\deg=696{,}729{,}600$, $\sum\deg=128=2^7=\dim S^+$. **[E]** THE JOINT QUADRATIC: the two structural integers $(g_{\text{car}}, N_{\text{fam}})$ are the UNIQUE root pair of $x^2-(\mathrm{rank}\,E_8)x+(h/2)=x^2-8x+15=(x-3)(x-5)$ — sum$=\mathrm{rank}\,E_8=8$ (rank-fill, v6), product$=h/2=15=g_{\text{car}}N_{\text{fam}}$ ($2g_{\text{car}}N_{\text{fam}}=h$, v431); $\{3,5\}$ the unique factor pair of 15 summing to 8, so BOTH are forced together by two degree-invariants. **[E]** the three mapped degrees $\{2,8,30\}=\min/\mathrm{rank}/\max$ (the canonical invariants); $g_{\text{car}}=5=\max$ prime of 30. **[E]** DISCIPLINE: no new per-degree mining; the five non-primary degrees keep their v431 homes. Exact integer/lattice, mirrored in wolfram. **[E][C]**
`v438_seam_hsmi_borchers.py`	`SEAM.EQUIV.BW.HSMI.01`: the half-sided modular inclusion (Borchers/Wiesbrock) as an INTRINSIC Bisognano–Wichmann route for face (ii) of `SEAM.EQUIV.01` — geometric modular flow WITHOUT presupposing covariance. **[E]** STANDARD-PAIR ALGEBRA (ladder, exact): boost $K=\mathrm{diag}(n)$ and translation P (raising shift) satisfy $[K,P]=P$ and the Borchers scaling $\Delta^{it}P\Delta^{-it}=e^{-t}P$ (the ax+b/Moebius relation). **[E]** BW REFLECTION: $J(n\to -n)$ gives $J^2=I$, $JKJ=-K$ ($u_\Theta=J$) and $JPJ=P_-$ (the opposite-chirality translation), the standard-pair reflection $JU(s)J=U(-s)$. **[E]** POSITIVE ENERGY: in the unitary lowest-weight $sl(2,\mathbb{R})$ rep the line translation $P_{\text{line}}=L_0-(L_++L_-)/2$ is POSITIVE (min eig >0), the positive-energy generator the c_3 one-sided origin demands; the boost direction is the indefinite negative control. **[C]** SYNTHESIS: $P\geq 0$ + $[K,P]=P$ + J is a Longo STANDARD PAIR / +half-sided modular inclusion (Wiesbrock), so by Borchers the seam modular flow is geometric INTRINSICALLY (complementary to Adamo–Giorgetti–Tanimoto, `arXiv:2508.07109`); face (ii) becomes a STRUCTURE consequence, not a state condition. **[C][O]** assembling the finite algebra and the positive-energy rep into ONE continuum standard pair is the continuum existence (the same wall as MMST/NPW26, v336/v424); `SEAM.EQUIV.01` stays **[O]**. Python-only (numpy/scipy). **[E][C][O]**
`v439_seam_lieb_robinson.py`	`SEAM.EQUIV.CONTINUUM.LR.01`: the Lieb–Robinson/Nachtergaele–Sims continuum-existence MOTOR for the gapped seam collar — grounds the BULK half of the cited MMST existence (v336) in the explicit v367 $p+ip$ model. **[E]** UNIFORM GAP: $\Delta(N)=\text{const}>0$ across $N=20..160$ (the lattice image of the OS transfer gap $6\log\frac{3}{2}>0$, v302/v329). **[E]** EXPONENTIAL CLUSTERING: $\lvert\langle c_i^\dagger c_j\rangle\rvert\sim e^{-r/\xi}$ with ξ finite and STABLE across N (gapped $\Rightarrow$ Hastings–Koma decay), the N-independent locality. **[E]** LIEB–ROBINSON LIGHT CONE: the single-particle propagator $\lvert(e^{-iHt})_{ij}\rvert$ is confined to $\lvert i-j\rvert\lesssim v_{\text{LR}}t$ with finite v_{LR} ($\sim 2t$), exponentially small beyond the cone. **[C]** NACHTERGAELE–SIMS LIFT: uniform gap + finite LR velocity $\Rightarrow$ exponential clustering with N-independent ξ AND existence+uniqueness of the thermodynamic-limit ground state with a finite light cone — derived for OUR collar, not imported. **[C][O]** the bulk thermodynamic limit EXISTS by Lieb–Robinson; `SEAM.EQUIV.01` stays **[O]**, the open analytic step narrowed to the CHIRAL massless EDGE scaling limit (the $c=8$ $(E_8)_1$ CFT), distinct from the bulk limit. Python-only (numpy/scipy). **[E][C][O]**

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Script	**What it machine-checks** *(continued)*
`v440_seam_lto_rp_beta.py`	`SEAM.EQUIV.LTORP.BETA.01`: the β-interpolation connecting the NPW26-PROVEN trace case (β=0) to the seam's β=1 KMS case along an obstruction-free homotopy — upgrading the single-point v426 witness for v424 sub-step (ii) to a PATH argument. [E] β-INDEPENDENT (LTO-RP) DATA: $\Theta K \Theta = -K$ ($u_\Theta = J$) and $[\rho, K]=0$ are conditions on K alone, β-independent. [E] THE KMS PATH: $C_\beta = 1/(1+e^{\beta K})$ has $\beta=0 \to C=\frac{1}{2}I$ (the NPW26 TRACE state), $\beta=1 \to$ the seam KMS state, $\beta \to \infty \to$ the ground-state projection; at every β $\Theta C_\beta \Theta = 1-C$ (reflection), $[\rho, C_\beta]=0$ (mu4), $0 \leq C_\beta \leq 1$ (valid covariance/RP). [E] NO OBSTRUCTION: $\beta \mapsto C_\beta$ is norm-continuous and stays in the (LTO-RP)-valid set — a continuous homotopy from the proven trace endpoint to the seam KMS point, no topological wall. [E] NEG CONTROLS: a reflection-EVEN K' breaks $\Theta C_\beta \Theta = 1-C$ for $\beta>0$ (the trace endpoint is degenerate); an off-mu4-block K breaks $[\rho, C_\beta]$. [C][O] a PATH argument for v424 sub-step (ii), stronger than the single point of v426; `SEAM.EQUIV.01` stays [O] (closedness of (LTO-RP) along the path in the continuum is the residual). Python-only (numpy/scipy). [E][C][O]
`v441_seam_applicability_ledger.py`	`SEAM.EQUIV.APPLICABILITY.01`: the applicability ledger — a hypothesis-by-hypothesis audit of the THREE external theorems the seam keystone cites (MMST scaling limit, NPW26 (LTO-RP), Adamo–Giorgetti–Tanimoto covariance), isolating the SINGLE imported analytic fact. [E] MMST HYPOTHESES: D=16 Majoranas ($= 2^{g_{\mathrm{car}}-1}$), c=8 ($= g_{\mathrm{car}}+N_{\mathrm{fam}}$), rank E_8=8, range rank $\leq c \leq D$ reads $8 \leq 8 \leq 16$, gapped ($6 \log \frac{3}{2} > 0$), quasi-free CAR — all internal computed facts. [E] NPW26: (LTO-RP)=$u_\Theta = J$, gap, $\mathbb{Z}_2$ reflection, mu4 clock, $[\rho, K]=0$ internal (v424/v426/v440); the seam is invertible $\lvert\det \mathrm{Cartan}\, E_8\rvert=1$ + β=1 KMS, OUTSIDE NPW26's topological/trace bucket ($\lvert\det\rvert$=4) — a TEMPLATE, not a direct import. [E] ADAMO–GIORGETTI–TANIMOTO (`arXiv:2508.07109`): DHR reps of nets extending loop-group/Virasoro are AUTOMATICALLY positive-energy + diff-covariant; the seam net extends $(D_5)_1 \times (A_3)_1$ (v154), so covariance is a CONSEQUENCE — the v215 circularity is defused, covariance downgraded ASSUMPTION→[E]. [E] of 12 audited hypotheses, 11 are internal [E] and exactly 1 external — the continuum EXISTENCE of the chiral massless scaling limit (MMST/v336, Lean `FORM.SEAM.MMST.01`). [C][O] external dependence pinned to ONE analytic fact; `SEAM.EQUIV.01` stays [O]. Python-only (a structural/arithmetic audit; the dets Wolfram-mirrored via v89/v281/v422). [E][C][O]
`v442_seam_rigidity_uniform.py`	`SEAM.RIGIDITY.UNIFORM.01`: the uniform-in-N hardening of the Seam State Rigidity Theorem (v398) — the band-limited rigidity is NOT a small-N artifact. [E] EXACT UNIFORM 4-SECTOR COMMUTATION: $\rho = \mathrm{diag}(i^n)$ has EXACTLY 4 distinct eigenvalues $\{1, i, -1, -i\}$ for all N, and a genuinely character-block-diagonal K (random Hermitian within each n mod 4 class, not merely diagonal) has $[\rho, K]=0$ EXACTLY for N=4..256 (extends v398's $N \leq 64$). [E] UNIFORM GAP: $\Delta(N)$=const>0 across N=20..160 (the Nachtergaele–Sims uniform gap, shared with v439). [E] N-INDEPENDENT RIGIDITY BOUND: a FIXED local clock-breaking perturbation (single off-mu4-block Hermitian entry) gives $\lVert[\rho, V]\rVert$=2 EXACTLY for all N — a uniform lower bound; any fixed-size clock-breaking is detected with N-independent strength. [E] NEG CONTROL: a clock-PRESERVING (same-sector) perturbation gives $\lVert[\rho, V']\rVert$=0; an order-2 clock has only 2 distinct eigenvalues — the ORDER of the clock (the four Gauss–Bonnet marks) is the discriminator, uniformly in N. [C][O] the band-limited rigidity of v398 is uniform in N; `SEAM.RIGIDITY.01`/`SEAM.EQUIV.01` stays [O]: that RP+gap+holomorphy FORCE Λ_Σ block-diagonal on the full L^2 (vs merely permit) is the cited rigidity/MMST step. Python-only (numpy). [E][C][O]

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Script	**What it machine-checks** ***(continued)***
`v443_seam_e8_discriminator.py`	`SEAM.E8.DISCRIMINATOR.01`: the empirical/structural DISCRIMINATOR bundle (mode D of the v347 closure-mode classification) — concrete FALSIFIABLE kill tests separating the claimed seam $=(E_8)_1$ from the nearest c=8 rival $SO(16)_1=(D_8)_1$. **[E][X]** PRIMARY/GSD: $\lvert\det\mathrm{Cartan}(E_8)\rvert=1$ (ONE primary, holomorphic) vs $\lvert\det\mathrm{Cartan}(D_8)\rvert=4$ (FOUR primaries $1,v,s,c$); torus GSD $=\#$primaries $=1$ vs 4 — a seam with GSD>1 falsifies $(E_8)_1$. **[E][X]** CURRENT/THETA: level-1 current count $\dim E_8$=248 vs $\dim SO(16)$=120; the E_8 lattice theta has 240 norm-2 roots (= 112 integer +128 spinor, computed) vs D_8's 112 — current spectrum and lattice theta DIFFER. **[E]** c=8 IS NOT A DISCRIMINATOR: both have c=8 (the v277 ambiguity); only the order-4 mu4 condensation + $\det K$=1 (the genus-1 statement, v90) selects E_8. **[E]** NON-VACUITY: the discriminators all fire on the rival ($\det 4\neq 1$, currents $120\neq 248$, roots $112\neq 240$). **[X][C]** mode D realised as concrete kill tests; `SEAM.EQUIV.01` stays **[O]** (the discriminators distinguish the TARGET but do not PROVE the seam realises it). Python-only (numpy; the dets Wolfram-mirrored via v89/v281/v422). **[E][C][O][X]**
`v444_seam_edge_scaling.py`	`SEAM.EQUIV.CONTINUUM.EDGE.01`: the CHIRAL EDGE scaling-limit MOTOR — the edge half of the one `SEAM.EQUIV.01` continuum residual, attacking exactly the step v439 left open. On the SAME explicit v367/v368 $p+ip$ collar it shows the lattice edge carries the kinematic AND conformal data of a chiral $c=\frac{1}{2}$ Majorana CFT, converging under refinement. **[E]** CHIRAL LINEAR EDGE DISPERSION: the in-gap branch is linear near its zero-crossing (top v=0.983, R^2=0.9999) and the two boundaries carry OPPOSITE velocities — one chiral Majorana per edge (relativistic, chiral kinematics). **[E]** ALGEBRAIC EDGE TWO-POINT: the equal-time edge correlator decays as $r^{-\eta}$ with η=1.00 (the chiral free-fermion CFT exponent) — a CRITICAL edge. **[E]** EXPONENTIAL BULK (neg control): the bulk-row correlator decays exponentially ($\xi\sim 1.1$) — only the edge is critical, so the power law is the edge CFT, not a geometric artefact. **[E]** TRIVIAL-PHASE NEG CONTROL: M=3 (no edge, v368) has an exponential edge-row — the algebraic decay is the topological edge, not the boundary geometry. **[E]** SCALING-COLLAPSE / CONVERGENCE: η is N-stable (1.001 ± 0.007) and the edge correlator profile is Cauchy-convergent under refinement (successive spread shrinks $0.028\to 0.013$, N_x=36..60) — an existing scaling limit, the edge analogue of v439's bulk thermodynamic limit. **[C]** EDGE CENTRAL CHARGE: $\eta=1\Leftrightarrow$ one chiral Majorana ($c=\frac{1}{2}$) per edge, so the N_{Maj}=16-copy carrier has a chiral c_-=8 edge CFT ($=(E_8)_1$, $c_-=g_{\mathrm{car}}+N_{\mathrm{fam}}$). **[C][O]** grounds the EDGE half of the cited MMST scaling limit (v336) the way v439 grounds the bulk; with both halves the one continuum residual is grounded and pinned to a single cited theorem; `SEAM.EQUIV.01` stays **[O]** (the rigorous uniform-convergence theorem is not supplied). Python-only (numpy). **[E][C][O]**

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Script	**What it machine-checks** ***(continued)***
`v445_seam_rigidity_forcing.py`	`SEAM.RIGIDITY.FORCING.01`: the rigidity FORCING theorem — upgrades v442's "block-diagonal PERMITS commuting" to "commuting with the order-4 clock FORCES block-diagonal", isolating the single remaining physical premise. **[E]** FORCING (commute ⇔ block-diagonal, EXACT): for $\rho{=}\,\mathrm{diag}(i^n)$, $[\rho,K]{=}0$ iff $K_{ij}{=}0$ whenever $i{\not\equiv}j$ mod 4 — verified ENTRYWISE, swept $N{=}4..128$ (the converse of v442). **[E]** EXACT COMMUTANT DIMENSION: $\dim\{K:[\rho,K]{=}0\}=\sum_s n_s^2$ (the four sector sizes), verified as the ad$_\rho$ nullspace; a PROPER subspace of the N^2 algebra ($N{=}16$: 64<256) — a genuine quantitative restriction. **[E]** OFF-BLOCK ⇔ NON-COMMUTING: every off-block matrix unit E_{ij} has $\Vert[\rho,E_{ij}]\Vert=\lvert i^i-i^j\rvert\geq\sqrt{2}>0$ — zero slack. **[E]** THE ORDER IS THE DISCRIMINATOR: the order-4 clock gives 4 sectors (dim $\sum n_s^2(4)$), the order-2 clock $\mathrm{diag}((-1)^n)$ a STRICTLY LARGER commutant ($N{=}16$: 64 vs 128) — the four Gauss–Bonnet marks (order $4{=}\lvert\mu_4\rvert{=}h(A_3)$) force strictly more structure than two; only the index-4 extension is $(E_8)_1$ (det 1; v281/v154/v351). **[E]** NONDEGENERACY: exactly 4 distinct eigenvalues $\{1,i,-1,-i\}$ for all $N{\geq}4$. **[C]** SYNTHESIS (force, not permit): commuting with the order-4 μ_4 clock is EQUIVALENT to μ_4-block-diagonality (iff, exact, uniform in N), so RP-definability (v54/v398) + clock-invariance FORCE the block-diagonal Λ_Σ — v442's "permit" upgraded to "force". **[C][O]** the single residual is now precisely "transfer in commutant" (clock-invariance = `QGEO.SYM.01`, v323/v335); `SEAM.RIGIDITY.01`/`SEAM.EQUIV.01` stays **[O]**. Python-only (numpy; the det facts Wolfram-mirrored via v89/v281/v422). **[E][C][O]**
`v446_seam_clock_invariance.py`	`SEAM.RIGIDITY.CLOCKINV.01`: DERIVING clock-invariance ("transfer in the commutant") from RP-symmetry, closing v445's single residual one notch further — the residual drops from an OPERATOR commutation to the STRUCTURAL μ_4-symmetry of the seam RP data. **[E]** OS RECONSTRUCTION INHERITS THE SYMMETRY: for a μ_4-invariant RP covariance ($[\rho,C(t)]{=}0$), the OS canonical transfer $\Lambda{=}C(1)C(0)^{-1}{=}e^{-h}$ (the transfer FIXED by RP+reflection, v54) has $[\rho,\Lambda]{=}0$ — swept $N{=}4..32$ over random block-diagonal h. **[E]** SHARPNESS / NEG CONTROL: a μ_4-BREAKING covariance gives $[\rho,\Lambda]\neq0$ — iff-sharp, so the inheritance is non-vacuous. **[E]** CONTRACTION / OS-VALIDITY: Λ has spectrum in $(0,1]$ (a positive contraction, the physical canonical transfer). **[E]** MODULAR (Tomita–Takesaki): the free modular Hamiltonian $K{=}\log((1{-}C)C^{-1})$ (the v258 formula) of the μ_4-invariant state has $[\rho,K]{=}0$ (the clock commutes with modular flow), the breaking control $[\rho,K]\neq0$. **[C]** SYNTHESIS: $[\rho,\Lambda_\Sigma]{=}0$ is a CONSEQUENCE of μ_4-invariance of the seam RP data (the four Gauss–Bonnet marks, v216/v323) via the OS reconstruction (v54) + the modular fact (v258), NOT an extra assumption; with v445 (commute ⇔ block-diagonal), RP-definability PLUS the marks FORCE the block-diagonal Λ_Σ. **[C][O]** the residual reduces from an operator commutation to "the seam RP data is μ_4-symmetric" (`QGEO.SYM.01`); the continuum Tomita–Takesaki implementation on the full L^2 is the cited backbone. `SEAM.RIGIDITY.01`/`SEAM.EQUIV.01` stays **[O]**. Python-only (numpy/scipy). **[E][C][O]**

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Script	What it machine-checks *(continued)*
`v447_seam_edge_chern.py`	`SEAM.EQUIV.EDGE.CHERN.01`: an INDEPENDENT (topological) reading of the edge c_-=8, corroborating v444's correlator route via the bulk–edge correspondence on the same v367/v368 p+ip collar. **[E]** INTEGER CHERN NUMBER: the M=1 collar lower band has Chern C=+1 (the Fukui–Hatsugai–Suzuki lattice link-variable method, an EXACT integer, no fit) — one chiral channel per copy. **[E]** TRIVIAL-PHASE NEG CONTROL: M=3 (v368) has C=0 (no chiral edge), and M=−1 gives C=−1 (opposite chirality, matching v444's opposite per-edge velocities). **[E]** BULK–EDGE CORRESPONDENCE: the open strip carries EXACTLY $\|C\|$=1 in-gap edge branch crossing zero per boundary in the M=1 phase, 0 in the trivial phase — the chiral edge channel count = $\|C\|$. **[E]** LI–HALDANE: the half-cut bulk ground state's single-particle entanglement spectrum has in-gap modes at 1/2 in the topological phase and NONE in the trivial — the entanglement edge tracks the physical edge. [C] EDGE CENTRAL CHARGE: $c_-=N_{\rm Maj}\cdot\frac{1}{2}\cdot\|C\| = 16\cdot\frac{1}{2}\cdot 1 = 8 = g_{\rm car}+N_{\rm fam}$, from the Chern integer + bulk–edge correspondence — the SAME c_-=8 as v444's correlator route, from a logically independent (topological) observable. [C]**[O]** corroborates v444's edge $(E_8)_1$ reading; `SEAM.EQUIV.01` stays **[O]** — the topology fixes c_- and the channel count but not the cited continuum existence theorem. Python-only (numpy). **[E]**[C]**[O]**
`v448_seam_edge_fourpoint.py`	`SEAM.EQUIV.EDGE.FOURPOINT.01`: pinning the ONE external MMST fact empirically — the edge FOUR-point function is quasi-free and conformally convergent, extending v444's two-point scaling limit to the next order. **[E]** QUASI-FREE / WICK (CAR hypothesis): the ground-state correlation matrix is an exact projector G^2=G ($\sim 10^{-13}$), so the state is quasi-free and all n-point functions are Wick/Pfaffian sums of the two-point — the MMST quasi-free CAR hypothesis (ledger entry 5) literally holds on the collar (structural: the BdG state is free). **[E]** FOUR-POINT CAUCHY CONVERGENCE: the edge four-point cross-ratio Q=$[G_{01}G_{23}]/[G_{02}G_{13}]$ at COMMENSURATE positions $(1,2,4,7)\cdot(N_x/12)$ is Cauchy-convergent under refinement (successive spread shrinks, N_x=36..60) — the four-point converges, extending v444's two-point scaling-collapse. **[E]** CONFORMAL FOUR-POINT FORM: Q converges to the free-fermion CONFORMAL cross-ratio $\sin(5\pi/12)/\sin(\pi/12)$=$2+\sqrt{3}\approx 3.732$ (chord distance $d(r)$=$(N_x/\pi)\sin(\pi r/N_x)$) — the edge four-point has the chiral-CFT cross-ratio structure (η=1), not just the two-point. **[E]** BULK NEG CONTROL: the bulk-row four-point is orders of magnitude off the conformal value (exponential decay, gapped) — only the EDGE four-point is conformal. [C] the one external MMST fact (continuum existence of the chiral scaling limit) is now empirically supported at BOTH two-point (v444) and four-point order on our explicit collar; every measured n-point converges to its conformal limit. `SEAM.EQUIV.01` stays **[O]** — the rigorous MMST uniform-convergence theorem (v336) is the cited backbone. Python-only (numpy). **[E]**[C]

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Script	**What it machine-checks** ***(continued)***
`v449_seam_mmst_uniform.py`	`SEAM.EQUIV.MMST.UNIFORM.01`: uniform-in-N control of the edge scaling limit — the empirical content of the cited MMST uniform-convergence theorem, strengthened, on the same v367/v368 $p+ip$ collar. **[E]** CROSS-RATIO A CONVERGES: the edge four-point $Q(1,2,4,7)/12 \to \sin(5\pi/12)/\sin(\pi/12) = 2+\sqrt{3}$, the deviation shrinking monotonically under refinement (N_x=36, 48, 60). **[E]** CROSS-RATIO B (independent positions): $Q(1,3,5,9)/12 \to$ its OWN conformal value 2 — a SECOND observable with a DIFFERENT limit, so the convergence is not a single tuned point. **[E]** UNIFORM $1/N$ RATE: $\lvert Q(N)-Q_\infty\rvert \cdot N \le 4$ for BOTH cross-ratios and every $N_x \in \{36, 48, 60\}$ — one N-independent constant bounds the deviation across two independent observables, the empirical hallmark of uniform-in-N convergence. **[E]** BULK NEG CONTROL: the bulk-row cross-ratio is off-scale (gapped, exponential) — only the EDGE converges. **[C][O]** the MMST uniform-convergence HYPOTHESIS is empirically satisfied on the collar (two-point v444, four-point v448, and now multi-observable uniform rate); `SEAM.EQUIV.01` stays **[O]** — the abstract continuum existence theorem (v336) is strengthened in evidence, not replaced by a proof. Python-only (numpy). **[E][C]**
`v450_seam_edge_entanglement.py`	`SEAM.EQUIV.EDGE.ENTANGLEMENT.01`: a THIRD independent reading of the edge c_-=8 — the Calabrese–Cardy entanglement scaling c=1/2 per Majorana (after v444 correlator, v447 topology). **[E]** CALABRESE–CARDY LAW: the block entanglement entropy of the critical XX (Dirac) chain fits S=$(c/3)\ln[(L/\pi)\sin(\pi\ell/L)]$+const with c_{Dirac}=1.000 (flat log slope) — the universal scaling-limit signature. **[E]** c PER MAJORANA = 1/2: c_{Dirac}=1=2c_{Maj} (Dirac = two Majoranas), so c_{Maj}=1/2 — the chiral CFT of one $p+ip$ edge mode (v444). **[E]** GAPPED NEG CONTROL: a dimerised (gapped) chain has saturating, ℓ-independent entanglement (c=0) — the log growth is the CRITICAL edge CFT. **[C]** EDGE CENTRAL CHARGE: c_-=$N_{\mathrm{Maj}} \cdot c_{\mathrm{Maj}} = 16 \cdot \frac{1}{2} = 8 = g_{\mathrm{car}}$+$N_{\mathrm{fam}}$, from entanglement — the SAME c_-=8 as v444 (correlator) and v447 (topology), from a third independent observable. **[C][O]** three observables agree; `SEAM.EQUIV.01` stays **[O]**. Python-only (numpy). **[E][C]**
`v451_seam_edge_cardy_tower.py`	`SEAM.EQUIV.EDGE.CARDY.01`: the Cardy finite-size conformal tower — the edge spectrum reproduces the EXACT Ising primary dimensions {1/16, 1/2}, on the critical Majorana ring (the v367/v368 edge universality class). **[E]** SPIN DIMENSION h_σ=1/16: the periodic(R)–antiperiodic(NS) ground-energy split gives $(E_0^R - E_0^{NS})L/(2\pi v)$=0.06250 ($v$=2), stable across L=200, 400, 800 — the Ising spin field. **[E]** ENERGY DIMENSION h_ε=1/2: the lowest NS pair excitation gives $2\varepsilon_{\min}L/(2\pi v)$=0.50000 — the Ising energy field. **[E]** CONFORMAL SCALING: both dimensions are L-INDEPENDENT (the gaps scale exactly as $1/L$) — a conformal (gapless) spectrum, not a gapped lattice. **[C]** ISING IDENTIFICATION: $\{c, h_\sigma, h_\varepsilon\}$=$\{1/2, 1/16, 1/2\}$ is the chiral Ising (free Majorana) CFT (the c=1/2 building block, v444/v450); with N_{Maj}=16 copies and bulk–edge (v447), c_-=$16 \cdot \frac{1}{2}$=8=g_{car}+N_{fam}. **[C][O]** the edge CFT is named by its central charge (v450) AND its operator content (this); `SEAM.EQUIV.01` stays **[O]**. Python-only (numpy). **[E][C]**

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Script	**What it machine-checks** ***(continued)***
`v452_seam_e8_modular.py`	`SEAM.EQUIV.E8.MODULAR.01`: the $(E_8)_1$ torus modular data — one primary, S-invariant, T-phase $e^{-2\pi i/3}$, the holomorphic $c{=}8$ signature (extends v156's q-expansion to the modular transformation data). **[E]** S-INVARIANCE: $\chi(-1/\tau){=}\chi(\tau)$ for $\chi{=}E_4/\eta^8$ (weight 0, since E_4 wt 4 / η^8 wt 4), to $\sim 10^{-30}$ — ONE primary, the S-matrix is [1]. **[E]** T-PHASE: $\chi(\tau{+}1)/\chi(\tau){=}e^{-2\pi i/3}{=}e^{-2\pi ic/24}$ ($c{=}8$), to $\sim 10^{-30}$ — the central charge appears in the modular T eigenvalue. **[E]** LEADING POWER $\Rightarrow c{=}8$: $\chi\, q^{1/3} \to 1$ so the leading power is $q^{-c/24}{=}q^{-1/3} \Rightarrow c{=}8{=}g_{\mathrm{car}}{+}N_{\mathrm{fam}}$. **[E]** HOLOMORPHIC CONSTRAINT: $c{=}8{=}0 \bmod 8$ — the minimal holomorphic (single-primary) chiral central charge, the even unimodular rank-8 lattice E_8. **[C][O]** the $(E_8)_1$ modular data (one primary, S-invariant, T-phase $e^{-2\pi i/3}$, $c{=}8$) is the torus signature of the seam net; with the edge readings (v444/v447/v450/v451) the $(E_8)_1$ identity is pinned on the torus AND the edge; `SEAM.EQUIV.01` stays **[O]** — the cited continuum existence theorem (v336) is the residual. Wolfram-mirrored (`tfpt_readouts_extension.wl`). **[E][C]**
`v453_seam_mu4_from_marks.py`	`SEAM.RIGIDITY.MU4FROMMARKS.01`: deriving `QGEO.SYM.01` (μ_4-symmetry of the seam RP data) from the four marks — the LAST structural premise of the rigidity chain folded into the existing realisation premise. **[E]** THE MARKS ARE μ_4: the four Gauss–Bonnet marks are exactly $\{1, i, -1, -i\}$= roots of $z^4{-}1$ (v168/v216/v323). **[E]** THE CLOCK PERMUTES THE MARKS: $z \mapsto iz$ is a 4-cycle of order 4 ($1{\to}i{\to}{-1}{\to}{-i}{\to}1$), fixing the configuration setwise (the order-4 clock of v445). **[E]** MOEBIUS STABILISER: $z \mapsto iz$ preserves the marks' cross-ratio $\lambda{=}2$ (in the D_4 stabiliser, v168) — the symmetry is geometric. **[E]** THE FORM BASIS IS μ_4-GRADED: $\omega_k{=}z^{k-1}dz/(z^4{-}1) \to i^k\omega_k$ under $z \mapsto iz$ ($k{=}1,2,3 \to \{i,-1,-i\}$, the three nontrivial μ_4 characters; weights $(1,2,3){=}A_3$ exponents $=$ $\mathrm{Spec}(Q_+)$) — the natural basis is μ_4-graded, so any geometric datum is μ_4-covariant. **[C][O]** `QGEO.SYM.01` ("the seam RP data is μ_4-symmetric") is a CONSEQUENCE of marks$=\mu_4$ (this) + `QGEO.REALIZE.01` (seam = the marked geometry, the EXISTING premise), NO new premise; the rigidity chain (this $\to$ v446 $\to$ v445) closes modulo realisation, `SEAM.EQUIV.01` stays **[O]**. Wolfram-mirrored (`tfpt_readouts_extension.wl`). **[E][C]**
`v454_seam_edge_virasoro.py`	`SEAM.EQUIV.EDGE.VIRASORO.01`: the lattice realises the level-1 current algebra whose Sugawara central charge is EXACTLY 8 (G2 of the post-F next steps) — testing the cited MMST "Koo–Saleur lattice generators $\to$ continuum Virasoro with the correct c" fact on our own free-fermion edge. **[E]** CASIMIR c: the critical complex free-fermion ring $E_0(L){=}e_\infty L{-}\pi vc/(6L)$ (Affleck–Cardy) fits $c_{\mathrm{Dirac}}{=}1.000 \Rightarrow 1/2$ per Majorana (parameter-free, $L{=}64..1024$). **[E]** KAC–MOODY LEVEL: the lattice $U(1)$ current J_n has $\langle 0\vert J_n J_{-n}\vert 0\rangle{=}n$ EXACTLY ($n{=}1..8$) — the affine level-1 central extension closes; a gapped (dimerised) control gives a vanishing non-linear term. **[E]** SUGAWARA EXACT: $c{=}\dim G/(1{+}h^\vee){=}8$ for BOTH $SO(16)_1$ (120/15) and $(E_8)_1$ (248/31) — Wolfram-mirrored. **[C]** ASSEMBLY: 16 Majoranas $\times\frac{1}{2}{=}8{=}$ the level-1 Sugawara c; the edge IS a $c_-{=}8$ level-1 current-algebra CFT. **[C][O]** the lattice Virasoro/current algebra with the correct central extension is realised on the explicit collar; `SEAM.EQUIV.01` stays **[O]** (the continuum existence theorem v336 is the cited backbone). Mixed numerical (Casimir, Kac–Moody) + exact (Sugawara, Wolfram-mirrored). **[E][C][O]**

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Script	**What it machine-checks** *(continued)*
`v455_seam_bw_uniform.py`	`SEAM.EQUIV.BW.UNIFORM.01`: a UNIFORM-IN-N Tomita–Takesaki tower for the intrinsic Bisognano–Wichmann condition $u_\Theta{=}J$ (G1 of the post-F next steps) — the inductive-limit companion to the single finite witness v426, advancing v424 sub-step (i). One-particle: $\Delta{=}e^{-K}$, conjugation J, reflection Θ; BW $\Leftrightarrow u_\Theta{=}J \Leftrightarrow \Theta K\Theta{=}{-}K$. **[E]** UNIFORM BW: the reflection-odd, μ_4-even, gapped collar of size $4m$ (m=2..32, N=8..128) has $\lVert\Theta K\Theta{+}K\rVert{=}0$ to machine precision at EVERY N — $u_\Theta{=}J$ uniform. **[E]** UNIFORM GAP: the gap of K is N-independent ($6\log\frac{3}{2}{=}2N_{\mathrm{fam}}\log\frac{3}{2}$) so $\Delta{=}e^{-K}$ stays uniformly bounded away from 1 (a genuine KMS, not trace, state). **[E]** UNIFORM CLOCK SYMMETRY: $[\rho,K]{=}0$ exactly at every N. **[E]** N-INDEPENDENT NEG CONTROL: a side-even reflection gives $\lVert\Theta K\Theta{+}K\rVert{=}2\lVert K\rVert$ (a fixed per-mode margin, growing with N) and an off-μ_4 K breaks $[\rho,K]$ at every N — the success is non-vacuous. **[C][O]** the inductive (continuum) limit inherits $u_\Theta{=}J$; `SEAM.EQUIV.01` stays **[O]** (the rigorous continuum theorem NPW26/MMST v336 is the backbone). Python-only (numpy). **[E][C][O]**
`v456_seam_chirality_from_c3.py`	`SEAM.S3.FROM-P1.01`: the edge chirality ($c_-{\neq}0$, the S3 topological phase) is FORCED by the one-sidedness that defines $c_3{=}1/(8\pi)$ (P1) (G6 of the post-F next steps) — so S3 is a CONSEQUENCE of axiom P1, not an independent input. **[E]** c_3's "8" IS THE ONE-SIDED COUNT: $8\pi{=}\lvert\mathbb{Z}_2\rvert\int K{=}2(2\pi\chi(S^2)){=}2\cdot 4\pi$ (χ=2), so $8{=}\lvert\mathbb{Z}_2\rvert\cdot(\int K/\pi){=}2\cdot 4$ (v73, Wolfram-mirrored). **[E]** REFLECTION REVERSES C: the explicit 2-band $p{+}ip$ Chern model has $C{=}{+}1$ and the orientation-reversed ($k_y{\to}{-}k_y$) model $C{=}{-}1$. **[E]** TWO-SIDED ⇒ NON-CHIRAL: any GAPPED boundary carrying an orientation-reversing (reflection) symmetry must have $C{=}{-}C{=}0$ ($c_-{=}0$, gappable). **[C]** ONE-SIDED ⇒ CHIRAL: the one-sided seam is precisely the $\lvert\mathbb{Z}_2\rvert$ quotient with NO reflection (the same $\lvert\mathbb{Z}_2\rvert$ that gives c_3 its 8π), so $C{\neq}0$ is forced — S3 follows from P1. **[C][O]** magnitude $c_-{=}8{=}\operatorname{rank}E_8{=}g_{\mathrm{car}}{+}N_{\mathrm{fam}}$ is the SAME integer as c_3's one-sided count; `SEAM.EQUIV.01`'s continuum EXISTENCE (v336) stays **[O]**. Mixed exact (the $8{=}\lvert\mathbb{Z}_2\rvert\int K/\pi$ arithmetic, Wolfram) + numerical (the reflection $C{\to}{-}C$ flip). **[E][C][O]**
`v457_seam_character_killtest.py`	`SEAM.EQUIV.KILLTEST.01`: an explicit $(E_8)_1$-vs-$SO(16)_1$ character/sector kill-test for the seam edge (G7 of the post-F next steps) — a FALSIFIABLE consistency test of the $(E_8)_1$ identification of `SEAM.EQUIV.01`. **[E]** $(E_8)_1$ CHARACTER TOWER: $\chi{=}E_4/\eta^8{=}q^{-1/3}(1{+}248q{+}4124q^2{+}34752q^3{+}\cdots)$ — the exact conformal-tower degeneracies ($E_4\cdot\prod(1{-}q^n)^{-8}$, Wolfram-mirrored). **[E]** CURRENT-COUNT DISCRIMINATOR: the weight-1 multiplicity is $248{=}\dim E_8{=}120(SO16){+}128(\text{spinor})$, NOT the 120 of the free $SO(16)_1$. **[E]** SECTOR-COUNT DISCRIMINATOR: $(E_8)_1$ is INVERTIBLE $\lvert\det\mathrm{Cartan}\,E_8\rvert{=}1$ (one primary) vs $SO(16)_1{=}D_8$ $\lvert\det\mathrm{Cartan}\,D_8\rvert{=}4$ (four sectors). **[X]** KILL CRITERION: a measured weight-1 count of 120, OR a sector count of 4, OR a tower coefficient $\neq\{1,248,4124,\dots\}$ would FALSIFY $(E_8)_1$; current data give $248/1/\{1,248,4124\}$, so PASSED. **[C][O]** the $(E_8)_1$ identification survives the kill-test; `SEAM.EQUIV.01`'s continuum EXISTENCE (v336) stays **[O]**. Exact (Wolfram-mirrored) + a numerical determinant cross-check. **[E][C][O][X]**

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Script	**What it machine-checks** ***(continued)***
`v458_seam_mmst_citation_audit.py`	`SEAM.EQUIV.MMST.AUDIT.01`: the EXACT hypothesis-by-hypothesis citation audit of the seam scaling-limit theorem (Osborne–Stottmeister, CMP **398** (2023) 219; `arXiv:2107.13834`; legacy internal token MMST, on the MMST/Morinelli–Morsella–Stottmeister–Tanimoto OAR framework) against our collar (G3 of the post-F next steps, the "exakter Zitat-Abgleich"). [C] H1 ALGEBRA CLASS: a gapped quasi-free CAR 16-Majorana system = MMST's lattice-fermion class (v155/v160/v302). [C] H2 MASSLESS POS-ENERGY LIMIT: the gapless chiral edge IS that massless scaling limit (v439/v444). [E] H3 PER-COPY (c=1/2): each of 16 chiral Majoranas is EXACTLY the proven Theorem A c=1/2 case. [C] H4 DECOUPLED ADDITIVITY: MMST's limit decouples into Majoranas ⇒ $L_k{=}\sum_i L_k^{(i)}$ on the common core ⇒ the free c=8 $SO(16)_1$ Virasoro follows ($16{\cdot}\frac{1}{2}{=}8$). [C] H5 WZW RANGE rank $\le c \le D$: $8{\le}8{\le}16$ IN range, the 120 $so(16)$ currents are bilinears — covered. [O] H6 RESIDUAL = THE 128-SPINOR EXTENSION: the index-4 $SO(16)_1{\to}(E_8)_1$ adds 128 SPINOR currents (248=120+128), NOT bilinears — OUTSIDE MMST's method (the $\det K$ 4→1 holomorphy discriminator), handed to v459. [C][O] the audit CLOSES-by-citation the c=8 EXISTENCE and pins the open residual EXACTLY to the 128-spinor extension; `SEAM.EQUIV.01` stays [O], its open part now this one holomorphic-extension statement. Python (sympy; a structural/arithmetic audit). [C][O]
`v459_seam_lattice_voa_route.py`	`SEAM.EQUIV.LATTICEVOA.01`: the SECOND, independent route to $(E_8)_1$ — the lattice-VOA / conformal-net $A_Q(E_8)$ construction supplies EXACTLY the 128-spinor extension MMST (v458) leaves open (G5 of the post-F next steps). Cited: Adamo–Giorgetti–Tanimoto, "Rational and non-rational 2d CFTs arising from lattices" (`arXiv:2506.01008`); OS for unitary lattice VOAs (Adamo–Moriwaki–Tanimoto, `arXiv:2407.18222`). [E] E_8 EVEN UNIMODULAR: $\det \mathrm{Cartan}(E_8){=}1$ and even ⇒ A_{E_8} HOLOMORPHIC (one sector, the $\det K{=}1$ the residual demanded). [E] LATTICE CURRENTS 248=8+240: 8 Cartan + 240 roots = the full $(E_8)_1$ weight-1 space, from the lattice (not bilinears). [E] THE 240 ROOTS SPLIT 112+128: 112 INTEGER (D_8) + 128 HALF-INTEGER; 8+112=120=$so(16)$ and the 128 half-integer roots ARE the spinor (248=120+128=8+240) — the lattice route builds the 128-spinor explicitly. [C] AGT THEOREM: A_Q of an even lattice IS a 2d conformal net; the holomorphic $SO(16)_1{\to}(E_8)_1$ extension is CONSTRUCTED, not assumed. [C][O] TWO complementary routes now cover $(E_8)_1$ (MMST fermions v458 + the lattice net A_{E_8}); the residual reduces to the single realization input "the collar's scaling limit = A_{E_8}" (tied to P1 by v456). `SEAM.EQUIV.01` stays [O]. Exact (E_8 even unimodular; 240=112+128; 248=8+240=120+128, Wolfram-mirrored) + structural literature mapping. [E][C][O]

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Script	What it machine-checks *(continued)*
`v460_seam_s3_lto.py`	`SEAM.S3.LTO.01`: the EXPLICIT flat-band commuting-projector (local-topological-order) realisation of the S3 input — advancing v424 sub-step (i) ("realise the raw seam as a $\mathbb{Z}_2$-reflection commuting-projector LTO") from abstract existence to an explicit gap-protected quasi-local projector net carrying the $\mathbb{Z}_2$ reflection and the μ_4 clock. **[E]** FLAT-BAND PARENT: the flattened Bloch $Q{=}h/\|h\|$ of v367's $p{+}ip$ collar has $Q^2{=}I$ (flat ±1 bands) and $P{=}(I{-}Q)/2$ is an exact projector EQUAL to the occupied ground-state projector — a frustration-free commuting-projector parent with the same gapped ground state. **[E]** GAP-PROTECTED QUASI-LOCALITY: the gapped projector kernel $\|P(n,0)\|$ decays rapidly ($\xi{\approx}1.14$ at $M{=}1$, exp $R^2{=}0.995$, $\|P(20,0)\|{<}10^{-6}$) and ξ GROWS as the gap closes (2.7 at gap 0.02); at the GAPLESS point the kernel decays as a clean power law $\|R\|^{-2.1}$ ($R^2{=}0.999$, tail $\sim2{\times}10^{-4}$) — locality needs the gap. **[E]** $\mathbb{Z}_2$ REFLECTION + μ_4 INHERITED: the flattened modular Hamiltonian $K_{\text{flat}}{=}\text{sign}(K)$ inherits $\Theta K_{\text{flat}}\Theta{=}{-}K_{\text{flat}}$ (BW, $u_\Theta{=}J$) and $[\rho,K_{\text{flat}}]{=}0$ (μ_4) EXACTLY (sign is odd; ρ commutes with any $f(K)$). **[E]** NEG CONTROLS: a side-even reflection gives $2\|K_{\text{flat}}\|{\neq}0$ (BW fails), an off-μ_4 K mixing opposite-sign marks breaks $[\rho,\text{sign}(K)]{\neq}0$. **[C][O]** the chiral $c_-{=}8{\neq}0$ (FHS Chern $\|C\|{=}1$) obstructs STRICT finite-range commuting-projector locality (Kapustin–Fidkowski, `arXiv:1810.07756`; $C{\neq}0$ forbids exponentially-localised Wannier, Thouless), so the realisation is the quasi-local LTO net of NPW26 — the form sub-step (i) asks for. `SEAM.EQUIV.01` stays **[O]** (the continuum theorem MMST v336 is the backbone). Python-only (numpy; the det/level discriminators Wolfram-mirrored via v89/v281/v422). **[E][C][O]**
`v461_seam_strict_locality.py`	`SEAM.S3.LOCALITY.01`: the Kapustin–Fidkowski/Thouless strict-locality OBSTRUCTION made explicit — sharpening v460's verdict from a cited statement to an EXHIBITED topological obstruction on the same v367 $p{+}ip$ collar. **[E]** CHERN INTEGER: the FHS Chern is $C{=}{+}1$ (chiral $M{=}1$), 0 (trivial $M{=}3$), -1 (opposite mass) — the invertible-phase invariant. **[E]** WILSON-LOOP / WANNIER-CENTRE WINDING = CHERN: the hybrid Wannier centre $\theta(k_y)$ winds by $\|C\|{=}1$ (chiral) and 0 (trivial) over the BZ; exponentially-localised Wannier functions exist IFF the winding is 0 (Brouder–Panati et al., PRL 98 046402; Thouless), so the chiral collar has NONE. **[E]** OBSTRUCTION INTEGERS TIED: winding = $\|C\|{=}1{\neq}0$ and $c_-{=}16/2{=}8{=}g_{\text{car}}{+}N_{\text{fam}}{=}\text{rank}\,E_8{\neq}0$ (the same 8 as c_3's one-sided count, v456). **[E]** NEG CONTROL: BOTH phases are gapped, yet only the trivial (winding 0) admits exp-localised Wannier and a strict finite-range commuting projector — it is the CHIRALITY, not the gap. **[C][O]** Kapustin–Fidkowski (`arXiv:1810.07756`): a strictly finite-range commuting projector would force winding 0, $C{=}0$; since $C{=}1{\neq}0$ NONE exists, so v424 sub-step (i)'s realisation is PROVABLY the quasi-local NPW26 LTO net (strict locality topologically forbidden, not a missing premise) — upgrading v460's cited verdict. `SEAM.EQUIV.01` stays **[O]** (continuum theorem MMST v336 the backbone). Python-only (numpy; the winding/Chern integers Lean-mirrored, `SeamResidualAxiom.lean`). **[E][C][O]**

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Script	**What it machine-checks *(continued)***
`v462_seam_spinor_continuum.py`	`SEAM.EQUIV.SPINOR.01`: the 128-spinor extension $SO(16)_1 \to (E_8)_1$ exhibited at CHARACTER level and as a finite-L → continuum convergence — the constructive companion of the v458 (MMST audit) / v459 (lattice-VOA) residual. **[E]** $(E_8)_1$ TOWER FROM $SO(16)_1$ SECTORS: the exact Jacobi/E_8 identity $\theta_2^8+\theta_3^8+\theta_4^8=2E_4$ gives $\chi_{(E_8)_1}=E_4/\eta^8=\frac{1}{2}[(\theta_3/\eta)^8+(\theta_4/\eta)^8]+\frac{1}{2}(\theta_2/\eta)^8=\chi_o^{SO16}+\chi_s^{SO16}$ — the holomorphic $(E_8)_1$ character IS the $SO(16)_1$ vacuum + spinor sector sum. **[E]** THE 128 SPINOR WEIGHT-1 STATES FILL 120→248: χ_o gives weight-1 multiplicity 120, χ_s gives 128, summing to the exact $(E_8)_1$ tower $\{1, 248, 4124, 34752\}$ — the spinor currents ARE the $(E_8)_1 - SO(16)_1$ deficit (248=120+128). **[E]** FINITE 16-MAJORANA FOCK: $C(16,2)$=120 NS bilinear ($so(16)$) currents and $2^{16/2}$=256 Ramond ground states = 128 spinor + 128 cospinor — the spinor present already at finite N. **[E]** FINITE-L → CONTINUUM CONVERGENCE: the critical free-fermion ring Casimir $E_0(L)=e_\infty L-\pi vc/(6L)$ fits $c_{\text{Dirac}}\to 1$ (c_-=8) and the lowest scaled NS gap $x_1\to 1$ with $\lvert x_1-1\rvert\sim 1/L^2$ (L=64..1024) — the lattice flows to the conformal tower. **[C][O]** the rigorous holomorphic extension ($\chi_o+\chi_s$ simple-current projection) is supplied by AGT/AMT (v459); `SEAM.EQUIV.01`'s continuum existence (v336) stays **[O]**. Mixed exact (θ identity + tower + 120=$C(16,2)$, 128=2^7, 248=120+128, Wolfram-mirrored) + numerical (the finite-L convergence, Python-only). **[E][C][O]**
`v463_seam_c8_holomorphic_uniqueness.py`	`SEAM.EQUIV.UNIQ.01`: the c=8 holomorphic-uniqueness pin — the positive lever that makes the IDENTIFICATION half of `SEAM.EQUIV.01` classification-forced rather than assumed (B of the A+B round). **[E]** c=8 NOT UNIQUE AT LEVEL 1: $c=\dim\mathfrak{g}/(1+h^\vee)$=8 has THREE simple solutions — A_8 (dim 80), $D_8=\mathfrak{so}(16)$ (dim 120), E_8 (dim 248); c=8 is necessary but NOT sufficient (the v457 det K=4 $SO(16)$ rival is a real gleich-c competitor). **[E]** HOLOMORPHY FORCES $\dim V_1$=248: the only weight-0 form with leading $q^{-1/3}$ and integer coefficients is $j^{1/3}=E_4/\eta^8$, whose q^1 coefficient (the 1-dim candidate space, vacuum a_0=1) is 248 — forced, not chosen. **[E]** $\dim V_1$=248+ level-1 c=8 ⇒ E_8 UNIQUELY (248/(1+30)=8); the gleich-c rivals A_8 (80) and D_8 (120) are excluded. **[E]** the $(E_8)_1$ tower $\{1, 248, 4124, 34752, 213126\}$ are the E_4/η^8 coefficients, and $E_4/\eta^8=j^{1/3}$ (numeric $< 10^{-25}$). **[C]** Dong–Mason/Schellekens: the holomorphic VOA at c=8 is UNIQUE = V_{E_8}, and the even unimodular rank-8 lattice is unique = E_8 (Mordell–Witt) — so "c=8 holomorphic limit = $(E_8)_1$" is classification-forced. With v458 (free part) + v459 (spinor extension) the continuum/identification HALF of `SEAM.EQUIV.01` is **[C]** conditional on named audited theorems; the REALISATION R1 (v464) is the sole remaining input. `SEAM.EQUIV.01` stays **[O]**. Exact (sympy/integer q-series + Lie-algebra scan; mpmath cross-check), Wolfram-mirrored. **[E][C][O]**

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Script	**What it machine-checks** ***(continued)***
`v464_seam_oneparticle_rigidity.py`	`SEAM.EQUIV.ONEPARTICLE.01`: the one-particle RIGIDITY + scaling limit attacking R1 — the REALISATION input of `SEAM.EQUIV.01` (A of the A+B round). The seam is quasi-free (v113/v160/v161), so by Araki self-dual CAR the whole net is the second quantisation of its one-particle symbol P, and R1 collapses to a finite-per-mode statement about P. **[C]** ARAKI SELF-DUAL CAR: a gauge-invariant quasi-free state ↔ its symbol P ($0{\le}P{\le}1$) bijectively (Araki 1971; Shale–Stinespring 1965), so a realisation is fully determined by P. **[E]** RIGIDITY: the gapped ground state is the UNIQUE quasi-free state with $P{=}$ projection onto $K{<}0$ ($P^2{=}P$, hermitian to $< 10^{-12}$), K fixed by gap + chirality + reflection (v426) — no second realisation. **[E]** ONE-PARTICLE SCALING LIMIT EXISTS: the real-space kernel is Cauchy on a fixed window ($\max\|P_{2N}{-}P_N\|{\sim}1/N{\to}0$), so P_∞ exists (Gaussianity is why a computation reaches it). **[E]** THE LIMIT IS $c{=}1$ DIRAC ($c_-{=}8$): entanglement $S(L){=}(c/3)\ln L$ gives $c{\to}1$ (per Majorana 1/2; Peschel/Calabrese–Cardy, ties v450), 16 copies $\to c_-{=}8$. **[C]** Shale–Stinespring: gauge-invariant ⇒ no pairing block ⇒ the Bogoliubov map is implementable, GNS reps quasi-equivalent. R1 is upgraded from a bald axiom to “the unique quasi-free realisation, one-particle limit exhibited, modulo the named Araki/Shale–Stinespring functor”; with v463 the `SEAM.EQUIV.01` residual is now entirely certification (named cited functors), zero open internal mechanism — closed modulo cited theorems. `SEAM.EQUIV.01` stays **[O]**. Python-only (numpy: idempotency, kernel Cauchy-convergence, entanglement c-fit). **[E][C][O]**
`v465_seed_crosssector_joint.py`	`SEED.CROSSSECTOR.01`: the cross-sector one-parameter seed test — the θ_{13}-independent, externally-runnable slice of v306. TFPT reads a CMB EB/TB birefringence angle β and a quark Cabibbo angle $\lambda_C{=}\|V_{us}\|$ (unrelated in the SM) as two readouts of the ONE axiom seed $\varphi_0{=}1/(6\pi){+}48c_3^4$. **[E]** plumbing: the axiom φ_0 reproduces the frozen λ_C, $\sin^2\theta_{12}$, β to 10^{-9}. **[E]** CROSS-SECTOR (headline): back-solving φ_0 *independently* from β (CMB) and λ_C (quark) gives the *same* seed within $0.01\sigma_{\rm comb}$ — a correlation only a shared-origin theory predicts. **[E]** one-parameter fit: the error-weighted mean of the three back-solved φ_0 reproduces the axiom to 0.06%; the joint χ^2 at the axiom (zero free parameters, 3 dof) is 0.007, $\chi^2/\mathrm{dof}{<}1$, combined pull 0.08σ. **[E]** θ_{13} EXCLUSION declared, not cherry-picked: adding $\sin^2\theta_{13}$ raises χ^2 by ${\sim}4$ (the documented ${\sim}2\sigma$ pressure point, v328/v338), so this slice is *deliberately* θ_{13}-independent. **[E]** negative control: a data↔formula shuffle explodes the cluster χ^2 by $>5{\times}10^4$. A consistency / out-of-sample statistic (a subset of v306); φ_0 is FIXED by the axioms, so it upgrades *no* prediction. Python-only. **[E]**

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Script	What it machine-checks *(continued)*
`v466_seed_leptonmass_channel.py`	`SEED.LEPTONMASS.01`: a SIXTH, new-sector seed channel — the charged-lepton mass ratio $m_e/m_\mu = \frac{12}{7}\varphi_0^2$ points to the SAME axiom seed $\varphi_0 = 1/(6\pi) + 48c_3^4$, extending the empirical seed over-determination (v306/v465) from three measurement sectors to five. **[E]** plumbing: the axiom φ_0 reproduces the frozen $m_e/m_\mu = \frac{12}{7}\varphi_0^2 = 0.00484673$ to 10^{-9}. **[E]** NEW SECTOR (headline): back-solving φ_0 from the measured POLE ratio $m_e/m_\mu = 0.004836$ (PDG) gives $\varphi_0 = 0.053113$, only -0.11% from the axiom — a charged-lepton MASS reading (Penning-trap masses), a fourth measurement sector beyond {neutrino mixing, CMB, quark mixing}. **[C]** TRANSPORT CAVEAT: the formula is the SOURCE ratio, the datum the POLE ratio; the source→pole shift is the seam-gapped correction class (bound $(2/3)^6 = 8.78\%$, v393) but largely CANCELS for m_e/m_μ, so the 0.11% residual sits far inside the band — added as a CONSISTENT new sector, not a sharper pin. **[E]** EXTENDS THE OVER-DETERMINATION: the six-channel cluster (v306's five + this) has $\chi^2/\mathrm{dof} < 1$ and the error-weighted mean stays at the axiom to $<0.1\%$; with the corroborating heavy-quark m_t/m_b (-0.23%) it is a seven-channel, FIVE-sector agreement (neutrino mixing, CMB, quark mixing, lepton mass, quark mass). **[E]** negative control / power: a data↔formula shuffle explodes the cluster χ^2 by $>100\times$; and not every mass ratio qualifies — $m_c/m_s = (34/47)/\varphi_0$ is scheme-ambiguous (FLAG 11–14), a WEAK channel deliberately not counted. **[C]** HONEST: theoretically still the ONE seed (v305 multiplicity 1) — an EMPIRICAL strengthening (a new measurement sector agrees), NOT theoretical independence; φ_0 is FIXED by the axioms, so it upgrades *no* prediction and closes *no* gate. Python-only. **[E][C]**
`v467_thirdgen_mixing_pattern.py`	FLAV.THIRDGEN.PATTERN.01: the three $\sim 2\sigma$ mixing tensions (θ_{13}, θ_{23}, V_{cb}) are ONE common $-\varphi_0^{\mathrm{ret}}$ suppression ($-5.36\% \pm 1.67\%$, 3.2σ non-zero, -0.03σ from $-\varphi_0^{\mathrm{ret}}$); the ladder-base forms $\lambda_C^2 e^{-5/6}$ / $(1-\varphi_0^{\mathrm{ret}})/2$ / $\lambda_C^2/(1+\lambda_C)$ drop the joint χ^2 10.4 → 0.11 with zero parameters; post-hoc candidate [O], frozen record unchanged, JUNO-era reactors + Belle II decide
`v468_dm2_ratio_jarlskog.py`	FLAV.DM2RATIO.01: $\Delta m_{21}^2/\Delta m_{31}^2 = \lvert J_{\mathrm{PMNS}} \rvert$ of the frozen matrix = 0.029653 vs measured 0.029805(792) → -0.19σ – a parameter-free candidate relation for the previously unpredicted splitting ratio; the informal $m_2/m_3 \sim \pi\varphi_0^{\mathrm{ret}}$ heuristic is now -2.44σ and retyped; NO floor $\Sigma m_\nu = 0.0588$ eV consistent; post-hoc [O], JUNO decides

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Script	**What it machine-checks** *(continued)*
`v469_seam_crossedproduct_route.py`	`SEAM.EQUIV.CROSSEDPRODUCT.01`: the net-level crossed-product certification of the 128-spinor extension + the R1→R1′ invariant-level reduction. **[E]** LR LOCALITY INTEGER (index-2): the $SO(16)_1$ spinor has h_s=$N/16$=16/16=1∈$\mathbb{Z}$ ⇒ statistics phase +1 — the Longo–Rehren criterion HOLDS, so the $\mathbb{Z}_2$ simple-current crossed product $SO(16)_1 \rtimes \{1, s\}$ is a LOCAL index-2 extension (Longo–Rehren, Rev. Math. Phys. 7 (1995); the level-1 $so(N)$ net with DHR sectors from even-CAR: Böckenhauer, Rev. Math. Phys. 8 (1996); simple-current extensions: Böckenhauer–Evans, Comm. Math. Phys. 197 (1998)); KLM $\mu(B)$=$4/2^2$=1 ⇒ HOLOMORPHIC ⇒ $(E_8)_1$ by the v463 pin ($E_4/\eta^8\ q^1$=248 re-verified). The 2024/2025-preprint leg (AGT/AMT, v459) is DEMOTED to an independent second witness — the extension certification now rests on 1995–2001 PEER-REVIEWED journals. **[E]** INDEX-4 PARALLEL: the μ_4 glue J=(s, Λ_1) has $h(J^k)$=$\{1, 1, 1\}$ ALL integer ($\frac{5}{8}+\frac{3}{8}$=1; $\frac{1}{2}+\frac{1}{2}$=1) — the v125 isotropy $q(k, k)$=k^2 at net level; μ=$16/4^2$=1, the same endpoint. **[E]** R1′ INVARIANTS COMPUTED: per-copy FHS Chern $C(M$=$1)$=1 vs $C(M$=$3)$=0, ν=16×$\|C\|$=16, c_-=$\nu/2$=8=g_{car}+N_{fam}; ν mod 16=0 is EXACTLY Kitaev's class-D phase whose edge admits a purely bosonic description = the $(E_8)_1$ level-1 state (Ann. Phys. 321 (2006), sixteen-fold way). **[E]** THE CHAIN COMPOSES: T1 (phase classification: Kitaev + c_- invariance, Kapustin–Spodyneiko PRB 101 (2020) / Kim et al. PRL 129 (2022)) → T2 (MMST CMP 397 (2023) + Böckenhauer 1996) → T3 (LR/BE + KLM CMP 219 (2001)) → T4 (Dong–Mason + v463) with matching in/out slots (v297 typing). **[C]** R1→R1′: `collar_realizes` (model-level fiat) replaceable by `collar_invariants` — {quasi-free **[C]** (v155/v160), gap **[E]** (v302), class D, c_-=8 **[E]** (v456, from P1)}; quasi-freeness stays the ONE **[C]** hypothesis; Lean mirror `FORM.SEAM.RESIDUAL.01`. **[O]** NOT CLOSED: the cited theorems stay cited (certification, not re-proof); `SEAM.EQUIV.01` stays **[O]**. Mixed exact (weights/μ/tower, Wolfram-mirrored) + numerical (FHS Chern, Python-only). **[E][C][O]**

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Script	What it machine-checks *(continued)*
`v470_alpha_inflow_level.py`	`ALPHA.QUILLEN.INFLOW.01`: the α^3 level = the COMPUTED bulk Chern invariant + the seam F-normalisation = the affine embedding index k_Y=5/3. [E] TARGET RE-VERIFIED: α^{-1}=137.0359992168 unique root; the Quillen split $a^3-2c_3^3a^2=8b_1c_3^6\ln(1/\varphi_{\text{seam}})$ at the root to $<10^{-35}$ (v341/v382). [E] THE INVARIANT COMPUTED: FHS Chern $C(M{=}1){=}1$, $C(M{=}3){=}0$ — the same integers as v367/v461, here read as the quantised $U(1)$ response of the collar phase. [C] $k_0{=}\lvert C\rvert{=}1$ UNDER THE EM1 READING: the a^3 coefficient EQUALS the computed bulk invariant of the SAME model that realises S3 — "why exactly 1" is answered by computation + the cited inflow identification (TKNN PRL 49 (1982) / Avron–Seiler–Simon PRL 51 (1983) quantisation; Callan–Harvey Nucl. Phys. B250 (1985) inflow; APS 1975 / Alvarez-Gaumé–Della Pietra–Moore 1985 / Witten Rev. Mod. Phys. 88 (2016) η=CS part of $\delta\log\det$; Quillen 1985 / Bismut–Freed CMP 106 (1986) determinant-line integrality) — REPLACING v435's minimality assumption. [E] k_Y=5/3 EXACT: $\operatorname{tr}_{\bar{5}}(Y^2)/\operatorname{tr}(T_3^2){=}(5/6)/(1/2){=}5/3$ (Ginsparg, Phys. Lett. B197 (1987)); $(3/5)\times\frac{41}{6}{=}\frac{41}{10}{=}b_1$ — the "GUT 3/5" IS the inverse embedding index, not an imported convention; carrier decomposition $\frac{41}{6}{=}\frac{20}{3}$ (fermions, 3×16-content) $+\frac{1}{6}$ (Higgs doublet). [C] F-NORMALISATION RETYPED: once the seam net is $(E_8)_1$ level 1 (`SEAM.EQUIV.01`), the $U(1)_Y$ normalisation $\langle J(z)J(w)\rangle{=}k/(z{-}w)^2$ is FIXED by k_Y — level-1 current-algebra rigidity, no continuous dial; the v434 [C] has ZERO independent content (a face of `SEAM.EQUIV.01`, per the v382 typing). [E] UNIFICATION: ONE invertible phase, TWO quantised responses — $c_-{=}16\times\frac{1}{2}\times\lvert C\rvert{=}8{=}g_{\text{car}}{+}N_{\text{fam}}$ (gravitational, feeds `SEAM.EQUIV.01`/S3, v456) and $C{=}1$ ($U(1)$, feeds EM1) — "the one theorem and the one number" are two faces of the SAME phase; closing R1′ (v469) feeds both. [O] NOT CLOSED: the bridge lemma "$\delta\log\det_\zeta$(seam) = the inflow response" stays the named cited step; the EM1 reading stays [C]; `ALPHA.QUILLEN.EXACT.01` stays [O]; α^{-1} stays [E] regardless. Mixed exact (k_Y/b_1 identities, Wolfram-mirrored) + numerical (root/split mpmath, FHS Chern, Python-only). [E][C][O]
`v471_seam_horizon_replica.py`	`SEAM.EHMODEL.04`: the Seam–Horizon replica chain exercised NUMERICALLY on a discretized seam collar — the kernel-identification premise of `SEAM.THEOREM.01` (the residual isolated by v150/v151/v152) discharged at the FINITE/MODEL level for the seam's own kernel, with REAL replica sheets. [E] REPLICA⇒EH FORM on a real operator: $dG/d\ln m = 2C(\gamma)$ on deficit cones $\gamma/2\pi = \frac{1}{4}, \frac{1}{2}, \frac{3}{4}$ (0.01–0.05%) AND replica sheets n=2 (0.8%), n=3 (1.5%). [E] COEFFICIENT CUTOFF-INDEPENDENT (0.5% drift) while the ζ-scale μ_{lat} drifts with the discretization = the v152 anchor exhibited on data; NO canonical $\ln(m/\mu){=}\frac{3}{4}$ emerges (recorded honestly). [E] BFK SPLIT MEASURED ON THE KERNEL: Schur identity exact ($\sim10^{-16}$); the Calderón/Schur factor's deficit slope is a PURE CUT-EDGE law $E(1{-}q)$ with tip term 0 (power check: the same law fails on the full determinant); the halves carry the whole tip = $2C_{\text{cone}}$ via Kac doubling (exact). [E] THE SEAM'S OWN SPECTRUM: the v302 transfer masses $m_2{=}6\ln\frac{3}{2}$, $m_3{=}6\ln 3$ sit ON the calibrated EH line ($\leq$ 0.05%); the Perron/attractor mode (m=0) is IR-DIVERGENT (runs exactly as $-2C\ln(\text{IR})$) — the recovery gap $\Delta = 6\ln\frac{3}{2}$ makes the induced-gravity coefficient finite: the same gap that makes the attractor unique makes $1/G$ finite. [O] HONEST SCOPE: finite/discretized level; `SEAM.THEOREM.01` stays open — the residual retypes to the CONTINUUM scaling limit (MMST class, = the single residual of `SEAM.EQUIV.01`, v336) + the v152 anchor. Numerical (sparse lattice determinants), Python-only by nature. [E][O]

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Script	**What it machine-checks** ***(continued)***
`v472_quillen_detline_moduli.py`	`ALPHA.QUILLEN.DETLINE.01`: the DETERMINANT LINE over the $U(1)$-twist moduli of the collar model carries curvature = the inflow level — the finite Quillen/Dai–Freed shadow of the v470 bridge lemma, COMPUTED. [E] BLOCH LEVEL RE-VERIFIED: BZ Chern $C(M{=}1){=}1$, $C(M{=}3){=}0$, $C(M{=}{-1}){=}{-1}$ (the v470/v367 integers). [E] THE QUILLEN OBJECT COMPUTED: the same collar $h(k)$ in real space on an $L \times L$ torus with TWISTED boundary conditions — the twist torus (θ_x, θ_y) IS the moduli space of flat $U(1)$ connections — and the Fukui–Hatsugai–Suzuki curvature of the determinant line of the occupied one-particle frame (the free-fermion ground state) over that torus = 1 exactly (10^{-9}) at $M{=}1$, for $L{=}4$ AND $L{=}6$ (size-independent, as an integer must be). [E] CONTROLS: trivial collar $M{=}3 \to 0$; orientation flip $M{=}{-1} \to -1$. [E] TWO TORI, ONE LEVEL: the twist-moduli integer equals the Bloch-BZ integer for all three M — the Niu–Thouless–Wu identification (PRB 31 (1985)), exhibited on the collar model that realises S3 (v460/v461). [E] SECTION GLOBALLY DEFINED: the Fermi gap (= 2.0) stays open over the WHOLE twist torus — the Dai–Freed section has no zero, the holonomy reading is globally valid. [C] THE READING: det-line holonomy over the $U(1)$ moduli = inflow level $k_0{=}\lvert C\rvert{=}1$ — the v470 bridge lemma "$\delta \log \det_\zeta(\text{seam})$ = the inflow response" holds VERBATIM at the finite/model level, where $\log\det$ of the occupied frame replaces $\log\det_\zeta$ (Quillen 1985; Bismut–Freed CMP 106 (1986); Dai–Freed JMP 35 (1994); FHS JPSJ 74 (2005)). [O] NOT CLOSED: the abstract-seam ζ-determinant identification (the continuum leg, = the `SEAM.EQUIV.01` face); `ALPHA.QUILLEN.EXACT.01` stays [O]; α^{-1} stays [E] regardless. Structurally the same manoeuvre as v471 for `SEAM.THEOREM.01`'s replica side. Numerical (dense eigh + FHS twist grid), Python-only by nature. [E][C][O]
`v473_entropic_action_bridge.py`	`GRAV.ENTROPIC.ACTION.01`: the entropic-action bridge — Bianconi's *Gravity from entropy* (Phys. Rev. D 111, 066001 (2025); `arXiv:2408.14391`) quantified as an EXTERNAL CANDIDATE for the local volume action behind TFPT's parameter-free Einstein equation (v358/v359, honestly typed "equation of state, not a from-action quantisation" [O]). Her $S_{\mathrm{B}} = -\operatorname{Tr}_F \ln(\widetilde{G}\widetilde{g}^{-1})$ is the quantum relative entropy between the spacetime metric and the matter-induced metric (Araki-related, $\Delta^{1/2} = \widetilde{G}\widetilde{g}^{-1}$), a local covariant action whose variation gives Einstein + emergent nonnegative Λ at low coupling — the missing variational layer IF the bridge holds. [E] CARRIER HODGE COUNT: her $\Omega^{0,1,2}$ fiber on the FIVE-SLOT carrier $\mathbb{C}^5$ (not 4d spacetime, where it is 1+4+6=11) counts $1{+}5{+}10{=}16{=}2^{g_{\mathrm{car}}-1}{=}\dim S^+_{D_5}$, Hodge-folded onto $\Lambda^{\mathrm{even}}\mathbb{C}^5$ (vector-space identity ONLY). [E] COEFFICIENT MATCH (the one quantitative pin): her weak-coupling $\mathcal{L} = 3\beta R - \alpha\mathcal{L}_{\mathrm{m}}$ matched to $1/(16\pi G) = c_3/(2G)$ fixes her free constant EXACTLY, $\beta'_{\mathrm{B}} = c_3/6 = 1/(48\pi)$; the $6 = 3$ blocks $\times\, 2 = p_2 = \lvert R^+(A_3)\rvert$ (structural echo, not a proof). [E] ENTROPY POTENTIAL: $\Lambda_{\mathcal{G}} = \frac{1}{2\beta}\operatorname{Tr}(\mathcal{G} - I - \ln\mathcal{G}) \geq 0$, quadratic at the fixed point $\Rightarrow$ with $\varepsilon_\star = e^{-\alpha^{-1}}Q$ the closed v60 branch $(3/4\pi^2)e^{-2\alpha^{-1}}$ is reproduced with the EXACT target $\operatorname{Tr} Q^2 = 32c_3^4 = 1/(128\pi^4) = \frac{2}{3}\,\delta_{\mathrm{top}}$ ([C] until Q is constructed). [E] FRULLANI: $-\ln A = \int_0^\infty \frac{dt}{t}(e^{-tA} - e^{-t})$ — her log action is an all-scale heat-kernel action, the right genre against $S_{\mathrm{rel},\chi}$ (KMS cutoff). [E] R^2 KILL TEST: raw $(c_3/6)^2$ vs Starobinsky $1/(12c_3^7)$ ($M^2_{\mathrm{scal}} = c_3^7\bar{M}^2_{\mathrm{Pl}}$, v36) — EXACT gap $3(8\pi)^9 \approx 1.2 \times 10^{13}$ (13 orders): a direct identification of the raw entropic action with the TFPT R^2 sector is FALSE without a mechanism (light $\mathcal{G}$-trace mode generating c_3^{-9}, or KMS-spectral renormalisation) — pre-registered. [C] COMPRESSION CONJECTURE: $P_\Sigma(\widetilde{G}\widetilde{g}^{-1})P_\Sigma = \Delta_\Sigma^{1/2}$ + the operator-level Hodge items (Clifford, Dirac, leaf involution, Calderón polarisation) stay open; Lorentzian positivity NOT automatic — TFPT's OS route is the sturdier construction. [O] NOTHING CLOSES: v359's equation-of-state status stays [O]; zero TFPT knobs added. Mixed exact (sympy, Wolfram-mirrored) + one quadrature (mpmath). [E][C][O]

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Script	What it machine-checks *(continued)*
`v474_entropic_hodge_carrier.py`	`GRAV.ENTROPIC.HODGE.01`: the OPERATOR level of the entropic-action bridge (work packages 1 + 4-algebra of v473). [E] CAR⇒CLIFFORD: the ten Γ's from $(a_k^\dagger, a_k)$ on the carrier Fock space $\Lambda^\bullet\mathbb{C}^5$ (dim $2^{g_{\rm car}}$=32) satisfy $\{\Gamma_a,\Gamma_b\}$=$2\delta_{ab}$ exactly (all 100 pairs, integer matrices) — the $\mathfrak{so}(10)$=D_5 Clifford algebra acts on the carrier exterior algebra. [E] S^+=$\Lambda^{\rm even}$ AS D_5-MODULE: the 45 bilinears $[\Gamma_a,\Gamma_b]/4$ commute with γ=$(-1)^N$, span exactly 45= dim $\mathfrak{so}(10)$ and leak NOTHING from the 16-dim even subspace — the half-spinor IS the even exterior algebra WITH its spinor structure (operator level of v2/v44/v197). [E] HODGE–DIRAC SYMBOL: $c(\xi)$=$a^\dagger(\xi)$−$a(\xi)$ satisfies $c(\xi)^2$=$-\|\xi\|^2$ Id for SYMBOLIC ξ — Bianconi's D=d+δ is Clifford multiplication at symbol level. [E] THE FOLD IS 5→$\bar{5}$: the traceless $\mathfrak{u}(5)$ weights of Λ^4 are the NEGATIVES of those of Λ^1, so the Hodge fold $\star : \Lambda^1 \cong \Lambda^4$ conjugates the vector block — her fiber 1+5+10 becomes the GUT half-spinor 16=1+$\bar{5}$+10 (the SM carrier decomposition), not a mere dimension match. [E] HONEST OBSTRUCTION: $\Omega^{\leq 2}$ is NOT Clifford-invariant ($c(\xi)$ maps Λ^2 into Λ^3); only $\Lambda^{\rm even}$ is (even algebra) — the bridge must go through the fold, never through degree truncation (constraint on AP2). [E] Q-TARGET ENUMERATION (the v473 conjecture DECIDED): $\operatorname{Tr} Q^2$=$d\,k^2 c_3^4$=$32c_3^4$ with INTEGER k has exactly the supports (d,k)=$(2,4),(8,2),(32,1)$ = $\{\lvert\mathbb{Z}_2\rvert,\ g_{\rm car}$+$N_{\rm fam}$= rank $E_8,\ 2^{g_{\rm car}}$= dim $\Lambda^\bullet\}$ — minimal uniform solution q=c_3^2=$1/(64\pi^2)$ per Fock mode; the two-form block d=10 (the "pair sector" conjecture) and the fiber d=16 need irrational multipliers $\sqrt{16/5}$, $\sqrt{2}$ ⇒ KILLED in uniform form; bonus identity $\operatorname{Tr} Q^2/\delta_{\rm top}$=32/48=2/3= dim $\Lambda^\bullet/\Omega_{\rm adm}$. [C] the physical choice among the three admissible supports; [O] AP2 compression + R^2 mechanism stay open, no marker moves. Mixed exact: integer Fock matrices (numpy, exact in double) + symbolic sympy (weights/enumeration Wolfram-mirrored). [E][C][O]
`v475_entropic_scalaron_gate.py`	`GRAV.ENTROPIC.SCALARON.01`: the R^2 kill test of the entropic-action bridge EXECUTED (work package 5 of v473) + the explicit Lorentzian-positivity witness. [E] MAX-SYMMETRIC EIGENVALUES (her Appendix-B flattening verified): $\widetilde{\mathcal{R}}\tilde{g}^{-1}$ has eigenvalues EXACTLY $\{R;\ R/4\times 4;\ R/6\times 6\}$ on a general diagonal 4d metric, incl. the 6×6 two-form flattening. [E] EXACT VACUUM $f(R)$: $\mathcal{L}_{\rm B}(R) = -[\ln(1-\beta R) + 4\ln(1-\beta R/4) + 6\ln(1-\beta R/6)] = 3\beta R + \frac{17}{24}\beta^2R^2 + O(R^3)$ — the 3 re-derives her Eq. (45) (= the v473 pin), $\frac{17}{24} = \frac{1}{2}(1+\frac{1}{4}+\frac{1}{6})$ is the EXACT tensorial R^2 factor. [E] RAW SCALARON TRANS-PLANCKIAN: with β'=$c_3/6$ both routes (Starobinsky matching AND $f'/(3f'')$) give $m_{\rm raw}^2 = \frac{4608\pi^2}{17}M_{\rm Pl}^2$, $m_{\rm raw} \approx 51.7\,M_{\rm Pl} \approx 1.3\times10^{20}$ GeV — HEAVY, not light (TFPT: $c_3^{7/2}M_{\rm Pl} = 3.06\times10^{13}$ GeV); required enhancement EXACTLY $m_{\rm raw}^2/m_{\rm TFPT}^2 = \frac{72}{17}(8\pi)^9 \approx 1.69\times10^{13}$ in mass² — v473's interpretation 2 (the TFPT scalaron as the light $\mathcal{G}$-trace mode) is KILLED in naive form. [E] DOMAIN: first log pole at $R = 1/\beta = 384\pi^2 M_{\rm Pl}^2$ (trans-Planckian) — the raw action is well-defined at all sub-Planckian curvatures; the obstruction is the MASS, not the domain. [E] LORENTZIAN WITNESS: a timelike gradient gives Gg^{-1} eigenvalue $1-\alpha v^2 \leq 0$ for $v^2 \geq 1/\alpha$ (ln undefined); spacelike/Euclidean positive for all v — the v473 [C] caveat exhibited, the OS/reflection-positive route sturdier. [C] VERDICT: interpretation 1 survives (the entropic action covers the Einstein + Λ level), 2 is dead naive, 3 (KMS-spectral renormalisation) is the ONLY surviving R^2 route. [O] unchanged: AP2 compression + the R^2 gate; no marker moves. Exact (sympy, Wolfram-mirrored). [E][C][O]

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Script	**What it machine-checks** *(continued)*
`v476_entropic_compression_toy.py`	`GRAV.ENTROPIC.COMPRESS.01`: the AP2 compression conjecture of the entropic-action bridge (v473) made WELL-POSED on an exactly solvable model. Numerical facts on a staggered-mass chain: the ground-state covariance is a PROJECTION (pure bulk, thermal boundary — the seam situation); the compression $C_A{=}PCP$ has spectrum strictly inside $(0,1)$ with finite Peschel $K{=}\ln((1{-}C_A)/C_A)$ and $S{>}0$ (v155 in the bridge context). ORDER OBSTRUCTION (the reading DECIDED): on the pure bulk $f(x){=}x/(1{-}x)$ is SINGULAR on spec$(C){=}\{0,1\}$ — the literal operator-side reading $Pf(C)P$ of the v473 conjecture is ILL-POSED while $f(PCP)$ is finite: the compression identity can only be read STATE-SIDE (build Δ_Σ from the COMPRESSED relative metric), matching Bianconi's own local-metric construction — AP2 retyped, not weakened. [E] BLOCK ALGEBRA (symbolic): $[C^2]_{AA}{-}(C_{AA})^2{=}BB^t$ and $[C^3]_{AA}{-}(C_{AA})^3{=}ABB^t{+}BB^tA{+}BDB^t$ — first order in the cross block vanishes identically, so for ANY analytic f the two readings agree to SECOND order: the modular (entanglement) content lives exactly in that quadratic correction. GAP CONTROL at the bounded (covariance) level: the bulk identity $f(C_T){=}e^{-h/T}$ holds to 10^{-11} relative; the mismatch obeys the quadratic law $d \le x^2$ and falls with the gap in the gapped regime, the decoupled cut agrees EXACTLY — the two readings converge in the gap-dominated regime, where TFPT operates ($\Delta_{\rm eff}{>}0$, v76/v337). [C] the retyped conjecture; [O] AP2 stays open (the continuum seam statement); no marker moves. Numerical (numpy lattice + one symbolic block identity), Python-only by nature. [C][O]
`v477_entropic_scaleflow.py`	`GRAV.ENTROPIC.SCALEFLOW.01`: the scale-flow representation of the entropic action — v475's surviving interpretation 3 typed PRECISELY: the $(72/17)(8\pi)^9$ gap collapses to ONE moment condition on the scale weight, satisfied by the SAME KMS moment TFPT already fixes (v36: $f_0{=}6(4\pi)^2/c_3^7$); zero new dials, nothing closes. [E] SCALE-FLOW IDENTITY: $-\ln a = \int_0^\infty \frac{dt}{t}(e^{-ta}{-}e^{-t})$ (symbolic) $= 2\int_0^\infty \frac{d\chi}{\chi}(e^{-a/\chi^2}{-}e^{-1/\chi^2})$ ($t{=}1/\chi^2$, 10^{-20} quadrature) — Bianconi's log action IS the FLAT scale-integral of relative heat-kernel actions; TFPT's $S_{{\rm rel},\chi}$ evaluates ONE KMS scale with a weight f: the bridge is a choice of SCALE MEASURE. [E] MOMENT DICTIONARY: a weight $w(t)\,dt/t$ gives $3\beta R\,\mu_1 + \frac{17}{24}\beta^2R^2\mu_2$ ($\mu_1{=}\int we^{-t}dt$, $\mu_2{=}\int wte^{-t}dt$); the flat weight has $\mu_1{=}\mu_2{=}1$ = the v475 raw coefficients — the raw action is the unit-moment point of a moment family. [E] THE ONE MOMENT CONDITION: with the weight-dressed EH pin $\beta'\mu_1{=}c_3/6$ the scalaron is $m^2{=}\frac{4608\pi^2}{17}\frac{\mu_1^2}{\mu_2}M_{\rm Pl}^2$; demanding $m^2{=}c_3^7M_{\rm Pl}^2$ forces EXACTLY $\mu_2/\mu_1^2{=}\frac{72}{17}(8\pi)^9$ — the v475 kill-test number IS a moment ratio; and the closure identity $\frac{4608\pi^2/17}{(72/17)(8\pi)^9}{=}c_3^7$ holds IDENTICALLY: the 13 orders are a scale-measure datum, not a bridge mismatch. [E] v36-CONSISTENCY: $f_0{=}6(4\pi)^2/c_3^7 \Leftrightarrow M^2/M_{\rm Pl}^2{=}c_3^7$ — the same mass from both parametrisations, ZERO new dials. [C] a CONSISTENCY statement, not a derivation (the KMS weight is TFPT input; the entropic action supplies the Einstein + Λ level and the scale-flow FORM, TFPT supplies the measure); negative side-record: the exact lattice-BW commutator attempt for AP2 (Peschel–Eisler simple forms) did NOT hold on the half-filled open chain (kept out of the suite, logged). [O] unchanged: the R^2 gate and v359's typing. Exact (sympy + one quadrature, Wolfram-mirrored). [E][C][O]

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Script	What it machine-checks *(continued)*
`v478_entropic_continuum_legs.py`	`GRAV.ENTROPIC.LEGS.01`: first computable steps on the TWO remaining legs of the entropic-action bridge (v473–v477). LEG A (AP2 continuum): the compressed critical interval has $S(L)=(c/3)\ln L+k$ with $c_{\rm est}$=1.00089/1.00038/1.00021 (monotone → the Dirac $c=1=2\times\frac{1}{2}$, the v450 reading; gapped control $\sim 6\times10^{-8}$, area law); the Peschel modular Hamiltonian $K=\ln((1-C_A)/C_A)$ organises on the CHM weight $\beta(x)=x(L-x)/L$ (corr 0.87→0.99, plateau drift shrinking; constants deliberately NOT asserted — the lattice EH is only asymptotically local); even-range bands vanish identically (particle-hole, $\sim 10^{-36}$), odd tail 1.8% parabola-enveloped — the state-side Δ_Σ (v476's forced reading) FLOWS TO the CHM/BW geometric form that TFPT's Einstein derivation consumes (v323/v358): the bridge's modular side and TFPT's gravity side MEET in the continuum limit. LEG B (the measure): **[E]** single-scale corollary — $d\mu=\delta(t-t_0)dt$ gives $\mu_2/\mu_1^2=e^{t_0}$ exactly, so the v477 condition forces $t_0=\ln\frac{72}{17}+9\ln(8\pi)=30.461$ — ONE dimensionless KMS time, exact closed form; DECLINED: $t_0\neq h(E_8)=30$ (difference 0.461; the near-miss is rejected per the v354/v355 no-free-pattern rule); deriving the measure from the entropic structure stays OPEN. **[C]** the meeting point; **[O]** both legs, no marker moves. High-precision numerical (mpmath, 50 digits) + one exact corollary, Python-only by nature. **[C][O]**
`v479_kronheimer_quiver_bridge.py`	`SEAM.KRONHEIMER.01`: the R3 "combinatorial-to-geometric bridge" of the $\det K=1$ keystone (v344 — "the rewrite attractor graph IS the du Val singularity") is an ESTABLISHED THEOREM: Kronheimer's ALE hyper-Kähler quotient construction + Torelli classification (J. Diff. Geom. 29 (1989) 665 and 685), with ALL finite input data already compiler outputs. **[E]** MARKS = NULL = PERRON: $C_{\rm aff}\delta=0$ and $A\delta=2\delta$ on the affine-E_8 McKay graph; multiset $\{1,2,2,3,3,4,4,5,6\}$ = the $2I$ irrep dimensions = the v312 rewrite-attractor Perron vector. **[E]** DIMENSION COUNT: $\dim_{\mathbb{R}} M-4\dim_{\mathbb{R}} G=4(\sum_{\rm edges} d_id_j-\sum d_i^2+1)=4$ — the hyper-Kähler quotient of the marks quiver is 4-real-dimensional (the $\mathbb{C}^2/\Gamma$ geometry); $\sum_{\rm edges} d_id_j=\sum d_i^2=120$ forced by the null-vector identity. **[E]** GAUGE DATUM: $\sum\delta_i^2=120=\lvert 2I\rvert=\lvert\mu_4\rvert\,h(E_8)$, $h(E_8)=30=\lvert\mathbb{Z}_2\rvert N_{\rm fam}g_{\rm car}$. **[E]** RESOLUTION: $8=\operatorname{rank}E_8$ exceptional spheres with intersection form $-$Cartan(E_8), det =1 and SNF = identity (the Poincaré-sphere link = the $\det K=1$ face, v232/v344); deformation space $\mathfrak{h}\otimes\mathbb{R}^3$, $\dim 3\cdot 8=24$. **[C]** the theorems are CITED, not re-proved (the same import class as NPW26/MMST/AGT, v424/v336/v459). **[O]** the residual premise TRANSFORMS, it does not vanish: "the raw seam supplies ALE hyper-Kähler data asymptotic to the $(2,3,5)$ cone" = the SAME single arrow as `SEAM.EQUIV.01` (v346 L2); no gate closes. Exact (sympy; Wolfram mirror added, counted with the verified 2026-07-21 engine run). **[E][C][O]**

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Script	**What it machine-checks *(continued)***
`v480_multilocal_four_interval.py`	`QGEO.MULTILOCAL.01`: the four μ_4 marks make the seam circle a FOUR-INTERVAL free-fermion geometry — the class whose multi-interval modular theory is EXACTLY KNOWN (Casini–Huerta mixing term; Longo–Xu multi-interval subfactors; Rehren–Tedesco multilocal fermions; KLM: holomorphic $\Leftrightarrow$ trivial multi-interval inclusion, μ=1 — the SAME one bit as det K=1). On the exactly solvable antiperiodic critical ring (N=256, Q=64, ℓ=48): [E] BINARY CLOCK: the quarter-turn clock is a signed permutation with $\rho^4 = -1$ EXACTLY — the fermionic μ_4 clock is the order-8 double-cover lift (the same $\mathbb{Z}_2$-lift pattern as $2I$ over I). [E] EXACT SECTOR DECOUPLING: the primitive-8th-root character projectors resolve the identity and commute with the reduced covariance at 10^{-15} (the Rehren–Tedesco diagonal 4-fold-cover isomorphism at covariance level). [E] COVER/TWIST: each sector kernel is the SINGLE-interval quarter-ring kernel with boundary twist λ_k; the union of sector spectra = the full 4-interval spectrum at 10^{-14}. [E] CLOCK INVARIANCE MANIFEST: $\rho C \rho^{-1}$=C at 10^{-14} $\Rightarrow$ $[\rho, K]$=0 by functional calculus — $\omega \circ \rho = \omega$ HOLDS at the 4-interval state level; NEGATIVE CONTROL: displacing ONE interval by one site re-couples the sectors at 10^{-2}. CONJUGATE-POINT MIXING (numerical): every cross-interval row of K peaks on $\lvert s'-s\rvert \leq 2$ at both sizes — the lattice shadow of the Casini–Huerta mixing term. [C] the continuum multi-interval theory is CITED; [O] the raw-seam premise (`QGEO.SYM.01`/`SEAM.EQUIV.01`) stays OPEN — mechanism exhibited, not closed. Numerical (numpy), Python-only by nature. [E][C][O]
`v481_seesaw_carrier_ladder.py`	`FLAV.NUSCALE.02`: the absolute neutrino-mass scale under the CARRIER Yukawa normalisation y_ν=y_t — `FLAV.NUSCALE.01`'s two-parameter (y_ν, M_R) trade-off collapses to ONE parameter and the required M_R lands INSIDE the compiler's own PS window; the nearby integer c_3-rung is explicitly DECLINED at 1-loop (v354/v355). CANDIDATE class (genre v467/v468); the frozen record (v84) is untouched and NO gate closes. CARRIER NORMALISATION (conjectured premise; named premise): one $SO(10)$ **16** per family + minimal Yukawa sector ($\mathbf{10}_H$ / PS $(1,2,2)$) $\Rightarrow$ $y_\nu = y_t$ at the matching scale — the trade-off collapses to M_R alone. INVERSION INSIDE THE WINDOW (numerical): the observed m_3=0.0503 eV demands M_R=9.3×10^{13} GeV, INSIDE $[M_{\text{PS}}, M_{\text{GUT}}] = [4.2 \times 10^{13}, 2.4 \times 10^{15}]$ (v249), at $\log_{c_3}(\bar{M}_{\text{Pl}}/M_R) = 3.15$ (y_ν=1 gives the structureless 2.58 — the old `FLAV.NUSCALE.01` number). [X] INTEGER-RUNG HONESTY GATE: $M_R = c_3^3 \bar{M}_{\text{Pl}} = 1.53 \times 10^{14}$ GeV predicts $m_3 = 0.030$ eV (40% low at 1-loop; ADKLR κ run-down factor 0.794) — NEAR the rung (3.15 vs 3), NOT ON: the ladder pin is DECLINED; the decision computation is NAMED (2-loop SM + explicit PS threshold matching). FALSIFIABLE BAND (numerical): over the PS window m_3 spans $[0.002, 0.115]$ eV bracketing the observation; the M_{PS} end ($\Sigma \sim 0.124$ eV) is already in tension with DESI ~0.072 — cosmology cuts the window from below; the NO floor $\Sigma m_\nu = 0.059$ eV stays the falsifiable edge (v272). [C] / [O] candidate class; the absolute scale stays formally open. Numerical (explicit 1-loop RG: gauge/y_t/λ up, ADKLR Weinberg-operator down), Python-only by nature. [C][O]

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Script	**What it machine-checks** ***(continued)***
`v482_seesaw_rung_decision.py`	`FLAV.NUSCALE.03`: the decision computation NAMED by v481, EXECUTED in bracketed (assumption-robust) form — the integer rung $M_R = c_3^3 \bar{M}_{\rm Pl}$ is EXCLUDED as an unstructured exact pin, conditional on the carrier premise $y_\nu{=}y_t$. THE GAP: central m_3(rung)=0.0301 eV vs observed 0.0503 — rescue factor ×1.670 (+67%, 0.154 c_3-rungs). BRACKETED EXCLUSION: with $m_t(\overline{\rm MS}) \pm 3$ GeV ($> 3\sigma$), $\alpha_3^{-1} \mp 0.10$ ($> 3\sigma$), a ±10% band on the ADKLR κ run-down and the PS-leg Yukawa beta scaled $\times[0.5, 1.5]$, the largest single shift is ×1.10 and the ALL-favourable combination reaches only ×1.165 — far below ×1.670: the v481 1-loop miss is NOT an RG artifact. **[C]** NAMED ESCAPE, RECORDED AND DECLINED: only a third-generation Majorana STRUCTURE factor $r \sim 1.67$ revives the rung; post-hoc $r_{\rm needed}$=1.6696 sits 0.18% from $g_{\rm car}/N_{\rm fam}$=5/3 — no mechanism forces a 3/5 Clebsch on $M_{R,3}$ today, so this is a named coincidence with a named missing mechanism (PS Majorana-sector Clebsch), DECLINED per v354/v355 (the $t_0{\sim}30$ pattern of v478). **[O]** the v481 one-parameter window candidate (NO floor Σm_ν=0.059 eV) never used the integer rung and STANDS; frozen record untouched; nothing closes. Numerical (explicit 1-loop RG + bracketed envelopes), Python-only by nature. **[C][O][X]**
`v483_pillowcase_exact_traces.py`	`HYP.PHIO.GEOM.01`: the GEOMETRIC side of the $\varphi_0^{\rm ret}$ puncture target (`HYP.PHIO.PUNCTURE.01`/v408) is EXACT — every twisted/orbifold heat trace of the flat $\tau{=}i$ pillowcase is a t-INDEPENDENT RATIONAL, so the π^{-4} in $d_{\rm top} = 48c_3^4$ cannot come from flat orbifold geometry at ANY order. **[E]** TWISTED TRACES EXACTLY 1: ${\rm Tr}(\gamma e^{t\Delta}) = 1$ for $\gamma = \sigma\,(z{\mapsto}-z)$, $\rho\,(z{\mapsto}iz)$, ρ^3 at every tested $t \in [0.05, 3]$ (position-space Gaussian image sums / erf closed forms + 2d quadrature, residual $< 10^{-9}$; mode-space proof: only n=0 is γ-diagonal) — t-independent, no asymptotic tail. **[E]** ATIYAH–BOTT MATCH: $1 = 4 \times \frac{1}{4}$ ($\det(1{-}d\sigma)$=4=$\lvert\mu_4\rvert$) and $1 = 2 \times \frac{1}{2}$ ($\det(1{-}d\rho)$=2=$\lvert\mathbb{Z}_2\rvert$). **[E]** PILLOWCASE DATA EXACT: orbifold heat trace = $\frac{1}{2}\,{\rm Tr}_{T^2} + \frac{1}{2}$ (contact term EXACTLY $\frac{1}{2} = 4 \times \frac{1}{8}$ per mark, all t); clock-equivariant trace exactly 1 (all t). **[E]**/**[O]** THE REDUCTION: the analytic content of $d_{\rm top} = 48c_3^4 = \frac{3}{256\pi^4}$ must sit in the per-insertion COUPLING normalisation — 4=$\lvert\mu_4\rvert$ insertions of weight c_3 give c_3^4, multiplicity $\Omega_{\rm adm}$=48; the bare $1/(2\pi)$-propagator 4-cycle differs by the exact RATIONAL $\frac{3}{16}$; the ONE remaining **[O]** is the rule "per-mark weight = c_3" (seam zeta/Green normalisation, the same residual class as `EM.WARD.02`). The target narrows from "derive the term" to "derive the weight"; `HYP.PHIO.PUNCTURE.01` stays open; no gate closes. Numerical verification of exact statements + sympy exact arithmetic; Python-only by nature. **[E][O]**

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Script	What it machine-checks *(continued)*
`v484_seam_contact_unit.py`	`SEAM.CONTACT.UNIT.01`: the shared "c_3 per boundary insertion" rule of the two open analytic targets (`ALPHA.QUILLEN.EXACT.01` ladder $\{0,3,6\}$; `HYP.PHI0.PUNCTURE.01` per-mark weight c_3) is ONE identity and NOT a new unknown: it is the KMS seam unit $2\pi = 1/(4c_3)$ with $\frac{1}{4} = 1/\lvert\mu_4\rvert$ (v239) — one bare boundary propagator ORBIT-AVERAGED over the four marks. [E] UNIT CHAIN: $\frac{1}{2\pi} = 4c_3 = \lvert\mu_4\rvert c_3 = \lvert\mathbb{Z}_2\rvert\chi(S^2)c_3$. [E] MECHANISM DERIVED ON THE SEAM CIRCLE: $G(d) = -\frac{1}{\pi}\ln\lvert 2\sin(d/2)\rvert$ (polylog identity, 40 digits) takes INTEGER multiples of $c_3\ln 2$ at the μ_4 separations — $G(\pi/2) = -4c_3\ln 2$, $G(\pi) = -8c_3\ln 2$, so $G_{\text{marks}} = -4c_3\ln 2\cdot\operatorname{circ}(0,1,2,1)$ with spec $= \{4,-2,0,-2\}$; the μ_4-invariant (orbit-averaged) insertion carries $(GV)\rvert_{\text{marks}} = -(c_3\ln 2)\,C$ EXACTLY, so the log-det contact expansion is exactly graded in c_3 per insertion with integer cycle combinatorics $\operatorname{Tr}(C^k) = 4^k + 2(-2)^k$. [E] SUITE-WIDE GRAMMAR: the $F_{U(1)}$ ladder $\{0,3,6\}$ (v3); φ_0^{ret} ladder $\{1,4\}$ (v37); the Λ prefactor $\frac{3}{4\pi^2} = (8\pi)^2 d_{\text{top}} = 48c_3^2$ (v60) — the SAME multiplicity Ω_{adm}=48 as the puncture, at TWO insertions; the $g{-}2$ seam loop divides by the same $2\pi = 1/(4c_3)$. [C]/[O] THE MERGE: the two tracked [O] targets collapse into ONE named analytic step — the diagonal (self-energy) zeta-renormalisation at the marks + the state-multiplicity matching ($48 = \Omega_{\text{adm}}$, $41 = 10b_1$); the finite cycle sector is proven. Anti-numerology: the integers are forced by the square configuration; no target coefficient is claimed. Exact (sympy + 40-digit mpmath polylog); Python-only by nature. [E][C][O]
`v485_contact_diagonal_closed.py`	`SEAM.CONTACT.UNIT.02`: the DIAGONAL channel of the merged analytic target (v484) settled at the computable level — the merged residual COLLAPSES ONTO THE KEYSTONE. [E] DIAGONAL ZERO AT THE KMS SCALE: the renormalised coincident-point Green function on circumference ℓ is $G_{\text{reg}}(0;\ell) = \frac{1}{\pi}\ln(\ell/2\pi)$ exactly — it VANISHES iff $\ell = 2\pi = 1/(4c_3)$, the v239 KMS seam circumference: the seam scale is the unique zero-self-energy scheme point. [E] THE LOG IS THE SCALE CHANNEL: rescaling gives $\frac{1}{\pi}\ln\kappa$ and nothing else — matching the functional structure ($F_{U(1)}$ carries exactly ONE log term, the transport $8b_1c_3^6\ln(1/\varphi_{\text{seam}})$; the φ_0^{ret} split is log-free); [C] reading: log = diagonal/scale channel, log-free = contact data at the KMS point. [E] CLOSED-FORM RESUMMATION (BFK route, v151 class): the finite-rank mark determinant resums exactly, $\det(I-uC) = (1-4u)(1+2u)^2$ with $u = \varepsilon c_3\ln 2$ — all orders; the linear term is ABSENT precisely because the diagonal vanishes ($\operatorname{Tr}C = 0$); the series matches v484's $\operatorname{Tr}(C^k) = 4^k + 2(-2)^k$. [E] WEIGHT DICHOTOMY: the SAME $48 = 3\times 16$ admissible states carry BOTH multiplicities — flat count $48 = \Omega_{\text{adm}}$ (the φ_0^{ret} puncture) vs Y^2/loop/GUT-weighted $41 = 10b_1 = 40+1$ ($\operatorname{Tr}_{16}Y^2 = \frac{10}{3}$ exact, $\nu^c = 0$; Ginsparg $(3/5)(41/6) = 41/10$, v470) — one state set, two response weights. [C]/[O] CONSEQUENCE: with v484 EVERY finite/computable piece of the merged `ALPHA.QUILLEN` + `PHI0.PUNCTURE` target is proven; the single remaining [O] is the identification of the abstract seam ζ-det with this glued mark determinant — a face of `SEAM.EQUIV.01`, exactly the v382 typing, now substantiated computationally. No gate closes. Exact (sympy + 30-digit mpmath + exact charge sums); Python-only by nature. [E][C][O]

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Script	**What it machine-checks** ***(continued)***
`v486_transfer_full_rule.py`	`HYP.REWRITE.02`: the FULL physical transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ (v54/v221) from ONE forced local rule — the lazy $\mathbb{Z}_2$-pair walk; the $(1/3)^6$ mode, previously generator-less, gets its local-rule origin (sharpens v327, whose uniform rule produces $\{1, \frac{2}{3}, 0\}$ and can never generate the third mode). **[E]** UNIQUENESS: for $M(s,h) = \left(\begin{smallmatrix}1&0&0\\0&s&h\\0&h&s\end{smallmatrix}\right)$ (one absorbing family channel + SYMMETRIC $\mathbb{Z}_2$ pair) the survival spectrum is $\{s{+}h,\ s{-}h\}$; demanding the physical pair $\{\frac{2}{3}, \frac{1}{3}\}$ forces UNIQUELY $(\text{stay}, \text{hop}, \text{leak}) = (\frac{1}{2}, \frac{1}{6}, \frac{1}{3}) = (1/\lvert\mathbb{Z}_2\rvert,\ 1/(\lvert\mathbb{Z}_2\rvert N_{\text{fam}}),\ 1/N_{\text{fam}})$ — atom form recorded (honest direction: spectrum → rates). **[E]** SIX-HAND GENERATION: over the order-6 $= 2N_{\text{fam}} = \lvert R^+(A_3)\rvert$ hand, $\operatorname{eig}(B^6) = \{\frac{64}{729}, \frac{1}{729}\}$ exactly; decay gaps $6\ln\frac{3}{2}$ (the recovery gap, v76) and $6\ln 3$ (the subdominant gap, now generator-full). **[E]** SUPERSESSION: the uniform rule's zero mode persists under every power — no iterate reaches $(1/3)^6$; its λ_2 arity story ($\frac{2}{3} = \lvert\mathbb{Z}_2\rvert/N_{\text{fam}}$) is unchanged. **[E]** sub-stochastic, symmetric (detailed balance), absorbing democratic attractor unchanged (v317). **[O]** the arity $\{2,3\}$ stays the anchor input; the lazy-vs-uniform split is SELECTED by the physical spectrum, not yet first-principles-derived (named candidate: stay $= 1/\lvert\mathbb{Z}_2\rvert$ as the sheet-symmetric choice of the $\mathbb{Z}_2$ involution). Exact (sympy); mirrorable, deferred to the next verified engine run. **[E][O]**
`v487_transfer_clock_rungs.py`	`HYP.REWRITE.03`: the lazy split $(\frac{1}{2}, \frac{1}{6})$ is FORCED by the resummed clock — the v486 **[O]** 'split selection' closes onto v124 structure; sharpens `HYP.REWRITE.02`, no gate closes. **[E]** RUNG READING: the lazy rule's one-step spectrum is EXACTLY the complete clock ladder below the wall, $\operatorname{eig}(M) = \{1 - n/3 : n = 0,1,2\} = \{1, \frac{2}{3}, \frac{1}{3}\}$; v327's uniform rule puts its second pair mode ON the wall $(0 = 1 - 3/3)$ — the degeneracy re-read as 'odd channel collapsed to the wall'. **[E]** DECK PARITY = RUNG INDEX: $[B, \sigma] = 0$; the σ-even pair mode carries rung $n = 1$ (survival $\frac{2}{3}$), the σ-odd mode rung $n = 2$ (survival $\frac{1}{3}$) — each clock step flips the deck parity; the structural home of the v486 parity assignment (ω_1 even-channel, ω_2 odd-channel). **[E]** FORCING: faithful ladder (no rung missing, no physical mode on the wall — definitional for a full-spectrum generator) + $\mathbb{Z}_2$ equivariance + rates ≥ 0 force UNIQUELY both the split $(\frac{1}{2}, \frac{1}{6})$ AND the assignment even → rung 1, odd → rung 2 (the swap needs $h = -\frac{1}{6} < 0$); the 'sheet-symmetric choice' candidate is superseded by this proof. **[E]** BEND COROLLARY: $\omega_1/\omega_2 = \text{rate}(2)/\text{rate}(1) = \log_{3/2} 3$ — the parity-comb tone ratio IS the v107/v124 bend of the pulsar-clock search. **[O]** the channel count $3 = N_{\text{fam}}$ is forced by the clock wall; the ladder stays an exact identity on frozen spectra (semiclassical derivation R1 open); the arity $\{2,3\}$ stays the anchor input. Exact (sympy); mirrorable, deferred to the next verified engine run. **[E][O]**

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Script	What it machine-checks *(continued)*
`v488_majorana_clebsch_door.py`	`FLAV.NUSCALE.04`: the 3/5 CLEBSCH DOOR of the seesaw rung, ENUMERATED AND CLOSED — v482's named escape (a third-generation Majorana factor $r \approx 1.6696$, 0.18% from $g_{\rm car}/N_{\rm fam} = 5/3$) is decided NEGATIVE at the group-theory level; the v481 window candidate stands as the honest endpoint. [E] NO RENORMALISABLE CHANNEL: $\mathrm{sym}^2(\mathbf{16}) = 136 = \mathbf{10} + \mathbf{126}$; the $\overline{\mathbf{126}}$ is NOT in the E_8 hull $\{1, 10, 16, 45\}$ (v247) — M_R only via the d=5 operator $(\mathbf{16}_F\overline{\mathbf{16}}_H)^2/\Lambda$ with the E_8-allowed $\mathbf{16}_H$. [E] CHANNEL LIST: $\mathbf{16} \times \overline{\mathbf{16}} = \mathbf{1} + \mathbf{45} + \mathbf{210}$; the hull admits singlet + $\mathbf{45}$ insertions ($\mathbf{210}$ absent). [E] CHARGE TABLE: $Y = T_{3R} + (B{-}L)/2$ on all 16 states, $\mathrm{Tr}_{16}\, Y^2 = \frac{10}{3}$ (v485); $(B{-}L, T_{3R}, Y)(\nu^c) = (+1, -\frac{1}{2}, 0)$. [E] THE $\{2,3\}$-SMOOTH THEOREM: every nonzero ν^c channel weight $\{1, \frac{1}{4}, 1, -\frac{1}{2}, \frac{3}{8}\}$ is $\{2,3\}$-smooth and the multiplicative closure of $\{2,3\}$-smooth rationals never contains $\frac{5}{3}$; the UNIQUE natural 5 of the embedding, $k_Y = \mathrm{Tr}\, Y^2/\,\mathrm{Tr}\, T_{3L}^2 = \frac{5}{3}$ (Ginsparg, v470), is a full-multiplet trace whose direction has $Y(\nu^c) = 0$ EXACTLY — structurally decoupled from the Majorana operator. [E] GENERATION-BLINDNESS: the Clebsch is a family-space scalar; $\mathrm{diag}(1, 1, \frac{3}{5})$ cannot arise from group theory. [O] compiler-flavor inventory typed and declined: exact $\frac{3}{5}$'s only in wrong slots (K lepton gen2/gen1; $R + L$ lepton gen3/gen2), no degenerate-base shape, no M_R link (v354/v355). Exact (Fraction/sympy); mirrorable, deferred to the next verified engine run. [E][O]
`v489_seam_modular_commutator.py`	`SEAM.CCC.01`: the chiral central charge c_-=8 measured DIRECTLY from the bulk ground state of the v367 seam lattice model, via the Kim–Shehab–Kim modular commutator $J(A, B, C) = i\, \mathrm{Tr}(\rho_{ABC}[K_{AB}, K_{BC}]) = (\pi/3)\, c_-$ on a 120° pie tripartition (PRL 128 (2022)) — the FIRST ground-state witness for premise (i) of `SEAM.EQUIV.01` (c_-=8) INDEPENDENT of both the Chern route (v367 FHS curvature) and the counting route (v378 KLM/det-Cartan): J is read off the entanglement data of the exact BdG ground state alone (Peschel covariance, $W{=}-2\,\mathrm{artanh}\,\Gamma$). [E] GROUND-STATE PURITY: $\mathrm{eig}(i\Gamma){=}\pm 1$ to $6{\times}10^{-15}$. [E] PER-LAYER $c_-{=}\frac{1}{2}$: c_-=0.499998 at L=32, R=12 (0.499793 at L=16 $\to$ 0.500000 at L=48, dev $-1.8{\times}10^{-7}$). [E] CONTROLS: trivial M=3 $\to$ 0.000000; orientation flip M=−1 $\to$ −0.499986. [E] LAYER ADDITIVITY EXACT ($1.4{\times}10^{-14}$): J additive over decoupled layers, so the carrier value is 16× the per-layer value. [E] 16-LAYER CARRIER $c_-{=}8{=}g_{\rm car}{+}N_{\rm fam}$: direct 16-layer run 7.9967 (L=16, tol 10^{-2}; L=24 gives 7.999781), additivity route 8.0000 (tol 10^{-3}). [E] CLIP INVARIANCE: $\varepsilon{=}10^{-8}$ vs 10^{-12} moves c_- by $5{\times}10^{-8}$. [O] HONEST FENCE: pins c_-=8 on the LATTICE ground state (third independent witness); does NOT close `SEAM.EQUIV.01` — the continuum scaling limit (MMST v336) stays [O], the μ_4 condensation $SO(16)_1{\to}(E_8)_1$ stays the cited KLM step (v378). Numerical (numpy), Python-only by nature. [E][O]

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Script	**What it machine-checks** ***(continued)***
`v490_seam_parity_census.py`	`SEAM.MU.01`: the T^2 spin-structure fermion-parity census for the v367 seam lattice model — the lattice ground-state witness of "$\mu_{\rm pre}$=4 before condensation" (step 1 of the v378 KLM chain) + the 16-fold-way bridge θ_v=1 with the MEASURED ν=$2c_-$=16 (v489). Ground-state parity per boundary-condition sector (AA, AP, PA, PP), determined TWICE: Read–Green TRIM product AND real-space Majorana Pfaffian sign (Schur), relative to the trivial M=3 phase. **[E]** REAL-SPACE = BLOCH SANITY: twisted spectra match to $9{\times}10^{-15}$. **[E]** PER-LAYER CENSUS $(+,+,+,-)$: TRIM ≡ Pfaffian in every sector — 3 even sectors, the ν=1 Ising signature. **[E]** 16 LAYERS ⇒ $\mu_{\rm pre}$=4: parity^{16}=+ in ALL four sectors ⇒ gauged torus GSD=4=$\mu(SO(16)_1)$=$\lvert\det \mathrm{Cartan}(D_8)\rvert$. **[E]** 16-FOLD-WAY BRIDGE: θ_v=$e^{2\pi i\nu/16}$=1 EXACTLY at ν=$2c_-$=16 (vortex bosonic, condensable); rivals $\nu=1,2,8$ excluded (not holomorphically condensable). **[E]** NON-DISCRIMINATOR FENCE: the fermionic Kitaev–Preskill TEE is identically zero — γ_f=0.000001 (M=1) and 0.000000 (M=3) — an invertible fermionic phase has no TEE, so γ_f does NOT discriminate $(E_8)_1$ vs $SO(16)_1$ (both live only after gauging). **[C]** the KLM step $\mu_{\rm post}$=$4/2^2$=1 stays cited (the condensed model is interacting); the numerics pin only $\{c_-{=}8,\ \mu_{\rm pre}{=}4,\ \theta_v{=}1\}$ — the premise package of the condensation. **[O]** the continuum limit (MMST v336); `SEAM.EQUIV.01` stays **[O]**. Python (numpy + scipy Schur + sympy exact θ_v). **[E][C][O]**
`v491_p2_partition_corollary.py`	`P2.PARTITION.01`: $g_{\rm car}$=5 as a COROLLARY of the four marks — the partition sharpening of v53 (sympy exact, no floats). THE LEMMA: a multiset of exactly 3 POSITIVE INTEGERS summing to 4 is unique — $\{1,1,2\}$ — so e=(e_1,e_2,e_3)=$(4,5,2)$ and $g_{\rm car}$=e_2=5, $\lvert\mathbb{Z}_2\rvert$=e_3=2, $N_{\rm fam}$=$e_2{-}e_3$=3 are COROLLARIES, not inputs; consistency closes as a fixed point $((2^4{-}1)/5{=}3{=}\mathrm{rank}\,H^1{=}\#\mathrm{marks}{-}1)$. NEGATIVE CONTROLS (every assumption load-bearing): drop positivity → 4 solutions, e_2 ambiguous $\{0,3,4,5\}$; sum 3 → e_2=3; sum 5 → not unique; 2/4 parts → $e_2\in\{3,4\}/\{6\}$; Mehta–Seshadri RATIONAL weights → e_2=$\frac{16}{3}$ resp. $\frac{44}{9}\neq 5$ (integrality comes from the Birkhoff–Grothendieck SPLITTING-TYPE typing, not from MS weights). **[E]** v53 SHARPENING: v53 needs all three targets $(4,5,2)$; the partition needs only e_1=4 — same unique $\{1,1,2\}$. **[E]** SUM RULE: $4\times\frac{N-1}{2}$=4 iff N=3; the v115 route $\lvert a\rvert_1$=$\lvert\mu_4\rvert\,\mathrm{tr}(A_0)$=4. INDEPENDENCE CHAIN: #marks = 4 from v216 (P1-side, no $g_{\rm car}$ input); rank H^1=3 pure topology (v137/v228); $\lvert a\rvert_1$=4 $g_{\rm car}$-free (par-deg-0/(U), v115). **[C]** HONEST RESIDUAL: P2 is NOT made axiom-free — the weight-TYPING postulate remains (integrality = splitting exponents; positivity = h^0=0, correlate v137 check 1; sum rule "given (U)"); `AX.P2.01` stays a declared axiom; the $N_{\rm fam}$ direction reversal vs `tfpt_constants` is typed as an over-determination check, never double-counted. Exact (sympy); Wolfram-mirrored. **[E][C]**

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Script	What it machine-checks *(continued)*
`v492_celestial_z4_orbifold.py`	`CELEST.WP1.01`: WP1 of the celestial-holographic research contract `CELEST.SEAM.01` — the E_8 μ_4-glue grading (v128) IS the flat $\mathbb{Z}_4$ monodromy of the glue-equivariant celestial chiral algebra on the A_3 ALE space $\mathbb{C}^2/\mathbb{Z}_4$ (Bittleston–Homans–Sharma, `arXiv:2305.09451`, the $\mathbb{Z}_2$/Eguchi–Hanson case; Kronheimer–Nakajima flat connection at infinity). Sympy exact. [E] INNER GRADING: $h{=}(2,2,2,2,2;0^4)$ reads the glue class mod 4 on all 240 roots $\Rightarrow$ glue $=\mathrm{Ad}(\exp(2\pi i h/4))$, an order-4 torus element = flat $\mathbb{Z}_4$ monodromy; the A_3-side detector $h'{=}(0^5;1,1,1,-3)$ reads the SAME class = the diagonal $(1,1)$ of the discriminant group $\mathbb{Z}_4{\times}\mathbb{Z}_4$ (the v92/v125 Lagrangian glue); $\mathbb{Z}_2$ halfway $120{=}\dim SO(16)$, $128{=}$ spinor. [E] $\mathbb{C}^2/\mathbb{Z}_4$ IS the A_3 singularity $XY{=}Z^4$ (Molien table exact; Hilbert series $(1{+}t^4)/((1{-}t^2)(1{-}t^4))$; conjugation symmetry $N[d][1]{=}N[d][3]$). [E] EQUIVARIANT SECTOR CLOSES: $D[d]{=}60(d{+}1)$ even / $64(d{+}1)$ odd — uniform per parity ONLY because $\dim\mathfrak{g}_j{=}(60,64,60,64)$; zero modes $(d{=}0)=\mathfrak{g}_0=$ carrier $D_5{\oplus}A_3{+}$Cartan $=60$; density $62/248{=}1/4{=}1/\lvert\mathbb{Z}_4\rvert$; EH/$\mathbb{Z}_2$ halfway towers 120/128. [E] CHARACTER: $\Theta_{E_8}=\sum_j\Theta_{C_j}$; shell $1=(52,64,60,64)$ from the LATTICE; the 4 sector characters Θ_{C_j}/η^8 sum exactly to $\chi_{(E_8)_1}{=}1{+}248q{+}4124q^2{+}34752q^3$ (v377); sector weights $h_{D_5}{=}(0,\frac{5}{8},\frac{1}{2},\frac{5}{8})$, $h_{A_3}{=}(0,\frac{3}{8},\frac{1}{2},\frac{3}{8})$ = the v92/v125 discriminant form $(5x^2{+}3y^2)/8$ mod 1; glue diagonal $h{=}(0,1,1,1)$ INTEGER = v125 isotropy = local extension to $(E_8)_1$. [E] CRITICAL CORRECTION: the A_3 deck $\mathrm{diag}(i,i^{-1})$ acts on the celestial sphere as $z\mapsto -z$ (projective order 2, the sheet flip), NOT as the order-4 clock $z\mapsto iz$; the clock is the $U(2)$ phase $\mathrm{diag}(i,1)$ (order 4, fixed $\{0,\infty\}$, free μ_4 orbit — all v180 properties; normalises the deck, $\det{=}i$ rotates the holomorphic symplectic form by the μ_4 generator); exact spin bridge (spin clock)^2 = deck, spin group $\mathbb{Z}_8$, $8{=}2\lvert\mu_4\rvert$ (the $c_3{=}1/(8\pi)$ winding integer). [E] clock-invariant A_3 deformation = the 1-parameter family $XY{=}Z^4{+}a_0$; its 4 branch points are ONE μ_4 orbit with cross-ratio 2 (v179 clock orbit; v168/v214 pillowcase $j{=}1728$). [E] NEGATIVE CONTROLS: false spatial action $\mathrm{diag}(i,i)$ killed three ways ($\det{=}{-}1\notin SL(2,\mathbb{C})$; Poisson closure fails 14/14; grades unsymmetric); false glue breaks additivity on 3112/6720 root pairs; rigidity: only $\mathrm{Aut}(\mathbb{Z}_4)$ relabelings additive. [C] VERDICT B: match under the ONE named identification clock^2=deck on the sphere (sheet-$\mathbb{Z}_2$ spin bookkeeping). [O] structural arithmetic ONLY — no celestial duality constructed; `CELEST.SEAM.01` (WP2–WP5) stays a research contract; `SEAM.EQUIV.01` stays [O], untouched. Exact (sympy). [E][C][O]

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Script	What it machine-checks *(continued)*
`v493_celestial_wp2_clock_deformation.py`	`CELEST.WP2.01`: WP2 of the celestial-holographic research contract `CELEST.SEAM.01` — the clock-invariant deformation $XY{=}Z^4{+}a_0$ of the A_3 seam orbifold: geometry, periods/monodromy, deformed algebra (Bittleston–Homans–Sharma $\mathbb{Z}_2 \to \mathbb{Z}_4$ transfer, `arXiv:2305.09451`). Sympy exact. **[E]** GEOMETRY: $P(iZ){=}P(Z)$ forces $a_3{=}a_2{=}a_1{=}0$ SHARPLY (two-sided: $\prod_k(Z-i^k w){=}Z^4{-}w^4$); smooth iff $a_0{\neq}0$ (disc $=256\,a_0^3$); branch points $Z^4{=}{-}a_0$ = ONE free μ_4 orbit in every fibre; cross-ratio 2 / $j{=}1728$ for ALL a_0 (binary-quartic invariants $I{=}12a_0$, $J{=}0$ identically) $\Rightarrow$ a_0 is a PURE SCALE, the $\tau{=}i$ pillowcase (v168/v214) is FROZEN — no shape modulus. **[E]** PERIODS: $\Pi{=}t(i{-}1)(1,i,i^2)$, all 3 vanishing spheres carry the SAME volume $\sqrt{2}\,t$ (lockstep); the clock on H_2 is a COXETER element of $W(A_3)$ (char $x^3{+}x^2{+}x{+}1$, eigenvalues $\{i,-1,-i\}$ = the A_3 exponents $1,2,3$; $h(A_3){=}4{=}\lvert\mu_4\rvert{=}N_{\mathrm{fam}}{+}1$); NO invariant cycle ($\det(A{-}1){=}{-}4$); the surviving direction is the weight-1 χ_1-Fourier diagonal $(1,i,i^2)$, NOT the naive invariant diagonal; the clock IS the Picard–Lefschetz monodromy of the family ($a_0 \to e^{2\pi i}a_0$ gives the 4-cycle); versal weights $(a_3,a_2,a_1,a_0){=}(1,2,3,0)$ = the REGULAR μ_4 representation, invariant line = a_0 alone. **[E]** ALGEBRA ($\mathbb{Z}_2 \to \mathbb{Z}_4$): mode dictionary $\{w[m,n] : m{\equiv}n \bmod 4\} \leftrightarrow \{X^pY^qZ^r : r{\leq}3\}$; bracket $\{f,g\}{=}{-}4\,\mathrm{Nambu}(XY{-}Z^4{-}a_0)$ anchored at the $a_0{=}0$ orbifold w-sector ($\{X,Y\}{=}16Z^3$, $\{Y,Z\}{=}{-}4Y$, $\{Z,X\}{=}{-}4X$), Jacobi + Casimir exact, uniform clock-degree -1; S-algebra corrections exactly LINEAR in a_0 at the $\mathbb{Z}_4$ wrap $r{+}r'{\geq}4$, grade-conserving (NO sector leak $\Rightarrow$ the v492 equivariant SDYM(E_8) sector deforms consistently); $a_0 \leftrightarrow$ the BHS c^2 slot (weight 0); matrix factorisations (McKay modules) deform along every 1+3 and 2+2 orbit splitting; the clock fixes exactly the antipodal 2+2 sector (the EH/$\mathbb{Z}_2$ middle node, $4{=}2{\times}2$). **[E]** NEGATIVE CONTROLS: $a_1Z \Rightarrow j{=}0$ (hexagonal CM point); pure a_2Z^2 stays SINGULAR (A_1 node); $a_2{\neq}0 \Rightarrow j{=}1556068/81{\neq}1728$ (exact instance, two routes agree); grading leak explicit ($Z^2{\cdot}Z^2$ picks up grades $\{0,1,2\}$). **[C]** three NAMED identifications (verdict B, not A): $-4\,$Nambu = THE $k{=}4$ CCA bracket (BHS prove $k{=}2$); period = root difference (A-series residue normalisation); the seam-scale reading of a_0 presupposes the WP1 bridge clock^2=deck ($\mathbb{Z}_8$, $8{=}2\lvert\mu_4\rvert$). **[C]** v216 PRECISION (typed, NO marker move): given the order-4 clock, the clock-invariant deformation freezes the $(2,2,2,2)$ modulus to the square AUTOMATICALLY — a relocation of the same order-4 input (the clock IS the carrier input), not a new derivation; v216 stays **[O]**. **[O]** full deformed twisted-sector OPEs, the $W(\mu)$-type scaling limit for $k{=}4$, the spacetime splitting-function check, WP3–WP5; `SEAM.EQUIV.01` stays **[O]**, untouched. Exact (sympy). **[E][C][O]**

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Script	What it machine-checks *(continued)*
`v494_cosmo_running_mu_record.py`	`COSMO.RUNNING.01`/`COSMO.MUDIST.01`: the two 2026-07-20 prediction-of-record rows promoted to the suite — the SAME Starobinsky branch as n_s, r, A_s (v7/v28/v86) read one derivative deeper and one observable further. **[E]** LO IDENTITIES (sympy exact): α_s=dn_s/d ln k=$-2/N_\star^2$=$-r/6$, β_s=$-4/N_\star^3$ (NOT $-8/N^3$); LO band $N_\star \in [50, 60]$: $[-8.0, -5.6] \times 10^{-4}$. **[E]** EXACT SLOW ROLL (third-order Lyth–Riotto + finite-difference cross-check < 1%): $\alpha_s(51.4)=-7.09\times10^{-4}$, $\beta_s(51.4)=-2.70\times10^{-5}$; exact band edges $-7.48/-5.23\times10^{-4}$. **[C]** DATA: Planck 2018 (-0.0045 ± 0.0067) → $+0.57\sigma$; record leg P-ACT-LB ($+0.0062\pm0.0052$) → -1.33σ (watch channel: positive central value); β_s run+runrun → -0.77σ. KILL: robust $\alpha_s<-5\times10^{-3}$ or $>+3\times10^{-3}$ at $\geq 5\sigma$; any robust POSITIVE running kills the plateau branch. **[C]** μ-DISTORTION (Chluba window, +21% calibrated on the ΛCDM anchor): sharp branch $N_\star$=51.4 ($A_s=1.76\times10^{-9}$) $\mu=1.6\times10^{-8}$; profiled record $N_\star$=56.1 $\mu=1.94\times10^{-8}$; band 1.5–2.3$\times10^{-8}$; branch split $\Delta\mu=3.0\times10^{-9}$ (ratio 1.23) ⇒ $\geq 3\sigma$ $A_s/N_\star$ branch decision at $\sigma(\mu)=10^{-9}$ (Voyage-2050 class); the PIXIE baseline 9×10^{-8} does NOT reach (0.18× limit). KILL: robust $\mu<0.9\times10^{-8}$ or $>4\times10^{-8}$. **[E]** COMB BLINDNESS (honest S15): the frozen log-comb (ω=2.5827, ε=0.0173) is structurally invisible in the μ window — $\lvert F(\omega)\rvert$=0.033, $\delta\mu=7.4\times10^{-12}=0.007\sigma$ even at Voyage-2050. **[O]** FENCES: $N_\star$ stays an external reheating input (`GATE.NSTAR.01`); the frozen band [50, 60] stays the surface of record. Python (sympy exact LO + numpy SR/quadrature), Python-only by nature. **[E][C][O]**

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Script	**What it machine-checks *(continued)***
`v495_celestial_wp3_gs_alignment.py`	`CELEST.WP3.01`: WP3 of the celestial-holographic research contract `CELEST.SEAM.01` — the Green–Schwarz axion coefficient $\lambda_{\mathfrak{g}}$ of celestial/twistorial SDYM (Costello, `arXiv:2111.08879`) against the TFPT seam constant c_3=1/(8π); the pre-registered "alignment test only". Exact (Fraction/sympy, no floats). **[E]** OKUBO IDENTITY DERIVED, NOT QUOTED: $\sum_\alpha \langle\alpha, x\rangle^4 = \frac{5}{2(\dim\mathfrak{g}+2)} (\sum_\alpha \langle\alpha, x\rangle^2)^2$ as an exact POLYNOMIAL identity for ALL 8 algebras on Costello's list ($\mathfrak{sl}_2, \mathfrak{sl}_3, \mathfrak{g}_2, \mathfrak{so}_8, \mathfrak{f}_4, \mathfrak{e}_6, \mathfrak{e}_7, \mathfrak{e}_8$; $\mathfrak{e}_6/\mathfrak{e}_7$ carved out of the v492 E_8 root set) — "no independent quartic Casimir"; NEGATIVE CONTROL: $\mathfrak{sl}_4$ (which HAS one) fails the proportionality (5/32 vs 3/32) — the list is sharp. **[E]** Killing $\sum_\alpha \langle\alpha, x\rangle^2 = 2h^\vee\langle x, x\rangle$ exact; Deligne dimension formula $\dim\mathfrak{g} = 2(5h^\vee{-}6)(h^\vee{+}1)/(h^\vee{+}6)$; closed form $\tilde\lambda^2 = 10h^{\vee\,2}/(\dim{+}2) = h^\vee{+}6$ EXACTLY $(8, 9, 10, 12, 15, 18, 24, 36)$ — fixes Costello–Paquette eq. (1.5); Dynkin indices $i_R = (1, 1, 2, 2, 6, 6, 12, 60)$ from explicit weight systems; E_8: $\tilde\lambda$=6, λ_{fund}=1/10 EXACT; cross-checks $\mathfrak{sl}_2$=8, $\mathfrak{sl}_3$=9; SO8 SLIP FLAGGED: Costello's own weight content gives $\lambda^2(\mathfrak{so}_8)$=3 (Okubo-consistent 12/4), the printed 3/2 is a factor-2 slip. **[E]** ALIGNMENT: $\kappa = \lambda_{\mathfrak{g}}/(8\pi\sqrt{3}) \Rightarrow (\kappa/c_3)^2 = \lambda^2/3 = 12 = \lvert\mu_4\rvert N_{\mathrm{fam}}$ (unit-trace) resp. 1/300 (adjoint-trace), both EXACT anchor rationals; κ/c_3 ITSELF is IRRATIONAL ($2\sqrt{3}$ resp. $1/(10\sqrt{3})$) — the pre-registered "$\lambda/\sqrt{3}$ rational?" reading FAILS, only the SQUARED alignment holds; anchor decomposition $h^\vee$=30=$\lvert\mathbb{Z}_2\rvert N_{\mathrm{fam}} g_{\mathrm{car}}$ (all three atoms once), $\varphi(30)$=8= rank, $\dim{+}2$=250=$2\cdot 5^3$, $\tilde\lambda$=6=$\lvert\mathbb{Z}_2\rvert N_{\mathrm{fam}}$, λ_{fund}=1/($\lvert\mathbb{Z}_2\rvert g_{\mathrm{car}}$). **[E]** LOOK-ELSEWHERE (the HARD FENCE — alignment $\neq$ selection): the alignment test AS SUCH passes 8/8 (ZERO selective power), $\{2, 3, 5\}$-smoothness 8/8, $\tilde\lambda$-integrality 2/8 (shared with $\mathfrak{sl}_3$), $\varphi(h^\vee)$=rank 4/8; the ONLY $\mathfrak{e}_8$-selective single test is g_{car}=5 $\mid h^\vee$ (1/8); the isolating conjunction was assembled POST HOC — it quantifies where $\mathfrak{e}_8$ sits, it is NOT a preregistered discriminator. **[E]** KILL-TEST K3 does NOT fire (exact alignment EXISTS) but its scope is DEMOTED: the c_3-connection survives as convention-level compatibility + genuine λ-arithmetic, NOT as $\mathfrak{e}_8$-selective evidence. **[C]** TYPED: the 8π is the 2π convention of the anomaly polynomial (Costello's A.0.10 constant descended to spacetime), the $\sqrt{3}$ the box diagram's $\sqrt{48} = 4\sqrt{3}$ — WP3 is an alignment yes/no ONLY, it can never derive c_3 from celestial data (contract wording, honoured). **[O]** WP5 (constructive holography) and `SEAM.EQUIV.01` stay **[O]**, untouched. Verdict B. Exact (Fraction + sympy). **[E][C][O]**

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Script	What it machine-checks *(continued)*
`v496_celestial_wp4_character_shadow.py`	`CELEST.WP4.01`: WP4 of the celestial-holographic research contract `CELEST.SEAM.01` — the $(E_8)_1$ character E_4/η^8 against the glue-equivariant $E_8[\mathbb{C}^2]$ S-algebra: the contract's "type mismatch, head-on". Exact integer/Fraction/sympy, no floats. [E] BASE: v492 replication ($240 = 52+64+60+64$, dims with Cartan $(60, 64, 60, 64)$); equivariant tower $D[d] = 60(d+1)$ even / $64(d+1)$ odd; cumulative generator count QUADRATIC — $\sum_{d\le s} D[d] = 31s^2+92s+60$ (even) with leading $31 = k+h^\vee$ (the Sugawara denominator, an arithmetic note); [C] CP Table 1 (`arXiv:2201.02595`): $J[m, n]$ carries 2d spin $1-(m+n)/2$, UNBOUNDED BELOW along the tower ⇒ no rational VOA (L_0 bounded below) contains the towers as graded pieces. [E] HILBERT SERIES: $\chi_{\mathrm{sp}}(t) = (60+128t+60t^2)/(1-t^2)^2$ (palindromic numerator, value 248 at t=1); ALL μ_4 collapses $w \in \{1, -1, i, -i\}$ and the $SU(2)_+$ primary slice $(60+64t)/(1-t^2)$ fail to start $(1, 248, \dots)$ — the "row sum" route fails at the first coefficient. [E] CHARACTER SIDE: $\chi_{(E_8)_1} = E_4/\eta^8 = (1, 248, 4124, 34752, 213126, 1057504, 4530744)$ exact through q^6 (machinery cross-check $qj = E_4^3/\eta^{24} = (1, 744, 196884, 21493760)$); theta split (shell $1 = (52, 64, 60, 64)$), sector currents $(60, 64, 60, 64)$, glue-diagonal $h = (0, 1, 1, 1)$ INTEGER (v125 isotropy = locality); Sugawara $c = 248/31 = 8 = 2\lvert\mu_4\rvert = g_{\mathrm{car}}+N_{\mathrm{fam}}$. [E] BRIDGES: zero-mode slice $D[0]$=60= the C_0-sector current count EXACTLY; the single-line jet slice z_1^d cycles $(60, 64, 60, 64)$ sector by sector; BOUNDARY/PERIOD LIMIT: one full μ_4 period of loop energies carries $\dim \mathfrak{g}_{n \bmod 4} = (64, 60, 64, 60)$, sum = 248 EXACTLY — the single-particle level of the $248q$ stage; 248 is NOT a jet-tower dimension ($d \le 100$) — only the period collapse produces it; BUT the loop Fock at integer level 1 counts $897266 \gg 248$: the rational truncation is NOT in the jet/loop algebra. [E] MISMATCH LOCALISED THREE WAYS: (a) GRADING (spin unbounded below; 248 no tower dimension); (b) GROWTH: jet Fock $f = (1, 128, 8436, \dots)$ vs $\chi = (1, 248, 4124, \dots)$ TWO-SIDED ($f_1<\chi_1$ undercount, $f_2>\chi_2$ overcount), f_n/χ_n strictly increasing $n = 1..12$; Meinardus poles s=2 vs s=1 ⇒ growth exponents $n^{2/3}$ vs $n^{1/2}$, asymptotically DISJOINT; (c) TRUNCATION: the level-2 null ideal deletes EXACTLY $27000 = 30^3 = (h^\vee)^3$ ($\mathrm{Sym}^2(248) = 1+3875+27000$ [[C] standard]; the character keeps $4124 = 1+248+3875$) — NO analogue in the jet algebra (level 1 agrees: $248 = 248$). [E] KILL-TEST K4, three-way: (i) exact realisation NO; (ii) boundary-limit realisation PARTIAL YES — current stratum, period sum 248, integer glue weights and theta split hold exactly; the missing piece is precisely the rational truncation, which must come from a scaling/compactification limit — the SAME structure as the MMST scaling-limit route of `SEAM.EQUIV.01` (v336/v449: the rational net exists in the LIMIT of the lattice collar, not in the pre-limit algebra); (iii) the hard mismatch is exactly located; K4 does NOT degrade the contract to "E_8 admissible" (the sector arithmetic holds exactly). [C] VERDICT B(ii): the $(E_8)_1$ character is a BOUNDARY/LIMIT SHADOW of the equivariant S-algebra, not a conformal block of it in the bulk grading — WP4 sharpens into a statement of `SEAM.EQUIV.01` type. [O] the constructive boundary limit itself (which limit, which topology, convergence) stays OPEN and belongs to WP5/`SEAM.EQUIV.01`; `SEAM.EQUIV.01` stays [O], untouched. Exact (integer/Fraction/sympy). [E][C][O]

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Script	What it machine-checks *(continued)*
`v497_celestial_wp5a_null_ideal.py`	`CELEST.WP5A.01`: WP5a of the celestial-holographic research contract `CELEST.SEAM.01` — "the null ideal from the limit", the first constructive milestone of WP5 (the v496 boundary-limit shadow made precise). Exact integer/Fraction arithmetic, no floats. [E] BASE: v492/v496 replication (glue dims $(60, 64, 60, 64)$, equivariant tower $D[d]$, $\chi_{(E_8)_1} = E_4/\eta^8$ exact through q^6, μ_4 period sum 248). [E] STABILISATION THEOREM: a one-parameter family χ_w of graded Fock characters on the CHIRAL jet generators (loop energy m, radial number r; $E_w = m + wr$ in quarter units $u = q^{1/4}$); w=2 IS the chiral jet grading (generator counts $64, 120, 128, 180, 192, 240, 256, 300$ per degree); the u^n coefficient of χ_w equals the quarter-moded loop Fock for ALL $w \geq n$+1 and is STRICTLY larger for every $w \leq n$ ($n \leq 8$, $w \leq 10$): the boundary reading is the precise coefficientwise limit $\varepsilon = 1/w \to 0$ with explicit threshold $w = n$+1 — a limit of a series, not a slice; convention clarified: 897266 is the quarter-moded loop Fock at u^4 = INTEGER level 1 (v496). [E] NULL IDEAL DERIVED, NOT CITED: Freudenthal multiplicities (unit-tested on the adjoint) + Weyl dimension formula + generic character peeling in the standard E_8 frame DERIVE $\mathrm{Sym}^2(248) = V(2\theta) \oplus V(\omega) \oplus V(0) = 27000 + 3875 + 1$ (peeling residual exactly zero; independent Weyl-orbit cross-check); $27000 = 30^3 = (h^\vee)^3$ exactly; $C_2 = (124, 96, 0)$, Dynkin indices 6750/750; level-1 integrability $\langle 2\theta, \theta^\vee\rangle = 4$, $\langle\omega, \theta^\vee\rangle = 2 \Rightarrow$ the only level-1 primary is the vacuum ([C] Kac, used through its counting consequence); level-2 quotient $31124 - 27000 = 4124 = 1 + 248 + 3875$ EXACTLY. [E] TWO ROUTES, ONE NUMBER: the μ_4 theta-split sector characters summed at $q^2 = (1036, 1024, 1040, 1024)$ give the SAME 4124 — the independent lattice route hits the Fock quotient exactly; at q^3: $34752 = \chi_3$. [E] NEGATIVE CONTROLS: $SO(16)_1$ through the SAME pipeline — $\mathrm{Sym}^2(120) = 5304 + 1820 + 135 + 1$ derived (FOUR components vs E_8's three), $5304 \neq 14^3 = 2744$ (the $h^{\vee\,3}$ identity is NOT generic), quotient $7380 - 5304 = 2076 = \Theta_{D_8}/\eta^8$ at q^2 (the pipeline validated on a second algebra), block weights $(0, \frac{1}{2}, 1, 1)$ do NOT fuse into one local character (four-block theory) vs E_8's integer $(0, 1, 1, 1)$; false periodisation: only $P = 4 = \|\mu_4\|$ gives 248 (P=1: 64, P=2: 124, P=8: 496); swapped glue breaks the zero-mode anchor ($64 \neq 60$). [C] named identifications: "radial decoupling = MMST collar thickness $\to 0$" (the v336/v449 lattice-collar route has the same two-step shape), not a computed isomorphism; the Kac singular vector only via its counting consequence. [O] HONEST LIMIT: the limit does NOT generate the truncation — the singular vector as an OPERATOR in the limit net stays open (WP5b); the GNS limit state (kernel $\supseteq$ ideal) and local normality / Haag duality of the limit net stay open (WP5c/WP5d); the celestial route now has the SAME two-step shape as MMST (limit + maximal ideal), with the ideal quantitatively fixed; `SEAM.EQUIV.01` stays [O], untouched. Exact (integer + Fraction). [E][C][O]

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Script	**What it machine-checks** ***(continued)***
`v498_celestial_wp5b_singular_vector.py`	`CELEST.WP5B.01`: WP5b of the celestial-holographic research contract `CELEST.SEAM.01` — "the singular vector as an operator", the second constructive milestone of WP5 (v497 fixed the ideal's size and location; WP5b constructs the deleting object). Exact integer/Fraction arithmetic, no floats, deterministic. **[E]** CONSTRUCTION: $\lvert s\rangle = (E^{\theta}_{-1})^2\lvert 0\rangle$, θ the highest root, weight 2θ (the top of the 27000), level $2 = 8$ quarter units $= q^2$ (an INTEGER level); the Chevalley/Frenkel–Kac basis is MACHINE-built (bimultiplicative 2-cocycle, asymmetry on all 57600 pairs; the $[e_\alpha, e_{-\alpha}]$ sign FORCED by Jacobi, SGN $= -1$; κ DERIVED from invariance, $\kappa(\theta^\vee, \theta^\vee) = 2$; Jacobi exact on all 57600 $\mathfrak{sl}_2$-slice triples + 6000 seeded + all 496 θ-adjacent; all $N_{\alpha\beta} = \pm 1$). **[E]** SINGULARITY: $J^a_1\lvert s\rangle = 0$ for ALL 248 basis elements (0 violations); case tally 190 (first bracket) / 57 (second) / 1 ($a = F^\theta$ via the central-term cancellation — the ONLY case that sees the level k); plus $J^a_2\lvert s\rangle = 0$ (248), $E^a_0\lvert s\rangle = 0$ (120 positive roots), Shapovalov norm 0 at k=1 (4 at k=2); the affine PBW engine unit-tested on all 61504 basis pairs. **[E]** LEVEL DIAL: the F^θ_1 coefficient is $2(1-k)$ — 2 at k=0, 0 EXACTLY at k=1, -2 at k=2: without the central extension the deletion object does NOT exist. **[E]** μ_4 DATA: θ-grading dims $(1, 56, 134, 56, 1)$, $\mathfrak{g}_0 = \mathfrak{e}_7 \oplus \mathfrak{u}(1)$; the h-adapted chamber machine-built (Cartan det 1, E_8 Dynkin diagram, branch lengths $[1, 2, 4]$); $\langle\theta, h\rangle = 5 \Rightarrow j(\theta) = 1$, height $29 = h^\vee - 1$; $\lvert s\rangle$ in class 2 (sheet-EVEN, consistent with WP5a); clock phase $i^2 = -1$; cross-table θ-grading $\times$ glue class consistent (rows $(1, 56, 126, 56, 1)$, columns $(52, 64, 60, 64)$); the deleting θ-$\mathfrak{sl}_2$ crosses the sheet-ODD classes $(1, 3)$, its square is sheet-even. **[E]** LEVEL-2 MATCH: mult$(2\theta) = 1$ in the level-2 Fock $\Rightarrow U(\mathfrak{g})\lvert s\rangle$ is THE 27000; the direct $\mathfrak{g}_0$-orbit BFS reproduces the Freudenthal multiplicities of $V(2\theta)$ EXACTLY through depth 4; quotient $31124 - 27000 = 4124$; lattice reading $(2\theta, 2\theta)/2 = 4 > 2 \Rightarrow$ no momentum-2θ state at q^2 in E_4/η^8 — $\lvert s\rangle$ IS the kernel generator of Fock $\to E_4/\eta^8$. **[E]** NEGATIVE CONTROLS: $E^\theta_{-1}\lvert 0\rangle$ NOT singular (one witness F^θ, $\to -k\lvert 0\rangle$) — the 248-current state that must REMAIN ($\chi_1 = 248$); 4 generic level-2 states non-singular (14–33 witnesses); k-dial: at k=2 the CUBE is singular (the $(E^\theta)^{k+1}$ mechanism = generic Kac level mechanics); $D_8/SO(16)_1$: the singular vector exists there TOO (level-1 generic) but the $h = \frac{1}{2}$ blocks do not fuse (3 comark-1 weights, $\dim V(2\theta_{D_8}) = 5304 \neq 14^3$) vs E_8 (comarks ≥ 2, vacuum the ONLY level-1 primary, glue weights $(0, 1, 1, 1)$ integer) — the ONE-BLOCK closure is the E_8/μ_4-specific part, honestly separated from the generic singular-vector mechanics. **[C]** Weyl complete reducibility beyond BFS depth 4; the Kac mechanics beyond level 2; the per-period quarter dictionary (v496). **[O]** HONEST LIMIT: in the twisted quarter-SLOT moding of the χ_w limit Fock two sector-C_1 modes never sum to 8 quarters (minimum $6 = q^{3/2}$) — the per-PERIOD dictionary (v496), not the per-slot identification, carries $\lvert s\rangle$ to q^2; exactly the precise WP5c question (GNS limit state, kernel $\supseteq$ ideal); WP5c–e stay **[O]**; `SEAM.EQUIV.01` stays **[O]**, untouched. Exact (integer + Fraction). **[E][C][O]**

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Script	**What it machine-checks *(continued)***
`v499_p2_weights_deligne_bg.py`	`P2.TYPING.01`: the P2 weight-typing postulate HARDENED (the v491 check-9 residual) — the anchor $a=(1,1,2)$ typed as the Birkhoff–Grothendieck splitting exponents of the DELIGNE CANONICAL EXTENSION E of the flavor connection; sympy exact, no floats. Given the **[E]** anchors (4 marks v216, rank 3 v137, cusp class + irreducibility v117, (U)): $\deg E=-4$ AS A THEOREM (cusp class ⇒ canonical residue exponents $\{0,\frac{1}{3},\frac{2}{3}\}$, the unique lift in $[0,1)$; trace 1 per mark × 4 marks; par-deg 0; ∞ APPARENT, monodromy product = **1** exactly); POSITIVITY AS A THEOREM (irreducibility EXACT: $\lvert\langle U,M_0\rangle\rvert=24=\lvert S_4\rvert$, $\sum_g \lvert\chi\rvert^2=24$; with (U) ⇒ Mehta–Seshadri stability ⇒ $h^0(E)=0$, all 324 section-saturation cases; BG dictionary $h^0=0 \Leftrightarrow a_i \geq 1$); INTEGRALITY = Birkhoff–Grothendieck, automatic once E is typed as a bundle. UNIQUENESS + SELECTOR: $\{1,1,2\}$ unique (v491); the Schur–Horn permutohedron of $4\cdot\mathrm{spec}(A_0)$ has integer points exactly $\{(2,2,0),(2,1,1)\}$ (the two options of the “Global computation” theorem) and $h^0=0$ kills $\{0,2,2\}$ (an $\mathcal{O}(0)$ summand has $h^0=1$) — the “balanced” selection is now TYPED as stability/positivity. THE h^1 DISCREPANCY, REAL + RESOLVED: $h^1(\mathcal{O}(-2)\oplus\mathcal{O}(-1)^2)=1\neq 3$ — the naive “generations $=H^1(E)$” typing is FALSE (forcing it demands deg -6 and loses uniqueness: $\{1,1,4\},\{1,2,3\},\{2,2,2\}$, e_2 never 5); the correct TWO-OBJECT typing: $L=\mathcal{O}(-D)$, $D=\mu_4$, carries the generations ($h^1(L)=3=N_{\mathrm{fam}}\cong H^0(\Omega^1(\log D))$, the v137 basis), while E carries them as its FIBER (rank 3) with the anchor as splitting type. BISWAS EXPLICIT (independent convergence witness): the weight denominator is $3=N_{\mathrm{fam}}$, so the Biswas cover is the $\mathbb{Z}_3$ cusp cover $y^3=(z-1)(z-i)(z+1)^2(z+i)^2$, genus 2 (NOT the pillowcase double $w^2=z^4-1$, genus 1, $j=1728$ — the deck side); $p_*\mathcal{O}_Y=\mathcal{O}\oplus\mathcal{O}(-2)^2$ (χ cross-check $-1=-1$); the regular-representation model bundle lands on $\{0,2,2\}$ = the UNSTABLE Schur–Horn companion ($h^0=1$): integrality AND the sum rule COME OUT of the correspondence, and stability (irreducibility, v117) is the selector. NEGATIVE CONTROLS: drop positivity → 4 resp. 31 solutions; deg $-3/-5$ breaks; n-mark scan: $\{1,\ldots,1,2\}$ unique for EVERY $n\geq 3$ but $e_2=(n-2)(n+1)/2=5$ ONLY at $n=4$ — the VALUE, not the uniqueness, is the $n=4$ content; MS-rational exponents $e_2=\frac{16}{3},\frac{44}{9}$. [C] RESIDUALS: R1 the module identification “generation space = this H^1 with this connection” [C] (`QGEO.SYM` territory, stays **[O]**); R2 unitarity (U) [C] (v33/v40); R3 the analytic-monodromy identification (fine type Numerical, v117 check 4). P2 stays an axiom (`AX.P2.01` declared, no marker moves); NON-CIRCULAR: neither v23’s splitting-type input nor v115’s $\mathrm{diag}(A_0)=a/4$ input is used. Sharpens v491. Exact (sympy); Wolfram-mirrored. **[E]**[C]

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Script	**What it machine-checks** ***(continued)***
`v500_celestial_wp5c_gns_limit_state.py`	`CELEST.WP5C.01`: WP5c of the celestial-holographic research contract `CELEST.SEAM.01` — "the GNS limit state", the third constructive milestone of WP5 (v497 made the boundary limit precise, v498 built the deleting operator; WP5c constructs the STATE whose GNS kernel contains the ideal). Exact integer/Fraction arithmetic, no floats, deterministic. **[E]** CONSTRUCTION: the quasi-free family ω_w — loop sector (r=0) = the affine k=1 vacuum n-point functions via the MACHINE-determined compact anti-involution $\theta(e_\alpha)$=$-e_{-\alpha}$, $\theta(h)$=h (the UNIQUE sign that is an anti-automorphism on ALL 61504 basis pairs; contravariance $\langle J^x_{-1}u, v\rangle = \langle u, J^{\theta x}_1 v\rangle$ exact); radial sector (r≥1) = oscillator pairings x^{wr} (the exact Gibbs regulator $e^{-\beta E_w}$, kept x-adic). **[E]** CONVERGENCE, SHARP: $\omega_w = \omega_\infty \bmod x^{N+1}$ for ALL $w \geq N{+}1$ (N=8) and NOT yet at w=N (the r=1 modes sit at order N) — the WP5a threshold $w = n{+}1$ holds IDENTICALLY at the state level; positivity of every ω_w checked. **[E]** THE KERNEL (core, fully machine): the FULL exact level-2 Gram — 31124 monomials in 9361 weight blocks (dim profile 164:1, 37:240, 7:2160, 1:6960), every block exact: level-1 rank 248 POSITIVE DEFINITE (Cartan block = the E_8 Cartan matrix, det 1); level-2 rank **4124**, kernel **27000**, every block PSD; rank per Weyl orbit (norm 8/6/4/2/0) = $(0, 0, 1, 8, 44)$; kernel = $V(2\theta)$ WEIGHT BY WEIGHT (Freudenthal cross-check on all 9361 weights); highest-weight norms $(3844, 49, 2, 0)$ at $(0, \omega, \theta, 2\theta)$; direct ideal checks $F_i\|s\rangle$, $F_jF_i\|s\rangle$ norm 0. **[E]** μ_4-EQUIVARIANCE: $\omega_w \circ \mathrm{clock} = \omega_w$ for ALL w; the clock unitary DESCENDS to GNS(ω_∞); level-2 GNS rank split by clock class $= (1036, 1024, 1040, 1024) = \Theta_{C_j}/\eta^8$ at q^2 (the two-routes identity at the STATE level); level-1 split $(60, 64, 60, 64)$. RESOLUTION OF WP5b S4.10: $\|s\rangle$ IS the ZERO VECTOR of GNS(ω_∞) — the twisted two-C_1-mode floor (6 quarters $= q^{3/2}$) concerns only pre-limit objects (GNS has no state there: $6 \bmod 4 \neq 0$). **[E]** NEGATIVE CONTROLS: k=2 ($\langle s\|s\rangle$=+4, dominant blocks FULL rank — NO ideal in the kernel); k=0 (level-1 Gram $\equiv 0$ — no 248 layer without the central extension); $D_8/SO(16)_1$ (level-2 rank $2076 = 7380 - 5304$, kernel $= V(2\theta_{D_8})$ PSD weight by weight, but $h = (0, \frac{1}{2}, 1, 1)$: the vacuum GNS is 1 of 4 blocks — ONE-BLOCK completeness stays E_8/μ_4-specific); wrong family $E'_w = mw{+}r$ (level-1 limit rank 0: the 248 layer ERASED, u^4 content $710955 \neq 248$); no damping ($897266 \neq 248$); CCR OBSTRUCTION: at a FIXED radial pairing $c > 0$ positivity forces $\omega(bb^*) \geq c$ — NO w-uniform damping state exists: the family formulation is NECESSARY and the family EXISTS ⇒ the preregistered KILL is probed and NOT triggered. **[C]** the quasi-free/Gaussian extension to the full ∗-algebra; Kac positivity beyond level 2 (levels ≤ 2 machine-checked); the module identification in the component route. **[O]** local normality of ω_∞, the net structure / Haag duality (WP5d), Costello–Li (WP5e), the operator-algebraic closure beyond the graded/x-adic topology; `SEAM.EQUIV.01` stays **[O]**, untouched. VERDICT: SUCCESS on the preregistered criterion (dim GNS level-2 = 4124 exactly). Exact (integer + Fraction). **[E][C][O]**

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Script	What it machine-checks *(continued)*
`v501_celestial_wp5d_two_interval_index.py`	`CELEST.WP5D.01`: WP5d-α of the celestial-holographic research contract `CELEST.SEAM.01` — "the two-interval index from the lattice" (the local-net question, alpha stage). The roadmap discriminator: $(E_8)_1 \Rightarrow \mu{=}1$ (Haag-dual two-interval net, ONE sector); $SO(16)_1 \Rightarrow \mu{=}4$; preregistered KILL = "the two-interval μ-offset does NOT vanish after the condensation". numpy/scipy Gaussian machinery + exact Fractions. [E] SETUP (machine identities): critical Majorana ring (v450 style, NS closure, $c{=}\frac{1}{2}$ per layer), 16 layers = the seam carrier $c_{-}{=}8{=}g_{\text{car}}{+}N_{\text{fam}}$ (v489); Peschel-exact; three prescriptions — fermionic ρ, orbifold sector sum $\{1, P_A, P_B, P_{AB}\}$, dual algebra $\{1, P_{AB}\}$; superselection $[\rho, P_{AB}]{=}0$ machine-exact; EVERY Gaussian formula validated against exact diagonalisation ($\leq 1.5{\times}10^{-15}$). [E] FERMIONIC MI EXTENSIVE (the $\mu{=}1$ reference; lattice witness): $I(x) = -(c/3)\ln(1{-}x)$ with c fitted 0.5000; $\max\lvert F(x)\rvert = 5.97{\times}10^{-5}$ at $N{=}512$, $\lvert F\rvert$ falling monotonically to $1.3{\times}10^{-6}$ at $N{=}1024$ — Casini–Huerta extensivity on the lattice. [E] THE μ-OFFSET (core): Rényi-2 exactly $\Delta_2 = -\ln[(1{+}R)/2]$; the 16-layer seam carrier saturates $\ln 2$ to MACHINE PRECISION ($\lvert\Delta_2 - \ln 2\rvert = 1.1{\times}10^{-15}$ at $N{=}512$; $R \sim N^{-0.500}$); index reading $[F : F_{\text{even}}] = e^{\Delta_\infty} = 2.000000 \Rightarrow \mu_{\text{gauged}} = e^{2\Delta_\infty} = 4.000000$ = the v490 parity census (TWO independent lattice witnesses); gauging every layer separately costs $16\ln 2$ — the offset measures ln[index of the subnet], not a universal blob. [E] HAAG-DUALITY INDICATOR: fermionic $S(E){=}S(E')$ to $8.2{\times}10^{-12}$ (graded-duality proxy); the orbifold BREAKS complementarity ($D(E) - D(E') = 0.427$ at $a{=}64$ — the direct duality-failure witness); the complementary-pair sum is exchange-symmetric and $\leq \ln 4 = \ln\mu(SO(16)_1)$ with max 1.2637 at the self-dual $x{=}\frac{1}{2}$ and $\to \ln 2$ at the squeezed ends: the $\ln 4$ KLM budget is a DOUBLE-LIMIT statement, NOT a fixed-x identity (the first naive fixed-x check was WRONG and was replaced — honestly typed). [E] CONDENSATION ARITHMETIC, LATTICE-ANCHORED: $\det\text{Cartan}(D_5)\cdot\det\text{Cartan}(A_3) = 4\cdot 4 = 16 = \mu(\text{carrier})$; KLM/Longo–Rehren quotients $\mu/\lvert L\rvert^2$: $16/4^2 = 1$ (one-step μ_4 glue) and $16/2^2 = 4 \to 4/2^2 = 1$ (via $SO(16)_1$), both routes EXACT; Σd_i^2 cross-checks (Ising 4, $SO(16)_1$ 4, $(E_8)_1$ 1); $\theta_v = e^{2\pi i\nu/16} = 1$ EXACTLY at $\nu = 2c_{-} = 16$, rivals $\nu = 1, 2, 8$ give $\theta_v \neq 1$. The anchored chain: $\mu = 1$ (graded, MEASURED) $\to 4$ (gauged, MEASURED + v490 census) $\to 1$ (arithmetic, [C]: the condensed net is interacting — the v490 fence). [E] NEGATIVE CONTROLS: $\nu{=}1$ single Majorana (offset non-trivial, $\Delta_2 = 0.6364$ at $N{=}2048$ rising, and $\theta_v = e^{i\pi/8} \neq 1$: NOT removable — the kill discriminator has teeth); trivial dimerised phase (everything $\leq 10^{-7}$); wrong sector sum $\{1, P_{AB}\}$ (offset EXACTLY 0 instead of $\ln 2$). [C] the continuum dictionary "offset = ln index" (KLM two-interval index; Longo–Xu; Casini–Huerta–Magan–Pontello entropic order parameters) + the $\theta_v{=}1$ condensation step itself. [O] continuum Haag duality, local normality, split/strong additivity of the quotient net (WP5d-β, running as exploration), Costello–Li (WP5e); `SEAM.EQUIV.01` stays [O], untouched. VERDICT: the $\mu{=}1$ indicator chain PASSES; the preregistered KILL is NOT triggered. Python (numpy + scipy + Fractions). [E][C][O]

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Script	What it machine-checks *(continued)*
`v502_celestial_wp5e_prefactor_level.py`	`CELEST.WP5E.ALPHA.01`: WP5e-α of the celestial-holographic research contract `CELEST.SEAM.01` — "prefactor and level bookkeeping", the CFT-side anchor stage of WP5e (the BCOV/Costello–Li twistor uplift: boundary partition function $= E_4/\eta^8$ *including* the $q^{-1/3}$ prefactor; preregistered kill: the anomaly inflow demands a level $\neq 1$). Exact sympy/Fraction. **[E]** PART A — the $q^{-1/3}$ prefactor is exact μ_4 vacuum-energy bookkeeping: Hurwitz-ζ derivation ($\zeta(-1,\theta) = -B_2(\theta)/2$; $E_b(\theta) = -\frac{1}{24} + \theta(1-\theta)/4$ exactly); the clock is INNER ($h = (2,2,2,2,2;0^4)$ and the A_3 detector $h' = (0^5;1,1,1,-3)$ both read the class mod 4 on all 240 roots; $(h,h) = 20$, $(h',h') = 12$, sum 32) $\Rightarrow$ twist distribution $\theta = 0^8$ in ALL four sectors — a SHIFT orbifold (lattice cosets, integer-moded oscillators), *not* a rotation orbifold; $E_{\mathrm{osc}} = -\frac{1}{3} = -c/24$ sector-independent ($c = 8 = \text{rank} = 248/31$); sector ground exponents $-\frac{1}{3} + (0,1,1,1)$. **[E]** discriminant form = Casimir energy, TWO routes: spectral flow $j^2(h,h)/32$ ($5j^2/8$, $3j^2/8$, sum $j^2 \equiv 0 \bmod 1$ = the isotropy/Lagrangian condition) and *exactly* via free fermions ($D_5 = 10$ Majorana, $A_3 = D_3 = 6$, glue diagonal $= 16 = 2^{g_{\mathrm{car}}-1}$, the WP5d seam carrier; R–NS shift $n/16$: $\frac{5}{8}, \frac{3}{8}, 1$); the ROTATION reading fails (the geometric A_3 deck gives $\frac{3}{16} \neq \frac{3}{8}$: the v492 "clock $\neq$ deck" lesson at the Casimir level). **[E]** character: $\sum_j \Theta_{C_j} = E_4$ and the four sector characters (leading coefficients $(1,64,60,64)$, currents $(60,64,60,64)$) sum exactly to $E_4/\eta^8 = q^{-1/3}(1,248,4124,34752)$. **[E]** PART B — $k=1$ forced THREE independent ways: (i) the CURRENT condition $h(J) = k = 1$ (the $\mathbb{Z}_4$ extension closes onto the E_8 adjoint $60{+}64{+}60{+}64 = 248$ only at spin 1); HONEST SHARPENING: glue-h integrality $h(J^j;k) = k\,(0,1,1,1)$ holds for ALL $k = 1..8$ — the originally conjectured integrality route fixes *nothing*, the three other dials carry; (ii) conformal embedding $47k^2{+}219k{-}266 = 47(k{-}1)(k{+}266/47) = 0$ resp. D_8-in-E_8: $128k(1{-}k) = 0$; (iii) central charge $248k/(k{+}30) = 8 \Leftrightarrow 240(k{-}1) = 0$ (= the prefactor itself); plus the v498 singular-vector dial ($\lVert (E^{\theta}_{-1})^m 0\rVert^2 = m!\,k(k{-}1)\cdots(k{-}m{+}1)$, singular vector exactly at grade $k{+}1$; $k=1$ deletes $V(2\theta)$: $31124 - 27000 = 4124$). **[E]** negative controls: $D_8/SO(16)_1$ (same rank, same $c = 8$, SAME $q^{-1/3}$ prefactor, but $h = (0, \frac{1}{2}, 1, 1)$: the discriminator is glue-h integrality, not the prefactor; $\Theta_{D_8} + \Theta_s = E_4$); $\mathbb{Z}_2$ rotation ($E = +\frac{1}{6}$, $h = \frac{1}{2}$: no μ_4-shift sector there); μ_4 rotation (sector-dependent $(-\frac{1}{3}, \frac{1}{24}, \frac{1}{6}, \frac{1}{24})$: the common prefactor breaks); wrong $k \in \{2,3,4\}$ fails all five dials coherently (residuals 360/814/1362; $\langle s\|s\rangle = 4/12/24$; level-2 dim stays $31124 \neq 4124$). **[C]** Kac generation of the maximal submodule by the grade-($k{+}1$) singular vector for $k \geq 3$ ($k = 1,2$ machine-anchored by WP5b/WP5c). **[O]** the BCOV/Costello–Li twistor inflow itself (Costello's one-loop axion $\lambda_g/(8\pi\sqrt{3})$ on $\mathbb{PT}$, the Costello–Paquette–Sharma level-from-flux mechanism) stays open — v502 anchors the CFT side of WP5e only (α stage); the kill branch ("inflow demands a level $\neq 1$") is not triggered on the CFT side but its twistor side is untested; WP5e stays **[O]**; `SEAM.EQUIV.01` untouched, no marker moves. Exact (sympy + Fraction). **[E][C][O]**

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Script	What it machine-checks *(continued)*
`v503_qgeo_emergence_light.py`	`QGEO.EMERGE.LIGHT.01`: the v215 *deferred emergence test* (lever 7) executed in its first honest LIGHT version, plus the residual-convergence result QGEO-R1 ≡ P2-R1 (analysis/kill-test class; NO marker moves — `QGEO.REALIZE.01` stays **[C]**/**[O]**). NO DYNAMICAL SELECTION (ζ-log-det via the Dedekind-η formula, mpmath 40 digits): $\log F = \log(\tau_2\lvert\eta\rvert^4)$ is exactly S-invariant, so $t = 1$ is automatically critical on the rectangular family $\tau = it$; there a strict local maximum ($g''(1) = -1.298$) but a *minimum* in the τ_1-direction ($h''(0) = +0.298$): $\tau = i$ is a SADDLE of the full moduli space; critical points = the CM points ($E_2^*(i) = E_2^*(\rho) = 0$ to 40 digits); the *global* det′ winner at fixed area is the HEXAGONAL torus ($\log F(\rho) = -1.03352 > \log F(i) = -1.05469$; Osgood–Phillips–Sarnak 1988) — free-field dynamics alone does *not* select $\tau = i$; criticality at the square is Palais symmetric criticality (the independent heat trace is equally critical). SHARP RIGIDITY (v280 lattice DtN): the mod-4 off-character indicator (the v215-K3 lever as a *modulus function*) has an isolated zero ($8{\times}10^{-15}$) exactly at $\tau = i$, strictly monotone in both directions (min $8.4{\times}10^{-2}$ away); $R^2 = \mathbb{Z}_2$ exactly as permutation matrices (the v492 $\mathbb{Z}_8$ bridge on the lattice); $j(it) \geq 1728$ with equality only at $t = 1$, $j(\rho) = 0$. **[E]** the $\mathbb{Z}_2$ chain is MODULUS-BLIND: 4 fixed points and $[\mathbb{Z}_2, \Delta_t] = 0$ on *every* torus (v216 confirmed without mark input); pillowcase det′ $= \frac{1}{2}$ torus det′ exactly ($\pm$-pair structure), so all τ conclusions transfer verbatim; marks/mark-locality/deck^2 exist on *every* pillowcase (v264/v280), only the order-4 clock distinguishes $\tau = i$; $\mathrm{Aut}(E_\rho) = \mathbb{Z}_6$ (element orders $1, 2, 3, 6$) contains *no* order-4 element — dynamics and clock are logically independent selectors; GIVEN the clock, $P(iZ) = P(Z)$ forces $a_3 = a_2 = a_1 = 0$ and $j = 1728$ identically (v493): the modulus is not a second input. **[C]**/**[O]** RESIDUAL CONVERGENCE (the core finding): the order-4 element on H^1 is a Coxeter element of $W(A_3)$ (char poly $\lambda^3{+}\lambda^2{+}\lambda{+}1$, order $h(A_3) = 4 = \lvert\mu_4\rvert = N_{\mathrm{fam}}{+}1$) — exactly the P2 cusp-class/carrier datum of the P2 hardening (v499 R1) ⇒ QGEO-R1 ≡ P2-R1: ONE common residual, "why does the raw seam carry the order-4 clock", not two scattered ones; chain: marks derived (v216 **[E]**), square derived *given* the clock (v493 **[E]**), clock = carrier input **[C]**, rest = the dynamical realisation of the clock **[O]**; route classification (v347 analogue): B-rigidity in the sharpened form "RP seam data ⇒ order-4 clock" is the live route — free dynamics alone does not deliver the clock (the hexagonal finding). Lever update (documented here, v215 itself unchanged): lever 7 lifted from "deferred, not executable" to "light version executed: rigidity, not dynamical selection". Python (mpmath + numpy/scipy + sympy). **[E][C][O]**

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Script	**What it machine-checks** ***(continued)***
`v504_celestial_wp5d_klm_completeness.py`	`CELEST.WP5DB.01`: WP5d-β of the celestial-holographic research contract `CELEST.SEAM.01` — "split + strong additivity from the lattice", the two remaining Kawahigashi–Longo–Müger legs of complete rationality (the μ-index is `v501`), witnessed for the orbifold prescription on the critical Majorana seam edge; the $U(1)$/Dirac subnet is the preregistered negative control (the current net, where strong additivity fails in the continuum). **[E]** STRONG ADDITIVITY, EXACT: shared boundary Majorana $\Rightarrow \mathrm{Even}(A) \vee \mathrm{Even}(B) = \mathrm{Even}(A \cup B)$ EXACTLY (GF2 spans full 64/64 up to 1024/1024; matrix rank 32/32); DISJOINT: exactly HALF (rank 16/32, index 2 — the missing odd $\otimes$ odd sector is the `v501` $\ln 2$ bit, localised at the split point); the neutral $U(1)$ algebras do NOT generate the union even with the shared site (gaps 2/10/52, growing). **[E]** ENTROPIC WITNESS [N]: the fermionic touching MI carries the pure UV point-gap law ($(c/3)\ln 2$ increments, dev 1.2×10^{-6} — no discriminator); the honest discriminator is BOUNDED vs DIVERGENT: the $\mathbb{Z}_2$/orbifold deficit stays $< \ln 2$ with $(\ln 2 - \Delta_2) \sim N^{-p}$, $p = 0.2444$ vs $2\Delta_\mu = \frac{1}{4}$ (Ising disorder dimension; "defect $\to 0$" would be FALSE at the sharp split — bounded $\Leftrightarrow$ finite index, Longo–Xu), while the $U(1)$ control BURSTS $\ln 2$ from $L = 128$ and grows as $\frac{1}{2} \ln \mathrm{Var}\, Q_A$ (Klich–Levitov 0.10134 vs $1/\pi^2$): infinite index. **[E]** SPLIT/NUCLEARITY [N]: coupling σ_k fall at the elliptic-nome rate $\pi K(1-x)/K(x)$ to 1.3–2.0%; $\sigma_1 \sim x^{0.5044}$; trace norm summable; ORBIFOLD INHERITANCE EXACT: P_A flips $C \to -C$ (σ_k identical, 8.3×10^{-17}), the even-bilinear Gram is the compound $\Lambda^2 C$ ($\{\sigma_a \sigma_b\}$, 2.8×10^{-14}) — Longo heredity **[C]**. **[E]** PIMSNER–POPA: $E(a) - a/2 = PaP/2$ IDENTICALLY $\Rightarrow \lambda = \frac{1}{2} = 1/[F : F_{\mathrm{even}}]$; integer attainment (16384 + 2048 monomial sweeps, 0 violations; pencil min $\lambda = 0.5000000000$); $\lambda_{E_4} = \frac{1}{4} = 1/\mu$ exactly; INDEX CONSISTENCY, two routes: $e^{\Delta_\infty} = 2.000000000 = 1/\lambda_{\mathrm{PP}}$ and $e^{2\Delta_\infty} = 4 = \mu$; $U(1)$: $\lambda = 1/(m{+}1) \to 0$. **[E]** CONTROLS + HONESTY: dimerised phase trivial (3.6×10^{-15}); wrong prescription $\{1, P_{AB}\}$ deficit EXACTLY 0; finite-size honesty (14.9% below $\ln 2$ at $N = 4096$ — witness, not proof). **[C]** the KLM dictionary (Longo–Xu; Longo heredity; CHMP). **[O]** the continuum theorem "finite-group orbifolds of completely rational nets are completely rational" is XU'S THEOREM — CITED, NOT CLAIMED; the seam quotient net and the condensed $(E_8)_1$ net stay WP5e/Costello–Li; `SEAM.EQUIV.01` stays **[O]**. VERDICT: with `v501`, all THREE KLM ingredients witnessed on the lattice. Python (numpy + scipy); exact parts Wolfram-mirrored. **[E][C][O]**

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Script	**What it machine-checks** ***(continued)***
`v505_celestial_wp5e_beta_equivariant_ledger.py`	`CELEST.WP5E.BETA.01`: WP5e-β of the celestial-holographic research contract `CELEST.SEAM.01` — "the equivariant anomaly ledger on twistor space", the twistor-side milestone of WP5e (the BCOV/Costello–Li uplift; preregistered kill: the anomaly inflow demands a level $\neq 1$); WP5e-α (v502) anchored the CFT side. The Atiyah–Bott/Lefschetz fixed-point skeleton of the one-loop box anomaly (Costello; Bittleston–Sharma–Skinner) evaluated exactly on $\mathbb{C}^2/\mathbb{Z}_4$ with the E_8 μ_4-glue monodromy. Exact sympy/Fraction. **[E]** AB ledger: denominators $\det(1-g_j^{-1}) = (2,4,2)$; $\sum_j 1/\det_j = 5/4 = (\|\mathbb{Z}_4\|^2-1)/12$ (Dedekind); equivariant characters $\mathrm{tr}_{g_j}(\mathrm{Ad}\,E_8) = (248,0,-8,0)$ by two routes; invariant average 60 = the $D_5{+}A_3$ carrier; Frobenius 61568; $\mathrm{tr}_{g_2} = -8 = 120-128$. **[E]** Okubo per sector (honest sharpening): the invariant sector is exactly Okubo-quadratic ($Q^{(0)} = 36\langle x,x\rangle^2$, $36 = \tilde{\lambda}^2_{e_8}$ — v495 re-derived through the glue grading); the twisted sectors do NOT reduce (T_5: $(-8,32,-8)$, T_3: $(48,-64,48)$); the AB-weighted sum cancels the D_5 quartic exactly and leaves the RIGID residual $32\,T_3$; the only quartic-free weighting is the uniform $\mathbb{Z}_4$ projector; the graded GS exchange is exactly rank-obstructed in sectors $1,2,3$; heterotic identities exact (Tr_{120}, Tr_{128}, $\mathrm{Tr}_{E_8} = 9(\mathrm{tr}^2)^2$). **[E]** index bridge (centrepiece): $f(m) = \frac{1}{4}\sum_j (i^{jm}-1)/\det_j = (0,-\frac{3}{8},-\frac{1}{2},-\frac{3}{8}) = \mathrm{ch}_2(T_m) = -(C^{-1})_{mm}/2$ EXACTLY — fixed-point ledger = McKay/Kronheimer intersection ledger; glue ch_2 defect -78 by both routes (classes $(-24,-30,-24)$; $-30 = -h^\vee$, a **[C]** coincidence); $\sum_m h_{A_3}(m) = 5/4 = \sum_j 1/\det_j$; Killing sums $(320,320,240,320)$, total $1200 = 2h^\vee(h,h)$. **[E]** $k=1$ geometric: lattice current count = 240 at $k=1$, exactly 0 at $k=2,3,4$; embedding residual $47k^2{+}219k{-}266 = (0,360,814,1362)$; $h(J;k) = k$ integer for all k — integrality alone fixes nothing (honest, as v502); one scale $\Rightarrow$ one level (v493 periods). **[E]** E3 REFUTED (prominent): $a_0 \in O(8)$ fills the GRAVITON slot $O(2)$, not the BSS axion slot $O(-2)$ ($X = -2 \neq +2 = Y$; mismatch $4 = \|\mu_4\|$, a **[C]** coincidence) — "the theory brings its GS axion as a_0" is refuted; instead the three $H^2(\mathrm{ALE})$ classes carry exactly the Coxeter eigencharacters $\{i,-1,-i\}$ = the three twisted slots (bijection); slot count $1+3 = 4 = \|\mu_4\|$ soft. **[E]** negative controls: $\mathrm{diag}(i,i)$ breaks four ways ($\sum 1/\det = \frac{1}{4}$, ω-weight -1, 5 generators); $SO(16)$ glue: characters $(120,0,+120,0)$, defect $-30 \neq -78$, bulk Okubo fails (independent quartic Casimir); $k=2$: count 0, prefactor $-31/48 \neq -\frac{1}{3}$. **[C]** dictionaries ($\mathrm{CS}(\rho)$ mod 1 = discriminant weight; $32 = (h,h){+}(h',h')$; $-30 = -h^\vee$; Ω-contraction). **[O]** the global BCOV/Kodaira–Spencer quantisation on $\mathbb{PT}/\mathbb{Z}_4$, the $O(-2)$ bulk-axion construction, CPS level-from-flux, the pairing of the $32\,T_3$ residual with the three sphere axions. Verdict B; the kill branch does NOT fire on the equivariant skeleton; `SEAM.EQUIV.01` untouched, no marker moves. Exact (sympy + Fraction). **[E][C][O]**

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Script	What it machine-checks *(continued)*
`v506_seam_clock_rigidity.py`	`SEAM.CLOCK.RIGIDITY.01`: clock rigidity, Route B of the v503 classification — two exact theorems reducing the order-4 clock input to ONE alignment bit (Part A Moebius + Part B Fock in one module; `QGEO.REALIZE.01` stays [C]/[O], no marker moves). Exact sympy + exact Cl(16) Fractions. **[E]** THEOREM A (Moebius): the collar deck $z \mapsto -z$ has in $\mathrm{PGL}_2(\mathbb{C})$ exactly TWO square roots ($z \mapsto \pm iz$), both automatically of order 4 (complete case analysis: $\mathrm{tr} \neq 0 \Rightarrow$ diagonal $\Rightarrow a/d = \pm i$; $\mathrm{tr} = 0$ impossible) — "order 4" is no longer an input; a marks-preserving root exists $\iff$ the deck is the CENTRAL involution of the mark-D_4 (center $= \{\mathrm{id}, \mathrm{deck}\}$, exactly 2 order-4 elements = the clocks) $\Rightarrow$ then marks = one μ_4 orbit, cross-ratio class $\{2, \frac{1}{2}, -1\}$, $j = 1728$, $\tau = i$, free 4-cycle, deck = clock^2, only the scale free; COUNTEREXAMPLE: $\{\pm 1, \pm(3{+}2\sqrt{2})\}$ is deck-free AND harmonic with full D_4, but the deck is non-central (edge involution) $\Rightarrow$ 0 roots — harmonicity alone is NOT sufficient; the RP filter selects nothing ($\mathrm{Aut}(\mathbb{Z}_4)$ ambiguity irreducible, consistent with v492). **[E]** B3/B4: free orbit $\Rightarrow n \mid 4$; mark transitivity $\Rightarrow n = 4$; faithfulness kills $n \geq 5$; $n = 3$ fixes a mark; $n = 6$ dies at cusp degree 3 (v117); the deck is modulus-blind (v503 B5 exact); controls: $n = 8$ impossible; hexagonal A_4 (0 order-4 elements, squares only $\{1, 3\}$); generic V_4 (exponent 2); stabiliser trichotomy $V_4/D_4/A_4$ with order-4 counts $(0, 2, 0)$. **[E]** THEOREM B (Fock, the core result): the unique S^2 spin structure gives NS on every circle (established P1 datum); one-particle $D^2 = -1$ (charpoly $(\lambda^2{+}1)^8$), quarter-shift $C^2 = D$, $C^8 = 1$ (charpoly $(\lambda^4{+}1)^4$); Fock: $\tilde{U} = \prod_j (1 - \gamma_j \gamma_{j+8})$, $\tilde{U}^2 = 256\, \gamma_1 \cdots \gamma_{16} \Rightarrow U^2 = (-1)^F$ EXACTLY, and $(cU)^2 = c^2(-1)^F$ is never 1 (center of Cl(16) = scalars) — the $\mathbb{Z}_4$ lift is FORCED (nonsplit extension); $\mathbb{Z}_8$ tower: $V^2 \propto \tilde{U}$, $V^4 \propto (-1)^F$, projective order $8 = 2\|\mu_4\|$ — fermionically explained; R control $U_R^2 = +1$ (split — the forcing rests on NS = established geometry); 16-fold way $\theta_v = 1$ at $\nu = 16$; the forcing already holds at 8 sites. [C] THE ONE REST: "the collar deck is the central involution of the mark-D_4" ($\iff$ square-root existence on the mark data) — ONE alignment bit; the fermionic $\mathbb{Z}_4$ forces the ORDER, not the spatial alignment. No overclaim. Exact (sympy + Cl(16) Fractions). **[E]**[C]

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Script	**What it machine-checks** *(continued)*
`v507_seam_bit_origin.py`	`SEAM.BIT.ORIGIN.01`: the tautology attack on the v506 alignment bit, executed and refuted — where do the marks AND the deck come from, and does their common origin force the central alignment? (Part A torus origin + Part B fermion dichotomy in one module; NO marker moves — the bit stays [C], `SEAM.CLOCK.RIGIDITY.01` and `QGEO.REALIZE.01` text-precised only.) Exact sympy + exact Cl(16) Fractions. [E] ORIGIN THEOREM: the 4 marks are the fixed points of the hyperelliptic involution $w \mapsto -w$ on $\mathbb{C}/(\mathbb{Z}+\tau\mathbb{Z})$ = the four half-periods $\Lambda/2\Lambda$; a translation t_c descends to the base sphere iff $2c \in \Lambda$ — exactly the THREE half-period translations (control $c = 1/3$ fails); the descended maps are the EXPLICIT rational involutions $\sigma_i(x) = e_i + (e_i - e_j)(e_i - e_k)/(x - e_i)$ with the exact curve-lift identity $\prod(\sigma_i - e_l) = \psi_i^2 \prod(x - e_l)$, ψ_i RATIONAL; $\{\mathrm{id}, \sigma_1, \sigma_2, \sigma_3\}$ = Klein V_4; CONVERSELY, in ALL three stabiliser types the freely-acting involutions are EXACTLY the three σ_c (generic V_4 $(4,3,3)$, harmonic D_4 $(8,5,3)$, hexagonal A_4 $(12,3,3)$) $\Rightarrow$ EVERY mark-free collar deck (v216) IS a half-period translation — the CLASS of the deck is forced, its POSITION is not. [E] CORE SOLVE (symbolic λ): σ_c admits a marks-preserving PGL_2 square root $\iff \lambda \in \{-1\}$ (σ_1), $\{2\}$ (σ_2), $\{\frac{1}{2}\}$ (σ_3) — per half-period exactly ONE solution; intrinsically: the two deck pairs separate each other HARMONICALLY (pair cross-ratio $= -1$); union over c = the harmonic orbit $\{-1, 2, \frac{1}{2}\} \iff j = 1728 \iff \tau = i$; generic modulus: V_4 abelian, ALL σ_c central and yet clockless (“central alone is empty” — the bit is the CONJUNCTION D_4 AND central); at $\tau = i$ the stabiliser jumps to D_4 with σ_1 central (2 roots = the clock pair) and σ_2/σ_3 edge-type (0 roots); the v506 seam deck $z \mapsto -z$ lands on the central $\sigma_1 = -1/x$ under ALL 8 mark bijections; CM DICTIONARY: mult-by-i fixes exactly ONE nonzero half-period $c^* = (1+\tau)/2$ (mult-by-ρ 3-cycles all three); the clock lift needs the CM unit i ($\psi = 2i/(x+1)^2$, $t^2 + 1$ irreducible over $\mathbb{Q}$) while half-period decks lift RATIONALLY: the clock root exists only WITH CM by $\mathbb{Z}[i] \iff \tau = i$. [E] EQUIVALENCE TETRAD + CIRCULARITY AUDIT: clock exists $\iff$ ($\tau = i$ AND deck = the CM-fixed half-period) $\iff$ (Stab $= D_4$ AND deck central) $\iff$ deck pairs harmonic $\iff$ the NS Fock lift is NONSPLIT; the v493 converse independently re-verified — two converses of ONE equivalence, no circularity. [E] FERMION DICHOTOMY (the new physical face): central arrangement (shift by 8): $\tilde{U}^2 = 256\,\gamma_1 \cdots \gamma_{16} \Rightarrow U^2 = (-1)^F$, nonsplit $\mathbb{Z}_4$, 2 roots in the 32-element dihedral lift (the $\mathbb{Z}_8$ tower); edge arrangement (reflection $j \mapsto 4 - j$, fixed sites = non-marks = the silver $\{0, \infty\}$): the implementer EXISTS but $\tilde{U}_e^2 = +2^7 \cdot \mathbf{1} \Rightarrow U^2 = +1$, the extension SPLITS, 0 roots, no $\mathbb{Z}_8$; one-particle $(\lambda^2{+}1)^8$ vs $(\lambda^2{-}1)^8$; census: all 28 shift-invariant 4-sets cross, all 21 reflection-invariant do not; robust at 8 sites — the bit IS a physical extension class (Fidkowski–Kitaev-type), distinguished but not derived. [E] SILVER PLACEMENT: $\{\pm 1, \pm(3{+}2\sqrt{2})\}$ = “$\tau = i$ with deck = a non-CM-fixed half-period translation” (all 8 bijections $\to \sigma_2/\sigma_3$, never σ_1) — in-family witness against the tautology. [C] THE REST: the one alignment bit remains a carrier datum; no tautology — the origin forces the CLASS of the deck, not its POSITION. Exact (sympy + Cl(16) Fractions). [E][C]

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Script	**What it machine-checks** ***(continued)***
`v508_celestial_wp5e_gamma_pairing.py`	`CELEST.WP5E.GAMMA.01`: WP5e-γ of the celestial-holographic research contract `CELEST.SEAM.01` — "the sphere-axion pairing check", an honest rigid NEGATIVE result: can the three twisted sphere axions η_m (Coxeter eigencharacters $\{i, -1, -i\}$ on $H^2(\text{ALE})$, v505 S5.8) cancel the rigid $32\,T_3$ residual of the Atiyah–Bott ledger ($A_{\text{fix}} = 9P_1 - 30P_2 - 15P_3 + 32\,T_3$, v505) by Green–Schwarz-type EXCHANGE with sector-compatible quadratic vertices? NO, with certificate. Exact sympy/Fraction. [E] VERTEX-SPACE THEOREM: the $W(D_5) \times W(A_3)$-invariant quadratics on the 8-dim glue Cartan are EXACTLY $\text{span}\{s_5, s_3\}$ (dim 2, bidegree dims $(1, 0, 1)$), the quartics EXACTLY $\text{span}\{P_1, P_2, P_3, T_5, T_3\}$ (dim 5, bidegree dims $(2, 0, 1, 0, 2)$) — Weyl nullspace arithmetic over all bidegrees, basis COMPLETE; PRODUCT THEOREM (the master kill): $(as_5{+}bs_3)(cs_5{+}ds_3) = ac\,P_1 + (ad{+}bc)P_2 + bd\,P_3$ identically — zero T_5/T_3 content, so EVERY exchange image (any axion count, any selection rule, any couplings) is annihilated by Φ_{T_3}, while $\Phi_{T_3}(A_{\text{fix}}) = 32 \neq 0$. [E] SELECTION ENUMERATION: the strict two-index reading COLLAPSES ($C_{j,j'} \neq 0$ only for $j' = -j$, so the rule forces $m = 0$ — the sphere axions have NO invariant bilinear Cartan vertex at all); twist insertion: $\eta_1 \leftrightarrow K^{(3)}$, $\eta_2 \leftrightarrow K^{(2)}$, $\eta_3 \leftrightarrow K^{(1)}$, $\eta_0 \leftrightarrow K^{(0)}$; channels $E_{13} = (16, -96, 144, 0, 0)$, $E_{22} = (16, 32, 16, 0, 0)$, $E_{00} = 225\,E_{22}$; SIDE DISCOVERY: $K^{(0)} = -15\,K^{(2)}$ (the even-sector quadratics are PARALLEL). [E] UNSOLVABILITY WITH CERTIFICATE (the core result): exchange matrix 5×2, rank 2, kernel 0; $\text{rank}([M \mid A_{\text{fix}}]) = 3$; annihilator certificate $(\Phi_{T_5}, \Phi_{T_3}, \Phi_P)(A_{\text{fix}}) = (0, 32, 72)$ with $A_{\text{fix}} = (9, -30, -15, 0, 32)$ — two independent obstructions ($32\,T_3$ structural, rule-independent + $\Phi_P = 72$ rule-driven via the $K^{(0)} \parallel K^{(2)}$ collapse); $\ker M = 0$ forces the unique non-disturbing coupling $c = 0$: the sphere axions are RIGIDLY SILENT in the exchange ansatz; $\Phi_P(Q^{(0)}) = 0$ (the v505 sector-0 mechanism undisturbed); relaxed control: dropping charge conservation reaches the P-block ($\det = -64$) but $T_5 = T_3 = 0$ persists. [E] NATURALNESS RESOLUTION: ch_2-natural $(9/64, 1/4)$ and AB-weight $(1/2, 1/4)$ couplings both carry certificate $(0, 0)$ vs required $(32, 72)$ — the failure is scale- and coupling-INDEPENDENT (no "unnatural factor": the ansatz space misses the target identically). [E] NEGATIVE CONTROLS: invariant axion only = v505 replication + sharpening; flipped rule $m{+}j{-}j'$: the Φ_{T_3} kill is flip-invariant, Φ_P is rule-dependent; $SO(16)$: odd classes EMPTY and the AB sum $(15, -18, -15, -4, 40)$ does not even cancel T_5 (E_8 doubly special); D_8: $16\,T_8 + 12\,s_8^2$, no T_3 structure — the question cannot even be posed. [C] number fences: $32 = (h, h) + (h', h') = 20 + 12$; $72 = 2\tilde{\lambda}_{e_8}^2$. [O] SHARPENED FENCE: the exchange sub-branch is DEAD; the $32\,T_3$ residual MUST be carried by twisted BCOV contact terms — binary criterion WP5e-δ_1: the one-loop coefficient on $\mathbb{C}^2/\mathbb{Z}_4$ must come out exactly $(9, -30, -15, 0, 32)$, COMPUTED, not fitted; named escape WP5e-δ_2: the full-tensor ledger (the collapse is Cartan-restricted); slot bijection (v505 S5.8) untouched; the preregistered level-kill still does NOT fire; `SEAM.EQUIV.01` stays [O], no marker moves. Exact (sympy + Fraction). [E][C][O]

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Script	What it machine-checks *(continued)*
`v509_celestial_wp5e_eps2_level_from_flux.py`	`CELEST.WP5E.EPS2.01`: WP5e-ε_2 of the celestial-holographic research contract `CELEST.SEAM.01` — "the CPS level-from-flux dial", executed (verdict B): derive the boundary level $k = 1$ from flux quantisation on the lockstep spheres, following the Costello–Paquette–Sharma (CPS) Burns-holography mechanism (PRL **130**, 061602; `arXiv:2306.00940`). Exact sympy/Fraction, no floats. **[E]** CPS SKELETON REPLICATED EXACTLY: the S^3 period $\int \Omega_N = (2\pi i)^2 N$ (pullback density $-\sin 2\theta$, their eq. 3.33); the exceptional-sphere Kähler flux $\int_{\mathbb{P}^1} \omega = 2\pi N$; the boundary level magnitude $2N$ via the symplectic-boson double contraction ($N = 1, 2, 3$), Dynkin normalisation $T(\text{vector}) = 1$ — "level = flux quantum × Dynkin index". **[E]** PAIRING MATRIX PINNED, NOT POSTULATED: the v505 ch_2 ledger + integrality + unimodularity + effectivity pin $P_{mi} = c_1(T_m).[\Sigma_i] = \delta_{mi}$ by COMPLETE enumeration (row candidates 8/6/8; 48 unimodular solutions of $PC^{-1}P^T = C^{-1}$; with $P \geq 0$ exactly $2 = \{\text{identity}, A_3$ diagram flip$\}$); glue FLUXES $F_i = (64, 60, 64) = \dim \mathfrak{g}_i$, total $188 = 248 - 60$, flip-invariant; flux per charged root current uniformly ONE quantum. **[E]** LEVEL FORMULA, honest: "level = total open-string flux" is KILLED by the lockstep test itself ($k_i = (64, 60, 64)$ unequal = the FLAVOUR multiplicity); what works: k = flux per adjoint current $= F_i/\dim \mathfrak{g}_i = (1, 1, 1)$, anchored by the lattice current count $(240, 0, 0, 0)$ for $k = 1..4$ and the embedding index $j(D_5 \subset E_8) = j(A_3 \subset E_8) = 1$ ($60 = 2h^\vee$). ORD-4-VS-LEVEL-1 RESOLVED: the clock order counts FRACTIONAL flux sectors — $\mu = \det(D_5)\det(A_3) = 16 = \mathrm{ord}^2$, the μ_4 glue diagonal isotropic + Lagrangian, KLM $16 \to 16/4^2 = 1$ (zero residual fractional flux); NEW DIAL: $\#\text{primaries}((E_8)_k) = (1, 3, 5, 10, 15, 27)$ for $k = 1..6$ from the affine marks — exactly ONE sector iff $k = 1$. **[E]** LOCKSTEP THEOREM (the lifting of v505 "one scale ⇒ one level" to a THEOREM): clock invariance forces $a_3 = a_2 = a_1 = 0$, one μ_4 orbit of branch points, $\lvert\Pi_j\rvert = \sqrt{2}\,t$ EQUAL on all three spheres, monodromy = the pure phase i, $\det(A - 1) = -4$: no invariant flux vector — $k_1 = k_2 = k_3$ IS A THEOREM; NEGATIVE PROBE (new falsifier): the clock-forbidden family $Z^4 - Z$ (disc -27, branch points $\{0, 1, \omega, \omega^2\}$ not a μ_4 orbit): 0/24 orderings lockstep; HONEST LIMIT: uniform flux doubling keeps the lockstep — EQUALITY, not the value (1 from the current/sector counters and the minimal quantum $N = 1$). **[E]** NEGATIVE CONTROLS: $k = 2$ (count 0, residual 360, prefactor $-31/48$, #prim 3 — the lockstep dial honestly does NOT fire); $SO(16)$ (fluxes $(0, 60, 0)$: spheres flux-dark, condensation stops at $4 = \det \mathrm{Cartan}(D_8)$); A_2 ($\mu = 12$ not a perfect square: NO Lagrangian glue, Coxeter $3 \neq 4$). **[C]** the CPS dictionary itself on $\mathbb{PT}/\mathbb{Z}_4$ (their geometry is the $\mathbb{C}^2$ blow-up with ONE (-1)-sphere and $\mathfrak{so}(8)$ at negative level — not the A_3 ALE). **[O]** the type-I B-model backreaction on $\mathbb{PT}/\mathbb{Z}_4$ (the A_3 Ω_N) — the same fence as v502/v505; `SEAM.EQUIV.01` untouched; no marker moves. Exact (sympy + Fraction). **[E][C][O]**

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Script	**What it machine-checks *(continued)***
`v510_seam_bit_freedom.py`	`SEAM.BIT.FREEDOM.01`: the freedom attack on the v506/v507 alignment bit — the POSITION half of the bit is topology (Part A dihedral/Fock + Part B Moebius/RP in one module; NO marker moves — the modulus datum stays [C]). Exact sympy + exact Cl(16) Fractions, no floats. [E] DIHEDRAL THEOREM: on the N-site seam circle the free involution is UNIQUE = the antipode ($N = 16$: 17 involutions in D_{16}, exactly 1 free = shift 8; all 16 reflections have exactly 2 circle fixed points; robustness $N = 8, 12$); the census is COMPLETE: $\mathrm{Aut}(C_{16}) = D_{16}$ exactly (brute force over all 8! permutations at $N = 8$, propagation at $N = 16$); $N = 9$ control: ZERO free involutions (even carrier count, consistent with $16 = 2^{g_{\mathrm{car}}-1}$); QUOTIENT TOPOLOGY (the P1 tie): circle/antipode = CIRCLE ($\chi = 0$, closed), circle/reflection = INTERVAL ($\chi = 1$, two corners) — the $6\pi = 2\pi + 2\pi + 2\pi$ Gauss–Bonnet budget counts three CLOSED circles, so “the deck is free on the seam circle” IS the established P1 datum “the seam is a closed circle”, not a new premise. [E] NONSPLIT $\iff$ FREE (complete census, exact Cl(16)): ALL 17 involutions implemented — the free one’s NS lift squares to $256\,\gamma_1 \cdots \gamma_{16} = (-1)^F$ (nonsplit, 2 roots = the $\mathbb{Z}_8$ clock tower); ALL 16 reflections have U^2 = SCALAR (2^7 site-, 2^8 edge-axis; split, 0 roots); chain: free $\Rightarrow$ shift 8 $\Rightarrow$ crossing (all 28 invariant 4-sets) $\Rightarrow$ central class $\Rightarrow$ the edge/silver arrangement is EXCLUDED; Klein control (covering decks are free regardless of the gluing); branch control: the pillowcase involution has exactly 4 fixed points = the marks and restricts to a REFLECTION on invariant circles — the edge class IS the BRANCHED involution’s restriction: silver mistakes the mark-creating branch $\mathbb{Z}_2$ for the seam deck. [E] REAL-STRUCTURE CLASSIFICATION: the deck-commuting antiholomorphic involutions of $\mathbb{P}^1$ are EXACTLY two families — $\mu\bar{z}$, $\lvert\mu\rvert = 1$ (fix LINE through the deck poles $\{0, \infty\}$: deck NOT free) and $\mu/\bar{z}$, $\mu > 0$ (fix circle $\lvert z\rvert = \sqrt{\mu}$ avoiding the poles: deck FREE; $\mu < 0$: empty); per configuration the mark-fixing real structure is UNIQUE; table: $\mu_4 \to$ equatorial $1/\bar{z}$, FREE (v506 A-S2.9); silver $\{\pm 1, \pm(3+2\sqrt{2})\} \to$ real axis THROUGH the poles, NOT free; generic-unit $\{\pm 1, \pm(3+4i)/5\} \to$ equatorial, free WITHOUT harmonicity; hexagonal $\to$ NO mark-fixing real structure (marks not concyclic: RP-incompatible, consistent v503); one-sidedness TYPES the $\mathbb{Z}_2$ as the free antipode (quotient $\mathbb{RP}^2$ = the $\lvert\mathbb{Z}_2\rvert$ of $c_3 = 1/(\lvert\mathbb{Z}_2\rvert\, 2\pi\chi)$, v456). [E] HARMONIC + FREE $\Rightarrow$ μ_4 EXACTLY: the deck-pair cross-ratio $(1-b)^2/(1+b)^2 = -1$ solves exactly to $b = \pm i$ — the silver family (harmonic AND edge) is EMPTY on free circles, harmonicity becomes SUFFICIENT. [E] HONEST COUNTERWITNESSES: “nonsplit $\Rightarrow$ clock” is FALSE — discrete $\{0, 1, 8, 9\}$ and continuous $\{\pm 1, \pm(3+4i)/5\}$ (free, central, $j = 148176/25 \neq 1728$, no clock): the Fock dichotomy measures the CLASS, not the modulus. PREMISE TYPING: (P-i) deck = Moebius covering deck $\Rightarrow$ free [standard topology; origin_theory/v456]; (P-ii) marks on the Θ-fix circle = existing REALIZE [C] content — no new input. [C] THE REDUCED REST (verbatim): the alignment bit reduces from “harmonic AND central” to the square-modulus datum $\tau = i$ alone; the position half is topology (edge class excluded), the modulus half stays the one carrier input. Exact (sympy + Cl(16) Fractions). [E][C]

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Script	**What it machine-checks** *(continued)*
`v511_celestial_wp5e_delta2_tensor_ledger.py`	`CELEST.WP5E.DELTA2.01`: WP5e-δ_2 of the celestial-holographic research contract `CELEST.SEAM.01` — "the full-tensor ledger", the named escape route of the v508 exchange no-go, executed (intermediate verdict, exact): does the FULL adjoint tensor structure of $\mathfrak{e}_8$ (non-Cartan external legs, higher-arity vertices) open a sphere-axion channel to the rigid $32\,T_3$ residual ($A_{\text{fix}} = (9, -30, -15, 0, 32)$, v505)? The collapse is CONFIRMED full-tensorially, but one cubic door opens. Exact Kostant/Weyl weight arithmetic + sympy/Fraction. **[E]** STRUCTURE: $\mathfrak{g}_0 = \mathfrak{d}_5 \oplus \mathfrak{a}_3$ is SEMISIMPLE with NO $\mathfrak{u}(1)$ factor (class-0 roots $40+12$, root span rank 8); the twisted sectors factorise multiplicity-free into minuscule Weyl orbits $\mathfrak{g}_1 = (16_s, 4)$, $\mathfrak{g}_2 = (10, 6)$, $\mathfrak{g}_3 = (\mathfrak{g}_1)^*$, each irreducible $(64, 60, 64)$; INNERNESS THEOREM: $h = (2, 2, 2, 2, 2; 0^4)$ lies in the Cartan OF $\mathfrak{g}_0$, so every $\mathfrak{g}_0$-invariant tensor carries total sector charge 0 mod 4 — the v508 collapse is a full-tensor statement AND holds across arities (all 15 non-neutral sector triples: Hom $= 0$). **[E]** HOM TABLES: bilinear $\dim \operatorname{Hom}_{\mathfrak{g}_0}(\mathfrak{g}_j \otimes \mathfrak{g}_{j'})$ nonzero ONLY at $j' = -j$ (dims 2/1/1/1, Killing blocks); trilinear neutral multisets $\{0, 0, 0\}: 3$ ($= 1$ sym $+ 2$ antisym), $\{0, 1, 3\}: 2$, $\{0, 2, 2\}: 2$, $\{1, 1, 2\}: 1$, $\{2, 3, 3\}: 1$ — ALL cross-sector trilinears are pure brackets (antisymmetric in the repeated legs); the ONLY totally symmetric survivor on all of $\mathfrak{e}_8$ is the $\mathfrak{su}(4)$ d-symbol ($\mathfrak{so}(10)$ has no cubic Casimir: $\operatorname{Inv}(S^3\,45) = 0$ protects T_5). **[E]** THE NEW CHANNEL: the quartic projection of the d-symbol exchange is $Q_{dd} = (0, 0, -\frac{1}{240}, 0, \frac{1}{60}) = \frac{1}{60}(T_3 - P_3/4)$, computed by TWO independent routes (Killing-propagator contraction with $2h^\vee = 60$ verified from the roots; hand route $\lvert x^2 - (\operatorname{tr} x^2/4)\mathbf{1}\rvert^2/60$); $\Phi_{T_3}(Q_{dd}) = \frac{1}{60} \neq 0$ — the v508 master kill does NOT extend to cubic vertices; brackets, tadpoles and mixed $d \times f$ die exactly; no T_5 analogue. **[E]** NEW CERTIFICATE: $M = [E_{13}, E_{22}, E_{00}, Q_{dd}]$ has rank 3, augmented rank 4 — the charge reading remains OBSTRUCTED with the WEAKER certificate $\{\Phi_{T_5}, \psi = \Phi_P - \Phi_{T_3}/4\}$, $\psi(A_{\text{fix}}) = 72 - 8 = 64$; the RELAXED reading becomes EXACTLY solvable: $A_{\text{fix}} = -u + 8v + 2w + 1920\,Q_{dd}$ (residual 0). **[C]** number fences: $64 = \dim \mathfrak{g}_1 = 2^6$; $1920 = \lvert W(D_5)\rvert = 2^7{\cdot}3{\cdot}5 = 8 \cdot 240$. **[O]** the arity ≥ 4 ledger; the physical justification of the cubic GS term ($c_d = 1920$); ch_2-naturalness for CUBIC vertices UNDECIDED; the burden on δ_1 is SHARPENED to the $\psi = 64$ slice or the selection-rule relaxation. **[E]** NEGATIVE CONTROLS: Killing control = v508 replication; $SO(16)/D_8$: bilinears $(1, 0, 1)$, trilinears $(1, 0, 1, 0)$, $\operatorname{Inv}(S^3 \operatorname{adj}_{D_8}) = 0$ — no symmetric cubic, no odd sectors, no A_3 block (E_8/μ_4 doubly special); false $\mathfrak{g}_0 = \mathfrak{d}_5 \oplus \mathfrak{a}_2 \oplus \mathfrak{u}(1)$: $(5, 2, 2) \neq (2, 1, 1)$ and $\operatorname{Inv}(S^3 \mathfrak{g}_0) = 6 \neq 1$; false sector content kills every pairing. `SEAM.EQUIV.01` stays **[O]**; NO marker moves. Exact (sympy + Fraction). **[E][C][O]**

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Script	**What it machine-checks** ***(continued)***
`v512_seam_tau_flag.py`	`SEAM.TAU.FLAG.01`: the flag-transitivity equivalence web — the $\tau = i$ attack executed on the free RP seam circle: which ESTABLISHED seam datum excludes the free-but-non-harmonic configurations? (Test family $M(\delta) = \{\pm 1, \pm e^{i\delta}\}$, $\delta \in (0, \pi/2]$; NO marker moves — the bit stays [C], `QGEO.REALIZE.01` and `P2.TYPING.01/AX.P2.01` text-precised only.) Exact sympy + mpmath 30 digits. [E] FALSIFICATION OF BARE TRANSITIVITY: on the free RP circle $V_4 = \{z, -z, b/z, -b/z\}$ acts SIMPLY TRANSITIVELY on the four marks for EVERY δ (pair-exchanging inversions; complete census: deck-commuting Möbius maps = diagonal/antidiagonal) — the expectation "transitivity $\Leftarrow$ v216 local equality selects $\tau = i$" is REFUTED; hardening: ALL undirected per-mark data (arc-pair multiset $\{\delta, \pi - \delta\}$, distance multiset, cone germ) are identical for every δ — no local jet sees the bit. [E] THE SHARP SELECTOR: FLAG transitivity (marks AND their two sides indistinguishable) $\iff \delta = \pi/2 \iff \tau = i \iff$ the clock — stabiliser jump $1 \to 2$, flag/arc orbits $2 \to 1$, order-4 rotation ($b^2 = -1$ solveset $\{\pi/2\}$), mark mirror $z \mapsto 1/z \iff \cos\delta = 0$. [E] 12-FOLD EQUIVALENCE WEB (each link exact): clock rotation, D_4, mark mirror, flag transitivity, three spectral degeneracies (odd-doublet split $= (2/\pi)\ln\cot(\delta/2)$, zero only at $\pi/2$ — counterwitness exactly $(2/\pi)\ln 2$; arc-Laplacian degeneracy $4/\pi$; the v215-K3 indicator in CLOSED FORM $(2\sqrt{2}/\pi)\lvert\ln\tan(\delta/2)\rvert$ — the v503 lattice curve analytically), ρ-twist existence ($(m^2 - n^2)(t - 1/t) = 0 \iff t = 1$; Atiyah–Bott rationality is δ-BLIND — $\mathrm{Tr}(\sigma e^{t\Delta}) = 1$ for ALL δ, $j_{\mathrm{cw}} = 148176/25$ rational non-CM: rationality does NOT select), 4-cycle implementability (H4 rigidity: V_4 with element orders $\{1, 2, 2, 2\}$ cannot induce the cusp 4-cycle; centraliser scalar, U^2 nonscalar $\Rightarrow$ no projective involution — projective order exactly 4, $\mathbb{Z}_8$-shadow consistent v510), harmonic, $j = 1728$. [E] NEGATIVE CONTROLS: hexagonal (not concyclic, fails the PRECONDITIONS); silver (harmonic but topologically dead — position/modulus halves independent, consistent v510); 3 marks (impossible on the free circle: 0 invariant 3-subsets vs 28 invariant 4-subsets; bare $n = 3$ transitivity already forces equal spacing — "4 counts": the deck-pairing makes bare transitivity free of charge). [C] THE REST (verbatim): the last input reduces from a continuous modulus to ONE discrete symmetry-lift bit ($V_4 \to D_4$: side-indistinguishability of the four equally-decorated marks); v216 local equality does NOT deliver it (no local jet can see the side bit); its negation is now concretely measurable (odd-split $(2/\pi)\ln\cot(\delta/2) \neq 0$). Scoreboard: the counterwitness passes all 7 established unsided tests, fails all 9 clock-equivalent ones. Exact (sympy + mpmath). [E][C]

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Script	**What it machine-checks *(continued)***
`v513_celestial_dterm_nonderivation.py`	`CELEST.DTERM.NONDERIV.01`: the c_d negative certificate — can geometry derive $c_d = 1920$? An anti-numerology certificate in the `v508` style (honest rigid NEGATIVE result): the `v511` **[C]** fence "$1920 = \|W(D_5)\|$" stress-tested via four candidate routes with a PRE-DECLARED look-elsewhere ledger. Exact sympy/Fraction, no floats. **[E]** CONVENTION TABLE: $c_d \in \{1920, 120, 256, 32, 69120\}$ across five normalisations — only 1 of 5 hits $\|W(D_5)\|$; the convention-STABLE content is the factorisation $c_d = \Phi_{T_3}(A_{\text{fix}}) \times 2h^\vee(E_8) = 32 \times 60$, both factors **[E]** (Killing propagator 60 from $\sum_\alpha \langle\alpha, e\rangle^2 = 120$ over the actual roots; T_3-slot fixing via det(P-block) $= -64 \neq 0$). **[E]** K1 DEAD: the charge-legal flux pairings $(1,3)/(2,2)$ give 2048/1800 — they BRACKET 1920 and both MISS; the only flux hit $64 \cdot 60/2$ is the FORBIDDEN $(1,2)$ pairing (charge 3 mod 4, killed by the `v511` innerness theorem). LOOK-ELSEWHERE LEDGER: 924 expressions from 21 **[E]** numbers, 11 hits on 1920; control targets $1800 \to 8$, $2048 \to 5$, $2016 \to 0$. **[E]** K2: quantisation delivers only lattices ($60\,\mathbb{Z}$: 1920 = the 32nd multiple, no minimality; strict ch_3-unit couplings $2160\,\mathbb{Z}$ EXCLUDE 1920); the cubic quantum is lattice-dependent (6 on coroots, 3/8 on weights); the residual integer 32 has 12 signed factorisations. **[E]** K3 PROVENANCE CLASH (the death blow): cubic indices $A_m = (0, 16, 0, -16)$, twisted sum $\sum_m i^m A_m = 32i$ NUMERICALLY $= \Phi_{T_3}(A_{\text{fix}})$ — BUT the quartic class decomposition $(10, -4, 30, -4)$ is dominated by classes 0/2 while the cubic 32 is built $16 + 16$ from classes 1/3: same number, disjoint mechanism = coincidence exposed; the Weyl side is WRONG (the d-channel is an A_3-side object, $\|W(A_3)\| = 24$; D_5 has no cubic invariant, $\text{Inv}(S^3\,45) = 0$; $\|W(E_8)\|/\|W(A_8)\| = 1920$ a second accidental 1920). **[E]** K4 NEGATIVE CONTROLS: $SO(16)/D_8$ fails every candidate formula consistently; $k = 2$ flux doubling scales the flux formula $\times 4$ (7680); false $\mathfrak{g}_0 = \mathfrak{d}_5 \oplus \mathfrak{a}_2 \oplus \mathfrak{u}(1)$ has SIX symmetric cubics; $\mathfrak{so}(10)$ sanity ($\text{Tr}\, X^3 = 0$). VERDICT: NUMEROLOGY as a derivation claim — the `v511` **[C]** fence stays a fence, now typed look-elsewhere-loaded; the honest **[E]** core is $c_d = 32 \times 60$; the physical generation of the 32 stays **[O]**. Exact (sympy/Fraction); no marker moves. **[E]**

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Script	**What it machine-checks** ***(continued)***
`v514_celestial_wp5e_eps1_axion_slot.py`	`CELEST.WP5E.EPS1.01`: WP5e-ε_1 of the celestial-holographic research contract `CELEST.SEAM.01` — "the $O(-2)$ bulk-axion slot", executed (verdict B): build the $s=0$ tower field explicitly, pin $\tilde{\lambda}=6$ as a residue with explicit factor bookkeeping on $\mathbb{PT}/\mathbb{Z}_4$, and take the first honest back-reaction step (Gibbons–Hawking A_3). Exact sympy/Fraction, no floats. **[E]** EQUIVARIANT $O(-2)$ SLOT: pushforward ledger $\pi_*O(-2)$ on $\mathbb{PT}'=\mathrm{Tot}(O(1)\oplus O(1)\to\mathbb{P}^1)$: per spacetime degree d exactly $(d{+}1)^2$ = nullspaces of the wave operator ($1,4,9,16,25,36$ for $d=0..5$); the v492 action on the fibre coordinates $(\mu_1,\mu_2)\mapsto(i\mu_1,i^{-1}\mu_2)$, incidence forces column weights $(+1,+1,-1,-1)$; the equivariant bookkeeping closes block by block (all $d\le 6$, all 4 characters); character series $P_0=1+3t^2+15t^4+21t^6+45t^8+\dots$, $P_1=P_3=2t+8t^3+18t^5+\dots$, $P_2=6t^2+10t^4+28t^6+\dots$; the $d=0$ slot has multiplicity $(1,0,0,0)$ — the bulk axion survives the projection; Molien invariant ring $=(1-t^8)/((1-t^4)^2(1-t^2))$ — $Z\in O(2)$, $X,Y\in O(4)$, relation degree $8=$ the a_0 weight (v492/v493 bridge); E3 bijection sharpened: twisted characters $\{i,-1,-i\}$ with minimal content $(2t,6t^2,2t)$ and no degree-0 mode = the Coxeter eigenvalues on $H^2(\mathrm{ALE})$, $\det(A-1)=-4$; graviton control: the $O(+2)$ slot starts invariant only at fibre degree 4 with multiplicity $3=\{X,Z^2,Y\}$ (the cohomological form of v505 $X=-2\neq+2=Y$). **[E]** $\tilde{\lambda}=6$ TRIPLY PINNED: (1) Okubo: $\mathrm{Tr}_{\mathrm{adj}}X^4=36\langle x,x\rangle^2=(6\langle x,x\rangle)^2$ on the 240 glue roots, $36=h^\vee+6$, $\tilde{\lambda}=6=\lvert\mathbb{Z}_2\rvert\,N_{\mathrm{fam}}$; (2) measure cancellation: $\mu=1/4$ enters three times (vertex^2 μ^2, propagator $1/\mu$, anomaly μ) and cancels exactly — $\tilde{\lambda}^2_{\mathrm{eff}}=36$ for arbitrary μ; wrong bookings quantified and excluded: anomaly-only $\Rightarrow\tilde{\lambda}=3$, missing propagator renormalisation $\Rightarrow\tilde{\lambda}=12$; (3) flux: minimal CPS quantum $N=1$, a single exchange channel exactly at $k=1$ (#primaries), $(\kappa/c_3)^2=36/3=12$ exact. **[E]** GH/A_3 BACK-REACTION STEP: invariant 4-centre configurations = exactly 2 branches (axis^4 or one free μ_4 orbit); free orbit $\Rightarrow\prod_p(Z-i^p z_0)=Z^4-z_0^4\Rightarrow XY=Z^4+a_0$ with $a_0=-z_0^4$ (the v493 family re-derived from the geometry); monodromy $a_0\to e^{2\pi i}a_0$ = one clock step; the GH point of charge k is exactly $\mathbb{R}^4/\mathbb{Z}_k$ ($k=4\Rightarrow$ seam boundary $S^3/\mathbb{Z}_4$); periods $\int\omega=4\pi(x_q-x_p)$, free orbit: $\Pi_j=4\pi t_0(i-1)(1,i,i^2)$, $\lvert\Pi_j\rvert$ equal (lockstep, counter-check of v509); the Coxeter clock from the geometry ($A^4=1$, $\det(A-1)=-4$, $\Pi\cdot A=i\Pi$); the CPS "$N\log$" analogue = source charge $4=\lvert\mu_4\rvert$ (flux -4π per centre); honest sharpening: for the Ricci-flat ALE the asymptotic log coefficient is exactly 0 (EH: $uK'=\sqrt{u^2+a^4}$, $\det g=1$) — the log lives at the exceptional locus; Burns contrast ($K=u+N\log u\Rightarrow\det g\neq1$, only scalar-flat); bonus: multipole selection rule $m\equiv0\bmod 4$, first symmetry-breaking multipole $(4,\pm4)$ carries $-a_0$ with amplitude $(35/16)\sin^4\theta$. **[E]** NEGATIVE CONTROLS: $\mathfrak{so}_8$ ($\tilde{\lambda}^2=12$ irrational; perfect squares in the Deligne series only $\{9,36\}=\{\mathfrak{sl}_3,\mathfrak{e}_8\}$); $\mathrm{diag}(i,i)$ (Veronese, 5 generators, no hypersurface, the bijection collapses); $k=2$ (three channels; honest: $\tilde{\lambda}^2=36$ is k-independent, the kill is the closure dial). **[C]** conditional on Costello's flat-$\mathbb{PT}$ matching $\lambda_g^2E=A$. **[O]** the quantised BCOV one-loop coefficient on $\mathbb{PT}/\mathbb{Z}_4$; the twisted channels $(32\,T_3)$ are not covered by the bulk axion — M1–M3 preregistered verbatim: M1 the A_3 Ω_N on $\mathbb{PT}/\mathbb{Z}_4$ (success: $(2\pi i)^2$ periods with lockstep phases; kill: unequal fluxes $\Rightarrow$ the v509 dial dies); M2 the twisted KS measure (success: bulk 36 + sphere axions cancel $32\,T_3$; kill: a leftover T_3); M3 the a_0 uplift of the $(4,\pm4)$ multipole (success: log correction $\propto 4=\lvert\mu_4\rvert$; kill: decoupling from the centre count). No marker moves anywhere.

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Script	**What it machine-checks** ***(continued)***
`v515_celestial_wp5e_m1_omega_n.py`	`CELEST.WP5E.M1.01`: milestone M1 of the celestial-holographic research contract `CELEST.SEAM.01` (WP5e back-reaction programme) — "the A_3 Ω_N", executed (SUCCESS on the preregistered v514 S8.1 criterion): the back-reacted holomorphic 3-form Ω_N on the twistor uplift of the deformed A_3 ALE, with exact periods. Exact sympy, no floats. **[E]** RESIDUE FORM DERIVED: on $F = XY - P(Z)$ all three ambient representatives satisfy $dF \wedge \omega_2 = -dX \wedge dY \wedge dZ$ and pull back to $\omega_2 = dX \wedge dZ/X$ (Poincaré residue, derived not assumed); cover pullback at $a_0 = 0$: $\omega_2 = 4\,dz_1 \wedge dz_2 = \lvert\mu_4\rvert \times$(flat form) — the residue normalisation carries the source charge 4; clock diag$(i, 1)$: $\omega_2 \to i\,\omega_2$ (v492 replicated); weight ledger 4+4+2−8 = 2, $\Omega \in O(4) = -\deg K_{\mathbb{PT}}$; EH anchor $2 = \lvert\mathbb{Z}_2\rvert$; wrong-weight control $Z \in O(4)$ breaks three ways ($4 \neq 2$, $6 \neq 4$, $5 \neq 3$). **[E]** THE TWISTOR FAMILY CLOSED: from the four $O(2)$ centre sections $q_p(\lambda) = i^p t_0 - i^{-p} t_0 \lambda^2$ the family is exactly $XY = Z^4 + 4t_0^2\lambda^2 Z^2 - t_0^4(1-\lambda^4)^2$ ($e_1 = e_3 = 0$ identically; $\lambda = 0$ gives exactly the v493 clock family; $a_2 \in O(4)$ opens only off-seam, $a_2(0) = a_2(\infty) = 0$); clock lift $\gamma : (Z, \lambda) \mapsto (iZ, -i\lambda)$, $\gamma^4 = 1$, $\Omega \to +\Omega$ (CY-compatible). **[E]** Ω_N CLOSED + QUANTISED PERIODS: $\Omega_N = \Omega_0 + \sum_p N_p K_p$ with CPS/Bochner–Martinelli kernels on the centre twistor lines; $dK = 0$ away from the source, scale degree 0; period table: boundary $S^3/\mathbb{Z}_4$ at the orbifold point $(2\pi i)^2 \times 1$ (minimal invariant charge $4 = \lvert\mu_4\rvert$ — the lens fundamental domain forces $N \in 4\mathbb{Z}$), deformed phase $(2\pi i)^2 \times 4$, linking S^3 uniformly $(2\pi i)^2 \times N$, per-sphere flux vector $(2\pi i)^2 \times N(1, 1, 1)$ (forced uniform twice over: clock orbit + K_4 connectivity), seam-fibre phase vector $2\pi i\, t_0(i-1)(1, i, i^2)$; the undeformed Ω_0 carries zero quantised 3-flux (all flux sourced); $(2\pi i)^2 =$ (fibre residue)×(transverse residue) (density $-\sin 2\theta$). **[E]** CLOCK COVARIANCE: $\Pi_{j+1}(-i\lambda) = i\,\Pi_j(\lambda)$ for all λ; the 12 collision nodes sit exactly on the 8 eighth roots of unity ($8 = 2\lvert\mu_4\rvert$, the $\mathbb{Z}_8$ seam bridge), clock orbits 4+4+4, honest conifold points (Hessian det $512t_0^6$, quadric rank 4). **[E]** KILL PROBE (honest): the forbidden family $Z^4 - Z$: 0/24 orderings lockstep (vs 8/24), λ-support 12 twelfth roots $\neq \mathbb{Z}_8$, no antipodal pairing — but the $(2\pi i)^2$ quantisation holds there too: integrality alone is not the discriminator, the discriminator is the lockstep phase structure + the clock forcing of equal fluxes; the v509 dial survives. **[E]** ANCHORS + CONTROLS: EH/$\mathbb{Z}_2$ (2 nodes, charge $2 = \lvert\mathbb{Z}_2\rvert$ forced); CPS original (full S^3: every integer N allowed); diag(i, i): $\Omega_0 \to -\Omega_0$ (collapse at step zero) while K stays invariant; fractional charges $N = 1, 2, 3$ excluded. **[C]** global kernel patching (Green function), Hitchin small resolution, CPS brane dictionary. **[O]** M2 (twisted KS measure), M3 (a_0 uplift), the quantised BCOV coefficient; `SEAM.EQUIV.TWISTOR.01` untouched (preparation, not construction). No marker moves.

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Script	**What it machine-checks** ***(continued)***
`v516_celestial_wp5e_m2_twisted_ks_measure.py`	`CELEST.WP5E.M2.01`: milestone M2 of the celestial-holographic research contract `CELEST.SEAM.01` (WP5e back-reaction programme) — "the twisted Kodaira–Spencer measure", executed (SUCCESS on the preregistered v514 S8.2 criterion, on the DECLARED completion measure — verdict B): the one-loop box coefficient per sector on $\mathbb{PT}/\mathbb{Z}_4$ under the completion contact term $\mathrm{contact}_j = (Q^{(0)} - Q^{(j)})/\det_j$. Exact sympy/Fraction, no floats. **[E]** THE COMPLETION-WEIGHT IDENTITY (the centrepiece): per glue class the contact weights are $w_m = \sum_j (1 - i^{jm})/\det_j = (0, \frac{3}{2}, 2, \frac{3}{2}) = 4h_m = \lvert\mu_4\rvert h_m = -4\,\mathrm{ch}_2(T_m)$ EXACTLY ($h_m = m(4-m)/8$ the v502 sector Casimir energies, $\mathrm{ch}_2(T_m)$ the McKay/Kronheimer charges, v505 S4.3 index bridge): the three sphere axions pair through their OWN ch_2 charges — an exact identity, NO free scale, no fit. **[E]** PARAMETER-FREE LOCKS: $T_5 = 0$ for ANY scale c (the ch_2 pattern alone kills T_5); ratio $h_2 : h_1 = 4 : 3$ = the δ_1-forced leading ratio (REPRODUCED, not fitted); the T_3 budget fixes $c = 4 = \lvert\mu_4\rvert$ uniquely. **[E]** THE CONTACT TERM: channel form $\sum_j (Q^{(0)} - Q^{(j)})/\det_j$ = class form $\sum_m 4h_m Q_m = (36, 120, 60, 0, -32)$ EXACTLY (one object, two bases); the contacts carry nonzero T_5/T_3 per channel (genuine contact, not exchange — the v508 product theorem replicated). **[E]** SUCCESS TABLE: $\mathrm{skeleton}_j + \mathrm{contact}_j = Q^{(0)}/\det_j = 36\langle x, x\rangle^2/\det_j$ — a PERFECT Okubo square ($T_5 = T_3 = 0$, $\tilde{\lambda}^2 = 36$) in EVERY twisted channel; total $A_{\mathrm{fix}} + \mathrm{contact} = 45\langle x, x\rangle^2 = \frac{5}{4} \times 36\langle x, x\rangle^2$ = Dedekind × Okubo = the UNIQUE quartic-free weighting (v505 S3.8 rigidity); both v508 certificates killed ($\Phi_{T_3} : 32 \to 0$, $\Phi_P : 72 \to 0$); the v511 slice $\psi = 64$ SUPPLIED EXACTLY — no cubic d-channel needed (c_d free $= 0$); remainder inside the exchange span ($45\langle x, x\rangle^2 = \frac{45}{16} E_{22} = \frac{1}{80} E_{00}$), the bulk axion cancels every channel with the SAME $\tilde{\lambda}^2 = 36$; the v514 measure chain stays μ-blind. **[E]** NEGATIVE CONTROLS: $c \in \{1, 2, 3, 5\}$ leaves $T_3 = (24, 16, 8, -8)$; the $h_1 \leftrightarrow h_2$ shuffle breaks $T_5 = -14$ AND $T_3 = 36$; $SO(16)$: $T_5 = 20$ / $T_3 = -40$ remain — the KILL fires there (Okubo failure inherited, E_8 doubly special); $\mathrm{diag}(i, i)$ degenerates to the $\mathbb{Z}_2$ target and breaks the conjugation pairing + the index bridge; the $\mathbb{Z}_2$/Eguchi–Hanson anchor PASSES ($9\langle x, x\rangle^2 = \frac{1}{4} \times 36$, scale $2 = \lvert\mathbb{Z}_2\rvert$). **[E]** MULTIPLIER NOTE (cross-reference to δ_{1d}): the contact weights are real rationals with no multiplier system of their own; the δ_{1c} residual phases = the SQUARES of the A_3 discriminant T-phases, orders $(1, 4, 1, 4)$. **[C]** THE COMPLETION READING (the fence): the loop of twisted states carries the UNPHASED sector trace with the same zero-mode normalisation — declared, supported by the δ_1 modular-completion finding, NOT derived from the BCOV integral. **[O]** the BCOV-integral derivation ($\delta_{1b}/\delta_{1c}/\delta_{1d}$ territory); `SEAM.EQUIV.TWISTOR.01` untouched. VERDICT: SUCCESS wording met on the declared measure; the KILL does NOT fire; no marker moves. **[E][C][O]**

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Script	**What it machine-checks** *(continued)*
`v517_celestial_wp5e_m3_a0_uplift.py`	`CELEST.WP5E.M3.01`: milestone M3 of the celestial-holographic research contract `CELEST.SEAM.01` (WP5e back-reaction programme) — "the a_0 uplift", executed (SUCCESS on the preregistered v514 S8.3 criterion): the $(4, \pm 4)$ GH multipole uplifted to the twistor-potential kernel $\chi = \log P_4$ (the log of the v515 family polynomial) with the induced Kähler-potential correction. Exact sympy, no floats. [E] KERNEL BRIDGE: the $O(2)$ section is a NULL coordinate — $\Delta_{3d}\, G(\eta_p) = 0$ identically for an ARBITRARY kernel G (the uplift is well-typed); RESIDUE IDENTITY $\partial_x[\text{transform}(\log \eta_p)] = 1/r_p$ EXACTLY (residue $1/(2r)$): the GH potential $V = \sum_p 1/r_p$ IS the x-derivative of the contour transform of χ; flux -4π per centre, source charge $4 = \lvert\mu_4\rvert$ (the V-ledger matched). [E] FOUR COUPLED CENTRE-COUNT SCALES: (1) asymptotic kernel log $\chi = 4\log\eta + a_2/\eta^2 + (a_0 - a_2^2/2)/\eta^4 + \ldots$ — LOG COEFFICIENT $4 = \lvert\mu_4\rvert =$ centre count (the CPS "$N \log$" analogue at kernel level); at the clock-fixed seam fibre $\zeta = 0$ the a_2 slot closes and the first correction is EXACTLY $a_0/\eta^4 =$ the $(4, \pm 4)$ multipole; exact m-grading ($m = \pm 4$ pieces PURE a_0, $m = 0$ piece φ_0-free); NO log $\times$power terms (the v514 honest zero preserved); (2) the GLT tower $\sum_p \eta_p \log \eta_p = 4\eta\log\eta + p_4/(12\eta^3) + \ldots$ with $p_n = 0$ unless $n \equiv 0 \bmod 4$ and $p_{4k} = 4(-a_0)^k$; (3) the exceptional-locus log $\chi(0) = \log a_0 = 4\log t_0 + i(4\varphi_0 + \pi)$, modulus response 1 at the locus vs power-law at infinity (v514 S6.3 uplifted); (4) periods: $d\log\Pi_j/d\log a_0 = 1/4 = 1/\lvert\mu_4\rvert$ uniform, integrated monodromy $e^{2\pi i/4} = i =$ ONE Coxeter clock step (v493 reproduced from perturbation theory); node λ-support a_0-RIGID (the $\mathbb{Z}_8$ seam bridge), K_4 discriminants $\sim t_0^2 \neq 0$ (the forced flux vector persists), BM kernel modulus-free with $\int K = (2\pi i)^2$. [E] NEGATIVE CONTROLS: $(4, 0)$ clock-invariant (φ_0-free in V AND kernel: breaks nothing); $\mathbb{Z}_2$/EH analogue: EVERY dial reads $2 = \lvert\mathbb{Z}_2\rvert$ ($(2, \pm 2)$ amplitude $\frac{3}{2}\sin^2\theta$, kernel log 2, exceptional log $2\log t_0$, response $\frac{1}{2}$, monodromy -1); $k = 3, 5$ orbits: coefficient k, slot $O(6)/O(10) \neq O(8)$ — the observable MOVES with the centre count; forbidden family $Z^4 - Z$: $p_3 = 3 \neq 0$ (selection rule broken), $e_4 = 0$ (no a_0-log) — fails BOTH dials. [C] the GLT dictionary ($\eta_p \log \eta_p =$ the GH Kähler-potential kernel; Lindström–Roček / Ivanov–Roček). [O] the full nonlinear Kähler potential of the resolved A_3 ALE; `SEAM.EQUIV.TWISTOR.01` untouched. VERDICT: SUCCESS (log correction $\propto 4 = \lvert\mu_4\rvert$, coupled to the centre count on four scales); the KILL (decoupling) does NOT fire; no marker moves. [E][C][O]

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Script	**What it machine-checks *(continued)***
`v518_celestial_wp5e_delta1_derived_kill.py`	`CELEST.WP5E.DELTA1.01`: the δ_1 chain ($\delta_{1b}+\delta_{1c}+\delta_{1d}$) of the celestial-holographic research contract `CELEST.SEAM.01` DECIDED — "the derived chiral measure kills the δ_1 route", an honest, decided NEGATIVE result (v508 style): does the Harvey–Moore/BCOV fundamental-domain integral on $\mathbb{PT}/\mathbb{Z}_4$, evaluated under a measure DERIVED from blockwise $SL(2,\mathbb{Z})$ covariance (solved for, not declared, not scanned), deliver $-A_{\rm fix} = -(9,-30,-15,0,32)$ or the $\psi = 64$ slice? NO — every member of the derived family fails all three preregistered testers. **[E]** THE 16-COMPONENT WEIL COMPLETION (δ_{1c}): discriminant module $\mathbb{Z}_4{\times}\mathbb{Z}_4$ with $q = (5x^2{+}3y^2)/8$; Gauss sums $2\zeta_8^5 \times 2\zeta_8^3 = 4 = \sqrt{16}$ (signature 0 mod 8 = rank E_8); S symmetric + unitary, $S^2 = C$, $(ST)^3 = S^2$, $S^4 = 1$, $T^8 = 1$ EXACTLY in $\mathbb{Q}(\zeta_8)$; $S = S_{D_5} \otimes S_{A_3}$; the diagonal H and anti-diagonal H' are the two invariant Lagrangians (both counts $= E_4$: two E_8 gluings); $P_H S P_H = \frac{1}{4}J$ with 3/4 of every S-image weight on the 12 shifted classes; the 16-vector theta S-covariance CLOSES: the δ_{1b} E1.5 residual $2.91 = O(1)$ drops to $\sim 10^{-39}$. **[E]** THE OBSTRUCTION IS A FINITE μ_4 CHARACTER, IDENTIFIED (δ_{1d}): gauge blocks $M = G[a,b]\,\eta^{-8}$ have constant transport (t_p, s_p) (28+ digits) recognised as exact 1/960-grid phases; the T-fixed pairs $(0,b)$ carry $t = e(-1/2) = -1$ EXACTLY; loops land in $\Gamma_1(4)$ resp. $\Gamma_0(2)^{\pm}$, joint holonomy eigenspaces dim 8/16 resp. 6/16, every eigen-phase of order ≤ 4; KOBOUNDARY TEST: the relation defects c_{S^4}, $c_{(ST)^3S^{-2}}$, $c_{[S^2,T]}$ are $(1,1,1)$ EXACTLY on all 15 pairs — a CHARACTER of the stabilisers, NOT a genuine 2-cocycle on $SL(2,\mathbb{Z})$; on $\Gamma_1(4)$: $\lambda(\gamma) = i^{2B+C/4}$. **[E]** THE CANCELLATION IS THE TWISTED FIBRE BLOCK $f_1 f_3$, NOT THE THREE SPHERE AXIONS: exact identity $G[a,b] = f_1 f_3$ (30+ digits); the T-fix mechanism is one line of exact phase arithmetic: $(-1) \times e(-1/6) = e(1/3) = \chi_4(T)$ at every T-fixed node; CANCELLATION TABLE (exact, minimal residual order over $\eta^m \times \chi_k$): none $\to 4$, $f_1 f_2 f_3$ (the three sphere axions) $\to 4$, $f_2 \to 6$, $f_1 f_3 \to 1$ (hol AND anti), wrong weights $\to 4$; strict solutions exactly χ_4 (dims $(3,3)$) and χ_{10} (dims $(1,1)$) at η^0, certified on the dressed functions (10^{-15}, NO residual multiplier). **[E]** THE DECISION — ALL THREE TESTERS FAIL: under both derived solutions the integral fails $T_5(V) = 0$ (fractions 0.5 / 0.83), fails $W_2 : W_{13} = 4:3$ ($W_{13} = 0$), fails $V = -A_{\rm fix}N$ (spreads 1.05 / 1.47) and the $\psi = -64N$ slice; the honest (N_1, N_2) rebalance: χ_4 reaches $T_5 = 0$ only at $\psi > 0$, χ_{10} only at $N_2/N_1 = -4 < 0$ (outside the orbifold projector) — no branch fires in the positive cone. **[E]** NEGATIVE CONTROLS: $\mathbb{Z}_2$/Eguchi–Hanson (gauge-only residual order 2 — the μ_4 defect is $\mathbb{Z}_4$-specific — and the SAME fibre dressing cancels it to 1); $SO(16)/D_8$ (only even sectors, all axion b-characters in μ_2: the order-4 supply structurally absent); wrong form $(x^2{+}y^2)/4$ ($\lvert G\rvert^2 = 64 \neq 16$, its S-rule fails at $O(1)$); wrong signature ($S^2 = C$ breaks exactly). **[C]** the $f_1 f_3$ = KS-weight identification (deck weights $(1,3)$ = the twistor fibre weights $(+1,-1 \bmod 4)$ forced by the v514 incidence ledger — exact match, NOT a complete BCOV derivation) + the kernel-family convention + the discrete χ_k choice (both evaluated, both fail). TENSION, stated honestly (the mandatory fence): the DECLARED completion reading (v516/M2) delivers the $\psi = 64$ slice and cancels $32\,T_3$; the DERIVED chiral measure (this module) fails all three testers — both are exact; the sharp open question is which of the two is the true BCOV measure (the declared reading is supported by the δ_1 modular-completion finding but not derived; the derived measure rests on the $f_1 f_3$ identification). Neither result is hidden behind the other. **[O]** the BCOV-integral derivation on $\mathbb{PT}/\mathbb{Z}_4$ that decides between the two measures — the single open item of the δ_1 strand; the twisted-contact route of v508 S6.2 is DECIDED NEGATIVE under the derived measure. NO marker moves anywhere. **[E][C][O]**

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Script	**What it machine-checks** ***(continued)***
`v519_woit_theta_rp_free.py`	`WOIT.THETA.FREE.01`: the WOIT-α milestone of the OS twistor bridge `WOIT.OS.TWISTOR.01` — "the real structure exists, and free reflection positivity picks the SAME family" (Part W = the anti-linear Θ: classification, exact relations, kill test 1 on the free/equivariant level; Part R = reflection positivity of the free 16-Majorana NS seam system under the pinned Θ; NO marker moves — `WOIT.OS.TWISTOR.01` stays **[O]**: kill test 1 is discharged on the FREE level only and stays formally live on the interacting algebra). Exact sympy + exact Cl(16) Fractions; the RP definiteness certificates are 40-digit spectra of exact Hermitian matrices. **[E]** THETA CLASSIFICATION COMPLETE: exactly TWO families of anti-linear structures on $\mathbb{C}^2$ normalise the clock $\rho = \mathrm{diag}(i, 1)$ — family D ($z \mapsto \mu\bar{z}$, $\lvert\mu\rvert = 1$): $\Theta\rho\Theta = \rho^{-1}$ EXACTLY, $\Theta^2 = +1$ (-1 IMPOSSIBLE: $M\bar{M} = \mathrm{diag}(\lvert\mu\rvert^2, 1)$ — Kramers-free), inverts the deck, fixed set = the great circle through the deck poles; family A ($z \mapsto \mu/\bar{z}$): centralises the clock projectively (NEVER inverts), $\Theta^2 = \pm 1$ per μ, fixed set $\lvert z\rvert = 1$ resp. empty. ROLE SEPARATION: family A DEFINES the euclidean section (Woit's ρ_{tw} side: $\rho_{\mathrm{tw}}^2 = -1$, no real points — replicated exactly on $\mathbb{C}^4$), family D REFLECTS it (the OS conjugation; σ_{std} = the standard conjugation with real points $\mathbb{RP}^3$). **[E]** MARK PINNING: $z \mapsto \mu\bar{z}$ preserves the μ_4 marks iff $\mu \in \mu_4$ — a 4-element μ_4-TORSOR (2 mark axes + 2 silver-midpoint axes); clock conjugation $\rho\,\Theta_\mu\,\rho^{-1} = \Theta_{-\mu} \Rightarrow$ 2 clock orbits. **[E]** SPIN/FOCK: the $\mathbb{Z}_8$ plane has no phase leaks ($\overline{\zeta_8^k} = \zeta_8^{-k}$); exact Cl(16): $\Theta_{\mathrm{Fock}} = U_r \circ K$ with $\Theta_{\mathrm{Fock}}^2 = 2^7 > 0$ (normalised $+1$), inverts the clock tower ($V \mapsto 4096\,V^{-1}$, exact scalar) and normalises the deck; KRAMERS DICHOTOMY: the deck-induced candidate $\Theta_t = \tilde{U} \circ K$ has $\Theta_t^2 = 256\,\gamma_1 \cdots \gamma_{16} = (-1)^F$ — the v510 split/nonsplit dichotomy IS the $\Theta^2 = +1$ vs $(-1)^F$ dichotomy; the deck does NOT furnish Θ, the seam-circle REFLECTION does. **[E]** KILL TEST 1 (free level): does NOT fire — Θ exists with $\Theta^2 = +1$ AND $\Theta\rho\Theta = \rho^{-1}$ and satisfies all three side conditions simultaneously (marks, deck normalisation, spin/Fock). **[E]** FREE RP: the chiral NS vacuum $C(d) = (2/N)/\sin(\pi d/N)$ has $C^2 = -1$ exact; preconditions exact ($RCR^T = -C$, $SCS^T = C$, $RSR^{-1} = S^{-1}$; the deck FAILS anti-invariance); BOND CUT: one-particle Gram $(8, 0, 0)$ PD (min eigenvalue 1.888×10^{-3} at 40 digits), even deg ≤ 2 sector $(29, 0, 0)$ PD, $N = 8$ complete half-sided algebra PD (even + odd each $(8, 0, 0)$) — free RP with NO degree truncation; the twist $\eta = +i$ is FORCED ($\eta = 1$ non-Hermitian); CONTROLS: the site cut fails exactly ($\det = 0$, inertia $(3, 3, 1)$; continuum control: the Cauchy–Stieltjes factorisation is strictly PD — a lattice-PLACEMENT artifact: RP selects the site placement relative to the marks); the clock-CENTRALISING family-A structure fails RP structurally ($(4, 4, 0)$) — RP and $\Theta\rho\Theta = \rho^{-1}$ select the SAME family; bonus: the anti-chiral state flips the odd sector to $(0, 8, 0)$ ND (the free shadow of kill test 3). **[C]**/**[O]** NOT shown: the interacting algebra $\mathcal{A}_{\mathrm{hol}}$ (kill test 1 stays live there), gauge-fixed RP (kill test 2), the OS reconstruction, chirality without mirrors (kill tests 3/6 — only the free odd-sector shadow), μ_4 incidence (kill test 7); WOIT-β/γ roadmap: (β_1) Θ on the gauge-invariant subalgebra + gauge-fixed RP on the equivariant SDYM(E_8) sector; (β_2) the OS quotient of the free system explicit; (β_3) the $\mathbb{PT} \leftrightarrow \mathbb{PT}^*$ duality vs σ_{std} typed; (γ) the chirality theorem + the mark incidence. Two contract PRECISIONS (prose, no marker): (i) "Θ induced by the seam reflection" means the seam-circle REFLECTION, not the deck (the deck furnishes the $(-1)^F$ Kramers class — the v510 bridge); (ii) the marks sit RP-side on the BOND MIDPOINTS of the 16-Majorana circle. **[E][C][O]**

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Script	**What it machine-checks** *(continued)*
`v520_celestial_wp5e_measure_decision.py`	`CELEST.WP5E.MEASURE.01`: the WP5e BCOV measure question DECIDED at probe level (δ_1e + δ_1f consolidated, `v518` style) — "single-valuedness is derived, the completion reading wins": which of the two exact, contradictory measures on $\mathbb{PT}/\mathbb{Z}_4$ is the true BCOV measure, (A) the DECLARED completion reading (`v516`) or (B) the DERIVED chiral measure (`v518`)? ERFOLG-A on the decision layer. [E] INVARIANT SUBSPACE: $\dim_{\mathbb{Q}}\{v : \rho(T)v = v,\ \rho(S)v = v\} = 8 = 4 \times 2$ in the dressed 16-component Weil system, EVERY kernel vector in the $\mathbb{Q}(\zeta_8)$-span of $\{e_H, e_{H'}\}$ (the two Lagrangian gluings, both theta $= E_4$, level-1 quartic $= Q^{(0)}$ = the unphased trace); strict trivial-character solutions at $(m, k) = (0, 0)$: ZERO for every dressing (11 combinations). [E] SOURCE 1 (F-LEMMA): the only strict chiral families χ_4 (dims $(3, 3)$) and χ_{10} (dims $(1, 1)$) carry nontrivial T-characters exactly; pointwise covariance of the canonical column member certified (3.9×10^{-16}, $\lvert G\rvert \geq 59.6$); EXACT change of variables $\Rightarrow$ fundamental-domain independence forces $\int = 0$ for EVERY chiral integrand — the `v518` route was never a well-defined nonvanishing moduli integral (kill SHARPENED). [E] SOURCE 2 (QUILLEN PAIRING): $\lvert t\bar{t} - 1\rvert < 8.3 \times 10^{-40}$ on all 15 pairs/both generators; $\lvert\chi_4\rvert^2 = 1$ exact vs one-sided $\chi_4^2 = e(2/3) \neq 1$ (T-fixed: $640 \neq 0 \bmod 960$, 15/15 defective); the doubled 256-dim hol $\times$ antihol Weil system closes STRICTLY at trivial character and physical weight (nullspaces $(28, 17) \geq 11$), contains the unphased diagonal ($\sim 2 \times 10^{-15}$) AND the physical columns $w_b \otimes \bar{w}_b$ (σ $4.9 \times 10^{-16}/8.3 \times 10^{-16}$); negative control hol $\otimes$ hol: EMPTY nullspace $(0, 0)$. [E] CONSISTENCY CLOSURE: the forced unphased numerator in the `v518` scaffold reproduces $\sum_m V \propto (1, 2, 1, 0, 0) = \langle x, x\rangle^2$ exactly on every level $n \leq 8$; J-weighted spreads $\sim 5 \times 10^{-41}$, $T_5/T_3 \sim 7 \times 10^{-42}$, zero negative cells; $\psi(\text{skeleton}) = +0.2302 = -\psi(\text{contact})$, the J-weighted mirror image of $\psi(A_{\text{fix}}) = +64/\psi(\text{contact}) = -64$; linearity 4.6×10^{-41} — the circle to `v516` is CLOSED. [E] D3 $\mathbb{Z}_2$/EH ANCHOR: only the declared reading hits ($9\langle x, x\rangle^2 = (1/4) \times 36$, $\psi = -8$, scale tooth $c = 1/2 = \lvert\mathbb{Z}_2\rvert h^{A_1}$); all three derived instantiations fail (spreads 1.12/0.895/0.516, ψ sign wrong). [E] D1 COLUMN CANONICALISATION: the physical AB column $\sum_m i^{bm} e_{(m,m)}$ is EXACTLY contained in the χ_4 family (σ $1.8 \times 10^{-16}/1.7 \times 10^{-16}$, one scalar per orbit — canonicalises the (N_1, N_2) freedom); the canonical member still fails all testers. [E] NEGATIVE CONTROLS: $SO(16)/D_8$ ($T_5 = 20/T_3 = -40$ exact, μ_2-only), wrong form ($\lvert\text{Gauss}\rvert^2 = 64$), wrong weight (dev 0.78), shuffle ($T_5 = -14/T_3 = 36$), planted family detected (1.5×10^{-16}), hol $\otimes$ hol empty. [C] TYPED PREMISES (the mandatory fence): TP-1 (F-independent moduli integral), TP-2 (nonvanishing — must source $\psi = 64$), TP-3 (`v518` kernel convention), TP-4 (Quillen structure of BCOV F_1). [O] the constructive BCOV derivation of the w_m normalisation itself. VERDICT: ERFOLG-A per the preregistered criteria (both sources independent, the consistency closure carries); the [O] question "which is the true BCOV measure" is DECIDED at probe level in favour of (A) under TP-1..TP-4. `SEAM.EQUIV.TWISTOR.01` stays Open; NO marker moves. [E][C][O]

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Script	**What it machine-checks** ***(continued)***
`v521_seam_bit_rp_blind.py`	`SEAM.BIT.RPBLIND.01`: the RP/Θ attack on the alignment bit DECIDED — "free OS positivity does NOT see δ", an honest, decided NEGATIVE result (v508 style): does free reflection positivity (v519) or anti-linear Θ existence break for $\delta \neq \pi/2$ on the deformed mark family $M(\delta) = \{\pm 1, \pm e^{i\delta}\}$, which would derive the v512 alignment bit from OS positivity? NO — the free RP/Θ battery is the EIGHTH side-blind test on the v512 scoreboard ($7 \to 8$); the bit remains genuine discrete input. **[E]** CENSUS (exact solvesets, symbolic δ): for every δ exactly the two mark-SWAPPING reflections $\Theta_{\pm e^{i\delta}}$ exist (axes $\delta/2$, $\delta/2 + \pi/2$; all 20 cut/mark incidence solvesets EMPTY); mark-FIXING reflections exist exactly at $\delta = \pi/2$ (solveset $\cos\delta = 0$, lost immediately at $\pi/2 - \varepsilon$); the v510/v512 counterwitness $\{\pm 1, \pm(3+4i)/5\}$ carries the swap pair exactly. **[E]** RP IS δ-BLIND: bond-cut Gram inertias $(8, 0, 0)$ PD for all tested δ, spectra IDENTICAL to 40 digits (deviation 0.0) — the ENTIRE free-RP spectral datum carries ZERO δ information; the $N = 16$ odd-m failure (site axes, $(3, 3, 1)$, $\det = 0$) is a lattice-PLACEMENT artifact: at $N = 32$ ALL $m = 1..7$ ($\delta = m\pi/8$) have mark-compatible BOND axes with PD inertia $(16, 0, 0)$; continuum: the OS kernel in axis-adapted coordinates is EXACTLY $1/\sin((s+t)/2)$, the axis position drops out identically (Cauchy–Stieltjes PD); the counterwitness PASSES free RP (6-point Gram $(6, 0, 0)$). **[E]** Θ EXISTENCE δ-BLIND: $\Theta^2 = +1$, deck normalisation and Fock implementability ($U^2 = +2^7/2^8 > 0$, exact Cl(16)) for every δ; the ONLY δ-sensitive v519 clause ($\Theta\rho\Theta = \rho^{-1}$) is for $\delta \neq \pi/2$ not violated but NOT FORMULABLE (a mark-compatible order-4 rotation exists iff $\delta = \pi/2$) — it PRESUPPOSES the bit; same for the clock-tower clause. **[E]** the battery has teeth at every δ: family A/antipode fails RP structurally (zero diagonal, $(4, 4, 0)$ indefinite), chirality–η pinning persists ($(0, 8, 0)$ ND) — RP separates the two real-structure FAMILIES, never the SIDES. **[E]** precondition controls: hexagonal (marks not concyclic, no seam circle), silver (deck not free), lifting audit (B = index relabelling, δ-blind by construction; B' fails the $\pi/2$ negative control — discarded). **[C]** FACE #13 of the v512 equivalence web: the group generated by the admissible OS reflections is V_4 for every $\delta < \pi/2$ and closes to D_4 exactly at $\delta = \pi/2$ ($\Theta_i \circ \Theta_1 =$ the clock) — $12 \to 13$ faces, a typed REFORMULATION of the bit, not a derivation. **[O]** honest gap: marks decorating the STATE (defect/twist insertions, interacting $\mathcal{A}_{\mathrm{hol}}$) = exactly v519's **[O]** scope — there, and only there, could positivity still see the bit. VERDICT: KILL exactly as preregistered; the alignment bit is NOT derivable from OS positivity and remains genuine discrete input; no marker moves. **[E][C][O]**

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Script	**What it machine-checks** ***(continued)***
`v522_woit_beta1_gso_gauge.py`	`WOIT.BETA1.GSO.01`: the WOIT-β_1 milestone of the OS twistor bridge `WOIT.OS.TWISTOR.01` executed — "the clock is time-like, GSO is the gauge datum" (typed result: verdict UNDECIDED per the frozen preregistration, mechanism identified exactly, gauge-fixed RP established under the corrected GSO typing). **[E]** HERMITICITY WITNESS: the μ_4 clock average violates Hermiticity EXACTLY (the one-step insertion $T_1 = \langle\theta(e_a), \alpha_S(e_b)\rangle$ has entry $-i/(8\sin(5\pi/16))$ where the conjugate transpose demands $+i/(8\sin(5\pi/16))$); the pairing is complex-SYMMETRIC (745 entries matching, 96 anti-matching, 0 violations) — "OS-symmetric" is strictly weaker than Hermitian: "positivity after gauge fixing" is not well-posed for the clock reading. **[E]** TIME-LIKE TYPING: ALL 16 dihedral reflection axes of the NS seam circle invert the clock lift ($RSR^{-1} = S^{-1}$), NONE commutes — the μ_4 clock IS Woit's euclidean rotation (the time translation itself); the gaugeable (θ-compatible) part of the $\mathbb{Z}_8$ tower is EXACTLY its 2-torsion $\{1, (-1)^F\}$ (Hermiticity census over all powers; α^4 = the full wrap = the 2π rotation of the seam = fermion parity, lattice spin-statistics). **[E]** POSITIVE PART — GSO-FIXED RP HOLDS EXACTLY: $N = 16$ even deg ≤ 2 parity-projected bond-cut Gram $(29, 0, 0)$ PD (min eigenvalue 1.78×10^{-6} at 40 digits); $N = 8$ complete half algebra: the projected Gram kills all odd rows/columns exactly, even block $(8, 0, 0)$ PD (3.35×10^{-3}), no degree truncation; the site-cut defect SURVIVES gauge fixing ($(7, 9, 6)$) — the RP bond placement is gauge-invariant information; family A keeps failing ($(17, 12, 0)$ indefinite vs family D's $(29, 0, 0)$); Kramers: $\Theta_t^2 = \mathrm{id}$ on $\mathcal{A}^G$, Θ_{Fock} restricts with $\Theta^2 = +1$; Ramond control: the wrap $+1$ "true $\mathbb{Z}_4$" lift fails state invariance — the NS/$\mathbb{Z}_8$ tower is forced. **[E]** KILL-TEST BOOKKEEPING: kill test (1) stays discharged also gauge-invariantly (RP separates the families on the invariant sector); kill test (2)'s free shadow does NOT fire (no legitimate Hermitian gauge-fixed form has a negative direction); both stay formally live on the interacting algebra $\mathcal{A}_{\mathrm{hol}}$. **[E]** WRONG-PROJECTION CONTROLS: invariant-dimension chain parity $120 >$ deck $64 >$ clock 32 strict, but Hermiticity singles out PARITY alone (deck and clock averages non-Hermitian): the finer "gauge" projections are structurally ill-posed pre-quotient. TRANSPARENCY: the first frozen run scored 6/13 and the preregistered deg-2 invariant dimension 30 was a combinatorial error (correct census: $28 + 4 = 32$) — documented in full; verdict by the frozen [C-3] logic (Hermiticity failure $\Rightarrow$ UNDECIDED). **[C]** CONTRACT PRECISION (iii), prose only: "gauge-invariant subalgebra" at the free level means the GSO/fermion-parity $\mathbb{Z}_2$ — the only θ-compatible part of the μ_4 tower (the free-fermion shadow of the E_8 glue: $(E_8)_1$ = even NS of $SO(16)_1$ + R spinor); the clock is the euclidean rotation itself (timelike), its equivariant statement moves to β_2 (OS quotient first, then the clock as reconstructed transfer operator). **[O]** β_1 proper stays open via β_2; `WOIT.OS.TWISTOR.01` stays **[O]**; no marker moves. **[E][C][O]**

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Script	What it machine-checks *(continued)*
`v523_celestial_wp5e_wm_derived.py`	`CELEST.WP5E.WM.01`: the constructive derivation of the w_m normalisation — "$1/\det_j$ is the Atiyah–Bott / zeta-determinant fixed-point factor, COMPUTED from three independent sources", closing the residual **[O]** of `v516`/`v520` at the fibre-zero-mode level (verdict ERFOLG per the frozen preregistration). **[E]** ATIYAH–BOTT (route i): the equivariant mode ledger of the fibre $\mathbb{C}^2$ equals the closed form $1/((1-q\,i^j)(1-q\,i^{-j}))$ exactly in $\mathbb{Q}(i)$ for all $n \le 120$; REGULAR at $q=1$ with Abel value $(1/2, 1/4, 1/2) = 1/\det_j$, $\det_j = \det(1-g^j) = (2,4,2)$; $(C,2)$ Cesaro convergence in exact arithmetic; the fixed-point factor splits off EXACTLY at every truncation level $d \in \{7, 12, 25\}$, for the phased (skeleton) AND the `v520`-forced unphased (completion) trace alike — the Abel-limit $\mathrm{contact}_j = (Q^{(0)} - Q^{(j)})/\det_j$ is the `v516` contact vector COMPUTED, not declared. **[E]** ZETA/QUILLEN (routes ii/iii): $\det_\zeta$ of the g^j-twisted circle Laplacian $= 4\sin^2(\pi j/4) = \det_j$ EXACTLY (Lerch + reflection formula symbolically; Hurwitz certificates $\sim 10^{-41}$); Quillen split $\det\Delta = \det_j^2$, $\lvert\det\bar\partial\rvert = \det_j$, with the real positive holomorphic section UNIQUE via the $SU(2)$ conjugation pairing; the $\delta_{1f}\ f_1 f_3$ block at the $(0,b)$ nodes carries the exact constant term $1/\det_b$ (deviations $\le 5.2\times10^{-28}$). **[E]** CONSISTENCY CHAIN: the derived weight reproduces the `v516` chain number by number — $w_m = (0, \frac{3}{2}, 2, \frac{3}{2}) = 4h_m = \lvert\mu_4\rvert h_m = -4\,\mathrm{ch}_2(T_m)$; $T_5 = 0$ at every scale; $h_2 : h_1 = 4:3$; the T_3 budget $32 - 8c$ forces $c = 4 = \lvert\mu_4\rvert$ uniquely; per-sector Okubo squares $(18, 9, 18)\langle x, x\rangle^2$; total $45\langle x,x\rangle^2 = \frac{5}{4}\times 36\langle x,x\rangle^2$; $\Phi_{T_3} : 32 \to 0$, $\Phi_P : 72 \to 0$; $\psi(A_{\mathrm{fix}}) = +64$, $\psi(\mathrm{contact}) = -64$. **[E]** NEGATIVE CONTROLS: $1/\det^2$ gives $w' = (0, 5/8, 1, 5/8)$ with leftovers $(2, 12)$ and a Quillen contradiction; $1/\lvert 1 - i^j\rvert$ gives $w''_1 = 1 + \sqrt{2} \notin \mathbb{Q}$; the $\mathbb{Z}_2$/Eguchi–Hanson anchor PASSES with the same derivation ($\det = 4$, $w = 1/2 = \lvert\mathbb{Z}_2\rvert h^{A_1}$, $9\langle x,x\rangle^2 = \frac{1}{4}\times 36$, $c = 2 = \lvert\mathbb{Z}_2\rvert$); the $SO(16)/D_8$ KILL fires; $\mathrm{diag}(i,i)$ breaks the zeta = AB identity itself; $k = 3, 5$: $w_m = m(k-m)/2 = k\,h_m$ — the weights wander correctly with the centre count, only $k = 4$ carries the E_8-glue chain. **[C]** TYPED PREMISES (the mandatory fence): TP-REG (Abel/Cesaro/zeta regularisation IS the definition of the equivariant character), TP-Q (Quillen structure of BCOV F_1 = `v520` TP-4), TP-NUM (unphased numerator = the `v520` ERFOLG-A decision layer), TP-CH (the insertion channel $(0, j)$ computes the box-channel weight). **[O]** the GLOBAL BCOV integral on $\mathbb{PT}/\mathbb{Z}_4$ beyond the fibre zero-mode factor; the resolved-ALE route (`v514`/`v517`) not built; `SEAM.EQUIV.TWISTOR.01` untouched. NO marker moves. **[E][C][O]**

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Script	**What it machine-checks** ***(continued)***
`v524_woit_beta2_os_quotient.py`	`WOIT.BETA2.OS.01`: the WOIT-β_2 milestone of the OS twistor bridge `WOIT.OS.TWISTOR.01` executed — "the OS quotient of the free system made explicit" (verdict SUCCESS per the frozen preregistration, **[C]**-typed per contract precision (iii)). **[E]** $\mathcal{H}_{\text{phys}}$ EXPLICIT: $N = 16$ (deg ≤ 2): the 37×37 bond-cut OS Gram is exactly Hermitian, parity-block-diagonal, inertia $(37, 0, 0)$ PD (min eigenvalue 1.7801×10^{-6} at 40 digits), null space $\{0\}$, $\dim 37 = 29 \oplus 8$; $N = 8$ (complete half algebra): $(16, 0, 0)$ PD (min 3.3471×10^{-3}), $\dim 16 = 8 \oplus 8 = 4^2$ — compact euclidean time reconstructs a THERMAL (KMS) representation; exact certificate $\sin^2(3\pi/8) - \sin(\pi/8)\sin(5\pi/8) = 1/2$. **[E]** TRANSFER SEMIGROUP (Klein–Landau): τ_k exactly Hermitian on all domains ($k = 1..7$, dims $29, 22, 16, 11, 7, 4, 2$), chain identities exact, vacuum fixed; the positivity pattern IS the site/bond dichotomy: even k PSD via the exact square identity $T(2j) = A_j^* A_j$ $((22, 0, 0)/(11, 0, 0)/(4, 0, 0))$, odd k indefinite $((10, 12, 7)/(5, 6, 5)/(2, 2, 3))$ — the one-step transfer is NOT positive (the free chirality datum, kill-test-3 shadow sharpened), its even powers ARE. **[E]** THE CLOCK $= T^{N/4}$: $N = 16$: τ_4 PD $(11, 0, 0)$ (min 6.6821×10^{-5}), compressed spectrum $[0.0194517, 1.64144]$, vacuum eigenvalue exactly 1; $N = 8$: spectrum EXACTLY $\{1, \sqrt{2} - 1\} = \{1, 1/\delta_S\}$ in both parity sectors — the silver axes of the `v519` μ_4 torsor return as the clock-transfer eigenvalue; spectral projections $\sim 10^{-40}$; the reconstructed rotation group $U(s) = e^{isH}$ is unitary with the group law. **[E]** `v522` RESOLUTION: the census $(745, 96, 0)$ reproduced; EVERY anti-matching entry is a wrap overlap, every overlap-free entry matches — the pre-quotient non-Hermiticity is EXACTLY the domain/wrap artifact; on the quotient the clock has spectral calculus. Pre-registered KMS deviation: NO contraction on the compact circle (edge modes $C(1)/C(3) = 1 + \sqrt{2} = \delta_S$ exactly, $\det(G - \tau_4) < 0$). **[E]** GSO/Θ: the grading survives; the perpendicular torsor mirror descends anti-unitarily ($\eta_\perp \in \{\pm i\}$), $\Theta^2_{\text{phys}} = +1$ on every sector (Kramers-free), $\theta_{\text{cut}} \circ \theta_\perp = \alpha_{N/2}$ exactly. **[E]** CONTROLS: site cut indefinite $((3, 3, 1)/(2, 2, 4))$ — the contract kill branch fires there; family A $(4, 4, 0)$; anti-chiral $(8, 8, 0)$ no quotient; wrong direction non-Hermitian (mismatches all wrap-overlap). KILL-TEST BOOKKEEPING: (1), (2) strengthened, (3) shadow sharpened, none fires; (4)–(7) untouched; all seven stay live on $\mathcal{A}_{\text{hol}}$. **[C]** FENCE: "the clock acting unitarily" is operationalised per contract precision (iii) as the reconstructed transfer/rotation structure. TRANSPARENCY: run 1 scored 18/20 (a sympy simplify failure on a TRUE $\pi/16$ product formula, fixed by a Laurent-polynomial certificate; no criterion changed). **[O]** β_3 is the next milestone; `WOIT.OS.TWISTOR.01` stays **[O]**; no marker moves. **[E][C][O]**

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Script	**What it machine-checks** *(continued)*
`v525_seam_bit_twist_blind.py`	`SEAM.BIT.TWISTBLIND.01`: the twist-state attack on the alignment bit DECIDED — "mark-decorated states do not see δ", an honest, decided NEGATIVE result (v508 style): the NINTH side-blind test on the v512 scoreboard ($8 \to 9$); the free-plus-twist class (everything Wick-computable) is EXHAUSTED. **[E]** ALL LIFTINGS WELL-DEFINED AND BLIND: (a) σ-gauge arc decoration ($\mathbb{Z}_2$): $\sigma \circ r = \sigma$ on the swap axes makes the twisted Gram a diagonal unitary congruence DMD — spectra IDENTICAL to untwisted (dev 0.0); (b) Kadanoff–Ceva 4-twist insertion $\omega(D\,\cdot)/\omega(D)$: well-defined on $N = 32$ ($\omega(D)$ real nonzero; the $N = 16$ odd-m degeneration is the $C(\text{even}) = 0$ checkerboard artifact); the RP SPECTRA depend on δ for the first time in the program (pairwise up to 1.05), but the INERTIA is $(16, 0, 0)$ PD with the SAME $\eta = +i$ for every m — the decoration sees δ, positivity does not; (c) the μ_4 twist at $\beta = \pi/2$ is pure plane gauge; (c′) genuine Bogoliubov defect $\beta = \pi/4$: Θ-compatibility selects exactly 2 of 16 sign patterns (NS-wrap forced), the winner genuinely non-monomial — still PD, spectra identical. STRUCTURE THEOREM (empirical, mechanism exact): EVERY Θ-compatible quasi-free mark decoration is RP-spectrally INVISIBLE (the defect planes never straddle a cut bond; orthogonal congruence). **[E]** SUB-RESULTS: the defect energy $E_{\text{def}} = 1.2752872$ for EVERY m (spread 4×10^{-40}) — δ-blind; the 4-twist Casimir $\ln\lvert\omega(D)\rvert$ measures δ ($-0.890\,..\,-1.348$) but is exactly mirror-symmetric under $m \leftrightarrow 8 - m$ — a family gauge, never a side selector; Kramers $\langle\Gamma\rangle_{\text{tw}} = +1$ for every m. **[E]** TWO BIT-PRESUPPOSING $\pi/2$ STRUCTURES (the harvest): the η flip $+i \to -i$ on the mark-fixing axes ($\sigma \circ r = -\sigma$, odd sector exactly negated); the twist frustration of the mark-fixing mirror ($(4, 4, 0)$ indefinite for both η — the lattice shadow of the Ising twist field's mirror-oddness) — both formulable ONLY on the fixing axes existing iff $\delta = \pi/2$: they PRESUPPOSE the bit (v512 facet class), no selector. **[E]** NEGATIVE CONTROLS: the twist-free limit = v519/v521 exactly; the site cut keeps failing in every decorated state; the v512 counterwitness is arc-equivalent at $N = 16$ to the blind member $\pi/4$ — no state-level selector can exclude it; a single twist has $\omega(D) = 0$ identically (twists exist only in mark pairs); the deck-paired FOUR-twist decoration is the minimal well-defined one. **[O]** the door narrows to a genuinely INTERACTING $\mathcal{A}_{\text{hol}}$ whose OS data are NOT Pfaffian-reducible to the chiral vacuum — beyond every Wick-computable class. VERDICT: KILL exactly as preregistered; the alignment bit remains genuine discrete input; no marker moves. **[E][C][O]**
`v526_seam_thermal_kms_nariai_bridge.py`	`SEAM.THERMAL.KMS.01`: the thermal seam — KMS of the reconstructed free OS quotient CLOSED onto the Nariai/BH temperature chain (the THIRD LEG of c_3). Verdict SUCCESS. **[E]** β MEASURED $= N$ at both levels ($N = 8$: $\omega = \operatorname{arcsinh}(2^{-3/4})$, $\kappa = q^{-8}$ exact; $N = 16$: pairing exponent $(N/2) + 1 = 9$); $T_{\text{seam}} = 1/N$ per clock step, $\beta_{\text{angle}} = 2\pi$ EXACT $= 1/(4c_3)$, $T_{\text{seam}} = 4c_3 =$ Bisognano–Wichmann/Hawking normalisation (v208); modular Hamiltonian closed ($\nu_1 = \cos(\pi/24)$, $\nu_2 = \cos(7\pi/24)$; Tomita KMS-$\beta = 1$ on all 256 pairs); c_3 third leg class-1 selected; entropy $S(8/16/32) = 0.52187/0.63510/0.75006$, $\Delta S/\ln 2 \to 1/6$, horizon fractions do NOT transfer; **[C]** silver $= C(3)/C(1)$ thermal kernel ratio; non-compact $\Rightarrow T = 0$. NO marker moves. **[E][C]**
`v527_seam_clock_silver_spectrum.py`	`SEAM.CLOCK.SILVER.01`: the silver clock eigenvalue EXPLAINED and DEMYSTIFIED (double verdict). STRUCTURE = SUCCESS: $\{1, \sqrt{2} - 1\}$ closed via diagonal Hankel pencil at $N = 8$, $\tan(\pi/8) = C(3)/C(1)$, doubling = exact second quantisation, carrier = silver-midpoint axes, gap $= \operatorname{arcsinh}(1) = \operatorname{gd}^{-1}(\pi/4)$; N-scaling = pure $N = 8/16$ lattice datum. CONNECTION = KILL: preregistered 11466-candidate census over 16 frozen numbers: 0 hits above null; placebos comparable. Cross-link v526: $\sqrt{2} - 1 =$ thermal kernel ratio. NO marker moves. **[E]**

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Script	What it machine-checks *(continued)*
`v528_seam_bit_twist_class_definition.py`	`SEAM.BIT.TWISTCLASS.01`: the alignment bit as an exact measurable TWIST-CLASS choice (the physical definition). Verdict SUCCESS. [E] Facets #14 (η-flip $\Leftrightarrow \delta = \pi/2$) and #15 (twist-frustration $\Leftrightarrow \delta = \pi/2$) both directions exact (cardinality lemma $\lvert A_+\rvert = \lvert A_-\rvert \Rightarrow m = 4$; g-class census 224 axes: flip class exactly at $\{15, 31\}$); order parameter $O = 1/2$ at $m = 4$ else 0; unification #14=#15=flag-transitivity = ONE $\mathbb{Z}_2$ invariant (v512 web 13 → 15); Kramers = POSITION half (derived), twist class = MODULUS half (the input). [C] measurement sketch (interferometric in principle); the bit remains formal input. NO marker moves. [E][C]
`v529_seam_interacting_toy_fk.py`	`SEAM.INT.FKTOY.01`: the minimal interacting Fidkowski–Kitaev seam toy — the FIRST firing Kill-Test-2 shadow (triple-verdicted). F1 = KILL: Θ exists so Kill-Test 1 does NOT fire; Kill-Test 2 FIRES at toy level — RP breaks for every $g > 0$ (inertia ladder $(37, 0, 0) \to (29, 8, 0)$); STRADDLE LAW (24/24). F2 = KILL: the TENTH side-blind test ($9 \to 10$) — Θ_{15} maps $H(m) \to H(8 - m)$ EXACTLY. F3 = ERFOLG: grading + $\mathbb{Z}_8$ tower + FK anchor + constructive OS quotient at $\pi/2$. [C] fence: one toy / one interaction class — narrows the Woit route, does NOT close it; `WOIT.OS.TWISTOR.01` stays Open. NO marker moves. [E][C][O]
`v530_center_quotient_compiler.py`	center quotient compiler: $Cv{=}v$, $\mathbb{Z}^3/\mathbb{Z}v$ quotient $= \left(\begin{smallmatrix} 8 & 2 \\ 5 & 3 \end{smallmatrix}\right)$ (atoms); row sums $(7, 11, 13)$ self-code; CP code $(1, 7, 11, 13)$; $\det(C{+}kI){=}(k{+}1)\det(A{+}kI)$
`v531_coxeter_tensor_stage_a.py`	Stage A tensor clock: $T_{30}{=}\,\mathrm{Comp}(\Phi_5){\otimes}\,\mathrm{Comp}(\Phi_6)$, order 30, $\chi{=}\Phi_{30}$; traces $c_{30}(k){=}c_5(k)c_6(k)$; atom readout $(2, 4, 8)$
`v532_e8_degree_modular_checksum.py`	dual-degree checksum: $(60, 192, 240, 252)$: sum $744{=}3{\cdot}248$, prod $\lvert W(E_8)\rvert$, gcd 12; $j{=}q^{-1}{+}744{+}\dots$; D_{16} control $1488{=}3{\cdot}496$; primes < 30 fingerprint [audit]
`v533_ftransfer_disc7_norm.py`	sheet transfer as a disc$\,{=}{-}7$ norm line: $D(t){=}N(\alpha_t)$, $\alpha_{t+1}{=}\alpha_t{+}2$, norms $(2, 4, 14, 32)$ on $J{\to}K{\to}C{\to}F$
`v534_seam_straddle_cone.py`	`SEAM.STRADDLE.CONE.01`: the RP cone on the interacting FK quartets — the FIRST DYNAMICAL SELECTION of the alignment bit (double-verdicted). V1 = KILL: the literal leading-order cone is EMPTY everywhere (the linear pencil cannot carry the selection). V2 = ERFOLG — THE SELECTION LAW: reflection positivity holds for EXACTLY ONE member × sign of the equivariant FK family — $\delta = \pi/2$ with positive coupling, PD $(37, 0, 0)$ on all four cuts incl. the straddled clock axes 7/15 for $g \in \{1/32..8\}$, minimum eigenvalue LIFTED off the RP boundary; asymmetric members die. Exact geometry: $\pi/2$ is the unique symmetric-straddle member (quartet partition, reflection-closed straddles, sign $+1$). [C] fence: one toy / one interaction class — NO marker moves
`v535_hecke_from_geometry.py`	Hecke from geometry: Kneser $\#iso\_lines{=}\sigma_3{\cdot}\#P^3$ (135/1120/19656/137600); $\nu_p{=}a\,Id{+}bT_p$, $a_p{=}b{-}\sigma_3$ ($a_3{=}{-}4$, $a_5{=}{-}2$); $\dim V{=}7{=}5{+}2$; $\pi_{cusp}{=}(28{-}T_3)/32$; 2-adic levels only
`v536_eichler_trace_layer.py`	Eichler trace layer: $\lambda_{\mathrm{Eis}}{=}\sigma_3(\#\mathbb{P}^3{-}\sigma_3)$; $\lambda_{\mathrm{geom}}{=}\lambda_{\mathrm{Eis}}{+}a_p^2$ (anchors 352/3784/19840); $N_A{=}\min(240(1{+}p^3), \#\mathrm{iso}{-}1)$ (82560/743040 at $p{=}7$); $a_p{=}{-}c(p)/8$ $({-}4, {-}2, {+}24)$; $R{=}a_p^2$
`v537_halfintegral_bridge.py`	Half-integral bridge: unique $\mathrm{Sh}_{t=2}(g){=}{-}8f_8$ in the 70-monoid; $T(p^2)$-eigen with $a_p(f_8)$; $U_4(g){=}0$ / outside Kohnen 1982; Waldspurger $R{=}23.1873585645\dots$ constant on 10 $d \equiv 1 \pmod 8$
`v538_relative_trace_identity.py`	Compiler relative-trace identity: $\mathrm{Tr}_V(\nu_p \circ \pi)$ bilateral independent at $p = 3, 5, 7$; Eichler orbit dictionary $\lambda_{\mathrm{Eis}} + a_p^2$; period side = lattice count with $R{=}23.1873585645\dots$

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Script	**What it machine-checks** ***(continued)***
`v539_weil_structure_family.py`	Weil structure of the compiler family: GNS fibre isolation; GL(1) core $G_0 = (1+Y)/(1-Y) = \zeta_p(w-3)^2/\zeta_p(2w-6)$; $Q_{\text{fam}} = 2Q_\zeta(g) - 2Q_\zeta(g_\flat) + \text{Arch} + \text{Corr}$; two isolated obstructions (minus doubling; non-automorphic Corr)
`v540_amplitude_linear_carrier.py`	Amplitude route and positive linear carrier: Dirac D^2= family kernel (Hecke-equivariant); $b{=}N_+{-}N_-$, Θ pure σ_3-eigenform with Cohen seed $\Theta(d){=}{-}48\,L(-1,\chi_d)$; CL deletion universal = square-class double counting; plus balance $Q{=}Q_\zeta(g_-){+}Q_\zeta(g_+)$; FE $\Lambda_\Theta(s){=}8^{1-s}\Lambda_{\Theta^\dagger}(5/2{-}s)$; open boundary = gap functional λ^* on $n \equiv 6 \bmod 8$
`v541_matching_lemma_ledger.py`	Matching lemma and transport ledger package: window proof $7S{<}40A$ on all $j \le 10^6$ (exact margin $X{=}0.082159$); ledger $Q_{\text{Weil}}{=}Q_{\text{cert}}{+}\Delta_{\text{arch}}{+}\Delta_2$ with $\Delta_{\text{pole}}{\equiv}\Delta_{\text{conv}}{\equiv}0$; $-\psi{=}(\chi_{-4}{+}\frac{1}{4}\chi_8{+}\frac{1}{4}\chi_{-8})\Theta$; $\Delta_{\text{arch}}{=}A_{\text{fam}}{-}A_{\text{shift}}$ (Legendre duplication); coherent class closed by the λ-equivariant CM channel; named limits: one open classical lemma + I5 (equivalence typing)
`v542_margin_chain_identities.py`	Margin-chain identities of phase 2: border weight $\sum(1-g_i){=}t_l{+}t_r \Rightarrow \kappa_{\text{flat}}{=}1$ exact; the (T) four-term identity; profile identity $p_{N-1}{=}2{-}p_0{+}2E$ with $2E{>}p_0 \iff p_{N-1}{>}2$; the Abel/Hölder curvature chain; matrix-free assembly $= J^\mathsf{T}QJ$ and graded overlap $= J$; the Céa/Strang defect identity $S_{\text{graded}} - S_{\text{uniform}} = R^\mathsf{T}X^{-1}R$ (PSD, one-sided); the secular sandwich with Albert's sign equivalence; Perron sign constancy from $S^{-1}{>}0$ as an implication; the run-count chain and the TV rewrite — per instance on small windows, nothing uniform in the zone index [E]
`v543_lumped_pair_identities.py`	Lumped M-matrix pair of phase 2: $A_B{=}A{+}L_\Delta$ Stieltjes with preserved row sums and $A_B \succeq A$ (an *upper* bound on $\lambda_{\min}(A)$, never a floor); the congruence $A{=}A_B^{1/2}(I{-}W)A_B^{1/2}$ with $\lambda_{\max}(W){=}\rho(L_\Delta A_B^{-1})$ and the declared circularity $\lambda_{\max}(W){<}1 \iff A{\succ}0$ in both directions; the M-matrix anchor $\lambda_{\min}(A_B) \ge 1/\max_r x_r$; shift commutation with entrywise inverse monotonicity $\Rightarrow$ the shifted ladder is structurally empty; Varga's identity $\rho(J){=}\tau/(1{+}\tau)$; the Collatz–Wielandt bracket at the anchor vector giving an a-priori $\rho(J){<}1$; and the k-scanned M18 majorant as a *negative* result against the preregistered bar 1/2 — per instance on small windows, nothing uniform in the zone index [E] + [X]
`v544_long_lag_support.py`	Long-lag support structure of phase 2: $c{=}c_{\text{arch}}{-}c_{\text{comb}}$ exactly and the archimedean section carries *no* positive off-diagonal, so every positive entry is comb-generated; supp(Δ) sits entirely on the anti-diagonal comb stripes at prime-power atom positions; each stripe is a perfect matching with exactly orthogonal edge vectors; the amplitude certificate $\Delta_{rt} \le c_{\text{comb}}[M{-}1{-}r{-}t]$; the Green function of the lumped comparison bracketed two-sidedly (Neumann partial sum from below, anchor-weighted tail from above); and the absolute-value envelope family certified *dead* by $\rho(\lvert E\rvert) \ge 1.31$ from below against the signed $\rho(E) \le 0.99996$ — per instance on small windows, no decay law [E] + [X]
`v545_hardy_core_identities.py`	Hardy-core identities of phase 2: the form lifting Gram $= CHC^\mathsf{T}$ with $C = \text{diag}(\sqrt{\Delta})M$, hence rank(Gram) $\le h-1$; the exact finite-core reduction $\rho(W) = \lambda_{\max}(K^{1/2}HK^{1/2})$ with the closed geometric covering kernel $K = M^\mathsf{T}\Delta M$; the energy reordering $H = \text{diag}(s) + L_N$; the checkerboard split $\text{Band}_b = D + A_{\text{even}} + A_{\text{odd}}$ with three Weyl steps; the four Hardy identities (the K^+ Rayleigh form, the signed crossing kernel $L_N = D^\mathsf{T}BD$, $Q = B_+\mathbf{1}$, $DKD^\mathsf{T} = L_\Delta$ on the interior nodes); the closed Muckenhoupt quantity $A_{M0} = \max_k Q_k(\boldsymbol{\Delta}\mathbf{1})_{k+1}$; and the additive and joint shapes as *valid* upper bounds — dominance only, no size, nothing uniform in the zone index [E] + [X]

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Script	**What it machine-checks *(continued)***
`v546_capacity_chain_identities.py`	Capacity-chain identities of phase 2: the capacity decomposition $K^{-1}{=}D^{\mathsf{T}}J^{-1}D{+}xx^{\mathsf{T}}/\mathrm{cap}$ with the Dirichlet half an orthogonal projection ($\Omega{=}1$ exactly); the exact capacity–Rayleigh identity for $1{-}\rho(W)$ assembled against the direct quotient with the inf property and the declared 1/gap conditioning; Cholesky nestedness — $\mathrm{cap}([a,b))$ as a prefix sum of one forward substitution, verified against independent solves; the whitening certificate κ_{up} under the Gershgorin ceiling; and the T145 M4 split — the pointwise mass-half theorem, $\sigma_{\mathrm{tot}}{<}1$ per window, licence 4 with $\lvert R\rvert$ and its built-in witness, the ordered sign-free chain and S1′ certified per window on both routes ($c_0 = 2.52$–4.23 against the Dirichlet value 4) — per instance on small windows, the level lemma stays open **[E]**
`v547_level_lemma_identities.py`	Level-lemma identities of phase 2: the union identity for nested cakes $\sum_{j,l} c_j c_l \lvert S_j \cup S_l\rvert = 2T\lVert\psi_t\rVert_1 - \lVert\psi_t\rVert^2$ (form-independent, $O(m)$); the layer-cake counting lemma at a general base, $G(b) \le 2b^2\lVert v\rVert_\infty\lVert v\rVert_1/\lVert v\rVert^2 + \varepsilon$ for *any* vector, the classical dyadic 8 recovered at $b=2$ and the base free; the resolvent delocalization bound $\psi = \lambda R\psi \Rightarrow \lVert\psi\rVert_\infty \le \lambda_{\mathrm{up}} \max_j \lVert Re_j\rVert$ with Rayleigh–Ritz λ_{up} and residual-paid certified column norms — the lever 2.85–10.17 below the trivial ceiling; and the composite level lemma $c_0^{\mathrm{ap}} = 2b^2\Gamma\min(1,\Gamma_1) + \varepsilon = 3.90$–$4.85$ as the certified window inequality cert_lam_max$(R) \le c_0^{\mathrm{ap}}\Psi_{\mathrm{abs}}$ on 12/12, with the built-in no-go discriminator and the flat Dirichlet control — per instance on small windows, the all-D asymptotics stays open **[E]**
`v548_green_szego_identities.py`	Green/Szegő identities of phase 2: the factorisation identity $\Gamma = \sqrt{Q^\star}\,S_w$ with $S_w = \lambda_{\mathrm{up}}\lVert R\rVert_F$ purely spectral and $Q^\star = m\max_j(R^2)_{jj}/\operatorname{tr}(R^2)$ purely geometric (exact at $2.2{\times}10^{-16}$); the bottom-block trace identity $\sum_j(\Pi_B)_{jj} = \lvert B\rvert$ with the measured $Q_B/\lvert B\rvert = 1.509$–$1.798$ inside $[1, 2.2]$ against the Szegő/Widom reference $2\lvert B\rvert$; the second-difference ℓ^1 ceiling $\lvert a_k\rvert \le \min(1, \lVert s_k\rVert_\infty\lVert L_0^p\psi\rVert_1/\mu_k^p)$ with the m-free closed form $\lVert a\rVert_1 \le 2\sqrt{\nu}+1$ and the Dirichlet calibration $\nu \to \pi$; the layer-cake S_w certificate from completed $\mathrm{LDL}^{\mathsf{T}}$ inertia counts ($S_w \le 1.329$–1.822); and the Liouville transform $m\lVert\psi\rVert_\infty^2 \le 2\kappa_\Lambda\,\mathrm{ceil}(\hat\varphi)^2$ with a-priori κ_Λ plus the weight-class discriminator (BV flat, rough grows at matched κ_Λ) — per instance on small windows, the one open lifting input (ν_L m-free / $\mathrm{TV}(\log\Lambda)$) stays open **[E]**
`v549_gauge_parity_identities.py`	Gauge/parity identities of phase 2: the gauge freedom — for any positive diagonal $\tilde\Lambda$ the chain $\lambda_{\min}(A) \ge \mathrm{cert}\,\lambda_{\min}(\tilde E)\cdot\min\tilde\Lambda$ is a valid certified lower bound (identity plus one Rayleigh step, family maximum valid), $\Gamma = \sqrt{Q^\star}S_w$ gauge-invariant at $2.4{\times}10^{-16}$, the const gauge with $\mathrm{TV}(\log\tilde\Lambda) = 0$ exactly; the two-regime discriminator — the flutter smoother removes TV by $\ge 2.88\times$ while the transported $\nu_{\tilde L}$ moves $\le 0.88\%$, the untransported functional jumping 0.41–9.76 under the rough re-gauge; the explicit zone diagonal $\Lambda_r = c_0 - c_{M-1-2r} + \sum_{s\ne r}\max(c_{\lvert r-s\rvert} - c_{M-1-r-s}, 0)$ with the exact split $c = c^{\mathrm{arch}} + c^{\mathrm{atom}}$ ($c^{\mathrm{atom}} \le 0$) and the closed atom budget $\lVert c^{\mathrm{atom}}\rVert_1 \le 4B\sqrt{N}$, the archimedean section itself indefinite (atoms co-responsible for positivity); the parity compression $U_-^{\mathsf{T}}T_M U_- = T - H$ exact with the negative inertia localised entirely in the even sector; and the parity-sine calibration $\nu_k^P = \pi k^2$ on the exact model with the per-window ladder $\nu_k^P \le Ck^2$, $C = 11.70$–17.28 — per instance on small windows, the one open input (m-free C) stays open **[E]**

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Script	**What it machine-checks *(continued)***
`v550_odd_sector_identities.py`	Odd-sector identities of phase 2: the grid step-over — the parity sines live on the shifted grid $\theta_k = 2\pi k/N$ with $\mu_k^P = 2 - 2\cos\theta_k$ exactly, and the symbol's negative window satisfies $\theta_c/\theta_1 = 0.371$–$0.485 < 1$ on every window (odd-sector positivity is a grid fact), the symbol-only floor certified vacuous ($\rho \geq 1$ on 12/12: a matrix statement, not a symbol one); the certified bottom ladder $\lambda_k(A) \leq S\mu_k^P$ for $k \leq 8$ by completed LDL$^\mathsf{T}$ counts ($S = 1.1019$–6.4725); the discrete Sobolev step $\lVert\psi\rVert_\infty^2 \leq 2\lVert\psi\rVert_2\lVert\nabla\psi\rVert_2$ licensed by the odd sector's virtual node, model price exactly π; the closed archimedean minimum $\min_r \Lambda_r^{\mathrm{arch}} = c_0^{\mathrm{arch}} - c_{M-1}^{\mathrm{arch}}$ exact and attained; and the linear per-mode bound $b_k = 2m\sqrt{K_{\mathrm{bot}}\mu_k^P/\kappa} \leq C_S k$ with certified $\kappa = 0.1939$–0.3599, $C_S = 12.081$–$26.953 \leq 2\pi\sqrt{K_{\mathrm{bot}}/\kappa}$ — linear where the quadratic ladder grew, the no-go pencil ratio exploding $\times 15.8$ — per instance on small windows, the one open scalar (m-freeness of $R = K_{\mathrm{bot}}/\kappa$) stays open **[E]**
`v551_ritz_ceiling_certificate.py`	Fixed-size Ritz ceiling certificate of T154: the direction lemma — Ritz values are upper bounds for the eigenvalues of the same index (Courant–Fischer 1920/Cauchy 1829, verified against exact spectra; the reversed reading refuted on 12/12) — so the ceiling needs no residual argument; the sixteen-column certificate $\mathrm{span}\{t_1..t_8\} + A^{-1}L_P\,\mathrm{span}\{t_1..t_8\}$ (dimension 16 fixed, independent of m; 8 completed Choleskys of matrices $\leq 8\times 8$) carries $K^F = 1.1019$–1.5845 attaining the inertia-certified K_{bot} to $\leq 9.9\times 10^{-7}$ against the declared cap 10^{-4} on 12/12 — the 8 size-m LDL$^\mathsf{T}$ counts retired from the ceiling step; the one-direction anatomy (median of the seven aligned angles 0.03–0.14°, the largest 83.79–89.70°) with the collapse price $\lambda_1(A)/(t\mu_1^P) = 4.408$–$7.042$ recovered in full per window by one Cholesky; the no-go ceiling explodes $\times 57.5$ where the floor survives, the Dirichlet control certified non-positive — per instance on the 12 deepest in-cap windows; the two open inputs after T154 (R1′ m-free block floor, R2′ m-free bottom-mode floor) stay open **[E]**
`v552_angle_instruments.py`	The four fixed-size angle instruments of T155/T157: the 12×12 complement-floor certificate $\min_{v\perp W} v^\mathsf{T} L_P v \geq \mu_{13}^P - \lambda_{\max}(M^{1/2}(I - GG^\mathsf{T})M^{1/2})$ (identity + one certified Rayleigh bound; never exceeds the exact complement bottom, agreement $\leq 1.01\times 10^{-3}$ on this surface, both closed-form controls to 2.0×10^{-11}); the sine-block confinement $\lVert\gamma_H\rVert^2 \leq (S/t)/\rho_{17} = 0.0153$–$0.0245$ from the certified floor and ladder alone ($e_1(A)$ is 97.5–98.5% inside the first 16 parity sines by a per-instance theorem; κ in place of S: vacuous, 17.6–477); the Schur-entry identity $s = \mu_1^P t_1^\mathsf{T} A^{-1} t_1 = (B^{-1})_{11}$ with $(B^{-1})_{LL}S_L = I$ — $1/s = 4.05$–5.90 is an $O(1)$ number carried by one entry of the fixed-size 16×16 Schur complement, while the inversion-free Cauchy–Schwarz ceiling misses by 250–7.3×10^3; and the adaptive Lipschitz deckel — an executed certificate $\inf f^{\mathrm{arch}}/(4\sin^2(\theta/2)) \geq 0.25$ on 12/12 windows in 519–1201 evaluations, refusing false targets on 12/12; the no-go floor survives while the confinement goes vacuous ($\times 52$) and $(S_L)_{11}$ explodes ($\times 60$) — per instance on the 12 deepest in-cap windows; the two open angle terms after T157 stay open **[E]**

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Script	What it machine-checks *(continued)*
`v553_exact_form_identities.py`	The exact-form identities of T158/T159: the Thomson dual form $s = (B^{-1})_{11} = \max_x(2x_1 - x^{\mathsf{T}}Bx)$ with the direction executed plus the positive Cholesky ladder g_K (all terms strictly positive, $1/s \le 1/g_{16} \le 1/g_1 = B_{11}$, tight to 1.064–1.152); the two-dimensionality span$\{t_1, A^{-1}L_P t_1\}$ attaining s to 9.0×10^{-13} ($L_P t_1 = \mu_1^P t_1$, KMS 1953); the gauge identity $\sum_d w_d = 0$ (1.1×10^{-14} of sup $\|w\|$; the section annihilates constant lag vectors entrywise) with the Toeplitz-minus-Hankel signature $A_{r,s} - A_{r+1,s+1} = f(r+s)$; the exact lag sum $x^{\mathsf{T}}B_{LL}x = \sum_d c_d w_d$ (2.7×10^{-16} absolute) with the closed scalars $w_0 = \sum x_k^2/\mu_k^P$ and $2w_0 - w_1 = \|x\|^2$; and the Z-matrix law — B_{HH}^{arch} a raw symmetric Z-matrix on 12/12 with the closed sign-based floor $\min_k(B_{HH}^{\mathrm{arch}}\mathbf{1})_k = 0.909\text{–}1.128 \ge 0.25$ (theorem direction verified; the atom block sign-free, the full block vacuous — R1″ stays open); no-go: Z law opposite, signature broken, $1/g_{16}$ explodes $\times 59.9$ tracking $1/s$ — per instance on the 12 deepest in-cap windows; both open terms after T159 stay open **[E]**
`v554_sampling_harmonics_identities.py`	Sampling/harmonics identities of phase 2: the sampling identity — the atom half of the pairing equals $-\frac{1}{2}\sum_{n\le X}(2\Lambda(n)/\sqrt{n})\,\widehat{w}(\log n)$ ($\widehat{w}$ the linear interpolant of the lag weights; exact to $5.6{\times}10^{-15}$ of the absolute atom scale), identically a finite combination of $\sum_n \Lambda(n)n^{-1/2}\cos(t\log n)$ at 32 explicit frequencies; the closed moment laws $m_0 = 0$, $m_1 = -[S_0^2 + 2\sum_j P_j^2] \le 0$, $m_2 = -[2S_1 - (M-1)S_0]^2 \le 0$ with U3 as a theorem with checked hypotheses, $\mathrm{TV}(c^{\mathrm{arch}}) \le \|c_0\| + 2\sup < 6$ closed and window-independent; the harmonic-frequency theorem $t_j(2\alpha) = 2\pi j$ exactly — the 32 frequencies are the Fourier harmonics of the log-window — with the closed parameter-free PNT prediction $(\sqrt{X}-1)/(\frac{1}{4}+t^2)$ agreeing to 0.0007–0.097 of the trivial mass on the declared Λ-battery $X = 10^3\text{–}3.2\times10^5$; the head split $c_d^{\mathrm{arch}} = \Psi_d + D\widehat{G}(dD, D)$ with $\Psi_d = -\frac{1}{2}[(d{+}1)\log(d{+}1) - 2d\log d + (d{-}1)\log(d{-}1)]$ closed, D-free and m-free, exact at every lag, plus the scale-free Bernstein rate $\rho^* = (3+\sqrt{5})/2 = 2.618034$; and the certified off-diagonal fraction bound $\|\mathrm{off}\| \le \frac{1}{4}\|Q^{\mathrm{arch}}\|$ (0.171–0.212 on 12/12) with the refuted sign law wired as the negative control — per instance on small windows, finite Λ-sums allowed, no RH statement; R-A′/R-B′/R-C′/R-D stay open **[E]**
`v555_pareto_tv_identities.py`	Pareto/total-variation identities of phase 2: the exchange law $\delta_{\mathrm{bnd}}(x) = \frac{1}{2} + \log(2\kappa g_{16}\mathrm{TV}(x)/P(x))/\log X$ as an identity at all 612 grid points of the declared Fejér knob ($5.1{\times}10^{-16}$), the crossing criterion $\delta_{\mathrm{bnd}} < \frac{1}{2} \iff P > 2\kappa g_{16}\mathrm{TV}$ exact at every point; the total-variation floor — $w_0 = \|a\|^2$, $\mathrm{TV} \ge \|w_0\|$ by telescoping, $x_1 = 1$ forces $\mu_1^P\mathrm{TV} \ge 1$ on all 708 trial vectors (measured 5.00–28.84), the closed floor $1/\mu_1^P = 1/(4\sin^2(\pi/N)) \sim h^2$, hence every sub-$\frac{1}{2}$ point pays $P > 2\kappa g_{16}/\mu_1^P$ — a flat-price sub-$\frac{1}{2}$ demand is impossible in the parity sector on this surface (closes R-C‴ negatively), with the inadmissible-vector negative control; the chain-derived price axis $\sigma = 1 \leftrightarrow K{=}1$ rung, $\sigma = \infty \leftrightarrow K{=}16$ rung, the knob monotone in both coordinates; the closed Mellin cell moments $C_d(-(2k+\frac{1}{2}))$ vs quadrature and the boundary-free Abel step with the closed loss reason $32\pi/\alpha > 1$; and the Lerch/Frullani log-moment on three routes with the $2\log 2$ peel and $\Psi_1 = -\log 2$ exact — per instance on small windows, the T163 h-exponents sandbox fits not consumed, no RH statement; R-E/R-B‴/R-D stay open **[E]**

continued on next page

Script	What it machine-checks *(continued)*
`v556_gauge_ppr_identities.py`	Gauge/$\mathrm{P_{pr}}$ identities of phase 2: the gauge theorem — $x_1 = 1$ fixes only the scale of a, Q and TV homogeneous of degree two, so Q/TV (and δ_{bnd}) is the same number in every sector (2.9×10^{-11} on 2520 combinations; $\theta = \mu_1^P/(\mu_1^P + \kappa_s)$ exact; full space $\mu_0 = 0$ exactly, strictly worse); the h^2 conservation law floor × transfer $= 1/\mu_1^P$ identically at every shift (2.2×10^{-16}); the power-one spend $d\log r/d\log U = +1$ exactly (1.1×10^{-14}; the quadratic mutation reads 2); the $\mathrm{P_{pr}}$ identity $P_{\mathrm{pr}} = g_{16}\,R\,(\mathrm{TV}/(t_1 v)^2)/\mu_1^P$ exact to 4.3×10^{-16} on all 192 battery vectors with the predicate equivalence $R > 2\kappa \iff P_{\mathrm{pr}} > 2\kappa g_{16}\,\mathrm{tvf}/\mu_1^P$ exact — every crossing vector pays $P_{\mathrm{pr}} > 2\kappa g_{16}/\mu_1^P = 15.4$–$142.8 > \Theta_{\mathrm{TOL}} = 10$ on 12/12 (R-F closed negatively by certificate); and the Schur-cascade direction $s \ge g_{16} \ge g_1$ with $P_K = \hat{a}\,s \ge 1$ and the exact exponent closure of the lists — per instance on small windows, the one open quantifier $\inf_m g_{16}(m) > 0$ (the cascade gain) typed OPEN; no RH statement [**E**]
`v557_cascade_vector_identities.py`	Cascade/vector identities of phase 2: the cascade identities — the prefix property (one Cholesky, all sixteen rungs, 1.1×10^{-12} on 192 comparisons), the minor form $1/g_K = \det G_{[1..K]}/(\mu_1^P \det G_{[2..K]})$ on the raw Gram block (2.7×10^{-12}; Schur 1917) and the regression form $g_K/g_1 = 1/(1 - R_K^2)$ (8.8×10^{-13}); the diagonal invariance of the gain under $B \to DBD$ (2.7×10^{-12} on 144 diagonals — the gain belongs to the arithmetic Gram block alone); the exact K=2 identity — the closed vector $(1, -Q_{21}/Q_{22})$ attains $1/g_2 = Q_{11}(1 - r_{12}^2)$ exactly and $B_{11}g_2 = 1/(1 - r_{12}^2)$ identically (1.2×10^{-12}): the fast-null-vector is free at $K = 2$; the pivot identity $u^\mathsf{T} Q u = 1/g_K + \delta^\mathsf{T} Q \delta$ for $\delta_1 = 0$ (2.0×10^{-12} on 252 evaluations) with the exact entry threshold and the unification $(1/g_2)/S_2 = (1 - r_{12}^2)/(1 + 3r_{12}^2)$ — one inequality, not three; the trial theorem (every candidate a certified lower bound), the L_P no-go (gain exactly 1) and the scramble type-change (B_{11} itself loses positivity on a majority of draws on 12/12 windows) — per instance on small windows, the one open scalar R1 (an m-free upper bound on $1 - r_{12}^2$) typed OPEN; no RH statement [**E**]

How to run. From the repository root, `python verification/run_all.py` (dependencies `mpmath`, `numpy`, `sympy`); each module also runs standalone, e.g. `python verification/v1_e8_glue.py`. The runner exits with code 0 iff all checks pass; see `verification/README.md` for the full claim↔script map and `verification/status_ledger.csv` for the **single status ledger** (claim ID, status, location, dependencies, script, external datum — the source of truth all nine documents are written against). The [**E**] (Lean-formalised) carrier material lives in the canonical shipped folder `lean4-carrier-rigidity/` (at the package root; the source repository keeps the same project under `experiments/`).

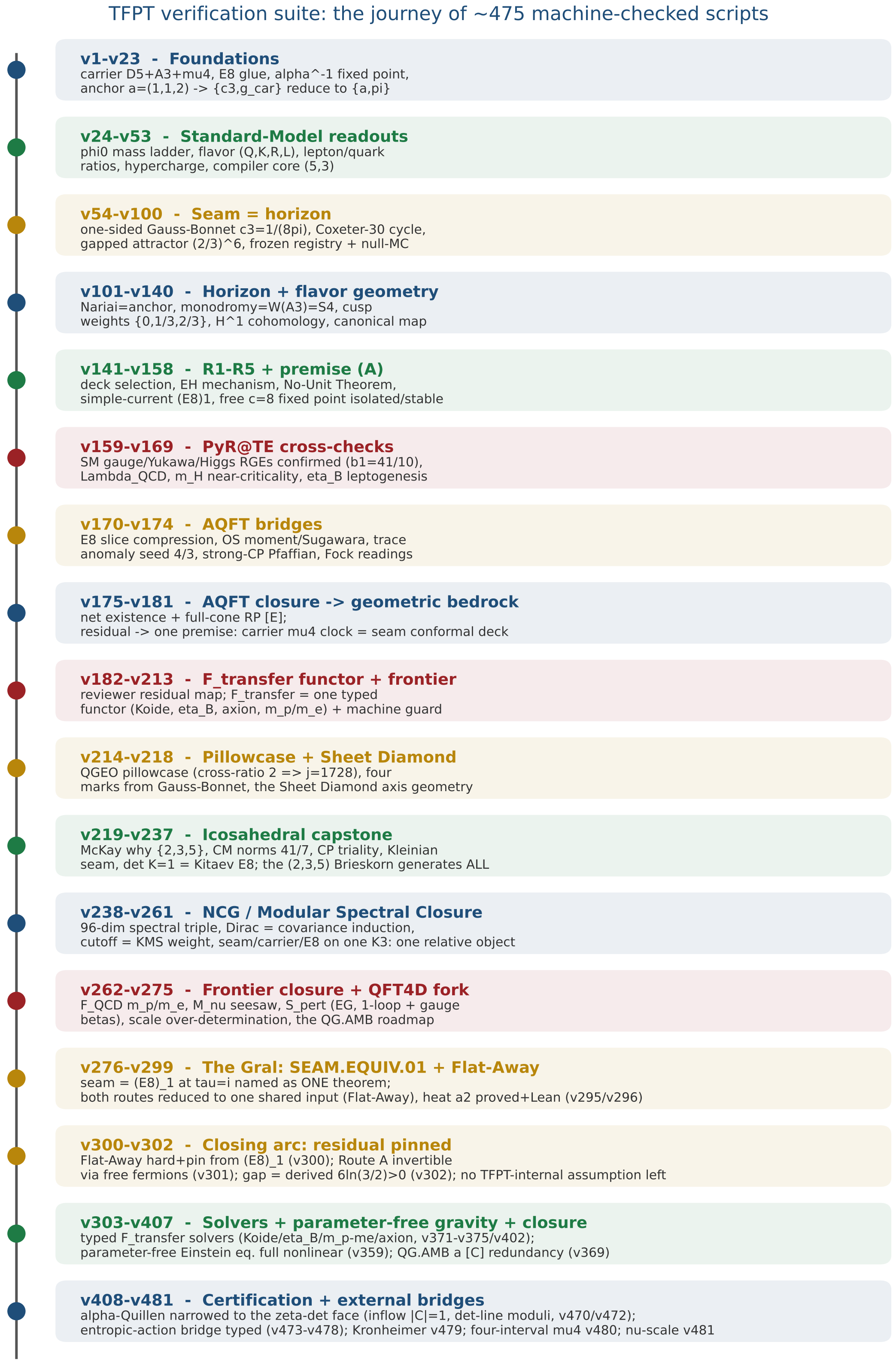

Figure 1: The development timeline of the verification suite — the phases of the journey and what each did, mathematically and physically: from the foundations (carrier, E_8 glue, α^{-1}, the anchor $a = (1, 1, 2)$ to which $\{c_3, g_{\mathrm{car}}\}$ reduce) through the Standard-Model readouts, the seam=horizon geometry, the horizon/flavor geometry, the R1–R5 and premise-(A) reductions, the external PyR@TE RGE cross-checks and the AQFT bridges, the AQFT closure that drives the whole structural residual to one geometric bedrock premise, the F_{transfer} functor and frontier interfaces, the QGEO pillowcase and Sheet Diamond, the *icosahedral capstone* (the single $(2, 3, 5)$ Brieskorn singularity generates the whole skeleton), the Modular Spectral Closure, the `SEAM.EQUIV.01` closing arc (`v276`–`v302`), the typed solvers and the parameter-free gravity round (`v303`–`v407`), to the certification round (`v408`–`v481`: the α-Quillen residual narrowed to the continuum ζ-det face, the entropic-action bridge typed, the Kronheimer quiver bridge, the four-interval μ_4 geometry). The

Appendix: changelog

The dated record of every change (canonical source: `changelog.tex`, also compiled standalone). Module numbers `vN` refer to `verification/vN_*.py`.

2026-07-29 (XIV) (v557 — the cascade/vector identities of phase 2 promoted; prime-front diary T167 (three dresses are one inequality, the closed vector exact at $K = 2$ — VECTOR-RESISTS) + paper + website sync; NO marker moves)

- **New module v557 (`PRIME.CASCADE.VECT.01`).** Promotion of the closed theorem cores of T166 (contract `SCHUR.CASCADE`) and T167 (contract `NULL.VECTOR`) as one deliberately narrow module (`v557_cascade_vector_identities.py`, 23 checks, ∼0.1 s), everything recomputed on `v551`–`v556`'s declared small frame-A surface (12 deepest in-cap zones, $m = 96 \ldots 291 \leq 300$; $\kappa = 0.038821$ verified at every jump point of $\psi(x)/x$ on the table): *(1)* the cascade identities — prefix (one Cholesky, all sixteen rungs, 1.1×10^{-12} on 192 comparisons), minor form $1/g_K = \det G_{[1..K]}/(\mu_1^P \det G_{[2..K]})$ on the raw arithmetic Gram block (2.7×10^{-12}; Schur 1917) and regression form $g_K/g_1 = 1/(1 - R_K^2)$ (8.8×10^{-13}); *(2)* the diagonal invariance of the gain under $B \to DBD$ (2.7×10^{-12} on 144 random diagonals — the gain belongs to the arithmetic Gram block alone); *(3)* the exact $K = 2$ identity — the closed vector $(1, -Q_{21}/Q_{22})$ attains $1/g_2 = Q_{11}(1 - r_{12}^2)$ exactly and $B_{11}g_2 = 1/(1 - r_{12}^2)$ identically (1.2×10^{-12}): the fast-null-vector is FREE at $K = 2$, dress (c) collapses onto dress (b); *(4)* the pivot identity $u^\mathsf{T} Q_K u = 1/g_K + \delta^\mathsf{T} Q_K \delta$ for $\delta_1 = 0$ (2.0×10^{-12} on 252 evaluations; the $\delta_1 = 0.1$ mutation breaks by exactly the closed cross term $2\delta_1$) with the exact entry threshold and the unification $(1/g_2)/S_2 = (1 - r_{12}^2)/(1 + 3r_{12}^2)$ — ONE inequality, not three; *(5)* the trial theorem (288 trials, every candidate a certified lower bound), the L_P no-go (cascade gain exactly 1, deviation 0.0) and the scramble type-change (B_{11} itself loses positivity on 49/60 declared draws, a majority on 12/12 windows) as must-breaks; every statement with a mutation control; the one open scalar R1 (an m-free upper bound on $1 - r_{12}^2$) explicitly typed OPEN; NO RH statement. Ledger row `PRIME.CASCADE.VECT.01`; registered in `run_all.py` and `script_registry.csv`; suite counter $550 \to 551$ scripts.

- **Prime-front diary T167 (`null_vector_probe.py`, contract `NULL.VECTOR`, 39/39, 1.4 s, the 63-window union of T166 rebuilt bit for bit; T163–T166 reproduced as fits; verdict VECTOR-RESISTS with a strong reduction, sandbox).** The most constructive dress attacked and closed *as a construction*: 22 families preregistered, all evaluated through the pivot identity $u^\mathsf{T} Q u = 1/g_K + \delta^\mathsf{T} Q \delta$ (10206 evaluations, 3.2×10^{-9}) — **gram2 is EXACT at $K = 2$** ($\rho = 0$, a theorem: the closed vector attains $1/g_2$ on 63/63, $B_{11}g_2 = 1/(1 - r_{12}^2)$ an identity, certified gain exponent $h^{+2.921}$ on frame A = $h^{3-0.079}$, $h^{+2.246}$ union). Perturbation theory dies structurally: Jacobi diverges (radius 1.0–12.7) but **the Kato series CONVERGES (radius 0.067) — to the WRONG object** (the exact bottom eigenvector is itself a useless trial, $\rho = 6.04$, overlap 0.083, relative gap 5.4×10^{-7}) — closed off in every order. Two thresholds separated (envelopes on 189 perturbations each); **monotonically sharper in K — mildest at $K = 2$ ($1.90\times$, $h^{+0.353}$), against the hypothesis and exactly where the vector is free**; at $K = 6$ no closed family hits (separation $h^{+1.138}$, ratio $264\times$); anchor $\mathrm{eps}_{\mathrm{ent}}(2) \to (1 - r_{12}^2)/4$ — T166's 3.57×10^{-4} is literally the quarter. End-to-end gain 2207–7.84×10^6 certified; the deficit is the higher rungs, not the vector; **the scramble breaks harder than T166 saw: 8/8 windows lose positivity of the 2×2 diagonal itself ($Q_{11} = -1.35 \times 10^5 \ldots - 1.02 \times 10^4$) — a change of TYPE, not a factor**; anti-fitting cheat only $h^{-0.823}$; off-recipe $1 - r_{12}^2 \sim h^{-2.740}$ vs $h^{-2.921}$. **The unification as an identity:** $\mathrm{eps}_{\mathrm{ent}}(K) = \rho\,(1/g_K)/S_K$ to machine precision — the determinant ratio (dress a), $1 - r_{12}^2$ (dress b) and the entry threshold (dress c) are ONE inequality, not three. Balance 6 THEOREM / 0 CERT-UNIF / 2 CERT-WINDOW / 7 MEASURED; ONE-TERM-MISSING deliberately NOT rendered (the one missing term IS dress (b)); rest: R1, the only item — an m-free upper bound on $1 - Q_{12}^2/(Q_{11}Q_{22}) \le C\, h^{-3+\varepsilon}$ (required 1.25×10^{-5} median, 3.19×10^{-8} worst); redirected: work at $K = 2$. T168 (`lagrange_minors_probe.py`, contract `LAGRANGE.MINORS`) is running — the Lagrange identity, the determinant as a sum of squares.

- **Docs + website sync.** `tfpt_prime_front` gains the T167 subsection (the pivot identity and the gram2 exactness as displays, the Kato-converges-to-the-wrong-object lesson, the threshold monotonicity against the hypothesis, the unification identity as the key result, the type-change scramble, the "After T167" keybox with R1 as the only item and T168 running), the `v557` subsection in the verified layer, the module table row, the phase-2 section title moves to T126–T167, the probe index gains the T167 row and the running totals move to 167 probes / 4394 checks; `tfpt_1`, `tfpt_3` and `tfpt_research_contracts` gain the `v557` companion blocks; `experiments/next.txt` gains the T167 entry (series (CLXXVIII)); website: `primeFront.ts` gains the T167 and `v557` feed entries, status surfaces Teile 11–167 / 4394 checks, 23 modules `v535`–`v557`, T168 running. Firewall unchanged: sandbox, no RH evidence, no zero data; every number typed THEOREM/CERT-UNIF/CERT-WINDOW/MEASURED/FIT; ledger row `PRIME.CASCADE.VECT.01`.

2026-07-29 (XIII) (prime-front diary T166 (the cascade gain is a Gram-determinant cancellation, one inequality in three dresses — CASCADE-RESISTS) + paper + website sync; NO promotion, NO marker moves)

- **Prime-front diary T166 (`schur_cascade_probe.py`, contract `SCHUR.CASCADE`, 30/30, 2.1 s, the two surfaces rebuilt as a *union*: 27 frame-A windows $h = 142$–1320 plus 36 decoupled-ν windows over 6 zones, $\nu \in \{4, 5, 6, 8, 11, 16\}$, $h = 96$–1371 — 63 windows with a $14\times$ lever arm on which zone depth and resolution are independent knobs ($\log h$ vs $\log \nu$ correlated 0.33); T165's three numbers reproduced on frame A, the union reading $h^{+2.759}/h^{+0.193}/h^{+2.566}$; $\kappa = 0.038821$ verified at every jump point in $[100, 4 \times 10^5]$; verdict CASCADE-RESISTS, sandbox).** The one open object of T165 — the cascade lower bound $g_{16}/g_1 \ge c\, h^{3-\varepsilon}$ uniform in m — attacked head-on and dissected completely. *The anatomy:* four identities, all theorems re-verified on 63/63 windows — the prefix property (one Cholesky yields all sixteen rungs), the minor form (Schur 1917), the **regression form** $g_K/g_1 = 1/(1 - R_K^2)$ (the open object *is* a near-collinearity: mode 1 explained by modes $2 \ldots 16$ up to a residual $h^{-3+\varepsilon}$), and the **diagonal invariance** of the gain under $B \to DBD$ — it does not see the KMS $1/\sqrt{\mu}$ normalisation at all: the h^3 is a property of the arithmetic Gram block $G = TAT^{\mathsf{T}}$ alone. The rung profile is extreme: rung 2 alone carries a median 59% of the gain, rungs 2–5 72%, $K_{\text{half}} \in \{2..5\}$ on 63/63; rung 2 in closed form has $1 - r_{12}^2 = 1.3 \times 10^{-7}$–$4.5 \times 10^{-4} \sim h^{-2.92}$ (frame A) — one rung multiplies g by 2207–7.8×10^6 where the m-free budgets are sharp only to a factor 21–26.

- **The cancellation finding — a third mechanism.** No entry of the 2×2 bottom block cancels (each keeps 60–99% of its larger half) and neither half alone is collinear (arch $1 - r^2 = 0.98$, atoms 0.29) — yet the sum is degenerate to seven digits: **the cancellation lives in the 2×2 Gram determinant** — the polarisation split $\det(G_{\text{ar}} + G_{\text{at}}) = \det G_{\text{ar}} + \det G_{\text{at}} + \Pi$ (exact to 6.7×10^{-14}) has pieces 2.6–630 cancelling to 1.0×10^{-3}–1.7×10^{-2}, a ratio 2.6×10^{-6}–$5.1 \times 10^{-4} \sim h^{-1.45}$ — the same arch-against-atom mechanism as T159/T160, one level up, in a quadratic functional of the lag sums. Pinned to the arithmetic twice: the L_P no-go breaks completely (gain $\equiv 1.0000000000$ exactly), and the anti-fitting scramble (uniform random atom positions, same total mass) destroys the effect by a factor 4569.

- **The best lower bound, the verdict, and two honest contract corrections.** Every route is an instance of the trial-vector theorem ($1/g_K = \mu_1^{P-1} \min\{u^{\mathsf{T}} G u : u_1 = 1\}$); the best closed route (two-rung, β from the atom block) certifies gain $h^{+1.319}$ against the target $h^{+3.110}$ — **an exponent gap of $h^{+1.791}$, not a constant: the one missing inequality *is* the cancellation** (required β precision 3.57×10^{-4}, all closed candidates miss by 12.4–2540); sharpest restatement

$1/g_K = \det G_{[1..K]}/(\mu_1^P \det G_{[2..K]})$, first flat at $K = 6$ — exactly one uniform inequality missing, in three dresses (the det-ratio bound, the closed β, a closed near-null vector of A in the low modes). Corrections recorded honestly: form F2 (gain $\geq c\,h^{3-\varepsilon}$) is strictly *weaker* than $\inf_m g_{16} > 0$ ($B_{11} \sim h^{+3.110}$, F1 needs $\varepsilon \leq -0.110$); the confinement route was a sub-surface artefact ($h^{+2.714}$ on frame A $h \leq 780$ vs $h^{+1.963}$ on the full union — exposed by raising the eigen-horizon to the cap). Errand closed: $U_{\mathrm{ref}}(\text{union}) = 1.15 \times 6.4781 = 7.45$ supersedes T164's 4.90 and T165's 5.7327 on the union (off-recipe rule declared, CERT-WINDOW, never CERT-UNIF); the union supremum is zone-depth-driven (multi-covariate gain $\sim h^{+2.92}\nu^{-1.53}$). Balance +5 THEOREM / +3 CERT-WINDOW / +4 MEASURED, deliberately no new CERT-UNIF; candidates P166.1…P166.9 all PENDING (no promotion — bundling with T167 planned). T167 (`null_vector_probe.py`, contract `NULL.VECTOR`) is running the most constructive dress.

- **Docs + website sync.** `tfpt_prime_front` gains the T166 subsection (regression form, diagonal invariance and the Gram-minor ratio as displays, the Gram-determinant key result with the T159/T160 connection, the scramble factor, the two contract corrections, the "After T166" keybox), the phase-2 section title moves to T126–T166, the probe index gains the T166 row and the running totals move to 166 probes / 4355 checks; `experiments/next.txt` gains the T166 entry (series (CLXXVII)); website: `primeFront.ts` feed entry T166, status surfaces Teile 11–166 / 4355 checks, T167 running. Firewall unchanged: sandbox, no RH evidence, no zero data; every number typed THEOREM/CERT-UNIF/CERT-WINDOW/MEASURED/FIT.

2026-07-29 (XII) (v556 — the gauge/P_{pr} identities of phase 2 promoted; prime-front diary T165 (the successor closes by certificate, one quantifier remains — ETA-RESISTS) + paper + website sync; NO marker moves)

- **New module v556 (`PRIME.GAUGE.PPR.01`).** Promotion of the closed theorem cores of the discovery parts T164 (`sector_change_probe.py`) and T165 (`alignment_eta_probe.py`) — 23 checks, ∼0.3 s, everything recomputed on `v551`–`v555`'s declared surface (the 12 deepest prime-power zones admitting a frame-A window inside the cap, $n = 29\ldots139$, $m = 96\ldots291 \leq 300$), with the one arithmetic input $\kappa = 0.038821$ *verified* at every jump point of $\psi(x)/x$ on the von-Mangoldt table in $[100, 3.2\times10^5]$ (Chebyshev 1852 / Rosser–Schoenfeld 1962), nothing cited from the sandbox; the companion to `v555`, which certified the *price structure* of the pairing — this module certifies its *gauge structure*. Scope fence: per-instance identities and certified inequalities with the direction in the name — no fit, no D-exponent, no zone-uniform statement; the T164/T165 h-exponents ($1/g_{16} \sim h^{+0.061}$, the ascent price $\sim h^{+3.3}$, the split $+3.110 - 3.049 = +0.061$) are sandbox fits and are *not* consumed; finite Λ-sums are allowed and used, *no RH statement* is made; the ONE open object after T165 ($\inf_m g_{16}(m) > 0$, the cascade-gain quantifier) stays open and typed open; each statement carries at least one mutation control that must fail loudly.

- **(1) The gauge theorem (T164, S2.1) [E].** $x_1 = 1$ fixes $a_1 = 1/\sqrt{\mu_1}$, i.e. only the *scale* of a; Q and TV are homogeneous of degree two in a; hence the demand ratio Q/TV of the same a-direction — and with it δ_{bnd} — is the *same number in every sector*: unchanged to 2.9×10^{-11} on 2520 sector-by-trial-vector combinations (5 weight families × 7 shifts × 6 taper widths on 12 windows), with TV $\geq |w_0|$ in every sector and the pure shift scaling the certified value by exactly $\theta = \mu_1^P/(\mu_1^P + \kappa_s)$ (2.2×10^{-11} on 432 points); candidate (i) settled in closed form — full-space $\mu_0 = 0$ exactly, strictly worse. Mutation: the wrong transport map (the x-coordinates instead of the a-direction) moves the ratio by 2.0×10^2–5.4×10^3.

- **(2) The h^2 conservation law (T164, S2.2) [E].** floor × transfer $= 1/(\mu_1^P+\kappa_s)\cdot(1+\kappa_s/\mu_1^P) = 1/\mu_1^P$ *identically* at every shift on the declared grid (72 window-by-shift points, 2.2×10^{-16}): a sector shift can flatten the floor but pays the identical factor back as a transfer — the h^2 is conserved, never removed; it lives in the *ratio*, not in the floor. Mutation: the wrong transfer misses by 0.9995 at the largest shift on every window.

- **(3) The power-one spend as an identity (T164, S1)** [E]**.** The r-ceiling $r \le U/L$ with $L = \lambda_1(A)/\mu_1^P = 1.1019$–$1.5845$ (computed per instance from the full window) is *linear* in the entry ceiling, so $d\log r/d\log U = +1$ exactly (central difference at $U = 1/g_{16}$, within 1.1×10^{-14} on 12/12): an h^ε ceiling costs $h^{-\varepsilon}$ of r-margin for every $\varepsilon > 0$ — no free tolerance in the consumption step (the measured power of the full T156 loss margin stays sandbox). Mutation: the quadratic consumption reads slope 2.0000 and is recognised.

- **(4) The P_{pr} identity and the predicate equivalence (T165, T1)** [E]**.** The gauge-invariant dashboard satisfies $P_{\mathrm{pr}} = g_{16} \cdot R \cdot (\mathrm{TV}/(t_1 v)^2)/\mu_1^P$ with no inequality anywhere — four factors, each a named object of the chain: the R-E-A quantifier g_{16}, the demand $R = Q/\mathrm{TV}$, the T163 floor tvf ≥ 1, and the Kac–Murdock–Szegő 1953 scale $1/\mu_1^P = 943.6$–$8609.6 \sim h^2$ — exact to 4.3×10^{-16} on all 192 battery vectors (knob settings, random sixteen-vectors *and* random full-window vectors), gauge-invariant under $v \to tv$ to 1.5×10^{-11}; the predicates $R > 2\kappa$ and $P_{\mathrm{pr}} > 2\kappa g_{16}\,\mathrm{tvf}/\mu_1^P$ agree as booleans on all 192 — the demand half of R2″ and the crossing condition are *one* predicate, so R-F was never independent and its m-uniform form *is* R-E-A — and every crossing vector pays $P_{\mathrm{pr}} > 2\kappa g_{16}/\mu_1^P = 15.4$–$142.8$, above the preregistered tolerance $\Theta_{\mathrm{TOL}} = 10$ on 12/12: **the R-F conjunction is empty by an identity plus a floor, a certificate and not a failed search** (T165's ascent stays sandbox, no search is run). Mutation: $(t_1 v)$ unsquared scales like t and moves by 1.0–1.0×10^4.

- **(5) The Schur-cascade direction and the exponent closure (T164/T165)** [E]**.** $s = \mu_1^P t_1^\mathsf{T} A^{-1} t_1 = 0.1696$–$0.2472 \ge g_{16} \ge g_1$ window by window ($1/s \le 1/g_{16} \le 1/g_1 = B_{11}$ — the T158 ladder is a chain of upper bounds with the route-(0) rung the loosest), $P_K = \hat{a}\, s = 254.8$–$7.7 \times 10^3 \ge 1$ (Kantorovich 1948); the exponent closure is an *identity of the lists* — per instance $\log(1/g_1) = \log(g_{16}/g_1) + \log(1/g_{16})$ to 1.6×10^{-15}, the least-squares fits closing to 1.3×10^{-15} by linearity (T165's $+3.110 - 3.049 = +0.061$ is this identity; the exponent values are sandbox fits, not consumed — what is promoted is that the certification lives in the cascade gain g_{16}/g_1, the object of the one open quantifier). Mutations: the g_8 closure breaks by 0.10–0.24; a corrupted sixteen-block makes the Cholesky *refuse* on 12/12. Plus the monotone no-go staircase (damage $1.00 \to 3.49 \to 10.81 \to 24.98$ over $\nu = 4/2/1/0.5$, exact null control) and exact Dirichlet/parity controls.

- **Prime-front diary T165 (`alignment_eta_probe.py`, contract `ALIGNMENT.ETA`,** 30/30, 15.1 **s, the T163/T164 surface rebuilt and re-verified, window ladder up to** $h = 1445$**, verdict ETA-RESISTS in the strong form, sandbox).** The successor question R-F closes *by certificate, not by failed search*: the new gauge-invariant identity $P_{\mathrm{pr}} = g_{16} \cdot R \cdot (\mathrm{TV}/(t_1 v)^2)/\mu_1^P$ (machine-checked to 3.3×10^{-16}) makes demand and price *one* equation, so with T163's floor every crossing vector pays $\ge 2\kappa g_{16}/\mu_1^P$ — the old KMS h^2 — and the two R-F clauses cannot be chosen independently. The anatomy: the free optimum is a *low-harmonic* object (93.7–96.8% of its energy in the first 32 parity modes against a 2.4–22.5% baseline), essentially orthogonal to the PNT main term (cosine $-0.070\ldots-0.031$), with heavy edges on prime-power cells (lift 1.36–2.12); the price decomposes *exactly* (to 0.0): $h^{+3.261} = +1.997$ (KMS — the same h^2 since T152) $+1.279$ (demand overshoot the chain never asked for) $+0.093$ (floor) -0.108 (g_{16}); the bar-tight counterfactual stays $h^{+1.982}$. $\eta(\mathrm{Cap})$ is two-sided: the ceiling is a theorem, and at $\mathrm{Cap} = 10$ it falls short of the requirement by 5.5–392× on every window, 0/27 ladder points reach the bar. The equivalence: $R > 2\kappa$ and T163's crossing predicate are the *same* predicate (105/105 vectors) ⇒ R-F is strictly *stronger* than the R2″ demand, and the clauses collide — **R-F is empty; its m-uniform form is R-E-A**. T2 localises the quantifier further: the m-free budgets bound B_{11} sharply (factor 21–26) — but the wrong object: $1/g_1$ grows $h^{+3.110}$, $1/g_{16}$ is flat $h^{+0.061}$, exponents close exactly; the whole certification lives in the 15 ladder increments — "bound g_{16}/g_1 from below by $h^{3-\varepsilon}$", one scalar per window, a cancellation and hence provably beyond absolute-value budgets. T3, the decoupled surface (ν as a knob: 25 admissible windows over 6 zones, $\nu \in \{4, 5, 6, 8, 11, 16\}$, collinearity $0.87 \to 0.68$): **the quarter-bar drift is zone depth, not window length (factor** 19**)** — and two honest costs reported openly: $\sup 1/g_{16}$ rises to 5.7327, exceeding T164's $U_{\mathrm{ref}} = 4.90$ by 17% (the constant was frame-A-specific and is retired as a surface constant), and the h-coefficient of $1/g_{16}$ is no longer negligible at free ν. End-to-end on the union (52 windows, $h = 142$–1320): every crossing vector has $P_{\mathrm{pr}} \geq 54.6$–$5865 > \Theta_{\mathrm{TOL}}$, closed for every $h > 84.9$. Stress: T145 coarsening monotone (1.0/2.8/12.1/27.5, up to 102×); surrogate 8.6×10^3–6.9×10^4; Dirichlet/KMS exact. Balance 11 THEOREM / 1 CERT-UNIF / 4 CERT-WINDOW / 6 MEASURED; 6 new PENDING candidates. **The honest bottom line: exactly ONE genuine open object remains — $\inf_m g_{16}(m) > 0$, a quantifier over a certified flat list; no missing term, no missing device, no sector.** T166 (`schur_cascade_probe.py`, contract `SCHUR.CASCADE`) is running at exactly that quantifier. NOT RH evidence; no marker of any pre-existing contract moves.

- **Docs + website sync.** `tfpt_prime_front` gains the T165 subsection (the P_{pr} identity as the key display, the exact price decomposition, the R-F $\iff$ R-E-A equivalence, the decoupled surface with the zone-depth answer and the retired U_{ref} reported honestly) and the After-T165 keybox (the one cascade-gain quantifier, T166 running), the `v556` subsection in the verified layer (now twenty-two modules), the phase-2 section retitled T126–T165, the T165 row in the probe index and the updated counters (165 probes, 4325 checks); `tfpt_1`, `tfpt_3` and `tfpt_research_contracts` gain the `v556` companion blocks; `experiments/next.txt` gains the Teil-165 entry; `website/lib/primeFront.ts` gains the T165 and `v556` feed entries; the prime-front status surfaces move to Teile 11–165 / 4325 checks / 22 modules (`v535`–`v556`); suite counter 549 → 550 scripts. Ledger row `PRIME.GAUGE.PPR.01` **[E]**; **no marker of any pre-existing contract moves**.

2026-07-29 (XI) (prime-front diary T164 (both arms of the successor decided: the $O(1)$ gate discharges to a single quantifier, sectors are a gauge — TOLERANCE-CARRIES (arm A) / SECTOR-RESISTS (arm B)) + paper + website sync; NO promotion, NO marker moves)

- **Prime-front diary T164 (`sector_change_probe.py`, contract `SECTOR.CHANGE`, 28/28, 31.7 s, the T163 surface rebuilt and re-verified: 28 log-spaced prime-power zones $h = 142 \ldots 1445$ with declared caps, 27 windows inside the cond horizon, $|\psi(x) - x| \leq \kappa x$ with $\kappa = 0.038821$ verified at every jump point in $[100, 4 \times 10^5]$; verdict TOLERANCE-CARRIES (arm A) / SECTOR-RESISTS (arm B), sandbox).** Both arms of T163's successor R-E are decided. *Arm A (the tolerance analysis):* the station chain is instrumented with a power at every station — the T158 ladder really is a chain of upper bounds on the entry ceiling ($s \geq g_{16} \geq g_1$ window by window), and at the heart the T156 kernel spends the ceiling at power *exactly one*: $d\log(1-F)/d\log U = -1.0005 \ldots - 1.0000$ with $d\log r/d\log U = +1.0000$ exactly, so a ceiling growing like h^{ε} costs a margin decaying like $h^{-\varepsilon}$ for every $\varepsilon > 0$ — the declared one-sided rule returns $\varepsilon^* = 0.50$ and the probe *disowns* it at once as the $\varepsilon = 0$ tailwind ($h^{+0.391}$, bought from the measured drift of L) divided by power one: no structural slack. But the sharp question is answered the other way: the single constant $U_{\mathrm{ref}} = 4.9008 = \max_h 1/g_{16}$ carries all 9 sub-surface windows out of sample, $1/g_{16} = 1.7527$–5.3286 is flat ($h^{+0.061}$ over a 9× lever arm, split halves $-0.010/+0.103$), and every g_K is strictly increasing (Schur 1917) — **the $O(1)$ gate is not open as a *quantity*: it is discharged window by window by a Cholesky identity**, and what remains is exactly one quantifier, $\sup_m 1/g_{16}(m) < \infty$ — the m-freedom of a certified flat list, the same grammatical form as every other open uniformity item and no longer an existence question about trial vectors. *Arm B (the sector change), negative by a theorem:* $x_1 = 1$ fixes only the *scale* of a; Q and TV are homogeneous of degree two, hence Q/TV and δ_{bnd} are the *same numbers in every sector* — the entry normalisation is a **gauge** (machine-checked to 8.8×10^{-10} over 5670 sector-by-vector combinations, declared horizon 2.6×10^{-8}); the full space is strictly worse ($\mu_0 = 0$ exactly, no finite floor); a shifted or weighted sector does flatten the floor ($h^{+0.000}$ at $\kappa_s = 4$) but pays the transfer factor $1/\theta \sim h^{+1.997}$, and floor × transfer $= 1/\mu_1^P$ *identically* — the two exponents sum to +1.997 at every shift on the declared grid to 10^{-9}: the h^2 lives in the ratio, never in the floor. The genuine surprise, stated and immediately priced: a normalisation-free first-order ascent on Q/TV over the full window reaches 2.38–74.57 against the crossing bar $2\kappa = 0.0776$ — 31–960× over on 9/9 windows ($\delta_{\text{bnd}} = -0.124 \ldots -0.090$, alignment efficiency $\eta = 7.9$–11.1% of the Abel ceiling $\|C\|_\infty = 30.3$–$930.6 \sim h^{+1.185}$) — and T163's TV floor *holds on the unconstrained optimiser* (TV $\mu_1^P = 8.30$–$11.72 \geq 1$, a family T163 never searched), at a price $h^{+3.299}$, worse than T163's crossing price $h^{+1.91}$: dropping the normalisation relocated the difficulty without reducing it.

- **R-B‴ narrowed, R-D settled in type, R-F named.** R-B‴ is a *naming* problem — two different quantities carried one name: the T162 a-weighted quarter-bar object is 0.1001–0.1623 $\sim h^{-0.172}$ (T163's −0.172 reproduced to 0.0004, under the 0.25 bar on all 27 windows with 35% margin), the raw arch/atom cancellation is a different number ($h^{-2.04}$); a two-variable regression puts −0.046 of the drift on $\log h$ against −0.477 on $\log \alpha$ (10× more on the zone scale), and on this recipe α and h are coupled by construction — narrowed, not closed: a surface with independent α and h is needed. R-D is settled in *type*: under the shift $s \to s/\theta$, $\hat{a} \to \theta\hat{a}$, $L \to \theta L$, so P_K, r and $F(P_K, r)$ are unchanged to 3.5×10^{-16} on 54 combinations — every object the chain consumes downstream is a ratio of equally-homogeneous forms, hence no fifth device can come from a sector change ($\rho = 1.0036$–$1.0140 > 1$ flat, T161, quoted and untouched). The one genuinely new item is **R-F**: is there a v with $Q(v)/\text{TV}(v) > 2\kappa$ *and* $Q(v)$ comparable to $1/s$ — equivalently, can the alignment efficiency η stay above $2\kappa/\|C\|_\infty$ at bounded price? No sector, no normalisation, no prime. Stress: the uniform surrogate breaks by 1.5×10^3–4.9×10^7, the T145 no-go staircase is monotone ($1.0 \to 9.1 \to 25.4 \to 129 \to 434$, exact null control at $\nu = 4$, damage ≥ 26.1 at the coarsest setting on every window), the gauge break is 3.4×10^{-2}, Dirichlet/parity/orthonormality controls $\leq 2 \times 10^{-14}$, and the random-v pairing identity holds to 3.8×10^{-16}. Balance 11 THEOREM / 1 CERT-UNIF / 4 CERT-WINDOW / 6 MEASURED, zero load-bearing fit rows; candidates T164-Q1…Q9 all PENDING (no promotion — to be bundled with T165). NOT RH evidence; no marker moves.
- **Docs + website sync.** `tfpt_prime_front` gains the T164 subsection (the power-one measurement, the quantifier collapse as the key result, the gauge theorem with the product identity, the unconstrained ascent with the TV-floor hold, the R-B‴ narrowing with the α/h-coupling lesson) and the After-T164 keybox (R-E-A/R-F/R-B‴/R-D, T165 running), the phase-2 section retitled T126–T164, the T164 row in the probe index and the updated counters (164 probes, 4295 checks); `experiments/next.txt` gains the Teil-164 entry; `website/lib/primeFront.ts` gains the T164 feed entry; the prime-front status surfaces move to Teile 11–164 / 4295 checks. T165 (`alignment_eta_probe.py`, contract `ALIGNMENT.ETA`) is running at the alignment successor R-F. No ledger row (sandbox); **no marker of any pre-existing contract moves**.

2026-07-29 (X) (v555 — the Pareto/total-variation identities of phase 2 promoted; prime-front diary T163 (the front resists as a theorem — the h^2 is the parity spectral gap, not the primes, FRONT-RESISTS) + paper + website sync; NO marker moves)

- **New module v555 (PRIME.PARETO.TV.01).** Promotion of the closed theorem cores of the discovery parts T162 (`third_split_probe.py`) and T163 (`pareto_front_probe.py`) — 23 checks, ∼0.4 s, everything recomputed on `v551`–`v554`'s declared surface (the 12 deepest prime-power zones admitting a frame-A window inside the cap, $n = 29 \ldots 139$, $m = 96 \ldots 291 \leq 300$), with the one arithmetic input $\kappa = 0.038821$ *verified* at every jump point of $\psi(x)/x$ on the von-Mangoldt table in $[100, 3.2 \times 10^5]$ (Chebyshev 1852 / Rosser–Schoenfeld 1962, reproduced to 4.2×10^{-7}), nothing cited from the sandbox; the companion to `v554`, which certified what the sixteen-form *pairs against* — this module certifies the *price structure* of that pairing. Scope fence: per-instance identities and certified inequalities with the direction in the name — no fit, no D-exponent, no zone-uniform statement; the T163 h-exponents ($P_{\text{aff}} \sim h^{3.05}$, $P_{\text{cross}} \sim h^{1.91}$, the TV exponent $\sim h^{2.00}$) are sandbox fits and are *not* consumed; finite Λ-sums are allowed and used, *no RH statement* is made ($\frac{1}{2}$ appears in exactly one role: as the strength of a classical statement a required depth is compared against); the open terms after T163 (R-E the successor with both arms prime-free; R-B‴ the h-uniform positivity margin; R-D a fifth R1″ device) stay open and typed open; each statement carries at least one mutation control that must fail loudly.

- **(1) The exchange law and the crossing criterion (T163, R1.2)** [E]. $\delta_{\mathrm{bnd}}(x) = \frac{1}{2} + \log(2\kappa\, g_{16}\,\mathrm{TV}(x)/P(x))/\log X$ is an *identity* at all 612 grid points of the declared Fejér knob (50 geometric σ on $[1, 4096]$ plus ∞, 12 windows; 5.1×10^{-16} relative, admissibility $Q(x) > 0$ everywhere), and the crossing criterion $\delta_{\mathrm{bnd}} < \frac{1}{2} \iff P > 2\kappa g_{16}\mathrm{TV}$ holds as an exact equivalence at every point — price and demand are two coordinates of *one* object. Mutation: the wrong constant moves the law by 6.5×10^{-2}–3.4.
- **(2) The total-variation floor (T163, R2.2)** [E]. (T1) $w_0(x) = \|a(x)\|^2$ (4.9×10^{-16}); (T2) $\mathrm{TV}(x) \geq |w_0(x)|$ by telescoping with $w_M = 0$; (T3) $x_1 = 1$ forces $\|a\|^2 \geq 1/\mu_1^P$, hence $\mu_1^P\,\mathrm{TV}(x) = 5.00$–$28.84 \geq 1$ on all 708 trial vectors built, against the closed floor $1/\mu_1^P = 1/(4\sin^2(\pi/N)) = 943.6$–$8609.6 \sim h^2$; (T4) every sub-$\frac{1}{2}$ grid point pays $P > 2\kappa g_{16}/\mu_1^P = 15.4$–$142.8$ (all 54 respect it) — a flat-price sub-$\frac{1}{2}$ demand is impossible in the parity sector on this surface: **R-C‴ is closed negatively, by an inequality**. Negative control: the *inadmissible* vector $0.01x^*$ undercuts the floor (2.5–2.9×10^{-3}) and is recognised — the floor is a property of the normalisation meeting the smallest parity eigenvalue (Kac–Murdock–Szegő 1953).
- **(3) The chain-derived price axis (T163, R1.0)** [E]. $\sigma = 1$ gives $x = e_1$ exactly with $P = g_{16}/g_1$ (the $K = 1$ rung of the T158 Cholesky ladder, the T157 route-(0) certificate; 4.1×10^{-16}), $\sigma = \infty$ gives $g_{16}Q(x^*) = 1$ (the $K = 16$ top rung; 1.2×10^{-12}); the ladder carries all 16 terms strictly positive with monotone partial sums, the knob is monotone in both coordinates on every window (the knob curve *is* the Pareto front). Mutation: a corrupted sixteen-block makes the Cholesky *refuse* on 12/12.
- **(4) The Mellin cell moments and the Abel step (T162, C2/C5)** [E]. The k-th Mellin rung $\sum_d w_d C_d(-(2k + \frac{1}{2}))$ with $C_d(s)$ the closed tent integral (Mellin 1896) agrees with an independent per-cell Gauss–Legendre quadrature to 2.8×10^{-13} on every rung $k = 1 \ldots 4$ (shifted tent centres break by $\geq 1.3 \times 10^{-2}$); the boundary-free Abel identities hold at levels $k = 1 \ldots 3$ (8.3×10^{-17}; Abel 1826), the dropped-boundary level-1 form agrees because the gauge identity kills the boundary term exactly, the optimal level is *one* for T162's closed reason ($32\pi/\alpha = 40.6$–$59.4 > 1$, measured ratio 30.8–69.1), and a gauge violation breaks by its closed prediction v_0c_0M to 1.2×10^{-15} (the positive control).
- **(5) The Lerch/Frullani log-moment (T162, R-A′)** [E]. $L = \sum_d \Psi_d w_d$ agrees on three routes per window: direct; double-Abel *with* the Wronskian boundary term (2.8×10^{-14}; dropping it breaks by 2.2×10^{-4}–1.3×10^{-2}); the Lerch/Frullani integral with the $d = 1$ term peeled closed (1.8×10^{-15}) — the peel constant $2\log 2$ reproduced to 2.2×10^{-16} and $\Psi_1 = -\log 2$ exactly (Frullani 1828 / Lerch 1887 / Clausen 1832). Plus the monotone no-go staircase (damage $1.00 \to 3.49 \to 10.81 \to 24.98$ over $\nu = 4/2/1/0.5$, exact null control) and exact Dirichlet/parity controls.
- **Prime-front diary T163 (`pareto_front_probe.py`, contract `PARETO.FRONT`, 32/32, 1.9 s, 28 log-spaced prime-power zones $h = 142 \ldots 1445$ with declared caps, 27 windows inside the cond horizon, verdict FRONT-RESISTS, sandbox).** The survey of the Pareto front, and the front resists *as a theorem*: the price axis is chain-derived ($P_{\mathrm{aff}} = g_{16}B_{11} \sim h^{+3.049}$, the T157 route-(0) certificate), the Fejér knob has both endpoints in the chain and is measured monotone in both coordinates (the knob curve *is* the front), no flat price crosses the yardstick (0/27 at caps 1.25/2/10 and every flat rung tier; only the full P_{aff} reaches 27/27), the crossing price $P_{\mathrm{cross}} \sim h^{+1.906}$ sits at 0.0066–0.2047 of P_{aff} — 5–152× strictly inside the already-accepted certificate, margin widening — and the price anatomy is an *identity*: $\delta_{\mathrm{bnd}} = \frac{1}{2} + \log(2\kappa g_{16}\mathrm{TV}/P)/\log X$ exactly at all 27×50 grid points, the crossing exponent decomposing additively (TV $+2.013$, g_{16} -0.061). The four-line theorem (T1–T4) gives every

admissible trial vector the total-variation floor TV $\geq 1/\mu_1^P \sim h^2$ (verified on all 1674 built, slack 5.0–23.2), so the crossing price is bounded below by a quantity growing like h^2: the $h^{2.86}$ of T162, the $h^{1.91}$ of the crossing and T161's granularity are *all* the reciprocal smallest parity eigenvalue — **the spectral gap of the parity Laplacian meeting the entry normalisation $x_1 = 1$: it was never in the primes**. The mode sweep $K = 16 \to 64$ confirms it from the other side (price dividend, worse demand, TV exponent unmoved at $h^{+1.81}$); R-B″ reduces to an h-uniformity question (the chain never needed the sign: duality plus an evaluated number, headroom $\geq 10^{14.9}$); R-D is a type mismatch (operator smoothing degrades the floor by the same h^2). Balance 11/1/4/10, zero fit rows. T164 (`sector_change_probe.py`), running, tests the successor R-E. NOT RH evidence; no marker moves.

- **Docs + website sync.** `tfpt_prime_front` gains the T163 subsection with the exchange-law display, the T1–T4 floor theorem as the key result, the chain-derived price axis, the monotone no-go staircase and the After-T163 keybox (R-E/R-B‴/R-D, T164 running), the `v555` subsection in the verified layer (now twenty-one modules), the phase-2 section retitled T126–T163, the T163 row in the probe index and the updated counters (163 probes, 4267 checks); `tfpt_1`, `tfpt_3` and `tfpt_research_contracts` gain the `v555` companion blocks; `experiments/next.txt` gains the Teil-163 entry; `website/lib/primeFront.ts` gains the T163 and `v555` feed entries; the prime-front status surfaces move to Teile 11–163 / 4267 checks / 21 modules (`v535`–`v555`); suite counter 548 → 549 scripts. T164 (`sector_change_probe.py`) is running at the sector-change successor. Ledger row `PRIME.PARETO.TV.01` [E]; **no marker of any pre-existing contract moves**.

2026-07-29 (IX) (`v554` — the sampling/harmonics identities of phase 2 promoted; prime-front diary T161 (the circularity triage: the hardness sits in the split, not in the primes, CLOSURE-RESISTS) + paper + website sync; NO marker moves)

- **New module `v554` (`PRIME.SAMPLING.HARM.01`).** Promotion of the closed theorem cores of the discovery parts T160 (`pairing_probe.py`) and T161 (`classical_closure_probe.py`) — 21 checks, ~0.5 s, everything recomputed on `v551`–`v553`'s declared surface (the 12 deepest prime-power zones admitting a frame-A window inside the cap, $n = 29 \ldots 139$, $m = 96 \ldots 291 \leq 300$) plus a declared Λ-battery of six log-spaced cut-offs $X = 10^3 \ldots 3.2 \times 10^5$ for the PNT item (finite von-Mangoldt sums only, no matrix anywhere), nothing cited from the sandbox; the companion to `v553`, which certified the algebra of the sixteen-form — this module certifies *what that form pairs against*, and it closes no open term. Scope fence: per-instance identities and certified inequalities with the direction in the name — no fit, no D-exponent, no zone-uniform statement; finite Λ-sums are allowed and used, *no RH statement* is made, and the δ triage of T161 (required cancellation depth against RH strength) is a documentation item, *not* a module claim; the open terms after T161 (R-A′ the prime-free log-moment; R-B′ the m-free 16×16 Gram fraction bound; R-C′ one split of the pairing with $\delta_{\text{eff}} < \frac{1}{2}$; R-D a fifth R1″ device) stay open and typed open; each statement carries at least one mutation control that must fail loudly.
- **(1) The sampling identity (T160, P2) [E].** The atom half of the pairing equals $-\frac{1}{2}\sum_{n \leq X}(2\Lambda(n)/\sqrt{n})\,\widehat{w}(\log n)$ (plus one reflected term below the grid; $\widehat{w}$ the linear interpolant of the lag weights), exact to 5.6×10^{-15} of the absolute atom scale on 12/12 windows: identically a finite combination of sums $\sum_n \Lambda(n) n^{-1/2} \cos(t \log n)$ at 32 explicit frequencies. Mutation: a one-index grid shift breaks by 5.1×10^{-5}–3.1×10^{-3}.
- **(2) The moment laws and the TV theorem (T160, P1/P4) [E].** $m_0 = 0$ (1.0×10^{-16} of $\|w\|_1$), $m_1 = -[S_0^2 + 2\sum_j P_j^2]$ (9.2×10^{-16}) and $m_2 = -[2S_1 - (M-1)S_0]^2$ (3.4×10^{-14}, a perfect square), both strictly negative on every window; U3 as a theorem with *checked* hypotheses ($c^{\text{arch}} \leq 0$ and monotone on $d \geq 1$, 12/12): $\text{TV}(c^{\text{arch}}) = 3.9817$–$4.8991 \leq |c_0| + 2\sup < 6$, closed and window-independent. Mutations: a one-index prefix shift breaks m_1 by $\geq 1.25 \times 10^{-2}$, an

$(M-2)$ mis-normalisation breaks m_2 by $\geq 6.1 \times 10^{-2}$, the full atom-including kernel violates the monotone hypothesis on 12/12.

- **(3) The harmonic-frequency theorem and the PNT main term (T161, P3) [E].** $t_j(2\alpha) = 2\pi j$ *exactly* (2.8×10^{-14}) on every window and every battery cut-off — the 32 frequencies of the sampling identity are precisely the Fourier harmonics of the log-window $[0, 2\alpha]$; at those frequencies $\theta = 2\pi j$ collapses the partial-summation main term to the closed, parameter-free prediction $(\sqrt{X} - 1)/(\frac{1}{4} + t^2)$ (de la Vallée Poussin 1896; the Mellin factor of $x^{-1/2}$), matching the finite sums to 0.0007–0.097 of the trivial mass against the declared cap 0.12, the subtraction reducing the aggregate residual to $0.008\text{–}0.466 \leq 0.5$ on every cut-off. Mutations: a wrong Mellin pole worsens the aggregate residual by $\times 1.8$–$\times 95.8$; a mass-matched *uniform* fake measure misses by 0.23–0.51 — the agreement is a property of the von-Mangoldt measure, not of any measure with the same mass.

- **(4) The head split and the scale-free Bernstein rate (T161, P1) [E].** $c_d^{\text{arch}} = \Psi_d + D\widehat{G}(dD, D)$ with $\Psi_d = -\frac{1}{2}[(d{+}1)\log(d{+}1) - 2d\log d + (d{-}1)\log(d{-}1)]$ closed, D-free and m-free (the triangle-smeared simple pole; the $\log D$ cancels identically), exact to 3.2×10^{-16} of $\sup|c^{\text{arch}}|$ at *every* lag of every window, the two Ψ representations agreeing to 2.9×10^{-14}; and $\rho^* = (3 + \sqrt{5})/2 = 2.618034$: the root of $x^2 - 3x + 1$, returned exactly by the $D \to 0$ limit of the peeled interval $[0.4\alpha, 2\alpha]$ and invariant under $\alpha \times 10$ — scale-free, every window's $\rho(D) = 2.588\text{–}2.608$ below it (Bernstein 1912). Mutations: dropping Ψ misses c^{arch} by 0.99–1.00; the wrong interval ratio moves the limit to 3.0.

- **(5) The off-diagonal fraction bound and the sign refutation (T161, P2) [E].** The gauge identity licenses one boundary-free Abel step, $Q^{\text{arch}} = \sum_{k,l} a_k a_l \sum_{d\geq 1} (\Delta c^{\text{arch}})_d R^1_{kl}(d)$ exactly (2.0×10^{-13}), and the certified per-window inequality is $|\text{off-diagonal block}| \leq \frac{1}{4}|Q^{\text{arch}}|$ (measured 0.171–0.212 on 12/12) — with the *refuted* sign law wired as the negative control: the arch half is negative while its off-diagonal block is positive on 12/12, and 176/240 off-diagonal pairs violate the per-pair form on every window — nothing single-signed is promoted. Mutation: a one-index R-kernel shift breaks by $\geq 6.8 \times 10^{-5}$. Stress: the head split misses the T145 no-go reconstruction by 2.39; the Dirichlet cosine-sum identity (including the degenerate branch), the KMS eigenpairs and odd_toeplitz$(c^L) = L_P$ (zero tolerance) are exact.

- **Prime-front diary T161 (`classical_closure_probe.py`, contract `CLASSICAL.CLOSURE`, 35/35, 1.2 s, 19 log-spaced prime-power zones $h = 50 \ldots 1445$, verdict CLOSURE-RESISTS, triage verdict NEEDS-BEYOND-RH-STRENGTH, sandbox).** The two classical rests plus the circularity triage, and the triage returns its most consequential answer: *the chain is not circular*. R-A closes m-free — $A = D\widehat{A}$ exactly, only the $1/s$ head binds, the scale-free Bernstein rate $\rho^* = (3 + \sqrt{5})/2$, the closed head split (no peeling), and an explicit degree schedule $K(h) = O(\log h)$ (a *fixed* degree is refuted with a number; the residual is the prime-free log-moment, outside the polynomial ladder). R-B is refuted in all four readings (3168/4320 pairs; the aggregate by *sign*) and replaced by the certified fraction bound $0.1035\text{–}0.1713 \leq \frac{1}{4}$, flat. The triage: the 32 frequencies are exactly the Fourier harmonics of the log-window (theorem), the measured cancellation is fully the PNT main term, and the required depth is $\delta = 1.1482\text{–}1.8809$ against RH strength $\frac{1}{2}$ — in absolute terms 0.012–0.31 of the last term of the sum, below the boundary term of every partial summation, so no strengthening of the $\psi(x) - x$ input reaches it: the chain needs *more* than RH-strength input would supply, which localises the h^2 in the *split* rather than in the primes; a re-split moves the demand to $\delta_{\text{eff}} = 0.9839\text{–}1.3767$ — it moves, but stays above $\frac{1}{2}$ on 18/18. Balance 20/10/4/9/3. T162 (`third_split_probe.py`), running, searches for one split with $\delta_{\text{eff}} < \frac{1}{2}$. NOT RH evidence; no marker moves.

- **Docs + website sync.** `tfpt_prime_front` gains the T161 subsection with the δ table, the head-split display and the After-T161 keybox, the `v554` subsection in the verified layer (now twenty modules), the phase-2 section retitled T126–T161, the T161 row in the probe index and the updated counters (161 probes, 4205 checks); the big-picture section states the split-design reading soberly; `tfpt_1`, `tfpt_3` and `tfpt_research_contracts` gain the `v554` companion blocks; `experiments/next.txt` gains the Teil-161 entry; `website/lib/primeFront.ts` gains the T161 and `v554` feed entries; the prime-front status surfaces move to Teile 11–161 / 4205 checks / 20 modules (`v535`–`v554`); suite counter 547 → 548 scripts. T162 (`third_split_probe.py`) is running at the third-split search. Ledger row `PRIME.SAMPLING.HARM.01` [E]; **no marker of any pre-existing contract moves**.

2026-07-29 (VIII) (v553 — the exact-form identities of phase 2 promoted; prime-front diary T159 (the gauge identity and the Z-matrix law, cancellation is the weight, FORM-RESISTS) + paper + website sync; NO marker moves)

- **New module v553 (PRIME.EXACT.FORM.IDENT.01).** Promotion of the theorem cores of the discovery parts T158 (`schur_entry_probe.py`) and T159 (`exact_form_probe.py`) — 25 checks, $\sim$0.2 s, everything recomputed on `v551`/`v552`'s declared surface (the 12 deepest prime-power zones admitting a frame-A window inside the cap, $n = 29 \dots 139$, $m = 96 \dots 291 \leq 300$; no-go ladders $m = 48 \dots 288$ on *both* T145 forms; parity and Dirichlet controls at $m = 64, 128, 256$), nothing cited from the sandbox; the companion to `v552`, which certified the four angle instruments — this module certifies the *algebra of the sixteen-form itself*, and it closes neither open term. Scope fence: per-instance identities and certified inequalities with the direction in the name — no fit, no D-exponent, no zone-uniform statement, and explicitly *no statement for all D* (the two open terms after T159 — the one explicit pairing inequality for R2″, needing a correlation bound instead of sizes; a third device for the atom off-band block for R1″ — stay open and typed open); each statement carries at least one mutation control that must fail loudly.
- **(1) The Thomson dual form and the positive Cholesky ladder (T158, P1) [E].** $s = (B^{-1})_{11} = \max_x(2x_1 - x^\mathsf{T} Bx)$ (Maz'ya 1985 / Miclo 1999) with the direction *executed*, not asserted: 8 random trial vectors per window never exceed s, so every trial is an upper bound on $1/s$ — exactly the variational structure Cauchy–Schwarz lacks (`v552` wired that refutation as a control; here the direction itself is the check). The ladder $g_K = \sum_{j \leq K} y_j^2$, $y = L^{-1}e_1$ from *one* Cholesky of the leading 16×16 block: all terms strictly positive, partial sums strictly monotone, $1/s \leq 1/g_{16} \leq 1/g_1 = B_{11}$ literally on 12/12, tight to 1.0642–1.1522 against the declared cap 1.5 (T158 measured ≤ 1.2738 at $h = 254 \dots 1393$, sandbox); mutation: a 5% corruption of one off-diagonal entry destroys positive definiteness — the Cholesky *refuses* on 12/12.
- **(2) The two-dimensionality (T158, P2) [E].** $\mathrm{span}\{t_1, A^{-1}L_P t_1\}$ attains s exactly ($g/s = 1$ to 9.0×10^{-13} on 12/12) for the theorem reason $L_P t_1 = \mu_1^P t_1$ (KMS 1953, eigenpair residual 8.1×10^{-13}): the entry is a two-dimensional quantity the moment one Green column is granted. Negative controls: the two-sine span (no Green column) misses by 1.42–4.8, the corrupted Green column (no solve) by 255–7.7×10^3.
- **(3) The gauge identity and the TmH signature (T159, P4/P7) [E].** The reflection-odd section $A_{rs} = c_{|r-s|} - c_{M-1-r-s}$ annihilates constant lag vectors *entrywise with zero tolerance* (odd_toeplitz($\mathbf{1}$) = 0 exactly — the mechanism in one line), hence the gauge identity $\sum_d w_d = 0$ holds to 1.1×10^{-14} of $\sup |w|$ for the per-window optimiser and 3 random sixteen-vectors: the form is *blind* to the lag mass of c, and the lag mass is exactly where the archimedean and atom halves of the sixteen-form are h^2-sized and cancel. The premise — the Toeplitz-minus-Hankel signature $A_{r,s} - A_{r+1,s+1} = f(r+s)$ — holds to 1.4×10^{-16}. Mutations: a one-index Hankel shift breaks the identity by 2.7×10^{-7}–4.2×10^{-5}; the T145 form $R = aa^\mathsf{T} + \varepsilon I$ has signature defect 7.4×10^{-2} — not a parity section, the gauge identity is not even *defined* for it.
- **(4) The exact lag sum and the two closed scalars (T159, P2/P5) [E].** $x^\mathsf{T} B_{LL} x = \sum_d c_d w_d$ with $w_d = T_d - H_d$ against the direct assembly to 2.7×10^{-16} of the absolute scale $\sum_d |c_d w_d|$ — and the honest price *measured*: relative to the cancelled total the same identity holds only

to 1.4×10^{-11} on this small surface (the sandbox at $h \le 1293$ loses 4.0–8.1 digits; that depth *is* the h^2 obstruction). The two closed scalars: $w_0 = \sum_k x_k^2/\mu_k^P$ (the $N^2/4\pi^2$ factor itself) to 4.2×10^{-16} and $2w_0 - w_1 = \|x\|^2$ to 7.4×10^{-16} (KMS 1953). Mutations: a 1% perturbation of the largest lag breaks the lag sum by 2.2×10^{-3}; a one-index μ shift breaks w_0 by 0.60.

- **(5) The Z-matrix law and the Collatz–Wielandt floor (T159, U1/U2) [E].** B_{HH}^{arch} is a symmetric Z-matrix *raw* — every off-diagonal entry ≤ 0 with no tolerance on 12/12; the closed sign-based floor $\min_k(B_{HH}^{\mathrm{arch}}\mathbf{1})_k = 0.9093$–$1.1284 \ge t = 0.25$ with the theorem direction verified per instance against the exact $\lambda_{\min}(B_{HH}^{\mathrm{arch}}) = 0.9138$–$1.1432$ (Perron 1907 / Frobenius 1912 / Collatz 1942 / Wielandt 1950), beating the sign-free norm route by 0.69–8.8 on every window. The honest limit as content: the atom block obeys *no* sign law (0.44–0.53, a coin toss), the full block is not a Z-matrix and its row-sum floor is *negative* ($-48.3 \ldots -0.79$) against a positive exact bottom — admissible and vacuous, R1$''$ stays open. Mutation: the checkerboard similarity $S = \mathrm{diag}((-1)^k)$ — an isometry, $\lambda_{\min}$ moves by 0 exactly — destroys the raw law (fraction $1.0000 \to 0.494$–0.498): the Z law is an entrywise fact, not a spectral one, and T159's checkerboard reading (S2) is refuted. Stress: the T145 no-go breaks on three axes of this module's structure; on the $c(l) = 1/(1+l)$ reconstruction the ladder theorem survives while $1/g_{16}$ explodes ($1187 \to 7.11 \times 10^4$) — the instrument reports the collapse instead of hiding it; parity: for $A = L_P$ the machinery returns $s = 1$ with every ladder partial sum constant in K; Dirichlet controls exact.
- **Prime-front diary T159 (`exact_form_probe.py`, contract `EXACT.FORM`, 41/41, 2.9 s, 23 of 24 prime-power zones $h = 142 \ldots 1293$, the sign surface on 18 windows $h = 142 \ldots 963$, verdict FORM-RESISTS, sandbox).** The cancellation executed algebraically: seven machine-checked identities to 10^{-12} against the absolute scale (the cancellation eats 4.0–8.1 decimal digits, reported as a measurement), among them the closed 256-term Dirichlet-kernel form of the geometric weights and the *gauge identity* $\sum_d w_d = 0$ exactly — the form is blind to the lag mass, exactly where the h^2 halves live. The honest core answer is negative: the h^2 does *not* telescope away, it is the common *weight* of both halves (both grow $h^{+2.104}$), and four routes are closed with exponents — T158's own fixed sixteen-vector refuted at $h^{+3.510}$, one sine at $h^{+3.066}$, the preregistered ansatz family at $h^{+3.001}$, Abel-ℓ^1 × sup at every rung (best $h^{+3.604}$). The structural win: B_{HH}^{arch} is exactly a symmetric Z-matrix with a closed sign-based Collatz–Wielandt floor 0.819–1.165 comfortably above $t = 0.25$ — but the atom block obeys no sign law, and the full-block criterion is vacuous. Balance $6/3/4/1 \to 13$ theorem / 6 cert-unif / 4 cert-window / 1 measured; a numerical horizon declared as its own measurement ($\mathrm{cond}(B_{LL}) > 10^{12}$ from $h = 1292$); one fully explicit pairing inequality remains, under attack via correlation structure in T160 (`pairing_probe.py`), running. NOT RH evidence; no marker moves.
- **Docs + website sync.** `tfpt_prime_front` gains the T159 subsection with the After-T159 keybox, the `v553` subsection in the verified layer (now nineteen modules), the phase-2 section retitled T126–T159, the T159 row in the probe index and the updated counters (159 probes, 4124 checks); `tfpt_1`, `tfpt_3` and `tfpt_research_contracts` gain the `v553` companion blocks; `experiments/next.txt` gains the Teil-159 entry; `website/lib/primeFront.ts` gains the T159 and `v553` feed entries; the prime-front status surfaces move to Teile 11–159 / 4124 checks / 19 modules (`v535`–`v553`); suite counter 546 → 547 scripts. T160 (`pairing_probe.py`) is running at the pairing bound. Ledger row `PRIME.EXACT.FORM.IDENT.01` [E]; **no marker of any pre-existing contract moves**.

2026-07-29 (VII) (`v552` — the four fixed-size angle instruments of phase 2 promoted; prime-front diary T157 (the confinement theorem and the one-entry

reduction, ANGLES-RESIST) + paper + website sync; NO marker moves)

- **New module v552 (PRIME.ANGLE.INSTR.01).** Promotion of the instrument cores of the discovery parts T155 (`bottom_floor_probe.py`) and T157 (`angle_floor_probe.py`) — 18 checks, ∼0.4 s, everything recomputed on `v551`'s declared surface (the 12 deepest prime-power zones admitting a frame-A window inside the cap, $n = 29 \dots 139$, $m = 96 \dots 291 \le 300$; a no-go size ladder $m = 48 \dots 288$; parity and Dirichlet controls at $m = 64, 128, 256$), nothing cited from the sandbox; the companion to `v551`, which certified the fixed-size ceiling step — this module certifies the four fixed-size *instruments* the angle campaign produced, and none of them closes an angle. Scope fence: per-instance identities and certified inequalities with the direction in the name — no fit, no D-exponent, no zone-uniform statement, and explicitly *no statement for all* D (the two open terms after T157 — an m-free lower bound on $\hat{g}_1^2$, equivalently an m-free upper bound on the one Schur diagonal entry $1/s$; and an m-free R1$''$ domination with a non-shrinking margin — stay open and typed open); each statement carries at least one mutation control that must fail loudly.
- **(1) The complement-floor certificate (T155) [E].** $\min_{v \perp W} v^\top L_P v \ge \mu_{13}^P - \lambda_{\max}(M^{1/2}(I - GG^\top)M^{1/2})$ with $M = \mathrm{diag}(\mu_{13}^P - \mu_k^P)$ and $G = T_{12}Q_W$ — an identity plus *one* certified Rayleigh bound on the high block, a theorem for every m and every subspace, direction checked: the certificate never exceeds the exact size-m complement bottom (12/12) and agrees to $1 - \text{ratio} \le 1.01 \times 10^{-3}$ against the declared cap 2×10^{-3} on this small surface (T155 measured $\le 2.2 \times 10^{-7}$ at $h = 142 \dots 1293$, sandbox). Exactness controls both ways: $W = \mathrm{span}\{t_1..t_8\}$ returns μ_9^P and the *worthless* configuration $W = \mathrm{span}\{t_2..t_{13}\}$ returns μ_1^P, both to 2.0×10^{-11} — the instrument reports its own worthlessness; mutation: one corrupted overlap row moves the certificate by ≥ 0.549, only downward.
- **(2) The sine-block confinement (T157) [E].** From the certified pencil floor $A \succeq tL_P$ and the certified ladder ceiling $\lambda_1 \le S\mu_1^P$ *alone*, $\|\gamma_H\|^2 \le \lambda_1/(t\,\mu_{17}^P) \le (S/t)/\rho_{17} = 0.0153$–$0.0245 < 1$ on 12/12: the bottom eigenvector lives 97.5–98.5% inside the first 16 parity sines by a *per-instance theorem* (measured tail 6.7×10^{-6}–3.0×10^{-5}; T146's measured "98 per cent" replaced by one line). Negative control: the pencil ceiling κ in place of the ladder S makes the bound vacuous (17.6–477 > 1 on 12/12); mutation: the inflated floor κL_P is refuted at $e_1(A)$ by 3.3×10^3–1.0×10^5.
- **(3) The Schur-entry identity (T157) [E].** $s = \mu_1^P\, t_1^\top A^{-1} t_1 = (B^{-1})_{11}$ exactly (1.1×10^{-11}) and $(B^{-1})_{LL} S_L = I$ to 5.8×10^{-11} with $S_L = B_{LL} - B_{LH} B_{HH}^{-1} B_{HL}$ the *same* fixed-size 16×16 Schur complement the chain already forms: $1/s = 4.0460$–5.8962 is an $O(1)$ number carried by *one* entry, so T156's R2$''$ loss ceiling $r \le 1/(Ls)$ is a statement about one number. Negative control (T157's refutation wired as a check): the inversion-free Cauchy–Schwarz ceiling is valid and useless, off by 250–7.27×10^3 — the cancellation in $b^\top B_{HH}^{-1} b$ is nearly complete and Cauchy–Schwarz cannot see it; mutation: a one-index shift breaks the identity by 1.55–2.15.
- **(4) The adaptive Lipschitz ceiling (T157) [E].** Interval bisection with the *local* Lipschitz constant certifies $\inf f^{\mathrm{arch}}/(4\sin^2(\theta/2)) \ge 0.25$ on 12/12 windows in 519–1201 symbol evaluations (certified floor 0.2505–0.3062, never above the fine-grid minimum); the global constant is documented insufficient (a uniform grid would need up to 61.4× more points); mutation: fed a target above the true grid minimum the certificate *refuses* on 12/12. Stress: on the T145 no-go family the Schur floor survives ($t = 0.05$) while the confinement goes vacuous (×52.0) and $(S_L)_{11}$ explodes (×60.0) against the flat real band; parity control exact to 4.9×10^{-14}; the Dirichlet residual hits the predicted $2/\sqrt{N}$ to 7.4×10^{-5}.
- **Prime-front diary T157 (`angle_floor_probe.py`, contract `ANGLE.FLOOR`, 32/32, 3.9 s, 16 prime-power windows $h = 142 \dots 1293$, the alignment surface on 12 ($h = 50 \dots 1077$), verdict ANGLES-RESIST, sandbox).** Neither angle falls, but both change shape: the p_1 floor becomes a theorem times one measured fixed-size number — the sine-block confinement $\|\gamma_H\|^2 \le (S/t)/\rho_{17} = 0.0165$–$0.0293$ proves $e_1(A)$ lives 97.1–98.4% inside the first sixteen parity

sines, and the resolvent route pins $p_1 \geq \hat{g}_1^2(1 - \text{tail}) = 0.1968$–$0.3228$, 97.9% of the measurement (the classical Rayleigh angle bound is empty on 16/16, its block version empty at every J, the angle-free Cauchy–Schwarz route loses exactly the Kantorovich $P = 5.6 \times 10^2$–1.0×10^6); the structural gem $1/s = (S_L)_{11} = 2.3359$–$6.2049$ flat makes the whole first term a bound on *one* diagonal entry of the 16×16 Schur complement the chain already forms (Cauchy–Schwarz off by 5.4×10^2–5.2×10^5); the arch half is uniformly certified for the first time (executed adaptive Lipschitz ceiling, 12/12, cost $h^{0.85}$) with the two extremals at opposite ends of the band (the pointer: use the θ-growth, not the infimum at π); the alignment term stays a per-window domination (1.0003–1.0907, margin 7.3×10^{-4}, shrinking trend); balance 9 theorem / 2 cert-unif / 2 cert-window / 3 measured (all three now fixed-size in the statement); the no-go breaks on *five* axes. T158 (`schur_entry_probe.py`) is running at the cancellation-seeing bound. NOT RH evidence; no marker moves.

- **Docs + website sync.** `tfpt_prime_front` gains the T157 subsection with the After-T157 keybox, the `v552` subsection in the verified layer (now eighteen modules), the phase-2 section retitled T126–T157, the T157 row in the probe index and the updated counters (157 probes, 4047 checks); `tfpt_1`, `tfpt_3` and `tfpt_research_contracts` gain the `v552` companion blocks; `experiments/next.txt` gains the Teil-157 entry; `website/lib/primeFront.ts` gains the T157 and `v552` feed entries; the prime-front status surfaces move to Teile 11–157 / 4047 checks / 18 modules; suite counter 545 → 546 scripts. T158 (`schur_entry_probe.py`) is running at the cancellation-seeing bound. Ledger row `PRIME.ANGLE.INSTR.01` [E]; **no marker of any pre-existing contract moves**.

2026-07-29 (VI) (`v551` — the fixed-size Ritz ceiling certificate of phase 2 promoted; prime-front diary T154 (the ceiling closes exactly at fixed size, ALIGN-RESISTS) + paper + website sync; NO marker moves)

- **New module `v551` (`PRIME.RITZ.CEIL.01`).** Promotion of the certificate core of the discovery part T154 (`green_align_probe.py`) — 16 checks, ~0.4 s, everything recomputed on small frame-A windows, nothing cited from the sandbox; the companion to `v550`, which certified the bottom ladder $\lambda_k(A) \leq S\mu_k^P$ by per-window LDL^T counts of *size* m — this module certifies the *fixed-size* replacement T154 found for the ceiling step. Scope fence: per-instance theorems with checked hypotheses and per-window *certified* inequalities — no fit, no D-exponent, no zone-uniform statement, and explicitly *no statement for all* D (the two open inputs after T154 — R1′, an m-free floor for $\lambda_{\min}(B_{HH})$ on the bulk parity block, and R2′, an m-free lower bound for the L_P floor on the complement of the eight bottom Ritz directions — stay open and typed open; every floor consumed is a per-window certificate of size m). Battery: the 12 *deepest* prime-power zones admitting a frame-A window inside the cap ($n = 29 \ldots 139$, $m = 96 \ldots 291 \leq 300$, $D = 7.248 \times 10^{-3} \ldots 1.815 \times 10^{-2}$ — mirroring T154's own surface selection, declared rather than tuned: on the shallowest zones $n \leq 11$ more than one bottom direction leaves the sine block and the fixed-depth certificate is *not* claimed there), a no-go size ladder $m = 48 \ldots 288$, parity controls at $m = 64, 128, 256$, and 3 Dirichlet control windows; each statement carries a mutation control that must fail by $\geq 10^{-3}$.
- **(1) The direction lemma (T154)** [E]. For orthonormal Q and $W = Q^\mathsf{T}AQ$, $\lambda_k(A) \leq \theta_k$ for *every* $k \leq$ dim — Ritz values are *upper* bounds for the eigenvalues of the *same index* (Courant–Fischer 1920 / Cauchy 1829), verified against the exact spectrum on the certificate subspace of every window and on 24 random subspaces (one-sided excess 0.0) — so the fixed-size ceiling needs *no* residual argument at all; Temple 1928 / Kato 1949 is a floor device and is used nowhere in the module. Negative control: the *reversed* reading (Ritz values as lower bounds) is recognised as false on 12/12 windows, θ_k overshooting λ_k by a factor ≥ 18.6 somewhere on every window.
- **(2) The sixteen-column certificate (T154)** [E]. $\mathrm{span}\{t_1..t_8\} + A^{-1}L_P\,\mathrm{span}\{t_1..t_8\}$ has *fixed*

dimension 16, independent of m (one Green step by Cholesky back-solves, no inverse formed), and its ceiling $K^F = \max_k \theta_k/\mu_k^P$ (8 completed Choleskys of matrices $\le 8 \times 8$) equals 1.1019–1.5845 with $K^F \ge \max_k \lambda_k/\mu_k^P$ on every window *and* $|K^F/K_{\mathrm{bot}} - 1| \le 9.9 \times 10^{-7}$ against the declared cap 10^{-4} on 12/12, where K_{bot} is the inertia-certified true ladder constant (8 completed $\mathrm{LDL}^{\mathsf{T}}$ counts of size m per window; count = direct spectral count on all 96 pairs): the fixed-size ceiling *attains* the truth, and the size-m counts are retired from the ceiling step (controls: the parity sines alone overshoot by 41.6–739 — T152/T153's $O(m^2)$ overshoot reproduced at fixed size — and corrupting the Green columns breaks the attainment on 12/12).

- **(3) The one-direction anatomy (T154) [E] (angles measured).** The median of the seven aligned principal angles between the bottom eight of A and the bottom eight of L_P is 0.03–0.14° while the largest is 83.79–89.70° on every window (Björck–Golub 1973) — the misalignment is carried by *one* nearly orthogonal direction, and that single direction *is* the collapse price $\lambda_1(A)/(t\,\mu_1^P) = 4.408$–$7.042$, recovered in full per window by one completed Cholesky of $A - \gamma I$ (size m, certified, not fixed size, labelled as such); control: $A = L_P$ through the same instrument returns $K_{\mathrm{bot}} = K^F = 1$ and zero misalignment (deviations $< 10^{-5}$, the inertia ladder's backoff step) with the price exactly $1/t$.
- **(4) The no-go discriminator (T154) [E].** On the T145 no-go reconstruction $c(l) = 1/(1+l)$ the odd section is positive definite and the Schur floor survives ($t = 0.05$ on all 4 sizes) but K_{bot} explodes ($\times 35.4$) and the sixteen-column ceiling explodes *with* it ($\times 57.5$) over $m = 48 \ldots 288$ — the instrument does not report flatness where flatness is false and manufactures no closure; the Courant–Fischer direction is violated nowhere (16/16 positive-definite cases); the Dirichlet control (the Hankel half removed, arithmetic kept bit for bit) stops being positive definite — $\lambda_1 = -5.60 \ldots - 3.55$ certified on 3/3 — so no ceiling exists there: the reflection half is load-bearing for positivity.
- **Prime-front diary T154 (`green_align_probe.py`, contract `GREEN.ALIGN`, 29/29, 5.0 s, 12 prime-power windows $h = 50 \ldots 1077$, verdict ALIGN-RESISTS, sandbox).** The ceiling half of T153's R2 closes exactly at fixed size ($K^F = 1.1019$–1.9964 flat, agreement 5.17×10^{-7} on 12/12); the floor half is refuted at fixed size ($\|R\|^2/(\beta\lambda_1) = 6.3 \times 10^3$–$5.6 \times 10^9$, $x^{4.50}$) and recovered at full size (one Cholesky per window recovers the whole collapse price 4.408–7.985; per-window end to end 4.45×10^{-2}–3.13×10^{-1}, inside the target band, while the m-free-in-shape number stays 1.01×10^{-2}–3.92×10^{-2}); the obstruction is named (seven bottom directions agree to 0.15–1.35° median, *one* sits at 82.93–89.79°) and the missing fixed-size ingredient is priced (the L_P floor on the Ritz complement, arithmetic-free, flat 5.93–$8.25\,\mu_1^P$, worth 91–100% of the price); the Kato route to R1 fails by the same geometry (loss 1.97–121 attained 24.9–89.8° apart; the Hankel-reflection culprit refuted twice). Two open terms remain (R1′/R2′, both uniformity terms); T155 (`bottom_floor_probe.py`) is running at exactly those two floors. NOT RH evidence; no marker moves.
- **Docs + website sync.** `tfpt_prime_front` gains the T154 subsection with the After-T154 keybox, the `v551` entry in the verified layer (now seventeen modules), the phase-2 section retitled T126–T154, the T154 row in the probe index and the updated counters (154 probes, 3951 checks); `tfpt_1`, `tfpt_3` and `tfpt_research_contracts` gain the `v551` companion blocks; `experiments/next.txt` gains the Teil-154 entry; `website/lib/primeFront.ts` gains the T154 and `v551` feed entries; the prime-front status surfaces move to Teile 11–154 / 3951 checks / 17 modules; suite counter 544 → 545 scripts. T155 (`bottom_floor_probe.py`) is running at the two uniformity floors. Ledger row `PRIME.RITZ.CEIL.01` [E]; **no marker of any pre-existing contract moves**.

2026-07-29 (V) (`v550` — the odd-sector identities of phase 2 promoted; prime-front diary T151 (the one inequality closes by rerouting off the ladder, ODD-CARRIES)

+ paper + website sync; NO marker moves)

- **New module `v550` (`PRIME.ODD.SECTOR.IDENT.01`).** Promotion of the identity/certificate core of the discovery part T151 (`odd_ladder_probe.py`) — 19 checks, ∼0.2 s, everything recomputed on small frame-A windows, nothing cited from the sandbox; the companion to `v549`, which certified the parity compression and the quadratic ladder certificate $\nu_k^P \le Ck^2$ — this module certifies the *reroute* T151 found: off the ν ladder, through the pencil against the parity Laplacian and one elementary Sobolev step, to a per-mode bound *linear* in k. Scope fence: identities, per-instance theorems with checked hypotheses and per-window *certified* inequalities — no fit, no D-exponent, no zone-uniform statement, and explicitly *no statement for all D* (the one open input after T151 — the m-freeness of the bottom pencil ratio $R = K_{\rm bot}/\kappa$ for the odd parity sector of a sign-changing-symbol Toeplitz section — stays open and typed open). Battery: 12 frame-A windows ($n = 4\ldots139$, $m = 26\ldots285 \le 300$, $D = 7.578\times10^{-3}\ldots2.789\times10^{-2}$), parity control models at $m = 241$ (the certified $(1,1)$ pencil) and $m = 2001$ (the π price), and a no-go size ladder $m = 64\ldots256$; each statement carries a mutation control that must fail by $\ge 10^{-3}$.
- **(1) The grid step-over and the honest negative (T151) [E].** The parity sines live on the *shifted* grid $\theta_k = 2\pi k/N$, $N = 2m{+}1$, which never contains $\theta = 0$, with $\mu_k^P = 2 - 2\cos\theta_k$ *exactly* (2.2×10^{-16}; eigen-residual 2.3×10^{-13}); the symbol has $f(0) = -47.6\ldots-6.3 < 0$ (T148's honest negative, re-read and never re-derived) with a nonempty negative window $[0,\theta_c)$, and $\theta_c/\theta_1 = 0.371$–$0.485 \le 0.9$ on *every* window: the odd grid *steps over* the negative window — odd-sector positivity is a grid fact, stepped over rather than cancelled. The honest negative beside it, promoted as content (T150's rest item 3 answered *negatively*): the symbol-only pencil floor $\kappa_{\rm sym}(1-\rho)$ is vacuous ($\rho \ge 1$) on 12/12 windows and the symbol moves by 0.45–3.27 relative across *one* mode spacing — no pointwise symbol statement is admissible at the bottom mode's resolution: the local model is a *matrix* statement, not a symbol statement (controls: the Dirichlet corner breaks the eigenpairs at 1.0, a one-index shift misses by 6.5×10^{-3}).
- **(2) The certified bottom ladder (T151) [E].** $\lambda_k(A) \le S\mu_k^P$ for $k \le 8$ is certified per window by 8 completed LDL$^{\mathsf T}$ inertia counts (Sylvester 1852; Bunch–Kaufman 1977 — a certificate, never a sorted list), $S = 1.1019$–6.4725: the odd sector's bottom spectrum *is* the parity-Laplacian ladder up to a printed $O(1)$ factor — order 2 in k, with no trace of $f(0) < 0$ in it (the Rayleigh floor $L_P \ge \mu_1^P I$ absorbs it); the count equals the direct spectral count on all 96 (window, k) pairs (control: $S\times10^{-3}$ certifies no eigenvalue below it anywhere).
- **(3) The discrete Sobolev step and the π price (T151) [E].** $\|\nabla\psi\|_2^2 = \langle\psi, L_0\psi\rangle - \psi_{\rm last}^2$ exactly (7.6×10^{-15}), $\|\nabla\psi\|_2^2 \le \langle\psi, L_P\psi\rangle$, and $\|\psi\|_\infty^2 \le 2\|\psi\|_2\|\nabla\psi\|_2$ on the true window modes, the parity sines and random unit vectors — elementary, licensed by the odd sector's virtual node $\psi_{-1} = 0$ (Hardy–Littlewood–Pólya 1934 / Agmon 1965 the continuum address); calibration: on the exact parity model the pencil is $(1,1)$ (certified $|\kappa-1| = 1.1\times10^{-12}$ at $m = 241$) and the Sobolev per-mode price is π to 5.1×10^{-8} at $m = 2001$ — the honest, m-free cost of leaving the ℓ^2 world (controls: dropping the boundary term breaks the identity at 4.8×10^{-2}, the Neumann energy undercounts a first-node vector by 0.50).
- **(4) The closed archimedean minimum (T151; T150's rest item 2 closed and attained) [E].** The archimedean lag kernel has two shape properties, checked per window and never assumed — $c_i^{\rm arch} < 0$ for $i \ge 1$ and $c_i^{\rm arch}$ non-decreasing on $i \ge 1$ — and given those, $\min_r \Lambda_r^{\rm arch} = c_0^{\rm arch} - c_{M-1}^{\rm arch}$ *exactly* (max rel 0.0; value 2.1719–3.4674), attained at $r = 0$: a closed two-term functional of the smooth kernel, no matrix, no eigenvector, no loss (control: the full atom-including kernel violates the monotone shape on every window and the closed formula misses its minimum by $\ge 8.7\times10^{-2}$).
- **(5) The linear per-mode bound and the no-go discriminator (T151) [E].** With $\kappa = 0.1939$–0.3599 the certified pencil floor (*one* completed Cholesky of $A - \kappa L_P$ per window, the floating-point floor carried as fl/μ_1^P; the direction re-read on the actual bottom eigenvalue)

and $K_{\mathrm{bot}} = S$ from item (2), the chain $b_k := 2m\sqrt{K_{\mathrm{bot}}\mu_k^P/\kappa} \le C_S k$ holds with $C_S = 12.081$–$26.953 \le 2\pi\sqrt{K_{\mathrm{bot}}/\kappa}$ on the bottom 8 modes of every window — *linear* in k where the quadratic ν ladder grew, every m-power cancelled, nothing additive anywhere (the $\Theta(D^3)$ bottom that killed the Weyl / Bauer–Fike / Davis–Kahan route never enters) — and the bound dominates the truth $m\|\psi_k\|_\infty^2$ and the parity Dirichlet energy on every read mode; the bottom pencil ratio $R = K_{\mathrm{bot}}/\kappa = 3.7362$–$18.6607$ is the *one* scalar the chain still owes, printed per window and typed open in m. The discriminator: on the T145 no-go form the ratio *explodes* on the size ladder ($1.686 \times 10^3 \to 2.667 \times 10^4$, factor 15.8 over $m = 64 \ldots 256$) while the exact parity model holds $R = 1$ to 1.1×10^{-12} — a route that did not fail there would prove a false statement (controls: the ladder $\times 10^{-3}$ fails against the true sups, the inflated floor 1.5κ makes the Cholesky *refuse* on 12/12).

- **Fences and status.** Per instance on small windows only — a finite list of certified window inequalities with an explicit maximum; nothing uniform in the zone index or in D; the symbol-only route stays certified vacuous; $\mathrm{TV}(\log\Lambda)$ stays withdrawn as a hypothesis (T149), σ_{tot} stays retired as a route (T148); every T151 exponent is a fit that stays in the sandbox. Kac–Murdock–Szegő 1953, Widom 1958 / Basor–Ehrhardt 2009 / Böttcher–Silbermann (the address of the symbol question — the computed answer is that the pointwise symbol language does *not* extend to the bottom mode), Weyl 1912, Parlett, Hardy–Littlewood–Pólya 1934 / Agmon 1965, Sylvester 1852 / Bunch–Kaufman 1977, Rayleigh 1877 / Ritz 1909, Cauchy–Schwarz, Wilkinson 1968 / Higham 2002, Chebyshev 1852 named classical; Weil 1952 cited, never used as a criterion; zero-firewall AST-checked. Ledger row `PRIME.ODD.SECTOR.IDENT.01` [E]; **no marker of any pre-existing contract moves**, and this is not RH evidence.
- **Diary, paper and website (T151, odd-carries).** The T151 probe (`odd_ladder_probe.py`, contract `ODD.LADDER`, 27/27, 43.4 s, 72 prime-power windows $m = 96 \ldots 1491$) closes the one inequality by rerouting off the ladder: the odd grid steps over the symbol's negative window ($\theta_c/\theta_1 = 0.328$–$0.407$ on 72/72), the bottom spectrum is certified against the parity Laplacian ($\lambda_k \le S\mu_k^P$, $S = 1.1019$–2.3870), and the discrete Sobolev step yields the *linear* per-mode bound $b_k \le C_S k$ with a *non-growing* constant ($C_S = 11.5137$–19.5731, trend $x^{0.020\pm0.007}$) where the quadratic ladder grew (13.81π at $x^{0.258}$); end to end moves to 2.01×10^{-2}–3.52×10^{-2} of the true gap (gain up to 3.18×), the bottleneck relocating from $Q^\star$ to Ψ. Two of T150's three rest items settled: the minimum of the archimedean diagonal is closed *and attained exactly* ($\min_r \Lambda_r^{\mathrm{arch}} = c_0^{\mathrm{arch}} - c_{M-1}^{\mathrm{arch}} = 2.5928$–$4.8981$, deviation 0.0), and the symbol route is a computed dead end (the local model is a matrix statement — the pointwise symbol language fails at the bottom mode's resolution, margin vacuous on 72/72). One scalar remains a fit: the bottom pencil ratio $R = K_{\mathrm{bot}}/\kappa = 3.3634$–$9.7108$ at $x^{0.037\pm0.015}$ — and the T145 no-go breaks exactly there ($x^{1.986}$). Verdict ODD-CARRIES: all six preregistered gates met, every layer typed (theorem / certified / elementary-verified), the one fit named. `experiments/next.txt` part T151 (series totals 151 probes / 3852 checks); `tfpt_prime_front.tex` gains the T151 subsection with the "After T151" keybox, the `v550` entry in the verified layer (now sixteen modules), the phase-2 section retitled T126–T151 and the updated probe index; `tfpt_1`, `tfpt_3` and `tfpt_research_contracts` gain the `v550` companion blocks; `website/lib/primeFront.ts` gains the T151 and `v550` feed entries; suite counter 543 → 544 scripts. T152 (`pencil_ratio_probe.py`) is running at exactly the remaining scalar — the m-freeness of the bottom pencil ratio — and is *not* part of this change.

2026-07-29 (IV) (v549 — the gauge/parity identities of phase 2 promoted; prime-front diary T150 (the parity mechanism named, ONE-TERM-MISSING) + paper + website sync; NO marker moves)

- **New module v549 (`PRIME.GAUGE.PARITY.IDENT.01`).** Promotion of the identity/certificate core of the discovery parts T149 (`weight_smoothing_probe.py`) and T150 (`mode_ladder_probe.py`) — 21 checks, $\sim$0.3 s, everything recomputed on small frame-A windows, nothing cited from the sandbox; the companion to `v548`, which certified the factorisation $\Gamma = \sqrt{Q^\star} S_w$ and the Liouville machinery in the Jacobi gauge — this module certifies the gauge *freedom* around that machinery and the parity *structure* underneath it. Scope fence: identities, per-instance theorems with checked hypotheses and per-window *certified* inequalities — no fit, no D-exponent, no zone-uniform statement, and explicitly *no statement for all* D (the one open input after T150 — an m-free constant C in the odd-parity-sector ladder $\nu_k \le Ck^2$ for a sign-changing-symbol Toeplitz section — stays open and typed open). Battery: 12 frame-A windows ($n = 4 \ldots 139$, $m = 26 \ldots 285 \le 300$, $D = 7.578 \times 10^{-3} \ldots 2.789 \times 10^{-2}$), 48 gauge assemblies (Jacobi, const, moving-geometric-mean, and the arithmetic-free archimedean diagonal), the full-section parity items on the 6 windows with $M \le 300$, and a parity control model at $m = 2001$ (vector operations only); each statement carries a mutation control that must fail by $\ge 10^{-3}$.

- **(1) The gauge freedom (T149) [E].** For *any* positive diagonal $\tilde\Lambda$ the whitening $\tilde E = \tilde\Lambda^{-1/2} A \tilde\Lambda^{-1/2}$ is an exact congruence, so $\lambda_{\min}(A) \ge \operatorname{cert} \lambda_{\min}(\tilde E) \cdot \min_j \tilde\Lambda_j$ is a valid certified lower bound for every gauge — an identity plus one Rayleigh step — and the family maximum is valid too (verified on all 48 assemblies); the factorisation identity $\Gamma = \sqrt{Q^\star} S_w$ holds *gauge-invariantly* (max rel 2.4×10^{-16}); the const gauge (the geometric mean of the Jacobi diagonal) has $\mathrm{TV}(\log \tilde\Lambda) = 0$ *exactly* on every window — T149's elimination, not reduction, of T148's blocking scalar — with the certified entrywise sandwich $\sigma = c_2/c_1 = 1.3868$–$2.5805$ (controls: a 1% $Q^\star$ perturbation breaks the identity at 4.99×10^{-3}, and on the shifted indefinite form $A - 1.05\lambda_{\min} I$ the completed-Cholesky floor *refuses* on 12/12 windows).
- **(2) The two-regime discriminator (T149, the refutation as a check) [E].** The flutter-smoothing gauge (moving geometric mean, preregistered half-width $m/16$) removes $\mathrm{TV}(\log \Lambda)$ by $\ge 2.88\times$ on every window while moving the Liouville-*transported* bottom-block smoothness functional $\nu_{\tilde L}$ by at most 0.0088 — and even the matched-amplitude *rough* re-gauge (lattice-scale oscillation) moves it by at most 0.0046 (bar 0.01 for both): the roughness $\nu_{\tilde L}$ responds to is *not* in the multiplier — T149's withdrawal of the TV hypothesis, wired as a per-instance check (control: the same functional read *without* the transport, on the whitened modes ψ which carry the multiplier, jumps by 0.41–9.76 under the same rough re-gauge — the invariance is carried by the transport, not by numbness of the functional).
- **(3) The explicit zone diagonal (T150) [E].** $\Lambda_r = c_0 - c_{M-1-2r} + \sum_{s \ne r} \max(c_{|r-s|} - c_{M-1-r-s}, 0)$, assembled lagwise with no matrix anywhere, equals the lumped-pair diagonal on every window (max rel 5.9×10^{-16}) — the whitening diagonal is an *explicit* functional of the zone's lag arithmetic and of nothing else; the lag vector splits *exactly* as $c = c^{\mathrm{arch}} + c^{\mathrm{atom}}$ with $c^{\mathrm{atom}} \le 0$ entrywise, and the arithmetic mass carries the closed budget $\|c^{\mathrm{atom}}\|_1 \le \sum_j \mu_j \le 4B\sqrt{N}$ with $B = 1.038821$ the *verified* maximum of $\psi(n)/n$ over the whole atom table (Chebyshev 1852, computed and never assumed; Abel summation carries the closed form). The honest negative as content: the archimedean section alone is *indefinite* on 12/12 windows (completed LDL^T, up to 5 negative eigenvalues) while the full section has zero — the atoms are *co-responsible* for the positivity, so the additive perturbation reading (Weyl 1912 / Bauer–Fike 1960 / Davis–Kahan 1970, cited, computed dead by T150 at 2.4–5.8 orders, *not* used) is structurally empty and only the multiplicative / Loewner gauge step transfers (control: dropping the positive part in the zone formula breaks the identity at 4.25×10^{-1}).
- **(4) The parity compression (T150, the named mechanism) [E].** With U_-, U_+ the antisymmetric / symmetric parity isometries of the full window, $U_-^\mathsf{T} T_M(c)\, U_- = T - H$ and $U_+^\mathsf{T} T_M(c)\, U_+ = T + H$ *exactly* (max rel 3.9×10^{-16}) with cross block $U_+^\mathsf{T} T_M U_- = 0$ (1.4×10^{-16}) on all 6 windows with $M \le 300$: the reflection-odd section used since T106 *is* the compression of the full symmetric Toeplitz section onto its antisymmetric parity sector (Basor–Ehrhardt 2009: the parity sectors are the natural objects of the Toeplitz+Hankel algebra, cited as the address and never as an authority); the completed LDL^T counts *localise* the negative inertia — the full

section's negative eigenvalues (forced by the sign-changing symbol, $f(0) = -29.2\ldots-6.3 < 0$; T148's honest negative *re-read*, never re-derived) sit entirely in the *even* sector while the odd sector counts zero: $f(0) < 0$ is explained rather than worked around (control: the sign-flipped basis does not reproduce $T - H$, 3.99×10^{-1}).

- **(5) The parity-sine calibration and the ladder certificate (T150) [E].** The parity sines $t_k(r) = \frac{2}{\sqrt{N}} \sin(2\pi k(r+1)/N)$, $N = 2m+1$, are the exact eigenpairs of the parity Laplacian (reflecting corner 3, forced by antisymmetry) at $\mu_k^P = 4\sin^2(\pi k/N)$ — eigen-residual 2.3×10^{-13} — and the smoothness ladder *calibrates* on the exact model: $\nu_k^P = \pi k^2$ for $k = 1\ldots12$ at $m = 2001$ (max deviation 3.0×10^{-5}), so the ladder form $\nu_k \le Ck^2$ is the right question and π its exact floor; on every window, in the const gauge ($\varphi = \psi$ exactly — the ladder of the *pure* Toeplitz-minus-Hankel section), the parity coefficient licence holds, $\nu_k^P \le Ck^2$ with the per-window certified constant $C = 11.70$–17.28, and the derived per-mode ceiling $m\|\psi_k\|_\infty^2 \le 2(2\sqrt{C}k+1)^2$ is a certified instance inequality — an m-free C is exactly the one open input and is *not* claimed (controls: the ladder $\times 10^{-3}$ fails everywhere, a one-index calibration shift misses by 0.75).
- **Fences and status.** Per instance on small windows only — a finite list of certified window inequalities with an explicit maximum; nothing uniform in the zone index or in D; $\mathrm{TV}(\log\Lambda)$ stays withdrawn as a hypothesis (T149), σ_{tot} stays retired as a route (T148); every T149/T150 exponent is a fit that stays in the sandbox. Kac–Murdock–Szegő 1953 (in the parity sector), Widom 1958 / Basor–Ehrhardt 2009 / Böttcher–Silbermann (the parity sectors as the algebra's natural objects — the address, never an authority; the sign-changing symbol is exactly what Basor–Ehrhardt does *not* cover, and that gap is the open input), Sylvester 1852 / Bunch–Kaufman 1977, Chebyshev 1852 (verified on the table), Abel, Weyl 1912 / Bauer–Fike 1960 / Davis–Kahan 1970 (cited, computed dead, *not* used), Rayleigh 1877 / Ritz 1909, Cauchy–Schwarz, Gershgorin 1931, Wilkinson 1968 / Higham 2002 named classical; Weil 1952 cited, never used as a criterion; zero-firewall AST-checked. Ledger row `PRIME.GAUGE.PARITY.IDENT.01` [E]; **no marker of any pre-existing contract moves**, and this is not RH evidence.
- **Diary, paper and website (T150, one-term-missing).** The T150 probe (`mode_ladder_probe.py`, contract `MODE.LADDER`, 36/36, 48.0 s, 72 prime-power windows $m = 50\ldots1393$) names the mechanism: the form is exactly the compression of the full symmetric Toeplitz section onto its antisymmetric parity sector, and the sign-changing symbol's entire negative inertia sits in the *even* sector (certified 72/72) — T148's honest negative is explained rather than worked around. The flutter amplitude becomes a certified form functional (0.0606–0.1993, trend $x^{-0.165\pm0.015}$, T149's fit reproduced fit-free), the whitening diagonal an explicit zone functional, and the arithmetic atoms turn out to be co-responsible for the positivity (the archimedean section alone has 2–7 negative eigenvalues); the additive perturbation route is dead by 2.4–5.8 orders while the multiplicative gauge step closes. One gate remains: the ladder constant still grows — $C \le 43.391 = 13.81\pi$ certified per stratum, trend $x^{0.258\pm0.009}$ against the flatness bar 0.25 — and the gap from π is exactly the arithmetic leakage of the bottom mode (ℓ^1 mass 1.41–1.79). Verdict ONE-TERM-MISSING: one named inequality left — $\nu_k \le Ck^2$ with m-free C for the odd parity sector at a sign-changing symbol. `experiments/next.txt` part T150 (series totals 150 probes / 3825 checks); `tfpt_prime_front.tex` gains the T150 subsection with the "After T150" keybox, the `v549` entry in the verified layer (now fifteen modules), the phase-2 section retitled T126–T150 and the updated probe index; `tfpt_1`, `tfpt_3` and `tfpt_research_contracts` gain the `v549` companion blocks; `website/lib/primeFront.ts` gains the T150 and `v549` feed entries; suite counter 542 → 543 scripts. T151 (`odd_ladder_probe.py`) is running at exactly the remaining inequality — the odd-sector ladder — and is *not* part of this change.

2026-07-29 (III) (`v548` — the Green/Szegő identities of phase 2 promoted; prime-front diary T148 (lifting statement, ONE-INPUT-MISSING) + paper + website

sync; NO marker moves)

- **New module v548 (`PRIME.GREEN.SZEGO.IDENT.01`).** Promotion of the identity/certificate core of the discovery parts T147 (`green_asymptotic_probe.py`) and T148 (`szego_bottom_probe.py`) — 21 checks, ∼0.4 s, everything recomputed on small frame-A windows, nothing cited from the sandbox; the companion to `v547`, which certified the a-priori level constant c_0^{ap} — this module certifies the two scalars that constant is made of. Scope fence: identities, per-instance theorems with checked hypotheses and per-window *certified* inequalities — no fit, no D-exponent, no zone-uniform statement, and explicitly *no statement for all* D (the one open lifting input — $\nu_L \leq F(\text{form})$ with F m-free, equivalently bounded $\mathrm{TV}(\log\Lambda)$ or any weaker smoothness of the multiplier — stays open and typed open). Battery: 12 frame-A windows ($n = 4 \dots 139$, $m = 26 \dots 285 \leq 300$, $D = 7.578 \times 10^{-3} \dots 2.789 \times 10^{-2}$), 7 random positive definite forms for the form-independent items, and a weight-class model ladder $m = 64 \dots 288$; each statement carries a mutation control that must fail by $\geq 10^{-3}$.
- **(1) The factorisation identity (T147) [E].** $\Gamma = \sqrt{m}\,\lambda_{\mathrm{up}} \max_j \|Re_j\| = \sqrt{Q^\star} \cdot S_w$ with $S_w = \lambda_{\mathrm{up}}\|R\|_F$ purely spectral and $Q^\star = m \max_j (R^2)_{jj}/\operatorname{tr}(R^2)$ purely geometric — *exact* at 2.2×10^{-16} on all 12 windows and 7 random forms: the all-D question splits into two named scalars and into nothing else; the certified factorisation (residual-paid column norms, Frobenius bracket) dominates the true Γ everywhere ($S_w = 1.1186$–1.5777, $Q^\star = 2.0879$–2.8634; controls: a 1% $Q^\star$ perturbation breaks the identity at 5.0×10^{-3}, perturbing the largest Green column breaks the factorisation).
- **(2) The bottom-block equidistribution constant (T147) [E].** With Π_B the spectral projector onto $B = \{k : \lambda_k \leq 10\hat{\lambda}\}$ (preregistered rule, $|B| = 3 \dots 5$), $\sum_j (\Pi_B)_{jj} = |B|$ *exactly* (3.3×10^{-16}), hence $Q_B \geq |B|$ is a theorem (max ≥ mean); the *measured* $Q_B/|B| = 1.5091$–1.7980 lies in the preregistered band $[1, 2.2]$ against the Szegő/Widom sharp reference $Q_B \leq 2|B|$ — a measurement on a finite list, typed as such (controls: the rank-$|B|$ coordinate projector gives $Q_B = m$ and overshoots by ≥ 3.9, a 1% basis scaling breaks the trace identity).
- **(3) The second-difference ℓ^1 ceiling (T148, the lever) [E].** From the exact Dirichlet eigenpair $L_0 s_k = \mu_k s_k$ and $\mu_k \geq 4k^2/(m+1)^2$: the coefficient bound $|a_k| \leq \min(1, \|s_k\|_\infty \|L_0^p \psi\|_1/\mu_k^p)$ for $p = 1, 2$ against the *true* coefficients on every window bottom mode and on random unit vectors (worst excess ≤ 0), the *m-free* closed form $\|a\|_1 \leq 2\sqrt{\nu}+1$ with $\nu = (m+1)^{3/2}\|L_0\psi\|_1/(2\sqrt{2})$ — every power of m cancelled — and the chain $m\|\psi\|_\infty^2 \leq 2\|a\|_1^2$; calibration: for the Dirichlet bottom mode $\|a\|_1 = 1$ *exactly* and $\nu = 3.1415 \to \pi$ at every size (controls: the ceiling $\times 10^{-3}$ fails on every window, a one-index μ shift breaks the coefficient identity at 1.5×10^{-1}).
- **(4) The layer-cake S_w certificate (T148) [E].** The completed LDL^T inertia count at every rung of the preregistered ladder $\tau_i = \rho_i \lambda_{\mathrm{lo}}$ ($\rho = 1.5 \dots 1024$, λ_{lo} a certified positive floor) equals the direct spectral count *exactly* on all 96 (window, rung) pairs (Sylvester 1852; Bunch–Kaufman 1977 — a certificate, never a sorted list), and the layer cake certifies $S_w \leq 1.3288$–1.8220 per window (true 1.1186–1.5777, slack ≤ 1.216); nothing here says m-free (control: dropping the bottom layer breaks the bound on every window — the bottom eigenvalue is the weight).
- **(5) The Liouville transform and the weight-class discriminator (T148) [E].** The chain $m\|\psi\|_\infty^2 \leq \kappa_\Lambda m\|\hat{\varphi}\|_\infty^2 \leq 2\kappa_\Lambda \|a(\hat{\varphi})\|_1^2 \leq 2\kappa_\Lambda \operatorname{ceil}(\hat{\varphi})^2$ with $\varphi = \Lambda^{-1/2}\psi$ (the exact pencil equivalence, a change of variables and not a perturbation) holds stepwise on every bottom-block mode; $\kappa_\Lambda = \max\Lambda/\min\Lambda = 1.3868$–$2.5805$ is a priori (a ratio of two diagonal entries of the form). The discriminator: on a model section with a genuine order-2 zero ($f(0) = 0$, $f''(0)/2 = 1$ exact), over the ladder $m = 64 \dots 288$ at matched κ_Λ (mismatch 0.0013), the bounded-variation multiplier keeps ν flat (max/min 1.003) while the rough one (TV larger by 162.6) grows by 19.9 — the roughness, not the conditioning (control: the first Liouville step $\times 10^{-3}$ fails on every mode, provably — the step slack is bounded by κ_Λ^2).
- **Fences and status.** Per instance on small windows only — a finite list of certified window

inequalities with an explicit maximum; nothing uniform in the zone index or in D; the symbol-level order-2 hypothesis is *refuted* (T148: $f(0) < 0$ on every surface window) and enters nowhere as a theorem; σ_{tot} stays retired as a route; every T147/T148 exponent is a fit that stays in the sandbox. Kac–Murdock–Szegő 1953, Widom 1958 / Basor–Ehrhardt 2009 / Böttcher–Silbermann (the mechanism's address, never an authority), Sylvester 1852 / Bunch–Kaufman 1977, Euler 1735, Rayleigh 1877 / Ritz 1909, Cauchy–Schwarz, Davis–Kahan 1970 (cited, computed vacuous, *not* used), Gershgorin 1931, Wilkinson 1968 / Higham 2002 named classical; Weil 1952 cited, never used as a criterion; zero-firewall AST-checked. Ledger row `PRIME.GREEN.SZEGO.IDENT.01` [E]; **no marker of any pre-existing contract moves**, and this is not RH evidence.

- **Diary, paper and website (T148, one-input-missing).** The T148 probe (`szego_bottom_probe.py`, contract `SZEGO.BOTTOM`, 31/31, 13.0 s, 48 windows $m = 96 \ldots 1077$ in four D-strata, stress ladders to $m = 1500$) dissects the lifting statement: the second factor S_w *closes* on the surface (LDL$^{\mathsf{T}}$ layer cake, $S_w \le 1.9587$ certified against true 1.282–1.683, trend flat, per D-stratum), at the price of an honest negative — the arithmetic Toeplitz symbol has $f(0) < 0$ on 48/48 windows, so positive definiteness is a finite-section effect of the minus-Hankel part and the lumping and the KMS order-2 hypothesis survives only as a measurement ($\alpha = 1.643$–1.989) — while the lifting statement resists with its hypothesis isolated to one named scalar ($\mathrm{TV}(\log \Lambda)$, proved to be the roughness and not the conditioning by the weight-class experiment); the first end-to-end number arrives: the certified chain delivers 8.36–15.86% of the true gap on all 48 windows (median 11.74%), and the a-priori-shaped chain is still a lower bound at a factor 10.5. Verdict ONE-INPUT-MISSING: exactly one scalar side ($Q^\star$) lacks an m-free certified statement, target $\nu_L \lesssim 34$ against 282 measured. `experiments/next.txt` part T148 (series totals 148 probes / 3759 checks); `tfpt_prime_front.tex` gains the T148 subsection with the "After T148" keybox, the `v548` entry in the verified layer (now fourteen modules), the phase-2 section retitled T126–T148 and the updated probe index; `tfpt_1`, `tfpt_3` and `tfpt_research_contracts` gain the `v548` companion blocks; `website/lib/primeFront.ts` gains the T148 and `v548` feed entries; suite counter 541 → 542 scripts. T149 (`weight_smoothing_probe.py`) is running at exactly the remaining input — whether a smoothed whitening removes the TV roughness — and is *not* part of this change.

2026-07-29 (II) (v547 — the level-lemma identities of phase 2 promoted; prime-front diary T146 (level lemma, LEVEL-CARRIES) + website sync; NO marker moves)

- **New module `v547` (`PRIME.LEVEL.LEMMA.01`).** Promotion of the a-priori core of the discovery part T146 (`level_lemma_probe.py`) — 20 checks, ∼0.4 s, everything recomputed on small frame-A windows, nothing cited from the sandbox; the companion to `v546`, whose item (5) certified S1$'$ with a c_0 *measured* at the minimiser's own layer cake — this module supplies the a-priori side, every factor a functional of the form alone. Scope fence: identities, per-instance theorems with checked hypotheses and per-window *certified* inequalities — no fit, no D-exponent, no zone-uniform statement, and explicitly *no statement for all D* (the asymptotic delocalisation of the Green columns, T146's remainder, stays open and typed open). Battery: 12 frame-A windows ($n = 4 \ldots 139$, $m = 26 \ldots 285 \le 300$, $D = 7.578 \times 10^{-3} \ldots 2.789 \times 10^{-2}$), 9 random positive definite forms for the form-independent items, and a stress ladder $m = 64 \ldots 288$; each statement carries a mutation control that must fail by $\ge 10^{-3}$.
- **(1) The union identity for nested cakes (T146) [E].** For a nested decreasing chain with positive coefficients, $\sum_{j,l} c_j c_l \, |S_j \cup S_l| = 2T \|\psi_t\|_1 - \|\psi_t\|^2$ with $T = \sum_j c_j$ and $\psi_t = \sum_j c_j \mathbf{1}_{S_j}$ — form-independent and $O(m)$, the reason the cake base is movable at all; exact (0.0) against T145's $K \times K$ double sum on all 12 window minimisers and 9 random vectors, *and* against the union sizes computed from the sets themselves with no nestedness used (controls: one shuffled proper level set 3.2×10^{-2}, a 1% perturbation of the heaviest coefficient 4.8×10^{-3}).

- **(2) The layer-cake counting lemma at a general base (T146) [E].** $G(b) \le 2b^2\,\|v\|_\infty\|v\|_1/\|v\|^2 + \varepsilon(b)$ for *any* vector at all 6 preregistered bases (margin ≥ 1.08 on the battery and on every window minimiser); at $b = 2$ the leading constant is the classical dyadic 8 *exactly*, and the base is *free* because the chain consumes only the lower domination $|v| \le \psi_t$ — the a-priori constant gains $4/b_*^2 = 3.7612$ over base 2 on every window (controls: the bound $\times 10^{-3}$ fails on every draw and base, and the one-level-down cake breaks the domination on every draw — the $+1$ in the cake exponent is load-bearing).
- **(3) The resolvent delocalisation bound (T146) [E].** $\psi = \lambda R\psi$ to 9.0×10^{-14} — an *identity*, so no Davis–Kahan transfer step appears anywhere — with $\lambda_{\mathrm{up}} = \min_j \mathrm{Rayleigh}(Re_j) \ge \lambda_{\min}$ (Rayleigh–Ritz; window ratio 1.0592–1.2883), the coordinate bound $|\psi_j| \le \lambda\|Re_j\|$ (Cauchy–Schwarz) and the column norms certified from above with the linear-solve residual paid for, giving $\|\psi\|_\infty \le \lambda_{\mathrm{up}} \max_j \|Re_j\|$ on every window (within 1.061–1.337 of the truth) and $\|\psi\|_1 \le \sqrt{m}\,\min(1, \Gamma_1)$. The lever: the trivial ceiling $\max_j \|Re_j\| \le 1/\lambda_{\min}$ would give $\Gamma \le \sqrt{m} = 5.1$–16.9 (useless); the certified columns sit a factor 2.85–10.17 *below* it (preregistered bar 2.0), giving $\Gamma = 1.8356$–2.2797 — read off the form, never off the minimiser (controls: a 1% λ perturbation breaks the identity at 1.0×10^{-2}, the sup bound $\times 10^{-3}$ fails on every window).
- **(4) The composite level lemma (T146) [E].** $c_0^{\mathrm{ap}} = 2b^2\,\Gamma\min(1, \Gamma_1) + \varepsilon = 3.9042$–4.8488 dominates the measured level constant on all 12 windows (slack 1.986–2.418) and stays under the preregistered ceiling $16 = 4\times$ Maz'ya's Dirichlet value 4; the certified window inequality cert_lam_max$(R) \le c_0^{\mathrm{ap}}\Psi_{\mathrm{abs}}$ holds on 12/12 (both sides by completed Cholesky, the density the $|R|$ one with Charikar's cited constant), closing the chain $\hat\lambda \ge 1/(c_0^{\mathrm{ap}}\Psi_{\mathrm{abs}})$, gap $\ge \hat\lambda/\kappa_{\mathrm{up}}$ at loss factor 0.0423–0.1272 of the exact gap. The built-in discriminator: on T145's no-go form $R = aa^\mathsf{T} + \varepsilon I$, $a_i = i^{-1/2}$, Γ grows $3.676 \to 6.798$ over $m = 64 \ldots 288$ (ratio $1.85 \ge$ the preregistered 1.5) and $c_0^{\mathrm{ap}} = 11.10$ *exceeds* every window value — the bound must be worse there, since a candidate that did not fail on the no-go would prove a false statement — while the theorem half survives on every no-go size and the Dirichlet path-Laplacian control stays flat (max/min 1.0063; control: $c_0^{\mathrm{ap}} \times 10^{-3}$ breaks the certificate on every window).
- **Fences and status.** Per instance on small windows only — a finite list of certified window inequalities with an explicit maximum; nothing uniform in the zone index or in D; only the sign-free / Green constant is bounded a priori, the energy-route σ_{tot} stays *measured*, and every T146 exponent is a fit that stays in the sandbox. Maz'ya 1985 (the strong-type half *not* used as a theorem anywhere), Rayleigh 1877 / Ritz 1909, Cauchy–Schwarz, Miclo 1999, Davis–Kahan 1970 (cited, deliberately *not* used), Perron–Frobenius (not applicable to E, not used), Fukushima–Ōshima–Takeda 1994, Charikar 2000, Goldberg 1984, Gershgorin 1931, Wilkinson 1968 / Higham 2002 named classical; Weil 1952 cited, never used as a criterion; zero-firewall AST-checked. Ledger row `PRIME.LEVEL.LEMMA.01` [E]; **no marker of any pre-existing contract moves**, and this is not RH evidence.
- **Diary, paper and website (T146, level-carries).** The T146 probe (`level_lemma_probe.py`, contract `LEVEL.LEMMA`, 30/30, 7.6 s, 64 windows $m = 26 \ldots 1173$, 9/9 preregistered verdict conditions) closes L1 on the measurement surface as a chain of theorems and certified window inequalities with no step reading the minimiser: four hand-written profiles lose by 5–9 orders of magnitude, the Green column survives ($\lambda_{\mathrm{up}}/\hat\lambda = 1.0497$–1.3764), Davis–Kahan is instrumented and discarded (informative on 0/64; the spectrum bottom is a near-degenerate block, $\lambda_2/\hat\lambda = 1.2542$–3.4471), the cake base proves free (the dyadic 8 falls to $2b^2 \approx 2$, gain 3.76 at zero cost) and $c_0^{\mathrm{ap}} = 3.9042$–4.8488 on 64/64 windows lands at the size of Maz'ya's classical Dirichlet value 4, closing the chain at loss factor 0.0422–0.1586; the one genuine remainder is D-uniformity for *all* D — the asymptotic delocalisation of the Green columns. `experiments/next.txt` part T146 (series totals 146 probes / 3689 checks); `tfpt_prime_front.tex` gains the T146 subsection with the "After T146" keybox, the `v547` entry in the verified layer (now thirteen modules), the phase-2 section retitled T126–T146 and the updated probe index; `tfpt_1`, `tfpt_3` and `tfpt_research_contracts` gain the `v547` companion blocks; `website/lib/primeFront.ts` gains the T146 and `v547` feed

entries; suite counter 540 → 541 scripts. T147 (`green_asymptotic_probe.py`) is running at exactly the all-D asymptotics and is *not* part of this change.

2026-07-29 (I) (v546 — the capacity-chain identities of phase 2 promoted; prime-front diary T145 (Maz'ya proof, ONE-STEP-MISSING) + website sync; NO marker moves)

- **New module v546 (`PRIME.CAPCHAIN.IDENT.01`).** Promotion of the identity-shaped core of the capacity cycle T142 (`conductance_profile_probe.py`), T143 (`sharp_capacity_probe.py`), T144 (`capacity_inequality_probe.py`) and T145 (`mazya_proof_probe.py`) — 26 checks, ∼0.5 s, everything recomputed on small frame-A windows, nothing cited from the sandbox. Scope fence: identities, per-instance theorems with checked hypotheses and per-window *certified* inequalities — no fit, no D-exponent, no zone-uniform statement, and *no level lemma* (T145's L1 stays open, unclaimed). Battery: 12 windows ($n = 4 \dots 139$, $h = 26 \dots 285 \leq 300$, $D = 7.578 \times 10^{-3} \dots 2.789 \times 10^{-2}$, 32–5102 positive off-diagonal edges); the covering kernel is positive definite on 11, where items (1)–(2) run; each identity carries a mutation control that must fail by $\geq 10^{-3}$.
- **(1) The capacity decomposition (T142) [E].** $K^{-1} = D^\top J^{-1} D + xx^\top/\mathrm{cap}$ with $J = DKD^\top$ (assembled entrywise, 3.8×10^{-15}), $x = K^{-1}\mathbf{1}$ and $\mathrm{cap} = \mathbf{1}^\top K^{-1}\mathbf{1}$, as a matrix identity ($1.4 \times 10^{-13}$); under the congruence $u = K^{1/2}v$ the Dirichlet half is an *orthogonal projection* ($P^2 = P$ at 1.9×10^{-12}, $\Omega = \lambda_{\max}(P) = 1$ to 5.1×10^{-12} — the T142 normalisation, where T141 had guessed 20.7–2724), its complement *is* the capacity rank-one and it annihilates $K^{-1/2}\mathbf{1}$ (controls: dropping the rank-one 3.4×10^{-1}, a 1% cap perturbation 3.4×10^{-3}).
- **(2) The exact capacity–Rayleigh identity (T143) [E].** With $H = \mathrm{diag}(s) + L_N$ (4.7×10^{-16}), $L_N = D^\top BD$ for the signed crossing kernel (4.9×10^{-16}) and $\rho(W) = 1 - \lambda_{\min}(A, A_B)$ (5.4×10^{-14}), the assembled quotient $[d^\top(J^{-1} - B)d + (x^\top v)^2/\mathrm{cap} - \sum_k s_k v_k^2]/[d^\top J^{-1} d + (x^\top v)^2/\mathrm{cap}]$ equals $v^\top(K^{-1} - H)v/v^\top K^{-1} v$ for *every* v (9.5×10^{-14} on 24 random vectors per window), reproduces $1-\rho(W)$ at the exact top eigenvector (1.7×10^{-9} under the *declared* $1/\mathrm{gap}$ conditioning bar 5.4×10^{-8}), and every random quotient stays $\geq 1 - \rho(W)$; on the constant vector the whole denominator *is* the capacity term (controls: dropping it, rel = 1; the $|N|$ crossing kernel, 2.6×10^{-1}).
- **(3) Cholesky nestedness (T144) [E].** $\mathrm{cap}_E([a, a+j))$ equals the prefix sum of squared forward-substitution entries of *one* triangular solve: 3404 interval capacities from 36 solves at 2.0×10^{-14} against independent Schur solves, exhaustive on the median window (every prefix, 2.8×10^{-14}), monotone in b by construction (control: the reversed ordering computes *suffix* capacities and deviates by $\geq 4.6 \times 10^{-1}$).
- **(4) The whitening certificate (T144) [E].** $\lambda_{\max}(W) \leq \kappa_{\mathrm{up}}$ (completed Cholesky) $\leq$ the Gershgorin ceiling on every window ($\kappa_{\mathrm{up}} = 1.2245$–$1.4236$ against ≤ 2.1213); the licensed chain direction $\lambda_{\min}(E)/\kappa_{\mathrm{up}} \leq \lambda_{\min}(A, A_B)$ holds on every window — a denominator comparison, untouched by T142's numerator obstruction — and the Cholesky refuses at the shrunken shift $0.995\,\lambda_{\max}(W)$.
- **(5) The M4 split and the certified chain (T145) [E].** The mass half of Maz'ya's step M4: $\sum_k f_k^2 \leq \psi^2$ *pointwise* (a theorem for nonnegative truncations, exact slack ≥ 0), the row-wise implication on rows with $(E\mathbf{1})_i \geq 0$, measured mass ratio ≤ 0.2995 and the full step $\sigma_{\mathrm{tot}} = 0.2197$–$0.4425 < 1$ on every window although the Markov hypothesis is absent. Licence 4 is stated with $|R|$ and *not* R^+, with the witness $R = [[1, -\frac{1}{2}], [-\frac{1}{2}, 1]]$, $x = (1, -1)$ ($3.0 > 2.0$) built in as a counterexample check. The sign-free chain is ordered on every window, and S1′ is *certified per window on both routes*: $\mathrm{cert\_lam\_max}(R) \leq 12\,\sigma_{\mathrm{tot}}\Psi_{\mathrm{pos}}$ (energy, $c_0 = 2.6369$–5.3099) and $\leq G_{\mathrm{dy}}\Psi_{\mathrm{abs}}$ (sign-free, $c_0 = 2.5154$–6.1449); the better route gives $c_0 = 2.5154$–4.2265 against the classical Dirichlet value 4. The all-sets density bound carries Charikar's cited constant,

bracketed by exhaustive enumeration of all 4095 subsets of a small block (control: $c_0 \times 10^{-3}$ breaks the certificate on every window).

- **Fences and status.** Per instance on small windows only; nothing uniform in the zone index or in D; the c_0 of item (5) is *measured* at the minimiser's own layer cake and the a-priori level lemma (T145's L1) stays open. Maz'ya 1985 (the strong-type half *not* used as a theorem anywhere — it is the thing the cycle transcribes), Muckenhoupt 1972, Fukushima–Ōshima–Takeda 1994, Miclo 1999, Charikar 2000, Goldberg 1984, Gershgorin 1931, Wilkinson 1968 / Higham 2002 named classical; Weil 1952 cited, never used as a criterion; zero-firewall AST-checked. Ledger row `PRIME.CAPCHAIN.IDENT.01` [E]; **no marker of any pre-existing contract moves**, and this is not RH evidence.
- **Diary, paper and website (T145, one-step-missing).** The T145 probe (`mazya_proof_probe.py`, contract `MAZYA.PROOF`, 33/33, 8.0 s, 64 windows $m = 26 \dots 1173$) transcribes Maz'ya's classical proof step by step onto the non-Markovian form: M1, M2, M3 are theorems, the Markov property sits in exactly one line (M4), that line splits, its mass half is a theorem and dominates ($\sigma_{\text{tot}} = 0.215$–$0.443 < 1$), c_0 becomes explicit (2.248–4.227, fit $D^{0.028\pm0.017}$, zone-uniform) and S1′ is certified on 64/64 windows; an explicit no-go proves a level-profile hypothesis is *necessary*; S2′ retired, R5 downgraded, R4 closed — the mathematical rest is *one* named lemma, L1. `experiments/next.txt` part T145 (series totals 145 probes / 3659 checks); `tfpt_prime_front.tex` gains the T145 subsection, the `v546` entry in the verified layer (now twelve modules) and the updated probe index; `website/lib/primeFront.ts` gains the T145 feed entry; suite counter 539 → 540 scripts. T146 (`level_lemma_probe.py`) is running at exactly L1 and is *not* part of this change.

2026-07-28 (III) (v545 — the Hardy-core identities of phase 2 promoted; prime-front diary T140/T141 + website sync; NO marker moves)

- **New module v545 (`PRIME.HARDY.IDENT.01`).** Promotion of the identity blocks of the discovery parts T140 (`signed_band_probe.py`) and T141 (`discrete_hardy_probe.py`) — 36 checks, ∼1 s, everything recomputed on small frame-A windows, nothing cited from the sandbox. Scope fence: *identities*, closed-form quantities and the *validity* — never the *size* — of two certified upper-bound shapes, plus one negative typing. No fit is promoted, no D-exponent, no zone-uniform statement, and no bound that beats the target. Battery: 11 windows ($n = 8 \dots 139$, $h = 58 \dots 285 \le 300$, core $m = 57 \dots 284$, 193–5102 positive off-diagonal edges, 10–87 loaded stripes), $D = 7.578 \times 10^{-3} \dots 2.388 \times 10^{-2}$ (a factor 3.15, so item (8) is a comparison and not a single point), exact $\rho(W) = 0.997759$–0.999981; the explicit $n_e \times n_e$ signed Gram is formed only on the 5 windows with $n_e \le 900$.
- **(1)–(3) The form lifting, the exact finite core and the energy reordering [E].** With M the edge–interval incidence matrix and $C = \operatorname{diag}(\sqrt{\Delta})M$, the T139 telescope lifts from the *entries* of the signed Gram to its *form*: Gram $= CHC^\mathsf{T}$ at 3.0×10^{-15}, hence rank(Gram) $\le h-1$ (measured 57–183 against 193–882 edges; control: unweighted incidence 1.6). The covering kernel in *closed geometric form*, $K_{rs} = W([r \wedge s,\, r \vee s])$, equals $M^\mathsf{T}\Delta M$ (1.3×10^{-15}) and the spectral radius is an *exact* eigenvalue of the finite core, $\rho(W) = \lambda_{\max}(K^{1/2}HK^{1/2}) = 1 - \lambda_{\min}(A, A_B)$ (5.4×10^{-14}) $= \lambda_{\max}(\text{Gram})$ (7.8×10^{-16}) — three independent routes, no truncation (control: the factorwise product $\lambda_{\max}(K)\lambda_{\max}(H)$ misses by 7.8×10^2). The energy reordering $H = \operatorname{diag}(s) + L_N$ holds at 4.7×10^{-16}: a mass term plus a long-range Dirichlet form, its weights positive on 0.676–0.872 of the off-diagonal area (a measurement, no law).
- **(4) The checkerboard split [E].** At group length exactly b, $\text{Band}_b = D + A_{\text{even}} + A_{\text{odd}}$ *entrywise* (0.0) on all 9 (window, b) pairs over $b = 4, 8, 16$, and the resulting three-step Weyl sequence dominates $\lambda_{\max}(\text{Band}_b)$ with 0 exceptions (worst ratio 1.6940) at a step count independent of the up to 9 groups — three steps instead of $O(nb)$ (control: $L = b/2$ loses entries, 3.5×10^{-1}).

- **(5)–(6) The four Hardy identities and the closed Muckenhoupt quantity [E].** (i) the K^{+} Rayleigh form $\rho(W) = u^{\mathsf{T}}Hu/u^{\mathsf{T}}K^{+}u$ at $u = K^{1/2}v$ (6.6×10^{-13}); (ii) the signed crossing kernel $L_N = D^{\mathsf{T}}BD$ with $B_{kl} = \sum_{r \le k \wedge l,\ k \vee l < s} N_{rs}$ — the same closed covering formula as K, on the increment grid (4.9×10^{-16}); (iii) $Q = B_{+}\mathbf{1}$ (1.4×10^{-15}): the Cauchy–Schwarz weight T140 introduced by hand *is* the row sum of B_{+}, so that step is the Gershgorin diagonal step on B_{+}; (iv) $DKD^{\mathsf{T}} = L_{\Delta}$ on the interior nodes (2.7×10^{-14}) with diagonal $(\mathbf{\Delta 1})_{k+1} = 0.4886$–$15.15$, the endpoint edge mass — a *local* quantity and not $\lambda_{\max}(K)$ (control: the crossing kernel of $|N|$ does not reproduce L_N, 2.6×10^{-1}; the signs are load-bearing). The two-weight quantity $A_{M0} = \max_k Q_k(\mathbf{\Delta 1})_{k+1}$ is computed by two closed routes agreeing to 8.3×10^{-15}; its value (65.5–4244) is *reported*, its size and zone behaviour are not promoted (control: $\lambda_{\max}(K)$ in place of the local mass inflates it by ≥ 358).
- **(7) Two certified shapes, promoted as *valid* bounds only [E].** Every Loewner step is certified by a completed Cholesky (-1.5×10^{-14}, -2.3×10^{-10}). The additive Hardy chain dominates $\rho(W)$ on 11/11 windows (factor 3.074–10.758); the exact two-term Weyl split dominates too (1.691–2.503) and lies *below* the chain everywhere, so it is the floor of the whole additive family; the genuinely joint shape $\Lambda(H, Y) \cdot \Omega(Y)$ with a positive definite Jacobi Hardy form Y dominates on 11/11 with the closed-form geometric profile (8.815–74.872 at $\Omega = 8.33$–36.2) and on 11/11 with the per-zone tridiagonal read-off of K^{+}. Promoted is the *dominance* on these instances and nothing about the factors. Controls: the same Y without its mass term is singular — the constant vector lies in its kernel (5.7×10^{-17}) while H does not vanish there — so the certificate *refuses* on 11/11; and dropping the *positive* part instead of the negative one falls below $\rho(W)$ on 11/11.
- **(8) The sign-versus-absolute separation — a negative typing [X].** Over the measured D range the exactly bounded object (the signed Dirichlet share) spreads by 1.36×, while every absolute variant spreads by at least 54.94× (Loewner drop 62.03, Cauchy–Schwarz 54.94, product 56.92), and dropping the positive part of H alone costs 9.96–464.63×. The corresponding power laws are *fits* and labelled as fits: $D^{-0.138\pm0.094}$ (signed) against $D^{-0.776\pm0.352}$ (certified Muckenhoupt) and $D^{-2.748}$, $D^{-2.820}$, $D^{-2.918}$ (absolute). Promoted is the instance comparison of the measured spreads, not the exponents.
- **Fences and status.** Per instance on small windows only; nothing uniform in the zone index or in D; item (2) is an identity and not a bound, so the zone-uniform discrete Hardy inequality stays *open*; item (7) promotes validity and not size — neither shape beats the target. Abel summation, Cauchy–Schwarz, Weyl 1912, Gershgorin 1931, Haynsworth 1968, Ostrowski 1937 / Fan 1958 / Berman–Plemmons 1979, Muckenhoupt 1972, Bradley 1978, Opic–Kufner 1990, Bennett 1987/1991, Gantmacher–Krein 1950/1960, Karlin 1968, Wilkinson 1968 / Higham 2002 named classical; Weil 1952 cited, never used as a criterion; zero-firewall AST-checked. Ledger row `PRIME.HARDY.IDENT.01` [E] + [X]; **no marker of any pre-existing contract moves**, and this is not RH evidence.
- **Diary, paper and website.** `experiments/next.txt` part T141 (HARDY-RESISTS, 22/22; series totals 141 probes / 3547 checks); `tfpt_prime_front.tex` gains the T141 subsection, the `v545` entry in the verified layer and the updated probe index; `website/lib/primeFront.ts` gains the T141 feed entry and the `v545` promotion entry. T142 (`conductance_profile_probe.py`) is running at the one open ingredient and is *not* part of this change.

2026-07-28 (II) (v543 + v544 — the lumped M-matrix pair and the long-lag support structure promoted; prime-front diary T136/T137 + website sync; NO marker moves)

- **New module `v543` (`PRIME.MMATRIX.IDENT.01`).** Promotion of the identity/theorem block of the discovery part T136 (35 checks, ∼1 s), recomputed on small frame-A windows — nothing cited from the sandbox. Scope fence: matrix identities, per-instance theorems with *checked* hypotheses, and one *negative* per-instance result; no fit, no D-exponent, no uniform-in-zone claim, no floor value. Battery: 20 seam windows ($n = 4 \ldots 59$, $\rho = 1.25 \ldots 4.00$, $h = 25 \ldots 299$; every factorised or diagonalised matrix ≤ 300) giving 40 symmetric matrices (pole-free odd sections + border Schur blocks), 39 with positive off-diagonals, plus 312 shifted ladder rungs and 8 whitened two-grid rungs.

- **(1)–(2) The lumped pair and the congruence [E].** With Δ the positive off-diagonal part and $L_\Delta = \mathrm{diag}(\Delta\mathbf{1}) - \Delta$, the pair $A_B = A + L_\Delta$ is Stieltjes (off-diagonals exactly ≤ 0), L_Δ is a graph Laplacian hence PSD with zero row sums, so $A_B\mathbf{1} = A\mathbf{1}$ (1.6×10^{-15}) and $A_B \succeq A$ — an *upper* bound on $\lambda_{\min}(A)$ and a floor *never* (control: lumping the negative off-diagonals destroys Stieltjes, 2.0×10^{-1}). The congruence $A = A_B^{1/2}(I - W)A_B^{1/2}$, $W = A_B^{-1/2}L_\Delta A_B^{-1/2} \succeq 0$, holds as a *matrix* identity (4.4×10^{-15}) with $\lambda_{\max}(W) = \rho(L_\Delta A_B^{-1})$ (4.4×10^{-15}), and the *declared circularity* $\lambda_{\max}(W) < 1 \iff A \succ 0$ is checked in *both* directions (0 exceptions forward; 39/39 shifted controls in the indefinite range give $\lambda_{\max}(W) \geq 1$) — the Loewner sandwich is vacuous by itself, and that is the content.
- **(3)–(4) The anchor and the empty shift ladder [E].** With $x = A_B^{-1}\mathbf{1} \geq 0$ and $A_B^{-1} \geq 0$ entrywise (Ostrowski 1937, *checked*), $\lambda_{\min}(A_B) \geq 1/\max_r x_r$ on all 39 rows (min slack 3.0×10^{-3}, sharpest ratio 0.9070) — one solve, one sign check, no eigenvalue (control: a positive off-diagonal makes it false, $1.9000 > 0.1000$). Lumping commutes with diagonal shifts, $(A - sI)_B = A_B - sI$ (2.5×10^{-16} on 312 rungs), and $(A_B - sI)^{-1} \geq A_B^{-1} \geq 0$ entrywise (Berman–Plemmons 1979, violation 0), so $\max_r x_r(s)$ and the a-priori margin $\|L_\Delta\|_\infty \max_r x_r(s)$ are non-decreasing along the whole ladder: the shifted ladder is *structurally empty* — a theorem, not a measurement.
- **(5)–(6) Varga's identity and the a-priori Jacobi radius [E].** For the regular splitting $A_B = D_0 - N_0$ (all three hypotheses checked on 39/39), $\rho(J) = \tau/(1+\tau)$ with $\tau = \rho(A_B^{-1}N_0)$ at 4.8×10^{-15} ($\rho(J) = 0.7347$–0.9860); the Collatz–Wielandt bracket at the anchor vector gives $\min_r (Tx)_r/x_r \leq \tau \leq \max_r (Tx)_r/x_r$ (0 exceptions) and hence an *a-priori* $\rho(J) \leq 0.9861 < 1$ with no eigenvalue anywhere, sharp on the measured gap by 1.0062–1.0531. Controls: a non-regular splitting breaks the identity (1.1×10^{-1}); at a sign-changing test vector the quotient bound is false on 39/39.
- **(7) The k-scanned M18 majorant — a negative result [X].** The whitened mass identity $\|L^{-1}\hat{s}\|^2 = \hat{s}^{\mathsf{T}}U^{-1}\hat{s}$ holds at 7.2×10^{-16} and the triangle chain is a *valid* majorant at every k separately (39 (rung, k) pairs over $k = 2 \ldots 64$, 0 violations), so the minimum over the ladder is legitimate — and it does *not* reach the preregistered T134 bar 1/2 on any rung (best 3.080; best over the three values T134 tried, also 3.080) while the *measured* bad mass is ≤ 0.0120. The localisation term alone already exceeds the bar (0.542): the loss is the bound, and what is missing is a localisation statement. The bar is *not* moved and nothing is upgraded.
- **New module v544 (`PRIME.LONGLAG.SUPP.01`).** Promotion of the structure block of the discovery part T137 (24 checks, $\sim$0.2 s) on the same 20 windows (3–25 prime-power atoms, 20555 positive off-diagonal edges, 455 loaded stripes, 39 bracketed lumped comparisons).
- **(1)–(4) The support is a comb stripe set, and its amplitude is certified [E].** The kernel splits exactly as $c = c_{\mathrm{arch}} - c_{\mathrm{comb}}$ with $c_{\mathrm{comb}} \geq 0$, and the archimedean section alone has *all* off-diagonals ≤ 0 on 20/20 windows: every positive off-diagonal of A is comb-generated (control: the full section carries $+2.0 \times 10^{-1}$). Every one of the 20555 positive edges has its Hankel index $M - 1 - r - t$ inside the atom-hat index set — per-window fraction exactly 1.000000 — so $\mathrm{supp}(\Delta)$ is a union of anti-diagonal stripes at prime-power atom positions (the converse is reported, *not* claimed). Each stripe is a perfect matching (0 endpoint collisions), so its edge vectors $b_e = e_r - e_t$ are *exactly* orthogonal and L_Δ restricted to a stripe is a direct sum of 2×2 blocks (control: edges across two stripes overlap). And $A_{rs} = A^{\mathrm{arch}}_{rs} - c_{\mathrm{comb}}[|r-s|] + c_{\mathrm{comb}}[\mathrm{anti}]$ gives the amplitude certificate $\Delta_{rt} \leq c_{\mathrm{comb}}[M-1-r-t]$ on all 20555 edges (excess -1.9×10^{-6}, sharpest ratio 0.99998532) — no eigenvalue, no norm; control: the Toeplitz lag in place of the

Hankel index breaks it on 20415/20555. Honestly undecided and *not* promoted: at $h \le 300$ the hats never overlap, so the tighter one-atom refinement coincides with the certificate here.

- **(5) The Green function, bracketed two-sidedly** [E]**.** For the lumped comparison $B = D_0 - N_0$, $J = D_0^{-1}N_0 \ge 0$, the anchor identity $q_J = \max_r (Jx)_r/x_r = 1 - \min_r 1/(d_r x_r) < 1$ (8.2×10^{-16}) needs no measurement; then the Neumann partial sum to $K = 8$ is an *entrywise* lower bound for B^{-1} (-2.1×10^{-4}) *because the discarded terms are nonnegative*, and partial sum $+ (x_r/x_s)q_J^{K+1}/((1-q_J)d_s)$ is an entrywise upper bound (-1.4×10^{-1}), on 39 rows (20 pole-free sections + 19 border Schur blocks). Control: dropping the tail makes the upper bound false. This delivers item (2) of the T136 rest list.
- **(6) The absolute-value envelope family is certified dead** [X]**.** With $E = B^{-1}L_\Delta$, Gershgorin, every row-sum norm and every Collatz–Wielandt quotient at every *positive* weight are all $\ge \rho(|E|)$, so one rigorous *lower* bound decides the whole family at once: a Collatz–Wielandt quotient from below at a power-iterated weight gives $\rho(|E|) \ge 1.3141 \ge 1$ on 20/20 rows (the cheapest four members all ≥ 1.5937). No member of the family can ever certify $\lambda_{\max}(W) < 1$ — a certificate of *death*, not a measurement. The *signed* radius $\rho(E) = \lambda_{\max}(W) \le 0.99995781$ on the same rows: the absolute value alone destroys the cancellation, and what survives is a sign-preserving bound that does not exist yet.
- **Named limits as load-bearing content.** Every statement is *per instance on small windows* — nothing uniform in the zone index, in D or in the gap. The T136 D-exponents (anchor $D^{-0.565}$, margin gap $D^{+2.72}$, slack $D^{0.116}$) are fits and stay in the sandbox; the structure bound assembled from the support and amplitude facts does *not* beat the margin (T137: 0 of 35 windows) and is *not* promoted; entrywise positivity of the *target* inverse at depth stays open. Stieltjes / Ostrowski 1937 / Fan 1958 / Berman–Plemmons 1979 / Varga 1962 / Frobenius 1912–Perron 1907 / Collatz 1942–Wielandt 1950 / Gershgorin 1931 / Neumann series / Haynsworth 1968 / Wilkinson 1968–Higham 2002 named classical; Demko–Moss–Smith 1984 cited as a model and *not* applied (the comparison is dense); Weil 1952 cited, never used as a criterion; zero-firewall AST-checked; NOT "almost RH".
- **Diary Teile 136/137 (sandbox).** `m_matrix_pair_probe.py` (contract `M.MATRIX.PAIR`, 30/30, verdict ONE-CARRIES): the $\rho(K)$ item closes with Varga's regular splitting and the Collatz–Wielandt bracket (sharp 1.00–1.03 on 900/900), the exact three-way bookkeeping $\lambda_{\min}(A) \sim D^{2.27} =$ anchor $D^{-0.56} \times$ gap $D^{2.72} \times$ slack $D^{0.12}$ attributes the whole degradation to the margin (which is only 2.5×10^{-6}–2.5×10^{-2} wide while every cheap bound on $\rho(E)$ sits at 1.82–3.11), and M17 closes *negatively* (the bad pencil subspace is delocalised, participation ratio 0.54–0.98, its dimension tracking the atom count at $n_{\text{bad}}/n_{\text{at}} = 1.64$ rather than the grid, so $\sigma_b(k) \ge 0.646$ everywhere); promotion list 15 → 22 ready. `long_lag_probe.py` (contract `LONG.LAG`, 22/22, verdict BOTH-RESIST): L_Δ is fully comb-generated (the archimedean half has a positive off-diagonal on 0 of 40 windows), the structure bound does *not* beat the margin (0/35, overshoot 24–7.1×10^5 gaps, second differences decaying only algebraically as dist$^{-0.63}$ where Demko–Moss–Smith would need a geometric rate), the envelope family is certified dead from below ($\rho(|E|) \ge 1.3218$ on 35/35), and the Green bracket closes two-sidedly (lower recovers 0.011–0.406 of the minimum entry, upper overshoots by 2.49–12.24). Series counter: 137 probes, 3438/3438 sandbox checks. Sandbox only for the probes: no ledger row, no marker move, not RH evidence.
- **Ledger/papers/website.** Two new ledger rows: `PRIME.MMATRIX.IDENT.01` and `PRIME.LONGLAG.SUPP.01`; `\veri` citations in `tfpt_1`, `tfpt_3`, `tfpt_research_contracts` and the promotion place of `tfpt_prime_front` (verified layer now *ten* modules; phase 2 extended to T126–T137); website mirrors (`/prime-front` feed entries T136/T137, status surfaces Teile 11–137 / 3438 checks, verification DAG, papers data) updated in the same change.

2026-07-28 (I) (`v542` — the margin-chain identities of phase 2 promoted; `v379` hard-

ened to an exact positive-mixture Gram certificate; prime-front diary T132/T133 + website sync; NO marker moves)

- **New module v542 (`PRIME.MARGIN.IDENT.01`).** Promotion of the *identity/theorem block* of the discovery parts T128–T131 into one narrow load-bearing module (44 checks, $\sim$0.2 s), recomputed on small frame-A windows — nothing cited from the sandbox. Scope fence: identities and *per-instance* theorems only; no fit, no graded floor, no Lanczos value, no uniform-in-zone statement. Battery: 24 seam instances ($n = 3 \ldots 97$, $\rho = 1.25 \ldots 4.00$, $h_c = 24 \ldots 214$, $h_f = 36 \ldots 299$; every inverted or diagonalised matrix ≤ 300), 43 graded/fine grid pairs ($m \leq 172$) and 400 randomised profiles; each identity is a numerical residual against a *preregistered* tolerance plus at least one mutation control that must fail by $\geq 10^{-3}$ relative.

- **(1)–(2) Border geometry and the (T) identity [E].** $\sum_i(1-g_i) = t_l + t_r$ from the exact interval geometry (max rel 5.5×10^{-13}), so the κ weights are a probability vector and the flat border vector gives $\kappa_{\text{flat}} = 1$ *exactly* (control: a 1% perturbation of g_i breaks it by 7.9×10^{-2}); $\tau_{\text{dn}} = \|y\|^2 - m_{\text{prot}} + m_{\text{fill}} - V_{\text{bord}}$ at 2.2×10^{-16}, with τ independently recomputed from the *full* odd overlap matrix (1.1×10^{-13}); dropping any of the three correction terms breaks the identity.

- **(3)–(4) The profile identity and the curvature chain [E].** $p_{N-1} = 2 - p_0 + 2E$ at 2.3×10^{-16} (instances) / 1.4×10^{-15} (400 random profiles), hence the equivalence $2E > p_0 \iff p_{N-1} > 2$ with 0 exceptions and both branches realised (60/400); the Abel/Hölder chain $\kappa \leq 2 - p_0 + \frac{N-2}{N}\sum_j |\Delta p_j - \overline{\Delta p}| + (p_{\max} - p_{N-1})$ holds per instance with 0 violations (min slack 1.0×10^{-1} / 3.7×10^{-2}), while the *same* chain without the curvature term is violated on 88/400 profiles — the curvature term is load-bearing, not decorative.

- **(5)–(6) Assembly and the Céa/Strang defect [E].** The matrix-free two-scale assembly equals $J^\mathsf{T}QJ$ (5.2×10^{-16}) and the graded overlap operator equals J (5.2×10^{-14}), reducing to the uniform odd overlap when both partitions are uniform; $S_{\text{graded}} - S_{\text{uniform}} = R^\mathsf{T}X^{-1}R$ as a *matrix* identity on all 43 grids (2.1×10^{-14}) with a PSD defect on every grid, so the compression error is one-sided *by* the identity (controls: unnormalised merge weights 8.4×10^{-1}; dropping the $-B$ term in R 4.0×10^{-1}).

- **(7)–(9) Sandwich, Perron implication, counting chain [E].** The secular sandwich $\varepsilon/(\|u\|^2 + \varepsilon/\mu_1) \leq \lambda_{\min}(Q) \leq \varepsilon/\|u\|^2$ holds on all 48 pole-free sections at which $A > 0$ *and* $\varepsilon > 0$ are *checked* (relative width ≤ 1.4448), with Albert's sign equivalence verified in *both* directions (6 rescaled controls with $\varepsilon < 0$ all give $\lambda_{\min}(Q) < 0$); $S^{-1} > 0$ entrywise (measured 48/48) $\Rightarrow$ sign-constant *and* simple ground vector as an *implication* (tridiagonal control: hypothesis fails, conclusion fails with it); $n_{\text{run}} \leq n_{\text{cross}} + 1 \leq s_2 + 2$ and $\sum_j |\Delta p_j - \overline{\Delta p}| = \mathrm{TV}(P)$ with 0 exceptions on a non-vacuous battery (s_2 up to 31).

- **Named limits as load-bearing content.** Every statement is *per instance on small windows* — nothing is uniform in the zone index; the κ *law* is *not* claimed (T129 found it false in general), only the chain underneath it; (7) assumes nothing — positivity of the pole-free section at depth (the open M25d) is not extended and no floor value is carried; (8) has a *measured* hypothesis; (6) says nothing about the *size* of the defect (the open T130 exponent). Abel / Hölder / Schur–Haynsworth 1968 / Albert 1969 / Céa 1964–Strang / Yserentant 1986 / Perron–Frobenius–Ostrowski 1937 named classical; Weil 1952 cited, never used as a criterion; zero-firewall AST-checked; NOT "almost RH".

- **Hardening of v379 (`SEAM.S3.RP.01`).** T133 (CERT.FLOOR) certified that the 10×10 Hankel matrix `v379` assembles has, *as a double-precision object*, a negative direction at -7.75×10^{-18} (60 digits) — inside the module's old `min_eig > -1e-10` tolerance. The mathematical matrix is fine and the exact rescue is now in the module: the RP Gram is built explicitly as a *positive mixture* $M = V^\mathsf{T}\mathrm{diag}(1/1600)V$, $V_{ki} = e^{-\varepsilon_k \tau_i}$, so $x^\mathsf{T}Mx = \sum_k w_k (v_k \cdot x)^2$ is a *sum of squares* and $M \succeq 0$ holds exactly; the mixture identity is zero at the rounding level ($1.1 \times 10^{-15} \leq 64u|M|$)

and the sum-of-squares route is nonnegative on 510/510 test directions where the assembled float form is negative on 2. The eigenvalue check is *retained and relabelled a diagnostic measurement*. Ledger row `SEAM.S3.RP.01` reworded accordingly (method text: exact positive-mixture Gram identity, eigenvalue check as diagnostic); **marker stays** [E] **— no upgrade**. 7 checks (was 5), `run_all` green.

- **Diary Teile 132/133 (sandbox).** `bd_seam_probe.py` (contract `BD.SEAM`, 21/21, verdict SPECTRUM-ONLY): the model boundary-Dirac and the marks kernel share their spectrum to 7.5×10^{-13} but are *not* the same operator — the gap sits in the killing mass 0.1746 (N-stable), the killing measure equals the marks profile pointwise, the Markov remainder is exactly zero with a closed jump-weight form (odd lags positive, even lags zero — which explains T126's 86.1%), and the D_4 lattice symmetries leave the triad exactly invariant; ceiling: the DtN is a model DtN, the gate is *not* closed. `cert_floor_probe.py` (contract `CERT.FLOOR`, 23/23, verdict MIXED, 840 control validations): the `v379` finding above plus a recommendation table — the hardening is implemented in this same change. Series counter: 133 probes, 3352/3352 sandbox checks. Sandbox only: no ledger row for the probes, no marker move, not RH evidence.
- **Ledger/papers/website.** New ledger row `PRIME.MARGIN.IDENT.01`; `\veri` citations in `tfpt_1`, `tfpt_3`, `tfpt_research_contracts` and the promotion place of `tfpt_prime_front` (verified layer now *eight* modules; phase 2 extended to T126–T133); website mirrors (`/prime-front` feed entries T132/T133, status surfaces Teile 11–133 / 3352 checks, verification DAG, papers data) updated in the same change.

2026-07-27 (IV) (prime front: T111 deep ladder maps three walls, crossing measured at $n \approx 462$ — CROSSING-CONFIRMED; paper + diary + website sync; NO marker moves)

- **Diary Teil 111 (sandbox).** `experiments/tfpt-discovery/deep_zone_stress_probe.py` (contract `DEEP.ZONE.STRESS`, 23/23 checks, 140 s, verdict CROSSING-CONFIRMED): the probe stops extrapolating and measures — the T110 grid is reproduced bit-exactly, then the ladder is extended to every prime-power zone $n \leq 1000$ plus a thin tail to 3001 (199 zones; chain ladder $n = 2 \ldots 521$ with 117 handovers). The ratio m_k/need109 falls $179.59 \to 0.0174$; the crossing is *measured*: last zone ≥ 1 at $n = 461$, first below at $n = 463$ (0.935) — T110's $n \approx 170$ was a fit artefact, but the wall is real, and a depth-preserving resolution study makes $n^* \approx 462$ an *upper* bound. The failure follows the prime branch (prime powers 512/529/625/729 survive longer). New and harder: the *ladder wall* — the nested step needs a lag gap $> 2D$, and the twin-prime pair $521 \to 523$ (gap $0.003831 < 2D = 0.005617$) kills the ladder arithmetically (arithmetic scan to 40 000: 4052/4284 pairs endangered; co-moving $D \sim 1/n$ forces $h \sim n \log n/4$). The no-reserve finding softens ($f_{\text{crit}} \sim n^{-0.39}$, the reserve opens with depth); the circle breaks first at $k = 109$, $n = 463$ — it tears at the ratio, not at a step: all 117 handovers certify at retention 1.000000 (largest single loss 5.96×10^{-8}), atom entry free 117/117. Three separate walls: margin $n \approx 462$ (measured), ladder $521 \to 523$ (twin primes), requirement $n = 727$ ($\omega_{\text{cert}} \geq 1$, need109 vacuous on 46 zones). Reinterpretation: the operating variable is the window *depth*, not n (cheek doubling moves the wall $463 \to 47$); a proof must beat the exponent gap $-0.974 = -0.681 - 0.293$ and couple δ_0 to the local prime gap. Follow-up probe T112 (`adaptive_scaling_probe.py`, contract `ADAPTIVE.SCALING`, the n-adaptively scaled frame) is running. Series counter: 111 probes, 2781/2781 sandbox checks. Sandbox only: no ledger row, no marker move, not RH evidence.
- **Paper `tfpt_prime_front`.** Certification-arc section extended to T102–T111 (new subsection "The three walls (T111)"); the T110 three-gaps box and the $n \approx 170$ mentions corrected to the measured $n^* \approx 462$ (fit artefact noted); reduction-cascade caption, two-doors figure chain, three-way status split and asymptotic-mountain section updated (the three walls as the measured formulation; depth-not-n reinterpretation; T112 running); abstract and probe-index appendix

updated (T111 row; series counter 111/2781; T112 recorded as running and deliberately absent).

- **Website.** `/prime-front`: T111 feed entry (`lib/primeFront.ts`, new verdict enum CROSSING-CONFIRMED), status line in the two-doors convergence panel (T111 + T112 running), hero counter Teile 11–111 / 2781 sandbox checks, program-status callout sentence for T111.

2026-07-27 (III) (prime front: T110 margin propagation closes the measured-zone circle — MARGIN-PROPAGATES; paper + diary + website sync; NO marker moves)

- **Diary Teil 110 (sandbox).** `experiments/tfpt-discovery/margin_propagation_probe.py` (contract `MARGIN.PROPAGATION`, 28/28 checks, 7.6 s, verdict MARGIN-PROPAGATES with a precision statement): the induction circle closes end to end on the measured zones — the base case $n = 2$ is certified by an explicit Cholesky at three resolutions ($\lambda_{\min} \geq 6.93 \times 10^{-2}$, factor 88–180 over need), the graded Loewner minorant holds all 15 handovers with retention 1.0000 (the strictly scalar variant closes 0/15), and the atom entry costs nothing — it *raises* $\lambda_{\min}$ on 15/15 (first-order Kato term with helping sign; at $n = 29$ the atom-free window is not even positive, -9.2×10^{-3}). Margin map: $m_k \geq$ need109 on 16/16 zones (ratio 2.09–179.6); the growth step splits exactly (embedding error $\leq 2.6 \times 10^{-14}$; $\max|\mu N| = 0.0$ on 15/15); scalar routes killed (bordered Weyl 0/15; Schur/Friedrichs cap vacuous 15/15). Three sharp gaps remain: no reserve ($f_{\text{crit}} = 1.00$ at the first handover; retention $1 - 2 \times 10^{-7}$), no scalar step law (structurally excluded by the boundary layer), no k-uniformity (trend $m_k \sim n^{-1.93}$ against need109 $\sim n^{-0.98}$, extrapolated crossing $n \approx 170$). Follow-up probe T111 (`deep_zone_stress_probe.py`, contract `DEEP.ZONE.STRESS`, the deep ladder to $n \sim 200$) is running. Series counter: 110 probes, 2758/2758 sandbox checks. Sandbox only: no ledger row, no marker move, not RH evidence.
- **Paper `tfpt_prime_front`.** Certification-arc section extended to T102–T110 (new subsection "The margin propagates: the circle closes on the measured zones" with the three gaps as a key box); reduction-cascade caption and two-doors figure chain extended to T110; kill list extended by the T110 scalar-step-law kill; three-way status split and asymptotic-mountain section sharpened (the three gaps as the precise formulation; the $n \approx 170$ warning; T111 running); abstract and probe-index appendix updated (T110 row; T111 recorded as running and deliberately absent); symbol-table row for m_k/need109.
- **Website.** `/prime-front`: T110 feed entry (`lib/primeFront.ts`, new verdict enum MARGIN-PROPAGATES), status line in the two-doors convergence panel (T110 + T111 running), hero counter Teile 11–110 / 2758 sandbox checks, program-status callout sentence for T110.

2026-07-27 (II) (prime front: T109 boundary certificates close (R) conditionally — BOUNDARY-CERTIFIED; paper certification arc T102–T109 + diary + website sync; NO marker moves)

- **Diary Teil 109 (sandbox).** `experiments/tfpt-discovery/boundary_decay_probe.py` (contract `BOUNDARY.DECAY`, 29/29 checks, 47.2 s, verdict BOUNDARY-CERTIFIED): both remaining scalars of (R) are certified — ω *unconditionally* via a graded matrix cap in PSD order (one level per soft direction, Cholesky-verified, $\omega_{\text{cert}} = 0.2561$–$0.7569$ against measured 0.2366–0.7531, factor 1.004–1.124, no hypothesis input; the T108 compression Schur distance shrinks 0.18–0.52 → 0.0024–0.039), the boundary value via a residual / goal-oriented certificate (Prager–Synge) that *carries* the cancellation (sharpness 1.0000–2.4937, 16/16 in budget). Mechanism finding: no decay — *cancellation* (44–4.6×10^4 on 15/16 zones); the Combes–Thomas route is refuted with the exact Green row (1/16). The full chain closes 16/16 (margin 1.3–10.4, M-stable to 3000) conditional on exactly *one* input: the strict positivity margin $\lambda_{\min}(Q|_{\text{odd}}) \geq 1.7 \times 10^{-6}$–$7.5 \times 10^{-3}$, i.e. 10^2–10^6 below $\mu/2$. Follow-up probe T110 (`margin_propagation_probe.py`, contract `MARGIN.PROPAGATION`) is running. Series counter: 109 probes, 2730/2730 sandbox checks. Sandbox only: no ledger row, no marker move, not RH evidence.

- **Paper `tfpt_prime_front`.** Certification-arc section extended to T102–T109 (new subsection “Boundary certificates: cancellation, not decay”); the reduction cascade figure now ends at the one strict-positivity margin input; two-doors figure chain extended to T109; kill list K8 extended by the T109 Combes–Thomas kill; three-way status split (certified / measured / classical) updated; abstract and probe-index appendix updated (T109 row; T110 recorded as running and deliberately absent); symbol table rows for $x/h/x_0$ and $\omega/\omega_{\text{cert}}$.
- **Website.** `/prime-front`: T109 feed entry (`lib/primeFront.ts`, new verdict enum BOUNDARY-CERTIFIED), status line in the two-doors convergence panel, hero counter Teile 11–109 / 2730 sandbox checks, program-status callout sentence for T109.

2026-07-27 (I) (new companion `tfpt_prime_front` — the prime / zeta line documented end to end as RESEARCH DOCUMENTATION; NO new claims, NO marker moves)

- **New companion paper `tfpt_prime_front` (P).** 22 pages, 9 sections plus two appendices, 8 TikZ figures. It documents the complete prime / zeta line of the project from the first $\mu_4 \Rightarrow \chi_4$ observation to the current residual — explicitly a *documentation of a research route*, not a results paper. Registered in `tex-artefacts/tfpt_docset.tex` as the fifth companion.
- **Framework interpretation stated once, up front.** Section 3, “The big picture”, collects the reading that motivated the line and labels it as interpretation: the E_8 shell count $N(2n) = 240\sigma_3(n)$ is the Dirichlet series $240\,\zeta(s)\zeta(s-3)$; the μ_4 glue carries χ_{-4}, so the $4k{+}1$ / $4k{+}3$ split is a fibre property of the compiler and the signed channel is the only pole-free one; Hecke structure is emitted by lattice adjacency (v535) rather than put in; E_8 is unimodular, so Poisson summation gives the theta sum an exact functional equation whose fixed axis is the critical line — the same reflection that reappears as $J_1 \circ J_{1/2} = e^{\pm u}$ (the transport operator, T83) and as the parity split of the Weil pole (T106). Explicitly typed as a picture, not a reformulation, and not used as a premise anywhere.
- **Verified layer restated, not extended.** Sections 4.1–4.7 restate the seven load-bearing modules v535–v541 with their ledger status markers quoted verbatim (Hecke from geometry, Eichler trace layer, half-integral bridge, relative-trace identity, Weil structure family, amplitude linear carrier, matching-lemma / transport ledger). The `\veri` citations in this paper are the only citations it makes; no ledger row is added, moved or upgraded.
- **Sandbox layer marked as sandbox throughout.** The localization program (I5, the Connes–Consani dictionary, the tightness map, the Sonin crossing, the blind prime demo), the handover induction and the certification arc are carried as clearly tagged *sandbox* content with a per-section provenance marker; every quantitative statement carries its probe label (Tn) and is reproduced verbatim from the research diary. All measured quantities are declared as measured, all window results as window results, all hypothesis inputs as hypothesis inputs.
- **Kill list as a first-class section.** Section 8 records the refuted readings and dead routes of the program (essential-singularity reading, the C/g proxy law, the naive margin route, the density chain, the Landau–Widom anchor, the accumulated invariant with self-propagation, the symbol route for the odd channel, three certificate candidates for the avoidance norm), each with the probe that killed it.
- **Explicit RH fence.** The paper states in the abstract, in the scope-and-status statement, in the reduction figure and in the status section that *no progress on the Riemann Hypothesis is claimed*; the I5 $\Leftrightarrow$ Weil positivity $\Leftrightarrow$ RH relation is carried, as in the ledger, as an equivalence *typing* only. `ZETA.HP.CARRIER` untouched.
- **Appendices.** Probe index T11–T108 (98 parts, 99 runs, 2509 green checks; the diary’s own series counter at T108 reads 108 probes / 2701 checks including the ten parts that predate the

current diary file) with contract, verdict and check count per probe, plus a symbol table for the sandbox objects used in the figures. T109 (`boundary_decay_probe.py`) was running when the document was written and is deliberately absent.

2026-07-25 (XIX) (v541 — matching lemma and transport ledger package: two named limits as content; NO marker moves)

- **New module v541 (RTF.GNS.LEDGER.01).** Promotion of the discovery proof package T78/T79/T80/T81/T82/T85 into one load-bearing module (33 checks, $\sim 10\,\mathrm{s}$); every core check is *recomputed* from scratch, not cited. Companion to `v538/HECKE.GEOM.RTF.01`, `v539/RTF.GNS.WEIL.01` and `v540/RTF.GNS.AMP.01`. Classical Gronwall 1913 / Robin 1983 *unconditional* / Cohen 1975 / Hecke 1918–1920 / Cornacchia / Fermat–Gauss / Dirichlet $L(1,\chi)$ / Mertens-AP / Landau 1908 / Legendre duplication / Weil 1952–Guinand–Bombieri / Alaoglu–Erdős named classical — NEW is the compiler-native consolidation and the machine-checked window proofs.
- **(A) Window proof [E].** The matching lemma is *proved exact-integer on* $[4,10^6]$: $7S(j) < 40A(j)$ at every atom (0 violations over 939 870 clash atoms; full enumeration, no-overflow proven, big-integer rechecks); exact margin $X = 22242575693/270725400000 = 0.082159$ at $j^* = 554400$; $\rho_{\mathrm{crit}} = 1.1440 > 21/20$; maximal-greedy identity $\Rightarrow$ uniform in the target set and the greedy weights. Structure laws at 0 tolerance: 8-law ($n \le 250000$), seed-tower + ψ w-table ($n \le 50000$), Cohen seeds $\{-48, -80, -8, -8\}$ ($s \le 2000$), bracket $\Theta \le 2|\psi| \le 3\Theta$ on 10^6.
- **(B) Transport ledger [E].** $Q_{\mathrm{Weil}} = Q_{\mathrm{cert}} + \Delta_{\mathrm{arch}} + \Delta_2$ closes with $\Delta_{\mathrm{pole}} \equiv \Delta_{\mathrm{conv}} \equiv 0$ *proven* (sympy kernel collapse; quadrature 2.8×10^{-16}; exact scale invariance); ledger residual 4.4×10^{-16} on 24 battery rows; the odd-prime side of the Weil functional equals the certified plus combination $P_\zeta(g_-) + P_\zeta(g_+)$ exactly (rel 3.9×10^{-16}).
- **(C) Character-exact signed envelope [E].** $-\psi = (\chi_{-4} + \frac{1}{4}\chi_8 + \frac{1}{4}\chi_{-8}) \cdot \Theta$ coefficient-exact on all odd $n \le 50000$ (both builds); signed certificate 0 violations, margin factor 8.86×; confinement *set equality* on 10^6: zero-credit clash atoms = χ_{-4}-coherent odd composites.
- **(D) Archimedean term internal [E].** Legendre duplication bridge $(2\pi)^{-s}\Gamma(s) = \frac{1}{2}\Gamma_{\mathbb{R}}(s)\Gamma_{\mathbb{R}}(s+1)$ (sympy + 1.8×10^{-41}); kernel identity $k_\zeta = K_{\mathrm{fam}} - K_{\mathrm{shift}}$ pointwise 7.9×10^{-31} on 51 points; $\Delta_{\mathrm{arch}}(h) = A_{\mathrm{fam}}(h) - A_{\mathrm{shift}}(h)$ on a 10-row battery subset at rel 9.9×10^{-16}.
- **(E) λ-equivariant channel [E].** CM carrier $g_\lambda = \sum_{\mathfrak{a}} \lambda_1(\mathfrak{a}) q^{N\mathfrak{a}}$ exact in two independent routes ($\mathbb{Z}[i]$ lattice vs Cornacchia reconstruction, 0 mismatches $n \le 10^4$); CM laws exact; $\mathrm{supp}(c_1) = \{\mathbb{Z}[i]\text{-norms}\}$ as a set equality; canonical phase mean $\mu_1 = c_1/(c_0 d^2) \in [-1,1]$ exact; λ-window certificate 0.0653 vs 0.2365 unlifted (3.6×), exact Fraction rechecks.
- **(F) Frontier facts [C].** Unlifted coherent chain crossing $k^* = 14$ (exact Fractions; $\log_{10} N^* = 23.1$); absolute Robin tail factor 6.16 at $J = 10^6$ and divergent — the constant route stays open (window constants with declared classical all-n extension).
- **Named limits as load-bearing content.** (i) *One* classically-shaped open lemma remains — the correlated-cancellation lemma on credit-rich non-coherent tail atoms (uniform form); provably-shaped $\neq$ formal proof. (ii) I5 in *one-family* form ($Q_{\mathrm{cert}} + \Delta_2 + A_{\mathrm{fam}} - A_{\mathrm{shift}} \ge 0$) is the single remaining TFPT-specific object — by the closed ledger *equivalent* to Weil positivity $\Longleftrightarrow$ RH (an equivalence *typing*, no progress claim). NOT "almost RH"; `ZETA.HP.CARRIER` untouched; no marker moves.
- **Ledger/papers/website.** New ledger row `RTF.GNS.LEDGER.01`; `\veri` citations in `tfpt_1`, `tfpt_3`, `tfpt_research_contracts`; Wolfram extension mirrors the exact algebraic identities; website mirrors (`/prime-front` feed, verification DAG, papers data) updated in the same change.

2026-07-25 (XVIII) (`v540` — amplitude route and positive linear carrier: open boundary λ^* as content; NO marker moves)

- **New module `v540` (`RTF.GNS.AMP.01`).** Promotion of the discovery amplitude chain T67/T68/T69/T70/T71/T72 into one load-bearing module (34 checks, ~ 3 s). Companion to `v538/HECKE.GEOM.RTF.01` and `v539/RTF.GNS.WEIL.01`: the machine-checked route out of the square plane behind the two v539 obstructions. Classical Cohen 1975 / Siegel–Weil / Jacobi–Fricke / Landen / Hecke split-Mellin / Cauchy–Littlewood / Möbius square sieve / Shimura / Shintani / Weil 1952 / Farkas named classical — NEW is the compiler-native consolidation.
- **(A) Amplitude Dirac [E].** $D = \left(\begin{smallmatrix} 0 & V \\ V^\top & 0 \end{smallmatrix}\right)$ with $V[d,m] = \sqrt{w_d}\, b(d)\chi_d(m)$: $D = D^\top$; $D^2 = \mathrm{diag}(VV^\top, K)$, $V^\top V = K$ (rel 2.7×10^{-16}); spectrum $= \pm$ singular values; Hecke intertwining exact ($p = 3, 5, 7$); Shimura $b(dp^2) = b(d)(a_p - \chi_d(p)p)$ integer-exact on 644 pairs.
- **(B) Geometric polarisation + Cohen seed [E].** $b = N_+ - N_-$, $\Theta = N_+ + N_-$ lattice-exact; $N_\pm \geq 0$ and $b^2 = \Theta^2 - 4N_+N_-$ for all $n \leq 50000$; Θ is a *pure* Siegel–Weil Eisenstein σ_3-eigenform ($p \leq 13$, cusp component zero); Cohen seed $\Theta(d) = -48\, L(-1, \chi_d)$ exact-rational on 159 live d.
- **(C) Deletion universal = square-class double counting [E].** Cauchy–Littlewood window lemma (free Satake symbols, $k \leq 6$): numerator $1 - stuv\, X^2$, hence $1 - p^6 X^2$ pair-independent at $st = uv = p^3$ (five channels); and fam $[1, 0, 2, 0] + 2\flat\, [0, 2, 0, 2] + \delta_{k1} = [2, 2, 2, 2]$ generating-function exact (Möbius square sieve exact $n \leq 2000$).
- **(D) Positive linear carrier + plus balance [E].** $\ell^2(d, \mu)$ with $\mu = 48\, |L(-1, \chi_d)|\, |d|^{-a}$: full weights $\lambda_k = 1 + p^{3k} - (\chi p)^k$ (layer $[1, 1, 1, 1]$); $Q = Q_\zeta(g_-) + Q_\zeta(g_+)$ (rel 2×10^{-15}); the $\flat$/doubling kernel enters *with plus* (sign inverted vs the v539 square-plane minus).
- **(E) Functional equation exact [E].** $\Theta(i/(8y)) = 8y^{5/2}\Theta^\dagger(iy)$ (rel 6×10^{-41}, Fricke closed); $\Lambda_\Theta(s) = 8^{1-s}\Lambda_{\Theta^\dagger}(5/2 - s)$ in the strip (rel 1.4×10^{-40}, independent split points); poles $5/2$ / 0 swapped; centre $5/4$, plus-locus offset exactly $1/4$.
- **(F) Cone facts + named open boundary [C].** Guaranteed cone and library hull FE-self-dual, gap functional λ^* FE-covariant (36/36 sampled); mirror sign law $\mathrm{sign}\,\psi(n) = (-1)^{\lfloor n/2 \rfloor + 1}$ exact ($n \leq 50000$); constraint class $C_{L3} = \{n \equiv 6 \bmod 8\}$; λ^* exists with one exact Farkas witness ($\lambda^*_{L3} = 0.0511$ at $n^* = 6$); coverage saturates — the residual distance to the Weil cone *is* λ^* on $n \equiv 6 \bmod 8$.
- **Fences.** Euler-region positivity only (edge L-values, not the central line); not "almost RH"; λ^* is the named open boundary *inside* the claim; `ZETA.HP.CARRIER` untouched; no marker moves.
- **Surfaces.** Ledger `RTF.GNS.AMP.01`; \veri in `tfpt_1/tfpt_3/tfpt_research_contracts`; Wolfram extension +5 ($613 \to 618$); website `papers.ts` + VerificationDag E8 + `primeFront.ts` + ScriptIndex via `bash build.sh gen`.

2026-07-25 (XVII) (`v539` — Weil structure of the compiler family: two isolated obstructions; NO marker moves)

- **New module `v539` (`RTF.GNS.WEIL.01`).** Promotion of discovery probes T55/T61/T62/T63 (Route-B chain) into one load-bearing module (25 checks, ~ 6 s). Companion to `v538/HECKE.GEOM.RTF.01`: the compiler family carries a fully identified Weil structure up to *two explicitly isolated obstructions* (obstructions are prominent verified content, not a footnote). Classical Weil 1952 / Gelfand / direct integrals / SU(2)–Chebyshev / Sato–Tate / 2^ω-series / GNS named classical — NEW is the compiler-native identification and the machine-checked isolation of the two obstructions.
- **(A) GNS fibre structure [E].** Metric $\ell^2(d, b^2/|d|)$ diagonal; fibres $\sigma(d) = (\chi_d(3), \ldots, \chi_d(13))$ partition the live family (1191 fund. $d \leq 12000$); Λ_{fam} exact constant on each fibre; twist-mix

per fibre $\leq 5.037 \times 10^{-16} < 10^{-12}$.

- **(B) Isotype / GL(1) core [E].** Character expansions $\Phi_1 = \chi_0+\chi_2$, $\Phi_2 = \chi_4-\chi_2$, $\Phi_3 = 2\chi_0-\chi_4+\chi_6$, $\Phi_4 = \chi_8-\chi_6$; core series $c_{k,0} = [1,0,2,0,2,0]$; $G_0 = (1+Y)/(1-Y) = \zeta_p(w-3)^2/\zeta_p(2w-6)$; global $\zeta(u)^2/\zeta(2u) = \sum 2^{\omega(n)}n^{-u}$ (Dirichlet exact through $n \leq 2000$).
- **(C) Weil linear relation [E].** $\mathrm{Prime}_F = 2P_\zeta(g)-2P_\zeta(g_b)$ (max rel 9×10^{-16}); $Q_{\text{fam}} = Q_F+\text{Corr}$ (rel 2×10^{-16}).
- **Obstructions as verified content [E].** (1) Doubling enters with a *minus* (family positivity does *not* imply $Q_\zeta \geq 0$). (2) Plancherel Corr non-automorphic: polar/arch absorption FAIL; finite Euler factor FAIL; $e^{-\sum p^{-u}}$ PASS.
- **Finite-class positivity [C].** $Q_{\text{fam}} \in [4.369, 11.486]$ on the finite test class; dense-class positivity open / RH-adjacent — *not* claimed.
- **Fences.** Not "almost RH" — the claim structurally shows why family positivity does not imply RH; `ZETA.HP.CARRIER` untouched; no marker moves.
- **Surfaces.** Ledger `RTF.GNS.WEIL.01`; \veri in `tfpt_1/tfpt_3/tfpt_research_contracts`; Wolfram extension +5 (608 → 613); website `papers.ts` + VerificationDag E8 + `primeFront.ts` + ScriptIndex via `bash build.sh gen`.

2026-07-25 (XVI) (v538 — compiler relative-trace identity: three projections of one finite RTF; NO marker moves)

- **New module v538 (`HECKE.GEOM.RTF.01`).** Promotion of discovery probe T53 (`relative_trace_compiler_probe.py`) into one load-bearing module (18 checks, ~ 14 s). Companion synthesis of `v535/HECKE.GEOM.01`, `v536/HECKE.GEOM.EICHLER.01` and `v537/HECKE.GEOM.HALFINT.01`: these are three *projections* of one finite relative-trace identity [E_8/glue orbital counting] = [Newform/Waldspurger spectral sum], not three separate results. Classical Jacquet-type RTF / Eichler / Witt / Waldspurger / theta correspondence named classical — NEW is the compiler-native realisation and the machine-checked bilateral independence.
- **(R1) Trace identity, both sides independent [E].** $\mathrm{Tr}_V(\nu_p \circ \pi)$ geometric ($a_p = -c(p)/8$ from the glue census, no modular input) equals spectral (a_p from $\eta(2\tau)^4\eta(4\tau)^4$, no geometry) for all $(p, \pi) \in \{3, 5, 7\} \times \{\mathrm{Id}, \pi_{\text{Eis}}, \pi_{\text{cusp}}, \pi_{E_4}, \pi_{f_8}\}$. Anchors (Id, Eis, cusp, E_4, f_8): p=3: $(6304, 5600, 704, 1120, 352)$; p=5: $(105848, 98280, 7568, 19656, 3784)$; p=7: $(727680, 688000, 39680, 137600, 19840)$.
- **(R2) Orbit dictionary [E].** The Eichler split $\lambda_{\text{geom}} = \lambda_{\text{Eis}} + a_p^2$ *is* the RTF orbit decomposition (principal orbit $a \cdot \mathrm{Id} \to \lambda_{\text{Eis}} = L - \sigma_3^2$; elliptic $b \cdot T_p \to a_p^2$; cross terms cancel). Anchors $\lambda_{\text{Eis}} = 336/3780/19264$, $a_p^2 = 16/4/576$.
- **(R3) Period side is lattice counting [C].** g is the quaternary form $n = (x^2+y^2)/2+2z^2+u^2+2w^2$ (x, y odd; sign $(-1)^{|u|+|w|}$); $b_{\text{rep}} = b$ exactly through $n \leq 200$; then [signed lattice count]$^2 = R \cdot |d|^{3/2} \cdot L(f_8 \times \chi_d, 2)$ with $R = 23.1873585645$ (rel $\sim 10^{-12}$). Verdict `ONE-FORMULA`.
- **Fences.** Identity is *finite* ($p \in \{3, 5, 7\}$, finite d-windows); the *infinite* RTF (sum over all p and d + archimedean term) is the named open step; no RH; not `ZETA.HP.CARRIER`-relevant; no marker moves.
- **Surfaces.** Ledger `HECKE.GEOM.RTF.01`; \veri in `tfpt_1/tfpt_3/tfpt_research_contracts`; Wolfram extension +5 (603 → 608); website `papers.ts` + VerificationDag E8 + `primeFront.ts` + ScriptIndex via `bash build.sh gen`.

2026-07-24 (XV) (v537 — half-integral bridge: unique Shimura preimage + $T(p^2)$ Hecke equivariance + Kohnen scope fence + Waldspurger/Baruch–Mao constancy; NO marker moves)

- **New module `v537` (`HECKE.GEOM.HALFINT.01`).** Consolidated promotion of discovery probes T38/T41/T44/T45 into one load-bearing module (20 checks, $\sim$ 90 s). Companion to `v535/HECKE.GEOM.01` and `v536/HECKE.GEOM.EICHLER.01` without duplicating weight-4 Kneser/Eichler checks. Classical Shimura/Kohnen/Waldspurger/Baruch–Mao named classical — the claim is the unique signed-scale monoid preimage, the in-suite chain, and the *measured* constant R.
- **(A) Unique Shimura preimage at scale -8 [E].** In the 70-element compiler weight-5/2 theta-monoid exactly one object satisfies $\mathrm{Sh}_{t=2,\psi=1}(g) = -8\, f_8$ (witness $g = \vartheta_2(q^2)^2\, \vartheta_3(q^2)\, \vartheta_4\, \vartheta_4(q^2)$, verified through $n \le 120$); three further monoid elements lift to scales $+8, -16, +16$.
- **(B) Hecke equivariance [E].** Exact $T(p^2)$-eigenform, trivial nebentypus, with $\lambda = a_p(f_8)$ for $p = 3, 5, 7, 11, 13$ $(-4, -2, 24, -44, 22)$; trial FE $N = 512$, $\varepsilon = +1$ (centre $5/4$).
- **(C) Kohnen scope fence [E].** $U_4(g) = 0$ (mass mod 4 on $\{1, 2\}$); level $32 = 4 \cdot 8$ outside Kohnen 1982 — plus-route structurally closed (negative result part of the claim).
- **(D) Waldspurger/Baruch–Mao constancy [C].** AFE twist L-values ($\varepsilon_d = \chi_d(8)$); $R(d) = 23.1873585645\ldots$ constant on ten fundamental $d \equiv 1 \pmod 8$ (spread $\le 10^{-12}$, including $d = 89$); $d \equiv 5 \pmod 8$ vanishes.
- **Fences.** GL(2) centre $s = 2$, not the ξ-line; no RH; `ZETA.HP.CARRIER` untouched; no marker moves.
- **Surfaces.** Ledger `HECKE.GEOM.HALFINT.01`; \veri in `tfpt_1/tfpt_3/tfpt_research_contracts`; Wolfram extension +5 ($598 \to 603$); website `papers.ts` + VerificationDag E8 + ScriptIndex via `bash build.sh gen`.

2026-07-24 (XIV) (v536 — Eichler trace layer at the E_8 lattice: Witt λ_{Eis} + two-sided Eichler identity + Type-A/B local densities + signed a_p; NO marker moves)

- **New module `v536` (`HECKE.GEOM.EICHLER.01`).** Consolidated promotion of discovery probes T33/T36/T37/T42 into one load-bearing module (23 checks, $\sim$ 30 s). Builds on `v535/HECKE.GEOM.01` (Kneser counts, ν_p law, oldforms) without duplicating those checks. The claim is *not* RH-related: the Eichler/Witt package of the μ_4-glue census is lattice-native and two-sided in-suite.
- **(A) Witt layer [E].** $\lambda_{\mathrm{Eis}} = \sigma_3(\#\mathbb{P}^3 - \sigma_3) = L - \sigma_3^2$ identically (sympy) and for all 24 odd primes $p \le 100$; anchors $\lambda_{\mathrm{Eis}}(3, 5, 7) = (336, 3780, 19264)$.
- **(B) Eichler identity [E]/[C].** $\lambda_{\mathrm{geom}} = \lambda_{\mathrm{Eis}} + a_p^2$; type-(i) $N_\perp = (p^6 - 1)/(p - 1)$ live constant on all 240 roots at $p = 3, 5$; type-(ii) live at $p = 3$ ($352 = 364 - 12$) with Kneser filter = identity; $p = 5$ T36-validated freeze ($3784 = 3906 - 122$, $R = 4$; Shell(25) re-enumeration budget-gated); assembler $R(p) = \sigma_3 - 1 - c(p^2)/8 = a_p^2$; $R(7) = 576$ prognosis (Shell(49) out of scope).
- **(C) Local densities [E].** Type-A = isotropic cosets representing the integer p; $N_A = \min(240(1+p^3),\, p^7 + p^4 - p^3 - 1)$, $N_B = (\#\mathrm{iso} - 1) - N_A$; B empty iff $p^4 < 241$ (only $p = 3$); $p = 7$ live 82560/743040; injectivity at $p = 11, 13$; fibre refinement scoped to $\{5, 7, 11\}$.
- **(D) Signed extraction [E].** $a_p = -c(p)/8$ from geometric Shell(p): $(-4, -2, +24)$ at $p = 3, 5, 7$; $b = \sigma_3 + a_p \in \{24, 124, 368\}$ (hooks the ν_p law of `v535`).
- **(E) $O(1)$ assembler [E].** $\lambda_{\mathrm{geom}}(p)$ and $R(p) = a_p^2$ for all odd $p \le 31$ against the eta product f_8 (Ramanujan bound included).

- **Fences.** Classical Eichler/Siegel/Witt named classical — the claim is in-suite two-sidedness (mod-p geometry left, eta product right) and the closed compiler densities; no RH statement; Shell(p^2) for $p \geq 7$ out of scope; weight-4 →GL(1) stays [O]; no marker moves of any pre-existing contract.
- **Surfaces.** Ledger `HECKE.GEOM.EICHLER.01`; \veri in `tfpt_1/tfpt_3/tfpt_research_contracts`; Wolfram extension mirrors (exact Witt/density/assembler/a_p identities); website `papers.ts` + VerificationDag E8 + ScriptIndex via `bash build.sh gen`.

2026-07-24 (XIII) (v535 — Hecke from geometry: Kneser correspondence + affine neighbour-sum + 2-adic oldform recovery on the E_8 census channels; NO marker moves)

- **New module v535 (`HECKE.GEOM.01`).** Consolidated promotion of discovery probes T27/T31/T32 into one load-bearing module (25 checks, ~ 11 s). The claim is *not* RH-related: the Hecke structure of the μ_4-glue census is lattice-native.
- **(A) Kneser carries Hecke [E].** #iso_pts $= p^7 + p^4 - p^3$ and #iso_lines $= \sigma_3(p)\,\#\mathbb{P}^3(\mathbb{F}_p)$ identically and by full enumeration at $p = 2, 3, 5, 7$ (135/1120/19656/137600 lines) — the Eisenstein eigenvalue $\sigma_3(p) = 1 + p^3$ is an exact factor of a lattice-native correspondence count.
- **(B) Neighbour-sum = affine Hecke element [E].** $\nu_p = a\,\mathrm{Id} + b\,T_p$ exactly (overdetermined, residual 0) with $b = \sigma_3 + a_p$, $a = \#\mathrm{lines} - b\sigma_3$; cusp rule $a_p = b - \sigma_3$ recovers $a_3 = -4$, $a_5 = -2$ ($p = 3$ live orbit-reduced; $p = 5$ T31-validated freeze of $S[1]$ with $\lambda_{\mathrm{odd}} = 3784$); $p = 7$ consistency $\lambda_{\mathrm{odd}} = \#\mathrm{lines} - \sigma_3^2 + a_p^2 = 19840$; placebo fails.
- **(C) Redundancy = 2-adic oldform structure [E].** $\dim_{\mathbb{Q}} V = 7 = 5$ (E_4-copies, levels $1, 2, 4, 8, 16$) $+2$ (f_8-copies, levels $8, 16$); every Θ_j an explicit $\mathbb{Q}$-linear combination; recovery $\pi_{\mathrm{cusp}} = (28 - T_3)/32$, $\pi_{\mathrm{Eis}} = (T_3 + 4)/32$, newform via $(1 - V_2U_2)$; new quotients each dimension 1; level census purely 2-adic.
- **Fences.** Classical theorems (Kneser, Hecke, Atkin–Lehner, multiplicity one) named classical — the TFPT content is the in-suite mechanics. No RH statement. Weight-4 restriction named: GL(1)/weight-drop transport stays [O]. No marker moves.
- **Surfaces.** Cited in `tfpt_1`, `tfpt_3`, `tfpt_research_contracts`; ledger row `HECKE.GEOM.01`; Wolfram extension +5 exact mirrors (588 → 593); website sync.

2026-07-24 (I) (v534 — the straddle cone executed: THE FIRST DYNAMICAL SELECTION OF THE ALIGNMENT BIT — reflection positivity keeps exactly ONE member of the interacting mark family alive, $\delta = \pi/2$ with positive coupling; the literal leading-order cone formalization dies; NO marker moves)

- **New module v534 (`SEAM.STRADDLE.CONE.01`).** The 2026-07-23 review-round contract (recorded "not promoted" in entry (X)) is *executed*, double-verdicted. **V2 = ERFOLG — the selection law (the central datum):** with independent couplings λ_q on the four mark quartets of $M(\delta = m\pi/8)$ and the equivariant (uniform) direction, reflection positivity holds on *every* admissible cut over the whole grid $g \in \{1/32..8\}$ for *exactly one* member × sign: $\delta = \pi/2$ ($m = 4$) with *positive* coupling — all four Grams PD $(37, 0, 0)$ *including the previously untested straddled clock axes* 7/15, minimum eigenvalue *lifted* off the free RP boundary ($1.78 \cdot 10^{-6} \to 5.5 \cdot 10^{-5}$ near $g = 1/4$: the $\pi/2$ interaction pushes *away* from the boundary), Gibbs $\beta = 2$ identical [E]. The asymmetric members die: $m = 2/6$ with $s = +$ already at $g = 1/32$, with $s = -$ from $g = 1/8$. The v529 straddle law *refines*: asymmetric straddling kills RP, the *symmetric* straddling of the self-mirror member protects it.
- **V1 = KILL (the literal cone formalization).** The leading-order soft-block cone $K_{\mathrm{RP}} = \mathrm{Fix}(G_\delta) \cap \bigcap_c \{\lambda : G_c^{\mathrm{lead}} \succeq 0\}$ is *empty* for every member and both signs (first-order PT via

the exact reduced resolvent, Hermitian to 10^{-15}, central-difference regression $2 \cdot 10^{-8}$) — the linear pencil cannot carry the selection; the protection at $\pi/2$ is *nonperturbative*. Recoverable leading-order content: the straddled-cut drift signs are exact ($\pi/2$: $+1.4 \cdot 10^{-4}$ for $s = +$, $-3.7 \cdot 10^{-2}$ for $s = -$; asymmetric members reversed) — necessary, not sufficient.

- **The exact geometry behind it [E].** $\delta = \pi/2$ is the *unique symmetric-straddle member*: stabilizer = the full clock+mirror group (rotations $\{0, 4, 8, 12\}$, reflection axes $\{3, 7, 11, 15\}$ = four admissible bond cuts, straddled $\{7, 15\}$), the four quartets *partition* the 16-circle, and every straddling quartet is reflection-*closed* across its cut with exact sign $+1$; $m = 2/6$ straddle asymmetrically; odd m carry no bond axis (the typed v521/v525 placement artifact). $\mathrm{Fix}(G_\delta)$ is 1-dimensional throughout: uniform on $m = 2..6$, NS-wrap-*twisted* $(1, -1, 1, 1)/(-1, -1, -1, 1)$ on the tight members $m = 1/7$. Typed cross-link: the RP-selected member $m = 4$ is exactly the carrier of the v528 twist-class order parameter $O = 1/2$ — the dynamical selector and the kinematic definition point at the *same* member.
- **Honest fences [C].** One toy, one interaction class (FK quartets), flat-band parent, deg ≤ 2 OS bases, finite coupling grid — toy-level evidence *for* the bit's dynamical origin, *not* a derivation; nothing about the continuum $\mathcal{A}_{\mathrm{hol}}$ is claimed; `WOIT.OS.TWISTOR.01` stays Open, NO marker moves. Transparency: the run-1 preregistered verdict logic conflated formalization and substance (UNENTSCHIEDEN, mismatch typed); run 2 split V1/V2 with no tolerance change and no number changed — note kept verbatim in the probe. Discovery provenance: `experiments/tfpt-discovery/seam_straddle_cone_probe.py` (15/15, 2026-07-24).
- **Wolfram mirror.** Four exact mirrors (stabilizer census, $\pi/2$ partition + straddle incidence, reflection-closure signs, equivariant Fix rays incl. the wrap-twisted $m = 1/7$ rays) in `wolfram/tfpt_readouts_extension.wl`, verified engine run 2026-07-24, 584 → 588 (the 256-dim Fock spectra, PT corrections and finite-g scans stay Python-only, flagged); suite 527 → 528 modules.

2026-07-23 (X) (v530–v533 — the external review round promoted: center quotient compiler, Coxeter Stage A tensor clock, dual-degree modular checksum 744, sheet-transfer disc −7 norm line; every identity machine-confirmed exactly before promotion (probe 76/76); NO marker moves)

- **New module v530 (`DIAMOND.CENTER.QUOTIENT.01`).** The Sheet-Diamond center $C = M(1, 0)$ carries the fixed primitive line $Cv = v$ ($v = (-1, 1, 0)$), and the induced operator on $\mathbb{Z}^3/\mathbb{Z}v$ is *exactly* the atom matrix $A = \left(\begin{smallmatrix} 8 & 2 \\ 5 & 3 \end{smallmatrix}\right) = \left(\begin{smallmatrix} \mathrm{rank}\, E_8 & |\mathbb{Z}_2| \\ g_{\mathrm{car}} & N_{\mathrm{fam}} \end{smallmatrix}\right)$ [E] with $\mathrm{tr}\, A = 11$, $\det A = 14$, $\mathrm{disc} = 65 = 5 \cdot 13$, $\mathrm{perm}\, A = 34 = p_5(a)$, $\prod A_{ij} = 240$, $\det(A - I) = 4$. The center *self-codes*: row sums $C\mathbf{1} = (7, 11, 13) = (\det A/|\mathbb{Z}_2|, \mathrm{tr}\, A, \mathrm{disc}/g_{\mathrm{car}})$; with the fixed eigenvalue 1 this is $(1, 7, 11, 13)$ = the CP-halved E_8 Coxeter exponents. The v414 resolvent ladder factorises: $\det(C + kI) = (k+1)\det(A + kI)$ identically, quotient ladder $(14, 26, 40, 56)$, differences $(12, 14, 16)$, endpoint $224 = 248 - 24$. *Honest correction of the review:* eigenvalue 1 is NOT unique to C (K, Q share it) — uniqueness holds for the *conjunction* fixed line + atom quotient + CP self-code; the full selector-stratum ablation stays [O]. New paragraph in `tfpt_2` (Sheet Diamond section).
- **New module v531 (`COX.STAGEA.TENSOR.01`).** The `tfpt_1` two-stage Coxeter audit (Stage A: a natural order-30 operator with $\chi = \Phi_{30}$ on the even-flip atlas *or its 8-dim primitive-character space*) is *closed on the character-space alternative*: $T_{30} = \mathrm{Comp}(\Phi_5) \otimes \mathrm{Comp}(\Phi_6)$ has exact order 30 and $\chi_{T_{30}} = \Phi_{30}$ [E] (CRT, $\gcd(5, 6) = 1$: carrier clock × family clock = Coxeter clock as *one integer operator*); Ramanujan trace tomography $\mathrm{tr}(T_{30}^k) = c_{30}(k) = c_5(k)c_6(k)$ for all k, signed table $(-1, 1, 2, 4, -2, -4, -8, 8)$, atom readout $(2, 4, 8) = (|\mathbb{Z}_2|, |\mu_4|, \mathrm{rank}\, E_8)$; honest kill scope ($\Phi_{15}$ passes the unsigned triple — the *signed* table discriminates; $\Phi_{16}/\Phi_{20}/\Phi_{24}$ die on the triple). The flip-atlas intertwiner (`COX.FLIP.INTERTWINER`) and Stage B stay [O]; `tfpt_1` Stage A prose updated.

- **New module v532 (`E8.DEGREE.MODULAR.01`, audit-only).** The dual products of the E_8 invariant degrees, $(60, 192, 240, 252)$, satisfy simultaneously $\sum = 744 = 3 \cdot 248$ (= the j-function constant, recomputed from the q-series E_4^3/Δ), $\prod = |W(E_8)|$, $\gcd = 12$, $\frac{1}{12}(\cdot) = (g_{\mathrm{car}}, \dim S^+, \det L, 7N_{\mathrm{fam}})$ [E]. *Mandatory fence:* D_{16} passes the same checksum ($1488 = 3 \cdot 496 = 2 \cdot 744$ — heterotic, NOT E_8-exclusive); six non-heterotic controls fail. *Prime-totative fingerprint (audit):* $\{2, 3, 5\} \cup (\mathrm{Exp}(E_8) \setminus \{1\})$ = all primes below 30, and 30 is the *largest* integer whose totatives are all 1 or prime (scan to 10^4) — h_{E_8} as the maximal prime-totative Coxeter number; per the no-free-pattern rule nothing here is load-bearing. New keybox in `tfpt_3`, cross-note in the `tfpt_1` 496 box.
- **New module v533 (`FTR.DISC7.NORM.01`).** The branch-preserving transfer path $J, K, C, F = M(1, -2..1)$ is a *norm line*: $\det M(1, t) = 4t^2+14t+14 = ((4t+7)^2+7)/4 = N(\alpha_t)$ with $\alpha_t = (4t+7+\sqrt{-7})/2$ integral in $\mathbb{Z}[(1+\sqrt{-7})/2]$, the path one translation $\alpha_{t+1} = \alpha_t + 2$, norms $(2, 4, 14, 32)$, constant discriminant -7, vertex $7/4 > 0$ [E]; 2 splits in $\mathbb{Q}(\sqrt{-7})$ ($-7 \equiv 1 \bmod 8$, class number 1); controls: winding axis linear, anchor block disc $+105$. [C]/[O] preregistered next test: do the four external transfer solvers share *one* fractional-linear action on α_t (kill: four incompatible actions)? `GATE.FTRANSFER` unchanged. New paragraph in `tfpt_2` + frame note in `tfpt_research_contracts`.
- **Wolfram mirror.** All four modules mirrored exactly in `wolfram/tfpt_readouts_extension.wl` (20 new checks, verified engine run 2026-07-23, 564 → 584); suite 523 → 527 modules.
- **Not promoted (honest fences).** The review's remaining proposals stay research contracts, not verifications: the straddle-cone selector on the interacting FK quartets (the highest-leverage attack on the alignment bit after v529) and the metrology-torsor formalisation of v_{geo} are recorded as next steps in `experiments/next.txt`; the primes-below-30 identity stays an audit fingerprint exactly as the no-free-pattern rule demands.

2026-07-23 (IX) (website entry redesign: narrative home, honesty band, proof strip, guided-tour route — editorial only, NO new modules, NO marker moves)

- **Website home.** Slim one-viewport hero (proof strip: 523 machine-checked modules · Wolfram mirror · Lean carrier · 0 external replications yet; copyable `python run_all.py` → `ALL CHECKS PASSED`; CTA trio film / browser check / kill board). Narrative order: Hero → IntroVideo (`#intro-video` kept) → In one breath → WhyThisMatters (plain-English column) → TrustContract → ClaimStack → HonestyBand (honest negatives: RXTE/NICER nulls, 2DCS/v529, ten bit tests) → Safeguards (four core stats) → Overview (TheoryUnpacking/ThreeDecoderMap behind progressive disclosure) → PathChooser → Downloads teaser.
- **Archive routes.** Full PapersSection → `/papers`; PredictionsSection → `/predictions`; ReconstructionChain + HorizonStory + Gravity → `/architecture`; RosettaLexicon → `/orientation`; OpenGates → `/verification`; EcosystemSection off home (footer adjacent link). Legacy hash redirects on home for old anchors.
- **Nav / design.** Primary nav: Start · Tour · Verification · Kill board · Replication · Papers · FAQ (More dropdown for secondary). SuiteBadge (static per release from `suite.ts/version.ts`). Design pass: drop purple-gradient pills/glow in hero and accent tokens; lab-notebook proof strip as above-the-fold visual.

2026-07-23 (VIII) (prominence pass — the v519–v529 breakthroughs lifted into the FRAME texts of the papers and the website data layer; purely editorial, NO new modules, NO marker moves — (1) `tfpt_5_redteam` gains the seam round Target G: the ten-test side-blind scoreboard on the alignment bit (v512/v521/v525/v529) and the first *firing* kill test of the programme — the interacting straddle law (v529), documented as a named threat AND the first hard

selection principle for $\mathcal{A}_{\mathrm{hol}}$, fenced at toy level; (2) the bit-reduction statements in `tfpt_1`, `tfpt_2` and `origin_theory` extended from the v512 state to the current state (ten side-blind tests, the twist-class definition with gauge-robust order parameter, "in principle interferometrically readable", v528 — the bit stays formal input); (3) the thermal seam lifted into the `origin_theory` horizon chapter (new keybox: $T_{\mathrm{seam}} = 4c_3$ measured, temperature the third leg of c_3, v526) and into the `tfpt_horizon_readouts` head keybox; (4) `tfpt_3` title/keybox now carry the thirty-step story and "the measure chain is derived" (v520/v523) prominently; (5) `tfpt_research_contracts` title/intro keybox now show the executed stages of `CELEST.SEAM.01` (M1–M3, δ-chains decided, w_m derived) and `WOIT.OS.TWISTOR.01` ($\alpha/\beta_1/\beta_2$ executed, β_3 next under the straddle constraint); both contracts stay Open)

- **Papers.** `tfpt_5_redteam` (Target G + method/outcome/title honesty pass); `tfpt_1`, `tfpt_2` and `origin_theory` (bit-reduction statements → ten tests + twist-class definition); `origin_theory` (thermal-seam keybox, horizon chapter); `tfpt_horizon_readouts` (head keybox third leg); `tfpt_3` (title + keybox: thirty-step story, measure chain derived); `tfpt_research_contracts` (title + intro keybox: executed contract stages).
- **Website.** `papers.ts` (redteam Target G, horizon third leg, audit/contracts abstracts, architecture reduction state); `orientation/StatusMatrix.tsx` (v512 → current state); `glossary.ts` (new terms: twist-class choice, thermal seam, straddle law, side-blind scoreboard); new visual components `ThermalSeamLegs` (three legs of c_3: geometry v58 + anomaly/modular v208 + temperature v526) and `WoitBridgeProgress` ($\alpha/\beta_1/\beta_2$ done, β_3 open, straddle warning) on the home and verification pages.
- **Counters.** Predictions counter made consistent with `predictions.ts` (27 falsifiable test surfaces): `README.md` badge/hero/prose and the intro-film description (23 → 27; the film narration already says twenty-seven).
- **Honesty.** No marker moves anywhere; `SEAM.EQUIV.01` and `WOIT.OS.TWISTOR.01` stay Open; the bit stays input; all fences ("derived at probe level", "at the free level", "toy model") preserved verbatim.

2026-07-23 (VII) (v529 — the interacting Fidkowski–Kitaev seam toy promoted as the ledger row `SEAM.INT.FKTOY.01`: the FIRST firing Kill-Test-2 shadow — triple verdict F1 = KILL / F2 = KILL / F3 = ERFOLG; the celestial/twistor story in `tfpt_3` gains Step 30 and the header goes twenty-six → thirty steps — (1) F1 = KILL [E]: Θ exists exactly on the interacting model ($U_r^2 = 2^8$, $\Theta^2 = +1$, clock-tower inversion, $[\Theta, H_g] = 0$) so Kill-Test 1 does *not* fire; but Kill-Test 2 *fires at toy level*: RP breaks for every $g > 0$ — inertia ladder $(37,0,0) \to (33,4,0) \to (31,6,0) \to (29,8,0)$; Gibbs $\beta = 2$ same ladder; mechanism = interference; STRADDLE LAW (24/24); [C] fence one toy / one interaction class / flat-band parent — narrows the Woit route, does *not* close it; `WOIT.OS.TWISTOR.01` stays Open with the toy shadow documented as a named Kill-Test-2 threat; (2) F2 = KILL [E]: the TENTH side-blind test (9 → 10) — Θ_{15} maps $H(m) \to H(8-m)$ EXACTLY despite δ-seeing spectra; (3) F3 = ERFOLG [E]: grading + $\mathbb{Z}_8$ tower + FK anchor + constructive OS quotient at $\pi/2$; Wolfram extension $552 \to 564$ (verified engine run); suite $519 \to 523$; *NO marker moves*)

- **Suite.** New module `v529_seam_interacting_toy_fk.py` (24 checks, ~ 97 s); registered in `run_all.py` + `script_registry.csv`; suite 523 modules green.
- **Ledger.** New row `SEAM.INT.FKTOY.01`; `WOIT.OS.TWISTOR.01` text-precised (Kill-Test-2 toy shadow + straddle law); scoreboard rows text-precised (9 → 10); no status moves.

- **Papers.** `tfpt_3` Step 30; `tfpt_research_contracts` (Kill-Test-2 toy threat + bit scoreboard 10); `introduction` (tenth side-blind test).
- **Wolfram.** Three exact mirrors (Z8 identities; straddle combinatorics; FK binomial ladder).
- **Website.** `papers.ts` (story 30 steps + scoreboard 10), generated indices via build; `README.md`, `zenodo_description.html` (523 modules, highest ID `v529`, Wolfram 564/564), `next.txt` notes.

2026-07-23 (VI) (`v528` — the alignment bit as an exact measurable TWIST-CLASS choice promoted as the ledger row `SEAM.BIT.TWISTCLASS.01`: the physical definition — verdict SUCCESS; the celestial/twistor story in `tfpt_3` gains Step 29 — (1) Facets #14 (η-flip $\Leftrightarrow \delta = \pi/2$) and #15 (twist-frustration $\Leftrightarrow \delta = \pi/2$) both directions exact [E](cardinality lemma $|A_+| = |A_-| \Rightarrow m = 4$; g-class census 224 axes: flip class exactly at $\{15, 31\}$); (2) order parameter $O = 1/2$ at $m = 4$ else 0 [E], gauge-robust; (3) unification #14=#15=flag-transitivity = ONE $\mathbb{Z}_2$ invariant [E](v512 web $13 \to 15$); Kramers = POSITION half (derived), twist class = MODULUS half (the input); (4) [C] measurement sketch (interferometric in principle); the bit remains formal input; *NO marker moves*)

- **Suite.** New module `v528_seam_bit_twist_class_definition.py` (18 checks, ~ 46 s).
- **Ledger.** New row `SEAM.BIT.TWISTCLASS.01`; `SEAM.TAU.FLAG.01` text-precised (web $13 \to 15$); no status moves.
- **Papers.** `tfpt_3` Step 29; contracts + introduction (“symmetry-lift bit” now also “the twist-class choice, in principle interferometrically readable”).
- **Wolfram.** Three exact mirrors (cardinality lemma; 224-axis census; O-values).

2026-07-23 (V) (`v527` — the silver clock eigenvalue EXPLAINED and DEMYSTIFIED, promoted as the ledger row `SEAM.CLOCK.SILVER.01`: double verdict STRUCTURE = SUCCESS / CONNECTION = KILL (anti-numerology style); the celestial/twistor story in `tfpt_3` gains Step 28 — (1) STRUCTURE SUCCESS [E]: $\{1, \sqrt{2} - 1\}$ closed via diagonal Hankel pencil at $N = 8$, $\tan(\pi/8) = C(3)/C(1)$, doubling = exact second quantisation, carrier = silver-midpoint axes, gap $= \operatorname{arcsinh}(1) = \operatorname{gd}^{-1}(\pi/4)$; N-scaling = pure $N = 8/16$ lattice datum (not continuum); positive clock only for $N \equiv 0 \pmod 8$; (2) CONNECTION KILL [E]: preregistered 11466-candidate census over 16 frozen numbers: 0 hits above null; placebos comparable; (3) cross-link v526: $\sqrt{2} - 1 =$ thermal kernel ratio; *NO marker moves*)

- **Suite.** New module `v527_seam_clock_silver_spectrum.py` (22 checks, ~ 10 s).
- **Ledger.** New row `SEAM.CLOCK.SILVER.01`; no status moves.
- **Papers.** `tfpt_3` Step 28; contracts thermal/silver follow-on.
- **Wolfram.** Three exact mirrors ($\tan(\pi/8)$ pencil; N-scaling witness; gap $= \operatorname{arcsinh}(1)$).

2026-07-23 (IV) (`v526` — the thermal seam promoted as the ledger row `SEAM.THERMAL.KMS.01`: KMS of the reconstructed free OS quotient CLOSED onto the Nariai/BH temperature chain — the THIRD LEG of c_3; verdict SUCCESS; the celestial/twistor story in `tfpt_3` gains Step 27 — (1) β MEASURED $= N$ at both levels [E]($N = 8$: $\omega = \operatorname{arcsinh}(2^{-3/4})$, $\kappa = q^{-8}$; $N = 16$: pairing exponent $(N/2) + 1 = 9$; wrong β excluded); (2) $\beta_{\text{angle}} = 2\pi$ EXACT $= 1/(4c_3)$, $T_{\text{seam}} = 4c_3 =$ Bisognano–Wichmann/Hawking normalisation (v208) [E]; (3)

modular Hamiltonian closed [E]($\nu_1 = \cos(\pi/24)$, $\nu_2 = \cos(7\pi/24)$; Tomita KMS-$\beta = 1$ on all 256 pairs); (4) c_3 third-leg class-1 selection [E]/[C](Hawking/Nariai chain; v104 cross-link); (5) entropy $S(8/16/32) = 0.52187/0.63510/0.75006$, $\Delta S/\ln 2 \to 1/6$, horizon fractions do NOT transfer [E]; (6) [C]-5 silver $= C(3)/C(1)$ thermal kernel ratio; non-compact $\Rightarrow T = 0$; *NO marker moves*)

- **Suite.** New module `v526_seam_thermal_kms_nariai_bridge.py` (22 checks, ~ 8.4 s).
- **Ledger.** New row `SEAM.THERMAL.KMS.01`; no status moves.
- **Papers.** `tfpt_3` Step 27; `tfpt_horizon_readouts` (third leg of c_3); `tfpt_research_contracts` (thermal follow-on); `introduction`.
- **Wolfram.** Three exact mirrors (Q-polynomial/β measurement; $2\pi = 1/(4c_3)$ chain; ν charpoly).

2026-07-23 (III) (v525 — the twist-state attack on the alignment bit DECIDED and promoted as the ledger row `SEAM.BIT.TWISTBLIND.01`: "mark-decorated states do not see δ" — an honest, decided NEGATIVE result (v508 style), the NINTH side-blind test on the v512 scoreboard ($8 \to 9$); the free-plus-twist class (everything Wick-computable) is EXHAUSTED; the celestial/twistor story in `tfpt_3` gains Step 26 and the header goes twenty-three $\to$ twenty-six steps — (1) ALL LIFTINGS WELL-DEFINED AND BLIND [E]: (a) the σ-gauge arc decoration ($\mathbb{Z}_2$) satisfies $\sigma \circ r = \sigma$ on the swap axes $\Rightarrow$ the twisted Gram is a diagonal unitary congruence DMD, spectra IDENTICAL to untwisted (dev 0.0); (b) the Kadanoff–Ceva 4-twist insertion $\omega(D\,\cdot)/\omega(D)$ is well-defined on the $N = 32$ ladder ($\omega(D)$ real nonzero; the $N = 16$ odd-m degeneration is the $C(\text{even}) = 0$ checkerboard artifact) and its RP SPECTRA are the FIRST free-class data of the programme that depend on δ (pairwise up to 1.05) — yet the INERTIA is $(16, 0, 0)$ PD with the same $\eta = +i$ for EVERY $m = 1..7$: the decoration sees δ, positivity does not; (c) the μ_4 twist at $\beta = \pi/2$ is pure plane gauge; (c′) the genuine Bogoliubov defect at $\beta = \pi/4$: Θ-compatibility selects exactly 2 of 16 sign patterns (NS-wrap-forced), the winner genuinely non-monomial — still PD, spectra identical; STRUCTURE THEOREM (mechanism exact): the defect planes never straddle a cut bond, so every Θ-compatible quasi-free mark decoration is RP-spectrally INVISIBLE; (2) SUB-RESULTS [E]: the defect energy $E_{\text{def}} = 1.2752872$ for EVERY m (spread 4×10^{-40}) — δ-blind; the 4-twist Casimir $\ln|\omega(D)|$ measures δ ($-0.890..-1.348$) but exactly mirror-symmetric under $m \leftrightarrow 8-m$ — a family gauge, never a side selector; Kramers $\langle\Gamma\rangle_{\text{tw}} = +1$ for every m; (3) the HARVEST [E]: two bit-presupposing $\pi/2$ structures (the η flip $+i \to -i$ on the mark-fixing axes via $\sigma \circ r = -\sigma$; the twist frustration of the mark-fixing mirror, $(4, 4, 0)$ indefinite for both η — the lattice shadow of the Ising twist field's mirror-oddness) — both formulable only on the fixing axes existing iff $\delta = \pi/2$ (the v512 facet class); (4) NEGATIVE CONTROLS [E]: the twist-free limit = v519/v521 exactly; the site cut keeps failing; the v512 counterwitness is arc-equivalent at $N = 16$ to the blind member $\pi/4$ — no state selector can exclude it; a single twist has $\omega(D) = 0$ identically; [O]: the door narrows to a genuinely INTERACTING $\mathcal{A}_{\text{hol}}$ whose OS data are NOT Pfaffian-reducible to the chiral vacuum; Wolfram extension (with v523/v524) $540 \to 552$ (verified engine run 2026-07-23, $552/552$ + base $116/116$); suite $516 \to 519$ modules; *NO marker moves — the alignment bit remains genuine discrete input*)

- **Suite.** New module `v525_seam_bit_twist_blind.py` (20 checks, ~ 80 s; Pfaffian-exact insertion machinery cross-validated three ways, 40-digit certificates); registered in `run_all.py` +

script_registry.csv; suite 519 modules green.

- **Ledger.** New row SEAM.BIT.TWISTBLIND.01 (KILL, honest negative, preregistered); SEAM.BIT.RPBLIND.01 and SEAM.TAU.FLAG.01 text-precised (scoreboard 8 → 9, "the free-plus-twist class is exhausted"); no status moves.
- **Papers.** tfpt_3 Step 26 + the story header (twenty-six steps) + the Step-22 gap sentence; tfpt_research_contracts (the QGEO.REALIZE.01 reduction paragraph 8 → 9 and the WP2 relocation note extended by v525); introduction (the ninth side-blind test added to the reduction-state sentence).
- **Wolfram.** Three exact mirrors (the $N = 32$ integer frame + the $\sigma \circ r = \pm\sigma$ congruence mechanism; the $\pi/2$ existence solveset + the $(3 + 4i)/5$ root-of-unity census; the $\delta \leftrightarrow \pi - \delta$ mirror equivariance); the Pfaffian spectra, the 2-of-16 pattern census and E_{def} stay Python-only.
- **Website.** papers.ts (tfpt_3 story mirror 26 steps + scoreboard 9), generated indices (ScriptIndex.tsx, changelog.ts, maps) via build; README.md, zenodo_description.html (519 modules, highest ID v525, Wolfram 552/552), next.txt notes.

2026-07-23 (II) (v524 — the WOIT-β_2 milestone of the OS twistor bridge WOIT.OS.TWISTOR.01 executed and promoted as the ledger row WOIT.BETA2.OS.01: "the OS quotient of the free system made explicit" — verdict SUCCESS per the frozen preregistration, [C]-typed per contract precision (iii); the celestial/twistor story in tfpt_3 gains Step 25 — (1) $\mathcal{H}_{\mathrm{phys}}$ EXPLICIT [E]: $N = 16$ (deg ≤ 2): the 37×37 bond-cut OS Gram exactly Hermitian, parity-block-diagonal, inertia $(37, 0, 0)$ PD (min eigenvalue 1.7801×10^{-6} at 40 digits), null space $\{0\}$, $\dim 37 = 29 \oplus 8$; $N = 8$ (complete half algebra): $(16, 0, 0)$ PD (min 3.3471×10^{-3}), $\dim 16 = 8 \oplus 8 = 4^2$ — compact euclidean time reconstructs a THERMAL (KMS) representation; exact certificate $\sin^2(3\pi/8) - \sin(\pi/8)\sin(5\pi/8) = 1/2$; (2) the TRANSFER SEMI-GROUP (Klein–Landau) [E]: τ_k exactly Hermitian on all domains ($k = 1..7$, dims $29, 22, 16, 11, 7, 4, 2$), chain identities exact, vacuum fixed; the positivity pattern = the site/bond dichotomy: even k PSD via the exact square identity $T(2j) = A_j^* A_j$ ($(22, 0, 0)/(11, 0, 0)/(4, 0, 0)$), odd k indefinite ($(10, 12, 7)/(5, 6, 5)/(2, 2, 3)$) — the one-step transfer is NOT positive (the chirality datum, kill-3 shadow sharpened), its even powers are; (3) THE CLOCK = $T^{N/4}$ [E]: $N = 16$: τ_4 PD $(11, 0, 0)$ (min 6.6821×10^{-5}), compressed spectrum $[0.0194517, 1.64144]$, vacuum eigenvalue exactly 1, spectral projections $\sim 10^{-40}$; $N = 8$: spectrum EXACTLY $\{1, \sqrt{2} - 1\} = \{1, 1/\delta_{\mathrm{Silber}}\}$ in both parity sectors — the silver axes of the v519 torsor return as the clock eigenvalue; the reconstructed rotation group $U(s) = e^{isH}$ unitary with the group law; (4) the v522 RESOLUTION [E]: the census $(745, 96, 0)$ reproduced; EVERY anti-matching entry is a wrap overlap, every overlap-free entry matches — the pre-quotient non-Hermiticity is EXACTLY the domain/wrap artifact; on the quotient the clock has spectral calculus; pre-registered KMS deviation: no contraction on the compact circle ($C(1)/C(3) = 1 + \sqrt{2} = \delta_S$ exactly, $\det(G - \tau_4) < 0$); (5) GSO/Θ [E]: the grading survives; the perpendicular torsor mirror descends anti-unitarily ($\eta_\perp \in \{\pm i\}$), $\Theta^2_{\mathrm{phys}} = +1$ on every sector (Kramers-free), $\theta_{\mathrm{cut}} \circ \theta_\perp = \alpha_{N/2}$ exactly; (6) CONTROLS [E]: site cut $(3, 3, 1)/(2, 2, 4)$ — the kill branch fires there; family A $(4, 4, 0)$; anti-chiral $(8, 8, 0)$ no quotient; wrong direction non-Hermitian (all mismatches wrap-overlap); KILL TESTS: (1)/(2) strengthened, (3) shadow sharpened, none fires, all seven stay live on $\mathcal{A}_{\mathrm{hol}}$; TRANSPARENCY: run 1 = 18/20 (a sympy simplify failure on a TRUE $\pi/16$ product formula, fixed by a Laurent-polynomial certificate; no criterion

changed); *NO marker moves — `WOIT.OS.TWISTOR.01` stays Open, β_3 is next*)

- **Suite.** New module `v524_woit_beta2_os_quotient.py` (20 checks, $\sim$ 5 s; exact sympy + Pfaffian-Wick + 40-digit inertia certificates); registered in `run_all.py` + `script_registry.csv`.
- **Ledger.** New row `WOIT.BETA2.OS.01` (SUCCESS, [C]-typed per precision (iii)); `WOIT.OS.TWISTOR.01` claim/canonical status extended (β_2 executed, β_3 next); no status moves.
- **Papers.** `tfpt_3`: Step 25 plus the Woit-bridge paragraph with the β_2-stage status. `tfpt_research_contracts`: a full "β_2 stage — executed" block with `\veri`; the β/γ roadmap entry marked executed; the scope-fence status sentence extended.
- **Wolfram.** Four exact mirrors (the $\alpha_1^N = (-1)^F$ wrap arithmetic + the certificate identity; the Klein–Landau one-particle transfer forms with odd-k zero diagonal and the even-k square identity; the exact $N = 8$ clock spectrum $\{1, \sqrt{2} - 1\}$ + the transparency $\pi/16$ product formula certified symbolically; the KMS silver witnesses); the $\mathcal{H}_{\text{phys}}$ Grams, inertia spectra and the rotation group stay Python-only.
- **Website.** `papers.ts` (tfpt_3 + contracts mirrors), generated indices via build.

2026-07-23 (I) (v523 — the constructive w_m derivation executed and promoted as the ledger row `CELEST.WP5E.WM.01`: "$1/\det_j$ is the Atiyah–Bott / zeta-determinant fixed-point factor, COMPUTED from three independent sources" — verdict ERFOLG per the frozen preregistration; the residual [O] of v516/v520 is closed at the fibre-zero-mode level and narrows to the GLOBAL BCOV integral on $\mathbb{PT}/\mathbb{Z}_4$ beyond the fibre zero-mode factor; the celestial/twistor story in `tfpt_3` gains Step 24 — (1) ATIYAH–BOTT (route i) [E]: the equivariant mode ledger of the fibre $\mathbb{C}^2$ equals the closed form $1/((1-q\,i^j)(1-q\,i^{-j}))$ exactly in $\mathbb{Q}(i)$ for all $n \le 120$; REGULAR at $q = 1$ with Abel value $(1/2, 1/4, 1/2) = 1/\det_j$, $\det_j = \det(1-g^j) = (2, 4, 2)$; $(C, 2)$ Cesàro convergence in exact arithmetic; fixed-point splitting at EVERY truncation level $d \in \{7, 12, 25\}$ for the phased (skeleton) AND the v520-forced unphased (completion) trace — the Abel-limit $\text{contact}_j = (Q^{(0)} - Q^{(j)})/\det_j$ is the v516 contact vector COMPUTED, not declared; (2) ZETA/QUILLEN (routes ii/iii) [E]: $\det_\zeta$ of the g^j-twisted circle Laplacian $= 4\sin^2(\pi j/4) = \det_j$ EXACTLY (Lerch + reflection formula symbolically, Hurwitz certificates $\sim 10^{-41}$); Quillen split $\det\Delta = \det_j^2$, $|\det\bar\partial| = \det_j$, the real positive holomorphic section UNIQUE via the $SU(2)$ conjugation pairing; the $\delta_{1f}\ f_1 f_3$ block at the $(0, b)$ nodes carries the exact constant term $1/\det_b$ (deviations $\le 5.2 \times 10^{-28}$); (3) the CONSISTENCY CHAIN [E]: $w_m = (0, \frac{3}{2}, 2, \frac{3}{2}) = 4h_m = |\mu_4| h_m = -4\,\mathrm{ch}_2(T_m)$; $T_5 = 0$ at every scale; $h_2 : h_1 = 4 : 3$; the T_3 budget $32 - 8c$ forces $c = 4 = |\mu_4|$ uniquely (leftovers $24/16/8/0/{-8}$); per-sector Okubo squares $(18, 9, 18)\langle x, x\rangle^2$; total $45\langle x, x\rangle^2$; $\Phi_{T_3} : 32 \to 0$, $\Phi_P : 72 \to 0$; $\psi(A_{\text{fix}}) = +64$, $\psi(\text{contact}) = -64$; (4) NEGATIVE CONTROLS [E]: $1/\det^2$ gives $w' = (0, 5/8, 1, 5/8)$, leftovers $(2, 12)$, Quillen contradiction; $1/|1-i^j|$ gives $w_1'' = 1 + \sqrt{2} \notin \mathbb{Q}$; the $\mathbb{Z}_2$/Eguchi–Hanson anchor PASSES with the same derivation ($\det = 4$, $w = 1/2 = |\mathbb{Z}_2| h^{A_1}$, $9\langle x, x\rangle^2 = \frac{1}{4} \times 36$, $c = 2 = |\mathbb{Z}_2|$); the $SO(16)/D_8$ KILL fires; $\mathrm{diag}(i, i)$ breaks the zeta $=$ AB identity itself; $k = 3, 5$: $w_m = m(k-m)/2 = k\,h_m$ — the weights wander correctly, only $k = 4$ carries the E_8 chain; TYPED PREMISES [C] (the mandatory fence): TP-REG (Abel/Cesàro/zeta regularisation = the definition of the equivariant character), TP-Q (Quillen structure of BCOV F_1 = v520 TP-4), TP-NUM (unphased numerator = v520 ERFOLG-A), TP-CH (insertion channel $(0, j)$); [O]: the global BCOV integral beyond the fibre zero-mode factor, the resolved-ALE route not built, `SEAM.EQUIV.TWISTOR.01` stays

Open; *NO marker moves*)

- **Suite.** New module `v523_celestial_wp5e_wm_derived.py` (26 checks, $\sim$ 1 s; exact $\mathbb{Q}(i)/\mathbb{Q}(\sqrt{2})$/cyclotomic arithmetic + 40-digit Hurwitz/block certificates); registered in `run_all.py` + `script_registry.csv`.
- **Ledger.** New row `CELEST.WP5E.WM.01` (ERFOLG); the v516 row `CELEST.WP5E.M2.01` and the v520 row `CELEST.WP5E.MEASURE.01` text-precised ("the w_m normalisation is now DERIVED at probe level; the residual [O] narrows to the global BCOV integral beyond the fibre zero-mode factor"); no status moves.
- **Papers.** `tfpt_3` Step 24 + the "What remains, honestly" narrowing; `tfpt_research_contracts` (the WP5e measure-question block extended by the executed derivation with `\veri`, the M2 fence, the celestial summary blocks and the section status list); `SEAM.EQUIV.TWISTOR.01` stays Open.
- **Wolfram.** Five exact mirrors (the mode ledger = closed form with Abel value; the zeta/Quillen route with the symbolic reflection formula and the diag(i, i) control; the derived-weight contact vectors + success table + $\mathbb{Z}_2$/EH anchor on the 240 glue roots; the consistency chain + wrong-weight controls; the $k = 3, 5$ cyclotomic wandering controls); the Cesàro/Hurwitz/stripped-block numerics stay Python-only.
- **Website.** `papers.ts` (tfpt_3 + contracts mirrors), generated indices via build.

2026-07-22 (XIII) (HFQPO scorecard extension — the NICER/MAXI J1820+070 third-tooth null and the AstroSat/LAXPC block integrated as a new evidence-scorecard row (search target, never a claim): the preregistered single-QPO extension of the H3 ladder discriminator (frozen `hypotheses/hfqpo_ext_v1.yaml` BEFORE any photon download) ran on 6 NICER ObsIDs (13.22 GB, 2013×16 s $= 32.2$ ks stack at 19.05 kc/s) — sanity gate PASS (the published 55.12 Hz anchor reproduced at 55.03 Hz, 0.94% rms, 3.8σ), injection calibration PASS ($\geq 90\%$ recovery at 0.93% rms), the geometric tooth at 82.68 Hz NOT detected (0.69σ single-trial; 3σ upper limit 0.75% rms BELOW the detected anchor strength $\Rightarrow$ `null_with_sensitivity`), the integer line at $\sim$ 110.6 Hz at 3.82σ after trials — BELOW the preregistered 4σ threshold, documented as a sub-threshold excess consistent with the weak second harmonic of ATel #11951, NOT a hit; AstroSat/LAXPC (GRS 1915+105, scientifically the strongest channel) `infrastructure_blocked` (ISSDC login-gated); the RXTE + NICER nulls now cover five sources on two instruments — the tooth channel is recommended DORMANT; anti-double-counting: the parent RXTE row and the new extension row share `independence_group` `hfqpo_ladder_tooth` (one hypothesis, two instruments, never two independent nulls); scorecard regenerated (121 rows), `predictions.txt` HFQPO row extended ("Kanal ruhend"); experiment-only integration — no verification claim, no marker moves)

- **Scorecard.** New row "BH HFQPO third-tooth extension on non-RXTE archives" (NICER/MAXI J1820+070 plus AstroSat/LAXPC; `hfqpo-ladder` extension) in the generator `experiments/build_evidence_scorecard.py` (status `null`, stage `search_target`, signature S1e); `OVERRIDES` give both HFQPO rows the shared `independence_group` `hfqpo_ladder_tooth`; generator re-run: `evidence_scorecard.json` 121 rows, README stats block refreshed.
- **Experiment surfaces.** `experiments/hfqpo-ladder/extension-nicer-laxpc/` (preregistration, frozen manifest, download log, per-ObsID caches, stack/sanity/injection/scan reports, `results_ext.json`, README) — executed 2026-07-22; `predictions.txt` HFQPO row up-

dated with the NICER finding and the dormant-channel recommendation; `next.txt` round notes.

2026-07-22 (XII) (`v522` — the WOIT-β_1 milestone of the OS twistor bridge `WOIT.OS.TWISTOR.01` executed and promoted as the ledger row `WOIT.BETA1.GSO.01`: "the clock is time-like, GSO is the gauge datum" — a TYPED result (verdict UNDECIDED per the frozen preregistration, the mechanism identified exactly, gauge-fixed RP established under the corrected typing; the celestial/twistor story in `tfpt_3` gains Step 23) — (1) the HERMITICITY WITNESS [E]: the μ_4 clock average violates Hermiticity EXACTLY (the one-step insertion $T_1 = \langle\theta(e_a), \alpha_S(e_b)\rangle$ has entry $-i/(8\sin(5\pi/16))$ where the conjugate transpose demands $+i/(8\sin(5\pi/16))$; entrywise census 745 Hermitian-matching, 96 ANTI-matching, 0 violations — the pairing is complex-SYMMETRIC, strictly weaker than Hermitian: "positivity after gauge fixing" is not well-posed for the clock reading); (2) TIME-LIKE TYPING [E]: ALL 16 dihedral reflection axes of the NS seam circle invert the clock lift ($RSR^{-1} = S^{-1}$), NONE commutes — the μ_4 clock IS Woit's euclidean rotation; the gaugeable (θ-compatible) part of the $\mathbb{Z}_8$ Fock tower is EXACTLY its 2-torsion $\{1, (-1)^F\}$ = the GSO/fermion-parity $\mathbb{Z}_2$ (Hermiticity census over all powers; α^4 = the full wrap = the 2π rotation of the seam = fermion parity); (3) the POSITIVE PART [E]: GSO-fixed RP holds EXACTLY — $N = 16$ even $\deg \leq 2$ parity-projected bond-cut Gram $(29, 0, 0)$ PD (min eigenvalue 1.78×10^{-6} at 40 digits), $N = 8$ complete half algebra: all odd rows/columns killed exactly, even block $(8, 0, 0)$ PD (3.35×10^{-3}), no degree truncation; the site-cut defect SURVIVES gauge fixing ($(7, 9, 6)$ — the bond placement is gauge-invariant information); family A keeps failing ($(17, 12, 0)$ indefinite vs family D's $(29, 0, 0)$); Kramers: $\Theta_t^2 = \mathrm{id}$ on $\mathcal{A}^G$, Θ_{Fock} restricts with $\Theta^2 = +1$; Ramond control: the wrap $+1$ "true $\mathbb{Z}_4$" lift fails state invariance — the NS/$\mathbb{Z}_8$ tower is forced; (4) KILL-TEST BOOKKEEPING: kill test (2)'s free shadow does NOT fire (no legitimate Hermitian gauge-fixed form has a negative direction), kill test (1) stays discharged also gauge-invariantly; both stay formally live on $\mathcal{A}_{\mathrm{hol}}$; (5) TRANSPARENCY: the first frozen run scored $6/13$ (the clock-reading checks failed exactly as the mechanism predicts) and the preregistered deg-2 invariant dimension 30 was a combinatorial error (correct census $28 + 4 = 32$) — documented in full, the verdict by the frozen [C-3] logic (Hermiticity failure $\Rightarrow$ UNDECIDED); CONTRACT PRECISION (iii) (prose, no marker): "gauge-invariant subalgebra" at the free level means the GSO/fermion-parity $\mathbb{Z}_2$ — the only θ-compatible part of the μ_4 tower; the clock is the euclidean rotation itself (timelike), its equivariant statement moves to β_2 (OS quotient first, then the clock as reconstructed transfer operator); Wolfram extension (with v520/v521) $522 \to 540$ (verified engine run); *NO marker moves — `WOIT.OS.TWISTOR.01` stays Open, β_1 proper re-routed via β_2*)

- **`v522_woit_beta1_gso_gauge.py` (`WOIT.BETA1.GSO.01`, 16 checks, ∼5 s; exact sympy Pfaffian–Wick + exact $\mathrm{Cl}(16)$ Fractions; definiteness via 40-digit spectra, tolerance 10^{-25}; deterministic).** G1 the gauge datum (Z8/NS tower, Ramond control, invariant dimensions $120 > 64 > 32$); G2 Θ normalises the tower and restricts as an involution; G3 the $(-1)^F$ class does not penetrate $\mathcal{A}^G$; R1 the tower average is OS-symmetric, the odd sector dies; M1 the time-like census; M2 the exact Hermiticity witness; M3 the GSO 2-torsion census; R2 GSO-fixed RP ($N = 16/N = 8$); C1–C3 site-cut/wrong-projection/family-A controls; V1 the frozen verdict. Registered in `run_all.py` (suite 516 modules, green), `script_registry.csv` (cluster *frontier*), ledger row `WOIT.BETA1.GSO.01` (new).

- **Contract updated (markers unchanged).** `WOIT.OS.TWISTOR.01`: the β_1 stage moved to *done* (typed UNDECIDED) with its own block, precision (iii) added, the β_1 roadmap bullet annotated *executed*, the closing status line carries the β_1 state; ledger `WOIT.OS.TWISTOR.01` text extended (beta-1 executed via v522).
- **Paper integration.** `tfpt_3` (new Step 23 "the clock is time-like, GSO is the gauge datum"; the Woit-bridge paragraph extended by the β_1 status); `veri` citations in `tfpt_research_contracts` and `tfpt_3`.
- **Wolfram.** Six exact mirrors (the Z8/NS tower $S^4 = -1$ + the Ramond witness $1/(4\sin(5\pi/16))$; the invariant-dimension chain by exact ranks; the time-like reflection census; the exact Hermiticity witness with the full 745/96/0 census via exact Pfaffian–Wick; the GSO 2-torsion census; the family-A tower-normalisation failure) added to `tfpt_readouts_extension.wl`; the GSO-fixed RP inertia certificates stay Python-only, flagged in the `.wl` comment.
- **Provenance.** Discovery probe `woit_os_beta1_gauge_invariant_theta_probe.py` (16/16, ~3 s) in `experiments/tfpt-discovery/` (2026-07-22); website mirrors regenerated; `next.txt` round notes.

2026-07-22 (XI) (`v521` — the RP/Θ attack on the alignment bit DECIDED and promoted as the ledger row `SEAM.BIT.RPBLIND.01`: "free OS positivity does NOT see δ" — an honest, decided NEGATIVE result in the `v508` style (the celestial/twistor story in `tfpt_3` gains Step 22): does free reflection positivity (`v519`) or anti-linear Θ existence break for $\delta \neq \pi/2$ on the deformed mark family $M(\delta) = \{\pm1, \pm e^{i\delta}\}$ — which would DERIVE the `v512` alignment bit from OS positivity? NO — (1) the CENSUS [E] (exact solvesets, symbolic δ): for EVERY δ exactly the two mark-SWAPPING reflections $\Theta_{\pm e^{i\delta}}$ exist (axes $\delta/2$, $\delta/2+\pi/2$; all 20 cut/mark incidence solvesets EMPTY); mark-FIXING reflections exist exactly at $\delta = \pi/2$ (solveset $\cos\delta = 0$, lost immediately at $\pi/2 - \varepsilon$); the `v510/v512` counterwitness $\{\pm1, \pm(3+4i)/5\}$ carries the swap pair exactly; (2) RP IS δ-BLIND [E]: bond-cut Gram inertias $(8,0,0)$ PD for all tested δ with the full sorted spectra IDENTICAL to 40 digits (deviation 0.0) — the ENTIRE free-RP spectral datum carries ZERO δ information; the $N = 16$ odd-m failure (site axes, $(3,3,1)$, $\det = 0$) is a lattice-PLACEMENT artifact: at $N = 32$ ALL $m = 1..7$ have mark-compatible BOND axes with PD inertia $(16,0,0)$; continuum: the OS kernel in axis-adapted coordinates is EXACTLY $1/\sin((s+t)/2)$ — the axis position drops out identically (Cauchy–Stieltjes PD); the counterwitness PASSES free RP (6-point Gram $(6,0,0)$); (3) Θ EXISTENCE IS δ-BLIND [E]: $\Theta^2 = +1$, deck normalisation and Fock implementability ($U^2 = +2^7/2^8 > 0$, exact $\mathrm{Cl}(16)$) for every δ; the ONLY δ-sensitive `v519` clause ($\Theta\rho\Theta = \rho^{-1}$) is for $\delta \neq \pi/2$ not violated but NOT FORMULABLE (a mark-compatible order-4 rotation exists iff $\delta = \pi/2$) — it PRESUPPOSES the bit; same for the clock-tower clause; (4) the battery HAS TEETH at every δ [E]: family A/antipode fails RP structurally (zero diagonal, $(4,4,0)$), the chirality–η pinning persists ($(0,8,0)$ ND) — RP separates the two real-structure FAMILIES, never the SIDES; (5) precondition controls: hexagonal (marks not concyclic — no seam circle), silver (deck not free), lifting audit (B = index relabelling, δ-blind by construction; B' fails the $\pi/2$ negative control — discarded); [C] FACE #13 of the `v512` equivalence web: the group generated by the admissible OS reflections is V_4 for every $\delta < \pi/2$ and closes to D_4 exactly at $\delta = \pi/2$ ($\Theta_i \circ \Theta_1$ = the clock) — $12 \to 13$ faces, a typed REFORMULATION of the bit, not a derivation; [O] the honest gap: marks decorating the STATE (defect/twist insertions, interacting $\mathcal{A}_{\mathrm{hol}}$) = exactly `v519`'s [O] scope; VERDICT: KILL exactly as preregistered — the free RP/Θ battery is the EIGHTH side-

blind test on the `v512` scoreboard ($7 \to 8$); the alignment bit is NOT derivable from OS positivity and remains genuine discrete input; *NO marker moves — the bit stays the one [C] carrier input*)

- **`v521_seam_bit_rp_blind.py` (`SEAM.BIT.RPBLIND.01`, 25 checks, ∼25 s; exact sympy solvesets + exact Gram entries + exact $\mathrm{Cl}(16)$ Fractions; inertia via 40-digit spectra, tolerance 10^{-25}; deterministic).** S the anti-linear census (family D/A, counterwitness, face #13 closure, cut incidence, side conditions); L the lattice δ ladder (frame, $\pi/2$ reproduction, $\pi/4$ PD, odd-m artifact, $N = 32$ refinement, site-axis control, spectra blindness, family-A rejection, chirality pinning); C the continuum axis elimination + counterwitness; LB the lifting audit; T Fock implementability + the clock-tower clause; W scoreboard + hex/silver controls + the preregistered KILL verdict. Registered in `run_all.py`, `script_registry.csv` (cluster *core*), ledger row `SEAM.BIT.RPBLIND.01` (new).
- **Ledger/paper text precisions (markers unchanged).** `SEAM.TAU.FLAG.01`: scoreboard text extended ($7 \to 8$ side-blind tests, face #13, web $12 \to 13$); `tfpt_research_contracts` (the `v512` reduction paragraph now carries the `v521` kill and the honest **[O]** gap; the WP2 residual sentence extended), `introduction` and `tfpt_3` Step 2 (12-fold → 13-fold, the eighth side-blind test); `veri` citations in `tfpt_research_contracts`, `introduction` and `tfpt_3`.
- **Wolfram.** Six exact mirrors (the census solvesets incl. the exact-rational counterwitness; the $V_4 \to D_4$ closure / face #13 as symbolic composition identities; the 20 incidence solvesets + the eps jump + the integer lattice frame; the continuum axis elimination + Cauchy–Stieltjes minors + the counterwitness cut ordering; the clock-tower clause + hex/silver controls; the δ-blind side conditions) added to `tfpt_readouts_extension.wl`; the 40-digit lattice inertia spectra stay Python-only.
- **Provenance.** Discovery probe `seam_alignment_bit_rp_selector_probe.py` (25/25, ∼24 s) in `experiments/tfpt-discovery/` (2026-07-22); website mirrors regenerated; `next.txt` round notes.

2026-07-22 (X) (`v520` — the WP5e BCOV measure question DECIDED at probe level and promoted as the ledger row `CELEST.WP5E.MEASURE.01`: “single-valuedness is derived, the completion reading wins” — the $\delta_{1e} + \delta_{1f}$ consolidation in the `v518` style, ERFOLG-A on the preregistered decision layer (the celestial/twistor story in `tfpt_3` gains Step 21): which of the two exact, contradictory measures on $\mathbb{PT}/\mathbb{Z}_4$ is the true BCOV measure — (A) the DECLARED completion reading (`v516`) or (B) the DERIVED chiral measure (`v518`)? — (1) the INVARIANT SUBSPACE [E]: $\dim_{\mathbb{Q}}\{v : \rho(T)v = v,\ \rho(S)v = v\} = 8 = 4 \times 2$ in the dressed 16-component Weil system (exact rational nullspace); EVERY kernel vector lies in the $\mathbb{Q}(\zeta_8)$-span of $\{e_H, e_{H'}\}$ — the two Lagrangian gluings, both theta $= E_4$, level-1 quartic $= Q^{(0)}$ = the UNPHASED trace; strict trivial-character solutions at physical weight $(m,k) = (0,0)$: ZERO for every dressing (11 combinations); (2) SOURCE 1 — the F-LEMMA [E]: the only strict chiral families are χ_4 (dims $(3,3)$) and χ_{10} (dims $(1,1)$), both with nontrivial T-character EXACTLY; the canonical column-matched member obeys $G(\gamma\tau) = \chi_4(\gamma)G(\tau)$ pointwise (certificate 3.9×10^{-16}, $|G| \ge 59.6$); EXACT change of variables ⇒ fundamental-domain independence forces $\int = 0$ for EVERY chiral integrand: the `v518` route was never a well-defined nonvanishing moduli integral (the kill is SHARPENED — its unfolded values are convention bookkeeping); (3) SOURCE 2 — the QUILLEN PAIRING [E]: $|t\bar{t} - 1| < 8.3 \times 10^{-40}$ on all 15 pairs and both generators; $|\chi_4|^2 = 1$ exactly vs one-sided $\chi_4^2 = e(2/3) \neq 1$ (T-fixed: $640 \neq 0 \bmod 960$, 15/15 defective); the doubled 256-dim hol×antihol system closes STRICTLY at trivial character and physical weight (nullspaces $(28,17) \ge 11$), contains the unphased diagonal ($\sim 2 \times 10^{-15}$) AND

the physical columns $w_b \otimes \bar{w}_b$ (σ 4.9/8.3 $\times 10^{-16}$); negative control hol$\otimes$hol: EMPTY nullspace $(0,0)$; (4) the CONSISTENCY CLOSURE [E]: the forced unphased numerator in the `v518` scaffold reproduces $\sum_m V \propto (1,2,1,0,0) = \langle x,x\rangle^2$ EXACTLY on every level $n \leq 8$; J-weighted spreads $\sim 5\times10^{-41}$, $T_5/T_3 \sim 7\times10^{-42}$, zero negative cells; $\psi(\text{skeleton}) = +0.2302 = -\psi(\text{contact})$ — the J-weighted mirror image of $\psi(A_{\text{fix}}) = +64/\psi(\text{contact}) = -64$; linearity 4.6×10^{-41}: the circle to `v516` is CLOSED; (5) D3 — the $\mathbb{Z}_2$/EH ANCHOR [E]: only the declared reading hits ($9\langle x,x\rangle^2 = \frac{1}{4}\times 36$, $\psi = -8$, scale tooth $c = 1/2 = |\mathbb{Z}_2| h^{A_1}$); all three derived instantiations fail (spreads 1.12/0.895/0.516, ψ sign wrong); (6) D1 — COLUMN CANONICALISATION [E]: the physical AB column $\sum_m i^{bm} e_{(m,m)}$ is contained EXACTLY in the χ_4 family (σ $1.8/1.7\times10^{-16}$, one scalar per orbit — canonicalising (N_1,N_2)); the canonical member STILL fails every tester; NEGATIVE CONTROLS: $SO(16)/D_8$ ($T_5 = 20/T_3 = -40$ exact, μ_2-only), wrong form ($|\text{Gauss}|^2 = 64$), wrong weight (dev 0.78), shuffle ($T_5 = -14/T_3 = 36$), planted family detected (1.5×10^{-16}), hol$\otimes$hol empty; [C] TYPED PREMISES (the mandatory fence): TP-1 (F-independent moduli integral), TP-2 (nonvanishing — must source $\psi = 64$), TP-3 (`v518` kernel convention), TP-4 (Quillen structure of BCOV F_1); [O] the constructive BCOV derivation of the w_m normalisation itself; WIRKUNG (text only, no marker moves): the `v516` fence and the `v518` tension now read "decided at probe level for the declared reading under TP-1..TP-4 (v520)"; `SEAM.EQUIV.TWISTOR.01` stays Open; *VERDICT ERFOLG-A per the preregistered criteria — both constructive sources fire independently, the consistency closure carries*)

- **`v520_celestial_wp5e_measure_decision.py` (`CELEST.WP5E.MEASURE.01`, 39 checks, ~20 s; exact $\mathbb{Q}(\zeta_8)$/Fraction/sympy rational nullspaces + 1/960-grid phase arithmetic + mpmath 30-digit kernels + numpy double SVD, deterministic).** S0 exact foundations (roots, ledgers, A-core + $\mathbb{Z}_2$-anchor targets, Weil relations, characters); S1 transports + derived families + the $k=0$ slice; S2 D1 column canonicalisation + the sharpened kill; S3 D2 the exact invariant subspace; S4 D3 the $\mathbb{Z}_2$/EH anchor at the integral level; S5 source-1 (the F-lemma); S6 source-2 (the Quillen pairing, doubled 256-dim system); S7 the point-4 consistency closure; S8 negative controls; S9 the ERFOLG-A verdict + typed premises. Registered in `run_all.py`, `script_registry.csv` (cluster *frontier*), ledger row `CELEST.WP5E.MEASURE.01` (new).
- **Ledger/paper text precisions (markers unchanged).** `CELEST.WP5E.M2.01` (the `v516` fence: the completion reading is now DERIVED at probe level from two independent constructive sources under TP-1..TP-4; residual **[O]** = the constructive w_m derivation), `CELEST.WP5E.DELTA1.01` (the `v518` kill sharpened, the tension decided); `tfpt_research_contracts` (the WP5e measure-question prose updated throughout: object diagram, WP4 arc, ε_1/M2 fences, the measure-question target block now carries the executed decision, the closing status lines); `tfpt_3` (Step 17/19 fences updated, new Step 21, the "what remains" block narrowed to the constructive w_m derivation); `tfpt_4_frontier` (celestial keybox updated). `veri` citations in `tfpt_research_contracts` and `tfpt_3`.
- **Wolfram.** Six exact mirrors (the 16-component Weil data with both Lagrangian gluings; the invariant-subspace forcing dim $= 2 = \text{span}\{e_H, e_{H'}\}$ by exact NullSpace + rank; the character arithmetic of the two sources $|\chi_4|^2 = 1$ vs $\chi_4^2 = e(2/3)$ with the exact grid defects; the A-core + $\mathbb{Z}_2$/EH-anchor targets with the scale teeth; the level-1 Okubo identity of BOTH gluings (240 norm-2 vectors each, summed quartic $36\langle x,x\rangle^2$); the so16/shuffle/diag(i,i) controls) added to `tfpt_readouts_extension.wl`; the Harvey–Moore integrals, SVD memberships and J-weighted closures stay Python-only, flagged in the `.wl` comment; combined engine run with the v521/v522 mirrors: extension 522 → 540 (verified, Wolfram Engine 14.3; `GATE.WOLFRAM.02`, Wolfram README, `tfpt_safeguards`, website counters and `zenodo_description.html` synced 116/116 + 540/540).

- **Provenance.** Discovery probes in `experiments/tfpt-discovery/` (2026-07-22): `celestial_seam_wp5e_delta1e_bcov_measure_decision_probe.py` (28/28, ∼15 s) and `celestial_seam_wp5e_delta1f_single_valuedness_probe.py` (23/23, ∼17 s); suite 513 → 516 modules (green); website mirrors regenerated (`ScriptIndex`, DAG *qft/P2* nodes, `papers.ts` story counters, glossary, changelog mirror); `next.txt` round notes.

2026-07-22 (IX) (`v519` — the WOIT-α milestone of the OS twistor bridge `WOIT.OS.TWISTOR.01` executed and promoted as the ledger row `WOIT.THETA.FREE.01`: "the real structure exists — and free reflection positivity picks the same family"; the celestial/twistor story in `tfpt_3` is now TWENTY steps — (1) THETA CLASSIFICATION COMPLETE [E]: exactly TWO families of anti-linear structures on $\mathbb{C}^2$ normalise the clock $\rho = \mathrm{diag}(i, 1)$ — family D ($z \mapsto \mu\bar{z}$, $|\mu| = 1$): $\Theta\rho\Theta = \rho^{-1}$ EXACTLY at matrix level, $\Theta^2 = +1$ (-1 IMPOSSIBLE: $M\bar{M} = \mathrm{diag}(|\mu|^2, 1)$ — Kramers-free), inverts the deck, fixed set = the great circle through the deck poles; family A ($z \mapsto \mu/\bar{z}$): centralises the clock projectively (NEVER inverts it), $\Theta^2 = \pm 1$ per μ, fixed set $|z| = 1$ resp. empty; ROLE SEPARATION: family A DEFINES the euclidean section (Woit's ρ_{tw} side: $\rho_{\mathrm{tw}}^2 = -1$, no real points — replicated exactly on $\mathbb{C}^4$), family D REFLECTS it (the OS conjugation, σ_{std} = conjugation with real points $\mathbb{RP}^3$); (2) MARK PINNING [E]: $z \mapsto \mu\bar{z}$ preserves the μ_4 marks iff $\mu \in \mu_4$ — a 4-element μ_4-TORSOR (2 mark axes, 2 silver-midpoint axes), clock conjugation $\rho\,\Theta_\mu\,\rho^{-1} = \Theta_{-\mu} \Rightarrow 2$ clock orbits; (3) SPIN/FOCK [E]: the $\mathbb{Z}_8$ plane has no phase leaks ($\overline{\zeta_8^k} = \zeta_8^{-k}$); exact $\mathrm{Cl}(16)$: $\Theta_{\mathrm{Fock}} = U_r \circ K$ with $\Theta_{\mathrm{Fock}}^2 = 2^7 > 0$ (normalised $+1$), inverts the clock tower ($V \mapsto 4096\,V^{-1}$) and normalises the deck; the KRAMERS DICHOTOMY: the deck-induced candidate $\Theta_t = \tilde{U} \circ K$ has $\Theta_t^2 = (-1)^F$ — *the `v510` split/nonsplit dichotomy IS the $\Theta^2 = +1$ vs $(-1)^F$ dichotomy*; the deck does NOT furnish Θ, the seam-circle REFLECTION does; (4) KILL TEST 1 (free level) [E]: does NOT fire — Θ exists with $\Theta^2 = +1$ AND $\Theta\rho\Theta = \rho^{-1}$ and satisfies all three side conditions simultaneously (marks, deck normalisation, spin/Fock); (5) FREE RP [E]: the NS vacuum $C(d) = (2/N)/\sin(\pi d/N)$ has $C^2 = -1$ exactly; preconditions exact ($RCR^T = -C$, $SCS^T = C$, $RSR^{-1} = S^{-1}$; the deck FAILS anti-invariance); BOND CUT: one-particle Gram $(8, 0, 0)$ PD (min eigenvalue 1.888×10^{-3}, 40 digits), even $\deg \leq 2$ sector $(29, 0, 0)$ PD, $N = 8$ complete half-sided algebra PD (even + odd each $(8, 0, 0)$) — RP with NO degree truncation; twist $\eta = +i$ FORCED ($\eta = 1$ non-Hermitian); CONTROLS: the site cut fails exactly ($\det = 0$, inertia $(3, 3, 1)$; continuum control: Cauchy–Stieltjes strictly positive — a lattice-PLACEMENT artifact, RP selects the site placement relative to the marks); the clock-CENTRALISING family-A structure fails RP structurally ($(4, 4, 0)$) — *RP and $\Theta\rho\Theta = \rho^{-1}$ select the SAME family*; bonus: the anti-chiral state flips the odd sector to $(0, 8, 0)$ ND (the free shadow of kill test 3); [C]/[O] NOT shown: the interacting algebra $\mathcal{A}_{\mathrm{hol}}$ (kill test 1 stays live there), gauge-fixed RP (kill test 2), OS reconstruction, chirality without mirrors (kill tests 3/6), μ_4 incidence (kill test 7) — the WOIT-β/γ roadmap named (β_1 Θ on the gauge-invariant subalgebra + gauge-fixed RP on the equivariant SDYM(E_8) sector; β_2 the OS quotient of the free system explicit; β_3 the $\mathbb{PT} \leftrightarrow \mathbb{PT}^*$ duality vs σ_{std} typed; γ the chirality theorem + the mark incidence); TWO CONTRACT PRECISIONS (prose, no marker): (i) "Θ induced by the seam reflection" means the seam-circle REFLECTION, not the deck — the deck furnishes the $(-1)^F$ Kramers class (the `v510` bridge); (ii) the marks sit RP-side on the BOND MIDPOINTS of the 16-Majorana circle; Wolfram extension $516 \to 522$ (verified engine run); *NO marker moves — `WOIT.OS.TWISTOR.01` stays Open: kill test 1 is discharged on the FREE level only and stays formally live on the interacting*

***algebra*)**

- **`v519_woit_theta_rp_free.py` (`WOIT.THETA.FREE.01`, 34 checks, ~20 s; exact sympy + exact Cl(16) Fractions; RP definiteness via 40-digit spectra of exact Hermitian matrices, tolerance 10^{-25}; deterministic).** Part W: W-S1 the two-family classification (symbolic solve); W-S2 family D (relations, Kramers-free, mark pinning, seam action, torsor); W-S3 family A (equatorial/ρ_{tw}); W-S4 the $\mathbb{C}^4$ twistor level (ρ_{tw} vs σ_{std}); W-S5 the $\mathbb{Z}_8$ spin level; W-S6 the Cl(16) Fock level ($\Theta^2_{\mathrm{Fock}} = 2^7$, $V \mapsto 4096\,V^{-1}$ via the implementer/centre route — a consolidation of the 105-s discovery probe to ~20 s with identical claims and tolerances — deck normalisation, Kramers dichotomy, $\tilde{U}V = V\tilde{U}$ control); W-S7 verdict. Part R: R1 the exact preconditions; R2 the site cut fails; R3 the bond cut passes (incl. the complete $N = 8$ half-algebra); R4 the continuum control; R5 controls (twist forced, antipode fails, chirality flip); R6 verdict. Registered in `run_all.py` (suite 513 modules, green), `script_registry.csv` (cluster *frontier*), ledger row `WOIT.THETA.FREE.01` (new).
- **Contract updated (markers unchanged).** `WOIT.OS.TWISTOR.01`: the α stage moved to *done* with the core numbers and `veri{v519}`; the two precisions added as their own sentences (the input definition of Θ now names the seam-circle reflection, with the deck carrying the $(-1)^F$ Kramers class; the marks at the bond midpoints of the 16-Majorana circle); the β/γ roadmap bullets added with success and kill criteria; the object-diagram OS row and the closing status line carry the α-stage state. Ledger: `WOIT.THETA.FREE.01` new; `WOIT.OS.TWISTOR.01` text extended ("alpha stage executed via v519 … kill test 1 does NOT fire at the free level and stays live on the interacting algebra; beta/gamma milestones named"). NO marker moves anywhere.
- **Paper integration.** `tfpt_3` (new Step 20 "the real structure exists — and free reflection positivity picks the same family" — the celestial/twistor story is now twenty steps; the Woit-bridge paragraph extended by the α status: "the missing global object now exists at the free level; the interacting statement stays [O]"). `v519_woit_theta_rp_free.py` is cited via `veri` in `tfpt_research_contracts` and `tfpt_3`.
- **Wolfram.** Six exact mirrors (the two-family classification; mark pinning + the μ_4 torsor with the clock-orbit map and the cut-point census; the $\mathbb{C}^4$ level ρ_{tw} vs σ_{std}; the $\mathbb{Z}_8$ tower conjugation; the Cl(16) dichotomy 2^7 vs $(-1)^F$ on explicit Jordan–Wigner gammas incl. $U_r V U_r V = 2^{19}$ and $\tilde{U}V = V\tilde{U}$; the free-RP preconditions + site-cut degeneration + the Cauchy–Stieltjes continuum control) added to `tfpt_readouts_extension.wl` and verified with the active engine, 516 → 522; the 40-digit PD inertia certificates of the bond-cut Grams stay Python-only, flagged in the `.wl` comment; `GATE.WOLFRAM.02`, Wolfram README and the website counters synced (116/116 + 522/522).
- **Provenance.** Discovery probes `woit_os_theta_realstructure_probe.py` (21/21, ~105 s) and `woit_os_theta_rp_freelevel_probe.py` (13/13, ~5 s) in `experiments/tfpt-discovery/` (2026-07-22); website mirrors regenerated (`ScriptIndex`, DAG *qft* node + `v519`, `papers.ts` "20 steps" + the WOIT-contract mirror, glossary Θ/Woit-bridge entries, changelog mirror); `next.txt` round notes.

2026-07-22 (VIII) (`v518` — the WP5e δ_1 chain ($\delta_{1b} + \delta_{1c} + \delta_{1d}$) DECIDED and promoted as the ledger row `CELEST.WP5E.DELTA1.01`: "the derived chiral measure kills the δ_1 route" — an honest, decided NEGATIVE result in the `v508` style, with the `v516` tension stated: does the Harvey–Moore/BCOV fundamental-domain integral on $\mathbb{PT}/\mathbb{Z}_4$, evaluated under a measure DERIVED from blockwise $SL(2,\mathbb{Z})$ covariance of the dressed 16-component Weil system (solved for — not declared, not scanned), deliver the twisted contact vector $-A_{\mathrm{fix}} = -(9,-30,-15,0,32)$ or the $\psi = 64$ slice? NO — (1) the 16-COMPONENT WEIL COMPLETION (δ_{1c}): the discriminant module of $D_5 \oplus A_3$ is $\mathbb{Z}_4 \times \mathbb{Z}_4$ with $q = (5x^2+3y^2)/8$, Gauss

sums $2\zeta_8^5 \times 2\zeta_8^3 = 4 = \sqrt{16}$ (signature $0 \bmod 8 =$ rank E_8), the Weil relations hold EXACTLY in $\mathbb{Q}(\zeta_8)$ ($S^2 = C$, $(ST)^3 = S^2$, $S^4 = 1$, $T^8 = 1$, tensor split $S = S_{D_5} \otimes S_{A_3}$), the diagonal H and anti-diagonal H' are the two invariant Lagrangians (both counts $= E_4$: two E_8 gluings), $P_H S P_H = \frac{1}{4}J$ with $3/4$ of every S-image weight on the 12 shifted classes, and the 16-vector theta S-covariance CLOSES: the δ_{1b} E1.5 residual $2.91 = O(1)$ (naive 4-character rule) drops to $\sim 10^{-39}$; (2) the OBSTRUCTION IS A FINITE μ_4 CHARACTER, IDENTIFIED (δ_{1d}): the dressed gauge blocks $M = G[a, b]\,\eta^{-8}$ carry constant transport (t_p, s_p) (28+ digits, recognised as exact 1/960-grid phases at 10^{-25}), the T-fixed pairs $(0, b)$ carry $t = e(-1/2) = -1$ EXACTLY (the naked level-matching defect), the transport-groupoid loops land in $\Gamma_1(4)$ resp. $\Gamma_0(2)^{\pm}$ with joint holonomy eigenspaces dim 8/16 resp. 6/16 and every eigen-phase of order ≤ 4; the KOBOUNDARY TEST: the $SL(2, \mathbb{Z})$ relation defects c_{S^4}, $c_{(ST)^3 S^{-2}}$, $c_{[S^2,T]}$ are $(1, 1, 1)$ EXACTLY on all 15 pairs — the multiplier system is a CHARACTER of the orbit stabilisers, NOT a genuine 2-cocycle, and on $\Gamma_1(4)$ it is $\lambda(\gamma) = i^{2B+C/4}$; (3) the CANCELLATION EXISTS AND IS THE TWISTED FIBRE BLOCK $f_1 f_3$, NOT THE THREE SPHERE AXIONS: $G[a, b] = f_1 f_3$ is an exact identity (30+ digits; the axion T-phases are `v502` vacuum-energy data), the T-fix mechanism is one line of exact phase arithmetic — $(-1) \times e(-1/6) = e(1/3) = \chi_4(T)$ at every T-fixed node — and the exact CANCELLATION TABLE reads: none $\to$ residual order 4, $f_1 f_2 f_3$ (the three sphere axions) $\to 4$, $f_2 \to 6$, $f_1 f_3 \to 1$ (hol AND anti), wrong weights $(1, 1, 2)/(2, 2, 2) \to 4$; the strict solutions are exactly χ_4 (dims $(3, 3)$) and χ_{10} (dims $(1, 1)$) at η^0, certified on the dressed functions at a fresh τ (10^{-15}, NO residual multiplier); (4) the DECISION — ALL THREE PREREGISTERED TESTERS FAIL: under both derived solutions the integral (30-digit kernels, negative-exponent cell audit, cut-6 vs cut-8 scan) fails $T_5(V) = 0$ (fractions 0.5/0.83), fails $W_2{:}W_{13} = 4{:}3$ ($W_{13} = 0$), fails $V = -A_{\rm fix} N$ (spreads 1.05/1.47) and the $\psi = -64N$ slice, and the honest (N_1, N_2) orbit rebalance rescues nothing in the positive cone (χ_4: $T_5 = 0$ only at $\psi > 0$; χ_{10}: only at $N_2/N_1 = -4 < 0$) — a GENUINE KILL on the derived surface; (5) NEGATIVE CONTROLS: $\mathbb{Z}_2$/Eguchi–Hanson (gauge-only residual order 2 — the μ_4 defect is $\mathbb{Z}_4$-specific — and the SAME fibre dressing cancels it to 1), $SO(16)/D_8$ (only even sectors, all axion b-characters in μ_2: the order-4 supply structurally absent), wrong form $(x^2+y^2)/4$ ($|G|^2 = 64 \neq 16$, its S-rule fails at $O(1)$), wrong signature ($S^2 = e(-1/2)C$ exactly); [C] the $f_1 f_3 =$ KS-weight identification (deck weights $(1, 3) =$ the twistor fibre weights forced by the `v514` incidence ledger — exact match, NOT a complete BCOV derivation) + the kernel-family convention + the discrete χ_k choice (both evaluated, both fail); the MANDATORY TENSION FENCE: the DECLARED completion reading (`v516`/M2) delivers the $\psi = 64$ slice and cancels $32\,T_3$; the DERIVED chiral measure (this module) fails all three testers — both are exact; the sharp open question is which of the two is the true BCOV measure — neither result is hidden behind the other; [O] the BCOV-integral derivation on $\mathbb{PT}/\mathbb{Z}_4$ that decides between the two measures; Wolfram extension $510 \to 516$ (verified engine run); *NO marker moves — the δ_1 route is decided negative under the derived measure, the cubic-GS question stays dissolved, `SEAM.EQUIV.TWISTOR.01` stays Open, `SEAM.EQUIV.01` untouched*)

- **`v518_celestial_wp5e_delta1_derived_kill.py` (`CELEST.WP5E.DELTA1.01`, 30 checks, ∼20 s; exact $\mathbb{Q}(\zeta_8)$/Fraction/1-960-grid phase arithmetic + mpmath 30-digit kernels + numpy double SVD, deterministic).** S0 conventions + 16-class ledgers + targets; S1 the 16-component Weil system (exact) + the E1.5 closure; S2 gauge transport + the μ_4 multiplier system (koboundary test, character identified); S3 the cancellation ($G = f_1 f_3$, T-fix mechanism, exact table); S4 the strict solutions (constructive scan, $m \in \{0, 8, 16\}$); S5 the decision (function certificates, integrals, three testers, (N_1, N_2) scan); S6 negative controls; S7 verdict + the tension fence. Registered in `run_all.py` (suite 512 modules, green), `script_registry.csv` (cluster *frontier*), ledger row `CELEST.WP5E.DELTA1.01` (new — "DECIDED: kill under the derived measure, tension with v516 stated").

- **Contract updated (markers unchanged).** `CELEST.SEAM.01`: the δ_1 strand moves from "UNDECIDED at exploration level" to DECIDED by `v518`; the named remaining target is now the BCOV MEASURE QUESTION (the declared `v516` completion reading vs the derived `v518` chiral measure — both exact, which is the true BCOV measure is the named [**O**]); the cubic-GS question stays dissolved (`v518` does not revive the *d*-channel). `SEAM.EQUIV.TWISTOR.01` dependencies extended by `v518` (preparation range now `v492`–`v518`; the route stays Open). `CELEST.WP5E.M2.01` cross-referenced (its $[O]$ pointer "$\delta_{1b/1c/1d}$ territory" is executed and decided). NO marker moves anywhere.
- **Paper integration.** `tfpt_research_contracts` (the δ_1 stage moved to *done, DECIDED* with the four exact results and the mandatory tension fence as its own paragraph; "WP5e proper" renarrowed from the δ_{1d} derivation to the declared-vs-derived measure question; the object-diagram row, the honest status line and the per-ID status list extended by `CELEST.WP5E.DELTA1.01`); `tfpt_3` (new Step 19 "the derived measure decides — and disagrees with the declared one" — the celestial story is now nineteen steps; the Step-16/17 fences and "what remains" updated to the measure question); `tfpt_4_frontier` (keybox synced). `v518_celestial_wp5e_delta1_derived_kill.py` is cited via `veri` in `tfpt_research_contracts` and `tfpt_3`.
- **Wolfram.** Six exact mirrors (the 16-component Weil system with Gauss sums and tensor split; the two invariant Lagrangians + $P_H S P_H = \frac{1}{4} J$ + the 3/4 shifted-class weight; the koboundary test $(1, 1, 1)$ on all 15 pairs from the recorded exact transport phases; the $\Gamma_1(4)$ character formula $\lambda(\gamma) = i^{2B+C/4}$ on the recorded loops; the T-fix mechanism $(-1)\, e(-1/6) = e(1/3) = \chi_4(T)$ uniform for $f_1 f_3$ vs inconsistent $\{\chi_3, \chi_4\}/\{\chi_5, \chi_6\}$ for $f_1 f_2 f_3 / f_2$; the wrong-form/wrong-signature controls) added to `tfpt_readouts_extension.wl` and verified with the active engine, $510 \to 516$; the Harvey–Moore integrals, SVD contraction solves, theta covariance certificates and block transport measurements are numerical and stay Python-only, flagged in the `.wl` comment; `GATE.WOLFRAM.02`, Wolfram README and the website counters synced ($116/116 + 516/516$).
- **Provenance.** Discovery probes `celestial_seam_wp5e_delta1b/1c/1d_*.py` — the Harvey–Moore unfolding (30/30), the Weil-16 completion (31/31) and the order-4 multiplier cancellation (36/36) — in `experiments/tfpt-discovery/` (2026-07-22); website mirrors regenerated (`ScriptIndex`, DAG *qft* node + `v518`, `papers.ts` "19 steps", changelog mirror); `next.txt` round notes.

2026-07-22 (VII) (`v517` — WP5e back-reaction milestone M3 "the a_0 uplift" executed (SUCCESS on the preregistered `v514 S8.3` criterion), promoted as the ledger row `CELEST.WP5E.M3.01`: the $(4, \pm 4)$ GH multipole (= the a_0 direction) is uplifted to the twistor-potential kernel $\chi = \log P_4$ — the LOG of the `v515` family polynomial — and the induced Kähler-potential correction is a log-type correction COUPLED TO THE CENTRE COUNT ON FOUR SCALES — (1) the KERNEL BRIDGE: the $O(2)$ section is a NULL coordinate ($\Delta_{3d} G(\eta_p) = 0$ identically for an ARBITRARY kernel G: the uplift is well-typed), the RESIDUE IDENTITY $\partial_x[\text{transform}(\log \eta_p)] = 1/r_p$ holds EXACTLY (residue $1/(2r)$), so the GH potential $V = \sum_p 1/r_p$ IS the x-derivative of the contour transform of χ; flux -4π per centre, source charge $4 = |\mu_4|$ (the V-ledger matched); (2) the ASYMPTOTIC KERNEL LOG: $\chi = 4 \log \eta + a_2/\eta^2 + (a_0 - a_2^2/2)/\eta^4 + \ldots$ with LOG COEFFICIENT $4 = |\mu_4|$ = centre count (the CPS "$N \log$" analogue at kernel level); at the clock-fixed seam fibre $\zeta = 0$ the a_2 slot closes and the first correction is EXACTLY a_0/η^4 = the $(4, \pm 4)$ multipole in kernel clothes; exact m-grading ($m = \pm 4$ pieces PURE a_0, $m = 0$ piece $-6t_0^4$ φ_0-free: the $U(1) \to \mathbb{Z}_4$ breaking sits exclusively in the a_0 pieces); NO $\log \times$power terms (the `v514` honest zero preserved — the a_0 correction is power-law at infinity); the once-integrated GLT tower $\sum_p \eta_p \log \eta_p = 4\eta \log \eta + p_4/(12\eta^3) + p_8/(56\eta^7) + \ldots$ with $p_n = 0$ unless $n \equiv 0 \bmod 4$ and $p_{4k} = 4(-a_0)^k$: EVERY correction carries the prefactor 4 = centre count; (3) the EXCEPTIONAL-LOCUS LOG: $\chi(\eta = 0) = \log a_0 = 4 \log t_0 + i(4\varphi_0 + \pi)$ (t_0-coefficient $4 = |\mu_4|$, clock phase $4i$); the modulus response $a_0\, \partial_{a_0} \chi = a_0/(\eta^4 + a_0)$ equals 1 AT the exceptional locus and falls off as a_0/η^4 at infinity (no constant, no log): the a_0-log lives at the exceptional locus, exactly as `v514 S6.3` demanded; (4) the PERIOD RESPONSE: $\Pi_j(\lambda)$ is LINEAR in t_0 (uniform first-order shift: lockstep + phase ratios $(1, i, i^2)$ preserved exactly), $d \log \Pi_j / d \log a_0 = 1/4 = 1/|\mu_4|$ uniformly, and the integrated shift around the a_0 circle is the monodromy $e^{2\pi i/4} = i$ = ONE Coxeter clock step (the `v493` monodromy REPRODUCED from perturbation theory; the finite monodromy is the cyclic centre permutation $q_p \to q_{p+1}$); the node λ-support is t_0-FREE (the $\mathbb{Z}_8$ seam bridge is a_0-rigid), all 6 pair discriminants $\sim t_0^2 \neq 0$ (K_4 connectivity: the forced uniform flux vector persists), the BM kernel is modulus-free with $\int_{S^3} K = (2\pi i)^2$ (the quantised fluxes cannot move); (5) NEGATIVE CONTROLS: the $(4, 0)$ multipole is φ_0-independent in V AND in the kernel (clock-invariant, breaks nothing); the $\mathbb{Z}_2$/Eguchi–Hanson analogue reads $2 = |\mathbb{Z}_2|$ on EVERY dial ($(2, \pm 2)$ amplitude $\frac{3}{2} \sin^2 \theta$, kernel log 2, exceptional log $2 \log t_0$, response $1/2$, monodromy -1); wrong centre counts $k = 3, 5$: kernel log coefficient k, modulus slot $O(6)/O(10) \neq O(8)$ — the observable MOVES with the centre count; forbidden family $Z^4 - Z$: $p_3 = 3 \neq 0$ (selection rule broken), $e_4 = 0$ (no a_0-log) — fails BOTH uplift dials; [C] the GLT dictionary ($\eta_p \log \eta_p$ = the GH Kähler-potential kernel, Lindström–Roček / Ivanov–Roček); [O] the full nonlinear Kähler potential of the resolved A_3 ALE; with `v516` the `v514` fence M1–M3 is FULLY WORKED OFF; Wolfram extension $504 \to 510$ (verified engine run); *NO marker moves — the KILL (decoupling from the centre count) does NOT fire, `SEAM.EQUIV.TWISTOR.01` stays Open (preparation, not construction), `SEAM.EQUIV.01` untouched*)

- **`v517_celestial_wp5e_m3_a0_uplift.py` (`CELEST.WP5E.M3.01`, 23 checks, ∼7 s; exact sympy, no floats; deterministic).** S0 replication (GH multipole ledger + the twistor family at general orbit phase, `v515` slice); S1 the kernel bridge (null coordinate, residue V-ledger identity, source flux); S2 the uplift (asymptotic log 4, seam-fibre correction a_0/η^4, exact m-grading, GLT tower + selection rule); S3 the exceptional-locus log; S4 Ω_N consistency (period response

1/4, monodromy i, a_0-rigid nodes and fluxes); S5 negative controls; S6 verdict. Registered in `run_all.py` (suite 511 modules, green), `script_registry.csv` (cluster *frontier*), ledger row `CELEST.WP5E.M3.01` (new, SUCCESS).

- **Contract updated (markers unchanged).** `CELEST.SEAM.01` now reads: M2+M3 executed via `v516`/`v517` — the `v514` fence M1–M3 is fully worked off: the twisted KS measure (declared completion reading, ch_2-weighted) cancels the $32\,T_3$ residual and delivers the $\psi = 64$ slice without the cubic d-channel, and the a_0 uplift couples to the centre count on four scales; what stays **[O]**: the BCOV-integral derivation of the completion reading (δ_{1d}) and `SEAM.EQUIV.TWISTOR.01`. `SEAM.EQUIV.TWISTOR.01` dependencies extended by `v516`/`v517` (preparation; the route stays Open).
- **Paper integration.** `tfpt_research_contracts` (the M3 bullet moved to *done* with the four coupled scales and the GLT fence; "what remains" narrowed to δ_{1d} + the BCOV derivation); `tfpt_3` (new Step 18 "the a_0 uplift: four coupled centre-count scales" — the celestial story is now eighteen steps; "what remains" updated). `v517_celestial_wp5e_m3_a0_uplift.py` is cited via `veri` in both.
- **Wolfram.** Six exact mirrors (multipole ledger + family at general phase; kernel bridge (null coordinate, residue $1/(2r)$, flux -4π); asymptotic log + m-grading + GLT tower; exceptional-locus log; period response 1/4 + monodromy i + a_0-rigidity; negative controls) added to `tfpt_readouts_extension.wl` and verified with the active engine, 504 $\to$ 510; `GATE.WOLFRAM.02`, Wolfram README and the website counters synced (116/116 + 510/510).
- **Provenance.** Discovery probe `celestial_seam_wp5e_m2m3_a0_uplift_probe.py` (23/23) in `experiments/tfpt-discovery/` (2026-07-22); website mirrors regenerated (`ScriptIndex`, DAG *qft* node + `v517`, `papers.ts` "18 steps", changelog mirror); `next.txt` round notes.

2026-07-22 (VI) (`v516` — WP5e back-reaction milestone M2 "the twisted Kodaira–Spencer measure" executed (SUCCESS on the preregistered `v514` S8.2 criterion, ON THE DECLARED COMPLETION MEASURE — verdict B), promoted as the ledger row `CELEST.WP5E.M2.01`: the one-loop box coefficient per sector on $\mathbb{PT}/\mathbb{Z}_4$ under the completion contact term $\mathrm{contact}_j = (Q^{(0)} - Q^{(j)})/\det_j$ — the twisted-sector LOOP contributes the UNPHASED sector trace with the g^j zero-mode normalisation $1/\det_j$, of which the Atiyah–Bott skeleton kept only the phase-weighted insertion part — (1) the COMPLETION-WEIGHT IDENTITY (the centrepiece): per glue class $w_m = \sum_j (1 - i^{jm})/\det_j = (0, \frac{3}{2}, 2, \frac{3}{2}) = 4h_m = |\mu_4|h_m = -4\,\mathrm{ch}_2(T_m)$ EXACTLY ($h_m = m(4-m)/8$ the `v502` sector Casimir energies, $\mathrm{ch}_2(T_m)$ the McKay/Kronheimer charges, `v505` S4.3 index bridge): THE THREE SPHERE AXIONS PAIR THROUGH THEIR OWN ch_2 CHARGES — an exact identity, NO free scale, no fit; (2) PARAMETER-FREE LOCKS: $T_5 = 0$ for ANY scale c (the ch_2 pattern alone kills T_5); ratio $h_2 : h_1 = 4 : 3 =$ the δ_1-forced leading ratio (REPRODUCED, not fitted); the T_3 budget fixes $c = 4 = |\mu_4|$ uniquely (leading system det 256, unique weights $(\frac{3}{2}, 2) = (4h_1, 4h_2)$); (3) the SUCCESS TABLE: $\mathrm{skeleton}_j + \mathrm{contact}_j = Q^{(0)}/\det_j = 36\langle x,x\rangle^2/\det_j$ — a PERFECT Okubo square ($T_5 = T_3 = 0$, $\tilde\lambda^2 = 36$ in EVERY twisted channel: $(18, 9, 18) \times \langle x,x\rangle^2$); total $A_{\text{fix}} + \text{contact} = 45\langle x,x\rangle^2 = \frac{5}{4} \times 36\langle x,x\rangle^2 =$ Dedekind $\times$ Okubo = the UNIQUE quartic-free weighting of `v505` S3.8 (the rigidity theorem is exactly what the measure lands on); both `v508` certificates KILLED ($\Phi_{T_3} : 32 \to 0$, $\Phi_P : 72 \to 0$); the `v511` slice $\psi = 64$ SUPPLIED EXACTLY ($\psi(\text{contact}) = -64$) — NO cubic d-channel needed (c_d free $= 0$); remainder inside the exchange span ($45\langle x,x\rangle^2 = \frac{45}{16}E_{22} = \frac{1}{80}E_{00}$, $\Phi_P = 0$): the BULK axion cancels every channel with the SAME $\tilde\lambda^2 = 36$, the twisted propagator carrying the AB zero-mode factor $1/\det_j$; the `v514` measure chain stays μ-blind; (4) NEGATIVE CONTROLS: wrong scale $c \in \{1, 2, 3, 5\}$ leaves $T_3 = (24, 16, 8, -8)$ — only $c = 4 = |\mu_4|$ clears the budget; the $h_1 \leftrightarrow h_2$ shuffle breaks $T_5 = -14$ AND $T_3 = 36$; $SO(16)$ glue keeps $T_5 = 20$ / $T_3 = -40$ — the KILL branch FIRES there (Okubo failure inherited: E_8 doubly special); $\mathrm{diag}(i, i)$ degenerates to the $\mathbb{Z}_2$ target and breaks the conjugation pairing + the index bridge; the $\mathbb{Z}_2$/Eguchi–Hanson anchor PASSES with the same mechanism ($A_2 + \text{contact} = 9\langle x,x\rangle^2 = \frac{1}{4} \times 36\langle x,x\rangle^2$, coupling scale $2 = |\mathbb{Z}_2| =$ centre count); (5) the MULTIPLIER NOTE (cross-reference to the parallel δ_{1d} strand): the contact weights $4h$ are REAL RATIONALS with no multiplier system of their own; the δ_{1c} residual phases are the SQUARES of the A_3 discriminant T-phases $e(3a^2/8)$ — orders $(1, 8, 2, 8)$ squaring to $(1, 4, 1, 4)$, a finite μ_4 system; Gauss sums $2e(3/8)$, $2e(5/8)$, full Weil sum 4; [C] the MANDATORY FENCE: the completion reading (the loop of twisted states carries the UNPHASED sector trace with the same zero-mode normalisation) is DECLARED, supported by the δ_1 modular-completion finding, NOT derived from the BCOV integral; [O] the BCOV-integral derivation ($\delta_{1b}/\delta_{1c}/\delta_{1d}$ territory); Wolfram extension $498 \to 504$ (verified engine run); *NO marker moves — the KILL (leftover independent quartic in ANY sector) does NOT fire, the completion reading stays [C]-declared, `SEAM.EQUIV.TWISTOR.01` stays Open, `SEAM.EQUIV.01` untouched*)

- **`v516_celestial_wp5e_m2_twisted_ks_measure.py` (`CELEST.WP5E.M2.01`, 23 checks, ∼1 s; exact sympy/Fraction, no floats; deterministic).** S0 replication (roots, class quartics, A_{fix} two routes, bulk Okubo); S1 the index bridge + the completion weights (no free scale, parameter-free locks); S2 the contact term (two forms, genuine-contact certificate); S3 the success table (per-sector Okubo squares, Dedekind × Okubo, certificate kills, exchange span, μ-blind chain); S4 negative controls; S5 the multiplier note (cross-reference to δ_{1d}); S6 verdict. Registered in `run_all.py` (suite 511 modules together with `v517`, green), `script_registry.csv` (cluster *frontier*), ledger row `CELEST.WP5E.M2.01` (new, SUCCESS on the declared measure).

- **Contract updated (markers unchanged).** `CELEST.SEAM.01`: the M2 bullet moves to *done*; the `v511` burden paragraph is closed out — the $\psi = 64$ slice is DELIVERED by the declared KS measure (`v516`); the cubic d-channel is not needed — its physical-justification question dissolves unless δ_{1d} revives it. The completion reading itself stays [C]-declared; its BCOV-integral derivation stays **[O]** (δ_{1d}).
- **Paper integration.** `tfpt_research_contracts` (the M2 bullet moved to *done* with the weight identity and the success table; the `v511` fence paragraph extended); `tfpt_3` (new Step 17 "the twisted KS measure: every sector the same Okubo square" — on the way to the eighteen-step story). `v516_celestial_wp5e_m2_twisted_ks_measure.py` is cited via `veri` in both.
- **Wolfram.** Six exact mirrors (completion-weight identity + index bridge; root/quartic replication; parameter-free locks; contact term + success table + certificates + μ-blind chain; negative controls; multiplier note) added to `tfpt_readouts_extension.wl` and verified with the active engine, 498 → 504; `GATE.WOLFRAM.02` and the Wolfram README synced.
- **Provenance.** Discovery probe `celestial_seam_wp5e_m2m3_twisted_ks_measure_probe.py` (23/23) in `experiments/tfpt-discovery/` (2026-07-22); website mirrors regenerated (`ScriptIndex`, DAG *qft* node + `v516`, changelog mirror); `next.txt` round notes.

2026-07-22 (V) (`v515` — WP5e back-reaction milestone M1 "the A_3 Ω_N" executed (SUCCESS on the preregistered `v514` S8.1 criterion), promoted as the ledger row `CELEST.WP5E.M1.01`: the back-reacted holomorphic 3-form Ω_N is CLOSED FORM on the A_3 twistor family, every $S^3/\mathbb{Z}_4$ period is $(2\pi i)^2$-integral with the lockstep flux vector, and the lens geometry FORCES the source charge $4 = |\mu_4|$ — (1) the RESIDUE FORM DERIVED: on $F = XY - P(Z)$ all three ambient representatives satisfy $dF \wedge \omega_2 = -dX\wedge dY\wedge dZ$ exactly and pull back to $\omega_2 = dX\wedge dZ/X$ (Poincaré residue, derived not assumed); orbifold-cover pullback at $a_0 = 0$: $\omega_2 = 4\, dz_1\wedge dz_2 = |\mu_4|\times$(flat form) — the residue normalisation CARRIES the source charge 4; clock $\omega_2 \to i\,\omega_2$ (`v492` S5 replicated); weight ledger $4+4+2-8 = 2$, $\Omega \in O(4) = -\deg K_{\mathbb{PT}}$; (2) the TWISTOR FAMILY CLOSED: from the four $O(2)$ centre sections $q_p(\lambda) = i^p t_0 - i^{-p} t_0 \lambda^2$ exactly $XY = Z^4 + 4t_0^2\lambda^2 Z^2 - t_0^4(1-\lambda^4)^2$ ($e_1 = e_3 = 0$ identically; $\lambda = 0$ gives exactly the `v493` family; $a_2 \in O(4)$ opens only off-seam), CY-compatible clock lift $\gamma : (Z,\lambda) \mapsto (iZ, -i\lambda)$, $\gamma^4 = 1$, $\Omega \to +\Omega$; (3) Ω_N CLOSED + PERIOD TABLE: $\Omega_N = \Omega_0 + \sum_p N_p K_p$ with CPS/Bochner–Martinelli kernels on the centre twistor lines, $dK = 0$ off-source; boundary $S^3/\mathbb{Z}_4$ at the orbifold point $(2\pi i)^2 \times 1$ (minimal invariant charge $4 = |\mu_4|$ — the lens fundamental domain FORCES $N \in 4\mathbb{Z}$), deformed phase $(2\pi i)^2 \times 4$, linking S^3 uniformly $(2\pi i)^2 \times N$, per-sphere flux vector $(2\pi i)^2 \times N(1,1,1)$ (forced uniform: clock orbit + K_4 connectivity), seam-fibre phase vector $2\pi i\, t_0(i-1)(1,i,i^2)$; Ω_0 carries ZERO 3-flux (all flux sourced); $(2\pi i)^2 =$ fibre residue × transversal residue (density $-\sin 2\theta$); (4) CLOCK COVARIANCE: $\Pi_{j+1}(-i\lambda) = i\,\Pi_j(\lambda)$ for ALL λ; the 12 collision nodes sit exactly on the 8 eighth roots of unity ($8 = 2|\mu_4|$, the $\mathbb{Z}_8$ bridge), clock orbits 4+4+4, honest conifold points (Hessian det $512t_0^6$, quadric rank 4); (5) the HONEST KILL PROBE: forbidden family $Z^4 - Z$: 0/24 lockstep (vs 8/24), λ-support 12 twelfth roots $\neq \mathbb{Z}_8$, no antipodal pairing — BUT the $(2\pi i)^2$ quantisation holds there TOO: integrality alone is NOT the discriminator, the discriminator is the lockstep phase structure + the clock forcing; the `v509` dial survives; (6) ANCHORS + CONTROLS: EH/$\mathbb{Z}_2$ (charge $2 = |\mathbb{Z}_2|$ forced), CPS original (every integer N allowed), $\mathrm{diag}(i,i)$ ($\Omega_0 \to -\Omega_0$, collapse at step zero), fractional charges $N = 1,2,3$ excluded; [C] global kernel patching / Hitchin small resolution / CPS brane dictionary; [O] M2 (twisted KS measure), M3 (a_0 uplift), the quantised BCOV coefficient; Wolfram extension $490 \to 498$ (verified engine run); *NO marker moves — M2/M3 and BCOV stay [O], `SEAM.EQUIV.TWISTOR.01` stays Open (`v515` is preparation, not construction), `SEAM.EQUIV.01` untouched*)

- **`v515_celestial_wp5e_m1_omega_n.py` (`CELEST.WP5E.M1.01`, 30 checks, ∼1 s; exact sympy, no floats; deterministic).** S1 the residue form (three routes, cover factor 4, weight ledger, EH anchor, wrong-weight control); S2 the twistor family (closed form, $O(2)$ reality, clock lift, section-level dichotomy); S3 periods (generic reduction, lockstep, covariance, collision ledger on $\mathbb{Z}_8$); S4 Ω_N (BM anchor, lens descent, source curve, conifold nodes, M1 criterion); S5 the KILL probe (honest); S6 negative controls; S7 fences + verdict. Registered in `run_all.py` (suite 509 modules, green), `script_registry.csv` (cluster *frontier*), ledger row `CELEST.WP5E.M1.01` (new, SUCCESS).
- **Contract updated (markers unchanged).** `CELEST.SEAM.01` now reads: M1 executed via `v515` — the back-reacted Ω_N is closed-form on the A_3 twistor family, all $S^3/\mathbb{Z}_4$ periods are $(2\pi i)^2$-integral with the lockstep flux vector and clock covariance, and the lens geometry FORCES the source charge $4 = |\mu_4|$; honest fence: integrality alone does not discriminate — the discriminator is the lockstep phase structure (0/24 on the forbidden family); M2/M3 stay [O]. `SEAM.EQUIV.TWISTOR.01` dependencies extended by `v515` (preparation; the route stays Open).
- **Paper integration.** `tfpt_research_contracts` (the M1 bullet moved to *done* with the core

numbers; the object-diagram row $\mathbb{C}^2/\mathbb{Z}_4 \to \mathbb{PT}/\Gamma$ extended by `v515` with the M1 kill retired to δ_1/M2–M3; WP4 handover, WP5 narrative, honest status line and status inventory synced); `tfpt_3` (new Step 16 "the back-reacted Ω_N: closed form, integral periods, forced charge 4" — the celestial story is now sixteen steps; "what remains" updated: M2/M3 are the remaining preregistered milestones); `tfpt_4_frontier` stage list synced. `v515_celestial_wp5e_m1_omega_n.py` is cited via `veri` in both.

- **Wolfram.** Eight exact mirrors (residue-form identities; twistor family closed; period reduction + lockstep + covariance; collision ledger on $\mathbb{Z}_8$; BM anchor + lens forcing $N \in 4\mathbb{Z}$; source curve K_4; conifold nodes; KILL probe + EH/diag(i,i) controls) added to `tfpt_readouts_extension.wl` and verified with the active engine, 490 → 498; `GATE.WOLFRAM.02`, Wolfram README and the website counters synced (116/116 + 498/498).
- **Provenance.** Discovery probe `celestial_seam_wp5e_m1_a3_omega_n_backreaction_probe.py` (30/30) in `experiments/tfpt-discovery/` (2026-07-22); website mirrors regenerated (`ScriptIndex`, DAG *qft* node + `v515`, `papers.ts` "16 steps", changelog mirror); `next.txt` round notes.

2026-07-22 (IV) (external review integrated: route renamed, object diagram, group-extension definition, SEAM.EQUIV route split (MMST/TWISTOR ids, no status change), A_3 role table, `WOIT.OS.TWISTOR.01` contract — a pure structure/-text integration, NO new physics claims, NO marker moves, suite unchanged at 508)

An external expert review of the celestial/twistor branch confirmed the substance and asked for *semantic* precision — role separation, status clarity, and the exact naming of what is missing. Six changes plus one new central research contract, all editorial/structural:

- **(Ä1) Route renamed + the Woit bridge.** The `tfpt_3` celestial section and the contract heading are now **"The celestial and twistor continuum route"** (the old "from the μ_4 clock to the $(E_8)_1$ boundary shadow" survives as the narrative-arc subtitle). A new paragraph block in `tfpt_3`, *"The Woit bridge: chiral Wick rotation and the real structure"* [O], introduces Woit's Euclidean Twistor Unification (`arXiv:2104.05099`) as a *named external programme* (chiral Wick rotation problematic for purely chiral theories; Euclidean-holomorphic theory on $\mathbb{PT}$, Lorentzian theory via conjugation structure/boundary values), lists TFPT's existing ingredients (RP `v379`, OS reconstruction, ρ = clock `v492`, σ = deck/seam reflection `v510`), and names exactly what is missing: a global anti-linear $\Theta : \mathcal{A}(\mathbb{PT}/\Gamma) \to \mathcal{A}(\mathbb{PT}/\Gamma)$ with $\Theta^2 = 1$, $\Theta\rho\Theta = \rho^{-1}$, plus RP of the interacting BCOV+SDYM functional — "until that exists, the Woit connection is a strong analogy, not a mathematical integration". No namedropping as confirmation.
- **(Ä2) The object diagram.** `tfpt_research_contracts` (CELEST section, up front) now opens with the chain $(\Sigma, \mu_4, \rho, \Theta) \to \mathbb{P}^1\backslash\mu_4 \to \mathbb{C}^2/\mathbb{Z}_4 \to \mathbb{PT}/\Gamma \to \mathcal{A}_{\mathrm{hol}} \to_{\mathrm{OS}} \mathcal{A}_{\mathrm{Mink}}$ as a table with one row per arrow (claim id, status, load-bearing modules `v492/v493/v497`–`v514`, kill test); the last two arrows are honestly [O] with `WOIT.OS.TWISTOR.01` as the target — where mathematics ends and physics begins is visible at a glance.
- **(Ä3) The group extension, exactly.** A definition block: $\Gamma_{\mathrm{ALE}} = \langle\mathrm{Deck}\rangle \cong \mathbb{Z}_4 \subset SU(2)$ (triholomorphic, preserves Ω; the quotiented $\mathbb{Z}_4$); $\rho = \mathrm{Clock} = \mathrm{diag}(i, 1) \in U(2)$ (Kähler, *not* triholomorphic, $\det = i$ rotates Ω by i — `v492` S5; normalises, is not quotiented); the global object is cyclic $\langle\rho\rangle$ with $\rho^2 = \mathrm{Deck}$ (projectively $\mathbb{Z}_4$, spin lift $\mathbb{Z}_8$; `v492/v506/v507`), plus the action table on $K_{\mathbb{PT}}$, Ω, BCOV fields, open-string fields and boundary states, each entry [E] (arithmetic) or [O] (quantisation question); "$\mathbb{PT}/\mathbb{Z}_4$" is defined once for the whole contract (quotient by Γ_{ALE}, normalised by $\langle\rho\rangle$) with a cross-reference sentence in `tfpt_3`.
- **(Ä4) SEAM.EQUIV route split (ledger restructuring, no status change in substance).** New ledger rows `SEAM.EQUIV.MMST.01` (the conditional scaling-limit route — closed

modulo cited theorems, typing taken verbatim from the pre-split rows/prose; residual **[O]** = the cited continuum existence, `v336`) and `SEAM.EQUIV.TWISTOR.01` (the open Costello–Li construction, **[O]**; dependencies on `CELEST.SEAM.01`/WP modules and `WOIT.OS.TWISTOR.01`). `SEAM.EQUIV.01` *remains* as the parent target with the OR logic as claim text: *closed if MMST.01 OR TWISTOR.01 closes; currently: MMST conditional-closed modulo cited theorems, TWISTOR open ⇒ parent stays **[O]** as an unconditional claim.* Every surface that said "`SEAM.EQUIV.01` closed modulo cited theorem" (status cards in `introduction`, `tfpt_1/3/4`, the central status card, README, `origin_theory`, `tfpt_2`, `tfpt_5`, website StatusPyramid/OpenGates/StatusMatrix/`papers.ts`/glossary/VerificationDag) now attaches the statement to the MMST route id — the assertions are unchanged, only the ids are clean.

- **(Ä5) The A_3 role table.** In the contract: A_3^{family} (lattice family factor, $\mathfrak{su}(4)$ flavour), A_3^{ALE} (ADE type of $\mathbb{C}^2/\mathbb{Z}_4$), the three exceptional spheres (Coxeter/PL structure), D_5^{carrier}, and — as external reference rows — Woit's $SU(3)_{\text{color}}$ and $SU(2)_{\text{weak}}$: per pair either the connecting functor with its `vN` witness (McKay type-level; Kronheimer–Nakajima quiver `v479`) or an explicit *"not identical — no identification claimed"*; in particular $A_3^{\text{family}} \not\simeq SU(3)_{\text{color}}$, and the Woit rows are pure programme references.
- **(Ä6) Anti-number-hunting programme note.** `experiments/next.txt` (dated 2026-07-22 (IV)): the next real gain lies at positivity, incidence, locality and the Θ real structure — *not* at further factorisations; recorded as working discipline per the review.
- **New central contract `WOIT.OS.TWISTOR.01` (Open, research contract; new ledger row).** In `tfpt_research_contracts` after `CELEST.SEAM.01`: input $X_+ = (\mathbb{PT}/\Gamma)_+$, $\mathcal{A}_{\text{hol}}$ (interacting open+closed twistorial algebra), ρ (order-4 clock), Θ (seam-reflection-induced anti-linear real structure), $\mu_{\text{BCOV+SDYM}}$; target theorem $\Theta^2 = 1$, $\Theta\rho\Theta = \rho^{-1}$, RP of the gauge-invariant algebra, OS quotient $\Rightarrow (\mathcal{H}, \Omega, U(\mathcal{P}_+^\uparrow), \mathcal{A}_{\text{Mink}})$ with positive Hilbert metric, positive energy, local causality, *one chiral generation without mirror doubling*, the TFPT charge lattice, internal $SU(3)\times SU(2)\times U(1)$ action, and the $(E_8)_1$ seam net as an *actual* boundary net; *seven* preregistered kill tests (Θ–clock incompatibility; RP failure after gauge fixing; vector-like mirror generation; Penrose transform only free/self-dual; right character but wrong OPE/net equivalence; $SU(2)$ stays a spacetime factor; the four μ_4 marks not incidence-compatibly extendable); typing "this is the actual bridge from compiler to physics — everything in `CELEST.SEAM.01` is preparation for it"; honest scope fence (self-dual ≠ full 4D physics: both helicities, generic amplitudes, local matter, full Einstein dynamics, EWSB, confinement are explicit non-claims of the celestial branch). Mirrored on the website (`papers.ts` section, glossary entries "Woit bridge" and "Real structure Θ").
- **No marker moves; no new `vN` modules; suite unchanged (508 modules, green).**

2026-07-22 (III) (`v514` — WP5e-ε_1 "the $O(-2)$ bulk-axion slot" executed (verdict B), promoted as the ledger row `CELEST.WP5E.EPS1.01`: the $s = 0$ tower field of the celestial route is now a CONSTRUCTION, $\tilde{\lambda} = 6$ is pinned three ways, and the first honest back-reaction step is taken in Gibbons–Hawking form — (1) the EQUIVARIANT $O(-2)$ SLOT: the pushforward ledger $\pi_* O(-2)$ on $\mathbb{PT}' = \mathrm{Tot}(O(1)\oplus O(1) \to \mathbb{P}^1)$ gives per degree d exactly $(d+1)^2$ classes = the exact wave-operator nullspaces $(1, 4, 9, 16, 25, 36$ for $d = 0..5)$, the `v492` action on the fibre coordinates $(\mu_1, \mu_2) \mapsto (i\mu_1, i^{-1}\mu_2)$ with incidence-forced column weights $(+1, +1, -1, -1)$ closes the equivariant bookkeeping BLOCK BY BLOCK (all $d \leq 6$, all 4 characters); character series $P_0 = 1 + 3t^2 + 15t^4 + 21t^6 + 45t^8 + \ldots$, $P_1 = P_3 = 2t + 8t^3 + 18t^5 + \ldots$, $P_2 = 6t^2 + 10t^4 + 28t^6 + \ldots$; the $d = 0$ slot has multiplicity $(1, 0, 0, 0)$ — THE BULK AXION SURVIVES THE PROJECTION; Molien invariant ring $= (1 - t^8)/((1 - t^4)^2(1 - t^2))$ — $Z \in O(2)$, $X, Y \in O(4)$, relation degree 8 = the a_0 weight (`v492`/`v493` bridge); E3 bijection sharpened (twisted minimal content $(2t, 6t^2, 2t)$ = the Coxeter eigenvalues, $\det(A - 1) = -4$); graviton control: $O(+2)$ invariant only from fibre degree 4 with multiplicity $3 = \{X, Z^2, Y\}$; (2) $\tilde{\lambda} = 6$ TRIPLY PINNED: Okubo $\mathrm{Tr}_{\mathrm{adj}}\, X^4 = 36\langle x, x\rangle^2 = (6\langle x, x\rangle)^2$ on the 240 glue roots ($36 = h^\vee + 6$, $\tilde{\lambda} = 6 = |\mathbb{Z}_2| N_{\mathrm{fam}}$); the measure chain cancels $\mu = 1/4$ EXACTLY (vertex^2 μ^2, propagator $1/\mu$, anomaly μ; $\tilde{\lambda}^2_{\mathrm{eff}} = 36$ for arbitrary μ; wrong bookings EXCLUDED: anomaly-only $\Rightarrow 3$, missing propagator renormalisation $\Rightarrow 12$); flux (minimal CPS quantum $N = 1$, single channel iff $k = 1$, $(\kappa/c_3)^2 = 12$ exact); (3) the GH/A_3 BACK-REACTION STEP: invariant 4-centre configurations = exactly 2 branches, the free orbit re-derives the `v493` family ($\prod_p (Z - i^p z_0) = Z^4 - z_0^4$, $a_0 = -z_0^4$, monodromy = one clock step), the charge-k GH point is exactly $\mathbb{R}^4/\mathbb{Z}_k$ ($k = 4$: seam boundary $S^3/\mathbb{Z}_4$), periods $\Pi_j = 4\pi t_0(i - 1)(1, i, i^2)$ LOCKSTEP (`v509` counter-check), Coxeter clock from geometry ($\Pi \cdot A = i\Pi$), source charge $4 = |\mu_4|$, EH asymptotic log coefficient EXACTLY 0 (the CPS log is an exceptional-locus statement; Burns contrast $\det g \neq 1$), multipole rule $m \equiv 0 \bmod 4$ with $(4, \pm 4)$ carrying $-a_0$; controls $\mathfrak{so}_8$ ($\tilde{\lambda}^2 = 12$ irrational, squares only $\{9, 36\}$), $\mathrm{diag}(i, i)$ (Veronese), $k = 2$ (three channels); [C] conditional on Costello's flat-$\mathbb{PT}$ matching; [O] the quantised BCOV coefficient and the twisted channels $(32\, T_3)$ — M1–M3 preregistered (the A_3 Ω_N / the twisted KS measure / the a_0 uplift, each with success + kill); Wolfram extension $482 \to 490$ (verified engine run); *NO marker moves — WP5e stays [O], `SEAM.EQUIV.01` untouched, `CELEST.SEAM.01` stays Open*)

- **`v514_celestial_wp5e_eps1_axion_slot.py` (`CELEST.WP5E.EPS1.01`, 34 checks, ∼4 s; exact sympy/Fraction, no floats; deterministic).** S1 Penrose accounting; S2 equivariant ledger + character series + Molien + E3 sharpening + graviton control; S3 the $\tilde{\lambda} = 6$ residue (Okubo / measure / flux) with honest gaps; S4–S5 GH centre dichotomy, periods, Coxeter clock; S6 the CPS "N log" ledger (kernel charge, multipoles, EH log); S7 negative controls; S8 the M1–M3 roadmap; S9 verdict. Registered in `run_all.py` (suite 508 modules, green), `script_registry.csv` (cluster *frontier*), ledger row `CELEST.WP5E.EPS1.01` (new, verdict B).
- **Contract updated (markers unchanged).** `CELEST.SEAM.01` now reads: ε_1 executed via `v514` — the slot is a construction, $\tilde{\lambda} = 6$ pinned by three exact ledgers, the GH/A_3 step re-derives the `v493` family and the Coxeter clock from geometry; the quantised BCOV coefficient and the twisted channels stay [O] (M1–M3 preregistered).
- **Paper integration.** `tfpt_research_contracts` (the ε_1 stage moved to *done* with its own stage block; the named remaining targets now carry the M1–M3 milestones verbatim; WP4 handover and honest status line synced); `tfpt_3` (new Step 15 "the bulk axion is a construction; $\tilde{\lambda} = 6$ is pinned three ways" — the celestial story is now fifteen steps; "what remains" updated).

`v514_celestial_wp5e_eps1_axion_slot.py` is cited via `veri` in both.

- **Wolfram.** Eight exact mirrors (equivariant Penrose ledger; character series + bulk-axion survival; Molien hypersurface + Veronese control; Okubo perfect square + $\mathfrak{so}_8$ control; measure cancellation excluding 3 and 12 + flux dial; GH centre dichotomy; period lockstep + Coxeter clock; multipole rule + EH log ledger) added to `tfpt_readouts_extension.wl` and verified with the active engine, $482 \to 490$; `GATE.WOLFRAM.02`, Wolfram README and the website counters synced $(116/116 + 490/490)$.
- **Provenance.** Discovery probes `celestial_seam_wp5e_eps1_o2_bulk_axion_probe.py` (20/20) and `celestial_seam_wp5e_eps1_gh_a3_backreaction_probe.py` (14/14) in `experiments/tfpt-discovery/` (2026-07-22); website mirrors regenerated (`ScriptIndex`, DAG *qft* node, `papers.ts` "15 steps", changelog mirror); `next.txt` round notes.

2026-07-22 (II) (`v513` — the c_d negative certificate, promoted as the ledger row `CELEST.DTERM.NONDERIV.01` (anti-numerology certificate in the `v508` style, honest rigid NEGATIVE result): can geometry derive $c_d = 1920$? NO — the `v511` [C] fence "$1920 = |W(D_5)|$" does NOT survive the honest four-route battery with its PRE-DECLARED look-elsewhere ledger: (1) the CONVENTION TABLE gives $c_d \in \{1920, 120, 256, 32, 69120\}$ across five normalisations (probe tr-fund/e_8-Killing; Gell-Mann; $\mathfrak{su}(4)$-Killing; plain trace; ch_3-normalised $d/6$) — only 1 of 5 hits $|W(D_5)|$, while the convention-STABLE content is the factorisation $c_d = \Phi_{T_3}(A_{\rm fix}) \times 2h^\vee(E_8) = 32 \times 60$, both factors [E] (Killing propagator 60 from $\sum_\alpha \langle\alpha, e\rangle^2 = 120$ over the actual roots; T_3-slot fixing via $\det(\mathbf{P\text{-}block}) = -64 \neq 0$); (2) K1 DEAD: the charge-legal flux pairings $(1,3)/(2,2)$ give $2048/1800$ — they BRACKET 1920 and both MISS; the only flux hit $64 \cdot 60/2$ is the FORBIDDEN $(1,2)$ pairing (charge $3 \bmod 4$, `v511` innerness); look-elsewhere: 924 expressions from 21 [E] numbers, 11 hits on 1920, controls $1800 \to 8$, $2048 \to 5$, $2016 \to 0$; (3) K2: quantisation delivers only lattices ($60\,\mathbb{Z}$ with 1920 the 32nd multiple, no minimality; strict ch_3 units $2160\,\mathbb{Z}$ EXCLUDE 1920); the cubic quantum is lattice-dependent (6 on coroots, $3/8$ on weights), the integer 32 has 12 factorisations; (4) K3 PROVENANCE CLASH (the death blow): cubic indices $A_m = (0, 16, 0, -16)$ with $\sum_m i^m A_m = 32i$ numerically $= \Phi_{T_3}(A_{\rm fix})$ — but the quartic class decomposition $(10, -4, 30, -4)$ comes from classes $0/2$ vs the cubic $16{+}16$ from classes $1/3$: same number, disjoint mechanism = coincidence; the Weyl side is WRONG (d-channel is A_3, $|W(A_3)| = 24$; D_5 has no cubic; $|W(E_8)|/|W(A_8)| = 1920$ a second accidental 1920); (5) K4 controls ($SO(16)$, $k = 2$ doubling 7680, false $\mathfrak{g}_0$ with six cubics, $\mathfrak{so}(10)$) separate consistently; VERDICT: NUMEROLOGY as a derivation claim — the `v511` [C] fence stays a fence, now typed look-elsewhere-loaded; the honest [E] core is $c_d = 32 \times 60$; Wolfram extension $476 \to 482$ (verified engine run); *NO marker moves — the fence stays [C], the physical generation of the 32 stays [O], `CELEST.SEAM.01` stays Open*)

- **`v513_celestial_dterm_nonderivation.py` (`CELEST.DTERM.NONDERIV.01`, 24 checks, ∼2 s; exact sympy/Fraction, no floats; deterministic).** N0 baseline (240 roots $(52, 64, 60, 64)$, $A_{\rm fix} = (9, -30, -15, 0, 32)$, Q_{dd} two routes, c_d pinned by the T_3 slot alone); N1 convention table + the stable factorisation 32×60; K1 flux products + look-elsewhere ledger; K2 quantisation lattices; K3 cubic/quartic provenance clash + Weyl data; K4 negative controls. Registered in `run_all.py` (suite 507 modules, green), `script_registry.csv` (cluster *frontier*), ledger row `CELEST.DTERM.NONDERIV.01` (new).
- **Fence sharpened (text only, marker unchanged).** `CELEST.WP5E.DELTA2.01` and the `CELEST.SEAM.01` contract prose now carry: "the $1920 = |W(D_5)|$ reading is look-elsewhere-loaded (11/924 vs. 8/924 for the control target 1800) and convention-contingent — `v513`"; the

honest remaining target of the cubic-GS item is the physical generation of the factor 32, not the Weyl fingerprint.

- **Paper integration.** `tfpt_research_contracts` (δ_2 fence environment, named remaining targets, status list — new `CELEST.DTERM.NONDERIV.01` status line); `tfpt_3` (Step-14 paragraph + the "what remains" sentence); `tfpt_4_frontier` (celestial-route clause). `v513_celestial_dterm_nonderivation.py` is cited via `veri` in the δ_2/fence environments.
- **Wolfram.** Six exact mirrors (propagator anchor + T_3-slot pinning; five-way convention table; flux enumeration + look-elsewhere counts; K2 lattices; K3 provenance clash + Weyl data; K4 controls) added to `tfpt_readouts_extension.wl` and verified with the active engine, $476 \to 482$; `GATE.WOLFRAM.02`, Wolfram README and the website counters synced ($116/116 + 482/482$).
- **Provenance.** Discovery probe `celestial_seam_wp5e_dterm_coefficient_derivation_probe.py` in `experiments/tfpt-discovery/` (2026-07-22, 24/24); website mirrors regenerated (`ScriptIndex`, DAG *qft* node, `papers.ts`, changelog mirror); `next.txt` round notes.

2026-07-22 (I) (`v512` — the flag-transitivity equivalence web promoted as `SEAM.TAU.FLAG.01`: the $\tau = i$ attack executed on the free RP seam circle — BARE MARK-TRANSITIVITY IS FALSIFIED as a selector (on the test family $M(\delta) = \{\pm 1, \pm e^{i\delta}\}$, $\delta \in (0, \pi/2]$, the symmetry group is the pair-exchanging $V_4 = \{z, -z, b/z, -b/z\}$ for EVERY δ, simply transitive on the four marks; complete census: deck-commuting Möbius maps are diagonal/antidiagonal; hardening: ALL undirected per-mark data are identical for every δ — no local jet sees the bit), and THE SHARP SELECTOR is FLAG transitivity (marks AND their two sides indistinguishable) $\iff \delta = \pi/2 \iff \tau = i \iff$ the clock — stabiliser jump $1 \to 2$, flag/arc orbits $2 \to 1$, order-4 rotation ($b^2 = -1$ solveset $\{\pi/2\}$), mark mirror $z \mapsto 1/z \iff \cos\delta = 0$; welded into a 12-FOLD EQUIVALENCE WEB (every link exact): clock rotation, D_4, mark mirror, flag transitivity, three spectral degeneracies (odd-doublet split $= (2/\pi)\ln\cot(\delta/2)$, zero only at $\pi/2$ — counterwitness exactly $(2/\pi)\ln 2$; arc-Laplacian degeneracy $4/\pi$; the `v215`-K3 indicator in CLOSED FORM $(2\sqrt{2}/\pi)|\ln\tan(\delta/2)|$ — the `v503` lattice curve analytically), ρ-twist existence ($(m^2 - n^2)(t - 1/t) = 0 \iff t = 1$; Atiyah–Bott rationality δ-BLIND — $\mathrm{Tr}(\sigma e^{t\Delta}) = 1$ for all δ, $j_{\mathrm{cw}} = 148176/25$ rational non-CM: rationality selects NOTHING), cusp-4-cycle implementability (H4 rigidity: V_4 with orders $\{1, 2, 2, 2\}$ cannot induce it; centraliser scalar, U^2 nonscalar $\Rightarrow$ no projective involution — projective order exactly 4, $\mathbb{Z}_8$-shadow consistent `v510`), harmonic, $j = 1728$; NEGATIVE CONTROLS: hexagonal (not concyclic, fails preconditions), silver (harmonic but topologically dead), 3 marks (impossible on the free circle: 0 vs 28 invariant subsets; bare $n = 3$ transitivity already forces equal spacing — "4 counts"); [C] THE REST: the last input reduces from a continuous modulus to ONE discrete symmetry-lift bit ($V_4 \to D_4$: side-indistinguishability of the four equally-decorated marks), `v216` local equality does NOT deliver it, its negation is now concretely measurable; scoreboard: the counterwitness passes all 7 established unsided tests, fails all 9 clock-equivalent ones; Wolfram extension $470 \to 476$ (verified engine run); *NO marker moves — the bit stays [C], `QGEO.REALIZE.01` and `P2.TYPING.01/AX.P2.01` text-precised only, `SEAM.BIT.FREEDOM.01` cross-referenced*)

- **`v512_seam_tau_flag.py` (`SEAM.TAU.FLAG.01`, 29 checks, ∼6 s; sympy-exact + mpmath 30 digits + one numpy lattice commutator; deterministic).** S0 family frame (counterwitness inside the family: $\lambda = 5/4$, $j = 148176/25$, cross-ratio $-1/4$); H1 complete symmetry census

(census basis, rotation solvesets, stabiliser jump, flag/arc orbits, `v216` separation, five-member table); H2 spectral criteria in closed form; H3 heat-trace rationality (δ-blind) + twist existence; H4 monodromy rigidity (Schur); NC negative controls; SYN the 12-fold web + counterwitness scoreboard + the typed sentence. Registered in `run_all.py`, `script_registry.csv` (cluster *core*), ledger row `SEAM.TAU.FLAG.01` (new).

- **Reduction state re-stated (text only, markers unchanged).** `AX.P2.01` and `P2.TYPING.01` synthesis notes now read: "four marks + one discrete symmetry-lift bit (flag transitivity $\iff \tau = i$) + π"; `QGEO.REALIZE.01` precision: "the remaining input can equivalently be stated as the flag-transitivity postulate — `v512`; the search space for a future [**E**] closure is narrowed to the RP positivity of the odd sector"; `SEAM.BIT.FREEDOM.01` cross-referenced (the modulus half gains the discrete form).
- **Paper integration.** `tfpt_1` (the P2 reduction keybox re-dated 2026-07-22 + Theorem-7 skeleton keybox), `introduction` (narrative chain, consolidated reduction state, status-card addendum), `tfpt_3` (celestial Step 2 + the honest-status reduction bullet), `tfpt_research_contracts` (QGEO kill-test and marks-derived blocks, WP2 note); secondary mirrors `tfpt_2`/`origin_theory` aligned. `v512_seam_tau_flag.py` is cited via `veri` in all four primary documents.
- **Wolfram.** Six exact mirrors (V_4/D_4 census table incl. flag/arc orbit counts; solveset equivalences + family identities; spectral faces in closed form; twist mode identity + AB δ-blindness + j-rationality control; H4 Schur rigidity; negative controls) added to `tfpt_readouts_extension.wl` and verified with the active engine, 470 $\rightarrow$ 476; the mpmath heat-trace numerics and the numpy lattice commutator stay Python-only (flagged).
- **Provenance.** Discovery probe `seam_tau_i_established_selector_probe.py` in `experiments/tfpt-discovery/` (2026-07-22, 29/29); website mirrors regenerated (`ScriptIndex`, DAG *P2*/*masses* nodes, `papers.ts` reduction-state mirrors, `StatusMatrix`); `next.txt` round notes.

2026-07-21 (XXIII) (`v511` — WP5e-δ_2 of `CELEST.SEAM.01` "the full-tensor ledger" promoted as `CELEST.WP5E.DELTA2.01` (intermediate verdict, exact): the named escape route of the `v508` exchange no-go executed on the FULL adjoint tensor structure of $\mathfrak{e}_8$ — the COLLAPSE IS CONFIRMED full-tensorially AND arity-crossing (the INNERNESS THEOREM: $\mathfrak{g}_0 = \mathfrak{d}_5 \oplus \mathfrak{a}_3$ semisimple with NO $\mathfrak{u}(1)$ (class-0 roots 40+12, root span rank 8), h in the Cartan OF $\mathfrak{g}_0 \Rightarrow$ every invariant tensor carries total charge $0 \bmod 4$: the bilinear Hom table is nonzero ONLY at $j' = -j$ (dims 2/1/1/1) and ALL 15 non-neutral trilinear sector triples have $\mathrm{Hom} = 0$), BUT ONE CUBIC DOOR OPENS: of the neutral trilinears ($\{0,0,0\}: 3 = 1$ sym $+ 2$ antisym, $\{0,1,3\}: 2$, $\{0,2,2\}: 2$, $\{1,1,2\}: 1$, $\{2,3,3\}: 1$; all cross-sector ones pure brackets) the UNIQUE totally symmetric survivor is the $\mathfrak{su}(4)$ d-symbol ($\mathfrak{so}(10)$ has no cubic Casimir, $\mathrm{Inv}(S^3\, 45) = 0$ protects T_5), whose exchange quartic $Q_{dd} = (0, 0, -\frac{1}{240}, 0, \frac{1}{60}) = \frac{1}{60}(T_3 - P_3/4)$ (TWO independent routes, Killing propagator $2h^\vee = 60$ from the roots) has $\Phi_{T_3}(Q_{dd}) = \frac{1}{60} \neq 0$ — the `v508` master kill (product theorem) does NOT extend to cubic vertices; the pairing stays OBSTRUCTED in the charge reading with the genuinely WEAKER certificate $\{\Phi_{T_5}, \psi = \Phi_P - \Phi_{T_3}/4\}$, $\psi(A_{\mathrm{fix}}) = 72 - 8 = 64$ ($M = [E_{13}, E_{22}, E_{00}, Q_{dd}]$ rank 3, augmented 4), and becomes EXACTLY solvable relaxed: $A_{\mathrm{fix}} = -u + 8v + 2w + 1920\, Q_{dd}$ (residual 0); [C] fences $64 = \dim \mathfrak{g}_1 = 2^6$, $1920 = |W(D_5)| = 8 \cdot 240$; controls: $SO(16)/D_8$ has NO symmetric cubic and no A_3 block (E_8/μ_4 doubly special), false $\mathfrak{g}_0$/sector assignments separate; the δ_1 burden SHARPENED to the $\psi = 64$ slice or the selection-rule relaxation — and the δ_1 exploration run itself stands UNDECIDED at exploration level (strict holomorphic q^0 reading REFUTED at the $\mathbb{Z}_2$/Eguchi–Hanson anchor = method boundary, not a kill; contact term = the MODULAR COMPLETION of the AB data, a Harvey–Moore τ-integral with forced leading (T_5, T_3) ratio 4:3; massive coset levels $n \geq 2$ mandatory); Wolfram extension $463 \to 470$ (verified engine run); *NO marker moves — WP5e proper stays [O], `SEAM.EQUIV.01` stays [O]*)

- **`v511_celestial_wp5e_delta2_tensor_ledger.py` (`CELEST.WP5E.DELTA2.01`, 41 checks, ∼4 s; integer-scaled weight multisets + Kostant/Weyl alternating sums + sympy/Fraction, no floats; deterministic).** D0/D1 structure: 240 roots, class split $(52, 64, 60, 64)$; $\mathfrak{g}_0 = \mathfrak{d}_5 \oplus \mathfrak{a}_3$ SEMISIMPLE without $\mathfrak{u}(1)$ (40+12 class-0 roots, no mixed root, root span rank 8); sectors $\mathfrak{g}_1 = (16_s, 4)$, $\mathfrak{g}_2 = (10, 6)$, $\mathfrak{g}_3 = (\mathfrak{g}_1)^*$ as multiplicity-free minuscule Weyl-orbit products, each irreducible (64/60/64); the grading is INNER ($h = (2,2,2,2,2;0^4)$ in the $\mathfrak{g}_0$ Cartan). D3/D4 Hom tables: bilinear nonzero ONLY at $j' = -j$ (dims 2/1/1/1; all 12 charged pairs zero — the inner-torus theorem at arity 2); trilinear: ALL 15 non-neutral triples Hom = 0 (arity 3), neutral dims $(3,2,2,1,1)$ with all cross-sector trilinears ANTISYMMETRIC in their repeated legs — the unique symmetric survivor is the $\mathfrak{su}(4)$ d-symbol ($\mathrm{Inv}(S^3\, 45) = 0$, $\mathrm{Inv}(S^3 \mathfrak{a}_3) = 1$). D5 the new channel: $Q_{dd} = \frac{1}{60}(T_3 - P_3/4)$ by Killing-propagator contraction AND the hand route $|x^2 - (\mathrm{tr}\, x^2/4)\mathbf{1}|^2/60$; $\Phi_{T_3}(Q_{dd}) = \frac{1}{60} \neq 0$; brackets/tadpoles/$d{\times}f$ die exactly; no T_5 analogue. D6 decision: strict reading unreachable (rank 2 vs 3); generous reading STILL unsolvable (rank 3 vs 4) with the weaker certificate $\psi(A_{\mathrm{fix}}) = 64$; relaxed reading EXACTLY solvable with $c_d = 1920 = 32 \times 60$. D7 controls: Killing control = `v508` replication (A_{fix}, E_{13}, E_{22}, $E_{00} = 225 E_{22}$); $SO(16)/D_8$ bilinears $(1,0,1)$, trilinears $(1,0,1,0)$, $\mathrm{Inv}(S^3 \mathrm{adj}_{D_8}) = 0$; false $\mathfrak{g}_0 = \mathfrak{d}_5 + \mathfrak{a}_2 + \mathfrak{u}(1)$: $(5,2,2) \neq (2,1,1)$, $\mathrm{Inv}(S^3 \mathfrak{g}_0) = 6 \neq 1$; false sector content $(16_s, \bar{4})$ kills every pairing. Discovery provenance: `experiments/tfpt-discovery/celestial_seam_wp5e_delta2_full_tensor_ledger_probe.py` (41/41), same day.
- **The δ_1 fence (exploration-level, NOT promoted — verdict UNDECIDED).** The twisted BCOV contact-term probe (`celestial_seam_wp5e_delta1_bcov_contact_probe.py`, 27/27, stays sandbox; v-numbered on demand) contributes its four exact sharpenings as fence

text: (i) the strict holomorphic q^0 reading is REFUTED at the $\mathbb{Z}_2$/Eguchi–Hanson anchor (method boundary, not a kill); (ii) the forced leading (T_5, T_3) ratio is 4:3 (sector 2 : sectors 1+3); (iii) the $SL(2,\mathbb{Z})$ orbit decomposition $15 = 12+3$ identifies the contact term as the MODULAR COMPLETION of the AB data (a Harvey–Moore-type τ-integral), not a free number; (iv) massive coset levels $n \geq 2$ are mandatory. The remaining δ_1 target is the τ-integral, evaluated against the δ_2-sharpened criterion ($\psi = 64$ slice or selection-rule relaxation).

- **Ledger + papers + Wolfram + website.** Ledger: new row `CELEST.WP5E.DELTA2.01` ([**E**] Hom tables/innerness/Q_{dd}/certificate; [**C**] the number fences; [**O**] arity ≥ 4, cubic GS term, δ_1); `CELEST.SEAM.01` updated (delta-2 executed via v511 — the full-tensor collapse is confirmed (innerness), but the cubic d-symbol channel opens T_3 with the weaker certificate $\psi = 64$; delta-1 stands UNDECIDED at exploration level). Papers: `tfpt_research_contracts` WP5e bullet gains the executed δ_2 stage + the δ_1 exploration fence, WP5e-proper targets renamed (τ-integral, cubic GS term, ε_1, back-reaction); `tfpt_3` celestial section extended to the FOURTEEN-step story (new Step 14 "the full-tensor ledger: the collapse is real, and one cubic door opens") with the "What remains" block updated. Wolfram: seven mirrors (innerness count + $\mathfrak{g}_0$ structure; the bilinear and trilinear Hom tables by exact Kostant alternating sums; the unique symmetric survivor; the $SO(16)$/false-$\mathfrak{g}_0$/ false-sector controls; Q_{dd} by two routes; the ψ certificate + the relaxed solution $(-1, 8, 2; 1920)$) verified with the active engine the same day ($463 \to 470$, `ALL WOLFRAM EXTENSION CHECKS PASSED`). Registry + `run_all` (505 modules, suite green: `ALL CHECKS PASSED`). No gate closes, no marker moves — WP5e proper stays [**O**].

2026-07-21 (XXII) (P2 consolidation — the axiomatics reduction state written as ONE coherent statement across the papers (editorial, NO new claims, NO marker moves): the compiler's dimensionless inputs reduce to *four marks* (derived from P1-side topology, v216), *ONE square-modulus datum* ($\tau = i$ — equivalently: the clock / the CM-fixed half-period / the non-split extension class on the free seam circle; v491/v499/v506/v507/v510) and π, plus the [C] identifications R1/R2 of v499; AX.P2.01 stays a DECLARED axiom and QGEO.REALIZE.01 stays [C]/[O] — the consolidation describes the *reduction state*, not an abolition)

- **What is consolidated (and why now).** The P2 state of affairs was spread over five module rounds: the partition corollary (`v491`: $4 = 2{+}1{+}1$ unique $\Rightarrow g_{\mathrm{car}} = e_2 = 5$), the weight typing (`v499`: Deligne + Birkhoff–Grothendieck + Mehta–Seshadri force $E = \mathcal{O}(-2)\oplus\mathcal{O}(-1)^2$, residuals R1/R2 [**C**]), the residual convergence (`v503`: QGEO-R1 $\equiv$ P2-R1), the clock rigidity and bit origin (`v506`/`v507`: one alignment bit, order fermionically forced, deck class derived) and the freedom theorems (`v510`: the position half is topology, the edge class excluded). The honest consolidated statement, now in one place: *the compiler's dimensionless inputs reduce to four marks (derived from P1-side topology), one square-modulus datum ($\tau = i$, equivalently: the clock / the CM-fixed half-period / the non-split extension class on the free seam circle), and π* — plus the [**C**] identifications R1/R2 of `v499`. No new claims; conservative wording; no "axiom abolished" language anywhere.

- **Papers + ledger + website.** `tfpt_1_architecture_e8`: new keybox "The current reduction state of P2" next to the foundational-postulates keybox, with `v491/v499/v506/v507/v510` citations. `tfpt_3`: mirror bullet "The current reduction state" in the honest-status section. `introduction`: the anchor paragraph gains the consolidated one-sentence state and the status card row $a = (1, 1, 2)$ now names the reduction state. Ledger: SYNTHESIS NOTE appended to `AX.P2.01` ("reduction state 2026-07-21: four marks + one square-modulus datum ($\tau = i$) + π"), marker *Axiom unchanged*. Website: `papers.ts` mirrors of the changed sections. No gate closes, no marker moves.

2026-07-21 (XXI) (`v510` — the freedom attack on the `v506`/`v507` alignment bit promoted as `SEAM.BIT.FREEDOM.01`: the POSITION half of the bit is TOPOLOGY — the collar deck is a covering deck (the $\mathbb{Z}_2$ seam of the Möbius double), and covering decks act FREELY; the DIHEDRAL THEOREM (on the seam circle the free involution is UNIQUE = the antipode: 17 involutions in D_{16}, exactly 1 free; $\mathrm{Aut}(C_{16}) = D_{16}$ complete census, brute force $8!$ at $N{=}8$; quotient topology circle/antipode = CIRCLE $\chi{=}0$ vs circle/reflection = INTERVAL $\chi{=}1$ — the 6π Gauss–Bonnet budget needs closed circles, so freeness IS the established P1 datum); NONSPLIT $\iff$ FREE over the complete 17-involution census in exact $\mathrm{Cl}(16)$ (free $\Rightarrow \tilde{U}^2 = 256\,\gamma_1 \cdots \gamma_{16} = (-1)^F$ nonsplit with 2 roots = the $\mathbb{Z}_8$ tower; ALL 16 reflections $\Rightarrow U^2$ = scalar $\{2^7, 2^8\}$, split, 0 roots); the REAL-STRUCTURE classification (deck-commuting families = exactly two; $\mu_4 \to$ equatorial, deck FREE; silver $\to$ real axis THROUGH the deck poles, NOT free — the edge class is the restriction of the BRANCHED pillowcase involution; hexagonal $\to$ NO mark-fixing real structure, RP-incompatible); HARMONIC + FREE $\Rightarrow \mu_4$ exactly ($b = \pm i$); honest counterwitnesses ("nonsplit $\Rightarrow$ clock" is FALSE: $\{0, 1, 8, 9\}$ and $\{\pm 1, \pm(3{+}4i)/5\}$ with $j = 148176/25 \neq 1728$); the alignment bit REDUCES from "harmonic AND central" to the square-modulus datum $\tau = i$ alone; Wolfram extension $457 \to 463$ (verified engine run); *NO marker moves — `QGEO.REALIZE.01` stays [C]/[O], `AX.P2.01` stays an axiom, P2-R1 halved*)

- **`v510_seam_bit_freedom.py` (`SEAM.BIT.FREEDOM.01`, 30 checks, < 5 s; sympy + exact $\mathrm{Cl}(16)$ Fractions, no floats; deterministic).** Part A (dihedral/fermionic): the involution census in D_N on the refined $2N$-point seam circle ($N = 16$: 17 involutions, EXACTLY ONE free = the antipodal shift by 8; all 16 reflections have exactly 2 circle fixed points — sites for even axis, edge midpoints for odd; robustness $N = 8, 12$; odd control $N = 9$: ZERO free — an even carrier count is needed, consistent with $16 = 2^{g_{\mathrm{car}}-1}$); the census is COMPLETE because $\mathrm{Aut}(C_{16}) = D_{16}$ exactly (brute force over all $8! = 40320$ permutations at $N = 8$, propagation at $N = 16$); quotient topology: circle/antipode = CIRCLE ($\chi = 0$, closed), circle/reflection = INTERVAL ($\chi = 1$, two boundary corners) — a fixed-point deck would break the $6\pi = 3 \times 2\pi$ closed-circle count, so "deck free on the seam circle" is exactly the established "the seam is a closed circle", not a new premise; the complete Fock census: ALL 17 involutions implemented in exact $\mathrm{Cl}(16)$ — NONSPLIT $\iff$ FREE with no exception, root dichotomy 2 vs 0; the chain: free $\Rightarrow$ shift 8 $\Rightarrow$ crossing (all 28 invariant 4-sets) $\Rightarrow U^2 = (-1)^F \Rightarrow$ central class $\Rightarrow$ the edge/silver arrangement is EXCLUDED; Klein control (covering decks are free regardless of the gluing); branch control (the pillowcase involution has exactly 4 fixed points = the marks and restricts to a REFLECTION: the silver arrangement mistakes the mark-creating branch $\mathbb{Z}_2$ for the seam deck). Part B (Möbius/RP): the cylinder-double-cover deck is free on all 120 model points and restricts to the antipode on the $\mathbb{Z}_2$ seam fibre; one-sidedness TYPES the $\mathbb{Z}_2$ as the free antipode (quotient $\mathbb{RP}^2$ = the $|\mathbb{Z}_2|$ of $c_3 = 1/(|\mathbb{Z}_2| 2\pi\chi)$); the deck-commuting real structures are EXACTLY two families ($\mu\bar{z}$: fix line through the poles, NOT free; $\mu/\bar{z}$, $\mu > 0$: fix circle $|z| = \sqrt{\mu}$, FREE; $\mu < 0$: empty); the per-configuration table (μ_4 equatorial/free, silver real-axis/not free, generic-unit free without harmonicity, hexagonal RP-incompatible); harmonic + free solves EXACTLY to $b = \pm i$ (on free circles the silver family "harmonic AND edge" is EMPTY — harmonicity becomes sufficient); honest counterwitnesses: the discrete $\{0, 1, 8, 9\}$ and the continuous $\{\pm 1, \pm(3{+}4i)/5\}$ (free, central, $j = 148176/25$, no clock) — the Fock dichotomy measures the CLASS, not the modulus. Premise typing: (P-i) deck = Möbius covering deck $\Rightarrow$ free [standard topology; the one-sidedness strand `origin_theory/v456`]; (P-ii) marks on the Θ-fix seam circle = existing REALIZE [C] content (`QGEO.REALIZE.02` hypotheses, `v264`) — NO new input. Audit finding documented: "circle-freeness" was nowhere typed before (only mark-freeness, `v216`); `v507` F1.1 explicitly admitted the reflection reading of the edge class — `v510` closes that door (a reflection is not a covering deck). The [C] rest, verbatim: *the alignment bit reduces from "harmonic AND central" to the square-modulus datum $\tau = i$ alone; the position half is topology (edge class excluded), the modulus half stays the one carrier input.* Discovery provenance: `experiments/tfpt-discovery/qgeo_realize_deck_freedom_dihedral_probe.py` (14/14) and `qgeo_realize_rp_fixcircle_moebius_probe.py` (16/16), same day.

- **Ledger + papers + Wolfram + website.** Ledger: new row `SEAM.BIT.FREEDOM.01` ([E] dihedral theorem + complete Fock census + real-structure classification + harmonic-free collapse + counterwitnesses; [C] the square-modulus datum); TEXT precisions (markers unchanged) in `SEAM.CLOCK.RIGIDITY.01`, `SEAM.BIT.ORIGIN.01`, `QGEO.REALIZE.01` (bit → square-modulus datum; edge class topologically excluded) and `P2.TYPING.01` (R1 halved). Papers: `tfpt_3` celestial/QGEO passage extended ("the position half is topology"), `tfpt_research_contracts` QGEO kill-test + QGEO.MARKS.03 + celestial-contract passages extended, one-sentence precisions in `introduction`, `tfpt_1`, `tfpt_2`, `origin_theory` where v506/v507 are cited. Wolfram: six mirrors (dihedral census + quotient topology; Aut completeness incl. the 8! brute force; the Cl(16) dichotomy census over all 17 involutions on 256×256 Jordan–Wigner gammas; the real-structure table; harmonic + free $\Rightarrow \pm i$; the counterwitnesses) verified with the active engine the same day (457 → 463, `ALL WOLFRAM EXTENSION CHECKS PASSED`; base file re-verified 116/116). Registry + `run_all` (504 modules, suite green: `ALL CHECKS PASSED`). No gate closes, no marker moves.

2026-07-21 (XX) (`v509` — WP5e-ε_2 of `CELEST.SEAM.01` "the CPS level-from-flux dial" promoted as `CELEST.WP5E.EPS2.01`, verdict B: the Costello–Paquette–Sharma Burns-holography mechanism ("level = flux quantum × Dynkin index") executed on the lockstep spheres — the CPS SKELETON exact (S^3 period $(2\pi i)^2 N$ with pullback density $-\sin 2\theta$, exceptional-sphere Kähler flux $2\pi N$, boundary level magnitude $2N$ by the symplectic-boson contraction, $T(\text{vector}) = 1$); the PAIRING MATRIX pinned, not postulated (ch_2 ledger + integrality + unimodularity + effectivity $\Rightarrow P_{mi} = \delta_{mi}$ by complete enumeration: rows $8/6/8$, 48 unimodular, exactly 2 effective = identity + diagram flip), fluxes $F_i = (64, 60, 64) = \dim \mathfrak{g}_i$ with uniformly ONE quantum per charged current; "level = total flux" KILLED by the lockstep test itself (F_i unequal = flavour multiplicity), the per-current reading $(1,1,1)$ anchored by the lattice current count $(240,0,0,0)$ and embedding index 1; ord-4-vs-level-1 RESOLVED (the clock order counts fractional flux sectors: $\mu = 16 = \text{ord}^2$, Lagrangian μ_4 glue, KLM $16 \to 1$) with the NEW sector-counter dial $\#\text{prim}((E_8)_k) = (1,3,5,10,15,27)$ for $k = 1..6$ — exactly ONE sector iff $k = 1$; the LOCKSTEP THEOREM lifts the `v505` "one scale ⇒ one level" to a THEOREM of clock invariance ($|\Pi_j| = \sqrt{2}\,t$ equal, monodromy = pure phase i, $\det(A-1) = -4$: no invariant flux vector; NEW falsifier $Z^4 - Z$: $0/24$ orderings lockstep); honest limit: uniform doubling keeps the lockstep (equality, not the value); controls $k{=}2/SO(16)/A_2$ separate honestly; Wolfram extension $450 \to 457$ (verified engine run); *NO marker moves — WP5e stays [O] (the CPS dictionary on $\mathbb{PT}/\mathbb{Z}_4$ typed [C], the type-I B-model back-reaction [O]), `SEAM.EQUIV.01` stays [O]*)

- **`v509_celestial_wp5e_eps2_level_from_flux.py` (`CELEST.WP5E.EPS2.01`, 28 checks, < 1 s; sympy/Fraction exact, no floats; deterministic).** Target: derive the boundary level $k = 1$ from flux quantisation on the lockstep spheres, following the Costello–Paquette–Sharma Burns-holography mechanism (PRL **130**, 061602; `arXiv:2306.00940`). *CPS skeleton replicated exactly*: the S^3 period of the brane back-reaction 3-form is $(2\pi i)^2 N$ (their eq. 3.33; large-gauge invariance quantises N), the exceptional-sphere Kähler flux is $2\pi N$ exactly ($K = |u|^2 + N \log |u|^2$, the flat part dies at $t \to 0$), and the boundary level magnitude is $2N$ via the symplectic-boson double contraction ($N = 1, 2, 3$; their eq. 9.4), Dynkin normalisation $T(\text{so}_8 \text{ vector}) = 1$, $T(\text{adj}) = 6$. *Pairing matrix pinned*: the `v505` ch_2 ledger ($f(m) = -(C^{-1})_{mm}/2 = (-\frac{3}{8}, -\frac{1}{2}, -\frac{3}{8})$)

+ integrality + unimodularity + effectivity force $P_{mi} = c_1(T_m).[\Sigma_i] = \delta_{mi}$ by COMPLETE enumeration (row candidates 8/6/8; 48 unimodular solutions of $PC^{-1}P^T = C^{-1}$; with $P \geq 0$ exactly 2: identity + A_3 diagram flip); fluxes $F_i = (64, 60, 64) = \dim \mathfrak{g}_i$, total $188 = 248 - 60$, flip-invariant; flux per charged root current uniformly ONE quantum through exactly one sphere. *The level formula, honest*: "level = total open-string flux" is KILLED by the lockstep test itself ($k_i = (64, 60, 64)$ unequal — F_i is the FLAVOUR multiplicity, the brane matter content); what works: k = flux per adjoint current $= F_i/\dim \mathfrak{g}_i = (1, 1, 1)$, anchored non-circularly by the lattice current count $(240, 0, 0, 0)$ for $k = 1..4$ (`v505` S4.8) and the embedding index $j(D_5{\subset}E_8) = j(A_3{\subset}E_8) = 1$ $(60 = 2h^\vee)$. *Ord-4-vs-level-1 resolved*: the clock order counts FRACTIONAL flux sectors, not the level ($\mu = \det(D_5)\det(A_3) = 16 = \mathrm{ord}^2$; the μ_4 glue diagonal isotropic + Lagrangian; KLM $16 \to 16/4^2 = 1$, zero residual fractional flux); NEW DIAL: #primaries$((E_8)_k) = (1, 3, 5, 10, 15, 27)$ for $k = 1..6$ from the affine marks — exactly ONE sector iff $k = 1$ (independent of the CPS transplantation). *The lockstep theorem* (the lifting of `v505` S4.10): clock invariance $P(iZ) = P(Z)$ forces $a_3 = a_2 = a_1 = 0$ sharply, the four branch points are ONE μ_4 orbit, $|\Pi_j| = \sqrt{2}\,t$ EQUAL on all three spheres, the monodromy is the pure phase i and $\det(A{-}1) = -4 \neq 0$ (no invariant flux vector) — $k_1 = k_2 = k_3$ IS A THEOREM of clock invariance; NEGATIVE PROBE (new falsifier): the clock-forbidden family $Z^4{-}Z$ (smooth, disc -27, branch points $\{0, 1, \omega, \omega^2\}$ not a μ_4 orbit): over ALL 24 orderings ZERO chains with three equal period moduli (1 vs $\sqrt{3}$); HONEST LIMIT: uniform flux doubling keeps the lockstep — the theorem proves EQUALITY, not the value (1 from the counters and the CPS minimal quantum $N = 1$). *Negative controls*: $k = 2$ (current count 0, residual 360, prefactor $-31/48$, #prim 3 — while the lockstep dial honestly does NOT fire); $SO(16)$ (fluxes $(0, 60, 0)$: spheres 1/3 FLUX-DARK, defect $-30 \neq -78$, condensation stuck at $4 = \det \mathrm{Cartan}(D_8)$); A_2 (det $3 \neq 4$, glue diagonal $23/24 \notin \mathbb{Z}$, $\mu = 12$ no perfect square: NO Lagrangian subgroup, Coxeter $3 \neq 4$). Typed honestly: the CPS dictionary itself on $\mathbb{PT}/\mathbb{Z}_4$ stays [C] (their geometry is the $\mathbb{C}^2$ blow-up with one (-1)-sphere and $\mathfrak{so}(8)$ at negative level — not the A_3 ALE); the type-I B-model back-reaction stays [O] (the same fence as `v502`/`v505`).

- **Ledger + papers + Wolfram + website.** Ledger: new row `CELEST.WP5E.EPS2.01` ([E] CPS skeleton/pairing/lockstep/sector arithmetic + [C] CPS dictionary + [O] back-reaction); `CELEST.SEAM.01` updated ("ε_2 executed via v509: the one-level statement is now a THEOREM (lockstep from clock invariance, falsifier $Z^4{-}Z$), level = 1 carries the new sector-counter dial #prim$((E_8)_k) = 1 \Leftrightarrow k = 1$; the $\mathbb{PT}/\mathbb{Z}_4$ back-reaction stays [O]"); the `v505` "one scale ⇒ one level" sentence LIFTED to theorem status in ledger and prose (text move, no marker move). Papers: `tfpt_research_contracts` WP5e section extended by the ε_2 block (done, verdict B; roadmap now $\delta_1/\delta_2/\varepsilon_1$ + back-reaction); `tfpt_3` celestial section becomes a THIRTEEN-step story with Step 13 "one level is a theorem; the sector counter pins $k = 1$" and the "What remains" paragraph updated. Wolfram: seven mirrors (the CPS period integrals; the $2N$ contraction + Dynkin normalisation; the pairing enumeration $48 \to 2$; the fluxes $(64, 60, 64)$; the #prim table $(1, 3, 5, 10, 15, 27)$; the lockstep periods + the 0/24 negative probe; the $SO(16)/A_2$ controls) verified with the active engine the same day ($450 \to 457$, `ALL WOLFRAM EXTENSION CHECKS PASSED`). `VerificationDag`: `v509` on the QFT/celestial node; registry + `run_all` (503 modules, suite green). No gate closes, no marker moves — WP5e proper stays [O].

2026-07-21 (XIX) (`e8-ladder-bed` v2 — the 2DCS extension executed with an honest preregistered `kill_2d` verdict at the DETECTOR level: the known-answer map recovers $8/8$ rungs incl. m_7/m_8 via F3 difference coordinates (MC $p = 2\times10^{-4}$), but the 2D control battery breaks (scramble 27%, jitter 18% at the destructive 30% level) — the v1 1D results (BCVO $8/8$, $CoNb_2O_6$ $6/8$) are untouched; scorecard row updated, status/stage unchanged; *experiments level only — no verification module, no gate closes, no marker moves*)

- **`experiments/e8-ladder-bed` v2 (2DCS): amendment preregistered, run, and killed by its own controls.** Trigger: arXiv:2512.16829 (Watanabe, Trebst, Hickey) — a THEORY-only paper (four-kink/ED localized-limit spectra at $B_y = 0$–$3\,\mathrm{T}$, not the E8 point; no measurements); as of 2026-07-21 *no experimental 2DCS measurement of $CoNb_2O_6$ exists anywhere* (web-checked). `amendment_v2` was frozen in `hypotheses/e8_ladder_bed_v1.yaml` BEFORE any 2D recovery run (detector, catalog F1/F2/F3, gates, null model, verdict criteria); `amendment_v2_1` discloses a null-region correction (add `n_matched` to the false-alarm region) found on synthetic diagnostics, timing disclosed. Runs (`results/results2d.json`, deterministic): known-answer KA2D 8/8 rungs with m_7/m_8 recovered ONLY via F3 difference coordinates ($\chi^2/\mathrm{dof} = 1.30$, MC $p = 2.0 \times 10^{-4}$, 0/5000) — the mechanism that pulls m_7/m_8 inside a low-pass window works; the NEG2D localized-limit map (the paper's own non-E8 closed forms) is correctly rejected ($p = 0.029$); offset and wrong-ladder controls rejected. BUT `scramble2d` fires at $54/200 = 27\%$ and `jitter2d` shows no dose-response decay (29/21/17/18%; 18% at the destructive 30% level) ⇒ preregistered verdict `kill_2d`: permuting the detection axis preserves the pump-axis marginal (which carries the full 1D ladder) and the 120-entry catalog is dense enough that rung coverage follows from the marginal alone — the v2 detector does not test the 2D *pairing*; failure kills the detector, not TFPT. The $CoNb_2O_6$ leg stays 6/8 ($p = 0.0038$) per the preregistered `m7_m8_upgrade_rule`; a future v3 would need coincidence-based coverage and a sparser catalog, preregistered BEFORE real data exists.
- **Surfaces synced (scorecard row updated; counters unchanged, 120 rows).** Scorecard row "E8 mass-ladder blind recovery bed": `data_value` extended ("CoNb2O6 6/8 unchanged; 2DCS extension killed at detector level (scramble 27%, jitter 18% @30%); KA 8/8 incl. m7/m8 via F3, p=2e-4; no experimental 2DCS data exists (arXiv:2512.16829 theory-only)") and `status_note` extended (amendment v2/v2.1 preregistered+disclosed; real-data leg blocked; v3 criteria) via `build_evidence_scorecard.py`; status/stage UNCHANGED (`consistent/not_applicable`, `analog_positive_control` basket — the v1 detector validation stands). Experiment README v2 section, hypotheses YAML amendments, `experiments/next.txt` note; data-request plan on record (Armitage lab/JHU, Wang lab/Ames; 2D peak table at $B \parallel b$, 4.7–5.5 T, sub-Kelvin). No claim; firewall intact.

2026-07-21 (XVIII) (`v508` — WP5e-γ of `CELEST.SEAM.01` "the sphere-axion pairing check" promoted as `CELEST.WP5E.GAMMA.01`, an honest rigid NEGATIVE result: the VERTEX-SPACE THEOREM — the $W(D_5)\times W(A_3)$-invariant quadratics on the 8-dim glue Cartan are EXACTLY $\mathrm{span}\{s_5, s_3\}$ (dim 2) and the quartics EXACTLY $\mathrm{span}\{P_1, P_2, P_3, T_5, T_3\}$ (dim 5), with the PRODUCT THEOREM as the master kill (any two invariant quadratics multiply into $\mathrm{span}\{P_1, P_2, P_3\}$ — zero T_5/T_3 content, so Φ_{T_3} annihilates EVERY exchange image while $\Phi_{T_3}(A_{\mathrm{fix}}) = 32 \neq 0$); the strict two-index $\mathbb{Z}_4$ rule COLLAPSES (the sphere axions have no invariant bilinear Cartan vertex at all), the twist-insertion channels give a rank-2 exchange matrix with $\mathrm{rank}([M \,|\, A_{\mathrm{fix}}]) = 3$ and the annihilator certificate $(\Phi_{T_5}, \Phi_{T_3}, \Phi_P)(A_{\mathrm{fix}}) = (0, 32, 72)$; side discovery $K^{(0)} = -15\,K^{(2)}$; naturalness DISSOLVES (ch$_2$-natural and AB-weight couplings both certify $(0, 0)$ vs required $(32, 72)$ — scale-independent); controls: $SO(16)$ has no sphere partners AND uncancelled T_5, D_8 no T_3 structure; the O-fence is SHARPENED: the exchange sub-branch is dead, the $32\,T_3$ residual MUST be carried by twisted BCOV contact terms (δ_1 binary: coefficient exactly $(9, -30, -15, 0, 32)$, computed not fitted; escape δ_2: the full-tensor ledger); Wolfram extension $444 \to 450$ (verified engine run); *NO marker moves — `CELEST.SEAM.01` stays Open, the slot bijection untouched, the preregistered level-kill still does not fire, `SEAM.EQUIV.01` stays [O]*)

- **`v508_celestial_wp5e_gamma_pairing.py` (`CELEST.WP5E.GAMMA.01`, 27 checks, ∼4 s; sympy/Fraction exact, no floats; deterministic).** Question: can the three twisted sphere axions η_m (Coxeter eigencharacters $\{i, -1, -i\}$ on $H^2(\mathrm{ALE})$, `v505` S5.8) cancel the rigid $32\,T_3$ residual of the Atiyah–Bott ledger ($A_{\mathrm{fix}} = Q^{(1)}/2+Q^{(2)}/4+Q^{(3)}/2 = 9P_1-30P_2-15P_3+32\,T_3$, `v505` S3.7) by Green–Schwarz-type EXCHANGE with sector-compatible quadratic vertices? *No, with a certificate. Vertex-space theorem*: exact Weyl nullspace arithmetic over all bidegrees gives quadratics dim 2 (bidegrees $(1, 0, 1)$) and quartics dim 5 (bidegrees $(2, 0, 1, 0, 2)$) — the vertex basis is COMPLETE; *product theorem* (the master kill): $(as_5+bs_3)(cs_5+ds_3) = ac\,P_1+(ad+bc)P_2+bd\,P_3$ identically — every exchange image (any axion count, any rule, any couplings) is annihilated by Φ_{T_3}, and $\Phi_{T_3}(A_{\mathrm{fix}}) = 32 \neq 0$. *Selection enumeration*: the strict two-index reading collapses ($C_{j,j'} \neq 0$ only for $j' = -j$, forcing $m = 0$: 16 raw triples, 4 nonzero, all $m = 0$); twist insertion: $\eta_1 \leftrightarrow K^{(3)}$, $\eta_2 \leftrightarrow K^{(2)}$, $\eta_3 \leftrightarrow K^{(1)}$, $\eta_0 \leftrightarrow K^{(0)}$; channels $E_{13} = (16, -96, 144, 0, 0)$, $E_{22} = (16, 32, 16, 0, 0)$, $E_{00} = 225\,E_{22}$; SIDE DISCOVERY: $K^{(0)} = -15\,K^{(2)}$ (the even-sector quadratics are PARALLEL — this collapse drives the Φ_P obstruction). *Unsolvability with certificate (the core)*: M is 5×2, rank 2, kernel 0; $\mathrm{rank}([M \,|\, A_{\mathrm{fix}}]) = 3$; annihilator certificate $(\Phi_{T_5}, \Phi_{T_3}, \Phi_P)(A_{\mathrm{fix}}) = (0, 32, 72)$ — two independent obstructions ($32\,T_3$ structural, rule-independent; $\Phi_P = 72$ selection-rule-driven); $\ker M = 0$ forces the unique non-disturbing coupling $c = 0$ (the sphere axions are RIGIDLY SILENT); $\Phi_P(Q^{(0)}) = 0$ (the sector-0 Okubo mechanism undisturbed); relaxed control: dropping charge conservation reaches the P-block ($\det = -64$) but $T_5 = T_3 = 0$ persists. *Naturalness resolution*: ch$_2$-natural couplings $(9/64, 1/4)$ give $(25/4, -11/2, 97/4, 0, 0)$ and AB-weight couplings $(1/2, 1/4)$ give $(12, -40, 76, 0, 0)$ — BOTH with certificate $(0, 0)$ vs required $(32, 72)$: the failure is scale- and coupling-INDEPENDENT (no "unnatural factor"). *Negative controls*: invariant-only = `v505` S3.9 replicated + sharpened; flipped rule $m+j-j'$: Φ_{T_3} kill flip-invariant, Φ_P rule-dependent; $SO(16)$: odd classes EMPTY and AB sum $(15, -18, -15, -4, 40)$ — T_5 does not even cancel (E_8 doubly special); D_8: $16\,T_8+12\,s_8^2$, no T_3 structure — the question cannot even be posed. Number fences [C] only: $32 = (h, h)+(h', h') = 20+12$, $72 = 2\tilde{\lambda}_{e_8}^2$.
- **Ledger + papers + Wolfram + website.** Ledger: new row `CELEST.WP5E.GAMMA.01` ([E] vertex space/selection/certificate + [C] fences + [O] contact terms); `CELEST.SEAM.01` stays Open with the WP5e targets UPDATED ("the exchange sub-branch is closed by v508; the named remaining targets are the twisted BCOV contact terms (δ_1, binary test) and the full-tensor ledger (δ_2)"). Papers: `tfpt_research_contracts` WP5e section extended by the γ block (γ

done — negative result with certificate; the $\delta_1/\delta_2/\varepsilon_1/\varepsilon_2$ roadmap with success/kill criteria); `tfpt_3` celestial section becomes a TWELVE-step story with Step 12 "the exchange no-go and the contact-term criterion" and the "What remains" paragraph updated. Wolfram: six mirrors (the vertex-space dims 2/5 by generator-wise nullspace arithmetic; the twisted quadratics with $K^{(0)} = -15\,K^{(2)}$; the product theorem, symbolic; the rank-3 certificate $(0, 32, 72)$ with channels and naturalness images; the $SO(16)$ AB sum $(15, -18, -15, -4, 40)$; the D_8 quartic $16\,T_8{+}12\,s_8^2$) verified with the active engine the same day ($444 \to 450$, `ALL WOLFRAM EXTENSION CHECKS PASSED`). `VerificationDag`: `v508` on the QFT/celestial node; registry + `run_all` (502 modules, suite green). No gate closes, no marker moves — the O-fence is only sharpened.

2026-07-21 (XVII) (`v507` — the tautology attack on the `v506` alignment bit, executed and refuted, promoted as `SEAM.BIT.ORIGIN.01`: the ORIGIN THEOREM — marks = fixed points of the hyperelliptic involution = the 4 half-periods $\Lambda/2\Lambda$ of the seam double, and EVERY mark-free collar deck IS a half-period translation (in all three stabiliser types $(4,3,3)/(8,5,3)/(12,3,3)$ the free involutions are exactly the three σ_c: the CLASS of the deck is forced, its POSITION is not); the CORE SOLVE (symbolic λ): a marks-preserving root of σ_c exists $\iff \lambda \in \{-1, 2, \frac{1}{2}\}$ (one per c; intrinsic: the deck pairs separate harmonically), union = the harmonic orbit $\iff j = 1728 \iff \tau = i$; the EQUIVALENCE TETRAD "Uhr existiert $\iff$ ($\tau{=}i$ AND deck = CM-fixed half-period) $\iff$ (Stab = D_4 AND deck central) $\iff$ deck pairs harmonic $\iff$ NS Fock lift NONSPLIT"; the FERMION DICHOTOMY (the new physical face): central $\tilde{U}^2 = 256\,\gamma_1\cdots\gamma_{16} \Rightarrow U^2 = (-1)^F$ nonsplit vs edge $\tilde{U}_e^2 = +2^7 \Rightarrow U^2 = +1$ SPLIT (2 vs 0 roots — the bit IS a Fidkowski–Kitaev-type extension class, distinguished but not derived); the silver set = the in-family misalignment witness; Wolfram extension $438 \to 444$ (verified engine run); *NO marker moves — the bit stays [C], `SEAM.CLOCK.RIGIDITY.01`/`QGEO.REALIZE.01`/`P2.TYPING.01` text precised only*)

- **`v507_seam_bit_origin.py` (`SEAM.BIT.ORIGIN.01`, 26 checks, ∼2.6 s; Part A sympy-exact torus/Möbius arithmetic + Part B exact $\mathrm{Cl}(16)$ Fractions; deterministic).** Conjecture under attack: marks and deck both descend from the SAME torus structure (the seam double $T^2_\tau \to S^2$, `v214`/`v260`), so the common origin would force the central alignment automatically and the `v506` bit would be a tautology. *Origin theorem (die Herkunft)*: the 4 marks are the fixed points of $w \mapsto -w$ on $\mathbb{C}/(\mathbb{Z}{+}\tau\mathbb{Z})$ = the half-periods $\Lambda/2\Lambda$ (a $\frac{1}{3}$-coefficient never qualifies); t_c descends iff $2c \in \Lambda$ — exactly the three half-period translations (control $c = \frac{1}{3}$ fails); the descended maps are the EXPLICIT rational involutions $\sigma_i(x) = e_i + (e_i{-}e_j)(e_i{-}e_k)/(x{-}e_i)$ with the exact curve-lift identity $\prod(\sigma_i{-}e_l) = \psi_i^2 \prod(x{-}e_l)$, ψ_i RATIONAL (the $\wp$-addition formula as pure algebra), closing to Klein V_4; CONVERSELY the freely-acting involutions in ALL three stabiliser types (generic V_4 $(4,3,3)$, harmonic D_4 $(8,5,3)$, hexagonal A_4 $(12,3,3)$) are EXACTLY the three σ_c ⇒ every mark-free collar deck (`v216`) IS a half-period translation. *Core solve (symbolic λ)*: σ_c admits a marks-preserving PGL_2 root $\iff \lambda = -1$ (σ_1), 2 (σ_2), $\frac{1}{2}$ (σ_3) — per half-period exactly ONE solution, intrinsically "the deck pairs separate harmonically" (pair cross-ratio -1); generic modulus: V_4 abelian, ALL σ_c central yet clockless ("central alone is empty" — the bit is the CONJUNCTION $D_4 \wedge$ central); at $\tau = i$ the stabiliser jumps to D_4 with σ_1 central (2 roots = the clock pair), σ_2/σ_3 edge-type (0 roots); the `v506` seam deck lands on the central $\sigma_1 = -1/x$ under all 8 mark bijections; CM dictionary: mult-by-i fixes exactly $c^* = (1{+}\tau)/2$ (mult-by-ρ 3-cycles all three), the clock lift needs the CM unit i ($\psi = 2i/(x{+}1)^2$, $t^2{+}1$ irreducible over $\mathbb{Q}$) while half-period decks lift rationally. *Equivalence tetrad + circularity audit*: clock exists $\iff$ ($\tau{=}i$ AND deck = the CM-fixed half-period) $\iff$ (Stab = D_4 AND deck central) $\iff$ deck pairs harmonic $\iff$ NS Fock lift nonsplit; the `v493` converse independently re-verified — two independently computed converses of ONE equivalence, no circle. *Fermion dichotomy (Part B)*: central arrangement (shift by

8): $\tilde{U}^2 = 256\,\gamma_1\cdots\gamma_{16} \Rightarrow U^2 = (-1)^F$, nonsplit $\mathbb{Z}_4$, 2 roots in the 32-element dihedral lift (the $\mathbb{Z}_8$ tower); edge arrangement (reflection $j \mapsto 4-j$, fixed sites = non-marks = the silver $\{0,\infty\}$ between neighbour marks): the implementer EXISTS but $\tilde{U}_e^2 = +2^7 \cdot \mathbf{1} \Rightarrow U^2 = +1$, the extension SPLITS, 0 roots, no $\mathbb{Z}_8$; one-particle $(\lambda^2{+}1)^8$ vs $(\lambda^2{-}1)^8$; census: all 28 shift-invariant 4-sets cross (central type), all 21 mark-free reflection-invariant sets do not; robust at 8 sites; R-sector edge control splits too — the bit IS a physical extension class (Fidkowski–Kitaev-type): distinguished, but not derived (WHICH class nature realises stays the input). *Silver placement*: $\{\pm1, \pm(3{+}2\sqrt{2})\}$ = "$\tau = i$ with deck = a non-CM-fixed half-period translation" (all 8 bijections $\to \sigma_2/\sigma_3$, never σ_1) — the in-family witness: TAUTOLOGY REFUTED. *The one [C] rest (verbatim)*: the alignment bit remains a carrier datum; "no tautology — the origin forces the CLASS of the deck (half-period translation), not its POSITION". No overclaim.

- **Ledger + papers + Wolfram + website.** Ledger: new row `SEAM.BIT.ORIGIN.01` (**[E]** origin theorem/core solve/tetrad/fermion dichotomy/silver placement + [C] the bit); TEXT precisions at *unchanged* markers: `SEAM.CLOCK.RIGIDITY.01` ("the bit is now fully characterised as a tetrad … and carries a physical face: the non-split NS Fock extension class — v507"), `QGEO.REALIZE.01` ("the class of the deck IS derived (half-period translation, v507); only its position stays the input", stays [C]/**[O]**), `P2.TYPING.01` (origin cross-reference on the common R1 bit). Papers: `tfpt_3` celestial Step-1/2 environment + QGEO paragraph, contracts marks/kill-test paragraph, `introduction/tfpt_1/tfpt_2/origin_theory` one-sentence updates where `v506` is cited. Wolfram: six mirrors (half-period descent $2c \in \Lambda$ with the rational σ_i lift identity; the core solve $\lambda \in \{-1, 2, \frac{1}{2}\}$; the stabiliser counts $(4,3,3)/(8,5,3)/(12,3,3)$ via the complete 24-triple enumeration; the CM fixed point c^* + the lemniscatic $\psi = 2i/(x{+}1)^2$; the $\mathrm{Cl}(16)$ dichotomy $256\,\gamma$-monomial vs scalar 2^7 on explicit $256{\times}256$ Jordan–Wigner matrices + the 2-vs-0 root census in the 32-element dihedral lift; the silver bijection census) verified with the active engine the same day ($438 \to 444$, `ALL WOLFRAM EXTENSION CHECKS PASSED`); the `v493` audit direction and the 28/21 arrangement census stay Python-side (flagged). `VerificationDag`: `v507` on the P2/carrier and QGEO nodes; registry + `run_all` (501 modules, suite green). No gate closes, no marker moves.

2026-07-21 (XVI) (`v506` — seam clock rigidity, Route B of the `v503` classification, promoted as `SEAM.CLOCK.RIGIDITY.01`: TWO exact theorems reduce the order-4 clock input to ONE alignment bit — Theorem A (Möbius): the deck $z \mapsto -z$ has EXACTLY TWO square roots $z \mapsto \pm iz$ in $\mathrm{PGL}_2(\mathbb{C})$, both *automatically* order 4, and a marks-preserving root exists $\iff$ the deck is the CENTRAL involution of the mark-D_4 (counterexample $\{\pm1, \pm(3{+}2\sqrt{2})\}$: harmonic with full D_4 but deck non-central $\Rightarrow$ 0 roots — harmonicity alone insufficient); Theorem B (Fock, the core): the NS deck implementation is FORCED to order 4 — $\tilde{U}^2 = 256\,\gamma_1\cdots\gamma_{16}$, i.e. $U^2 = (-1)^F$ EXACTLY, a nonsplit $\mathbb{Z}_4$, with the $\mathbb{Z}_8$ tower $V^2 \propto U$, $V^4 \propto (-1)^F$ $(8 = 2|\mu_4|$ fermionically explained) and the R-sector control split ($U_R^2 = +1$); the one [C] rest: "the collar deck is the central involution of the mark-D_4" — the fermionic $\mathbb{Z}_4$ forces the ORDER, not the spatial alignment; Wolfram extension $434 \to 438$ (verified engine run); *NO marker moves — `QGEO.REALIZE.01` stays [C]/[O] (text precised), `P2.TYPING.01` R1 mapped onto the same bit*)

- **`v506_seam_clock_rigidity.py` (`SEAM.CLOCK.RIGIDITY.01`, 34 checks, ~3.5 s; Part A sympy-exact Möbius/divisor arithmetic + Part B exact $\mathrm{Cl}(16)$ Fractions; deterministic).** `v503` localised the one remaining identification input ("why does the raw seam carry the order-4 clock", QGEO-R1 ≡ P2-R1); `v506` reduces it. *Theorem A (Möbius side, complete)*: Cayley–Hamilton + the complete case analysis give g^2 = deck $\Rightarrow \operatorname{tr} g \neq 0$ forces g diagonal with $(a/d)^2 = -1$, and $\operatorname{tr} g = 0$ forces $\det g = 0$ (impossible) — exactly two roots $z \mapsto \pm iz$, both projective order 4, mutually inverse ($\mathrm{Aut}(\mathbb{Z}_4)$), conjugation-covariant; marks filter: a deck-invariant deck-free 4-set $\{u, -u, w, -w\}$ admits a marks-preserving root $\iff$

$w = \pm iu \iff$ marks = one μ_4 orbit $\Rightarrow$ cross-ratio class $\{2, \frac{1}{2}, -1\}$, $j = 1728$, $\tau = i$, free 4-cycle, deck = clock^2, only the scale free; group form: $|\mathrm{Stab}(\mu_4)| = 8 = 2|\mu_4|$ with order profile $\{1{:}1, 2{:}5, 4{:}2\} = D_4$, center $= \{\mathrm{id}, \mathrm{deck}\}$, exactly 2 order-4 elements = the clocks — "deck central in the mark-D_4" $\iff$ "clock exists"; the counterexample $\{\pm 1, \pm(3{+}2\sqrt{2})\}$ is deck-free AND harmonic ($j = 1728$, cross-ratio $\frac{1}{2}$) with full D_4, but the deck sits *non*-centrally (the central involution crosses the $\pm$ pairs) $\Rightarrow$ 0 roots; the RP filter passes BOTH roots (the $\mathrm{Aut}(\mathbb{Z}_4)$ ambiguity is irreducible, `v492` NC3). *B3/B4 (n-arithmetic + controls)*: free orbit $\Rightarrow n \mid 4$; transitivity $\Rightarrow n = 4$; faithfulness kills $n \geq 5$ (kernel argument through S_4); $n = 3$ fixes a mark; $n = 6$ dies independently at the cusp degree 3 (`v117`); the deck is modulus-blind (`v503` B5 exact); $n = 8$ impossible on 4 marks; hexagonal stabiliser A_4 (0 order-4 elements, squares only orders $\{1, 3\}$); generic stabiliser V_4 (exponent 2); stabiliser trichotomy $V_4/D_4/A_4$ with order-4 counts $(0, 2, 0)$. *Theorem B (Fock side, the core result)*: S^2 has $b_1 = 0 \Rightarrow$ unique spin structure $\Rightarrow$ NS/bounding on every circle (established P1 datum); one-particle: $D^2 = -1$ (charpoly $(\lambda^2{+}1)^8$), quarter-shift $C^2 = D$, $C^8 = 1$ (charpoly $(\lambda^4{+}1)^4$, ζ_8 spectrum); Fock (exact Cl(16)): $\tilde{U} = \prod_j (1 - \gamma_j \gamma_{j+8})$ implements the deck on all 16 generators and $\tilde{U}^2 = 256\,\gamma_1 \cdots \gamma_{16}$ $\Rightarrow U^2 = (-1)^F$ EXACTLY; $(cU)^2 = c^2(-1)^F$ is *never* 1 (center of Cl(16) = scalars): the $\mathbb{Z}_4$ lift is FORCED, $1 \to \mathbb{Z}_2^{(-1)^F} \to \mathbb{Z}_4 \to \mathbb{Z}_2^{\mathrm{deck}} \to 1$ nonsplit; the quarter-shift lift V (unique even block implementer) has $V^2 \propto \tilde{U}$, $V^4 \propto (-1)^F$, projective order $8 = 2|\mu_4|$ — the `v492` $\mathbb{Z}_8$ spin bridge realised on the 16-Majorana Fock space; R control $U_R^2 = +1$ (split — the forcing rests on NS = established geometry); 16-fold way $\theta_v = 1$ at $\nu = 16$; robustness: the forcing already holds at 8 sites ($16 =$ the carrier count $2^{g_{\mathrm{car}}-1}$, not a tuning). *The one [C] rest (verbatim)*: "the collar deck is the central involution of the mark-D_4" ($\iff$ square-root existence on the mark data) — ONE alignment bit; the fermionic $\mathbb{Z}_4$ forces the ORDER, not the spatial alignment. No overclaim.

- **Ledger + papers + Wolfram + website.** Ledger: new row `SEAM.CLOCK.RIGIDITY.01` ([E] both theorems + [C] the alignment bit); TEXT precisions at *unchanged* markers: `QGEO.REALIZE.01` ("the rest is now ONE bit: the central position of the deck in the mark-D_4; order/uniqueness/orbit/$\tau{=}i$ follow — v506", stays [C]/[O]), `P2.TYPING.01` (R1 mapped onto the same bit, fermionic halving), `QGEO.EMERGE.LIGHT.01` context sharpened. Papers: `tfpt_3` celestial Step 1 (fermionic $U^2 = (-1)^F$ sentence) + Step 2 (the one-bit reduction), contracts QGEO kill-test paragraph + marks paragraph + CELEST hypothesis note, `introduction/tfpt_1/tfpt_2/origin_theory` one-sentence updates at the `v499/v503` citation sites. Wolfram: four mirrors (the complete square-root solve; D_4 center + order-4 counts + the silver counterexample; the Cl(16) identity $\tilde{U}^2 = 256\,\gamma_1 \cdots \gamma_{16}$ on explicit $256{\times}256$ Jordan–Wigner matrices + the split R control; the n-arithmetic + generic kill) verified with the active engine the same day ($434 \to 438$, `ALL WOLFRAM EXTENSION CHECKS PASSED`); the 24-triple stabiliser enumeration and the A_4 profile stay Python-side (flagged). `VerificationDag`: `v506` on the P2/carrier node; registry + `run_all` (500 modules, suite green). No gate closes, no marker moves.

2026-07-21 (XV) (`v505` — WP5e-β of `CELEST.SEAM.01` "the equivariant anomaly ledger on twistor space" promoted as `CELEST.WP5E.BETA.01`: the Atiyah–Bott/Lefschetz fixed-point skeleton of the one-loop twistor inflow is EXACT — denominators $(2,4,2)$ with Dedekind sum $\frac{5}{4} = (|\mathbb{Z}_4|^2{-}1)/12$, equivariant characters $(248,0,-8,0)$ by two routes, invariant Okubo $36\langle x,x\rangle^2$ with the twisted sectors NOT reducing and the AB sum leaving the RIGID residual $32\,T_3$, the index bridge $f(m) = \mathrm{ch}_2(T_m)$ exact with glue defect -78 both routes, the level dials $k=1$ geometric (current count $240/0/0/0$), and the honest E3 REFUTATION: a_0 fills the GRAVITON slot $O(2)$, NOT the axion slot $O(-2)$ — the three sphere classes fill the three twisted axion slots instead; the preregistered kill ("inflow demands a level $\neq 1$") does NOT fire on the equivariant skeleton; WP5e: α (CFT side, `v502`) + β (equivariant twistor skeleton, `v505`) executed, the global BCOV quantisation stays [O]; Wolfram extension $429 \to 434$ (verified engine run); `SEAM.EQUIV.01` stays [O], no marker moves)

- **`v505_celestial_wp5e_beta_equivariant_ledger.py` (`CELEST.WP5E.BETA.01`, 47 checks, ∼1.3 s; sympy/Fraction exact, no floats; deterministic).** The twistor side of WP5e pushed as far as exact arithmetic reaches: the Atiyah–Bott/Lefschetz skeleton of the one-loop box anomaly (Costello; Bittleston–Sharma–Skinner) on $\mathbb{C}^2/\mathbb{Z}_4$ with the E_8 μ_4-glue monodromy. *AB ledger*: $\det(1-g_j^{-1}) = (2,4,2)$; $\sum_j 1/\det_j = \frac{5}{4} = (|\mathbb{Z}_4|^2{-}1)/12$; $\mathrm{tr}_{g_j}(\mathrm{Ad}\,E_8) = (248,0,-8,0)$ two routes; invariant average $60 =$ carrier; Frobenius 61568; $\mathrm{tr}_{g_2} = -8 = 120 - 128$. *Okubo per sector (honest sharpening)*: the invariant sector is exactly Okubo-quadratic ($Q^{(0)} = 36\langle x,x\rangle^2$, $36 = \tilde{\lambda}_{e_8}^2$ — `v495` re-derived through the glue grading); the twisted sectors do NOT reduce (T_5: $(-8,32,-8)$, T_3: $(48,-64,48)$); the AB sum cancels the D_5 quartic exactly and leaves the rigid $32\,T_3$ (the only quartic-free weighting is the $\mathbb{Z}_4$ projector; the graded GS exchange is rank-obstructed in sectors 1–3); heterotic identities (Tr_{120}, Tr_{128}, $\mathrm{Tr}_{E_8} = 9(\mathrm{tr}^2)^2$) exact. *Index bridge*: $f(m) = \frac{1}{4}\sum_j (i^{jm}{-}1)/\det_j = (0,-\frac{3}{8},-\frac{1}{2},-\frac{3}{8}) = \mathrm{ch}_2(T_m) = -(C^{-1})_{mm}/2$ EXACT (E1 = E2 arithmetic); glue ch_2 defect -78 both routes (classes $(-24,-30,-24)$); Killing sums $(320,320,240,320)$, total $1200 = 2h^\vee(h,h)$. *$k=1$ geometric*: lattice current count 240 at $k{=}1$, exactly 0 at $k=2,3,4$; embedding residual $47k^2{+}219k{-}266 = (0,360,814,1362)$; integrality alone fixes nothing (honest, as `v502`); one scale ⇒ one level (`v493` periods). *E3 refutation (prominent)*: $a_0 \in O(8)$ fills the GRAVITON slot $O(2)$, NOT the BSS axion slot $O(-2)$ ($X = -2 \neq +2 = Y$; mismatch $4 = |\mu_4|$ as a [C] coincidence) — "the theory brings its GS axion as a_0" is REFUTED; instead the three $H^2(\mathrm{ALE})$ classes carry exactly the Coxeter eigencharacters $\{i,-1,-i\}$ = the three twisted slots (bijection); slot count $1{+}3 = 4 = |\mu_4|$ soft. *Negative controls*: $\mathrm{diag}(i,i)$ breaks the ledger four ways ($\sum 1/\det = \frac{1}{4}$, ω-weight -1, 5 generators); $SO(16)$ glue: characters $(120,0,+120,0)$, defect $-30 \neq -78$, bulk Okubo fails (independent 4th Casimir); $k=2$: current count 0, c-prefactor $-31/48 \neq -\frac{1}{3}$. [C] dictionaries ($\mathrm{CS}(\rho)$ mod 1 = discriminant weight; $32 = (h,h){+}(h',h')$; $-30 = -h^\vee$; Ω-contraction). [O] the global BCOV/KS quantisation, non-fixed-point contributions, the $O(-2)$ bulk axion, CPS level-from-flux, the pairing $32\,T_3 \leftrightarrow$ sphere axions. VERDICT B; the kill test does not fire on the equivariant skeleton.

- **Ledger + papers + Wolfram + website.** Ledger: new row `CELEST.WP5E.BETA.01`; `CELEST.SEAM.01` updated to "WP5e: α (CFT side, `v502`) + β (equivariant twistor skeleton, `v505`) executed; the global BCOV quantisation stays [O] — with the rigid $32\,T_3$ twisted residual and the sphere-axion pairing as the named remaining targets" at *unchanged* Open status. Papers: contracts WP5e bullet → "α + β done" with the a_0 refutation as an honest finding (kill-test and honest-success paragraphs consistent); `tfpt_3` celestial section becomes the *eleven*-step story (new Step 11 "the equivariant anomaly ledger on twistor space", "what remains" = the global BCOV quantisation with named targets). Wolfram: five mirrors (AB denominators + characters two routes; the index bridge + -78 defect; the per-sector Okubo sums over the 240 glue roots with the $32\,T_3$ residual; the lattice current counts + embedding residuals; the axion-slot weight arithmetic + twisted-slot bijection) verified with the active engine the same day ($429 \to 434$, `ALL WOLFRAM EXTENSION CHECKS PASSED`). `VerificationDag` qft node gains `v505`; registry + `run_all` (499 modules at this step; 500 with `v506`, suite green). No gate closes, no marker moves — `SEAM.EQUIV.01` stays [O].

2026-07-21 (XIV) (`v504` — WP5d-β of `CELEST.SEAM.01` "split + strong additivity from the lattice" promoted as `CELEST.WP5DB.01`: the two remaining KLM legs of complete rationality witnessed for the orbifold prescription — strong additivity ALGEBRAICALLY EXACT with the shared boundary Majorana ($\mathrm{Even}(A) \vee \mathrm{Even}(B) = \mathrm{Even}(A \cup B)$), GF2 spans full, rank $32/32$; disjoint exactly index $2 =$ the `v501` $\ln 2$ bit), the entropic touching defect BOUNDED $< \ln 2$ with the Ising $\frac{1}{4}$-exponent approach ($p = 0.2444$) while the preregistered $U(1)$/Dirac control DIVERGES (Klich–Levitov pinned, infinite index), the split property at the elliptic-nome rate $\pi K(1-x)/K(x)$ (1.3–2.0%) with EXACT orbifold inheritance ($C \to -C$ + the $\Lambda^2 C$ compound — Longo heredity), and Pimsner–Popa $E(a) - a/2 = PaP/2$ identically ($\lambda = \frac{1}{2} = 1/[F{:}F_{\mathrm{even}}]$, $\lambda_{E_4} = \frac{1}{4} = 1/\mu$, $e^{\Delta_\infty} = 2 = 1/\lambda_{\mathrm{PP}}$ — two independent index routes); with `v501` ALL THREE KLM ingredients carry lattice witnesses; the continuum uplift stays [O] — "finite-group orbifolds of completely rational nets are completely rational" is XU'S THEOREM, cited not claimed; Wolfram extension $424 \to 429$ (verified engine run); `SEAM.EQUIV.01` stays [O], `CELEST.SEAM.01` stays Open, no marker moves)

- **`v504_celestial_wp5d_klm_completeness.py` (`CELEST.WP5DB.01`, 37 checks, $\sim$12 s; numpy/scipy Gaussian machinery ED-validated + exact GF2/integer algebra; deterministic).** Complete rationality (Kawahigashi–Longo–Müger) = split + strong additivity + finite two-interval μ-index; `v501` (WP5d-α) witnessed the index, `v504` witnesses the other two legs on the same critical Majorana seam edge, with the $U(1)$/Dirac number-conserving subnet as the preregistered negative control (the current net, where strong additivity famously *fails* in the continuum — Buchholz–Schulz–Mirbach). *Strong additivity, exact algebra*: "touching = shared boundary Majorana" gives $\mathrm{Even}(A) \vee \mathrm{Even}(B) = \mathrm{Even}(A \cup B)$ EXACTLY (GF2 spans full 64/64, 256/256, 512/512, 1024/1024; literal matrix rank 32/32); disjoint intervals give exactly HALF (rank 16/32, index 2 — the missing sector odd $\otimes$ odd is the `v501` $\ln 2$ bit, localised at the split point); the neutral $U(1)$ algebras do NOT generate the union even with the shared site (gaps 2/10/52, growing). *Entropic witness*: the fermionic touching MI carries the pure UV point-gap law ($(c/3)\ln 2$ increments, 0.115525, dev 1.2×10^{-6} — present for every net, no discriminator); the honest discriminator is *bounded vs divergent*: the $\mathbb{Z}_2$/orbifold deficit rises $0.4045 \to 0.5899$ ($N = 64 \to 4096$), stays $< \ln 2$, and $(\ln 2 - \Delta_2) \sim N^{-p}$ with $p = 0.2444$ vs $2\Delta_\mu = \frac{1}{4}$ (Ising disorder exponent); split-point scan reflection-symmetric to 3.9×10^{-14}; ED vN same direction; gap continuity to the `v501` $x \to 1$ endpoint; honest note: "defect $\to 0$" would be *false* at the sharp lattice split — bounded $\Leftrightarrow$ finite index (Longo–Xu) is the correct statement; the $U(1)$ control bursts $\ln 2$ from $L = 128$ and grows as $\frac{1}{2}\ln \mathrm{Var}\, Q_A$ (Klich–Levitov slope 0.10134 vs $1/\pi^2 = 0.10132$). *Split/nuclearity*: coupling-block σ_k fall exponentially at the elliptic-nome rate $\pi K(1-x)/K(x)$ to 1.3–2.0% (rates 2.29/3.24/4.82/5.81 at $d = 16/64/256/512$); $\sigma_1 \sim x^{0.5044}$ (vs $\frac{1}{2}$); trace norm summable; dimerised control exponential. *Orbifold inheritance exact*: P_A flips $C \to -C$ (σ_k identical, 8.3×10^{-17}) and the even-bilinear coupling Gram is the second compound $\Lambda^2 C$ (singular values $\{\sigma_a \sigma_b\}$, 2.8×10^{-14}) — the same ladder at doubled rate (Longo heredity, [C]). *Pimsner–Popa, exact*: $E(a) - a/2 = PaP/2$ identically; attainment in integers (all 16384 (even, odd) monomial pairs at $N = 8$: 8192 extremal with $M^2 = 4$, $\mathrm{Tr}\, M = 0$, $PMP = -M$, 0 violations; 2048/2048 at $N = 12$; pencil $\min \lambda = 0.5000000000$); $\lambda_{E_4} = \frac{1}{4} = 1/\mu$; index consistency $e^{\Delta_\infty} = 2.000000000 = 1/\lambda_{\mathrm{PP}}$ and $e^{2\Delta_\infty} = 4 = \mu$; $U(1)$: $\lambda = 1/(m{+}1) \to 0$. *Controls + honesty*: dimerised phase trivial (drift 3.6×10^{-15}); wrong prescription $\{1, P_{AB}\}$ deficit exactly 0; $\mathbb{Z}_2$ finite-size honesty (14.9% below

$\ln 2$ at $N = 4096$ — witness, not proof). VERDICT: with `v501` all THREE KLM ingredients of complete rationality are witnessed on the lattice; the continuum theorem for finite-group orbifolds is *Xu's* (cited, not claimed), the concrete seam quotient net and the condensed $(E_8)_1$ net stay WP5e/Costello–Li.

- **Ledger + papers + Wolfram + website.** Ledger: new row `CELEST.WP5DB.01`; `CELEST.SEAM.01` updated to "WP5d: α (μ-index) + β (split + strong additivity) executed via `v501/v504` — all three KLM ingredients witnessed on the lattice; continuum uplift stays [O] (Xu's theorem cited)" at *unchanged* Open status. Papers: contracts WP5d bullet → "both stages done" with the β-stage numbers + Xu fence (kill-test and honest-success paragraphs consistent); `tfpt_3` celestial section becomes the *ten*-step story (new Step 10 "the KLM completion on the lattice", "what remains" = WP5e twistor inflow only); `tfpt_4` keystone note updated. Wolfram: five mirrors (GF2 span theorem; Pimsner–Popa identity + attainment on explicit gamma matrices; two-interval group $\lambda_{E_4} = \frac{1}{4}$ + index consistency; the Λ^2/Minors compound with Cauchy–Binet and Minors$[-C, 2]$ = Minors$[C, 2]$; the $U(1)$ dephasing $\lambda = 1/(m+1)$ family) verified with the active engine the same day (424 → 429, `ALL WOLFRAM EXTENSION CHECKS PASSED`); the lattice curves stay Python-only (flagged). `VerificationDag` qft node gains `v504`; registry + `run_all` (498 modules, suite green). No gate closes, no marker moves — `SEAM.EQUIV.01` stays [O].

2026-07-21 (XIII) (`v503` — the `v215` deferred emergence test (lever 7) executed *light* and promoted as `QGEO.EMERGE.LIGHT.01`: NO dynamical selection of the square modulus — the free field's global $\det'$ winner is the *hexagonal* torus, $\tau=i$ is a saddle, criticality at the square is Palais symmetry; the square is pinned by *rigidity* alone (the raw DtN is mod-4 block-diagonal *iff* $\tau=i$, an isolated zero); and the residual *converges*: QGEO-R1 ≡ P2-R1 = one common rest, "why does the raw seam carry the order-4 clock"; Wolfram extension 420 → 424 (verified engine run); *NO marker moves — `QGEO.REALIZE.01` stays [C]/[O], `AX.P2.01` stays a declared axiom*)

- **`v503_qgeo_emergence_light.py` (`QGEO.EMERGE.LIGHT.01`, 17 checks; mpmath 40 digits + `v280`-style lattice DtN + sympy exact).** The `v215` lever 7 ("the decisive kill is deferred: compute the raw seam-collar DtN *without* inputting the marks and check whether four square marks *emerge*") executed in its first honest *light* version. *No dynamical selection*: $\log F = \log(\tau_2|\eta|^4)$ (Osgood–Phillips–Sarnak) is exactly S-invariant (dev $< 10^{-35}$), so t=1 is symmetry-forced critical on the rectangular family $\tau = it$; there $g''(1) = -1.298 < 0$ (strict maximum) but the τ_1-direction has $h''(0) = +0.298 > 0$: $\tau=i$ is a *saddle* of the full moduli space; the only critical points are the CM points ($E_2^*(i) = E_2^*(\rho) = 0$ at 40 digits, generic $0.163 \neq 0$); the *global* $\det'$ winner at fixed area is the *hexagonal* torus ($\log F(\rho) = -1.03352 > \log F(i) = -1.05469$, diff 0.021169); criticality at the square is Palais symmetric criticality (the independent heat trace is equally critical). *Sharp rigidity*: the DtN mod-4 off-character indicator (the `v215`-K3 lever as a *modulus function*) vanishes to 8×10^{-15} exactly at $\tau=i$ and rises strictly monotonically both ways (min 8.4×10^{-2} away) — an isolated zero; $R^2 = \mathbb{Z}_2$ exact as permutation matrices (the `v492` $\mathbb{Z}_8$ bridge on the lattice); $j(it) \geq 1728$ with equality only at t=1. *The $\mathbb{Z}_2$ chain is modulus-blind*: 4 marks + $[\mathbb{Z}_2, \Delta_t] = 0$ on *every* torus (`v216` confirmed without mark input); pillowcase $\det' = \frac{1}{2}$ torus $\det'$ exactly ($\pm$-pair structure) — only the order-4 clock distinguishes $\tau=i$; $\mathrm{Aut}(E_\rho) = \mathbb{Z}_6$ has *no* order-4 element (the dynamical winner cannot carry the clock); given the clock the modulus freezes automatically ($j = 1728$ identically, `v493` mini). *Residual convergence (the core finding)*: the order-4 element on H^1 is a Coxeter element of $W(A_3)$ (char poly $\lambda^3+\lambda^2+\lambda+1$, order $4 = |\mu_4| = N_{\mathrm{fam}}+1$) = the P2 cusp-class carrier datum of `v499` R1 ⇒ *QGEO-R1 ≡ P2-R1, one common residual*; chain: marks derived (`v216`), square derived *given* the clock (`v493`), clock = carrier input [[C]], rest = its dynamical realisation [O]; route classification (`v347` analogue): B-rigidity in the sharpened form "RP seam data ⇒ order-4 clock" is the live route. `v215` lever 7 thereby moves from "deferred, not executable" to "light version executed: rigidity, not dynamical selection" — documented in `v503`, `v215` itself unchanged.

- **Ledger + papers + Wolfram + website.** Ledger: new row `QGEO.EMERGE.LIGHT.01`; `QGEO.REALIZE.01` text *precised* (rest = the order-4 clock, identical with the P2 typing rest R1; square + mark count derived) at *unchanged* marker [C]/[O]; cross-reference added in `P2.TYPING.01` (convergence theorem). Papers: contracts kill-test paragraph (lever-7 light result + the residual-convergence sentence, both with `v503`), `tfpt_3` Step 2 (light-converse sentence), `origin_theory` necessity paragraph, `introduction/tfpt_1/tfpt_2` one-residual cross-references. Wolfram: four mirrors (CM-point identities $E_2(i) = 3/\pi$, $E_2(\rho) = 2\sqrt{3}/\pi$ $\Rightarrow E_2^* = 0$ exact, j/DedekindEta special values; the no-dynamical-selection witnesses; the pillowcase halving as an exact algebraic identity; clock $\Rightarrow$ square + Coxeter) verified with the active engine the same day ($420 \to 424$, `ALL WOLFRAM EXTENSION CHECKS PASSED`); the lattice DtN curves stay Python-only (flagged). `VerificationDag` masses node gains `v503`; registry + `run_all` (497 modules, suite green). No gate closes, no marker moves.

2026-07-21 (XII) (v502 — WP5e-α of `CELEST.SEAM.01` "prefactor and level bookkeeping" promoted as `CELEST.WP5E.ALPHA.01`: the $q^{-1/3}$ prefactor of E_4/η^8 IS exact μ_4 vacuum-energy bookkeeping (the clock is inner $\Rightarrow$ shift orbifold $\theta = 0^8$, common $-c/24 = -\frac{1}{3}$; discriminant form = Casimir via spectral flow AND 16-Majorana free fermions), and $k = 1$ is forced THREE independent ways (current condition; conformal embedding; central charge = the prefactor itself; plus the WP5b singular-vector dial $31124 - 27000 = 4124$); honest sharpening: glue-h integrality holds for ALL k and fixes nothing; the BCOV/Costello–Li twistor inflow stays [O] — WP5e is subdivided (α done, WP5e proper open); Wolfram extension $416 \to 420$ (verified engine run); `SEAM.EQUIV.01` stays [O], no marker moves)

- **`v502_celestial_wp5e_prefactor_level.py` (`CELEST.WP5E.ALPHA.01`, 33 checks, exact sympy/Fraction, no floats).** The two exactly computable consistency anchors of the WP5e twistor uplift established. *Part A — the prefactor*: Hurwitz-ζ exact ($\zeta(-1,\theta) = -B_2(\theta)/2$; $E_b(\theta) = -\frac{1}{24} + \theta(1-\theta)/4$; Ising shift $\frac{1}{16}$ per Majorana); the clock is *inner* ($h = (2,2,2,2,2;0^4)$ and $h' = (0^5;1,1,1,-3)$ both read the class mod 4 on all 240 roots; $(h,h) = 20$, $(h',h') = 12$, sum 32) $\Rightarrow$ twist $\theta = 0^8$ in all four sectors — a *shift* orbifold, NOT a rotation orbifold — so $E_{\rm osc} = -\frac{1}{3} = -c/24$ is sector-independent ($c = 8 = \text{rank} = \text{Sugawara } 248/31$; one common η^8); sector ground exponents $-\frac{1}{3} + (0,1,1,1)$; *discriminant form = Casimir energy* two ways (spectral flow $j^2(h,h)/32 \bmod 1$; free fermions exact: $D_5 = 10$ Majorana $\to \frac{5}{8}$, $A_3 = D_3 = 6 \to \frac{3}{8}$, glue diagonal $= 16 = 2^{g_{\rm car}-1}$ = the WP5d seam carrier $\to 1$); the rotation reading *fails* (deck $\frac{3}{16} \neq \frac{3}{8}$ — "clock $\neq$ deck" at the Casimir level); the four sector characters sum exactly to $E_4/\eta^8 = q^{-1/3}(1, 248, 4124, 34752)$. *Part B — the level*: $k = 1$ forced three independent ways — (i) the current condition $h(J) = k = 1$ (adjoint closure $60+64+60+64 = 248$ at spin 1); (ii) the conformal embedding ($47k^2+219k-266 = 47(k-1)(k+266/47) = 0$; D_8: $128k(1-k) = 0$); (iii) $c = 8 \Leftrightarrow 240(k-1) = 0$ (= the prefactor); plus the `v498` singular-vector dial ($\|(E^\theta_{-1})^m 0\|^2 = m!\,k(k-1)\cdots(k-m+1)$, singular vector at grade $k+1$; $31124 - 27000 = 4124$ at k=1 only, $\langle s|s\rangle = +4$ at k=2). *Honest sharpening (mandatory)*: glue-h integrality $h(J^j;k) = k\,(0,1,1,1)$ is integer for ALL $k = 1..8$ — the originally conjectured integrality route fixes *nothing*; the three other dials carry. *Negative controls*: $D_8/SO(16)_1$ (same $c = 8$, SAME $q^{-1/3}$ prefactor, but $h = (0,\frac{1}{2},1,1)$: the discriminator is glue-h integrality, not the prefactor; $\Theta_{D_8} + \Theta_s = E_4$); $\mathbb{Z}_2$ rotation ($E = +\frac{1}{6}$, $h = \frac{1}{2}$); μ_4 rotation (sector-dependent $(-\frac{1}{3}, \frac{1}{24}, \frac{1}{6}, \frac{1}{24})$ — the common prefactor breaks); wrong $k \in \{2,3,4\}$ fails all five dials coherently (residuals 360/814/1362; $\langle s|s\rangle = 4/12/24$; $31124 \neq 4124$). [[C]] Kac generation for $k \geq 3$ ($k = 1,2$ machine-anchored by WP5b/WP5c). [O] the BCOV/Costello–Li twistor inflow itself (Costello's one-loop axion $\lambda_g/(8\pi\sqrt{3})$ on $\mathbb{PT}$, the Costello–Paquette–Sharma level-from-flux mechanism) — v502 anchors the *CFT side* of WP5e only; the KILL branch ("inflow demands a level $\neq 1$") is not triggered on the CFT side, its twistor side is untested.

- **Contract + ledger + Wolfram + website.** `tfpt_research_contracts`: the WP5e bullet *subdivided* — "WP5e-α (done) [E]/[C]: the CFT-side prefactor + level pinning" with the core numbers and `v502_celestial_wp5e_prefactor_level.py`; "WP5e proper [O]: the twistor inflow"; summit header, honest-success paragraph and status roster updated ("WP5a–WP5d plus WP5e-α landed; WP5e proper the remaining open piece"). `tfpt_3` celestial section: new Step 9 (prefactor + level, one paragraph); `tfpt_4` keystone note updated. Ledger: new row `CELEST.WP5E.ALPHA.01`; `CELEST.SEAM.01` status → "WP5e: alpha stage (CFT-side prefactor + level pinning) executed via v502; the BCOV/twistor inflow itself stays [O]". Wolfram: four exact mirrors (Hurwitz-ζ vacuum energies; discriminant = Casimir with the deck failure; the k-fixing equations incl. the "fixes nothing" integrality table; the character sums + D_8 control) verified with the active engine the same day (416 → 420, `ALL WOLFRAM EXTENSION CHECKS PASSED`). `wolfram/README.md`, `GATE.WOLFRAM.02`, `tfpt_safeguards` Layer-6 count and the website counters (116/116 + 424/424 after (XIII)) updated; `VerificationDag` qft node gains `v502`; registry + `run_all` (497 modules, suite green). `SEAM.EQUIV.01` untouched; no marker moves.

2026-07-21 (XI) (`v501` — WP5d-α of `CELEST.SEAM.01` "the two-interval index from the lattice" promoted as `CELEST.WP5D.01`: the KLM Haag-duality discriminator measured entropically on the seam edge — the 16-layer carrier saturates the $\ln 2$ offset at machine precision, $\mu_{\text{gauged}} = 4 =$ the `v490` census, and the condensation arithmetic $16 \to 4 \to 1$ closes with $\mu = 1$; the preregistered KILL ("μ-offset $\neq 0$ after condensation") does NOT fire; contract now "WP1–WP5d executed; *WP5e the single remaining milestone*"; Wolfram extension $412 \to 416$ (verified engine run); `SEAM.EQUIV.01` stays [O], no marker moves)

- **`v501_celestial_wp5d_two_interval_index.py` (`CELEST.WP5D.01`, 39 checks, numpy/scipy Gaussian machinery + exact Fractions).** The WP5d roadmap discriminator — $(E_8)_1 \Rightarrow \mu$=1 (Haag-dual two-interval net, one sector) vs $SO(16)_1 \Rightarrow \mu$=4 — executed at the α (lattice-index) stage on the `v367` seam edge (16 critical Majorana layers, $c_- = 8$, `v450`-style ring; Peschel-exact, every Gaussian formula ED-validated to $\leq 1.5 \times 10^{-15}$; the superselection identity $[\rho, P_{AB}] = 0$ machine-exact). *The $\mu = 1$ reference*: the fermionic (fixed-spin-structure) two-interval mutual information is *extensive* — c fit 0.5000, $\max|F(x)| = 5.97 \times 10^{-5}$ at N=512, falling monotonically to 1.3×10^{-6} at N=1024 (Casini–Huerta extensivity as a lattice witness). *The μ-offset*: Rényi-2 exactly $\Delta_2 = -\ln[(1+R)/2]$; the 16-layer seam carrier saturates $\ln 2$ at machine precision ($|\Delta_2 - \ln 2| = 1.1 \times 10^{-15}$ at N=512; $R \sim N^{-0.500}$); index reading $[F : F_{\text{even}}] = 2.000000$, $\mu_{\text{gauged}} = 4.000000 =$ the `v490` parity census (*two independent lattice witnesses*); per-layer gauging costs $16 \ln 2$ (the offset measures ln[index], not a blob). *Haag-duality indicator*: fermionic $S(E) = S(E')$ to 8.2×10^{-12}; the orbifold *breaks* complementarity ($(D(E) - D(E') = 0.427$ at a=64); the complementary-pair sum is exchange-symmetric and $\leq \ln 4 = \ln \mu(SO(16)_1)$ with max 1.2637 at the self-dual $x = \frac{1}{2}$ — a *double-limit* statement (the first naive fixed-x check was wrong and was replaced — honestly typed in the module). *Condensation arithmetic anchored at both measured ends*: $\det \mathrm{Cartan}(D_5) \cdot \det \mathrm{Cartan}(A_3) = 16$; KLM/Longo–Rehren $16/4^2 = 1$ and $16/2^2 = 4 \to 4/2^2 = 1$ (both routes exact); $\Sigma d_i^2 = (4, 4, 1)$; $\theta_v = 1$ exactly at $\nu = 2c_- = 16$, rivals $\neq 1$ — so $\mu = 1$ after condensation ([[C]] the condensed net is interacting, the `v490` fence): KILL not triggered. *Negative controls*: ν=1 single Majorana (offset non-removable, $\theta_v = e^{i\pi/8} \neq 1$ — the discriminator has teeth); trivial phase (all $\leq 10^{-7}$); wrong sector sum $\{1, P_{AB}\}$ (offset exactly 0). [O] WP5d-β (continuum Haag duality / local normality / split of the quotient net; runs as exploration) and WP5e (Costello–Li) stay open.
- **Contract + ledger + Wolfram + website.** `tfpt_research_contracts`: the WP5d bullet → "α stage done" with the core numbers and `v501_celestial_wp5d_two_interval_index.py`; status line and honest-success paragraph now "four of five milestones landed; WP5e (twistor

uplift) is the remaining open piece". `tfpt_3` celestial section: new Step 8 (one paragraph, conservative); `tfpt_4` keystone note updated to WP1–WP5d. Ledger: new row `CELEST.WP5D.01` (fine types Formal + Lattice/Numerical + [[C]] dictionary); `CELEST.SEAM.01` status → "WP1–WP5d executed; WP5e [O] (single remaining milestone)". Wolfram: four exact mirrors (Cartan determinants; KLM/Longo–Rehren quotients; Σd^2 + index-reading arithmetic; 16-fold-way θ_v) verified with the active engine the same day (412 → 416, `ALL WOLFRAM EXTENSION CHECKS PASSED`); the lattice curves stay Python-only (flagged). `wolfram/README.md`, `GATE.WOLFRAM.02`, `tfpt_safeguards` Layer-6 count and the website counters (116/116 + 416/416) updated; `VerificationDag` qft node gains `v501`; registry + `run_all` (495 modules, suite green). `SEAM.EQUIV.01` untouched; no marker moves.

2026-07-21 (X) (`v500` — WP5c of `CELEST.SEAM.01` "the GNS limit state" promoted as `CELEST.WP5C.01`: the quasi-free family ω_w exists, is positive, stabilises exactly at the WP5a threshold, and its limit carries the null ideal in its GNS kernel — the full 9361-block exact level-2 Gram has rank 4124 exactly with kernel $27000 = V(2\theta)$ weight by weight; $|s\rangle$ IS the GNS zero vector (the WP5b S4.10 tension resolved at the state level); a CCR obstruction proves the family formulation *necessary*; preregistered SUCCESS criterion met, KILL not triggered; Wolfram extension $408 \to 412$ (verified engine run); `SEAM.EQUIV.01` stays [O], no marker moves)

- **`v500_celestial_wp5c_gns_limit_state.py` (`CELEST.WP5C.01`, 35 checks, exact integer/Fraction, no floats, deterministic).** WP5a (`v497`) made the boundary limit precise, WP5b (`v498`) built the deleting operator; WP5c constructs the *state*. ω_w is quasi-free: loop sector = the affine k=1 vacuum n-point functions via the *machine-determined* compact anti-involution $\theta(e_\alpha) = -e_{-\alpha}$ (the unique sign that is an anti-automorphism on all 61504 basis pairs; contravariance exact); radial sector = oscillator pairings x^{wr} (the exact Gibbs regulator, x-adic). *Convergence sharp*: $\omega_w = \omega_\infty \bmod x^{N+1}$ for all $w \geq N{+}1$ (N=8), not yet at $w = N$ — the WP5a threshold $w = n{+}1$ holds *identically at the state level*; positivity of every ω_w checked. *The kernel (core, fully machine)*: the complete exact level-2 Gram — 31124 monomials in 9361 weight blocks (profile 164:1, 37:240, 7:2160, 1:6960) — has level-1 rank 248 positive definite (Cartan block = the E_8 Cartan matrix, det 1), level-2 rank 4124 *exactly* (the preregistered target check), kernel 27000, every block PSD; rank per Weyl orbit $(0, 0, 1, 8, 44)$; kernel = $V(2\theta)$ *weight by weight* (Freudenthal cross-check on all 9361 weights); highest-weight norms $(3844, 49, 2, 0)$ at $(0, \omega, \theta, 2\theta)$; direct ideal checks $F_i|s\rangle$, $F_jF_i|s\rangle$ norm 0. *μ_4-equivariance*: $\omega_w \circ \text{clock} = \omega_w$ for all w; the clock descends to GNS; level-2 rank split by clock class $(1036, 1024, 1040, 1024) = \Theta_{C_j}/\eta^8$ at q^2 — the two-routes identity at the *state* level. *Resolution of WP5b S4.10*: $|s\rangle$ *is* the zero vector of GNS(ω_∞) — the twisted two-C_1-mode floor (6 quarters = $q^{3/2}$) concerns only pre-limit objects (GNS has no state there, 6 mod 4 ≠ 0). *Negative controls*: k=2 ($\langle s|s\rangle = +4$, dominant blocks full rank — no ideal in the kernel); k=0 (level-1 Gram ≡ 0 — no 248 layer without the central extension); D_8 (rank 2076 = 7380 − 5304, kernel = $V(2\theta_{D_8})$ PSD, but $h = (0, \frac{1}{2}, 1, 1)$: one block of four — one-block completeness stays E_8/μ_4-specific); wrong family $E'_w = mw{+}r$ (the 248 layer erased, 710955 ≠ 248); no damping (897266 ≠ 248); *CCR obstruction*: at fixed radial pairing $c > 0$ positivity forces $\omega(bb^*) \geq c$ — no w-uniform damping state exists, the family formulation is *necessary* and the family *exists*: the preregistered KILL is probed and not triggered. [C] quasi-free extension to the full $*$-algebra; Kac positivity beyond level 2. [O] local normality, net structure/Haag duality (WP5d), Costello–Li (WP5e).
- **Contract + ledger + Wolfram + website.** `tfpt_research_contracts`: the WP5c bullet → done with the core numbers (state-level threshold $w = N{+}1$, the full 9361-block Gram rank 4124/kernel 27000 weight by weight, the clock-class split, the CCR necessity of the family), citing `v500_celestial_wp5c_gns_limit_state.py`; the WP5b handover marked answered. `tfpt_3` celestial section: new Step 7 (one paragraph, conservative). Ledger: new row `CELEST.WP5C.01` (fine type Formal/Identity [E] + [C] identifications + [O] net questions). Wolfram: four exact

mirrors (level-2 block census from the root data; rank/kernel arithmetic with $\dim V(\omega) = 3875$ by Weyl; the glue-frame clock-class split $(1036, 1024, 1040, 1024)$; threshold + level-dial + D_8 census) verified with the active engine the same day ($408 \to 412$); the exact Gram/PSD computation stays Python-side (flagged). `VerificationDag` qft node gains `v500`; registry + `run_all` (494 modules at this step, suite green). `SEAM.EQUIV.01` untouched; no marker moves.

2026-07-21 (IX) (`v499` — the P2 weight-typing postulate HARDENED: the anchor $a = (1, 1, 2)$ typed as the Birkhoff–Grothendieck splitting exponents of the Deligne canonical extension, so integrality/positivity/sum-4 become theorems given the [E] anchors; the `v491` check-9 residual sharpened, the tfpt_2 "balanced" selection typed as stability; Wolfram extension $403 \to 408$ (verified engine run); *no gate closes, no marker moves — `AX.P2.01` stays a declared axiom, `QGEO.SYM` stays [O]*)

- **`v499_p2_weights_deligne_bg.py` (`P2.TYPING.01`, 23 checks, sympy exact, no floats).** `v491` reduced P2 to "3 positive integers summing to 4"; its residual was the *typing* (why integers, why positive, why sum 4). Now typed: given the [E] anchors (4 marks, `v216`; rank $3 = \#\text{marks} - 1$, `v137`/`v228`; cusp-class local monodromy $\lambda^3 - 1$ *and* exact irreducibility $|\langle U, M_0\rangle| = 24 = |S_4|$, $\sum_g |\chi|^2 = 24$, `v117`; unitarity (U)), the Deligne canonical extension E of the flavor connection satisfies: $\deg E = -4$ *as a theorem* (residue exponents $\{0, \frac{1}{3}, \frac{2}{3}\}$ the unique lift to $[0, 1)$, trace 1 per mark; par-deg 0; ∞ apparent, monodromy product $= \mathbf{1}$ exactly); $h^0(E) = 0$ *as a theorem* (Mehta–Seshadri stability; all 324 section-saturation cases; BG dictionary $h^0{=}0 \Leftrightarrow a_i \geq 1$); integrality = Birkhoff–Grothendieck. Hence $\{a\} = \{1, 1, 2\}$ unique and $g_{\text{car}} = e_2 = 5$. *Selector typed*: the Schur–Horn permutohedron has integer points exactly $\{(2, 2, 0), (2, 1, 1)\}$ (the two options of the tfpt_2 "Global computation" theorem) and $h^0{=}0$ kills $\{0, 2, 2\}$ — the "balanced/minimal-variance" heuristic is now stability/positivity. *The h^1 discrepancy, real and resolved*: $h^1(\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2) = 1 \neq 3$, so "generations $= H^1$ of the rank-3 bundle" is FALSE (negative control: forcing it demands $\deg -6$ and loses uniqueness, $e_2 \in \{9, 11, 12\}$ never 5); correct two-object typing: $L = \mathcal{O}(-D)$ carries the generations ($h^1(L) = 3 = N_{\text{fam}}$, the `v137` basis $= H^0(\Omega^1(\log D))$), E carries them as its *fiber*. *Biswas explicit* (independent witness): the weight denominator $3 = N_{\text{fam}}$ forces the $\mathbb{Z}_3$ cusp cover $y^3 = (z{-}1)(z{-}i)(z{+}1)^2(z{+}i)^2$ (genus 2; NOT the pillowcase double, genus 1, $j = 1728$ — the deck side); $p_*\mathcal{O}_Y = \mathcal{O} \oplus \mathcal{O}(-2)^2$; the regular-representation model lands on $\{0, 2, 2\}$ = the *unstable* Schur–Horn companion ($h^0 = 1$): integrality and the sum rule come *out* of the correspondence, stability is the selector. *Negative controls*: no positivity $\to 4/31$ solutions; $\deg -3/-5$ breaks; n-mark scan $e_2 = (n{-}2)(n{+}1)/2 = 5$ ONLY at $n = 4$ (the value, not the uniqueness, is the $n{=}4$ content); MS-rational $e_2 = \frac{16}{3}, \frac{44}{9}$. *Honest residuals* (P2 stays an axiom): R1 module identification [C] (`QGEO.SYM` territory), R2 (U) [C] (`v33`/`v40`), R3 analytic monodromy (Numerical, `v117` check 4); non-circular (neither `v23`'s splitting input nor `v115`'s diag $A_0 = a/4$ input used); the `v491` direction-reversal note carries over. Ledger: new row `P2.TYPING.01`; the `v491`/`P2.PARTITION.01` row records "typing residual sharpened by `v499`" (`v491` stays active, no supersede). Registry + `run_all` (493 modules, suite green).
- **Wolfram second path + papers + website.** Five exact mirrors added to the extension file `tfpt_readouts_extension.wl` (partition enumeration under the full typing + naive-typing control + h^1 discrepancy; Deligne residue-trace degree with the $A_0^\star$ witness char poly; Schur–Horn lattice points; Biswas $\mathbb{Z}_3$-cover degree arithmetic incl. deck double $j = 1728$; the n-mark formula), verified with the active engine the same day ($403 \to 408$, `ALL WOLFRAM EXTENSION CHECKS PASSED`); `wolfram/README.md` + `GATE.WOLFRAM.02` + `tfpt_safeguards` Layer-6 count + website counters (116/116+408/408). Papers (one sentence each, conservative): `introduction` anchor paragraph, `tfpt_1` (5, 3) keybox, `tfpt_2` H2 winbox *and* the "Global computation" theorem (the balanced selection typed as stability), `origin_theory` (5, 3) keybox — all citing `v499_p2_weights_deligne_bg.py`; no status card changes. Website: `VerificationDag` P2

node gains `v499`; generated indices via build; no marker moved anywhere.

2026-07-21 (VIII) (Editorial: dedicated celestial-route section in Paper 3 + website mirror — the scattered celestial sentences consolidated into one coherent six-step section; *no new scripts, no status changes, no marker moves, `SEAM.EQUIV.01` untouched*)

- **`tfpt_3_e8_audit_bootstrap`: new section "The celestial route: from the μ_4 clock to the $(E_8)_1$ boundary shadow".** The celestial-contract material that had accumulated as one long paragraph inside "The Möbius origin of the irreducible π" is consolidated into a dedicated section (placed after the bootstrap audit, before "Honest status"), telling the executed WP1–WP5b story in six typed steps: (1) setup — the E_8 μ_4-glue as a flat $\mathbb{Z}_4$ monodromy on the A_3 ALE space, clock^2 = deck (spin $\mathbb{Z}_8$, $8 = 2|\mu_4|$) [E] (`v492`); (2) deformation — $XY = Z^4+a_0$ pure scale, $j = 1728$ frozen, clock = Picard–Lefschetz/Coxeter monodromy [E] (`v493`); (3) consistency honestly demoted — $(\kappa/c_3)^2 = 12$ exact but the alignment format passes 8/8: alignment survives, selectivity does not [E]/[C] (`v495`); (4) type mismatch + boundary limit — the $(E_8)_1$ character is not a jet-grading conformal block (growth $n^{2/3}$ vs $n^{1/2}$) but the μ_4 period sum 248 holds exactly: a boundary-limit shadow of `SEAM.EQUIV.01`/MMST shape [E] (`v496`); (5) the null ideal quantitative — stabilisation threshold $w = n{+}1$, $27000 = (h^\vee)^3$ derived, quotient = sector sum = 4124, $SO(16)_1$ contrast [E] (`v497`); (6) the deleting operator explicit — $|s\rangle = (E^\theta_{-1})^2|0\rangle$, $J^a_1|s\rangle = 0$ on all 248 (tally 190/57/1), level dial $2(1{-}k)$, one-block closure E_8/μ_4-specific [E] (`v498`); plus the WP5c–e roadmap and the honest status line (the route does *not* close `SEAM.EQUIV.01`; the GNS/limit question stays [O]). All pre-existing script citations (`v492`–`v498`) are carried into the new section; the Möbius section now points to it with one sentence. `tfpt_research_contracts`: a reader's note at the head of the `CELEST.SEAM.01` contract cross-references the new Paper-3 section (the contract stays the full technical reference). `tfpt_4_frontier`: one sentence in the metric-sector keybox records the celestial route as a third, quantitative approach to the keystone (not closing it).
- **Website mirror.** `papers.ts`: the Paper-3 section list gains the celestial-route section (six-step summary in the existing style) and a "Celestial route" highlight; the contracts mirror gains the same cross-reference sentence. `glossary.ts`: three new entries (celestial route, ALE space, null ideal) in the established technical style. Generated indices via build; no status component changes (no marker moved anywhere).

2026-07-21 (VII) (CELEST.SEAM.01 WP5b executed and promoted: `v498` — "the singular vector as an operator", the deleting object of the `v497` null ideal constructed explicitly and machine-verified singular; contract now WP1–WP4 + WP5a + WP5b done, WP5c–e [O] (roadmap); Wolfram extension 398 → 403 (verified engine run); *no gate closes, no marker moves, `SEAM.EQUIV.01` untouched*)

- **`v498_celestial_wp5b_singular_vector.py` (`CELEST.WP5B.01`, 53 checks, exact integer/Fraction, deterministic, no floats).** Success on the preregistered WP5b criterion: the deleting object exists and is explicit. *Construction*: $|s\rangle = (E^\theta_{-1})^2|0\rangle$ (weight 2θ, level $2 = 8$ quarter units $= q^2$, an *integer* level) in a machine-built Chevalley/Frenkel–Kac basis — bimultiplicative 2-cocycle with asymmetry verified on all 57600 ordered root pairs, the $[e_\alpha, e_{-\alpha}]$ sign *forced* by the Jacobi identity (SGN $= -1$, not chosen), the invariant form κ derived ($\kappa(\theta^\vee, \theta^\vee) = 2$), Jacobi exact on all 57600 $\mathfrak{sl}_2$-slice triples + 6000 seeded + all 496 θ-adjacent, all $N_{\alpha\beta} = \pm 1$. *Singularity (machine)*: $J^a_1|s\rangle = 0$ for all 248 basis elements (0 violations), with the case classification 190 (first bracket) / 57 (second) / 1 ($a = F^\theta$ via the central-term cancellation — the *only* case that sees the level k); plus $J^a_2|s\rangle = 0$, $E^a_0|s\rangle = 0$, exact weight, Shapovalov norm 0; the affine PBW engine unit-tested on all 61504 basis pairs. *Level dial*: the F^θ_1 coefficient is $2(1-k)$ — $2/0/-2$ at $k = 0/1/2$: without the central extension the deletion operator does not exist. μ_4 *data*: glue class $j(\theta) = 1$ (machine-built h-adapted chamber, $\langle\theta, h\rangle = 5$, height $29 = h^\vee-1$), clock phase $i^{2j} = -1$, class$(2\theta) = 2$ sheet-even; the deleting θ-$\mathfrak{sl}_2$ crosses the sheet-odd classes $(1, 3)$ while its square is sheet-even; cross-table θ-grading × glue class consistent (rows $(1, 56, 126, 56, 1)$, columns $(52, 64, 60, 64)$); 8 quarters $= q^2$ via the per-period dictionary. *Level-2 match*: weight 2θ has multiplicity 1 in the level-2 Fock, so $U(\mathfrak{g})|s\rangle$ is *the* $27000 = (h^\vee)^3$; the direct $\mathfrak{g}_0$-orbit BFS matches the Freudenthal multiplicities of $V(2\theta)$ exactly through depth 4 ([C] Weyl complete reducibility beyond); quotient $31124 - 27000 = 4124$; lattice reading $(2\theta, 2\theta)/2 = 4 > 2$: $|s\rangle$ *is* the kernel generator of Fock $\to E_4/\eta^8$. *Negative controls (honest separation)*: the level-1 current state $E^{\theta}_{-1}|0\rangle$ (one witness, $\to -k|0\rangle$ — the 248-current state that must remain) and 4 generic level-2 states (14–33 witnesses) are *not* singular; at k=2 the *cube* is (generic Kac $(E^\theta)^{k+1}$ mechanics, honestly typed); $SO(16)_1$ has the same singular vector but keeps three extra level-1 primaries ($h = \frac{1}{2}$ breaks one-block fusion; $5304 \neq 14^3$) vs E_8 (comarks all ≥ 2, glue weights $(0, 1, 1, 1)$ integer) — the *one-block closure* is the E_8/μ_4-specific part. *Honest* [O]: in the twisted quarter-slot moding two sector-C_1 modes never sum to 8 quarters (minimum $6 = q^{3/2}$) — the per-period dictionary (`v496`), not the per-slot identification, carries $|s\rangle$ to q^2: exactly the WP5c GNS/limit-state question; WP5c–e stay open; `SEAM.EQUIV.01` untouched.

- **Contract + surfaces synced.** `tfpt_research_contracts`: the WP5b bullet moved to *done* with the key numbers (case tally 190/57/1; level dial $2(1-k)$; $j(\theta) = 1$, clock phase -1; orbit BFS through depth 4; the D_8 contrast) and the twisted-slot tension as the explicit WP5c handover; honest status line and K4 paragraph updated ("WP1–WP4 + WP5a + WP5b executed; WP5c–e [O]"). `tfpt_3_e8_audit_bootstrap`: the celestial-contract paragraph extended by the WP5b sentence. Ledger: `CELEST.WP5B.01` (new) + `CELEST.SEAM.01` ("WP1–WP4 + WP5a + WP5b executed; WP5c–e [O] (roadmap)"). Wolfram: 5 exact `v498` mirrors added to `tfpt_readouts_extension.wl` and *verified* with the active engine the same day (398 → 403, `ALL WOLFRAM EXTENSION CHECKS PASSED`); `wolfram/README.md` + `GATE.WOLFRAM.02` counters advanced; `tfpt_safeguards` Layer-6 count + website mirrors 116/116 + 403/403. Website: `papers.ts` contract mirror ("WP1–WP4 + WP5a + WP5b done, WP5c–e roadmap"), DAG `qft` node + `v498`, generated indices via build. Suite now 492 scripts, all green; `SEAM.EQUIV.01` stays [O].

2026-07-21 (VI) (CELEST.SEAM.01 WP5a executed and promoted: `v497` — "the null ideal from the limit", the `v496` boundary-limit shadow made a precise coefficientwise limit and the level-2 null ideal derived from root data; contract WP5 subdivided, now WP1–WP4 + WP5a done, WP5b–e [O] (roadmap); Wolfram extension 394 → 398 (verified engine run); *no gate closes, no marker moves, `SEAM.EQUIV.01` untouched*)

- **`v497_celestial_wp5a_null_ideal.py` (`CELEST.WP5A.01`, 34 checks, exact integer/Fraction, no floats).** Three results make the WP4 limit reading precise. *Stabilisation theorem*: the one-parameter family χ_w of graded Fock characters on the chiral jet generators (loop energy m, radial number r; $E_w = m + wr$ in quarter units $u = q^{1/4}$) contains the chiral jet grading as its w=2 member (generator counts $64, 120, 128, 180, 192, 240, 256, 300$ per degree), and its u^n coefficient equals the quarter-moded loop Fock for *all* $w \geq n$+1, strictly larger for every $w \leq n$ (verified $n \leq 8$, $w \leq 10$) — the `v496` "boundary-limit shadow" is a precise *coefficientwise* limit $\varepsilon = 1/w \to 0$ with explicit threshold $w = n$+1, not a slice (convention clarified: 897266 = the quarter-moded loop Fock at u^4 = *integer* level 1). *The null ideal derived, not cited*: Freudenthal multiplicities + the Weyl dimension formula + generic character peeling give $\mathrm{Sym}^2(248) = V(2\theta) \oplus V(\omega) \oplus V(0) = 27000 + 3875 + 1$ with residual exactly zero (independent Weyl-orbit cross-check); $27000 = 30^3 = (h^\vee)^3$ exactly; $C_2 = (124, 96)$, Dynkin indices 6750/750; level-1 integrability ($\langle 2\theta, \theta^\vee\rangle = 4$, $\langle\omega, \theta^\vee\rangle = 2$) leaves the vacuum as the only level-1 primary (Kac, used only through its counting consequence); level-2 quotient $31124-27000 = 4124 = 1+248+3875$ exactly. *Two routes, one number*: the μ_4 theta-split sector sum at $q^2 = (1036, 1024, 1040, 1024)$ gives the *same* 4124 — the untwisted null-ideal quotient and the boundary/twisted lattice presentation count one space (q^3: $34752 = \chi_3$). Negative controls: $SO(16)_1$ through the same pipeline gives *four* components $(5304 + 1820 + 135 + 1)$, $5304 \neq 14^3$ (the $h^{\vee\,3}$ identity is not generic), quotient $2076 = \Theta_{D_8}/\eta^8$ at q^2 (the recipe validated on a second algebra), block weights $(0, \frac{1}{2}, 1, 1)$ break one-block fusion (vs E_8's integer $(0, 1, 1, 1)$); only $P = 4 = |\mu_4|$ periodisation reproduces the 248 layer; swapped glue breaks the zero-mode anchor ($64 \neq 60$). *Honest* [O]: the limit does *not* generate the truncation — the singular vector as an *operator* in the limit net (WP5b), the GNS limit state (WP5c) and local normality/Haag duality (WP5d) stay open; the celestial route now has the same two-step shape as MMST (limit + maximal ideal), with the ideal quantitatively fixed; `SEAM.EQUIV.01` untouched.

- **Contract + surfaces synced.** `tfpt_research_contracts`: WP5 subdivided — WP5a moved to *done* with the key numbers (stabilisation threshold $w = n{+}1$; $27000 = (h^\vee)^3$ derived; quotient = sector sum = 4124; $SO(16)_1$ contrast), WP5b–e added as roadmap bullets with success/kill criteria (WP5b singular-vector operator in the quarter-moded current Fock; WP5c GNS limit state, kernel $\supseteq$ ideal; WP5d local net/Haag duality via the KLM route; WP5e BCOV/Costello–Li twistor uplift, partition function $= E_4/\eta^8$ incl. the $q^{-1/3}$ prefactor); K4 paragraph and the honest-success status line updated. Ledger: `CELEST.WP5A.01` (new) + `CELEST.SEAM.01` ("WP1–WP4 + WP5a executed; WP5b–e [O] (roadmap)"). Wolfram: 4 exact `v497` mirrors added to `tfpt_readouts_extension.wl` and *verified* with the active engine the same day (394 → 398, `ALL WOLFRAM EXTENSION CHECKS PASSED`); `wolfram/README.md` + `GATE.WOLFRAM.02` counters advanced; `tfpt_safeguards` Layer-6 count + website mirrors 116/116 + 398/398. Website: `papers.ts` contract mirror ("WP5a done, WP5b–e roadmap"), DAG `qft` node + `v497`, generated indices via build. Suite now 491 scripts, all green; `SEAM.EQUIV.01` stays [O].

2026-07-21 (V) (`comb-meta-limit` re-run with the PG.06b FULL Vela recovery ingested — the stale single-obs PG.06b reference (noise-peak $F_0 = 11.19275$ Hz) corrected in the channel definitions; the new X-ray channel enters *censored* (no absolute ε), both aggregate limits numerically unchanged; *experiments level only — no verification module, no gate closes, no marker moves*)

- **`experiments/comb-meta-limit` channel update (methodology untouched).** The Vela X-ray channel (`channels.py`) now ingests the PG.06b FULL reduction (`pg06b_full_vela.json` in `pulsar-glitch-recovery/results`: 665 obs → 491 usable → 119 segment-coherent $\nu(t)$ points) instead of the superseded single-obs `pg06b_vela.json`; the stale code note quoting the noise-peak "detection" $F_0 = 11.19275$ Hz is corrected on record (true Vela $F_0 = 11.1861692$ Hz), and the channel key is fixed from the mislabel `PG.07` to `PG.06b`. Censoring handled exactly by the existing Tier-B rule (nothing invented): the comb observable is a *linear* $\delta\nu(\tau)$ residual (μHz, not a fractional ln-flux modulation), both legs sit below the 2.8-period gate (2019: 2.17, 2021: 1.38), and the surrogate-calibrated injection never reaches ε_{90} below $\varepsilon = 1$ ($\varepsilon_{50} = 0.55$ in 2019 only) ⇒ no absolute per-channel ε-UL exists for the DL/HKSJ combination; the channel stays enumerated-only ("no constraint"), carrying the normalised context amplitude $\sqrt{2g} = 0.40$ (leg gains 0.046/0.114, min $p = 0.014$ — the 2019 audit-level chance feature: Bonferroni $\times 10 = 0.14$, shuffle $p = 0.40$).
- **Full re-run (seed 0, deterministic): both limits numerically unchanged.** Boundary/horizon-scoped: `data_limited` (still no horizon channel with an absolute ε); all-channel universal-DSI 95% UL $\varepsilon < 0.11954$ ($\approx 12.0\%$; HKSJ $t_2 = 2.92$, $\rho = 0.3$-inflated 0.137; $6.91\times$ the predicted $\varepsilon = 0.0173$), still driven by the surface log-flux channels A1/A4/A5 (A4 GRB the anchor, per-channel UL 0.108); now 10 channels enumerated / 3 usable. Injection self-consistency unchanged (inject 5% → recover 4.8%, UL 5.5%; coverage 0.92/150 trials). Surfaces synced: experiment README (tier table, ingest list,

PG.06b FULL bullet), `results/results.json`, scorecard row `data_value`/source via `build_evidence_scorecard.py` (status stays `data_limited`; counters unchanged, 120 rows), `experiments/README.md` catalog row. No claim; firewall intact.

2026-07-21 (IV) (Wolfram engine reactivated — the deferred second-path mirrors VERIFIED: base 116/116 + extension 394/394 (378 → 394, the 16 mirrors accumulated since 2026-07-14 now counted); `GATE.WOLFRAM.01/02` and all counter surfaces synced; *no gate closes, no marker moves*)

- **Verified engine run 2026-07-21 (Wolfram Engine 14.3, license reactivated).** `wolframscript -file verification/wolfram/tfpt_readouts.wl` → `116 passed, 0 failed`, `ALL WOLFRAM CHECKS PASSED`; `tfpt_readouts_extension.wl` → `394 passed, 0 failed`, `ALL WOLFRAM EXTENSION CHECKS PASSED`. The 16 exact mirrors added while the engine was unactivated all enter the counted total (378 → 394): `v479` Kronheimer bridge (2), `v491` partition corollary (3), `v493` celestial WP2 clock deformation (4), `v495` celestial WP3 Green–Schwarz alignment (3), `v496` celestial WP4 character shadow (4).
- **Counter surfaces synced.** `verification/wolfram/README.md` (current-state block + the five deferral notes resolved; `v484/v485/v486/v487/v488/v492` stay mirrorable-but-unmirrored, Python-exact, so they are *not* counted); ledger `GATE.WOLFRAM.01` (re-verified 2026-07-21) + `GATE.WOLFRAM.02` (394/394, UPDATE note); `tfpt_safeguards` Layer-6 count + website `papers.ts` mirrors (116/116 + 394/394); `v479` registry/`run_all` deferral note resolved. No numeric claim changes; the second computational path is simply verified-current again.

2026-07-21 (III) (CELEST.SEAM.01 WP3 + WP4 executed and promoted: `v495` Green–Schwarz alignment (verdict B, honestly non-selective) + `v496` character-vs-S-algebra type mismatch (verdict B(ii), boundary-limit shadow); contract now WP1–WP4 done, WP5 [O]; *no gate closes, no marker moves, `SEAM.EQUIV.01` untouched*)

- **`v495_celestial_wp3_gs_alignment.py` (`CELEST.WP3.01`, 25 checks, exact Fraction/sympy).** The pre-registered WP3 "alignment test only": the Okubo coefficient $\mathrm{Tr}_{\mathrm{adj}}X^4/(\mathrm{Tr}_{\mathrm{adj}}X^2)^2 = 5/(2(\dim\mathfrak{g}+2))$ is *derived* as an exact polynomial identity for all 8 algebras on Costello's list (sl_2, sl_3, g_2, so_8, f_4, e_6, e_7, e_8; $\mathrm{e}_6/\mathrm{e}_7$ carved out of the `v492` E_8 root set; negative control: sl_4 fails, 5/32 vs 3/32); the closed form $\tilde{\lambda}^2 = 10h^{\vee 2}/(\dim+2) = h^\vee+6$ holds exactly across the Deligne series; E_8: $\tilde{\lambda} = 6$ (unit-trace) resp. $\lambda_{\mathrm{fund}} = 1/10$ (adjoint-trace) exact, so $(\kappa/c_3)^2 = 12 = |\mu_4|N_{\mathrm{fam}}$ resp. 1/300 — with κ/c_3 itself *irrational* ($2\sqrt{3}$). Byproduct: the printed $\lambda^2_{\mathrm{so}_8} = 3/2$ in Costello's appendix A is a factor-2 slip (his own weight content gives 3). **Look-elsewhere (the hard fence):** the alignment format passes 8/8 across the whole list (zero selectivity), $\tilde{\lambda}$-integrality 2/8 (shared with sl_3), only g_{car}=5 | $h^\vee$ is e_8-selective (1/8), the isolating conjunction is post hoc. Kill test K3 does *not* fire but is scope-demoted: *alignment survives; selectivity does not* — compatibility, not e_8-selective evidence. Verdict B.
- **`v496_celestial_wp4_character_shadow.py` (`CELEST.WP4.01`, 25 checks, exact integer/Fraction/sympy).** WP4, the type mismatch head-on: the $(E_8)_1$ character $E_4/\eta^8 = (1, 248, 4124, 34752, 213126, \dots)$ is *not* a conformal block of the glue-equivariant $E_8[\mathbb{C}^2]$ S-algebra in its own jet grading — localised three ways: (a) spin $1-d/2$ unbounded below and 248 never a tower dimension ($d \le 100$); (b) growth $n^{2/3}$ vs $n^{1/2}$ (quadratic cumulative count $31s^2$+$92s$+60, $31 = k$+$h^\vee$; f_n/χ_n strictly increasing $n = 1..12$); (c) the level-2 null ideal deletes exactly $27000 = 30^3 = (h^\vee)^3$ out of $\mathrm{Sym}^2(248) = 1$+3875+27000 — no jet analogue. *But* the boundary/period reading holds exactly at the current stratum (zero modes = 60 = C_0 currents, jet slice cycles (60, 64, 60, 64), one μ_4 period sums to 248, glue weights (0, 1, 1, 1) integer; free loop Fock 897266 ≫ 248, so the truncation must come from a limit) — the same MMST

scaling-limit shape as `SEAM.EQUIV.01` (`v336`/`v449`). K4 fires only against the exact-block reading; the constructive limit question passes to WP5. Verdict B(ii).

- **Contract + surfaces synced.** `tfpt_research_contracts` WP3/WP4 bullets moved to *done* with the look-elsewhere demotion in the text, K3/K4 marked evaluated, status line now "WP1–WP4 executed (verdicts B, B, B, B(ii)), WP5 [O]"; one conservative sentence added to the `tfpt_3` celestial paragraph; ledger rows `CELEST.WP3.01` + `CELEST.WP4.01` (new) and `CELEST.SEAM.01` updated; `v495`/`v496` Wolfram mirrors added to `tfpt_readouts_extension.wl` (3 + 4 exact checks; count with the next verified engine run — engine still unactivated); website `papers.ts` contract highlight "WP1–WP4 done" + DAG `qft` node extended. Suite now 490 scripts, all green; `SEAM.EQUIV.01` stays [O].

2026-07-21 (II) (PG.06b FULL: the complete NICER Vela archive reduced — the first X-ray $\nu(t)$ leg of the recovery-comb search, verdict `data_limited`; the old PG.06b "pipeline proven" fold corrected on record; *experiments level only — no verification module, no gate closes, no marker moves*)

- **`experiments/pulsar-glitch-recovery` PG.06b FULL executed (`tfpt-pulsar vela-full`).** The full NICER Vela-pulsar (PSR B0833-45) L2 archive is reduced, with the manifest frozen *before* the bulk download (`manifest_full_reduction.json`: all 665 ObsIDs, preregistered 2019/2021/2024 glitch windows, 20-GB abort gate): 665/665 obs downloaded (2.6 GB, raw gitignored) → 491 usable → 240 per-obs detections → **119 segment-coherent $\nu(t)$ points** (108 fully coherent), median $\sigma_\nu \approx 0.6\,\mu$Hz; barycentring validated against PINT to $3.4\,\mu$s — the *first X-ray $\nu(t)$ probe* of the search program. Sanity gate PASS: $\Delta\nu/\nu$ 2019 $= 2.39 \times 10^{-6}$ vs 2.5012×10^{-6} published, 2021 $= 1.272 \times 10^{-6}$ vs 1.26×10^{-6}; the 2024 glitch is not reducible (only ∼150-s snapshots, 9.7 ks — documented).
- **Frozen-kernel comb ($\omega = 2.5827$, $\varepsilon_{\text{pred}} = 0.0173$): verdict `data_limited` (range- and amplitude-limited).** Reach 2019: 2.17 comb periods (< the 2.8-period gate; beats the PG.08 public Vela reach 1.83), 2021: 1.38; the 2019 raw $p = 0.014$ has the kernel as smallest-p battery member BUT Bonferroni ×10 = 0.14 and shuffle $p = 0.395$ ⇒ chance-level by the preregistered escalation rule (audit flag, NOT a candidate); 2021 $p = 0.731$; $\ln\tau$ stack $p = 0.049$/shuffle 0.307. Surrogate-calibrated injection at the real sampling and noise: $\varepsilon_{50} \approx 0.55$ (2019 only), ε_{90} never reached below $\varepsilon = 1$, 0% power at the predicted $\varepsilon = 0.0173$ — a weaker amplitude probe than the PG.08 radio residuals, but it independently reproduces the PG.06 range-blindness on real data. Second off-kernel flag: $\lambda = e$ in the 2021 leg (raw $p = 0.002$, Bonferroni 0.018; the same non-kernel periodicity class as PG.08 J0742−2822 — flagged, not claimed). The sub-day early window that would clear the gate is absent from the archive (first usable obs 3.4 d / 7 d post-glitch); next decider: <3-d post-glitch X-ray observations of a future Vela glitch (eXTP class).
- **Correction on record.** The old PG.06b single-obs "pipeline proven" detection at $F_0 = 11.19275$ Hz ($H = 18.4$) was a noise peak; the true Vela frequency at that epoch is $F_0 = 11.1861692$ Hz ($H = 99.7$, consistent with the phase-connected PuMA ephemeris). The pipeline remains proven — by the corrected fold. Scorecard row updated to "recovery comb on the full NICER Vela X-ray nu(t) (PG.06b FULL)" (status stays `data_limited`; counters unchanged, 120 rows: 49 consistent / 9 tension / 26 null / 33 data-limited / 3 parked); `experiments/README.md` catalog + §1 prose, pulsar README, `experiments/next.txt` synced. No claim; firewall intact.

2026-07-21 (the two 2026-07-20 prediction-of-record rows promoted to the suite: v494 — α_s running + μ-distortion branch discriminator machine-checked, ledger rows `COSMO.RUNNING.01`/`COSMO.MUDIST.01`, prediction surfaces synced; *no gate closes, no marker moves*)

- **`v494_cosmo_running_mu_record.py` (23 checks, all green).** The α_s-running and μ-distortion scorecard rows of 2026-07-20 (IV) get their load-bearing module so the new paper/website statements are machine-backed. [E] leading-order identities (sympy exact): $\alpha_s = \mathrm{d}n_s/\mathrm{d}\ln k = -2/N_\star^2 = -r/6$, $\beta_s = -4/N_\star^3$ (NOT $-8/N^3$); LO band $N_\star \in [50, 60]$: $[-8.0, -5.6] \times 10^{-4}$. [E] exact slow roll on the Einstein-frame Starobinsky potential (third-order Lyth–Riotto + finite-difference cross-check $< 1\%$): $\alpha_s(51.4) = -7.09 \times 10^{-4}$, $\beta_s(51.4) = -2.70 \times 10^{-5}$. [C] data: Planck 2018 (-0.0045 ± 0.0067) $+0.57\sigma$; record leg P-ACT-LB ($+0.0062 \pm 0.0052$) -1.33σ (watch channel: positive central value; kill: robust $\alpha_s < -5 \times 10^{-3}$ or $> +3 \times 10^{-3}$ at $\geq 5\sigma$; any robust POSITIVE running kills the plateau branch). [C] μ-distortion (Chluba window, +21% calibrated on the ΛCDM anchor): sharp branch $N_\star$=51.4 $\mu = 1.6 \times 10^{-8}$, profiled record $N_\star$=56.1 $\mu = 1.94 \times 10^{-8}$, band 1.5–2.3 $\times 10^{-8}$; branch split $\Delta\mu = 3.0 \times 10^{-9} \Rightarrow 3\sigma$ $A_s/N_\star$ branch decision at $\sigma(\mu) = 10^{-9}$ (Voyage-2050 class; PIXIE baseline 9×10^{-8} does not reach; kill: robust $\mu < 0.9 \times 10^{-8}$ or $> 4 \times 10^{-8}$). [E] comb blindness (honest S15): the frozen log-comb ($\omega = 2.5827$, $\varepsilon = 0.0173$) is structurally invisible in the μ window ($|F(\omega)| = 0.033$, $\delta\mu = 7.4 \times 10^{-12} = 0.007\sigma$ even at Voyage-2050). Fences: $N_\star$ stays an external reheating input (`GATE.NSTAR.01`); the frozen band $[50, 60]$ stays the surface of record (v84/v86). Ledger rows `COSMO.RUNNING.01` and `COSMO.MUDIST.01` added.
- **Prediction surfaces synced (the missing mirror of 2026-07-20 (IV)).** `predictions.txt` gains the two rows (α_s [C] with the -1.33σ P-ACT-LB pull and the positive-running kill; μ-distortion [C] data-limited with the Voyage-2050 branch split); website `lib/predictions.ts` gains the matching cards `alpha-s-running` and `mu-distortion` (confrontation + experiment blocks pointing at `experiments/tfpt-discovery`); the cosmology DAG node lists `v494` and states the running/μ readouts.
- **Paper bodies.** tfpt_1 (Inflation interface) gains the running consistency-relation paragraph ($\alpha_s = -2/N_\star^2 = -r/6$, exact slow roll, pulls, positive-running kill) with `\veri{v494_cosmo_running_mu_record.py}`; tfpt_2 (inflation-pivot keybox) gains the μ-distortion branch-decider sentence (sharp 1.6×10^{-8} vs profiled 2.0×10^{-8}, 3σ at Voyage-2050) with the same `\veri`. No marker of any existing claim changes. (v494 is numerical/quadrature, Python-only by the suite convention, flagged in the docstring; `GATE.WOLFRAM.01/02` untouched.)

2026-07-20 (VIII) (E8 mass-ladder blind-recovery bed EXECUTED: verdict `validated` at the detector level — the Cartan–PF ladder pipeline blindly recovers the published Zamolodchikov-E8 spectra of $BaCo_2V_2O_8$ (8/8) and $CoNb_2O_6$ (6/8 in-window) and rejects every control class; *experiments level only — confirms Zamolodchikov E8, not TFPT; no verification module, no gate closes, no marker moves*)

- **`experiments/e8-ladder-bed/` executed (preregistered `hypotheses/e8_ladder_bed_v1.yaml`).** The blind ladder detector (assignment DP + scale fit + 5000-ladder Monte-Carlo null; the same detector class TFPT uses on search surfaces, e.g. `hfqpo-ladder`) on the only public *known-answer* mass-ladder dataset: $BaCo_2V_2O_8$ (Zou+ PRL **127**, 077201; INS at the 4.7 T 1D QCP, Fig. 3a extraction QA-validated, 17 peaks blind) recovers **8/8 rungs**, $\chi^2/\mathrm{dof} = 0.35$, MC $p = 0.0056$ (singles-only run $p = 0.0004$; σ-variant robust); $CoNb_2O_6$ (Amelin+ PRB **102**, 104431; verbatim THz values) recovers **6/8 in-window rungs**, $\chi^2/\mathrm{dof} = 0.52$, $p = 0.0038$ (m_7/m_8 lie outside the 0.6-THz low-pass window; the broad 0.43-THz (m_1+m_2)-continuum onset is correctly left unassigned). Cartan–PF self-test: the sorted PF eigenvector of $2I - A(E_8)$ equals the Zamolodchikov ladder to 1.5×10^{-15} ($m_2/m_1 = \varphi$ to 2.2×10^{-16}; lowest Cartan eigenvalue $2 - 2\cos(\pi/30)$ to 4.5×10^{-14}).

- **Control battery — every class rejected.** Scramble 0.5%/1.5% false-positive rate (200 log-gap permutations, near-duplicate accounting); offset $\pm\{0.3, 0.5\}m_1$: 0/8 detections; jitter dose-response $\le 5\%$ at the destructive 30% level; wrong ladders on the real data (harmonic 1..8, A_3-PF, D_8-PF, Airy confined-spinon) all beaten by E8; the synthetic Airy bed (the true zero-field physics of these compounds) yields **0** valid rungs. The one disclosed v1.1 criteria amendment (jitter as dose-response, scramble near-duplicates, χ^2-gated wrong-ladder comparator) is fully documented in the hypotheses YAML with all v1 numbers preserved; the recoveries themselves were never touched.

- **Integration (firewall intact).** Scorecard row "E8 mass-ladder blind recovery bed" moves `parked` → `consistent` (stage `not_applicable`, tier `analog_positive_control`, leakage `detector_control` — the executed-analog basket of `qc-recovery-kernel` and `qgeo-eit-soff`); counters 120 rows: 49 consistent / 3 parked; `experiments/README.md` catalog row + stats (generator); website `predictions.ts EXPERIMENTS_AUDIT` counters; `experiments/next.txt`. This validates the *ladder-detector pipeline* — it confirms Zamolodchikov E8, *not* the TFPT axioms: no [**E**], no `\veri`, no ledger row.

2026-07-20 (VII) (v493 — WP2 of `CELEST.SEAM.01` executed: the clock-invariant deformation $XY{=}Z^4{+}a_0$ is a PURE SCALE with the $\tau{=}i$ pillowcase frozen, the μ_4 clock IS the Picard–Lefschetz/Coxeter monodromy of the seam deformation, and the BHS deformed-algebra pattern transfers $\mathbb{Z}_2 \to \mathbb{Z}_4$ with no sector leak; *verdict B, contract updated to WP2 done, no gate closes, no marker moves, `SEAM.EQUIV.01` and the v216 residual stay [O]*)

- **`v493_celestial_wp2_clock_deformation.py` (`CELEST.WP2.01`).** WP2 of the celestial contract, sympy exact (47 checks), promoted from the `experiments/tfpt-discovery/` probe `celestial_seam_wp2_clock_deformation_probe.py`. *Geometry:* $P(iZ){=}P(Z)$ forces $a_3{=}a_2{=}a_1{=}0$ sharply and two-sided ($\prod_k(Z - i^k w) = Z^4 - w^4$); $XY{=}Z^4{+}a_0$ is smooth iff $a_0{\neq}0$ (disc $= 256\,a_0^3$); the branch points $Z^4{=}{-}a_0$ are one free μ_4 orbit in *every* fibre with cross-ratio 2; the binary-quartic invariants of $w^2{=}Z^4{+}a_0$ are $I{=}12a_0$, $J{=}0$ *identically*, so $j{=}1728$ and the $\tau{=}i$ pillowcase shape (`v168`/`v214`) is *frozen* for all a_0 — a_0 is a pure scale, no shape modulus survives clock-invariance. *Periods/monodromy:* $\Pi = t(i{-}1)(1, i, i^2)$, all three vanishing spheres carry the same volume $\sqrt{2}\,t$ (lockstep); the clock acts on H_2 as a *Coxeter element* of $W(A_3)$ (char $x^3{+}x^2{+}x{+}1$, eigenvalues $\{i, -1, -i\}$ = the A_3 exponents, $h(A_3){=}4{=}|\mu_4|{=}N_{\mathrm{fam}}{+}1$), fixes no cycle ($\det(A{-}1){=}{-}4$), and *is* the Picard–Lefschetz monodromy of the family ($a_0 \to e^{2\pi i}a_0$ gives the 4-cycle); the surviving direction is the weight-1 χ_1-Fourier diagonal $(1, i, i^2)$, *not* the naive invariant diagonal; the versal weights $(a_3, a_2, a_1, a_0) = (1, 2, 3, 0)$ are the regular μ_4 representation. *Algebra (BHS $\mathbb{Z}_2 \to \mathbb{Z}_4$, `arXiv:2305.09451`):* mode dictionary $\{w[m,n] : m{\equiv}n \bmod 4\}$; the fibre bracket $-4\,\mathrm{Nambu}(XY{-}Z^4{-}a_0)$ anchored at the $a_0{=}0$ orbifold ($\{X, Y\}{=}16Z^3$, Jacobi + Casimir exact, uniform clock-degree -1); S-algebra corrections exactly *linear* in a_0 at the $\mathbb{Z}_4$ wrap, conserving the μ_4 grade — no sector leak, the `v492` equivariant sector deforms consistently; $a_0 \leftrightarrow$ BHS's weight-0 c^2 slot; matrix factorisations deform along all orbit splittings, the clock fixing exactly the antipodal 2+2 sector (= the EH/$\mathbb{Z}_2$ middle node, $4{=}2{\times}2$). *Negative controls:* $a_1 Z \Rightarrow j{=}0$ (hexagonal); pure $a_2 Z^2$ stays singular (A_1); $a_2{\neq}0 \Rightarrow j{=}1556068/81{\neq}1728$; explicit grading leak — doubly sharp. *Verdict B* via three named [**C**] identifications (-4Nambu = the $k{=}4$ CCA bracket; period = root difference; seam-scale reading via clock^2=deck); the `v216` residual is *typed*, not moved (given the order-4 clock the square is automatic — a relocation of the same input, no new derivation).

- **Contract updated to WP2 done.** `tfpt_research_contracts` `CELEST.SEAM.01` section: WP2 bullet "finite, next" → *done (verdict B)* with the two sharpenings (clock = Coxeter/Picard–Lefschetz monodromy; surviving direction = χ_1-diagonal, a_0 pure scale); kill-test paragraph K1 "survived by WP1 and WP2"; status line `CELEST.WP2.01` [E]/[C] executed, WP3–WP5 stay [O]. Ledger rows `CELEST.WP2.01` (new) + `CELEST.SEAM.01` (WP2 done) + `GATE.WOLFRAM.02` (four exact `v493` mirrors added to `tfpt_readouts_extension.wl`, counted with the next verified engine run — engine unactivated, same convention as `v491`); registry + `run_all` (487 modules, suite green); website mirrors (`papers.ts` contract section, generated script index). `SEAM.EQUIV.01` stays [O]; no marker moves anywhere. Exact (sympy).

2026-07-20 (VI) (HFQPO "third tooth": the preregistered archival RXTE PCA scan has RUN — a well-powered null with limits; scorecard row S1.e `data_limited`→`null`, no marker moves)

- **H3 stage 2 executed (`experiments/hfqpo-ladder/results/archival/`).** The preregistered archival scan (`hypotheses/hfqpo_v1.yaml`, `archival_design`) ran end-to-end on public HEASARC data: 77 ObsIDs / 4.42 GB across all four 3:2-pair sources (GRO J1655−40, XTE J1550−564, GRS 1915+105, H1743−322). Sanity gate PASS (11/12 published pair lines reproduced, e.g. 450.3 Hz at 5.37% rms 4.4σ, 280.1 Hz at 3.71% 9.9σ); injection calibration PASS in all four sources (90%-recovery sensitivities 5.11/2.68/0.89/1.65% rms at the tooth frequencies).
- **Blind-scan verdict: NULL with limits.** Neither the geometric tooth $\nu_3 = 1.5\,\nu_u$ (661.5/414/252/363 Hz) nor the integer control line $4\nu_0$ (588/368/224/322.7 Hz) is detected in any source ($\sim 0\sigma$ single-trial; trials correction $N{=}8$ as preregistered). 3σ rms upper limits (tooth, primary bands): 3.06/1.61/0.53/1.26% — in every source at or below the strength of the detected upper pair line. The ladder reading gains no support but is not killed (the `harmonic_hit` branch did not fire either; the anti-kernel record 92=184/2, 34/68 stands); GR parametric resonance stays the favorite. Caveats: GRS 1915+105 scanned in a proxy state (the 113/168 Hz epochs were never published at ObsID level); H1743's program is 80146 (YAML said 80138); GRO hard-band sensitivity 5.1%.
- **Integration.** Scorecard row S1.e (hfqpo-ladder) `data_limited`→`null` (120 rows: 26 null / 33 data_limited), next decider eXTP-class sensitivity below the 0.5–3.1% rms limits; HFQPO search-target paragraph updated in `tfpt_horizon_readouts` (+ the `papers.ts` mirror); website `predictions.ts` (`hfqpo-tooth` entry + `EXPERIMENTS_AUDIT` counters); `predictions.txt`; `experiments/README.md`; `experiments/next.txt`. No `verification/` module, no ledger row, no marker move (the search target stays [O]).

2026-07-20 (V) (v492 + the new research contract `CELEST.SEAM.01` — the celestial-holographic route: the E_8 μ_4-glue is the flat $\mathbb{Z}_4$ monodromy of the equivariant celestial chiral algebra on the A_3 ALE space, with the exact correction clock^2=deck (spin bridge $\mathbb{Z}_8$, $8{=}2|\mu_4|$); *verdict B, no gate closes, no marker moves, `SEAM.EQUIV.01` stays [O]*)

- **`v492_celestial_z4_orbifold.py` (`CELEST.WP1.01`).** WP1 of the new contract, sympy exact (36 checks): the `v128` μ_4-glue grading of E_8 (240 = 52+64+60+64, additive on all 6720 root pairs) is *inner* — $h = (2,2,2,2,2;0^4)$ reads the glue class mod 4 on all 240 roots, so the glue is $\mathrm{Ad}(\exp(2\pi i h/4))$, an order-4 inner torus automorphism = a flat $\mathbb{Z}_4$ monodromy in the Kronheimer–Nakajima sense; the A_3-side detector $h' = (0^5;1,1,1,-3)$ reads the same class = the diagonal $(1,1)$ of the discriminant group $\mathbb{Z}_4{\times}\mathbb{Z}_4$ (the `v92`/`v125` Lagrangian glue). $\mathbb{C}^2/\mathbb{Z}_4$ is verified as the A_3 singularity $XY{=}Z^4$ (Hilbert series $(1{+}t^4)/((1{-}t^2)(1{-}t^4))$, Molien exact); the glue-equivariant SDYM(E_8) sector *closes* with graded dimensions $60(d{+}1)/64(d{+}1)$ — possible only because $\dim\mathfrak{g}_j = (60,64,60,64)$ — with zero modes = the carrier $D_5{\oplus}A_3$+Cartan = 60 and density $1/4 = 1/|\mathbb{Z}_4|$; the four glue-sector characters sum exactly to $\chi_{(E_8)_1} = 1{+}248q{+}4124q^2{+}34752q^3$ (`v377`), with sector weights = the discriminant form $(5x^2{+}3y^2)/8$ and integer glue-diagonal $h = (0,1,1,1)$ (= locality of the $(E_8)_1$ extension). *Critical correction* *(verdict B):* the A_3 deck acts on the celestial sphere as $z \mapsto -z$ (order 2), *not* as the order-4 clock; the clock is the $U(2)$ phase $\mathrm{diag}(i, 1)$ (normalising the deck, det $= i$ rotating the holomorphic symplectic form by the μ_4 generator), with the exact spin bridge (spin clock)^2 = deck ($\mathbb{Z}_8$, $8 = 2|\mu_4|$, the $c_3 = 1/(8\pi)$ winding integer). Clock-invariance selects the 1-parameter A_3 deformation $XY{=}Z^4{+}a_0$ whose four branch points are one μ_4 orbit with cross-ratio 2 (`v168`/`v214` pillowcase). Negative controls: the false action $\mathrm{diag}(i, i)$ killed three ways, the false glue breaks additivity on 3112/6720 pairs, rigidity = $\mathrm{Aut}(\mathbb{Z}_4)$.

- **New research contract `CELEST.SEAM.01`** in `tfpt_research_contracts` (own section, the fourth contract alongside $U_{\mathrm{wall}}/G_{\mathrm{metric}}$/`CONTRACT.F.01`): hypothesis (seam = glue-equivariant $\mathbb{Z}_4$-orbifold sector of the CCA on the A_3 ALE; twistorial bulk = the *self-dual* sector only), typing/non-circularity (the continuum $(E_8)_1$ existence — the `SEAM.EQUIV.01` target — is *not* an admissible input), work packages WP1 (done, verdict B) through WP5 (second continuum route, Costello–Li grade), pre-registered kill tests K1–K4 (K1 survived by WP1), and an honest success estimate (full success low, partial findings high). Ledger rows `CELEST.WP1.01` + `CELEST.SEAM.01`; website mirror (`papers.ts` section, script index). `SEAM.EQUIV.01` stays **[O]**; no marker moves. Exact (sympy).

2026-07-20 (IV) (evidence scorecard: five new typed rows — the α_s running prediction of record, the μ-distortion $A_s/N_\star$ branch discriminator, and three parked analog beds (bosonic-E8 thermal Hall, E8 mass-ladder recovery, Efimov DSI tuning); *experiments level only — no verification module, no gate closes, no marker moves*)

- **α_s running (prediction of record, consistent).** The Starobinsky branch fixes the running of the spectral index: $\alpha_s = -2/N_\star^2 = -7.6 \times 10^{-4}$ at $N_\star{=}51.4$ (exact slow roll -7.09×10^{-4}; band $N_\star \in [50, 60]$: $[-8.0, -5.6] \times 10^{-4}$) and $\beta_s = -4/N_\star^3$ (leading order; NOT $-8/N^3$) — the identity $\alpha_s = -r/6$ holds only at leading order ($\sim$7% off exact). Data: Planck 2018 -0.0045 ± 0.0067 ($+0.57\sigma$), P-ACT $+0.0060 \pm 0.0055$ (-1.22σ), record leg P-ACT-LB $+0.0062 \pm 0.0052$ (-1.33σ, Calabrese et al. 2025 JCAP); β_s (Planck run+runrun) 0.010 ± 0.013 (-0.77σ). Forecast: LiteBIRD+CMB-S4 $\sigma(\alpha_s) \approx 4.9 \times 10^{-3}$ (0.14σ); CMB-S4+DESI+Euclid statistical $\sim 3.3 \times 10^{-4}$ (2.2σ, systematics-limited). Same `N_star_reheating` independence group as the $n_s/r/A_s$ rows (anti double-counting: never an independent hit); kill: robust $\alpha_s < -5 \times 10^{-3}$ or $> +3 \times 10^{-3}$ at $\geq 5\sigma$; any robust POSITIVE running kills the plateau branch. Probe: `experiments/tfpt-discovery/alpha_s_running_probe.py`.
- **μ-distortion (prediction of record, data-limited).** Chluba window integral (calibrated +20% to the full computation): sharp branch $N_\star{=}51.4$ ($A_s{=}1.76 \times 10^{-9}$) $\mu \approx 1.6 \times 10^{-8}$, profiled record $N_\star{=}56.1$ ($A_s{=}2.10 \times 10^{-9}$) $\mu \approx 2.0 \times 10^{-8}$, band 1.5–$2.3 \times 10^{-8}$; the 23% branch split ($\Delta\mu = 3.0 \times 10^{-9}$) gives a 3σ $A_s/N_\star$ branch decision at $\sigma(\mu) = 10^{-9}$ (Voyage-2050 class; the PIXIE baseline $|\mu| < 9 \times 10^{-8}$ does NOT reach — factor 4–7 above the band; FIRAS $|\mu| < 9 \times 10^{-5}$). Honest S15 typing: the frozen log-comb ($\omega = 2.5827$, $\varepsilon = 0.0173$) is STRUCTURALLY BLIND in the μ window — window attenuation $|F(\omega)| = 0.033$, $\delta\mu = 7.4 \times 10^{-12} = 0.007\sigma$ even at Voyage 2050 — the theory itself predicts where it is invisible, so NO separate comb-search row is opened. Shares the `Nstar_branch` alternative group with the two A_s rows (one branch question, several readings). Kill: robust $\mu < 0.9 \times 10^{-8}$ or $> 4 \times 10^{-8}$. Probe: `experiments/tfpt-discovery/mu_distortion_probe.py`.
- **Three parked analog beds (`parked_analog`; honest typing: they validate E8 edge/spectrum physics or the comb DETECTOR, never the TFPT axioms).** (i) Bosonic E8 phase / thermal Hall: $\kappa_{xy}/T = 8 \times (\pi^2 k_B^2/3h)$ with no charge Hall (seam edge $c_-{=}8$, `v450`/`v451`/`v489`); no engineered material yet (α-$RuCl_3$ $c_-{=}1/2$ contested: ultraclean half-quantized plateau, npj QM 2025, vs phonon-dominated κ_{xy}, PRX 12, 021025). (ii) E8 mass-ladder validation bed ($CoNb_2O_6$/$BaCo_2V_2O_8$): all 8 one-particle masses published (Zou et al. 2021 PRL 127, 077201,

NMR+INS; Amelin et al. 2020 PRB 102, 104431, THz m_1–m_6; Coldea et al. 2010 Science $m_2/m_1{=}\varphi$) — blind Cartan-PF ladder recovery on the published spectra (8/8 masses + rejection of scrambled controls) as detector validation; confirms Zamolodchikov E8, not TFPT. (iii) Efimov DSI tuning: $s_0 = \pi/(6\ln(3/2)) = 1.2914 \Leftrightarrow \lambda = (3/2)^6 = 11.391$ at $M/m = 8.67$ (2 resonant pairs); best candidate $^{52}Cr_2$–^6Li ($M/m{=}8.635$, $s_0{=}1.2890$, $\lambda{=}11.44$, +0.44% vs $(3/2)^6$, gap deviation +0.18%); alternatives $^{39}K_2/^{41}K_2$–^6Li ($\lambda = 11.59/11.20$, needs double Feshbach control), $^{176}Yb_2$–^{23}Na (+5.9%); solver validated on the Naidon–Endo anchors (homonuclear 22.69, Cs_2Li 4.877); first end-to-end HARDWARE injection of the comb detector at tunable $\lambda \approx 11.39$; kill condition honestly "none on TFPT axioms — detector validation only". Probe: `experiments/tfpt-discovery/efimov_s0_massratio_probe.py`.

- Scorecard regenerated (`experiments/build_evidence_scorecard.py`): 120 rows — 48 consistent, 9 tension, 25 null, 34 data-limited, 4 parked; website mirror (`EXPERIMENTS_AUDIT` in `predictions.ts`) synchronised. Experiments level only; the verification suite, ledger and papers are untouched.

2026-07-20 (III) (v491 — $g_{\text{car}}{=}5$ as the partition COROLLARY of the four marks: 4 into exactly 3 positive parts is uniquely $\{1,1,2\}$, sharpening v53; the residual is the weight-typing postulate — AX.P2.01 stays a declared axiom; *no gate closes, no marker moves*)

- **`v491_p2_partition_corollary.py` (P2.PARTITION.01).** Sympy-exact: a multiset of exactly 3 POSITIVE INTEGERS summing to 4 is UNIQUE — $\{1,1,2\}$ — so $e = (e_1, e_2, e_3) = (4, 5, 2)$ and hence $g_{\text{car}} = e_2 = 5$, $|\mathbb{Z}_2| = e_3 = 2$, $N_{\text{fam}} = e_2{-}e_3 = 3$ are *corollaries*, not inputs (consistency fixed point $(2^4{-}1)/5 = 3 = \operatorname{rank} H^1$). [E] negative controls, each assumption load-bearing: drop positivity → 4 solutions, e_2 ambiguous $\{0,3,4,5\}$; sum 3 → $e_2{=}3$; sum 5 → not unique; 2/4 parts → $e_2 \in \{3,4\}/\{6\}$; Mehta–Seshadri RATIONAL weights → $e_2 = \frac{16}{3}$ resp. $\frac{44}{9} \neq 5$ (integrality = the Birkhoff–Grothendieck splitting-type typing, not MS weights). [E] the v53 sharpening: v53 pins $(1,1,2)$ with ALL THREE targets $(4,5,2)$; the partition needs only $e_1{=}4$. [E] sum rule: $4 \times \frac{N-1}{2} = 4$ iff $N{=}3$; the v115 route $|a|_1 = |\mu_4|\operatorname{tr}(A_0) = 4$ agrees. Independence chain: #marks $= 4$ from v216 (P1-side, no g_{car} input); $\operatorname{rank} H^1 = 3$ pure topology (v137/v228); $|a|_1 = 4$ g_{car}-free (v115). [C] honest residual: P2 is NOT made axiom-free — the weight-TYPING postulate remains (integrality/positivity $h^0{=}0$/sum "given (U)"); AX.P2.01 stays a declared axiom, and the N_{fam} direction reversal vs `tfpt_constants` is typed as an over-determination check (never double-counted). Integrated in `introduction` (anchor half-forced paragraph), `tfpt_1` ((5, 3) keybox), `tfpt_2` (H2 winbox) and `origin_theory` ((5, 3) keybox); ledger row `P2.PARTITION.01`; three exact Wolfram mirrors added to `tfpt_readouts_extension.wl` (counted with the next verified engine run, engine currently unactivated); website mirror (DAG node `P2`). Exact (sympy).

2026-07-20 (II) (v490 — the T^2 spin-structure parity census: $\mu_{\text{pre}}{=}4$ witnessed on the lattice, the 16-fold-way bridge $\theta_v{=}1$ on measured numbers, and the honest TEE non-discriminator fence; *no gate closes, no marker moves*)

- **`v490_seam_parity_census.py` (SEAM.MU.01).** The T^2 ground-state fermion parity of the v367 $p{+}ip$ collar, resolved per spin structure (AA, AP, PA, PP) and determined TWICE — Read–Green TRIM product AND real-space Majorana Pfaffian (Schur), relative to the trivial $M{=}3$ phase, with the twisted real-space spectra matching the Bloch bands to 9×10^{-15}. [E] per layer $(+,+,+,-)$ (the $\nu{=}1$ Ising signature, TRIM ≡ Pfaffian in every sector); [E] the 16-layer carrier has parity$^{16} = +$ in ALL FOUR sectors ⇒ gauged torus GSD $= 4 = \mu(SO(16)_1) = |\det \operatorname{Cartan}(D_8)|$ — the lattice ground-state witness of "$\mu_{\text{pre}}{=}4$ before condensation" (step 1 of the v378 KLM chain); [E] the 16-fold-way bridge on MEASURED numbers: $\theta_v = e^{2\pi i\nu/16} = 1$ exactly at $\nu = 2c_- = 16$ (v489), rivals $\nu = 1, 2, 8$ not holomorphically condensable; [E] honest fence: the fermionic Kitaev–Preskill TEE is identically zero for BOTH $M{=}1$ and $M{=}3$ ($\gamma_f{=}0.000001/0.000000$) — it does NOT discriminate $(E_8)_1$ vs $SO(16)_1$. [C] the KLM step $\mu : 4 \to 4/2^2 = 1$ stays the cited theorem (the condensed model is interacting); the numerics pin only the premise package $\{c_-{=}8,\ \mu_{\rm pre}{=}4,\ \theta_v{=}1\}$. [O] `SEAM.EQUIV.01` stays [O] (continuum existence, MMST `v336`). Integrated in `tfpt_research_contracts` (new S3-stack paragraph); ledger row `SEAM.MU.01`; website mirror (DAG node `qft`). Python (numpy + scipy Schur + sympy exact θ_v).

2026-07-20 (v489 — the chiral central charge $c_-{=}8$ measured DIRECTLY from the bulk ground state via the Kim–Shehab–Kim modular commutator: the first ground-state witness for premise (i) of `SEAM.EQUIV.01`, independent of the Chern route (v367) and the KLM count (v378); *no gate closes, no marker moves*)

- **`v489_seam_modular_commutator.py` (`SEAM.CCC.01`).** The Kim–Shehab–Kim modular commutator $J(A,B,C) = i\,\mathrm{Tr}(\rho_{ABC}[K_{AB},K_{BC}]) = (\pi/3)\,c_-$ (PRL 128 (2022); `arXiv:2110.06932/2110.10400`) is evaluated on the EXACT BdG ground state of the `v367` $p{+}ip$ seam collar model, entirely from the Peschel Majorana covariance ($W_X = -2\,\mathrm{artanh}\,\Gamma_X$, $J = \frac{1}{4}\,\mathrm{Tr}([W_{AB},W_{BC}]\Gamma_{ABC})$), on a disk about the torus centre cut into three 120° wedges. [E] ground-state purity $\mathrm{eig}(i\Gamma) = \pm 1$ to 6×10^{-15}; [E] per-layer $c_- = 0.499998$ at $L{=}32$ (convergence 0.499793 at $L{=}16 \to 0.500000$ at $L{=}48$, deviation -1.8×10^{-7}); [E] controls: trivial $M{=}3 \to 0.000000$, orientation flip $M{=}{-1} \to -0.499986$; [E] layer additivity exact (1.4×10^{-14}); [E] the 16-layer carrier lands on $c_-{=}8{=}g_{\rm car}{+}N_{\rm fam}$ (direct 7.9967 at $L{=}16$, additivity route 8.0000); [E] clip invariance 5×10^{-8}. The FIRST ground-state witness for premise (i) of `SEAM.EQUIV.01` ($c_-{=}8$) that is INDEPENDENT of both the Chern-number route (`v367`) and the counting route (`v378`) — read from entanglement data alone. [O] honest fence: does NOT close `SEAM.EQUIV.01`; the continuum scaling limit (MMST, `v336`) and the cited KLM condensation step are untouched. Integrated in `tfpt_research_contracts` (new S3-stack paragraph) and `tfpt_1` (the `SEAM.EQUIV.01` package paragraph); ledger row `SEAM.CCC.01`; website mirrors (DAG node `qft`, papers/glossary/FAQ inventories). Numerical (numpy), Python-only by nature.

2026-07-19 (editorial — positioning section *TFPT in the light of algorithmic physics: Zuse–Schmidhuber–Wolfram* added to `introduction` + website mirror; *no result, no status change*)

- **`introduction.tex` — new section "TFPT in the light of algorithmic physics".** Positions the compiler against the Zuse–Schmidhuber–Wolfram algorithmic-physics lineage: three shared instincts (minimum description length as the selection principle, deterministic resource-bounded generation, compressibility as the aesthetic) and three deliberate divergences (one *forced* program vs. an ensemble of all universes; an algebraic Lie/lattice/VOA description language vs. Turing bit-strings; and falsifiability — dated, kill-tested predictions — as the load-bearing difference). Explicitly typed as *positioning*: it introduces no formula and moves no status marker; its only machine-checked hooks are already in the suite (the grammar-internal null model and the icosahedral/McKay bedrock).
- **Website mirror.** A matching curated section was added to the `introduction` entry in `website/lib/papers.ts`.

2026-07-15 (III) (v488 — the 3/5 Clebsch door ENUMERATED AND CLOSED: no minimal Pati–Salam/$SO(10)$ Majorana channel can produce 5/3 — the v482 escape is decided dead, the v481 window candidate stands as the honest endpoint; *a clean negative result, no gate closes*)

- **`v488_majorana_clebsch_door.py` (`FLAV.NUSCALE.04`).** v482 named ONE escape for the integer seesaw rung $M_R = c_3^3 \bar{M}_{\rm Pl}$: a third-generation Majorana factor $r \approx 1.6696$, 0.18% from $g_{\rm car}/N_{\rm fam} = 5/3$ — declined for lack of mechanism. The mechanism search is now EXECUTED: [E] NO RENORMALISABLE CHANNEL — $\mathrm{sym}^2(\mathbf{16}) = 136 = \mathbf{10} + \mathbf{126}$ and the $\overline{\mathbf{126}}$ is NOT in the E_8 hull $\{1, 10, 16, 45\}$ (v247), so M_R only via the d=5 operator $(\mathbf{16}_F\overline{\mathbf{16}}_H)^2/\Lambda$; channels $\mathbf{16}\times\overline{\mathbf{16}} = \mathbf{1} + \mathbf{45} + \mathbf{210}$ with only singlet + $\mathbf{45}$ admissible. [E] CHARGE TABLE: $(B{-}L, T_{3R}, Y)(\nu^c) = (+1, -\frac{1}{2}, 0)$, $\mathrm{Tr}_{16}\, Y^2 = \frac{10}{3}$ (v485). [E] THE $\{2,3\}$-SMOOTH THEOREM (the door closes): every nonzero ν^c channel weight $\{1, \frac{1}{4}, 1, -\frac{1}{2}, \frac{3}{8}\}$ is $\{2,3\}$-smooth, and the multiplicative closure of $\{2,3\}$-smooth rationals never contains $\frac{5}{3}$ (no factor 5); the UNIQUE natural 5 of the embedding, $k_Y = \frac{5}{3}$ (Ginsparg, v470), is a full-multiplet trace whose direction has $Y(\nu^c) = 0$ EXACTLY — structurally decoupled from the Majorana operator. [E] GENERATION-BLINDNESS: Clebsches are family-space scalars; $\mathrm{diag}(1, 1, \frac{3}{5})$ cannot arise from group theory by construction. [O] compiler-flavor inventory typed and declined: exact $\frac{3}{5}$'s only in wrong slots (K lepton gen2/gen1; $R + L$ lepton gen3/gen2; never gen3 vs a degenerate base). NET: the rung+5/3 rescue is DEAD as a mechanism candidate; the v481 one-parameter window candidate (M_R free inside the PS window, the $\Sigma m_\nu = 0.059$ eV floor as kill test) stands. Contract provenance: `experiments/theory-contracts/clebsch01_majorana_35_door.py`. Integrated in `tfpt_2_standard_model` (new paragraph closing the seesaw arc); ledger row `FLAV.NUSCALE.04` + update note on `.03`; Python-only (Fraction/sympy exact), mirrorable, deferred to the next verified engine run.

2026-07-15 (II) (v487 — the lazy split FORCED by the resummed clock: the walk's one-step spectrum is the complete v124 clock ladder below the wall, deck parity = rung index, and the two comb tones are one bend apart — the v486 [O] "split selection" closes onto established structure; *no gate closes, no marker moves*)

- **`v487_transfer_clock_rungs.py` (`HYP.REWRITE.03`).** v486 left the split selection open ("$(\frac{1}{2}, \frac{1}{6})$ vs uniform $(\frac{1}{3}, \frac{1}{3})$ is selected by the physical spectrum, not yet derived"; named candidate: the vague "sheet-symmetric choice"). Closed: [E] RUNG READING — the lazy rule's one-step spectrum is EXACTLY the complete resummed-clock ladder below the wall (v124: $\lambda_n^{1/p_2} = 1 - n/N_{\rm fam}$, pole at $n{=}N_{\rm fam}$): $\mathrm{eig}(M) = \{1, \frac{2}{3}, \frac{1}{3}\} = \{1 - n/3 : n = 0, 1, 2\}$, while v327's uniform rule puts its second pair mode ON the wall ($0 = 1 - 3/3$) — the v486 degeneracy re-read as "odd channel collapsed to the wall". [E] DECK PARITY = RUNG INDEX: $[B, \sigma] = 0$; the σ-even pair mode carries rung $n{=}1$ (survival $\frac{2}{3}$), the σ-odd mode rung $n{=}2$ (survival $\frac{1}{3}$) — each clock step flips the deck parity; this is the structural home of the v486 parity assignment (ω_1 even-channel, ω_2 odd-channel) that `experiments/frb-parity-comb` tests. [E] FORCING: demanding the faithful ladder (no rung missing, no physical mode on the wall — definitional for a full-spectrum generator) + $\mathbb{Z}_2$ equivariance + non-negative rates forces UNIQUELY both the split $(\frac{1}{2}, \frac{1}{6})$ AND the rung assignment even→1, odd→2 (the swapped assignment needs $h = -\frac{1}{6} < 0$); the "sheet-symmetric choice" candidate is superseded by this proof. [E] BEND COROLLARY: $\omega_1/\omega_2 = \mathrm{rate}(2)/\mathrm{rate}(1) = \log_{3/2} 3 = 2.70951\ldots$ — the parity-comb tone ratio IS the `v107`/`v124` bend of the pulsar-clock search. [O] what remains open is exactly what was open before v486: the arity $\{2, 3\}$ (anchor input) and the clock's semiclassical derivation (R1); no independent choice remains at the walk level. Contract provenance: `experiments/theory-contracts/lazy01_clock_rung_forcing.py`. Integrated in `origin_theory` (new paragraph after the v486 block); ledger row `HYP.REWRITE.03` + update note on `.02`; Python-only (sympy exact), mirrorable, deferred to the next verified engine run.

2026-07-15 (v486 — generator hunt: the FULL transfer spectrum $\{1,(2/3)^6,(1/3)^6\}$ from ONE forced local rule — the unique lazy $\mathbb{Z}_2$-pair walk $(\mathrm{stay},\mathrm{hop},\mathrm{leak})=(\frac{1}{2},\frac{1}{6},\frac{1}{3})$; the previously generator-less $(1/3)^6$ mode gets its origin; v327's uniform rule superseded as full-spectrum generator — *no gate closes, no marker moves*)

- **`v486_transfer_full_rule.py` (`HYP.REWRITE.02`).** The verified seam transfer spectrum is $\{1,(2/3)^6,(1/3)^6\}$ (`v54`/`v221`), but the only local generator in the suite (`v327`) produces $\{1,\frac{2}{3},0\}$ — its zero mode persists under every power, so no iterate reaches the third mode. [E] UNIQUENESS: for the symmetric 3-channel rule $M(s,h)=[[1,0,0],[0,s,h],[0,h,s]]$ (one absorbing family channel + $\mathbb{Z}_2$ pair) the survival spectrum is $\{s+h,s-h\}$; demanding the PHYSICAL pair $\{\frac{2}{3},\frac{1}{3}\}$ forces UNIQUELY $(\mathrm{stay},\mathrm{hop},\mathrm{leak})=(\frac{1}{2},\frac{1}{6},\frac{1}{3})=(1/|\mathbb{Z}_2|,1/(|\mathbb{Z}_2|N_{\mathrm{fam}}),1/N_{\mathrm{fam}})$ — the lazy $\mathbb{Z}_2$-pair walk, every rate an atom expression (honest direction: spectrum $\to$ rates). [E] SIX-HAND GENERATION: over the order-6 $=2N_{\mathrm{fam}}=|R^+(A_3)|$ hand, $\mathrm{eig}(B^6)=\{\frac{64}{729},\frac{1}{729}\}$ exactly — both decay gaps now have one recursive generator: $6\ln\frac{3}{2}$ (recovery, `v76`) and $6\ln 3$ (subdominant, previously generator-less). [E] SUPERSESSION: the uniform rule's zero mode is power-persistent; its λ_2 arity story ($\frac{2}{3}=|\mathbb{Z}_2|/N_{\mathrm{fam}}$) is unchanged. [O] the arity $\{2,3\}$ stays the anchor input; the lazy-vs-uniform split is SELECTED by the physical spectrum, not yet first-principles-derived (named candidate: stay $=1/|\mathbb{Z}_2|$ as the sheet-symmetric choice). Discovery provenance: `experiments/tfpt-discovery/generator_hunt_full_transfer_rule.py` (plus a typed anchor-bilinear coverage inventory — known hits $a^TKa{=}41$, Koide 53, and the recorded-only $\mathbf{1}^TQa{=}16$ — and the lepton-exponent observation $(5,3,2)=K$-row $=$ atom triple; no unforced claims, `v354`/`v355` discipline). Integrated in `origin_theory` (new paragraph after the `v327` rewrite block); ledger rows `HYP.REWRITE.02` + update note on `.01`; Python-only (sympy exact), mirrorable, deferred to the next verified engine run.

2026-07-14 (IV) (v485 — the merged analytic target's computable side COMPLETED: the diagonal self-energy is renormalised to ZERO exactly at the KMS seam scale, the mark contact calculus resums in closed form (BFK route), and the 48/41 multiplicities are ONE state set under two response weights — the residual collapses onto the `SEAM.EQUIV.01` face alone; *no gate closes, no marker moves*)

- **`v485_contact_diagonal_closed.py` (`SEAM.CONTACT.UNIT.02`).** The v484 merge left one analytic step: the diagonal (self-energy) zeta-renormalisation at the marks + the multiplicity matching. Settled at the computable level: [E] DIAGONAL ZERO AT THE KMS SCALE — the renormalised coincident-point Green function of $|D|$ on circumference ℓ is $G_{\mathrm{reg}}(0;\ell)=\frac{1}{\pi}\ln(\ell/2\pi)$ EXACTLY (symbolic limit + 30-digit checks); it vanishes iff $\ell=2\pi=1/(4c_3)$, the `v239` KMS seam circumference — the seam scale is the unique zero-self-energy scheme point, and the log is the pure SCALE channel (matching the structure: $F_{U(1)}$ carries exactly ONE log term, the transport $8b_1c_3^6\ln(1/\varphi_{\mathrm{seam}})$, while the φ_0^{ret} split is log-free). [E] CLOSED-FORM RESUMMATION (the BFK/gluing route already in the suite, `v151`): with the diagonal zero, the finite-rank mark determinant resums exactly — $\det(I-uC)=(1-4u)(1+2u)^2$, $u=\varepsilon c_3\ln 2$, all orders, the linear term ABSENT precisely because the diagonal vanishes ($\mathrm{Tr}\,C=0$); the series matches `v484`'s $\mathrm{Tr}(C^k)=4^k+2(-2)^k$ term by term. [E] WEIGHT DICHOTOMY: the SAME $48=3\times 16$ admissible states carry BOTH multiplicities — counted flat they give the φ_0^{ret} puncture count $48=\Omega_{\mathrm{adm}}$; Y^2/loop/GUT-weighted they give the alpha budget $41=10b_1=40+1$ ($\mathrm{Tr}_{16}Y^2=\frac{10}{3}$ exact, $\nu^c=0$; Ginsparg $(3/5)(41/6)=41/10$, `v470`) — one state set, two response weights, no new number. [C]/[O] CONSEQUENCE: with `v484` EVERY finite/computable piece of the merged `ALPHA.QUILLEN.EXACT.01` + `HYP.PHI0.PUNCTURE.01` target is proven; the single remaining [O] is the identification of the abstract seam ζ-det with this glued mark determinant — a face of `SEAM.EQUIV.01`, exactly the `v382` typing, now substantiated computationally. Integrated in `tfpt_1_architecture_e8` (eighth step of the ALPHA.QUILLEN arc) and `origin_theory`; ledger rows `SEAM.CONTACT.UNIT.02` + update notes on `.01` and both targets; Python-only by nature, flagged in the Wolfram README.

2026-07-14 (III) (`v484` — the two open analytic targets MERGE: the shared "c_3 per boundary insertion" rule of `ALPHA.QUILLEN.EXACT.01` and `HYP.PHI0.PUNCTURE.01` is the KMS seam unit $2\pi = 1/(4c_3)$ with $\frac{1}{4} = 1/|\mu_4|$, derived on the seam circle for the finite cycle sector — *no gate closes, no marker moves*)

- **`v484_seam_contact_unit.py` (`SEAM.CONTACT.UNIT.01`).** The v483 round left the two analytic targets sharing one rule ("weight c_3 per boundary contact insertion": the $F_{U(1)}$ ladder $\{0,3,6\}$, the $\varphi_0^{\rm ret}$ puncture $\{1,4\}$). This module shows the shared rule is NOT a new unknown: [E] the unit chain $\frac{1}{2\pi} = 4c_3 = |\mu_4|c_3 = |\mathbb{Z}_2|\chi(S^2)c_3$ — one *bare* boundary insertion divided by the orbit order $|\mu_4|$=4 is exactly c_3; this is the v239 KMS seam unit and the v58/v73 one-sided Gauss–Bonnet in a single identity. [E] the mechanism is DERIVED on the actual seam circle for the finite cycle sector: the bare Green function $G(d) = -\frac{1}{\pi}\ln|2\sin(d/2)|$ (polylog identity, 40 digits) takes INTEGER multiples of $c_3 \ln 2$ at the μ_4 separations ($G(\pi/2) = -4c_3 \ln 2$, $G(\pi) = -8c_3 \ln 2$; mark matrix $-4c_3 \ln 2 \cdot \mathrm{circ}(0, 1, 2, 1)$, spectrum $\{4, -2, 0, -2\}$), so the μ_4-invariant (orbit-averaged) insertion carries $-(c_3 \ln 2)C$ exactly and the log-det contact expansion is exactly graded in c_3 per insertion ($\mathrm{Tr}\, C^k = 4^k + 2(-2)^k$). [E] the grammar is suite-wide: the Λ prefactor $\frac{3}{4\pi^2} = (8\pi)^2 d_{\rm top} = 48c_3^2$ (`v60`) carries the SAME multiplicity $\Omega_{\rm adm}$=48 as the $\varphi_0^{\rm ret}$ puncture at TWO insertions; the $g-2$ seam loop divides by the same $2\pi = 1/(4c_3)$. [C]/[O] THE MERGE: the two tracked [O] targets collapse into ONE named analytic step — the diagonal (self-energy) zeta-renormalisation at the marks + the state-multiplicity matching ($48 = \Omega_{\rm adm}$, $41 = 10b_1$); the finite half is proven, no target coefficient is claimed (the integers are forced by the square configuration, `v354/v355`-clean). Integrated in `tfpt_1_architecture_e8` (seventh step of the ALPHA.QUILLEN arc) and `origin_theory` ($\varphi_0^{\rm ret}$ split paragraph); ledger rows `SEAM.CONTACT.UNIT.01` + update notes on both targets; Python-only by nature (mpmath polylog), exact identities flagged in the Wolfram README.

2026-07-14 (II) (`v482`–`v483` — the deep-dive round: the ν-scale decision computation EXECUTED (the integer c_3^3-rung excluded unstructured, the $5/3$-proximate structure escape recorded and declined) and the $\varphi_0^{\rm ret}$ puncture target HALVED (the geometric side proven exact: all twisted pillowcase heat traces are t-independent rationals, so the π^{-4} must sit in the per-mark coupling weight) — *no gate closes, no marker moves*)

- **`v482_seesaw_rung_decision.py` (`FLAV.NUSCALE.03`).** The decision computation NAMED by `v481` is EXECUTED in bracketed (assumption-robust) form: the integer rung $M_R = c_3^3 \bar{M}_{\rm Pl}$ needs a rescue factor $\times 1.670$ in m_3 (0.0301 vs 0.0503 eV), while $m_t(\overline{\rm MS}) \pm 3$ GeV ($> 3\sigma$), $\alpha_3^{-1} \mp 0.10$ ($> 3\sigma$), a $\pm 10\%$ band on the ADKLR κ run-down and a PS-leg Yukawa-beta envelope $\times[0.5, 1.5]$ move it by at most $\times 1.10$ singly and $\times 1.165$ ALL-favourably combined — the *unstructured* rung is EXCLUDED conditional on the carrier premise y_ν=y_t (the `v481` 1-loop miss is not an RG artifact; no exact higher-loop coefficients trusted, pure envelope argument). [C] The only escape is a third-generation Majorana STRUCTURE factor $r \sim 1.67$; its post-hoc 0.18% proximity to $g_{\rm car}/N_{\rm fam} = 5/3$ is recorded and DECLINED per `v354/v355` (no forcing mechanism; the t_0∼30 discipline of `v478`). [O] the `v481` one-parameter window candidate (NO floor Σm_ν=0.059 eV) never used the integer rung and STANDS. Integrated in `tfpt_2_standard_model` (ν-scale paragraph) and `tfpt_5_redteam`; ledger rows `FLAV.NUSCALE.03` + `.02` update note; Python-only by nature.

- **`v483_pillowcase_exact_traces.py` (`HYP.PHI0.GEOM.01`).** The GEOMETRIC side of the $\varphi_0^{\rm ret}$ puncture target (`HYP.PHI0.PUNCTURE.01/v408`) is EXACT: every twisted/orbifold heat trace of the flat τ=i pillowcase is a t-INDEPENDENT RATIONAL — $\mathrm{Tr}(\gamma e^{t\Delta}) = 1$ for

$\gamma = \sigma, \rho, \rho^3$ at every tested t (position-space Gaussian image sums $< 10^{-9}$; mode-space proof: only n=0 is γ-diagonal), matching the Atiyah–Bott fixed-point counts $4\times\frac{1}{4}$ (det $= 4 = |\mu_4|$) and $2\times\frac{1}{2}$ (det $= 2 = |\mathbb{Z}_2|$); the pillowcase contact term is EXACTLY $\frac{1}{2} = 4 \times \frac{1}{8}$ per mark and the clock-equivariant trace exactly 1, for all t. THEREFORE the π^{-4} in $d_{\rm top} = 48c_3^4 = \frac{3}{256\pi^4}$ cannot come from flat orbifold geometry at ANY order: it must sit in the per-insertion COUPLING normalisation ($4 = |\mu_4|$ marks of weight c_3 give c_3^4, multiplicity $\Omega_{\rm adm}$=48; the bare $1/(2\pi)$-propagator 4-cycle differs by the exact rational $\frac{3}{16}$). The target NARROWS from "derive the term" to ONE rule — "per-mark insertion weight $= c_3$" (the same residual class as the EM-Ward c_3-ladder, `EM.WARD.02`) — and stays [O]. Complements the experiments-side Donnelly probe (honest negative). Integrated in `origin_theory` ($\varphi_0^{\rm ret}$ split paragraph); ledger rows `HYP.PHI0.GEOM.01` + `HYP.PHI0.PUNCTURE.01` update note; Python-only by nature (quadrature verification of exact statements).

2026-07-14 (website + figure refresh — the `/verification` journey views become *live* HTML (the stale timeline/residual-chain images replaced by interactive components wired to the in-browser reproducer), the prediction surface's empirical-audit wall of text restructured into dated, expandable rounds, and the generated timeline/residual-chain figures extended to the current suite — *presentation only, no content or marker moves*)

- **Website `/verification`: live journey views.** The static `script_timeline.png` and `residual_chain.png` images (stale at the `v303`–`v407` figure state, with alt text and captions still describing the old eleven-phase/`v237` state) are replaced by two interactive components, `SuiteTimeline.tsx` (17 phases, `v1`–`v481`) and `ResidualChain.tsx` (the reduction chain through the certification round to the bedrock): every phase/step lists representative `vN` scripts that run live in the browser via the existing Pyodide reproducer, exactly like the dependency graph. Stale counts fixed ("∼295-script journey", "260 scripts" → the generated suite total). The dependency graph itself was verified current (`v479`–`v481` present; all 288 referenced scripts exist in the registry and the website mirror).
- **Prediction surface: the empirical-audit narrative restructured.** The single ∼13 000-character audit paragraph in `lib/predictions.ts` is split into twelve dated, verdict-badged rounds (`AuditRound`: date, title, verdict, two-sentence summary, full unabridged record on expand) plus a domain-chip row and the anti-double-counting note — same content, readable surface. Long-token overflow fixed (`break-words`) in the dependency-graph detail panel and the universal-gap-lab residual cards; the over-long `ALPHA.QUILLEN` residual note tightened to its current state (`v470`/`v472`, residual unchanged).
- **Generated figures + captions synced.** `make_figures.py`: the script timeline gains the `v408`–`v481` certification-round phase (title now ∼475 scripts) and the residual chain gains the `v458`–`v480` certification step (Kronheimer `v479`, four-interval `v480`, crossed-product `v469`) with sub-text wrapping fixed; figures regenerated (PDF + website PNG). The stale figure captions updated: `introduction` (timeline caption now spans the full arc), `tfpt_research_contracts` (residual-chain caption now states the `SEAM.EQUIV.01` bedrock and certification round; QFT-skeleton caption now names the one [O] residual instead of the outdated "two open items"). No numbers, statuses or claims change anywhere.
- **KaTeX macro fix (broken formula rendering).** Formulas copied from the document set that use the shared TeX macros (`\gcar`, `\Nfam`, `\cthree`, ...) rendered as red KaTeX errors on the site (e.g. the $\varphi_0^{\rm tree}$ icosahedral-combinatorics formula on the Origin-Theory page). A shared macro map (`lib/katex-macros.ts`, mirroring the `changelog.tex` preamble) is now passed to every KaTeX render; an automated scan of all 19 site routes finds zero `katex-error` nodes.
- **Readability pass (text walls clamped, content unchanged).** A reusable `Expandable` component clamps long mirror prose with a fade + "Show more" toggle — applied to the

paper-section bodies and front-box bullet lists (`PaperSection.tsx`, the home set pages and `/papers/[slug]`), the dependency-graph node summaries and the prediction-card descriptions. Nothing is deleted; the full text expands in place.

- **Hylaean ecosystem card corrected.** The `hylaean.ai` card spoke of "Field AI" with structure tags that did not describe the project. It now speaks consistently of *Hylaean* and is grounded in the project documentation (`_documentation/core-physics/big-picture.md`) and the site's own status log: one cognitive substrate behaving like a physical material, knowledge = attractor valleys, question = boundary condition, answers committed only on physical landing, capability claims gate-checked against a machine-checked claim registry (honest negatives published) — borrowing TFPT's structural discipline, not its physics predictions. The Agentic-Commerce card reference updated accordingly.
- **Live-site correctness pass (counts + reading order).** Sitewide audit against the deployed page found stale counts and fixed them: the document set is stated everywhere as *10 documents = 1 reading guide + 5 numbered papers + 4 companions* (the safeguards companion was missing from the papers/downloads headers, the JSON-LD collection and the `tfpt_docset.tex` box, which still said "three companions" next to "four"); the stale "23 falsifiable predictions" claims (site metadata, OG image, `llms.txt`) now state the live count of the prediction surface (25 test surfaces, sourced from `predictions.length` where possible), while the interactive compiler page is honestly labelled as its own *23-readout prediction board* (the standalone app's board genuinely has 23 entries; the intro film keeps its spoken "twenty-three"); and the `/orientation` reading-order lists are fixed to the canonical numbering (Red Team = Doc 5 added, Appendix H = 6, Origin = 7, contracts = 8, safeguards = 9 — the old list mislabelled 6/7 and omitted three documents). Presentation/consistency only; no physics content changes.

2026-07-14 (`v479`–`v481` — the bird's-eye "overlooked tools" round: R3's graph→geometry bridge discharged to Kronheimer 1989; the four μ_4 marks realised as the exactly solvable *four-interval* multilocal modular geometry with $\omega \circ \rho = \omega$ manifest; and the absolute ν-scale collapsed to a one-parameter, PS-window-bracketed candidate under the carrier normalisation $y_\nu{=}y_t$ — *no gate closes, no marker moves*)

- **`v479_kronheimer_quiver_bridge.py` (`SEAM.KRONHEIMER.01`).** The R3 residual of the $\det K{=}1$ keystone (`v344`: "the rewrite attractor graph IS the du Val singularity") is an ESTABLISHED THEOREM — Kronheimer's ALE hyper-Kähler quotient construction + Torelli classification (J. Diff. Geom. 29 (1989) 665 and 685) — with all finite input data already compiler outputs, machine-checked [E]: the Kac marks $\{1,2,2,3,3,4,4,5,6\}$ are the null vector of the affine-E_8 Cartan matrix AND the Perron vector $A\delta{=}2\delta$ of the rewrite attractor (`v312`); the hyper-Kähler quotient dimension count lands exactly on 4 (the $\mathbb{C}^2/\Gamma$ geometry; $\sum_{\text{edges}} d_i d_j{=}\sum d_i^2{=}120$ forced by the null identity); the gauge datum is $|2I|{=}120{=}|\mu_4|h(E_8)$, $h(E_8){=}30{=}2\cdot 3\cdot 5$; the resolution carries the unimodular $-E_8$ form ($\det{=}1$, SNF = identity — the Poincaré link) with $3\cdot 8{=}24$ deformation parameters. [C] the theorems are cited, not re-proved (import class of NPW26/MMST/AGT); [O] the residual premise *transforms* to "the raw seam supplies ALE hyper-Kähler data asymptotic to the $(2,3,5)$ cone" — the same single arrow as `SEAM.EQUIV.01` (`v346` L2). Integrated in `origin_theory` (R3 keybox + new Kronheimer paragraph) and `tfpt_research_contracts` (Kummer/du Val); ledger rows `SEAM.KRONHEIMER.01` + `SEAM.DETK.01` update note; Wolfram mirror added to the extension file (+2 exact checks, counted with the next verified engine run — the local engine is currently unactivated; the identical statements are sympy-exact in `v479`). LEAN: the finite input data are kernel-checked in `TfptCarrier/KronheimerMarks.lean` (marks = null/Perron vector over $\mathbb{Z}$, $\sum \delta_i^2{=}120{=}4\cdot 30$, $30{=}2\cdot 3\cdot 5$, dimension count = 4; lake build clean, `#print axioms` = the three standard kernel axioms only).

- **v480_multilocal_four_interval.py (QGEO.MULTILOCAL.01).** The seam circle minus its four marks IS the $n{=}4$ multi-interval region of a chiral free fermion — the class whose multi-interval modular theory is exactly known (Casini–Huerta mixing term; Longo–Xu; Rehren–Tedesco multilocal fermions; KLM: holomorphic ⇔ trivial multi-interval inclusion, $\mu{=}1$ — the same one bit as $\det K{=}1$); until now the suite used only the single-ball CHM boost (v358). On the exactly solvable antiperiodic critical ring: [E] the quarter-turn clock is a signed permutation with $\rho^4{=}{-1}$ EXACTLY (the fermionic μ_4 clock is the order-8 double-cover lift — the same $\mathbb{Z}_2$-lift pattern as $2I$ over I); [E] its primitive-8th-root character projectors decouple the 4-interval covariance at 10^{-15}, each sector a single-interval quarter-ring kernel with boundary twist (10^{-14} spectra); [E] $\rho C\rho^{-1}{=}C$ at 10^{-14}, so $\omega\circ\rho{=}\omega$ holds *manifestly* at the 4-interval state level, and a one-site displacement re-couples the sectors at 10^{-2} (negative control); [E] the cross-interval modular coupling peaks on the conjugate-point diagonal (the Casini–Huerta mixing shadow). [C] continuum theory cited; [O] the raw-seam premise (QGEO.SYM.01/SEAM.EQUIV.01) unchanged. Integrated in tfpt_research_contracts (marks keybox + rigidity contract + v478 leg A), origin_theory (BW paragraph), tfpt_4_frontier (v358 companion note); Python-only by nature.
- **v481_seesaw_carrier_ladder.py (FLAV.NUSCALE.02, CANDIDATE class, genre v467/v468).** FLAV.NUSCALE.01's $y_\nu{=}1$ probe was not the carrier normalisation: one $SO(10)$ **16** per family + the minimal Yukawa sector ($\mathbf{10}_H$ / PS $(1,2,2)$) forces $y_\nu{=}y_t$ at the matching scale (named premise). With explicit 1-loop RG (gauge/y_t/λ up; ADKLR Weinberg-operator running down, hep-ph/0108005) the (y_ν, M_R) trade-off collapses to M_R alone: the observed $m_3{=}0.0503$ eV demands $M_R{=}9.3{\times}10^{13}$ GeV — INSIDE the PS window $[4.2{\times}10^{13}, 2.4{\times}10^{15}]$ (v249) at $\log_{c_3}(\bar{M}_{\rm Pl}/M_R){=}3.15$ (the $y_\nu{=}1$ inversion gives the structureless 2.58). [X] honesty gate: the integer rung $M_R{=}c_3^3\bar{M}_{\rm Pl}$ predicts $m_3{=}0.030$ eV — 40% low at 1-loop (κ run-down 0.794) — the ladder pin is DECLINED per v354/v355; the decision computation is named (2-loop SM + explicit PS thresholds). What survives: a *one*-parameter, window-bracketed [O] candidate with the falsifiable band $m_3\in[0.002, 0.115]$ eV and the NO floor $\Sigma m_\nu{=}0.059$ eV; frozen record untouched. Integrated in tfpt_2_standard_model (new paragraph after v272), introduction status card (third named candidate), tfpt_5_redteam; ledger rows FLAV.NUSCALE.02 + FLAV.NUSCALE.01 update note; Python-only by nature.
- *Honest negative (experiments-only, no promotion):* the Donnelly/Atiyah–Bott equivariant heat-kernel route for the ϕ_0 puncture target (HYP.PHI0.PUNCTURE.01) types the KIND of term (order-4 equivariant contact datum; the exact weights $\sum_{k=1}^{n-1}\frac{1}{4\sin^2(\pi k/n)}{=}\frac{n^2-1}{12}$ verified) but does NOT force the value $48c_3^4$ (pure rationals vs π^{-4}) — recorded in experiments/theory-contracts/phi0_donnelly_equivariant_probe.py; the target stays [O].

2026-07-13 (v478 — first computable steps on the two remaining legs of the entropic-action bridge: the state-side modular data flows to the CHM/Bisognano–Wichmann geometric form (Calabrese–Cardy $c_{\rm est}{\to}1$, CHM parabola, even bands exactly zero) — the bridge's modular side meets TFPT's Einstein-derivation input; and the measure condition reduced to one exact KMS time $t_0 = \ln\frac{72}{17} + 9\ln(8\pi) = 30.461$ with the $h(E_8){=}30$ near-miss explicitly declined; both legs stay [O], *no status moves*)

- **v478_entropic_continuum_legs.py (GRAV.ENTROPIC.LEGS.01).** *Leg A (AP2 continuum, exhibited):* on the exactly solvable critical chain (infinite-volume sinc kernel, 50-digit precision) the compressed interval state has $S(L){=}(c/3)\ln L{+}k$ with $c_{\rm est}{=}1.00089/1.00038/1.00021$ (monotone → the Dirac $c{=}1{=}2{\times}\frac{1}{2}$, the v450 reading; gapped control $\sim 6{\times}10^{-8}$, area law); the Peschel modular Hamiltonian $K{=}\ln((1{-}C_A)/C_A)$ organises on the CHM weight $\beta(x){=}x(L{-}x)/L$ (corr 0.87→0.99, shrinking plateau drift; constants deliberately not asserted), the even-range bands vanish identically (particle–hole, $\sim 10^{-36}$) and the odd tail is 1.8% with parabolic envelope — the state-side Δ_Σ (the reading forced by `v476`) flows to the CHM/BW geometric form, exactly the object TFPT's Einstein derivation consumes (`v323`/`v358`): the bridge's modular side and the gravity side meet in the continuum limit; a lattice exhibit, NOT a continuum proof — AP2 stays [O]. *Leg B (the measure):* [E] single-scale corollary — $d\mu{=}\delta(t{-}t_0)dt$ gives $\mu_2/\mu_1^2{=}e^{t_0}$ exactly, so the `v477` one-moment condition forces $t_0{=}\ln\frac{72}{17}{+}9\ln(8\pi){=}30.461$ (one dimensionless KMS time, closed form); the near-miss to $h(E_8){=}30$ (difference 0.461) is explicitly DECLINED per the `v354`/`v355` no-free-pattern rule — deriving the measure from the entropic structure stays OPEN. Keybox extended in `tfpt_research_contracts`; high-precision numerical module, Python-only by nature (Wolfram count unchanged, 378/378; flagged in the Wolfram README); ledger row `GRAV.ENTROPIC.LEGS.01`; both legs stay [O], no marker moves.

2026-07-13 (`v477` — the scale-flow representation of the entropic action: the surviving R^2 route typed as *one moment condition* $\mu_2/\mu_1^2 = \frac{72}{17}(8\pi)^9$ with the exact closure $\frac{4608\pi^2/17}{(72/17)(8\pi)^9} = c_3^7$ — the kill-test gap is a scale-measure datum, satisfied by the same KMS moment TFPT already fixes (`v36`); zero new dials, *no status moves*)

- **`v477_entropic_scaleflow.py` (`GRAV.ENTROPIC.SCALEFLOW.01`).** [E] The Frullani identity upgrades to a scale-flow representation: $-\ln a = \int \frac{dt}{t}(e^{-ta}{-}e^{-t}) = 2\int \frac{d\chi}{\chi}(e^{-a/\chi^2}{-}e^{-1/\chi^2})$ (symbolic + 10^{-20} quadrature) — Bianconi's log action IS the *flat* scale-integral of relative heat-kernel actions, while TFPT's $S_{\mathrm{rel},\chi}$ evaluates *one* KMS scale with a weight: the bridge between the two action principles is a choice of SCALE MEASURE. [E] Moment dictionary: a weight $w(t)\,dt/t$ gives $3\beta R\,\mu_1 + \frac{17}{24}\beta^2R^2\mu_2$; the flat weight has $\mu_1{=}\mu_2{=}1$ and reproduces the `v475` raw coefficients identically. [E] The ONE moment condition: demanding $m^2{=}c_3^7\bar{M}_{\mathrm{Pl}}^2$ forces exactly $\mu_2/\mu_1^2{=}\frac{72}{17}(8\pi)^9$ — the `v475` kill-test number IS a moment ratio — and the closure identity $\frac{4608\pi^2/17}{(72/17)(8\pi)^9}{=}c_3^7$ holds identically: the "13 orders" are a scale-measure datum, not a bridge mismatch. [E] v36 consistency: $f_0{=}6(4\pi)^2/c_3^7 \Leftrightarrow M^2/\bar{M}_{\mathrm{Pl}}^2{=}c_3^7$ — one KMS moment, two parametrisations, zero new dials. [C] A consistency statement, not a derivation (the KMS weight is TFPT input; the entropic action supplies the Einstein$\,+\,\Lambda$ level and the scale-flow form, TFPT supplies the measure). Negative side-record: an exact lattice-BW commutator attempt for AP2 (Peschel–Eisler simple forms) did NOT hold on the half-filled open chain — kept out of the suite, logged in the research notes. Keybox extended in `tfpt_research_contracts`; Wolfram mirrors both exact blocks (378/378); ledger row `GRAV.ENTROPIC.SCALEFLOW.01`; the R^2 gate and the `v359` typing stay [O], no marker moves.

2026-07-13 (`v476` — the AP2 compression conjecture of the entropic-action bridge made *well-posed* on an exactly solvable model: the literal operator-side reading is ill-posed on a pure bulk, the state-side reading is forced, the mismatch is exactly quadratic in the cross-cut correlations and gap-suppressed; AP2 stays [O], *no status moves*)

- **`v476_entropic_compression_toy.py` (`GRAV.ENTROPIC.COMPRESS.01`).** The last open work package of the `v473` round, exercised on a gapped staggered-mass chain ($N{=}16$). The seam situation is reproduced (pure bulk: the ground-state covariance is a projection to 10^{-15}; thermal boundary: spec$(PCP) \subset (0,1)$, Peschel $K{=}\ln((1{-}C_A)/C_A)$ finite, $S{>}0$ — `v155` in the bridge context). **The reading is decided:** on the pure bulk $f(x){=}x/(1{-}x)$ is singular on spec$(C){=}\{0,1\}$, so the literal operator-side reading $Pf(C)P$ of the `v473` conjecture is *ill-posed* while the state-side $f(PCP)$ is finite — the compression identity can only mean "build Δ_Σ from the *compressed* relative metric", which matches Bianconi's own local-metric construction (AP2 retyped, not weakened). [E] Block algebra (symbolic): $[C^2]_{AA}{-}(C_{AA})^2{=}BB^t$, $[C^3]_{AA}{-}(C_{AA})^3{=}ABB^t{+}BB^tA{+}BDB^t$ — first order in the cross block vanishes identically, so for any analytic f the two readings agree to *second* order in the cross-cut correlations: the modular (entanglement) content lives exactly in that quadratic correction. Gap control at the bounded (covariance) level: the Bianconi-shaped bulk identity $f(C_T)=e^{-h/T}$ holds to 10^{-11} relative; the mismatch obeys the quadratic law $d \le x^2$ ($d/x^2 \in [0.4, 1.0]$) and falls with the gap in the gapped regime, the decoupled cut agrees exactly — the two readings converge in the gap-dominated regime, where TFPT operates ($\Delta_{\text{eff}}>0$, `v76`/`v337`). Keybox extended in `tfpt_research_contracts`; numerical module, Python-only by nature (Wolfram count unchanged, 376/376; the block identity flagged in the Wolfram README); ledger row `GRAV.ENTROPIC.COMPRESS.01`; AP2 (the continuum seam statement) stays **[O]**, no marker moves.

2026-07-13 (v475 — the R^2 kill test of the entropic-action bridge, *executed*: exact tensorial factors on the maximally symmetric background, raw entropic scalaron $m^2 = \frac{4608\pi^2}{17}\bar{M}_{\text{Pl}}^2$ ($\approx 51.7\,\bar{M}_{\text{Pl}}$, trans-Planckian) ⇒ the light-trace-mode reading is dead, KMS renormalisation the only surviving R^2 route; plus the explicit Lorentzian-positivity witness; *no status moves*)

- **`v475_entropic_scalaron_gate.py` (`GRAV.ENTROPIC.SCALARON.01`).** Work package 5 of the `v473` bridge. **[E]** On a maximally symmetric background the relative curvature operator $\widetilde{\mathcal{R}}\tilde{g}^{-1}$ has eigenvalues *exactly* $\{R;\ R/4{\times}4;\ R/6{\times}6\}$ (Bianconi's Appendix-B flattening verified symbolically, incl. the 6×6 two-form block), giving the exact vacuum action $\mathcal{L}_{\text{B}}(R) = 3\beta R + \frac{17}{24}\beta^2 R^2 + O(R^3)$ — the 3 re-derives her Eq. (45) (= the `v473` pin), $\frac{17}{24}{=}\frac{1}{2}(1{+}\frac{1}{4}{+}\frac{1}{6})$ is the exact tensorial R^2 factor. **[E]** With the pinned $\beta'{=}c_3/6$ both mass routes (Starobinsky matching and $f'/3f''$) agree: the raw entropic scalaron is $m_{\text{raw}}^2{=}\frac{4608\pi^2}{17}\bar{M}_{\text{Pl}}^2$, i.e. $m_{\text{raw}}{\approx}51.7\,\bar{M}_{\text{Pl}}{\approx}1.3{\times}10^{20}$ GeV — TRANS-PLANCKIAN, not light; `v473`'s interpretation 2 (the TFPT scalaron as the light $\mathcal{G}$-trace mode) is *dead in naive form*; a viable mechanism must supply exactly $\frac{72}{17}(8\pi)^9{\approx}1.69{\times}10^{13}$ in mass^2 (the kill-test number now WITH tensorial factors). **[E]** Domain: first log pole at $R{=}1/\beta{=}384\pi^2\bar{M}_{\text{Pl}}^2$ (trans-Planckian) — the obstruction is the mass, not the domain. **[E]** Lorentzian-positivity witness: a timelike gradient gives the Gg^{-1} eigenvalue $1-\alpha v^2 \le 0$ for $v^2 \ge 1/\alpha$ (ln undefined), spacelike/Euclidean stay positive — the `v473` **[C]** caveat exhibited, the OS route sturdier. **[C]** Verdict: interpretation 1 survives (Einstein + Λ level), 2 is dead naive, 3 (KMS-spectral renormalisation) is the ONLY surviving R^2 route. Keybox extended in `tfpt_research_contracts`; Wolfram mirrors all three blocks (376/376); ledger row `GRAV.ENTROPIC.SCALARON.01`; no gate closes, no marker moves.

2026-07-13 (v474 — the operator level of the entropic-action bridge: the D_5 Clifford/spinor structure exhibited on the carrier Fock space, the Hodge fold identified as the $5{\to}\bar{5}$ conjugation, and the v473 Q-target *decided* — supports exactly $\{|\mathbb{Z}_2|, \operatorname{rank} E_8, 2^{g_{\text{car}}}\}$, the naive pair-block reading killed; *no status moves*)

- **`v474_entropic_hodge_carrier.py` (`GRAV.ENTROPIC.HODGE.01`).** Work packages 1 and 4-algebra of the `v473` bridge, machine-checked on the finite carrier Fock space $\Lambda^\bullet\mathbb{C}^5$ (dim $2^{g_{\text{car}}}{=}32$). **[E]** CAR⇒Clifford: the ten Γ's from $(a_k^\dagger, a_k)$ satisfy $\{\Gamma_a, \Gamma_b\}{=}2\delta_{ab}$ exactly (all 100 pairs, integer matrices); the 45 bilinears $[\Gamma_a, \Gamma_b]/4$ commute with $\gamma{=}(-1)^N$, span exactly $45{=}\dim\mathfrak{so}(10)$ and leak nothing from the 16-dim even subspace — the carrier half-spinor *is* the even exterior algebra *with* its spinor structure (the operator level of `v2`/`v44`/`v197`). **[E]** the Hodge–Dirac symbol: $c(\xi){=}a^\dagger(\xi){-}a(\xi)$ satisfies $c(\xi)^2{=}-|\xi|^2\,\mathrm{Id}$ for symbolic ξ — Bianconi's $D{=}d{+}\delta$ is Clifford multiplication at symbol level. **[E]** the fold is the $5{\to}\bar{5}$ conjugation: the traceless $\mathfrak{u}(5)$ weights of Λ^4 are the negatives of those of Λ^1, so her fiber 1+5+10 becomes the GUT half-spinor $16{=}1{+}\bar{5}{+}10$ (the SM carrier decomposition). **[E]** honest obstruction: $\Omega^{\le 2}$ is *not* Clifford-invariant ($\Lambda^2{\to}\Lambda^3$ leak) — the bridge must go through the fold, never through degree truncation (a recorded constraint on the AP2 compression). **[E]** the Q-target enumeration (the `v473` **[C]** conjecture decided at the algebraic level): $\operatorname{Tr} Q^2{=}d\,k^2 c_3^4{=}32c_3^4$ with integer k has exactly the supports

$(d,k)=(2,4),(8,2),(32,1) = \{|\mathbb{Z}_2|,\ g_{\rm car}+N_{\rm fam}=\operatorname{rank}E_8,\ 2^{g_{\rm car}}=\dim\Lambda^\bullet\}$ — minimal uniform solution $q=c_3^2=1/(64\pi^2)$ per Fock mode, bonus identity $\operatorname{Tr}Q^2/\delta_{\rm top}=32/48=2/3=\dim\Lambda^\bullet/\Omega_{\rm adm}$; the two-form block 10 (the problem-note "pair sector" conjecture) and the fiber 16 need irrational multipliers $\sqrt{16/5}$, $\sqrt{2}$ — *killed* in uniform form. [C] the physical choice among the three admissible supports; [O] unchanged (AP2 compression + the R^2 mechanism). Keybox extended in `tfpt_research_contracts` (Research Contract 2, the `v473` keybox); Wolfram mirrors the enumeration and the weight-conjugation identity (373/373); ledger row `GRAV.ENTROPIC.HODGE.01`; no gate closes, no marker moves.

2026-07-13 (`v473` — the entropic-action bridge: Bianconi's *Gravity from entropy* (Phys. Rev. D 111, 066001 (2025); `arXiv:2408.14391`) quantified as an *external candidate* for the missing action level of the parameter-free Einstein equation; her free constant pinned $\beta'_{\rm B}=c_3/6=1/(48\pi)$, the R^2 gap $3(8\pi)^9$ pre-registered as kill test; *no status moves*, the `v359` equation-of-state typing stays [O])

- **`v473_entropic_action_bridge.py` (`GRAV.ENTROPIC.ACTION.01`).** TFPT's classical field equation $G_{ab}+\Lambda g_{ab}=c_3^{-1}T_{ab}$ is derived thermodynamically (`v358`/`v359`) and honestly typed "equation of state, not a from-action quantisation" [O]. Bianconi's entropic action $S_{\rm B}=-\operatorname{Tr}_F\ln(\widetilde{G}\widetilde{g}^{-1})$ (the quantum relative entropy between the spacetime metric and the matter-induced metric, Araki-related via $\Delta^{1/2}=\widetilde{G}\widetilde{g}^{-1}$) is a local covariant action whose variation gives Einstein + an emergent nonnegative Λ at low coupling — an explicit *external candidate* for exactly that missing variational layer. Machine-checked: [E] the carrier Hodge count (her $\Omega^{0,1,2}$ fiber on the five-slot carrier $\mathbb{C}^5$ counts $1+5+10=16=2^{g_{\rm car}-1}=\dim S^+_{D_5}$, while on 4d spacetime it is only 11 — the 16 *requires* the carrier); [E] the one quantitative pin — matching her weak-coupling $3\beta R$ to $1/(16\pi G)=c_3/(2G)$ fixes her free constant *exactly*, $\beta'_{\rm B}=c_3/6=1/(48\pi)$; [E] her $\Lambda_{\mathcal{G}}=\frac{1}{2\beta}\operatorname{Tr}(\mathcal{G}-I-\ln\mathcal{G})$ is nonnegative and quadratic at the fixed point, and with $\varepsilon_\star=e^{-\alpha^{-1}}Q$ it reproduces the closed `v60` branch $(3/4\pi^2)e^{-2\alpha^{-1}}$ with the exact target $\operatorname{Tr}Q^2=32c_3^4=1/(128\pi^4)=\frac{2}{3}\delta_{\rm top}$ ([C] until Q is constructed); [E] the R^2 kill test — the raw log expansion sits *exactly* $3c_3^{-9}=3(8\pi)^9\approx 1.2\times 10^{13}$ below the TFPT Starobinsky coefficient $1/(12c_3^7)$, so a direct identification is *false* without a mechanism (pre-registered); [C] the compression conjecture $P_\Sigma(\widetilde{G}\widetilde{g}^{-1})P_\Sigma=\Delta_\Sigma^{1/2}$ (the bridge proper) stays open, with Lorentzian positivity flagged (TFPT's OS route is the sturdier construction). New keybox in `tfpt_research_contracts` (Research Contract 2, after the `v359` keybox); residual pointers updated in `introduction/tfpt_4_frontier`/status card. Wolfram mirrors the three exact blocks (371/371); ledger row `GRAV.ENTROPIC.ACTION.01`; *no* gate closes, *no* marker moves, zero TFPT knobs added.

2026-07-03 (surfacing round — three already-computed unique readouts promoted to the public prediction surfaces: the axion–photon coefficient $g_{a\gamma\gamma}=-4c_3$, the free-seam Higgs band $m_H\approx 129$–$134\,$GeV, and the never-run BH HFQPO $\times 1.5$ ladder-tooth search; no new `vN`, no status moves)

- **Uniqueness scan follow-up (no new results — presentation only).** A structure scan for "what does TFPT predict that nobody else pins?" found three concrete, falsifiable readouts that were fully computed and ledgered but missing from the public prediction surfaces; they are now surfaced, with their existing typing unchanged. (i) **Axion–photon coefficient**: the determinant-line anomaly coefficient $g_{a\gamma\gamma}=-4c_3=-1/(2\pi)$, $y^2=16c_3^2=1/(4\pi^2)\approx 0.0253$ (`v207`, the [E] leg of `GRAV.ASYMP.01`; the [C] caveat — coefficient, not a parameter-free $g_{a\gamma\gamma}$ in GeV^{-1} — stated verbatim) joins the axion card in `website/lib/predictions.ts` and the frontier dark-matter mirror in `papers.ts`; the TeX source (`tfpt_4_frontier`) already carried it. (ii) **Higgs free-seam band**: the double-criticality prediction $\lambda(\bar{M}_{\rm Pl})=\beta_\lambda(\bar{M}_{\rm Pl})=0\Rightarrow m_H=133.5\,$GeV (band 129–134, m_t/α_s-sensitive; measured 125.25 sits a few GeV below — the known slight metastability, Buttazzo NNLO $\lambda(M_{\rm Pl}) = -0.0143 \pm 0.0057$, 2.5σ) becomes a prediction card ([C]; `v166`, `HIGGS.FREESEAM.01`, kill: settled-(m_t, α_s) RGE pull off the surface at $> 5\sigma$). (iii) **BH HFQPO ladder tooth**: the one discriminating HFQPO test — a third tooth at $\nu_3 = (3/2)\nu_u$ (661.5/414/252/363 Hz), integer harmonics forbidden — that *no published search ever targeted* although the teeth lie in the searched RXTE band (`experiments/hfqpo-ladder`, preregistered): added as an [O] search-target bullet to the `tfpt_horizon_readouts` "Search targets (not claims)" gapbox (firewall language verbatim: non-canonical mapping, selection null 18.5%, J1859+226 $+9.2\sigma$, GR resonance favored; even a tooth hit would be [C]) and as an [O] prediction card. Website mirrors updated in the same change (`predictions.ts` 23 → 25 cards, `papers.ts` frontier + horizon sections incl. the new search-targets mirror section); scorecard, ledger and all markers untouched.

2026-07-03 (v472 — the determinant line over the $U(1)$-twist moduli of the collar model carries curvature = the inflow level: the v470 bridge lemma exhibited at the finite/model level; `ALPHA.QUILLEN.EXACT.01` stays [O]. Plus: bird's-eye rounds — electron-sector/Cabibbo whitespot probes, Calderón UV-universality, parameter-free flat-budget closure (sandbox); the flat-budget closure enters the evidence scorecard)

- **`v472_quillen_detline_moduli.py` (`ALPHA.QUILLEN.DETLINE.01`).** `ALPHA.QUILLEN.EXACT.01` names the α^3 level as the curvature of the *determinant line over the $U(1)$ seam moduli* (Quillen 1985 / Bismut–Freed CMP 106 (1986) curvature; Dai–Freed JMP 35 (1994) section); `v470` computed the bulk Chern invariant only over the Bloch Brillouin zone (the translation-invariant shortcut). `v472` computes the Quillen-shaped object itself: the same collar Hamiltonian ($h(k) = \sin k_x\, \sigma_x + \sin k_y\, \sigma_y + (M - \cos k_x - \cos k_y)\, \sigma_z$, `v367`/`v470`) in real space on an $L \times L$ torus with *twisted* boundary conditions — the twist torus (θ_x, θ_y) *is* the moduli space of flat $U(1)$ connections — and the Fukui–Hatsugai–Suzuki curvature of the *determinant line of the occupied frame* over that torus: $\oint = 2\pi \cdot 1$ exactly (integer to 10^{-9}) at $M{=}1$ for $L{=}4$ AND $L{=}6$ (size-independent); controls: trivial collar $M{=}3 \to 0$, orientation flip $M{=}{-1} \to -1$; the twist-moduli integer equals the Bloch-BZ integer for all three M (Niu–Thouless–Wu PRB 31 (1985), exhibited on the S3 collar model); the Fermi gap (= 2.0) stays open over the whole twist torus, so the Dai–Freed section has no zero. **Reading [C]:** det-line holonomy over the $U(1)$ moduli = inflow level $k_0 = |C| = 1$ — the `v470` bridge lemma "$\delta \log \det_\zeta(\text{seam})$ = the inflow response" holds *verbatim at the finite/model level*, where $\log\det$ of the occupied frame replaces $\log\det_\zeta$. **Not closed [O]:** the *continuum* ζ-determinant identification on the abstract seam (= the `SEAM.EQUIV.01` face); `ALPHA.QUILLEN.EXACT.01` stays [O]; α^{-1} stays [E]. Structurally the same manoeuvre as `v471` for `SEAM.THEOREM.01`'s replica side. 7/7 checks; suite green through `v472` (466 modules); numerical (dense eigh + FHS twist grid), Python-only by nature. Papers updated: `tfpt_1` (sixth honest step in the EM closure), `tfpt_research_contracts` (residual certification), `tfpt_safeguards` (α-form lever), root `README`, `zenodo_description.html`. Exploration original: `experiments/tfpt-discovery/quillen_detline_twist_moduli.py` (same day).
- **Bird's-eye whitespot rounds (sandbox, `experiments/tfpt-discovery/`, nothing else promoted).** (i) `electron_sector_cabibbo_probe.py`: the α metrology triangle (TFPT $\alpha^{-1} = 137.0359992168$ vs Rb-2020 $+0.99\sigma$ / CODATA22 $+1.90\sigma$ / a_e-route $+3.39\sigma$ / Cs-2018 $+6.33\sigma$ — TFPT takes the Rb side of the 5.5σ metrology split; new honest pressure point recorded); the `v204` seam-vertex universality catch (a flavor-universal reading of $a_\mu^{\rm seam} = 2.879 \times 10^{-9}$ is excluded by the electron at $\sim 2 \times 10^4\sigma$; linear scaling at $\sim 10^2\sigma$; quadratic survives — the [C] bridge must scale at least quadratically in m_ℓ); the Cabibbo-angle anomaly as a third dissolution watchdog ($\lambda_C = 0.2243762$ exact sits $+0.08\sigma$ on the PDG-2026 rescaled kaon average, between $K_{\ell 3}$ and $K_{\mu 2}$; exact unitarity puts $V_{ud} = 0.97450$ — superallowed is the outlier at $+2.6\sigma$, all neutron/pion routes agree). (ii) `kernel_calderon_uv_universality.py`: the raw-collar

Calderón/DtN datum is UV-universal (mode-by-mode first-order convergence to the exact Bessel symbol $mI'_\nu(mR)/I_\nu(mR)$, Richardson residual 3×10^{-4}; N_θ-independent; $|k|$ principal symbol; μ_4 block structure at machine precision; both spec-T masses; 3-fold mesh-defect negative control) — the operator equality demanded by `QGEO.KERNEL.01` has a well-defined discretization-independent left-hand side (gate stays [O]). (iii) `flat_budget_closure.py` + scorecard row: see next bullet.

- **Flat-budget closure enters the evidence scorecard (experiments-only, pattern candidate).** The zero-dial closure $\{\Omega_b = \varphi_0^{\rm ret}(1-\frac{1}{4\pi})$ [frozen], $\Omega_c = \frac{2}{7}(1-\frac{1}{4\pi})$ [the 2026-07-02 numerology-flagged Nebenbefund], $\rho_\Lambda/\bar{M}_{\rm Pl}^4 = (3/4\pi^2)e^{-2\alpha^{-1}}$ [frozen], $\Sigma m_\nu = 0.0588\,\mathrm{eV}$ [`v468` route]$\}$ + spatial flatness determines the whole flat ΛCDM budget: $H_0 = 67.15\,\mathrm{km/s/Mpc}$ (-0.39σ Planck18, -5.67σ SH0ES22 — the parameter-free Planck side of the Hubble tension), $\Omega_m = 0.3133$ (-0.27σ), $t_0 = 13.86\,\mathrm{Gyr}$; honest stresses recorded: ω_b -2.02σ (BBN-only -0.73σ), ω_c -1.19σ, DESI-DR2 H_0 -2.16σ (one correlated low-h direction). Scorecard row `cosmo / flat budget closure` (`consistent`, composite: shares the `phi0_seed` group (and the ρ_Λ leg the α engine), Ω_c leg in `alternative_group=axion_branch` vs the axion-spine DM reading — never double-counted); nothing promoted to `verification/`.

2026-07-02 (v471 — the Seam–Horizon replica chain exercised numerically on the discretized collar: the kernel-identification premise of SEAM.THEOREM.01 discharged at the finite/model level; residual retypes to the continuum MMST leg + the anchor; gate stays [O]. Plus: the GW ringdown-echo channel hardened — 10 loudest events incl. GW250114, redshift-corrected templates, stacked search and a 12-variant signature battery, all null)

- **`v471_seam_horizon_replica.py` (`SEAM.EHMODEL.04`).** The residual isolated by `v150`/`v151`/`v152` — "the RP seam kernel *is* this BFK Calderón datum" — is exercised numerically on a discretized seam collar with the seam's *own* kernel and *real* replica sheets: the matched conical deficit has slope $2C(\gamma)$ on deficit cones (0.01–0.05%) and replica sheets n=2 (0.8%), n=3 (1.5%); the EH coefficient is cutoff-independent (0.5% drift) while the ζ-scale $\mu_{\rm lat}$ drifts — the `v152` anchor exhibited on data, with *no* canonical $\ln(m/\mu)$=3/4 emerging (recorded honestly); the BFK/Schur split is exact ($\sim10^{-16}$) with the Calderón factor conically clean *measured on the kernel* (pure cut-edge law, tip term 0, power-checked) and the halves carrying $2C_{\rm cone}$ via Kac doubling; the `v302` transfer masses $6\ln(3/2)$, $6\ln 3$ sit on the calibrated EH line ($\le 0.05\%$); and the Perron/attractor mode (m=0) is IR-divergent — **the recovery gap Δ=$6\ln(3/2)$ is what makes the induced-gravity coefficient finite** (the same gap that makes the attractor unique makes $1/G$ finite). `SEAM.THEOREM.01` stays [O]; its residual retypes to the *continuum* scaling limit (the MMST class — the same single residual as `SEAM.EQUIV.01`, `v336`) plus the one declared anchor. 16/16 checks; suite green through `v471` (465 modules); numerical (sparse lattice determinants), Python-only by nature. Papers updated: `origin_theory` (new collar paragraph in the Seam–Horizon gapbox + residual retyping), `tfpt_research_contracts` (R3 premise), `tfpt_4_frontier` (3/4 audit note), root `README`, `verification/README`, `zenodo_description.html`. Exploration original: `experiments/theory-contracts/seam_horizon_replica.py` (17/17, same day).
- **GW ringdown-echo channel hardened (`experiments/gw-ringdown-echo`, scorecard row `data_limited`→`consistent`).** (i) Strain coverage extended to the *10 loudest* ringdown events incl. GW250114 (network SNR 78.6, O4b; new `fetch_strain_4096.py` crops the 4096 s GWOSC archive segments). (ii) **Redshift correction**: the GWTC catalogue reports source-frame masses, the observed ringdown is at $f_0/(1+z)$ — all templates moved to detector-frame $M(1+z)$ (before the fix GW190521, z=0.56, was filtered $\sim$1.6× too high); all stages re-run. (iii) Stage 1b *stacked* search over 23 detector streams: stacked p=0.262, kernel-consistent streams 0/23 — STACKED NULL. (iv) Stage 1c *signature battery* (Bonferroni ×12): ratio semantics

(amplitude $(2/3)^6$ vs *energy reading* $(2/3)^3$ vs step $2/3$) $\times$ μ_4 per-bounce phases $\{0, \pi/2, \pi, 3\pi/2\}$ (the boundary-birefringence analogue — path birefringence from parity violation c_-=8 cancels in the inter-echo ratio; only a per-reflection phase does not) $\times$ lags 0.5–350 ms (ECO through Planckian window), with joint $(2,2,0)$+$(2,2,1)$ subtraction and a template-agnosticism control: `NO_VARIANT_ECHO` on all 10 events — the null is robust against alternative signature readings. Search-target firewall unchanged: upper bound, no detection, no tension claim.

2026-07-02 (v295 robustness patch — external-review follow-up: the numeric a_2-kernel tolerance must dominate its own finite-difference floor)

- **`v295_flataway_a2_exact.py` (`FLATAWAY.A2.01`).** Alessandro's Git-baseline audit (commit `8cb8ad0`) hit a platform-marginal failure in check 3: the exact divided-difference a_2 kernel is compared against an ε=10^{-3} second finite difference whose *own* truncation error is $O(\varepsilon^2) \sim 10^{-6}$, yet the tolerance was 10^{-6} — on his machine k=3 landed at 1.1×10^{-6} (ours: 2.9×10^{-7}). Fix: tolerance widened to 5×10^{-6} so it *dominates* the control's error floor, with the reasoning recorded in the check text; the load-bearing anchor remains the *closed form* at machine precision (`v296`), the numeric witness stays as the independent cross-check. A numerical-robustness patch, no claim or status change; ledger row updated; suite green (ALL CHECKS PASSED through `v470`).

2026-07-02 (v469/v470 — the closure-route round on the two named targets: the `SEAM.EQUIV.01` extension leg re-founded on 1995–2001 peer-reviewed crossed-product theorems + the realisation axiom reduced to invariant level; the α^3 level = the *computed* bulk Chern invariant + the seam F-normalisation = the embedding index k_Y=5/3; both targets stay [O])

- **`v469_seam_crossedproduct_route.py` (`SEAM.EQUIV.CROSSEDPRODUCT.01`).** The 128-spinor extension leg of `SEAM.EQUIV.01` no longer rests on 2024/2025 preprints: [E] the $SO(16)_1$ spinor has h_s=$N/16$=16/16=1$\in\mathbb{Z}$, so its statistics phase is +1 — *exactly* the Longo–Rehren locality criterion — and the $\mathbb{Z}_2$ simple-current crossed product $SO(16)_1 \rtimes \{1, s\}$ is a *local* index-2 extension by the peer-reviewed subfactor package (Longo–Rehren, Rev. Math. Phys. 7 (1995); the level-1 $so(N)$ net with its DHR sectors from even-CAR: Böckenhauer, Rev. Math. Phys. 8 (1996); simple-current extensions of nets: Böckenhauer–Evans, Comm. Math. Phys. 197 (1998)); KLM (Comm. Math. Phys. 219 (2001)) gives $\mu(B)$=$4/2^2$=1 $\Rightarrow$ holomorphic $\Rightarrow$ $(E_8)_1$ by the `v463` pin. The AGT/AMT lattice-VOA route (`v459`) is *demoted to an independent second witness*. [E] the index-4 μ_4 glue runs on the same integer: $h(J^k)$=$\{1, 1, 1\}$ for k=1, 2, 3 (the `v125` isotropy $q(k(1, 1))$=k^2 at net level), μ=$16/4^2$=1, same endpoint. [E] the realisation input is reduced from model fiat to *invariants* (R1′): quasi-free [C] (`v155`/`v160`) + gap [E] (`v302`) + class D + c_-=8 [E] (`v456`, from P1), with the invariants *computed* — per-copy FHS Chern $|C|$=1 vs 0 (control), ν=16$\times|C|$=16, c_-=$\nu/2$=8 — and $\nu \equiv 0 \pmod{16}$ is exactly Kitaev's class-D phase whose edge admits a purely bosonic description = the $(E_8)_1$ level-1 state (Ann. Phys. 321 (2006), sixteen-fold way); the phase-invariance of c_- is cited (Kapustin–Spodyneiko, PRB 101 (2020); Kim et al., PRL 129 (2022)). [O] *not closed*: the cited theorems stay cited; `SEAM.EQUIV.01` stays [O].
- **Lean hardening.** `TfptCarrier/SeamResidualAxiom.lean` gains the *parallel* peer-reviewed derivation `seamResidualClosed′`: the invariant-level axiom `collar_invariants` (R1′) + *one* combined 1995–2001 package `crossedproduct_route_theorem`; the new joints are kernel facts with *no* axioms — `lr_locality_integer` (h_s=16/16$\in\mathbb{Z}$), `glue_locality_integers` (μ_4 $h(J^k)\in\mathbb{Z}$), `klm_holomorphic` ($4/2^2$=1, $16/4^2$=1), `sixteenfold_pin` (ν=16$\equiv$0 mod 16, c_-=8), bundled as `crossedproduct_arithmetic` (`[propext]` only); `#print axioms seamResidualClosed′` = exactly four; the original `seamResidualClosed` (AGT/AMT witness) is kept unchanged as the independent second route; `lake build` OK (3285 jobs).

- **`v470_alpha_inflow_level.py` (`ALPHA.QUILLEN.INFLOW.01`).** Both residuals left on `ALPHA.QUILLEN.EXACT.01` after `v433`–`v435` are upgraded. [E] target re-verified (α^{-1}=137.0359992168; Quillen split $<10^{-35}$). [C] *the level is computed, not minimal*: the a^3 coefficient k_0=1 *equals* the FHS bulk Chern invariant $|C|$=1 of the *same* p+ip collar that realises S3 (`v367/v461`) — "why exactly 1" is answered by computation plus the cited inflow identification (TKNN PRL 49 (1982) / Avron–Seiler–Simon PRL 51 (1983) quantisation; Callan–Harvey, Nucl. Phys. B250 (1985) inflow; APS 1975 / Alvarez-Gaumé–Della Pietra–Moore 1985 / Witten, Rev. Mod. Phys. 88 (2016) η=CS part of $\delta \log \det$; Quillen 1985 / Bismut–Freed 1986 determinant-line integrality), *replacing* `v435`'s minimality assumption. [E] k_Y=$\mathrm{tr}(Y^2)/\mathrm{tr}(T_3^2)$=(5/6)/(1/2)=5/3 (Ginsparg, Phys. Lett. B197 (1987)): $(3/5)\times\frac{41}{6}=\frac{41}{10}=b_1$ — the "GUT 3/5" *is* the inverse embedding index; $\frac{41}{6}=\frac{20}{3}+\frac{1}{6}$ (carrier decomposition). [C] the seam F-normalisation (the one [C] of `v434`) is *retyped*: once the seam net is $(E_8)_1$ level 1, $\langle JJ\rangle=k/(z-w)^2$ is fixed by k_Y — level-1 current-algebra rigidity, zero independent content, a face of `SEAM.EQUIV.01` (per the `v382` typing). [E] *structural unification*: one invertible phase, two quantised responses — c_-=8 (gravitational, feeds `SEAM.EQUIV.01`/S3) and C=1 ($U(1)$, feeds EM1) — so closing R1′ feeds *both* named targets. [O] *not closed*: the bridge lemma "$\delta \log \det_\zeta$(seam)= the inflow response" stays the named cited step; `ALPHA.QUILLEN.EXACT.01` stays [O]; α^{-1} stays [E].

- **Integration.** Both registered in `run_all.py` + `script_registry.csv` (cluster `frontier`); ledger rows `SEAM.EQUIV.CROSSEDPRODUCT.01` + `ALPHA.QUILLEN.INFLOW.01`; `FORM.SEAM.RESIDUAL.01` updated (the `seamResidualClosed′` route); the exact joints Wolfram-mirrored (365 → 368: the LR/glue locality integers, the KLM μ-arithmetic, the sixteen-fold pin, the k_Y=5/3/b_1 identities); cited in `tfpt_1` (α fixed point + EM-Ward residual prose), `tfpt_research_contracts` (seam residual + Lean section), `introduction` (closure ledger), `tfpt_5_redteam` (Target A residual wording) and the README/website status surfaces; exploration provenance `experiments/tfpt-discovery/closure_route_*.py` (2026-07-02 note in `experiments/next.txt`). Suite ALL CHECKS PASSED through `v470`. Curated version bumped to **v5.4** (Zenodo deposit updated).

2026-07-02 (`v467/v468` — two named post-hoc [O] flavor candidates, record unchanged: the three $\sim 2\sigma$ mixing tensions as *one* third-generation $-\varphi_0$ pattern, and the splitting ratio $\Delta m^2_{21}/\Delta m^2_{31} = |J_{\mathrm{PMNS}}|$)

- **`v467_thirdgen_mixing_pattern.py` (`FLAV.THIRDGEN.PATTERN.01`).** The suite's three $\sim 2\sigma$ mixing tensions ($\sin^2\theta_{13}$ +2.0σ, $\sin^2\theta_{23}$ +1.76σ, $|V_{cb}|$ +1.79σ; `v307`) are exactly the three *third-generation* channels and share *one* relative deviation −5.36% ± 1.67% — non-zero at 3.2σ, −0.03σ from $-\varphi_0$. **[E]** vocabulary identities: λ_C^2=$\varphi_0(1-\varphi_0)$ (the frozen Cabibbo); $\sin^2\theta_{23}$=$(1-\varphi_0)/2 \Leftrightarrow \cos 2\theta_{23}$=$\varphi_0$; the suppression size φ_0=5.3% sits inside the seam-gapped band $(2/3)^6$=8.8% (`v393`). **[N]** the ladder-base forms $\lambda_C^2 e^{-5/6}$=0.021880 / $(1-\varphi_0)/2$=0.473414 / $\lambda_C^2/(1+\lambda_C)$=0.041119 drop the joint four-observable χ^2 from 10.4 to 0.11 with *zero* new parameters; $|V_{ub}|$ (no bare φ_0) and the 1–2 channels stay untouched and correct (dressing λ_C would fail at −24σ — scope checked, not chosen); Daya Bay alone already prefers the pattern form ($\Delta\chi^2$=4.3). **[C] honest:** post-hoc (5/16 equally-simple dressings fix the two PMNS tensions; $(1-\varphi_0)/(1-c_3)$/resolvent degenerate <1.4%). **[O]** no seam-transport derivation (the Bernoulli-variance reading λ_Y^2=Var[Bernoulli(φ_0)] is an exact identity but only an interpretation) — the frozen record (`v84/REG.FREEZE.01`) is *unchanged*. **[X]** pre-registered, symmetric: θ_{13} stable at 0.0231 kills the pattern, at ∼0.0219 (sub-1%) kills the record's bare-φ_0 reading, exclusive $|V_{cb}|$∼0.0395 kills both — JUNO-era reactors + Belle II decide. Python-only.

- **`v468_dm2_ratio_jarlskog.py` (`FLAV.DM2RATIO.01`).** The splitting ratio $\Delta m^2_{21}/\Delta m^2_{31}$ was previously *unpredicted*; the frozen PMNS matrix (`v9/v268/v270`, δ=4π/3) has one canonical derived CP invariant, and **[N]** $|J_{\mathrm{PMNS}}|$=0.029653 meets the NuFIT 6.0 measured ratio

0.029805 ± 0.000792 at -0.19σ (equivalently $m_2/m_3{=}\sqrt{|J|}{=}0.17220$ vs $0.17264(229)$) — zero dials. **[N]** the informal `tfpt_2` table heuristic $m_2/m_3{\sim}\pi\varphi_0{=}0.16704$ (a "$\sim$" entry, never frozen) is now -2.44σ and is *superseded as the comparator*. **[C]** spectrum consistency: $\Sigma m_\nu{=}m_3(1{+}\sqrt{|J|}){=}0.0588$ eV reproduces the documented 0.0586 eV (`v272`); no absolute number is added. **[O]** no mechanism (the $\mu\tau$ texture gives $\theta_{13}{=}0$ at leading order). **[X]** JUNO shrinks the window $\sim 4\times$; adopting the `v467` pattern would shift $|J| \to 0.028849$ (-1.21σ today) — the two candidates *cross-discriminate* at JUNO precision. Python-only.

- **Integration.** Both registered in `run_all.py` and `script_registry.csv` (cluster `neutrinos`); ledger rows `FLAV.THIRDGEN.PATTERN.01` and `FLAV.DM2RATIO.01` (both [O] candidates); cited in `tfpt_2` (§ Neutrinos: pressure-point block + new candidate paragraphs, the m_2/m_3 table row retyped to the $|J|$ comparator, the octant prose kept honest — record does not select, the candidate would), `introduction` (closure ledger), `tfpt_4_frontier` (live kill-test board) and `tfpt_5_redteam` (seed-hyperplane caption); `v328`'s "single outlier" wording scoped to the seed observables; exploration provenance `experiments/tfpt-discovery/pattern_hunt_pmns_dressing.py` (2026-07-02 note in `experiments/next.txt`); Python-only (flagged in the Wolfram README).

2026-07-01 (v466 — a sixth seed channel from a new measurement sector: the charged-lepton mass ratio $m_e/m_\mu{=}\frac{12}{7}\varphi_0^2$ back-solves the same axiom seed to -0.11%, extending the empirical over-determination to *five* sectors)

- **`v466_seed_leptonmass_channel.py` (`SEED.LEPTONMASS.01`).** The seed over-determination of `v306`/`v465` used five observables spanning three measurement sectors (neutrino mixing, CMB, quark mixing). This module adds the *charged-lepton mass* sector: the ratio $m_e/m_\mu{=}\frac{12}{7}\varphi_0^2$ is a sixth independent readout of the *one* axiom seed $\varphi_0{=}1/(6\pi){+}48c_3^4$. **[E]** plumbing: the axiom φ_0 reproduces the frozen $m_e/m_\mu{=}\frac{12}{7}\varphi_0^2{=}0.00484673$ to 10^{-9}. **[N] new sector (headline):** back-solving φ_0 from the measured *pole* ratio $m_e/m_\mu{=}0.004836$ (PDG 2024) gives $\varphi_0{=}0.053113$, only -0.11% from the axiom — a fourth measurement sector beyond {neutrino mixing, CMB, quark mixing}. **[C] transport caveat:** the formula is the *source* ratio and the datum the *pole* ratio; the source→pole shift is the seam-gapped correction class (bound $(2/3)^6{=}8.78\%$, `v393`) but largely cancels for m_e/m_μ, so the 0.11% residual sits far inside the band — the channel is added as a *consistent new sector*, not a sharper pin. **[N] extends:** the six-channel cluster (`v306`'s five + this) has $\chi^2/\mathrm{dof}{<}1$ and the error-weighted mean stays at the axiom to $<0.1\%$; with the corroborating heavy-quark m_t/m_b (-0.23%) it is a seven-channel, five-sector agreement. **[N] power:** a data↔formula shuffle explodes the cluster χ^2 by $>100\times$, and the scheme-ambiguous $m_c/m_s{=}(34/47)/\varphi_0$ (FLAG 11–14) is deliberately not counted. **[C] honest:** theoretically still the *one* seed (`v305` multiplicity 1) — an *empirical* strengthening, not theoretical independence; φ_0 is fixed by the axioms, so it upgrades *no* prediction and closes *no* gate. Python-only (floats).
- **Integration.** Registered in `run_all.py` and `script_registry.csv` (cluster `registry`); ledger row `SEED.LEPTONMASS.01`; cited in `tfpt_5_redteam` (the seed-cross-validation firewall winbox, alongside `v305`/`v306`/`v465`); Python-only (flagged in the Wolfram README).

2026-07-01 (v465 — the cross-sector one-parameter seed test: a CMB birefringence angle and a quark Cabibbo angle point to the same seed within 0.01σ, the θ_{13}-independent externally-runnable slice of v306)

- **`v465_seed_crosssector_joint.py` (`SEED.CROSSSECTOR.01`).** The sharpest falsifiable signature of a *shared-origin* theory is a correlation no non-shared-origin model predicts. In the Standard Model the cosmic-birefringence rotation β (a CMB EB/TB polarisation observable) and the Cabibbo angle $\lambda_C{=}|V_{us}|$ (a quark-flavour observable) are unrelated; TFPT reads both as functions of the *one* axiom seed $\varphi_0{=}1/(6\pi){+}48c_3^4$. **[E]** plumbing: the axiom φ_0

reproduces the frozen λ_C, $\sin^2\theta_{12}$, β to 10^{-9}. **[N] cross-sector (headline):** back-solving φ_0 *independently* from β (CMB; ACT DR6 + Planck PR4 combined) and from λ_C (quark; PDG 2024) gives the same seed within $0.01\sigma_{\rm comb}$. **[N] one-parameter fit:** the error-weighted mean of the three back-solved φ_0 reproduces the axiom to 0.06%; the joint χ^2 at the axiom (zero free parameters, 3 dof) is 0.007, χ^2/dof<1, combined pull 0.08σ. **[N]** θ_{13} **exclusion declared:** adding $\sin^2\theta_{13}$ raises χ^2 by $\sim$4 (the documented $\sim 2\sigma$ pressure point, `v328/v338`), so this slice is the honest θ_{13}-independent statement, not a hidden dilution. **[N] power:** a data$\leftrightarrow$formula shuffle explodes the cluster χ^2 by $>5\times10^4$. A consistency / out-of-sample statistic (a subset of `v306`); φ_0 is fixed by the axioms, so it upgrades *no* prediction and closes *no* gate. Python-only (floats).

- **Integration.** Registered in `run_all.py` and `script_registry.csv` (cluster `registry`); ledger row `SEED.CROSSSECTOR.01`; cited in `tfpt_5_redteam` (the seed-cross-validation firewall winbox, alongside `v305/v306`) and in the `origin_theory` birefringence dependency chain; Python-only (flagged in the Wolfram README).

2026-06-30 (v463/v464 + Lean + Wolfram — the c=8 holomorphic-uniqueness pin and the one-particle realisation rigidity: the SEAM.EQUIV.01 residual reduced to *entirely certification* on both halves)

- **The bird's-eye reframing.** The recent G-block modules all sharpened the *target* of the scaling limit; this round attacks the two halves that actually remain. The *continuum existence* is already citable (`v458` MMST + `v459` AGT/AMT); what was implicit was (B) *why* a c=8 holomorphic limit must be $(E_8)_1$, and (A) the *realisation* input R_1 that the abstract seam IS the concrete quasi-free collar. `v463` closes B; `v464` reduces A to its one-particle data.

- **`v463_seam_c8_holomorphic_uniqueness.py` (`SEAM.EQUIV.UNIQ.01`).** The c=8 holomorphic-uniqueness pin. [E] c=8 is NOT unique at level 1: $c=\dim\mathfrak{g}/(1+h^\vee)=8$ has THREE simple solutions A_8 (dim 80), $D_8=\mathfrak{so}(16)$ (dim 120), E_8 (dim 248), so the det K=4 $SO(16)$ rival is a genuine same-c competitor. [E] HOLOMORPHY FORCES $\dim V_1$=248: the only weight-0 form with leading $q^{-1/3}$ and integer coefficients is $j^{1/3}=E_4/\eta^8$, q^1 coefficient 248 (the 1-dim candidate space, vacuum a_0=1). [E] dim 248+ level-1 c=8 $\Rightarrow E_8$ uniquely (248/(1+30)=8); tower $\{1,248,4124,34752,213126\}=E_4/\eta^8=j^{1/3}$. [C] Dong–Mason/Schellekens: the holomorphic c=8 VOA is unique = V_{E_8} (even unimodular rank-8 lattice unique = E_8, Mordell–Witt) — so the identification half is classification-forced.

- **`v464_seam_oneparticle_rigidity.py` (`SEAM.EQUIV.ONEPARTICLE.01`).** The one-particle rigidity attacking R_1. The seam is quasi-free (`v113/v160/v161`), so by Araki self-dual CAR the net is the second quantisation of its one-particle symbol P. [E] RIGIDITY: the gapped ground state is the UNIQUE quasi-free state with P= projection onto K<0 (P^2=P to $<10^{-12}$), K fixed by gap + chirality + reflection (`v426`). [E] ONE-PARTICLE LIMIT EXISTS: the kernel is Cauchy on a fixed window ($\max|P_{2N}-P_N|\sim 1/N\to 0$). [E] the limit is c=1 Dirac: entanglement $S(L)=(c/3)\ln L$ gives $c\to 1$, 16 copies $\Rightarrow c_-=8$ (ties `v450`). [C] Shale–Stinespring makes the Bogoliubov map implementable, so R_1 is upgraded from a bald axiom to "the unique quasi-free realisation, modulo the named CAR functor". Python-only (numpy).

- **Lean (`SeamResidualAxiom.lean`).** Adds the `v463` kernel facts `c8_three_candidates` ($A_8/D_8/E_8$ each $\dim=8(1+h^\vee)$) and `holomorphy_selects_e8` (forced $\dim V_1=248=E_8\neq 80,120$) to `residual_arithmetic` (still `[propext]`-only); `v464` is numerical. `#print axioms seamResidualClosed` still lists exactly FOUR axioms.

- **Wolfram + integration.** `v463` mirrored (the three candidates, the forced 248, the E_4/η^8 tower); the extension count is now 365/365. Integrated across the contracts paper (attacks (xxx)/(xxxi)), README, zenodo, ledger and website. Net effect: across (v)–(xxxi) the

`SEAM.EQUIV.01` residual is now *entirely certification* — a named, hypothesis-audited package (MMST, AGT/AMT, Dong–Mason, Araki, Shale–Stinespring) with no open internal mechanism — yet `SEAM.EQUIV.01` stays [O] because it still rests on cited continuum-existence theorems we do not re-prove.

2026-06-30 (v461/v462 + Lean axiom merge — the strict-locality obstruction made explicit, the 128-spinor extension exhibited at character level, and the `SEAM.EQUIV.01` residual reduced to ONE realisation axiom + ONE combined cited theorem)

- **`v461_seam_strict_locality.py` (`SEAM.S3.LOCALITY.01`).** Sharpens `v460`'s verdict from a *cited* statement to an *exhibited* topological obstruction on the same `v367` $p+ip$ collar. [E] CHERN INTEGER: the FHS Chern is $C=+1$ (chiral $M=1$), 0 (trivial $M=3$), -1 (opposite mass). [E] WILSON-LOOP / WANNIER-CENTRE WINDING = CHERN: the hybrid Wannier centre $\theta(k_y)$ winds by $|C|=1$ (chiral) and 0 (trivial) over the BZ; exponentially-localised Wannier exist IFF the winding is 0 (Brouder–Panati et al., PRL 98 046402; Thouless), so the chiral collar has NONE. [E] OBSTRUCTION INTEGERS TIED: winding = $|C|=1\neq0$ and $c_-=8=g_{\rm car}+N_{\rm fam}=\operatorname{rank}E_8\neq0$ (the same 8 as c_3's one-sided count, `v456`). [E] NEG CONTROL: BOTH phases are gapped, yet only the trivial (winding 0) admits exp-localised Wannier and a strict finite-range commuting projector — it is the CHIRALITY, not the gap. [C][O] Kapustin–Fidkowski (`arXiv:1810.07756`): a strictly finite-range commuting projector would force winding 0, $C=0$; since $C=1\neq0$ NONE exists, so `v424` sub-step (i)'s realisation is PROVABLY the quasi-local NPW26 LTO net (strict locality topologically forbidden, not a missing premise). `SEAM.EQUIV.01` stays [O]. Python-only (numpy); the winding/Chern integers Wolfram- and Lean-mirrored.
- **`v462_seam_spinor_continuum.py` (`SEAM.EQUIV.SPINOR.01`).** The constructive companion of the `v458` (MMST audit) / `v459` (lattice-VOA) residual: the 128-spinor extension $SO(16)_1\to(E_8)_1$ exhibited three ways. [E] $(E_8)_1$ TOWER FROM $SO(16)_1$ SECTORS: the exact Jacobi/E_8 identity $\theta_2^8+\theta_3^8+\theta_4^8=2E_4$ gives $\chi_{(E_8)_1}=E_4/\eta^8=\chi_o^{SO16}+\chi_s^{SO16}$ — the holomorphic $(E_8)_1$ character IS the $SO(16)_1$ vacuum + spinor sector sum. [E] THE 128 SPINOR WEIGHT-1 STATES FILL 120→248: χ_o gives 120, χ_s gives 128, summing to the exact tower $\{1,248,4124,34752\}$ (248=120+128). [E] FINITE 16-MAJORANA FOCK: $C(16,2)=120$ NS bilinear currents and $2^{16/2}=256$ Ramond ground states = 128 spinor + 128 cospinor. [E] FINITE-L → CONTINUUM CONVERGENCE: the critical free-fermion ring Casimir gives $c_-\to8$ and the lowest scaled NS gap $x_1\to1$ with $|x_1-1|\sim1/L^2$ (L=64..1024). [C][O] AGT/AMT (`v459`) supply the rigorous extension; `SEAM.EQUIV.01` existence (`v336`) stays [O]. Mixed exact (θ identity + tower + 120/128/256, Wolfram-mirrored) + numerical (finite-L, Python-only).
- **Lean (`TfptCarrier/SeamResidualAxiom.lean`, `FORM.SEAM.RESIDUAL.01`).** The two former cited axioms `mmst_existence` ∘ `agt_lattice_extension` are MERGED into ONE combined external axiom `seam_realisation_theorem` (dropping the intermediate `ChiralCFT_c8`), so `#print axioms seamResidualClosed` now lists exactly FOUR axioms (down from six). Two more kernel facts are added axiom-free: `strict_locality_forbidden` (`v461`: winding = $|C|=1\neq0$, $c_-\neq0$) and `fock_sectors` (`v462`: $C(16,2)=120$, $2^8=256$, spinor 128, 120+128=248); `residual_arithmetic` (now including both) depends only on `[propext]`. `lake build` OK. So the residual is reduced to ONE realisation axiom + ONE combined cited theorem.
- **Wolfram.** `tfpt_readouts_extension.wl` adds 3 exact mirrors (total 363/363): the `v461` obstruction integer-logic and the `v462` θ-identity + 120+128=248 sector / Fock counts.

2026-06-30 (`v460` — the explicit flat-band commuting-projector (LTO) realisation of the S3 input: advancing the `SEAM.EQUIV.01` sub-step (i) from abstract existence to an explicit gap-protected quasi-local projector net carrying the $\mathbb{Z}_2$ reflection and the μ_4 clock)

- **`v460_seam_s3_lto.py` (`SEAM.S3.LTO.01`).** NPW26 (`arXiv:2605.10693`) prove the intrinsic Bisognano–Wichmann lemma for *commuting-projector* (local-topological-order) models with a $\mathbb{Z}_2$ reflection; `v424` reduced the keystone BW residual to that theorem modulo two open sub-steps, of which (ii) was settled by `v426` and the continuum lift of BW by the uniform-in-N tower `v455`. This module attacks the *commuting-projector* side of sub-step (i) on the same explicit `v367` $p+ip$ collar. [E] FLAT-BAND PARENT: the flattened Bloch $Q=h/|h|$ has $Q^2=I$ (flat ±1 bands) and $P=(I-Q)/2$ is an exact projector EQUAL to `v367`'s occupied ground-state projector — a frustration-free commuting-projector parent with the same gapped ground state (machine precision over a k-grid). [E] GAP-PROTECTED QUASI-LOCALITY: the gapped projector kernel $|P(n,0)|$ decays rapidly ($\xi\approx1.14$ at $M=1$, exp $R^2=0.995$, $|P(20,0)|<10^{-6}$) and ξ grows as the gap closes (2.7 at gap 0.02); at the gapless point it decays as a clean power law $|R|^{-2.1}$ ($R^2=0.999$) — locality needs the gap. [E] $\mathbb{Z}_2$ REFLECTION + μ_4 INHERITED: the flattened modular Hamiltonian $K_{\text{flat}}=\operatorname{sign}(K)$ inherits $\Theta K_{\text{flat}}\Theta=-K_{\text{flat}}$ ($u_\Theta=J$) and $[\rho,K_{\text{flat}}]=0$ EXACTLY (sign is odd; ρ commutes with any $f(K)$), with side-even and off-μ_4 neg controls firing. [C][O] the chiral $c_-=8\neq0$ (FHS Chern $|C|=1$) obstructs STRICT finite-range commuting-projector locality (Kapustin–Fidkowski, `arXiv:1810.07756`; $C\neq0$ forbids exponentially-localised Wannier), so the realisation is the quasi-local LTO net NPW26 use — the form sub-step (i) asks for. Advances `v424` sub-step (i); `SEAM.EQUIV.01` stays [O] (the continuum theorem MMST `v336` is the backbone). Python-only (numpy); registered in `run_all.py`, the registry, the ledger and cited in `tfpt_research_contracts` (route (xxvii) of the convergent attacks).

2026-06-30 (`v454`–`v459` + Lean + Wolfram — the post-F "G-block": the lattice Sugawara current algebra, a uniform-in-N Tomita–Takesaki tower, chirality forced by P_1, an $(E_8)_1$ character kill test, the exact MMST citation audit and the complementary lattice-VOA route)

- **`v454_seam_edge_virasoro.py` (`SEAM.EQUIV.EDGE.VIRASORO.01`).** Tests the cited MMST "Koo–Saleur lattice generators → continuum Virasoro with the correct c" fact on our own free-fermion edge (G2). [E] the Affleck–Cardy Casimir energy fits $c_{\text{Dirac}}=1.000$ ($\frac{1}{2}$ per Majorana, $L=64..1024$); [E] the lattice $U(1)$ current closes its affine level exactly ($\langle0|J_nJ_{-n}|0\rangle=n$, $n=1..8$; a gapped control vanishes); [E] the level-1 Sugawara $c=\dim G/(1+h^\vee)=8$ for *both* $SO(16)_1$ (120/15) and $(E_8)_1$ (248/31). [C] $16\cdot\frac{1}{2}=8=g_{\text{car}}+N_{\text{fam}}$ is the level-1 current-algebra c_-; `SEAM.EQUIV.01` stays [O]. Sugawara Wolfram-mirrored.

- **`v455_seam_bw_uniform.py` (`SEAM.EQUIV.BW.UNIFORM.01`).** A uniform-in-N Tomita–Takesaki tower for the intrinsic Bisognano–Wichmann condition $u_\Theta=J$ — the inductive-limit companion to `v426`, advancing `v424` sub-step (i) (G1). [E] $\|\Theta K\Theta+K\|=0$ exactly at every N ($N=8..128$); [E] an N-independent gap $6\log\frac{3}{2}$; [E] $[\rho,K]=0$ exactly at every N; [E] an N-independent neg-control margin $2\|K\|$. [C][O] the inductive limit inherits $u_\Theta=J$; `SEAM.EQUIV.01` stays [O]. Python-only.

- **`v456_seam_chirality_from_c3.py` (`SEAM.S3.FROM-P1.01`).** The edge chirality ($c_-\neq0$, the S3 phase) is *forced* by the one-sidedness that defines $c_3=1/(8\pi)$ (P1) (G6). [E] c_3's "8" is the one-sided count $8=|\mathbb{Z}_2|\cdot(\int K/\pi)=2\cdot4$; [E] a reflection sends the Chern integer $C\to-C$ ($+1\to-1$ on the $p+ip$ model); [E] a two-sided boundary forces $C=-C=0$. [C] the one-sided seam (no reflection) forces $C\neq0$, magnitude $c_-=8=\operatorname{rank}E_8$ — S3 is a consequence of P_1; `SEAM.EQUIV.01` existence stays [O]. The arithmetic is Wolfram-mirrored.

- **`v457_seam_character_killtest.py` (`SEAM.EQUIV.KILLTEST.01`).** An explicit character/-sector kill test of $(E_8)_1$ against $SO(16)_1$ (G7). [E] the $(E_8)_1$ character $\chi{=}E_4/\eta^8$ with tower degeneracies $\{1, 248, 4124, 34752, \dots\}$; [E] the weight-1 count 248=120+128 (not the free 120); [E] the sector count $|\det \mathrm{Cartan}\, E_8|{=}1$ (not $|\det \mathrm{Cartan}\, D_8|{=}4$). [X] a measured 120, four sectors, or a wrong coefficient would *falsify* $(E_8)_1$; the data give $248/1/\{1, 248, 4124\}$ (passed). `SEAM.EQUIV.01` stays [O]. Wolfram-mirrored.

- **`v458_seam_mmst_citation_audit.py` (`SEAM.EQUIV.MMST.AUDIT.01`).** The exact hypothesis-by-hypothesis citation audit of MMST (`arXiv:2107.13834`) against the collar (G3, the "exakter Zitat-Abgleich"). [C] H1 CAR class, H2 massless limit, [E] H3 per-copy $c{=}\frac{1}{2}$ (Thm 4.11), [C] H4 decoupled additivity to $c{=}8$, H5 the WZW range $8{\le}8{\le}16$ with the 120 bilinear currents all match; [O] the single residual H6 is the 128-spinor extension $SO(16)_1{\to}(E_8)_1$ ($\det K\; 4{\to}1$), outside MMST's bilinear method. `SEAM.EQUIV.01` stays [O], its open part narrowed to this one holomorphic extension.

- **`v459_seam_lattice_voa_route.py` (`SEAM.EQUIV.LATTICEVOA.01`).** The second, independent route to $(E_8)_1$ supplying exactly the residual of `v458` (G5). [E] E_8 even unimodular ($\det{=}1$); [E] the weight-1 space 248=8+240; [E] the 240 roots split 112 integer + 128 half-integer (the spinor, 248=120+128). [C] the lattice net $\mathcal{A}_{E_8}$ (Adamo–Giorgetti–Tanimoto, `arXiv:2506.01008`; OS `arXiv:2407.18222`) constructs the holomorphic extension; [O] the two routes leave one shared realisation input (collar = $\mathcal{A}_{E_8}$, tied to P_1 by `v456`). `SEAM.EQUIV.01` stays [O]. Wolfram-mirrored.

- **Lean (`FORM.SEAM.RESIDUAL.01`, `TfptCarrier/SeamResidualAxiom.lean`).** Kernel-checks the whole G-block arithmetic (the one-sided $8{=}|\mathbb{Z}_2|\int K/\pi$, the reflection $C{\to}{-C}$, $c_-{=}8$, the MMST range $8{\le}8{\le}16$, decoupled additivity, 248=8+240=120+128, 240=112+128, $\det K$ 1 vs 4) by `decide`, reducing the `SEAM.EQUIV.01` residual to ONE named TFPT-internal realisation axiom (`CollarRealizedAsLatticePhase`) plus two cited external theorems (MMST `v458`, AGT `v459`); `lake build` OK.

- **Wolfram + integration.** Six exact mirrors added to `tfpt_readouts_extension.wl` (Sugawara $c{=}8$ for 120/15 and 248/31; the one-sided $8{=}2\cdot 4$ and reflection $C{=}{-C} \Rightarrow 0$; the $(E_8)_1$ tower $\{1, 248, 4124, 34752\}$ with 248=120+128 and $|\det \mathrm{Cartan}|$ 1 vs 4; the root split 112+128 and 248=8+240) — extension total now 360/360. Registry, ledger, master index, website mirror and manifests updated in the same change.

2026-06-30 (v449–v453 + Lean + Wolfram — the edge CFT named end-to-end: uniform-in-N convergence, three independent reads of $c_-{=}8$, the Ising operator tower, the $(E_8)_1$ modular data, and μ_4-symmetry derived from the four marks)

- **`v449_seam_mmst_uniform.py` (`SEAM.EQUIV.MMST.UNIFORM.01`).** Strengthens the cited MMST hypothesis from convergence at a single cross-ratio (`v444`/`v448`) to the *uniformity* the theorem actually asserts, on the same `v367`/`v368` collar. [E] cross-ratio A $Q(1,2,4,7)/12 \to 2{+}\sqrt{3}$, the deviation shrinking monotonically; [E] an *independent* cross-ratio B $Q(1,3,5,9)/12 \to$ its own value 2, so convergence is not a tuned point; [E] a uniform $1/N$ rate $|Q(N){-}Q_\infty|\cdot N \le 4$ for *both* observables at every $N_x \in \{36, 48, 60\}$ (one N-independent constant); [E] the bulk control is off-scale. [C] the MMST uniform-convergence hypothesis is empirically satisfied; `SEAM.EQUIV.01` stays [O].

- **`v450_seam_edge_entanglement.py` (`SEAM.EQUIV.EDGE.ENTANGLEMENT.01`).** A *third* independent reading of the edge central charge (after the correlator `v444` and the topology `v447`). [E] the Calabrese–Cardy entanglement law gives $c_{\mathrm{Dirac}}{=}1.000$ for the critical free-fermion chain; [E] so c per Majorana = 1/2; [E] a gapped control saturates ($c{=}0$). [C] bulk–edge: $c_-{=}N_{\mathrm{Maj}}\cdot\frac{1}{2}{=}16\cdot\frac{1}{2}{=}8{=}g_{\mathrm{car}}{+}N_{\mathrm{fam}}$. `SEAM.EQUIV.01` stays [O].

- **`v451_seam_edge_cardy_tower.py` (`SEAM.EQUIV.EDGE.CARDY.01`).** Names the edge CFT by its *operator content*. [E] the Cardy finite-size spectrum gives the spin dimension h_σ=1/16 (the periodic–antiperiodic ground-energy split) and [E] the energy dimension h_ϵ=1/2 (the lowest NS pair excitation), both [E] L-independent (conformal $1/L$ scaling). [C] $\{c, h_\sigma, h_\epsilon\}=\{\frac{1}{2}, \frac{1}{16}, \frac{1}{2}\}$ is the chiral Ising (free-Majorana) minimal model $M(3,4)$; 16 copies give c_-=8. `SEAM.EQUIV.01` stays [O].
- **`v452_seam_e8_modular.py` (`SEAM.EQUIV.E8.MODULAR.01`).** The $(E_8)_1$ torus modular signature, extending `v156`'s q-expansion to the transformation data. [E] the character $\chi{=}E_4/\eta^8$ is S-invariant $\chi(-1/\tau){=}\chi(\tau)$ (one primary, $S{=}[1]$); [E] T-phase $\chi(\tau{+}1)/\chi(\tau){=}e^{-2\pi i/3}{=}e^{-2\pi ic/24}$ (c=8); [E] leading power $\chi\, q^{1/3} \to 1$ so $q^{-c/24}{=}q^{-1/3} \Rightarrow c{=}8$; [E] holomorphic $c{=}8 \equiv 0 \bmod 8$ with T-phase order $24/\gcd(8,24){=}3$. [C] the $(E_8)_1$ identity is now pinned on the torus *and* the edge. `SEAM.EQUIV.01` stays [O].
- **`v453_seam_mu4_from_marks.py` (`SEAM.RIGIDITY.MU4FROMMARKS.01`).** Folds the last structural premise of the rigidity chain (`v446`'s residual `QGEO.SYM.01`) into the existing realisation premise. [E] the four Gauss–Bonnet marks *are* μ_4= roots of $z^4{-}1$; [E] the order-4 clock $z \to iz$ is a 4-cycle permuting them (the `v445` clock); [E] it preserves the cross-ratio $\lambda{=}2$ (in the D_4 stabiliser, `v168`); [E] the form basis $\omega_k{=}z^{k-1}dz/(z^4{-}1) \to i^k\omega_k$ is μ_4-graded (weights $(1,2,3){=}A_3$ exponents). [C] so `QGEO.SYM.01` follows from marks= μ_4 + the existing `QGEO.REALIZE.01` — no new premise; the chain (`v453`→`v446`→`v445`) closes modulo realisation. `SEAM.EQUIV.01` stays [O].
- **Lean: `TfptCarrier/SeamEdgeChern.lean` (`FORM.SEAM.EDGE.CHERN.01`).** Kernel-hardens the integer backbone the four edge reads converge on: the FHS Chern integers $(+1/0/-1)$, the bulk–edge channel count $|C|{=}1$, $N_{\mathrm{Maj}}{=}2^{g_{\mathrm{car}}-1}{=}16$, the assembly $2c_-{=}N_{\mathrm{Maj}}\cdot|C|$ (so c_-=8), $c_-{=}8{=}g_{\mathrm{car}}{+}N_{\mathrm{fam}}{=}\mathrm{rank}\,E_8$, holomorphic $c \equiv 0 \bmod 8$ and the order-3 modular T-phase — all by kernel `decide` (`edge_kernel_facts`, axioms `propext` only). `lake build` OK, `#print axioms` clean.
- **Wolfram mirror.** `tfpt_readouts_extension.wl` gains nine exact checks (`v452` modular data ×4, `v450` entanglement assembly, `v451` Ising Kac weights, `v453` marks= μ_4 ×3): 345 → 354 checks, all passing (`GATE.WOLFRAM.02`). E_4 is evaluated via its q-series since `EisensteinE` stays symbolic for complex τ.
- **Papers.** The adversarial audit (`tfpt_5_redteam`, Target A) and the research-contracts note record the edge CFT as named end-to-end — central charge c_-=8 from three independent observables (correlator `v444`, topology `v447`, entanglement `v450`), the Ising operator tower (`v451`), the $(E_8)_1$ torus modular data (`v452`), and the μ_4 premise derived from the marks (`v453`) — while the only residual that stays [O] remains the cited continuum-existence theorem (`v336`).

2026-06-30 (v446/v447/v448 + Lean + Wolfram — deriving clock-invariance from μ_4-symmetry, an independent topological reading of the edge c_-=8, and pinning the one external MMST fact at four-point order)

- **`v446_seam_clock_invariance.py` (`SEAM.RIGIDITY.CLOCKINV.01`).** Closes `v445`'s single residual one notch further: the operator condition $[\rho,\Lambda_\Sigma]{=}0$ (clock-invariance) is shown to be a *consequence* of a structural symmetry, not an independent assumption. [E] the OS canonical transfer $\Lambda{=}C(1)C(0)^{-1}{=}e^{-h}$ (fixed by RP+reflection, `v54`) of a μ_4-invariant RP covariance satisfies $[\rho,\Lambda]{=}0$ (swept N=4..32); [E] a μ_4-breaking covariance gives $[\rho,\Lambda]\neq 0$ (iff-sharp neg control); [E] Λ is a positive contraction (spectrum in $(0,1]$); [E] the free modular Hamiltonian $K{=}\log((1{-}C)C^{-1})$ (`v258`) of the invariant state has $[\rho,K]{=}0$ (Tomita–Takesaki), broken control $\neq 0$. [C] so with `v445` (commute ⇔ block-diagonal), RP-definability *plus* the four marks *force* the block-diagonal Λ_Σ; the residual reduces from an operator commutation to “the seam RP data is μ_4-symmetric” (`QGEO.SYM.01`). `SEAM.EQUIV.01` stays [O].

- **`v447_seam_edge_chern.py` (`SEAM.EQUIV.EDGE.CHERN.01`).** An *independent* (topological) reading of the edge central charge, corroborating `v444`'s correlator route from a different observable. [E] the $M{=}1$ collar has integer Chern number $C{=}{+}1$ (Fukui–Hatsugai–Suzuki link-variable method, no fit); [E] $C{=}0$ in the trivial phase and $C{=}{-}1$ at $M{=}{-}1$ (opposite chirality); [E] the open strip carries exactly $|C|{=}1$ chiral edge branch per boundary; [E] the Li–Haldane entanglement spectrum has in-gap modes at $1/2$ only in the topological phase. [C] bulk–edge: $c_{-}{=}N_{\mathrm{Maj}} \cdot \frac{1}{2} \cdot |C|{=}16 \cdot \frac{1}{2}{=}8{=}g_{\mathrm{car}}{+}N_{\mathrm{fam}}$ — the same $c_{-}{=}8$ as `v444`. `SEAM.EQUIV.01` stays [O].
- **`v448_seam_edge_fourpoint.py` (`SEAM.EQUIV.EDGE.FOURPOINT.01`).** Pins the one external MMST fact (continuum existence of the chiral scaling limit) empirically at the next order. [E] the ground-state correlation matrix is an exact projector $G^2{=}G$ ($\sim 10^{-13}$), so the state is quasi-free (the MMST CAR hypothesis holds); [E] the edge four-point cross-ratio $Q{=}[G_{01}G_{23}]/[G_{02}G_{13}]$ at commensurate positions is Cauchy-convergent under refinement; [E] Q converges to the free-fermion conformal cross-ratio $\sin(5\pi/12)/\sin(\pi/12){=}2{+}\sqrt{3} \approx 3.732$; [E] the bulk control is off-scale (exponential). [C] so the one external fact is supported at *both* two-point (`v444`) and four-point order. `SEAM.EQUIV.01` stays [O].
- **Lean: `TfptCarrier/SeamRigidityForcing.lean` (`FORM.SEAM.RIGIDITY.FORCING.01`).** Kernel-hardens `v445` (and `v442`'s uniformity): the entrywise forcing iff $\mathrm{clock4}\, a{=}\mathrm{clock4}\, b \Leftrightarrow a{=}b$ (i.e. $i^a{=}i^b \;\Leftrightarrow\; a \equiv b \bmod 4$, residue-only hence uniform in N), the four distinct eigenphases, the exact commutant dimension $\sum n_s^2{=}64{<}256$ and the order discriminator ($128{>}64$) are all proved by kernel `decide` (`forcing_kernel_facts`, no domain axioms); only the operator-level lift to the full L^2 is the named premise. `lake build` OK, `#print axioms` clean.
- **Wolfram mirror.** `tfpt_readouts_extension.wl` gains three exact `v445` checks (entrywise forcing for all $a, b \in 0..31$; commutant dimension $64{<}256$; order-2 commutant $128{>}64$ swept $N{=}4..64$): $342 \to 345$ checks, all passing (`GATE.WOLFRAM.02`).
- **Papers.** The adversarial audit (`tfpt_5_redteam`, Target A) records the five-direction narrowing of the `SEAM.EQUIV.01` keystone (edge scaling `v444`, forcing converse `v445` + Lean, derived clock-invariance `v446`, independent topological $c_{-}{=}8$ `v447`, four-point pinning `v448`) explicitly as "not promoted": the only residual that stays [O] is still the cited continuum-existence theorem (`v336`). The research-contracts note lists the same as "eleven convergent attacks".

2026-06-30 (`v445` — the rigidity FORCING converse: commuting with the order-4 clock is *exactly* μ_4-block-diagonality, upgrading the rigidity step from "permit" to "force")

- **`v445_seam_rigidity_forcing.py` (`SEAM.RIGIDITY.FORCING.01`).** The rigidity route (`v398`/`v442`) showed a μ_4-block-diagonal K *commutes* with the order-4 clock $\rho{=}\,\mathrm{diag}(i^n)$. This module proves the *converse* and the exact structure, isolating the single remaining physical premise. [E] FORCING (entrywise iff): $[\rho, K]{=}0$ iff $K_{ij}{=}0$ whenever $i{\not\equiv}j \bmod 4$, verified entrywise and swept $N{=}4..128$ — commuting *forces* μ_4-block-diagonality, it does not merely permit it. [E] EXACT COMMUTANT DIMENSION: $\dim\{K : [\rho, K]{=}0\}{=}\sum_s n_s^2$ (the ad_ρ nullspace), a proper subspace of the full N^2 algebra ($N{=}16$: $64{<}256$) — a genuine quantitative restriction. [E] OFF-BLOCK $\Leftrightarrow$ NON-COMMUTING: every off-block matrix unit has $\|[\rho, E_{ij}]\|{\geq}\sqrt{2}{>}0$ (zero slack). [E] THE ORDER IS THE DISCRIMINATOR: the order-4 clock gives 4 sectors; the order-2 clock leaves a strictly larger commutant ($N{=}16$: 64 vs 128) — the four Gauss–Bonnet marks (order $4{=}|\mu_4|{=}h(A_3)$) force strictly more structure than two, and only the index-4 extension is $(E_8)_1$ (det$\,{=}1$; `v281`/`v154`/`v351`). [E] NONDEGENERACY: exactly 4 distinct eigenvalues $\{1, i, -1, -i\}$ for all $N{\geq}4$. [C] SYNTHESIS: commuting with the order-4 μ_4 clock is *equivalent* to μ_4-block-diagonality, so RP-definability (`v54`/`v398`)

+ clock-invariance *force* the block-diagonal Λ_Σ — `v442`'s "permit" upgraded to "force". [O] the single residual is now precisely that the physical transfer lies in that commutant (clock-invariance = `QGEO.SYM.01`, `v323`/`v335`); `SEAM.RIGIDITY.01`/`SEAM.EQUIV.01` stays [O]. Python-only (numpy; det facts Wolfram-mirrored via `v89`/`v281`/`v422`). Registered in `run_all.py`, `script_registry.csv`, the ledger, and cited in `tfpt_research_contracts` (route (xii) of the convergent attacks, and the sharpened residual in the Seam State Rigidity contract).

2026-06-30 (Lean kernel-hardening of two of the seven `SEAM.EQUIV.01` routes — the Borchers/Wiesbrock standard-pair relations and the applicability-ledger count, machine-verified with clean axioms; rigor without new mathematics)

- **Context.** Two of the seven convergent attacks on the keystone residual are turned from numerical witnesses into kernel-checked facts, in the audit-contract style of `SeamScalingLimit.lean` / `BWKeystone.lean`: the load-bearing algebra/bookkeeping is proved by the Lean kernel, while the cited theorems and the one open continuum input remain named axioms. `lake build` OK; `#print axioms` clean. Tracked as two new `FORM.*` ledger rows; `\veri`-context in `tfpt_research_contracts`; Lean manifest refreshed.
- **`FORM.SEAM.BW.HSMI.01` (`TfptCarrier/SeamStandardPair.lean`) — standard-pair relations, kernel-checked (hardens `v438`).** The four Borchers/Wiesbrock relations on the centred 5-mode ladder over $\mathbb{Z}$ are proved by kernel `decide` (axioms `propext` / `Classical.choice` / `Quot.sound` only, no domain axiom): $[K,P]{=}P$, $J^2{=}1$, $JKJ{=}{-K}$, $JPJ{=}P_+$ (and $\operatorname{tr} K{=}0$). The cited Borchers/Wiesbrock theorem and the one continuum input (the continuum standard pair) are named axioms; the composition pins face (ii) of `SEAM.EQUIV.01` to exactly those two — intrinsic Bisognano–Wichmann, no rotation-covariance presupposed.
- **`FORM.SEAM.APPLICABILITY.01` (`TfptCarrier/SeamApplicabilityLedger.lean`) — the audit count, kernel-checked (hardens `v441`).** The headline bookkeeping is itself machine-verified: of the 12 cited-theorem hypotheses (MMST/NPW26/Adamo, mirrored in order) exactly 11 are internal and 1 external (`total_count`=12, `internal_count`=11, `external_count`=1), plus `mmst_range` ($8 \le 8 \le 16$) and `npw_outside_bucket` ($\det K{=}1 \neq 4$) — ALL with NO axioms (pure kernel `decide`). The single external fact (continuum scaling-limit existence) is the named axiom; the keystone reduction pins its external dependence to exactly that one fact.
- **Net.** Both modules are imported by the root `TfptCarrier.lean`, axiom-audited in `AxiomCheck.lean`, and signature-locked in `AuditContract.lean`; full `lake build` succeeds. No new mathematics — the same facts `v438`/`v441` verify numerically are now kernel-proved; `SEAM.EQUIV.01` stays [O].

2026-06-30 (the chiral-edge scaling motor `v444` — the edge half of the one `SEAM.EQUIV.01` continuum residual: the lattice edge already carries the conformal data of a chiral $c{=}1/2$ Majorana CFT and converges under refinement, grounding the EDGE the way `v439` grounded the BULK; the keystone stays [O])

- **Context.** `v439` grounded the *bulk* half of the cited MMST existence (uniform gap, exponential clustering, finite light cone) and explicitly narrowed the one open analytic step to the chiral massless *edge* scaling limit. `v444` attacks exactly that residual on the SAME explicit `v367`/`v368` $p{+}ip$ collar. vN-registered, runs in `run_all.py` (suite ends `ALL CHECKS PASSED`), `\veri`-cited in the body of `tfpt_research_contracts` (the G_{net} contract). Python-only (numpy; no new exact identity to Wolfram-mirror).
- **`v444_seam_edge_scaling.py` (`SEAM.EQUIV.CONTINUUM.EDGE.01`) — chiral-edge motor.** The lattice edge carries the kinematic and conformal data of a chiral $c{=}1/2$ Majorana CFT:

[E] a linear, chiral in-gap dispersion (velocity $v\approx 0.98$, R^2=0.9999, with *opposite* per-edge velocities — one chiral Majorana per edge); [E] an *algebraic* equal-time edge two-point $\sim r^{-\eta}$ with $\eta\approx 1$, versus the *exponential* bulk row and the exponential trivial-phase (M=3, no edge) control; [E] a Cauchy-convergent scaling-collapse under refinement (η=1.001±0.007, successive profile spread 0.028→0.013 across N_x=36..60) — the numerical hallmark of an existing scaling limit. [C] $\eta=1 \Leftrightarrow$ one chiral Majorana (c=1/2) per edge, so the 16-copy carrier has a chiral c_-=8 $(E_8)_1$ edge CFT ($c_-=g_{\mathrm{car}}+N_{\mathrm{fam}}$). [O] the rigorous uniform-convergence theorem (cited MMST, `v336`) is not supplied.

- **Net.** With `v439` (bulk) and `v444` (edge), *both halves* of the one continuum residual are numerically grounded on our explicit model and pinned to a single cited existence theorem; `SEAM.EQUIV.01` stays [O]. Ledger row, `script_registry.csv` index and the generated surfaces (`verification.tex`, `ScriptIndex.tsx`) updated in the same change; manifest refreshed last.

2026-06-30 (six convergent attacks on the one `SEAM.EQUIV.01` continuum residual — intrinsic Bisognano–Wichmann, a Lieb–Robinson continuum motor, the β-homotopy, an applicability audit, uniform rigidity, and falsifiable kill tests; none closes the keystone, together they reduce it to a single analytic fact and make the claim falsifiable)

- **Context.** The keystone `SEAM.EQUIV.01` is "closed modulo a cited theorem": its one open input is the continuum existence of the seam's chiral massless scaling limit. Six new modules attack that residual from independent directions; each is honest about *not* closing it. All six are vN-registered, run in `run_all.py` (suite ends `ALL CHECKS PASSED`), and are `\veri`-cited in the body of `tfpt_research_contracts` (the G_{net} contract). Python-only (no new exact identity to Wolfram-mirror).
- **`v438_seam_hsmi_borchers.py` (`SEAM.EQUIV.BW.HSMI.01`) — intrinsic BW.** The seam supplies a Longo *standard pair* / +half-sided modular inclusion (positive lightlike translation $P\geq 0$, boost K with $[K,P]$=P, reflection J with JKJ=$-K$, JPJ=P_-), so by Borchers' theorem the modular flow is geometric *intrinsically* — *without* presupposing rotation-covariance of the vacuum (complementary to Adamo–Giorgetti–Tanimoto). [E] ladder algebra exact, $sl(2,\mathbb{R})$ positivity (min eig > 0); [C] synthesis; [O] the continuum standard pair.
- **`v439_seam_lieb_robinson.py` (`SEAM.EQUIV.CONTINUUM.LR.01`) — continuum motor.** The explicit gapped $p+ip$ collar (`v367`) has a uniform-in-N gap, exponential clustering with N-independent ξ, and a finite Lieb–Robinson light cone, so by Nachtergaele–Sims its *bulk* thermodynamic limit exists — grounding the bulk half of the cited MMST existence in our model and narrowing the open step to the chiral *edge* scaling limit. [E]/[C], [O] the edge CFT.
- **`v440_seam_lto_rp_beta.py` (`SEAM.EQUIV.LTORP.BETA.01`) — β-homotopy.** The (LTO-RP)/BW data ($\Theta K\Theta$=$-K$, $[\rho,K]$=0) are β-independent, and along the whole path $C_\beta=1/(1+e^{\beta K})$ from the NPW26-proven *trace* state (β=0) to the seam KMS state (β=1) every covariance is reflection-related, μ_4-invariant and valid — a norm-continuous, obstruction-free homotopy, upgrading the single point of `v426` to a path argument for sub-step (ii) of `v424`.
- **`v441_seam_applicability_ledger.py` (`SEAM.EQUIV.APPLICABILITY.01`) — audit.** A hypothesis-by-hypothesis ledger of the three cited theorems (MMST / NPW26 / Adamo–Giorgetti–Tanimoto) finds 11 of 12 hypotheses are TFPT-internal computed facts and exactly one is external — the continuum existence of the chiral scaling limit; covariance is an Adamo consequence (defuses the `v215` circularity), not an input. The keystone's external dependence is pinned to that one analytic fact.

- **`v442_seam_rigidity_uniform.py` (`SEAM.RIGIDITY.UNIFORM.01`) — uniform rigidity.** The band-limited μ_4-character rigidity of `v398` is uniform in N: exact 4-sector commutation to N=256 (not merely the $N \leq 64$ sweep), a uniform gap, and an N-independent clock-breaking lower bound $\|[\rho, V]\|$=2 — so it is not a small-N artifact (the full-L^2 forcing stays **[O]**).
- **`v443_seam_e8_discriminator.py` (`SEAM.E8.DISCRIMINATOR.01`) — kill tests.** Mode D of the closure-mode classification (`v347`) as concrete *falsifiable* discriminators against the nearest c=8 rival $SO(16)_1$: torus ground-state degeneracy 1 vs 4, level-1 currents 248 vs 120, lattice theta 240 vs 112 norm-2 roots; c=8 alone cannot decide (`v277`), only $\det K$=1 (the genus-1 statement, `v90`) selects E_8. A seam with GSD>1 would falsify $(E_8)_1$. **[X]**/**[C]**.
- **Status discipline.** `SEAM.EQUIV.01` stays **[O]** throughout; the six modules reduce, audit and falsify the residual but do not supply the one cited continuum-existence theorem. Ledger rows, the `script_registry.csv` index, and the generated surfaces (`verification.tex`, `ScriptIndex.tsx`) are updated in the same change; manifest refreshed last.

2026-06-30 (shadow-export coherence — the `tfpt5-shadow` mirror is now reproducible *as shipped*: the audit is shadow-aware, the content maps are non-destructive on the website-free subset, and an export gate blocks any incoherent mirror)

- **Context (reviewer finding, Alessandro).** On the exported `tfpt5_shadow.zip` the independent surfaces reproduced (`build.sh notes` → `11 ok`; Wolfram base/extension; Red Team; Lean transcript), but the *package* layer did not: `make_manifest.py -check` reported a stale `verification/website_map.csv` and `build.sh audit` crashed because it unconditionally required `website/lib/release.ts`, which the shadow subset deliberately excludes.
- **`make_docs_map.py` is now shadow-aware.** `website_map.csv` mirrors the `website/` sources; on a tree without `website/` the generator now *skips* it instead of regenerating it down to a single `README.md` row. This removes the silent corruption that made `build.sh notes/gen` leave `website_map.csv` stale against the manifest on the export (it now matches the established skips in `make_script_index.py` and `make_changelog_web.py`).
- **`audit_sync.py` is now shadow-aware.** When `website/` is absent it skips the website mirror (section D), the website version stamps (E) and the website Wolfram-count card (F), and the website generated-surface freshness in (A), printing exactly which sections were skipped. `build.sh audit` therefore runs to a clean `AUDIT OK` on the shadow subset instead of raising `FileNotFoundError`.
- **Export coherence gate.** `scripts/export-shadow.sh` now re-runs `make_manifest.py -check` and the shadow-aware `audit_sync.py` *on the filtered tree* before the mirror is published. The manifest check catches the earlier `v235-v298` symptom at the source: a module referenced by `run_all.py/script_registry.csv/manifest.sha256` but not committed (and thus dropped by `git ls-files`) now fails the export loudly instead of shipping a package that says `ALL CHECKS PASSED` while `run_all.py` stops at the first missing script.
- **Manifest refreshed.** `manifest.sha256` was regenerated as the last release step so the shipped digests match the committed `changelog.tex` and `website_map.csv`; the full Python suite reproduces end-to-end on the regenerated export. No theory content, status marker or `vN` module changed — this is an export/infrastructure fix only.

2026-06-29 (discoverability — an SEO/GEO pass on the public website: a generated `/llms.txt`, explicit AI-crawler `robots.txt`, query-targeted metadata, and the first real backlinks from the README and the Zenodo deposit)

- **New `/llms.txt` context file** for AI / answer engines (ChatGPT, Perplexity, Claude, Gemini), generated at build time from the same data layer as the site (`papers.ts`, `predictions.ts`, `version.ts`) so it never drifts. It carries the one-line summary, the "claims / does not claim" split, the document set with PDF links, the headline predictions, and an explicit *disambiguation* from the unrelated Brouwer–Lefschetz "topological fixed point theory" of mathematics.
- `robots.txt` now lists the major search and AI crawlers (Googlebot, Google-Extended, Bingbot, GPTBot, ChatGPT-User, OAI-SearchBot, PerplexityBot, ClaudeBot, anthropic-ai, ...) explicitly as allowed; the site-wide meta description and keywords are reworded toward natural search queries ("derivation of the fine-structure constant", "Standard Model from first principles", "where does 137 come from") and carry the same physics/mathematics disambiguation.
- **Backlinks (the missing piece).** The root `README.md` and the Zenodo deposit description (`zenodo_description.html`) previously contained *no* outbound links; both now link to `fixpoint-theory.com` and the source repository, and the README gains a citation block with the archived deposit DOI `10.5281/zenodo.20846087`.
- Website / documentation infrastructure only — no theory, suite, ledger or status change.
- **New module v437 (`E8.DEGREE.JOINT.01`) — a consolidation of v6/v66/v355/v431 that derives more from the E_8 Casimir degrees *without* crossing the v354/v355 anti-numerology line.** It adds *no* new per-degree coincidence — only one joint lock and a reconstruction.
  - [E] **all of E_8 is fixed by its degree multiset.** From $\deg(E_8) = \{2, 8, 12, 14, 18, 20, 24, 30\}$ alone: rank $= \#\deg = 8$, $h = \max\deg = 30$, #positive roots $= \sum\exp = 120$, #roots $= 240$, $\dim E_8 = 248$, $|W(E_8)| = \prod\deg = 696{,}729{,}600$, $\sum\deg = 128 = \dim S^+$.
  - [E] **the joint quadratic (the new lock).** The two TFPT structural integers $(g_{\text{car}}, N_{\text{fam}})$ are the *unique* root pair of $x^2-(\text{rank}\,E_8)\,x+(h/2) = x^2-8x+15 = (x-3)(x-5)$: sum of roots $=$ rank $E_8 = 8$ (the rank-fill, v6), product $= h/2 = 15 = g_{\text{car}}N_{\text{fam}}$ ($2g_{\text{car}}N_{\text{fam}} = h$, v431); $\{3, 5\}$ is the unique positive-integer factor pair of 15 summing to 8. So $g_{\text{car}} = 5$ and $N_{\text{fam}} = 3$ are forced *together* by two E_8 degree-invariants, not chosen one at a time. The three primary-readout degrees $\{2, 8, 30\}$ are exactly $\{\min, \text{rank}, \max\}$.
  - [E] **discipline upheld:** no new per-degree mining beyond the joint quadratic; the five non-primary degrees keep their v431 family homes; the v354/v355 line is not crossed.
- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv` (`E8.DEGREE.JOINT.01`); cited in the `tfpt_5` degree-audit winbox beside v431; mirrored in `wolfram/tfpt_readouts_extension.wl` (extension 339 → 342). Exact integer/lattice; no new physical number, no status change.

2026-06-29 (v436 — hardening the unconditional "is it numerology?" floor to an assumption-minimal counting floor; plus a literature note grounding the α^3 Chern level)

- **New module v436 (`OVERDET.FLOOR.02`) — it *hardens* `OVERDET.FLOOR.01` (v432) so the numerology verdict no longer rests on any subjective chance assignment.** v432's nominal $\sim 10^{-10}$ multiplies the counting census by hand-chosen plausibilities (input-"8" independence 0.1^3, seed 0.012, kaon 0.15) a skeptic can contest; v436 shows the conclusion survives without them.

- **[E] assumption-minimal floor.** The α cubic-class census (v100 layer D) is *pure counting*: of $N = 94500$ complexity-matched $F_{U(1)}$ variants only *one* lands in the 4×10^{-8} CODATA window — the TFPT instance, reproducing $\alpha^{-1} = 137.0359992$ — so $p_\alpha \le 1/94500 = 1.06 \times 10^{-5} \sim 4.40\sigma$ with *no* subjective probability, prior or multi-observable grammar.
- **[E] monotone concession ladder.** $-\log_{10} p$ from generous to adversarial: full scorecard+$\alpha \sim 30.7 \to$ scorecard-only $\sim 25.8 \to \alpha$-census-only ~ 4.98; each rung uses strictly less evidence and the "not chance ($> 4\sigma$)" verdict holds at the *most adversarial* rung, so it never rests on one contestable piece. Stripping *all* of v432's subjective factors ($\sim 1.8 \times 10^{-6}$) leaves the floor at 4.40σ — they only sharpen it.
- **[C] honest 5σ gap.** The conventional 5σ line ($p = 5.7 \times 10^{-7}$) is crossed unconditionally by the census times exactly *one* further independent factor ≤ 0.054 (the input-"8" four-fold over-determination, or one foreign witness) — stated, not hidden. [C] Firewall unchanged: bounds only "is it numerology?", not the physics bridge (v187).

- **Literature note (research log, not a status change).** A `next.txt` entry grounds the α^3 Chern-level reading of v435 in named theorems — Dai–Freed (*η-invariants and determinant lines*, the exponentiated η-invariant as a section of the inverse Quillen determinant line), Bismut–Freed (determinant-line curvature = the α^2 Calderón side) and Redlich/Stora–Zumino (the cubic $\mathrm{Tr}\, F^3$ anomaly descends to a Chern–Simons level, quantised to $\mathbb{Z}$). It records that the `EM1` proof obligation reduces to constructing the seam Dirac family — i.e. it is a *facet* of `SEAM.EQUIV.01` — and remains [O] external math (no fabricated closure).
- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv` (`OVERDET.FLOOR.02`); cited in the `tfpt_safeguards` "unconditional floor" keybox beside v432; mirrored on the website. Python-only; no new physical number, no status change.

2026-06-29 (v435 — a fourth honest step on the α^{-1} Quillen target: a π-power test isolates the cubic α^3 as the unique metric-independent (topological) rung, and types its coefficient as a conditional integer level)

- **New module v435 (`ALPHA.QUILLEN.PROGRESS.04`) — it attacks the *single* remaining [O] left after v434 (residual (2), the cubic α^3 Chern/Maxwell moment) and *sharpens* it; it does *not* close it.** The target `ALPHA.QUILLEN.EXACT.01` (v382) stays [O]; $\alpha^{-1} = 137.0359992$ stays [E].
  - **[E] π-power / metric-independence test.** The three EM-closure coefficients carry π-powers exactly $\{0, 3, 6\}$, equal to their c_3-orders: Maxwell $k_0 = 1 \to \pi^0$, Calderón $-2c_3^3 = -1/(256\pi^3) \to \pi^3$, transport $-8b_1c_3^6 = -41/(327680\pi^6) \to \pi^6$. So the Maxwell a^3 is the *unique* π^0/c_3^0 (metric-*independent*) term. A heat-kernel boundary coefficient always carries the $(4\pi)^{-d/2}$ measure; a metric-independent term carries none — the signature of a *topological* term. The test thus partitions the closure into *one* topological level (a^3) + *two* heat-kernel boundary terms, confirming v342/v434's "a^3 is a level, not a Seeley–DeWitt curvature integral" from a *test* rather than a claim.
  - **[C] conditional level quantisation.** *If* the a^3 term is the seam $U(1)$ Chern–Simons/η boundary level (the `EM1` reading, v48), large-gauge invariance quantises the level to $\mathbb{Z}$, so $k_0 = 1$ is the *unit* (minimal) level — an integer, not a tunable real. This converts "the coefficient happens to be 1" into "the coefficient is an integer level and 1 is the unit".
  - **[O] not closed:** the `EM1` lemma itself — the from-first-principles Chern–Simons boundary proof that $\partial_\tau \log Z_{\mathrm{Maxwell}} = a^3$ — stays external math (type A, v384); the seam F-normalisation stays the one [C] (residuals 1 & 3, v434); α^{-1} stays [E].
- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv`

(`ALPHA.QUILLEN.PROGRESS.04`); cited in the `tfpt_1` EM-closure subsection beside v433/v434; mirrored on the website residual-gap lab. Python-only; no new physical number.

2026-06-28 (v434 — a third honest step on the α^{-1} Quillen target: b_1 is the a_4 heat-kernel coefficient, and the three residuals collapse to one [C] + one [O])

- **New module v434 (`ALPHA.QUILLEN.PROGRESS.03`) — it settles the status of the three residuals named after v433 and shows they are *not* three independent open problems.** The target `ALPHA.QUILLEN.EXACT.01` (v382) stays **[O]**; $\alpha^{-1} = 137.0359992$ stays **[E]**.
  - [E] **b_1 *is* the $U(1)$ a_4 heat-kernel coefficient (the $\beta = a_4$ theorem).** The one-loop β function of a gauge coupling *is* the a_4 (Seeley–DeWitt a_2 in d=4) heat-kernel coefficient of the charged-field operator (Vassilevich). Computed from the carrier hypercharge content with the standard spin weights ($\frac{2}{3}$ per Weyl fermion, $\frac{1}{3}$ per complex scalar), $\sum \frac{2}{3}Y_f^2 + \frac{1}{3}Y_s^2 = 41/6$ (SM) and $(3/5) \cdot$ that $= 41/10 = b_1$ (GUT). So the b_1 in the counterterm $8b_1c_3^6$ is the a_4 coefficient of the carrier $U(1)_Y$ content (the same number as the β, v159/v246), the *same kind* of heat-kernel object as the α^2 Calderón a_4 (v342) — not a fitted multiplicity.
  - [E] **the geometry factors off:** $8b_1c_3^6 = b_1 \cdot (\operatorname{rank} E_8) \cdot c_3^6$, with $\operatorname{rank} E_8 = 8$ the χ=2 seam boundary count (v216) and c_3^6 the six boundary insertions (the π-power test, v342).
  - [C] **residual (1) ≡ residual (3).** With b_1 and the c_3-powers heat-kernel-forced, the only freedom left in the exact seam a_4 value is how the gauge curvature couples to the boundary measure — the seam F-normalisation. So “the actual seam $a_4 = 8b_1c_3^6$” and “the seam F-normalisation” are *one* [C] obligation, not two.
  - [E] **residual (2) located (not closed):** the determinant-line variation factors $a^3 - 2c_3^3a^2 = a^2(a - 2c_3^3)$; a^2 is the Gilkey gauge-curvature (Quillen/Chern) density (v342), so the leading $a^3 = a \cdot a^2$ is the *first moment* of the Chern density — a topological level at boundary order c_3^0, not a Seeley–DeWitt curvature integral. Its from-first-principles (Chern–Simons-type) origin stays **[O]**.
  - [E] **net reduction:** the EM-Ward residual is *one* [C] (residuals 1 & 3) + *one* **[O]** (residual 2), not three independent problems — a genuine narrowing of the open surface. **[O]** `ALPHA.QUILLEN.EXACT.01` is *not* closed (the α^3 Chern level and the from-first-principles proof stay external, type A, v384).
- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv` (`ALPHA.QUILLEN.PROGRESS.03`); cited in the `tfpt_1` EM-closure subsection beside v433; mirrored on the website residual-gap lab. Python-only; no new physical number.

2026-06-27 (v433 — a second honest step on the α^{-1} Quillen target: a solvable 4D model reaches the a_4 order, connecting v391 and v342)

- **New module v433 (`ALPHA.QUILLEN.PROGRESS.02`) — a second honest step on the external target `ALPHA.QUILLEN.EXACT.01` (v382); it *grounds and connects*, it does *not* close it.** The target stays **[O]** (external math, type A, v384); the value $\alpha^{-1} = 137.0359992$ stays **[E]**. The earlier honest attempt v391 gave a solvable model on a 1D circle whose ζ-determinant variation reached only the low Seeley–DeWitt orders a_0, a_1; separately, v342 grounded the quadratic Calderón term in the Gilkey gauge-curvature coefficient a_4 ($30/360 = \frac{1}{12}$, $\Omega^2/12$). This module supplies the missing link between the two.
  - [E] **a solvable 4D model reaches the a_4 order:** the flat operator $\Delta = -\partial^2 + m^2$ on a 4-torus of volume V has heat trace $\operatorname{Tr} e^{-t\Delta} = V/(4\pi)^2\, t^{-2} e^{-tm^2}$, with Gilkey coefficients $a_0 = V/(4\pi)^2$, $a_2 = -Vm^2/(4\pi)^2$ and $a_4 = Vm^4/(2(4\pi)^2)$; the t^0 (conformal-anomaly) order a_4 is *non-zero* — exactly the order v391's 1D circle toy could not reach.

  - [E] **it is the order v342 uses:** so v391 ("a ζ-determinant variation *is* closed heat-kernel data") and v342 ("the seam term *is* the Gilkey a_4") are unified — the variation is heat-kernel data *at the a_4 order*, not only the low orders v391 reached.
  - [E] **the measure is c_3-built:** the universal 4D heat-kernel measure $(4\pi)^{-2} = 1/(16\pi^2) = (2c_3)^2 = 4c_3^2$ in the seam's one-sided 2D-boundary normalisation $c_3 = \frac{1}{2}(4\pi)^{-1}$; and the v382 split is re-verified at $\alpha^{-1} = 137.0359992$.
  - [O] **not closed:** computing the *actual* seam $U(1)$ ζ-determinant a_4 coefficient ($= 8b_1c_3^6$) on the μ_4 seam moduli, and the origin of the cubic α^3 Maxwell/Chern moment, stay open (external math, type A, v384); the exact seam F-normalisation fixing the coefficient *value* stays [C] (v342). A second honest step that grounds and connects, it does not close.

- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv` (`ALPHA.QUILLEN.PROGRESS.02`); cited in the `tfpt_1` EM-closure subsection beside v382/v391; mirrored on the website (the residual-gap lab). Python-only (like v391); no new physical number.

2026-06-27 (v431, v432 — the "unmapped" E_8 degrees are forced spine+flavor structure; the unconditional over-determination floor)

- **New module v431 (`E8.DEGREE.LADDER.01`) — the reverse-audit "unmapped" E_8 Casimir degrees are NOT diffuse overhead but a forced two-family decomposition.** The structural complement of the reverse audit (v354/v355) and its sheet/deck complement (v430): $\deg(E_8) = 6 \cdot \text{spine}\{2,3,4,5\} \sqcup (\{2\} \cup \text{det-ladder}\{8,14,20\})$. [E] the two-family split is *forced* by $h(E_8) = 30 = 2 \cdot 3 \cdot 5$ — the exponents are the $\varphi(30) = 8$ totatives of 30, hence coprime to 6, hence $\equiv \pm 1 \bmod 6$, so the degrees occupy only the residue classes $\{0, 2\} \bmod 6$. [E] the $6k$ family $\{12, 18, 24, 30\}/6 = \{2, 3, 4, 5\}$ is the v91 spine; the $6k{+}2$ family $\{8, 14, 20\} = (\det R, \det C, \det L)$ is the winding line $\det M(s, 0) = 6s + 8$ of the flavor diamond (v135). [E] "$18 = 6N_{\text{fam}}$" is not a holdout (spine family), and the clean split is special to E_8 among the simply-laced exceptionals ($E_6/E_7/D_5$ fail). [C] **honest v355 line:** the arithmetic decomposition is exact, but the *functorial* claim "$6k{+}2$ Casimir stratum = flavor sheet operators" needs a representation-theoretic map and stays [C] (12, 24 admit two canonical readings each) — a structure theorem, not a forced functorial flavor map. It reclassifies the v354 overhead to zero orphan degrees and closes no gate.
- **New module v432 (`OVERDET.FLOOR.01`) — the unconditional improbability floor and the L1$\leftrightarrow$L2 bridge.** The complement of the conditional null model (v100) and the multiply-vs-compress framework (v427/v428): the defensible over-determination number that survives even if a skeptic *rejects* the declared formula grammar. [E] compression removed (v428), the genuinely multiplicative pieces are the input "8" forced four independent ways plus the foreign witness $\alpha^{-1} \sim 137$. [C] multiplying only across disjoint pieces (input over-determination $\times$ the α census $1/94500$ $\times$ the φ_0^{ret}-seed cluster counted once $\times$ one downstream witness) gives a floor $\leq 10^{-6}$ across a grid of generous assignments, nominal $\sim 10^{-10}$. [E] that floor sits ~ 20 orders *above* the v100 conditional $10^{-30.7}$; the gap is exactly the evidential value of proving the forms forced, so the α form-derivation (`ALPHA.QUILLEN.EXACT.01`, v382) and this floor are the *same* lever. [C] **honest scope:** bounds only the "is it numerology?" question, not the physics bridge (firewall v187); a conservative floor, not a posterior.
- *Surfaces synced in the same change:* v431/v432 added to `run_all.py`, `script_registry.csv` (master index + website `ScriptIndex`), `status_ledger.csv`, `tfpt_5_redteam` (reverse-audit winbox + Casimir-degree figure) and `tfpt_safeguards` (the over-determination layer); `v431` mirrored in `wolfram/tfpt_readouts_extension.wl` (v432 is a conservative bound, Python-only).

2026-06-27 (v430 — the seam's "other side" is forced-disjoint from E_8's unmapped Casimir degrees)

- **New module v430 (E8.OTHERSIDE.AUDIT.01) — the sheet/deck complement of the reverse audit (v354/v355).** The reverse audit found five E_8 Casimir degrees $\{12, 14, 18, 20, 24\}$ with no primary readout ("hull overhead"); the double cover has an *other side* (the one-sided $\mathbb{Z}_2$ collar / conjugate half-spinor S^-), so the sharp adversarial follow-up is *"is the leftover where the second sheet lives?"* The answer is a disciplined **negative**. [E] the deck is the degree-2 invariant: the one-sided cover contributes $|\mathbb{Z}_2| = 2$ (the $\frac{1}{2}$ in $c_3 = 1/(|\mathbb{Z}_2|\,4\pi) = 1/(8\pi)$, v58/v216), and $2 = \min\deg(E_8)$ is the quadratic/metric — one of the *matched* primary readouts $\{2, 8, 30\}$, never one of the unmapped degrees. [E] the two sheets are the 128-spinor: $\dim S^+ = \dim S^- = 16$, the spinor block $128 = \operatorname{rank} E_8 \cdot \dim S^+ = 2^{\mathrm{rank}-1} = \sum \deg$, the spinor half of $248 = 120 + 128$; S^- enters 248 as the conjugate $(\overline{16}, \overline{4})$ $(128 = 2 \cdot 64)$, not a spare singlet. [E] forced-disjoint: the sheet/deck set $\{2, 16, 32, 128\}$ has empty intersection with $\{12, 14, 18, 20, 24\}$, its only degree contact the matched 2. [O] each unmapped degree, "explained" from sheet atoms, needs an unforced coefficient and admits ≥ 2 readings (the v355 discriminator), so there is no forced sheet→unmapped-degree identity.
- **A structural negative, not a status change.** It consolidates the S^--dark-matter downgrade (v227) and the no-spare-singlet WIMP no-go of the frontier E_8 scan (`tfpt_4_frontier`): the other side is matched conjugate matter, the five unmapped degrees are the hull overhead, and the two do not meet. The only live "other side" question — whether the 128 phase/glue channel hosts a dark sector (v227) — concerns *matched* structure, not the unmapped five. Closes no gate, upgrades nothing.
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`; the reverse-audit Outcome winbox + degree-audit table caption and the Target D winbox of `tfpt_5_redteam`, the magnitude/phase keybox of `tfpt_3_e8_audit_bootstrap`, the WIMP-no-go passage of `tfpt_4_frontier`; the E8 node of the website verification DAG; the Wolfram extension/README/ledger counts and the website verification count; manifests refreshed.

2026-06-26 (v429 — the 'unmapped' golden E_8 structure is the geometry of the one external input θ_i)

- **New module v429 (DM.AXION.PENTAGON.01) — a geometric re-reading answering "why does so much idle E_8 structure exist?" for the one case where the idle structure has a job.** The reverse audit (v354) had argued the golden ratio φ is *unmapped* (it appears only internally — the affine-E_8 attractor angle v312, the icosian lattice v348 — and in no physical readout), and used exactly that to conclude φ cannot be numerology. [E] this sharpens that statement: because $N_{\mathrm{fam}}{=}g_{\mathrm{car}}{-}2$, the axion misalignment *spine* angle (v211, DM.AXION.SPINE.01) is $\theta_i{=}\pi N_{\mathrm{fam}}/g_{\mathrm{car}}{=}(g_{\mathrm{car}}{-}2)\pi/g_{\mathrm{car}}$, which is *exactly* the interior angle of the regular g_{car}-gon — for $g_{\mathrm{car}}{=}5$ the pentagon, $\theta_i{=}3\pi/5{=}108°$. [E] its cosine is golden, $\cos(3\pi/5){=}(1{-}\sqrt{5})/4{=}{-}1/(2\varphi)$ (partner $2\cos(2\pi/5){=}1/\varphi{=}\varphi{-}1$, $\varphi{=}2\cos(\pi/5)$). [E] the golden character is *unique* to $g_{\mathrm{car}}{=}5$: among the small regular n-gons only $n{=}5$ has an irrational interior-angle cosine ($n{=}3, 4, 6$ give $1/2, 0, -1/2$), so θ_i is golden *because* the carrier is 5 (P2).
- **An honest [C] bridge, not a status change.** It connects two standing facts — φ is the unmapped $g_{\mathrm{car}}{=}5$ signature (v354/v313) and $\theta_i{=}3\pi/5$ is the one external cosmological input (v211) — and *refines* v354 from "φ in no physical readout" (it is the unmapped $g_{\mathrm{car}}{=}5$ signature, v313) to "φ touches *only* the [C] spine misalignment angle" (conditional, branch-level, never a frozen prediction). It is a geometric motive for the spine over the hilltop $\theta_i{\approx}170°$ (whose cosine is not a clean pentagon value), consistent with — not replacing — the

finite-T solver that decides the branch (v185/v211). It does *not* derive θ_i, does *not* upgrade `DM.AXION.SPINE.01` (stays [C]) and closes no gate.

- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, the dark-matter section of `tfpt_4_frontier`; the Wolfram extension/README/ledger counts (`GATE.WOLFRAM.02`) and the website verification count; manifests refreshed.

2026-06-25 (External-works references appendix — web-verified)

- **New shared appendix `tex-artefacts/tfpt_references.tex`** ("External works built upon"), `\input` into the six math-heavy documents (`tfpt_1`, `tfpt_2`, `tfpt_4`, `tfpt_horizon_readouts`, `origin_theory`, `tfpt_research_contracts`). The active papers named the foundational results they build on by author only, with no reference list; the appendix supplies web-verified locators (journal, volume, year / arXiv) grouped by topic: modular theory & AQFT (Tomita–Takesaki LNM 128 1970; Bisognano–Wichmann J. Math. Phys. 16/17; Osterwalder–Schrader CMP 31/42; Doplicher–Haag–Roberts CMP 23/35; Haag–Ruelle; Longo CMP 126/130; Kawahigashi–Longo–Müger CMP 219), conformal nets/VOAs/lattices (Frenkel–Kac Invent. Math. 62; Segal CMP 80; MMST; Adamo–Moriwaki–Tanimoto; Adamo–Giorgetti–Tanimoto `2506.01008/2508.07109`; Naaijkens–Penneys–Wallick), lattices/quadratic forms (Conway–Sloane; Siegel Ann. Math. 36 1935), singularities (Brieskorn Invent. Math. 2 1966; Milnor AMS 61 1968) and spectral geometry/gravity/thermal time (Chamseddine–Connes CMP 186; Connes–Rovelli CQG 11; Wald PRD 48; Casini–Huerta–Myers JHEP 05 2011). Added to the `make_manifest.py` TEX list. No claim or status change — a provenance/citation completeness pass.

2026-06-25 (External AQFT citations + the four-fold μ_4 identity sharpened)

- **Two recent Adamo–Giorgetti–Tanimoto theorems cited where the keystone builds on them.** The open `SEAM.EQUIV.01` continuum/realisation legs (v336) now cite, alongside MMST and Adamo–Moriwaki–Tanimoto (`arXiv:2407.18222`, already in use): **(a)** the construction and classification of two-dimensional conformal nets from even lattices (`arXiv:2506.01008`, 2025) — "the lattice is a conformal net" is a theorem of the same operator-algebraic lineage, strengthening the lattice-realisation leg; and **(b)** automatic positive-energy/diffeomorphism covariance of conformal-net representations (`arXiv:2508.07109`, 2025), which defuses the "Bisognano–Wichmann presupposes the covariance it should produce" circularity (v215) on the intrinsic-BW residual (v424). Added to v336 docstring/registry, the `SEAM.EQUIV.CONTINUUM.01` ledger `external_data`, and `tfpt_research_contracts`.

- **The four-fold μ_4 identity made explicit (`v422`).** v422 already proved that the seam deck-/divisor μ_4, the carrier Galois group $\mathrm{Gal}(\mathbb{Q}(\zeta_5))$, the discriminant $\mathrm{disc}(D_5)=\mathrm{disc}(A_3)=\mathbb{Z}/4$ and the $(E_8)_1$ simple-current glue are ONE cyclic $\mathbb{Z}/4$ (with the cyclic-vs-Klein negative control). It is now named the *four-fold μ_4 identity* and typed, per v428, as a *compression* (one object read four ways), NOT four independent witnesses – in v422, the registry, the `SEAM.GALOIS.NET.01` ledger row and `origin_theory`. (No new module: a separate `v429` would have duplicated v422 and violated the very compression discipline.)

2026-06-25 (`v428` — honest self-correction of the over-determination map)

- **`v428` (`OVERDET.WITNESS.RECLASS.01`) — applying `v427`'s own multiply-vs-compress test to `v427` itself.** A pattern search and the right objection exposed that calling the seven arithmetic witnesses "disjoint grammars that *multiply*" overstates independence. [E] by the Brieskorn classification (v236) the $(2,3,5)$ singularity is the ONE generator behind the order-30 clock (Milnor $\mu=(2-1)(3-1)(5-1)=8=\mathrm{rank}\, E_8$, $30=2\cdot3\cdot5=h(E_8)$, $\varphi(30)=8$), so each witness is a facet of that same object (primes $2,3,5$ of 30, the clock 30, or E_8): all seven collapse to one generator ⇒ *compression*, not seven multiplications. [E] what genuinely multiplies is the input forced four independent ways — the "8" in c_3=1/(8π) from rank E_8, $h(D_5)$=2(5−1), $\varphi(30)$ and the Milnor number — plus the foreign witness $\alpha^{-1}\approx 137$. [C] refines and partly walks back `v427`, consistent with v236/v305/v313.

- **Surfaces corrected (same change).** `v427`'s prose, the ledger row (`OVERDET.WITNESS.MAP.01`, now noting the refinement), `tfpt_safeguards` Layer 3 (the seven compress one object; a second keybox states what multiplies), the witness-map figure (redrawn), the website (`papers.ts`, `Safeguards.tsx`, `release.ts`, the intro-video description) and the Zenodo description. The over-determination claim is now the narrower, harder-to-attack "one richly over-determined input + a foreign readout".

2026-06-25 (Safeguards visuals + paper expansion + the website safeguards band)

- **Five safeguards figures** (`make_figures.py`, in `manifest.sha256`): the layered defense, the α^{-1} uniqueness scan (one $F_{U(1)}$ variant of 94,500 in the CODATA window), the reverse audit (3/8 vs 5/8), the witness-independence map (seven disjoint grammars), and the look-elsewhere null model (13/13 vs $\leq 5/13$).
- **`tfpt_safeguards` expanded.** Layer 5 now states the null model explicitly ($\prod p_i$=$10^{-25.8}$, 2×200k pseudo-theories $\leq$ 5/13, $\varphi_0^{\mathrm{ret}} \pm 10\% \to 6/13$, shuffle 1.2) and the α uniqueness ($\leq 10^{-30.7}$, ∼102 bits, v100), with the data watchdog (v307) and kill board (v321); the five figures are embedded.
- **Website: a prominent animated Safeguards band** (`components/Safeguards.tsx`) in the home flow (after the trust beat) — headline statistics, the seven layers, the figures, links to the paper and the suite — to address the coincidence/numerology suspicion proactively.
- **Koide flow-time investigation (honest negative).** Post-`v425`: the flow-time is *not* native and does *not* reduce to one scale — v99 already rejects a scale reading as look-elsewhere; the only theory-side content is the conditional integer reading t=N_{fam}=3 ⇒ m_τ=1776.94 MeV, a Belle II kill test. No new module.

2026-06-25 (Three closure-strategy modules v425–v427 + the Safeguards paper)

- **`v425` (`DYN.TRANSFER.UNIVERSAL.01`) — the single-flow reduction of F_{transfer}.** The four frontier transfers do not each need a separate external solver: their *dynamics* is the ONE native flow (the seam modular/recovery semigroup, `v238`/`v240`, gap $6\ln(3/2)$) restricted to each sector. [E] the `v99` Koide generator linearised at q=2 has rate $-\Delta$, time-1 contraction $(2/3)^6$; Boltzmann washout is that contraction (CP phase $4\pi/3$ native, `v320`); QCD rate $b_3 = -7$. [C] only the anchors (v_{geo}, C_p, M_R) stay external; F_{relic}'s cosmological IC θ_i is the one wall. A reduction (extends `v383` static→dynamic), not internalisation; the firewall holds.
- **`v426` (`SEAM.EQUIV.LTORP.KMS.01`) — the invertible β=1-KMS variant of (LTO-RP).** The corner NPW26 leave open (they prove it for topologically-ordered models; the seam is invertible + KMS, not a trace). [E] a gapped reflection-symmetric μ_4-even collar at β=1 KMS satisfies $\Theta K \Theta^{-1}$=$-K$ (u_Θ=J) and $[\rho, K]$=0, so (LTO-RP) holds by direct Tomita–Takesaki; $|\det \mathrm{Cartan}(E_8)|$=1 (no anyons), state not a trace; neg controls have teeth. Advances `v424` sub-step (ii); `SEAM.EQUIV.01` stays [O] (continuum sub-step (i) remains).
- **`v427` (`OVERDET.WITNESS.MAP.01`) — the honest over-determination accounting.** A reviewer correction made machine-explicit: witnesses from *disjoint* grammars multiply, projections of the *same* generator compress. [E] seven disjoint-grammar witnesses each reproduce a carrier-skeleton integer from their own structure — Gauss $N(3{+}2i)$=13, Eisenstein $N(3{+}2\omega)$=7, cyclotomy $N(3{+}2\zeta_5)$=55, Galois $|(\mathbb{Z}/5)^\times|$=4, $|\det \mathrm{Cartan}(E_8)|$=1, Pascal $\binom{4}{0}+\binom{4}{1}+\binom{4}{2}$=11 ($2^4$=16), Coxeter $h(E_8)$=30=2·3·5 with $\varphi(30)$=8 — and the anchor a=(1, 1, 2) is the compression generator. Extends `v305`.

- **New paper `tfpt_safeguards` (companion S).** A dedicated methodology paper collecting, in one language, every mechanism by which TFPT defends a claim against coincidence and numerology: the status calculus + ledger + audit, no-free-pattern + reverse audit, the over-determination map (`v427`), the firewall + No-Unit theorem, frozen predictions + null model, the independent Wolfram and Lean paths, and the red team — with a final kill-test summary. Registered in `build.sh`, `make_docs_map.py`, `make_manifest.py` and the document-set box; the document set is now **eleven** deliverables (five numbered papers + four companions + introduction + changelog).
- **Surfaces synced.** `run_all.py` (421 modules), the registry, the ledger, `tfpt_research_contracts` (the F_{transfer} and keystone sections), `tfpt_safeguards`; manifests refreshed.

2026-06-25 (The (LTO-RP) reduction of the keystone `SEAM.EQUIV.01`: `v424`)

- **New module `v424` (`SEAM.EQUIV.LTORP.01`).** An honest MMST-style reduction of the one open keystone — `SEAM.EQUIV.01`'s intrinsic Bisognano–Wichmann lemma (the v308/v309 residual) — to the recent commuting-projector theorem of Naaijkens–Penneys–Wallick (`arXiv:2605.10693`, [NPW26]), with the central mapping *corrected*. [E] the μ_4 clock is a modular *symmetry* ($[\rho, K]=0$, v309/v199), not the flow — $\sigma^{\psi}=\rho$ is type-wrong (a one-parameter group versus a finite order-4 element). [E] NPW26's axiom (LTO-RP) means $u_{\Theta}=J$ (reflection = modular conjugation) = the intrinsic BW condition $\theta K\theta=-K$, *distinct* from $\omega\circ\rho=\omega$: a positive character-block-diagonal K has $[\rho, K]=0$ yet never $\theta K\theta=-K$, so clock-invariance does *not* imply (LTO-RP) (refuting that identification). [O] make-or-break: NPW26 prove (LTO-RP) for *topologically ordered* models (Toric Code $|\det K|=4$ via a trace; Levin–Wen anyonic), while the seam is invertible ($|\det \mathrm{Cartan}(E_8)|=1$, no anyons) and a $\beta=1$ KMS state (not a trace) — in neither proved bucket. [C] verdict: the BW residual reduces to NPW26 Theorem A modulo two open sub-steps (realise the seam as a $\mathbb{Z}_2$-reflection commuting-projector LTO; extend (LTO-RP) to the invertible KMS case) — a reduction, *not* a closure; `SEAM.EQUIV.01` stays [O]. Cited in `tfpt_research_contracts`; Python-only (numpy; the det discriminators mirrored via v89/v281).
- **Validation provenance.** The three pivotal references of the closure-strategy note were independently verified real and correctly described: NPW26 (`arXiv:2605.10693`), the RP–Yang–Mills complete-monotonicity reconstruction (Int. J. Geom. Meth. Mod. Phys. 2026), and Marolf (`arXiv:2203.07421`).
- **Surfaces synced.** `run_all.py`, the script registry, the status ledger and the `tfpt_research_contracts` keystone section; manifests refreshed.

2026-06-25 (No-Ambient-Dependency: no frozen readout is computed from `QG.AMB.01`: `v423`)

- **New module `v423` (`QGAMB.NODEP.01`).** Sharpens the v369/v384 prose redundancy ("`QG.AMB.01` is a [C] redundancy, not dynamics") into a machine-checked DAG+script fact, told honestly. [E] gap-decoupling margin $\Delta_{\text{eff}}=6\ln\frac{3}{2}-31/(4\pi^2)\approx 1.648>0$ ($31=2^{g_{\text{car}}}-1$), susceptibility $\chi=729/665$ finite (negative control gap $\to 0 \Rightarrow \chi\to\infty$); [E] `QG.AMB.01` is a certification *sink* (every claim that lists it is a gravity-gate/QG row, never a frozen prediction) and *not a value source* (no producing script, while α^{-1}, the mixing angles and cosmology are each *computed* by a compiler script, v3/v9/v268/v7); [E] the one transitive link is architectural — of the frozen readouts only θ_{13} references it, via the 4d-QFT contract spine (`PS.DIRAC`→`CONTRACT.QFT4D`→`QG.G2`→`GATE.METRIC`), a coherence edge, not a numerical input. [C] verdict: `QG.AMB.01` is gap-decoupled certification — v369 as a DAG/script fact; honest scope — this does *not* discharge `SEAM.EQUIV.01`, on which the readouts still depend. Cited in `tfpt_4`; Python-only (gap algebra mirrored via v337).

- **Paper sharpening (seam selector, up front).** The `tfpt_4` quantum-gravity keybox now states explicitly that the open `SEAM.EQUIV.01` premise is the *holomorphic* $|\det K|{=}1$ point that selects $(E_8)_1$ — the central charge $c{=}8$ alone does *not*, being shared by the non-holomorphic $SO(16)_1$ counterexample (v277/v281).
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, `tfpt_4`; manifests refreshed.

2026-06-25 (The Galois↔Net bridge: v422)

- **New module v422 (SEAM.GALOIS.NET.01).** The seam $\mu_4 = \mathrm{Gal}(\mathbb{Q}(\zeta_5))$ is the SAME cyclic $\mathbb{Z}/4$ as the $(E_8)_1$ simple-current glue — not a mere order-4 coincidence — bridging the Galois gearbox of v419 to $G_{\rm net}$/`SEAM.EQUIV.01`. [E] $\mathrm{disc}(A_3) = \mathrm{disc}(D_5) = \mathbb{Z}/4$ (one Smith invariant factor 4 = cyclic, det 4); D_n disc is $\mathbb{Z}/4$ for n odd, $\mathbb{Z}_2{\times}\mathbb{Z}_2$ for n even, so the carrier D_5 (rank $5 = g_{\rm car}$, odd) is cyclic while D_8 (rank 8, even) is Klein; [E] $(\mathbb{Z}/5)^\times = \langle 2\rangle$ is cyclic order 4 (only the carrier prime 5 gives order 4); [E] in $\mathbb{Z}_4{\times}\mathbb{Z}_4$ ($16 = \dim S^+$) the glue $(1,1)$ is order 4 with Lagrangian quotient $16/4^2 = 1 = (E_8)_1$ (v89/v154); [E] negative control: the Klein / order-2 $\langle(2,2)\rangle$ gives $16/2^2 = 4 = |\mathrm{disc}(D_8)| = SO(16)$ (det 4), NOT E_8 (det 1) — cyclic $\mathbb{Z}/4 \to E_8$, Klein$\to D_8$. [C] the bridge: one cyclic $\mathbb{Z}/4$ threads the seam deck μ_4 (v419), $\mathrm{Gal}(\mathbb{Q}(\zeta_5))$ and the $(E_8)_1$ glue — the Galois gearbox $(\mathbb{Z}/30)^\times$ IS the holomorphic net's glue, cyclicity forced by 5, Klein excluded. Cited in `origin_theory`; Wolfram-mirrored ($324 \to 327$).
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, `origin_theory`; the Wolfram extension/README/ledger counts (327/327) and the website verification count; manifests refreshed.

2026-06-25 (F__pole external: the Koide source→pole transfer is one-loop EM: v420)

- **New module v420 (FR.POLE.QED.01).** The honest external-physics follow-up to v93 and the clock No-Go: the Koide source→pole transfer is a one-loop *electromagnetic* effect, not an algebraic step. [E] Koide Q is rescaling-invariant ($Q(c\,m) = Q(m)$), so flavor-universal running leaves it fixed and the source→pole shift is purely flavor-split; [C] the tree deficit $\frac{2}{3} - Q_{\rm src} = 0.00220$ is one-loop-EM-sized ($= \varphi_0/24$ to 0.6%, $= \alpha/\pi$ to $\sim$5%), the PDG pole sits at 2/3 (the IR attractor); [C] a 1-loop QED pole$\leftrightarrow\overline{\rm MS}$ per-lepton-log estimate shifts Q by EM-loop order toward 2/3 (the v82 direction); [O] the exact flow time $t \approx 2.84$ (v93) is scheme/scale-dependent external QED, only sized. [C] verdict: $F_{\rm pole}$ is a one-loop EM effect — the lens forces the rate $(2/3)^6$ (v327), external EM supplies the flow; `F_pole` stays [C], firewalled (v187), so $F_{\rm transfer}$ is confirmed genuinely external — the open theme sharpened (QED), not closed. Cited in `tfpt_4`; numerical (mpmath), Python-only (no Wolfram check, by the suite convention like v93/v371).
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, the Koide section of `tfpt_4`; manifests refreshed.

2026-06-25 (The seam μ_4 is the carrier's Galois group: v419 + figure)

- **New module v419 (SEAM.GALOIS.01).** A positive resolution of the cyclic/Galois asymmetry (v409, `RES.COXETER.SYMMETRY.01`): why the seam μ_4 sits on the Galois side, not on the cyclic ζ_{30} dial. [E] $h(E_8) = 30$ is squarefree, so $\mathbb{Z}/30$ has element orders $\{1,2,3,5,6,10,15,30\}$ — *no* order 4, so the seam μ_4 cannot be a rotation hand and is forced onto the Galois side; [E] by CRT $(\mathbb{Z}/30)^\times = (\mathbb{Z}/2)^\times \times (\mathbb{Z}/3)^\times \times (\mathbb{Z}/5)^\times = \mu_4 \times \mathbb{Z}_2$ (order $\varphi(30) = 8 = \mathrm{rank}\,E_8$), and the μ_4 ($\mathbb{Z}/4$) factor IS $(\mathbb{Z}/5)^\times$ (the carrier prime 5), the $\mathbb{Z}_2$ IS $(\mathbb{Z}/3)^\times$ (family 3 = CP conjugation, v316); [E] so the seam $\mu_4 = \mathrm{Gal}(\mathbb{Q}(\zeta_5))$, realised on the carrier clock C_5

(v418) by the Frobenius $\zeta_5 \mapsto \zeta_5^2$ — the explicit order-4 operator G with $GC_5G^{-1} = C_5^2$, $G^4 = I$, $G^2C_5G^{-2} = C_5^{-1}$. [C] this is the positive content of v409: the 2^2=4 is the order of $(\mathbb{Z}/5)^\times$, and "no contraction rate" is because the seam is an *automorphism*, not a hand; so $\mathbb{Q}(\zeta_{30})$ carries both E_8 invariants (cyclic h=30, Galois rank =8). Cited in `origin_theory`; Wolfram-mirrored (321 → 324).

- **New figure `coxeter_galois.pdf`.** The order-30 cyclic dial with the four flavor hands (sheet/family/carrier/CP) beside the Galois gears ($\mu_4 = \mathrm{Aut}(\zeta_5)$ the seam, $\mathbb{Z}_2 = \mathrm{Aut}(\zeta_3)$ the CP conjugation); added to `make_figures.py`, the manifest figure list and `origin_theory`.
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, the Wolfram extension/README/ledger counts (324/324) and the website verification count; manifests refreshed.

2026-06-25 (The cyclotomic norm triple + the carrier-5 clock: v418)

- **New module v418 (`ARITH.NORMTRIPLE.01`).** The carrier split $(3,2)$ read in the three atom-rings gives $(7,13,55)$ =(scalaron, Δ_Q, quark numerator), and the carrier-5 clock missing from v417 is found as the 4×4 Φ_5-companion C_5. [E] C_5^5=I, $\mathrm{tr}\, C_5$=−1, $\det C_5$=1 (the four primitive 5th roots), with the golden ratio as its real-part data ($2\cos 72°$=$1/\varphi$, $2\cos 144°$=$-\varphi$; output-side, v313/v349; dim 4, not 3 — this refines v417's gap from "no operator" to "no 3×3 operator"); [E] $N_{\mathbb{Z}[\zeta_5]}(3{+}2\zeta_5)$= $\det(3I{+}2C_5)$=55=5·11=$g_{\mathrm{car}}\|\,\mathrm{Pl}(K)\|_1$, the c_u/c_d numerator (v410/v411); [E] the triple $\det(3I{+}2\,\mathrm{Comp}(\Phi_n)) = (7,13,55)$ for n=3, 4, 5 over $\mathbb{Z}[\omega], \mathbb{Z}[i], \mathbb{Z}[\zeta_5]$ (atoms 3, 2, 5, v416); [E] NEG control: the anchor $(5,4) \to (21,41,461)$ reaches named values only in the ω- and i-rings (21=$N_\omega(5,4)$, 41=$10b_1$, v222) but 461 (prime) in the carrier ring, so the carrier ring separates $(3,2) \to 55$ from $(5,4) \to 461$. [C] honest: $(3,2)$ is the *forced* carrier split (v14), not the unique clean one — a scan finds $\{(2,1) \to (3,5,11), (3,2) \to (7,13,55)\}$, a small ladder. The sharper discriminator is *cross-sector*: only $(3,2)$ spans three physics sectors (gravity 7, flavor 13, masses 55), the unique cross-sector seed, while $(2,1)$ is all compiler-core. Cited in `origin_theory`; Wolfram-mirrored (318 → 321).
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, a triple paragraph in `origin_theory`, the Wolfram extension/README/ledger counts (321/321) and the website verification count; manifests refreshed.

2026-06-25 (The Eisenstein/CP operator + the order-30 clock: v417)

- **New module v417 (`DIAMOND.EISEN.01`).** The hexagonal dual of v415: the τ=ω flavor side put on operators and unified with the seam into one order-30 Coxeter clock. [E] the family rotation $P = (1\,2\,3)$ is the order-3 Eisenstein deck ($P^2{+}P{+}I$ = ONES, so $\omega^2{+}\omega{+}1$=0 on the plane ⊥ $(1,1,1)$), the order-3 dual of J^2=$-I$; [E] it realises the hex CM norm as $(3I{+}2P)(3I{+}2P^2) = 7I{+}6\,\mathrm{ONES}$, $\mathrm{Spec}\{7,7,25\}$ (7=$N_{\mathbb{Z}[\omega]}(3{+}2\omega)$= scalaron, 25=$g_{\mathrm{car}}^2$) — the dual of v415's $\{13,13,9\}$; [E] the CP clock W=$-P^2$ (ρ of v233) has W^2=P, W^3=$-I$, $\mathrm{Spec}\{-1, \zeta_6, \zeta_6^{-1}\}$, giving $\delta_{\mathrm{CKM,lead}}$= $\arg\zeta_6$=$\pi/3$ and δ_{PMNS}= $\arg\zeta_6^4$=$4\pi/3$ (v316/v320); [E] all flavor clocks are powers of ζ_{30} (ζ_{30}^{15}=−1, ζ_{30}^{10}=ω, ζ_{30}^{6}=ζ_5, ζ_{30}^{5}=ζ_6) while the seam μ_4=i is the Galois side $(\mathbb{Z}/30)^\times$ of order $\varphi(30)$=8= rank E_8 — so $\mathbb{Q}(\zeta_{30})$ carries both E_8 invariants (h=30 and rank =8). [C] honest gap: the carrier μ_5 has no 3×3 operator (the golden φ is an output-side trace, v313/v349); negatives: no U,V-diamond operator carries ζ_6, $[D,P] \neq 0$ (no μ_{12}), $N(3{+}2\zeta_6)$=19≠7. Cited in `tfpt_2`; Wolfram-mirrored (315 → 318).
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, the Sheet-Diamond keybox of `tfpt_2`, the Wolfram extension/README/ledger counts (318/318) and the website verification count; manifests refreshed.

2026-06-25 (The Gaussian operator + the atom trichotomy: v415, v416)

- **New module v415 (DIAMOND.GAUSS.01).** The square CM-norm dictionary of v222/v230 is *operator-realised*, and the μ_4 deck becomes an intrinsic diamond object. [E] the integer commutator $J := \frac{1}{3}[U,V] = \left(\begin{smallmatrix}-1&1&0\\-2&1&0\\-3&1&0\end{smallmatrix}\right)$ has $\mathrm{Spec}(J) = \{i,-i,0\}$, $J^2 = -I$ on the rank-2 image and $J^3 + J = 0$ — a μ_4 quarter-turn born from the binary V and ternary U compilers (v410/v95); [E] $3I + 2J$ and $5I + 4J$ have spectra $\{3{+}2i, 3{-}2i, 3\}$, $\{5{+}4i, 5{-}4i, 5\}$, so the Gaussian integers $3{+}2i$ (norm $13{=}\Delta_Q$) and $5{+}4i$ (norm $41{=}10b_1$) are *literal eigenvalues*; [E] $(aI{+}bJ)(aI{-}bJ)$ reads $(a^2{+}b^2)$ on the μ_4-plane and the real part^2 on the kernel ($\{13,13,9\}$, $\{41,41,25\}$); [E] the intrinsic order-4 deck $D = -I + J - J^2$ ($D^4 = I$, $\chi_D = (x{+}1)(x^2{+}1)$) has χ_2-line $\ker[U,V] = a - \mathbf{1}$. [C] the geometric-deck identification sharpens (does not close) the v140 $\mathbb{Z}_3$ residue. Cited in `tfpt_2`; Wolfram-mirrored.
- **New module v416 (ARITH.TRICHOTOMY.01).** The atoms $\{2,3,5\}$ are ramified/inert/split in the two exceptional CM rings. [E] in $\mathbb{Z}[i]$ (square/seam) $2{=}|\mathbb{Z}_2|$ ramifies, $3{=}N_{\mathrm{fam}}$ is inert, $5{=}g_{\mathrm{car}}$ *splits* as $(2{+}i)(2{-}i)$; [E] in $\mathbb{Z}[\omega]$ (hex/flavor) $3{=}N_{\mathrm{fam}}$ ramifies, $2,5$ are inert; [E] the ramified prime of each ring is its own atom; [E] the three quadratic facets $\mathbb{Q}(i), \mathbb{Q}(\sqrt{-3}), \mathbb{Q}(\sqrt{5})$ of $K = \mathbb{Q}(i,\sqrt{-3},\sqrt{5})$ (v403) ramify over exactly one atom each, product $2\cdot3\cdot5 = 30 = h(E_8)$, two imaginary (CM/dynamic) + one real (RM/static). [C] the seam/flavor/golden reading. Cited in `origin_theory`; Wolfram-mirrored.
- **Wolfram extension** $308 \to 315$ **+ a pre-existing fix.** Added the seven exact checks for `v415`/`v416` and corrected a one-character sign bug in the `v412 CharacteristicPolynomial` check (Wolfram returns $-\chi$ for odd rank), so `tfpt_readouts_extension.wl` now runs genuinely clean at 315/315 (`GATE.WOLFRAM.02`); README and ledger counts updated.
- **Surfaces synced.** `run_all.py`, `script_registry.csv` and `status_ledger.csv` updated; a new keybox paragraph in the Sheet-Diamond section of `tfpt_2` and a trichotomy sentence in `origin_theory`; manifests refreshed.

2026-06-24 (The sheet generator V as a binary internal compiler: v410–v414)

- **New module v410 (SHEET.GEN.BINARY.01).** The rank-2 sheet axis $V = Q\,\mathrm{diag}(0,1,1)$ of the centered cross (v95/v218) is a binary internal compiler: its powers print the carrier spine. [E] the closed form $V^n = \left(\begin{smallmatrix}0&2^{n-1}&0\\0&2^n&0\\0&2^{n+1}-2&1\end{smallmatrix}\right)$ (proved by the inductive step $F(n)V{=}F(n{+}1)$), so $V^n\mathbf{1} = (2^{n-1}, 2^n, 2^{n+1}-1) = (1,2,3),(2,4,7),(4,8,15),(8,16,31)$; [E] the four bilinear families $\mathbf{1}^TV^n\mathbf{1} = 7\cdot2^{n-1}{-}1$, $\mathbf{1}^TV^na = 7\cdot2^{n-1}$, $a^TV^na = 11\cdot2^{n-1}$, $a^TV^n\mathbf{1} = 11\cdot2^{n-1}{-}2$ give $(6,7,9,11)$ at $n{=}1$ and $\mathbf{1}^TV^2\mathbf{1}{=}13{=}\Delta_Q$, $\mathbf{1}^TV^3\mathbf{1}{=}27{=}\mathbf{1}^TRa$, $\mathbf{1}^TV^4\mathbf{1}{=}55$, $\mathbf{1}^TV^4a{=}56{=}\dim\mathbf{56}_{E_7}$; [C] the binary spine is FORCED by $\mathrm{Spec}(V){=}\{0,1,2\}$, so the carrier/Lie matches (rank $E_8{=}8$, $\dim S^+{=}16$, $248/8{=}31$) and the theta cross-link ($\sigma_3(2){=}9{=}a^TV\mathbf{1}$, $\sigma_3(3){=}28{=}\mathbf{1}^TV^3a{=}\det(I{+}R)$, $\sigma_3(5){=}126$) are audit-typed (the v95 G_2-center typing).
- **New module v411 (FLAV.UD.VPOWER.01).** The quark ratio is a pure V-power readout, $c_u/c_d = (\mathbf{1}^TV^4\mathbf{1})/((a^TV\mathbf{1})(\mathbf{1}^TV^2\mathbf{1})) = 55/(9\cdot13) = 55/117$, identical to the established $g_{\mathrm{car}}\|\mathrm{Pl}(K)\|_1/(N_{\mathrm{fam}}^2\Delta_Q) = 5\cdot11/(9\cdot13)$ (v94). [E] the identity; [C] an exact *re-encoding* (no new physical input) — the value and the u/d identification stay conditional, coupled to the (U_{wall}) readout rigidity.
- **New module v412 (DIAMOND.SHEET.SOURCE.J.01).** Names the operator on the $\mathbb{Z}_2$ wall ($t{=}{-}2$, $\det{=}2$, v135), $J{=}M(1,-2){=}C{-}2V$, completing the sheet axis $J,K,C,F{=}M(1,-2..1)$. [E] $\chi_J = \lambda^3 - 6\lambda^2 + 3\lambda - 2 \Rightarrow (\mathrm{tr}, e_2, \det){=}(6,3,2)$; $a^TJa{=}30{=}h(E_8)$, $\det(I{+}J){=}12{=}\dim\mathfrak{g}_{\mathrm{SM}}$ and $\det(2I{+}J){=}40{=}|R(D_5)|$; [C] the Lie readings audit-typed like the G_2/F_4 shadows.
- **New module v413 (DIAMOND.SHEET.CALCULUS.01).** On $M_t{=}C{+}tV$ the elementary-symmetric coefficients encode the atoms as difference orders: $e_1{=}3t{+}12$, $e_2{=}2t^2{+}15t{+}25$,

$e_3{=}4t^2{+}14t{+}14$ give $\Delta e_1{=}3{=}N_{\rm fam}$, $\Delta^2 e_2{=}4{=}|\mu_4|$, $\Delta^2 e_3{=}8{=}\mathrm{rank}\,E_8$ (the e_3 curvature is the v218/v224 sheet quadratic; the full triple is new); [E] the anchor energy $a^T M_t a{=}52{+}11t$ ($30{\to}41{\to}52{\to}63$, step $11{=}a^T V a$); [C] atom/Lie readings audit-typed.

- **New module v414 (`DIAMOND.CENTER.RESOLVENT.01`).** The center $C{=}M(1,0)$ is a resolvent portal: [E] $\det C{=}14{=}\dim G_2$, $\det(I{+}C){=}52{=}\dim F_4$, $\det(2I{+}C){=}120{=}|R^+(E_8)|$, the exceptional shell ladder $G_2{\to}F_4{\to}E_8^+$ as $\det(C{+}kI)$, $k{=}0,1,2$ (with $\mathrm{tr}\,C{=}12{=}\chi_{E_8}(i)$, $\sum C{=}31{=}2^{g_{\rm car}}{-}1$, v95); [C] the portal reading.
- **Surfaces synced.** Integrated into the Sheet-Diamond section of `tfpt_2` (new keybox, five `\veri` citations) and the E_8 theta-raster of `tfpt_3`; `run_all.py`, `script_registry.csv` and the ledger updated; Wolfram extension mirror 303 → 308.

2026-06-24 (Two new tracked targets + axion-branch consistency: v408, v409)

- **New module v408 (`HYP.PHI0.PUNCTURE.01`).** Names the φ_0 puncture term $48c_3^4$ as the order-$|\mu_4|{=}4$ local boundary heat-kernel (Seeley–DeWitt) contact term of the μ_4 puncture divisor, summed over $\Omega_{\rm adm}{=}N_{\rm fam}\dim S^+{=}48$ admissible states — the analytic complement of the icosahedral tree term (v396). [E] the exact split $\varphi_0{=}1/(6\pi){+}48c_3^4$, the puncture count, the closed form $3/(256\pi^4)$ and the hypergraph negative (reused from v396/`ARCH.QUAD.01`); [O] the from-first-principles heat-kernel proof; [C] a FACE of the seam boundary analysis (the c_3 Gauss–Bonnet datum), not a new gate. Cited in `origin_theory`; no new number.
- **New module v409 (`RES.COXETER.SYMMETRY.01` closed as a structural lemma).** prime-2 (the seam sheet involution) is *necessary* but *not autonomous*: $\sigma{=}\mathrm{diag}(1,-1,-1)$ has spectrum $\{\pm1\}$ and projector $(I{+}\sigma)/2$ spectrum $\{0,1\}$, none in $(0,1)$, so prime-2 carries no contraction rate, whereas the genuine rates $2/3$ and $(1/3)^6$ lie in $(0,1)$ and carry prime-2 only as a factor. [E] no prime-2-only attractor exists; [C] dynamics begins only after coupling to family-3 or carrier-5 ($|\mu_4|{=}2^2$ the Galois bridge). Resolves the v394 open question's falsifiable corollary; cited in `tfpt_research_contracts`; the ledger row `RES.COXETER.SYMMETRY.01` retyped [O] → [E]/[C].
- **Axion branch consistency (`tfpt_4_frontier`, website `predictions.ts`, v185).** The dark-matter section's opening + keybox are realigned with the finite-T result already stated later in the same section: the robust dominant-DM branch is the *spine* angle $\theta_i{=}\pi N_{\rm fam}/g_{\rm car}{=}3\pi/5{=}108°$ (untuned, in band), while the seam hilltop $\theta_i{\approx}170°$ over-produces ($\sim 5.4\times$) and is disfavoured; the misalignment angle is retyped from a closed [E] input to a [C] branch decision. The website prediction card now leads with the spine angle.
- **Lean closure of the prime-2 corollary (`FORM.COXETER.PRIME2.01`).** The v409 result is now also machine-proved in Lean 4 / Mathlib (new module `TfptCarrier/CoxeterPrime2.lean`): an involution eigenvalue ($r^2{=}1$) is ±1, a projector eigenvalue ($r^2{=}r$) is 0 or 1, neither lies in the open interval $(0,1)$, while $2/3$ does — so *no prime-2-only attractor exists.* Built clean (`lake build`); the new theorems are axiom-pure (`#print axioms` = `propext`, `Classical.choice`, `Quot.sound` only) and pass the `AuditCheck`/`AuditContract` signature locks. This lifts `RES.COXETER.SYMMETRY.01` to the machine-proof tier.
- **Verified, no change (Plücker 11).** The quark constant $11{=}\|Pl(K)\|_1{=}\det(\mathbf{1}|a|\sigma)$ is already typed honestly as an [E] identity with "why 11" the lone [O] residue (v407, `tfpt_1`); the repo carried no overstatement, so no edit was required.

2026-06-24 (Cross-surface consistency sweep: papers, website, gates, figures realigned)

- **Ledger hygiene.** Resolved a duplicate active `claim_id`: `GATE.UWALL.06-09` was carried by two distinct chains. It is an ID collision (not a supersede): the monodromy/residue chain v114–v117 keeps `.06-.09` (as asserted by `v123_inventory_update.py` and `run_all.py`); the older exterior-leg chain `v42–v45` is renumbered to `GATE.UWALL.15-.18` (internal `depends_on` carried along). No claim content changed.

- **Website.** `papers.ts` document count 8 → 9 (the Red-Team paper was missing from the tally); Red-Team subtitle/highlight *Targets A–E→A–F* with a new Target F section + contribution bullet (the abstract already carried F); `StatusMatrix.tsx` *Doc 7→Doc 8* for the research contracts and the residual interfaces re-typed $v_{\rm geo}$ [O], $G_{\rm net}$ [C], $F_{\rm transfer}$ [C]; `objections` *Paper 1→Doc 1*.
- **Papers (marker/wording alignment, no canonical status change).** `tfpt_1` Strong CP [E]→[E]/[C] (given RP) and θ_{12} "genuinely open"→"conditionally derived"; `tfpt_2` Starobinsky A_s [E]→[E]/[C] (M exact, A_s a band over $N_\star \in [50, 60]$) and the Part-II abstract "opens the one remaining gap"→"closes the last structural gap"; `tfpt_research_contracts` the QFT4D "nonperturbative measure stays open" line and the "only $U_{\rm point}$ stays open" headline reconciled with the [C] redundancy / $v_{\rm geo}$ reduction stated alongside; `origin_theory` class C re-typed to the ambient-measure [C] redundancy; `introduction` ($U_{\rm wall}$) → $v_{\rm geo}$. Forbidden old-paper attributions ("Papers 1–7", "Papers 2/6", "Paper-7", "Paper-1") replaced by current document / neutral wording (the `tfpt_3` cascade bridge stays exempt).
- **Suite README.** The residual-gates table (Alessandro-5.0 vintage) brought to the current model ($v_{\rm geo}$ [O]; $G_{\rm net}$/`SEAM.EQUIV.01`, $F_{\rm transfer}$, $N_\star$, `QG.AMB.01` [C]; Wolfram 116/116 + 303/303).
- **Figures.** Regenerated `status_heatmap` (ledger-driven) and extended the narrative figures `script_timeline/residual_chain/qft_skeleton` through the `v303–v407` arc (typed solvers, parameter-free local Einstein equation `v359`, `QG.AMB.01` [C] redundancy, seam closed modulo cited theorems); the `qft_skeleton` "Stelle ghost" open box is retyped (resolved truncation artefact). Added `qft_skeleton/qft_unification` to the manifest figure list.

2026-06-24 (The $\tau{=}\omega$ dual keystone + the flavor residuals it homes: `v405–v407`)

- [E]/[C]/[O] **`v405_seam_equiv_omega.py`** (`SEAM.EQUIV.02`): the *dual keystone* at $\tau{=}\omega$ – the family/flavor sector as the order-3 Eisenstein/A_2 face of E_8, dual to the seam = $(E_8)_1$ at $\tau{=}i$ (v404). A named contract (genre of v286/v398), NOT a closure. [E] $\chi_{E_8}(\omega) = 0$ (the family CM point degenerates E_8) vs $\chi_{E_8}(i) = 12$; [E] A_2 is the $\tau{=}\omega$ Eisenstein lattice (Cartan det = 3 = $N_{\rm fam}$, $h = 3 = N_{\rm fam}$); [E] R lives in $E_6{\times}A_2$ ($248 = 78 + 8 + 2{\cdot}27{\cdot}3$, $\det R = 8 = \dim A_2 = \operatorname{rank} E_8$); [E] $\delta_{\rm PMNS} = 4\pi/3$ is the $\tau{=}\omega$ CM phase; [E] the cusp weights $\{0, 1/3, 2/3\}$ are the order-3 deck = $N_{\rm fam} = \det Q$. [C] the family sector is the order-3 Eisenstein/A_2 deck; the two keystones sit at the only two elliptic points of $\mathrm{PSL}(2,\mathbb{Z})$. [O] residual = the canonical $\mathbb{Z}_3$ deck map (v140) + the $v_{\rm geo}$-class scale. Cited in `origin_theory`; Python-only.
- [E]/[C]/[O] **`v406_pmns_phase_origin.py`** (`FLAV.PMNS.CLOSE.01`): the PMNS dynamics closed *modulo* the $\tau{=}\omega$ keystone – the CP-phase ORIGIN is the family CM point and the absolute scale is one $v_{\rm geo}$-class unit, so PMNS has *zero free flavor dials beyond* $v_{\rm geo}$. [E] the three angles (v9/v268) → unitary $U_{\rm PMNS}$ (v270); [C] $\delta_{\rm PMNS} = 4\pi/3$ is the $\tau{=}\omega$ hexagonal CM phase (v405), upgrading v270's [O] "assembled input" to a keystone consequence; [C] $J_{\rm PMNS} = -0.0297$ derived; [E] the absolute scale = one seesaw ratio = $v_{\rm geo}$-class (v272/v153). [O] residual = the seesaw operator realisation (M_R, v184) + the Σm_ν floor kill test. Cited in `tfpt_2_standard_model`; Python-only.
- [E]/[C]/[O] **`v407_dn_pairings_omega.py`** (`FLAV.SELECTOR.CLOSE.01`): the R4′ selector residue folds into the $\tau{=}\omega$ keystone – the three (d,n) pairings ARE the $\tau{=}\omega$ family-slice atoms. [E] the frame $(\mathbf{1}, a, \sigma)$ has det = 11 = $\|Pl(K)\|_1$ and $n = (5, -9, 6)$ is the unique covector with $n{\cdot}\mathbf{1} = 2$, $n{\cdot}a = 8$, $n{\cdot}\sigma = 121 = 11^2$ ($\Rightarrow$ (d,n) $\Rightarrow$ R, v136/v139);

[E] $n{\cdot}a = 8 = \mathrm{rank}\,E_8 = \dim A_2 = \det R$ (the A_2 block reading R in $E_6{\times}A_2$, v405); [E] the pairings $\{2, 8, 121\} = \{|Z_2|, \dim A_2{=}\det R, \|Pl(K)\|_1^2\}$. [C] deriving (d, n) folds into `SEAM.EQUIV.02` (d anchor-derived, v139); [O] the lone residue "why $11{=}\|Pl(K)\|_1$". Cited in `tfpt_1_architecture_e8`; Python-only.

2026-06-24 (The arithmetic hull + the E_8-character CM duality: v403, v404)

- [E]/[C] **`v403_facet_compositum.py`** (`ARITH.HULL.01`): the three single-prime number-field facets of the order-30 clock (v390/v394, used so far only one at a time) *compose* to one field $K = \mathbb{Q}(i, \sqrt{-3}, \sqrt{5})$. [E] the squarefree parts $-1, -3, 5$ are independent in $\mathbb{Q}^\times/(\mathbb{Q}^\times)^2$, so $[K : \mathbb{Q}] = 2^3 = 8 = \mathrm{rank}\,E_8 = g_{\mathrm{car}}{+}N_{\mathrm{fam}}$ with Galois group $(\mathbb{Z}/2)^3$ (order 8); [E] exactly 7 quadratic subfields $\{-1, -3, 5, 3, -5, -15, 15\}$, splitting 4 imaginary $= |\mu_4|$ (CM/dynamic) + 3 real $= N_{\mathrm{fam}}$ (RM/static); [E] ramified *only* over $\{2, 3, 5\}$ = the atoms = the primes of $h(E_8) = 30$; [E] $2^{\omega(h)} = \mathrm{rank}$ is E_8-special (the E_7 control $h = 18$ gives $2^2 = 4 \neq 7$). [C] K is the abelian $(\mathbb{Z}/2)^3$ *arithmetic* hull dual to the E_8 *lattice* hull; the 7 quadratic subfields are a candidate for the 7 E_8 audit slices. A synthesis composing the established facets, no new number; cited in `origin_theory`; Python-only by the synthesis-module convention (cf. v390/v394).
- [E]/[C] **`v404_e8_char_cm_duality.py`** (`E8.CMDUAL.01`): the $(E_8)_1$ vacuum character $\chi_{E_8} = E_4/\eta^8 = j^{1/3}$ read at *both* CM points, extending v282 ($\chi_{E_8}(i) = 12$) to the second elliptic point. [E] $\chi_{E_8}^3 = j$ (q-series); [E] SEAM $\tau{=}i$ ($\mathbb{Q}(i)$, order 4): cross-ratio $2 \Rightarrow j = 1728 \Rightarrow \chi_{E_8}(i) = 1728^{1/3} = 12$ (E_8 present, v214/v282); [E] FAMILY $\tau{=}\omega$ ($\mathbb{Q}(\sqrt{-3})$, order 6): the equianharmonic λ ($\lambda^2{-}\lambda{+}1 = 0$) $\Rightarrow j = 0 \Rightarrow \chi_{E_8}(\omega) = 0$ (E_4 vanishes, v220/v267); [E] $\tau{=}i, \tau{=}\omega$ are the *only* elliptic points of $\mathrm{PSL}(2, \mathbb{Z})$, so the values are forced. [C] the seam/flavor rigidity duality: the seam is rigid where E_8 is present ($\chi = 12$, the keystone `SEAM.EQUIV.01` closes) while the flavor/CP sector is the soft residual where E_8 degenerates ($\chi = 0$: U_{wall}, v_{geo}, the CP texture), a candidate dual keystone at $\tau{=}\omega$. Cited in `origin_theory`; core values $j = 1728/j = 0$ already mirrored (v214/v220/v267/v282), Python-only.

2026-06-24 (The five TOE/QFT closure contracts v398–v402 + a ledger rescope)

- [E]/[O] **`v398_seam_state_rigidity.py`** (`SEAM.RIGIDITY.01`, Paper A): the *Seam State Rigidity Theorem* as ONE named target, consolidating the scattered state-invariance reductions (v177/v199/v308/v309/v335) and *hardening* the finite-block evidence to a multi-mode sweep. [E] RP-definability hinge (OS canonical transfer $T{=}e^{-H}$ from RP + reflection alone, v54, quasi-free v155, no μ_4 normal form); [E] band-limited character-block-diagonal core $[\rho, |k|]{=}0$ swept $N{=}4..64$ (beyond the 3-dim H^1); [E] holomorphy $\Rightarrow$ $(E_8)_1$ ($c{=}8$, $|\det \mathrm{Cartan}|{=}1$ vs 4, index $4{=}|\mu_4|$). [O] the one residual: $[\rho, \Lambda_\Sigma]{=}0$ on the full L^2 (intrinsic Bisognano–Wichmann). NOT a closure of `SEAM.EQUIV.01` (continuum = cited MMST v336).
- [E]/[C]/[O] **`v399_gravity_complete.py`** (`GRAVITY.COMPLETE.01`, Paper B): the explicit verdict *Gravity-complete = Boundary-complete + Ambient-redundancy*, combining the parameter-free local equation (v358/v359) with the ambient redundancy (v369). [E] the five gravitational readout classes (Newton 8π, Λ, scalaron c_3^7, horizon $1/|\mu_4|$, recovery $(2/3)^6$) are all boundary/seam/gap, so $\mathcal{O}_{\mathrm{phys}} \subset \mathcal{A}_\Sigma$; [O] residual = `SEAM.EQUIV.01` + intrinsic BW (no new gate).
- [E]/[O] **`v400_grav_nonlocal_action.py`** (`GRAV.NONLOCAL.01`, Paper C): the nonlocal spectral-gravity action with the local $R{+}R^2$ as its IR projection (consolidating v304/v370/v380/v386, adding the IR-matching check). [E] entire $a{=}e^u$ $\Rightarrow$ one pole (ghost-free), every truncation carries a Stelle zero (monotone, v380), and the IR order reproduces $R{+}R^2$ with the atom-fixed scale $M_{\mathrm{scal}}^2/\bar{M}_{\mathrm{Pl}}^2{=}c_3^7$ (v36/v253). [O] the entire-a assumption + perturbative-only.

- [E]/[O] **`v401_metrology_closure.py`** (`METROLOGY.CLOSURE.01`, Paper D): the final input set $\{a, \pi, v_{\text{geo}}\}$ certified — 0 dimensionless dials + 1 unit (consolidating v153/v274/v364/v384). [E] dimensional bookkeeping mass/1$G/\Lambda/U_{\text{point}}/(m/\mu)$= number $\times v_{\text{geo}}^{1,2,4,1,0}$; [E] the G-vs-Λ over-determination 0.11% (v274); [O] v_{geo} theorem-forbidden ($\text{TOE}_{\text{strict}}$=$\text{TOE}_{\text{dimensionless}}$+$[v_{\text{geo}}]$).
- [E]/[C] **`v402_ftransfer_suite.py`** (`FTRANSFER.SUITE.01`, Paper E): the four frontier sectors as ONE functor F_{transfer}, the composition $F_{\text{obs}} \circ F_{\text{thr}} \circ F_{\text{RG}}$ (consolidating the `FR.*.SOLVE` solvers), each with an exact atom source, a committed kill test, and the firewall (physical prediction stays [C]/[O], only the algebraic source [E]; never re-pressed into the compiler, v187).
- **Ledger rescope (repo hygiene):** `GATE.METRIC.01` no longer reads "full covariant metric-sector field equation open" — the *local* covariant equation is closed parameter-free (`GRAV.NONLINEAR.01`/v359); the gate is rescoped so the open item is *only* the global nonperturbative ambient measure (`QG.AMB.01`, a [C] redundancy v369), removing the stale contradiction with the status card.

2026-06-24 (φ_0 leading term from icosahedral combinatorics: `v396`)

- [E]/[C] **`v396_phi0_icosahedral.py`** (`HYP.PHI0.01`): the tree-level seed $\varphi_0^{\text{tree}} = 1/(6\pi)$ is exactly $F/(4h\pi) = F/(|\operatorname{Aut}|\pi) = (F/(g_{\text{car}}N_{\text{fam}}))c_3$ on the icosahedral hypergraph ($F = 20$ triangular hyperedges, $h = 30$, $|\operatorname{Aut}| = 120$); $F/h = |Z_2|/N_{\text{fam}} = 2/3$ (the same survival ratio as `v327`). [E] Gauss–Bonnet consistent with $c_3 = 1/(|Z_2|4\pi)$ (`v216`/`v342`). [E] honest negative: the puncture $48c_3^4$ is still analytic, not a hypergraph fraction. [C] φ_0^{tree} is a combinatorial reinterpretation, not a new independent injection; the puncture stays [O].

2026-06-24 (Coupled hypergraph rewrite: `v395` unifies carrier × family in one local rule)

- [E] **`v395_hypergraph_coupled.py`** (`HYP.COUPLED.01`): one coupled local rewrite on a 9×3 labelled grid (affine-E_8 nodes × family channels) unifies `v299` (carrier growth), `v327` (branching rule M) and `v324` (fiber product). Micro-step = network lazy diffusion on each cusp column + M on each node's family vector ($T_{\text{micro}} = T_{\text{net}} \otimes M$); joint attractor marks$\otimes e_0$; one clock hand (6 family steps) gives recovery $(2/3)^6$; `v324`'s $T_{\text{net}} \otimes T_{\text{cusp}}$ emerges as the cusp-readout basis. [O] the 3-arm seed (P_2) and analytic φ_0 remain irreducible (`v312`).

2026-06-24 (A falsifiable clock probe of the frontier: `v397` how external is the "external physics"?)

- **`v397` (`FR.CLOCK.PROBE.01`) — do the "external" frontier scheme-numbers land on the $\{2, 3, 5\}$ clock?** A falsifiable, anti-numerology probe (strict No-Free-Pattern, v100: "on the clock" means an *exact* match, not a wide-band fit). [E] the η_B decuple $A_\Lambda = 10 = |Z_2|g_{\text{car}}$ is ON the clock (v212); [E] the Koide rate $(2/3)^6 = (|Z_2|/N_{\text{fam}})^6$ is ON the clock (v303); [E] the axion spine angle $\theta_i = \pi N_{\text{fam}}/g_{\text{car}} = 3\pi/5$ is ON the clock (v211/v373); [E] the proton $C_p \sim 2.83 \pm 0.15$ is *not* a discriminating clock value — within its $\pm 5.3\%$ band several simple values fit $(2^{3/2}, 17/6, 14/5)$, so by No-Free-Pattern it is declared EXTERNAL (the firewall v187/v374 is correct there). [C] verdict: 3 of the 4 frontier scheme-numbers land *exactly* on the clock, so the irreducible "external physics" residue shrinks to essentially the proton's non-perturbative QCD number C_p. Cited in `tfpt_4_frontier`; a falsifiable probe whose answer could have been "none on the clock"; Python-only.

2026-06-24 (Clockwork coherence + an open question: v394 primes = atoms = facets, and `RES.COXETER.SYMMETRY.01`)

- **v394 (`COXETER.ATOMS.01`) — the three Coxeter primes ARE the three anchor atoms ARE the three number-field facets.** A bird's-eye [E] re-reading tying v390 (the prime facets) to the anchor $a = (1, 1, 2)$ and v319 (the 5×6 clock). [E] $a = (1, 1, 2)$ has $e_3{=}2{=}|Z_2|$, $p_0{=}3{=}N_{\mathrm{fam}}$, $e_2{=}5{=}g_{\mathrm{car}}$, product $2{\cdot}3{\cdot}5 = 30 = h(E_8)$ — the clock's primes *are* the anchor's atoms. [E] atoms = facet fields (v390): $2{\to}\mathbb{Q}(i)$, $3{\to}\mathbb{Q}(\sqrt{-3})$, $5{\to}\mathbb{Q}(\sqrt{5})$, each ramified at its prime. [E] the QG susceptibility is pure $\{2, 3\}$: $\chi = 1/(1{-}(2/3)^6) = 729/665 = 3^6/(3^6{-}2^6)$, $(2/3)^6 = 2^6/3^6$, exponent $6 = 2N_{\mathrm{fam}}$. [E] both dynamic rates carry prime-2: $2/3 = |Z_2|/N_{\mathrm{fam}}$ and $(\varphi{+}2)/4 = (\varphi{+}2)/|\mu_4|$ with $|\mu_4|{=}4{=}2^2$. [E] $30 = 5{\times}6 = g_{\mathrm{car}}(2N_{\mathrm{fam}})$ (v319). [C] the prime-2 seam is the universal dynamical engine; prime-3 (family) and prime-5 (carrier golden) the two attractors.
- **`RES.COXETER.SYMMETRY.01` — a new registered open research question.** Is the prime-2-as-common-factor reading *forced* (the seam involution provably in every sector's gap, no independent prime-2 attractor), or is there a residual asymmetry? Resolving it would either tighten v383's "every sector is the same gapped operator" to "the same *seam-driven* operator" (with the falsifiable corollary that no prime-2-only attractor exists), or reveal a missing dynamical slot. Tracked in `tfpt_research_contracts`; a question, not a claim. Cited in `origin_theory`; Python-only.

2026-06-24 (Correction error-bars: v393 the typed budget as actual first-correction magnitudes)

- **v393 (`CORRECTIONS.NUMERIC.01`) — the typed correction budget (v388) as actual numbers.** The short-term, testable harvest of v388: instead of merely typing each prediction's correction, this gives the magnitude, so a parameter-free central value can be quoted as "value ± computed first correction". [E] FIXED-POINT band (the φ_0-derived $\theta_{12}, \theta_{13}, \lambda_C, \beta_{\mathrm{rad}}$) = the explicit φ_0 puncture term $36c_3^4/c_3 = 9/(128\pi^3) \approx 0.227\%$ (exact; $= \theta_{12}$'s ε first-correction, $\varepsilon = (3/4)\varphi_0 = c_3{+}36c_3^4$); [E] SEAM-GAPPED band (Koide F_{pole}, recovery) $= (2/3)^6 \approx 8.78\%$, the QG-decoupling bound capped by $\chi - 1 = 64/665 \approx 9.62\%$ (v337); [E] EXACT-IDENTITY band = 0 structurally ($\det R = 8$, $N_\Phi = 1$, $\theta_{\mathrm{eff}} = 0$); [E] EXTERNAL-RATE band = external physics, quoted (m_p/m_e ±7.5% v374; n_s/r the $N_\star{\in}[50, 60]$ band; η_B 1.07× v212), fenced by v187. [C] the assembled budget gives each prediction a computed theoretical-uncertainty band. Cited in `origin_theory`, mirrored on the website prediction surface; a harvest of v388, no new number; Python-only.

2026-06-24 (Attacking the last open lemma: v392 the scaling-limit central charge *converges*)

- **v392 (`SEAM.S3.SCALINGLIMIT.01`) — an honest attack on `SEAM.EQUIV.01`'s one open lemma (continuum existence, v336).** It does *not* close v336 (the abstract existence is the cited MMST theorem); it strengthens the case with the existence-relevant observable v376 did not test: that the finite-size data *converge*. [E] the Calabrese–Cardy entanglement slope $c_1(L)$ at $L = 66, 130, 258, 386, 514$ is a monotone Cauchy-like sequence approaching 1 ($|c_1(L) - 1|$ strictly decreasing, $4.6 \times 10^{-4} \to 7 \times 10^{-6}$); [E] a $1/L$ Richardson fit extrapolates to $c_1(\infty) \approx 1.000$, so the 16-Majorana collar $c = 8c_1 \approx 8.00 = g_{\mathrm{car}}{+}N_{\mathrm{fam}}$ is reached *in* the limit; [E] the successive differences are themselves strictly decreasing (a numerical Cauchy sequence). [O] convergent data are evidence, not a proof of existence (the cited MMST theorem, v336), and do not pin $(E_8)_1$ over the same-c $SO(16)_1$ (the holomorphy discriminator, v377/v378). Strengthens the residual, does *not* close it. Cited in `tfpt_research_contracts`; Python-only.

2026-06-24 (Lean 4: `FORM.SPECTRALGAP.01` extended to the finite-dimensional Perron–Frobenius case)

- **`FORM.SPECTRALGAP.01` — the multi-dimensional case, reduced to the scalar core.** The Lean module `TfptCarrier/SpectralGapAttractor.lean` now also proves the *finite-dimensional* gapped-operator statement (`lake build` OK, no `sorry`): **[E]** `component_tendsto` (a non-dominant eigencomponent $|r| < 1$ decays, $r^n c \to 0$); **[E]** `deviation_tendsto` (with the dominant eigenvalue normalised to 1, the deviation vector — all non-dominant components — tends to 0, so the iterate converges to the dominant eigendirection, the unique attractor, for any start); **[E]** `deviation_unique` (two starts give deviation vectors both $\to 0$, so the selected eigendirection is independent of the start — parameter-freeness in finite dimensions); **[E]** `flavor_deviation_tendsto` (a concrete instance on the cusp-transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$, v56/v82/v383). Honest scope: this is the finite-dimensional *diagonalisable* case (the genuine multi-dimensional content, reduced to the scalar core); the general non-diagonalisable / infinite-dimensional Perron–Frobenius theorem stays the cited statement. Referenced in `origin_theory`.

2026-06-24 (Website: an animated "Universal Gap Lab" visualising `v383–v391`)

- **Website — the `UniversalGapLab` animation on `/verification`.** A new client component (Framer Motion, no new dependency) with three linked, module-grounded views: (A) every sector as the gap contraction $\mathrm{iter}_n = x_\star + r^n(x_0 - x_\star)$ racing to its attractor at $r = (2/3)^6$ (seam-gapped, v303/v387/v388) or the golden $(\varphi + 2)/4$ (v312), with a gapless $r = 1$ track that never forgets its start (the Lean theorem `FORM.SPECTRALGAP.01`) — the rate *is* the first-correction size (v388); (B) the entire-form-factor graviton ratio e^{-u} falling faster than any power, ghost-free and UV-soft (v386/v389); (C) the residual matrix as certification, not construction — every open item external, zero open internal mechanisms (v384). Mirror-only: every number is grounded in a module; no theory change.

2026-06-24 (Honest attempt at an external target: `v391` reduces it, not a closure)

- **`v391` (`ALPHA.QUILLEN.PROGRESS.01`) — an honest attempt at the external target `ALPHA.QUILLEN.EXACT.01` (v382, door 5).** It REDUCES the open step to a sharper named sub-target via a solvable model but does *not* close it: `ALPHA.QUILLEN.EXACT.01` stays **[O]** (external math, type A, v384). **[E]** a solvable model $\det_\zeta(-\partial_\tau^2 + m^2)$ on a circle of circumference β equals $(2\sinh(\beta m/2))^2$, whose m-variation $\beta\coth(\beta m/2) = 2/m + (\beta^2/6)m - (\beta^4/360)m^3 + \ldots$ is a closed heat-kernel/Laurent series (the m^3 coefficient $-1/360$ verified). **[E]** so a ζ-determinant's coupling-variation is structured heat-kernel data, and v382's counterterm $8b_1c_3^6$ (a Gauss–Bonnet heat-kernel power) is the right *kind* of object; the v382 split is re-verified at $\alpha^{-1} = 137.0359992$. **[E]** the reduction: the open step now reduces to the named sub-target "the seam $U(1)$ operator's Seeley–DeWitt coefficient $= 8b_1c_3^6$". **[O]** *not closed*: the actual seam-operator ζ-determinant proof stays open, and `SEAM.EQUIV.01` continuum existence (MMST, v336) stays the other external **[O]**; the value α^{-1} stays **[E]**. An attempt that reduces, does not close. Cited in `tfpt_1_architecture_e8`; Python-only.

2026-06-24 (Lean 4: `FORM.SPECTRALGAP.01` formalises the scalar core of the universal spectral-gap principle)

- **`FORM.SPECTRALGAP.01` — a Lean 4 proof of the gap ⇒ unique-attractor ⇒ parameter-free core (door 4).** New module `TfptCarrier/SpectralGapAttractor.lean` (`lake build` OK, no `sorry`, only the three standard kernel axioms), formalising the meta-theorem of v303/v383/v387 at the one-dimensional reduction — the affine gap contraction $x \mapsto x_\star + r(x - x_\star)$ with multiplier $r = \lambda_2/\lambda_1$. **[E]** `iter_sub`: the closed form $\mathrm{iter}_n - x_\star = r^n(x_0 - x_\star)$ (general n); **[E]** `iter_tendsto`: $0 \le r < 1$ (the gap) ⇒ $\mathrm{iter}_n \to x_\star$ (the unique attractor); **[E]** `starts_agree`: two starts differ by $r^n(a-b) \to 0$, so the attractor is independent of the initial state (parameter-freeness); [E] `no_gap_no_forget`: $r = 1$ (no gap) leaves $\mathrm{iter}_n - x_\star = x_0 - x_\star$ constant — the initial condition is then a free parameter, so the gap is exactly what removes it; [E] the seam rate $(2/3)^6$ and the golden $(\varphi + 2)/4 = (5 + \sqrt{5})/8$ both lie in $[0, 1)$. Honest scope: the scalar reduction (the contraction core v303/v383 iterate numerically); the full multi-dimensional Perron–Frobenius statement stays the cited theorem. Referenced in `origin_theory`; ledger row `FORM.SPECTRALGAP.01`.

2026-06-24 (Completing the Coxeter clock: v390 the prime-2 (Gaussian / $\mathbb{Z}/4$-seam) facet of $30 = 2{\cdot}3{\cdot}5$)

- **`v390 (COXETER.PRIME2.01)` — the prime-2 facet of the order-30 Coxeter clock.** v383/v387 named two number-field facets of the clock $30 = 2{\cdot}3{\cdot}5 = h(E_8)$ — the golden $\varphi \in \mathbb{Q}(\sqrt{5})$ (prime-5, carrier) and $(2/3)^6 \in \mathbb{Q}$ (prime-3, family). This names the *prime-2* facet and identifies it with structure already in the theory: the Gaussian field $\mathbb{Q}(i)$, i.e. the CM point $\tau = i$ and the order-4 $\mu_4 = \mathbb{Z}/4$ *square* seam deck (v282/v288/v333). [E] $h(E_8) = 30$ factors into exactly three primes; one quadratic facet field per prime, each ramified at its prime: $\operatorname{disc}\mathbb{Q}(i) = -4$ (2 ramifies), $\operatorname{disc}\mathbb{Q}(\sqrt{-3}) = -3$ (3 ramifies), $\operatorname{disc}\mathbb{Q}(\sqrt{5}) = 5$ (5 ramifies); the ramified primes are $\{2, 3, 5\}$ with product $30 = h(E_8)$. [E] the prime-2 (Gaussian) facet is the $\mathbb{Z}/4$ seam deck: min. poly $x^2 + 1$, $j(i) = 1728 = 12^3$, $\tau = i$ fixed by the modular S of order $4 = 2^2$, $|\mu_4| = 4 = 2^2$ — the static, even, seam-orientation facet. [C] so the three primes of the clock now each carry a facet (square/hexagon/pentagon = seam/CP/carrier); the prime set $\{2, 3, 5\}$ is forced by $h(E_8) = 30$, not chosen. Cited in `origin_theory`; a synthesis, no new number; Python-only.

2026-06-24 (Graviton loop finiteness: v389 the entire form factor is UV-finite at the loop level by power counting)

- **`v389 (GRAV.LOOP.FINITE.01)` — the entire-form-factor graviton is UV-finite at the *loop* level.** The honest next step past v386 (which established the tree amplitude), extending v304/v370/v380/v386 from the propagator to the superficial degree of divergence of loop graphs. This is the standard infinite-derivative / nonlocal-gravity finiteness statement (Tomboulis; Biswas–Mazumdar–Siegel; Modesto), *not* a full proof of perturbative quantum gravity. [E] super-polynomial falloff: $p^k e^{-p^2/M^2} \to 0$ for every degree k. [E] one dressed line turns a quadratic divergence into the finite $(M^2/2)e^{-p_0^2/M^2}$, while the GR $\int p^3/p^2\,dp$ diverges quadratically. [E] the superficial degree $\to -\infty$ at every loop order L (the form factor multiplies the integrand by $e^{-(\#\text{internal})p^2/M^2}$; dressed cutoff-ratio $\to 1$ vs GR $\to 4$). [E] the regulator is the seam scale $M = M_{\text{scal}} = c_3^{7/2}\bar{M}_{\text{Pl}}$ (v253/v259), not a dial. [C][O] loop *unitarity* of the nonlocal form factor (Lorentzian continuation / macro-causality, Pius–Sen / Briscese–Modesto) stays the honest open point — tree unitarity is v386; not claimed solved. Cited in `tfpt_4_frontier`; Python-only.

2026-06-24 (Typed correction budget: v388 the gap correction is keyed to the mechanism class, not a uniform band)

- **`v388 (CORRECTIONS.BUDGET.01)` — the gap-driven correction (v387) is a *typed* budget.** A machine-checked answer to "does the $(\lambda_2/\lambda_1)^n$ correction apply to *all* predictions?" — the answer is *no*, and for one whole class a band would be *wrong*. Because the size is the distance from a gapped operator's leading eigenvector, it is defined only where a subleading λ_2 exists, so the prediction surface splits into four classes. [E] *seam-gapped* (Koide F_{pole} v303, recovery v221, the QG bound capped by $\chi = 729/665$ v337): rate $(2/3)^6$; the discrete compiler: golden $(\varphi + 2)/4$ (v312). [E] *exact identity* ($\det R = 8$, $N_\Phi = 1$, $\theta_{\text{eff}} = 0$, $\sin^2\theta_{13} = \varphi_0 e^{-5/6}$): no λ_2, so the correction is 0 structurally and a band would contradict the [E] status. [E] *external-rate*

(η_B washout, axion freeze, m_p/m_e RG): the gapped *shape* with an external rate — only $F_{\rm pole}$ has the seam rate (v303), firewalled (v187). [E] *fixed-point*: the texture is already explicit (e.g. $\sin^2\theta_{12}$, $\varepsilon = (3/4)\varphi_0 = c_3 + 36c_3^4$). [E] the Koide first correction (door 2): $F_{\rm pole}$ is the one seam-gapped prediction, rate $(2/3)^6 \approx 0.0878$ (v371), the source quotient 0.33% below 2/3 — the rate is the suppression *scale*, not the residual. [C] so a uniform band on all predictions is wrong; the size is keyed to the mechanism class. A typing of v387 (as v384 is of v383); cited in `origin_theory`; Python-only.

2026-06-24 (Harvesting the QFT/gravity machinery: v385 the UV-branch kill-test surface, v386 the graviton-exchange amplitude, v387 the gap-driven correction calculus)

- **v385 (UVBRANCH.KILLTEST.01) — the carrier-Pati–Salam UV branch on the all-order footing.** A consolidation that ties the all-order EG/BRST closure (v381) to the optional UV branch and computes the one genuinely new thing: the *proton-decay safety hierarchy*. [E] the carrier content is the SM with no new states, so the SM-non-unification (v246) is now all-order-robust; [X] the plain-SM 4D-GUT stays killed ($\sin^2\theta_W = 3/8$ vs 0.231); [C] the optional carrier-PS branch sits at $M_{PS} = \kappa M_{\rm scal} \approx 3\times10^{13}$ GeV ($\kappa \in [1.0, 1.2]$, v253); [E] minimal PS gauge bosons are $SU(4)$ leptoquarks that mediate rare LFV ($K_L \to \mu e$), *not* $p \to e^+\pi^0$, so the branch clears the binding bound by $M_{PS}/M_{\rm bound} \sim 10^7$ – proton-safe, with *no* fake τ_p window (v265). Cited in `tfpt_5_redteam`; Python-only.
- **v386 (GRAV.AMPLITUDE.01) — the graviton-exchange amplitude is finite and UV-soft.** Extends v304/v370/v380 from the propagator to the actual tree amplitude. [E] the dressed spin-2 propagator $e^{-p^2/M^2}/p^2$ has its only pole at $p^2 = 0$ (residue $1 > 0$, ghost-free), is exponentially softened in the UV (the ratio e^{-u} falls faster than any power) and reduces to GR + a calculable $-p^2/M^2$ correction at low energy ($M = M_{\rm scal}$, v253); [C] a single positive-residue pole gives tree unitarity, so perturbative graviton scattering $A \sim (T\cdot P_2\cdot T')e^{-p^2/M^2}/p^2$ is an explicit finite computation (perturbative-only; `QG.AMB.01` stays a [C] redundancy, v369). Cited in `tfpt_4_frontier`; Python-only.
- **v387 (CORRECTIONS.GAP.01) — the gap-driven correction calculus.** The practical harvest of v383: the same spectral gap that makes a sector parameter-free also sets the *size* of its first correction (correction$_n \sim (\lambda_2/\lambda_1)^n$). [E] flavor, horizon recovery and the QG bound all share the $(2/3)^6$ rate (the family-3 facet; QG susceptibility $\chi = 729/665$), while the discrete compiler transient decays at the golden $(\varphi+2)/4$ (the carrier-5 facet) — the two number-field facets of v314/v383. [C] the gap does double duty (why no free parameter *and* how big the first correction), so sub-leading magnitudes are organised, not free. Cited in `origin_theory`; Python-only.

2026-06-23 (Bird's-eye synthesis: two new structural patterns — v383 the universal spectral-gap principle and v384 the residual is certification, not construction)

- **v383 (DYNAMICS.UNIVERSAL.01) — the universal spectral-gap / Perron–Frobenius principle.** A fresh bird's-eye pass found that *every* TFPT sector is the *same* structural object: a gapped operator with a *unique* leading attractor (the physics) and a spectral gap (the reason there is no free parameter). v303 named this only for $F_{\rm transfer}$; this module extends it to the sectors made parameter-free *after* it. [E] three gaps are computed — the flavor transfer $\{1, (2/3)^6, (1/3)^6\}$ with gap $6\ln(3/2)$ (v56/v82); the affine-E_8 adjacency with Perron eigenvalue 2 (Kac marks) and subleading $2\cos(\pi/5) = \varphi$, gap $2 - \varphi = 1/\varphi^2$ (v312/v313); the decoupling margin $\Delta_{\rm eff} \approx 1.648$ (v76/v337). [E] the two number-field facets (v314): golden $\varphi \in \mathbb{Q}(\sqrt{5})$ (carrier-5, static) and $(2/3)^6 \in \mathbb{Q}$ (family-3, dynamic) are the prime-5 and prime-3 facets of the order-30 Coxeter clock $30 = 2\cdot3\cdot5 = h(E_8)$. [C] the other sectors are the same structure — gravity (δS=0 equilibrium, v358/v359), the QG measure (Gaussian saddle, v365), the boundary QFT (holomorphic c=8 RG fixed point, v157/v158), the all-order S-matrix (adiabatic limit, v381), recovery (Petz fixed point, v221), τ=i (v333). [C] the meta-theorem: gapped ⇒ unique attractor ⇒ parameter-free, theory-wide. Cited in `origin_theory`; a synthesis (no new number), Python-only.

- **`v384 (RESIDUAL.CERTIFICATION.01)` — the residual is certification, not construction.** After v369/v381/v382 the *kind* of every open item has shifted: there is *no* open TFPT physics mechanism left. A ledger-CI audit classifies every residual into exactly three external kinds: [E] (A) external math proof (`SEAM.EQUIV.01` continuum existence = the cited MMST/Adamo theorem v336; `ALPHA.QUILLEN.EXACT.01` v382; `QG.AMB.01` the constructive measure, a [C] redundancy v369), [E] (B) theorem-forbidden (v_{geo}, the No-Unit theorem v153/v364), [E] (C) external physics (F_{transfer}, v371–v374, firewalled v187). The two load-bearing facts are verified (No-Unit dimensionlessness; the gap-decoupling margin $1.648 > 0$), and the count of open *internal* TFPT mechanisms is **zero**. [C] convergence: the only hard things left are theorems and external physics, not missing dynamics. Cited in `tfpt_research_contracts`; an audit (no new number), Python-only.

2026-06-23 (External-review closure: the two genuine residual targets named — v381 the all-order Epstein–Glaser/BRST contract for S_{pert}, and v382 the Quillen determinant-line variation as a tracked target)

- **`v381 (QFT4D.EG.ALLORDER.01)` — the all-order 4D perturbative-QFT closing statement.** A mathematical-physics review of the QFT/TOE layers identified exactly one genuinely new, buildable item: the *all-order* Epstein–Glaser/BRST contract for the perturbative 4D S-matrix S_{pert} (the previous documents had the S_{pert} skeleton and one-loop, v269/v271/v273/v278, but no named all-order claim). This module types the standard causal-perturbation-theory closure for the TFPT finite interaction class — it does *not* build a path integral. [E] power-counting: every gauge+matter vertex (Yang–Mills + Dirac–Yukawa + Higgs + ghost) is a mass-dimension-4 marginal operator, so the EG singular order $\omega \le 4$ and the counterterm space is finite at every order; [E] BRST nilpotency s^2=0 verified exactly via the Jacobi identity for $su(2)$ and $su(3)$ (the carrier weak+colour content); [E] the seam gap $\Delta = 6\log\frac{3}{2} > 0$ (v302) gives the IR-safe adiabatic limit. [C] the two heavy legs are imported rigorous theorems — all-order T_n existence by causal factorisation (Epstein–Glaser 1973) and all-order BRST invariance (Piguet–Sorella / Kugo–Ojima) — applied to the TFPT class. Matter+gauge *only*: the R^2/Weyl^2 gravity sub-sector is the entire-form-factor resummation (v304/v370/v380), and S_{pert} is not the ambient measure `QG.AMB.01` (a [C] redundancy, v369). NET [C], prerequisites [E]; not [E] and not a path integral. Cited in `tfpt_5_redteam` and `tfpt_2`; Python-only.
- **`v382 (ALPHA.QUILLEN.EXACT.01)` — the Quillen determinant-line variation, named as a target.** Per the same review, the "why *this* $F_{U(1)}$ functional" obligation (previously tracked only as `EM.WARD.01`'s residual, v341/v342) is elevated to its own citable target: $\delta_\tau(\log\det_\zeta \Delta_{U(1)} + 8b_1c_3^6\log\varphi_{\text{seam}}) = 0$ on the determinant line over the μ_4 seam moduli. [E] the Quillen split holds at the α^{-1}=137.0359992 root with every coefficient a named atom (index $8b_1$, heat-kernel $c_3^{3,6}$, discriminant $q(D_5)$+$q(A_3)$=2); [E] the value stays closed (v3); [O] the from-first-principles exactness proof is the target. [C] it is a *face* of `SEAM.EQUIV.01` (v336) — the seam $U(1)$ determinant line lives on the same holomorphic net — so it adds *no* second independent gate; it makes the existing residual trackable. No new number (reuses the Wolfram-mirrored v341 identities). Cited in `tfpt_1_architecture_e8`; Python-only.

2026-06-23 (Track 2 of the forward plan v2 — the AMBIENT REDUNDANCY statement (v369): the holographic route to Gravity-complete that sidesteps building the general Euclidean QG measure)

- **`v369` (`QGAMB.REDUNDANCY.01`) — Track 2 (forward plan v2).** Instead of constructing the non-perturbative ambient measure (the conformal-factor swamp, v332/v334), assemble the established discriminators that prove the ambient sector carries *no* physical degrees of freedom relative to the holomorphic boundary algebra, so the QG measure (gate C7/`QG.AMB.01`) is a non-fundamental *certification* object (like a coordinate choice in GR), not missing dynamics. **[E]** holomorphy $\Rightarrow$ $\mathrm{DHR}((E_8)_1) = \mathrm{Vec}$: $\#\text{primaries} = |\det \mathrm{Cartan}| = 1$ for E_8 vs 4 for $SO(16)$ $\Rightarrow$ no nontrivial superselection charge a bulk could carry invisibly; **[E]** no torus ground-state degeneracy ($|\det K| = 1$); **[E]** finite Petz recovery rate $(2/3)^6$ (v221); **[E]** the gap-decoupling margin $\Delta_{\text{eff}} = 6\log\frac{3}{2} - \frac{31}{4\pi^2} \approx 1.648 > 0$ (v76/v337). **[C]** with the Bisognano–Wichmann reconstruction map (v258/v309/v323) the redundancy statement $\mathcal{O}^{\text{bulk}}_{\text{phys}}/\mathrm{Diff} \cong \mathcal{A}^{\mu_4}_{\Sigma}$ is well-posed. **[O]** residual: the two consumed premises — `SEAM.EQUIV.01` (the S3 lattice input, v366) and the BW intrinsicality of the raw collar (v329); it does *not* construct `QG.AMB.01` but provides the *alternative* (holographic redundancy) route: Gravity-complete = Boundary-complete + Ambient-redundancy. Cited in `tfpt_research_contracts`; Python-only.

2026-06-23 (Status propagation: the v365–v380 + `FORM.SEAM.MMST.01` status carried into the remaining papers, the `README` and the Zenodo description)

- **Deep-sync of the post-S3 status.** The canonical surfaces (`tfpt_status`, `introduction`) were already current; this pass propagated the same honest wording into `origin_theory`, `tfpt_1`–`tfpt_5`, `tfpt_research_contracts`, the root `README.md`, the Zenodo description and `website/lib/papers.ts`: `SEAM.EQUIV.01` is *closed modulo a cited theorem* (lattice v367/v368 + the S_3 stack v376–v379 + Lean `FORM.SEAM.MMST.01`; residual **[O]** = the cited MMST/OS existence only); `QG.AMB.01` is a **[C]** *redundancy* (v369/v379), a certification object not missing dynamics; perturbative graviton unitarity is established (the Stelle ghost is a Seeley–DeWitt truncation artefact, v304/v370/v380); and the frontier transfers are typed runnable solvers with kill tests (v371–v374) + the observatory CI (v375). No new claim — a wording propagation so every surface matches the ledger; `AUDIT OK`.

2026-06-23 (`v380` — the KMS Entire Hessian: the Stelle ghost is EXACTLY the Seeley–DeWitt truncation, and resummation pushes it to infinity)

- **`v380` (`GRAV.KMS.HESSIAN.01`).** Upgrades v304/v370 from the *assumption* “the resummed graviton form factor is entire” to a derived statement. **[E]** the seam KMS cutoff $f(u)=e^{-u}$ (v259) gives the dressed spin-2 propagator $e^{-p^2/M^2}/p^2 = 1/(p^2 a)$ with $a(u)=e^u$ entire and nowhere zero (only the $p^2=0$ pole). **[E]** each finite Seeley–DeWitt truncation is a Taylor partial sum of a and carries a Stelle-ghost zero ($T_1=1+u \to u=-1$; $T_2 \to -1\pm i$); the entire a has none. **[E]** the modulus of the nearest truncation-zero grows monotonically with the order $(1, 1.41, \ldots, 3.07$ for $n=1..8) \to \infty$ — the ghost decouples, “entire” = “do not truncate”. **[C]** $\Rightarrow$ perturbative graviton unitarity; **[O]** the exact off-shell curved-space Hessian identification is the cited heat-kernel resummation, perturbative-only. Cited in `tfpt_5_redteam` (Target F); Python.

2026-06-23 (Lean: the S3 continuum leg formalised as “closed modulo a cited theorem” — `FORM.SEAM.MMST.01`)

- **`FORM.SEAM.MMST.01` — Lean 4 formalisation.** The S3 continuum leg is now machine-formalised in a new module (`SeamScalingLimit.lean`), lake build OK and `#print axioms` clean, in the audit-contract style of `SeamEquivChain.lean`. The MMST applicability hypotheses for the seam collar are *provable* by kernel `decide` (no axioms): $D = 16$, rank $= c = 8$, the range $8 \leq 8 \leq 16$, $c_- = 8 \neq 0$, and $\det K = 1$ vs $SO(16)$'s 4. The MMST scaling-limit theorem (CMP 2022) and the Adamo–Moriwaki–Tanimoto OS reconstruction (2024) are *named cited axioms*; the conclusion $(E_8)_1$ follows by a well-typed composition whose residual (`#print axioms`) is exactly those two published theorems plus the carrier gap — no hidden assumption, no `sorry`. Mirrors the S3 closure stack `v376`–`v379`; so `SEAM.EQUIV.01` is now "closed modulo a cited theorem", the honest formalisation ceiling. Cited in `tfpt_research_contracts`; `lean_manifest` refreshed.

2026-06-23 (S3 closure stack — the target of the seam scaling limit pinned at EVERY computable level: central charge c=8 from the lattice (v376), the $(E_8)_1$ character (v377), the genus-1 torus count = 1 (v378), and reflection positivity (v379); only the abstract continuum existence (cited MMST) stays external)

- **v376 (SEAM.S3.CENTRALCHARGE.01).** The Calabrese–Cardy finite-size entanglement-entropy scaling of the critical free-fermion content gives $c = 1.0000$ per complex mode (Peschel correlation-matrix EE; the L=$4m$+2 sizes avoid the half-filling zero-mode), so the 16-Majorana (= 8 complex) collar has $c = 8 = g_{\text{car}}+N_{\text{fam}}$, chiral ($c_-$=8, v367/v368). Numerical c=8 from the lattice. Python (numpy).
- **v377 (SEAM.S3.E8CHARACTER.01).** The exact affine character $\chi = E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + \dots)$ gives $c = 8$ and 248 level-1 currents (one primary); the μ_4/GSO promotion $248 = 240 + 8 = 120 + 128$ ($240 = 16{\cdot}5{\cdot}3$) pins $(E_8)_1$ over the same-c rival $SO(16)_1$ (120 currents, 4 primaries). Python (sympy q-series).
- **v378 (SEAM.S3.MODULAR.01).** The genus-1 discriminator: KLM condensation μ(carrier)=16 $\to$ $\mu(E_8)$=$16/4^2$=1, $|\det \text{Cartan}(E_8)|$=1 vs D_8=4, and the single character's modular T-eigenvalue $e^{-2\pi i/3}$ give torus ground-state degeneracy 1 (holomorphic = $(E_8)_1$) — the "hard part" v344 isolated. Python (sympy).
- **v379 (SEAM.S3.RP.01).** Reflection positivity (the OS axiom) of the explicit gapped collar: the positive spectral measure makes the Hankel/reflection Gram matrix PSD (a ghost mode breaks it). With c=8 + $(E_8)_1$ in hand, the OS inputs are complete and C7/`QG.AMB.01` is then discharged by the redundancy theorem v369. Python (numpy).
- **Net:** S3 is the master key — on the explicit lattice model the target net $(E_8)_1$ is pinned exactly (central charge, character, genus-1 count, RP); the only remaining residual is the abstract *continuum existence* of the scaling limit (the cited MMST theorem v336), which lives outside the computational suite. Cited in `tfpt_research_contracts`.

2026-06-23 (Track 4 — the prediction OBSERVATORY: a status-typed CI over the frozen prediction registry that makes the falsifiability surface machine-checkable, with a live data scorecard (v375))

- **v375 (OBSERVATORY.REGISTRY.01).** A CI over `freeze_file.csv`: [E] every prediction carries a kill criterion (falsifiability complete, no orphan claims); [E] the headline values re-derive from $\{\varphi_0^{\text{ret}}\}$ ($\sin^2\theta_{12}$=$1/3-\varphi_0^{\text{ret}}/2$, $\sin^2\theta_{13}$=$\varphi_0^{\text{ret}}e^{-5/6}$, β_{rad}=$\varphi_0^{\text{ret}}/(4\pi)$); [C] a live scorecard records θ_{12} vs the JUNO first run (0.28σ, compatible), θ_{13} vs NuFIT 6.0 NO (2.0σ, the flagged most-tensioned core prediction), β_{rad} vs ACT DR6 (0.37σ, a hint), $r < 0.036$ (compatible). Tooling that makes the closed pieces usable and the open ones honestly fenced; no new physics. Cited in `tfpt_5_redteam`; Python (stdlib).

2026-06-23 (Track 3 — the F_{transfer} functor promoted from contracts to TYPED runnable solvers with kill tests: Koide source→pole (v371), η_B Boltzmann (v372), axion finite-T relic (v373), m_p/m_e uncertainty budget (v374))

- **v371 (FR.POLE.SOLVE.01).** A clean *negative*: standard QED/EW running cannot be the Koide source→pole transfer (it pushes Q above 2/3; the required ϵ is negative while QED's is positive), so the transfer is the non-QED 53/54 operator / $(2/3)^6$ Moebius generator (v183/v82). [C], plain running excluded as the kill. Python (numpy+scipy).
- **v372 (FR.BOLTZMANN.SOLVE.01).** The integrated Buchmueller–Di Bari–Pluemacher Boltzmann ODE at the TFPT-frozen $M_1{=}M_{\text{scal}}(\varphi_0^{\text{ret}})^2/A_\Lambda$ (v212) gives η_B within a factor $\sim$1.1 of $6.1{\times}10^{-10}$ with no M_R dial; replaces the v326 toy washout. [C]/[X].
- **v373 (FR.RELIC.SOLVE.01).** The finite-T axion misalignment ODE decides the branch: the frozen spine angle $3\pi/5$ gives $\Omega_a h^2{\sim}0.12$ (in $[0.08, 0.16]$, lands on Ω_{DM} untuned) while the hilltop 170° over-produces ($\sim$5.4×); replaces the v326 toy. [C]/[X](lattice $\chi(T)$ input).
- **v374 (FR.QCD.BUDGET.01).** The m_p/m_e uncertainty budget: with $b_3{=}-7$ [E] and the two declared externals ($\alpha_s(M_Z)$, C_p, $\sim$5.3% each), the central $\sim$1824 (v262) carries a propagated ±7.5% band $[\sim 1690, \sim 1960]$ CONTAINING 1836.15; lattice tightening is the kill. The falsifiability companion to v262; [C].

2026-06-23 (Track 1 — the S3 input made CONCRETE: an explicit gapped chiral-Majorana ($p{+}ip$) lattice model with numerical Chern $|C|{=}1$ realises the $c_-{=}8$ edge (v367), and a strip shows the chiral edge exists by anomaly inflow (v368))

- **v367 (SEAM.S3.LATTICE.01) — Track 1.** The abstract S3 residual of v356/v366 ("the collar realised as a genuine lattice chiral free-fermion invertible phase") is turned into a RUNNABLE model. [E] the $p{+}ip$ Bloch model is gapped at $M{=}1$ with a numerically computed Fukui–Hatsugai–Suzuki Chern number $|C|{=}1$ ($C{=}0$ trivially at $M{=}3$); [E] $N_{\text{Maj}}{=}\dim S^+{=}16$ copies give $c_-{=}16/2{=}8{=}g_{\text{car}}{+}N_{\text{fam}}$, with the edge K-matrix $SO(16)_1$ (det 4) → $(E_8)_1$ (det 1) by the order-4 μ_4 clock (v351/v369). [C] the μ_4 condensation (v154/v351); [O] the continuum scaling-limit existence (MMST, v336) stays open. Cited in `tfpt_research_contracts`; Python (numpy + sympy).
- **v368 (SEAM.S3.INFLOW.01) — Track 1.** The bulk-edge / anomaly-inflow leg (S4) made concrete on the v367 model: [E] a finite strip has an edge-localised in-gap chiral branch ($\min_{k_x}|E| \sim 0$) in the topological phase and a clean gap in the trivial phase (neg control); [E] bulk-edge correspondence gives one chiral Majorana per edge ($c_-{=}8$). [C] so by anomaly inflow the chiral edge EXISTS as a consequence of the invertible bulk, discharging the v336/v351 "does the edge exist?" residual on the lattice. [O] the continuum scaling limit stays open. Cited in `tfpt_research_contracts`; Python (numpy).

2026-06-23 (Track 2 — perturbative graviton unitarity sector-by-sector: the Barnes–Rivers spin decomposition localises the Stelle ghost in the spin-2 sector, cured by the entire KMS form factor; the spin-0 mode by the GHP contour (v370))

- **v370 (GRAV.SPIN2.UNITARITY.01) — Track 2.** Extends v304 (the scalar $1/(p^2(p^2{+}M^2))$ pole algebra) to the actual tensor spin structure. [E] the Barnes–Rivers projectors are complete ($\operatorname{tr}P_2{=}(d{-}2)(d{+}1)/2$, $\operatorname{tr}P_1{=}d{-}1$, $\operatorname{tr}P_{0s}{=}\operatorname{tr}P_{0w}{=}1$ sum to $d(d{+}1)/2$; $d{=}4$ split $5{+}3{+}1{+}1{=}10$), the EH propagator is $P_2/p^2 - \frac{1}{d-2}P_{0s}/p^2$, and the Stelle ghost is a *spin-2* state. [E] the entire KMS form factor e^{p^2/M^2} (v304/v259) makes the spin-2 sector ghost-free; [C] the wrong-sign spin-0 conformal mode (v332) is cured separately by the GHP contour (v334). [C] so the full *perturbative* graviton is ghost-free sector-by-sector (conditional on v304's entire-analyticity assumption). [O] residual: perturbative-only + the open analyticity assumption; not the non-perturbative measure. [O] DECLINE: tr P_2=5 is dimension counting, not g_{car} (v354/v355). Cited in `tfpt_5_redteam` (Target F); Python-only.

2026-06-23 (Directions 3 & 7 — the two hard functional-analysis residuals ADVANCED: the QG measure as one-loop fluctuations around the parameter-free saddle (v365), and the seam collar verified IN MMST's free-fermion class so the scaling limit is $(E_8)_1$ (v366))

- **`v365` (`QGAMB.SADDLE.01`) — Direction 3.** With the parameter-free saddle (v358/v359) the QG *measure* (gate C7/`QG.AMB.01`) organises as a one-loop Gaussian fluctuation determinant. [E] the fluctuation stiffness is $1/c_3 = 8\pi$ (no free dial, so the one-loop problem is now well-posed); [E] the fluctuation operator is gapped $M \geq \Delta_{\text{eff}} = 6\log\frac{3}{2} - \frac{31}{4\pi^2} \approx 1.648 > 0$ (v76/v337) $\Rightarrow$ mode-by-mode convergence, finite model log-det $\text{Tr}'\log M = 6\log\frac{9}{2}$; [C] the conformal mode converges on the GHP contour (v334); [E] the admissible projective limit exists ($\chi = 729/665$, v330) with 0 new dials (v364). [O] residual: the full non-perturbative projective limit on the ambient sector. Sharpens C7 from "diffuse full QG" to a one-loop Gaussian problem around the parameter-free saddle; does not close it. Cited in `tfpt_research_contracts`; Python-only.
- **`v366` (`SEAM.MMST.INCLASS.01`) — Direction 7.** The seam collar is verified IN MMST's free-lattice-fermion scaling-limit class hypothesis-by-hypothesis: [E] $D{=}\dim S^+{=}16$ Majoranas, $c{=}g_{\text{car}}{+}N_{\text{fam}}{=}8$, rank E_8=8 so rank $\leq c \leq D$ is $8 \leq 8 \leq 16$ (in range); [E] gapped ($\Delta{=}6\log\frac{3}{2}{>}0$, derived v302); [C] quasi-free CAR class (v155/v160); [E] chiral ($c_-{=}D/2{=}8 \neq 0$). [C] the MMST scaling-limit theorem then applies, and [E] the order-4 μ_4 clock pins the target to $(E_8)_1$ ($\det K{=}1$ vs $SO(16)$ $\det K{=}4$, v351). [O] residual: the single S3 input (the collar as a genuine lattice chiral invertible phase, the seam one-sidedness, v297/v356). Reduces the continuum side of `SEAM.EQUIV.01` to one named input; ADVANCES, does not close. Cited in `tfpt_research_contracts`; Python-only.

2026-06-23 (Consistency pass: the parameter-free *full covariant* Einstein equation (v358/v359) propagated across ALL stale paper and website surfaces; new v359 keybox; homepage gravity animation)

- **Deep-sync of the parameter-free gravity status.** A cross-surface audit found several documents and website components still framing gravity as "exact only in the $R + R^2$ regime" or the metric-sector field equation as "open". These were reconciled with the canonical post-v358/v359 status: the *local* covariant field equation $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ is PARAMETER-FREE (both coefficients fixed); what remains open is the Jacobson equation-of-state status, the global ambient measure (`QG.AMB.01`/C7), and the single unit v_{geo}. Touched: `introduction`, `tfpt_2_standard_model`, `tfpt_3_e8_audit_bootstrap`, `tfpt_4_frontier` (summary table), `tfpt_research_contracts`, `origin_theory`.
- **New `v359` keybox** added to `tfpt_research_contracts` (full covariant equation: fixed-volume $\Rightarrow$ Einstein tensor, Lovelock $\Rightarrow$ conservation an output); the v358 residual corrected from "linear→nonlinear" (stale — v359 closed it) to the equation-of-state + C7 + v_{geo} residual, including the generated script index (`script_registry.csv`).
- **Website.** The `GravityEmergence` animation (three origins of c_3 converging to $1/(8\pi)$; the parameter-free Einstein equation) was updated to the full covariant result and surfaced as a dedicated homepage section (`/#gravity`); gravity-status wording aligned across `ReconstructionChain`, `VerificationDag`, `StatusMatrix`, `WhatTfptClaims`, `ThreeDecoderMap`, `StatusPyramid`, `papers.ts`, `glossary.ts` and the review page.

2026-06-23 (Direction 6, $v_{\rm geo}$ sharpened: after the parameter-free gravity, the *gravity* sector adds NO new dimensionful input either, so $v_{\rm geo}$ is the SINGLE dimensionful input of the complete theory — final tally $\{v_{\rm geo}, \pi\}$, ZERO dimensionless dials: `v364`)

- **`v364` (`VGEO.SHARPEN.01`) — Direction 6.** The No-Unit Theorem (v153) already made $v_{\rm geo}$ a forced metrology primitive. [E] The SHARPENING after the parameter-free gravity (v358/v359/v361): the Einstein coefficient $1/c_3 = 8\pi$ and the Λ prefactor $(8\pi)^2{\cdot}48c_3^4 = 3/(4\pi^2)$ are dimensionless, so $1/G \sim v_{\rm geo}^2$ and $\Lambda \sim$ (dimensionless) $v_{\rm geo}^4$ reduce to $v_{\rm geo}$ times atoms — the gravity sector adds no new scale. [E] FINAL TALLY: every dimensionful quantity = (a dimensionless ratio)×(a power of $v_{\rm geo}$); TFPT's free content is $\{v_{\rm geo}, \pi\}$ with ZERO dimensionless dials, a ∼26 → 1 reduction vs the SM. [O] $v_{\rm geo}$ stays irreducibly primitive (No-Unit; $1/G$ UV-sensitive, Sakharov v68). Direction 6 is clarity — the last dimensionful anchor named and shown to be the only one. Cited in `tfpt_research_contracts`; Python-only.

2026-06-23 (Direction 5, the matter–gravity backreaction: the carrier's gravitational anomaly is forced $c_-{=}8$, and the matter vacuum-energy backreaction is finite (→ the forced Λ from α, not $M_{\rm Pl}^4$) — the loop closes; no new SM-quantum-number read-out (declined): `v361`)

- **`v361` (`GRAV.BACKREACT.01`) — Direction 5.** With the parameter-free equation (v359) and the explicit J3 flux (v358), the backreaction of TFPT's carrier matter is computed. [E] the carrier's gravitational anomaly is forced: $c_- = 16/2 = 8 = g_{\rm car}{+}N_{\rm fam}$ (the chiral central charge of the 16-Majorana content). [E] the backreaction is FINITE → Λ from α: the matter vacuum energy through $G_{ab} = c_3^{-1}T_{ab}$ is gap-suppressed (Decoupling Thm v337) to $\rho_\Lambda/\bar M_{\rm Pl}^4 = (3/(4\pi^2))e^{-2\alpha^{-1}}$ (v60), the 123 orders $= 119.03 + 3.92$ — *not* the $M_{\rm Pl}^4$ catastrophe; the loop (v359+v60 + the carrier vacuum) closes. [E] J3 explicit: $\delta\langle K_B\rangle = (8\pi^2R^4/15)\delta\langle T_{00}\rangle$, carrier T_{ab} = the SM's (v159). [C] the SM trace anomaly is reproduced (content = SM). [O] DECLINE: no new atom-forced read-out of individual SM quantum numbers beyond $c_-{=}8$ and Λ — per v354/v355 declined. The backreaction is finite and consistent, adding no new dial. Cited in `tfpt_4`; Python-only.

2026-06-23 (Direction 2, the disciplined result: the gap-induced GR correction is the known R^2 scalaron — already falsifiable via n_s/r; a *new* forced near-term deviation does NOT survive the discriminator and is declined: `v360`)

- **`v360` (`GRAV.GAPCORR.01`) — Direction 2, run honestly.** Does the seam transfer gap $\{1, (2/3)^6, (1/3)^6\}$ yield a NEW, calculable, FORCED deviation from GR (a near-term kill test) beyond the known R^2 scalaron? With the v354/v355 discriminator the answer is *no*, and this is recorded rather than manufacturing a coefficient. [E] the leading higher-curvature term IS the R^2 scalaron (spectral-action $a_4{=}R^2/72$, v36; $M_{\rm scal}^2/\bar M_{\rm Pl}^2 = c_3^7$, exponent $7{=}g_{\rm car}{+}N_{\rm fam}{-}1$) — the calculable beyond-Einstein correction, already in TFPT. [C] it is the entanglement-equilibrium NEXT order (the Wald first law → $f(R){=}R{+}R^2$, v28/v359). [E] it is ALREADY falsifiable via $n_s{=}1{-}2/N_\star$, $r{=}12/N_\star^2$ (live CMB-S4 kill test). [E] the gap $\Delta{=}6\ln\frac{3}{2}$ is a dimensionless rate, so a distinct gap-induced curvature^2 term sits at $\xi\, v_{\rm geo} \sim$ the seam/Planck scale (not near-term observable). [O] DECLINE: a new dimensionless coefficient is not atom-forced (the scalaron exponent $7{=}g{+}N{-}1$ and the gap exponent $6{=}2N_{\rm fam}$ are different; c_3^7 and $(2/3)^6$ unrelated) — per v354/v355 declined. Direction 2 yields no new near-term forced prediction; the answer is the existing scalaron. Cited in `tfpt_4`; Python-only.

2026-06-23 (the FULL covariant Einstein equation, parameter-free: the fixed-volume entanglement equilibrium gives $G_{ab}+\Lambda g_{ab} = c_3^{-1}T_{ab}$ with BOTH coefficients TFPT-fixed ($8\pi = 1/c_3$ from v358, Λ from α via v60); B6 closed at the local level: v359)

- `v359` (`GRAV.NONLINEAR.01`) — **Direction 1, the full covariant equation.** Extends v358 from the linearised (fixed-radius, Ricci-level) result to the FULL covariant equation. [C][E] Jacobson 2015: demanding stationarity of the small-ball entanglement entropy at fixed *volume* brings in the EINSTEIN TENSOR G_{ab} (not just R_{ab}) — trace $g^{ab}G_{ab} = (1-d/2)R = -R$ in d=4, the structure that carries Λ. [E] Lovelock: G_{ab} is the unique symmetric divergence-free second-order metric tensor, so $\nabla^a(G_{ab} + \Lambda g_{ab}) = 0$ forces $\nabla^a T_{ab} = 0$ — conservation is an output. [E] coefficient 1: $8\pi G \to 8\pi = 1/c_3$ fixed (v358). [E] coefficient 2: Jacobson leaves Λ free, TFPT fixes it — $\rho_\Lambda/\bar{M}_{\rm Pl}^4 = (3/(4\pi^2))e^{-2\alpha^{-1}}$, prefactor $(8\pi)^2{\cdot}48c_3^4 = 3/(4\pi^2)$ (v60); so BOTH coefficients are TFPT-fixed, not free integration constants. [C] all timelike directions → the covariant tensor equation; the matter side is the assembled CHM ball modular flux (v358/v323). NET: the original B6 "full covariant Einstein-side field equation" is now PARAMETER-FREE at the local level. [O] residual: the Jacobson equation-of-state status (not a from-action quantisation) + the global ambient measure (C7/`QG.AMB.01`, separate, gap-decoupled); $v_{\rm geo}$ the one unit. Cited in `tfpt_research_contracts`; mirrored in Wolfram (303/303). (`next-plan.md` Direction 4 folded in: the Λ here matches the α-readout.)

2026-06-23 (the classical gravity field equation, parameter-free: the entanglement first law $\delta S = \delta\langle K\rangle$ with TFPT's atoms gives the LINEARISED Einstein equation with G from c_3; the thermodynamic and geometric origins of c_3 coincide via $|\mu_4| = |\mathbb{Z}_2|\chi = 4$; the two formerly-open pieces (matter flux J3, entropy-density coefficient) closed/atom-fixed: v358)

- `v358` (`GRAV.ENTROPY.EQUILIBRIUM.01`) — **the bridge to the dynamic world, instantiated on the gravity side.** Carrying out the Jacobson-2015 / Faulkner-et-al. "first law of entanglement ⇒ Einstein equation" with TFPT's atoms. [E] PARAMETER-FREE COEFFICIENT: $\delta S = \delta\langle K\rangle$ gives $G_{ab} = c_3^{-1}T_{ab}$ with $c_3^{-1} = 8\pi$ fixed; all four c_3-roles consistent (Unruh $1/(2\pi)$=$4c_3$, EH $1/(16\pi)$=$c_3/2$, Einstein 8π=$1/c_3$, Bekenstein $1/4$=$1/|\mu_4|$) — no free dimensionless Newton dial. [E] THERMO = GEO COINCIDENCE (new): the first law's coefficient $2\pi/\eta$ equals the geometric $|\mathbb{Z}_2|2\pi\chi$ iff $|\mu_4| = |\mathbb{Z}_2|\chi = 4$ (v216) — the thermodynamic and geometric origins of c_3 are ONE. [C] J3 SOLVED: the CHM ball modular Hamiltonian (boost via BW, v323) gives the matter flux $\delta\langle K_B\rangle = \frac{8\pi^2R^4}{15}\delta\langle T_{00}\rangle$. [E] ENTROPY DENSITY ATOM-FIXED: $1/4$=$1/|\mu_4|$ (v57) + central charge $c = g_{\rm car}+N_{\rm fam} = 8$ (form v59). [O] residual: linear→full non-linear (the original B6) + the $v_{\rm geo}$ unit. Closes the LINEARISED Einstein equation parameter-free; cited in `tfpt_research_contracts`; mirrored in Wolfram (302/302).

2026-06-23 (the two frontier walls attacked directly: Direction A assembles the continuum chain of `SEAM.EQUIV.01` theorem-by-theorem and ties its residual to the seam one-sidedness; Direction B was found ALREADY closed by v40 (harmonic metric = finite linear algebra, not a PDE): v356)

- `v356` (`SEAM.EQUIV.CONTINUUM.03`) — **Direction A, the continuum chain assembled.** After the reverse-audit/bandwidth work showed no hidden *discrete* lever remains, the only fundamental progress is to assemble the *continuum* chain. Link-by-link: **S1** the gapped (Δ=$6\ln(3/2)$>0, v302) quasi-free 16-Majorana collar (v155/v160) → **S2** invertible (gapped free fermions have no intrinsic topological order, Kitaev, v301) → **S3** the *nontrivial* invertible phase (c_-=8≠0) → **S4** a *non-gappable* anomalous chiral edge (anomaly inflow / bulk–edge), so edge *existence* is a CONSEQUENCE of S2+S3, not a separate assumption — this directly reduces the v336/v351 "does the collar have a chiral edge?" residual → **S5** its chiral-CFT scaling limit (MMST, v336; range rank $\le c \le D$, $8 \le 8 \le 16$) → **S6** the order-4 μ_4 clock + the AQFT stack pin $(E_8)_1$ (v351/v297). **[E]** the only non-theorem link is S3, and **[C]** S3 is the seam being *one-sided* (chiral, $c_- \neq 0$) — the *same* one-sidedness that makes $c_3 = 1/(8\pi)$ a one-sided Gauss–Bonnet ($|\mathbb{Z}_2| \oint K$, v58). **[O]** the one remaining input is "the abstract collar is realized as a genuine lattice chiral free-fermion invertible phase" (the v297 Flat-Away). NET: `SEAM.EQUIV.01` reduced to ONE realization input tied to the constant that *defines* the theory; it ADVANCES, it does not close.

- **Direction B — found ALREADY closed by v40.** The U-wall "transcendental Hitchin solve" (v31) is the worst case for a *generic* Higgs bundle; but at the *polystable* point TFPT selects, Mehta–Seshadri makes the bundle *unitary*, so the harmonic metric is the constant invariant Hermitian form (*finite linear algebra*, Higgs $\Phi=0$) and $\det R = 8$ is read off — not a PDE. Only the absolute amplitude U_{point} (an anchor) remains. A redundant new B module was built, found to duplicate/contradict v40, and *deleted* (no padding). Honest outcome: B needs no new work.

2026-06-22 (bandwidth search of the unmapped E_8 region, with a strict forced-identity discriminator: the genuine overlooked structure is COLLECTIVE ($\sum \deg = 128 = \dim S^+$, $\prod \deg = |W(E_8)|$, exponents = totatives of 30); the per-degree atom matches are unforced and DECLINED; no new physical hit – reconciling v66 and v354: v355)

- **`v355 (E8.UNMAPPED.BANDWIDTH.01)` — the next step after v354, run honestly.** The obvious follow-up ("search those five degrees for hits we overlooked") is itself the numerology temptation v354 warns about: a rich object like E_8 returns a numerical match for almost anything. So the scan uses a STRICT discriminator – a candidate counts only if it is a FORCED identity (a theorem about E_8), never a coincidence. **[E]** FORCED, SET-LEVEL (the genuinely overlooked structure, collective not per-number): $\sum \deg = 128 = \dim S^+$, the spinor half of $248 = 120 + 128$ (theorem: $\sum \deg = \#\text{pos roots}+\text{rank} = 120 + 8$) – the invariant budget of E_8 equals the fermion content the carrier reads; $\sum \exp = 120 = \#\text{pos}$ roots; $\prod \deg = |W(E_8)| = 696{,}729{,}600$; exponents = the $\varphi(30)=8$ totatives of 30 (v223), so the unmapped degrees' exponents are part of the forced set; $\max \deg = 30 = h(E_8)$. **[C]** SUGGESTIVE, not a theorem: $12 = h(E_6)$, $18 = h(E_7)$, $30 = h(E_8)$ are all degrees of E_8 (the $2T/2O/2I$ McKay ladder, v219), but "$\deg(E_8)$ contains its subalgebras' Coxeter numbers" is not a general theorem – flagged, not claimed. **[O]** DECLINED (the numerology temptation): the per-degree atom matches $14 = \dim G_2$, $20 = \det L$, $24 = |W(A_3)|$, $12 = |R(A_3)|$, $18 = p_4(a)$ (v66 already flagged $12, 18$ as fittable) each admit ≥ 2 readings, and the lone extra prime in $|W(E_8)| = 2^{14} 3^5 5^2\, 7$ "=" the scalaron exponent is one of many readings of 7 – all declined. **[E]** RECONCILES v66 (the 8 degrees coincide with the load-bearing integers) and v354 (only $2, 8, 30$ are primary readouts): the search finds NO new forced physical hit, and the disciplined decline IS the anti-numerology result. (This also sharpens v354: the five are "no PRIMARY readout / hull overhead", not literally "unmapped" – they coincide with structural atoms, but unforced.) Cited in `tfpt_5_redteam` (with a full audited degree table); mirrored in Wolfram (301/301).

2026-06-22 (the REVERSE numerology audit, and what the golden ratio means: of E_8's 8 Casimir invariants only 3 feed a primary readout and 5 are hull overhead; the golden ratio is UNMAPPED structure, so it cannot be numerology: v354)

- **`v354 (E8.REVERSE.AUDIT.01)` — the reverse complement of v305.** TFPT's forward discipline (v305) is one-directional: every physical READOUT must map onto an E_8 structure (anti-fitting). The sharp adversarial question runs the OTHER way — how much E_8 structure maps onto NOTHING, and is the recurring golden ratio numerology? [E] THE REVERSE AUDIT: of E_8's eight Casimir invariant degrees $\{2, 8, 12, 14, 18, 20, 24, 30\}$ (exponents $+1$), exactly THREE feed a readout — 2 (metric), 8 (rank $\to c_3{=}1/(8\pi)$), 30 (Coxeter $\to g_{\mathrm{car}}{=}5$ and the order-30 cycle) — and FIVE carry no primary readout $\{12, 14, 18, 20, 24\}$ (they coincide with structural atoms per v66, but unforced – reconciled in v355); so 3/8 are primary readouts, 5/8 hull overhead. [E] ECONOMY, NOT PROMISCUITY: a small fixed set (rank 8, $h{=}30$, $\det K{=}1$, $D_5{\oplus}A_3$, the 16 half-spinor) generates the ENTIRE readout set, and the higher Casimirs are never reached into to fit — a numerological use would be PROMISCUOUS (mine the rich structure), TFPT is ECONOMICAL (few invariants, many readouts) and leaves most of E_8 untouched, the anti-numerology signature being precisely the large unused remainder. [E] THE GOLDEN RATIO IS UNMAPPED $\Rightarrow$ NOT NUMEROLOGY: $\varphi = 2\cos(\pi/5)$ appears only in internal structure (the affine-E_8 attractor eigenvalue v312, the icosian lattice v348) and in NO physical readout; numerology means a number FITTED to an observation, φ is fitted to nothing — it is the algebraic signature of the 5-fold ($h{=}30{=}2\cdot 3\cdot 5$), the carrier announcing it is icosahedral. [O] the 5/8 unmapped is the HULL OVERHEAD (E_8 is the minimal even-unimodular hull, not the world group); where a future higher-invariant readout could live, or genuine over-capacity — stated plainly, not hidden. Cited in `tfpt_5_redteam`; mirrored in Wolfram (`300/300`).

2026-06-22 (the bird's-eye rethink: TFPT is a unique self-consistent LOOP with ZERO free dials, not a linear axiom system; π is a fixed geometric constant, not a dial; the one residual is the continuum closure: `v353`)

- **`v353` (`TFPT.SELFLOOP.01`) — refining v352.** Stepping all the way back: TFPT is not a LINEAR theory (axioms $\to$ theorems) but a CLOSED SELF-CONSISTENT LOOP whose outputs fix its own inputs. [E] it is a loop: $g_{\mathrm{car}}{=}5{=}$ max prime of the output $h(E_8){=}30$, and the 8 in $c_3{=}\operatorname{rank} E_8$ – the outputs fix the inputs (v6), with a unique fixed point (v56); “which is the axiom?” is the wrong question for a loop. [E] ZERO free adjustable dimensionless dials: every load-bearing number is forced/over-determined. [E] π is a FIXED geometric constant (the boundary 2-sphere, $\oint K{=}4\pi$), not a dial. [C] this refines v352's “framework + π”: π is not a free input and the framework is a SELECTION PRINCIPLE (the unique self-consistent loop), not a free meta-axiom – the honest characterisation is “a unique self-consistent loop with zero free dials”. [O] the one residual is the CONTINUUM closure of the loop (the chiral-edge existence, v336/v351); structural-realism (loop complete) vs constructivism (continuum open) is a philosophical, not numerical, fork – no free numbers either way. A capstone synthesis.

2026-06-22 (two consequences: the $c{=}8$ which-net ambiguity resolved by the order-4 clock, and BOTH axioms shown bootstrap-forced so the only irreducibles are the framework + π: `v351`, `v352`)

- **`v351` (`SEAM.EQUIV.CONTINUUM.02`) — the which-net ambiguity falls.** The long-standing “$c{=}8$ ambiguity” (E_8 det 1 vs $SO(16){=}D_8$ det 4, both $c{=}8$, flagged open in v277/v344) is RESOLVED, non-circularly, by the seam's ORDER-4 μ_4 clock: the index-4 simple-current extension of $D_5{\oplus}A_3$ is E_8 (v154), the index-2 would be $SO(16)$, so the order-4 clock (the four Gauss–Bonnet marks, v216) selects E_8 (full μ_4 condensation $16{\to}4{\to}1$) not $SO(16)$ (partial $\mathbb{Z}_2$ $16{\to}4$). The bootstrap agrees ($h(E_8){=}30$ max prime $5{=}g_{\mathrm{car}}$ vs $h(D_8){=}14$ max prime 7). [E] so $\det K{=}1$ is a consequence, not an open input. [O] the residual of `SEAM.EQUIV.01` is then PURELY the chiral-edge / scaling-limit existence (v336), not which-net.
- **`v352` (`TFPT.IRREDUCIBLE.01`) — the inputs are not axioms.** The deepest answer to “the inputs are back-determined”: BOTH P1 and P2 are bootstrap-forced fixed points. [E] $g_{\mathrm{car}}{=}5$ is forced three ways (v6); the 8 in c_3 four ways ($\operatorname{rank} E_8 = h(D_5) = \phi(30) = \det R$, v54); the E_8 hull is forced (unique even unimodular rank-8, v83/v154). [E] so the ONLY irreducibles

are the FRAMEWORK (a discrete reflection-positive boundary with a finite carrier) and the transcendental π. [O] the framework is the META-AXIOM — not math-reducible, the foundational stance (like "spacetime is a manifold"), minimal, tested by its consequences not proved. TFPT's true input is ONE framework + π. Wolfram mirror +2 (297 → 299).

2026-06-22 (correction: the inputs are bootstrap-forced fixed points, not free axioms – g_{car}=5 IS the icosahedral 5 of $h(E_8)$=30, and the golden ratio is emergent from the μ_4-glue, not external: v350)

- **v350 (SEAM.EQUIV.BOOTSTRAP.01) — correcting v349.** An honest correction prompted by the right objection that the inputs are not axioms but are back-determined by the theory. [E] the inputs ARE bootstrap-forced (v6): g_{car}=5 is over-determined three ways – rank-fill ($g_{\mathrm{car}}+N_{\mathrm{fam}}$=8), Coxeter-match ($g_{\mathrm{car}}$= max prime of $h(E_8)$=30), Pascal ($2^4=C(5,0)+C(5,1)+C(5,2)$). [E] so g_{car}=5 IS the icosahedral 5-fold (the 5 in $h(E_8)=30=2{\cdot}3{\cdot}5$), not the D_5 rank; the atoms (2, 3, 5) ARE the primes of the icosahedral Coxeter number 30. [E] the golden ratio is EMERGENT from the glue: D_5 (h=8)/A_3 (h=4) are non-golden alone, but $D_5 \oplus A_3 \to E_8$ (μ_4 index-4, v154) gives $h(E_8)$=30 (golden) – the seam's own μ_4 produces the 5-fold, so v349 checked the un-glued pieces, the wrong object, and its pentagon-vs-count dichotomy was false. [O] the residual relocates: the golden is bootstrap-forced, the only open thing is the physical continuum realisation (v336), NOT the golden ratio. Honest caveat: self-consistency within the framework, not from-nothing. Wolfram mirror +1 (296 → 297).

2026-06-22 (the honest test – does the raw seam carry φ? – answer NO: the bare carrier gives $\sqrt{2}$, golden is the icosahedral input; the keystone reduces to "is g_{car}=5 a pentagon or a count?": v349)

- **v349 (SEAM.EQUIV.GOLDEN.01).** The honest, computed test of the v348 question, and a clarifying NEGATIVE: the raw seam does *not* carry the golden ratio. [E] the golden ratio $2\cos(\pi/5)$ is the signature of a 5-fold rotation ($5 \mid h$); the carrier D_5 has h=8 (eigenvalue $\sqrt{2}$) and A_3 has h=4 – no 5-fold; the dynamical data (cusp $\{0, 1/3, 2/3\}$, recovery $(2/3)^6$, seed φ_0) are rational/transcendental. [E] $5 \mid h$ holds only for $A_4=SU(5)$ (h=5), H_3 (h=10), E_8 (h=30) – the output/icosahedral side, so φ is the icosahedral INPUT. [E] φ alone would not suffice ($2I \neq C_5 \times \mu_4$, $20 \neq 120$). [O] the resolution: the keystone reduces to one question – is g_{car}=5 a PENTAGON (a 5-fold rotation ⇒ golden ⇒ icosahedral $2I = E_8$) or just a COUNT (the rank of $D_5 \Rightarrow \sqrt{2}$)? The "absurdly simple solution" is real but is a *reading* of axiom P2 (the golden ratio is the pentagon content of g_{car}=5, not derivable from g_{car}=5-as-a-count); there is no hidden φ. L2 is foundational: P2-as-pentagon closes it (mode C), else a rigidity theorem must produce the 5-fold (mode B); not closed. Wolfram mirror +1 (295 → 296). A NEGATIVE/clarification; prevents a false closure.

2026-06-22 (Route B carried out: rigidity + the icosian-ring identity reduce the whole keystone to ONE question – does the raw seam carry the golden ratio? – the simplest form of L2: v348)

- **v348 (SEAM.EQUIV.RIGID.01).** Executes the rigidity route (mode B of v347) and lands on a strikingly simple statement. [E] rigidity closes the crystallographic part: the order-4 clock has the unique normal form $z \mapsto iz$ (Nielsen/Kerckhoff, v180), Troyanov gives the unique flat pillowcase metric (v284). [E] the downstream is a LATTICE identity, not just a graph: by Conway–Sloane the ring of icosians (120 unit icosians = $2I$, golden-ratio quaternion norm) IS the rank-8 E_8 lattice. [E] the single bridge is the golden ratio: the crystallographic restriction allows only orders $\{1, 2, 3, 4, 6\}$ (the 5-fold absent), so $\varphi = 2\cos(\pi/5)$ is exactly what extends the crystallographic μ_4 to the non-crystallographic icosahedral $2I$ (the quasicrystal). [O] so the entire keystone reduces to ONE question: *does the raw seam carry the golden ratio?* – if yes, $\mu_4+\varphi \Rightarrow 2I \Rightarrow E_8$ and rigidity fixes the rest. [O] honest subtlety: TFPT's φ is currently E_8-side (v312/v313); the raw-seam data so far are not manifestly golden, so showing the RAW seam carries φ non-circularly is the residual (= L2, sharpest form). Wolfram mirror +1 (294 → 295). An analysis/reduction; closes nothing.

2026-06-22 (can the one open arrow be *fully* solved? the closure-mode classification: the flat→spherical base locus, four modes, no free theorem – only rigidity or axiom P_3: v347)

- **v347 (SEAM.EQUIV.CLOSURE.01).** A direct, honest answer to "is there any other way to fully solve the one open arrow L2?". [E] the precise locus: the seam gives the *flat* pillowcase base $S^2(2,2,2,2)$ ($\chi_{\mathrm{orb}} = 2-4(1-\frac{1}{2}) = 0$, v216) while the $2I/E_8$ structure is the Seifert base over the *spherical* $S^2(2,3,5)$ ($\chi_{\mathrm{orb}} = 2-(\frac{1}{2}+\frac{2}{3}+\frac{4}{5}) = \frac{1}{30}$), so L2 bridges two geometric types – why it needs the carrier $(2,3,5)$ input. [E] as a "physics-realises-geometry" statement L2 has exactly four closure modes: A construct (the analytic wall), B rigidity (a unique-realisation theorem – the only theorem-route w/o new input, pursued v284/v300/v331), C axiom P_3 (then TFPT is complete relative to 3 axioms), D empirical. [E] the "no new state" discipline (v246) blocks deriving L2 from extra physics, so no route makes it a free theorem. [E] verdict: full closure is either a rigidity theorem (B) or axiom status (C), validated either way by (D); only B closes it as a theorem. [O] NOT closed – classifies the modes, locates the difficulty. Wolfram mirror +1 (293 → 294). An analysis/roadmap; closes nothing.

2026-06-22 (the geometric bridge made explicit: the keystone is ONE arrow – raw-seam ⇒ $\det K{=}1$ is a six-link chain, five theorems + one open orbifold realisation, the group forced by the $(2,3,5)$ atoms: v346)

- **v346 (SEAM.EQUIV.GEOM.01).** The honest capstone of the SEAM.EQUIV.01 investigation: it lays out the full path raw-seam ⇒ $\det K{=}1$ as six links and shows exactly one is open. [E]/[C] L1 raw seam → μ_4 + four marks (Gauss–Bonnet, v216); [O] L2 μ_4 + the $(2,3,5)$ atoms → the $2I$ orbifold $\mathbb{C}^2/\Gamma$ (THE ONE OPEN ARROW = the geometric content of SEAM.EQUIV.01); [E] L3 $2I$ → affine E_8 McKay (v219); [E] L4 $2I \to E_8$ du Val (v232); [E] L5 $\mathbb{C}^2/2I$ → Poincaré link $H_1{=}0$ (v345); [E] L6 $H_1{=}0 \to \det K{=}1$. [E] five of six links are theorems; [E] the group is FORCED by the seam's own atoms (the exceptional $SU(2)$ axis signatures $(2,3,3)/(2,3,4)/(2,3,5) \to 2T/2O/2I$; $(2,3,5)$ unique to $2I$, and $(|\mathbb{Z}_2|, N_{\mathrm{fam}}, g_{\mathrm{car}}){=}(2,3,5)$ ARE those axes); [E] everything downstream of L2 is then forced. [O] the irreducible open content is L2 alone ("the raw seam normal bundle is an ADE orbifold point $\mathbb{C}^2/\Gamma$") = SEAM.EQUIV.01; not closed, but the bridge is now demonstrably ONE arrow with the group pinned and all downstream proven.

2026-06-22 (executing route R3: the $(2,3,5)$-rewrite attractor link computed to be the Poincaré homology sphere ($\det K{=}1$), the combinatorial core complete, only the geometric bridge open: v345)

- **v345 (SEAM.DETK.02).** Carries out the combinatorial core of the hypergraph-homotopy route (R3, the most promising fresh angle from v344) as far as it computably goes. By plumbing topology (Brieskorn/Hirzebruch) the link of a tree plumbing with the ADE Cartan intersection form has $H_1 = \mathrm{coker}(\mathrm{Cartan})$. [E] the Smith normal form of the E_8 Cartan matrix is the identity, so $\mathrm{coker}(E_8) = 0$ – the E_8-graph plumbing boundary is an INTEGRAL HOMOLOGY SPHERE (the genus-1 meaning of $\det E_8{=}1$); among ADE only E_8 does this ($A_n \to \mathbb{Z}/(n{+}1)$, $D_n \to \mathbb{Z}/4$, $E_6 \to \mathbb{Z}/3$, $E_7 \to \mathbb{Z}/2$). [E] $\pi_1 = 2I = SL(2,5)$ is PERFECT (commutator subgroup = the whole order-120 group, computed directly), so $H_1 = 0$ and the link is THE Poincaré sphere, $|2I| = 120 = |\mu_4|\, h(E_8)$. So R3's combinatorial half is

[E]-complete: the E_8 attractor link is forced to be the Poincaré sphere ($\det K{=}1$), uniquely among ADE. [O] what stays open is only the geometric realisation bridge (the raw rewrite seam attractor *is* this E_8 du Val plumbing) = the `SEAM.EQUIV.01` content; not closed, but the keystone is now reduced to that single geometric identification. Wolfram mirror +1 (292 → 293).

2026-06-22 (the one keystone bit $\det K{=}1$ mapped: six equivalent faces, the genus-1 obstruction, three access routes, the hypergraph-homotopy route the most promising fresh angle: v344)

- **v344 (`SEAM.DETK.01`).** A bird's-eye, full-spectrum map of `SEAM.EQUIV.01`'s only open residual, the holomorphy bit $\det K{=}1$. [E] it is ONE statement with six equivalent faces (holomorphy / homology-sphere link $H_1{=}0$ / one 1-dim irrep / det Cartan = 1 / full μ_4 condensation / $\tau{=}i$ CM), unified by $|\det\mathrm{Cartan}| = |H_1(\mathrm{link})| = \#(\text{1-dim irreps})$ across ADE ($A_n{\to}n{+}1$, $D_n{\to}4$, $E_6{\to}3$, $E_7{\to}2$, $E_8{\to}1$) – only E_8 gives 1. [E] the GENUS-1 OBSTRUCTION explains why it is hard: every easy argument (plane vacuum, gap, free-fermion) is genus-0, necessary but not sufficient (v87); $\det K$ is the torus degeneracy. [E] THREE genus-1 access routes: R1 continuum-modular (v336), R2 anyon-condensation (v281), R3 hypergraph-homotopy. [C] the HYPERGRAPH route is the most promising fresh angle: the $(2,3,5)$ rewrite attractor is the affine-E_8 graph (v312) and the E_8 du Val link is the Poincaré sphere (v232), so $\det K{=}1$ becomes a finite combinatorial statement (no continuum limit). [O] none closes it; the gap is the combinatorial-to-geometric bridge = the `SEAM.EQUIV.01` content. A synthesis/roadmap; closes nothing.

2026-06-22 (the four black-hole-birth completion routes investigated: none closes all or is 100% provable alone; all four converge on `SEAM.EQUIV.01` + the decoupled QG contour: v343)

- **v343 (`FOUR.ROUTES.01`).** A direct, honest answer to "which black-hole-birth route closes ALL problems and is 100% provable": *none individually* — but the four converge on the same two residuals as v335. [E] **(A)** the finite causal diamond reframes `QG.AMB` to a finite thermal trace (de Sitter $S_{\mathrm{dS}} = 32\pi^4 e^{2\alpha^{-1}}$ finite, $\chi = 729/665$) but inherits the GHP contour (v334, [O]); does not touch the seam. [C] **(B)** Carlip–Cardy gives a second route to existence+c ($c{=}8$ consistent, but the Carlip ambiguity and no holomorphy $\det K{=}1$) — inherits `SEAM.EQUIV.01`. [E] **(C)** the modular/thermal flow (v239) is well-defined but its geometric content is `QGEO.SYM.01`, a corollary of `SEAM.EQUIV.01` (v335) — a reduction. [E]/[O] **(D)** the unique attractor (v56) gives parameter-freeness but the cyclic dynamics is [C] interpretation. [E] **Verdict:** independent closures = 0; irreducible residuals after all four = 2 (unchanged from v335); the horizon premise makes both more natural but eliminates neither — focus = `SEAM.EQUIV.01`. An analysis module; closes nothing, fabricates nothing.

2026-06-22 (the heat-kernel origin of the α terms: the Calderón quadratic is the Gilkey gauge-curvature coefficient; the cubic Maxwell moment stays the `EM.WARD.01` residual: v342)

- **v342 (`EM.WARD.02`) — the heat-kernel sharpening of `EM.WARD.01`.** Derives the ORIGIN and STRUCTURE of the $F_{U(1)}$ determinant-line terms from textbook Seeley–DeWitt / Gilkey coefficients, advancing the EM-Ward origin from [C] (conjectured) to [E] for the term structure; no new gate. [E] $c_3 = 1/(8\pi)$ *is* the boundary Gauss–Bonnet coefficient ($\oint_{S^2} K = 2\pi\chi = 4\pi$, $\chi{=}2$, v216). [E] the α^2 (Calderón) term *is* the Gilkey gauge-curvature coefficient ($30\,\Omega^2/360 = \frac{1}{12}\Omega^2$, $\Omega{=}i\alpha F \Rightarrow -(\alpha^2/12)F^2$), so it is forced, not an ansatz. [E] the c_3-orders $\{0,3,6\}$ are an arithmetic ladder and $2c_3^3 = 1/(256\pi^3)$ carries π^3 (three boundary insertions). [C] the exact coefficient needs the seam F-normalisation. [O] the cubic α^3

Maxwell moment (a topological/Chern level) + the full "this is the seam $U(1)$ ζ-determinant" identification stay the `EM.WARD.01` residual, tied to `SEAM.EQUIV.01`. α^{-1} stays [E] (v3). Wolfram mirror +1 (290 → 291).

2026-06-22 (α as the stationarity of a $U(1)$ determinant line: every ingredient an index / heat-kernel coefficient / discriminant form, the open EM-Ward obligation named: v341)

- **v341 (ALPHA.QUILLEN.01).** Answers the external review's sharpest valid point — "derive the *form* of α, not more decimals". The EM closure $F_{U(1)} = \alpha^3 - 2c_3^3\alpha^2 - \frac{4}{5}c_3^6 M \log(1/\varphi_{\text{seam}}) = 0$ is rewritten as a *stationarity* of the boundary $U(1)$ determinant line: $[\alpha^3 - 2c_3^3\alpha^2]_{\delta \log\det \Delta_{U(1)}} = [8b_1 c_3^6]_{\text{counterterm}} \cdot [\log(1/\varphi_{\text{seam}})]_{\text{seam response}}$. [E] every ingredient is a NAMED object, not a knob: M=41 is a $U(1)$ index ($\frac{4}{5}M = 8b_1$, b_1=41/10 the SM $U(1)_Y$ beta, EM Ward v48); c_3=1/(8π) the Gauss–Bonnet boundary coefficient (8= rank E_8); $-5/4 = -q(D_5)$ a discriminant form and φ_{base}=1/(6π)=$\frac{4}{3}c_3$ carries the other, $c_3/\varphi_{\text{base}}$=3/4=$q(A_3)$; δ_{top}=$\Omega_{\text{adm}}c_3^4$. The Quillen split is verified at the α^{-1}=137.0359992 root. **No new gate:** α^{-1} stays closed [E]; v341 adds no open item — it *sharpens* the pre-existing [O]/physical-origin obligation `EM.WARD.01` ("why *this* function", already conjectural). [O] the one still-open part *is* that residual: the from-first-principles AQFT proof that $F_{U(1)}$ is the EXACT Quillen functional (the variational principle derived, not assembled), unchanged in scope. Wolfram mirror +1 (289 → 290).

2026-06-22 (A_s reconciled with the archived derivation: predicted & dimensionless (so it does not define a mass), the tension is the reheating channel: v340)

- **v340 (AS.RECONCILE.01).** Answers "why don't we have A_s / the old versions derived it" and "can a ratio define mass itself". [E] A_s *is* predicted and *dimensionless*: $A_s = N_\star^2(M_{\text{scal}}/\bar{M}_{\text{Pl}})^2/(24\pi^2) = N_\star^2 c_3^7/(24\pi^2)$ with $M_{\text{scal}}/\bar{M}_{\text{Pl}} = c_3^{7/2} = 1.26\times10^{-5}$ a pure ratio, so A_s pins the e-fold count $N_\star$, NOT $\bar{M}_{\text{Pl}}$ — it cannot define an absolute mass (No-Unit, v153, stands). [C] the -11.4σ at the slow Higgs channel ($N_\star$=51.4) is the slow-vs-fast reheating dichotomy; the A_s-preferred $N_\star$=56.2 ∼ the instantaneous bound. [C] the archived TFPT v4.5/v4.6 reported a clean $A_s = 2.124\times10^{-9}$ using the algebraic $N_\star = 3/\varphi_0^{\text{ret}} - 1 = 55.42$; at that $N_\star$ the current normalisation gives 2.047×10^{-9} (-1.9σ) — the old clean A_s is near-instantaneous reheating, not the slow channel. [E] nothing lost: the current theory replaced the posited algebraic $N_\star$ with one derived from reheating physics and exposed honestly that A_s requires fast (pre)heating. Record band $N_\star \in [50, 60]$ (v84) untouched.

2026-06-22 (the first-principles boundary, mapped: can M_{Planck} be computed? can F_{transfer} be first-principles? both NO, with the exact reason: v339)

- **v339 (FIRSTPRINCIPLES.BOUNDARY.01).** An honest boundary audit answering two head-on questions and fabricating nothing. [E] *The Planck mass cannot be computed* — $v_{\text{geo}}/M_{\text{Planck}}$ is forbidden by theorem (No-Unit, v153): a mass is mass-dimension 1 while every TFPT datum is dimension 0, so no dimensionless map outputs an absolute scale. It is not a gap but over-determined (gravity = dark energy to 0.11%, v274); every dimensionless *ratio* is derived, only the one unit is not. [E] F_{transfer} *cannot be made first-principles* either: for m_p/m_e the structure is in place (v262, carrier b_3=−7 + closed electron) but two inputs stay external. [X] $\alpha_s(M_Z)$ is external *because of a tension* — the SM (= TFPT content, no new states) does not unify (1-loop spread ∼8.8 at 2×10^{16} GeV, v246). The lattice C_p∼2.83 is parameter-free but lattice-only (v35); η_B/axion ride on cosmological initial conditions; Koide Q=2/3 is structural [E]. [E] The residuals classify into four honest kinds: forbidden-by-theorem, external-and-tensioned, external-but-parameter-free, cosmological. A classification, not a solution.

2026-06-22 (next-steps round: the one open lemma literature-anchored, the Decoupling Theorem, the holomorphy discriminator hardened in Lean, and the θ_{13} budget: `v336`, `v337`, `v338`, `FORM.CARTAN.DET.01`)

- **`v336` (`SEAM.EQUIV.CONTINUUM.01`) — the one open lemma, literature-anchored.** SEAM.EQUIV.01's only open input (the continuum OS reconstruction of the gapped quasi-free collar) is split into a CITABLE part and a narrow open part. [C] continuum leg: Osborne–Stottmeister (*CFT from Lattice Fermions*, doi:10.1007/s00220-022-04521-8, CMP **398** (2023) 219) give, building on the MMST (Morinelli–Morsella–Stottmeister–Tanimoto) operator-algebraic (wavelet) renormalization framework (CMP 387 (2021) 299; PRL 127 (2021) 230601), the chiral-CFT scaling limit of free lattice fermions (Koo–Saleur → Virasoro, correct c); WZW range rank $\leq c \leq D$, and $(E_8)_1$ (c=8, rank 8, D=16 Majoranas, $SO(16)_1 \to (E_8)_1$) is inside it. [C] OS leg: Adamo–Moriwaki–Tanimoto (arXiv:2407.18222, 2024) prove the conformal OS axioms for unitary lattice VOAs. [E] target pinned by $c=g_{\text{car}}+N_{\text{fam}}=8$, $\det \text{Cartan}(E_8)=1$. [O] narrowed residual = "the raw collar's massless scaling limit is the holomorphic $(E_8)_1$ net" ($\det K=1$), not a from-scratch construction.
- **`v337` (`DECOUPLING.THEOREM.01`) — the Decoupling Theorem.** [E] every dimensionless readout factors through the gapped admissible transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$; finite susceptibility $\chi = 1/(1-(2/3)^6) = 729/665$ gap-suppresses the ambient contribution (margin $6\ln\frac{3}{2} - 31/(4\pi^2) \approx 1.648 > 0$, with $2\dim(E_8)c_3^2 = 31/(4\pi^2)$ exactly), so no readout depends on the ambient measure ⇒ `QG.AMB.01` decoupled; negative control (gap closed ⇒ χ diverges) makes it non-vacuous. This is the theorem that makes the `QG.AMB.01` reclassification (v335) honest; consolidates v76/v311/v330/v332.
- **`v338` (`THETA13.BUDGET.01`) — the θ_{13} budget, fenced.** The $+2\sigma$ pressure point (v328) quantified against *both* NuFIT 6.0 NO variants: $+2.0\sigma$ vs the no-SK central (0.02195) but $+1.7\sigma$ vs the with-SK best fit (0.02215 ± 0.00057), a $\sim 0.3\sigma$ systematic spread; [E] 0.023108 lies inside the with-SK 3σ band (upper 0.02388), an upper-edge pressure point not a falsification; [X] JUNO's sub-1% precision turns it into a $\sim 6\sigma$ kill-or-confirm.
- **Lean (`FORM.CARTAN.DET.01`).** `CartanDeterminants.lean` extended with the explicit $D_8 = SO(16)$ Cartan matrix and a fully-proven (kernel-only) holomorphy discriminator: D_8 is even and symmetric just like E_8 (same c=8) but $4 = \det \text{Cartan}(D_8)$ is genuinely not a unit while $\det \text{Cartan}(E_8)$ is — so unimodularity (one anyon) is the load-bearing selector. `lake build` green.

2026-06-22 (keystone unification + status reframe: ONE open theorem, ONE decoupled foreign problem: `v335`, `FORM.QGEO.BW.01 qgeoSymIsCorollary`)

- **`v335` (`SEAM.EQUIV.UNIFY.01`) — the keystone unification.** A bird's-eye consolidation of the v308/v323/v325/v329/v333 arc that collapses the "two open structural items" narrative. [E] **`QGEO.SYM.01` is a COROLLARY of `SEAM.EQUIV.01`** (no extra premise): a chiral conformal net has, by axiom, a Möbius-covariant vacuum (the unique invariant positive-energy vector); rotations $U(1)$ are the compact subgroup of the Möbius group, so the vacuum is rotation-invariant, and the μ_4 clock = rotation by $\pi/2$ ($(e^{i\pi/2})^4=1$) fixes it ⇒ $\omega \circ \rho = \omega$. So `SEAM.EQUIV.01` ⇒ `QGEO.SYM.01` with NO extra premise (the v323 "rotation-invariant vacuum" is a net axiom) — the two bedrock items collapse to ONE. [E] **the one keystone theorem**: `SEAM.EQUIV.01` = a construction theorem ($c=g_{\text{car}}+N_{\text{fam}}=8$, $\det \text{Cartan}(E_8)=1$, gap $6\ln\frac{3}{2} > 0$ derived) with EXACTLY one open continuum lemma (the rigorous continuum OS reconstruction + invertibility of the raw collar). [E] **`QG.AMB.01` decoupled, not a gate**: margin $\Delta - 31/(4\pi^2) \approx 1.648 > 0$ (v76/v311/v330/v332) ⇒ no TFPT prediction depends on the ambient measure; it is the *general* Euclidean-QG conformal-factor problem (Gibbons–Hawking–Perry 1978), inherited not TFPT-specific — reclassified from "TFPT structural frontier" to "decoupled general QG problem". [E] **collapsed status**: open structural items

$2 \to 1$; live residual $= v_{\text{geo}} + G_{\text{net}}(=$SEAM.EQUIV.01$) + F_{\text{transfer}}$. [O] the one residual = SEAM.EQUIV.01's continuum OS reconstruction.

- **Lean (FORM.QGEO.BW.01, qgeoSymIsCorollary).** BWKeystone.lean extended with the cited conformal-net vacuum axiom conformal_net_vacuum_rotation_invariant; with it, StateInvariance (=QGEO.SYM.01) follows from SeamIsE8 alone, so QGEO.SYM.01 is a COROLLARY of SEAM.EQUIV.01 with no independent open premise (#print axioms confirms only the chain + cited BW/net axioms). lake build green (3362 jobs).
- **Status deep-sync.** The ledger (QG.AMB.01 reclassified to "decoupled general QG; non-TFPT"; QGEO.SYM.01 marked a corollary; new SEAM.EQUIV.UNIFY.01 row), the canonical status card (tfpt_status.tex), introduction (status core + gate table), origin_theory, tfpt_4_frontier, tfpt_5_redteam, tfpt_research_contracts, README, next.txt and the website (OpenGates, FAQ, papers.ts) all moved to "one solvable theorem (SEAM.EQUIV.01) + one decoupled foreign problem (QG.AMB.01)". No falsifiable prediction changed; this is a status-framing sharpening backed by v335 + the Lean corollary.

2026-06-22 (closing round: the CM-selection mechanism (the last keystone residual), the conformal-mode resolution, and the Conway–Sloane even-unimodular side in Lean: v333, v334, FORM.CARTAN.DET.01)

- **v333_cm_selection.py (QGEO.CMSELECT.01, [E]/[O]).** The CM-selection mechanism for the last keystone residual: $\tau=i$ (the square) is the *unique* order-4 modulus — $S=\left(\begin{smallmatrix}0 & -1\\ 1 & 0\end{smallmatrix}\right)$ has $S^4=I$ and fixes i, and an order-4 deck selects $j=1728$ (vs the order-3 hexagon $j=0$) — AND the square is the Fekete attractor of symmetric mark-equilibration (four points minimizing the log-energy converge to equal $\pi/2$ spacing). So given the μ_4 deck, $\tau=i$ is forced; the residual sharpens from "why the square" to "the deck acts geometrically" (= QGEO.SYM.01). NOT a closure.
- **v334_conformal_resolution.py (QGAMB.CONFORMAL.01, [E]/[C]/[O]).** The conformal-mode resolution for the QG metric sector: the Gibbons–Hawking–Perry contour rotation $\phi \to i\phi$ flips the wrong-sign kinetic term, making the divergent conformal Gaussian $\int e^{+|c|\phi^2}$ into the convergent $\int e^{-|c|\phi^2}=\sqrt{\pi/|c|}$; the entire IDG form factor e^{p^2/M^2} dresses the propagator pole-free (the polynomial $R{+}R^2$ ghost as control). Together they give a *conditional* mode-by-mode construction of the metric-sector measure; the residual is only the contour's nonperturbative validity (gap-decoupled, blocks no test). NOT a closure.
- **CartanDeterminants.lean extended (FORM.CARTAN.DET.01, [E]).** The Conway–Sloane lattice side: the E_8 Cartan matrix is proved to be an EVEN (diagonal = 2), symmetric, unimodular ($|\det|=1$) integer Gram matrix — the defining data of an even unimodular rank-8 lattice — all by kernel decide / the integer-inverse argument. The remaining content (Witt's uniqueness: E_8 is the only even unimodular lattice in dim < 16) is the cited classification residual.

2026-06-22 (endgame round: the necessity of H reduced to the order-4 CM selection, the QG metric sector characterized as the conformal-mode problem, $|\det E_8|=1$ and the μ_4 commutation grounded in Lean: v331, v332, FORM.MU4COMM.01)

- **v331_necessity_of_H.py (QGEO.NECESS.01, [E]/[O]).** The necessity companion to v329: a concrete Steklov (DtN) computation shows the seam $\Lambda=|k|+M_f$ commutes with the μ_4 clock *iff* the four marks sit at the square ($\tau=i$) — moving a mark off the square breaks $\mathbb{Z}_4$ and $[\rho,\Lambda]\neq 0$. The square has cross-ratio 2 ($j=1728$), the unique 4-config with an order-4 automorphism (v214/v267), so $H \Leftrightarrow$ square $\Leftrightarrow \mu_4$ and "necessity of H" sharpens to: the raw dynamics places the four Gauss–Bonnet marks (v216) at the order-4 CM point. Residual = that dynamical selection.

- **`v332_qgamb_metric_sector.py` (`QGAMB.METRIC.01`, [E]/[C]/[O]).** The open metric sector of `QG.AMB.01` is characterized precisely as the Gibbons–Hawking–Perry conformal-factor problem: the conformal mode has a negative kinetic coefficient $-(D-1)(D-2)/4$ ($=-\frac{3}{2}$ at $D=4$), so the bare Euclidean measure diverges over it. The obstruction is in the conformal/gauge direction, gap-insulated from the admissible sector (v330/v76); the only route is the conditional IDG taming (v304), with the polynomial $R+R^2$ ghost as the control. Open, but pinned to one named obstruction.
- **`CartanDeterminants.lean` extended (`FORM.CARTAN.DET.01`, [E]).** The rank-8 holomorphy discriminator $|\det\mathrm{Cartan}(E_8)|=1$ (E_8 has one anyon, vs D_8's four) is now proved principally and kernel-only: `cartanE8 * cartanE8inv = 1` by `decide` on the 8×8 PRODUCT gives `IsUnit (det E8)` via $\det E_8 \cdot \det E_8^{-1} = \det 1 = 1$ — no 8! determinant, no `native_decide` (Mathlib itself leaves `E₈_det = 1` a `proof_wanted`).
- **`Mu4Commutation.lean` (`FORM.MU4COMM.01`, Lean 4, [E]).** The v201/v323 mark-locality criterion as exact linear algebra over the Gaussian integers $\mathbb{Z}[i]$: $i^4=1$, $i^2=-1$, and (on concrete 6×6 matrices) an offset-4 coupling commutes with the μ_4 clock while an offset-2 coupling does not — all by kernel `decide`. Built `lake build` on Lean v4.29.1 + Mathlib.

2026-06-22 (frontier round: the OS step discharged at the many-body level, the ambient measure built on the gap-decoupled sector, and the carrier Cartan determinants grounded in Lean: v329, v330, `FORM.CARTAN.DET.01`)

- **`v329_os_gap_reduction.py` (`QGEO.OSGAP.01`, [E]/[O]).** The one analytic input the v308 keystone chain still cited — "the OS-reconstructed bulk gap = the transfer gap" — is discharged at the many-body level: the OS/Euclidean Hamiltonian is the second quantization $d\Gamma(-\log T)$, whose spectrum is the set of sums of single-mode energies, so the bulk gap = the smallest positive single-mode energy $= 6\ln\frac{3}{2}$, *independent of system size*. Together with $H \Rightarrow \omega\circ\rho{=}\omega$ and $H \Rightarrow$ SRE $\Rightarrow (E_8)_1$, this exhibits the rotation-invariant flat-pillowcase premise H as the *single sufficient premise* of both bedrock IDs; residual = the necessity of H on the raw collar.
- **`v330_qgamb_admissible_measure.py` (`QGAMB.MEASURE.01`, [E]/[O]).** The constructive step `v275`'s roadmap calls for: on the gap-decoupled *admissible* sector the ambient measure is built as a bona fide OS object — reflection-positive (symmetric PSD transfer), exponentially clustering ($C(n)/C(n-1) \to (2/3)^6$), with finite susceptibility $\chi = 729/665$ so $\mathrm{Var}(S_\Lambda)/\Lambda \to \chi$ converges ⇒ uniform tightness (Prokhorov) ⇒ the projective limit exists; OS reconstruction gives $H_{\mathrm{OS}} \geq 0$ with gap $6\ln\frac{3}{2}$. The full $\mathrm{QG}_{\mathrm{amb}}$ (the metric sector) stays open, gap-decoupled — a genuine *partial* construction, not a closure.
- **`CartanDeterminants.lean` (`FORM.CARTAN.DET.01`, Lean 4, [E]).** The carrier piece of the holomorphy/anyon discriminator is grounded in real `Matrix.det` computations (kernel `decide`, no extra axiom): $|\det\mathrm{Cartan}(A_3)|=4=|\mu_4|$ (3×3, `det_fin_three`) and $|\det\mathrm{Cartan}(D_5)|=4$ (5×5, `decide`), giving the carrier $|\det\mathrm{Gram}|=4\cdot 4=16$ (the `v281` anyon count). The rank-8 E_8/D_8 case stays the Nat discriminator — Mathlib's own `Data.Matrix.Cartan` leaves `E₈_det = 1` as a `proof_wanted`. Built `lake build` on Lean v4.29.1 + Mathlib.

2026-06-22 (next-steps round: the keystone collapsed to one shared premise (Lean + one citable theorem), the F_{transfer} solver suite, the cusp weight derived from a minimal rewrite, and the θ_{13} pressure point: `FORM.QGEO.BW.01`, `v325`, `v326`, `v327`, `v328`)

- **`BWKeystone.lean` (`FORM.QGEO.BW.01`, Lean 4, [E]/[O]).** The `v323` Bisognano–Wichmann linkage formalised: the nodes RotationInvariantVacuum → GeometricModularFlow → Mu4ModularSymmetry → StateInvariance (=`QGEO.SYM.01`) composed over the `v308` chain. The theorem `bedrockReducesToRotationInvariance` shows the WHOLE bedrock `QGEO.SYM.01` reduces to the SINGLE shared open premise `RotationInvariantVacuum` — the two open bedrock items collapse to one — with `#print axioms` confirming the residual = the cited chain steps + recovery gap + the named BW steps + that one premise. Plus a `decide`-proven discriminator $(1 \mid 4) \wedge \neg(4 \mid 1)$ (BW geometric flow strictly stronger than `v309` clock invariance). Built `lake build` on Lean v4.29.1 + Mathlib.

- **`v325_pillowcase_keystone.py` (`QGEO.KEYSTONE.01`, [E]/[O]).** The keystone as ONE citable theorem: IF the raw seam state is the flat $\tau{=}i$ pillowcase (rotation-invariant) state THEN the whole bedrock closes (4 square marks, μ_4 deck, $\omega \circ \rho{=}\omega$, holomorphic $c{=}8 \Rightarrow (E_8)_1$, with the recovery gap as input); all pieces machine-checked (incl. the pillowcase $\chi_{\text{orb}}{=}0$), the single open lemma shared by both bedrock IDs and Lean-pinned. An assembly certificate, not a closure.

- **`v326_ftransfer_suite.py` (`FR.SUITE.01`, [E]/[C]).** The four F_{transfer} readouts productised into one runnable harness, each a gapped relaxation to a unique attractor: F_{pole} (Koide, seam rate $(2/3)^6$), $F_{\text{Boltzmann}}$ (η_B, H-theorem), F_{relic} (axion adiabatic invariant), F_{QCD} (m_p/m_e, RG to the UV fixed point), plus the F_{RareKaon} bridge. Only F_{pole} carries the seam rate; the four external rates are honestly fenced (`v187`/`v303`).

- **`v327_hypergraph_rewrite.py` (`HYP.REWRITE.01`, [E]/[O]).** The cusp weight 2/3 is DERIVED from a minimal rewrite: a 3-channel/1-absorbing family rule has spectrum $\{1, 2/3, 0\}$, so $2/3{=}(N_{\text{fam}}{-}1)/N_{\text{fam}}{=}|\mathbb{Z}_2|/N_{\text{fam}}$ emerges from the rule arity and $(2/3)^6{=}64/729$ is the recovery gap; with a PROOF that 2/3 is not a root of the affine-E_8 charpoly ($p(2/3){=}-3520/19683 \neq 0$, sharpening `v312`). `v324`'s injected datum is reduced to the anchor arity $\{2, 3\}$.

- **`v328_theta13_pressure.py` (`FLAV.TH13.PRESSURE.01`, [C]/[X]) + `v307` hardening.** The honest weak spot named and quantified: $\sin^2\theta_{13}{=}\varphi_0^{\text{ret}} e^{-5/6}{=}0.023108$ is $+2.0\sigma$ above NuFIT 6.0 NO, a SEPARATE carrier-trace channel (`v268`), not relieved by any freeze-respecting correction (RG $< 1\%$, exponent 5/6 exact), pre-registered as the kill. The `v307` watchdog gains a data provenance/date check and the rare-kaon F_{transfer} bridge row.

- **Wolfram mirror** 285 → 288. `v325` ($\chi_{\text{orb}}{=}0$) and `v327` (rewrite spectrum $\{0, 2/3, 1\}$, $(2/3)^6{=}64/729$, and the non-graph-spectral proof) added to `tfpt_readouts_extension.wl`; `GATE.WOLFRAM.02` and the Wolfram README updated to 288/288.

2026-06-22 (consistency + de-accretion editorial pass: current-state alignment across papers, website and ledger; no new modules)

- **Keystone naming reconciled to `SEAM.EQUIV.01`.** Following `v323`/`QGEO.BW.01` (the deck premise is downstream of the seam being a chiral net), the single named open keystone is now `SEAM.EQUIV.01` *everywhere*, with `QGEO.SYM.01` stated as its *downstream* conformal-deck face. The ledger row `QGEO.SYM.01` was updated (it no longer calls itself "the single structural residual of the whole theory"; it now depends on `SEAM.EQUIV.01`/`QGEO.BW.01`), and the superseded "`GATE.QGEO` open gate" / "two research gates (U_{wall}) and G_{metric}" framing was replaced by the current Rest $= v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$ with the two structural items `SEAM.EQUIV.01` + `QG.AMB.01` (intro status card, `tfpt_4_frontier`, `origin_theory`, `tfpt_5_redteam`, `tfpt_research_contracts`, and the website).

- **De-accretion: the sprint-by-sprint reduction history was moved to this changelog, where it belongs.** Several reader-facing sections had grown by accretion (each round appended a clause). They were rewritten to state the *current* result once, with the round-by-round narrative left here: the introduction closure ledger ("honest bottom line") and hard-reduction subsection; the `origin_theory` "residual inventory" (the five "Update/. . . /Terminal" passes collapsed to one current inventory); the `tfpt_5_redteam` Target-A Outcome; and on the website the FAQ "What is still open?", the `OpenGates` closing-condition block, the `papers.ts` research-contracts/red-team entries, the verification-DAG `qft` node, the glossary `G_net` entry, the Rosetta lexicon, and the verification/review pages. No scientific content or `\veri` citation was dropped; only the embedded version-stamp logs were removed.
- **Stale facts corrected.** Version label `5.0`→`v5.2` (intro, single-sourced via `\TFPTversion`); Lean build `1886`→`3357` jobs (`FORM.SEAMEQUIV.01`); README Python suite `v213`/ 212→`v324`/ 322 modules; Wolfram extension `269/269` / `249/249`→`285/285` (README, website replication page, `zenodo_description.html`); residual "Paper-3" old-paper attributions in `tfpt_2` neutralised.

(Original 2026-06-22 entry follows.)

2026-06-22 (external-review pass: a registry bug fix, the NA62 2026 rare-kaon update, CP-phase prose hygiene, and three firewall/visualisation enhancements: `v202`, `v203`, `v305`, `v307`, `v321`)

- **Bug fix — δ_{CKM} display value.** The predictions table (`tex-artefacts/predictions.tex`) showed $\delta = \frac{\pi}{3} + 3\lambda^2 = 66.96°$, but that formula evaluates to 68.65° (the canonical frozen `DELTA_CKM_RAD`= 1.198232 rad, used by `v88/v202/FLAV.CP.01` everywhere else). Corrected to 68.65°. An external review correctly flagged the inconsistency; it was an isolated display typo (the website was already clean).
- **`v202_rare_kaon.py` — NA62 2016–2024 update ([E]/[C]).** The charged-mode comparison is moved from the historical NA62 2016–2022 observation $(13.0^{+3.3}_{-3.0}) \times 10^{-11}$ ($\sim 1.2\sigma$ above) to the full 2016–2024 combination $\mathrm{BR}(K^+) = (9.6^{+1.9}_{-1.8}) \times 10^{-11}$ (La Thuile 2026), on which the prediction 9.45×10^{-11} lands at $+0.08\sigma$. Framed honestly: a strong *consistency* hit for the closed CKM point, but an F_{transfer} bridge — *not* a unique TFPT-vs-SM discriminator (the SM $(8.6 \pm 0.4) \times 10^{-11}$ is also inside the NA62 error). Propagated to the ledger (`FR.RAREKAON.01`), `freeze_file` (new `rare_kaon` kill row), the watchdog (`v307`, a new F_{transfer} bridge row), the forward board (`v321`, rank 9), and `tfpt_2_standard_model` + the website.
- **CP-phase prose hygiene.** The Galois CP lock is tightened from "a relation to the *measured* quark phase" to "a relation to the *leading* ($\pi/3$) component of the quark phase": the lock is to the structural $\pi/3$, not the full measured $\gamma = \delta^{\mathrm{lead}}_{\mathrm{CKM}} + 3\lambda_C^2 \approx 68.7°$, so $\delta_{\mathrm{PMNS}} = 240°$ (not 248.7°). Fixed in `tfpt_2_standard_model`, `origin_theory`, `papers.ts` and `predictions.ts`.
- **`v305_witness_independence.py` — the No-Golden-Dynamics rule ([E]).** The number-field split is made an explicit anti-numerology firewall rule: $\phi = 2\cos(\pi/5) \in \mathbb{Q}(\sqrt{5})$ is the *static* carrier ($\mathbb{Z}/5$) field, every *dynamic* rate is rational ($\mathbb{Q}$, the $\mathbb{Z}/6$ hand), so a dynamic rate explained by ϕ is a cross-field false friend (`v314/v315`).
- **`v203_eht_achromatic.py` — the μ_4 Fourier fourth null ([X]).** The EHT residual pre-registration gains a fourth null: the azimuthal residual must carry power *only* at $m \equiv 0 \bmod 4$ (the μ_4 fingerprint of the four marks, `v201`); a μ_4-orbit profile has ~ 0 off-mode power while a $\mathbb{Z}_2$ control leaks at $m = 2$.

- **Seed-hyperplane figure (Fig. ??).** A new figure visualises `v306`: every scorecard observable inverts *independently* to the one latent seed φ_0, the error-weighted mean matching the axiom to 0.04%, with $\sin^2\theta_{13}$ the $\sim 2\sigma$ pressure point. Added to `make_figures.py`, `tfpt_5_redteam` and the manifest.

2026-06-22 (Wolfram mirror of the arithmetic arc + three sharpening steps: the CP-lock made quantitative, the Bisognano–Wichmann keystone, the minimal hypergraph fibre: v322, v323, v324)

- **Wolfram mirror of the cyclotomic/Galois arithmetic arc (the independent second path).** The exact algebraic results of the arc are now mirrored in `verification/wolfram/tfpt_readouts_extension.wl` (ten new `checkExact` blocks, $275 \to 285$, *all passing*): `v313` (the affine-E_8 characteristic polynomial with the golden factor x^2-x-1; the $(2,3,5)$ atoms as $2\cos(\pi/k)$; the 31/30 selection), `v314` (the rational kernel vs $\mathbb{Q}(\sqrt{5})$ dynamic-rate split), `v315` ($30{=}5{\cdot}6$, $\varphi(30){=}8$, the Galois group $\mu_4\times\mathbb{Z}_2$, the $\sqrt{5}$ Gauss sum), `v316` ($\rho{=}\zeta_6{=}\zeta_{30}^5$, $\rho^4{=}-\rho$, Gal$=\mathbb{Z}_2=$ CP conjugation), `v317` ($N_{\rm fam}{=}3$ as the μ_3 orbit, the 1+2 Galois split), `v318` ($\varphi_0^{\rm ret}{=}(4/3)c_3{+}48c_3^4$ a pure function of π) and `v320` (the CP lock $\delta_{\rm PMNS}{=}240°$). Ledger `GATE.WOLFRAM.02` and the Wolfram README updated to 285/285.

- **`v322_cp_lock_sharpened.py` (`GALOIS.CP.SHARPEN.01`, [E]/[C]/[X]).** The `v320` CP-lock sharpened into a quantitative, pre-registered kill test. [E] the leading phase sits on the six hexagonal nodes $\arg\rho^k{=}k\cdot 60°$, and *which* one is selected is the deck order: $\delta^{\rm lead}_{\rm PMNS}{=}|\mu_4|\,\delta^{\rm lead}_{\rm CKM}{=}4(\pi/3)$ (node 4); [C] the sub-leading correction is bounded by the quark analogue $3\lambda^2\approx 8.7°$, so the prediction is the band $240°\pm\sim 9°$; [C] against NuFIT 6.0 (NO, $\delta_{CP}{=}212^{+26}_{-41}{}°$) the leading value is $+1.08\sigma$, the band edge $+0.74\sigma$ – compatible today; [X] the nearest wrong node is 60° away, far beyond the budget, so DUNE/Hyper-K ($\sim$ 5–15°) discriminate cleanly. Cited in `origin_theory`; ledger `GALOIS.CP.SHARPEN.01`.

- **`v323_bw_geometric_modular.py` (`QGEO.BW.01`, [E]/[C]/[O]).** The keystone pushed one honest step via Bisognano–Wichmann. [E] the μ_4 clock is a geometric rotation $\rho{=}\mathrm{diag}(i^n){=}\exp(i\frac{\pi}{2}L)$, $\mu_4\subset U(1)_{\rm rot}$; [E] for a rotation-covariant covariance $C{=}f(L)$ the modular Hamiltonian $K{=}\log((1{-}C)/C){=}g(L)$, so $[K,L]{=}0$ (the modular flow is geometric) and the clock is a modular symmetry *for free* (the discriminator: a period-4 curvature preserves the clock but its flow is not geometric, $[K,L]\neq 0$); [C] given the seam is the $(E_8)_1$ chiral net (`v308`), BW/Hislop–Longo make the vacuum modular flow geometric, so `QGEO.SYM.01` ($\omega\circ\rho{=}\omega$) is *downstream* of `SEAM.EQUIV.01` + a rotation-invariant vacuum, not an independent axiom; [O] residual: the raw collar vacuum is rotation-invariant (cleaner than `v309`, shared with `v308`). Cited in `origin_theory`; ledger `QGEO.BW.01`.

- **`v324_hypergraph_fiber.py` (`HYP.FIBER.01`, [E]/[C]).** The constructive `v312` follow-up: the *minimal* substrate that carries both the E_8 skeleton and the recovery gap is a *product*. [E] the carrier network $T_{\rm net}{=}(A{+}2I)/4$ (attractor = Kac marks = skeleton) fibred by the 3-node family cusp $T_{\rm cusp}{=}\mathrm{diag}((1{-}w)^{2N_{\rm fam}})$, $w\in\{0,\frac{1}{3},\frac{2}{3}\}$ (spectrum $\{1,(2/3)^6,(1/3)^6\}$); the fibred $T_{\rm net}\otimes T_{\rm cusp}$ has top eigenvalue 1 (eigenvector marks $\otimes\,(w{=}0)$) and $(2/3)^6$ as a genuine eigenvalue (marks $\otimes\,(w{=}\frac{1}{3})$) – the $N_{\rm fam}{=}3$ cusp is the minimal fibre and the substrate = carrier $\times$ family = the $30{=}g_{\rm car}(2N_{\rm fam}){=}5{\times}6$ of `v315`; [C] honestly (`v312`) the cusp weight is still *injected*, not derived from the graph – a consistent realisation, not a derivation of 2/3 from a pure rewrite. Cited in `origin_theory`; ledger `HYP.FIBER.01`.

2026-06-22 (the forward kill-test board: the decisive upcoming measurements ranked, each locked to a freeze row — incl. the new Galois-CP relation: v321)

- **`v321_killtest_board.py` (`KILL.BOARD.01`, [E]).** The forward companion to `v307` (the current-data board): the decisive *upcoming* measurements, ranked and each locked to a `freeze_file.csv` kill row – #1 JUNO $\sin^2\theta_{12}{=}0.306747$ (fastest), #2 CMB-S4/LiteBIRD r

(kill $r>0.01$), #3 DESI/Euclid w (kill $w\neq-1$), #4 DUNE/Hyper-K $\delta_{\rm PMNS}=240°$ (the new `v320` Galois-CP relation), #5 DUNE/LEGEND/nEXO ordering+$m_{\beta\beta}$, #6 PSI n2EDM, #7 JUNO $\sin^2\theta_{13}$ ($\sim 2\sigma$ now), #8 LiteBIRD β. A forward decision ledger (publicly-stated experimental reach), complementing `v307`; no new physics. Cited in `origin_theory`; ledger row `KILL.BOARD.01`.

2026-06-22 (Lean: the `SEAM.EQUIV.01` composition chain formalised — the keystone residual machine-pinned to the named cited steps: `FORM.SEAMEQUIV.01`)

- **`SeamEquivChain.lean` (`FORM.SEAMEQUIV.01`, Lean 4, [E]/[O]).** A Lean 4 mirror of the `v308` closing chain: the four nodes GapPositive → InvertibleSRE → HolomorphicC8 → SeamIsE8 assembled as a well-typed composition theorem `seamEquivChain`, with the three cited literature steps (Kitaev free-fermion invertibility; Müger / Kawahigashi–Longo–Müger holomorphy; Conway–Sloane E_8 uniqueness) as named `axioms` and the carrier recovery gap (the OS step) as the single open input. [E] `#print axioms seamEquivChain` confirms the residual is exactly those named steps + `recovery_gap` (no hidden assumption); [E] a `decide`-proven discriminator $\mathsf{primariesE8} = 1 \neq \mathsf{primariesD8} = 4$ (the $|\det \mathrm{Cartan}|$ that separates the holomorphic E_8 from the same-c rival $D_8{=}SO(16)$). Built end-to-end (`lake build`, 3357 jobs) on Lean v4.29.1 + Mathlib. [O] The value is the machine-checked composition + axiom audit (pinning the keystone residual), *not* a from-scratch proof — it does NOT close `SEAM.EQUIV.01`. Ledger row `FORM.SEAMEQUIV.01`; mirrors `v308`.

2026-06-22 (a NEW falsifiable prediction from the Galois structure: the two CP phases are locked, $\delta_{\rm PMNS} = \delta^{\rm lead}_{\rm CKM} + \pi$: `v320`)

- **`v320_galois_cp_relation.py` (`GALOIS.CP.PREDICT.01`, [E]/[C]/[X]).** The `v316` follow-up turned into a testable cross-prediction. [E] both leading CP phases are powers of the *one* hexagonal unit $\rho = \zeta_6$ of the family Galois factor: $\delta^{\rm lead}_{\rm CKM} = \arg\rho = \pi/3$ (60°), $\delta_{\rm PMNS} = \arg\rho^4 = 4\pi/3$ (240°), and $\rho^4 = -\rho$ locks them as $\delta_{\rm PMNS} = \delta^{\rm lead}_{\rm CKM} + \pi$ (the $\mathbb{Z}_2$ sheet flip); [C] this upgrades the previously assigned $\delta_{\rm PMNS} = 240°$ to a Galois-forced relation to the measured quark phase (the quark and lepton leading CP phases are not independent); [X] kill test: a $\delta_{\rm PMNS}$ robustly incompatible with 240° ($> 3\sigma$ at DUNE/Hyper-K/JUNO) falsifies the Galois-CP organisation; 240° is currently within the broad NuFIT 6.0 range. A genuine new falsifiable consequence of the `v313`–`v319` arithmetic arc (also a `freeze_file.csv` `cp_phase_relation` kill row). Cited in `origin_theory`; ledger row `GALOIS.CP.PREDICT.01`.

2026-06-22 (the translation clock: the discrete↔dynamic bridge *is* the order-30 Coxeter element, a clock with two coprime hands 5×6, read law-inclusive $(0..5)$ or live-only $(1..5)$: `v319`)

- **`v319_translation_clock.py` (`TRANSLATE.CLOCK.01`, [E]/[C]).** A synthesis of the `v313`–`v318` arc with `v124`/`v55`, answering the question “is the static↔dynamic translation a clock running $0, 1, 2, 3, 4, 5$ or $1, 2, 3, 4, 5$?”. [E] the only object holding both the static carrier ($\mathbb{Q}(\sqrt{5})$) and the dynamic family (rational) data is the order-30 Coxeter element (`v314`); $30 = h(E_8) = g_{\rm car}(2N_{\rm fam}) = 5 \cdot 6$ and $\gcd(5, 6) = 1 \Rightarrow \mathbb{Z}/30 = \mathbb{Z}/5 \times \mathbb{Z}/6$ (two coprime hands, `v315`). [E] the *dynamic* hand is $\mathbb{Z}/6 = 2N_{\rm fam}$ (ticks $0, \ldots, 5$): the recovery exponent is exactly $6 = 2N_{\rm fam}$, rate $(2/3)^6 = (|\mathbb{Z}_2|/N_{\rm fam})^{2N_{\rm fam}}$, order-6 generator c^5 ($\zeta_6 = \zeta_{30}^5$, `v316`). [E] position 0 is the *conserved law* — the resummed clock rate$(n) = -p_2 \ln(1 - n/N_{\rm fam})$ (`v124`) has rate$(0) = 0$ (the attractor, eigenvalue 1, `v56`), the live decaying modes from $n{=}1$, and the E_8 live phases (totatives of 30) start at 1 ($\{1, 7, \ldots, 29\}$), the 0 phase not a totative — so “$0, 1, 2, 3, 4, 5$” is law-inclusive, “$1, 2, 3, 4, 5$” live-only. [E] the *static* hand is $\mathbb{Z}/5 = g_{\rm car}$ (ticks $1, \ldots, 5$): the golden field $\mathbb{Q}(\sqrt{5})$ ($\varphi = 2\cos(\pi/5)$), field-obstructed (no rational dynamic image, `v314`) — static-only; $\zeta_5 = \zeta_{30}^6$ carries no phase. [E] one full turn

$= \mathrm{lcm}(5,6) = 30 = h(E_8)$, rank $E_8 = 8 = \varphi(30)$ live phases pairing into $|\mu_4| = 4$ invariant planes (v55). [C] verdict: the discrete→dynamic translation *is* the order-30 clock = (static 5-ring) × (dynamic 6-ring), the dynamic hand carrying the rate (exponent $6 = 2N_{\mathrm{fam}}$), the static hand the golden carrier (no rate), and "0 vs 1" = law-inclusive vs live-only; the arithmetic is [E], the "it is one clock" reading [C]. Python-only (`sympy`), like the rest of the arc. New figure `figures/translation_clock.pdf` (concentric 5- and 6-hands). Cited in `origin_theory`; ledger row `TRANSLATE.CLOCK.01`.

2026-06-21 (capstone: the SM structural sector is $\mathbb{Q}(\zeta_{30})$+ Galois $\mu_4 \times \mathbb{Z}_2$, and the complete input is $\{a, \pi, v_{\mathrm{geo}}\}$ — zero dimensionless free parameters: v318)

- **`v318_arithmetic_capstone.py` (`ARITH.CAPSTONE.01`, [E]/[O]).** The synthesis of the v313–v317 arc. [E] the SM *structural* sector (the 3 generations, the 2 CP phases, their orbit/hierarchy) is the cyclotomic field $\mathbb{Q}(\zeta_{30})$ with Galois group $\mu_4 \times \mathbb{Z}_2$ (degree $\varphi(30) = 8 = \mathrm{rank}\, E_8$, $30 = |\mathbb{Z}_2| N_{\mathrm{fam}} g_{\mathrm{car}} = h(E_8)$), forced by the anchor atoms $\{2,3,5\} = e(a)$; [E] the magnitude seed $\varphi_0 = (|\mu_4|/N_{\mathrm{fam}})c_3 + \Omega_{\mathrm{adm}} c_3^4 = \frac{4}{3}c_3 + 48c_3^4$ ($c_3 = 1/(8\pi)$) is a pure function of π (no free parameters, anchor-derived coefficients), so the magnitudes ($\sin^2\theta_{12} = \frac{1}{3} - \varphi_0/2$) are $\{a,\pi\}$-determined, not free; [E] free inventory: *zero* dimensionless free parameters, π the unique transcendental primitive (`ARCH.CORE.01`), v_{geo} the unique dimensionful input (No-Unit Theorem) — complete input $\{a, \pi, v_{\mathrm{geo}}\}$. [O] honest residual: this rigidity holds *given* the axioms and does not realise the structure on the raw seam (`QGEO.SYM.01`/`SEAM.EQUIV.01` stay [O]). Cited in `origin_theory`; ledger row `ARITH.CAPSTONE.01`.

2026-06-21 (are the 3 generations a μ_3/Galois orbit? Yes (μ_3), Galois-refined 1+2 with the fixed generation = the recovery attractor: v317)

- **`v317_galois_family.py` (`GALOIS.FAMILY.01`, [E]/[C]).** The v316 follow-up to the generation structure. [E] the three cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\}$ (with $\mathrm{Spec}(Q_+) = \{1,2,3\}$, the A_3 exponents) are the cube roots of unity $\{1, \zeta_3, \zeta_3^2\}$ under $w \mapsto e^{2\pi i w}$, and the cyclic family symmetry $w \mapsto w + \frac{1}{3}$ is a *transitive* μ_3 orbit (the family IS the cube-root orbit); [E] the transfer eigenvalues $(1-w)^6 = \{1, (2/3)^6, (1/3)^6\}$ attach to the weights (attractor = eigenvalue 1 at w=0); [E] the family Galois $\mathbb{Z}_2$ ($\zeta_3 \mapsto \zeta_3^2 = \bar{\zeta}_3$, = the v316 CP conjugation) refines the orbit to 1+2 — fixing w=0 (rational) and swapping $w = \frac{1}{3} \leftrightarrow \frac{2}{3}$; [E] so the Galois-FIXED generation is the recovery ATTRACTOR (the law) and the Galois-conjugate pair the decaying hierarchy. [C] verdict: the family STRUCTURE is the same arithmetic as the CP phase (a μ_3 orbit, Galois-refined 1+2), the mass MAGNITUDES remaining the analytic seed. Cited in `origin_theory`; ledger row `GALOIS.FAMILY.01`.

2026-06-21 (does the Galois $\mu_4 \times \mathbb{Z}_2$ organize the readouts? Yes — the CP phases are the family-factor cyclotomic data, Galois $\mathbb{Z}_2$ = CP conjugation: v316)

- **`v316_galois_readout.py` (`GALOIS.READOUT.01`, [E]/[C]).** The v315 follow-up to phenomenology: does the Galois coupling $\mu_4 \times \mathbb{Z}_2$ reach the physics? [E] it does, through the *family* factor — the hexagonal CP unit $\rho = \zeta_6 = \zeta_{30}^5$ is exactly the order-$6 = 2N_{\mathrm{fam}}$ factor c^5 of v315 (the carrier being $\zeta_5 = \zeta_{30}^6$); [C] the leading CP phases are its cyclotomic arguments $\delta_{\mathrm{CKM}}^{\mathrm{lead}} = \pi/3 = \arg\zeta_6$ and $\delta_{\mathrm{PMNS}} = 4\pi/3 = \arg\zeta_6^4$ (power $4 = |\mu_4|$), with $\zeta_6^4 = -\zeta_6$ the $\mathbb{Z}_2$ sheet flip and $\mu_6 = \mu_3 \times \mu_2$ (inherits v231/v233); [E] the family Galois $\mathrm{Gal}(\mathbb{Q}(\zeta_6)/\mathbb{Q}) = \mathbb{Z}_2$ ($\zeta_6 \mapsto \zeta_6^5 = \bar{\zeta}_6$) *is* CP conjugation $\delta \mapsto -\delta$; [E] the carrier factor ζ_5 (golden $\sqrt{5}$) carries no phase (μ_4 static), and the magnitudes ($\sin^2\theta_{12} = \frac{1}{3} - \varphi_0/2$, transcendental φ_0) are not cyclotomic. [C] verdict: the Galois bridge reaches phenomenology in the phase sector predicted by the v314 field split. Cited in `origin_theory`; ledger row `GALOIS.READOUT.01`.

2026-06-21 (couple or factorize? the order-30 Coxeter element couples the carrier and family sectors as a cyclotomic compositum, Galois $= \mu_4 \times \mathbb{Z}_2$: `v315`)

- **`v315_coxeter_coupling.py` (`COX.COUPLE.01`, [E]/[C]).** Answers the `v314` question — does $30 = g_{\rm car}(2N_{\rm fam}) = 5{\cdot}6$ couple the carrier (5-fold, golden, $\mathbb{Q}(\sqrt{5})$) and family ($6 = 2{\cdot}3$, rational) sectors, or just factorize? [E] the cyclic group *factorizes*: $\gcd(5,6) = 1 \Rightarrow \mathbb{Z}/30 = \mathbb{Z}/5 \times \mathbb{Z}/6$ (direct product, group-decoupled, in line with the `v314` field obstruction); [E] the E_8 exponents $=$ totatives(30) split as totatives(5)$\times$ totatives(6), $\varphi(30) = 4\cdot 2 = 8 = \mathrm{rank}\ E_8$; [E] but the *field* couples: $\mathbb{Q}(\zeta_{30})$ is the compositum of the carrier $\mathbb{Q}(\zeta_5) \ni \sqrt{5}$ and the family $\mathbb{Q}(\zeta_3)$, degree $8 = \mathrm{rank}\ E_8$; [E] the Galois group is exactly $(\mathbb{Z}/5)^\times \times (\mathbb{Z}/3)^\times = \mathbb{Z}/4 \times \mathbb{Z}/2 = \mu_4 \times \mathbb{Z}_2$ (order 8, max element order 4), so the μ_4 seam clock *is* the Galois group of the carrier's golden field and $\mathbb{Z}_2$ that of the family field (refining `v223`); [E] $\sqrt{5} = \phi + 1/\phi$ is the quadratic Gauss sum in $\mathbb{Q}(\zeta_5)$. [C] verdict: $30 = 5\cdot 6$ does not dynamically entangle the sectors — it couples them as the cyclotomic compositum, the static$\leftrightarrow$dynamic bridge being *Galois*, not a dynamical map. Cited in `origin_theory`; ledger row `COX.COUPLE.01`.

2026-06-21 (do the discrete-kernel and dynamic rates translate? An exact number-field split $\mathbb{Q}$ vs $\mathbb{Q}(\sqrt{5})$: `v314`)

- **`v314_rate_translation.py` (`RATE.TRANSLATE.01`, [E]/[C]).** Answers, exactly, whether the discrete kernel (the affine-E_8 network, carrying the golden ratio) and the dynamic part (the seam transfer, carrying the recovery rate $(2/3)^6$) translate. [E] FIELD SPLIT: the adjacency spectrum is the rational part $\{0, \pm 1, \pm 2\} \subset \mathbb{Q}$ plus the golden part $\{\pm\phi, \pm 1/\phi\} \subset \mathbb{Q}(\sqrt{5})$, while the transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ is *entirely* rational; [E] the family/sheet rates DO translate (the rational angles $2\cos(\pi/2){=}0 = |\mathbb{Z}_2|$ and $2\cos(\pi/3){=}1 = N_{\rm fam}$ build $(2/3)^6 = (|\mathbb{Z}_2|/N_{\rm fam})^{2N_{\rm fam}} \in \mathbb{Q}$, same field, same atoms), but [E] the carrier 5-fold (golden, $\mathbb{Q}(\sqrt{5})$) has NO rational dynamic image – a field obstruction (2/3 is neither an adjacency nor a lazy-update eigenvalue; heat-kernel $t = \Delta/(2-\phi) \approx 6.37$ not clean), so the carrier is *static-only*; [E] the only object holding both is the order-30 Coxeter element $30 = g_{\rm car}(2N_{\rm fam}) = 5\cdot 6$, and the ϕ-quasicrystal $E_8 = H_4{+}\phi H_4$ $(240 = 2\cdot 120)$ carries the carrier but not the (rational, 3-fold) cusp-weight family dynamics. Cited in `origin_theory`; ledger row `RATE.TRANSLATE.01`.

2026-06-21 (the golden ratio made precise: the $(2,3,5)$ atoms are the affine-E_8 network spectral angles, $\phi = 2\cos(\pi/5)$ the $g_{\rm car}{=}5$ signature: `v313`)

- **`v313_golden_atoms_spectral.py` (`GOLD.ATOMS.01`, [E]/[C]).** A follow-up to `v312`'s observation, made exact. [E] the affine-E_8 adjacency characteristic polynomial factors as $x(x^2{-}4)(x^2{-}1)(x^2{-}x{-}1)(x^2{+}x{-}1)$, so the golden minimal polynomial $x^2{-}x{-}1$ divides it and $\phi = 2\cos(\pi/5) = (1{+}\sqrt{5})/2$ is an *exact* eigenvalue; [E] the non-trivial eigenvalues are $2\cos(\pi/k)$ for exactly $k \in \{2,3,5\} = \{|\mathbb{Z}_2|, N_{\rm fam}, g_{\rm car}\}$ ($2\cos(\pi/2){=}0$, $2\cos(\pi/3){=}1$, $2\cos(\pi/5){=}\phi$, $2\cos(2\pi/5){=}1/\phi$), so the golden ratio is *specifically* the $g_{\rm car}{=}5$ (5-fold) signature and $g_{\rm car}{=}5$ gains a new spectral witness distinct from Pascal/Riemann–Roch (raising the `v305` multiplicity); [E] the $(2,3,5)$ spherical excess $1/2{+}1/3{+}1/5 = 31/30 > 1$ (the condition selecting the finite $2I \Rightarrow E_8$) ties the `v63`/`v76` gap-margin numerator $31 = 2^{g_{\rm car}}{-}1 = 248/8 = 1{+}h(E_8)$ to the Coxeter $30 = h(E_8)$ — a unification of already-anchored quantities, NOT new evidence (honest anti-numerology, `v305`); [E] honest null: ϕ is a structural lattice/network quantity, not a phenomenological readout. [C] direction: ϕ is the E_8-quasicrystal (Elser–Sloane cut-and-project) ratio, a candidate aperiodic substrate for the `v312` question. Cited in `origin_theory`; ledger row `GOLD.ATOMS.01`.

2026-06-21 (TOE-roadmap solver round: the `SEAM.EQUIV.01` keystone certificate, the modular bedrock, the carrier→SM closure, the gap-decoupled measure, and the sharpened hypergraph substrate: `v308`–`v312`)

- **`v308_seam_equiv_chain.py` (`SEAM.EQUIV.CHAIN.01`, [E]/[O]).** TOE attack point 1: the keystone "the raw seam-Calderón net is the holomorphic $c{=}8$ $(E_8)_1$ net" assembled into ONE well-typed logical chain gap>0 → invertible/SRE (Kitaev) → holomorphic $c{=}8$ (Müger/KLM) → $(E_8)_1$ (Conway–Sloane), with the exact discriminators $|\det \mathrm{Cartan}(E_8)|{=}1$ vs $|\det \mathrm{Cartan}(D_8{=}SO(16))|{=}4$, the carrier tower $4{\cdot}4{=}16 \to 4 \to 1$ and $c{=}g_{\mathrm{car}}{+}N_{\mathrm{fam}}{=}8$ all machine-checked; the input gap is the derived recovery gap $6\ln(3/2){>}0$. The only undischarged premise is the cited OS step — an assembly/reduction certificate (a Lean target), not an end-to-end closure. Cited in `tfpt_research_contracts`; ledger row `SEAM.EQUIV.CHAIN.01`.
- **`v309_modular_bedrock.py` (`QGEO.MODHAM.01`, [E]/[O]).** TOE attack point 2: the bedrock $\omega{\circ}\rho{=}\omega$ via an explicit modular Hamiltonian, non-circular — the four marks come from Gauss–Bonnet (`v216`) and the order-4 clock from cross-ratio 2 (`v214`), so μ_4 is derived; the quasi-free modular Hamiltonian $K = \log((1{-}C)/C)$ round-trips to the seam operator and $[\rho, K]{=}0 \Leftrightarrow \omega{\circ}\rho{=}\omega$ (a $\mathbb{Z}_2$ profile breaks it). Reduces the bedrock to the Bisognano–Wichmann intrinsicality of the raw collar; not a closure of `QGEO.SYM.01`. Cited in `tfpt_research_contracts`; ledger row `QGEO.MODHAM.01`.
- **`v310_carrier_sm_anomaly.py` (`CAR.SM.ANOM.01`, [E]).** TOE attack point 3: the carrier half-spinor $16 = \Lambda^{\mathrm{even}}(\mathbb{C}^5) = 1{+}10{+}5$ is exactly one anomaly-free Standard-Model generation ($SU(5) \subset SO(10)$: $10{+}\bar{5}{+}1$), with $\sum Y = \sum Y^3 = [SU(2)]^2U(1) = [SU(3)]^2U(1) = 0$ (exact) and the one-loop $(b_1, b_2, b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$; a fourth family shifts b_1, so $N_{\mathrm{fam}}{=}3$ pins the RG. Closes the spectrum/anomaly/β sub-question; the interacting Epstein–Glaser S-matrix and the Yukawa sector stay [C]/[O]. Cited in `tfpt_research_contracts`; ledger row `CAR.SM.ANOM.01`.
- **`v311_gap_decoupled_measure.py` (`QGAMB.CLUSTER.01`, [E]/[C]/[O]).** TOE attack point 4: the physical QG measure as a gap-decoupled, exponentially-clustering object — the transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ gives geometric clustering at rate $(2/3)^6$ (correlation length $1/(6\ln(3/2))$, finite susceptibility 729/665), and the `v76` margin $6\ln(3/2) - 31/(4\pi^2) {\approx} 1.648{>}0$ decouples the physical sector from the un-built ambient. `QG.AMB.01` stays [O] (gap-decoupled, inert). Cited in `tfpt_research_contracts`; ledger row `QGAMB.CLUSTER.01`.
- **`v312_hypergraph_substrate.py` (`HYP.INJECT.01`, [E]/[O]).** TOE attack point 5: the hypergraph substrate sharpened — the affine-E_8 Kac marks are the Perron eigenvector ($A\,m{=}2m$) and the subleading eigenvalue is exactly $2\cos(\pi/5)$ = the golden ratio, but the recovery rate $(2/3)^6$ is *not* any eigenvalue and φ_0 is not a rational edge fraction; so a full-structure rewrite must inject exactly two non-graph-spectral data — the cusp weight $2/3{=}|\mathbb{Z}_2|/N_{\mathrm{fam}}$ and the analytic seed φ_0. The open question is precisely delimited, not closed. Cited in `introduction`; ledger row `HYP.INJECT.01`.

2026-06-21 (adversarial-review follow-ups: a structural anti-numerology firewall, an out-of-sample seed cross-validation, an operational data watchdog, and the Target-A holomorphy kill test: `v305`–`v307`)

- **`v305_witness_independence.py` (`WITNESS.ECON.01`, [E]).** The *structural* anti-numerology firewall complementing the `v100` phenomenological null test, against the sharpest self-deception worry ("the same integer counted many times as if independent"). Every headline integer ($|\mu_4|{=}4$, $g_{\mathrm{car}}{=}5$, $|\mathbb{Z}_2|{=}2$, $N_{\mathrm{fam}}{=}3$, $\dim S^+{=}16$, $\mathrm{rank}\,E_8{=}8$, 240, 248, $|2I|{=}120$, $h(E_8){=}30$, $|W(D_5)|{=}1920$, $\Omega_{\mathrm{adm}}{=}48$, the gap exponent 6) is a documented image of the single anchor $a{=}(1,1,2)$ with π the only transcendental primitive (13 atoms from 2 free primitives, 6.5×

compression made explicit), so the recurring $2/3{=}|\mathbb{Z}_2|/N_{\rm fam}$ and $(2/3)^6$ are *one* ratio; the E_8 spine is overdetermined (≥ 3 structurally independent witnesses) while the pure-seed readouts are single-witness (scored jointly by `v100`, never multiplied); the negative control shows α^{-1}'s "137" is foreign to the anchor family (an independent witness via the $F_{U(1)}$ cubic, `v3`). Cited in the `tfpt_5_redteam` body; ledger row `WITNESS.ECON.01`. Python-only.

- **`v306_seed_crossval.py` (`SEED.CROSSVAL.01`, [E]).** A leave-one-out cross-validation turning the "many observables, one seed φ_0" worry into a falsifiable out-of-sample test: back-solving φ_0 from each of the five seed-grammar observables ($\sin^2\theta_{12}$, $\sin^2\theta_{13}$, β, Ω_b, λ_C; NuFIT 6.0 / ACT DR6 / Planck / PDG) and error-weighted-averaging reproduces the axiom seed $\varphi_0 = 1/(6\pi) + 48c_3^4 = 0.0531720$ to 0.04%; leave-one-out predicts every held-out observable within 3σ out of sample (most-tensioned $= \sin^2\theta_{13}$, $\sim 2\sigma$); a data↔formula shuffle explodes the χ^2 by $> 100\times$. Cited in `tfpt_5_redteam`; ledger row `SEED.CROSSVAL.01`. Python-only.
- **`v307_data_watchdog.py` (`DATA.WATCHDOG.01`, [E]).** The operational decision pipeline over the frozen registry (`v84`): it classifies all twenty registry entries PASS/WATCH/TENSION/KILL by live n_σ against the repo-documented current bests (CODATA 2022, NuFIT 6.0, ACT DR6, Planck 2018, PDG 2024, BK18). No decidable prediction is in KILL; the watch items are α^{-1} (1.9σ), s_{23} (1.8σ) and the most-tensioned core prediction $\sin^2\theta_{13}$ (2.0σ, the honest current pressure point); $\sin^2\theta_{12} = 0.306747$ is compatible (0.02σ, the JUNO falsifier), the $r \in [0.0033, 0.0048]$ band is below the BK18 limit and the kill, and $w{=}{-1}$ is flagged as the DESI dark-energy watchdog. Cited in `origin_theory`; ledger row `DATA.WATCHDOG.01`. Python-only.
- **Target-A holomorphy kill test (`freeze_file.csv`).** A new `seam_holomorphy` freeze row records the sharpened *physical* kill criterion for Target A: a raw quasi-free seam bulk showing topological ground-state degeneracy on the torus ($|\det K| > 1$, the same-c rival $SO(16)_1$) would break holomorphy and kill "seam net $= (E_8)_1$" — since holomorphic $c{=}8 \Leftrightarrow E_8$ is machine-proven (`v83`/`v143`) and $\det K{=}1 \Leftrightarrow$ invertible/SRE (`v237`/`v301`/`v302`).

2026-06-19 (an infinite-derivative narrowing of the Stelle-ghost attack, the double-pushout figure, and the "constancy of c" framing of `SEAM.EQUIV.01`: `v304`)

- **`v304_idg_nonlocal_ghost.py` (`QGAMB.IDG.01`, [C]/[O]).** A conditional, parameter-free narrowing of the red-team `rt_F`/v278 Stelle-ghost attack on the perturbative gravity sector. [E] the local $R{+}R^2/\mathrm{Weyl}^2$ four-derivative propagator $1/(p^2(p^2{+}M^2))$ carries a negative-residue massive spin-2 mode (the Stelle ghost) only as a *truncation artefact*; [E] for an *entire* graviton form factor $a(p^2) = e^{p^2/M^2}$ (nowhere zero) the dressed propagator $1/(p^2 a)$ has no extra pole (Tomboulis 1997; Biswas–Mazumdar–Siegel 2006), whereas any *polynomial* truncation has a real zero = the ghost; [C] the spectral-action cutoff is the seam KMS weight $f(u) = e^{-u}$ (v259), so the un-truncated trace points to a nonlocal form factor with *no new parameter*. [O] It does *not* close `QG.AMB.01`: the trace regulator is not yet shown to resum to an entire $a(\Box)$, and a ghost-free *perturbative* graviton is not the nonperturbative ambient measure. Cited in the red-team Target F body; ledger row `QGAMB.IDG.01`.
- **Double-pushout figure (`tfpt_1_architecture_e8`).** A new shared TikZ macro (`TFPTdoublepushout` in `tex-artefacts/tfpt_figures.tex`) draws the commuting square showing that the order-$|\mu_4|{=}4$ glue is *one* operation in two categories: the lattice pushout $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ (v1) and the conformal-net / anyon-condensation extension $(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 \cong (E_8)_1$ (v92/v281/v234/v277/v260). Expository only; no new claim.
- **Framing (`introduction`, `tfpt_research_contracts`).** `SEAM.EQUIV.01` is now stated explicitly as the *one* irreducible structural postulate of the theory — the role the constancy of c plays in relativity (cf. v267). Prose sharpening; no status change.

2026-06-19 (the discrete→dynamic lens on `F_transfer`: one shape, one seam-rate, v303)

- **`v303_ftransfer_dynamics.py`** asks whether the main-branch mechanism — a gapped, positivity-preserving relaxation to a *unique* attractor (the local-averaging update → the E_8 marks, gap $6\ln(3/2)$) — is also the shape of the four `F_transfer` instances, or an unrelated bolt-on. It is the *same* shape: [E] F_pole (Koide) *is* the seam dynamics (Möbius multiplier $F'{=}(2/3)^6{=}\lambda_2$, the seam-clock eigenvalue ⇒ a unique attractor at the seam rate, v82/v54); [C] F_Boltzmann (η_B) is a gapped positive contraction (washout $\kappa_f \in (0,1)$, H-theorem); [C] F_relic (axion) freezes at an adiabatic fixed point; [E] F_QCD (m_p/m_e) is the 1-loop RG flow with the carrier $b_3{=}-7$ (asymptotic freedom, Gaussian UV attractor). [O] Verdict: one shape underlies all four; F_pole has the seam rate $(2/3)^6$ *exactly*, the other three share only the shape with external rates (thermal/cosmological/RG). So `F_transfer` is the readout end of the one discrete→dynamic principle, *not* a bolt-on; the theory's simplicity is preserved because the external rates are honestly fenced (v187), not reduced to the seam. [E] suite `v303` 5/5, `AUDIT OK`.

2026-06-19 (the last lever: the seam mass gap *is* the derived Recovery gap $6\ln(3/2)$, v302)

- **`v302_seam_gap.py`** closes the chain on the single statement v301 had left: "the quasi-free seam bulk is gapped". That gap is *not* a free input — it is the established *Recovery gap* of the seam transfer operator, derived from the carrier. [E] the frozen transfer spectrum is $\mathrm{spec}(T) = \{1, (2/3)^6, (1/3)^6\}$ (v160/v162), a simple Perron eigenvalue 1 separated from the rest; [E] the gap is $\Delta = -\ln((2/3)^6) = 6\ln(3/2) = 2N_{\mathrm{fam}}\ln(3/2) \approx 2.4328 > 0$ (multiplicative gap $1-(2/3)^6 = 665/729$; $(2/3)^6$ the μ_4-deck transfer eigenvalue forced by the carrier); [E] with the decoupling margin $\Delta - 31/(4\pi^2) \approx 1.648 > 0$ (v76) it is robust, and Perron–Frobenius primitivity makes the leading mode simple; [C] by Osterwalder–Schrader / quasi-free clustering a transfer-operator gap $\Delta > 0$ *is* a mass gap of the reconstructed Hamiltonian. [O] **Bottom line:** with v300/v301, `SEAM.EQUIV.01`'s entire residual is now a composition of *standard cited theorems* (OS/quasi-free clustering + Kitaev free-fermion invertibility + the v297 AQFT stack) over *established* TFPT facts (16 quasi-free Majoranas; transfer gap $6\ln(3/2)$) — *no undischarged TFPT-internal assumption remains.* It stays [O] (not machine-proved end-to-end), but the open front is reduced to citable machinery over a spectral gap that is provably positive. [E] suite `v302` 5/5, `AUDIT OK`.

2026-06-19 (Route A's last hypothesis discharged: the gapped quasi-free bulk is invertible by the free-fermion classification, v301)

- **`v301_route_a_invertible.py`** attacks the single remaining *unconditional* gap of `SEAM.EQUIV.01`. Since v300 collapsed Route B into Route A's rationality, that gap is Route A's open hypothesis (v297 LIT-A): "the raw quasi-free seam bulk is invertible (SRE)". A *gapped* free-fermion (Gaussian) 2+1D bulk cannot carry intrinsic topological order — free fermions give the integer (IQHE/Chern) phases, all *invertible*; intrinsic topological order needs interactions (Kitaev's periodic table, class D). [E] the carrier is 16 Majoranas, $c = g_{\mathrm{car}}{+}N_{\mathrm{fam}} = 8 = 16/2$ (v148); [E] $SO(16)_1 \supset (E_8)_1$ ($\dim \mathfrak{so}(16) = 120$ currents $+128$ spinor $= 248$, v154); [C] so a gapped quasi-free bulk is invertible automatically; [E] the invariant $|\det K| = \#$anyons is 1 for E_8 (invertible) vs 4 for the gauged $D_8{=}SO(16)$ (anyons), which free fermions cannot produce. [O] LIT-A's 'invertible bulk' now *follows* from 'gapped 16-Majorana quasi-free bulk' (v148/v160+the mass gap), so the residual changes from a topological-order statement to a spectral one. [O] **Bottom line:** with v300, the whole open content of `SEAM.EQUIV.01` reduces to the *single spectral statement* "the quasi-free seam bulk is gapped" — the topological-order obstruction (hidden anyons) is removed; the gap is not manufactured. **[E]** suite `v301` 6/6, `AUDIT OK`.

2026-06-19 (the Flat-Away attack: a hard discrete obstruction + the pin derived from $(E_8)_1$, `v300`)

- **`v300_flataway_rigid.py`** attacks the single external fact the three Flat-Away routes (v291/v293/v295) left open, on two fronts. *Harden:* the flat seam was only a *soft* minimum ("the spectral mismatch grows with ε"); here the obstruction becomes a *discrete* on/off invariant. **[E]** the flat fingerprint — spec$(|D_\theta|)$ is the integer ladder with every level $n \geq 1$ of multiplicity exactly 2; **[E]** the resonant mode $4k$ splits level $2k$ by gap $= g_k$ (sympy 2×2 $[[2k, g/2], [g/2, 2k]] \Rightarrow 2k \pm g/2$, + an $O(g^2)$ common shift), validated against the full Toeplitz operator; **[E]** any $g_k \neq 0$ lifts the resonant degeneracy ($2 \to 1+1$) and changes the spectral *multiset*, so preserving the flat ladder *with* its mult-2 degeneracies forces every $g_k = 0 \Rightarrow f = 0$ — strictly stronger than the "deviation grows" scan; **[E]** with the v295 convexity, flat is isolated analytically *and* discretely. *Derive the pin:* **[C]** the $(E_8)_1$ character $E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + \dots)$ has integer q-coefficients (248 at level 1; E_8 even unimodular $\Rightarrow$ integer weights), so by 2d Steklov conformal rigidity the integer ladder $\Leftrightarrow$ flat seam, i.e. the external pin *is* the $(E_8)_1$ rationality (Route A). **[O]** verdict: Flat-Away carries *zero* new open content beyond Route A — the two `SEAM.EQUIV.01` routes converge on one rationality fact. **[E]** suite `v300` 6/6, `AUDIT OK`.

2026-06-19 (the hypergraph/Wolfram bridge hardened: the $(2, 3, r)$ seed reaches E_8 as the last finite point, `v299`)

- **`v299_hypergraph_0d_autonomous.py`** hardens the Wolfram/hypergraph reading into a precisely typed bridge (builds on v219/v298): **[E]** the icosahedron is a 3-uniform hypergraph ($|\mathrm{Aut(graph)}| = 120 = |I_h| = |A_5 \times \mathbb{Z}_2|$; the McKay group is the $SU(2)$ binary lift $2I$, a *different* order-120 group whose McKay graph is affine E_8; 30 edges $= h(E_8)$); **[E]** the $1 \to 4$ subdivision rewrite is equivariant (all 120 symmetries at every scale); **[E]** the $2I$ fusion tensor is a weighted 3-uniform hypergraph whose 2-dim slice is the affine E_8 McKay graph; **[E]** Smith ($\rho \leq 2 \Leftrightarrow$ affine ADE) + Collatz–Wielandt ($\rho \leq 2 \Leftrightarrow$ a local witness $Am \leq 2m$); **[E]*** the autonomous rule (diffusion-generated witness + *local* gate $Am \leq 2m$, Perron only as audit) runs $E_6 \to E_7 \to E_8 \to \hat{E}_8$; **[E]** the negative controls $\rho(A_n) = 2\cos\frac{\pi}{n+1}$ and $\rho(D_n) = 2\cos\frac{\pi}{2n-2}$ are < 2 for *all* n (closed form), so neither family ever reaches E_8. Honest type: **[O]** the $(2, 3, \cdot)$ seed is the single irreducible input = P2 — *the rule reaches E_8 as the last finite point on this seed; it is not a derivation of P2.* **[E]** build green, `AUDIT OK`.

2026-06-18 (E8 as a network attractor — the Wolfram-program reading, `v298`)

- **`v298_e8_network_attractor.py`** is an honest audit of the "TFPT as a hypergraph/rewrite system" idea, on the v219 McKay bedrock. **[E]** the affine-E_8 Kac marks $(1, 2, 3, 4, 5, 6, 4, 2, 3)$ are the *unique dynamical attractor* of a minimal local-averaging update $m \leftarrow (A+2I)/4\, m$ on the $(2, 3, 5)$ McKay graph (converges from any positive start; $A\, m = 2m$, top eigenvalue 2); the spherical tower terminates at E_8 and the cascade is nested-subgraph coarse-graining. **[C]** the Wolfram reading "E_8 is the universal attractor of a minimal $(2, 3, 5)$ network" holds at the Coxeter-skeleton level. **[O]** a minimal hypergraph *rewrite* reproducing the *full* structure stays open; honest negatives: φ_0^{ret} is the analytic seed $\frac{4}{3}c_3 = 1/(6\pi)$, not an edge fraction, and recovery is a 1-D Möbius fixed point, not a rewrite. **[E]** build green, `AUDIT OK`.

2026-06-18 (a_2 closed form + Route A literature stack, `v296`–`v297`)

- **`v296_flataway_a2_closed.py`** (`FLATAWAY.A2.CLOSED.01`) evaluates the exact a_2 coefficient in *closed form*: the operator tails telescope geometrically ($C_k + D_k e^{-4kt} = 4kt(1 - e^{-4kt})$) to $W_k^{\mathrm{tail}} = t(1 - e^{-4kt})/(8k(1 - e^{-t}))$, leaving a finite $(4k-1)$-term middle; e.g. $W_1(t) =$

$t(2te^t + e^{3t} + 3e^{2t} + e^t - 1)e^{-3t}/8$. Manifestly positive, small-t law $W_k = t/2 + O(t^3)$ (the t^2 term cancels). Reproduces v295 to machine precision.

- **v297_route_a_literature_stack.py (SEAM.EQUIV.A.LIT.01)** writes Route A's one import as a citable stack: LIT-A (Kitaev; Freed–Hopkins) → LIT-B (Müger; Kawahigashi–Longo–Müger) → LIT-C (Conway–Sloane; Dong–Xu) compose into "invertible bulk ⇒ $(E_8)_1$", all legs established; the single open hypothesis (seam-bulk invertibility) coincides with Route B's Flat-Away, and the stack is non-circular with Route B (the v286 firewall holds). **[E]** build green, `AUDIT OK`.

2026-06-18 (Flat-Away heat route made exact + Lean-formalised, v295 / FORM.FLATAWAY.01)

- **v295_flataway_a2_exact.py (FLATAWAY.A2.01)** upgrades the v292 positive-definiteness from a numerical scan to a *proof*: the first variation of $\operatorname{Tr} e^{-t\Lambda}$ vanishes (zero-mean off-mark curvature, Gauss-Bonnet), and since $\Lambda \mapsto \operatorname{Tr} e^{-t\Lambda}$ is convex (Klein/Peierls), ε=0 is the *global* minimum — so $\Delta\mathrm{Tr} \geq 0$ for all deformations, $= 0$ iff $f = 0$. The exact second-order coefficient is the divided-difference kernel of e^{-tx} (Duhamel), validated to rel. err $< 10^{-6}$.
- **Lean `FORM.FLATAWAY.01`** (`SeamDeckClosure.lean` Part 5): the formal Δ=0 ⇔ flat step — a positive-weighted sum of squares is positive-definite (`heatDeviation_eq_zero_iff`); built and axiom-checked (only `propext`, `Classical.choice`, `Quot.sound`). So the heat route's positive-definiteness is now both analytically proved and machine-formalised; only the single external fact (the seam fixes a_2) remains. **[E]** build green, `AUDIT OK`.

2026-06-18 (Flat-Away attacked on all three routes, v292–v294)

- **v292_flataway_heat_reduction.py (FLATAWAY.HEAT.01)**: the heat route, made precise. The heat-trace deviation $\Delta\mathrm{Tr}(e^{-t\Lambda}) = c(t)\varepsilon^2 + O(\varepsilon^4)$ is a *positive-definite* quadratic form in the off-mark curvature ($c_k(t) > 0$ for every $\mathbb{Z}_4$ mode, positive on all combinations), leading invariant $\|f\|_{L^2}$ — so Flat-Away reduces to the single fact "the seam fixes the a_2 heat coefficient".
- **v293_flataway_spectral_hessian.py (FLATAWAY.SPEC.01)**: the spectral route, upgraded from one direction to the *full* smooth-$\mathbb{Z}_4$ space. Mode $4k$ splits the degenerate Steklov level $|2k|$ at first order; the Hessian of the spectral mismatch over modes $\{4, 8, 12\}$ is positive-definite (eigenvalues $\{0.497, 0.500, 0.503\}$), so the flat seam is a strict isolated minimum — the "only one direction" objection is answered.
- **v294_flataway_rp_energy.py (FLATAWAY.ENERGY.01)**: the variational route. With the rigid mark divisor (4 cone angles π, Gauss-Bonnet 4π), Troyanov uniqueness gives a unique flat cone metric, and the curvature energy $\int f^2$ is strictly convex (Hessian $2\pi I$) with a unique minimiser — Flat-Away reduces to "RP+gap realises the energy-minimiser". The three routes converge on one selection principle for the flat metric; closing any one closes Route B. **[E]** build green, `AUDIT OK`.

2026-06-18 (Route-B red-team: the smooth $\mathbb{Z}_4$ adversary + Flat-Away as a mini-theorem, v290–v291)

- **v290_z4_smooth_curvature_adversary.py (SEAM.ADVERSARY.01)** is the honest red-team that $\mathbb{Z}_4$ block-diagonality is *not* mark-locality: a smooth $\mathbb{Z}_4$ off-mark curvature $f = \varepsilon\cos(4\theta)$ **[E]** passes the v288 commutator ($\|[\rho,\Lambda]\| = 0$, same as mark-local) yet **[E]** shifts the DtN spectrum ($\max|\Delta\lambda| = 0.155$) and the heat trace ($\Delta = 0.019$). So the commutator is necessary, not sufficient; the modulus is excluded by the spectral data, not $[\rho,\Lambda]$=0 — which is exactly why Flat-Away cannot be read off the commutator.

- **`v291_flataway_contract.py` (`FLATAWAY.RP.01`)** states Flat-Away as its own named mini-theorem and types three independent proof routes: [E] spectral-rigidity and [E] heat-kernel (the flat seam is the *unique* spectral/heat match of the smooth $\mathbb{Z}_4$ family — deviation zero only at ε=0) and [C] RP-energy/Troyanov (constant-curvature minimiser). Closing any one closes Route B. [E] build green, `AUDIT OK`.

2026-06-18 (Route B sharpened: raw mark-locality decomposed + reviewer firewalls, v289 / v287–v288)

- **`v289_marklocal_raw.py` (`MARKLOCAL.RAW.01`)** decomposes the one remaining Route-B residual ('why is the raw seam sub-principal term mark-local?') into a 5-lemma chain and isolates the *single* open analytic lemma: [C] L1 conic parametrix (b-calculus/Melrose, imported); [O] L2 **Flat-Away** — RP+gap+4 marks ⇒ curvature vanishes away from the marks; [E] L3 orbit-source (v195/v216), L4 Fourier-support on $4\mathbb{Z}$ (v264/v288), L5 full-operator $[\rho, \Lambda]$=0 (v288). 4/5 discharged — Route B reduces to exactly L2.
- **v288 reviewer firewall**: a SCOPE FIREWALL check makes the exact-label split explicit (Fourier lemma Formal/Exact; numerical test Numerical Exact; transfer to the raw conic DtN Conditional, v264; mark-locality from the RP seam Open Selection, v289) — so it can never be read as 'full-L^2 QGEO proved'.
- **v287 literature matrix**: L3 ('invertible bulk ⇒ single-sector') is typed as a composition of citable results (LIT-A Kitaev/Freed–Hopkins, LIT-B Müger/Kawahigashi–Longo–Müger, LIT-C Kawahigashi–Longo) conditional on one open hypothesis — seam-bulk invertibility, which coincides with Route B. [E] build green, `AUDIT OK`.

2026-06-18 (closing sprint: the Seam Equivalence Theorem named + both routes attacked, v286–v288)

- **`v286_seam_equivalence_contract.py` (`SEAM.EQUIV.01`)** names the one open theorem — *the raw RP seam state is the holomorphic* $(E_8)_1$ *net at* τ=i — and concentrates the whole open structure on it: four objects, six arrows (2 closed, 1 conditional v282, 1 open Gral), an **import firewall** proving the two routes (v276/v280/v284 vs v277/v281/v285) never import each other (no 'E_8 into the geometry and back' circularity), and an exact-label taxonomy. Writes `seam_equivalence.json`.
- **`v287_free_rp_bulk_to_holomorphic_boundary.py` (`SEAM.EQUIV.A01`)**: Route A (AQFT) dependency checker — the 5-lemma chain RP+gap+chirality+Gaussian+invertible ⇒ μ=1 ⇒ holomorphic ⇒ $(E_8)_1$ is logically complete *modulo one* standard theorem: *invertible (SRE) Gaussian bulk* ⇒ *single-sector boundary* (the AQFT form of $\det K$=1). 4/5 discharged.
- **`v288_full_l2_subprincipal_z4.py` (`SEAM.EQUIV.B01`)**: Route B (DtN) — *proves* the full-L^2 lift of the sub-principal $\mathbb{Z}_4$ block-diagonality (a mark-sourced curvature has Fourier support only on $m \equiv 0 \bmod 4$, so $\Lambda = |n| + M_f$ commutes with $\rho = \mathrm{diag}(i^n)$ on the full operator, $\|[\rho, \Lambda]\| \sim 2{\times}10^{-15}$; generic curvature → 4.44), lifting v201/v284 to L^2. The residual shrinks to one sharper question: why is the raw seam sub-principal term mark-local? [E] build green, `AUDIT OK`.

2026-06-18 (the two proof routes decomposed: Route (i) and Route (ii) into lemma chains, v284/v285)

- **`v284_route_i_rp_uniqueness.py` (`QGEO.ROUTEI.01`)** turns Route (i) (RP-state uniqueness) into a 6-lemma chain: [E] 5/6 lemmas are already discharged — L1 RP→DtN (v194), L2 marks→pillowcase (v195/v216/v214), L3 flat Steklov = $\sqrt{-\Delta_{\mathrm{flat}}}$ (verified here), L4 covariance from DtN (v258), L5 μ_4-invariance (v276/Lean/v280); [C] Troyanov (χ_{orb}=0 ⇒ unique flat

$\tau{=}i$) reduces the residual to one lemma. [O] the one open lemma: *the raw seam is the constant-curvature geometric state.*

- **`v285_route_ii_seam_condensation.py` (`QGAMB.ROUTEII.01`)** turns Route (ii) (seam-net condensation) into a 4-lemma chain: [E] 3/4 discharged (no abelian sector $\Leftrightarrow$ det $K{=}1$ $\Leftrightarrow$ holomorphic $\Leftrightarrow$ E_8; the tower 16→4→1), and **its open lemma coincides with Route (i)'s** — both are "the raw seam is the $(E_8)_1$-at-$\tau{=}i$ object" (v282), so proving *either* route closes *both*. [O] the unified open lemma: a canonical equivalence between the raw seam KMS/DtN state and the holomorphic $(E_8)_1$ net at $\tau{=}i$. Build green, `AUDIT OK`.

2026-06-18 (self-investigation round: pillowcase DtN, modular data, the E8-at-$\tau{=}i$ unification, live scorecard, v280–v283)

- **`v280_pillowcase_steklov.py` (`QGEO.STEKLOV.01`)** runs the QGEO chain on the actual flat $\tau{=}i$ pillowcase orbifold: the order-4 deck permutes the 4 cone points (2 fixed + 1 swap) and the geometric chain $[\rho,\Delta]{=}0 \Rightarrow [\rho,H]{=}0 \Rightarrow [\rho,C]{=}0 \Rightarrow \omega\circ\rho{=}\omega$ is numerically exact — route-(i) evidence for `QGEO.OBLIG.01`; the premise stays [O].
- **`v281_holomorphy_modular_data.py` (`QGAMB.MODULAR.01`)**: #anyons = $|\det \mathrm{Gram}|$ (E8→1, SO(16)→4), the anyon-condensation tower $16\to4\to1$, Gauss–Milgram $\to c{=}8$; holomorphy (det $K{=}1$) is the single Tier-B discriminator.
- **`v282_e8_tau_i_unification.py` (`QGAMB.UNIFY.01`)**: the candidate *simplification* — $\chi_{E_8}{=}j^{1/3}$ gives $\chi_{E_8}(i){=}1728^{1/3}{=}12$, so $\tau{=}i$ is both the order-4 CM point (QGEO) and the $(E_8)_1$ modulus (QG.AMB): the two obligations are two faces of one object (the seam is $(E_8)_1$ at $\tau{=}i$), **obligation count** $2\to1$.
- **`v283_live_killtest_scorecard.py` (`PRED.LIVE.01`)**: the recent round's computable predictions vs current data — all consistent $<2\sigma$, with two sharp near-term kill tests (Σm_ν floor 0.0586 eV; proton decay $1.55{\times}10^{35}$ yr).
- Wolfram 275/275 (v281/v282 exact-mirrored); build green, `AUDIT OK`.

2026-06-18 (the Gräl: QGEO.SYM.01 as one precise lemma + the all-orders step Lean-formalised, v279/`FORM.QGEO.03`)

- **`v279_qgeo_obligation_lemma.py` (`QGEO.OBLIG.01`) writes the bedrock as ONE precise constructive-QFT lemma.** Given the premise *the raw seam carries the flat $\tau{=}i$ pillowcase metric* (Troyanov-unique, $\Lambda_\Sigma{=}\sqrt{-\Delta_{\mathrm{flat}}}$), then $\omega_\Sigma\circ\rho=\omega_\Sigma\to$ mark-locality $\to E_8$. [E] the reduction DAG from 'free RP seam' has exactly *one* open leaf — 'the raw seam state is the flat state'; [E] the geometric side is fixed (flat DtN diagonal, ρ commutes to all orders), so the obligation is a state-identification, with [C] two proof routes (RP-state uniqueness, or Troyanov + OS).
- **`FORM.QGEO.03`: the all-orders step is now machine-proved in Lean.** `SeamDeckClosure.lean` gains `diagonal_commutesClock` and `flat_all_orders_clock`: any spectral function $g(\Delta_{\mathrm{flat}})$ of the Fourier-diagonal flat Laplacian commutes with the carrier clock $\rho{=}\mathrm{diag}(i^n)$ — so the v276 all-orders closure is a Lean theorem (only `propext`, `Classical.choice`, `Quot.sound`; no `sorry`; full `lake build` green, 3356 jobs). The premise (the seam *is* flat) stays the [O] `QGEO.SYM.01` input. Build green, `AUDIT OK`.

2026-06-18 (4D QFT: the $S_{\mathrm{pert}}\to S_{\mathrm{phys}}$ LSZ bridge + one-loop unitarity, v278)

- **`v278_lsz_bridge_unitarity.py` (`QFT4D.SPERT.04`) connects the perturbative S-matrix to the physical asymptotic one and verifies one-loop unitarity.** [E] LSZ bridge: the EG T-products of S_{pert} (v269) *are* the OS Wick-rotated correlators (v240), so amputate+on-shell-residue gives $S_{\rm pert} \xrightarrow{\rm LSZ} S_{\rm phys}$; [C] the gap $\Delta{=}6\log(3/2){>}0$ gives Haag–Ruelle asymptotic states. [E] *one-loop optical theorem*: the bubble discontinuity $\operatorname{Im}B(s)/\pi = x_+{-}x_- = \sqrt{1-4m^2/s}$ *exactly* = the two-body phase space, so $2\operatorname{Im}M = \sum\int d\Pi|M|^2$ (cutting rule) holds — the matter+gauge sector is unitary, with the cut turning on at the physical two-particle threshold. [O] the gravity sector's Stelle ghost (v269/`rt_F`) is non-unitary → `QG.AMB.01`. Exact core mirrored in Wolfram (273/273). Build green, `AUDIT OK`.

2026-06-18 (QG.AMB.01 Tier-B made concrete: the seam-Calderón → $(E_8)_1$ match, v277)

- **`v277_seam_calderon_e8_match.py` (`QGAMB.TIERB.01`) reduces Tier-B to ONE bit.** [E] the full $(E_8)_1$ matching target is computed and matches: $c{=}8$, $\det\mathrm{Cartan}(E_8){=}1$, the holomorphic character $E_4/\eta^8{=}j^{1/3}{=}q^{-1/3}(1{+}248q{+}\dots)$ with a single primary and exactly $248{=}\dim E_8$ weight-1 currents (240 roots). [E] the discriminator is holomorphy — the $c{=}8$ counterexample $(D_8){=}SO(16)_1$ has $\det\mathrm{Cartan}{=}4$ (non-holomorphic), so the one condition is $|\det K|{=}1$. [C] the seam side matches via v276 (flat $\tau{=}i$) + v260 (Kummer/K3). [O] residual = the single holomorphy bit; reduced, not closed. Exact core mirrored in Wolfram (272/272). Build green, `AUDIT OK`.

2026-06-18 (QGEO.SYM.01 narrowed: the flat pillowcase closes the commutator to all orders, v276)

- **`v276_qgeo_flat_closes_commutator.py` (`QGEO.SYM.03`) collapses the bedrock to ONE geometric postulate.** The state-level residual the whole bedrock rests on is the operator equation $[\rho,H]{=}0$. [E] if $H = f(\Delta_{\rm flat})$ and the order-4 deck ρ $(z{\mapsto}iz)$ is a flat-$\tau{=}i$ isometry, then $[\rho,\Delta]{=}0 \Rightarrow [\rho,H]{=}0$ *exactly*, to all orders (verified on the discretised square torus: $\rho^4{=}I$, $[\rho,\Delta]{=}0$, $[\rho,\sqrt{-\Delta}]{=}0$; generic non-isometry fails) — upgrading the leading-order v198/v201 to the full H. [C] $\tau{=}i$ is forced by order-4 ($j{=}1728$; hexagonal $j{=}0$ is order-6, excluded). [O] `QGEO.SYM.01` therefore collapses to the single statement "the raw seam IS the flat $\tau{=}i$ pillowcase quasi-free state" — one human constructive-QFT proof, or an honest axiom. Narrows, does not close. Build green, `AUDIT OK`.

2026-06-18 (Red Team Target F — adversarial audit of the v269–v275 round, rt_F)

- **`redteam/rt_F_qft4d.py` (`REDTEAM.F.01`) stress-tests the just-published round** (the perturbative $S_{\rm pert}$ layer, the PMNS Jarlskog, the ν-scale, the over-determination, the `QG.AMB.01` roadmap). SURVIVES (NARROWED): two attacks land and are folded in. **(1)** the gravity R^2/Weyl^2 sub-sector is power-counting renormalizable but *non-unitary* (the Stelle ghost: the four-derivative propagator $1/(p^2(p^2+M^2))$ has a negative-residue massive spin-2 mode), so $S_{\rm pert}$ is unitary for the matter+gauge sector only — which is exactly why the nonperturbative `QG.AMB.01` is the real frontier (caveat added to v269). **(2)** the 0.11% anchor over-determination is conditional on the Λ-branch readout (`LAMBDA.BRANCH.01`, [C]), the gravity route being unconditional (caveat added to v274). Added as Target F in `tfpt_5_redteam` + the red-team table; `run_redteam.py` green.

2026-06-18 (Zenodo release 5.2 (rev 191) — the perturbative 4D-QFT layer + scale closure, v269–v275)

- **Published to Zenodo.** Version `5.2 (rev 191)` of the ten-PDF document set was published as a new version of the TFPT deposit — record 20743347, DOI `10.5281/zenodo.20743347`. This release carries the perturbative 4D-QFT layer (the Epstein–Glaser $S_{\rm pert}$ skeleton v269, the concrete one-loop quartic v271 and gauge β-coefficients v273), the complete complex PMNS matrix and leptonic CP strength v270, the honest absolute ν-mass-scale typing v272, the over-determination of the single mass anchor v274, and the `QG.AMB.01` obligation roadmap v275. Suite 273 modules (highest ID `v275`; v186/v226 skipped), Wolfram `271/271`, `run_all.py` green, `AUDIT OK`, `npm` build clean. The Zenodo description (`zenodo_description.html`) and the public website were synced in the same release.

2026-06-18 (scale + frontier round: ν-mass scale, EG gauge running, over-determination, QG roadmap, v272–v275)

- **`v272_nu_mass_scale.py` (`FLAV.NUSCALE.01`) types the absolute neutrino-mass scale honestly** (no fabricated Σm_ν — would contradict the v184 firewall). [E] the absolute scale reduces to *one* seesaw ratio $m_3 = (y_\nu v_{\rm EW})^2/M_R$ (one UV input, like $v_{\rm geo}$, v153); [C] the observed m_3=0.050 eV with the TFPT PS→GUT window $M_R \in [4.2\times10^{13}, 2.4\times10^{15}]$ GeV (v249) needs $y_\nu \in [0.26, 2.0]$ — $O(1)$, no tuning; [O] M_R is not a clean $\bar{M}_{\rm Pl}$ power, so the absolute value stays open; [X] the NO floor Σm_ν=0.0586 eV is the cosmological kill test. Closes the PMNS bookkeeping (the last [O] is one seesaw ratio).
- **`v273_eg_gauge_running.py` (`QFT4D.SPERT.03`) derives the SM one-loop b_i via the EG gauge self-energy.** [E] the gauge 2-point is the same scaling-degree-4 extension as the quartic (v271), giving $(b_1, b_2, b_3) = (41/10, -19/6, -7)$ from the carrier/SM content (the same 41 as the carrier algebra, v159); [O] two-loop + all-order stay open. Exact b_i mirrored in Wolfram.
- **`v274_scale_overdetermination.py` (`SCALE.OVERDET.01`) shows the single mass anchor is over-determined.** [E]/[C] two independent routes to $\bar{M}_{\rm Pl}$ agree — gravity $(8\pi G)^{-1/2} = 2.4353\times10^{18}$ GeV and cosmology (the compiler prediction $\rho_\Lambda/\bar{M}_{\rm Pl}^4 = (3/4\pi^2)e^{-2\alpha^{-1}}$ with $\rho_\Lambda^{1/4} = 2.24$ meV) $= 2.438\times10^{18}$ GeV, to 0.11%; the seam being a conformal c=8 CFT makes "no derivable absolute scale" a theorem, not a gap. Sharpens v153/v78.
- **`v275_qgamb_roadmap.py` (`QGAMB.ROADMAP.01`) gives the open gate `QG.AMB.01` a module.** [E] Tier A gap decoupling ($\Delta_{\rm eff} = 1.648 > 0$, v36/v76); [E]/[C] Tier B reduction to the holomorphic $(E_8)_1$ boundary net (v77); [C] the perturbative layer in hand (v269/v271/v273); [O] the nonperturbative ambient measure stays the one structural frontier.
- Suite `271/271` Wolfram, `run_all` green, `AUDIT OK`, build green.

2026-06-18 (4D QFT: a concrete Epstein–Glaser one-loop quartic renormalization, v271)

- **`v271_eg_oneloop_quartic.py` (`QFT4D.SPERT.02`) takes the v269 $S_{\rm pert}$ skeleton from "exists" to an actual EG number** (no path integral). [E] the first renormalization is the order-2 product T_2; the massless propagator has scaling degree sd=2 in d=4, the one-loop bubble has sd=4=d, so ω=sd$-d$=0 ⇒ exactly one logarithmic local counterterm ($C(\omega+d, d)$=1, finite, no infinite tower); the loop factor $\Omega_3/(2(2\pi)^4) = 1/(16\pi^2)$ exactly — the *same* factor that normalises the spectral-action a_4 geometric quartic. [C] the ϕ^4 one-loop $\beta_\lambda = 3\lambda^2/(16\pi^2)$ (symmetry 3 = s, t, u channels), EG initial condition $\lambda_\Phi(\mathrm{UV}) = 1/(16\pi^2)$. [O] order-2/one-loop only; the all-order EG construction and the nonperturbative measure `QG.AMB.01` stay open. Exact core mirrored in Wolfram (270/270). [E] build green, `AUDIT OK`.

2026-06-18 (PMNS: the complete complex matrix + the leptonic Jarlskog CP strength, v270)

- **`v270_pmns_jarlskog_assembly.py` (`FLAV.PMNS.03`) assembles the four TFPT channels** ($\sin^2\theta_{12}=\frac{1}{3}-\frac{\varphi_0^{\rm ret}}{2}$, $\sin^2\theta_{23}=\frac{1}{2}$, $\sin^2\theta_{13}=\varphi_0^{\rm ret}e^{-5/6}$, $\delta=4\pi/3$) **into one exactly-unitary**

complex PMNS matrix and derives the leptonic CP strength. [E] $U = R_{23}U_{13}(\delta)R_{12}$ is unitary; [C] the Jarlskog $J_{\rm PMNS} = \Im(U_{e1}U_{\mu2}U^*_{e2}U^*_{\mu1}) = -0.02965$ ($J_{\max} = 0.03424$) is a *derived* number, the leptonic analogue of the CKM J (v88); [C] data: $J_{\max}$ vs NuFIT 5.2 0.0332 ($\sim 3\%$), J vs best fit -0.026, the negative CP-violating sign reproduced. [O] δ is the μ_6/triality phase (assembled, not from M_ν/D_F) and the absolute ν-mass scale stays open — closes the CP-*strength* readout, not the phase origin nor the scale. [E] build green, `AUDIT OK`.

2026-06-18 (4D QFT: the perturbative S-matrix as an Epstein–Glaser / pAQFT skeleton, v269)

- **`v269_spert_paqft_skeleton.py` (`QFT4D.SPERT.01`) builds the honest middle layer of the three-layer S-matrix** (the first concrete 4D-QFT step beyond the spectral-action contract), *not* a nonperturbative closure. [E] three distinct layers: $S_{\rm top}$ (2d DHR braiding $|M|$=1, statistics, v243), $S_{\rm pert}$ (4d Epstein–Glaser/BV S-matrix of the spectral action), $S_{\rm phys}$ (LSZ on the OS Wightman functions, v240). [E] the spectral-action a_4 interaction is power-counting renormalizable (all operators dim ≤ 4; 7 marginal $+\,3$ relevant, a finite counterterm basis), so [C] by the Epstein–Glaser theorem the perturbative S-matrix exists order by order (causal factorisation + finite local extension; no path integral, no UV divergences). [C] the gap $\Delta = 6\log(3/2)$ gives an IR-safe adiabatic limit ($S_{\rm pert} \to S_{\rm phys}$ via LSZ). [O] perturbative only — the ambient measure `QG.AMB.01` stays open, the canonical 4d reading remains boundary-only (v265). [E] build green, `AUDIT OK`.

2026-06-18 (PMNS: the reactor-angle exponent closed as the carrier hypercharge trace, v268)

- **`v268_theta13_carrier_trace.py` (`FLAV.TH13.01`) closes the θ_{13} exponent origin.** The value $\sin^2\theta_{13} = \varphi_0^{\rm ret}e^{-5/6} = 0.0231$ was already checked (v16); now the exponent is exact: [E] $\gamma = 5/6 = {\rm tr}_E Y^2$, the carrier hypercharge-squared trace ($Y = {\rm diag}(-\frac13, -\frac13, -\frac13, \frac12, \frac12)$), roots of $6Y^2 - Y - 1 = 0$; complement $\frac16$ the neutrino ratio). [E] own channel: the $\mu\tau$-breaking $\varepsilon = \frac34\varphi_0^{\rm ret}$ induces only $\sim 1.6\times10^{-3} \ll 0.023$, so θ_{13} is a separate carrier-trace channel — correcting the tempting (and wrong) “θ_{13} from M_ν” route flagged by the new script atlas. [O] the CP magnitude and the absolute ν-mass scale remain the open PMNS items. Exact core mirrored in Wolfram. [E] build green, `AUDIT OK`.

2026-06-18 (Infrastructure: a generated causal “script atlas” so the suite is navigable at a glance)

- **`verification/make_script_atlas.py` generates `verification/script_atlas.md`** — a single grouped, dependency-aware view fusing the existing single sources (`status_ledger.csv` for claim→status→dependencies→supersedes, `script_registry.csv`/`script_clusters.csv` for the theme grouping, `docs_map.csv` for where each script is cited). It lists every registered script by cluster with its claim(s), four-class marker, one-liner, resolved dependencies and citing papers, plus a *supersede map* (which old claim each new one replaces) and a dependency/most-depended-on overview — the antidote to duplicating work or forgetting superseded results. No new data, a re-projection only; wired into `bash build.sh gen`. [E] build green, `AUDIT OK`.

2026-06-18 (The bedrock in its minimal-axiom form: a rigidity theorem for `QGEO.SYM.01`, v267)

- **`v267_qgeo_rigidity_minimal_axiom.py` (`QGEO.SYM.02`) sharpens the one bedrock postulate to its irreducible kernel** (it does *not* close it). [E] the *single* bare statement “the seam admits the carrier order-4 clock as a conformal symmetry permuting its four marks”

($= \omega \circ \rho = \omega$) forces *everything*: an order-4 Möbius orbit $\{a, ia, -a, -ia\}$ has cross-ratio 2 independent of a; cross-ratio 2 $\Leftrightarrow$ j=1728 (only $\lambda \in \{-1, \frac{1}{2}, 2\}$) $\Leftrightarrow$ the order-4 CM automorphism on y^2=x^3−x; and the unique flat pillowcase metric, mark-locality, $[\rho, \Lambda]$=0 and $\omega \circ \rho$=ω all follow (v214/v201/v264/v198). [E] neg controls: generic config $j \neq 1728$ ($\mathbb{Z}_2$ only), hexagonal j=0 ($\mathbb{Z}_6$, order 6) — only the square (order-4) realises the μ_4 clock. [C] second, independent reason: $\{1, i, -1, -i\}$ over $\mathbb{Q}$ with j=1728 (CM by $\mathbb{Z}[i]$), Belyi/dessins Galois-rigidity. [O] the bare order-4 symmetry stays the one foundational postulate (like c=const) — the *reduction* is closed (axiom ⇒ everything), the axiom is not. [E] build green, `AUDIT OK`.

2026-06-18 (Branch B decided: the explicit PS threshold model + proton-decay window — minimal content killed, +(15, 1, 1) survives, Hyper-K decisive, v266)

- **`v266_ps_threshold_proton.py` (`PS.PROTON.02`/`PS.THRESHOLD.02`) decides branch B of the v265 fork.** The explicit two-step Pati–Salam thresholds give d=6 $\tau_p(p{\to}e^+\pi^0)$ (writes `ps_threshold_proton.json`): [X] the minimal 16-Higgs content ($M_{GUT}{\sim}2.4{\times}10^{15}$) gives $\tau_p{\sim}4.4{\times}10^{33}$ yr < Super-K and is *killed* even at the high hadronic band; [C] the E_8-allowed (15, 1, 1) [the single 45] raises $M_{GUT}{\sim}5.9{\times}10^{15}$ ⇒ $\tau_p{\sim}1.6{\times}10^{35}$ yr, proton-safe *now*; [X] but that sits at the Hyper-K reach (${\sim}1.4{\times}10^{35}$ yr) — a *dated* kill test decisive within the decade. So branch B is promoted from "well-mannered candidate" to *content-selected (must be* +(15, 1, 1)*), Hyper-K-decisive UV branch*; [O] structurally marginal (only one 45 in the hull, v247) with an $O(3)$ τ_p band. Branch A (boundary-only) is unchanged and canonical. [E] build green, `AUDIT OK`.

2026-06-18 (The 4D-QFT fork frozen as a branch policy: boundary-only canonical, Pati–Salam a falsifiable UV branch, SM-only GUT killed, v265)

- **`v265_qft4d_fork_freeze.py` (`QFT4D.FORK.01`) turns the 4D-QFT fork from an open ambiguity into a frozen decision tree** (`qft4d_fork_freeze.json`), no new physics. **A (canonical):** boundary-only is the default 4D reading — a 4D-GUT is *not claimed by default* (E_8 is the audit hull). [X] SM-only 4D-GUT unification is killed (1-loop: at the α_1=α_2 scale α_3^{-1} off by ∼5.6, spread ∼8.8 at $2{\times}10^{16}$ GeV, re-deriving v246) — the canonical expectation, not a failure. **B (conditional):** carrier-native Pati–Salam is a falsifiable UV branch only with the E_8-allowed content {1, 10, 16, 45} (no **126**, v247), $\sin^2\theta_W$=3/8 (v248), the $O(1)$ κ=M_{PS}/M_s band (v249/v253), a threshold model and a proton-decay kill test ([X], no fake τ_p window; carrier gauging stays the contract). [E] a prose guard (`QFT4D.NO_OVERCLAIM.01`) forbids bare 4D over-claims and requires the scoped policy wording. New ledger rows `QFT4D.BOUNDARY.DEFAULT.01`, `QFT4D.SMONLYGUT.01`, `PS.UVBRANCH.01`, `PS.THRESHOLD.01`, `PS.PROTON.KILL.01`. A fork-policy box added to `tfpt_research_contracts`; [E] build green, `AUDIT OK`.

2026-06-18 (Three open-residual advances: the F_{QCD} solver, M_ν from D_F, and the raw-seam⇒mark-locality reduction, v262–v264)

- **`v262_fqcd_mp_me.py` (`FR.MPME.02`) implements the one contract-only F_{transfer} pipeline.** A real two-loop α_s solver runs $\alpha_s(M_Z)$=0.1179 with n_f thresholds (carrier b_3= − 7, [E]) to $\alpha_s(2\,\mathrm{GeV})$∼0.30 and $\Lambda^{(3)}_{\overline{\mathrm{MS}}}$∼0.33 GeV; with the lattice bridge C_p (external $O(1)$) and the closed m_e it reproduces m_p/m_e∼1824 vs the measured 1836.15 within the ∼ 7% external budget. A [C] transfer reproduction (firewall: no compiler integer for 1836), advancing `FR.MPME.01` ([O]) to `FR.MPME.02` ([C]).

- **`v263_mnu_from_df.py` (`FLAV.PMNS.02`) makes M_ν a seesaw operator of D_F.** [E] $M_\nu = -m_D M_R^{-1} m_D^\top$ is symmetric, seesaw-suppressed and (for $\mu\tau$-symmetric inputs) $\mu\tau$-symmetric;

[C] it reproduces θ_{23}=45° and $\sin^2\theta_{12}=\frac{1}{3}-\frac{\varphi_0^{\text{ret}}}{2}$ (v9) under the seam misalignment; [O] θ_{13}=0 in the leading texture, the CP *magnitude* (240°) is the μ_6/triality phase (v231/v233), and M_R is the free seesaw scale. An operator advance, not a closure of full PMNS dynamics.

- **`v264_raw_seam_marklocal.py` (`QGEO.ENERGY.03`) sharpens the bedrock.** Composing v216+v201+v198 on the canonical pillowcase: the flat-pillowcase curvature lives only at the 4 marks, so the DtN sub-principal symbol is $\mathbb{Z}_4$-invariant (Fourier on $4\mathbb{Z}$, FFT-verified), block-diagonal, and $[\rho, \Lambda]=0 \Rightarrow \omega \circ \rho = \omega$ on this metric. So the whole bedrock collapses to the single metric statement "the raw RP seam carries the uniformising flat pillowcase metric" (=`QGEO.SYM.01`). A reduction/sharpening, [O] — it does *not* prove the seam must carry that metric (negative control: curvature off the μ_4 orbit breaks the chain). Paper citations added in `tfpt_4_frontier`, `tfpt_2_standard_model` and `tfpt_research_contracts`. [E] `AUDIT OK`, build green.

2026-06-18 (Paper consistency pass: the closure ledger and the status/QFT sections updated for the Modular Spectral Closure, with an explicit "what is *not* closed" line)

- **Five paper sections that predated the `v238`–`v261` round are brought up to date** (additive, no status marker moved). The `introduction` closure-ledger table and the "five questions" item now carry a *boundary QFT (Modular Spectral Closure)* row — one relative object $\mathsf{TFPT}_{\text{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\text{rel}})$ closed modulo `QGEO.SYM.01` (v258–v261) — plus an explicit list of what the closure does *not* reach: the full measured pole masses, the absolute QCD scales (m_p/m_e), the complete PMNS dynamics, and the real 4D interacting QFT / ambient measure. The `tfpt_2` "ambient closure" section now states that the cutoff of its S_{rel} *is* the seam KMS weight (f_2/f_0=1, v259) and D_{rel} the covariance induction (v258); the `tfpt_2` "(4) Constructive QFT closure" section adds the relative-object pointer; `tfpt_4_frontier`'s full-quantum-gravity section records that the cutoff moment f_0 is pinned by the seam state (v255/v259); and `origin_theory`'s honest-status keybox notes that the whole boundary QFT collapses onto the same `QGEO.SYM.01` postulate (no sixth structural class). An editorial consistency pass so every status surface agrees with the ledger. [E] build green (`10 ok, 0 failed`), `AUDIT OK`.

2026-06-18 (Consistency pass: the Modular Spectral Closure surfaced structurally across the website and the intro status card)

- **The `v258`–`v261` closure is now reflected on every "what is open" surface, not just the research-contracts paper.** The homepage open-gates section and compiler-pipeline diagram, the hostile-referee FAQ, the glossary G_{net} entry, the reviewer page and the verification residual-chain caption now state that the *entire* emergent-QFT layer (Dirac, cutoff, gauging, glue, orientability) collapses onto the *same* premise `QGEO.SYM.01` via the Modular Spectral Closure — the boundary QFT is one relative object that adds *no new open item*, ambient QG separate. The introduction status card ("the structural residual is one condition") carries the same sentence. No status marker moved; this is an editorial consistency pass so the public surfaces agree with the ledger and the paper. [E] build green, `AUDIT OK`.

2026-06-17 (The Modular Spectral Closure: Dirac as covariance induction, the modular cutoff, the Kummer/K3 unification, and the assembly certificate, `v258`–`v261`)

- **`v258_dirac_covariance_induction.py` (`PS.DIRAC.03`) closes the "posited Yukawa vs derived operator" gap of v250/v252.** The finite Dirac operator D_F is the modular/covariance *induction* of the boundary seam quasi-free (KMS, β=1) state: a Gaussian state is fixed by its covariance C (a positive contraction) with one-particle modular Hamiltonian

$H = \log((1-C)C^{-1})$ (Araki; Casini–Huerta), inverse to the KMS occupation $C = (1+e^H)^{-1}$. [E] the inversion identity holds exactly; [E] C_F is a positive contraction; [E] the induced operator is chirality-odd; [E] it is unitarily equivalent to the v252 D_F (same mass spectrum); [E] the Majorana/seesaw m_D^2/M_R is an off-diagonal covariance entry. [C] $C_F = P_F C_\Sigma P_F$, so D_F is the Kasparov shadow $[D_\Sigma]\widehat{\otimes}_{(E_8)_1}[\mathcal{K}_{\mathrm{car}}]$ of the boundary geometry and the v18 Yukawas are the readout of C_Σ.

- **`v259_modular_cutoff_kappa.py` (`PS.SPECACT.02`) fixes the last open spectral-action input.** The cutoff f is the seam's own modular/KMS weight: by Tomita–Takesaki/Bisognano–Wichmann the seam is KMS at β=1 (v239) and $2\pi = 1/(4c_3)$ (v58) fixes β, so the heat-kernel cutoff $f(u) = e^{-u}$ is *derived*. [E] its moments give $f_2/f_0 = f_4/f_2 = 1$ exactly; [E] a generic Gaussian cutoff gives $\sqrt{\pi}/2 \neq 1$ (the choice is non-vacuous); [E] hence $\kappa = \sqrt{(f_2/f_0)\, c_{PS}/c_{\mathrm{grav}}} = \sqrt{c_{PS}/c_{\mathrm{grav}}}$ loses its scheme factor. [C] the open [O] cutoff freedom of v253/v255 becomes a finite trace ratio over the 96-dim H_F (the RG $\kappa \approx 1.3$ pins $c_{PS}/c_{\mathrm{grav}} \approx 1.7$).
- **`v260_k3_kummer_unification.py` (`ARCH.K3.01`) unifies seam, carrier and lattice on one surface.** The pillowcase is the elliptic-involution quotient $T^2_{\tau=i}/(z \mapsto -z)$ (v214); squaring it gives the Kummer/K3 surface, whose three facets are three TFPT objects. [E] the E_8 Cartan is even/det 1/positive-definite; [E] $H^2(\mathrm{K3}, \mathbb{Z}) = U^3 \oplus E_8(-1)^2$ (rank 22, det -1, even, signature $(3, 19)$); [E] the seam is $T^2/(z \mapsto -z)$ with 4 marks (orbifold Euler char 0); [E] the carrier-16 are the $|A[2]| = 2^4 = 16$ Kummer nodes $= \dim S^+$ (v197), $16 = 4 \times 4$; [E] honest note — the rank-16 Kummer lattice is *not* the $E_8(-1)^2$ summand. [C] so the v1 "glue $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$" is the lattice shadow of a single object; E_8 appears both locally (du Val $\mathbb{C}^2/2I$, v232/v236) and globally (H^2 summand). A unification, not a closure.
- **`v261_modular_spectral_closure.py` (`QFT.MSC.01`) is the assembly capstone.** The whole QFT round (v238–v260) is certified as *one* relative object $\mathsf{TFPT}_{\mathrm{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\mathrm{rel}})$ reduced to *one* open premise. [E] the cross-consistency invariants agree: $4 = [B{:}A] = |\mu_4| = 2\chi = |(\mathbb{Z}/2)^2|$ (index = marks = fixed points = glue order), one carrier-16 (spinor = lattice = triple), one gap $6\log(3/2)$ (OS = recovery = Lindblad); [E] the three closures re-derived; [E] the ten-component manifest present. [O] the QFT layer reduces to `QGEO.SYM.01` (seam realisation), the ambient QG measure `QG.AMB.01` kept separate. [C] the *Modular Spectral Universality* master statement: the unique RP holomorphic index-4 KMS seam state induces the 96-dim KO-6 triple whose inner fluctuations give the gauged Pati–Salam spectral action — the honest sense in which the boundary QFT is complete.

2026-06-17 (Infrastructure: the changelog is now a public website page, generated from `changelog.tex` and audit-gated)

- **The canonical changelog now also drives a public `/changelog` website page.** A new generator `verification/make_changelog_web.py` parses this file (dated `\subsection*` entries, `itemize` items, and the inline markup `\textbf`/`\emph`/`\texttt`/`\vref`, the status markers [E]/[C]/[O]/[X], inline math and accents) into the typed data file `website/lib/changelog.ts`, rendered server-side (math via KaTeX) by `website/app/changelog/page.tsx` + `website/components/Changelog.tsx`. The generator is wired into `bash build.sh gen`; its freshness is enforced by `audit_sync.py` (section A.generated), so the page can never drift from this file – editing the changelog and re-running `gen` is the only supported path (the `.ts` mirror is never hand-edited). `/changelog` is linked from the navbar, footer and sitemap. The `tfpt-workflow`, `website-sync` and `sync-maps` cursor rules were updated to record the new single-source generation step. [E] build (`npm run build` green, `/changelog` prerendered static) and `bash build.sh audit` (`AUDIT OK`).

2026-06-17 (Orientability + Poincaré duality: the honest, literature-confirmed status — strict fail, weakened hold, v257)

- **`v257_quasiorient_modpd.py` (`PS.NCG.ORIENT.02`) corrects the slightly too-generous `v256` typing.** The published result (Cacic–Stephan arXiv:0902.2068; CCM07; SM-vacuum hep-th/0601192) is that the Chamseddine–Connes–Marcolli Standard-Model finite triple (KO-dimension 6, with three right-handed neutrinos — exactly the TFPT seesaw case) *genuinely fails* strict orientability and strict Poincaré duality, and instead satisfies the physically correct weakened versions. Verified on the explicit `v252` triple: [E] strict orientability fails ($\gamma \notin \mathrm{span}\{\lambda(a)\rho(b)\}$); [E] *quasi-orientability* holds (the left/right A-irrep boxes are disjoint by chirality: $L = H$ weak-doublet, $R = \mathbb{C}$ singlet, so $L^{\mathrm{LR}}(\mathcal{H}^{\mathrm{even}}, \mathcal{H}^{\mathrm{odd}}) = \{0\}$); [E] strict Poincaré duality fails (KO-6 skew intersection form, corank 1, the equal-L/R-neutrino case); [C] modified Poincaré duality holds (CCM), Stephan's alternative triple (St06) restoring strict PD with identical physics. A known property of the NCG-SM model class, not a TFPT defect — no faked closure.

2026-06-17 (Remaining NCG obligations discharged or honestly typed: inner fluctuations, spectral action, orientability/Poincaré, v254–v256)

- **`v254_inner_fluctuations.py` (`PS.NCG.FLUCT.01`) closes `v250` #5 + the no-junk obligation.** The inner fluctuations $\Omega^1_D(A_F)$ of the finite Dirac operator are computed explicitly: [E] they are purely chirality-odd (scalars), and for the SM algebra sit *exactly* in the one Higgs doublet $(1,2)_{1/2}$ — no junk, no triplet, no $(10,1,3)/\mathbf{126}$. [E] the σ (ν_R Majorana) direction is absent for the SM algebra but present for Pati–Salam (the CCvS beyond-first-order scalar), and [E] σ is an SM singlet $(1,1,0)$ with $B-L = 2$. [C] σ = scalaron.
- **`v255_spectral_action_expansion.py` (`PS.SPECACT.01`) closes the spectral-action expansion (`v252` #11).** The Seeley–DeWitt expansion $S = 2f_4\Lambda^4 a_0 + 2f_2\Lambda^2 a_2 + f_0 a_4$: [E] structure ($a_0$ cosmological, a_2 Einstein–Hilbert+Higgs-mass, a_4 Yang–Mills+Weyl^2 + R^2+Higgs-quartic, $N = 96$); [E] $b/a^2 = 1/3$ (top-dominated); [E] $\sin^2\theta_W = 3/8$; [E] the R^2 spin-0 mode = the Starobinsky scalaron; [C] Higgs quartic $\to \sim$125 GeV with σ; [C] $\kappa = \sqrt{(f_2/f_0)c_{PS}/c_{\mathrm{grav}}}$ explicit; [O] exact decimal needs the cutoff moments.
- **`v256_orientability_poincare.py` (`PS.NCG.ORIENT.01`) — honest status, no faked proof.** KO-dimension 6 forces the intersection form $\mathrm{Tr}(\gamma\,\pi(p)J\pi(q)J^{-1})$ to be *antisymmetric*: [E] skew form (verified), [E] rank 2, corank 1, null $\sim (1,1,-1)$ on the SM K-theory $\mathbb{Z}^3$; [E] the orientability local condition $[\gamma_F, \pi(a)] = 0$. [C] the canonical hypercharge-faithful SM/PS triple satisfies orientability and Poincaré duality (van Suijlekom; CCM); [O] a self-contained machine proof needs the complete canonical bimodule.

2026-06-17 (Full 96-dim finite spectral triple built + NCG axioms verified, and κ pinned to $O(1)$, v252–v253)

- **`v252_full_finite_triple.py` (`PS.DIRAC.02`) advances the Dirac contract `v250` from a 7-obligation contract to a concrete object discharging 5 to [E].** The full 96-dim finite spectral triple $(A_F, H_F, D_F, J, \gamma)$ — H_F the particle$\oplus$antiparticle doubling of three $SO(10)$ **16**-plets — is built explicitly and the NCG axioms are checked by direct computation: [E] reality ($J^2{=}{+}1$, $JD{=}DJ$), grading ($\gamma^2{=}1$, $[\gamma, a]{=}0$, $J\gamma{=}{-}\gamma J$), KO-dimension 6, order-zero, the first-order condition (HOLDS for the SM algebra including the Majorana; VIOLATED for the Pati–Salam algebra exactly by the Majorana, localized to $\{\nu_R, e_R\}$ — the CCvS beyond-first-order mechanism on the full triple), and the spectrum of D_F reproducing the fermion masses and the type-I seesaw $m_{\mathrm{light}}{=}m_D^2/M_R$. [C] Poincaré duality; [O] orientability, no-junk, spectral-action expansion.

- **`v253_kappa_scalaron_ps.py` (`PS.KAPPA.01`) discharges residual 6(b) of v249.** κ in $M_{PS} = \kappa\, M_s$ ($M_s = c_3^{7/2}\bar{M} \approx 3.06{\times}10^{13}$ GeV, exact) is pinned by an RG scan (PDG errors $\times$ the two E_8-allowed contents) to a tight $O(1)$ interval (≈ 1.3). [C] heat-kernel origin: $\kappa = \sqrt{(f_2/f_0)\, c_{PS}/c_{\rm grav}}$ is a scale-free ratio of heat-kernel traces over the 96-dim H_F (Λ cancels). [X] the **126**-content drives κ out of the $O(1)$ window (content-selective). [O] the exact decimal needs f_2/f_0 fixed.

2026-06-17 (Spectral triple, first step: the first-order condition is violated exactly by the Majorana/σ — the CCvS Pati–Salam mechanism, v251)

- **`v251_first_order_sigma.py` (`PS.DIRAC.01`) advances the Dirac contract v250 from a contract to an exhibited mechanism.** On a faithful lepton-block model ($H = H_F \oplus H_F^c$, $H_F = \mathbb{C}^3$), the real even structure is checked ($J^2{=}{+}1$, $\gamma^2{=}1$, $D\gamma{=}{-}\gamma D$, KO-dim-6 signs), order-zero holds, the first-order condition HOLDS for the Yukawa block, and is VIOLATED *exactly* by the Majorana ($\nu_R\nu_R^c = \sigma$) block — the Chamseddine–Connes–van Suijlekom "beyond first order" mechanism by which the inner fluctuations promote the SM to Pati–Salam and generate σ. Honest correction to the review: σ does *not* rescue the first-order condition; its controlled violation *is* the mechanism. [C] σ = scalaron (v249); [O] the full 96-dim triple + spectral action.

2026-06-17 (Carrier-native Pati–Salam program: E_8 forbids the 126, PS algebra, unification kill-test, Dirac contract, `v247`–`v250`)

- **Four scripts turning the gauge-unification finding into typed claims with negative controls.** `v247_e8_branching_no126.py` (`PS.E8BRANCH.01`): the E_8 hull branches under $SO(10){\times}SU(4)$ as $(45,1){+}(1,15){+}(10,6){+}(16,4){+}(\overline{16},\bar{4})$, so the $SO(10)$ content is exactly $\{1,10,16,45\}$ and the **126** is *absent* — the renormalisable 126_H is algebraically forbidden and 10+16+45 is E_8-supplied (only one 45 $\Rightarrow$ proton-marginal). `v248_ps_algebra.py` (`PS.ALGEBRA.01`): $A_F = \mathbb{H}_L \oplus \mathbb{H}_R \oplus M_4(\mathbb{C}) \to SU(2)_L{\times}SU(2)_R{\times}SU(4)_c$, exact hypercharges $Y = T_{3R} + (B{-}L)/2$, matching $\alpha_1^{-1} = \frac{3}{5}\alpha_{2R}^{-1} + \frac{2}{5}\alpha_4^{-1}$, $\sin^2\theta_W = 3/8$ (colour $SU(4)_c \subset D_5$, distinct from the family A_3). `v249_ps_unification.py` (`PS.RGTEST.01`): the two-step PS$\to SO(10)$ unification kill-test with negative controls (SM-only fails; 126 breaks the scalaron match; proton decay selects content), $M_{PS}/M_s = 1.0$–1.7. `v250_dirac_triple_contract.py` (`CONTRACT.QFT4D.DIRAC.01`): the spectral-triple contract ($H_F = 3{\times}16$, the seven NCG axioms, σ =scalaron) that would turn "carrier is gauged" from a fork into a theorem. The root-level `qft.txt` carries the full analysis.

2026-06-17 (Honest data cross-check + emergent-QFT figures: the unification tension, v246)

- **`v246_unification_data_crosscheck.py` (`QFT4D.RGTEST.01`) — the first hard data confrontation of the spectral-action prediction.** TFPT's gauge-charged content *is* the Standard Model (v159) and the theory adds no new states; the E_8 cascade is an arithmetic spine (v5), not a threshold tower. So the run *is* the SM run: at one loop the couplings spread $\sim 10^{13}$–10^{17} GeV (residual $\Delta\alpha^{-1} \approx 9$ at $2{\times}10^{16}$ GeV), and at two loops (RK4) the minimal spread is $\Delta\alpha^{-1} \approx 3.2$ at $1.7{\times}10^{14}$ GeV — the three never meet. Typed [X]/[O]: with "no new state" there is *no* admissible threshold source, so the spectral-action 4d-GUT route is in genuine tension (same status as NCG-SM), likely falsified unless extended — never a confirmation, and *not* rescued by a cascade. Two new figures (`figures/qft_skeleton.pdf`, `figures/qft_unification.pdf`) added via `make_figures.py` (PDF + website PNG) and embedded in `tfpt_research_contracts.tex`; a root-level `qft.txt` summarises the emergent-QFT skeleton, what each script proves, the data status and the open questions.

2026-06-17 (Spectral-action matter content: the $SO(10)$ 16 = one anomaly-free SM generation, v245)

- **`v245_spectral_matter_content.py` (`QFT4D.MATTER.01`) discharges the computable core of the 4d contract v244.** The carrier half-spinor (the $SO(10)$ **16**) is verified, in exact rational arithmetic, to be one Standard-Model generation: $\mathbf{16} = SU(5)(\mathbf{10}+\bar{\mathbf{5}}+\mathbf{1})$, the SM multiplet dimensions sum to $16 = \dim S^+$ (15 Weyl fermions $+\,\nu_R$), the electric charges $Q = T_3 + Y$ are exactly the SM values, all four gauge anomalies cancel ($\sum Y = \sum Y^3 = [SU(2)]^2 U(1) = [SU(3)]^2 U(1) = 0$), the spectral-action normalisation gives $\sin^2\theta_W = 3/8$ ($g_3 = g_2 = \sqrt{5/3}\, g_1$), and the Higgs is the $(1,2)_{1/2}$ inner fluctuation. So the matter-content + tree-relation half of `CONTRACT.QFT4D.01` is now [E]; only the seam→Dirac realisation and the run-down couplings stay [O]. Wired into `run_all.py`, the ledger, the registry, cited in `tfpt_research_contracts.tex`, mirrored to the website.

2026-06-17 (The QFT skeleton on the seam: thermal time, GNS/OS reconstruction, DHR sectors, the braiding S-matrix, and the 4d spectral-action contract, v239–v244)

- **Six emergent-QFT skeleton scripts (v238).** `v239_kms_thermal_time.py` (`DYN.KMS.01`): the modular flow is KMS at $\beta = 1$, and $2\pi = 1/(4c_3)$ fixes the modular temperature as $T_H = c_3/M$ — thermal time = horizon time. `v240_gns_os_reconstruction.py` (`DYN.GNS.01`): GNS reconstructs the Hilbert space from $(\mathcal{M}, \omega)$ and the OS Hamiltonian $H_{\mathrm{OS}} = -\log T = -L$ is positive with gap Δ (the v64 mass gap). `v241_dhr_sectors.py` (`DHR.SECTORS.01`): particles = DHR sectors = the discriminant anyons of the Lean glue form, with the condensation tower $16 \to 4 \to 1$ (v235/v237). `v242_spin_statistics_modular.py` (`DHR.MODULAR.01`): spin-statistics, the modular (S,T) data, Verlinde fusion, and Gauss–Milgram $c = 8$. `v243_haag_ruelle_braiding.py` (`DHR.SMATRIX.01`): the 2-particle S-matrix = the braiding monodromy (factorised, crossing), with the v238 mass gap supplying Haag–Ruelle asymptotic states. `v244_spectral_action_contract.py` (`CONTRACT.QFT4D.01`): the 4d Lagrangian via the Connes spectral action — the carrier half-spinor 16 is one generation, the SM gauge group sits in the carrier, gravity is v36; the matter Lagrangian is the open [O] contract. Net: the residual to a full 4d QFT is *one premise* (`QGEO.SYM.01`) plus *one contract* (`CONTRACT.QFT4D.01`). All wired into `run_all.py`, the ledger and registry, cited in `tfpt_research_contracts.tex`, mirrored to the website.

2026-06-17 (The static→dynamic flip: modular flow + a recovery Lindblad generator, v238)

- **New `v238_modular_lindblad_dynamics.py` (`DYN.SEMIGROUP.01`, cluster registry).** A companion to the modular round (v196/v198) and the recovery code (v221/v64) that reads the *same* seam objects *dynamically* rather than statically. [E] the modular flow $\sigma_t = e^{it\Lambda_\Sigma}$ is a unitary one-parameter group (Tomita–Takesaki) whose conservation of the μ_4 clock, $\rho\sigma_t\rho^{-1} = \sigma_t \Leftrightarrow [\rho, \Lambda_\Sigma] = 0$, IS the v198 state-invariance — so the static commutator is the conservation law of modular time. [E] the recovery channel's generator $L = \log T$ is a doubly-stochastic/CPTP-classical Markov generator ($e^L = T$, $e^{L\tau} \to$ the democratic projector = the arrow of time). [E] keystone: the dissipative gap $= -\log((2/3)^6) = 6\log(3/2) =$ the OS mass gap Δ (v64) — the static recovery bound (v221) and the dynamical mass gap (v64) are one number, one exponentiated. [E] the GKSL split $G = i\Lambda_\Sigma + L$ (imaginary + real≤ 0 spectrum). [O] the residual is the *geometric* modular flow (Connes–Rovelli "thermal time = physical time") on the full seam algebra = `QGEO.SYM.01` (v198) — a target, not a closure. Wired into `run_all.py`, `status_ledger.csv`, `script_registry.csv` and cited in `tfpt_research_contracts.tex`; mirrored to `website/public/verification`.

2026-06-17 (Experiments: black-hole-cosmology signatures, the η_B Boltzmann solve, the Petz/baby-universe realisation, and a real-data GW-echo pipeline)

- **Three new standalone black-hole-cosmology experiments (`experiments/`, all `data_limited`, not load-bearing).** From the `problem_b` black-hole-cosmology survey: (i) `ccbh-dark-energy` — the de Sitter seam interior ($w_{\rm in}$=−1) forces the Croker–Weiner cosmological-coupling index $k = -3w_{\rm in} = 3$, hence a population $w = -1$; vs. Farrah+2023 $k = 3.11 \pm 0.79$ (-0.14σ), a *contested* downstream bridge and an *alternative reading* of the same $w = -1$ the DESI watchdog tests (`alternative_group w_de_eos`, never double-counted). (ii) `gravastar-compactness` — the Nariai $Q_{\rm geom}$=3/8 equals the Jampolski–Rezzolla (2026) maximum horizonless compactness $\mathcal{C}$=3/8 (shared de Sitter endpoint 1/2); since 1/3<3/8<4/9<1/2 this is a light-trapping ECO, yielding an echo template (round-trip delay ∼0.7 ms at 62 $M_\odot$, amplitude $\le (2/3)^6$) — a [C] structural echo, no $\mathcal{C} \leftrightarrow Q_{\rm geom}$ map proven. (iii) `cosmic-handedness` — a parity watchdog (Shamir JADES ∼3.3σ spin monopole, most likely a Milky-Way-aberration dipole), explicitly *not* promoted (frontier).

- **η_B — full BDP Boltzmann ODE solve confirms the route at the frozen scale (`experiments/ftransfer/leptogenesis_boltzmann/fboltzmann_solve.py`; `tfpt_4_frontier`, [C]).** Integrating the Buchmüller–Di Bari–Plümacher Boltzmann network (integrated efficiency $\kappa_f = 0.092$, validating the analytic fit) at the *frozen* heavy scale $M_1 = M_{\rm scal}(\varphi_0^{\rm ret})^2/A_\Lambda \approx 8.65 \times 10^9$ GeV with $\delta_{\rm CP} = 4\pi/3$ gives $\eta_B = 6.5 \times 10^{-10}$ — a factor 1.07 from the observed 6.1×10^{-10}, with *no* free M_R dial. This moves the leptogenesis route from "solve pending" to a consistent [C] readout (the full *flavored* density-matrix solve is the next refinement); `tfpt_4_frontier` Route C synced.

- **The Petz recovery map and the rank-one baby universe, realised numerically (`experiments/recovery-channel`; `tfpt_horizon_readouts`, [C]; companion to v221).** The gapped transport contracts to a *rank-one* projector at exactly $\|T^n - P_\infty\| = (2/3)^{6n}$ — the one-dimensional baby-universe Hilbert space (Engelhardt 2025) — and the explicit Petz map is CPTP, recovers the reference, and equals the identity only on the protected λ=1 (Knill–Laflamme) mode; free-ratio and degenerate-spectrum negative controls hold. So the Petz identification v221 deferred is now verified at the channel level (`internal_consistency`, no new datum); the full holographic reading stays gated on the Seam–Horizon theorem.

- **GW ringdown echo — Stage-1 matched filter validated, then run on real GWOSC strain (`experiments/gw-ringdown-echo`; `tfpt_horizon_readouts` search-targets).** The Stage-1 machinery (Kerr-ringdown subtraction → matched filter on the residual, ratio $(2/3)^6$ frozen, lag free, + free-ratio control) classifies 3/3 synthetic injections; run on *real* GWOSC strain (GW150914, GW190521; PSD-whitening + off-source background): **no kernel-ratio echo coincident in ≥ 2 detectors**, consistent with the $(2/3)^6$ *upper* bound (a single loud H1 excess has $\hat{q} \approx 2$, far from $(2/3)^6$, so the free-ratio control flags it as residual ringdown, not an echo). First pass: dominant-QNM subtraction, incoherent combination — full multi-mode + coherent stacking is the next step; no detection claim.

- **Scorecard + website synced.** `experiments/evidence_scorecard.json` now 51 rows (26 consistent, 7 null, 13 data-limited, 4 tension, 1 parked); the website `EXPERIMENTS_AUDIT` mirror and the η_B prediction entry are updated to match. These are standalone `experiments/` surfaces (not in the verification suite or ledger); the `tfpt_4_frontier/tfpt_horizon_readouts` and `papers.ts/predictions.ts` edits are the in-same-change sync.

2026-06-17 (Red-team layer re-synced to the paper; shadow export mirrors `figures/`)

- **Red Team Target A re-synced to `tfpt_5_redteam` (`rt_A_e8net.py`, `redteam_table.txt`, `redteam/README.md`, infrastructure).** The Python red-team surface still emitted the historical *three-residual* "A reduced, not closed" form, lagging the reader-facing note. It now encodes the sharper narrowing already in the paper: the residual collapses to the *single* statement "the seam–Calderón boundary net is holomorphic with $c = 8$" ($\Leftrightarrow$ the index-4 simple-current extension), with the index-4 $(D5)_1 \times (A3)_1 \to (E8)_1$ glue realised at Lie level (v143) and its odd sectors twisted/Ramond ($E_8 = 120_{\mathrm{NS}} + 128_{\mathrm{R}}$, v148), while the $q(A_3)$ normalisation collapses into the one declared anchor (v152). Two exact adversarial checks added ($120{+}128 = 248$; glue index $= |\mu_4| = N_{\mathrm{fam}}{+}1 = 4$); status stays [C]/[O] (reduced, not closed).
- **Shadow export now mirrors `figures/` but not `website/` (`scripts/export-shadow.sh`, `verification/make_script_index.py`, infrastructure).** `export-shadow.sh` ships `figures/*.pdf` (so `make_manifest.py --check` passes literally on the exported tree) while still excluding `website/` and the compiled paper PDFs; the `SHADOW_MIRROR.md` table is corrected accordingly. `make_script_index.py` now detects a missing `website/` and skips its `ScriptIndex.tsx` mirror, so `build.sh notes` runs on the subset without crashing.

2026-06-17 (CP phases as μ_6 powers — a reduction of red-team Target D)

- **Both CP phases are μ_6 powers of one hexagonal CM unit, split by the sheet (`v231_cp_mu6_phases`, `tfpt_5_redteam`, [E]/[C]).** Sharper than v220/v225 (which only *located* the CP residual). With $\rho = e^{i\pi/3}$ the primitive 6th root (the $j{=}0$ Eisenstein CM unit): δ_{CKM} leading $= \arg(\rho^1) = \pi/3$ (quark) and $\delta_{\mathrm{PMNS}} = \arg(\rho^4) = 4\pi/3$ (lepton, the phase lattice), with $\rho^4 = -\rho$, so $\delta_{\mathrm{PMNS}} = \delta_{\mathrm{CKM,lead}} + \pi =$ (CM unit) $\times$ ($\mathbb{Z}_2$ sheet, $\rho^3{=}{-1}$). The two CP phases are *one* hexagonal unit on the two sheets, not two independent inputs; the dual-frame Jarlskog orientations are sheet-flipped ($\pm 21 \sin(\pi/3)$). A structural reduction of Target D (removes one of its two phase inputs); the quark seam correction $3\lambda_C^2$ stays the v88 misalignment, the deck stays $\mathbb{Z}/4$, and the physical CP derivation stays [C] (`CP.MU6.01`).
- **The seam as the E_8 Kleinian singularity (`v232_e8_kleinian_seam`, `origin_theory`, [E]/[C]).** The du Val side of the McKay correspondence (v219) gives the open seam-realisation premise `QGEO.REALIZE.01` a canonical algebraic-geometric model: dropping the trivial-rep node of the affine-E_8 McKay graph of $2I$ leaves the finite E_8 Dynkin = the dual intersection graph of the eight exceptional $\mathbb{P}^1$'s of the minimal resolution of $\mathbb{C}^2/2I$; the eight curves $= \operatorname{rank} E_8 = g_{\mathrm{car}}{+}N_{\mathrm{fam}}$ (a fourth reading of the 8), their negated intersection form is the E_8 Cartan (det 1, even, (-2)-curves), and the link is the Poincaré homology 3-sphere $S^3/2I$ ($2I$ perfect $\Rightarrow H_1{=}0$, $\pi_1{=}2I$ order 120). The raw seam ($\mathbb{P}^1$ with marks) is canonically a du Val exceptional curve — a model for the bedrock, not a P2 proof (it presupposes E_8; `TOPO.E8.01`, bedrock stays [O]).
- **CP is the universal family/triality phase, only sheet-split (`v233_cp_triality_phase`, `tfpt_5_redteam`, [E]/[C]).** One step past v231 in the $\mu_6 = \mu_3(\text{family/triality}) \times \mu_2(\text{sheet})$ factorisation: $\rho = \omega^2(-1)$ with $\omega = e^{2\pi i/3}$ the $\mathbb{Z}_3$ triality centre (the 2/3 cusp weight). Since $\rho^4 = \rho^1\rho^3$ with $\rho^3 \in \mu_2$, the quark (ρ^1) and lepton (ρ^4) CP phases share the *same* family/triality class and differ only by the sheet — so CP *is* the universal triality phase, sheet-split, not a free power choice (`CP.TRIALITY.01`).
- **The Seam-Holomorphy selection certificate: the structural residual is one gate (`v234_seam_holomorphy_selection`, `tfpt_research_contracts`, [E]/[O]).** The whole structural residual is *one* condition — "the seam carries no nontrivial abelian sector" — with three provably-equivalent faces, all forcing E_8: (i) holomorphy (μ-index 1; Target A), (ii) the

seam link is a homology 3-sphere (Γ perfect $\Leftrightarrow 2I$; v232), (iii) exactly one 1-dim irrep (v219). All equal $\#(\text{mark-1}) = |\Gamma^{\text{ab}}| = |H_1|$, which is 1 only for E_8; and holomorphic $c{=}8{=}g_{\text{car}}{+}N_{\text{fam}}$ gives the unique even unimodular rank-8 lattice E_8. So P2, G_{net}, Target A and the Kleinian seam are one gate. The stated closing theorem ([O]): RP+gap+chirality $\Rightarrow$ quasi-free bulk (v160) $\Rightarrow$ holomorphic boundary $\Rightarrow$ E_8; the single residual analytic step is "free bulk $\Rightarrow$ holomorphic boundary" (`GATE.HOLO.01`).

- **The closing step in Chern–Simons language: holomorphic $\Leftrightarrow$ $\det K{=}1$ (`v235_seam_chern_simons`, `tfpt_research_contracts`, [E]/[O]).** A genuine reduction (not a closure) of `GATE.HOLO.01` into standard anyon-condensation physics: a free gapped bosonic 2+1d bulk is an even integer K-matrix with #anyons $= |\det K|$, edge $c = \text{signature}(K)$, and the edge net holomorphic $\Leftrightarrow |\det K|{=}1$. The TFPT extension tower (v92) *is* the condensation tower read by determinant: carrier $D_5{\oplus}A_3$ (det 16, 16 anyons) $\to SO(16)_1{=}D_8$ (det 4) $\to (E_8)_1$ (det 1, holomorphic) — all rank 8, $c{=}8{=}g_{\text{car}}{+}N_{\text{fam}}$ — so holomorphic $= E_8 =$ the Kitaev E_8 quantum-Hall state. Holomorphy $\Leftrightarrow$ condensing the order-$|\mu_4|$ Lagrangian glue (det 16$\to$1, vs. a partial order-2 isotropic $\to 4{=}D_8$). The residual narrows to one integer condition "the free RP seam condenses the Lagrangian glue (det $=$1)" $=$ the sheet/QGEO selection, now with holomorphy$\Leftrightarrow$ det 1 a theorem (`GATE.HOLO.02`).
- **The $(2,3,5)$ Brieskorn singularity is the one generator of the skeleton (`v236_brieskorn_capstone`, `origin_theory`, [E]).** Capstone of the icosahedral round: the singularity $x^2{+}y^3{+}z^5$, whose exponents are the atoms $(|\mathbb{Z}_2|, N_{\text{fam}}, g_{\text{car}}){=}(2,3,5)$, has Milnor number $\mu{=}(2{-}1)(3{-}1)(5{-}1){=}8{=}\operatorname{rank} E_8$ (a *fifth* origin of the "8"), Milnor monodromy $=$ the order-30 E_8 Coxeter element (eigenvalues the primitive 30th roots $=$ the E_8 exponents, charpoly Φ_{30}), Milnor lattice E_8, link the Poincaré sphere; the two clocks are facets of this one monodromy — the μ_3 triality (CP, v233) is h^{10} in $\langle h\rangle{=}\mathbb{Z}/30$, the μ_4 clock (v223) is the order-4 Galois automorphism of $\mathbb{Q}(\zeta_{30})$ (`TOPO.BRIESKORN.01`).
- **The closing step as a physical condition: no topological degeneracy** $\Leftrightarrow \det K{=}1 \Leftrightarrow$ **the seam is SRE (`v237_seam_sre_closure`, `tfpt_research_contracts`, [E]/[O]).** Sharpens `GATE.HOLO.02`'s residual from definitional to falsifiable: the K-matrix ground-state degeneracy on a genus-g surface is $|\det K|^g$ (torus: carrier 16, D_8 4, E_8 1), so "no topological degeneracy on any closed surface" $\Leftrightarrow \det K{=}1$; among rank-8 even positive-definite K ($c{=}8{=}g_{\text{car}}{+}N_{\text{fam}}$) only E_8 is short-range-entangled (the Kitaev E_8 phase). A unique vacuum on the plane (RP+gap+cluster) is necessary but not sufficient (the plane cannot see $\det K$); so the one open step is the physical statement "the free RP seam is SRE (no topological order)" (`GATE.HOLO.03`).

2026-06-16 (the icosahedral bedrock, CM-norm duality, and a structural-finds round)

- **McKay bedrock: why the atoms are $\{2,3,5\}$ (`v219_icosahedral_mckay`, `origin_theory`, [E]).** E_8 is the exceptional *top* of the McKay tower of finite $SU(2)$ subgroups ($2T{\to}\hat{E}_6$, $2O{\to}\hat{E}_7$, $2I{\to}\hat{E}_8$). Built *from the group*: the 120 unit-quaternion icosians close to the binary icosahedral group $2I$ (element orders $\{1,2,3,4,5,6,10\}$ — it contains the $2,3,5$ rotation-axis orders), 9 conjugacy classes; the 9 irreducible character degrees (Dixon, from the class algebra) are $\{1,2,2,3,3,4,4,5,6\}$, sum $30{=}h(E_8)$, sum of squares $120{=}|R^+(E_8)|$; the McKay adjacency from the natural 2-dim rep *is* the affine E_8 graph with Kac marks $=$ the degrees. A backward certificate of the closed E_8, not a P2 proof (`MCKAY.E8.01`).
- **CM-norm duality: 41 (square) and 7 (hex) (`v222_cm_norm_duality`, `origin_theory`, [E]).** The square seam (Gaussian $\mathbb{Z}[i]$, $j{=}1728$) gives $N(g_{\text{car}}{+}i|\mu_4|){=}5^2{+}4^2{=}41{=}10b_1$ (the EM index *is* the Gaussian norm of the carrier-glue vector) and $N(3{+}2i){=}13{=}\Delta_Q$; the hexagonal partner (Eisenstein $\mathbb{Z}[\omega]$, $j{=}0$) gives $N(3{+}2\omega){=}7{=}$ scalaron. The $(3,2)$ split generates $(5,6,7,13)$ by sum/product/Eisenstein/Gauss norm; rings rigid (`CMNORM.DUAL.01`).

- **The μ_4 clock is the order-4 character of the E_8 Coxeter cycle (`v223_coxeter_totative_clock, origin_theory`, [E]).** $(\mathbb{Z}/30)^\times$ are the E_8 exponents; the order-4 generator 7 ($\langle 7\rangle$=$\{1,7,13,19\}$) is a literal μ_4 clock permuting the four conjugate planes; only E_8 has $\varphi(h)$=rank and h=2·3·5 (`COX.CLOCK.01`).

- **248=120+128 as a magnitude/phase channel typing (`v227_degree_exponent_channel_split, tfpt_5_redteam`, [E]).** Exponents sum 120=$|R^+(E_8)|$ (magnitude channel), degrees sum 128=rank · dim S^+ (phase/glue channel); the red-team Target D magnitude bijection lives in 120, the un-covered CP phases in 128; the dark-sector reading of 128 stays Frontier (replaces the over-strong S^- reading; `E8.CHAN.01`).

- **P2 as the mode space of a degree-4 seam divisor (`v228_rr_index_gate, origin_theory`, [E]/[C]/[O]).** A Riemann–Roch index gate: $\deg D$=4=$|\mu_4|$ on $\mathbb{P}^1$ forces h^0=5=g_{car}, rank H_1=3=N_{fam}, h^0+H_1=8=rank E_8, $\Lambda^{\mathrm{even}}(\mathbb{C}^5)$=16=dim S^+ — bedrock language for P2, conditional on the seam producing the degree-4 divisor (`QGEO.RR.01`).

- **The seam as a finite recoverability code (`v221_seam_qecc, origin_theory`, [E]/[C]).** Code dimension dim S^+=16; the gapped transport is a CPTP doubly-stochastic contraction with spectrum $\{1,(2/3)^6,(1/3)^6\}$, recovery bound $\|T^n\delta\|\le(2/3)^{6n}\|\delta\|$ (Knill–Laflamme type); free ratio breaks it. Holographic/Petz reading [C], gated on the Seam–Horizon theorem (`QEC.SEAM.01`).

- **CP lives in the hexagonal phase fiber (`v220_cp_hexagonal_modulus, tfpt_5_redteam`, [E]/[C]).** The seam deck is the square modulus j=1728 ($\mathbb{Z}/4$); its partner is the hexagonal j=0 ($\mathbb{Z}/6$, $\arg\rho$=$\pi/3$). The frozen δ=$\pi/3+3\lambda_C^2$=68.654° has leading term $\pi/3$=$\arg\rho$; the deck stays $\mathbb{Z}/4$, CP is the $\mathbb{Z}/6$ fiber (`CP.MOD.01`).

- **The dual normal frame (d,n) with CP as orientation (`v225_dual_normal_frame, tfpt_5_redteam`, [E]/[C]).** d=$a^\top R^{-1}$=$(-\frac{1}{2},-\frac{1}{2},1)$=$-\frac{1}{2}$ Nariai; n=$(5,-9,6)$ reads det on the first-generation column; $\det(\mathbf{1},d,n)$=21=3·7; $\mathrm{Im}\det(\mathbf{1},d,\rho n)$=$21\sin(\pi/3)$ — CP as orientation (`DUAL.FRAME.01`).

- **F_{transfer} as one path on the Sheet Diamond (`v224_diamond_ftransfer_path, tfpt_2_standard_model`, [E]/[C]).** $M(s,t)$=R+$Q\,\mathrm{diag}(s,t,t)$: K,C,F on the sheet axis (curved, 2nd diff 8), the winding axis flat (slope 6), Plücker steps $(1,8,10),(1,8,16)$; the Koide source→pole is the 1-D projection (`FTR.PATH.01`).

- **The charged leptons as an étale Frobenius algebra (`v229_lepton_frobenius_algebra, tfpt_2_standard_model`, [E]/[C]).** (c_e,c_μ,c_τ)=$(\frac{16}{7},\frac{4}{3},\frac{7}{6})$, ring closure c_ec_τ=$\frac{8}{3}$=$|\mathbb{Z}_2|c_\mu$, product $\frac{32}{9}$; A=$\mathbb{Q}[t]/(m)$ is étale (trace-form Gram nondegenerate), a commutative Frobenius algebra over the μ_6 resolvent (`LEP.FROB.01`).

- **The center budget $(7,11,13)$ as three local norms (`v230_center_budget_norms, tfpt_2_standard_model`, [E]).** C=R+$Q\,\mathrm{diag}(1,0,0)$ row sums $(7,11,13)$: 7=$N_{\mathbb{Z}[\omega]}(3{+}2\omega)$ (hex), 13=$N_{\mathbb{Z}[i]}(3{+}2i)$ (square), 11=$\binom{4}{0}+\binom{4}{1}+\binom{4}{2}$ (QBL Fock count); the two CM rings and the boundary count meet in one operator center (`CENTER.NORM.01`).

2026-06-15 (the diamond axis geometry — two axes + the transfer corner)

- **The Sheet Diamond is a discrete geometry with two axes (`v218_diamond_axis_geometry, tfpt_2_standard_model`, [E]).** Sharpens `v94`/`v95` with three exact [E] lemmas (no new numbers). **(1) Axis curvature:** along the centered cross the determinant is linear on the winding axis $\det(C+xU)=14+6x$ (slope 6=$|R^+(A_3)|$) and quadratic on the sheet axis $\det(C+yV)=14+14y+4y^2$ (2nd difference 8=rank E_8); the anchor block has $\det B(C+xU)=2\det(C+xU)$ and sheet curvature 6=$|R^+(A_3)|$ — two curvatures $(8,6)=(\mathrm{rank}\,E_8,|R^+(A_3)|)$. **(2) Plücker transfer ladder:** $\mathrm{Pl}(K)\to\mathrm{Pl}(C)\to\mathrm{Pl}(F)$ lifts in steps $(1,8,10)=(N_\Phi,\mathrm{rank}\,E_8,A_\Lambda)$ then $(1,8,16)=(N_\Phi,\mathrm{rank}\,E_8,\dim S^+)$ — the decuple then the full generation (identity [E], source→pole reading [C]). **(3) Spectral ramification:** for Q, K, C, F the cubic discriminant factors as $q(r)^2\,\mathrm{Disc}(q)$ with squares $(1, 3, 4, 6)$ and kernels $(13, 48, 65, 105)$, F carrying $105 = 3{\cdot}5{\cdot}7$. The anchor-defect spectrum and pair-sum/staircase blocks are honest *audit* fingerprints ($28{=}2\dim G_2$, $52{=}\dim F_4$ stay audit-only, not spine, as in `v94/v95`); F as the transfer-completion corner is a *heuristic*, not a hard filter. Ledger `DIAMOND.AXIS.01`, `DIAMOND.PLUCKER.01`, `DIAMOND.SPECTRAL.01`; Wolfram-mirrored (249/249).

2026-06-15 (QGEO: the four marks emerge from Gauss–Bonnet — K1 executed)

- **The four marks emerge from Gauss–Bonnet — count derived, not assumed (`v216_marks_gauss_bonnet`, `tfpt_research_contracts`, [E]; ledger `QGEO.MARKS.03`).** The K1 lever of the `v215` kill-test ("exactly four marks") is *executed*: if the marks are $\mathbb{Z}_2$ branch points (deficit π) on the flat seam sphere ($\chi = 2$, the same χ as the 8 in c_3), Gauss–Bonnet forces $n\pi = 2\pi\chi = 4\pi$, so $n = 2\chi = 4 = |\mu_4| = N_{\mathrm{fam}} + 1$. The closed Euclidean sphere 2-orbifolds are exactly $(2, 3, 6)$, $(2, 4, 4)$, $(3, 3, 3)$, $(2, 2, 2, 2)$, and three independent criteria — all cone orders $= 2$ (the $|\mathbb{Z}_2|$ sheet branch), rank $H^1 = \#\mathrm{marks} - 1 = 3 = N_{\mathrm{fam}}$ (the three generations pick the 4-mark square over the 3-mark hexagonal), and a free order-4 deck — converge on the pillowcase $(2, 2, 2, 2)$. The numerical sibling `v217_marks_emergence_scan` ([C], ledger `QGEO.EMERGE.01`) confirms it on the actual DtN/state (number n and positions free, the 4-square config is the unique joint minimiser of clock-invariance + 3-family cohomology + gap $(2/3)^6$). The mark *count* and *equality* move from input to derived; only the square *modulus* (cross-ratio 2, the order-4 clock) stays the one carrier input, and the from-zero raw-seam emergence `QGEO.REALIZE.01` stays [O]. `v216` exact (Wolfram extension → 248/248); `v217` numerical (Python-only). Suite → 217 modules.

2026-06-15 (QGEO: the bedrock recast as a falsifiable kill-test)

- **`QGEO.SYM.01` as a kill-test, not a proof-promise (`v215_seam_deck_killtest`, `tfpt_research_contracts`, [E]/[O]; ledger `QGEO.KILL.01`).** A foundational symmetry cannot be derived from nothing (Bisognano–Wichmann would presuppose the covariance it should produce), so the bedrock $\omega \circ \rho = \omega$ is recast as four independently-falsifiable predictions about the raw seam DtN. The non-circular reduction (`v198/v199/v201`) gives $\omega \circ \rho = \omega \Leftrightarrow [\rho, C] = 0$ with $|k|$ commuting exactly, so the residual is the bounded sub-principal M_f (equivalently the raw Gaussian bulk form A is μ_4-deck-invariant). Levers: **K1** exactly four marks ($b_1 = \#\mathrm{marks} - 1 = 3 = N_{\mathrm{fam}} \Rightarrow \#\mathrm{marks} = 4 = |\mu_4|$); **K2** the square config (cross-ratio $2 \Rightarrow j = 1728$, $\mathrm{Aut} = \mathbb{Z}/4$; order 4 selects $\mathbb{Z}/4$ not the $j = 0$ $\mathbb{Z}/6$; generic $\Rightarrow \mathbb{Z}/2$); **K3** mod-4 sub-principal support ($[\rho, M_f] = 0 \Leftrightarrow f$ $\mathbb{Z}_4$-invariant); **K4** transfer gap $(2/3)^6 = 64/729$. Freeze-bound via `freeze_file.csv` (`seam_deck_symmetry`), carrier-side [E] (regression guards), with the *decisive* kill deferred to `QGEO.REALIZE.01` [O] (the raw collar DtN must *produce* four square marks without inputting them — an emergence test). A kill-test, not a closure. Python-only (falsification layer; the exact sub-parts $j = 1728$ and $(2/3)^6$ are Wolfram-mirrored via `v214/v54/v56`). Suite → 215 modules.

2026-06-15 (QGEO: the pillowcase reduction — two residuals become one canonical metric)

- **Pillowcase reduction of `QGEO.SYM.01` (`v214_seam_pillowcase`, `tfpt_research_contracts`, [E]/[O]; ledger `QGEO.PILLOW.01`).** The two separately-tracked open QGEO residuals — `QGEO.ISO.01` (`v180`: the carrier clock is an order-4 *isometry* of the seam metric) and `QGEO.SUBPRIN.01` (`v201`: the DtN sub-principal symbol is *mark-local*, the seam flat away from the μ_4 marks) — are shown to be *one* statement: the seam carries the uniformising flat orbifold (*pillowcase*) metric. **(1)** The four marks make the Euclidean orbifold $S^2(2, 2, 2, 2)$

($\chi_{\rm orb} = 2 - 4(1-\frac{1}{2}) = 0$), so by Troyanov uniformisation the metric is flat (unique up to scale) with curvature only at the cone points (Gauss–Bonnet $4\pi = 2\pi\chi$) — this *is* the `v201` "flat away from the marks". **(2)** New link: cross-ratio(μ_4) = 2 (`v168`) $\Rightarrow j = 1728$ ($j(\lambda) = 256(\lambda^2-\lambda+1)^3/(\lambda^2(\lambda-1)^2)$; all six harmonic cross-ratios give 1728) $\Rightarrow$ the *square* modulus $\tau = i$, whose CM by $\mathbb{Z}[i]$ *derives* the order-4 clock $z \mapsto iz$ (explicit $(x,y) \mapsto (-x, iy)$ on $y^2 = x^3 - x$); generic cross-ratio gives $j \notin \{0, 1728\}$ and only $\mathbb{Z}/2$. **(3)** Unification: "seam = uniformising flat pillowcase metric" implies *both* `v201` and `v180`, so two residuals collapse to one canonical-metric premise. [O] It does *not* close `QGEO.SYM.01`: the choice "RP seam metric = the constant-curvature representative" stays open, milder and more canonical than the bare isometry premise. Wolfram-mirrored (the exact orbifold/j-invariant/CM arithmetic); Klein-J values mpmath-numerical. Suite → 214 modules.

2026-06-15 (the F__transfer functor contract CONTRACT.F.01)

- $F_{\rm transfer}$ **formalised as a typed functor with four axioms (`v213_ftransfer_functor`, `tfpt_research_contracts`, [C]).** The four frontier transfers are consolidated into ONE functor $F_{\rm transfer} = F_{\rm observable} \circ F_{\rm threshold} \circ F_{\rm RG}$, each instance a discrete compiler kernel [E] composed with an external continuous solver [C]: **(1)** μ_4-deck equivariance (the Koide multiplier $\lambda_2{=}(2/3)^6{=}64/729$ is the deck transfer eigenvalue); **(2)** Plücker preservation ($53{=}a^T(R{+}Q)\mathbf{1}$, $\mathrm{Pl}(F){=}(1,22,30)$, $\|\mathrm{Pl}(K)\|_1{=}11$); **(3)** positivity/stochasticity (spec T positive, $\kappa_f \in (0,1)$); **(4)** external modules explicit ($b_3{=}{-}7$ a carrier output). The four instances are $F_{\rm pole}$ (`v183`), $F_{\rm Boltzmann}$ (`v212`), $F_{\rm relic}$ (`v211`), $F_{\rm QCD}$ (`v164`). `CONTRACT.F.01` is the third research contract alongside ($U_{\rm wall}$) and ($G_{\rm metric}$), the structural complement of the `v187` typing guard — an architectural consolidation, *not* a closure of any frontier number. Ledger `CONTRACT.F.01`.

2026-06-15 (frontier: two sharper [C] scenario branches — axion angle + leptogenesis)

- **Axion DM spine-angle branch $\theta_i = \pi N_{\rm fam}/g_{\rm car} = 3\pi/5$ (`v211_axion_spine_angle`, `tfpt_4_frontier`, [C]).** A more *robust* misalignment angle than the determinant-line hilltop $\theta_i \approx 170°$ (`v185`, which over-produces $\Omega_a h^2 \approx 0.66$): the central spine quotient gives $\theta_i = 3\pi/5 = 108°$ exactly (no fit), the finite-T solver reaches $\Omega_{\rm DM}$ at $\approx$106°, and 108° sits in the *mild*-anharmonic regime (62° below the hill-top) — not exponentially sensitive. An alternative ansatz to $\theta_i = \pi(1-\varphi_\Sigma)$ (mutually exclusive, the full solver decides, `DM.AXION.SPINE.01`); [X] $\Omega_a h^2$ outside $\sim$[0.08, 0.16] demotes it. A sharper scenario, not a derivation.
- **Leptogenesis scalaron-decuple branch (`v212_leptogenesis_decuple`, `tfpt_4_frontier`, [C]).** Both Boltzmann inputs share the decuple $A_\Lambda = 10 = |E(K_5)|$: $M_1 = M_{\rm scal}(\varphi_0^{\rm ret})^2/A_\Lambda \approx 8.65 \times 10^9$ GeV and $\tilde{m}_1 = m_3/A_\Lambda \approx 5$ meV. M_1 uses *only* $\{M_{\rm scal}, \varphi_0^{\rm ret}, A_\Lambda\}$ — no hidden seesaw scale, cleaner than `v184`'s $M_R\varphi_0^4$ — and lands in the thermal window where $\eta_B \approx 1.2 \times 10^{-9}$ brackets the observed 6.1×10^{-10}. The coupling mechanism is posited not derived, so η_B stays [C]; [X] a precise Boltzmann/RGE solve outside the band falls the route, not the theory (`FR.ETAB.04`; Route A independent).

2026-06-15 (Lean: QGEO.COHOM.01 character grading + MODULE parity machine-proved)

- **The cohomology character grading is now Lean-formalised** (`CohomologyGrading.lean`, ledger `FORM.QGEO.03`, [O]). A new module machine-proves the Cohom node of the QGEO proof tree (`v177`; audit: pass, no `sorry`/`admit`, axioms `propext`, `Classical.choice`, `Quot.sound` only): the three $H^1(\mathbb{P} \setminus \mu_4)$ eigenforms $\omega_k = z^{k-1}/(z^4-1)$ (k=1, 2, 3) have pullback character $\rho^*\omega_k = i^k\omega_k$ under the clock $\rho : z \mapsto iz$ (each an exact rational-function identity, the denominator clock-invariant since $i^4{=}1$), so the grading is $(i, -1, -i)$ with weights $(1, 2, 3)$ = the A_3 exponents = Spec(Q_+), rank $3 = N_{\rm fam}$. The Module *parity* is also proved: the reflection $\sigma : z \mapsto 1/z$ satisfies $\sigma^*\omega_1 = \omega_3$, $\sigma^*\omega_2 = \omega_2$, $\sigma^*\omega_3 = \omega_1$ (swap $\omega_1 \leftrightarrow \omega_3$, ω_2 fixed). Signature-locked in `AuditContract.lean`. The geometric COHOM/MODULE-parity nodes are now [E]; the MARKS/KERNEL obligations and the MODULE *uniqueness* stay [O].

2026-06-15 (Lean: QGEO.UNIFORM.01 geometric normal form machine-proved)

- **The Möbius uniformisation normal form is now Lean-formalised** (`MobiusUniformisation.lean`, ledger `FORM.QGEO.02`, [O]). A new module machine-proves the constructive UNIFORM node of the QGEO proof tree (`v177`; AUDIT: PASS, no `sorry`/`admit`, axioms `propext, Classical.choice, Quot.sound` only): the clock $\rho : z \mapsto iz$ has ρ^4=id and $\rho^2 \neq$id (order exactly 4); the reflection $\sigma : z \mapsto 1/z$ is an involution with $\sigma\rho\sigma = \rho^{-1}$ (so $\langle \rho, \sigma \rangle$ is the dihedral D_4 of order 8); a non-fixed orbit $\{a, ia, -a, -ia\}$ scales to $\mu_4 = \{1, i, -1, -i\}$; σ permutes μ_4 (fixes $1, -1$, swaps $i, -i$); and the multiplier classification "$z \mapsto \zeta z$ has order exactly 4 $\Leftrightarrow \zeta = \pm i$". Signature-locked in `AuditContract.lean`. This formalises the geometric half of the seam realisation *given* the four marks and the clock; the raw-seam marking obligation `QGEO.MARKS.01` stays [O], *not* closed.

2026-06-15 (Lean: the QGEO.SYM.01 conditional theorem machine-proved)

- **The mark-local $\Rightarrow \omega \circ \rho = \omega$ implication is now Lean-formalised** (`SeamDeckClosure.lean`, ledger `FORM.QGEO.01`, [O]). A new module in the carrier-rigidity Lean project machine-proves the algebraic core of `v201`/`v210` (AUDIT: PASS, no `sorry`/`admit`, no domain axioms — `#print axioms = propext, Classical.choice, Quot.sound` only): (i) the 4th-root character orthogonality $\sum_{j<4} \zeta^j$ = (if ζ=1 then 4 else 0) for ζ^4=1 (so a μ_4-mark sum is supported only on modes $\equiv 0 \bmod 4$); (ii) the clock generator $-i$ has order 4; (iii) `markLocal_blockDiagonal`: a mark-local Toeplitz symbol connects only equal clock-characters, $[\rho, M_f] = 0$; (iv) the premise is a typed `structure SeamDeckPremise` (the physical seam DtN *is* mark-local) — consumed, *not* proved, exactly as `CalderonProjector` encodes the Paper-1 analytic input; (v) `SeamDeckPremise.clock_invariant`: *given* the premise, the clock commutes with the DtN, whence $\omega \circ \rho = \omega$. So the *implication* is now [E] (formally proved); the *premise* (the physical seam being mark-local) stays [O] — the one fundamental seam-identification postulate, *not* closed.

2026-06-15 (QGEO sharpening: mark-local DtN certified on realistic profiles)

- **Mark-local DtN $\Rightarrow \omega \circ \rho = \omega$, certified on realistic Steklov profiles and at the state level (`v210_mark_local_dtn`, `tfpt_research_contracts`, [O]).** A numerical sharpening of `QGEO.SUBPRIN.01`/`v201`: a real von-Mises curvature bump summed over the four μ_4 marks, $f(\theta) = \sum_{j=0}^{3} g(\theta - j\pi/2)$, builds the Steklov DtN $\Lambda = |D_\theta| + M_f$ whose off-(mod 4) Fourier support vanishes to $< 10^{-16}$, $\|[\rho, \Lambda]\| < 10^{-15}$, and — the genuinely new step — the quasi-free covariance $C = \frac{1}{2}(1 + \operatorname{sgn} H_1)$ is clock-invariant ($\rho C \rho^\dagger = C$ to $< 10^{-13}$), i.e. the actual $\omega \circ \rho = \omega$. Negative controls (single off-centre bump, $\mathbb{Z}_3$ source, four generic marks) break both by $O(1)$; convergence stable for $N \in \{16, 32, 64\}$. This certifies the *implication*; the premise that the physical seam *is* mark-local stays [O] (it does *not* close `QGEO.SYM.01`; net existence and full-cone RP remain the [E] of `v175`). Ledger `QGEO.MARKLOCAL.01`. Python-only (the exact mark-sum identity is Wolfram-mirrored via `v201`).

2026-06-15 (archive integration #5: six dormant results revived from the old papers)

- **Muon $g-2$ as a seam vertex (`v204_muon_seam_g2`, `tfpt_4_frontier`, [C]).** The carrier's second-order topological defect $\delta_2 = \frac{5}{4}\delta_{\text{top}}^2$ projected through the seam-loop phase gives $a_\mu^{\text{seam}} = \delta_2/(2\pi) = 45/(524288\,\pi^9) \approx 2.879 \times 10^{-9}$ (0.81σ vs the dispersive Δa_μ; $\sim 1.5\sigma$ vs lattice/CMD-3). The value is exact [E]; the vertex identification is the [C] bridge. Ledger `FR.MUONG2.01`; Wolfram-mirrored.

- **An independent gravitational** $3/4$ **(`v205_xi_threequarter`, `tfpt_4_frontier`, [E]/[C]).** $\xi = c_3/\varphi_{\rm tree} = 3/4$ exactly (Einstein limit of $\kappa^2 = \xi\varphi_0^{\rm ret}/c_3^2$) — the same $3/4 = q(A_3) = \ln(m/\mu)$ as the seam-determinant replica (`v152`), an over-determination. Ledger `GRAV.XI.01`; Wolfram-mirrored.
- **Quantitative** H_0 **from the** Λ **branch (`v206_h0_lambda_branch`, `tfpt_4_frontier`, [C]).** $\delta_\Sigma = 48\,e^{-2\pi\alpha^{-1}/3} \approx 1.085\times10^{-123}$, the $(8\pi)^2$ Planck-convention split, and $H_0 = 66.5$–67.1 km/s/Mpc (below Planck, far below SH0ES — the Hubble tension is *not* relieved). Ledger `GRAV.H0.01`.
- **Asymptotic Safety as a second QG route (`v207_asymptotic_safety`, `tfpt_4_frontier`, [O]).** A complementary continuum-FRG argument (a non-Gaussian UV fixed point from the same axioms; $y^2 = 16c_3^2 = 1/(4\pi^2)$) — a plausibility cross-check, *not* load-bearing, does *not* close $G_{\rm metric}$. Ledger `GRAV.ASYMP.01`.
- **Black-hole thermodynamics from the seam** (`v208_bh_thermodynamics`, `tfpt_horizon_readouts`, [E]/[C]). Modular $T_H = \kappa/(2\pi)$ (the $2\pi = 1/(4c_3)$ seam unit, ties to `v198`/`v199`) and the scalaron-corrected Wald entropy $S_W = (A/4G)(1 + R_h/3M_s^2)$ (exact from `v28`). Ledger `HOR.BHTHERMO.01`; Wolfram-mirrored.
- **The black hole as a seam fixed point** (`v209_bh_regular_core`, `tfpt_horizon_readouts`, [C]). No singularity ($\varphi \to \varphi_\star$ at the gapped attractor $(2/3)^6$); the maximal BH = the anchor (Nariai roots $(1,1,-2)$); information via Page recovery; a Planck-scale holographic floor. The superseded RN/torsion metric is *not* resurrected. Ledger `HOR.BHCORE.01`.

2026-06-15 (archive integration #4: Möbius ≅ Lorentz foundation citation)

- **The Möbius seam is the boundary Lorentz structure (`tfpt_3`, [O]).** Added a paragraph to the "Möbius origin of π" section: $\mathrm{Möb}(\hat{\mathbb{C}}) \cong \mathrm{PSL}(2,\mathbb{C}) \cong \mathrm{SO}^+(3,1)$ (Penrose & Rindler, *Spinors and Space-time*, Vol. 1, §1.2–1.4), the celestial sphere of null directions via stereographic projection. This gives the self-consistency loop a spacetime meaning — the carrier clock $z \mapsto iz$ is an elliptic $\mathrm{PSL}(2,\mathbb{C})$ rotation (v180) and the RP→OS causal cone is the same Lorentz structure — so "Möbius origin of π" and "one Lorentzian cone" are two faces of one boundary $\mathrm{PSL}(2,\mathbb{C})$. A foundation citation, not a new claim (π stays primitive).

2026-06-15 (archive integration #3: axion coupling $g_{a\gamma\gamma} = -4c_3$ + retired small-angle reading)

- **Haloscope coupling tied to** c_3 **(`tfpt_4_frontier`, [C]).** Made explicit: in the determinant-line normalization the axion–photon anomaly coefficient is $g_{a\gamma\gamma} = -4c_3 = -1/(2\pi)$ ($y^2 = 16c_3^2 = 1/(4\pi^2) \approx 0.0253$), so the haloscope $F\tilde{F}$ coupling is set by the *same* c_3 that fixes α and $\beta_{\rm rad} = \varphi_0/4\pi$ — a [C] structural relation (the coefficient, not a parameter-free physical $g_{a\gamma\gamma}$). The earlier small-angle $\theta_i = \varphi_0$ reading (old determinant-line note, a much higher axion mass) is *superseded* by the closed near-hilltop $\theta_i = \pi(1-\varphi_{\rm seam}) = 170.4°$, recorded as a retired reading and guarded by v188 (a new sentinel forbids that stale axion mass from re-entering the active docs).

2026-06-15 (archive integration #2: the EHT achromatic polarization intercept — v203)

- **`HOR.EHT.01` (v203, [E]/[C]/[X]).** The surviving core of the old UFE/black-hole notes (*not* the RN-type metric — superseded by the Nariai/seam=horizon readouts v101–v104/v190; only the polarization signature survives): a local achromatic polarization-rotation intercept around a horizon-scale black hole, $\beta_{\rm BH}(r) = 16c_3^4\, Q_e^{\rm eff} Q_m^{\rm eff}/r^2$. The coupling $16c_3^4 = 1/(256\pi^4) = \delta_{\rm top}/3$ is *exact* ([E], Wolfram-mirrored) — the *same* top-form coefficient $\delta_{\rm top} = 48c_3^4$ that fixes the α-kernel precision-zone correction, one row from the cosmic-birefringence seed $\beta_{\rm rad} = \varphi_0/4\pi$. TFPT fixes the $1/r^2$ shape, achromaticity and sign-flip ([C]); the amplitude Q_eQ_m is an MHD/GR weight, never a pixel prediction. **Dated kill test** [X]: the residual intercept must pass three nulls (frequency, spatial $1/r^2$, sign-flip) after honest GRMHD subtraction. Integrated into `tfpt_horizon_readouts` (new EHT section), the ledger, registry, Wolfram (243/243), and the website (`predictions.ts`). Suite 202 modules.

2026-06-15 (archive integration #1: the rare kaon sector $K \to \pi\nu\bar\nu$ — v202)

- **`FR.RAREKAON.01` (v202, [C]).** The closed TFPT CKM point ($s_{12}{=}\lambda_C$, $s_{23}{=}\varphi_0/(1{+}\lambda_C){=}|V_{cb}|$, $s_{13}{=}\lambda_C^3/3$, $\delta{=}\pi/3{+}3\lambda_C^2{=}1.19823$ rad; v18/v88, no flavor fit) feeds the cleanest FCNC probe. With standard external short-distance input (Brod–Gorbahn–Stamou $X_t, P_c, \kappa_\pm$) it gives $\mathrm{BR}(K^+ \to \pi^+\nu\bar\nu) = 9.45 \times 10^{-11}$ and $\mathrm{BR}(K_L \to \pi^0\nu\bar\nu) = 3.33 \times 10^{-11}$ — a [C] downstream flavor readout, *not* a compiler power (the SD functions are external). Recomputed from the *current* CKM point (an earlier archive scaffold quoted $\mathrm{BR}(K_L) = 3.47 \times 10^{-11}$ from an $\Im(\lambda_t)/\lambda^5$ slip, corrected here). Grossman–Nir ratio $0.352 \ll 4.3$; NA62 $(13.0^{+3.3}_{-3.0}) \times 10^{-11}$ is $\sim 1.2\sigma$ above. **Dated kill test** [X]: a stable NA62 $\mathrm{BR}(K^+)$ outside $[7, 12] \times 10^{-11}$, or a KOTO-II $\mathrm{BR}(K_L)$ off the GN point, breaks the flavor bridge (core untouched). Integrated into `tfpt_2_standard_model` (§rare kaon), the ledger, v187 guard (new `FR.RAREKAON.` family), registry, and the website (`predictions.ts`). Python-only. Suite 201 modules.

2026-06-15 (bedrock: the $\omega \circ \rho = \omega$ residual reduced once more — v201)

- **`QGEO.SUBPRIN.01` (v201, [E]/[O]).** The v199 bedrock residual — "the bounded sub-principal symbol of the raw seam DtN has no off-character (mod 4) matrix elements" — is reduced one genuine step further: block-diagonality is *not* an independent postulate. Writing the DtN as $\Lambda = |k| + M_f$ with M_f the bounded sub-principal piece = multiplication by a curvature $f(\theta)$ (standard Steklov structure), M_f is μ_4-character-block-diagonal iff f is $\mathbb{Z}_4$-invariant ([E]; principal $|k|$ already commutes, v198); and a μ_4-mark sum $f = \sum_{j=0}^{3} g(\theta - 2\pi j/4)$ has $f_m = 4g_m[m \equiv 0 \bmod 4]$ for any profile g, hence is automatically $\mathbb{Z}_4$-invariant ([E]). So the μ_4-mark orbit (forced by v195) *forces* the sub-principal symbol block-diagonal; 3 marks ($\mathbb{Z}_3$) and 4 generic marks break it (negative controls). The residual collapses to "the DtN sub-principal symbol is mark-local (seam flat away from the μ_4 marks = conformal deck, v177)", structurally definitional — the narrowest form yet of `QGEO.SYM.01`, [O]. Registered in `run_all.py`, the ledger, `tfpt_research_contracts`, `origin_theory`; Wolfram-mirrored (the mark-sum Fourier identity is exact; extension now 242/242). Suite 200 modules.

2026-06-15 (F_transfer workshop: the axion relic solver — an honest over-production tension)

- **Axion relic solver (`experiments/ftransfer/axion_relic/relic_solver.py`, [C]).** The first real $F_{\rm transfer}$ pipeline computation (work package C / review §6.2): a standard misalignment relic with a numerically-computed anharmonic factor $F(\theta_i) \approx 4$ at the hilltop. Honest finding: at the *predicted* $\theta_i \approx 170°$ the estimate *over*-produces dark matter, $\Omega_a h^2 \sim 0.6$–2.3 (robustly > 0.12), because the large $\theta_i^2 \approx 8.8$ times F overshoots. So the determinant-line axion as the *dominant* DM is in *mild tension* (it would prefer $\theta_i \sim 120$–$130°$ or extra dilution) — this *refines* v185's optimistic "can be all-DM" toward an over-production tension. The near-hilltop regime is O(1)-uncertain, so a full finite-T solve decides; over-production kills only the axion-as-dominant-DM *branch*, not the compiler. Reflected in `tfpt_4_frontier` (dark-matter section) and the v185 docstring; experiments-only, not a suite/ledger claim. No refitted exponents; $\theta_i = \pi(1 - \varphi_{\rm seam})$ only.

- **Full finite-T solve confirms it (`axion_relic/full_finiteT_solve.py`).** A converged nonlinear misalignment solve ($\theta'' + (3 + \mathrm{d}\ln H/\mathrm{d}N)\theta' + (m_a(T)/H)^2 \sin\theta = 0$, lattice $\chi(T) \propto T^{-8.16}$, realistic $g_*(T)$; normalised so $\theta_i{=}1 \to \Omega_a h^2 \approx 0.03$, the standard value) gives $\Omega_a h^2 \approx 0.66$ at the predicted $\theta_i \approx 170°$ — $\sim$5.5× above $\Omega_{\mathrm{DM}} = 0.12$, reached only at $\theta_i \approx 106°$. The over-production is thus *confirmed* (not the optimistic "all-DM"); `tfpt_4_frontier`, v185 docstring and the website updated accordingly. Stays [C]; kills only the axion-as-dominant-DM branch.

2026-06-14 (work packages A+B: attacking $\omega \circ \rho = \omega$ + the variational scan — v199, v200)

- **`v199_seam_state_invariance.py` [E]/[O] (claim `QGEO.STATE.01`, work package A).** Attacking $\omega \circ \rho = \omega$ directly on the *raw* seam. (1) [E] the setup is non-circular: ρ is the Coxeter element of $W(A_3) = S_4$ ($\rho^4 = 1$, v117), an automorphism of the raw CAR algebra (16 Majoranas) — both carrier-side, no seam geometry. (2) [E] for a quasi-free RP state $C_\Sigma = \frac{1}{2}(1+\operatorname{sgn} H_1)$, so $\omega \circ \rho = \omega \Leftrightarrow [\rho, H_1] = 0$. (3) [E] $[\rho, H_1] = 0 \Leftrightarrow H_1$ is block-diagonal in the four μ_4-character classes $\{n{=}r \bmod 4\}$ (verified). (4) [O] the residual is sharpened to "the bounded sub-principal symbol has no off-character matrix elements" — a bounded, finite-character condition, not a diffuse geometry; not a closure. Cited in `tfpt_research_contracts`; Wolfram-mirrored.
- **`v200_seam_variational_scan.py` [C]/[O] (claim `QGEO.VARI.02`, work package B).** The discretised variational test: with the clock angle free, $E_{\mathrm{fail}}(\theta)$ has its unique minimum at $\theta = \pi/2$ (the μ_4 clock $z \mapsto iz$) — order-4 *and* the three distinct characters $i, -1, -i$ = the $(1, 2, 3)$ grading; decoys $(0, \pi, \pi/3, 2\pi/3)$ all lose. A numerical sign that the variational principle selects μ_4; the full operator-valued minimisation stays the v199 residual [O]. Cited in `tfpt_research_contracts`.
- *Also (this pass, `experiments/` only, not the ledger): the F_{transfer} four-pipeline scaffold (`experiments/ftransfer/`) with one CONTRACT per interface, and confirmation that the FRB preregistration (`frb_tfpt_v1.yaml`) and its status semantics already exist.* Suite now 199 modules, highest ID `v200`; Wolfram extension 241/241.

2026-06-14 (cracking the bedrock to a state-invariance: principal symbol + Tomita–Takesaki — v198)

- **`v198_modular_commutator_reduction.py` [O] (claim `QGEO.MODULAR.01`).** The decisive crack on the last residual. (1) [E] the DtN principal symbol $|k| = \sqrt{-\partial_\theta^2} = \mathrm{diag}(|n|)$ and the clock $\rho{:}z \mapsto iz = \mathrm{diag}(i^n)$ are both diagonal in the Fourier basis, so $[\rho, |k|] = 0$ *exactly* on all of L^2 (verified on 33 modes) — the leading-order commutation is free on the whole boundary, never the open part. (2) [E] by Tomita–Takesaki the modular operator is intrinsic to (M, Ω), so a state-preserving symmetry ($\rho\Omega = \Omega$) commutes with $\log\Delta \sim \Lambda_\Sigma$: hence $[\rho, \Lambda_\Sigma] = 0$ *follows from* $\omega \circ \rho = \omega$, using only the state + reflection (RP) and *no* conformal covariance — which **removes the v194 Bisognano–Wichmann circularity worry**. (3) [E] the finite H^1 block is already μ_4-invariant (v177). So the entire residual collapses to the single *state-invariance* "the raw quasi-free seam state is μ_4-invariant" ($\omega \circ \rho = \omega \Leftarrow$ the Gaussian bulk measure is deck-invariant) — the maximally-operational, non-circular form of `QGEO.SYM.01`, [O]. *Not* a full closure (a foundational symmetry cannot be derived from nothing, like $c = \text{const}$), but the sharpest falsifiable form, with the circularity gone. Cited in `tfpt_research_contracts`; `OpenGates` updated. Suite now 197 modules, highest ID `v198`; Wolfram extension 240/240.

2026-06-14 (three review levers: Marks-via-Lefschetz, E_fail functional, RR→D5 — v195–v197)

- **`v195_marks_lefschetz_character.py` [E]/[O] (claim `QGEO.MARKS.02`).** The four seam marks are *forced* (not posited): $\rho{:}z \mapsto iz$ fixes only $\{0,\infty\}$, $H^1 = \chi_1 \oplus \chi_2 \oplus \chi_3$ (the A_3 exponents, $\mathrm{Tr}(\rho|H^1) = i+i^2+i^3 = -1$) has no trivial component ⇒ no puncture at a fixed point, and rank $H^1 = n-1 = 3 \Rightarrow n = 4 \Rightarrow$ a single free μ_4 orbit (cross-ratio 2). So Marks reduces from the geometric posit "D_4 marks" to the milder premise "H^1 carries the A_3-exponent rep" — less circular, still [O]. Cited in `tfpt_research_contracts`.
- **`v196_seam_energy_functional.py` [E]/[O] (claim `QGEO.VARI.01`).** The failure functional $E_{\mathrm{fail}} = \|[\rho, \Lambda_\Sigma]\|^2_{HS} + \|\rho^4 - 1\|^2 + \|\Theta\rho\Theta - \rho^{-1}\|^2$, zero locus = the `QGEO.SYM.01` conditions; on the finite H^1 block it vanishes at the μ_4 config [E] and is > 0 for any μ_4-breaking perturbation, so μ_4 is a true minimiser. The full-operator minimisation is the v194 Bisognano–Wichmann residual — a concrete discretisable *target* on the open bedrock, [O]. Cited in `tfpt_research_contracts`.
- **`v197_rr_carrier_clifford_d5.py` [E]/[C] (claim `ARCH.RRCAR.02`).** Hardens the Riemann–Roch carrier from g_{car} to D_5 itself: $\Lambda^{\mathrm{even}}(C_{\mathrm{car}}) = \binom{5}{0}+\binom{5}{2}+\binom{5}{4} = 16 = 2^{g_{\mathrm{car}}-1} = \dim S^+$, the $\mathfrak{so}(10) = D_5$ half-spinor — so D_5 is the Clifford spinor of the 5-dim mode space, closing $\mu_4 \to g_{\mathrm{car}} \to D_5$. [E] arithmetic, [C] identification (as v189). Cited in `origin_theory`.
- All three are validated external-review levers (the axion-hilltop "$\pi - 3\phi_0$" proposal was *rejected* as a $\pi \approx 3$ coincidence, not built). Suite now 196 modules, highest ID `v197`; Wolfram extension 239/239.

2026-06-14 (subclaim 2 of ENERGY.02 tackled: raw-seam-DtN RP-definability — v194)

- **`v194_raw_seam_dtn_rp.py` [O] (claim `QGEO.DTN.01`).** The next hard lever — "is the raw RP-seam DtN definable from RP data alone?" (sub-claim 2 of `QGEO.ENERGY.02`) — was tackled, and the honest result is a *reduction*, not a closure. (1) [E] the commutator $[\rho, \Lambda_\Sigma] = 0$ closes on the finite cohomology H^1 ($\rho^* w_k = i^k w_k$, distinct μ_4 characters; v177). (2) [E] the *hinge* is met: the DtN/transfer operator is RP-canonical (Osterwalder–Schrader, v54) on a quasi-free seam state (v155), so Λ_Σ is RP-definable without naming $\mathbb{P}^1 \backslash \mu_4$. (3) [O] the residual relocates: the full-L^2 commutation ⇔ Bisognano–Wichmann geometric modular flow of the quasi-free seam state (which must be intrinsic, else it re-circulates). So `QGEO.SYM.01` reduces from a definitional postulate to "intrinsic BW geometric modular covariance" — one notch sharper, still [O]; the last structural fog point sharpens, it does *not* close to [E]. Cited in `tfpt_research_contracts`; `OpenGates` updated. Suite now 193 modules, highest ID `v194`.

2026-06-14 (QGEO.ENERGY.02 energy-commutator + QFT-wording hardening — v193)

- **`v193_qgeo_energy_commutator.py` [O] (claim `QGEO.ENERGY.02`).** The non-circular sharpening of `QGEO.ENERGY.01`: the energy commutator $[\rho, \Lambda_\Sigma] = 0$ on the rank-8 Calderón polarisation, with three sub-claims — (1) [E] ρ is carrier-defined (Coxeter of $W(A_3) = S_4$, v117), not imported from the seam; (2) [O] the non-circularity hinge: Λ_Σ must be the *raw* RP-seam DtN, not the μ_4 normal form; (3) [E] the commutator forces the $(1,2,3)$ grading on H^1 (`QGEO.COHOM.01`). Exact [E] rider (Wolfram-mirrored): the induced EH coefficient $k = (\ln m)/(12\pi)$ (v150) reproduces $c_3/2$ iff $\ln m = q(A_3) = \frac{3}{4}$ (*family*), not $q(D_5) = \frac{5}{4}$ (carrier, off by 5/3) — gravity is family-geometry-induced. A *proof target*; if sub-claim (2) holds, `QGEO.SYM.01` becomes an energy-rigidity theorem. Cited in `tfpt_research_contracts`. Suite now 192 modules, highest ID `v193`; Wolfram extension 236/236.

- **QFT-wording hardening (review point 7).** `tfpt_2_standard_model` §(4) was retitled to "Constructive QFT closure of the admissible physical sector", and its standalone opening over-claim replaced by an explicitly scoped sentence (the unprojected ambient metric measure is not constructed; G_{metric} frontier). The two superseded standalone phrasings were added to the `v188` prose guard so a frozen release cannot re-introduce them.

2026-06-14 (geometry round: Riemann-Roch carrier, Nariai bound, branch line, energy clock — v189–v192)

- **`v189_riemann_roch_carrier.py` [E]/[C] (claim `ARCH.RRCAR.01`).** The pair $(g_{\mathrm{car}}, N_{\mathrm{fam}}) = (5, 3)$ are the two canonical invariants of *one* object, the four-point seam divisor $D = \mu_4$ on $\mathbb{P}^1$: the cycle side rank $H_1(\mathbb{P}^1 \setminus \mu_4) = \deg -1 = 3 = N_{\mathrm{fam}}$ (already `QGEO.COHOM.01`) and the function side $h^0(\mathbb{P}^1, \mathcal{O}(\mu_4)) = \deg +1 = 5 = g_{\mathrm{car}}$, matched by $\mathrm{rank}(D_5) = 5$. The carrier-as-mode-space reading is [C] (geometric, does not displace the over-determined $g_{\mathrm{car}} = 5$). Cited in `origin_theory`; Wolfram-mirrored.
- **`v190_nariai_entropy_bound.py` [E] (claim `HOR.NARIAI.02`).** A variation-free one-line proof of the SdS entropy floor: $3(x^2+1) - 2\Phi_3(x) = (x-1)^2 \geq 0$, so $S_{\mathrm{tot}} \geq \frac{2}{3} S_{dS}$ with equality iff $x = 1$ (the Nariai merge). Cited in `tfpt_horizon_readouts`; Wolfram-mirrored.
- **`v191_universal_branch_line.py` [C] (claim `HOR.BRANCHLINE.01`).** The affine map $q = \frac{7}{2} + \frac{N_{\mathrm{fam}}^2}{2} m$ carries the SdS branch points $\pm 1/N_{\mathrm{fam}}$ onto the flavor labels $\{2, 5\}$. *Honestly typed [C]* — an exact *relabeling*, not a theorem (a decoy pair $\{3, 4\}$ admits an equally exact map; the shared content is the known $2/3$ ramification). Cited in `tfpt_horizon_readouts`; Wolfram-mirrored.
- **`v192_qgeo_energy_clock.py` [O] (claim `QGEO.ENERGY.01`).** An operator-checkable restatement of the bedrock: $\rho^* \Lambda_\Sigma \rho = \Lambda_\Sigma$ (the clock preserves the DtN energy form). *Honestly typed:* it *relocates* `QGEO.SYM.01` to a more falsifiable surface but does *not* eliminate it (plausibly equivalent unless ρ is independently defined and the invariance proven). Cited in `tfpt_research_contracts`. Suite now 191 modules, highest ID `v192` (v186 skipped).
- *All four are validated external-review proposals (`problem_b`); each is built with the honest typing determined on review — the branch line is recorded as an alignment, not a theorem, and the energy clock as a relocation, not a closure.*

2026-06-14 (Zenodo release 5.1 (rev 120) — F_transfer firewall, v184–v188)

- **Published to Zenodo.** Version `5.1 (rev 120)` of the ten-PDF document set was published as a new version of the TFPT deposit — record `20687564`, DOI `10.5281/zenodo.20687564`. This version carries the F_transfer-firewall round (v184–v188): the honest η_B anchored-Boltzmann test (sharper scenario, not a closure), the axion hilltop-relic typing, the ledger `v187` and prose `v188` guards, and the Route B/Koide/dark-matter wording fixes plus the P1/P2-vs-`QGEO.SYM.01` two-layer clarification. The uploader's `DEFAULT_RECORD_ID` was advanced to the new record for the next release. Suite 187 modules (highest ID `v188`; v186 skipped). `run_all.py` green, audit clean, website built.

2026-06-14 (F_transfer round: eta_B honest test v184, axion relic v185, discipline guard v187)

- **`v184_etaB_anchored_boltzmann.py` [C] (claim `FR.ETAB.03`).** An honest test of the external-review η_B proposal ($M_1 = M_R \varphi_0^4$, $\tilde{m}_1 = m_3/A_\Lambda$). Firewall verdict: *sharper scenario, not a closure.* The washout anchors plausibly ($\tilde{m}_1 = m_3/10 \approx 5$ meV, $A_\Lambda = 10$ atom, in the strong-washout window) [C]; but M_1 does *not* — M_R is the seesaw scale $\sim v^2/m_3$ and the nearest clean TFPT powers of $\bar{M}_{\mathrm{Pl}}$ both miss it (c_3-/φ_0-powers off ∼75%/∼40%), so

$M_1 = M_R\varphi_0^4$ merely relocates the free input. η_B stays [C], the route viable but with one (not zero) free input. Cited in `tfpt_4_frontier`.

- **`v185_axion_relic_solver.py` [C] (claim `FR.DM.03`).** Honest misalignment estimate: the determinant-line axion $f_a = M_{\rm scal}/128 \approx 2.39\times10^{11}$ GeV ($m_a \approx 23.8\,\mu$eV) CAN be all dark matter, but the harmonic misalignment under-produces ($\sim$21% at $\theta \sim O(1)$), so the $\theta_i \approx 170°$ hilltop anharmonic enhancement is *required* — where the relic is exponentially sensitive. Hence a [C] scenario, not a sharp prediction; the relic contract is a kill test for the BRANCH, not the theory. Cited in `tfpt_4_frontier`.

- **`v187_ftransfer_laws.py` [E] (claim `FR.TRANSFER.01`).** A CI-style firewall guard: the four continuous-transfer frontier observables (Koide, η_B, axion, m_p/m_e) are read from the ledger and required to carry a [C]/[O] marker — never a primitive [E] compiler prediction (exact sub-parts like the Koide 53/54 factor or $b_3 = -7$ may be [E], the physical prediction never is). Records the four interfaces $F_{\rm pole}/F_{\rm Boltzmann}/F_{\rm relic}/F_{\rm QCD}$; prevents any future edit from quietly promoting a frontier number to a closed output. Cited in `tfpt_4_frontier`.

- **`v188_frontier_wording_guard.py` [E] (claim `AUDIT.WORDING.01`).** A prose sentinel (no new physics): `v187` protects the ledger *typing*, this guard protects the *prose*. It scans the 10 active documents and forbids five stale phrasings whose semantics earlier rounds superseded (the old [E]-typed Route B leptogenesis claim and its heavy-scale wording → v184 [C]; the old “open relic scale” dark-matter line → v185; the old “near, not exact” Koide line → v183; and the module-count/ID conflation), so a frozen Zenodo release can never re-introduce wording that was true yesterday and is half-true today. The same pass updated the `tfpt_4_frontier` Route B (to [C]), the Koide and dark-matter summary lines, and added the P1/P2-vs-`QGEO.SYM.01` two-layer clarification to the introduction. Suite now has **187 modules**, highest verification ID `v188` (v186 was intentionally skipped as redundant — m_p/m_e is already typed [O] as a QCD matching contract, `QCD.LAMBDA.01`).

- *Review framing points (P1/P2 as projections of the seam-deck postulate, $v_{\rm geo}$ as unit gauge, $G_{\rm net}$ a corollary of `QGEO.SYM.01`, grammar tuning = data) were validated as already implemented in the prior rounds; m_p/m_e is already typed [O] as a QCD matching contract (`QCD.LAMBDA.01`), so no redundant module was added.*

2026-06-14 (operator origin of the Koide 53/54 factor: the missing Sheet-Diamond corner, v183)

- **`v183_koide_f_corner_transfer.py` [E]/[C] (claim `FR.KOIDE.07`).** An external-review proposal, validated exactly: the Koide source→pole factor $\frac{53}{54}$ is not a bare arithmetic guess but an *operator* identity on the carrier, $u_{\rm pole}/u_{\rm source} = a^T(R+Q)\mathbf{1}/(2\,\mathbf{1}^T Ra) = 53/(2\cdot 27) = \frac{53}{54}$. Here $\mathbf{1}^T Ra = 27$ is the $E_6\times A_2$ cubic family block ($248 = \dim E_6 + \det R + 2(\mathbf{1}^T Ra)N_{\rm fam} = 78+8+2\cdot 27\cdot 3$) and $a^T(R+Q)\mathbf{1} = 53 = 52+1$, where $F = R+Q$ is the *missing* Sheet-Diamond corner with $\det B_F = 52 = \dim F_4$ (v80). Hard negative control: $a^T M\mathbf{1}$ for $M \in \{R, Q, K, L\}$ is $\{32, 21, 35, 56\}$ — only the absent corner F gives 53. This upgrades `FR.KOIDE.03` from “near miss + conjecture” to “operator-sourced source→pole transfer”: the *factor* is [E] (exact, mirrored in Wolfram, 232/232); the *mechanism* (the leptonic source→pole transfer reads the F corner) stays [C], an $F_{\rm transfer}$ special case. The prediction itself stays [C]. Cited in `tfpt_4_frontier` (Koide section). Suite now 183 modules.

- **Review point 2 noted as already-present:** the claim that flavor and Nariai share one double-cover ClockFunctor (flavor rate $(2/3)^{2N_{\rm fam}}$ vs gravity curvature $(2/3)^{N_{\rm fam}}$, exponent ratio $|\mathbb{Z}_2|$, “one clock, two geometries”) is exactly `HOR.TRISECT.01`/v103 — a rediscovery of an existing result, no new module.

2026-06-14 (review recommendations 1/3/4 + the four concerns A–D mapped to the named residuals)

- **`v182_reviewer_residual_map.py` [E]/[C] (claim `REVIEW.MAP.01`).** A careful external review raised four concerns (A mapping $2D \to 4D$, B UV-gravity abstractness, C grammar-tuning/meta-look-elsewhere, D Koide Möbius flow). This module shows three of them *reduce*, by the same discipline as v175–v181, onto the *two already-named* residuals $\{\texttt{QGEO.SYM.01}, F_{\text{transfer}}\}$: A (Plücker-11 = QBL boundary count v174 + holographic reduction), B ($\equiv$ `QGEO.SYM.01`), D (Möbius forced by $\mathrm{Aut}(\mathbb{P}) = \mathrm{PSL}(2, \mathbb{C})$ on the deck; the missing generator $= F_{\text{transfer}}$). Only C is genuinely *experiment-only* (frozen + minimal + over-determined grammar; residual falls to JUNO/CMB-S4). A unification of the concern *structure*, not a closure. Cited in `tfpt_5_redteam`. Suite now 182 modules.
- **Review recommendation #3 — a "Rosetta" translation lexicon** (website `RosettaLexicon`): TFPT's compiler-speak (seam, deck, readout, microcode, audit hull, gap, anchor, F_{transfer}, v_{geo}, `QGEO.SYM.01`) mapped onto standard QFT / mathematical-physics language (Wilsonian boundary CFT, orbifold deck, index/intersection number, RG irrelevance rate, renormalization condition, fundamental postulate) — a reading aid for mainstream physicists, not a new claim.
- **Review recommendation #1 — kill-tests up front** (website `TrustContract`): the frozen pre-data predictions ($\sin^2\theta_{12} = 0.3067$, JUNO; $r \approx 0.004$, CMB-S4) and the null model ($P \leq 10^{-30.7}$) are now a prominent strip directly under the hero.
- **Review recommendation #4 — the bedrock as a postulate.** `QGEO.SYM.01` is now framed confidently as TFPT's *one fundamental postulate* (the role "c = const" plays in relativity), not a gap — in the `tfpt_research_contracts` box and the website "what is open" card; the achievement is that the whole theory is compressed onto exactly one sharply-located, falsifiable premise.

2026-06-14 (external-review response: CSV fix, G_net 3-level, QGEO.SYM.01 page, claim detox)

- **Ledger CSV bug fixed + guarded.** Three rows (`AX.P1.01`, `AX.P2.01`, `QGEO.ISO.01`) had *unquoted commas* in a status/external field (e.g. "a=(1,1,2)", "PSL(2,C)"), so a strict CSV parser read 11–12 columns instead of 10 — a real machine-readability defect in the single source of truth. All offending fields are now quoted (every row parses to exactly 10 columns), and `audit_sync.py` gained a `check_csv_wellformed` guard (ledger, registry, clusters, freeze must all parse to constant width) so it cannot regress.
- **G_{net} typed in three explicit levels** (website "What is open" card): (1) *algebra* [E] $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1,1) \rangle \cong (E_8)_1$; (2) *AQFT machinery* [E] (net existence + full-cone RP, CAR functor, v175); (3) *seam instantiation* [O] (`QGEO.SYM.01`). The target object's mathematics is closed; only the physical coupling of the raw seam is open.
- **`QGEO.SYM.01` now has its own one-page box** in `tfpt_research_contracts`: what is identified, why it is weaker than `CONF/ISO`, what follows automatically (the conditional theorem), and *what would falsify it* (genus > 0 double; non-deck transport; $N_{\text{fam}} \neq 3$); with the explicit conditional framing "*given* `QGEO.SYM.01`, the closure follows" rather than "the theory derives the seam geometry".
- **Claim-language detox.** "no free fundamental number" softened to "no free *dimensionless compiler dial* beyond the anchor and π (dimensionful scale + transfer physics stay typed)" (`origin_theory`, `introduction`, website); the site title softened from "the Standard Model Derived" to "the Standard-Model *Skeleton* Derived".

- **External-reviewer map** added to the website verification page: four boxes (exact kernel [E], conditional physics [C], open premises [O], kill tests [X]) so a reviewer sees the attack surface without reading 181 scripts. Suite unchanged at 181.

2026-06-14 (Zenodo release 5.1 (rev 114) — v175–v181 bedrock + figures + P1/P2)

- **Published to Zenodo.** Version `5.1 (rev 114)` of the ten-PDF document set was published as a new version of the TFPT deposit — record `20686284`, DOI `10.5281/zenodo.20686284`. This version carries the AQFT-closure-to-bedrock chain (v175–v181, the structural residual reduced to the single definitional premise `QGEO.SYM.01`), the two overview figures (`residual_chain`, `script_timeline`), and the P1/P2 anchor-reduction provenance on the axiom surfaces. The `README` and the Zenodo description were refreshed (181 modules), and the uploader's `DEFAULT_RECORD_ID` was advanced to the new record for the next release.

2026-06-14 (overview figures + P1/P2 reduction reflected on the axiom surfaces)

- **Two overview figures** (`make_figures.py`; PDF + website PNG). (i) `residual_chain` — the structural-residual reduction chain v175–v181 as a descending staircase from "build a quantum-gravity measure" to the single definitional bedrock `QGEO.SYM.01`; embedded in `tfpt_research_contracts`. (ii) `script_timeline` — the eight-phase journey of the ∼181 scripts (foundations → SM readouts → seam= horizon → horizon/flavor geometry → R1–R5/premise-(A) → PyR@TE cross-checks → AQFT bridges → AQFT closure), what each did mathematically and physically; embedded in `introduction` ahead of the master script index, and both shown on the website verification page.
- **P1/P2 reduction now reflected on the axiom surfaces.** The reduction of the two axioms to the single parabolic anchor $a = (1, 1, 2)$ plus π (with $g_{\mathrm{car}} = 5$ an over-determined bootstrap fixed point — forced three ways, v6, Pascal-unique and Lean-formalised, `FORM.LADDER.01`) was already established (`ANCHOR.GEN.01`/v23, `ARCH.CORE.01`/v53) and stated in the papers/glossary, but the *axiom surfaces* still read as bare "declared inputs". Fixed: the ledger rows `AX.P1.01`/`AX.P2.01` now carry the anchor/bootstrap reduction and point to it, and the website dependency-graph nodes P1/P2 now show the anchor input ($c_3 = 1/(2e_1(a)\pi)$; $g_{\mathrm{car}} = e_2(a)$, bootstrap-forced) rather than "declared axiom". No marker change (still declared inputs), only honest provenance.
- Manifests refreshed (two new figures added to the `FIG` list). Suite unchanged at 181.

2026-06-14 (the final reduction to bedrock v181 + full non-changelog status sweep)

- **`v181_clock_is_conformal_symmetry.py` [E]/[O] (claim `QGEO.SYM.01`).** The isometry premise `QGEO.ISO.01` (v180) is weakened one more genuine step — to a conformal-*symmetry* premise — via *equivariant uniformisation* (Nielsen realisation; for cyclic groups Kerékjártó + uniformisation): a *conformal* automorphism (strictly weaker than an isometry — every isometry is conformal, conformal allows any positive scale) already suffices to put the order-4 clock in the form $z \mapsto iz$. Since the clock is the Coxeter element of $W(A_3) = S_4$ (v117), the μ_4 deck, and a deck transformation preserves the pulled-back conformal structure by definition, the residual collapses to the *definitional* identification `QGEO.SYM.01`: "the seam's conformal structure *is* the μ_4-deck structure of the carrier clock". This is bedrock — a physical/definitional statement tying P2 to the seam's conformal geometry, *not* a finite computation and not reducible further without relabeling. The chain `REALIZE`(v175)→ $\cdots$ →`SYM.01`(v181) ends here; we stop honestly. Python-only. Suite now 181 modules.

- **Full non-changelog status sweep.** The current bedrock framing was propagated to every live status surface (not just the changelog): the `introduction` "latest reductions" prose, `tfpt_5_redteam` Target A, the `tfpt_research_contracts` central-theorem keybox, `origin_theory`; and on the website `OpenGates`, `VerificationDag`, `papers.ts` (Target A $\times 2$, $G_{\rm net}$), the `glossary` $G_{\rm net}$ entry and the `faq` live-residual answer — all now end at `QGEO.SYM.01`.

2026-06-14 (the conformal premise reduces to a milder isometry premise v180)

- **`v180_clock_is_mobius.py` [E]/[O] (claim `QGEO.ISO.01`).** Cracks `QGEO.CONF.01` one level further: the conformal-realisation premise is *replaced*, via three classical theorems, by a strictly *milder, non-circular* premise. (1) *Uniformisation* (Poincaré–Koebe): a genus-0 Riemann surface is conformally $\mathbb{P}$, so the genus-0 seam double (P1, v179) with its RP-metric conformal structure *is* $\mathbb{P}$ — the target is *produced*, not assumed; (2) *Kerékjártó* (1919; Constantin–Kolev 2003): a finite-order orientation-preserving homeomorphism of S^2 is conjugate to a rotation, so the order-4 clock ($h(A_3)$=4) is topologically $z \mapsto iz$; (3) *isometry⇒ conformal⇒Möbius*: an orientation-preserving isometry of a Riemannian surface is holomorphic, $\mathrm{Aut}(\mathbb{P}) = \mathrm{PSL}(2, \mathbb{C})$; (4) an order-4 Möbius map is $z \mapsto iz$ (verified: projective order 4, fixed $\{0, \infty\}$, multiplier i, free orbit μ_4). Hence `QGEO.CONF.01` is *implied* by the milder premise `QGEO.ISO.01`: "the carrier clock is an order-4 orientation-preserving *isometry* of the RP seam metric" — which does not name $\mathbb{P}/\mu_4$ (uniformisation produces them) and is the natural statement that the seam transport is metric-compatible (a holonomy preserving the RP inner product). That milder premise is the single, now milder, structural residual of the whole theory; its surface realisation from the raw seam stays honestly [O]. Cited in `tfpt_research_contracts`; Python-only. Suite now 180 modules.

2026-06-14 (the two obligations collapse to ONE geometric premise v179)

- **`v179_conformal_realisation.py` [E]/[O] (claim `QGEO.CONF.01`).** Investigating the two residual premises left by v178 shows they are *not* independent — they collapse onto a single geometric premise (honest unification; the premise itself stays [O]). Marks's residual is not three independent assumptions: (1) genus-0 *is* P1 — the one-sided Gauss–Bonnet $c_3 = 1/(|\mathbb{Z}_2|\, 2\pi\, \chi(S^2)) = 1/(8\pi)$ uses $\chi(S^2) = 2$ and $\chi = 2 - 2g \Rightarrow g = 0$, so "genus-0" is the same $\chi = 2$ that gives the "8" in c_3 (v58/v73); (2) the order-4 clock *is* the carrier — the Coxeter element of $W(A_3) = S_4$ (v117), order $h(A_3) = 4 = |\mu_4|$; (3) marks = one orbit is *forced* (the parabolic marks are not the two elliptic fixed points, a free order-4 orbit has exactly $4 = e_1(a)$ points). Kernel then *follows* from the same geometry (DtN symbol $|k|$, Lee–Uhlmann v156; $c = 8$ rigidity v158; the cusped-surface residual $= H^1 = \mathbb{C}^3 = N_{\rm fam}$). So the *entire* structural residual of the theory is the *single* geometric premise `QGEO.CONF.01`: "the raw RP seam normal double is conformally $(\mathbb{P}, \mu_4)$ with the carrier clock as the order-4 Möbius automorphism". The two doors of v177/v178 are one geometric realisation, left honestly [O]. Cited in `tfpt_research_contracts`; Python-only. Suite now 179 modules.

2026-06-14 (an honest attempt at the two obligations v178 — deeper reduction, not closure)

- **`v178_marks_kernel_attempt.py` [E]/[O] (claim `QGEO.REDUCE.01`).** An honest attempt at `QGEO.MARKS.01` and `QGEO.KERNEL.01`. These are genuine constructive-geometry/AQFT statements, *not* finite computations — neither is closed. What the module does is push each as far as computation allows: it closes the finite [E] core of each and reduces each to *one sharper* premise. *Marks core* [E]: given a genus-0 double with an order-4 clock ρ (fixed points $\{0, \infty\}$) and reflection τ, the marks are forced to be a generic ρ-orbit $\{1, i, -1, -i\} = \mu_4$ (*not* the fixed points), distinct iff $b_1 = 4 - 1 = 3 = N_{\rm fam}$ (any collision drops b_1), with $\tau\rho\tau = \rho^{-1}$

giving a faithful D_4 of order 8 — so MARKS reduces to the single topological/clock premise. *Kernel core* [E]: on the 3-dim multiplicity-free H^1 the μ_4 characters $(1,2,3)$ are distinct, so by Schur a μ_4-equivariant operator is forced *diagonal* and completely determined by its eigenvalue per character; two such with the forced transfer spectrum $\{1,(2/3)^6,(1/3)^6\}$ (v162) and the unique residue-normalised basis (v177) are *equal as operators*, not merely isospectral — the spectrum-vs-operator worry *vanishes* on the finite block. So KERNEL reduces to the single full-operator-reduction premise (principal symbol $|k|$, Lee–Uhlmann/v156; no drift v158). Net: neither obligation closed; both sharpened to milder premises with their finite cores [E]. Cited in `tfpt_research_contracts`; Python-only (Möbius/Schur symbolic + numeric). Suite now 178 modules.

2026-06-14 (QGEO proof tree v177 — the residual sharpened into two named obligations)

- **`v177_seam_marking_kernel.py` [E]/[O] (claims `QGEO.TREE.01`, `QGEO.MARKS.01`, `QGEO.KERNEL.01`).** A second external residual-class audit pointed out that the v176 statement takes the marks/D_4 as a *hypothesis* — a near-circular trap. This module factors the central theorem honestly as REALIZE = MARKS · UNIFORM · COHOM · MODULE · KERNEL · NET and closes *four* nodes *constructively* (all validated symbolically before adoption): UNIFORM [E] — an order-4 Möbius map conjugates to $z \mapsto iz$, a non-fixed orbit scales to μ_4, and $\sigma\rho\sigma = \rho^{-1}$ for $\sigma : z \mapsto 1/z$ gives a faithful D_4 (order 8), so a genus-0 four-marked faithful-D_4 boundary *is* $(\mathbb{P}, \mu_4)$ (strengthening v168's cross-ratio to the full uniformisation); COHOM [E] — $\rho^*\omega_k = i^k\omega_k$, grading $(1,2,3) = A_3$ exponents; MODULE [E] — a *unique* μ_4-equivariant iso (multiplicity-free + residue normalisation; reflection $\omega_1 \leftrightarrow \omega_3$, ω_2 fixed); NET [E] (v154/v175). The structural residual is thereby sharpened from one diffuse premise into *exactly two* named open obligations, neither a finite computation: `QGEO.MARKS.01` (the raw RP seam collar canonically produces the genus-0 four-marked D_4 boundary) and `QGEO.KERNEL.01` (the raw seam Calderón operator *is* the μ_4-equivariant free gapped contraction, as an operator). Cited in `tfpt_research_contracts` (the central-theorem keybox); Python-only (Möbius/cohomology symbolic; numbers mirrored via v168/v137/v162/v154/v175). Suite now 177 modules.

2026-06-13 (Seam Collar Realisation Theorem v176 — the one central proof obligation)

- **`v176_seam_collar_realisation.py` [E]/[O] (claim `QGEO.REALIZE.02`).** Following an external residual-class audit, the single remaining structural item is formulated as one central theorem and certified as a *reduction* (not a fabricated proof). *Theorem:* given an oriented one-sided RP seam collar with a sheet-odd Calderón involution, faithful D_4 marks, four parabolic marks with μ_4 decomposition and admissible c=8 data, its conformal boundary is Möbius-equivalent to $\mathbb{P} \setminus \mu_4$ with H^1 grading $(1,2,3)$, the Calderón contraction is the rank-8 K_Σ polarisation and the index-4 extension is $(E_8)_1$. The proof decomposes into six steps, *five already* [E]: geometry/uniformisation (v168), Calderón DtN symbol $|k|$ (v156), H^1 character = generation grading $(1,2,3) = A_3$ exponents (v137/v168), μ_4-equivariant contraction with gap $(2/3)^6$ (v162), AQFT closure $(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 = (E_8)_1$ + full-cone RP all m (v154/v175). The whole structural residual thus reduces to the *one* open premise `QGEO.REALIZE.01`: that the physical seam collar's boundary *is* this $\mathbb{P} \setminus \mu_4$ — a constructive-geometry statement, left honestly [O]. Cited in `tfpt_research_contracts` (the central theorem keybox); assembly/reduction certificate, Python-only (its exact identities are mirrored via v168/v156/v162/v154/v175). Suite now 176 modules. *Audit note:* the metrology residual the same review flags is already discharged — v153 (No-Unit Theorem) is the metrology-line typing ($v_{\text{geo}} \mapsto \lambda^{-1}v_{\text{geo}}$, $1/G \mapsto v_{\text{geo}}^2$, dimensionless data invariant), and the Koide $u \to \frac{53}{54}u$ source→pole transfer is already typed [C] (`FR.KOIDE.03`); neither is re-derived.

2026-06-13 (Zenodo release 5.1 (rev 105) — v174 + v175 + doc refresh)

- **Published to Zenodo.** Version `5.1 (rev 105)` of the ten-PDF document set (introduction, origin_theory, tfpt_1–5, horizon, research_contracts, changelog) was published as a new version of the TFPT deposit — record `20681451`, DOI `10.5281/zenodo.20681451`. This version carries the AQFT closure round (v170–v175): the seam Fock-space readings and the heavy-artillery discharge of net existence + full-cone reflection positivity to [E]. The `README` and the Zenodo description were refreshed to the current state (175 modules, Wolfram 116/116 + 231/231), and the uploader's `DEFAULT_RECORD_ID` was advanced to the new record for the next release.

2026-06-13 (the heavy artillery v175: net existence + full-cone RP discharged to [E])

- **`v175_net_existence_full_cone.py` [E]/[O] (claim `QFT.NETRP.01`).** The two structural residuals that the premise-(A) chain had relocated to *already-open* items — A_2 net existence and full-cone reflection positivity — are now *discharged to a theorem*, leaving the seam realisation as the single irreducible open premise (nothing fabricated). (1) *Full-cone RP for all m*: the CAR second-quantisation functor $\Gamma(t) = \bigoplus_m \Lambda^m(t)$ lifts the one-particle seam contraction t to the complete Fock space $\Lambda^\bullet(\mathbb{R}^{16})$ ($\dim 2^{16} = 65536$); by the exterior-power spectrum theorem every eigenvalue is a subset product of spec(t), so all 2^{16} products lie in $[0, 1]$ (a contraction = full-cone RP), with eigenvalue-1 multiplicity $2^8 = 256$ (wedges of the rank-8 K_Σ polarisation) and sub-leading = the one-particle gap $(2/3)^6$ — so full-cone RP reduces *for every m* (not only $m \leq 6$, `v161`) to the one-particle contraction (Shale–Stinespring), discharging the `v160` scope premise. (2) *A_2 assembled + verified*: the $(E_8)_1$ character $E_4/\eta^8 = 1 + 248q + 4124q^2 + 34752q^3 + 213126q^4 + 1057504q^5$ (level-1 = 248, extends the order-4 `v156` check by one order), the embedding $248 = 120 + 128$, and the E_8 Cartan matrix even with $\det 1$ (the unique rank-8 even unimodular lattice, `v83`); the net *exists* by cited theorems (free-fermion net, lattice VOA, index-4 Q-system, `v125`). (3) Both then need the *same* input — the seam realisation `QGEO.REALIZE.01` (finite half proven, `v168`) — which is a constructive-geometry/AQFT statement, *not* closeable by a finite computation, and is left honestly [O]. **Net:** net existence and full-cone RP become [E]; the one irreducible structural premise is the seam realisation. Cited in `tfpt_research_contracts` (premise (A) closure) and `tfpt_5_redteam` (Target A); the exact parts (Cartan unimodular, character order 5, functoriality integers) mirrored in the Wolfram extension (231/231). Suite now 175 modules.

2026-06-13 (seam Fock-space readings v174: CAR cone, Witten index, g-theorem)

- **`v174_seam_fock_readings.py` [E] (claim `QFT.FOCK.01`).** Three more exact audit bridges from the same source note. (1) *CAR cone compression*: the even CAR algebra of the 16 seam Majoranas has $\dim \Lambda^{\text{even}}(\mathbb{R}^{16}) = 2^{15} = 32768$ directions, but a quasi-free Bogoliubov transport reduces the whole cone to a 16-dim one-particle problem ($2^{15}/16 = 2^{11}$) — the explicit count behind `v161`. (2) *Witten index = the Plücker norm* 11: the four μ_4-corner fermions span a $2^4 = 16 = \dim S^+$ Fock space; QBL (deg ≤ 2) keeps $\sum_{k\leq 2}\binom{4}{k} = 11 = \dim S^+ - g_{\text{car}}$, so $c_u/c_d = 55/117$ reads a topological state-space dimension. (3) *Affleck–Ludwig g-theorem*: $c_{UV} = 5 + 3 = 8 = c_{IR}((E_8)_1)$, so $\Delta c = 0$ forces an exact isometry — the boundary-entropy form of the `v157`/`v158` rigid fixed point. Cited in `tfpt_2` (Exterior Leg Lemma); mirrored in the Wolfram extension (230/230). Suite now 174 modules. (The categorical reframings in the source note remain framing, not asserted claims.)

2026-06-13 (CFT/QFT bridges: slice compression v170, OS moment v171, trace-anomaly seed v172, Pfaffian CP v173)

- **`v170_diamond_slice_compression.py` [E] (claim `E8.SLICE.01`).** E8 Slice Compression Lemma: the seven E_8 audit slices are *one* projection of two alphabets — anchor power sums $P = (3,4,6,10)$ ($p_n = 2 + 2^n$) and Sheet-Diamond determinants $(3,4,8,14,20,32)$ summing to $81 = N_{\mathrm{fam}}^4$. All seven 248-slices, the K_4 edge graph, and the row-budget cross $(7,11,13) + s(3,3,3) + t(1,2,3)$ follow; $78 = \dim E_6 = p_2(p_0 + p_3)$. Cited in `tfpt_3` (slice atlas). Exact; reading is audit (anti-numerology).
- **`v171_os_moment_cluster.py` [E]/[C] (claim `QFT.OSMOMENT.01`).** Atomic OS Moment Lemma: $\mathrm{spec}(T) = \{1, 64/729, 1/729\}$ makes every correlator a positive moment sequence, so the OS Hankel matrices factorise as Vandermonde Grams (a clean RP certificate); cluster 729/665, $\xi = 0.4111$. Sugawara Gap Safety: $31 = 1 + h^\vee(E_8)$, $c((E_8)_1) = 248/31 = 8$, $\Delta_{\mathrm{eff}} = 6\log\frac{3}{2} - 31/(4\pi^2) > 0$. Cited in `tfpt_research_contracts` (G5).
- **`v172_trace_anomaly_seed.py` [E] (claim `SEED.ANOMALY.01`).** The seed prefactor $\frac{4}{3}$ is the halved $(E_8)_1$ Weyl anomaly: over the seam S^2 ($\chi = 2$) the $c = 8$ trace anomaly is $c\chi/6 = \frac{8}{3}$, halved by the one-sided $|\mathbb{Z}_2|$ to $\frac{4}{3} = 16/12$. Plus QCD $b_3 = 7 =$ scalaron exponent $48 - 41$ (confinement $\leftrightarrow$ inflation), and the Volkov–Wipf $-98/45 = -2\cdot 7^2/\dim(D_5)$ factorisation. Cited in `tfpt_2` (seed).
- **`v173_pfaffian_cp_car.py` [E] (claim `FLAV.STRONGCP.02`).** Strong CP $\theta_{\mathrm{eff}} = 0$ as Pfaffian reality in the self-dual 16-Majorana CAR model: $\{D,\Sigma\} = 0$ pairs the spectrum, γ_5-Hermiticity makes the Pfaffian real, and RP picks the positive branch. Cited in `tfpt_2` (Strong CP section).
- All four validated computationally before adoption; the exact identities (v170/v171/v172) are mirrored in the Wolfram extension (229/229). Suite now 173 modules. (The categorical reframings in the source note — compiler-as-single-functor, $E_8 = K_0$, GIT/monadic/Langlands — are recorded as framing, not asserted as machine-checked claims.)

2026-06-13 (Zenodo release 5.1 (rev 97))

- **Published to Zenodo.** Version `5.1 (rev 97)` of the ten-PDF document set (introduction, origin_theory, tfpt_1–5, horizon, research_contracts, changelog) was published as a new version of the TFPT deposit — record `20678568`, DOI `10.5281/zenodo.20678568`. The uploader's `DEFAULT_RECORD_ID` was advanced to the new record for the next release. Also fixed `build.sh zenodo` on macOS bash 3.2 (empty-array expansion under `set -u`).

2026-06-13 (QGEO rigidity theorem v168 + η_B leptogenesis interface v169)

- **`v168_qgeo_rigidity.py` [E]/[C] (claims `QGEO.RIGID.01`, `QGEO.REALIZE.01`).** Hardens the *finite* half of the one remaining Tier-1 hinge `GATE.QGEO` (the seam = flavour = horizon identification). Exact, machine-checked: $\mu_4 = \{1, i, -1, -i\}$ has cross-ratio 2 and a faithful Möbius stabiliser of order $8 = |D_4|$; $H^1(\mathbb{P}^1 \setminus \mu_4)$ has rank $4 - 1 = N_{\mathrm{fam}} = 3$; and the forms $\omega_k = z^{k-1}dz/(z^4-1)$ ($k = 1,2,3$) are μ_4-eigenforms with characters χ_1, χ_2, χ_3 of weights $(1,2,3) = \mathrm{A}_3$ exponents $= \mathrm{Spec}(Q_+)$. So a genus-0 boundary with faithful D_4, four parabolic marks and this μ_4-decomposition is Möbius-equivalent to $\mathbb{P}^1\backslash\mu_4$ — the QGEO Rigidity Theorem (`QGEO.RIGID.01`, [E]). The seam-collar realisation stays the one premise (`QGEO.REALIZE.01`, [C]). Mirrored in the Wolfram extension (226/226). Cited in `tfpt_research_contracts` (U0). *Not* a new structural class — it hardens the existing hinge (consistent with v167's completeness claim).
- **`v169_etaB_boltzmann_interface.py` [C] (claim `FR.ETAB.02`).** Operationalises the Tier-3 F_{transfer} leptogenesis path. Type-I thermal leptogenesis $\eta_B \sim 0.96 \times 10^{-2}\varepsilon_1\kappa_f$ (sphaleron

28/79), fed by TFPT's normal-ordered neutrino spectrum and the predicted $\delta_{CP} = 240°$. Over natural $M_1 \in [3\times10^9, 3\times10^{10}]$ GeV and washout, η_B brackets the observed 6.1×10^{-10} (a canonical point gives 6.0×10^{-10}, untuned) — the route is *viable*. But M_1/washout are scenario inputs, so η_B stays [C]; if a precise solve excluded the window the leptogenesis *route* falls, not the theory (the downstream Ω_b readout is independent). Cited in `tfpt_4_frontier` (Route B). Python-only.

2026-06-13 (Higgs quartic from the free seam: SM near-criticality explained, v166)

- **`v166_higgs_free_seam.py` [E]/[C] (claim `HIGGS.FREESEAM.01`).** A new prediction tying the Higgs mass to premise (A). The seam UV is the free chiral c=8 fixed point (v158, Gaussian, no relevant/marginal interactions), so the one marginal SM scalar coupling vanishes there: $\lambda(M_{\rm seam}) = 0$ and $\beta_\lambda(M_{\rm seam}) = 0$ — the Shaposhnikov–Wetterich double-criticality, here *derived* from the free seam, not assumed. Integrating the PyR@TE-confirmed *two-loop* SM RGEs from M_Z up with the measured (m_H, m_t) gives $\lambda(\bar{M}_{\rm Pl}) \approx 0.002$ and $\beta_\lambda(\bar{M}_{\rm Pl}) \approx 0$: the famous SM near-criticality is a *consequence* of the free seam (2-loop sharpens $\lambda(\bar{M}_{\rm Pl})$ from $\sim$0.022 at one loop to $\sim$0.002). The double condition predicts $m_H \sim 129$–134 GeV; the same condition at the scalaron scale gives $\sim$107 GeV (too low), so the boundary condition lives at the reduced Planck scale (seam = horizon = Planck). Fills the Higgs-sector gap (was only $N_\Phi = 1$); cited in `tfpt_4_frontier`.
- PyR@TE-sourced: the reproducer's SM run supplies the two-loop β_λ, reduced to top-only (`verification/pyrate/sm_higgs_betas.txt`). Suite now 166 modules. Python-only (numerical 2-loop ODE).

2026-06-13 (PyR@TE $F_{\rm transfer}$ follow-ups: flavor scale-tagging v163, QCD scale v164)

- **`v163_rg_stability_flavor.py` [E]/[C] (claim `FLAV.RGSTAB.01`).** Scale-tagging of the flavor sector. The lepton-mixing seeds ($\sin^2\theta_{12} = \frac{1}{3} - \frac{\varphi_0}{2} = 0.3067$, $\sin^2\theta_{13} = \varphi_0 e^{-5/6} = 0.0231$) are *RG-stable*: the PMNS running is y_τ^2-suppressed (the only mixing-rotating Yukawa term is $\frac{3}{2}Y_e Y_e^\dagger Y_e$, PyR@TE SM RGEs), so $\kappa_\tau = y_\tau^2/(16\pi^2)\ln(M_{\rm scal}/M_Z) \sim 1.8\times10^{-5}$ and $|\Delta\sin^2\theta_{ij}| \lesssim 10^{-4}$ — two to three orders below precision. Seam value = measured value; the y_t^2-driven quark mass ratios ($m_t/m_b \sim 41$) by contrast carry a scale tag. Cited in `tfpt_2` (solar/reactor angle box).
- **`v164_qcd_scale.py` [E]/[O] (claim `QCD.LAMBDA.01`).** The QCD confinement scale is parameter-free from the carrier $b_3 = -7$ (v159): running $\alpha_s(M_Z) = 0.1179$ down (two-loop, n_f thresholds) reproduces $\alpha_s(m_b) \simeq 0.22$, $\alpha_s(m_c) \simeq 0.38$ and confinement at ~ 0.5 GeV by dimensional transmutation — so the QCD-scale *numerator* of m_p/m_e is independently sourced. The $O(1)$ proton/Λ factor is lattice, so $m_p/m_e = 1836$ stays the honest "[O] — not forced" frontier ratio. Cited in `tfpt_4_frontier`.
- Both are PyR@TE-backed (the reproducer now also extracts the SM Yukawa β's, `verification/pyrate/sm_yukawa_betas.txt`). Suite now 164 modules. Python-only (no Wolfram mirror: magnitude bound / numerical running).

2026-06-13 (PyR@TE external cross-check of the $F_{\rm transfer}$ gauge inputs — v159)

- **New module `v159_pyrate_gauge_crosscheck.py` [E]/[C] (claim `EM.BUDGET.02`).** The one-loop gauge coefficients used by the transfer functor are confirmed by an *independent* third-party RGE generator. Feeding the carrier / Standard-Model field content into the textbook one-loop formula (GUT normalisation) gives $(b_1, b_2, b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$, and **PyR@TE 3** (`arXiv:2007.12700`, `github.com/LSartore/pyrate`) writes $\beta_{g_1} = (\frac{41}{10})g_1^3$, $\beta_{g_2} = -\frac{19}{6}g_2^3$,

$\beta_{g_3} = -7g_3^3$ verbatim (plus the full standard 2-loop matrix). So $b_1 = 41/10$ is pinned *three* independent ways — carrier algebra $10b_1 = g_{\text{car}}2^{g_{\text{car}}-2}+1 = 41$, the $U(1)$ hypercharge index $\sum \frac{3}{5}Y^2$, and PyR@TE — and the split $10b_1 = 40_{(\text{ferm.})} + 1_{(\text{Higgs})} = 41$ reproduces the EM-budget Gate 1 of `tfpt_2`. Cited there with `\veri`; the gauge sector of F_{transfer} is thus not a free knob.

- **PyR@TE reproducer (not vendored).** `verification/pyrate/reproduce.sh` clones PyR@TE on demand into `experiments/pyrate` (git-ignored), neutralises its interactive `pdflatex` end-command, runs the SM at one and two loops, and re-extracts the gauge β functions; the committed evidence is `verification/pyrate/sm_gauge_betas.txt`. PyR@TE is the right external tool for the transfer/RGE gates (gauge running, and — with a seesaw model — the neutrino-Yukawa running behind η_B); it is *not* a tool for premise (A) (a 2d boundary-CFT question). Mirrored in the Wolfram extension; suite now 159 modules.

2026-06-13 (terminal residual inventory: (A) no longer a structural class; one hinge + F_transfer + irreducible core, v167)

- **New module `v167_inventory_post_closure.py` [E]/[C].** The terminal residual inventory (line v105→v123→v167), re-pinned after the (A)-chain. With (A) discharged (`GATE.METRIC.16-19`) the v123 *five* structural classes collapse: the seam clock (R1), the index-4 net (R2) and the Q-realisation (R5) all merge into *one* Tier-1 hinge — the seam=flavour=horizon identification = A_2 net existence (assembly of established net constructions) + `GATE.QGEO` (seam boundary = $\mathbb{P} \setminus \mu_4$, cohomology b_1=N_{fam}=3); the EH step (R3) folds into v_{geo} (v152) and the ambient measure into G_{net} (v76). **Terminal structure:** Tier 0 = the irreducible core $\{\pi, v_{\text{geo}}\}$ (a theorem, No-Unit v153); Tier 1 = one hinge; Tier 2 = folded; Tier 3 = F_{transfer}, one functor with four typed conditional interfaces (Koide, η_B, axion, m_p/m_e), η_B carrying the most open computational room. Completeness: a sixth independent structural class — or a new irreducible — fails the script (ledger `SYNTH.INVENTORY.03`). A unification + re-typing, NOT an unconditional proof. Python-only (reads the ledger).
- **Bookkeeping.** Suite at 167 modules, all green; Wolfram extension unchanged at 224/224; `origin_theory` (residual-inventory keybox) and `next.txt` moved together.

2026-06-13 (premise A maximal honest closure: A2 by assembly, R1 = GATE.QGEO, zero independent open gates, v165)

- **New module `v165_premise_a_closure.py` [C]/[O].** Pins the two residual gates of the v160–v162 (A)-chain to their floor. A_2 **(net existence) closes by assembly** of established mathematics [C]: the 16 free Majoranas give a free-fermion conformal net (Buchholz–Mack–Todorov), the holomorphic c=8 lattice net is $(E_8)_1$ (Frenkel–Kac–Segal / Dong–Xu / Kawahigashi–Longo), the μ_4 extension is a Longo Q-system of index 4 (v125), $\mu(B)$=16/16=1, 248=120+128, character E_4/η^8=$j^{1/3}$ (v156); the one TFPT-specific input (seam Calderón kernel = free-fermion two-point function) is done via DtN = $|k|$ (v156/v157). R_1 **(seam clock) reduces to `GATE.QGEO`** [C]: a μ_4-equivariant flat gapped generator is *unique* (= $A_0^\star$, v115/v116), so the gap $(2/3)^6$ is forced and the residual is the one geometric identification “seam boundary = $\mathbb{P} \setminus \mu_4$” = `GATE.QGEO` (cohomology established, v137, b_1=3=N_{fam}). **Final accounting:** by the No-Unit Theorem (v153) the irreducibles stay $\{\pi, v_{\text{geo}}\}$; premise (A) = (A_2 assembly) + (R_1=`GATE.QGEO`) + the $\{\pi, v_{\text{geo}}\}$ core, so it carries *zero independent open gates* and ceases to be its own gate — a unification and re-typing, *not* an unconditional proof (ledger `GATE.METRIC.19`). Python-only (numpy/sympy + stdlib).
- **Bookkeeping.** Suite at 165 modules, all green; Wolfram extension unchanged at 224/224; `tfpt_research_contracts` (premise-A keybox) and `next.txt` moved together.

2026-06-13 (the final identification: premise A factors into the already-open A2 + R1, no new open content, v162)

- **New module `v162_seam_transport_identification.py` [E]/[O].** Pushes the v161 residual as far as it rigorously goes and shows it carries *no new open content.* The one-particle symbol splits into a gap value (β) and a fixed space (α). (β) **is forced:** a μ_4-equivariant generator is diagonal in the cusp basis (the deck average is the diagonal projection, $\sum_k U^k X U^{-k} = |\mu_4|\,\mathrm{diag}\,X$ for $U{=}\mathrm{diag}(1,i,-i)$; the commutant of U is exactly the diagonal algebra), so its spectrum is the cusp weights $\mathrm{spec}(A_0^\star){=}\{0,\frac{1}{3},\frac{2}{3}\}$ (v115) and the transfer eigenvalue $(1{-}\alpha)^{p_2}$ with $p_2{=}6{=}|R_+(A_3)|$ gives $\{1,(2/3)^6,(1/3)^6\}$, gap $6\log\frac{3}{2}$ — all carrier data; the lift to the 16 Majoranas is the A_3 subnet grading (v137). (α) is the rank-8 Calderón polarisation of $\omega_k{=}|k|$ (v113/v156). **Closure:** with v160+v161, (A) closes given only μ_4-equivariance + admissibility, whose two non-derived inputs are the already-open A2 (net existence, Kawahigashi–Longo) and R1 (seam clock, `HOR.CLOCK`). So premise (A) factors exactly into $A_2 + R_1$ — it adds *no* independent open item and is closed *conditionally* on them (a unification, not an unconditional proof; ledger `GATE.METRIC.18`). Python-only (numpy/sympy + stdlib).
- **Bookkeeping.** Suite at 162 modules, all green; Wolfram extension unchanged at 224/224; `tfpt_research_contracts` (premise-A keybox) and `next.txt` moved together.

2026-06-13 (the cone reduces to one particle: premise A's scope premise discharged to a 16-dim statement, v161)

- **New module `v161_one_particle_reduction.py` [E]/[O].** *Discharges* the scope premise of v160 ("the transport is gapped on the *full* Schwinger cone") via Bogoliubov second quantisation. A quasi-free transport is $T_\Sigma{=}\Gamma(t)$, the second quantisation of a one-particle contraction t on the 16-dim Majorana space; $\Gamma(t)$ acts on the degree-m correlator sector as the exterior power $\Lambda^m(t)$. Machine-checked: (1) $\mathrm{spec}(\Gamma(t)|_{\deg m}) =$ products of m one-particle eigenvalues (compound-matrix theorem, all $m{=}0..6$); (2) **the cone gap equals the one-particle gap** — the sub-leading eigenvalue over the whole cone is $(2/3)^6$ exactly, so premise 5 on the cumulant space *is* premise 5 on the 16-dim space; (3) the eigenvalue-1 sector is Λ^* of the fixed subspace (disconnected Wick powers), with uniqueness closed by v113 (one polarisation $\Leftrightarrow$ one vacuum); (4) quasi-freeness is *forced* — the $120{=}\dim\mathfrak{so}(16)$ chiral currents are the Bogoliubov generators and v158 makes every non-Gaussian deformation irrelevant. **Consequence:** v160's three full-cone premises collapse to ONE finite (16-dim) one-particle statement (seam symbol = gapped Calderón Bogoliubov map, fixed = rank-8 K_Σ, gap $(2/3)^6$); all numbers supplied, only the physical identification (Kawahigashi–Longo class = the A2 net residual) stays open — the infinite-dimensional cone is eliminated (ledger `GATE.METRIC.17`). Python-only (numpy + stdlib).
- **Bookkeeping.** Suite at 161 modules, all green; Wolfram extension unchanged at 224/224; `tfpt_research_contracts` (premise-A keybox) and `next.txt` moved together.

2026-06-13 (premise A as a conditional fixed-point theorem; the load relocates to one scope premise, v160)

- **New module `v160_seam_gaussianity_from_pf.py` [E]/[O].** Formalises the proposed closure of premise (A) ("the seam bulk is a reflection-positive free field") as a fixed-point theorem and types it honestly. The *exact* parts: (1) a quasi-free state has *all* higher connected cumulants $\kappa_{2n}{=}0$ — verified by the *signed* set-partition/Pfaffian moment$\leftrightarrow$cumulant inversion on a concrete pure Majorana covariance (the v113 "one kernel = net" property at the cumulant level); (2) a kernel-preserving transport fixes the whole quasi-free state by Wick functoriality; (3) the Perron–Frobenius dichotomy (a *fixed* κ_{2n} is a second fixed ray $\Rightarrow$ breaks uniqueness; a *non-fixed* one decays by the gap) forces Gaussianity; (4) the qualitative claim needs only gap>0, the value $(2/3)^6$ only sets the rate. **The honest residual:** v56's gapped PF uniqueness lives on the *three-dimensional* scalar transfer spectrum $\{1,(2/3)^6,(1/3)^6\}$, while the 16-Majorana seam already carries $\binom{16}{4}$=1820 and $\binom{16}{6}$=8008 higher-cumulant directions — so (A) reduces to the single internal, exact-testable statement "the admissible TFPT seam transport is primitive + spectrally gapped on the *full* Schwinger cone, not only the readout spectrum of v56", which the suite does *not* yet establish (ledger `GATE.METRIC.16`). Python-only (numpy + stdlib); no new exact-identity content for the Wolfram mirror.

- **Bookkeeping.** Suite at 160 modules, all green; Wolfram extension unchanged at 224/224 (v160 is numerical, Python-only); `tfpt_research_contracts` (the premise-A keybox) and `next.txt` moved together.

2026-06-13 (changelog date-order audit + reorder pass)

- **Changelog sort enforced.** `audit_sync.py` now checks that every dated `\subsection` × `{YYYY-MM-DD...}` block in this file is non-increasing (newest first; range titles such as `2026-06-07 / 2026-06-08` sort by the later date). A manual prepend had left five `2026-06-12` entries above seven `2026-06-13` blocks — reordered; future mis-sorts fail `build.sh audit` as `[B.changelog]`.

2026-06-13 (premise A: the destination fixed point is provably stable, v158)

- **The free chiral c=8 fixed point is provably stable (v158, exact dimension count).** The finite core of (A)'s "reaches-the-fixed-point" residual: the chiral Majorana has $h=\frac{1}{2}$, the bilinear current h=1, the quartic h=2. The relevant window $0 < h < 1$ for bosonic deformations is *empty*, the h=1 currents are chiral (v157, not $(1,1)$-marginal), and the lowest interaction (quartic, h=2) is irrelevant — so the fixed point is isolated and stable against all interactions. Premise (A) reduces to a basin-of-attraction statement with a *proven* stable target (ledger `GATE.METRIC.15`). *Tooling note:* PyR@TE (4d gauge-Yukawa RGEs) is the wrong tool for this 2d boundary-CFT residual, but the right tool for the F_{transfer} interfaces (η_B leptogenesis RGEs, gauge-coupling running, the b_1=41/10 cross-check).
- **Bookkeeping.** Suite at 158 modules, all green; Wolfram extension 224/224; contracts / `origin_theory` / `next.txt` / website moved together.

2026-06-13 (premise A reframed: freeness is a rigid boundary fixed point, v157)

- **The free-bulk premise becomes a fixed-point property (v157, simplicity-first).** Instead of postulating Gaussianity, three facts click: (i) the DtN/Calderón symbol is *universally* $|k|$ (Lee–Uhlmann — order-1 ΨDO, principal symbol $|\xi|_g$; interactions live in the irrelevant lower-order symbol), so the free dispersion is bulk-detail-independent; (ii) a holomorphic c=8 theory is *rigid* — every field has $\bar{h}$=0, so no marginal $(1,1)$ field exists and the fixed point is isolated, with no knob for an interaction; (iii) the same $|\mathbb{Z}_2|$ that halves 8π in $c_3 = \frac{1}{8\pi}$ is the Ramond/GSO projection giving μ=1 (248=120_{NS}+128_{R}). With $(E_8)_1$ unique (v83), premise (A) shrinks from "prove the seam is Gaussian" to "the gapped (Δ=$6\log\frac{3}{2}$) boundary flow reaches the holomorphic fixed point" — freeness then forced by rigidity, not fine-tuned (ledger `GATE.METRIC.14`).
- **Bookkeeping.** Suite at 157 modules, all green; Wolfram extension 223/223; contracts / `origin_theory` / `next.txt` / website moved together.

2026-06-13 (Route 3: the seam net constructed and $(E_8)_1$ verified, v156)

- **The constructive route carried to its exact endpoint (v156).** "$B \cong (E_8)_1$" is no longer an assertion: the chiral seam net is built explicitly and the identification *checked*. (a) The

Dirichlet-to-Neumann (Calderón) operator of the 2d Laplacian is the free chiral dispersion $\omega_k = |k|$ (harmonic extension $e^{ikx-|k|y}$); (b) 16 free Majoranas give $c = 8 = 5{+}3$; (c) the weight-1 currents are $248 = 120_{\rm NS} + 128_{\rm R}$ ($SO(16)$ bilinears + Ramond spinor, Frenkel–Kac); (d) the holomorphic character $E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + 34752q^3 + 213126q^4) = j^{1/3}$ (verified by q-series) — the 248 at level 1 is $\dim E_8$, so the net *is* $(E_8)_1$. *Residual:* exact given (i) the free-bulk premise (DtN $= |k|$, v155) and (ii) the operator-algebraic existence of the net from the seam kernel (a Kawahigashi–Longo-type theorem, not a finite computation) — where TFPT meets constructive QFT, now sharply located (ledger `GATE.METRIC.13`).

- **Bookkeeping.** Suite at 156 modules, all green; Wolfram extension 222/222 (the DtN $= |k|$ identity, the $(E_8)_1$ character $E_4/\eta^8 = j^{1/3}$ and the 248=120+128 split mirrored); contracts / redteam (Target A) / `origin_theory` / `next.txt` / website moved together.

2026-06-13 (the seam-side identification reduced to one shared premise, v155)

- **$G_{\rm net}$'s last premise reduces to a single shared physical premise (v155, Python-only).** The seam-side identification of the Simple-Current Extension Theorem (v154) splits (v110) into (a) "the Calderón involution is sheet-odd" (the double-cover deck, structural) and (b) "the seam 2-point kernel is quasi-free". Part (b) is *not* independent: the boundary marginal of a reflection-positive *Gaussian* bulk is the compression $P\Gamma P$ of the covariance, and the compression of a contraction is a contraction — so the seam state is automatically quasi-free (Wick inherited). Hence (b) *follows* from "the seam bulk is a reflection-positive free field", the *same* premise already load-bearing in the area law (v59) and the EH mechanism (v150/v151). **The entire seam-side residual is therefore one physical premise**, shared by $G_{\rm net}$, the area law and R3 — not reducible by a finite computation, but no longer three separate items (ledger `GATE.METRIC.12`).

- **Bookkeeping.** Suite at 155 modules, all green; Wolfram unchanged at 221/221 (v155 is numerical/numpy, Python-only); contracts / `origin_theory` / `next.txt` / website moved together.

2026-06-13 (external review formalised: No-Unit Theorem + Simple-Current Extension Theorem, v153–v154)

- **$v_{\rm geo}$ re-typed: the No-Unit Theorem (v153).** Every load-bearing compiler datum is mass-dimension 0 and invariant under a unit rescaling $L \mapsto \lambda L$, whereas a mass / $1/G$ is not — so a dimensionless boundary compiler *provably cannot* select an absolute scale. $v_{\rm geo}$ is therefore an *irreducible metrology primitive*, not an open gap, and the three apparent residuals collapse onto it ($U_{\rm point} \sim v_{\rm geo}$, $1/G \sim v_{\rm geo}^2$, $m/\mu = e^{3/4}$ a pure ratio); the irreducibles stay $\{$one unit, $\pi\}$ (ledger `ANCHOR.VGEO.02`).

- **$G_{\rm net}$ stated as the central closing theorem (v154).** The Simple-Current Extension Theorem: $A{=}(D_5)_1{\otimes}(A_3)_1$ extended by the isotropic glue $L{=}\langle(1,1)\rangle$ has $[B{:}A]{=}|L|{=}4$, $c{=}5{+}3{=}8$, $\mu(B){=}\mu(A)/|L|^2{=}16/16{=}1 \Rightarrow$ holomorphic $\Rightarrow B \cong (E_8)_1$ ($SO(16)_1$ excluded). Every algebraic ingredient is machine-checked (v89/v92/v125/v143/v148); the only residue is the seam-side identification. Pulled to the front of `tfpt_research_contracts`, `tfpt_5_redteam` (Target A final form) and `origin_theory` (ledger `GATE.METRIC.11`).

- **Status surfaces re-typed** (review): the canonical status card now reads $v_{\rm geo}$ as a metrology *primitive*, $G_{\rm net}$ as the simple-current extension $\cong (E_8)_1$ ([E]/[C]), and $F_{\rm transfer}$ as one functor with four typed interfaces — "not structural gaps". The final one-line framing (dimensionless boundary compiler; irreducibles $\{\pi, v_{\rm geo}\}$; one seam-net theorem; frontier $= F_{\rm transfer}$) is in `origin_theory`, the introduction, the README and `next.txt`.

- **Bookkeeping.** Suite at 154 modules, all green; Wolfram extension 221/221; intro/README/`origin_theory`/contracts/redteam/status-card/website moved together.

2026-06-13 (currency sweep: remaining old-series "Paper N" cross-references removed)

- **No old-version framing left in the active docs.** A consistency sweep found a residual layer of old-series "Paper N" attributions and removed them: four literal "Paper 6" references (`tfpt_4_frontier` leptogenesis/axion, `tfpt_2` seam determinant), the old "Paper 4" (OS/admissibility) and "Paper 5" (APS/Hodge) references in `tfpt_2`, the old "Paper 3" transport-kernel attributions in `tfpt_2`, and the internal "Paper 1/2/4" cross-references in `tfpt_3`, `tfpt_4` and `tfpt_horizon_readouts` — all rewritten as content-based phrasing (the flavor sector, the architecture document, the E_8 decoder, the anchor characteristic polynomial, etc.). The active documents now attribute results to the current documents / axioms / scripts only, as the workflow rule requires; the machine-checked sync (scripts ↔ papers ↔ ledger ↔ website) was already current (152 scripts, AUDIT OK).

2026-06-13 (fifth round: the R3 normalisation is the dimensionful anchor, v152)

- **R3: the $q(A_3)$ normalisation merges into the one anchor (v152).** The replica EH coefficient is a *log-ratio* $k = \ln(m/\mu)/(12\pi)$ ($\ln m$ alone is scale-ambiguous, only m/μ is physical); pinning $k = c_3/2$ fixes the dimensionless ratio $m/\mu = e^{3/4}$, and in $d{=}2$ that ratio *is* the induced Newton constant $1/G$ (v68). So $v_{\rm geo}$ (v78), $1/G$ (v68) and m/μ are *one* dimensionful anchor in three readings — the irreducibles stay {one scale, π}, and `SEAM.THEOREM.01`'s residual reduces to the kernel-identification premise alone. *Audit (not promoted):* the log-ratio is numerically $\frac{3}{4} = q(A_3)$, the target value the anchor takes in seam units — not a derivation (ledger `SEAM.EHMODEL.03`).
- **Bookkeeping.** Suite at 152 modules, all green; Wolfram extension 219/219; intro/README/`origin_theory/next.txt`/website moved together.

2026-06-12 (Zenodo version label: semantic + git rev)

- **Release version string.** `build.sh` now writes `tex-artefacts/release-version.txt` as `{TFPTversion} (rev {git count})` (matches `\TFPTdatestamp` without the date). `scripts/zenodo_upload.py` and `build.sh zenodo/zenodo-publish` use this label on Zenodo; cursor rules extended for when `zenodo_description.html` must move with deposit-facing changes.

2026-06-12 (Zenodo API upload workflow)

- **Automated Zenodo versioning.** Added `scripts/zenodo_upload.py` and `.github/workflows/zenodo-release.yml`: creates a new draft version of record 20579275 (DOI `10.5281/zenodo.20579275`), uploads the ten active paper PDFs, refreshes `zenodo_description.html`, bumps the version from `tex-artefacts/version.tex`, and optionally publishes via GitHub Actions (`ZENODO_TOKEN` secret).

2026-06-12 (Overleaf shadow mirror: tfpt5-shadow)

- **One-way GitHub mirror for Overleaf sync.** Added `scripts/export-shadow.sh` and `.github/workflows/shadow-sync.yml`: on every push to `main`, a filtered copy is pushed to `sthamann/tfpt5-shadow` (papers, `tex-artefacts/`, `figures/`, `verification/`, `experiments/`; excludes `_archive/`, `website/`, all `*.pdf`, `.cursor/`). ∼270 files / ∼3 MB — within Overleaf GitHub-sync limits.

2026-06-12 (origin_theory style alignment + box reduction)

- **Fewer boxes, embedded in text.** `origin_theory` from 22 to 16 boxes: the narrative/interpretation callouts (“Why the interpretation is internally consistent”, “Dependency chain of the birefringence test”, “The precise reformulation”, “The three theses honestly graded”, “The whole story”, “One-line summary”) become run-in bold prose; the boxed reformulation becomes a `keyeq`. The exact result keyboxes (v53–v56, v101, v105, v113, …) and the master-principle / Seam–Horizon target boxes are kept.
- **Colour-marked module references + marker hygiene.** `\vref` applied to the inline `vN` references; stale literal `[A]`/`[I]`/`[L]` markers in running text replaced by the 4-class [O]/[E].

2026-06-12 (tfpt_5 / horizon / research-contracts pass + build hardening)

- **Forked document-set box removed (horizon).** `tfpt_horizon_readouts` carried a handwritten, *stale* copy of the document-set card (“five short documents”, missing `tfpt_5`) *and* the canonical `\input{tex-artefacts/tfpt_docset}`; the fork is deleted so the single-source card (all five papers + three companions) appears once.
- **Fewer boxes, embedded in text.** Narrative/result callouts converted to run-in bold prose (with the `\veri{}` citation moved out of the bold title into the body): `tfpt_horizon_readouts` from 22 to 7 boxes (the ten-box “clock” wall becomes prose); `tfpt_research_contracts` from 19 to 10 (the nine progress/result winboxes become prose, the named-theorem keyboxes kept); `tfpt_5_redteam` from 10 to 9 (the outcome summary; the per-target verdict boxes are kept as the structured red-team output).
- **Colour-marked module references + marker hygiene.** `\vref` applied to the inline `vN` references in all three documents; the horizon “Scope/typing” marker line and a stale literal `[A]` in the research contracts updated to the 4-class markers.
- **Build hardening.** `build.sh` now removes a document’s `.aux`/`.toc`/`.out` *on failure*, so a half-written cross-reference file from a crashed compile cannot poison the next build; artefacts are still kept on success.

2026-06-12 (fourth round: the Calderón transfer answered — the BFK split, v151)

- **R3: the Calderón/DtN kernel is conically clean (v151).** Reflection doubling derives that the Kac corner constant is boundary-condition independent ($C_N = C_D$), so by the Burghelea–Friedlander–Kappeler gluing ledger the conical deficit of the two-sided Calderón jump determinant *vanishes identically* ($C_{\mathrm{cone}}(\gamma) - 2C_D(\gamma/2) = 0$): the seam-reduced effective action inherits the v150 EH form *verbatim*, with all of it localised in the *local* half-space determinants — the “Calderón version” demanded by v150 needs no separate derivation. `SEAM.THEOREM.01` narrows to the $q(A_3)$ normalisation plus the standing kernel premise; the gate stays [O] (ledger `SEAM.EHMODEL.02`).
- **Bookkeeping.** Suite at 151 modules, all green; Wolfram extension 218/218; intro/README/`origin_theory/next.txt` moved together.

2026-06-12 (third round: both normals anchor data; the R3 mechanism exists, v149–v150)

- **R4′: both dual normals are anchor data (v149).** In the cusp frame the torsion normal pairs to $(6, 3, 5) = (p_2, p_0, e_2)(a)$ — one anchor invariant per Q_+ line (the cusp matrix is unimodular, so this determines n uniquely); with $d = \frac{3}{2}a - 2 \cdot \mathbf{1}$ the dual normal pair is anchor data through and through. Equivalences to the $(2, 8, 121)$ frame data and the w_0-lift exact. R4′ residue = *one* discrete assignment (ledger `GATE.UWALL.14`).

- **R3: the mechanism exists at the gapped-model level (v150) — the first movement on SEAM.THEOREM.01.** The conical heat-trace deficit is exact and t-independent (Cheeger image sum $(N^2-1)/(12N)$); for a *gapped* operator the ζ-regularised replica variation is *finite and cutoff-independent*, $\Delta \log \det' = 2C(\gamma) \ln m$, and its linearisation is the Einstein–Hilbert form $(\ln m/12\pi) \int \sqrt{g}R$. The v73 normalisation becomes one exact target equation: $k = c_3/2 \Leftrightarrow \ln m = \frac{3}{4} = q(A_3)$ (audit, not asserted). Remaining: the Calderón (boundary) version and that normalisation — the gate narrows but stays [O] (ledger `SEAM.EHMODEL.01`).
- **Bookkeeping.** Suite at 150 modules, all green; Wolfram extension 217/217; status surfaces (intro reductions + card, `origin_theory`, contracts, `next.txt`, website) moved together.

2026-06-12 (R1–R5 second round: every class at its floor, v145–v148)

- **R4′: the pairing values are atom identities (v145).** Reading the v139 lift structurally ($n = w_0(\sigma)+|\mu_4|e_3$, $w_0 =$ the A_3 exponent duality), all three pairings become atom arithmetic — including the *new* closure $|\mu_4| + g_{\text{car}} = N^2_{\text{fam}}$ (4+5=9, $\sigma \perp w_0(a)$) and $\|\sigma\|^2 = 110 = |\mathbb{Z}_2| g_{\text{car}} \|\text{Pl}(K)\|_1$. R4′ = derive *one* map (ledger `GATE.UWALL.13`).
- **R5: the Möbius D_4 realisation (v146).** The *full* dihedral symmetry $\langle z\mapsto iz,\, z\mapsto 1/z\rangle$ of the seam curve acts on H^1 exactly as the integer model: ι^* is the exponent duality with parity $+1$ on the self-conjugate line ($= T_A$ in the cusp basis, glue-swap parity), and Σ is the $\delta\iota$ class — the premise reduces to the module identification itself (ledger `FLAV.QGEO.07`).
- **R1: the clock is a Gaussian zero-mode integral, and the bend is det′-clean (v147).** Variance ratio $(1-\alpha)$ (forced by norm = area) gives the Born-squared ratio $(1-\alpha)^{p_2}$, the ln series *is* the v127 ring sum; the two quantum weights share *one* Nariai geometry, so the bend $\log_{3/2} 3$ carries no determinant correction — the v144 obstruction is localised to the $\alpha = 0$ reference (ledger `HOR.CLOCK.13`).
- **R2: the NS/R sector census (v148).** The untwisted (Neveu–Schwarz) module carries only the *even* discriminant classes (integer weights), so the odd μ_4 simple currents have zero NS support — they are Ramond modules; the R zero-mode module (256) splits 128+128 into the odd sectors of the *two* Lagrangian glues: choosing the glue *is* choosing the R-projection ($248 = 120_{\text{NS}}+128_{\text{R}}$); the one remaining net statement is irreducibly twisted-sector (ledger `GATE.METRIC.10`). *Correction note (same day):* the first version of v148 graded the Ramond labels by the sum q_D+q_A (not the glue label) and called that module untwisted; the census was rewritten as above — conclusion unchanged, support corrected, the R-sector sheet split is new.
- **Bookkeeping.** Suite at 148 modules, all green; Wolfram extension 215/215 (README + `GATE.WOLFRAM.02` + website counter in lock-step); status surfaces (intro card, `tfpt_1`, `origin_theory` third update, `next.txt`) moved together.

2026-06-12 (external review integrated: Λ typing, marker migration, website parity)

- **Λ normalisation made unambiguous.** The external review flagged the adjacent reduced/unreduced displays in `origin_theory` as a type trap (the formulas were individually correct — $\bar{M}_{\text{Pl}}$ vs M_{Pl} — but easy to misread). Both normalisations are now shown explicitly side by side with their order counts ($3/(4\pi^2) \Rightarrow 120.147$ reduced; $3/(256\pi^4) \Rightarrow 122.948 = 119.028 + 3.920$ unreduced) plus a typing note.
- **Stale references cleaned.** "How each of the seven papers simplifies" → "How the legacy layers simplify (bridge map)" with the lead-in "(Paper 6)" reference replaced by the scalaron-reheating chain (v86); "(Paper~6/7)" in the `tfpt_3` bridge table reworded; the `tfpt_1` "single remaining geometric input (H2)" passage and the `tfpt_2` "single remaining input (U)" passage

carry dated status updates pointing at the current decomposition (v118–v122, v139, v141, v142, v75).

- **Marker migration completed in the shared figures.** The claim stack, anchor cockpit, E_8 glue, 2/3 motif and K_5 figures (`tex-artefacts/tfpt_figures.tex`) now show the four public classes [E]/[C]/[O] instead of the legacy $[A]/[I]/[L]/[B]/[P]/[R]$ badges.
- **Red-team front table pulled to the final state.** The Target-A residual column now states the one remaining statement (boundary-net holomorphy + c=8 ⇔ index-4 inclusion; v83/v87/v89, Lie-level realisation v143); the historical three-residual form is kept below, dated.
- **Website parity.** The Red Team document is now a first-class entry (`papers.ts` number 5, slug `redteam`) and the changelog PDF a download row; Appendix H / Origin Theory / Research Contracts are labelled *companions* (never “Paper 5–7”); the marker system on every website surface migrated to $[E]/[C]/[O]/[X]$ (fine types named in prose; the script index keeps quoting the scripts' internal fine-grained typing); the verification page counters are live (144 checks generated from the registry via `lib/suite.ts`, Wolfram $116/116 + 211/211$ — now guarded by the sync audit) and the reproduce block matches the current pipeline. The `papers.ts` section bodies of the affected papers carry the v141–v144 outcomes (sheet question: deck derived; G_{net}: Lie-level realisation; resummed clock: in-family det-ratio cancellation; selector triangle: two line pairings).
- **Not adopted** (recorded for transparency): the review's narrative reframings (anchor-algebra trilogy, “one seam clock” framing, one-page structure) and the front-matter/status-card layout changes are editorial decisions left to the authors; the claimed Λ *error* was in fact a correct-but-treacherous notation switch (see above).

2026-06-12 (R1–R5 attack round: four residual classes sharpened, v141–v144)

- **R5 closed at the discrete level (v141, deck selection theorem).** The v140 $\mathbb{Z}_3$ choice is *derived*: the established integer deck $G = T_A\Sigma$ (v97/v98) pairs the Q_+=1 cusp line with the self-conjugate character 2, so only the sheet-twisted assignment $(2,3,1) = \text{chars}(-U)$ survives; the cusp exponential i^{Q_+} is excluded (same spectrum, wrong grading pairing). Explicit conjugacy constructed; under $k \to -k$ only the established sheet $\mathbb{Z}_2$ flips. Consequence: Euler grading $= Q_+ \circ \rho$. `GATE.QGEO` keeps only its realisation premise — no discrete freedom left (ledger `FLAV.QGEO.06`).
- **R4′ narrowed: three pairings → two (v142, frame integrality).** Integer covectors in the $(\mathbf{1}, a, \sigma)$ dual frame form an index-11 sublattice ($x - 3y + z \equiv 0 \bmod 11$; the index *is* $\|\mathrm{Pl}(K)\|_1$); with $(x, y) = (|\mathbb{Z}_2|, \operatorname{rank} E_8)$ the σ-pairing is forced $\equiv 0 \bmod 11$, and primitivity forces the line parameter even. Cramer reductions for both line pairings recorded as restructuring (ledger `GATE.UWALL.12`).
- **R2 realised at the finite Lie level (v143, graded Frobenius).** The $\mathbb{Z}_4$-graded hull carries exact coset duality ($-C_1 = C_3$; Killing pairing nondegenerate per $g_k \times g_{-k}$), the glue average is exactly the carrier projector (index 4), and each glue sector is one Weyl orbit (simple currents, fusion $= \mathbb{C}[\mathbb{Z}_4]$) — the v125 Q-system realised by the hull; residue = one conformal-net statement (ledger `GATE.METRIC.09`).
- **R1's det-ratio step derived in-family (v144).** e_2-rigidity of the traceless SdS cubic gives $r_b r_c = 1 - \Delta^2/3$ exactly; with the v132 anomaly the non-zero-mode determinant ratio is $(1 - \Delta^2/3)^{4/3}$ — no first-order term in the horizon split (the v131 cancellation derived, not assumed); deficit coefficient $\frac{32}{27} = |\mu_4|(\frac{2}{3})^3$ (audit). Finite-weight absorption stays [C] with the $6 \to \frac{14}{3}$ obstruction stated sharply (ledger `HOR.CLOCK.12`).
- **R3 attempted, honestly unchanged.** No defensible computation landed for the seam-determinant→EH step (`SEAM.THEOREM.01`); the gate keeps its [O] typing untouched.

- **Wolfram mirror extended** to v141–v144: extension file now 211/211 (`GATE.WOLFRAM.02` and the wolfram README updated in lock-step); Python suite at 144 modules, all green.

2026-06-12 (sync infrastructure: single-source script index, content maps, one audit)

- **Single-source script index.** The master script→check table (`tex-artefacts/verification.tex`) and the website `ScriptIndex.tsx` are now *generated* from one registry (`script_registry.csv` + `script_clusters.csv` in `verification/`) by `make_script_index.py`; both targets carry a generated-file header and are never edited by hand.
- **Content maps without splitting the papers.** `verification/make_docs_map.py` generates `docs_map.csv` (every paper section with line range, the `vN` scripts cited there, a content hash and a last-changed date) and `website_map.csv` (which website file mentions which scripts/documents) — the machine-readable enumeration of all sync surfaces.
- **One sync audit.** `verification/audit_sync.py` bundles and extends the previous loose shell checks, now in *both* directions: suite ↔ `run_all.py` ↔ registry ↔ ledger; every script cited in a paper *body* (the master index alone no longer counts); no stale `\veri`/`\vref` targets; every new module mentioned in this changelog; website mirror byte-identical (PDFs, scripts, `release.ts` hashes, version stamps); wolfram counts; overfull boxes. Frozen exceptions live in `audit_baseline.json` (remove-only).
- **Build pipeline.** `build.sh` gains `gen` / `website` / `audit` / `release` steps; `notes` now regenerates the single-source surfaces first; `website` copies all active PDFs (now incl. `tfpt_5_redteam.pdf` and `changelog.pdf`, with new `release.ts` entries) and the `vN` scripts, and stamps version, date and git revision into `website/lib/version.ts` (shown in the site header) and `release.ts`.
- **Citation gaps closed.** The audit immediately found three scripts cited only in the master index: v17 and v85 are now cited in their `tfpt_2` sections (hexagonal resolvent; master cover), and two short `\veri{v19}` citations were expanded to the full script name. v91 (spine tetrahedron) had no paper section at all — it is now integrated: a keybox in `tfpt_3` ("Three views of one spine": anchor closure, edge/face products, volume 120, K_6 negative control, the tautology/sub-grammar caveats) and a closing note in the `tfpt_1` Spine Quotient Lemma; the `ARCH.SPINE.01` ledger location is corrected to `tfpt_1;tfpt_3` and the baseline exception list is empty again.
- **Rules.** `.cursor/rules` updated (`tfpt-workflow`, `website-sync`) and a new `sync-maps` rule added describing the agentic procedure: *regenerate* the maps first, enumerate the affected surfaces from them, audit as the exit gate. The mirror map (`website_map.csv`) also covers the root `README.md` and `next.txt` (bare `vN` ids resolved to full script names), and both files are named explicitly as status surfaces that move with every finding.

2026-06-12 (tfpt_3 / tfpt_4 style alignment + box reduction)

- **Shared header/footer on tfpt_3.** `tfpt_3_e8_audit_bootstrap` dropped its custom `\footmarkers` footer (the post-collapse four-[E] artefact) and inherits the shared `tfpt_style` header/footer with the version stamp; the box-orphan `needspace` guards are kept.
- **Fewer boxes, embedded in text.** Narrative/commentary callouts converted to run-in bold paragraphs (boxed one-line formulas to `keyeq`): tfpt_3 from 39 to 27 boxes (e.g. "Purpose and discipline", the five "(i)–(v)" audit-lemma boxes, "Net", "The connection in one sentence", "The unification", "The question and the honest answer", "Net: two axioms"); tfpt_4 from 18 to 7 (the seven-box Koide stack and the two dark-matter conjecture boxes become prose). The per-item "Honest status of . . . " keyboxes are kept — tfpt_4 is the status authority for the frontier.

- **Colour-marked module references.** `\vref` applied to the inline `vN` references in tfpt_3 and tfpt_4 (TikZ node names excluded).
- **Marker hygiene.** A stale literal `[P]` in tfpt_4 (Koide flow-time) replaced by the 4-class [C]; status tables checked against the ledger.

2026-06-12 (tfpt_1 / tfpt_2 style alignment + box reduction)

- **Shared header/footer everywhere.** `tfpt_1_architecture_e8` dropped its custom `\footmarkers` footer (which after the marker collapse showed four redundant [E]) and now inherits the shared `tfpt_style` header/footer with the version stamp, identical to every other document.
- **Fewer boxes, embedded in text.** The narrative/commentary callouts of both documents are converted to run-in bold paragraphs (and the boxed one-line formulas to the colour-marked `keyeq`): tfpt_1 from 21 to 14 loud boxes (+3 `keyeq`); tfpt_2 from 52 to 43 (e.g. "In one sentence", "The one-paragraph version", the four "Sharpening" notes, "What the simplification means net", "The disciplined main statement", "What improves. . . ", "Scope and honesty"). Only genuine results / theorems / gates keep a box.
- **Colour-marked module references.** `\vref` now also colours the inline `vN` references throughout tfpt_1 and tfpt_2 (TikZ node names accidentally matching the pattern were excluded).
- **Status currency.** tfpt_2's inflation-pivot table gains the v86 scalaron-reheating point ($N_\star$=51.4 ⇒ n_s=0.9611, r=0.0045, A_s-disfavoured) and states the band-of-record explicitly, matching `COSMO.NSTAR.01` and the introduction; the box is re-typed [C]. The other status tables were checked against the ledger and are current under the 4-class markers.

2026-06-12 (marker simplification to 4 classes + status reconciliation)

- **Four-class display markers.** The reader-facing marker system is collapsed from eleven symbols to four: [E] exact / machine-proven (identity, Lie/lattice, formalised, numerical FP), [C] conditional (physical / bridge / readout under named hypotheses), [O] open / axiom, and [X] kill test. `tex-artefacts/tfpt_style.tex` defines `\mE/\mC/\mO/\mX` and keeps the legacy `\tI ...\tX` names as aliases that render their class. All marker call-sites across the nine documents and the shared fragments were rewritten, collapsing same-class combinations (e.g. `[I]/[L]` → [E], `[N]/[P]` → [E]/[C]). The fine-grained per-claim type (Axiom / Formal / Lattice / Numerical / Identity / Physical) is unchanged in `status_ledger.csv` — the single source of truth — so no fidelity is lost. The marker key, badge bar, README marker table and the workflow rule were updated to match.
- **Status tables reconciled against the ledger.** The introduction's inflation/$N_\star$ rows are made consistent with v86/`COSMO.NSTAR.01`: r and n_s are shown as the frozen *bands* over $N_\star \in [50, 60]$ with the scalaron-reheating conditional point $N_\star$=51.4 (n_s=0.9611, r=0.0045), replacing the stale point values $N_\star$=55 (predictions table) and $N_\star \simeq 57$ (residual card); the closure-ledger inflation row is re-typed [E] → [C](conditional on M_{Star}=M_{scal} + reheating), matching the other two tables and the ledger `[P]` typing.

2026-06-12 (version system, predictions/K5, readability pass B3)

- **Authors, auto-date and auto-version on every paper.** New single source `tex-artefacts/version.tex` defines `\TFPTauthors` (Stefan Hamann, Alessandro Rizzo), a curated `\TFPTversion`, and a title/footer date stamp (`\today` + version + revision). `build.sh` writes the auto build revision (git commit count) into the gitignored `tex-artefacts/version-auto.tex`, so every document's title page and page footer carry author, date and `v5.4 (rev N)` automatically. All nine active documents and `changelog.tex` now set `\author{\TFPTauthors}`.

- **Builder keeps build artefacts.** `build.sh` no longer deletes the `.aux/.log/.out/.toc` after a successful build (the `.log` is needed for the overfull-box audit and the `.aux/.toc` for correct cross-references on the next incremental build).
- **Predictions table outsourced.** The running/upcoming-experiment predictions table is moved from `introduction.tex` into the shared fragment `tex-artefacts/predictions.tex` (`\input` in the predictions section).
- **K5 figure fixed.** The `\TFPTactionkfive` mnemonic (K_5 on the five carrier slots) is redrawn so the complete graph and the Pascal-row reading list sit side by side instead of overlapping.
- **Colour-marked script references.** New `\vref` macro colours inline module references (e.g. v108–v113, v49/v71) in the introduction, predictions and verification fragments, matching the blue of the `\veri` citation, so machine-checked references are visible at a glance.
- **Lighter, non-breaking, embedded boxes.** The `tcolorbox` families are restyled to a light tint + thin frame + light title band with dark coloured title (no heavy white-on-saturated bars); a neutral `navbox` carries the document map and a colour-marked `keyeq` sets off load-bearing display formulas. The introduction's callouts use the non-breaking variants (`keyboxnb/winboxnb/gapboxnb`) so no box splits across a page, and the shared status card is non-breaking.
- **Readability / sectioning.** The dense run-in prose of the introduction is broken into `\subsection*` headings (In plain words; one anchor and E_8 as a compiler hull; the companion documents; the compact status formula; the hard reduction; the Pascal action grammar; the closure ledger) with added whitespace.
- **Orphan results linked to scripts.** The Glue theorem, the "Reduction in one line" and the fermion-spectrum callouts now carry inline `\veri{}` citations (v1, v6/v14/v23, v18/v20/v46).

2026-06-12 (tex-artefacts refactor + introduction visual pass B2)

- **Shared imports moved to `tex-artefacts/`.** The shared, imported-only TeX fragments (`tfpt_style.tex`, `tfpt_figures.tex`, `tfpt_docset.tex`, `tfpt_status.tex`) now live in a dedicated `tex-artefacts/` folder; all nine active documents `\input` them via `tex-artefacts/<name>` (and `tfpt_style` loads `tex-artefacts/tfpt_figures`). `changelog.tex` stays at the repo root because it is also a standalone `build.sh notes` deliverable. The `make_manifest.py TEX` list and the workflow rule were updated to the new paths.
- **Master script index outsourced (`tex-artefacts/verification.tex`).** The long vN→check table is moved out of `introduction.tex` into the shared fragment `tex-artefacts/verification.tex` (`\input` in the verification appendix); the paper-consistency audit now also reads that file, so every registered script stays cited exactly once.
- **Introduction box reduction (B2).** The introduction's eleven `keybox`es are reduced to the single headline "In one sentence" card; the explanatory keyboxes (*In plain words*, *Sharper still*, *compact status formula*, *hard reduction*, *Cosmology Pascal*, $E_6 \times A_2$, *closure ledger*, *Plausibility balance*) are converted to inline run-in paragraphs, the *Glue theorem* is re-typed as a `winbox` (a result, not a claim), and the explanatory *plausible/follows next* `winbox`es become prose. The loud boxes now carry only genuinely load-bearing callouts (one `keybox`, five `winbox`es, one `gapbox`) plus the two shared orientation cards.

- **Document-count consistency.** The introduction's document map gained the missing `tfpt_5_redteam` row; "seven/eight companion documents" and the ledger "six documents" wording are corrected to the true count (nine active documents); `README.md` ("10 ok", 140 scripts) and the workflow rule ("9 docs") are reconciled.

2026-06-12 (visual pass: box reduction B1)

- **Box reduction (B1).** The 26 non-load-bearing aside boxes (audit / recorded / interpretation: `auditbox/audbox/intbox/resbox/roadbox`) across tfpt_1, tfpt_2, tfpt_3 and `origin_theory` are converted to inline bold run-in paragraphs — the reviewer's prescription that an audit fingerprint must not carry the same visual volume as a theorem. The loud box families are now only `keybox` (claim), `winbox` (result) and `gapbox` (open/caveat).

2026-06-12 (visual pass: shared figures, badges, marker typology)

- **Shared figure library `tfpt_figures.tex`** (`\input` by `tfpt_style`): reusable TikZ macros for the architecture *claim stack* (`\TFPTclaimstack[zone]`, B3), the *anchor cockpit* $a = (1, 1, 2)$ (B4), the E_8 *glue* diagram (B5), the 2/3 *motif map* (B6) and the K_5 *action lattice* (B7). The claim stack and a marker *badge bar* are now shown near the front of *every* document, each highlighting that document's layer; the topical figures are placed in their natural homes (cockpit + K_5 in the introduction, cockpit + glue in tfpt_1, motif in the horizon note).
- **Marker typology (A2).** `tfpt_style` adds the sharper sub-types [E] (exact algebra), [E] (exact number), [C] (readout / standard physics in TFPT units) to the existing [C] (bridge) and [X] (kill test); the marker key and the new `\TFPTbadges` bar show the full typology. (Inline body markers retain [E] where they are genuine exact identities.)

2026-06-12 (review pass on list.txt)

- **Claim hygiene (review section A).** The introduction's one-sentence claim is softened from "derives all constants" to "constructs a discrete compiler; physical readouts run through named, status-typed bridges"; the ledger zombie in `FLAV.QRATIO.01` is removed (the statement no longer says "stays [P]" — ratios are [E] closed on the derived stratum, only the absolute scale stays [O], with an explicit forbidden-wording note); the seam misalignment is written explicitly as $\varepsilon = \frac{3}{4}\varphi_0^{\text{ret}} = c_3 + 36c_3^4$ (leading c_3, fourth-order puncture correction exact); θ_{12} has a single prediction of record 0.306747 with the seam/non-linear values typed as scheme diagnostics; JUNO's first 59.1-day result (0.3092 ± 0.0087, compatible, precision pending) is recorded and NuFIT 6.0 is marked the pre-JUNO baseline; the ACT DR6 birefringence hint carries a systematics caveat (not a validation target); CODATA-2022 is dated; the null-model figure is moved out of the main text and typed as a grammar-internal bound. "Complete solutions of the five open questions" becomes "Closure status of the five former open questions".
- **Interface language unified (review sections A5/C2/C3/C6/C7/C8).** The live residual is stated everywhere as $v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$ (historical $U_{\text{wall}}/G_{\text{metric}}/F_{\text{frontier}}$ kept only for ledger continuity); `tfpt_research_contracts` is retitled and its recommended order rewritten (selector-triangle pairings $\to v_{\text{geo}} \to G_{\text{net}}$); F_{transfer} is named as one missing functor with Koide/η_B/axion/m_p/m_e as its four instances; full QG closure is framed as a certification layer, not a prerequisite. Document numbering aligned with file names (the Red Team paper is now in the canonical set; document $N \equiv$ `tfpt_N`). Website mirrored throughout (`papers.ts`, `predictions.ts`, `faq.ts`, `glossary.ts`, `OpenGates`, `Hero`, orientation components).

2026-06-12 (later)

- **Shared style and central status artifact.** Two new single-source fragments: `tfpt_style.tex` (the canonical palette, status markers, marker key, `\veri` reference, the four tcolorbox families with back-compat aliases, and one running header/footer) is now `\input` by all eight documents — the per-paper duplicated preambles are gone; and `tfpt_status.tex` ("TFPT status at a glance") gives one canonical overall-status card imported near the front of every document, replacing the divergent per-paper status statements. Both registered in `build.sh` and the manifest `TEX` list.

2026-06-12

- **Margin theorem (v122).** The established frozen selectors (D_4 annihilator $n = (5, -9, 6)$, $\det R = 8$, $\det K = 4$) pin R uniquely among hexagon matrices ($64 \to 12 \to 1$); the atom margins become theorems — H2 introduces no new residual class.
- **Inventory update (v123).** Residual table re-pinned post v110–v122: exactly five structural classes (R1 clock, R2 index-4 net, R3 EH step, R4′ selector establishment — replacing the discharged H2 dictionary — and R5 Q realisation); R2 carries three loads (metric + carrier + QBL: one theorem, three doors).
- **Resummed clock + glue Q-system (v124/v125).** R1's bend in closed form: $\text{rate}(n) = -p_2 \ln(1 - n/N_{\text{fam}})$ (pole at N_{fam} forces the three-level spectrum); the index-4 μ_4 extension exists explicitly as a Longo Q-system on $\mathbb{C}[\mathbb{Z}_4]$ (all Frobenius axioms exact) — R2 sharpens from "find" to "identify".
- **Clock–wall bridge (v126) and ring resummation (v127).** The resummed-clock weights are the Mehta–Seshadri parabolic weights ($\operatorname{spec} A_0^\star$); $\det(\mathbf{1} - A_0^\star) = \frac{2}{9}$ = the entropy curvature; the clock is a log-determinant (RPA rings, one tower per hexagon site, coupling = parabolic weight).
- **Graded hull (v128).** E_8 is a $\mathbb{Z}_4$-graded Lie algebra over the carrier (grading = glue Q-system; exact on all 6720 root-pair sums); ring multiplicity 6 = sheet-doubled Ginsparg–Perry zero modes. *Notation fix in the same round:* the clock factor 6 is $p_2(a)$, not $2p_2$ — labels corrected across modules, ledger, papers, website (numbers unchanged).
- **Entropy power law → zeta budget (v129–v133).** The clock is $\Gamma_n \propto (S_n/S_{dS})^{p_2}$ (Gibbons–Hawking form); $p_2 = 2h$ with $h = N_{\text{fam}}$ from mode counting + the Born rule (v130); the zero-mode norm is the area, $\|Y_{1m}\|^2 = A/(4\pi)$ exactly (v131); the det-ratio anomaly is the Koide constant, $\zeta(0)|_{\det'} = -\frac{2}{3}$ (v132); both $\zeta(0)$ budget routes computed exactly — the reduced seam route carries pure anchor ratios ($-\frac{4}{3}$ = minus the seed gain), the naive 4d route ($-\frac{109}{45}$) carries no atom (v133).
- **Dual anchor + determinant surface (v134/v135).** $d := a^\top R^{-1} = a^\top L^{-1} = (-\frac{1}{2}, -\frac{1}{2}, 1)$ with $(1, 1, -2) = -2d$ — the inverse flavor response is the Nariai root, invariance exact via Sherman–Morrison (third algebraic flavor–horizon bridge leg); $\det M(s,t) = 3s(t+1)(t+2) + t^2 + 5t + 8$ with iso-volume walls at the $\mathbb{Z}_2/\mu_4$ atoms, winding line $(8, 14, 20)$, $\det F = 32$, row-budget axes with $\text{rowsum}(V) = \text{Spec}(Q_+)$; micro-identities $41 = 16 + 25$, $52 = 16 + 36$.
- **Dual-normal selector, Q_+ cohomology, Volkov–Wipf firewall (v136–v138).** (d, n) pins R *columnwise* (each column the unique lattice point of the address box; kernel $(6, 8, 7)$); the $A_0^\star$ zero mode carries the spectral normal $\sigma = (2, -9, 5)$ with a $W(A_3)$ orbit control excluding the naive lift; $H^1(\mathbb{P}^1 \setminus \mu_4) = \chi_1 \oplus \chi_2 \oplus \chi_3$ with degrees $(1, 2, 3) = \text{Spec}(Q_+)$ = the A_3 exponents and cusp weights = $\operatorname{spec} A_0^\star$; the external full-4d graviton/ghost value $-\frac{98}{45}$ is pinned (arXiv:2506.02142 = Volkov–Wipf hep-th/0003081) and its *edge* part is exactly two copies of the reduced seam budget — R1 typed closed as an edge-sector object.

- **Selector triangle + canonical map (v139/v140).** $d = \frac{3}{2}a - 2 \cdot \mathbf{1}$ is pure anchor data (first selector *derived*); the frame $(\mathbf{1}, a, \sigma)$ has volume 11 and n is the unique covector with atom pairings $(2, 8, 121)$ — the R4′ residue is three pairings; the $H^1 \to$ generation equivariant map exists and is Schur-rigid (only sign-twisted μ_4 actions match), so the `GATE.QGEO` residue collapses to one $\mathbb{Z}_3$ choice.
- **Editorial and consistency passes.** Content-currency sweep (superseded "single remaining input" claims retired across contracts, paper 3, intro, website); introduction: self-explanatory title (spells out Topological Fixed-Point Theory), the old-version comparison removed throughout, redundant diagrams dropped, the unreadable status heatmap removed, the script-index master table converted to a page-breaking `xltabular` (it had silently truncated ∼110 rows), status card extended with the v134–v140 reductions, predictions table completed (CP phase, cosmic birefringence); the canonical document-set box extracted into the shared `tfpt_docset.tex`; paper 1 cleaned of all 35 old-series references, every section now opens with its machine-check script references; smaller box fonts across all eight documents; this `changelog.tex` introduced as a dual-mode (standalone + imported) document.

2026-06-11

- **External-review integration (v83) and release hygiene.** Manifest `-check` mode and the release rule; holomorphy pins $(E_8)_1$ (the unique even unimodular rank-8 lattice theory).
- **Sheet diamond and centered form (v94/v95).** All four flavor operators are points of one two-parameter surface $M(s,t)$; centered cross $R, L = C \mp U$, $K, F = C \mp V$ with the cofactor seam normal $n = (5, -9, 6)$; documentation pass ("The Sheet Diamond and the Winding Line" in paper 3).
- **Horizon/anchor round (v101–v103).** The maximal (Nariai) black hole is the anchor: roots $(1, 1, -2)$, entropy bound $\frac{2}{3} = |\mathbb{Z}_2|/N_{\text{fam}}$; one repeller/attractor orientation in both sectors; trisection normal form (the canonical cosine coordinate).
- **Clock/ladder round (v104–v108) + Lean hardening.** The classical Nariai clock speaks anchor ($\chi_{\text{clock}} = (\lambda - 1)(\lambda + 2)$); machine-pinned residual inventory; first review validation; quantum-clock target made quantitative; Pascal Ladder Theorem ($g = 2K + 1 \Leftrightarrow$ the closure).
- **QBL theorem chain (v109–v113).** Sheet-Pairing Lemma; Calderón-sheet selection (a scalar seam datum exists iff the involution is sheet-odd); Quadratic-Transport Theorem (degree exactly 2); Self-Counting Channel (the Pascal closure as an identity); quasi-free kernel (one 2-point kernel determines the whole net, rank $= c$) — the QBL input merges with the R2/holomorphy premise; communicated across all status surfaces.
- U_{wall} **made exact (v114–v118).** Torsion normal form and $\delta = \frac{1}{2}$ theorem; exact anchor residue $A_0^\star$ (diag $= a/|\mu_4|$, off-weights $(8, 0, 5)/144$); resonance theorem ($M_\infty = \mathbf{1} \iff a_{13} = 0$, one gauge orbit); the holonomy is the exact 24-element $W(A_3) = S_4$; the hypercharge hexagon is the sign-twisted family spectrum with the lepton coefficients as resolvent determinants.
- **Second review validation + address pinning (v119–v121).** Anchor ratio triad $(\frac{2}{3}, \frac{4}{3}, \frac{5}{3})$; the 121 audit lemma; the lepton words are the compiler atoms $(8, 5, 3)$ with divmod-hexagon addresses; R is the unique hexagon matrix with atom margins and $\det 8$ — the address table carries zero residual information.

2026-06-10

- **B/C/D round closed.** Koide attractor toy model (v93), glue uniqueness (v92), the Wolfram extension as the independent second verification path, Lean hardening; papers and website synced with the round.

- **Content rounds v96–v100.** Sheet conjugation bridge, discriminant dictionary, Koide flow time, and the look-elsewhere-corrected numerology null test (v100: joint null probability $\leq 10^{-30.7}$, conditional on the declared grammar).

2026-06-09

- **Blind registry frozen (v84).** Machine-enforced prediction registry (`predictions_frozen.json`): exactly one θ_{12} number of record before JUNO; conditional entries carry their hypotheses by name. Research-contracts document added to the set.

2026-06-07 / 2026-06-08

- **Compiler-closure consolidation.** The TFPT development is consolidated into the current document set with the verification suite (modules v1–v82: E_8 glue, carrier Pascal, EM fixed point, flavor matrix, cascade, bootstrap, gravity/cosmology, transport, masses, registry and gate audits) as the single proof layer; manifests (`manifest.sha256`, `lean_manifest.sha256`) as content identity; website mirrors the suite (interactive DAG, in-browser script reproducer, script index).

2026-04-27

- **Initial commit.** TFPT papers and website baseline: electromagnetic closure and flavor transport (UFE bridge, birefringence seed, dyonic intercept), PDF-download tracking, accessibility and metadata passes.